ILLINOIS ENVIRONMENTAL PROTECTION AGENCY

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JB PRITZKER, GOVERNOR JOHN J. KIM, DIRECTOR

217-782-9333

February 1, 2024

Mr. Joseph Weber
AFCEC/CZOM
701 Hanger Road, Bldg 531
Scott AFB, IL 62225-5035

RE: 1630105014 – St. Clair County
Scott Air Force Base
Superfund/Technical Reports

Draft Final 2023 Annual Monitored Natural Attenuation Report Site OT007 – Former Sludge Weathering Lagoon / Bulk Fuel Facility, Scott Air Force Base (AFB), IL

Dear Mr. Weber:

The Illinois Environmental Protection Agency (Illinois EPA or Agency) has reviewed the *Draft Final 2023 Annual Monitored Natural Attenuation Report Site OT007 – Former Sludge Weathering Lagoon / Bulk Fuel Facility, Scott Air Force Base (AFB), IL* (Report) for Site OT007 at Scott Air Force Base. The Report is dated December 2023 and was received by the Illinois EPA on December 21, 2023.

The Illinois EPA concurs with the finding in the Report and requests that the final version of the Report be submitted.

If you have any questions, then please contact me at Jordan.Simpkins@illinois.gov or at (217) 782-9333.

Sincerely,

Jordan Simpkins

Jordan Simpkins
Project Manager
Federal Site Remediation Section
Bureau of Land

CAH:NMW:JPS:P:\Projects\Scott's AFB\Operating units\OT007\OT007 2023 GW Monitoring Report\Scott AFB Site OT007 Draft Final 2023 Groundwater Monitoring Report.docx

DEPARTMENT OF THE AIR FORCE
AIR FORCE CIVIL ENGINEER CENTER
JOINT BASE SAN ANTONIO LACKLAND TEXAS

28 FEB 2024

MEMORANDUM FOR Illinois Environmental Protection Agency
ATTN: Mr. Jordan Simpkins
Bureau of Land
Federal Site Remediation Section
1021 North Grand Ave. East
Springfield, IL 62794-9276

FROM: AFCEC/CZOM
701 Hangar Road, Bldg 531
Scott AFB, IL 62225-5035

SUBJECT: *Final 2023 Monitored Natural Attenuation Report for Site OT007 – Former Sludge Weathering Lagoon/Former Bulk Fuel Facility, Scott Air Force Base (AFB), IL,* dated February 2024

1. The Air Force submits the *Final 2023 Monitored Natural Attenuation Report for Site OT007 – Former Sludge Weathering Lagoon/Former Bulk Fuel Facility, Scott Air Force Base (AFB), IL,* dated February 2024, for your records. The Illinois EPA approved the draft final version of this document via letter correspondence dated February 1, 2024.

2. If you have any questions, please contact me at 618-256-2125 or email joseph.weber.16@us.af.mil.

Joseph Weber

Joseph Weber
Remedial Project Manager

cc: Eliud Burgos, AFCEC/CZRE
Audrey Meads, Scott AFB RPM
Patrick Reilley, Plexus

FINAL

2023 Annual Monitored Natural Attenuation Report Site OT007 – Former Sludge Weathering Lagoon / Bulk Fuel Facility Scott Air Force Base, Illinois

Prepared for:

U.S. Air Force Civil Engineer Center

AIR FORCE CIVIL ENGINEER CENTER
2261 Hughes Avenue, Suite 155
Lackland AFB, Texas 78236-9853

Contract FA8903-14-C-0009

February 2024

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FINAL

2023 Annual Monitored Natural Attenuation Report
Site OT007 – Former Sludge Weathering Lagoon / Bulk Fuel Facility
Scott Air Force Base, Illinois

Prepared for:

U.S. Air Force Civil Engineer Center

AIR FORCE CIVIL ENGINEER CENTER
2261 Hughes Avenue, Suite 155
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Contract FA8903-14-C-0009

February 2024

Prepared by:

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ACRONYMS AND ABBREVIATIONS

#	Number
%	Percent
µg/kg	micrograms per kilogram
µg/L	micrograms per liter
AFB	Air Force Base
AFCEC	Air Force Civil Engineer Center
AOC	Area of Concern
AST	Aboveground Storage Tank
bgs	Below Ground Surface
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act
cis-1,2-DCE	cis-1,2-dichloroethene
CoC	Chain of Custody
COC	Contaminant of Concern
COPC	Contaminant of Potential Concern
CSIR	Comprehensive Site Investigation Report
CVOC	Chlorinated Volatile Organic Compound
DL	Detection Limit
DO	Dissolved Oxygen
DoD	Department of Defense
ERM	Environmental Resources Management, Inc.
F	result greater than or equal to the method detection limit but less than the reporting limit
ft/day	feet per day
ft/ft	feet per foot
ft/min	feet per minute
GRO	Groundwater Remediation Objective
HGL	HydroGeoLogic, Incorporated
IAC	Illinois Administrative Code
IDW	Investigation-Derived Waste
IEPA	Illinois Environmental Protection Agency
ILWATER	ISGS Water Well Database
ISGS	Illinois State Geological Survey
J	The positive result reported for this analyte is a quantitative estimate due to discrepancies in meeting certain analyte-specific quality control criteria (data designation) or detection between the method detection limit and reporting limit
LCS/LCSD	Laboratory Control Sample/Laboratory Control Sample Duplicate
LOD	Limit of Detection
LOQ	Limit of Quantitation
LUC	Land Use Control

ACRONYMS AND ABBREVIATIONS (CONTINUED)

M	For 2007, 2010, and 2011 datasets: Matrix effect was present, the concentration is estimated. All other datasets: Manually integrated compound (data designation)
mg/L	milligrams per liter
MNA	Monitored Natural Attenuation
MS/MSD	Matrix Spike/Matrix Spike Duplicate
mV	millivolt
MW	Monitoring Well
ORP	Oxidation Reduction Potential
OWS	Oil-Water Separator
PAGC	Pace Analytical Gulf Coast Laboratories
PAL	Project Action Limit
PID	photo-ionization detector
Plexus	Plexus Scientific Corporation
PPE	Personal protective equipment
QA	Quality Assurance
QC	Quality Control
QSM	Quality Systems Manual
R	Rejected Data (data designation)
RAP	Remedial Action Plan
ROR	Remediation Objectives Report
RPD	Relative Percent Difference
RSK	Robert S. Kerr Laboratory
RSL	regional screening level
SM	Standard Method
SOP	Standard Operating Procedure
SRO	Soil Remediation Objective
SRP	Site Remediation Program
SW	Solid Waste
TACO	Tiered Approach to Corrective Action Objectives
TAT	Turnaround Time
TCE	Trichloroethylene
TOC	Total Organic Carbon
U.S.	United States
UFP-QAPP	Uniform Federal Policy-Quality Assurance Project Plan
USEPA	United States Environmental Protection Agency
VC	Vinyl Chloride
VOC	Volatile Organic Compound

1.0 INTRODUCTION

This Annual Monitored Natural Attenuation (MNA) Report provides a summary of the 2023 groundwater monitoring activities. The work was performed under Contract Number FA8903-14-C-0009. The 2023 field event was conducted in accordance with the *Final Amended Remedial Action Plan for Site OT007* (*Final Amended RAP;* Plexus Scientific Corporation [Plexus], 2016a), the *Final Remedial Action Plan Addendum, Uniform Federal Policy-Quality Assurance Project Plan (UFP-QAPP), Site OT007 Former Sludge Weathering Lagoon/Bulk Fuel Facility*, (Plexus, 2019b), and the *Comprehensive Site Investigation Report (CSIR) / Remediation Objectives Report (ROR) / Remedial Action Plan (RAP) for Site OT007* (*CSIR/ROR/RAP;* HydroGeoLogic, Incorporated [HGL], 2013).

Site OT007 encompasses the two former 30,000-gallon jet fuel aboveground storage tanks (ASTs) (Tanks 8552 and 8554) previously known as Area of Concern (AOC) 22, and the former sludge weathering lagoon. The ASTs were used to store jet fuels from 1970 until 1994. In April 1996, the two 30,000-gallon ASTs were removed from the site in preparation for the construction of Taxiway G, which opened in 1998. Subsequent to the AST removal action, in July and August 1996, a remedial action was performed to remove contaminated soil from the former AST area (Burns & McDonnell, 1997). The excavation was as deep as 16 feet in some areas. A total of 6,063 tons of soil was removed from the site. Groundwater encountered during excavation activities limited the excavation depth. The sludge weathering lagoon, which measured 20 feet by 40 feet, was used for approximately two years in the mid-1970s to dry out the sludge removed from the bottoms of the adjacent ASTs. The lagoon was closed in 1981. Closure activities included removing soil to two feet below ground surface (bgs) and backfilling the excavation to grade with sand and gravel (ERM, 1992).

The United States (U.S.) Air Force Civil Engineer Center (AFCEC) contracted Plexus in 2014 to conduct additional investigation, potentially remove impacted soil, and continue annual groundwater monitoring and surface water sampling of the former OT007 Sludge Weathering Lagoon/Army Reserve Bulk Fuel Facility (Site OT007 or “the Site”), located at Scott Air Force Base (AFB), Illinois. In 2011, Scott AFB requested that Illinois Environmental Protection Agency (IEPA) allow the environmental investigation and any needed response action for Site OT007 to stop following the Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (CERCLA) statute and implementing regulations because of CERCLA's petroleum exclusion. The IEPA agreed on September 28, 2011 to let Site OT007 follow the Site Remediation Program (SRP) evaluation, cleanup, and reporting process (35 Illinois Administrative Code [IAC] Part 740.435).

Pursuant to SRP requirements, usable results from the investigations at Site OT007 were screened against remediation objectives found in Illinois' Tiered Approach to Corrective Action Objectives (TACO; 35 IAC Part 742). As presented in the *Final CSIR/ROR/RAP* (HGL, 2013), the selected remedial action for Site OT007 is MNA with land use controls (LUCs). This action was selected to address the groundwater contaminants of concern (COCs) of arsenic, iron, manganese, cis-1,2-dichloroethene (cis-1,2-DCE), and vinyl chloride (VC). The groundwater remediation objectives (GROs) selected for those COCs are presented in **Table 1-1**. For the metal COCs, the highest background concentrations were selected as their GROs. Background concentrations in groundwater for Scott AFB were established in the *Final Basewide Groundwater Dataset Evaluation Technical Memorandum* (HGL, 2012). The GROs for cis-1,2-DCE and VC are based on the Class I groundwater values in TACO (35 IAC 742) Tier 1, Appendix B, Table E. The *Final CSIR/ROR/RAP* (HGL, 2013) identified no surface water or soil COCs or surface water or soil remediation objectives (SROs).

Based on groundwater sampling events conducted at the Site since 2015, the chlorinated volatile organic compound (CVOC) biodegradation process appears to be effectively reducing the CVOC concentrations. As described in the *Final Amended RAP* (Plexus, 2016b), the current presence and extent of CVOCs and other volatile organic compounds (VOCs) within the soil at Site OT007 in the vicinity of monitoring wells (MWs) OT07-MW04 and OT07-MW06 and upgradient of both MWs, was evaluated in 2017. The soil investigation phase of the field work was postponed until June 2017 due to the runway closure and repaving effort, which had resulted in more traffic on Taxiway G.

Results of the investigation, along with recommendations, were reported in the *Final, Revision 1, Soil Characterization Report* (Plexus, 2018c). Several analytes were detected above project action limits (PALs) in saturated soil, including 1,2,4-trimethylbenzene, 1,3,5-trimethylbenzene, benzene, ethylbenzene, m,p-xylene, o-xylene, naphthalene, and VC (Plexus, 2018c). The PALs for the 2017 soil characterization effort were presented in Worksheet #15 of the *Final Amended RAP* (Plexus, 2016b). The PALs for the 2017 work are based on the lowest available TACO (35 IAC 742) Tier 1 SROs for residential properties (Appendix B, Table A), industrial/commercial properties (Appendix B, Table B), or the Soil Component of the Groundwater Ingestion Route for Class I groundwater values (IEPA, 2013b). If a TACO (35 IAC 742) Tier 1 SRO had not been promulgated for a given compound (e.g., 1,3,5-trimethylbenzene), then the PAL was set at the U.S. Environmental Protection Agency (USEPA) regional screening level (RSL) for residential exposures (USEPA, 2021) or at a Non-TACO calculated value protective of construction workers (Plexus, 2016b).

The 2017 soil investigation determined that residual contamination was present in the saturated capillary fringe above the groundwater table rather than in unsaturated soil. These results are consistent with historical groundwater data, which indicate higher groundwater contaminant concentrations when the water table is elevated. Additionally, benzene was added to the list of groundwater COCs for the Site after being detected above the GRO for the second time during 2017 (*Final 2017 Annual MNA Report*, Plexus, 2019a). Based on the results of the soil investigation, chemical oxidant injections within the capillary fringe and shallow groundwater were recommended to abiotically degrade CVOCs and petroleum hydrocarbons at the Site, followed by continuation of the MNA remedy (Plexus, 2018c). As a result, chemical oxidant injections into the capillary fringe were performed in May 2020, followed by the collection of post-injection soil characterization samples on an annual basis to assess the effectiveness of the injections on subsurface soils around the capillary fringe at that time. The PALs for the post-injection soil characterization work were presented in Worksheet #15 of the *Final Remedial Action Plan Addendum, Uniform Federal Policy Quality Assurance Project Plan* (*Final RAP Addendum*; Plexus, 2019b). They were selected following the same approach that was used for PAL selection in the *Final Amended RAP* (Plexus, 2016b), as described above. The PALs for the post-injection soil characterization work are presented in **Table 5-4** (see **Section 5.4**). Additional groundwater monitoring and soil characterization events were performed annually between 2021 and 2023.

In June 2023, an annual groundwater and surface water monitoring and soil characterization event was performed. The groundwater PALs for contaminants of potential concern (COPCs) are presented in **Table 5-3** (see **Section 5.2**). The groundwater PALs for COPCs are based on the Class I groundwater values in TACO (35 IAC 742) Tier 1, Appendix B, Table E. If such values do not exist for a given COPC, then the PALs are based on USEPA tap water RSLs (e.g., methyl tert-butyl ether) or USEPA Maximum Contaminant Levels (e.g., nitrite). The surface water PALs for surface water COPCs are based on TACO (35 IAC 302.210), Derived Water Quality Criteria List. No surface water samples were collected in June 2023.

1.1 Report Organization

This Annual MNA Report is organized as follows:

- **Section 1.0 Introduction –** provides a general background of the Site and describes the organization and intent of this monitoring report.
- **Section 2.0 Background Information –** summarizes previous investigations, physical setting, and current land use.
- **Section 3.0 Summary of Field Activities –** summarizes the field activities conducted in 2022.
- **Section 4.0 Data Quality Evaluation** - describes the data quality evaluation process used to review and evaluate laboratory analytical data collected.
- **Section 5.0 Summary of Analytical Results** - provides the results of the sampling and evaluation of the data.
- **Section 6.0 Conclusions and Recommendations –** presents the conclusions and recommendations for Site OT007.
- **Section 7.0 References –** lists the reference documents used to prepare this report.

1.2 Purpose of Report

The purpose of this report is to discuss the sampling activities and analytical results for the groundwater monitoring and the post-injection soil characterization sampling event that was conducted on June 1-2, 2023. Analytical results from 2023 were also compared to historical data to assess trends and develop a path forward for future sampling events.

1.3 Project Location

Scott AFB is located in St. Clair County, Illinois, approximately 25 miles southeast of St. Louis, Missouri, in southwestern Illinois (**Figure 1-1**). The city of Belleville, Illinois, is located approximately six miles southwest of Scott AFB, and the city of O'Fallon, Illinois, is located approximately 5 miles northwest of the base. Scott AFB property encompasses approximately 3,545 acres of U.S. Government-owned and easement land. Agricultural land, retail development, a commercial airport, and residential properties border the base.

Scott AFB was established in 1917 as an aircraft pilot training facility known as Scott Field. Between 1920 and 1937, dirigible airships were assigned to Scott Field. A base wide construction program began in 1938, and between 1940 and 1942, four concrete runways were constructed. During and following World War II, Scott AFB has been the headquarters for various military commands (Environmental Resources Management, Inc. [ERM], 1992). Today, headquarters for U.S. Transportation Command and Air Mobility Command, U.S. Army Military Surface Deployment and Distribution Command, a reserve aero-medical airlift wing, and an Illinois Air National Guard aviation support facility are stationed at the base. The primary mission of Scott AFB is global mobility support (Scott AFB Website, 2015).

Site OT007 is the former Sludge Weathering Lagoon/Bulk Fuel Facility. It is located in the east-central portion of Scott AFB within the controlled movement area of the flight line south of Taxiway G, as shown on **Figure 1-2**. The Site OT007 study area is largely encompassed by grass and Taxiway G.

2.0 BACKGROUND INFORMATION

The investigation area for Site OT007 encompasses the two former 30,000-gallon jet fuel aboveground storage tanks (ASTs; Tanks 8552 and 8554) previously known as Area of Concern (AOC)-22, and the former sludge weathering lagoon. The Oil-Water Separator (OWS)-14 site is located adjacent to the study area. OWS-14 is being investigated separately under Site SS028 (formerly AOC-6). The Site boundaries and historical features are depicted on **Figure 2-1**.

The ASTs were used to store jet fuels from 1970 until 1994. In April 1996, the two 30,000-gallon ASTs were removed from the Site in preparation for the construction of Taxiway G, which opened in 1998. Subsequent to the AST removal action, in July and August 1996, a remedial action was performed to remove contaminated soil from the former AST area (Burns & McDonnell, 1997). The approximate extent of the soil excavation is shown on **Figure 2-1.** The excavation was as deep as 16 feet in some areas. A total of 6,063 tons of soil was removed from the Site. Groundwater encountered during excavation activities limited the excavation depth.

A sludge weathering lagoon, which measured 20 feet by 40 feet, was located at Site OT007 (ERM, 1992). The lagoon was used for approximately two years in the mid-1970s to dry out the sludge removed from the bottoms of the adjacent ASTs. The lagoon was closed in 1981. Closure activities included removing soil to 2 feet below ground surface (bgs) and backfilling the excavation to grade with sand and gravel (ERM, 1992).

Based on the Tier 1 risk analysis completed as part of the *Final CSIR/ROR/RAP* (HGL, 2013), cis-1-2-DCE, VC, arsenic, iron, and manganese in the Aquifer Zone 1 groundwater (present from approximately 5 feet bgs to approximately 20 feet bgs) were identified as COCs at Site OT007.

No COCs were identified for soil, surface water, or Aquifer Zone 2 groundwater (present from approximately 20 feet bgs to approximately 35 feet bgs) at that time; however, CVOCs were identified as COPCs in the soil within the vicinity of OT07-MW06 (HGL, 2016). A 2017 Phase 1 delineation/characterization soil investigation conducted by Plexus confirmed that CVOCs were present in soil at the Site (Plexus, 2018c). Based on the results of the 2017 soil investigation, oxidant injections within the capillary fringe and shallow groundwater were recommended to abiotically degrade the CVOCs and petroleum hydrocarbons at the Site, followed by continuation of the MNA remedy (Plexus, 2019b). The recommended oxidant injections were conducted in May 2020.

2.1 Physical Setting

2.1.1 Site-Specific Geology and Hydrogeology

The boring logs for Site OT007 indicate that a brownish-gray silty clay/clayey silt is present from the ground surface to at least 13 feet bgs. A thin greenish-gray clay layer is also consistently present over most of the investigation area at depths ranging from 4.6 to 7.7 feet bgs, ranging in thickness from 0.3 foot to 0.5 feet thick. A dark gray to black, thin silt layer with nodules is encountered throughout the study area at approximately 12 feet bgs. At approximately 20 to 28 feet bgs, a sand unit with gravel is present. All of the permanent MW borings were terminated between 14 and 34 feet bgs (HGL, 2013).

No significant regional aquifers exist in the area of western Illinois where Scott AFB is located; however, groundwater located within unconsolidated units and bedrock is occasionally used locally as a source of potable water. No drinking water wells exist at the base (HGL, 2013); Scott AFB and nearby communities purchase water from a municipal water distribution system that obtains water from the Mississippi River.

During the 2010/2011 site investigation field activities, two hydrogeologic units, Aquifer Zone 1 and Aquifer Zone 2, were identified and investigated at Site OT007. Aquifer Zone 1 consists predominantly of clayey silts and extends from approximately 5 feet bgs to approximately 20 feet bgs. Aquifer Zone 2 underlies Aquifer Zone 1 and consists predominantly of medium fine sand to gravel of the Pearl Formation. The specific depth of Aquifer Zone 2 at Site OT007 is not well defined but based on regional geology observed at other Scott AFB sites, it may extend from approximately 20 feet bgs to 35 feet bgs (HGL, 2013; Plexus, 2016a). There is no aquitard present between Aquifer Zone 1 and Aquifer Zone 2. However, a distinction is made between the two aquifer zones because of the soil type transition and the higher hydraulic conductivity of Aquifer Zone 2.

Eleven permanent MWs, both within the Site limits and outside the Site boundary, have been installed at Site OT007, seven of which currently exist (**Figure 2-2**). One well (OT07-MW04D, screened from 22 to 32 feet bgs) was installed, and currently exists, to assess the groundwater conditions in Aquifer Zone 2. The 2010/2011 data indicated that groundwater in Aquifer Zone 1 flows toward the drainage ditch from both the north and the south with the ditch acting as a boundary for groundwater discharge. The drainage ditch was backfilled and converted to a stormwater line in 2014. During the 2010/2011 site investigation the groundwater gradient measured from the north side of the ditch is estimated to be 0.04 feet per foot (ft/ft). Based on the 1988 water levels measured from MW7-1 (abandoned), MW7-2 (abandoned), and MW7-3 (abandoned), flow in Aquifer Zone 2 occurred roughly parallel to the drainage ditch and ran from the northwest to southeast. The calculated gradient based on the 1988 data in Zone 2 is 0.006 ft/ft. Synoptic water level measurements for the MWs, as presented in the *Final CSIR/ROR/RAP* (HGL, 2013), are provided in **Table 2-1**.

Based on historical data, the groundwater elevation difference between OT07-MW04 and OT07-MW04D suggests a slightly downward vertical gradient between Aquifer Zones 1 and 2. In February 2011, the groundwater elevation at OT07-MW04 was 0.25 feet higher than the groundwater elevation at OT07-MW04D (HGL, 2013). Gauging data from June 2023 confirmed this downward gradient with a difference of 0.85 feet between OT07-MW04 and OT07-MW04D.

As part of the *Final CSIR/ROR/RAP* (HGL, 2013), seepage velocities in Zone 1 were estimated to range from 9.6 x 10^{-6} feet per minute (ft/min) to 2 x 10^{-4} ft/min, or 0.01 feet per day (ft/day) to 0.3 ft/day (HGL, 2013).

In 2014 the drainage ditch south of OT007 was backfilled. Surface water flow was altered as a result and is now towards the east southeast, ultimately making its way to Silver Creek via a drainage ditch south of Building 5032.

2.1.2 Land Use

The area encompassing Site OT007 is listed as aircraft operations and maintenance land use; however, the majority of the study area is currently covered by grass and Taxiway G. Future land use is expected to continue as aircraft operations and maintenance.

2.1.3 Groundwater Use

The base has been supplied by municipal water since 1917 for its drinking water (ERM, 1992). Scott AFB and the surrounding communities purchase municipal water supplies from the Illinois American Water Company, which obtains its water from the Mississippi River. No drinking water wells are known to be in use at Scott AFB.

Plexus completed a search of the Illinois State Geological Survey (ISGS) water well database (ILWATER, 2016). Six wells (ISGS Wells Number [#] 19, 20, 21, 27, 31, and 37) were noted to be within 2,500 feet of Site OT007. All six wells are listed as "Scott Field Test" water wells and were drilled in 1942. According to the *Final CSIR/ROR/RAP* (HGL, 2013), all of the "Scott Field Test" water wells have been abandoned and are no longer active. There are no known water wells on base used for public consumption (HGL, 2013). Well details for these water wells were provided in the *Final Amended RAP*, Appendix B (Plexus, 2016b).

2.2 Previous Investigations

Multiple site investigations have occurred at Site OT007 since 1988. Detailed results and discussion can be found in the *Final CSIR/ROR/RAP* (HGL, 2013) and the *Final Amended RAP* (Plexus, 2016b). It should be noted that both USEPA Solid Waste (SW) Test Methods 6010C and 6020A were used for metals analysis prior to 2018; Method 6020B has been used since 2018. Additionally, Method 9056A replaced Standard Methods (SM) 310 and 2320B for analysis of alkalinity. Method 9056A also replaced SM 310 and SM 300 for analysis of anions. The following is a summary of these findings.

2.2.1 2013 Comprehensive Site Investigation Report

Based on the Tier 1 risk analysis completed as part of the *Final CSIR/ROR/RAP* (HGL, 2013), cis-1,2-DCE, VC, arsenic, iron, and manganese in the Aquifer Zone 1 (present 5 to 20 feet bgs) groundwater at Site OT007 were identified as COCs requiring remediation to protect human health. The groundwater data demonstrated that the CVOCs at OT07-MW06 were attenuating naturally under the anaerobic conditions created by biological degradation of petroleum hydrocarbons. The analytical results also indicate that conditions at OT07-MW02 supported removal of arsenic, iron, and manganese via precipitation as sulfides. No COCs were identified for soil, surface water, or Aquifer Zone 2 (present 20 to 35 feet bgs) groundwater.

2.2.1.1 *Summary of Analytical Results in Groundwater and Surface Water*

In the 2007 groundwater sample collected from OT07-MW04, trichloroethylene (TCE) exhibited a concentration of 6.4 M (matrix effect was present, the concentration is estimated) micrograms per liter (µg/L), greater than the IEPA TACO (35 IAC 742) Tier 1, Class I, GRO of 5 µg/L. TCE was not detected in the 2010 and 2011 samples from OT07-MW04. In addition, TCE was not detected in the other piezometers or MWs installed at Site OT007. The constituents cis-1,2-DCE and VC, the daughter products of TCE degradation, were detected at concentrations greater than the Tier 1, Class I, GROs in piezometer AOC22-B15/PZ06 and OT07-MW06.

In 2011, the VC concentration at OT07-MW06 was 89.9 M µg/L, and the cis-1,2-DCE concentration was 71.1 µg/L. These exceedances are delineated in the downgradient direction by surface water sample OT07-SW01, collected from the drainage ditch in 2011, to which the Aquifer Zone 1 groundwater discharged at the time. Both concentrations detected in OT07-SW01, 1.2 M µg/L for VC and 1.1 µg/L for cis-1,2-DCE, were less than the surface water PALs (2 µg/L and 1,100 µg/L, respectively; IEPA, 2019). The surface water PALs are based on TACO (35 IAC 302.210), Derived Water Quality Criteria List.

In 2011, only samples from OT07-MW02, exceeded the IEPA TACO (35 IAC 742) Tier 1, Class I, GRO of 10 µg/L for arsenic (81.2 µg/L unfiltered), 5,000 µg/L for iron (40,700 µg/L unfiltered), and 150 µg/L for manganese (14,100 µg/L unfiltered). The surface water sample from location OT07-SW02, collected from the drainage ditch, bounded the extent of metals contamination to the south. The unfiltered arsenic result of 9.2 µg/L F (result greater than or equal to the method detection limit but less than the reporting limit), unfiltered iron (11,300 µg/L), and unfiltered manganese (1,470 µg/L) concentrations were all below the IEPA TACO (35 IAC 742) Tier 1, Class I, GRO or, in the case of manganese, background.

Figure 2-2 depicts the locations of the former piezometers, permanent MWs, surface water samples (2011) and groundwater flow direction discussed above. **Figure 2-3** presents the 2011 results for OT07-MW02, OT07-MW06, OT07-SW01 and OT07-SW02. The area of CVOC contamination was estimated to be 7,700 square feet (0.18 acre) surrounding OT07-MW06, and the area of metals contamination was estimated to be 15,600 square feet (0.36 acre) surrounding OT07-MW02.

2.2.1.2 *Selected Remedy*

According to the *Final CSIR/ROR/RAP* (HGL, 2013), the selected remedy for the groundwater COCs consists of MNA. During attenuation of the groundwater contamination, LUCs were implemented to (1) prevent installing water supply wells at the Site and, (2) require incorporating vapor intrusion controls into the building design in the unlikely situation that the taxiway is relocated and a building is constructed at the Site. The estimated remedial time frame was 27 years for attenuation of CVOCs and seven years for the metals (HGL, 2013).

In a letter to the Air Force dated October 3, 2013, the IEPA approved the remedial approach of MNA coupled with LUCs for the groundwater COCs (cis-1,2-DCE, VC, arsenic, iron, and manganese; IEPA, 2013a).

2.2.2 2013 Groundwater Monitoring Event

In December 2013, OT07-MW02, OT07-MW04, OT07-MW05, and OT07-MW06 were gauged and sampled in accordance with the *Final CSIR/ROR/RAP* (HGL, 2013) for the initial MNA monitoring event at Site OT007. Surface water was not present in the drainage ditch; therefore, surface water samples were not collected. Complete results can be found in the *Final Groundwater Monitoring Report* (HGL, 2014).

The COCs cis-1,2-DCE and VC were detected only in the groundwater sample collected from OT07-MW06. All other VOCs were detected at concentrations below detection limits or the Class I GROs. Total and dissolved arsenic, iron, and/or manganese exceeded one or more GROs in December 2013 in the wells OT07-MW02 and OT07-MW05.

Based on the data collected during the 2013 sampling event, recommendations from the *Final Groundwater Monitoring Report* (HGL, 2014) included use of low-flow sampling techniques rather than passive samplers for future monitoring events and elimination of carbon dioxide from the analyte list. These sampling recommendations were implemented during the 2015 groundwater monitoring events.

2.2.3 2015 Groundwater Monitoring Event

In March/April and September/October 2015, MWs OT07-MW02, OT07-MW04, OT07-MW05, and OT07-MW06 were gauged and sampled in accordance with the *Final CSIR/ROR/RAP* (HGL, 2013). Surface water was not present in the drainage ditch; therefore, surface water samples were not collected (HGL, 2016). Complete results can be found in the *Final Annual Groundwater Monitoring Report OT007 Former Sludge Weathering Lagoon/Bulk Fuel Facility* (*Final Groundwater Monitoring Report*, HGL, 2016).

Results from the March/April 2015 groundwater monitoring event indicated that both cis-1,2-DCE and VC were detected above the Class I GROs in OT07-MW06. The concentration of CVOCs in the remaining Site OT007 MWs were below detection limits (HGL, 2016). The concentration of metals detected in all samples were below their respective GROs.

Results from the September/October 2015 groundwater monitoring event indicated that the concentration of both cis-1,2-DCE and VC in OT07-MW06 were below detection limits (0.5 µg/L). Total and dissolved (field filtered) concentrations of arsenic and iron exceeded their respective GROs at OT07-MW02; however, concentrations of metals detected in the remaining three MWs were below the GROs. Manganese also exceeded the GRO in OT07-MW02, only in the dissolved (field filtered) sample.

To assess long-term concentration trends, HGL plotted the concentration of CVOCs and groundwater elevation over time. Although the September/October 2015 cis-1,2-DCE and VC concentrations were below the GROs, the fluctuations in cis-1,2-DCE and VC concentration over time may be a function of changes in the groundwater elevation near OT07-MW06. HGL concluded in the *Final Groundwater Monitoring Report* (HGL, 2016) that a direct relationship between CVOC concentrations and groundwater elevation is present, resulting in elevated cis-1,2-DCE and VC concentrations when relatively high groundwater elevations are observed (HGL, 2016).

HGL concluded in the *Final Groundwater Monitoring Report* (HGL, 2016) that the increase in metals concentrations in well OT07-MW02 was likely due to the dissolution of metals resulting from the anaerobic/reducing conditions present in this portion of the Site. The absence of ethene, ethane, and methane, the elevated oxidation reduction potential (ORP) measurements, the reduced total organic carbon (TOC) results, and the non-detect COC results indicated that chlorinated solvents near OT07-MW06 may have completely biodegraded, and groundwater was returning to natural, aerobic conditions. However, the decrease in cis-1,2-DCE and VC concentrations may also be correlated with the lower groundwater table observed in October 2015. Modifications to the sampling network, analytical suite, or sampling frequency were not recommended in the 2015 report. The sampling network and analytical suite remained unchanged for the 2016 annual MNA event. In accordance with the *Final CSIR/ROR/RAP* (HGL, 2013), MWs OT07-MW02, OT07-MW04, OT07-MW05, and OT07-MW06 were to be sampled semiannually for the first 2 years followed by annual sampling (HGL, 2013).

2.2.4 2016 Groundwater Monitoring Event

In December 2016, MWs OT07-MW02, OT07-MW04, OT07-MW05, and OT07-MW06 were gauged and sampled in accordance with the *Final Amended RAP* (Plexus, 2016b). Although only cis-1,2-DCE and VC have been identified as COCs for the Site, IEPA requested that the full VOC suite of compounds be analyzed (IEPA, 2016). Complete results can be found in the *Final 2016 Annual Monitored Natural Attenuation Report OT007 Former Sludge Weathering Lagoon/Bulk Fuel Facility* (*Final 2016 Annual MNA Report*, Plexus, 2018b).

Arsenic, iron, and manganese exceedances of the GRO were observed in December 2016 in OT07-MW02. Total and dissolved arsenic was detected above the GRO in OT07-MW02 and OT07-MW06; only the total arsenic concentration was detected above the GRO in OT07-MW04. Total and dissolved iron concentrations in OT07-MW02, and total iron concentrations in OT07-MW06 were also detected above the GRO. With the exception of OT07-MW04, the total and dissolved concentrations were similar, indicating that these metals are primarily present in the dissolved state and impacts due to sample turbidity were low. The TOC concentration of 9.8 milligrams per liter (mg/L) in well OT07-MW02 coupled with the observation of anaerobic/reducing geochemistry suggested that conditions remained favorable for microbial activity to continue and for metals to remain in solution.

Exceedances of cis-1,2-DCE, VC, and benzene were observed in OT07-MW06. Vinyl chloride was also detected above the GRO at OT07-MW04. The presence of dissolved gases, methane, reduced TOC, anaerobic conditions, and an increase in CVOCs at OT07-MW06 indicated that reductive dechlorination was still occurring at this portion of the Site. A significant increase in the concentration of CVOCs detected at OT07-MW06 was observed in 2016; these concentrations were over 300-fold times the concentrations observed in 2013. However, the increase in cis-1,2-DCE and VC may also be correlated with the higher groundwater table observed in December 2016. Benzene was also detected above the GRO for the first time in OT07-MW06 and was considered a potential COC. Vinyl chloride was also detected above the GRO at OT07-MW04, which historically has not had detections of chlorinated VOCs.

2.2.5 2017 Soil Characterization Investigation

On June 13, 2017, twelve soil borings (OT07-BO-26 through OT07-BO37) were installed to characterize the extent of CVOC contamination in accordance with the *Final Amended RAP* (Plexus. 2016b). The *Final Amended RAP* (Plexus, 2016b) indicated that soil borings would be terminated above the water table; however, based on the photo-ionization detector (PID) readings, a decision was made to terminate the borings in the water table at a depth of 10 feet bgs. At least one sample was collected from each boring location from the interval with the highest PID reading. Based on the initial sampling results, an additional ten step-out soil borings were installed on June 20, 2017. Step-out borings were installed to the north, west, and east of OT07-BO-34 and OT07-BO-36 to delineate the extent of VOC impacted soil. Boring locations to the north of OT07-MW04 were limited by the 250-foot taxiway buffer zone. Four (OT07-BO-40, OT07-BO-41, OT07-BO-42, and OT07-BO-46) out of the ten step-out borings sampled had detections of VOCs at concentrations greater than the PALs presented in Worksheet #15 of the *Final Amended RAP* (Plexus, 2016b). Chemicals detected at elevated levels were primarily petroleum constituents and VC. All four step-out samples that exceeded the PALs were collected in the capillary fringe zone, at depths of 6 to 8 feet bgs (Plexus, 2018c).

The 2017 soil investigation determined that residual contamination was present in the capillary fringe above the groundwater table rather than in unsaturated soil. The results were consistent with historical groundwater data, which indicated higher groundwater contaminant concentrations when the water table was elevated (Plexus, 2018c). Based on the results of the soil investigation, oxidant injections within the capillary fringe and shallow groundwater were recommended to abiotically degrade CVOCs and petroleum hydrocarbons at Site OT007, followed by MNA (Plexus, 2019b).

2.2.6 2017 Groundwater Monitoring Event

In November 2017, MWs OT07-MW02, OT07-MW04, OT07-MW05, and OT07-MW06 were gauged and sampled in accordance with the *Final Amended RAP* (Plexus, 2016b). Surface water was not present in the drainage ditch; therefore, surface water samples were not collected. Complete results can be found in the *Final 2017 Annual Monitored Natural Attenuation Report OT007 Former Sludge Weathering Lagoon/Bulk Fuel Facility* (Plexus, 2019a).

Arsenic, iron, and manganese exceedances were observed in November 2017 in OT07-MW02. Total and dissolved arsenic were detected above the GRO in OT07-MW02 and OT07-MW06; only total arsenic was detected above the GRO in the duplicate sample for OT07-MW06. Total and dissolved iron and manganese concentrations also exceeded the GROs in OT07-MW02. The total and dissolved concentrations were similar, indicating that these metals are primarily present in the dissolved state and impacts due to sample turbidity were low. The TOC concentration of 12.0 mg/L in well OT07-MW02 coupled with the observation of anaerobic/reducing geochemistry suggested that conditions remained favorable for microbial activity to continue and for metals to remain in solution.

Exceedances of cis-1,2-DCE, VC, and benzene were observed in OT07-MW06. The presence of dissolved gases, methane, reduced TOC, anaerobic conditions, and increase in CVOCs at OT07-MW06 indicated that reductive dechlorination was still occurring at this portion of the Site. An increase in the concentration of CVOCs detected at OT07-MW06 was observed again in 2017; these concentrations were over 400-fold times the concentrations observed in 2013. However, the increase in cis-1,2-DCE and VC may also have been correlated with the higher groundwater table observed in November 2017. Benzene was also detected above the GRO for the second time in OT07-MW06 and was added to the list of COCs.

A limited soil impact was identified during the June 2017 soil sampling event. VOCs were detected in the capillary fringe zone and were likely contributing to the concentrations observed in the groundwater at OT07-MW06. Results of the soil investigation are discussed in a separate technical memorandum (Plexus, 2018c).

2.2.7 2018 Groundwater Monitoring Event

In October 2018, MWs OT07-MW02, OT07-MW04, OT07-MW05, and OT07-MW06 were gauged and sampled in accordance with the *Final Amended RAP* (Plexus, 2016b). Surface water was not present in the drainage ditch; therefore, surface water samples were not collected. Complete results can be found in the *Final 2018 Annual Monitored Natural Attenuation Report OT007 Former Sludge Weathering Lagoon/Bulk Fuel Facility* (*Final 2018 Annual MNA Report*, Plexus, 2020a).

During the October 2018 event, arsenic, iron, and manganese were detected in each of the four wells. At OT07-MW02, total and dissolved manganese were detected at concentrations that exceeded the GRO; however, total and dissolved arsenic and iron were detected at concentrations below the GROs. Total and dissolved metals were detected at concentrations below the GROs for the other three sampled wells (OT07-MW04, OT07-MW05, and OT07-MW06). Elevated manganese concentrations in OT07-MW02 were 6,750 µg/L (total) and 6,350 µg/L (dissolved), slightly exceeding the GRO (6,120 µg/L). OT07-MW02 experienced a decrease in concentration when compared to total and dissolved manganese concentrations detected in 2017. The TOC concentration of 12.2 mg/L in well OT07-MW02 coupled with the observation of anaerobic/reducing geochemistry suggested that conditions remained favorable for microbial activity to continue and for metals to remain in solution.

Vinyl chloride was detected at 45 µg/L in OT07-MW06, exceeding the GRO of 2 µg/L; however, there was a significant decrease from concentrations detected in 2016 (860 µg/L) and 2017 (1,000 J [the positive result reported for this analyte is a quantitative estimate due to discrepancies in meeting certain analyte-specific quality control criteria] µg/L). The absence of dissolved gases, methane, reduced TOC, anaerobic conditions, and decrease in CVOCs at OT07-MW06 indicated that reductive dechlorination is no longer occurring at this portion of the Site. Benzene, added to the list of COCs in 2017, was not detected in 2018 (Plexus, 2020a).

2.2.8 2019 Groundwater Monitoring Event

In June 2019, MWs OT07-MW02, OT07-MW04, OT07-MW05, and OT07-MW06 were gauged and sampled in accordance with the *Final Amended RAP* (Plexus, 2016b). Surface water was not present in the drainage ditch; therefore, surface water samples were not collected. Complete and historical results can be found in the *Final 2019 Annual Monitored Natural Attenuation Report OT007 Former Sludge Weathering Lagoon/Bulk Fuel Facility* (*Final 2019 Annual MNA Report*, Plexus, 2020b).

During the June 2019 event, arsenic, iron, and manganese were detected in each of the four wells. OT07-MW02 experienced a decrease in concentration when compared to total and dissolved manganese concentrations detected in 2018 at concentrations below the GRO. Total and dissolved arsenic and iron continued to be detected at concentrations below the GROs at OT07-MW02, as well. The TOC concentration of 10.1 mg/L in well OT07-MW02 coupled with the observation of anaerobic/reducing geochemistry suggested that conditions remained favorable for microbial activity to continue and for metals to remain in solution.

Vinyl chloride was detected at 5.41 µg/L in OT07-MW06, exceeding the GRO of 2 µg/L; however, this was a significant decrease from concentration detected in 2017 (1000 J µg/L) and continued a decreasing trend from 2018 (45 µg/L). The absence of dissolved gases, methane, reduced TOC, anaerobic conditions, and decrease in CVOCs at OT07-MW06 indicate that reductive dechlorination was no longer occurring at this portion of the site. Benzene, added to the list of COCs in 2017, was not detected in 2019 (Plexus, 2020b).

2.2.9 2020 Oxidant Injection Event

Based on the 2017 soil characterization investigation and groundwater data collected since that investigation, oxidant (Persulfox®) injections were conducted at the Site in May 2020. Injection locations were chosen to address the residual petroleum hydrocarbons and VC in the saturated capillary fringe in the vicinity of OT07-BO-34, OT07-BO-36, OT07-BO-40, OT07-BO-41, OT07-BO-42, and OT07-BO-46. The oxidant injection was conducted in accordance with the *Final RAP Addendum* (Plexus, 2019b). In June 2020, six soil borings were advanced at the same location and depth interval where exceedances of petroleum hydrocarbons and CVOC concentrations were first observed in 2017 to evaluate the effectiveness of the injection event.

Following the oxidant injection event in 2020, no exceedances of the PALs were observed at boring locations OT07-BO-42 and OT07-BO-46; however, while generally lower than pre-injection concentrations, soil sampling results obtained during the 2020 site investigation indicated that VOC contamination remains present in the saturated soil of the capillary fringe. The VOC contamination continued to exceed the PALs in the capillary fringe at OT07-BO-34, OT07-BO-36, OT07-BO-40, and OT07-BO-41. The PALs for the post-injection soil characterization work were presented in Worksheet #15 of the *Final RAP Addendum* (Plexus, 2019b). They were selected following the same approach that was used for PAL selection in the *Final Amended RAP* (Plexus, 2016b).

2.2.10 2020 Groundwater Monitoring Event

In June 2020, MWs OT07-MW02, OT07-MW04, OT07-MW05, and OT07-MW06 were gauged and sampled in accordance with the *Final Amended RAP* (Plexus, 2016b). Surface water was not present in the drainage ditch; therefore, surface water samples were not collected. Complete and historical results can be found in the *Final 2020 Annual Monitored Natural Attenuation Report OT007 Former Sludge Weathering Lagoon/Bulk Fuel Facility (Final 2020 Annual MNA Report*, Plexus, 2020c).

During the June 2020 event, arsenic, iron, and manganese were detected in each of the four wells, however, none of the concentrations exceeded the GROs. The concentrations of arsenic, iron, and manganese showed dissolved arsenic with a decreasing trend and dissolved iron and dissolved manganese with a “probably decreasing” trend at OT07-MW02. The drainage ditch did not contain surface water during the 2020 sampling event, thus the presence of metals in groundwater in the downgradient direction could not be assessed in 2020.

While no metals were detected above the GROs in any of the four wells, the anaerobic conditions will persist if there is sufficient TOC to fuel microbial activity. In June 2020, the TOC concentrations in each well decreased compared to 2019 observation data.

Benzene and cis-1,2-DCE were only detected at OT07-MW06 in June 2020 at concentrations lower than their respective PALs. In OT07-MW06, VC exceeded the PAL of 2 µg/L at a concentration of 6.91 µg/L, which is similar to the concentration of VC in the well in 2019. Low levels of total cis-1,2-DCE and VC were detected in OT07-MW04 for the first time in 2016; these CVOCs have not been detected above the reporting limits since 2017 in this well.

2.2.11 2021 Groundwater Monitoring Event

In June 2021, MWs OT07-MW02, OT07-MW04, OT07-MW05, and OT07-MW06 were gauged and sampled in accordance with the *Final Amended RAP* (Plexus, 2016b). Surface water was not present in the drainage ditch; therefore, surface water samples were not collected. Complete and historical results can be found in the *Final 2021 Annual Monitored Natural Attenuation Report OT007 Former Sludge Weathering Lagoon/Bulk Fuel Facility (Final 2021 Annual MNA Report*, Plexus, 2022).

During the June 2021 event, arsenic, iron, and manganese were detected in each of the four wells, however, none of the concentrations exceeded the GROs.

While no metals were detected above the GROs in any of the four wells, the anaerobic conditions will persist as long as there is sufficient TOC to fuel microbial activity. In June 2021, the TOC concentrations either decreased or remained stable compared to 2020 observation data.

Benzene and cis-1,2-DCE were only detected at OT07-MW06 in June 2021 at concentrations lower than their respective PALs. In OT07-MW06, VC exceeded the PAL of 2 µg/L at a concentration of 18.1 µg/L. Low levels of total cis-1,2-DCE and VC were detected in OT07-MW04 for the first time in 2016; these CVOCs have not been detected above the reporting limits since 2017 and remained undetected in 2021 in this well.

2.2.12 2022 Groundwater Monitoring Event

In June 2022, MWs OT07-MW02, OT07-MW04, OT07-MW05, and OT07-MW06 were gauged and sampled in accordance with the *Final Amended RAP* (Plexus, 2016b). Complete and historical results can be found in the *Final 2022 Annual Monitored Natural Attenuation Report OT007 Former Sludge Weathering Lagoon/Bulk Fuel Facility (Final 2022 Annual MNA Report*, Plexus, 2023).

During the June 2022 event, arsenic, iron, and manganese were detected in each of the four wells, however, none of the concentrations exceeded the GROs.

While no metals were detected above the GROs in any of the four wells, the anaerobic conditions will persist as long as there is sufficient TOC to fuel microbial activity. In June 2022, the TOC concentrations either increased or remained stable compared to 2021 observation data.

Benzene was not detected in any wells in June 2022. Cis-1,2-DCE was only detected at OT07-MW06 in June 2022 at concentrations lower than its PAL. Vinyl chloride was previously identified at concentrations exceeding its PAL in OT07-MW06 since 2016, but in 2022 vinyl chloride was detected at a concentration lower than its PAL in this well.

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3.0 SUMMARY OF FIELD ACTIVITIES

This section summarizes the post-injection soil characterization sampling and groundwater sampling activities conducted at Site OT007 in 2023. All field work was performed in accordance with the *Final Amended RAP* (Plexus, 2016b), the *Final RAP Addendum* (Plexus, 2019b), and the *Final CSIR/ROR/RAP* (HGL, 2013). Field forms documenting the sampling activities are provided in **Appendix A**.

3.1 Soil Characterization Sampling

On June 2, 2023, one soil boring (identified as OT07-BO-40-4 to differentiate from the 2017, 2020, 2021, and 2022 borings) was installed at approximately the same locations as the 2017, 2020, 2021, and 2022 soil borings to characterize the extent of CVOC contamination following the oxidant injections in May 2020. This boring was advanced in the location where exceedances of CVOC concentrations persisted during the 2022 soil characterization sampling event. The 2017, 2020, 2021, 2022, and 2023 boring locations are shown on **Figure 3-1**.

The 2023 soil boring was installed using a hand auger which allowed for the collection of a disturbed grab sample. The boring was installed to a depth of 8 feet bgs, in accordance with the *Final Amended RAP* (Plexus, 2016b). Soil borings were continuously visually screened for staining and monitored using a PID with a 10.6 eV bulb calibrated to isobutylene gas. Soil samples were collected from the same interval where exceedances were observed in 2017, 2020, 2021, and 2022. Boring logs from the event are provided in **Appendix A**.

All samples were submitted to Pace Analytical Gulf Coast Laboratories (PAGC) in Baton Rouge, Louisiana, on a standard turnaround time (TAT) and analyzed for the full VOC list (Method 8260B) and moisture content. A data quality evaluation for all soil samples collected during the characterization event is provided in **Section 4.0**. The soil sample results are discussed in **Section 5.4**. The analytical data from the sampling event are provided in **Appendix B** and the data validation reports are provided in **Appendix C**.

3.1.1 Quality Assurance/Quality Control Samples

The following quality assurance (QA)/quality control (QC) samples were collected for soil sampling during the June 2023 sampling event:

- Field duplicate: sample OT07-BO-840-4-6-8 was collected as a field duplicate sample from OT07-BO-40-4-6-8 on 2 June 2023.
- Matrix spike (MS)/matrix spike duplicate (MSD): an MS/MSD sample was collected from OT07-BO-40-4-6-8 on 2 June 2023.
- Equipment blank: No equipment blank was taken as only one soil boring was advanced.

3.2 Groundwater Monitoring

All MWs within the OT007 MNA network (OT07-MW02, OT07-MW04, OT07-MW05, and OT07-MW06) were gauged and sampled as part of the June 2023 sampling event. The four wells were installed in Aquifer Zone 1 at depths ranging from 14.48 to 18.56 feet bgs; each well was screened with a 10-foot screen (ERM, 1992). Monitoring well construction details for Site OT007 are presented in **Table 5-1**. All groundwater samples were analyzed for VOCs (SW 8260B), arsenic, iron, and manganese (total and dissolved, SW 6020B), alkalinity (SW 2320B), TOC (SW 9060A), anions (SW 9056A), and dissolved gases (Robert S. Kerr Laboratory [RSK] 175).

3.2.1 Water Level Measurements

Groundwater level measurements were collected from the four MWs in the OT007 MNA network before purging and sampling activities began. Water levels were measured from the top of the well casing using an electronic water level indicator. The depth to water was measured to the nearest 0.01 foot and recorded on the groundwater MW purge and sampling data forms provided in **Appendix A**. Synoptic water level measurements were also recorded for MWs OT07-MW01, OT07-MW03, and OT07-MW04D in June 2023. A summary of the groundwater elevation data is presented in **Table 5-2**, and groundwater elevations are discussed further in **Section 5.1**.

3.2.2 Groundwater Sampling Activities

After gauging activities were complete, each MW was purged using low-flow techniques before groundwater samples were collected. Wells were purged until groundwater quality parameters stabilized, indicating that the groundwater sample collected is representative of the targeted hydrogeologic zone. Groundwater quality parameters were measured using a Horiba U-52 (or equivalent) water quality meter fitted with an in-line flow-through cell. The groundwater was purged until the following parameters were stable within:

- Temperature: ± 1.0 degrees Celsius
- pH: ± 0.1 standard units
- ORP: ± 10 millivolts (mV) or within 10 percent (%; whichever is less)
- Conductivity: ± 0.01 milli Siemens per centimeter or within 3% (whichever is less)
- Turbidity: less than or equal to 10 nephelometric turbidity units, or three consecutive readings within 10%
- Dissolved oxygen (DO): ± 0.3 mg/L or within 10 percent (whichever is less)

Dissolved metals samples have historically been collected using a 0.45-micron or 0.2-micron filter to filter out clay particles from groundwater. The *Final 2022 Annual MNA Report* (Plexus,

2023) recommended that analysis of dissolved metals be removed from the analytical suite as all metals COCs, both total and dissolved, had been below their respective GROs for four years. In 2023, no samples were collected for dissolved metals analysis. Total metals samples were collected unfiltered.

The wells were purged and sampled using a submersible bladder pump in accordance with the procedures specified in the *Final Amended RAP* (Plexus, 2016b) and the *Final UFP-QAPP for Performance-Based Remediation, Scott Air Force Base (*Plexus, 2018a). Groundwater samples were collected from each well after water quality parameters stabilized. Field groundwater quality parameters from 2013 to 2023 are presented in **Table 3-1**. The groundwater field sheets for the 2023 event are provided in **Appendix A**.

3.2.3 Quality Assurance/Quality Control Samples

The following QA/QC samples were collected for groundwater samples during the June 2023 sampling event:

- Field duplicate: sample OT07-MW806-060123 was collected from OT07-MW06 on 1 June 2023.
- MS/MSD: an MS/MSD sample was collected from OT07-MW04 on 1 June 2023.
- Trip blanks: trip blank TB-01 was collected on 1 June 2023 blanks and submitted to be analyzed for VOCs.
- Equipment blank: on 1 June 2023, one equipment rinsate blank was collected from the decontaminated bladder pump used to sample the MWs. This sample was used to test for potential cross contamination due to insufficient decontamination of the sampling equipment.

3.3 Sample Preservation and Shipping

Samples collected during the two-day field event were stored in coolers with wet ice to maintain a temperature of less than 4 degrees centigrade. At the conclusion of field work each day, the cooler was prepared for shipping. This included lining the cooler with a bag, packing the samples, and adding ice to maintain a temperature of less than 4 degrees centigrade. The chain of custody (CoC) was taped to the inside of the lid, and a custody seal sealed the cooler. Packing tape was placed over the custody seal and reaffirmed the integrity of the cooler for shipping. The coolers were then relinquished to FedEx (for the soil samples) or UPS (for the groundwater samples) and sent to PAGC in Baton Rouge, Louisiana, for analysis.

3.4 Investigation-Derived Waste Disposal

Investigation-derived waste (IDW) included:

- Soil cuttings and direct push technology sleeves generated during the post-injection soil characterization sampling.

- Purge water generated during groundwater sampling activities.
- Decontamination water generated during soil and groundwater sampling activities.
- Used personal protective equipment (PPE, i.e., nitrile/latex disposable gloves) and miscellaneous trash.

Disposable PPE and miscellaneous trash were treated as municipal waste. Following generation, all other solid and liquid IDW was placed in 55-gallon drums on a secondary containment pallet located at the Centralized Accumulation Site, Building 3306. One solid drum and one liquid drum were labelled non-hazardous based off of previous Site conditions. The IDW was picked up by Illini Environmental Inc on 5 October 2023 for off-site disposal. Waste manifest forms are presented in **Appendix D**.

3.5 Surface Water Monitoring

In accordance with the *Final CSIR/ROR/RAP* (HGL, 2013), surface water samples are to be collected during the MNA sampling events to determine whether Zone 1 groundwater contamination has an impact on surface water in the drainage ditch. The drainage ditch was backfilled in 2014 and a surface water conveyance is no longer present at the site.

3.6 Work Plan Deviations

The following deviations from the *Final RAP Addendum* (Plexus, 2019b) were made during the 2023 event:

- The addendum called for the resampling of BO-34, BO-36, BO-41, and BO-46. The 2023 event collected soil samples from BO-40 because of soil PAL exceedances that were detected in this location during the 2022 event. Locations BO-34, BO-36, BO-41, and BO-46 were not resampled in 2023 because no PALs were exceeded in the last samples collected at each location.
- The addendum called for the collection of surface water samples from the drainage ditch. The drainage ditch was backfilled in 2014 and surface water samples were not collected from the former drainage ditch location during the 2023 event because it was dry.

No other work plan deviations were made during the 2023 event.

4.0 DATA QUALITY EVALUATION

This section describes the data quality evaluation process used to review and evaluate laboratory analytical data collected at the Site during the 2023 monitoring event. The data QA/QC procedures are detailed in the *Final Amended RAP* (Plexus, 2016b), *UFP-QAPP* (Plexus, 2018a), *Final RAP Addendum* (Plexus, 2019b), and *Department of Defense (DoD) Quality Systems Manual (QSM) for Environmental Laboratories, Version 5.3* (DoD, 2019).

Environmental samples collected were analyzed by PAGC of Baton Rouge, Louisiana, which holds current DoD Environmental Laboratory Accreditation Program and State of Illinois National Environmental Laboratory Accreditation Program certification. Data from this event are presented in two sample delivery groups, identified by PAGC under the project ID of "Scott AFB" and report numbers 223060502 (soil samples) and 223060210 (groundwater samples). Samples were analyzed as follows:

2023 Soil Characterization samples:

- VOCs (USEPA 8260B)
- Moisture content

2023 Groundwater MNA samples:

- VOCs (USEPA 8260B)
- Total arsenic, iron, and manganese (USEPA 6020B)
- Total alkalinity (SM 2320 B-2011)
- TOC (USEPA 9060A)
- Ethane, ethene, and methane (RSK-175)
- Chloride, sulfate, nitrate, and nitrite anions (USEPA 9056A)

The 2023 analytical data reports are provided in **Appendix B**.

4.1 Quality Assurance/Quality Control Sampling

The following field QA/QC samples were collected to ensure that the measurement systems were maintained within prescribed limits and the sample results were of acceptable quality:

2023 Soil Characterization samples:

- Field duplicate: sample OT07-BO-840-4-6-8 was collected as a field duplicate sample from OT07-BO-40-4-6-8 on 2 June 2023.
- MS/ MSD: an MS/MSD sample was collected from OT07-BO-40-4-6-8 on 2 June 2023.
- Equipment blank: No equipment blank was taken as only one soil boring was advanced.

2023 Groundwater MNA samples:

- Field duplicate: sample OT07-MW806-060123 was collected from OT07-MW06 on 1 June 2023.
- MS/MSD: an MS/MSD sample was collected from OT07-MW04 on 1 June 2023.
- Trip blanks: trip blank TB-01 was collected on 1 June 2023 blanks and submitted to be analyzed for VOCs.
- Equipment blank: on 1 June 2023, one equipment rinsate blank was collected from the decontaminated bladder pump used to sample the MWs. This sample was used to test for potential cross contamination due to insufficient decontamination of the sampling equipment.

4.2 Data Validation

Plexus performed data validation on the analytical results. All analytes were validated to Level III (corresponding to USEPA Stage 2A/2B data review). The validator compared the performance of the QC elements presented in the laboratory analytical reports to the criteria presented in the *Final Amended RAP* (Plexus, 2016b), *UFP-QAPP* (Plexus, 2018a), *Final RAP Addendum* (Plexus, 2019b), and *QSM* v5.3 (DoD, 2019) to evaluate the quality of the laboratory results.

The data reports were validated by the Project Chemist at Plexus. The validator examined the laboratory signed CoC records and sample receiving documentation to verify sample receipt and condition. Data points associated with non-conformances were qualified in accordance with the *UFP-QAPP* (Plexus, 2018a) using the data qualifiers presented in the validation review report glossary. The validator also compared the laboratory method detection and reporting limits to ensure the *UFP-QAPP* (Plexus, 2018a) was followed, and the project-specific data quality objectives were met. The final validation reports are provided in **Appendix C**.

4.3 Data Quality Assessment and Usability

Data quality and usability is assessed in terms of precision, accuracy, representativeness, comparability, completeness, and sensitivity.

4.3.1 Precision

Precision is evaluated using the relative percent difference (RPD) by evaluating the reproducibility of field duplicate samples, MS/MSD samples, laboratory control sample (LCS)/laboratory control sample duplicate (LCSD) and laboratory duplicate sample. RPDs are calculated when both the parent and duplicate results exceed the limit of detection (LOD). Results with RPDs that exceed the acceptance criteria defined in the *Final Amended RAP* (Plexus, 2016b) are reviewed and qualified. Systemic RPD exceedances are investigated to ensure that procedural errors are not present.

Precision indicators were within acceptance criteria with the following exceptions:

2023 Groundwater MNA samples:

- **USEPA 8260B:** The LCS/LCSD RPD for 1,2-dibromo-3-chloropropane, 2-butanone, 4-methyl-2-pentanone, and tert-butyl methyl ether were above the control limit. No qualification was necessary.
- **USEPA 8260B:** The MS/MSD RPD was above acceptance criteria for dichlorodifluoromethane. No qualifications were necessary.

Except for the variances noted above, the precision indicators are within acceptable parameters. No systematic trends or biases in precision were observed.

4.3.2 Accuracy

Accuracy is evaluated by reviewing the surrogate recoveries, MS/MSD recoveries, LCS/LCSD recoveries, and initial and continuing calibration recoveries.

Notable accuracy concerns are presented below:

2023 Soil Characterization samples:

- **USEPA 8260B:** The LCS recovery for chloroethane and bromobenzene for analytical batch 767415 were below control limits. The samples were qualified as “UJ” for chloroethane and bromobenzene.
- **USEPA 8260B:** The MS and MSD recoveries were below the control limits for sample OT07-BO-40-4-6-8-MS and OT07-BO-40-4-6-8-MSD for 1,2,4-trimethylbenzene. The LCS had an acceptable recovery. The results for parent sample OT07-BO-40-4-6-8 were qualified as “J” for 1,2,4-trimethylbenzene.

2022 Groundwater MNA samples:

- **USEPA 8260B:** In the analysis for analytical batches 767028 and 767081, the ICV recovery is outside the control limits for acetone and chloromethane. In the analysis for analytical batch 766942, the ICV recoveries were outside the control limits for acetone, 2-butanone, 2-hexanone and chloromethane. In the analysis for analytical batch 767028, the %Drift is outside ± 20% for dichlorodifluoromethane and bromomethane in the opening CCV. In the analysis for analytical batch 767081, the %Drift is outside ± 50% for acetone in the closing CCV. Samples OT07-EB-060123, OT07-MW02-060123, OT07-MW04-060123, OT07-MW05-060123, and OT07-MW06-060123 were qualified as “UJ” for acetone, 2-butanone, 2-hexanone and chloromethane. Sample OT07-MW-806-060123 was qualified as “UJ” for acetone, chloromethane, dichlorodifluoromethane and bromomethane. Sample OT07-TB-01-060124 was qualified as “J” for acetone and “UJ” for chloromethane.
- **USEPA 8260B:** The surrogate dibromofluoromethane was above the upper control limit for sample OT07-MW04-060123. The sample was non-detect for all compounds; no qualification was necessary.

Except for the variances noted above, the accuracy indicators are within acceptable parameters. No systemic trends or biases in accuracy were observed.

4.3.3 Representativeness

Representativeness is demonstrated through the review of sample documentation and the evaluation of field, laboratory, and equipment blanks.

There were no variances on the CoC reported upon sample check-in at the laboratories. Samples were collected according to the procedures outlined in the *Final Amended RAP* (Plexus, 2016b) and the most current version of the *QSM* v5.3 (DoD, 2019).

Laboratory and equipment blanks were all intact and non-detect with the following exceptions:

2023 Soil Characterization samples:

- **USEPA 8260B:** The method blank contained a detection of m,p-Xylene at a concentration less than the limit of quantitation (LOQ). All results are more than 10 times the blank result; no qualifiers were applied.

2023 Groundwater MNA samples:

- **USEPA 8260B:** The equipment blank contained detections of acetone, methylene chloride, and chloride at concentrations less than the LOQ. The method blank contained a detection of manganese at a concentration less than the LOQ. The trip blank contained acetone at a concentration less than the LOQ. All acetone, methylene chloride, chloride, and manganese results were non-detect or detections greater than 10 times the blank result and no qualifiers were necessary.

Except for the variances noted above, the representativeness indicators are within acceptable parameters.

4.3.4 Comparability

Comparability is evaluated by ensuring that the data set collected during the investigation event is comparable or agrees with the data set collected during a previous sampling event. A review of the sampling, preparation, and analytical methods is performed to verify consistency between events.

All samples were collected, prepared, and analyzed in accordance with the standard operating procedures (SOPs) and methodologies presented in the most current version of the *QSM* v5.3 (DoD, 2019) and the *Final Amended RAP* (Plexus, 2016b). No issues were observed in comparability evaluations to the extent possible.

4.3.5 Completeness

Completeness is calculated as the ratio of usable data to all analytical data collected or planned to be collected. For completeness requirements, usable results are all results not qualified with an

"R" (rejected data) qualifier during data verification/validation. The completeness goal, as defined in the *Final Amended RAP* (Plexus, 2016b), is 90% of each parameter. The following equation is used to calculate analytical completeness:

% Analytical Completeness = (number of non-rejected results/number of reported results) * 100

The analytical completeness goal for this sampling event is 90%. All samples were collected during this sampling effort and no data points were rejected. The calculated analytical completeness attained for this sampling effort is 100% for all parameters.

4.3.6 Sensitivity

Sensitivity is evaluated by comparing the reporting limits to the PALs. The instrument or method should provide an accurate analyte concentration that is not greater than the applicable standard and/or screening level. Sample results have been reported using the most current version of the *QSM* v5.3 (DoD, 2019). The reporting requirement for this version of the *QSM* (DoD, 2019) is to report for each compound, results in three levels (high level – LOQ, middle level – LOD, and low level – detection limit [DL]). As required by *QSM* v5.3 (DoD, 2019) protocol, all compounds, which were qualitatively identified at concentrations below their respective LOQ but above the DL, have been marked with "J" qualifiers to indicate that they are quantitative estimates. Non-detect results have been reported to the LOD.

All 2023 samples were analyzed at a 1x dilution with the following exceptions:

2023 Soil Characterization samples:

- **USEPA 8260B:** Samples were analyzed at a dilution of 250x in all samples. Dilutions result in elevated detection limits and should be considered when evaluating data usability.

2023 Groundwater MNA samples:

- **Method 9056A:** Method 9056A: Samples were analyzed for sulfate at a dilution of 2x (OT07-MW05-060123), 5x (OT07-MW02-060123, OT07-MW06-060123, and OT07-MW-806-060123) and 50x (OT07-MW04-060123, OT07-MW04-060123 MS, and OT07-MW04-060123 MSD).

Dilutions result in elevated detection limits and should be considered when evaluating data usability and are presented in the data validation report provided in **Appendix C**. The dilution analyses were performed because of the suspected presence of high levels of target compounds and/or interferences. The LODs/LOQs are elevated by the dilution factor for target compounds that were not detected. Several non-detect results for the soil sample OT07-BO-40-4-6-8 were elevated above their respective PALs as a result of the dilutions, including: 1,1,2-trichloroethane, 1,1-dichloroethene, 1,2,3-trichloropropane, 1,2-dibromo-3-chloropropane, 1,2-dibromoethane, 1,2-dichloroethane, 1,2-dichloropropane, benzene, bromoethane, carbon tetrachloride, methylene chloride, tetrachloroethene, trichloroethene, and vinyl chloride. There were no non-detect results in the groundwater samples elevated above the PALs as a result of the dilutions.

4.4 Data Quality Conclusions

All data generated during the sampling event were evaluated followed the requirements of the *Final Amended RAP* (Plexus, 2016b). All results for QC samples, including field duplicates, equipment blanks, LCSs, method blanks, and MS/MSD analyses, met the *Final Amended RAP* (Plexus, 2016b) control limits, with the exceptions discussed in **Section 4.3**. All data were determined to be usable. Comparability was achieved by using standard methods for sampling and analysis, reporting data in standard units, and using standard reporting formats. The overall data quality appears to be acceptable. The quality of the data is sufficient for qualitative and quantitative evaluations.

5.0 SUMMARY OF ANALYTICAL RESULTS

This section summarizes the hydrogeologic observations and analytical results from the June 2023 groundwater sampling activities. Note that MWs OT07-MW01, OT07-MW03, and OT07-MW04D are only gauged and not sampled. OT07-MW05 is the most upgradient well at the Site, OT07-MW04 is located in the center of the Site, and OT07-MW02 and OT07-MW06 are downgradient wells located directly upgradient of the former drainage ditch.

5.1 Zone 1 Groundwater Flow Direction and Hydraulic Gradient

The MWs at Site OT007 were gauged before sample collection activities began. According to the *Final CSIR/ROR/RAP* (HGL, 2013), groundwater in Zone 1 flows toward the drainage ditch (backfilled in 2014) both from the north and the south, with the drainage ditch acting as a groundwater discharge boundary for Zone 1 (HGL, 2013). Monitoring wells on the north and south side of the former drainage ditch were gauged as part of the sampling event. In 2023, the groundwater flow direction observed for MWs north of the former drainage ditch was generally toward the southwest and consistent with the groundwater flow direction described in the *Final CSIR/ROR/RAP* (HGL, 2013). The groundwater flow direction observed for the MWs south of the former drainage ditch (OT07-MW01 and OT07-MW03) was generally to the southeast and parallel to the former drainage ditch in 2023.

The estimate of the hydraulic gradient between wells OT07-MW05 and OT07-MW02 was measured as 0.01 ft/ft in June 2023. This hydraulic gradient estimate is generally consistent with estimates from past sampling events conducted by Plexus, along with the Zone 1 hydraulic gradient of 0.04 ft/ft estimated within the *Final CSIR/ROR/RAP* (HGL, 2013). The groundwater potentiometric surface map for June 2023 is presented on **Figure 5-1**. Groundwater elevations are presented in **Table 5-2**.

5.2 Groundwater Analytical Data

The analytical data from the June 2023 groundwater sampling event were compared to the GROs presented in the *Final CSIR/ROR/RAP* (HGL, 2013) to assess the status of the ongoing MNA remedy and are discussed in the sections below. The groundwater quality parameters measured during sample collection are presented in **Table 3-1**. A summary of the analytical data, including all COCs, COPCs, and natural attenuation parameters from 2013 to 2023, is presented in **Table 5-3**. Selected results for the Site COCs and benzene from 2018 to 2023 are also shown on **Figure 5-2**. The 2023 laboratory analytical data reports are provided in **Appendix B**. The data validation report for the analytical data are provided in **Appendix C**.

5.2.1 Metals Results

Arsenic, iron, and manganese were identified as metal COCs in the Zone 1 groundwater (HGL, 2013). GROs for these COCs were identified in the *Final CSIR/ROR/RAP* (HGL, 2013) and are based on the Class I groundwater values in TACO (35 IAC 742) Tier 1, Appendix B, Table E, and maximum background concentrations. Background concentrations in groundwater for Scott

AFB were established in the *Final Basewide Groundwater Dataset Evaluation Technical Memorandum* (HGL, 2012).

During the June 2023 event, arsenic, iron, and manganese were detected in each of the four wells; however, none of the concentrations exceeded their respective GROs.

Detected arsenic concentrations in 2023 ranged from 0.85 J µg/L in OT07-MW05 to 10.1 µg/L in OT07-MW02. However, arsenic was 11.7 J µg/L in the OT07-MW06 duplicate (4.36 J µg/L in the OT07-MW06 primary). All four wells generally remained stable in detected concentrations from 2022 to 2023 and remained below the GRO (19.1 µg/L) in 2023.

Detected iron concentrations in 2023 ranged from 151 µg/L in OT07-MW04 to 7,940 µg/L in OT07-MW02. All four wells generally remained stable in detected concentrations from 2022 to 2023, with the exception of OT07-MW05 which exhibited a decrease from 1,890 µg/L to 246 µg/L. All four wells remained below the GRO (26,000 µg/L) in 2023.

Detected manganese concentration in 2023 ranged from 66.5 µg/L in OT07-MW04 to 1,650 µg/L in OT07-MW02. All four wells generally remained stable in detected concentrations from 2022 to 2023 and remained below the GRO (6,120 µg/L) in 2023.

In comparison to the elevated concentrations of arsenic, iron, and manganese observed in OT07-MW02 between 2013 and 2018, the concentrations of these metals decreased to below their respective GROs in June 2019. Since 2020 the concentrations of these metals have remained generally stable. The 2023 event marks the fifth consecutive event where no exceedances were noted for any of the three metal COCs in OT07-MW02.

Since 2017, arsenic, iron, and manganese were either non-detect or detected at concentrations below their respective GROs at OT07-MW04. Exceedances of these metal COCs were only observed in OT07-MW04 during one sampling event in 2016.

Since 2015, arsenic, iron, and manganese were either non-detect or detected at concentrations below their respective GROs at OT07-MW05. Exceedances of these metal COCs were only observed in OT07-MW05 during one sampling event in 2013.

Since 2018, arsenic, iron, and manganese were either non-detect or detected at concentrations below their respective GROs at OT07-MW06. Exceedances of these metal COCs were only observed in OT07-MW06 during sampling events in 2016 and 2017.

As part of the *Final CSIR/ROR/RAP* (HGL, 2013), exceedances of arsenic, iron, and manganese in OT07-MW02 were intended to be delineated in the downgradient direction by surface water results from the drainage ditch (HGL, 2013). However, the drainage ditch has not contained surface water during the sampling events since 2011 and was backfilled in 2014. Thus, surface water samples recommended in the *Final 2022 Annual MNA Report* (Plexus, 2023) could not be collected.

According to the *Final CSIR/ROR/RAP* (HGL, 2013), the metals contamination in groundwater is a result of microbial degradation of petroleum hydrocarbons. This microbial degradation caused the groundwater to become anaerobic, dissolving iron-, arsenic-, and manganese-bearing minerals. The anaerobic conditions will remain for as long as the TOC concentration at OT07-MW02 (11 mg/L in December 2013) remains elevated. The TOC concentration in OT07-MW02 was 5.95 mg/L in 2023, which is an increase from a low of 1.78 J mg/L in 2021 and 5.63 mg/L in 2022. When the residual TOC has been degraded, the groundwater is expected to return to its natural aerobic state. Under aerobic conditions, iron and manganese oxidizes and precipitates as iron oxides/hydroxides, manganese oxide, and arsenic will co-precipitate with iron (HGL, 2013).

5.2.2 Volatile Organic Compounds Results

The CVOCs cis-1,2-DCE and VC were identified as COCs in the Zone 1 groundwater (HGL, 2013). Benzene was added to the list of COCs in 2017 (Plexus, 2019a).

Historically (April 2015, December 2016, and November 2017), cis-1,2-DCE has only exceeded the GRO of 70 µg/L (116 µg/L, 360 µg/L, and 420 J µg/L, respectively) at well OT07-MW06. However, as in October 2015 and all sampling events since October 2018, the concentration of cis-1,2-DCE was below the GRO value in June 2023 with a concentration of 14.3 µg/L (13.5 µg/L in the duplicate sample). The compound cis-1,2-DCE was not detected in the remaining three wells.

While VC has previously exceeded the GRO of 2 µg/L in OT07-MW06 since 2016 with a maximum concentration of 1,000 µg/L in 2017, the concentration fell below the GRO to a concentration of 1.22 µg/L (0.985 J µg/L in the duplicate) in 2022. In 2023, VC was again detected at OT07-MW06 at a concentration (94.6 µg/L in the primary, 88.5 µg/L in the duplicate) above its GRO. This marks the highest concentration of VC detected in this well since 2017. As with samples collected from 2017 to 2022, VC was not detected in the remaining three wells (OT07-MW02, OT07-MW04, and OT07-MW05) in 2023.

Additionally, although not identified as a COC in the *Final CSIR/ROR/RAP* (HGL, 2013), benzene was detected at 16 µg/L in 2016 and 28 µg/L in 2017, above the TACO (35 IAC 742), Appendix B, Table E, Class I value of 5 µg/L at OT07-MW06. The benzene concentration dropped below that value (to a concentration of 3.14 µg/L) in 2018 and has not exceed that value since. Benzene was detected in 2023 at 1.71 µg/L (1.57 µg/L in the duplicate sample) in OT07-MW06. Benzene has not been detected in the remaining three wells (OT07-MW02, OT07-MW04, and OT07-MW05) since 2018, as shown in **Table 5-3**.

To assess long-term concentration trends, a trend graph was developed for OT07-MW06, which presents both VOC concentrations and groundwater elevations over time. The trend graph is presented as **Figure 5-3**. Though historically there appeared to be a direct relationship between VOC concentration and groundwater elevation near OT07-MW06, the sampling events from December 2016 through June 2023 seem to support an inverse relationship between the two parameters. As depicted on **Figure 5-3**, these opposing trends seem to suggest something else is influencing the rise and fall of VOC concentration in OT07-MW06. As discussed in **Section 2.1.5**, a soil investigation was completed at Site OT007 in June 2017. During the soil investigation, VOCs were detected in the capillary fringe zone and may have contributed to the concentrations observed in the groundwater at OT07-MW06 at that time.

Note that CVOC trend graphs are not presented for the other three wells at the Site. Wells OT07-MW02 and OT07-MW05 have not contained detectable concentrations of CVOCs since monitoring began in 2013. Low levels of total cis-1,2-DCE and VC were detected in OT07-MW04 for the first time in 2016; these CVOCs were not detected above the reporting limits in any groundwater sampling events conducted from 2017 through 2023 so trend analyses would be inherently inconclusive.

5.2.3 Monitored Natural Attenuation Parameter Results

To assess the progress of the MNA remedy, samples collected during the June 2023 event were analyzed for the natural attenuation parameters ethane, ethene, methane, chloride, nitrate, nitrite, sulfate, TOC, and alkalinity. In addition, the field parameters of DO and ORP were measured during low-flow sampling. The natural attenuation parameter results are summarized in **Table 5-3**, and the field measurements of DO and ORP are summarized in **Table 3-1**. An evaluation of these results is provided below.

- For chlorinated solvents to follow reductive pathways, the ideal TOC concentration is greater than 20 mg/L (Wiedemeier at al., 1998). TOC results have fluctuated in all wells during the eleven sampling events reported on **Table 5-3**; however, as shown on **Figure 5-4a** (OT07-MW02), **Figure 5-4b** (OT07-MW04), and **Figure 5-4d** (OT07-MW06), the fluctuations have been slight and are below the ideal concentration. The TOC concentrations in OT07-MW05 (**Figure 5-4c**) fluctuated between 2.3 mg/L and 5.3 mg/L between 2011 and 2017; however, in 2018, TOC increased to 12.3 mg/L before declining to 5.0 mg/L in 2019. The increase in TOC in 2018 corresponds to a decrease in sulfate concentration (from 13.0 mg/L in 2017 to 4.29 mg/L in 2018) and a subsequent increase in 2019 (9.5 mg/L), indicating that the TOC was likely being used for microbial activity during the 2018 timeframe. The TOC concentration continued to decrease to low of 0.345 J mg/L in 2020; however, it increased to 2.9 mg/L in 2021 and has remained stable since (2.7 mg/L in 2023).
- For chlorinated solvents to follow reductive pathways, the ideal sulfate concentration is less than 20 mg/L; higher concentrations are thought to interfere with natural attenuation (Wiedemeier et al., 1998). In June 2023, the sulfate concentrations ranged from 10.9 mg/L in OT07-MW05 to 225 mg/L in OT07-MW04, with sulfate concentrations in OT07-MW04 and OT07-MW06 (33.5 mg/L in the primary, 37.9 mg/L in the duplicate) over 20 mg/L. These concentrations of sulfate suggest the

reductive pathway of VOC contamination is inhibited, especially at OT07-MW04 which has recently remained elevated compared to the 2019 concentration of 115 mg/L. It should also be noted that no VOCs have been detected above the GROs in any well since 2017, except for VC at OT07-MW06 (which has ranged between 0.985 J µg/L in 2022 and 94.6 µg/L [88.5 µg/L duplicate] in 2023 during this time). From 2018 to 2019, there were both decreases (OT07-MW02 and OT07-MW06) and increases (OT07-MW04 and OT07-MW05) in sulfate concentrations, which is a reversal of the trends observed between 2017 and 2018. Sulfate concentrations continued to fluctuate in 2023, with increases at OT07-MW05 and OT07-MW06 and decreases at OT07-MW02 and OT07-MW04. Trend graphs depicting TOC and sulfate concentrations over time are presented for each MW on **Figures 5-4a** through **5-4d**.

- Methane concentrations in OT07-MW02 declined from a high of 8,600 µg/L in 2016 to 161 J µg/L (236 J µg/L, duplicate) in 2019. The concentration of methane in OT07-MW02 has fluctuated since 2019 and went from 1,360 µg/L in 2021 to 137 µg/L in 2022, and then increased again to 1,480 µg/L in 2023. In OT07-MW04, methane concentrations also peaked in 2016, but at a much lower level (61 J µg/L; 97 µg/L [duplicate sample]) and then were either below detection limits or observed at low concentrations between 2018 and 2022. In 2023, methane concentrations in OT07-MW04 increased to 32.8 µg/L, the highest concentration observed since 2017. A similar trend was observed in OT07-MW05, with a small peak of 24 µg/L in 2016 and no detections since 2018, including the 2023 event. In OT07-MW06, methane was detected at a concentration of 3,900 µg/L in 2016 but decreased to 2,800 µg/L in 2017. From 2018 to 2020, the concentration of methane in OT07-MW06 was below detection limits, but in 2021 it was detected at a concentration of 166 µg/L (164 µg/L in the duplicate sample) before decreasing again to low levels in 2022. In 2023, methane was detected in OT07-MW06 at 327 µg/L (501 µg/L in the duplicate), the highest concentration since 2017. For OT07-MW04 and OT07-MW06, the elevated levels of methane in 2016 correspond with elevated levels of cis-1,2-DCE (22 J µg/L; 23 µg/L [duplicate sample] in OT07-MW04 and 360 µg/L in OT07-MW06). A similar correlation between an elevated methane level and an elevated level of cis-1,2-DCE (420 J µg/L) was observed in OT07-MW06 in 2017. In addition, detections of benzene (16 µg/L in 2016 and 28 µg/L in 2017) and VC (860 µg/L in 2016 and 1,000 J µg/L in 2017) corresponded to the peak in methane in OT07-MW06. Though still exceeding the GRO of 2 µg/L, only lower concentrations of VC (45 µg/L in 2018, 5.41 µg/L in 2019, 6.91 µg/L in 2020, and 18.1 µg/L [23.2 µg/L duplicate] in 2021) were detected in OT07-MW06 between 2018 and 2021, with no exceedance observed in 2022. VC was detected at OT07-MW06 in 2023 at 94.6 (88.5 in the duplicate), the highest concentration observed since 2017.
- Ethane and ethene have either been not detected or detected at very low levels in

OT07-MW02, OT07-MW04, and OT07-MW05 since 2013, which is to be expected as no CVOCs were present in groundwater at these locations. Ethane was detected in OT07-MW06 in 2017 (200 µg/L) and 2018 (210 µg/L), as was ethene, detected in OT07-MW06 in 2017 (21 µg/L) and 2018 (28 µg/L). These trends are consistent with the degradation of cis-1,2-DCE and VC in 2016 and 2017; however, ethane and ethene were not detected in OT07-MW06 in 2018, 2019, or 2020. In 2021, low levels of ethane (1.12 µg/L and 0.901 J µg/L in the duplicate) and ethene (0.346 J µg/L and 0.309 J µg/L in the duplicate) were detected in OT07-MW06. There were no detections of ethane or ethene in OT07-MW06 in 2022 and 2023. The low levels suggest that the reductive dechlorination process may be nearing completion.

- Nitrate and nitrite have been largely absent from the groundwater in all four wells since 2013. Nitrate is an electron acceptor that may have been present prior to 2013, but it is typically one of the earlier electron acceptors depleted during the degradation of petroleum hydrocarbon contamination (USEPA, 2017).
- For reductive pathways to occur, the ideal range for DO is less than 0.5 mg/L (Wiedemeier et al, 1998). In June 2023, the measured DO value was 0.00 mg/L in all wells except for OT07-MW02, in which it was 0.07 mg/L. This indicates reductive pathways are likely occurring across the Site. ORP values ranged from -54 mV (OT07-MW06) to 105 mV (OT07-MW04).

In 2016 and 2017, the presence of dissolved gases and methane, reduced TOC, anaerobic conditions, and elevated concentrations of CVOCs at OT07-MW06 indicated that reductive dechlorination was occurring at this portion of the Site. However, by October 2018, dissolved gases and methane were not detected, TOC slightly increased, CVOCs decreased, and the presence of aerobic conditions were observed; these trends continued in 2019. While anaerobic conditions were again observed in 2021, CVOC concentrations remained low, and it appeared that the reductive dechlorination process has addressed most of the CVOC contamination at OT07-MW06 given the lack of CVOC exceedances at this well in 2022. The increased concentrations of CVOCs and dissolved gases such as methane observed in this well in 2023, however, suggest that reductive dechlorination is still occurring.

Benzene was also detected above the GRO (5 µg/L) in OT07-MW06 in 2016 and 2017, although the concentration decreased to below the GRO in 2018, and it was not detected in 2019. Benzene was again detected at OT07-MW06 in June 2020 and June 2021, but at an estimated concentration significantly lower than the GRO. In June 2022, there was no detection of benzene in OT07-MW06. In June 2023, benzene was detected at a concentration (1.71 µg/L [1.57 µg/L in the duplicate]) below the GRO.

The historically high concentrations of metals in well OT07-MW02 were likely due to the dissolution of metals resulting from the anaerobic/reducing conditions present in this portion of the Site. The aquifer appears to be reverting to more aerobic/oxidizing conditions as the VOC contamination has been mostly addressed, and metals concentrations are decreasing as the metals precipitate out of solution.

The ORP values measured during the 2023 event were spatially varied. The groundwater is becoming less reducing on the north and central portions of the Site, as evidenced by ORP measurements that are positive for OT07-MW04 and OT07-MW05. Conversely, the groundwater is more reducing on the southern portion of the Site, as evidenced by ORP values that are negative for OT07-MW02 and OT07-MW06. A summary of all groundwater quality parameters is presented in **Table 3-1**.

5.3 Mann-Kendall Statistical Evaluation

For this report, the Mann-Kendall Toolkit from GSI Environmental (GSI, 2012) was used to statistically evaluate COC concentration trends in groundwater at individual MW locations. Concentration data for VOCs (VC, cis-1,2-DCE, and benzene) and dissolved metals (arsenic, iron, and manganese) from 2013 through 2023 was entered into the Mann-Kendall Toolkit worksheet. One MNA parameter, TOC, was also evaluated as a measure of microbial degradation activity at OT007. Only data from 2013 to 2023 was included in the statistical analysis as 2013 was the first year of implementation of the selected MNA remedy. The GSI Mann-Kendall User's Manual does not recommend including wells with all non-detect results and thus wells which had all non-detect results for a specific constituent were not included in the Mann-Kendall analysis as the true concentration trend for the specific chemical is considered stable at a concentration below the detection limit (GSI, 2012). Wells with fewer than four detections for each COC were also not included in the Mann-Kendall statistical analysis. Wells that were not dropped from the statistical analyses are discussed and identified below.

The worksheets associated with the Mann-Kendall Toolkit analysis are presented in **Appendix E**. Conclusions of the Mann-Kendall Toolkit analysis for VC, cis-1,2-DCE, and benzene are summarized as follows:

- One MW was observed to have no trend for VC, cis-1,2-DCE, and benzene post-2013: OT07-MW06. This lack of trend was also observed in the 2019, 2021, 2022, and 2023 Mann-Kendall Toolkit Analyses for OT07-MW06 and is due to data variability in the COC concentrations as evidenced by a coefficient of variation value of greater than 1 with a confidence factor less than 90 percent for all constituents even though the S-statistic was negative for benzene.
- Two MWs were excluded from the Mann-Kendall analysis due to having only non-detect VC, cis-1,2-DCE, and benzene concentrations post-2013: OT07-MW02 and OT07-MW05. OT07-MW04 was also excluded from the Mann-Kendall analysis as this well had fewer than four detections for the specific VOCs.

Conclusions of the Mann-Kendall Toolkit analysis for total arsenic, iron, and manganese are summarized as follows:

- Arsenic concentration trends ranged from no trend (OT07-MW05) to stable (OT07-MW06) to decreasing (OT07-MW02 and OT07-MW04).
- Iron concentration trends ranged from no trend (OT07-MW05 and OT07-MW06) to decreasing (OT07-MW02 and OT07-MW04).
- Manganese concentration trends ranged from stable (OT07-MW04 and OT07-MW06) to decreasing (OT07-MW02 and OT07-MW05).

Conclusions of the Mann-Kendall Toolkit analysis for TOC are summarized as follows:

- OT07-MW04 and OT07-MW05 continued to exhibit a stable trend in 2023.
- The concentrations of TOC in OT07-MW02 are probably decreasing.
- The concentrations of TOC in OT07-MW06 is probably increasing.

5.4 Soil Analytical Data

As described in **Section 3.1**, one hand-augered boring was advanced on June 2, 2023 to again evaluate the extent of CVOC contamination in the capillary fringe following the injection of oxidant at the Site in May 2020. The boring was advanced in the location where exceedances of CVOC concentrations persisted during the 2022 soil characterization sampling event after first being observed during the 2017 soil characterization sampling event, and soil samples were collected from the same intervals where exceedances were observed in 2017, 2020, 2021, and 2022. A summary of 2023 soil detections is included in **Table 5-4** and are shown on **Figure 5-5**. The PALs shown in **Table 5-4** are consistent with the PALs selected for the post-injection soil characterization in Worksheet #15 of the *Final RAP Addendum* (Plexus, 2019b). The following bullet points outline the results of the investigation (e.g., 2017-2022).

- **OT07-BO-40**

  In 2017, benzene was the only analyte detected above its PAL in boring OT07-BO-40.

  Following the oxidant injection event in 2020, benzene was detected above the PAL and was an increase compared to the concentration observed in 2017. Additionally, VC was detected above the PAL of 10 micrograms per kilogram (µg/kg). During the June 2021 sampling event, benzene concentration decreased when compared to the concentration observed in 2020. Additionally, VC was again detected at a concentration above the PAL of 10 µg/kg. During the June 2022 sampling event, exceedances of 1,2,4-trimethylbenzene, 1,2-dichloroethane, benzene, m,p-xylene, and methylene chloride were observed. Benzene increased to 556 J µg/kg, which was higher than the previous peak in 2020.

  During the June 2023 sampling event, exceedances of 1,2,4-trimethylbenzene (1,960 J µg/kg [28,500 µg/kg in the duplicate]), benzene (136 U µg/kg [337 J µg/kg in the duplicate]), m,p-xylene (310 J µg/kg [4,960 µg/kg in the duplicate]), and naphthalene (316 J µg/kg [4,830 µg/kg in the duplicate]) were observed; the exceedances of benzene, m,p-xylene, and naphthalene were only observed in the duplicate sample. With the exception of naphthalene, the max concentrations of these compounds detected in June 2023 were lower than the maximums detected in June 2022.

The PAL for benzene (30 µg/kg) is equal to the TACO (35 IAC 742) Tier 1 Soil Component of the Groundwater Ingestion Route SRO for Class 1 groundwater. Because the concentration of benzene in OT07-BO-40 was 337 J µg/kg in June 2023, benzene may continue to leach into groundwater at this boring location and impact Zone 1 groundwater at the Site.

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6.0 CONCLUSIONS AND RECOMMENDATIONS

This report presents the results of groundwater monitoring and the post-injection soil characterization sampling conducted in June 2023 at Site OT007. Groundwater sampling activities included water level gauging of all seven site MWs and groundwater sample collection from four MWs in the OT007 MNA network. Groundwater samples were analyzed for the groundwater COCs (arsenic, iron, manganese, cis-1,2-DCE, benzene, and VC) and natural attenuation parameters. Soil sampling activities included the advancement and sampling of a hand-augered soil boring to continue to evaluate the extent of VOC contamination in the capillary fringe after the oxidant injection event, further detailed in the *Final 2020 Injection Completion Report* (Plexus, 2020c). Soil samples were analyzed for VOCs. No surface water samples were collected as the former drainage ditch was backfilled in 2014. The conclusions from each respective sampling activity are summarized in the sections below.

6.1 Groundwater MNA Conclusions

All seven site monitoring wells were gauged, and four of the six site monitoring wells were sampled in accordance with the recommendations of the *Final 2021 Annual MNA Report* (Plexus, 2022). After all groundwater COCs were observed below their respective GROs for the first time in 2022, in 2023 an exceedance of VC was observed in OT07-MW06 and marked the highest concentration of this compound observed since 2017. While the presence of dissolved gasses such as methane in this well indicate reductive dechlorination is ongoing, the process may be nearly complete given the low concentrations of VC observed over the last several years and that VC is the last daughter product of reductive dechlorination before ethane and ethene. The increased concentrations of CVOCs and dissolved gases such as methane observed in this well in 2023, however, suggest that reductive dechlorination is still occurring. Mann-Kendall concentration trend analyses for the VOC COCs (VC, cis-1,2-DCE, and benzene) in OT07-MW06 indicates no trend due to data variability; however, qualitative assessment of the historical data shows a clear decreasing trend for cis-1,2-DCE and benzene since the annual 2017 event. Qualitative assessment shows a clear decreasing trend for VC between 2018 and 2022 before the increase in concentrations observed in 2023.

Groundwater conditions remain anaerobic at the site due to the degradation of petroleum hydrocarbons. However, TOC concentrations are decreasing or stable at relatively low levels (around 10 mg/L or less). While anaerobic conditions responsible for metals contamination in groundwater persist, concentrations of total metals remained below their respective GROs in all wells for the fifth consecutive year. Mann-Kendall trend analyses of total metals indicate no trend, stable, or decreasing trends depending on the specific metal and well.

6.2 Soil Characterization Sampling Conclusions

One hand-augered soil boring (OT07-BO-40) was advanced and sampled for VOCs during the 2023 event as recommended in the *Final 2021 Annual MNA Report* (Plexus, 2022).

Concentrations of 1,2,4-trimethylbenzene, benzene, m,p-xylene, and naphthalene exceeded their respective PALs at OT07-BO-40, though the exceedances of benzene, m,p-xylene, and naphthalene were only observed in the duplicate sample. With the exception of naphthalene, the max concentrations of these compounds detected in 2023 were lower than the maximums detected in 2022.

The PAL for benzene (30 µg/kg) is equal to the TACO (35 IAC 742) Tier 1 Soil Component of the Groundwater Ingestion Route SRO for Class 1 groundwater. Because the concentration of benzene in OT07-BO-40 was 337 J µg/kg in June 2023, benzene may continue to leach into groundwater at this boring location and impact Zone 1 groundwater at the Site.

6.3 Recommendations:

Based on the conclusions of this annual report, the following recommendations are made:

1. Annual groundwater gauging of all site wells and groundwater sampling of monitoring wells OT07-MW02, OT07-MW04, OT07-MW05 and OT07-MW06 is recommended to continue in 2024.
2. In monitoring wells OT07-MW02, OT07-MW04, OT07-MW05 and OT07-MW06, all total metals COCs have been below their respective GROs for at least five years (OT07-MW02) and as many as nine years (OT07-MW05). Additionally, in OT07-MW02, the well in which metal exceedances were most recently detected, Mann-Kendall analysis indicates that the total concentrations of all three total metals COCs are decreasing. Therefore, it is recommended that analysis of total metals be removed from the analytical suite for 2024.
3. Concentrations of VOCs in soil continue to exceed PALs at OT07-BO-40, therefore annual sampling of soil at OT07-BO-40 is recommended to continue.

Table 6-1 presents a summary of the sampling network wells, analytical suite, and sampling frequency. No other modifications to the annual MNA monitoring are recommended.

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FIGURES

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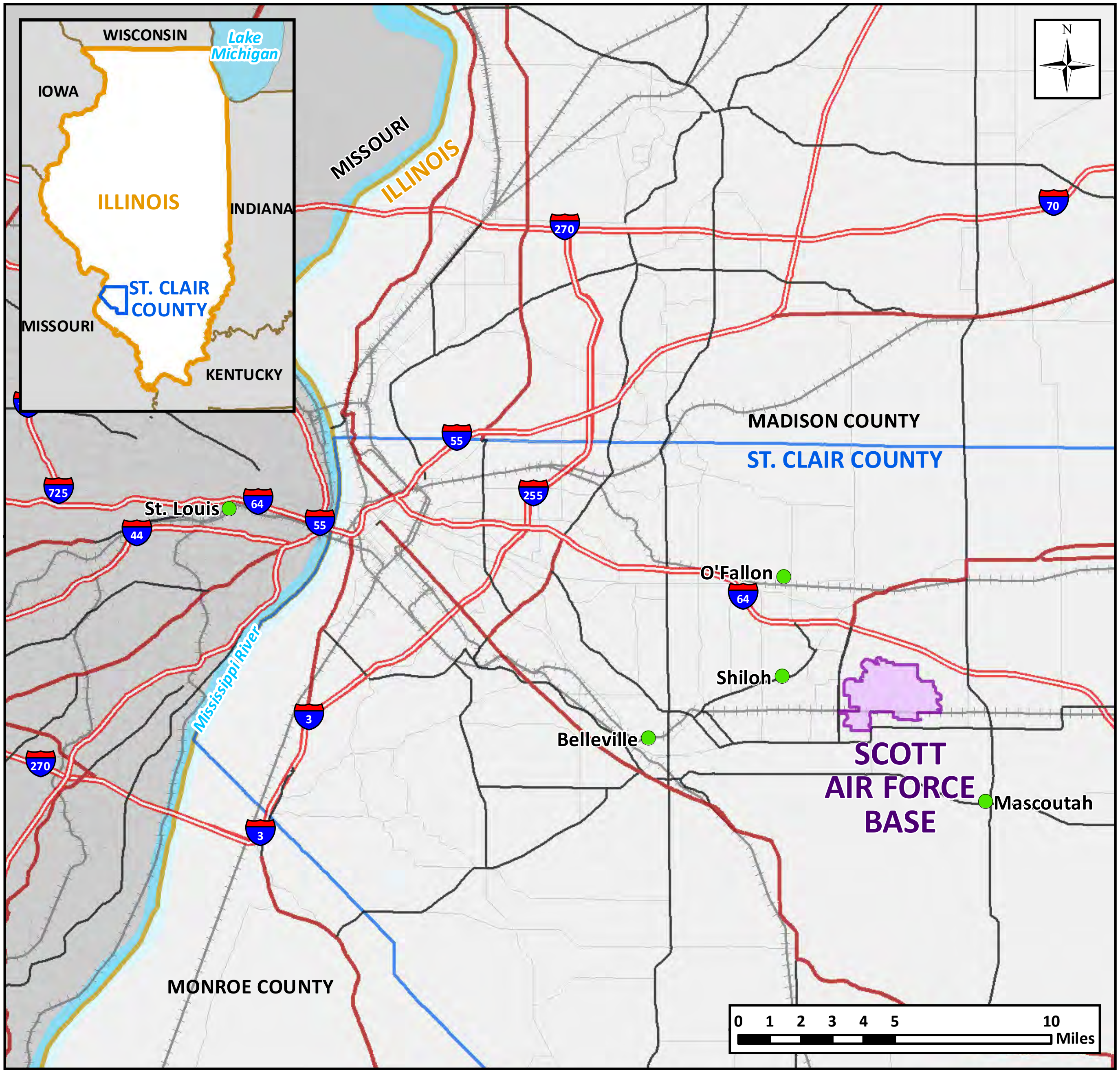

Map Key:

- State Boundary
- County Boundary
- Scott Air Force Base Boundary
- Lake Michigan
- Mississippi River
- Cities
- Interstate
- Highway
- Major Road
- Local Road
- Railroad

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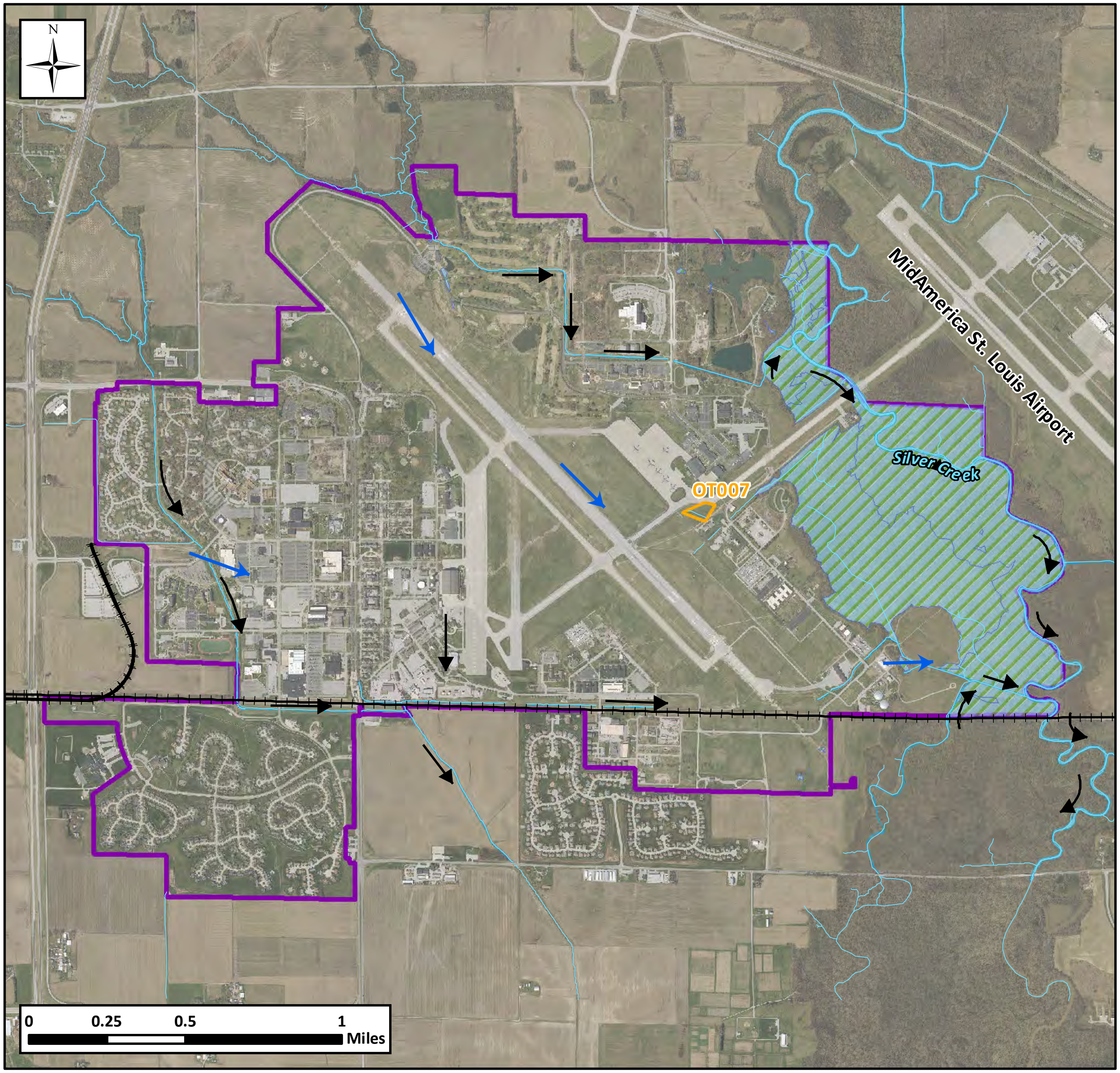
Date: August 2023

FIGURE 1-1

Scott AFB Location Map

2023 Annual MNA Report
Site OT007
Scott Air Force Base, Illinois

C:\Users\bbouton\OneDrive - Plexus Scientific Corp\Documents\GIS_PRJ_BU\Scott AFB PBR 2015\Maps\OT007\2023 MNA Report\Figure 1-1 - Scott AFB Location Map.mxd *08/01/2023 12:13*



Map Key:

- Scott Air Force Base Boundary
- Site Boundary
- Railroad
- Surface Water
- Wetlands/Silver Creek Lowlands
- Generalized Groundwater Flow Direction
- Stormwater/Surface Water Flow Direction

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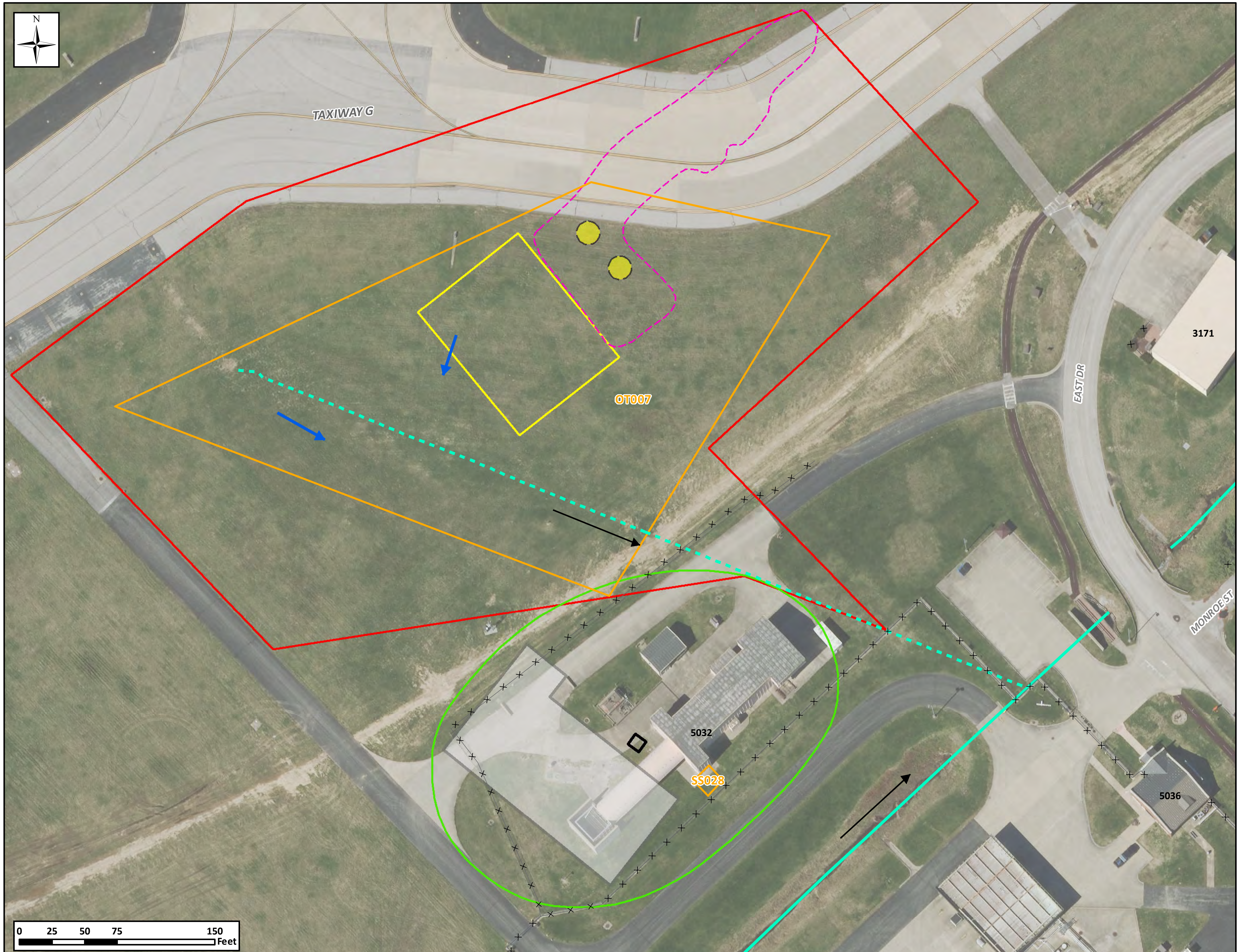
FIGURE 1-2

Site Vicinity Map

2023 Annual MNA Report
Site OT007
Scott Air Force Base, Illinois



Map Key:

- Site Boundary
- 1996 AST Soil Excavation (Burn & McDonnell, 1997)
- CSIR Study Area
- OWS-14 Study Area
- Forner OWS-14
- Approximate Extent of the Former Sludge Weathering Lagoon Remedial Excavation (ERM, 1992)
- Former AST
- Former Concrete Pad
- Drainage Ditch
- Historical Drainage Ditch
- Stormwater/Surface Water Flow Direction
- Shallow Groundwater Flow Direction
- Fence

Abbreviation Key:
AST = Aboveground storage tank
CSIR = Comprehensive Site Investigation Report
OWS = Oil-Water Separator

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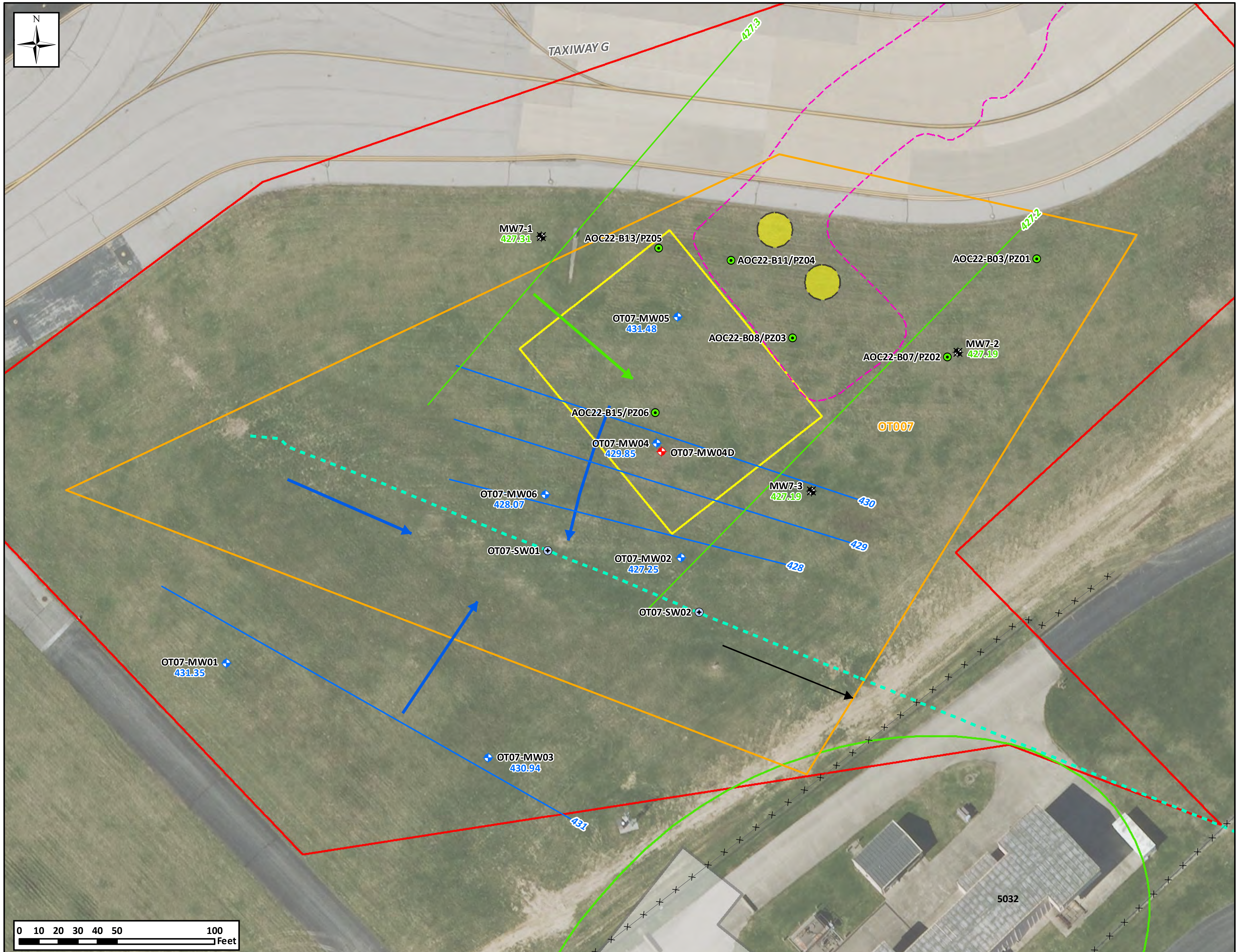
FIGURE 2-1

Site Location Map

2023 Annual MNA Report
Site OT007
Scott Air Force Base, Illinois



Map Key:

- Site Boundary
- Shallow Monitoring Well
- Deep Monitoring Well
- 1988 Abandoned Deep Monitoring Well (Not Surveyed)
- 2005 Soil Sample/Temporary Monitoring Well (Not Surveyed)
- 2011 Surface Water Sample Location
- 2011 Shallow Potentiometric Contour (ft msl)
- 1988 Deep Potentiometric Contour (ft msl)
- Shallow Groundwater Flow Direction
- Deep Groundwater Flow Direction
- Stormwater/Surface Water Flow Direction
- 1996 AST Soil Excavation (Burn & McDonnell, 1997)
- CSIR Study Area
- OWS-14 Study Area
- Approximate Extent of the Former Sludge Weathering Lagoon Remedial Excavation (ERM, 1992)
- Former AST
- Former Concrete Pad
- Historical Drainage Ditch
- Fence

Abbreviation Key:
AST = Aboveground storage tank
CSIR = Comprehensive Site Investigation Report
ft = ft
msl = Mean sea level
OWS = Oil-Water Separator

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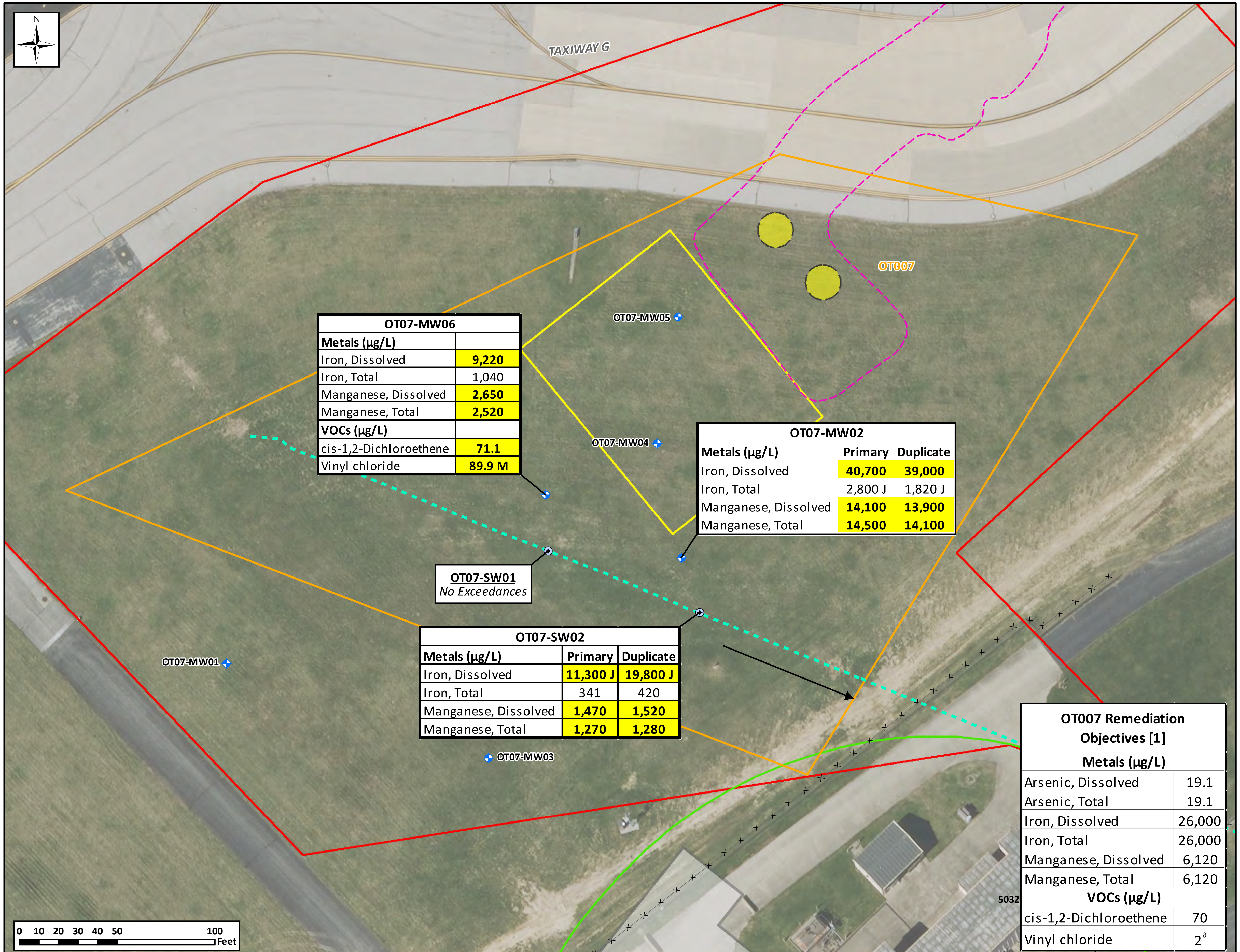
FIGURE 2-2

March 2011 Zone 1 and December 1988 Zone 2 Potentiometric Map and Historical Sample Locations

2023 Annual MNA Report
Site OT007
Scott Air Force Base, Illinois



OT07-MW06	
Metals (µg/L)	
Iron, Dissolved	9,220
Iron, Total	1,040
Manganese, Dissolved	2,650
Manganese, Total	2,520
VOCs (µg/L)	
cis-1,2-Dichloroethene	71.1
Vinyl chloride	89.9 M

OT07-MW02		
Metals (µg/L)	Primary	Duplicate
Iron, Dissolved	40,700	39,000
Iron, Total	2,800 J	1,820 J
Manganese, Dissolved	14,100	13,900
Manganese, Total	14,500	14,100

OT07-SW01
No Exceedances

OT07-SW02		
Metals (µg/L)	Primary	Duplicate
Iron, Dissolved	11,300 J	19,800 J
Iron, Total	341	420
Manganese, Dissolved	1,470	1,520
Manganese, Total	1,270	1,280

OT007 Remediation Objectives [1]	
Metals (µg/L)	
Arsenic, Dissolved	19.1
Arsenic, Total	19.1
Iron, Dissolved	26,000
Iron, Total	26,000
Manganese, Dissolved	6,120
Manganese, Total	6,120
VOCs (µg/L)	
cis-1,2-Dichloroethene	70
Vinyl chloride	2[a]

Map Key:

- Site Boundary
- Shallow Monitoring Well
- 2011 Surface Water Sample Location
- 1996 AST Soil Excavation (Burn & McDonnell, 1997)
- CSIR Study Area
- OWS-14 Study Area
- Approximate Extent of the Former Sludge Weathering Lagoon Remedial Excavation (ERM, 1992)
- Former AST
- Former Concrete Pad
- Historical Drainage Ditch
- Stormwater/Surface Water Flow Direction
- Fence

Notes:

Shaded results indicate concentrations above the Project Action Limit (PAL) or OT007 Remediation Objective (RO).

[1] = Remediation objectives for cis-1,2-dichloroethene and vinyl chloride are based on the Class I groundwater values in Tiered Approach to Corrective Action Objectives (TACO) (35 Illinois Administrative Code 742) Tier 1, Appendix B, Table E. Remediation objectives for arsenic, iron, and manganese are based on the highest maximum background values in groundwater for Scott AFB established in the Final Basewide Groundwater Dataset Evaluation Technical Memorandum (HGL, 2012).

a = Illinois Administrative Code Title 35, Part 742, TACO, Appendix B, Table E, Tier I, Groundwater Remediation Objectives for the Groundwater Component of the Groundwater Ingestion Route (IEPA, 2013).

- Volatile organic compounds (VOCs) analyzed by method SW 8260B.
- Metals analyzed by method SW 6020B.

Abbreviation Key:

µg/L = micrograms per liter
AST = Aboveground storage tank
CSIR = Comprehensive Site Investigation Report
J = The positive result reported for this analyte is a quantitative estimate.
M = A matrix effect was present.
OWS = Oil-Water Separator

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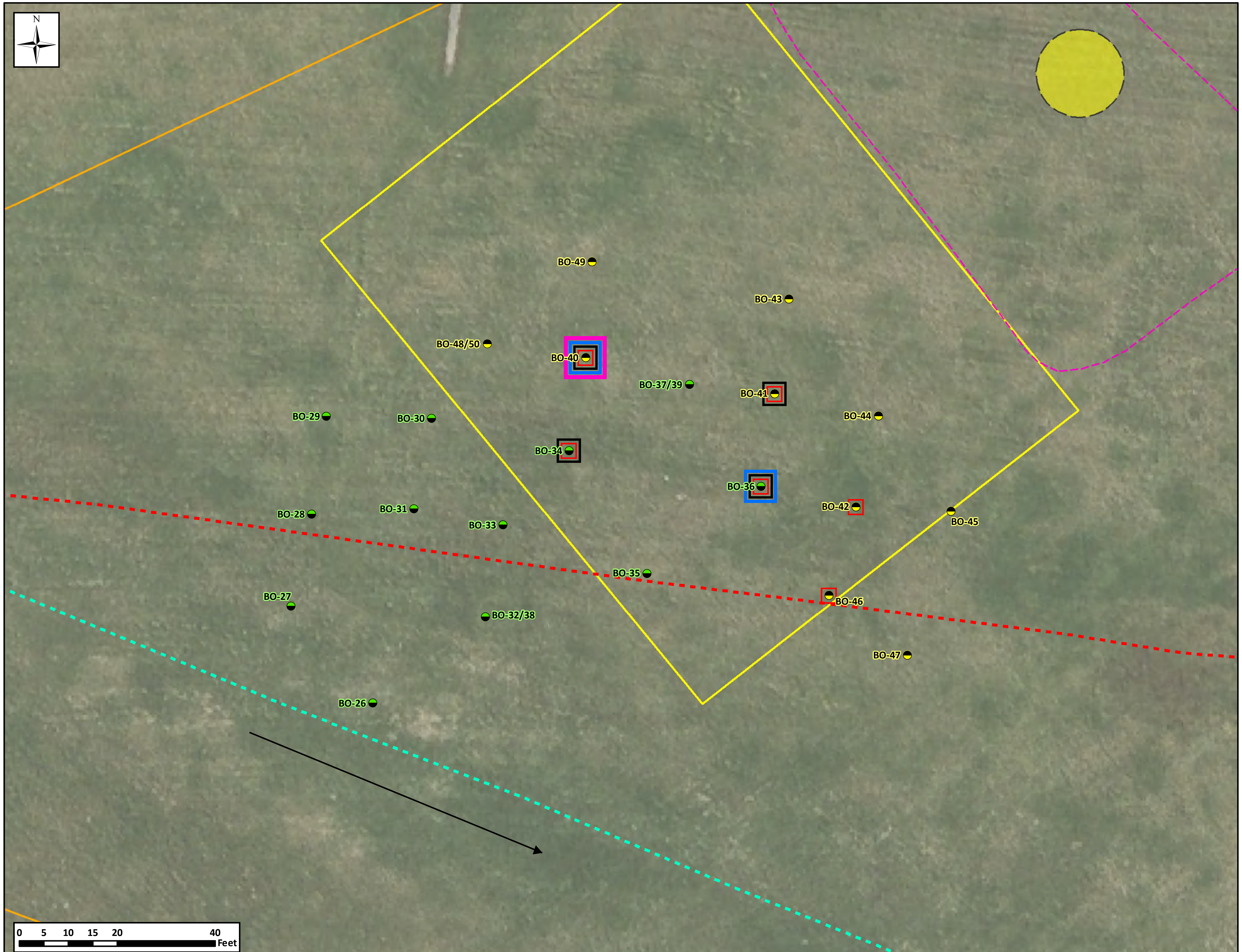
FIGURE 2-3

March 2011 Groundwater and Surface Water Sample Exceedances

2023 Annual MNA Report
Site OT007
Scott Air Force Base, Illinois



Map Key:
Site Boundary
2017 Soil Borings:
Initial Soil Boring Location
Stepout Soil Boring Location
2020 Soil Boring Location
2021 Soil Boring Location
2022 Soil Boring Location
2023 Soil Boring Location
200 Foot Taxiway Perimeter
1996 AST Soil Excavation (Burn & McDonnell, 1997)
Approximate Extent of the Former Sludge Weathering Lagoon Remedial Excavation (ERM, 1992)
Former AST
Historical Drainage Ditch
Stormwater/Surface Water Flow Direction

Note:
Site OT007 is located within the Controlled Movement Area of Scott Air Force Base. Borings located within 200 feet of the centerline of Taxiway G will require notification to the Scott Air Force Base Tower prior to initiation.
Abbreviation Key:
AST = Aboveground storage tank

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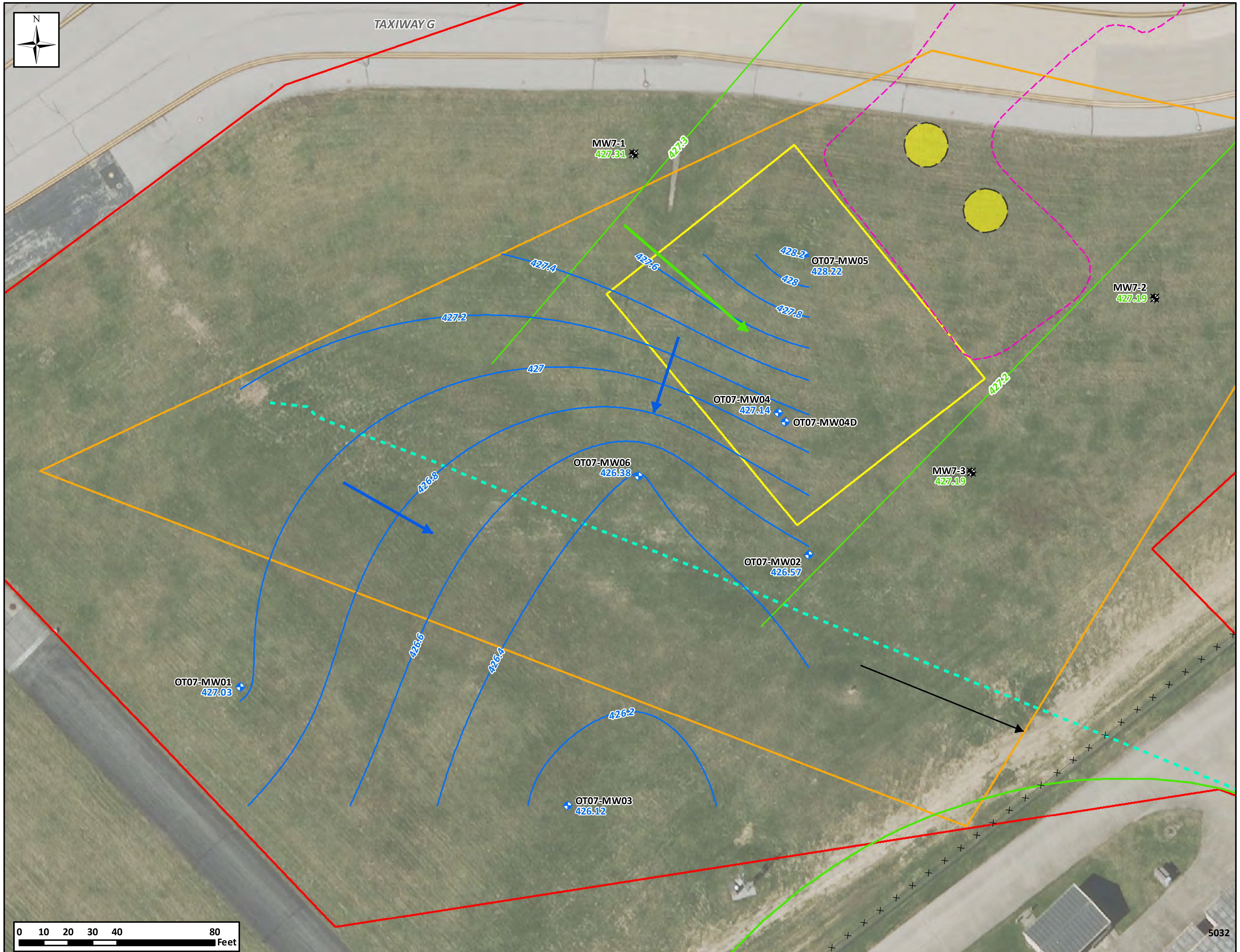
FIGURE 3-1

Soil Boring Locations June 2017, June 2020, June 2021, June 2022, and June 2023

2023 Annual MNA Report
Site OT007
Scott Air Force Base, Illinois



Map Key:

- Site Boundary
- Shallow Monitoring Well
- 1988 Abandoned Deep Monitoring Well (Not Surveyed)
- 2023 Shallow Potentiometric Contour (ft msl)
- 1988 Deep Potentiometric Contour (ft msl)
- Shallow Groundwater Flow Direction
- Deep Groundwater Flow Direction
- Stormwater/Surface Water Flow Direction
- 1996 AST Soil Excavation (Burn & McDonnell, 1997)
- CSIR Study Area
- OWS-14 Study Area
- Approximate Extent of the Former Sludge Weathering Lagoon Remedial Excavation (ERM, 1992)
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FIGURE 5-1

Potentiometric Surface Map
June 2023

2023 Annual MNA Report
Site OT007
Scott Air Force Base, Illinois

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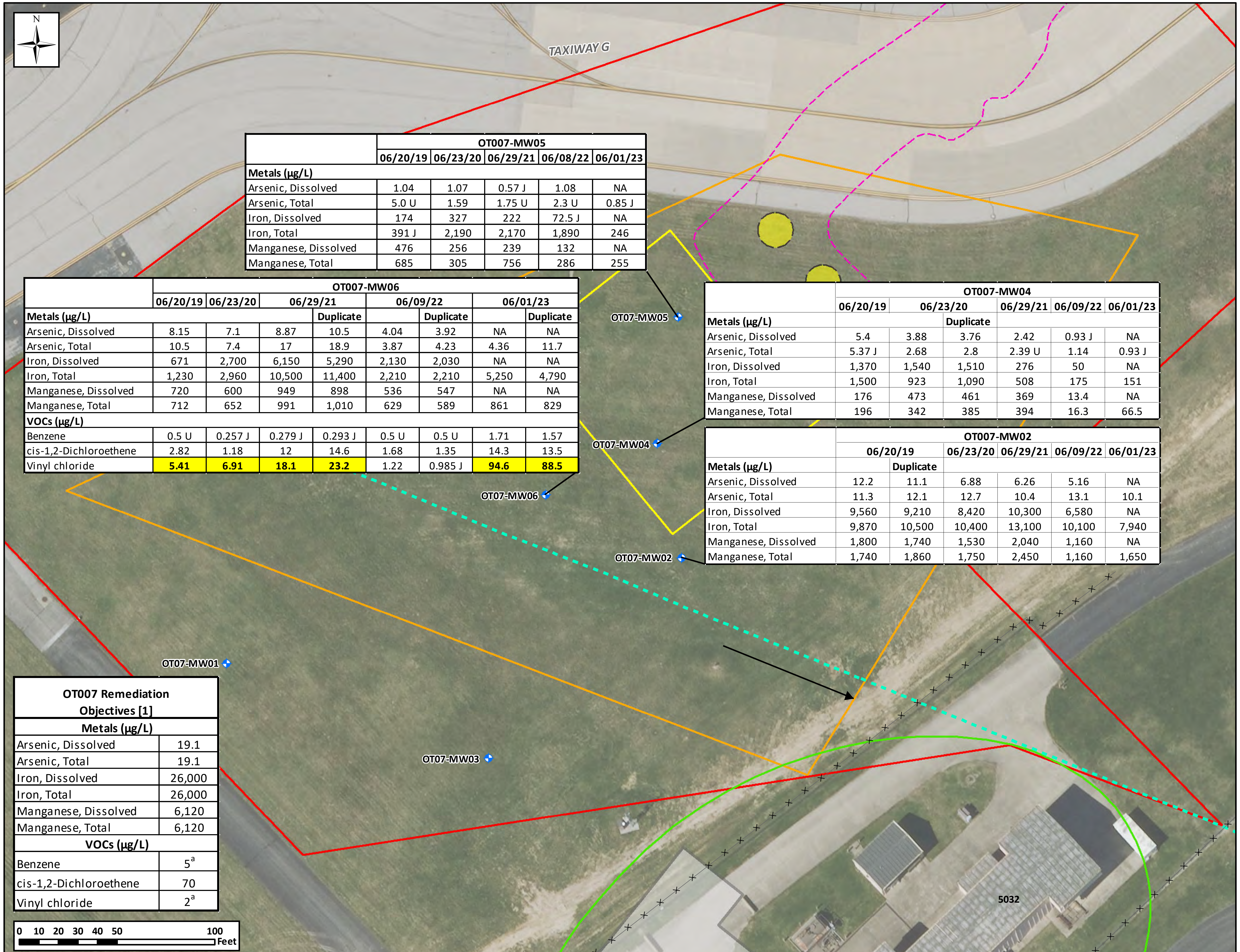

	OT007-MW05				
	06/20/19	06/23/20	06/29/21	06/08/22	06/01/23
Metals (µg/L)					
Arsenic, Dissolved	1.04	1.07	0.57 J	1.08	NA
Arsenic, Total	5.0 U	1.59	1.75 U	2.3 U	0.85 J
Iron, Dissolved	174	327	222	72.5 J	NA
Iron, Total	391 J	2,190	2,170	1,890	246
Manganese, Dissolved	476	256	239	132	NA
Manganese, Total	685	305	756	286	255

	OT007-MW06							
	06/20/19	06/23/20	06/29/21		06/09/22		06/01/23	
Metals (µg/L)				Duplicate		Duplicate		Duplicate
Arsenic, Dissolved	8.15	7.1	8.87	10.5	4.04	3.92	NA	NA
Arsenic, Total	10.5	7.4	17	18.9	3.87	4.23	4.36	11.7
Iron, Dissolved	671	2,700	6,150	5,290	2,130	2,030	NA	NA
Iron, Total	1,230	2,960	10,500	11,400	2,210	2,210	5,250	4,790
Manganese, Dissolved	720	600	949	898	536	547	NA	NA
Manganese, Total	712	652	991	1,010	629	589	861	829
VOCs (µg/L)								
Benzene	0.5 U	0.257 J	0.279 J	0.293 J	0.5 U	0.5 U	1.71	1.57
cis-1,2-Dichloroethene	2.82	1.18	12	14.6	1.68	1.35	14.3	13.5
Vinyl chloride	5.41	6.91	18.1	23.2	1.22	0.985 J	94.6	88.5

	OT007-MW04					
	06/20/19	06/23/20		06/29/21	06/09/22	06/01/23
Metals (µg/L)			Duplicate			
Arsenic, Dissolved	5.4	3.88	3.76	2.42	0.93 J	NA
Arsenic, Total	5.37 J	2.68	2.8	2.39 U	1.14	0.93 J
Iron, Dissolved	1,370	1,540	1,510	276	50	NA
Iron, Total	1,500	923	1,090	508	175	151
Manganese, Dissolved	176	473	461	369	13.4	NA
Manganese, Total	196	342	385	394	16.3	66.5

	OT007-MW02					
	06/20/19		06/23/20	06/29/21	06/09/22	06/01/23
Metals (µg/L)		Duplicate				
Arsenic, Dissolved	12.2	11.1	6.88	6.26	5.16	NA
Arsenic, Total	11.3	12.1	12.7	10.4	13.1	10.1
Iron, Dissolved	9,560	9,210	8,420	10,300	6,580	NA
Iron, Total	9,870	10,500	10,400	13,100	10,100	7,940
Manganese, Dissolved	1,800	1,740	1,530	2,040	1,160	NA
Manganese, Total	1,740	1,860	1,750	2,450	1,160	1,650

OT007 Remediation Objectives [1]	
Metals (µg/L)	
Arsenic, Dissolved	19.1
Arsenic, Total	19.1
Iron, Dissolved	26,000
Iron, Total	26,000
Manganese, Dissolved	6,120
Manganese, Total	6,120
VOCs (µg/L)	
Benzene	5[a]
cis-1,2-Dichloroethene	70
Vinyl chloride	2[a]

Map Key:

- Site Boundary
- Shallow Monitoring Well
- 1996 AST Soil Excavation (Burn & McDonnell, 1997)
- CSIR Study Area
- OWS-14 Study Area
- Approximate Extent of the Former Sludge Weathering Lagoon Remedial Excavation (ERM, 1992)
- Former AST
- Former Concrete Pad
- Historical Drainage Ditch
- Stormwater/Surface Water Flow Direction
- Fence

Notes:
Shaded results indicate concentrations above the Project Action Limit (PAL) or OT007 Remediation Objective (RO).
[1] = Remediation objectives for cis-1,2-dichloroethene and vinyl chloride are based on the Class I groundwater values in Tiered Approach to Corrective Action Objectives (TACO) (35 Illinois Administrative Code 742) Tier 1, Appendix B, Table E. Remediation objectives for arsenic, iron, and manganese are based on the highest maximum background values in groundwater for Scott AFB established in the Final Basewide Groundwater Dataset Evaluation Technical Memorandum (HGL, 2012).
a = Illinois Administrative Code Title 35, Part 742, TACO, Appendix B, Table E, Tier I, Groundwater Remediation Objectives for the Groundwater Component of the Groundwater Ingestion Route (IEPA, 2013).
• Volatile organic compounds (VOCs) analyzed by method SW 8260B.
• Metals analyzed by method SW 6020B.
Abbreviation Key:
µg/L = micrograms per liter
AST = Aboveground storage tank
CSIR = Comprehensive Site Investigation Report
J = The positive result reported for this analyte is a quantitative estimate.
NA = not analyzed
OWS = Oil-Water Separator
U = This analyte was not detected in the sample. The numeric value represents the sample Limit of Detection (LOD)."

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Date: October 2023

FIGURE 5-2

Selected Results and Groundwater Exceedances June 2019 - June 2023

2023 Annual MNA Report
Site OT007
Scott Air Force Base, Illinois

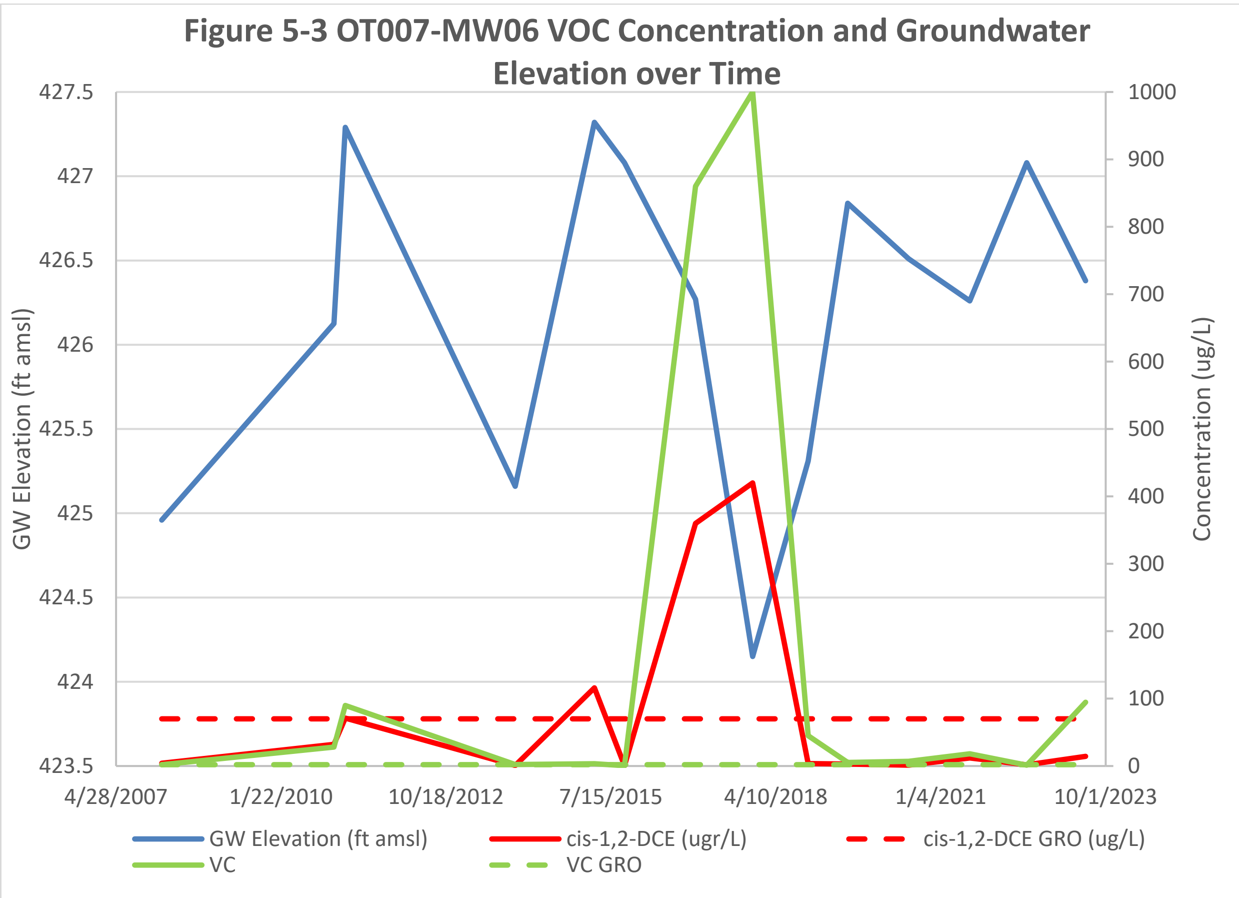
Figure 5-3 OT007-MW06 VOC Concentration and Groundwater Elevation over Time
GW Elevation (ft amsl)
Concentration (ug/L)
427.5
427
426.5
426
425.5
425
424.5
424
423.5
1000
900
800
700
600
500
400
300
200
100
0
4/28/2007
1/22/2010
10/18/2012
7/15/2015
4/10/2018
1/4/2021
10/1/2023
GW Elevation (ft amsl)
cis-1,2-DCE (ugr/L)
cis-1,2-DCE GRO (ug/L)
VC
VC GRO

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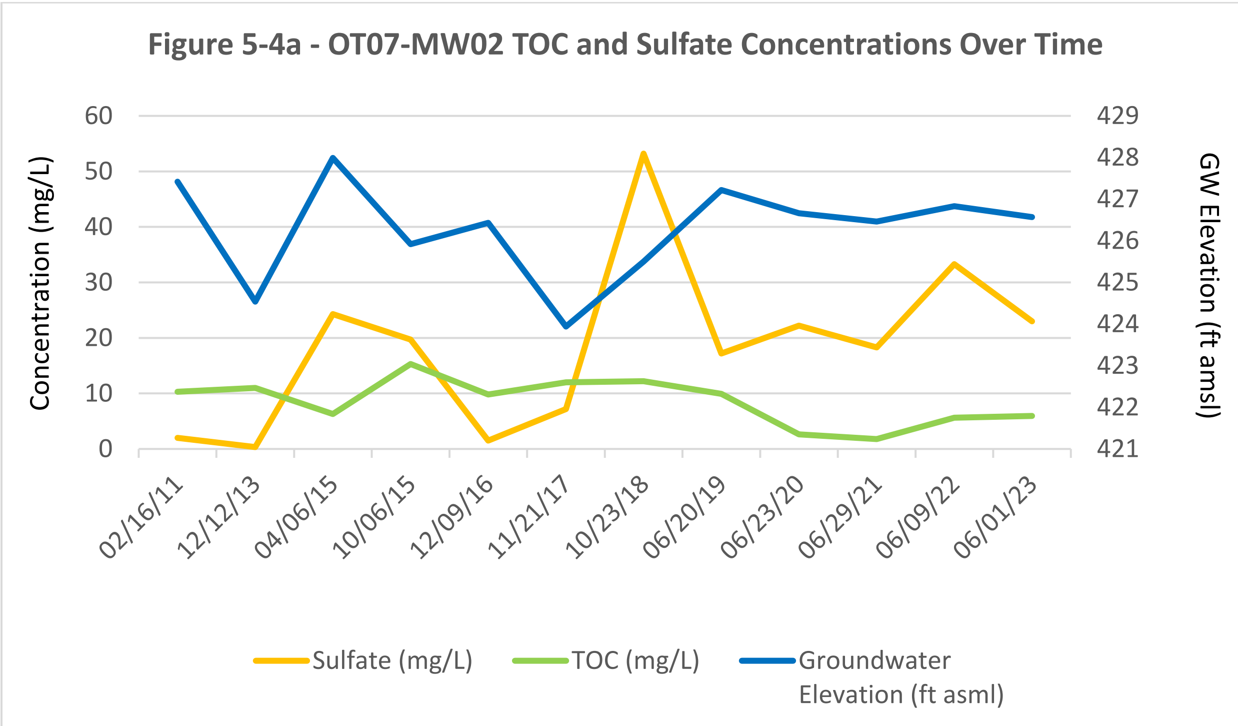
Figure 5-4a - OT07-MW02 TOC and Sulfate Concentrations Over Time
Concentration (mg/L)
GW Elevation (ft amsl)
60
50
40
30
20
10
0
429
428
427
426
425
424
423
422
421
02/16/11
12/12/13
04/06/15
10/06/15
12/09/16
11/21/17
10/23/18
06/20/19
06/23/20
06/29/21
06/09/22
06/01/23
Sulfate (mg/L)
TOC (mg/L)
Groundwater Elevation (ft asml)

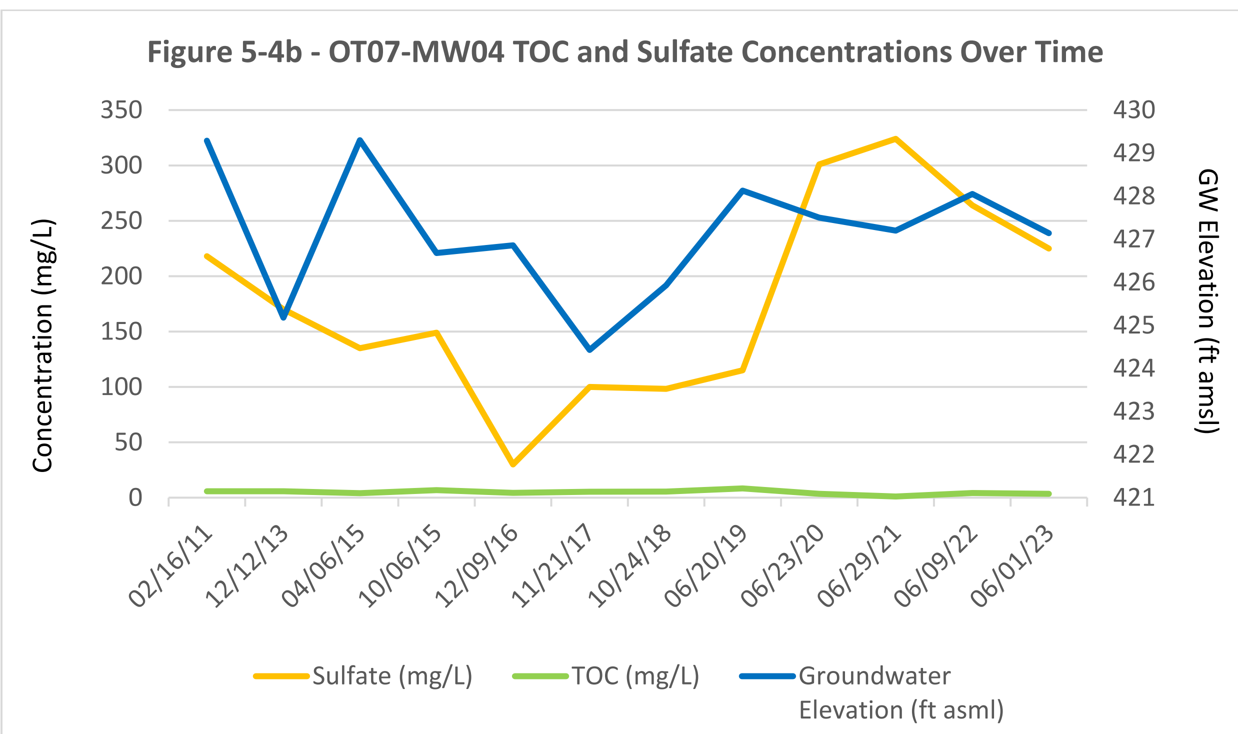
Figure 5-4b - OT07-MW04 TOC and Sulfate Concentrations Over Time
Concentration (mg/L)
GW Elevation (ft amsl)
350
300
250
200
150
100
50
0
430
429
428
427
426
425
424
423
422
421
02/16/11
12/12/13
04/06/15
10/06/15
12/09/16
11/21/17
10/24/18
06/20/19
06/23/20
06/29/21
06/09/22
06/01/23
Sulfate (mg/L)
TOC (mg/L)
Groundwater Elevation (ft asml)

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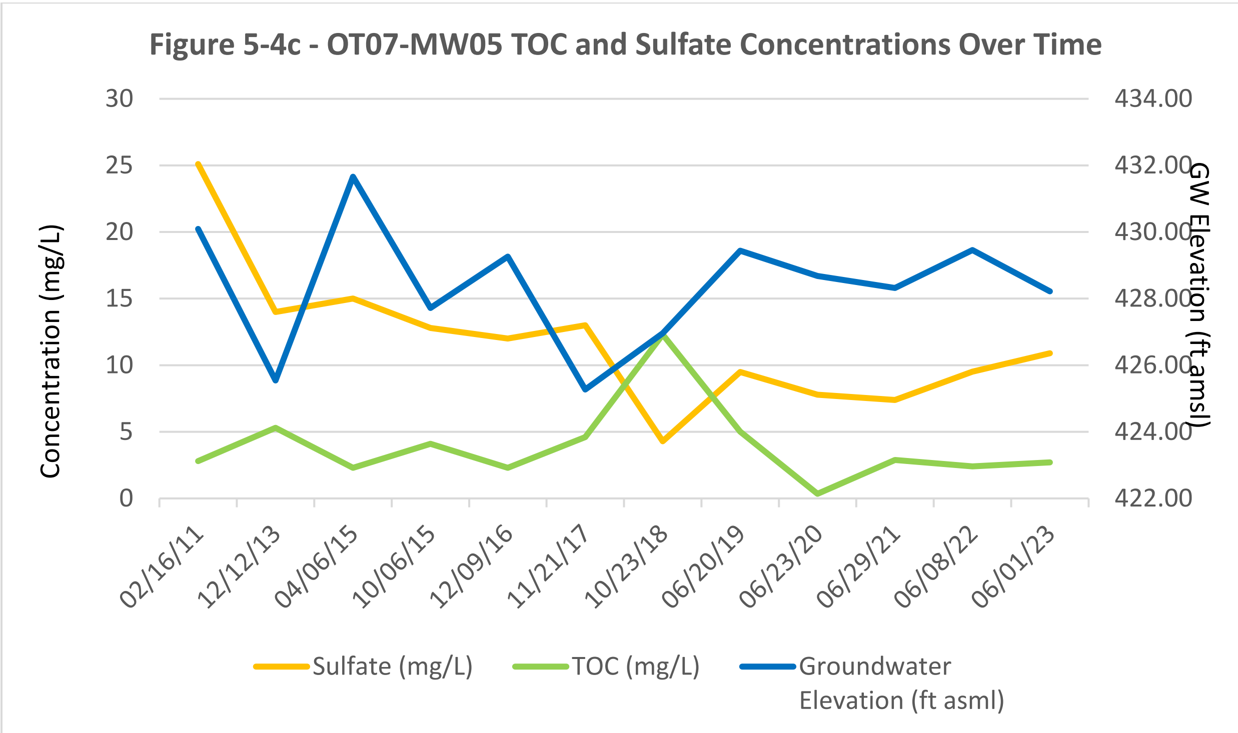
Figure 5-4c - OT07-MW05 TOC and Sulfate Concentrations Over Time
Concentration (mg/L)
GW Elevation (ft amsl)
30
25
20
15
10
5
0
434.00
432.00
430.00
428.00
426.00
424.00
422.00
02/16/11
12/12/13
04/06/15
10/06/15
12/09/16
11/21/17
10/23/18
06/20/19
06/23/20
06/29/21
06/08/22
06/01/23
Sulfate (mg/L)
TOC (mg/L)
Groundwater Elevation (ft asml)

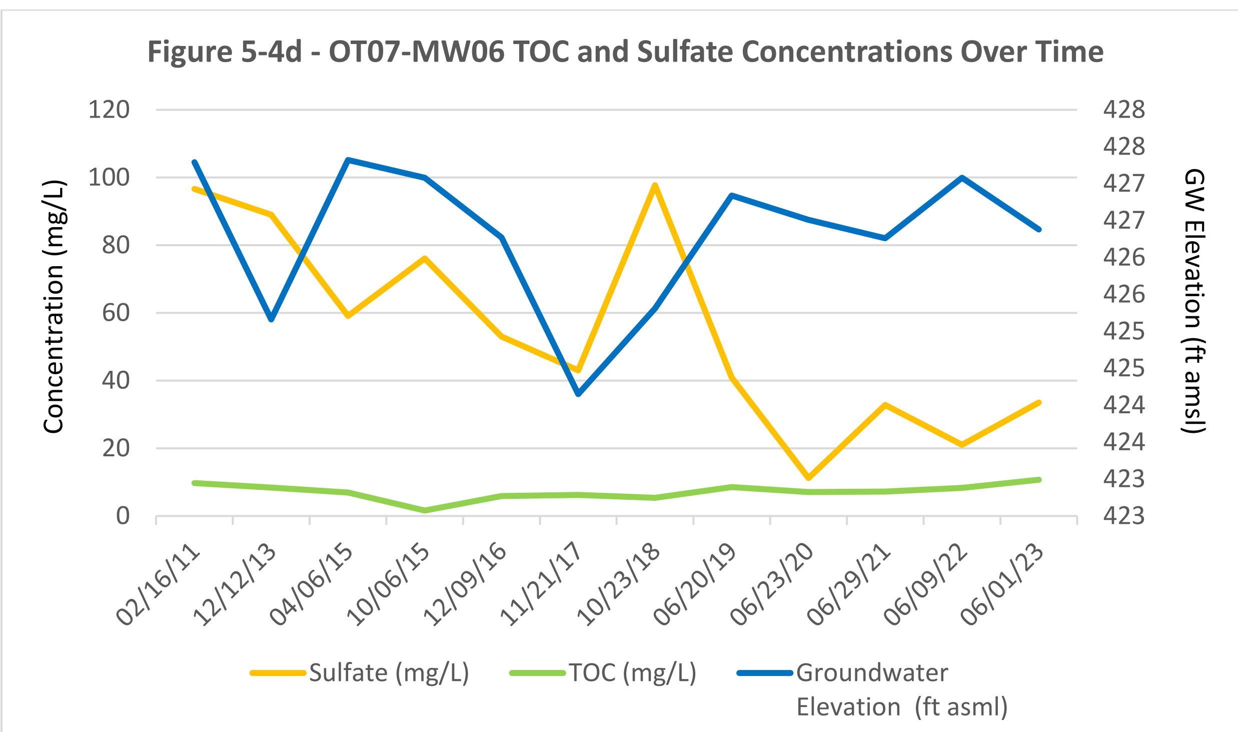
Figure 5-4d - OT07-MW06 TOC and Sulfate Concentrations Over Time
Concentration (mg/L)
GW Elevation (ft amsl)
120
100
80
60
40
20
0
428
428
427
427
426
426
425
425
424
424
423
423
02/16/11
12/12/13
04/06/15
10/06/15
12/09/16
11/21/17
10/23/18
06/20/19
06/23/20
06/29/21
06/09/22
06/01/23
Sulfate (mg/L)
TOC (mg/L)
Groundwater Elevation (ft asml)

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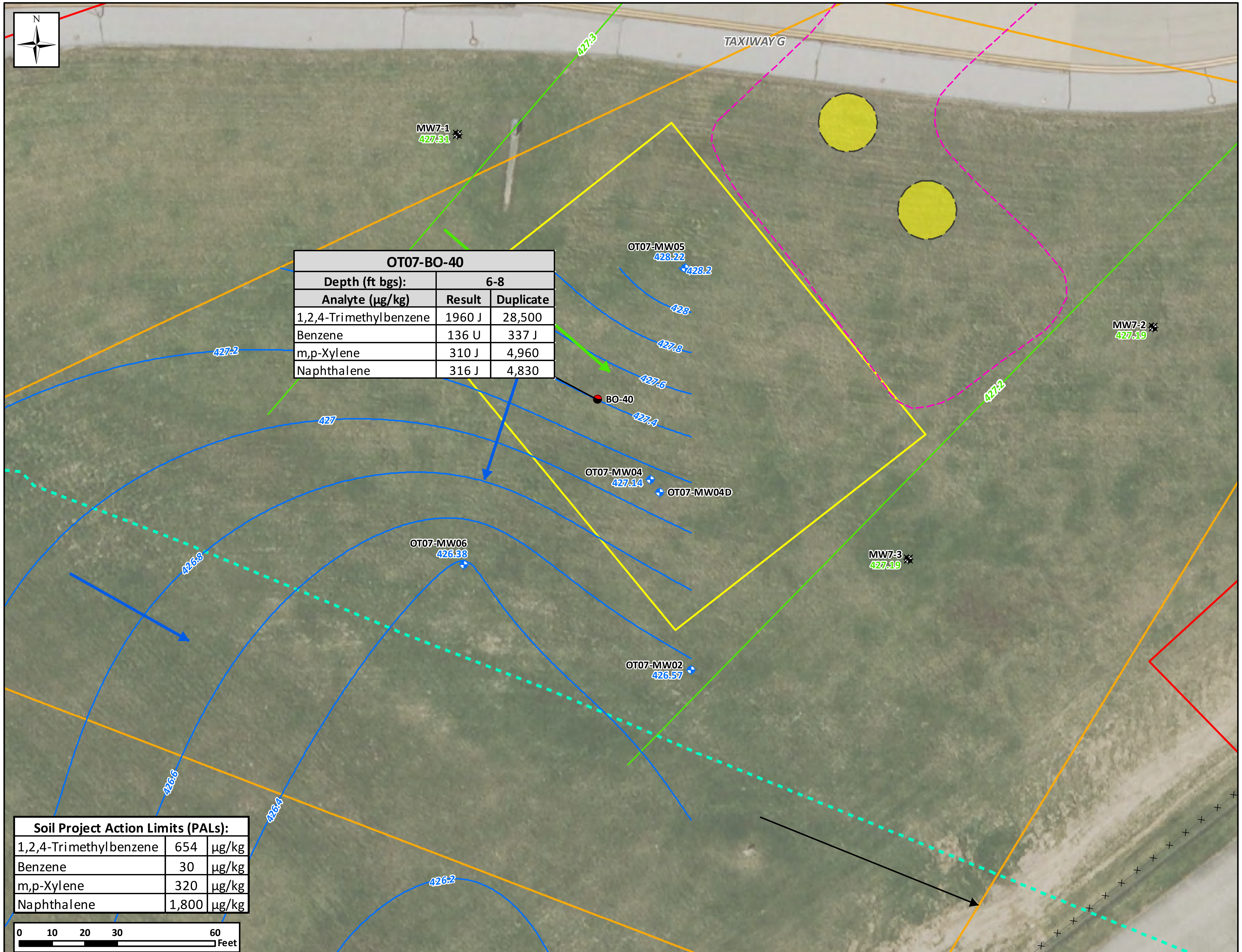

OT07-BO-40		
Depth (ft bgs):	6-8	
Analyte (µg/kg)	Result	Duplicate
1,2,4-Trimethylbenzene	1960 J	28,500
Benzene	136 U	337 J
m,p-Xylene	310 J	4,960
Naphthalene	316 J	4,830

Soil Project Action Limits (PALs):		
1,2,4-Trimethylbenzene	654	µg/kg
Benzene	30	µg/kg
m,p-Xylene	320	µg/kg
Naphthalene	1,800	µg/kg

Map Key:

- Site Boundary
- 2023 Soil Boring Location
- Shallow Monitoring Well
- 1988 Abandoned Deep Monitoring Well (Not Surveyed)
- 2022 Shallow Potentiometric Contour (ft amsl)
- 1988 Deep Potentiometric Contour (ft amsl)
- Shallow Groundwater Flow Direction
- Deep Groundwater Flow Direction
- Stormwater/Surface Water Flow Direction
- 1996 AST Soil Excavation (Burn & McDonnell, 1997)
- CSIR Study Area
- Approximate Extent of the Former Sludge Weathering Lagoon Remedial Excavation (ERM, 1992)
- Former AST
- Historical Drainage Ditch
- Fence

Notes:
- Soil results are only shown for those analytes that exceed the Project Action Limits (PALs).
- Volatile organic compounds (VOCs) analyzed by method SW 8260B.

Abbreviation Key:
µg/kg = micrograms per kilogram
AST = Aboveground storage tank
CSIR = Comprehensive Site Investigation Report
ft amsl = feet above mean sea level
ft bgs = feet below ground surface
J = The positive result reported for this analyte is a quantitative estimate

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Date: October 2023

FIGURE 5-5

Soil Boring Results June 2023

2023 Annual MNA Report
Site OT007
Scott Air Force Base, Illinois

C:\Users\bbouton\OneDrive - Plexus Scientific Corp\Documents\GIS_PRJ_BU\Scott AFB PBR 2015\Maps\OT007\2023 MNA Report\Figure 5-5 - June 2023 Soil Boring Results.mxd

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TABLES

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Table 1-1 - Site OT007 Groundwater Remediation Objectives

Analyte	Illinois Class I Groundwater Value[1] (µg/L)	Range of Scott AFB Basewide Concentrations[2] (µg/L)	Groundwater Remediation Objective[3] (µg/L)
cis-1,2-Dichloroethene	70	N/A	70
Vinyl Chloride	2	N/A	2
Arsenic	10	2.5 – 19.1	19.1
Iron	5,000	33.7 – 26,000	26,000
Manganese	150	1.9 – 6,120	6,120

Notes:

1) Tiered Approach to Corrective Action Objectives (35 Ilinois Administrative Code 742) Tier 1, Appendix B, Table E

2) Final Groundwater Background Dataset Evaluation Technical Memorandum (HGL, 2012).

3) Final CSIR/ROR/RAP OT-07 Former Sludge Weathering Lagoon Bulk Fuel Facility (HGL, 2013).

µg/L = micrograms per liter

N/A = not applicable

Table 2-1 – Site OT007 Historical Synoptic Water Levels

Monitoring Well	DTW 12/1998	GWE 12/1998	DTW 2/2011	GWE 2/2011	DTW 3/2011	GWE 3/2011
OT07-MW01	8.42	427.10	NM	NM	4.17	431.35
OT07-MW02	7.75	426.45	6.72	427.48	6.95	427.25
OT07-MW03	5.77	428.89	NM	NM	3.72	430.94
OT07-MW04	7.75	427.70	5.99	429.26	5.60	429.85
OT07-MW04D	8.90	426.30	5.99	429.21	6.55	428.65
OT07-MW05	7.30	428.59	5.80	430.09	4.41	431.48
OT07-MW06	8.55	427.05	8.01	427.59	7.53	428.07

Notes:
All elevations measured in feet below ground surface (bgs) from the top of the well casing.
DTW = depth to water
GWE = groundwater elevation
NM = not measured

Table 3-1 - Site OT007 Water Quality Parameters

Sample Location	Date Sampled	pH	Temperature (°C)	Specific Conductance (µS/cm)	ORP (mV)	DO (mg/L)	Turbidity (NTU)[2]
OT07-MW02	12/12/2013[1]	6.81	4.03	1750	221	4.00	53.8
	4/6/2015	5.09	20.01	608	137	7.83	211
	10/5/2015	6.49	19.83	1020	-145	0.00	0.2
	12/9/2016	6.6	14.54	1500	-127	0.93	55.8
	11/21/2017	6.80	16.07	1350	-119	0.00	54.7/72.6
	10/23/2018	6.84	20.81	1320	-113	5.36	45.1/44.9
	6/20/2019	6.57	15.11	256	-68	1.16	0.5/10.53
	6/23/2020	6.10	15.1	338	-95	0.00	15.9
	6/29/2021	6.99	18.76	395	-124	0.00	29.8/17.9
	6/9/2022	6.25	19.69	323	-69	0.00	70.6/41.00
	6/1/2023	6.34	18.51	337	-36	0.07	10/8.1
OT07-MW04	12/12/2013[1]	6.95	6.05	859	115	13.54	650
	4/6/2015	6.75	10.58	722	125	3.53	15.7
	10/5/2015	6.42	18.29	799	126	0.00	47
	12/9/2016	6.15	8.15	881	9	0.71	0.0
	11/21/2017	6.61	14.57	745	15	0.84	55.0/70.7
	10/24/2018	6.74	16.45	796	-64	0.00	18.3/9.7
	6/20/2019	6.75	15.26	449	58	0.00	0.0/3.57
	6/23/2020	6.07	14.33	830	75	0.00	3.4
	6/29/2021	6.69	20.07	880	90	0.14	77.9/36.0
	6/9/2022	6.31	15.50	763	109	0.00	11.9/3.69
	6/1/2023	6.86	18.54	739	105	0.00	5.14/2.9
OT07-MW05	12/12/2013[1]	6.67	16.17	403	187	9.35	47.6
	4/8/2015	6.21	12.15	381	246	2.94	8.6
	10/6/2015	5.64	19.26	309	84	0.00	0
	12/9/2016	5.74	13.2	241	118	0.80	0.8
	11/21/2017	6.01	15.16	341	0	0.00	14.5/27.2
	10/23/2018	6.25	17.22	308	72	5.26	9.6/6.48
	6/20/2019	6.60	15.01	190	75	3.11	0.0/6.96
	6/23/2020	6.03	13.77	270	130	0.00	21
	6/29/2021	6.57	21.00	260	115	3.29	32.3/25.0
	6/8/2022	5.89	15.50	266	116	0.00	30.0/24.63
	6/1/2023	6.28	17.45	226	89	0.00	4.97/1.6
OT07-MW06	12/12/2013[1]	6.86	5.14	860	63	5.00	31.7
	4/6/2015	5.88	11.49	419	78	2.33	10.3
	10/6/2015	5.79	21.15	389	159	0.00	0
	12/9/2016	6.31	15.85	749	-29	0.00	17.2
	11/21/2017	6.56	17.24	829	-60	0.00	2.0/0
	10/23/2018	6.67	19.15	627	-91	3.82	4.9/2.60
	6/20/2019	6.06	16.70	315	61	0.00	0.0/4.73
	6/23/2020	6.03	15.66	272	-50	0.00	4.1
	6/29/2021	6.40	27.55	281	-43	0.28	18.2/9.01
	6/9/2022	5.92	22.26	195	32	0.00	0.0/2.08
	6/1/2023	6.56	20.54	376	-54	0.00	5.51/4.8

Notes:

Parameters after purging stabilization at time of sample collection
[1] ORP and turbidity measured on February 19, 2014
[2] 2017-2021 Turbidity readings from the Lamotte 2020WE/Horiba U-52
°C = degrees Celsius
µS/cm = micro-Siemens per centimeter
DO = dissolved oxygen
mg/L = milligrams per liter
mV = millivolts
NTU = nephelometric turbidity unit
ORP = oxidation-reduction potential

Table 5-1 - Site OT007 Monitoring Well Construction Details

Well ID	Date Installed	Status	Top of Casing Elevation (ft MSL)	Depth to Water (ft BTOC)	Groundwater Elevation (ft msl)	Total Depth (ft btoc)	Well Diameter (inches)	Screen Length (ft)	Screen Zone (ft btoc)	Screen Zone Elevation (ft MSL)
MW7-1	Oct-88	Abandoned	438.26	NA	N/A	32.10	2	10	22.10 - 32.10	416.16 - 406.16
MW7-2	Oct-88	Abandoned	437.54	NA	N/A	32.17	2	10	22.17 - 32.17	415.37 - 405.37
MW7-3	Oct-88	Abandoned	437.43	NA	N/A	31.88	2	10	21.88 - 31.88	415.55 - 405.55
OT07-MW01	Dec-06	Active	435.52	7.09	428.43	18.55	2	10	8.55 - 18.55	426.97 - 416.97
OT07-MW02	Dec-06	Active	434.20	7.31	426.89	18.56	2	10	8.56 - 18.56	425.64 - 415.64
OT07-MW03	Dec-06	Active	434.66	7.64	427.02	19.12	2	10	9.12 - 19.12	425.54 - 415.54
OT07-MW04	Feb-07	Active	435.45	7.23	428.22	17.00	2	10	7.00 - 17.00	428.45 - 418.45
OT07-MW04D	Oct-10	Active	435.20	7.87	427.33	32.00	2	10	22 - 32	413.40 - 403.40
OT07-MW05	Jan-08	Active	435.89	6.43	429.46	14.48	2	10	4.48 - 14.48	431.41 - 421.41
OT07-MW06	Jan-08	Active	435.60	8.28	427.32	15.34	2	10	5.34 - 15.34	430.26 - 420.26

Notes:

1) Well completed as a flush-mount. Ground surface elevation likely within 1 foot of top of casing elevation.

ft = feet

ft btoc = feet below top of casing

ft MSL = feet Mean Sea Level

N/A = not available

Table 5-2 - Groundwater Elevations at Site OT007

Monitoring Well Identification	Date Measured	Top of Casing Elevation (ft amsl)	Depth to Water (ft btoc)	Groundwater Elevation (ft amsl)
OT07-MW01	3/1/2011	435.43	4.17	431.26
	12/12/2013		NM	NM
	3/29/2015		NM	NM
	9/10/2015		NM	NM
	12/9/2016		8.78	426.65
	11/21/2017		10.94	424.49
	10/23/2018		9.63	425.80
	6/20/2019		7.69	427.74
	6/23/2020		8.10	427.33
	6/29/2021		8.47	426.96
	6/8/2022		7.09	428.34
	6/1/2023		8.40	427.03
OT07-MW02	2/16/2011	434.14	6.72	427.42
	12/3/2013		9.60	424.54
	3/29/2015		6.15	427.99
	9/10/2015		8.22	425.92
	12/9/2016		7.71	426.43
	11/21/2017		10.2	423.94
	10/23/2018		8.64	425.50
	6/20/2019		6.92	427.22
	6/23/2020		7.48	426.66
	6/29/2021		7.68	426.46
	6/8/2022		7.31	426.83
	6/1/2023		7.57	426.57
OT07-MW03	3/1/2011	434.57	3.72	430.85
	12/12/2013		NM	NM
	3/29/2015		NM	NM
	9/10/2015		NM	NM
	12/9/2016		8.35	426.22
	11/21/2017		12.29	422.28
	10/23/2018		9.26	425.31
	6/20/2019		6.09	428.48
	6/23/2020		6.75	427.82
	6/29/2021		7.51	427.06
	6/8/2022		7.64	426.93
	6/1/2023		8.45	426.12
OT07-MW04	2/16/2011	435.28	5.99	429.29
	12/3/2013		10.1	425.18
	3/29/2015		5.98	429.30
	9/10/2015		8.6	426.68
	12/9/2016		8.42	426.86
	11/21/2017		10.85	424.43
	10/24/2018		9.35	425.93
	6/20/2019		7.15	428.13
	6/23/2020		7.78	427.50
	6/29/2021		8.08	427.20
	6/8/2022		7.23	428.05
	6/1/2023		8.14	427.14

Table 5-2 - Groundwater Elevations at Site OT007

Monitoring Well Identification	Date Measured	Top of Casing Elevation (ft amsl)	Depth to Water (ft btoc)	Groundwater Elevation (ft amsl)
OT07-MW04D	2/16/2011	435.20	7.75	427.45
	12/3/2013		11.87	423.33
	3/29/2015		NM	NM
	9/10/2015		NM	NM
	12/9/2016		NM	NM
	11/21/2017		11.72	423.48
	10/23/2018		10.15	425.05
	6/20/2019		7.65	427.55
	6/23/2020		8.65	426.55
	6/29/2021		8.34	426.86
	6/8/2022		7.87	427.33
	6/1/2023		8.91	426.29
OT07-MW05	2/16/2011	435.89	5.80	430.09
	12/3/2013		10.35	425.54
	3/29/2015		4.23	431.66
	9/10/2015		8.17	427.72
	12/9/2016		6.63	429.26
	11/21/2017		10.62	425.27
	10/23/2018		8.93	426.96
	6/20/2019		6.45	429.44
	6/23/2020		7.21	428.68
	6/29/2021		7.57	428.32
	6/8/2022		6.43	429.46
	6/1/2023		7.67	428.22
OT07-MW06	2/16/2011	435.36	8.07	427.29
	12/3/2013		10.20	425.16
	3/29/2015		8.04	427.32
	9/10/2015		8.28	427.08
	12/9/2016		9.09	426.27
	11/21/2017		11.21	424.15
	10/23/2018		10.05	425.31
	6/20/2019		8.52	426.84
	6/23/2020		8.85	426.51
	6/29/2021		9.10	426.26
	6/8/2022		8.28	427.08
	6/1/2023		8.98	426.38

Notes:

amsl = above mean sea level
ft = feet

btoc = below top of casing
NM = not measured

Table 5-3 - OT007 Summary of June 2023 Groundwater Results

		Sample Location	OT07-MW02									
		Sample Date	12/12/2013	4/8/2015	10/5/2015	12/9/2016	11/21/2017	11/21/2017	10/23/2018	6/20/2019	6/20/2019	6/23/2020
Analyte	Scott AFB Project Action Limit	Sample Type / OT007 Remediation Objectives[1]	Normal	Normal	Normal	Normal	Normal	Duplicate	Normal	Normal	Duplicate	Normal
Metals (SW 6020B) (µg/L)												
Arsenic, Dissolved	--	19.1	95	14.9	40.4	89	71	61	17.5	12.2	11.1	6.88
Arsenic, Total	--	19.1	95	13.6	35.7	79	61	66	17.1	11.3	12.1	12.7
Iron, Dissolved	--	26,000	38,000	11,400	33,400	35,000	37,000	37,000	24,100	9,560	9,210	8,420
Iron, Total	--	26,000	46,000	19,200	36,900	39,000	35,000	37,000	25,500	9,870	10,500	10,400
Manganese, Dissolved	--	6,120	12,000	2,940	6,170	8,400	8,200	7,700	6,350	1,800	1,740	1,530
Manganese, Total	--	6,120	12,000	2,320	5,850	8,500	7,200	7,500	6,750	1,740	1,860	1,750
VOCs (SW 8260B) (µg/L)												
1,2-Dichloroethene, Total	--	--	NA	NA	NA	0.2 U	0.2 U	0.2 U	1 U	1.0 U	1.0 U	1.0 U
Acetone	6300[a]	--	10 U	5 U	5 U	6.4 U	6.4 U	6.4 U	1 U	1.0 U	1.0 U	1.0 U
Benzene	5[a]	--	0.4 U	1 U	1 U	0.4 U	0.4 U	0.4 U	0.5 U	0.5 U	0.5 U	0.5 U
cis-1,2-Dichloroethene	--	70	1 U	1 U	1 U	0.4 U	0.4 U	0.4 U	0.5 U	0.5 U	0.5 U	0.5 U
Methylene Chloride	5[a]	--	0.38 U	5 U	5 U	0.8 U	0.8 U	0.8 U	0.5 U	0.5 U	0.5 U	0.703 U
Methyl tert-butyl ether	14[b]	--	0.39 F	1 U	1 U	0.8 U	0.8 U	0.8 U	0.5 U	0.5 U	0.5 U	0.5 U
Toluene	1000[a]	--	1 U	1 U	1 U	0.4 U	0.4 U	0.4 U	0.5 U	0.5 U	0.5 U	0.5 U
trans-1,2-Dichloroethene	100[a]	--	1 U	1 U	1 U	0.4 U	0.4 U	0.4 U	0.5 U	0.5 U	0.5 U	0.5 U
Trichloroethene	5[a]	--	1 U	1 U	1 U	0.4 U	0.4 U	0.4 U	0.5 U	0.5 U	0.5 U	0.5 U
Vinyl chloride	--	2	1 U	1 U	1 U	0.2 U	0.2 U	0.2 U	0.5 U	0.5 U	0.5 U	0.5 U
Dissolved Gases (RSK-175) (µg/L)												
Ethane	--	--	5 U	1 U	1 U	1.5 U	1.5 U	1.5 U	0.22 U	0.22 U	0.22 U	0.22 U
Ethene	--	--	5 U	1 U	1 U	1.4 U	1.4 U	1.4 U	0.29 U	0.29 U	0.29 U	0.29 U
Methane	--	--	6,200	2 U	4,970	8,600	7,800	8,600	2,260 J	161 J	236 J	786
Propylene	--	--	NA	23.4	23	NA	NA	NA	58	NA	NA	NA
MNA Parameters (mg/L)												
Alkalinity (SM 2320B)	--	--	1,200	307	628	950 B	820	850	638	178	162	185
Chloride (SW 9056A)	200[a]	--	NA	13.1	27.9	27	27	26 U	17.3	5.87 J	5.72 J	8.77 J
Nitrate as N (SW 9056A)	10[a]	--	1 R	0.128	0.2 U	0.1 U	0.1 U	0.1 U	0.112 J	0.105 J	0.128 J	0.15 U
Nitrite as N (SW 9056A)	1[b]	--	1 R	0.2 U	0.2 U	0.1 U	0.1 U	0.1 U	0.15 U	0.15 U	0.15 U	0.15 U
Sulfate (SW 9056A)	400[a]	--	0.34 F	24.3	19.7	1.5 J	7.2	4.9 F	53.2	16.6	17.7	22.2
TOC (SW 9060A)	--	--	11	6.3	15.3	9.8	12	12	12.2	10.1	9.78	2.63

Table 5-3 - OT007 Summary of June 2023 Groundwater Results

Analyte	Scott AFB Project Action Limit	OT007 Remediation Objectives[1]	OT07-MW02			OT07-MW04							
		Sample Date	6/29/2021	6/9/2022	6/1/2023	12/12/2013	12/12/2013	4/6/2015	10/5/2015	12/9/2016	12/9/2016	11/21/2017
		Sample Type	Normal	Normal	Normal	Normal	Duplicate	Normal	Normal	Normal	Duplicate	Normal
Metals (SW 6020B) (µg/L)												
Arsenic, Dissolved	--	19.1	6.26	5.16	NA	30 U	30 U	0.76	0.95	9.5 J	10 J	15 U
Arsenic, Total	--	19.1	10.4	13.1	10.1	30 U	30 U	0.8	3.21	32	21 J	15 U
Iron, Dissolved	--	26,000	10,300	6,580	NA	540	530	100	40.1	4,700	5,000	2,300
Iron, Total	--	26,000	13,100	10,100	7940	2,700 J	1,600 J	250	6,070	33,000	22,000	3,700
Manganese, Dissolved	--	6,120	2,040	1,160	NA	480	470	7.34	717	270 J	280	800
Manganese, Total	--	6,120	2,450	1,160	1,650	440 J	270 J	11.4	1,110	480 J	430	870
VOCs (SW 8260B) (µg/L)												
1,2-Dichloroethene, Total	--	--	1.0 U	1.0 U	1 U	NA	NA	NA	NA	22 J	23	0.2 U
Acetone	6300[a]	--	1.0 U	1.0 U	1 UJ	1.9 F	2.3 F	5 U	5 U	6.4 U	3.7 J	6.4 U
Benzene	5[a]	--	0.5 U	0.5 U	0.5 U	0.4 U	0.4 U	1 U	1 U	0.4 U	0.4 U	0.4 U
cis-1,2-Dichloroethene	--	70	0.5 U	0.5 U	0.5 U	1 U	1 U	1 U	1 U	22 J	23	0.4 U
Methylene Chloride	5[a]	--	0.5 U	0.5 U	0.5 U	0.42 U	0.4 U	5 U	5 U	0.8 U	0.8 U	0.8 U
Methyl tert-butyl ether	14[b]	--	0.5 U	0.5 U	0.5 U	5 U	5 U	1 U	1 U	0.8 U	0.8 U	0.8 U
Toluene	1000[a]	--	0.5 U	0.5 U	0.5 U	1 U	1 U	1 U	1 U	0.4 U	0.21 J	0.4 U
trans-1,2-Dichloroethene	100[a]	--	0.5 U	0.5 U	0.5 U	1 U	1 U	1 U	1 U	0.17 J	0.18 J	0.4 U
Trichloroethene	5[a]	--	0.5 U	0.5 U	0.5 U	1 U	1 U	1 U	1 U	0.17 J	0.29 J	0.4 U
Vinyl chloride	--	2	0.5 U	0.5 U	0.5 U	1 U	1 U	1 U	1 U	3.2 J	3.3	0.2 U
Dissolved Gases (RSK-175) (µg/L)												
Ethane	--	--	0.22 U	0.22 U	10 U	5 U	NA	1 U	1 U	1.5 U J	1.5 U	1.5 U
Ethene	--	--	0.29 U	0.29 U	10 U	5 U	NA	1 U	1 U	0.43 M J	0.69 J M	1.4 U
Methane	--	--	1,370	137	1,480	1.7	NA	2 U	2 U	61 J	97	87
Propylene	--	--	NA	NA	NA	NA	NA	26.6	22.8	NA	NA	NA
MNA Parameters (mg/L)												
Alkalinity (SM 2320B)	--	--	332	148	175	380	NA	246	333	360 B	360 B	340
Chloride (SW 9056A)	200[a]	--	7.43	5.22	4.08	NA	NA	2.08	8.75	7.5	7.7	9.1
Nitrate as N (SW 9056A)	10[a]	--	0.1 J	0.15 U	0.15 U	1 R	NA	0.17	0.2 U	0.1 U	0.1 U	0.1 U
Nitrite as N (SW 9056A)	1[b]	--	0.15 U	0.15 U	0.15 U	1 R	NA	0.2 U	0.2 U	0.1 U	0.1 U	0.1 U
Sulfate (SW 9056A)	400[a]	--	18.3	33.3	23	170	NA	135	149	30	29	190
TOC (SW 9060A)	--	--	1.78 J	5.63	5.95	5.8	NA	4.1	6.8	4.4	5.1	5.3 J

Table 5-3 - OT007 Summary of June 2023 Groundwater Results

Analyte	Scott AFB Project Action Limit	OT007 Remediation Objectives[1]	OT07-MW04								OT07-MW05	
Sample Location			OT07-MW04								OT07-MW05	
Sample Date			10/24/2018	10/24/2018	6/20/2019	6/23/2020	6/23/2020	6/29/2021	6/9/2022	6/1/2023	12/12/2013	4/8/2015
Sample Type			Normal	Duplicate	Normal	Normal	Duplicate	Normal	Normal	Normal	Normal	Normal
Metals (SW 6020B) (µg/L)												
Arsenic, Dissolved	--	19.1	11.5	11.4	5.4	3.88	3.76	2.42	0.93 J	NA	30 U	0.86
Arsenic, Total	--	19.1	11.4	11.6	5.37 J	2.68	2.8	2.39 U	1.14	0.93 J	58	1.36
Iron, Dissolved	--	26,000	10,100	10,300	1,370	1,540	1,510	276	50	NA	4,700	100
Iron, Total	--	26,000	9,960	10,400	1,500	923	1,090	508	175	151	99,000	734
Manganese, Dissolved	--	6,120	1,130	1,120	176	473	461	369	13.4	NA	15,000	12.2
Manganese, Total	--	6,120	1,200	1,180	196	342	385	394	16.3	66.5	31,000	76.7
VOCs (SW 8260B) (µg/L)												
1,2-Dichloroethene, Total	--	--	1 U	1 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U	NA	NA
Acetone	6300[a]	--	24.3 J	13.9 J	1.0 U	0.615 U	1.0 U	2.3 J	1.0 U	1.0 UJ	2.1 F	5 U
Benzene	5[a]	--	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.4 U	1 U
cis-1,2-Dichloroethene	--	70	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	1 U	1 U
Methylene Chloride	5[a]	--	0.5 U	1.19 U	0.5 U	0.806 U	0.446 U	0.5 U	0.5 U	0.5 U	0.42 U	5 U
Methyl tert-butyl ether	14[b]	--	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	5 U	1 U
Toluene	1000[a]	--	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	1 U	1 U
trans-1,2-Dichloroethene	100[a]	--	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	1 U	1 U
Trichloroethene	5[a]	--	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	1 U	1 U
Vinyl chloride	--	2	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	1 U	1 U
Dissolved Gases (RSK-175) (µg/L)												
Ethane	--	--	0.22 U	0.22 U	0.22 U	0.22 U	0.22 U	0.22 U	0.22 U	10 U	5 U	1 U
Ethene	--	--	0.29 U	0.29 U	0.29 U	0.29 U	0.29 U	0.29 U	0.29 U	10 U	5 U	1 U
Methane	--	--	0.455 U	1.09 U	1.56 U	4.49	5.12	1.1 U	0.883 J	32.8	18	2 U
Propylene	--	--	63	74	NA	NA	NA	NA	NA	NA	NA	24.6
MNA Parameters (mg/L)												
Alkalinity (SM 2320B)	--	--	367	375	232	240	233	255	284	279	220	187
Chloride (SW 9056A)	200[a]	--	8.64	2.56	4.98	4.14	4.04	3.5 J	2.48	2.37		3.42
Nitrate as N (SW 9056A)	10[a]	--	0.15 U	0.15 U	0.104 J	0.3 U	0.3 U	0.3 U	0.3 U	0.158 J	0.054 F	4.31
Nitrite as N (SW 9056A)	1[b]	--	0.15 U	0.15 U	0.15 U	0.3 U	0.3 U	0.3 U	0.3 U	0.15 U	1 R	0.074
Sulfate (SW 9056A)	400[a]	--	98.2	37.2	115	300	301	324	264	225	14	15
TOC (SW 9060A)	--	--	5.43	1 U	8.36	2.99	3.5	1.06 J	4.13	3.53	5.3	2.3

Table 5-3 - OT007 Summary of June 2023 Groundwater Results

Analyte	Scott AFB Project Action Limit	OT007 Remediation Objectives[1]	OT07-MW05									
		Sample Date	4/8/2015	10/6/2015	12/9/2016	11/21/2017	10/23/2018	6/20/2019	6/23/2020	6/29/2021	6/8/2022	6/1/2023
		Sample Type	Duplicate	Normal	Normal	Normal	Normal	Normal	Normal	Normal	Normal	Normal
Metals (SW 6020B) (µg/L)												
Arsenic, Dissolved	--	19.1	0.92	1.39	6 J	6.9 F	1.54	1.04	1.07	0.57 J	1.08	NA
Arsenic, Total	--	19.1	1.25	1.43	15 U	15 U	1.75	5 U	1.59	1.75 U	2.3 U	0.85 J
Iron, Dissolved	--	26,000	100	193	120	5,600	800	174	327	222	72.5 J	NA
Iron, Total	--	26,000	695	384	2,500	6,700	1,050	391 J	2,190	2,170	1,890	246
Manganese, Dissolved	--	6,120	12.9	893	520	850	545	476	256	239	132	NA
Manganese, Total	--	6,120	71.1	1,010	960	890	600	685	305	756	286	255
VOCs (SW 8260B) (µg/L)												
1,2-Dichloroethene, Total	--	--	NA	NA	0.2 U	0.2 U	1 U	1.0 U	1.0 U	1.0 U	1.0 U	1.0 U
Acetone	6300[a]	--	5 U	5 U	6.4 U	6.4 U	1 U	1.0 U	1.0 U	0.7 J	1.0 U	1.0 UJ
Benzene	5[a]	--	1 U	1 U	0.4 U	0.4 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
cis-1,2-Dichloroethene	--	70	1 U	1 U	0.4 U	0.4 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
Methylene Chloride	5[a]	--	5 U	5 U	0.8 U	0.8 U	0.5 U	0.5 U	0.587 U	0.5 U	0.5 U	0.5 U
Methyl tert-butyl ether	14[b]	--	1 U	1 U	0.8 U	0.8 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
Toluene	1000[a]	--	1 U	1 U	0.4 U	0.4 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
trans-1,2-Dichloroethene	100[a]	--	1 U	1 U	0.4 U	0.4 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
Trichloroethene	5[a]	--	1 U	1 U	0.4 U	0.4 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
Vinyl chloride	--	2	1 U	1 U	0.2 U	0.2 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
Dissolved Gases (RSK-175) (µg/L)												
Ethane	--	--		1 U	1.5 U	1.5 U	0.22 U	0.22 U	0.22 U	0.22 U	0.22 U	10 U
Ethene	--	--		1 U	1.4 U	1.4 U	0.29 U	0.29 U	0.29 U	0.29 U	0.29 U	10 U
Methane	--	--		2.84	24	22	1.09 U	1.09 U	1.1 U	1.1 U	1.1 U	6 U
Propylene	--	--	NA	28.3	NA	NA	88	NA	NA	NA	NA	NA
MNA Parameters (mg/L)												
Alkalinity (SM 2320B)	--	--		148	140 B	200	124	134	155	161	166 U	145
Chloride (SW 9056A)	200[a]	--		4.42	2.7 J	4	9.89	2.22	1.9	3.54	1.84 U	3.5
Nitrate as N (SW 9056A)	10[a]	--		0.2 U	0.26 J	0.1 U	0.15 U	0.15 U	0.15 U	0.991	0.167 J	0.186 J
Nitrite as N (SW 9056A)	1[b]	--		0.2 U	0.1 U	0.1 U	0.15 U	0.15 U	0.15 U	0.15 U	0.15 U	0.15 U
Sulfate (SW 9056A)	400[a]	--		12.8	12	13	4.29	9.49	7.79	7.39	9.52 U	10.9
TOC (SW 9060A)	--	--		4.1	2.3	4.6 J	12.3	5.01	0.345 J	2.89	2.4	2.7

Table 5-3 - OT007 Summary of June 2023 Groundwater Results

Analyte	Scott AFB Project Action Limit	OT007 Remediation Objectives[1]	OT07-MW06								
		Sample Location	OT07-MW06								
		Sample Date	12/12/2013	4/6/2015	10/6/2015	10/6/2015	12/9/2016	11/21/2017	10/23/2018	6/20/2019	6/23/2020
		Sample Type	Normal	Normal	Normal	Duplicate	Normal	Normal	Normal	Normal	Normal
Metals (SW 6020B) (µg/L)											
Arsenic, Dissolved	--	19.1	8.4 F	5.84	1	1	29	8.4 F	9.98	8.15	7.1
Arsenic, Total	--	19.1	15 F	4.62	1	1	20 J	21 F	11.1	10.5	7.4
Iron, Dissolved	--	26,000	6,400	3,190	100	100	4,900	1,600	8,860	671	2,700
Iron, Total	--	26,000	11,000	1,220	34.5	42.5	5,600	4,300 J	10,900	1,230	2,960
Manganese, Dissolved	--	6,120	2,300	922	9.3	9.41	2,100	1,300	2,020	720	600
Manganese, Total	--	6,120	2,700	374	20.5	19.9	2,300	2,300 J	2,210	712	652
VOCs (SW 8260B) (µg/L)											
1,2-Dichloroethene, Total	--	--	NA	NA	NA	NA	360	420	3.58	2.82 J	1.18 J
Acetone	6300[a]	--	3.3 F	5 U	5 U	5 U	32 U	13 U	18.1	1.0 U	1.0 U
Benzene	5[a]	--	0.58	0.725	1 U	1 U	16	28	3.14	0.5 U	0.257 J
cis-1,2-Dichloroethene	--	70	0.69 F	116	1 U	1 U	360	420 J	3.58	2.82 J	1.18
Methylene Chloride	5[a]	--	0.49 U	5 U	5 U	5 U	4 U	1.6 U	1.53 U	0.542 J	0.618 U
Methyl tert-butyl ether	14[b]	--	5 U	1 U	1 U	1 U	4 U	1.6 U	0.5 U	0.5 U	0.5 U
Toluene	1000[a]	--	1 U	1 U	1 U	1 U	2 U	0.89 J	1.15	0.5 U	0.5 U
trans-1,2-Dichloroethene	100[a]	--	1 U	1 U	1 U	1 U	20 U	8 U	0.5 U	0.5 U	0.5 U
Trichloroethene	5[a]	--	1 U	1 U	1 U	1 U	2 U	0.8 U	0.5 U	0.5 U	0.5 U
Vinyl chloride	--	2	2.2	34	1 U	1 U	860	1,000 J	45	5.41	6.91
Dissolved Gases (RSK-175) (µg/L)											
Ethane	--	--	5 M	1 U	1 U		21 D M	28	0.22 U	0.22 U	0.22 U
Ethene	--	--	1.7 FMJ	1 U	1 U		200	210	0.29 U	0.29 U	0.29 U
Methane	--	--	1,800	0.544	2 U		3,900	2,800 J	1.09 U	1.09 U	1.1 U
Propylene	--	--	NA	28.6	32.3	NA	NA	NA	61	NA	NA
MNA Parameters (mg/L)											
Alkalinity (SM 2320B)	--	--	570	250	120		530 B	510	356	160	153
Chloride (SW 9056A)	200[a]	--		6.58	3.03		11	12	8.53	2.9	2.61
Nitrate as N (SW 9056A)	10[a]	--	0.043 F	0.098	0.234		0.1 U	0.1 U	0.15 U	0.15 U	0.15 U
Nitrite as N (SW 9056A)	1[b]	--	1 R	0.2 U	0.2 U		0.1 U	0.1 U	0.15 U	0.15 U	0.15 U
Sulfate (SW 9056A)	400[a]	--	89	59.1	76.1		53	43	97.7	40.9	11.2
TOC (SW 9060A)	--	--	8.4	6.9	1.6		5.9	6.2 J	5.35	8.53	7.08

Table 5-3 - OT007 Summary of June 2023 Groundwater Results

Analyte	Scott AFB Project Action Limit	OT007 Remediation Objectives[1]	OT07-MW06					
		Sample Date	6/29/2021	6/29/2021	6/9/2022	6/9/2022	6/1/2023	6/1/2023
		Sample Type	Normal	Duplicate	Normal	Duplicate	Normal	Duplicate
Metals (SW 6020B) (µg/L)								
Arsenic, Dissolved	--	19.1	8.87	10.5	4.04	3.92	NA	NA
Arsenic, Total	--	19.1	17	18.9	3.87	4.23	4.36 J	11.7 J
Iron, Dissolved	--	26,000	6,150	5,290	2,130	2,030	NA	NA
Iron, Total	--	26,000	10,500	11,400	2,210	2,210	5,250	4,790
Manganese, Dissolved	--	6,120	949	898	536	547	NA	NA
Manganese, Total	--	6,120	991	1010	629	589	861	829
VOCs (SW 8260B) (µg/L)								
1,2-Dichloroethene, Total	--	--	12	14.6	1.68 J	1.35 J	14.3	13.5
Acetone	6300[a]	--	1.0 U	1.0 U	1.0 U	1.0 U	1.0 UJ	1.0 UJ
Benzene	5[a]	--	0.279 J	0.293 J	0.5 U	0.5 U	1.71	1.57
cis-1,2-Dichloroethene	--	70	12	14.6	1.68	1.35	14.3	13.5
Methylene Chloride	5[a]	--	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
Methyl tert-butyl ether	14[b]	--	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
Toluene	1000[a]	--	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
trans-1,2-Dichloroethene	100[a]	--	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
Trichloroethene	5[a]	--	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U
Vinyl chloride	--	2	18.1	23.2	1.22	0.985 J	94.6	88.5
Dissolved Gases (RSK-175) (µg/L)								
Ethane	--	--	0.346 J	0.309 J	0.22 U	0.22 UJ	10 U	10 U
Ethene	--	--	1.12	0.901 J	0.29 U	0.29 UJ	10 U	10 U
Methane	--	--	166	164	0.918 J	1.1 UJ	327	501
Propylene	--	--	NA	NA	NA	NA	NA	NA
MNA Parameters (mg/L)								
Alkalinity (SM 2320B)	--	--	251	235	110	113	146	168
Chloride (SW 9056A)	200[a]	--	3.68	3.44	1.85	1.58	1.5	2.11
Nitrate as N (SW 9056A)	10[a]	--	0.15 U	0.15 U	0.15 U	0.15 U	0.135 J	0.15 U
Nitrite as N (SW 9056A)	1[b]	--	0.15 U	0.15 U	0.15 U	0.15 U	0.15 U	0.15 U
Sulfate (SW 9056A)	400[a]	--	32.8	26.8	21	20.9	33.5	37.9
TOC (SW 9060A)	--	--	7.19	7.59	8.28	9.2	10.7	8.93

Table 5-3 - OT007 Summary of June 2023 Groundwater Results

Notes:

Shaded results indicate concentrations above the Project Action Limit (PAL) or OT007 Remediation Objective (RO).

[1] = Remediation objectives for arsenic, iron, manganese, cis-1,2-dichloroethene, and vinyl chloride in groundwater were identified in the Final CSIR/ROR/RAP (HGL, 2013). They are based on the Class I groundwater values in TACO (35 IAC 742) Tier 1, Appendix B, Table E, and maximum background concentrations.

a = Illinois Administrative Code Title 35, Part 742, Tiered Approach to Corrective Action (TACO), Appendix B, Table E, Tier I, Groundwater Remediation Objectives for the Groundwater Component of the Groundwater Ingestion Route (IEPA, 2013).

b = EPA Regional Screening Level (RSL) for tapwater or maximum contaminant level (MCL), May 2021.
https://www.epa.gov/risk/regional-screening-levels-rsls-generic-tables

"--" = Scott AFB PAL or OT007 RO has not been established.

µg/L = micrograms per liter

B = Blank contamination: The analyte was detected above one-half the reporting limit in an associated blank.

D = The reported value is from a dilution.

F - The analyte was positively identified but the associated numerical value is below the reporting limit.

J = The positive result was detected between the Detection Limit (DL) and the Limit of Quantitation (LOQ) and has been flagged with a "J" as estimated.

M = For 2007, 2010, and 2011 datasets: Matrix effect was present, the concentration is estimated. All other datasets: Manually integrated compound.

mg/L = milligrams per liter

NA = Not analyzed

R = This data was rejected as it failed one or more quality control/quality assurance parameters.

U = This analyte was not detected in the sample. The numeric value represents the sample Limit of Detection (LOD).

The performing laboratory is Pace Gulf Coast in Baton Rouge, Louisiana.

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Table 5-4 - Site OT007 Summary of June 2023 Soil Results

Analyte	PAL	PAL Source	Sample ID	OT07-BO-40-4-6-8	OT07-BO-840-4-6-8
			Location	OT07-BO-40	OT07-BO-40 (FD)
			Depth	6-8'	6-8'
			Date	6/2/2023	6/2/2023
VOCs (SW 8260B)					
1,1,1,2-Tetrachloroethane	2,000	c	ug/Kg	136 U	129 U
1,1,1-Trichloroethane	2,000	a	ug/Kg	136 U	129 U
1,1,2,2-Tetrachloroethane	600	c	ug/Kg	136 U	129 U
1,1,2-Trichloroethane	20	a	ug/Kg	136 U	129 U
1,1-Dichloroethane	23,000	a	ug/Kg	136 U	129 U
1,1-Dichloroethene	60	a	ug/Kg	136 U	129 U
1,1-Dichloropropene	NE	NA	ug/Kg	136 U	129 U
1,2,3-Trichlorobenzene	6,300	c	ug/Kg	272 U	259 U
1,2,3-Trichloropropane	5.1	c	ug/Kg	272 U	259 U
1,2,4-Trichlorobenzene	5,000	a	ug/Kg	272 U	259 U
1,2,4-Trimethylbenzene	654	d	ug/Kg	**1960 J**	**28500**
1,2-Dibromo-3-chloropropane	2	a	ug/Kg	545 UJ	517 UJ
1,2-Dibromoethane	0.4	a	ug/Kg	545 U	517 U
1,2-Dichlorobenzene	17,000	a	ug/Kg	136 U	129 U
1,2-Dichloroethane	20	a	ug/Kg	136 U	129 U
1,2-Dichloroethene(Total)	NE	NA	ug/Kg	272 U	259 U
1,2-Dichloropropane	30	a	ug/Kg	136 U	129 U
1,3,5-Trimethylbenzene	27,000	c	ug/Kg	**306 J**	**6080**
1,3-Dichlorobenzene	NE	NA	ug/Kg	136 U	129 U
1,3-Dichloropropane	160,000	c	ug/Kg	136 U	129 U
1,4-Dichlorobenzene	2,600	c	ug/Kg	136 U	129 U
2,2-Dichloropropane	NE	NA	ug/Kg	136 U	129 U
2-Butanone	720,125	d	ug/Kg	545 U	517 U
2-Chlorotoluene	160,000	c	ug/Kg	136 U	129 U
2-Hexanone	4711	d	ug/Kg	545 U	517 U
4-Chlorotoluene	160,000	c	ug/Kg	136 U	129 U
4-Isopropyltoluene	NE	NA	ug/Kg	**147 J**	**2430**
4-Methyl-2-pentanone	374,479	d	ug/Kg	136 U	**7460**
Acetone	25,000	a	ug/Kg	545 U	517 U
Benzene	30	a	ug/Kg	136 U	**337 J**
Bromobenzene	5930	d	ug/Kg	136 UJ	129 UJ
Bromochloromethane	1,692	d	ug/Kg	272 U	259 U
Bromodichloromethane	600	a	ug/Kg	136 U	129 U
Bromoform	800	a	ug/Kg	272 U	259 U
Bromomethane	200	a	ug/Kg	545 U	517 U
Carbon disulfide	9,000	b	ug/Kg	136 U	129 U
Carbon tetrachloride	70	a	ug/Kg	136 U	129 U
Chlorobenzene	1,000	a	ug/Kg	136 U	129 U
Chloroethane	152,637	d	ug/Kg	136 UJ	129 UJ
Chloroform	300	a	ug/Kg	136 U	129 U
Chloromethane	1,247	d	ug/Kg	545 U	517 U
cis-1,2-Dichloroethene	400	a	ug/Kg	136 U	129 U
cis-1,3-Dichloropropene	NE	NA	ug/Kg	136 U	129 U
Dibromochloromethane	400	a	ug/Kg	136 U	129 U
Dibromomethane	267	d	ug/Kg	136 U	129 U
Dichlorodifluoromethane	987	d	ug/Kg	136 U	129 U
Ethylbenzene	13,000	a	ug/Kg	**241 J**	**3150**
Hexachlorobutadiene	1,200	c	ug/Kg	272 U	259 U
Isopropylbenzene (Cumene)	29,317	d	ug/Kg	**128 J**	**1880**
m,p-Xylene	320	a	ug/Kg	**310 J**	**4960**
Methylene chloride	20	a	ug/Kg	545 U	517 U
Naphthalene	1,800	b	ug/Kg	**316 J**	**4830**
n-Butylbenzene	390,000	c	ug/Kg	**146 J**	**2820**
n-Propylbenzene	82,586	d	ug/Kg	**315 J**	**4170**
o-Xylene	6,500	b	ug/Kg	136 U	**443 J**
sec-Butylbenzene	780,000	c	ug/Kg	**239 J**	**3250**
Styrene	4,000	a	ug/Kg	136 U	129 U
tert-Butyl methyl ether (MTBE)	320	a	ug/Kg	136 U	129 U
tert-Butylbenzene	780,000	c	ug/Kg	272 U	259 U
Tetrachloroethene	60	a	ug/Kg	272 U	259 U
Toluene	12,000	a	ug/Kg	136 U	129 U
trans-1,2-Dichloroethene	700	a	ug/Kg	136 U	129 U
trans-1,3-Dichloropropene	NE	NA	ug/Kg	136 U	129 U
Trichloroethene	60	a	ug/Kg	136 U	129 U
Trichlorofluoromethane	2,300,000	c	ug/Kg	136 U	129 U
Vinyl chloride	10	a	ug/Kg	136 U	129 U

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Table 5-4 - Site OT007 Summary of June 2023 Soil Results

Notes:

Bold values are detected concentrations

Highlighted values exceed PAL

μg/kg = micrograms per kilogram

PAL = Project Action Limit

VOC = Volatile Organic Compound

FD = Field Duplicate

NA = not applicable

NE = not established. Non-TACO values could not be calculated as toxicity and chemical specific parameters were not provided in the EPA RSL tables.

J = Estimated result.

U = Indicates the analyte was analyzed for but not detected, value shown is the limit of detection (LOD).

UJ = Analyte not detected in the sample. Value represents the LOD. Recovery in the MS/MSD or calibration verification outside acceptance criteria.

a = Lowest screening value from the Illinois Administrative Code Title 35, Part 742 (35 IAC 742), Tiered Approach to Corrective Action (TACO), Appendix B, Table A, Tier I Soil Remediation Objectives for Residential Properties.

b = 35 IAC 742 TACO, Appendix B, Table B, Soil Remediation Objectives for Industrial/Commercial Properties, lower of the ingestion/inhalation exposure routes for the construction worker receptor.

c = EPA Regional Screening Level (RSL) for residential exposures in soil, May 2021. https://www.epa.gov/risk/regional-screening-levels-rsls-generic-tables

d = Non-TACO calculated value for construction worker, lowest of inhalation, ingestion and fugitive dust pathways, carcinogenic and non-carcinogenic chemicals.

The performing laboratory is Pace Gulf Coast in Baton Rouge, Louisiana.

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Table 6-1 - Annual Groundwater Sample and Analysis[1,2]

Sample Locations	Sampling SOP Reference	VOCs	Total Organic Carbon	Alkalinity, Total as CaCo3	Sulfate	Chloride	Nitrate/ Nitrite	Ethene/ Ethane/ Methane
		SW 8260B	SW9060	SM2320B	SW 9056A			EPA Method RSK175
OT07-MW02	SESDPROC-301-R3	X	X	X	X	X	X	X
OT07-MW04	SESDPROC-301-R3	X	X	X	X	X	X	X
OT07-MW05	SESDPROC-301-R3	X	X	X	X	X	X	X
OT07-MW06	SESDPROC-301-R3	X	X	X	X	X	X	X

Notes/Abbreviations:
1 = Additional field quality control samples will be collected as specified in the UFP-QAPP Worksheet #21.
2 = All samples will be analyzed on a standard TAT, which is equivalent to 15 business days.
$CaCO_3$ = calcium carbonate
SESDPROC-301-R3 = *U.S. Environmental Protection Agency, Region 4 Science and Ecosystem Support Division Procedure for Groundwater Sampling (USEPA SESD, 2013b).* Copies of all applicable SOPs are provided in the *UFP-QAPP, Appendix C*
SESDPROC-201-R3 = *U.S. Environmental Protection Agency, Region 4 Science and Ecosystem Support Division Procedure for Surface Water Sampling (USEPA SESD, 2013b).* Copies of all applicable SOPs are provided in the *UFP-QAPP, Appendix C*
SESDPROC = U.S. Environmental Protection Agency, Region 4 Science and Ecosystem Support Division Procedure
SOP = Standard Operating Procedure
SW = USEPA Solid Waste Test Method
VOCs = volatile organic compounds

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APPENDIX A

Field Forms and Daily Quality Control Reports

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Horiba Calibration Form

June 1, 2023

General

Project Name	Scott AFB
Date	June 1, 2023
Time of Calibration	06:58
Calibrating Personnel	L Revels
Instrument Name/ Model	Horiba U52 #23066
Weather Conditions	Clear, 67
Barometric Pressure (inches Hg)	30.01

pH (4.01)

pH Initial Reading	4.29
pH Value Entered	4
pH Calibrated Reading	3.99

Conductivity (mS/m)

Conductivity Initial Reading	4.46
Conductivity Value Entered	4.49
Conductivity Calibrated Reading	4.48

Turbidity (1.0 NTU)

Turbidity Initial Reading	5.1
Turbidity Value Entered	0
Turbidity Calibrated Reading	0

DO (mg/L)

DO Initial Reading	11.84
DO Calibrated Reading	9.58

ORP (mV)

ORP Initial Reading	286
ORP Calibrated Reading	289

Water Temperature (Deg C)

Water Temperature Initial Reading	21.74
Water Temperature Calibrated Reading	21.73

Horiba Calibration Form

June 1, 2023

General

Project Name	Scott AFB
Date	June 1, 2023
Time of Calibration	06:43
Calibrating Personnel	L Revels
Instrument Name/ Model	Horiba U52 #24742
Weather Conditions	Clear, 66
Barometric Pressure (inches Hg)	30.01

pH (4.01)

pH Initial Reading	4.05
pH Value Entered	4
pH Calibrated Reading	4

Conductivity (mS/m)

Conductivity Initial Reading	4.53
Conductivity Value Entered	4.49
Conductivity Calibrated Reading	4.48

Turbidity (1.0 NTU)

Turbidity Initial Reading	0.2
Turbidity Value Entered	0
Turbidity Calibrated Reading	0.1

DO (mg/L)

DO Initial Reading	12.32
DO Calibrated Reading	9.27

ORP (mV)

ORP Initial Reading	260
ORP Calibrated Reading	263

Water Temperature (Deg C)

Water Temperature Initial Reading	21.19
Water Temperature Calibrated Reading	21.19

Lamotte Calibration Form

General

Project Name	Scott AFB
Date	June 1, 2023
Time of Calibration	07:09
Calibrating Personnel	L Revels
Instrument Name/ Model	LaMotte 2020t #801219
Weather Conditions	Clear, 69
Barometric Pressure (inches Hg)	30.01
Barometric Pressure (mm Hg)	

Calibration Event

1.0 NTU

1.0 Initial Reading	1.54
1.0 Value Entered	1
1.0 Calibrated Reading	1.32
1.0 NTU Comments	

10.0 NTU

10.0 Initial Reading	9.7
10.0 Value Entered	10
10.0 Calibrated Reading	10.2
10.0 NTU Comments	

Lamotte Calibration Form

General

Project Name	Scott AFB
Date	June 1, 2023
Time of Calibration	07:01
Calibrating Personnel	John Filoon
Instrument Name/ Model	LaMotte 2020we / A02733
Weather Conditions	Sunny, 70s
Barometric Pressure (inches Hg)	30
Barometric Pressure (mm Hg)	

Calibration Event

1.0 NTU

1.0 Initial Reading	2.05
1.0 Value Entered	1
1.0 Calibrated Reading	1.8
1.0 NTU Comments	

10.0 NTU

10.0 Initial Reading	7.11
10.0 Value Entered	10
10.0 Calibrated Reading	7.4
10.0 NTU Comments	

TAILGATE H&S MEETING

PROJECT	Scott AFB
PROJECT NUMBER	108272.
CLIENT	United States Air Force
DATE	June 1, 2023
TIME	06:38
REPORT NUMBER	3
ON-SITE LOCATION(S) OF TODAY'S TASK	OT007
TYPE OF TASK(S) BEING PERFORMED	Groundwater sampling, soil sampling
LEVEL OF PPE REQUIRED FOR TASK(S)	Modified D
SITE-SPECIFIC CHEMICAL HAZARDS	Bottleware preservatives; Chlorinated VOCs and petroleum at OT 007 and metals (manganese SS029, manganese and sulfate at SA026)
SITE-SPECIFIC PHYSICAL HAZARDS	Airfield traffic
LOCATION OF EMERGENCY EQUIPMENT, TELEPHONE NUMBERS, AND ROUTE TO HOSPITAL	911; in cars; on phones
AREA OF RETREAT IN EVENT OF A WEATHER EMERGENCY	Visitor’s center
TODAY'S WEATHER FORECAST	High 90; sunny; chance of showers
TODAY'S HEALTH AND SAFETY TOPIC	Heat Stress
Toolbox safety topic resources:	

LIST OF MEETING ATTENDEES

John Filoon

Pat Reilly

Locke Revels

PID CALIBRATION FORM

June 2, 2023

Project	Scott AFB
Site Number	OT007
Serial Number	24762
Date	June 2, 2023
Time	15:00
Model Number	MiniRae 3000+
Battery Check	Good
Introduce calibration gas to OVM.	
Calibration Gas Type / Concentration	Isobutylene - 100ppm
Initial Reading (ppm)	98.9
Zero Calibration	0
Adjusted Span?	No
Final Reading (ppm)	100
Calibrated by	John Filoon

TAILGATE H&S MEETING

PROJECT	Scott AFB
PROJECT NUMBER	108272.xxx
CLIENT	United States Air Force
DATE	June 2, 2023
TIME	06:35
REPORT NUMBER	4
ON-SITE LOCATION(S) OF TODAY'S TASK	Soil boring installation at OT007, groundwater sampling at SS029 and SA026.
TYPE OF TASK(S) BEING PERFORMED	Soil boring and groundwater sampling.
LEVEL OF PPE REQUIRED FOR TASK(S)	Mod level D
SITE-SPECIFIC CHEMICAL HAZARDS	Chlorinated VOCs and petroleum at OT 007 and metals (manganese SS029, manganese and sulfate at SA026)
SITE-SPECIFIC PHYSICAL HAZARDS	Hydration and heat exhaustion.
LOCATION OF EMERGENCY EQUIPMENT, TELEPHONE NUMBERS, AND ROUTE TO HOSPITAL	On phones and in safety plan.
AREA OF RETREAT IN EVENT OF A WEATHER EMERGENCY	Civil engineering office building.
TODAY'S WEATHER FORECAST	Sunny, mid 90's
TODAY'S HEALTH AND SAFETY TOPIC	Fatigue Management
Toolbox safety topic resources:	

LIST OF MEETING ATTENDEES

Patrick Reilley		**Locke Revels**	
John Filoon			

Static Water Level Record

Site ID: Scott AFBOT007

Plexus Personnel: L Revels

Frequency: Semi-annual

Equipment Make/Model/Serial: Interface probe

Weather Conditions: Sunny 80

Equipment Make/Model/Serial:

PLEXUS scientific — Celebrating 35 years

Well ID	Well Diameter	Date	Time	PID at Wellhead (ppmv)	Depth to Product (ft btoc)	Depth to Water (ft btoc)	Depth to Bottom (ft btoc)	Bottom Hard or Soft?	Bolts Present or Bollard Condition	Working Lock Present	Working Cap Present	Well Labeled	Pad Condition	Sample Required? (Y/N)	Comments
OT07-MW01	2"	6/1/2023	7:42	0	NP	8.4	18.32	SOFT	1 of 2 bolt	yes	yes	yes	Good		
OT07-MW02	2"	6/1/2023	7:51	0.1	NP	7.57	18.65	HARD	One bolt	yes	yes	yes	OK	yes	
OT07-MW03	2"	6/1/2023	7:43	0	NP	8.45	19.77	HARD	0 of 2 bolt	yes	yes	yes	OK	no	
OT07-MW04	2"	6/1/2023	7:52	0.3	NP	8.14	17.1	HARD	1 of 2 bolt	yes	yes	yes			
OT07-MW04D	2"	6/1/2023	7:52	0.1	NP	8.91	32.12	SOFT	1 of 2 bolt	yes	yes	yes	Good		
OT07-MW05	2"	6/1/2023	8:02	0	NP	7.67	14.59	HARD	One bolt	yes	yes	yes	OK		
OT07-MW06	2"	6/1/2023	7:47	0	NP	8.98	15.49	HARD	1 of 2 bolt	yes	yes	yes	Good.		

Notes:
btoc = below top of casing
ft = feet
NA = not applicable
ppmv = parts per million by volume

Well Pad Condition:
G = Good Condition
B = Broken or cracked
F = Floating
NV = Not visible (pad is present)

Bollard Condition:
G = Good Condition
L = Leaning/tilted
D = Damaged or loose
M = Missing (but needed)
NR - Not required (no bollards necessary)

Illinois Environmental Protection Agency

Field Boring Log

County: St Clair

Site ID No. OT007 Federal ID No. ______

Boring No. SO-40-4 Monitoring Well No. ______

Site Name Scott AFB OT007

Surface Elevation: ______ Completion Depth: 8'

Quadrangle ______ Sec. ______ T. ______ R. ______

Auger Depth: 8'

Rotary Depth: ______

UTM (or State Plane) Coord. N. (X) ______ E. (Y) ______

Date: Start: 6/2/23 7:50 Finish: 6/2/23 8:45

Latitude: ° ' " Longitude: ° ' "

Boring Location: ______

Drilling Equipment: Hand Auger

Personnel:
G - John Filan
D - Locke Revels
H -
H -

Elev.	Description of Material	Graphic Log	Depth In Feet	Sample No.	Sample Type	Sample Recovery (X)	Penetrometer	N Values (Blow Counts)	OVA or HNU Readings	REMARKS
	Topsoil								0	
	Dry, loose, reddish brown silty clay								0	
			2						0.1	
	Moist, medium-dense, dark reddish brown silty clay, some darker red & tan mottling								0.3	Some gravel at 3.5'
			4			100%			1.9	
	Moist, medium-dense orangeish tan silty clay									Some gray staining at 4.5 - 6' slight odor at 4.5'
	Moist, medium-dense orangeish and tanish brown silty clay								15.1	
			6							More prominent staining 6-8'
									413	Stronger odor at 6'
	Moist to wet, medium-dense, orangeish brown silty clay			6-8	SS				808	
									538	
									68	
			8							
			10							

Field Boring Log (revised 02/02/04)

Groundwater Monitoring Well Purging and Sampling

Celebrating 35 years PLEXUS scientific

Site Name: Scott AFB

Project #: OT007

Well ID: OT07-MW06

Initial Water Level: 8.83 **(ft btoc)**

Well Construction: FM 2" PVC

Sampling Method: Low Flow (Submersible) - Bladder Pump

Pump Inlet: 10 **(ft btoc)**

Flow Rate: 200 **mL/min**

Personnel: LR, PR, JF

Date: 6/1/2023

Time	Temp +/- 3% (°C)	pH +/- 0.1	ORP +/- 10	Conductivity +/- 3% (mS/cm)	Turbidity +/-10% (NTU)	Turbidity +/10% (NTU) LaMotte	DO +/- 10% (mg/L) or <0.5	Volume Removed (Liters)	Depth to Water +/- 0.3 (ft)
10:30	25.76	6.67	-25	0.402	17	13.6	0	0.5	9.08
10:35	23.66	6.67	-41	0.422	15.8	12.5	0	1	9.14
10:40	21.79	6.66	-52	0.428	13.3	11.6	0	1.5	9.18
10:45	20.98	6.63	-55	0.409	9.9	9.2	0	2	9.36
10:50	20.73	6.62	-57	0.386	8.2	7.45	0	2.5	9.45
10:55	20.65	6.61	-55	0.379	6.1	6.12	0	3	9.53
11:00	20.54	6.56	-54	0.376	4.8	5.51	0	3.5	9.6

QC SAMPLE(S): OT07-MW806-060123 **MNA SAMPLES:**

Notes: Sample ID: OT07-MW06-060123

Groundwater Monitoring Well Purging and Sampling

Site Name: Scott AFB

Project #: OT007

Well ID: OT07-MW02

Initial Water Level: 7.41 **(ft btoc)**

Well Construction: FM 2" PVC

Sampling Method: Low Flow (Submersible) - Bladder Pump

Pump Inlet: 13.5 **(ft btoc)**

Flow Rate: 90 **mL/min**

Personnel: LR, PR, JF

Date: 5/31/2023

Time	Temp +/- 3% (°C)	pH +/- 0.1	ORP +/- 10	Conductivity +/- 3% (mS/cm)	Turbidity +/-10% (NTU)	Turbidity +/10% (NTU) LaMotte	DO +/- 10% (mg/L) or <0.5	Volume Removed (Liters)	Depth to Water +/- 0.3 (ft)
11:35	21.71	7.12	-85	0.401	15.8	22.5	0.65	0.8	7.69
11:40	18.60	6.71	-65	0.384	18.6	23.5	0.27	1.4	7.86
11:45	18.08	6.48	-57	0.376	24.6	27.5	0.28	2	8
11:50	18.18	6.43	-55	0.367	27.5	27	0.43	2.5	8.14
11:55	18.19	6.41	-53	0.354	28.8	24.4	0.69	3.1	8.25
12:00	17.94	6.40	-48	0.357	22.3	20.6	0.51	3.6	8.35
12:05	17.71	6.37	-42	0.348	14.3	14.4	0.43	4.2	8.51
12:10	18.09	6.35	-38	0.341	11.4	12.4	0.38	4.7	8.65
12:15	18.17	6.34	-37	0.338	9	11.42	0.16	5.3	8.72
12:20	18.27	6.34	-36	0.338	8.8	10.45	0.1	5.8	8.79
12:25	18.51	6.34	-36	0.337	8.1	10	0.07	6.4	8.84

QC SAMPLE(S):

MNA SAMPLES:

Notes: Sample ID: OT07-MW02-053123

Groundwater Monitoring Well Purging and Sampling

Site Name: Scott AFB
Project #: OT007
Well ID: OT07-MW04
Initial Water Level: 8.06 **(ft btoc)**
Well Construction: FM 2" PVC

Sampling Method: Low Flow (Submersible) - Bladder Pump
Pump Inlet: 12 **(ft btoc)**
Flow Rate: 250 **mL/min**
Personnel: LR, PR, JF
Date: 5/31/2023

Time	Temp +/- 3% (°C)	pH +/- 0.1	ORP +/- 10	Conductivity +/- 3% (mS/cm)	Turbidity +/-10% (NTU)	Turbidity +/10% (NTU) LaMotte	DO +/- 10% (mg/L) or <0.5	Volume Removed (Liters)	Depth to Water +/- 0.3 (ft)
8:55	18.38	6.48	96	0.755	56.9	32.3	0	1.2	8.41
9:00	18.61	6.55	97	0.76	48.4	26.3	0	1.6	8.51
9:05	18.44	6.71	99	0.754	29.5	17.9	0.25	2.1	8.63
9:10	18.44	6.76	101	0.75	20.6	13.2	0.24	2.6	8.66
9:15	18.42	6.82	103	0.744	12.8	9.65	0.13	3.1	8.75
9:20	18.45	6.84	103	0.74	6.7	6.01	0	3.6	8.78
9:25	18.54	6.86	105	0.739	2.9	5.14	0	4.1	8.85

QC SAMPLE(S): OT07-MW04-053123 MS, OT07-MW04-053123 MSD **MNA SAMPLES:**

Notes: Sample ID: OT07-MW04-053123

Groundwater Monitoring Well Purging and Sampling

Site Name: Scott AFB
Project #: OT007
Well ID: OT07-MW05
Initial Water Level: 8.51 **(ft btoc)**
Well Construction: FM 2" PVC

Sampling Method: Low Flow (Submersible) - Bladder Pump
Pump Inlet: 10 **(ft btoc)**
Flow Rate: 200 **mL/min**
Personnel: LR, PR, JF
Date: 6/1/2023

Time	Temp +/- 3% (°C)	pH +/- 0.1	ORP +/- 10	Conductivity +/- 3% (mS/cm)	Turbidity +/-10% (NTU)	Turbidity +/10% (NTU) LaMotte	DO +/- 10% (mg/L) or <0.5	Volume Removed (Liters)	Depth to Water +/- 0.3 (ft)
8:45	16.75	6.38	135	0.26	20	11.2	0	1	8.03
8:50	15.78	6.30	100	0.235	10.1	6.55	0	2	8.29
8:55	16.05	6.26	86	0.231	6.1	4.88	0	2.5	8.39
9:00	16.55	6.26	84	0.228	3.3	8.64	0	3	8.64
9:05	17.15	6.27	86	0.227	2.8	3.03	0	3.5	8.75
9:10	17.45	6.28	89	0.226	1.6	4.97	0	4	8.85

QC SAMPLE(S):
MNA SAMPLES:

Notes: Sample ID: OT07-MW05-060123

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APPENDIX B

Analytical Reports
(See Attached CD)

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LELAP Certificate Number: 01955
A2LA Accredited (DoD ELAP-QSM 5.4) Certificate Number: 6429.01

ANALYTICAL RESULTS

PERFORMED BY

Pace Analytical Gulf Coast
7979 Innovation Park Dr.
Baton Rouge, LA 70820
(225) 769-4900

Report Date 09/25/2023

Report # 223060210

Project Scott AFB

Samples Collected 6/1/23

Deliver To	***Additional Recipients***
Patrick Reilley Plexus Scientific Corporation 5510 Cherokee Ave. Suite 350 Alexandria, VA 22312 (703) 989-8405	NONE

ACCREDITED
CERT # 6429.01

Laboratory Endorsement

Sample analysis was performed in accordance with approved methodologies provided by the Environmental Protection Agency or other recognized agencies. The samples and their corresponding extracts will be maintained for a period of 30 days unless otherwise arranged. Following this retention period the samples will be disposed in accordance with Pace Gulf Coast's Standard Operating Procedures.

Common Abbreviations that may be Utilized in this Report

ND	Indicates the result was Not Detected at the specified reporting limit
NO	Indicates the sample did not ignite when preliminary test performed for EPA Method 1030
DO	Indicates the result was Diluted Out
MI	Indicates the result was subject to Matrix Interference
TNTC	Indicates the result was Too Numerous To Count
SUBC	Indicates the analysis was Sub-Contracted
FLD	Indicates the analysis was performed in the Field
DL	Detection Limit
LOD	Limit of Detection
LOQ	Limit of Quantitation
RE	Re-analysis
CF	HPLC or GC Confirmation
00:01	Reported as a time equivalent to 12:00 AM

Reporting Flags that may be Utilized in this Report

J or I	Indicates the result is between the MDL and LOQ
J	DOD flag on analyte in the parent sample for MS/MSD outside acceptance criteria
U	Indicates the compound was analyzed for but not detected
B or V	Indicates the analyte was detected in the associated Method Blank
Q	Indicates a non-compliant QC Result (See Q Flag Application Report)
*****	Indicates a non-compliant or not applicable QC recovery or RPD – see narrative
E	Organics **-** The result is estimated because it exceeded the instrument calibration range
E	Metals - % diference for the serial dilution is > 10%
L	Reporting Limits adjusted to meet risk-based limit.
P	RPD between primary and confirmation result is greater than 40
DL	Diluted analysis – when appended to Client Sample ID

Sample receipt at Pace Gulf Coast is documented through the attached chain of custody. In accordance with NELAC, this report shall be reproduced only in full and with the written permission of Pace Gulf Coast. The results contained within this report relate only to the samples reported. The documented results are presented within this report.

This report pertains only to the samples listed in the Report Sample Summary and should be retained as a permanent record thereof. The results contained within this report are intended for the use of the client. Any unauthorized use of the information contained in this report is prohibited.

I certify that this data package is in compliance with The NELAC Institute (TNI) Standard 2009 and terms and conditions of the contract and Statement of Work both technically and for completeness, for other than the conditions in the case narrative. Release of the data contained in this hardcopy data package and in the computer readable data submitted has been authorized by the Quality Assurance Manager or his/her designee, as verified by the following signature.

Estimated uncertainty of measurement is available upon request. This report is in compliance with the DOD QSM as specified in the contract if applicable.

Authorized Signature
Pace Gulf Coast Report 223060210

Certifications

Certification	Certification Number
A2LA Accredited (DoD ELAP-QSM 5.4)	6429.01
Alabama	01955
Arkansas	88-0655
Colorado	01955
Delaware	01955
Florida	E87854
Georgia	01955
Hawaii	01955
Idaho	01955
Illinois	200048
Indiana	01955
Kansas	E-10354
Kentucky	95
Louisiana	01955
Maryland	01955
Massachusetts	01955
Michigan	01955
Mississippi	01955
Missouri	01955
Montana	N/A
Nebraska	01955
New Mexico	01955
North Carolina	618
North Dakota	R-195
Oklahoma	9403
South Carolina	73006001
South Dakota	01955
Tennessee	01955
Texas	T104704178
Vermont	01955
Virginia	460215
Washington	C929
USDA Soil Permit	P330-16-00234

Case Narrative

Client: Plexus Consultants **Report:** 223060210

Pace Analytical Gulf Coast received and analyzed the sample(s) listed
on the Report Sample Summary page of this report. Receipt of the sample(s) is documented
by the attached chain of custody. This applies only to the sample(s) listed in this report.
No sample integrity or quality control exceptions were identified unless noted below.

This report was completed in accordance with DOD QSM 5.3 as specified in the contract.

This report resubmitted 09/25/23 to include the subcontracted RSK-175 data. This report supersedes and replaces any prior report issued for this workorder.

VOLATILES MASS SPECTROMETRY

In the EPA 8260B DOD analysis for analytical batch 767028, the %D/%Drift is outside ± 20% for Dichlorodifluoromethane and Bromomethane in the opening CCV. The closing CCV had all compounds within the control limits.

In the EPA 8260B DOD analysis for analytical batch 767081, the %D/%Drift is outside ± 50% for Acetone in the closing CCV. Acetone was within limits in the opening CCV.

In the EPA 8260B DOD analysis, the recovery for the surrogate Dibromofluoromethane was above the upper control limit for sample 22306021003 (OT07-MW04-060123). The sample was non-detect for all compounds, therefore no further corrective action was taken.

In the EPA 8260B DOD analysis for analytical batch 766942, the MS/MSD exhibited RPD failure for Dichlorodifluoromethane.

In the EPA 8260B DOD analysis for analytical batch 767081, the LCS/LCSD RPD are above the control limit for 2-Butanone, 4-Methyl-2-pentanone, 1,2-Dibromo-3-chloropropane and tert-Butyl methyl ether (MTBE).

In the EPA 8260B DOD analysis for analytical batches 767028 and 767081, the ICV recovery is outside the control limits for Acetone and Chloromethane. These compounds are defined as poor performers by the method 8260.

In the EPA 8260B DOD analysis for analytical batch 766942, the ICV recoveries were outside the control limits for Acetone, 2-Butanone, 2-Hexanone and Chloromethane. These compounds are defined as poor performers by the method 8260.

MISCELLANEOUS

See subcontract laboratory report case narrative.

Sample Summary

Lab ID	Client ID	Matrix	Collect Date	Receive Date
22306021001	OT07-EB-060123	Water	6/01/23 12:14	6/02/23 10:17
22306021002	OT07-MW02-060123	Water	6/01/23 12:30	6/02/23 10:17
22306021003	OT07-MW04-060123	Water	6/01/23 09:35	6/02/23 10:17
22306021004	OT07-MW05-060123	Water	6/01/23 09:10	6/02/23 10:17
22306021005	OT07-MW06-060123	Water	6/01/23 11:00	6/02/23 10:17
22306021006	OT07-MW04-060123MS	Water	6/01/23 09:55	6/02/23 10:17
22306021007	OT07-MW04-060123MSD	Water	6/01/23 10:15	6/02/23 10:17
22306021008	OT07-MW-806-060123	Water	6/01/23 11:00	6/02/23 10:17
22306021009	OT07-TB-01-060124	Water	6/01/23 07:30	6/02/23 10:17

Test Summary

EPA 6020B

Lab ID	Client ID	Matrix	Batch	Instru	RunDate
22306021001	OT07-EB-060123	Water	767033	ICPMS2	6/05/23 11:54
22306021002	OT07-MW02-060123	Water	767033	ICPMS2	6/05/23 11:57
22306021003	OT07-MW04-060123	Water	767033	ICPMS2	6/05/23 12:01
22306021004	OT07-MW05-060123	Water	767033	ICPMS2	6/05/23 12:04
22306021005	OT07-MW06-060123	Water	767033	ICPMS2	6/05/23 12:08
22306021006	OT07-MW04-060123MS	Water	767033	ICPMS2	6/05/23 12:11
22306021007	OT07-MW04-060123MSD	Water	767033	ICPMS2	6/05/23 12:15
22306021008	OT07-MW-806-060123	Water	767033	ICPMS2	6/05/23 12:25

EPA 8260B DOD

Lab ID	Client ID	Matrix	Batch	Instru	RunDate
22306021001	OT07-EB-060123	Water	766942	MSV11	6/02/23 14:46
22306021002	OT07-MW02-060123	Water	766942	MSV11	6/02/23 15:32
22306021003	OT07-MW04-060123	Water	766942	MSV11	6/02/23 15:56
22306021004	OT07-MW05-060123	Water	766942	MSV11	6/02/23 16:19
22306021005	OT07-MW06-060123	Water	766942	MSV11	6/02/23 16:42
22306021006	OT07-MW04-060123MS	Water	766942	MSV11	6/02/23 18:38
22306021007	OT07-MW04-060123MSD	Water	766942	MSV11	6/02/23 19:02
22306021008	OT07-MW-806-060123	Water	767028	MSV15	6/05/23 14:30
22306021009	OT07-TB-01-060124	Water	767081	MSV15	6/06/23 13:42

EPA 9056A

Lab ID	Client ID	Matrix	Batch	Instru	RunDate
22306021001	OT07-EB-060123	Water	767085	BRWA13	6/02/23 13:34
22306021002	OT07-MW02-060123	Water	767085	BRWA13	6/02/23 13:51
22306021002DL	OT07-MW02-060123DL	Water	767223	IC4	6/07/23 14:45
22306021003	OT07-MW04-060123	Water	767085	BRWA13	6/02/23 12:07
22306021003DL	OT07-MW04-060123DL	Water	767281	IC4	6/08/23 11:00
22306021004	OT07-MW05-060123	Water	767085	BRWA13	6/02/23 11:50
22306021004DL	OT07-MW05-060123DL	Water	767223	IC4	6/07/23 15:03
22306021005	OT07-MW06-060123	Water	767085	BRWA13	6/02/23 12:59
22306021005DL	OT07-MW06-060123DL	Water	767223	IC4	6/07/23 16:12
22306021006	OT07-MW04-060123MS	Water	767085	BRWA13	6/02/23 12:25
22306021006DL	OT07-MW04-060123MSDL	Water	767281	IC4	6/08/23 11:17
22306021007	OT07-MW04-060123MSD	Water	767085	BRWA13	6/02/23 12:42
22306021007DL	OT07-MW04-060123MSDDL	Water	767281	IC4	6/08/23 11:35
22306021008	OT07-MW-806-060123	Water	767085	BRWA13	6/02/23 13:17
22306021008DL	OT07-MW-806-060123DL	Water	767223	IC4	6/07/23 15:20

EPA 9060A

Lab ID	Client ID	Matrix	Batch	Instru	RunDate
22306021001	OT07-EB-060123	Water	767077	TOC6	6/06/23 13:36
22306021002	OT07-MW02-060123	Water	767077	TOC6	6/06/23 14:12
22306021003	OT07-MW04-060123	Water	767167	TOC6	6/08/23 00:53
22306021004	OT07-MW05-060123	Water	767077	TOC6	6/06/23 14:40
22306021005	OT07-MW06-060123	Water	767077	TOC6	6/06/23 15:07
22306021006	OT07-MW04-060123MS	Water	767167	TOC6	6/08/23 01:19
22306021007	OT07-MW04-060123MSD	Water	767167	TOC6	6/08/23 01:48
22306021008	OT07-MW-806-060123	Water	767077	TOC6	6/06/23 16:51

SM 2320 B-2011					
Lab ID	**Client ID**	**Matrix**	**Batch**	**Instru**	**RunDate**
22306021001	OT07-EB-060123	Water	767172	TITR1	6/07/23 11:04
22306021002	OT07-MW02-060123	Water	767075	TITR1	6/06/23 10:38
22306021003	OT07-MW04-060123	Water	767075	TITR1	6/06/23 10:38
22306021004	OT07-MW05-060123	Water	767075	TITR1	6/06/23 10:38
22306021005	OT07-MW06-060123	Water	767075	TITR1	6/06/23 10:38
22306021008	OT07-MW-806-060123	Water	767075	TITR1	6/06/23 10:38

Manual Integrations

Manual Integrations for LC and IC (if performed) are documented in the raw data.
No other manual integrations were performed by Pace Gulf Coast.

Q Flag Summary

Client Sample ID: **OT07-MW-806-060123** Lab Sample ID: **22306021008**

EPA 8260B DOD Analysis Date: 6/5/2023 2:30:00 PM						
CAS	**Analyte**	**CCV**	**LCS/D**	**SURR**	**ISTD**	**CLCCV**
74-83-9	Bromomethane	X				
75-71-8	Dichlorodifluoromethane	X				

Client Sample ID: **OT07-TB-01-060124** Lab Sample ID: **22306021009**

EPA 8260B DOD Analysis Date: 6/6/2023 1:42:00 PM						
CAS	**Analyte**	**CCV**	**LCS/D**	**SURR**	**ISTD**	**CLCCV**
96-12-8	1,2-Dibromo-3-chloropropane		X			
78-93-3	2-Butanone		X			
108-10-1	4-Methyl-2-pentanone		X			
1634-04-4	tert-Butyl methyl ether (MTBE)		X			

CCV=Initial or Continuing CCV out of limits
LCS/D=LCS/LCSD out of limits
SURR=Surrogate out of limits
ISTD=Internal Standard out of limits
CLCCV=Closing CCV out of limits

Metals

Form I

Sample Results

I

INORGANIC ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-EB-060123
Collect Date:	06/01/23 Time: 1214	GCAL Sample ID:	22306021001
Matrix:	Water % Solids: NA	Instrument ID:	ICPMS2
Sample Amt:	50 mL	Lab File ID:	2230605a_MS2.b\1218SMPL.d
Prep Vol.:	50 (mL)	Dilution Factor:	1 Analyst: TDM
Prep Date:	06/05/23	Analysis Date:	06/05/23 Time: 1154
Prep Batch:	766941	Analytical Batch:	767033
Prep Method:	3010A	Analytical Method:	EPA 6020B

ANALYTE	***RESULT***	***UNITS***	***Q***	***DL***	***LOD***	***LOQ***
Arsenic	0.50	ug/L	U	0.25	0.50	1.00
Iron	50.0	ug/L	U	25.0	50.0	100
Manganese	2.50	ug/L	U	1.25	2.50	5.00

FORM I - IN

Sample Report

Sample Name	22306021001	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1218SMPL.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 11:54:17	**Total Dilution**	1.0000
Sample Type	Sample	**Sample Pass/Fail**	Pass
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Pass

Units : ppb

QC Analyte Table

Name	Mass	ISTD	Mode	MeasValue	Final Conc	RSD	CPS	LDR	QC Flag
Li	7	6	No Gas	0.277	0.277	0.8	94515.07	500	
Be	9	6	No Gas	0.015	0.015	11.5	104.00	1000	
B	11	6	No Gas	19.707	19.707	1.0	42637.52	500	
Sr	88	72	No Gas	0.152	0.152	5.1	12305.79	1000	
Zr	90	72	No Gas	0.007	0.007	4.2	2032.40	100	
Mo	95	115	No Gas	0.244	0.244	4.7	5244.35	1000	
Ag	107	115	No Gas	0.008	0.008	12.1	378.90	100	
Cd	111	115	No Gas	0.014	0.014	12.0	257.78	1000	
Sb	121	115	No Gas	0.144	0.144	0.0	7553.22	1000	
Ba	137	115	No Gas	0.244	0.244	1.7	3195.96	1000	
Tl	205	209	No Gas	-0.035	-0.035	1.7	3110.49	1000	
Pb	208	209	No Gas	0.061	0.061	2.7	3627.30	1000	
Na	23	45	He	42.403	42.403	2.2	32950.08	100000	
Mg	24	45	He	7.199	7.199	12.3	2036.89	100000	
Al	27	45	He	4.483	4.483	11.7	378.01	20000	
Si	29	45	He	-35.438	-35.438	35.1	3009.68	10000	
K	39	45	He	3.218	3.218	1.6	29897.36	100000	
Ca	44	45	He	37.497	37.497	9.8	1088.39	500000	
Ti	47	45	He	0.168	0.168	25.5	22.67	1000	
V	51	72	He	0.026	0.026	5.9	406.68	1000	
Cr	52	72	He	0.256	0.256	4.5	3305.96	1000	
Mn	55	72	He	0.367	0.367	7.4	870.04	5000	
Fe	57	72	He	3.347	3.347	14.5	616.70	100000	
Co	59	72	He	0.016	0.016	9.7	227.78	1000	
Ni	60	72	He	0.132	0.132	7.4	505.57	2000	
Cu	63	45	He	0.141	0.141	7.5	1492.32	1000	
Zn	66	72	He	12.21	12.210	2.2	11971.65	20000	
As	75	72	He	0.021	0.021	5.0	142.67	1000	
Se	78	72	He	-0.047	-0.047	2.1	23.79	50	
Sn	120	115	He	0.053	0.053	9.2	640.03	1000	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	5740672.00	6090049.07666667	94.26	
Ge	72	He	175094.58	183172.586666667	95.59	
In	115	He	1750337.95	1893460.06333333	92.44	
Lu	175	He	4116938.27	4453666.28666667	92.44	
Rh	103	He	5622420.06	5805949.63	96.84	
Sc	45	He	198143.93	210499.653333333	94.13	
Tb	159	He	6928591.15	7379981.76666667	93.88	

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
(Li)	6	No Gas	1407280.44	1489089.40333333	94.51	
Bi	209	No Gas	13296692.72	12052546.07	110.32	
Ge	72	No Gas	2442656.01	2410488.1	101.33	
In	115	No Gas	17140816.95	15897428.5	107.82	
Lu	175	No Gas	20278491.78	18289690.1433333	110.87	
Rh	103	No Gas	16478219.90	15416412.72	106.89	
Sc	45	No Gas	9652281.56	9412079.90333333	102.55	
Tb	159	No Gas	21420670.51	19489435.1266667	109.91	

I

INORGANIC ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW02-060123
Collect Date:	06/01/23 Time: 1230	GCAL Sample ID:	22306021002
Matrix:	Water % Solids: NA	Instrument ID:	ICPMS2
Sample Amt:	50 mL	Lab File ID:	2230605a_MS2.b\1219SMPL.d
Prep Vol.:	50 (mL)	Dilution Factor:	1 Analyst: TDM
Prep Date:	06/05/23	Analysis Date:	06/05/23 Time: 1157
Prep Batch:	766941	Analytical Batch:	767033
Prep Method:	3010A	Analytical Method:	EPA 6020B

ANALYTE	RESULT	UNITS	Q	DL	LOD	LOQ
Arsenic	10.1	ug/L		0.25	0.50	1.00
Iron	7940	ug/L		25.0	50.0	100
Manganese	1650	ug/L		1.25	2.50	5.00

FORM I - IN

Sample Report

Sample Name	22306021002	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1219SMPL.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 11:57:47	**Total Dilution**	1.0000
Sample Type	Sample	**Sample Pass/Fail**	Fail
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Pass

Units : ppb

QC Analyte Table

Name	Mass	ISTD	Mode	MeasValue	Final Conc	RSD	CPS	LDR	QC Flag
Li	7	6	No Gas	5.096	5.096	1.4	129959.74	500	
Be	9	6	No Gas	0.035	0.035	16.4	156.00	1000	
B	11	6	No Gas	49.85	49.850	0.7	99536.33	500	
Sr	88	72	No Gas	126.241	126.241	0.5	9282570.49	1000	
Zr	90	72	No Gas	0.337	0.337	4.6	17395.42	100	
Mo	95	115	No Gas	1.04	1.040	2.5	16307.17	1000	
Ag	107	115	No Gas	0.008	0.008	10.4	358.90	100	
Cd	111	115	No Gas	0.034	0.034	5.3	405.57	1000	
Sb	121	115	No Gas	0.265	0.265	0.9	11466.99	1000	
Ba	137	115	No Gas	118.017	118.017	0.6	1451596.71	1000	
Tl	205	209	No Gas	-0.048	-0.048	1.8	2153.59	1000	
Pb	208	209	No Gas	0.341	0.341	5.0	16203.53	1000	
Na	23	45	He	32522.764	32522.764	0.6	15985847.26	100000	
Mg	24	45	He	12327.732	12327.732	0.7	2651031.73	100000	
Al	27	45	He	326.025	326.025	10.8	20959.91	20000	
Si	29	45	He	12273.741	12273.741	4.1	34794.18	10000	>LDR
K	39	45	He	1811.727	1811.727	0.3	439108.30	100000	
Ca	44	45	He	30914.111	30914.111	0.5	331071.24	500000	
Ti	47	45	He	8.64	8.640	8.1	846.72	1000	
V	51	72	He	2.7	2.700	2.3	10817.42	1000	
Cr	52	72	He	0.741	0.741	3.6	5781.41	1000	
Mn	55	72	He	1652.257	1652.257	1.1	3431272.36	5000	
Fe	57	72	He	7940.148	7940.148	0.7	766581.63	100000	
Co	59	72	He	1.206	1.206	2.0	11154.33	1000	
Ni	60	72	He	2.562	2.562	1.9	6828.33	2000	
Cu	63	45	He	2.662	2.662	1.3	19615.44	1000	
Zn	66	72	He	41.58	41.580	1.0	40149.45	20000	
As	75	72	He	10.132	10.132	1.3	10101.51	1000	
Se	78	72	He	0.196	0.196	8.7	36.26	50	
Sn	120	115	He	0.161	0.161	1.6	1128.95	1000	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	5530314.50	6090049.07666667	90.81	
Ge	72	He	175341.76	183172.586666667	95.72	
In	115	He	1706284.29	1893460.06333333	90.11	
Lu	175	He	4097254.11	4453666.28666667	92	
Rh	103	He	5357743.26	5805949.63	92.28	
Sc	45	He	197687.67	210499.653333333	93.91	
Tb	159	He	6889555.10	7379981.76666667	93.35	

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
(Li)	6	No Gas	1354993.42	1489089.40333333	90.99	
Bi	209	No Gas	12696709.81	12052546.07	105.34	
Ge	72	No Gas	2435360.25	2410488.1	101.03	
In	115	No Gas	16557326.33	15897428.5	104.15	
Lu	175	No Gas	19920319.70	18289690.1433333	108.92	
Rh	103	No Gas	15521521.05	15416412.72	100.68	
Sc	45	No Gas	9574156.01	9412079.90333333	101.72	
Tb	159	No Gas	21089099.69	19489435.1266667	108.21	

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INORGANIC ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123
Collect Date:	06/01/23 Time: 0935	GCAL Sample ID:	22306021003
Matrix:	Water % Solids: NA	Instrument ID:	ICPMS2
Sample Amt:	50 mL	Lab File ID:	2230605a_MS2.b\1220SMPL.d
Prep Vol.:	50 (mL)	Dilution Factor:	1 Analyst: TDM
Prep Date:	06/05/23	Analysis Date:	06/05/23 Time: 1201
Prep Batch:	766941	Analytical Batch:	767033
Prep Method:	3010A	Analytical Method:	EPA 6020B

ANALYTE	RESULT	UNITS	Q	DL	LOD	LOQ
Arsenic	0.93	ug/L	J	0.25	0.50	1.00
Iron	151	ug/L		25.0	50.0	100
Manganese	66.5	ug/L		1.25	2.50	5.00

FORM I - IN

Sample Report

Sample Name	22306021003	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1220SMPL.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 12:01:16	**Total Dilution**	1.0000
Sample Type	Sample	**Sample Pass/Fail**	Pass
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Pass

Units : ppb

QC Analyte Table

Name	Mass	ISTD	Mode	MeasValue	Final Conc	RSD	CPS	LDR	QC Flag
Li	7	6	No Gas	1.826	1.826	1.9	99873.05	500	
Be	9	6	No Gas	0.017	0.017	13.2	100.67	1000	
B	11	6	No Gas	163.051	163.051	2.5	307863.83	500	
Sr	88	72	No Gas	374.552	374.552	1.0	27329911.26	1000	
Zr	90	72	No Gas	0.182	0.182	30.7	10125.85	100	
Mo	95	115	No Gas	1.236	1.236	0.3	18579.93	1000	
Ag	107	115	No Gas	0.008	0.008	2.3	362.23	100	
Cd	111	115	No Gas	0.041	0.041	14.8	443.35	1000	
Sb	121	115	No Gas	0.351	0.351	2.3	14069.35	1000	
Ba	137	115	No Gas	60.012	60.012	1.3	718454.69	1000	
Tl	205	209	No Gas	-0.042	-0.042	7.4	2376.97	1000	
Pb	208	209	No Gas	0.125	0.125	8.3	6121.72	1000	
Na	23	45	He	41848.294	41848.294	1.5	20405585.95	100000	
Mg	24	45	He	44406.491	44406.491	0.7	9472186.11	100000	
Al	27	45	He	109.809	109.809	8.1	7071.45	20000	
Si	29	45	He	5603.486	5603.486	5.7	17409.48	10000	
K	39	45	He	2672.921	2672.921	0.6	628863.30	100000	
Ca	44	45	He	113695.813	113695.813	0.9	1205943.89	500000	
Ti	47	45	He	2.877	2.877	18.0	285.34	1000	
V	51	72	He	0.933	0.933	3.6	3876.11	1000	
Cr	52	72	He	0.407	0.407	2.5	4015.06	1000	
Mn	55	72	He	66.525	66.525	1.3	136008.21	5000	
Fe	57	72	He	150.784	150.784	4.1	14585.31	100000	
Co	59	72	He	0.13	0.130	9.0	1255.62	1000	
Ni	60	72	He	1.474	1.474	3.7	3930.57	2000	
Cu	63	45	He	1.585	1.585	1.2	11773.70	1000	
Zn	66	72	He	12.665	12.665	1.9	12216.29	20000	
As	75	72	He	0.925	0.925	1.0	1015.71	1000	
Se	78	72	He	0.583	0.583	2.8	55.07	50	
Sn	120	115	He	0.07	0.070	4.5	688.92	1000	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	5321088.15	6090049.07666667	87.37	
Ge	72	He	172614.76	183172.586666667	94.24	
In	115	He	1666857.94	1893460.06333333	88.03	
Lu	175	He	3983149.32	4453666.28666667	89.44	
Rh	103	He	5196053.13	5805949.63	89.5	
Sc	45	He	196304.90	210499.653333333	93.26	
Tb	159	He	6727255.11	7379981.76666667	91.16	

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
(Li)	6	No Gas	1306846.22	1489089.40333333	87.76	
Bi	209	No Gas	12177267.32	12052546.07	101.03	
Ge	72	No Gas	2417397.26	2410488.1	100.29	
In	115	No Gas	16126609.14	15897428.5	101.44	
Lu	175	No Gas	19362187.21	18289690.1433333	105.86	
Rh	103	No Gas	15223806.62	15416412.72	98.75	
Sc	45	No Gas	9425276.02	9412079.90333333	100.14	
Tb	159	No Gas	20618467.19	19489435.1266667	105.79	

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INORGANIC ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW05-060123
Collect Date:	06/01/23 Time: 0910	GCAL Sample ID:	22306021004
Matrix:	Water % Solids: NA	Instrument ID:	ICPMS2
Sample Amt:	50 mL	Lab File ID:	2230605a_MS2.b\1221SMPL.d
Prep Vol.:	50 (mL)	Dilution Factor:	1 Analyst: TDM
Prep Date:	06/05/23	Analysis Date:	06/05/23 Time: 1204
Prep Batch:	766941	Analytical Batch:	767033
Prep Method:	3010A	Analytical Method:	EPA 6020B

ANALYTE	***RESULT***	***UNITS***	***Q***	***DL***	***LOD***	***LOQ***
Arsenic	0.85	ug/L	J	0.25	0.50	1.00
Iron	246	ug/L		25.0	50.0	100
Manganese	255	ug/L		1.25	2.50	5.00

FORM I - IN

Sample Report

Sample Name	22306021004	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1221SMPL.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 12:04:46	**Total Dilution**	1.0000
Sample Type	Sample	**Sample Pass/Fail**	Pass
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Pass

Units : ppb

QC Analyte Table

Name	Mass	ISTD	Mode	MeasValue	Final Conc	RSD	CPS	LDR	QC Flag
Li	7	6	No Gas	1.129	1.129	0.9	93632.58	500	
Be	9	6	No Gas	0.013	0.013	14.2	89.33	1000	
B	11	6	No Gas	74.61	74.610	0.6	141138.18	500	
Sr	88	72	No Gas	202.938	202.938	0.3	14829164.78	1000	
Zr	90	72	No Gas	0.102	0.102	14.5	6427.89	100	
Mo	95	115	No Gas	0.937	0.937	1.8	14834.43	1000	
Ag	107	115	No Gas	0.005	0.005	27.8	243.34	100	
Cd	111	115	No Gas	0.058	0.058	8.4	586.69	1000	
Sb	121	115	No Gas	0.126	0.126	0.9	6682.77	1000	
Ba	137	115	No Gas	139.751	139.751	0.3	1716424.65	1000	
Tl	205	209	No Gas	-0.046	-0.046	6.7	2270.29	1000	
Pb	208	209	No Gas	0.122	0.122	3.8	6261.78	1000	
Na	23	45	He	3123.324	3123.324	1.4	1536602.48	100000	
Mg	24	45	He	14157.954	14157.954	1.1	3025951.72	100000	
Al	27	45	He	86.371	86.371	8.1	5578.74	20000	
Si	29	45	He	6675.092	6675.092	5.5	20224.29	10000	
K	39	45	He	3025.688	3025.688	1.5	709480.87	100000	
Ca	44	45	He	41327.498	41327.498	1.2	439681.07	500000	
Ti	47	45	He	2.552	2.552	10.9	253.00	1000	
V	51	72	He	1.325	1.325	3.9	5382.16	1000	
Cr	52	72	He	0.416	0.416	2.6	4060.60	1000	
Mn	55	72	He	255.447	255.447	1.4	522499.23	5000	
Fe	57	72	He	245.575	245.575	3.1	23629.79	100000	
Co	59	72	He	0.301	0.301	2.9	2798.07	1000	
Ni	60	72	He	1.867	1.867	3.4	4945.34	2000	
Cu	63	45	He	1.307	1.307	3.9	9807.82	1000	
Zn	66	72	He	38.066	38.066	2.6	36221.39	20000	
As	75	72	He	0.852	0.852	6.4	946.70	1000	
Se	78	72	He	1.467	1.467	3.7	99.50	50	
Sn	120	115	He	0.038	0.038	12.9	540.02	1000	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	5344548.88	6090049.07666667	87.76	
Ge	72	He	172659.18	183172.586666667	94.26	
In	115	He	1658418.00	1893460.06333333	87.59	
Lu	175	He	3878600.26	4453666.28666667	87.09	
Rh	103	He	5214581.18	5805949.63	89.81	
Sc	45	He	196471.85	210499.653333333	93.34	
Tb	159	He	6591001.78	7379981.76666667	89.31	

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
(Li)	6	No Gas	1295688.96	1489089.40333333	87.01	
Bi	209	No Gas	12763536.48	12052546.07	105.9	
Ge	72	No Gas	2420329.69	2410488.1	100.41	
In	115	No Gas	16532181.88	15897428.5	103.99	
Lu	175	No Gas	19757732.21	18289690.1433333	108.03	
Rh	103	No Gas	15592744.94	15416412.72	101.14	
Sc	45	No Gas	9401097.13	9412079.90333333	99.88	
Tb	159	No Gas	21026288.02	19489435.1266667	107.89	

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INORGANIC ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW06-060123
Collect Date:	06/01/23 Time: 1100	GCAL Sample ID:	22306021005
Matrix:	Water % Solids: NA	Instrument ID:	ICPMS2
Sample Amt:	50 mL	Lab File ID:	2230605a_MS2.b\1222SMPL.d
Prep Vol.:	50 (mL)	Dilution Factor:	1 Analyst: TDM
Prep Date:	06/05/23	Analysis Date:	06/05/23 Time: 1208
Prep Batch:	766941	Analytical Batch:	767033
Prep Method:	3010A	Analytical Method:	EPA 6020B

ANALYTE	***RESULT***	***UNITS***	***Q***	***DL***	***LOD***	***LOQ***
Arsenic	4.36	ug/L		0.25	0.50	1.00
Iron	5250	ug/L		25.0	50.0	100
Manganese	861	ug/L		1.25	2.50	5.00

FORM I - IN

Reference Sample Report

Sample Name	22306021005
File Name	1222SMPL.d
Data Path Name	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
Acq Time	6/5/2023 12:08:16
Sample Type	AllRef
Total Dilution	1.0000
Comment	ICPMS-2,TDM
ISTD Ref FileName	004CALB.d
Sample QC Pass/Fial	Pass
ISTD QC Pass/Fail	Pass

QC Analyte Table

Name	Mass	ISTD	Tune Mode	Conc.	Conc. RSD	CPS	LDR	QC Flag
Li	7	6	No Gas	1.812	15.5	89562.84	500	
Be	9	6	No Gas	0.032	10.8	120.67	1000	
B	11	6	No Gas	73.965	1.4	116671.56	500	
Sr	88	72	No Gas	162.828	0.4	11682679.00	1000	
Zr	90	72	No Gas	0.583	2.6	28167.72	100	
Mo	95	115	No Gas	0.555	5.1	9177.49	1000	
Ag	107	115	No Gas	0.007	14.8	310.01	100	
Cd	111	115	No Gas	0.051	12.8	522.24	1000	
Sb	121	115	No Gas	0.223	4.5	9753.47	1000	
Ba	137	115	No Gas	52.858	0.7	630910.86	1000	
Tl	205	209	No Gas	-0.053	N/A	1763.52	1000	
Pb	208	209	No Gas	0.519	1.4	23618.51	1000	
Na	23	45	He	20670.986	1.6	10018690.48	100000	
Mg	24	45	He	13507.453	0.6	2863108.50	100000	
Al	27	45	He	163.057	7.0	10361.88	20000	
Si	29	45	He	8260.505	5.9	24081.97	10000	
K	39	45	He	5049.661	1.0	1155120.22	100000	
Ca	44	45	He	36914.779	1.2	389563.91	500000	
Ti	47	45	He	7.617	4.7	737.02	1000	
V	51	72	He	2.429	2.7	9585.43	1000	
Cr	52	72	He	1.113	5.3	7535.32	1000	
Mn	55	72	He	860.711	1.3	1755190.62	5000	
Fe	57	72	He	5254.845	1.7	498270.74	100000	
Co	59	72	He	1.276	1.5	11581.35	1000	
Ni	60	72	He	6.448	1.4	16631.77	2000	
Cu	63	45	He	1.865	1.3	13680.93	1000	
Zn	66	72	He	22.994	0.3	21928.71	20000	
As	75	72	He	4.358	1.9	4334.65	1000	
Se	78	72	He	0.348	47.0	43.23	50	
Sn	120	115	He	0.067	19.4	667.80	1000	

QC ISTD Table

Name	Mass	Tune Mode	CPS	CPS RSD	Ref CPS	% Rec	%QC Low	%QC High	QC Flag
(Li)	6	No Gas	1192399.94	2.4	1489089.40333333	80.08	70	120	
Sc	45	No Gas	9164630.19	0.2	9412079.90333333	97.37	70	120	
Ge	72	No Gas	2376384.84	0.2	2410488.1	98.59	70	120	

Reference Sample Report

Name	Mass	Tune Mode	CPS	CPS RSD	Ref CPS	% Rec	%QC Low	%QC High	QC Flag
Rh	103	No Gas	15176847.73	0.3	15416412.72	98.45	70	120	
In	115	No Gas	16100838.60	0.2	15897428.5	101.28	70	120	
Tb	159	No Gas	20419203.45	0.2	19489435.1266667	104.77	70	120	
Lu	175	No Gas	19371807.21	0.7	18289690.1433333	105.92	70	120	
Bi	209	No Gas	12308795.23	1.3	12052546.07	102.13	70	120	
Sc	45	He	194853.82	1.3	210499.653333333	92.57	70	120	
Ge	72	He	172185.00	1.7	183172.586666667	94	70	120	
Rh	103	He	5227608.68	2.0	5805949.63	90.04	70	120	
In	115	He	1651544.43	2.0	1893460.06333333	87.22	70	120	
Tb	159	He	6575204.28	1.7	7379981.76666667	89.1	70	120	
Lu	175	He	3883991.71	2.3	4453666.28666667	87.21	70	120	
Bi	209	He	5254889.40	2.1	6090049.07666667	86.29	70	120	

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INORGANIC ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123MS
Collect Date:	06/01/23 Time: 0955	GCAL Sample ID:	22306021006
Matrix:	Water % Solids: NA	Instrument ID:	ICPMS2
Sample Amt:	50 mL	Lab File ID:	2230605a_MS2.b\1223SMPL.d
Prep Vol.:	50 (mL)	Dilution Factor:	1 Analyst: TDM
Prep Date:	06/05/23	Analysis Date:	06/05/23 Time: 1211
Prep Batch:	766941	Analytical Batch:	767033
Prep Method:	3010A	Analytical Method:	EPA 6020B

ANALYTE	RESULT	UNITS	Q	DL	LOD	LOQ
Arsenic	52.0	ug/L		0.25	0.50	1.00
Iron	5270	ug/L		25.0	50.0	100
Manganese	113	ug/L		1.25	2.50	5.00

FORM I - IN

Matrix Spike Sample (MS) Report

Sample Name 22306021006
File Name 1223SMPL.d
Data Path Name C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
Acq Time 6/5/2023 12:11:45
Sample Type MS
Total Dilution 1.0000
Comment ICPMS-2,TDM
ISTD Ref FileName 004CALB.d
Sample QC Pass/Fial Fail
ISTD QC Pass/Fail Pass
QC Ref File Name 1222SMPL.d

QC Analyte Table

Name	Mass	Tune	Conc.	Conc. RSD	CPS	Reference Conc	Spk Amt	% Rec	%Low	%High	Flag
Li	7	No Gas	299.320	1.1	2037027.89	1.81187143512997	250	119	80	120	
Be	9	No Gas	58.280	1.4	129641.22	0.0316349780495815	50	116.5	80	120	
B	11	No Gas	480.628	0.6	758711.78	73.9645540902783	250	162.67	80	120	> +/- 20%
Sr	88	No Gas	412.694	0.5	29743108.72	162.828467037716	50	499.73	80	120	> +/- 20%
Zr	90	No Gas	10.516	0.8	482351.80	0.582540159720245	10	99.34	80	120	
Mo	95	No Gas	52.488	0.8	709082.02	0.554598515148509	50	103.87	80	120	
Ag	107	No Gas	51.264	1.0	1778606.53	0.00699813446788482	50	102.51	80	120	
Cd	111	No Gas	50.879	1.0	376094.02	0.0513613853824138	50	101.66	80	120	
Sb	121	No Gas	106.915	0.5	3515257.50	0.223371263913561	100	106.69	80	120	
Ba	137	No Gas	109.766	0.2	1287789.57	52.8577925988333	50	113.82	80	120	
Tl	205	No Gas	50.637	1.0	3059777.14	-0.0528443679417813	50	101.38	80	120	
Pb	208	No Gas	50.739	1.2	2210173.09	0.519339892532968	50	100.44	80	120	
Na	23	He	45242.166	0.2	21730464.68	20670.9858945964	5000	491.42	80	120	> +/- 20%
Mg	24	He	46698.123	1.5	9813901.94	13507.4529866846	5000	663.81	80	120	> +/- 20%
Al	27	He	1214.449	1.5	76000.63	163.056829490888	1000	105.14	80	120	
Si	29	He	10585.682	4.2	29754.78	8260.50489149299	5000	46.5	80	120	> +/- 20%
K	39	He	7740.176	1.5	1740477.58	5049.66117579662	5000	53.81	80	120	> +/- 20%
Ca	44	He	133181.906	0.7	1391904.35	36914.779350816	25000	385.07	80	120	> +/- 20%
Ti	47	He	55.866	2.7	5320.39	7.61707827383935	50	96.5	80	120	
V	51	He	54.023	0.6	205237.16	2.42900783792457	50	103.19	80	120	
Cr	52	He	53.124	0.1	265648.89	1.11280393902529	50	104.02	80	120	
Mn	55	He	113.313	0.6	229334.80	860.711422393556	50	-1494.8	80	120	> +/- 20%
Fe	57	He	5265.622	0.9	495372.58	5254.84502002217	5000	0.22	80	120	> +/- 20%
Co	59	He	51.842	0.8	463683.66	1.27574605730146	50	101.13	80	120	
Ni	60	He	104.039	0.5	263800.24	6.44769558018653	100	97.59	80	120	
Cu	63	He	53.827	1.3	378760.14	1.86460390848303	50	103.92	80	120	
Zn	66	He	1019.398	0.4	952498.09	22.9944155358434	1000	99.64	80	120	
As	75	He	52.039	0.8	50050.59	4.3583260798696	50	95.36	80	120	
Se	78	He	10.662	2.7	555.12	0.348258735316304	10	103.14	80	120	
Sn	120	He	54.197	0.5	237024.87	0.067008715233113	50	108.26	80	120	

QC ISTD Table

Matrix Spike Sample (MS) Report

Name	Mass	Tune Mode	CPS	CPS RSD	Ref CPS	% Rec	%QC Low	%QC High	QC Flag
(Li)	6	No Gas	1214899.26	1.8	1489089.40333333	81.59	70	120	
Sc	45	No Gas	9123909.64	0.8	9412079.90333333	96.94	70	120	
Ge	72	No Gas	2387291.09	1.2	2410488.1	99.04	70	120	
Rh	103	No Gas	14838813.30	0.5	15416412.72	96.25	70	120	
In	115	No Gas	15826855.57	0.7	15897428.5	99.56	70	120	
Tb	159	No Gas	20593555.11	0.3	19489435.1266667	105.67	70	120	
Lu	175	No Gas	19332672.21	0.3	18289690.1433333	105.7	70	120	
Bi	209	No Gas	12114553.15	1.0	12052546.07	100.51	70	120	
Sc	45	He	193205.20	0.7	210499.653333333	91.78	70	120	
Ge	72	He	170795.76	0.7	183172.586666667	93.24	70	120	
Rh	103	He	5103259.24	1.1	5805949.63	87.9	70	120	
In	115	He	1598579.15	1.6	1893460.06333333	84.43	70	120	
Tb	159	He	6582842.40	1.8	7379981.76666667	89.2	70	120	
Lu	175	He	3873628.48	0.9	4453666.28666667	86.98	70	120	
Bi	209	He	5148095.55	0.8	6090049.07666667	84.53	70	120	

I

INORGANIC ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123MSD
Collect Date:	06/01/23 Time: 1015	GCAL Sample ID:	22306021007
Matrix:	Water % Solids: NA	Instrument ID:	ICPMS2
Sample Amt:	50 mL	Lab File ID:	2230605a_MS2.b\1224SMPL.d
Prep Vol.:	50 (mL)	Dilution Factor:	1 Analyst: TDM
Prep Date:	06/05/23	Analysis Date:	06/05/23 Time: 1215
Prep Batch:	766941	Analytical Batch:	767033
Prep Method:	3010A	Analytical Method:	EPA 6020B

ANALYTE	RESULT	UNITS	Q	DL	LOD	LOQ
Arsenic	51.8	ug/L		0.25	0.50	1.00
Iron	5210	ug/L		25.0	50.0	100
Manganese	115	ug/L		1.25	2.50	5.00

FORM I - IN

Matrix Spike Duplicate (MSD) Sample Report

Sample Name	22306021007
File Name	1224SMPL.d
Data Path Name	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
Acq Time	6/5/2023 12:15:15
Sample Type	MSD
Total Dilution	1.0000
Comment	ICPMS-2,TDM
ISTD Ref FileName	004CALB.d
Sample QC Pass/Fial	Pass
ISTD Ref FileName	Pass
QC Ref File Name	1223SMPL.d

QC Analyte Table

Name	Mass	Tune Mode	Conc.	Conc. RSD	CPS	Reference Conc	RPD	RPD Max	Flag
Li	7	No Gas	295.958	1.7	1909738.83	299.319922482179	1.13	20	
Be	9	No Gas	57.947	1.4	122172.33	58.2796096354296	0.57	20	
B	11	No Gas	492.141	2.2	736134.39	480.628231636701	2.37	20	
Sr	88	No Gas	416.174	0.9	29763677.06	412.694315278725	0.84	20	
Zr	90	No Gas	10.430	0.7	474775.95	10.5164054823397	0.82	20	
Mo	95	No Gas	51.976	0.4	701476.71	52.4884684545942	0.98	20	
Ag	107	No Gas	50.419	0.9	1747371.04	51.2636197635508	1.66	20	
Cd	111	No Gas	50.323	1.5	371570.23	50.8792860553532	1.1	20	
Sb	121	No Gas	105.703	1.1	3471937.57	106.914552096389	1.14	20	
Ba	137	No Gas	111.902	0.5	1311495.36	109.765872843894	1.93	20	
Tl	205	No Gas	50.246	0.4	3012315.89	50.6374775461998	0.78	20	
Pb	208	No Gas	50.402	1.2	2178304.29	50.7389278767018	0.67	20	
Na	23	He	45412.513	1.1	21800374.68	45242.1655477931	0.38	20	
Mg	24	He	47357.033	1.5	9946696.93	46698.1233087174	1.4	20	
Al	27	He	1166.414	1.1	72959.05	1214.44868163217	4.04	20	
Si	29	He	10377.844	3.9	29224.39	10585.6820501386	1.98	20	
K	39	He	7750.628	0.8	1741983.15	7740.17606420886	0.13	20	
Ca	44	He	134062.403	0.8	1400398.83	133181.905583705	0.66	20	
Ti	47	He	53.714	2.8	5111.89	55.8663230615002	3.93	20	
V	51	He	53.465	0.3	202475.01	54.0232565471463	1.04	20	
Cr	52	He	52.680	1.0	262621.13	53.1235827525945	0.84	20	
Mn	55	He	115.315	0.5	232640.06	113.312735208285	1.75	20	
Fe	57	He	5205.790	0.1	488179.59	5265.62218587004	1.14	20	
Co	59	He	51.746	0.9	461322.60	51.8424844206753	0.19	20	
Ni	60	He	103.387	0.3	261312.09	104.039240219527	0.63	20	
Cu	63	He	53.291	2.2	374758.70	53.8268649792443	1	20	
Zn	66	He	1018.378	0.3	948513.37	1019.39753133315	0.1	20	
As	75	He	51.790	0.5	49651.23	52.0386757561472	0.48	20	
Se	78	He	9.866	1.0	513.95	10.6622589854251	7.76	20	
Sn	120	He	54.046	0.2	233343.49	54.1968893786676	0.28	20	

QC ISTD Table

Matrix Spike Duplicate (MSD) Sample Report

Name	Mass	Tune Mode	CPS	CPS RSD	Ref CPS	% Rec	%QC Low	%QC High	QC Flag
(Li)	6	No Gas	1151389.41	1.0	1489089.40333333	77.32	70	120	
Sc	45	No Gas	8971499.09	1.1	9412079.90333333	95.32	70	120	
Ge	72	No Gas	2368979.98	1.1	2410488.1	98.28	70	120	
Rh	103	No Gas	14771102.19	0.9	15416412.72	95.81	70	120	
In	115	No Gas	15810525.84	0.9	15897428.5	99.45	70	120	
Tb	159	No Gas	20531535.94	0.9	19489435.1266667	105.35	70	120	
Lu	175	No Gas	19396716.79	1.1	18289690.1433333	106.05	70	120	
Bi	209	No Gas	12019575.65	1.3	12052546.07	99.73	70	120	
Sc	45	He	193110.59	0.8	210499.653333333	91.74	70	120	
Ge	72	He	170254.72	0.8	183172.586666667	92.95	70	120	
Rh	103	He	5050750.49	1.7	5805949.63	86.99	70	120	
In	115	He	1578071.53	0.8	1893460.06333333	83.34	70	120	
Tb	159	He	6448322.19	1.1	7379981.76666667	87.38	70	120	
Lu	175	He	3802790.36	0.9	4453666.28666667	85.39	70	120	
Bi	209	He	5096241.49	1.0	6090049.07666667	83.68	70	120	

I

INORGANIC ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW-806-060123
Collect Date:	06/01/23 Time: 1100	GCAL Sample ID:	22306021008
Matrix:	Water % Solids: NA	Instrument ID:	ICPMS2
Sample Amt:	50 mL	Lab File ID:	2230605a_MS2.b\1227SMPL.d
Prep Vol.:	50 (mL)	Dilution Factor:	1 Analyst: TDM
Prep Date:	06/05/23	Analysis Date:	06/05/23 Time: 1225
Prep Batch:	766941	Analytical Batch:	767033
Prep Method:	3010A	Analytical Method:	EPA 6020B

ANALYTE	***RESULT***	***UNITS***	***Q***	***DL***	***LOD***	***LOQ***
Arsenic	11.7	ug/L		0.25	0.50	1.00
Iron	4790	ug/L		25.0	50.0	100
Manganese	829	ug/L		1.25	2.50	5.00

FORM I - IN

Sample Report

Sample Name	22306021008	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1227SMPL.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 12:25:42	**Total Dilution**	1.0000
Sample Type	Sample	**Sample Pass/Fail**	Pass
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Fail

Units : ppb

QC Analyte Table

Name	Mass	ISTD	Mode	MeasValue	Final Conc	RSD	CPS	LDR	QC Flag
Li	7	6	No Gas	2.064	2.064	0.2	73948.50	500	
Be	9	6	No Gas	0.034	0.034	9.4	106.00	1000	
B	11	6	No Gas	69.834	69.834	2.1	96992.10	500	
Sr	88	72	No Gas	168.575	168.575	0.1	11791742.74	1000	
Zr	90	72	No Gas	0.64	0.640	3.4	30001.13	100	
Mo	95	115	No Gas	0.902	0.902	1.8	13644.37	1000	
Ag	107	115	No Gas	0.008	0.008	8.7	347.79	100	
Cd	111	115	No Gas	0.02	0.020	4.2	283.34	1000	
Sb	121	115	No Gas	0.335	0.335	2.0	13212.99	1000	
Ba	137	115	No Gas	47.403	47.403	0.6	553980.20	1000	
Tl	205	209	No Gas	-0.054	-0.054	4.8	1640.17	1000	
Pb	208	209	No Gas	0.254	0.254	2.7	11385.34	1000	
Na	23	45	He	26204.146	26204.146	0.6	12412917.32	100000	
Mg	24	45	He	14970.601	14970.601	0.6	3101815.06	100000	
Al	27	45	He	61.978	61.978	3.3	3907.25	20000	
Si	29	45	He	7270.986	7270.986	6.5	21072.15	10000	
K	39	45	He	5052.027	5052.027	0.7	1129753.19	100000	
Ca	44	45	He	37941.05	37941.050	0.1	391380.69	500000	
Ti	47	45	He	6.043	6.043	9.7	573.01	1000	
V	51	72	He	2.143	2.143	0.5	8399.14	1000	
Cr	52	72	He	0.879	0.879	1.1	6300.28	1000	
Mn	55	72	He	829.008	829.008	0.7	1672190.18	5000	
Fe	57	72	He	4792.027	4792.027	1.0	449506.59	100000	
Co	59	72	He	1.349	1.349	1.9	12103.98	1000	
Ni	60	72	He	6.812	6.812	2.2	17371.52	2000	
Cu	63	45	He	0.899	0.899	5.1	6687.16	1000	
Zn	66	72	He	26.878	26.878	1.3	25305.31	20000	
As	75	72	He	11.682	11.682	1.9	11295.38	1000	
Se	78	72	He	0.401	0.401	10.5	45.30	50	
Sn	120	115	He	0.047	0.047	3.9	544.46	1000	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	4896718.16	6090049.07666667	80.41	
Ge	72	He	170298.88	183172.586666667	92.97	
In	115	He	1559296.70	1893460.06333333	82.35	
Lu	175	He	3613846.92	4453666.28666667	81.14	
Rh	103	He	5031021.47	5805949.63	86.65	
Sc	45	He	190490.63	210499.653333333	90.49	
Tb	159	He	6177180.95	7379981.76666667	83.7	

Name	Mass	Mode	CPS	Ref CPS	Rec
(Li)	6	No Gas	950021.09	1489089.40333333	63.8
Bi	209	No Gas	11810143.99	12052546.07	97.99
Ge	72	No Gas	2316836.37	2410488.1	96.11
In	115	No Gas	15730797.30	15897428.5	98.95
Lu	175	No Gas	18805913.47	18289690.1433333	102.82
Rh	103	No Gas	14854772.74	15416412.72	96.36
Sc	45	No Gas	8734028.26	9412079.90333333	92.8
Tb	159	No Gas	19990755.12	19489435.1266667	102.57

Flag
<70% or >120%

Metals

Form II

Calibration Verifications

II

INITIAL CALIBRATION VERIFICATION (ICV) STANDARD

Report No:	223060210	GCAL QC ID:	1600
Instrument ID:	ICPMS2	Lab File ID:	2230605a_MS2.b\010_ICV.d
Analyst:	TDM	Analytical Batch:	767033
Analysis Date:	06/05/23 Time: 1115	Analytical Method:	EPA 6020B

ANALYTE	***TRUE***	***FOUND***	***%RECOVERY***	***Q***	***UNITS***
Arsenic	50.0	51.1	102		ug/L
Iron	5000	5190	104		ug/L
Manganese	50.0	51.2	102		ug/L

CONTROL LIMITS 90-110%

FORM II - IN

Initial Calibration Verification (ICV) Report

Sample Name	1600	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	010_ICV.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 11:15:55	**Total Dilution**	1.0000
Sample Type	ICV	**Sample Pass/Fail**	Pass
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Pass

Units : ppb

QC Analyte Table

Recovery Limits: 90-110%

Name	Mass	ISTD	Mode	Conc	RSD	CPS	ExpValue	Rec	QC Flag
Li	7	6	No Gas	266.675	1.2	2508358.82	250	106.67	
Be	9	6	No Gas	53.061	0.9	162678.54	50	106.12	
B	11	6	No Gas	271.625	0.4	592172.15	250	108.65	
Sr	88	72	No Gas	50.985	0.5	3763106.09	50	101.97	
Zr	90	72	No Gas	47.874	1.3	2242211.37	50	95.75	
Mo	95	115	No Gas	49.602	1.4	695099.31	50	99.2	
Ag	107	115	No Gas	52.106	1.5	1875015.48	50	104.21	
Cd	111	115	No Gas	50.670	0.7	388466.52	50	101.34	
Sb	121	115	No Gas	52.897	0.9	1805072.29	50	105.79	
Ba	137	115	No Gas	50.307	0.8	612209.26	50	100.61	
Tl	205	209	No Gas	50.418	1.5	3107488.39	50	100.84	
Pb	208	209	No Gas	49.647	1.2	2205885.33	50	99.29	
Na	23	45	He	5203.381	0.7	2727056.63	5000	104.07	
Mg	24	45	He	5150.498	0.3	1176603.03	5000	103.01	
Al	27	45	He	1029.677	0.7	70033.67	1000	102.97	
Si	29	45	He	5240.031	6.9	17668.46	5000	104.8	
K	39	45	He	5164.121	1.0	1272158.05	5000	103.28	
Ca	44	45	He	25588.777	1.1	291189.20	25000	102.36	
Ti	47	45	He	51.582	2.7	5337.97	50	103.16	
V	51	72	He	51.226	0.4	208352.74	50	102.45	
Cr	52	72	He	52.125	1.1	279094.20	50	104.25	
Mn	55	72	He	51.176	0.8	110944.06	50	102.35	
Fe	57	72	He	5185.607	1.0	522263.52	5000	103.71	
Co	59	72	He	52.583	1.2	503492.20	50	105.17	
Ni	60	72	He	105.028	1.4	285098.85	100	105.03	
Cu	63	45	He	54.557	0.8	417130.60	50	109.11	
Zn	66	72	He	1052.905	0.5	1053179.51	1000	105.29	
As	75	72	He	51.051	1.1	52566.71	50	102.1	
Se	78	72	He	25.201	1.2	1367.30	25	100.8	
Sn	120	115	He	51.485	1.3	263665.03	50	102.97	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	5998674.29	6090049.07666667	98.5	
Ge	72	He	182853.46	183172.586666667	99.83	
In	115	He	1866245.11	1893460.06333333	98.56	
Lu	175	He	4461226.18	4453666.28666667	100.17	
Rh	103	He	5730135.75	5805949.63	98.69	
Sc	45	He	209936.57	210499.653333333	99.73	
Tb	159	He	7397051.76	7379981.76666667	100.23	

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
(Li)	6	No Gas	1514339.61	1489089.40333333	101.7	
Bi	209	No Gas	12357422.73	12052546.07	102.53	
Ge	72	No Gas	2444106.43	2410488.1	101.39	
In	115	No Gas	16379244.41	15897428.5	103.03	
Lu	175	No Gas	19363192.63	18289690.1433333	105.87	
Rh	103	No Gas	15551450.49	15416412.72	100.88	
Sc	45	No Gas	9577507.68	9412079.90333333	101.76	
Tb	159	No Gas	20389875.53	19489435.1266667	104.62	

II

LOW LEVEL CONTINUING CALIBRATION VERIFICATION (LLCCV) STANDARD

Report No:	223060210	GCAL QC ID:	1803
Instrument ID:	ICPMS2	Lab File ID:	2230605a_MS2.b\1211CCV1.d
Analyst:	TDM	Analytical Batch:	767033
Analysis Date:	06/05/23 Time: 1129	Analytical Method:	EPA 6020B

ANALYTE	***TRUE***	***FOUND***	***%RECOVERY***	***Q***	***UNITS***
Arsenic	1.00	0.940	94		ug/L
Iron	100	102	102		ug/L
Manganese	5.00	5.21	104		ug/L

CONTROL LIMITS 80-120%

FORM II - IN

Low Level Continuing Calibration Verification(LLCCV) Report

Sample Name	1803	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1211CCV1.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 11:29:53	**Total Dilution**	1.0000
Sample Type	LLCCV1	**Sample Pass/Fail**	Fail
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Pass

Units : ppb

QC Analyte Table

Recovery Limits: Initial 6020B DOD 80-120% / 70-130% 6020B and 200.8

Name	Mass	ISTD	Mode	Conc	RSD	CPS	ExpValue	Rec	QC Flag
Li	7	6	No Gas	5.104	1.4	143138.76	5	102.08	
Be	9	6	No Gas	1.040	2.9	3204.36	1	104	
B	11	6	No Gas	11.147	0.8	26911.40	10	111.47	
Sr	88	72	No Gas	1.008	1.7	74986.34	1	100.8	
Zr	90	72	No Gas	0.927	0.9	44816.31	1	92.7	
Mo	95	115	No Gas	0.885	1.0	14309.46	1	88.5	
Ag	107	115	No Gas	0.979	0.9	36126.94	1	97.9	
Cd	111	115	No Gas	0.975	2.5	7794.43	1	97.5	
Sb	121	115	No Gas	1.968	0.4	71046.74	2	98.4	
Ba	137	115	No Gas	0.946	2.9	11865.16	1	94.6	
Tl	205	209	No Gas	0.885	2.3	61482.67	1	88.5	
Pb	208	209	No Gas	0.965	3.3	45013.89	1	96.5	
Na	23	45	He	103.462	1.2	66142.25	100	103.46	
Mg	24	45	He	102.156	3.5	23615.85	100	102.16	
Al	27	45	He	20.905	5.9	1501.42	20	104.52	
Si	29	45	He	150.736	29.1	3669.17	200	75.37	> +/- 20%
K	39	45	He	96.534	1.2	53597.14	100	96.53	
Ca	44	45	He	493.656	2.7	6273.09	500	98.73	
Ti	47	45	He	1.084	20.5	117.67	1	108.4	
V	51	72	He	1.001	4.8	4375.14	1	100.1	
Cr	52	72	He	0.990	1.3	7341.90	1	99	
Mn	55	72	He	5.207	2.7	11371.13	5	104.14	
Fe	57	72	He	101.507	2.2	10507.41	100	101.51	
Co	59	72	He	1.046	2.4	10082.46	1	104.6	
Ni	60	72	He	2.074	1.8	5786.75	2	103.7	
Cu	63	45	He	1.060	1.6	8518.12	1	106	
Zn	66	72	He	20.694	1.8	20951.78	20	103.47	
As	75	72	He	0.943	2.8	1094.38	1	94.3	
Se	78	72	He	0.983	3.1	79.48	1	98.3	
Sn	120	115	He	0.992	2.4	5445.55	1	99.2	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	5967130.54	6090049.07666667	97.98	
Ge	72	He	182554.46	183172.586666667	99.66	
In	115	He	1853898.36	1893460.06333333	97.91	
Lu	175	He	4350165.98	4453666.28666667	97.68	
Rh	103	He	5743717.27	5805949.63	98.93	
Sc	45	He	207969.52	210499.653333333	98.8	
Tb	159	He	7252578.23	7379981.76666667	98.27	

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
(Li)	6	No Gas	1491290.54	1489089.40333333	100.15	
Bi	209	No Gas	12777285.23	12052546.07	106.01	
Ge	72	No Gas	2428394.97	2410488.1	100.74	
In	115	No Gas	16768992.33	15897428.5	105.48	
Lu	175	No Gas	19181129.71	18289690.1433333	104.87	
Rh	103	No Gas	16053934.36	15416412.72	104.14	
Sc	45	No Gas	9649365.17	9412079.90333333	102.52	
Tb	159	No Gas	20621883.86	19489435.1266667	105.81	

II

LINEAR DYNAMIC RANGE (LDR) STANDARD

Report No:	223060210	GCAL QC ID:	2500
Instrument ID:	ICPMS2	Lab File ID:	2230605a_MS2.b\1214_QC1.d
Analyst:	TDM	Analytical Batch:	767033
Analysis Date:	06/05/23 Time: 1140	Analytical Method:	EPA 6020B

ANALYTE	***TRUE***	***FOUND***	***%RECOVERY***	***Q***	***UNITS***
Arsenic	1000	974	97		ug/L
Iron	100000	97000	97		ug/L
Manganese	5000	4880	98		ug/L

CONTROL LIMITS 90-110%

FORM II - IN

Linear Dynamic Range Check (LDR) Report

Sample Name	LDR	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1214_QC1.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 11:40:21	**Total Dilution**	1.0000
Sample Type	QC1	**Sample Pass/Fail**	Pass
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Pass

Units : ppb

QC Analyte Table

Recovery Limits: 90-110%

Name	Mass	ISTD	Mode	Conc	RSD	CPS	ExpValue	Rec	QC Flag
Be	9	6	No Gas	1024.122	0.6	2923281.83	1000	102.41	
Sr	88	72	No Gas	960.834	0.4	67036462.34	1000	96.08	
Mo	95	115	No Gas	991.739	1.2	12768500.32	1000	99.17	
Cd	111	115	No Gas	953.404	1.1	6727536.26	1000	95.34	
Sb	121	115	No Gas	976.863	0.5	30654102.12	1000	97.69	
Ba	137	115	No Gas	960.401	0.4	10759231.52	1000	96.04	
Tl	205	209	No Gas	950.504	1.1	53227304.21	1000	95.05	
Pb	208	209	No Gas	963.945	0.8	38962181.92	1000	96.39	
Na	23	45	He	96977.173	0.7	48806402.61	100000	96.98	
Mg	24	45	He	98290.699	1.7	21649978.84	100000	98.29	
Al	27	45	He	19454.859	0.7	1274678.46	20000	97.27	
K	39	45	He	97808.002	0.4	22705652.99	100000	97.81	
Ca	44	45	He	495078.483	0.5	5421276.58	500000	99.02	
Ti	47	45	He	974.688	0.5	97174.93	1000	97.47	
V	51	72	He	1008.866	1.1	3941858.18	1000	100.89	
Cr	52	72	He	986.101	1.1	5043092.44	1000	98.61	
Mn	55	72	He	4880.290	0.4	10167328.48	5000	97.61	
Fe	57	72	He	97017.564	0.5	9394073.82	100000	97.02	
Co	59	72	He	983.217	1.2	9055033.25	1000	98.32	
Ni	60	72	He	1925.577	1.1	5025370.77	2000	96.28	
Cu	63	45	He	944.828	1.1	6960137.78	1000	94.48	
Zn	66	72	He	18803.898	0.7	18089246.50	20000	94.02	
As	75	72	He	973.931	1.0	962515.02	1000	97.39	
Sn	120	115	He	1015.748	1.0	4792486.47	1000	101.57	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	5522930.34	6090049.07666667	90.69	
Ge	72	He	175896.18	183172.586666667	96.03	
In	115	He	1721843.46	1893460.06333333	90.94	
Lu	175	He	4303347.23	4453666.28666667	96.62	
Rh	103	He	5274765.76	5805949.63	90.85	
Sc	45	He	202497.97	210499.653333333	96.2	
Tb	159	He	7082290.10	7379981.76666667	95.97	

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
(Li)	6	No Gas	1410406.19	1489089.40333333	94.72	
Bi	209	No Gas	11244831.50	12052546.07	93.3	
Ge	72	No Gas	2311170.95	2410488.1	95.88	
In	115	No Gas	15080757.39	15897428.5	94.86	
Lu	175	No Gas	18377140.98	18289690.1433333	100.48	
Rh	103	No Gas	14311466.66	15416412.72	92.83	
Sc	45	No Gas	9341395.74	9412079.90333333	99.25	
Tb	159	No Gas	19475915.54	19489435.1266667	99.93	

II

CONTINUING CALIBRATION VERIFICATION (CCV) STANDARD

Report No:	223060210	GCAL QC ID:	1800
Instrument ID:	ICPMS2	Lab File ID:	2230605a_MS2.b\1228_CCV.d
Analyst:	TDM	Analytical Batch:	767033
Analysis Date:	06/05/23 Time: 1229	Analytical Method:	EPA 6020B

ANALYTE	***TRUE***	***FOUND***	***%RECOVERY***	***Q***	***UNITS***
Arsenic	10.0	10.5	105		ug/L
Iron	1000	1060	106		ug/L
Manganese	50.0	53.8	108		ug/L

CONTROL LIMITS 90-110%

FORM II - IN

Continuing Calibration Verification (CCV) Report

Sample Name	1800	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1228_CCV.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 12:29:10	**Total Dilution**	1.0000
Sample Type	CCV	**Sample Pass/Fail**	Fail
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Fail

Units : ppb

QC Analyte Table

Recovery Limits: 90-110%

Name	Mass	ISTD	Mode	Conc	RSD	CPS	ExpValue	Rec	QC Flag
Li	7	6	No Gas	56.850	1.5	406899.42	50	113.7	> +/- 10%
Be	9	6	No Gas	11.606	0.7	23658.37	10	116.06	> +/- 10%
B	11	6	No Gas	71.561	2.5	105127.98	50	143.12	> +/- 10%
Sr	88	72	No Gas	10.986	0.7	773928.30	10	109.86	
Zr	90	72	No Gas	11.049	0.5	494643.31	10	110.49	> +/- 10%
Mo	95	115	No Gas	10.047	0.6	139219.86	10	100.47	
Ag	107	115	No Gas	10.506	1.0	370491.30	10	105.06	
Cd	111	115	No Gas	10.427	0.9	78440.47	10	104.27	
Sb	121	115	No Gas	20.612	1.2	690652.53	20	103.06	
Ba	137	115	No Gas	10.350	0.7	123488.43	10	103.5	
Tl	205	209	No Gas	10.106	0.4	618016.12	10	101.06	
Pb	208	209	No Gas	10.229	1.2	448608.94	10	102.29	
Na	23	45	He	1163.999	1.3	571092.33	1000	116.4	> +/- 10%
Mg	24	45	He	1109.560	0.4	233855.29	1000	110.96	> +/- 10%
Al	27	45	He	214.128	2.5	13486.10	200	107.06	
Si	29	45	He	2222.343	13.4	8645.89	2000	111.12	> +/- 10%
K	39	45	He	1113.975	0.4	275114.15	1000	111.4	> +/- 10%
Ca	44	45	He	5363.111	1.4	56746.49	5000	107.26	
Ti	47	45	He	10.805	2.0	1034.71	10	108.05	
V	51	72	He	10.835	1.7	41551.52	10	108.35	
Cr	52	72	He	11.004	0.6	56788.54	10	110.04	> +/- 10%
Mn	55	72	He	53.758	0.5	109265.39	50	107.52	
Fe	57	72	He	1060.853	2.5	100411.65	1000	106.09	
Co	59	72	He	11.101	1.9	99720.80	10	111.01	> +/- 10%
Ni	60	72	He	21.845	0.8	55722.83	20	109.22	
Cu	63	45	He	11.323	0.8	80113.94	10	113.23	> +/- 10%
Zn	66	72	He	216.126	0.6	202916.58	200	108.06	
As	75	72	He	10.521	0.8	10252.28	10	105.21	
Se	78	72	He	10.441	1.7	546.13	10	104.41	
Sn	120	115	He	10.863	0.9	48330.36	10	108.63	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	5119036.70	6090049.07666667	84.06	
Ge	72	He	171446.09	183172.586666667	93.6	
In	115	He	1612270.24	1893460.06333333	85.15	
Lu	175	He	3663830.15	4453666.28666667	82.27	
Rh	103	He	5221876.46	5805949.63	89.94	
Sc	45	He	193389.02	210499.653333333	91.87	
Tb	159	He	6226322.82	7379981.76666667	84.37	

Name	Mass	Mode	CPS	Ref CPS	Rec
(Li)	6	No Gas	1005407.44	1489089.40333333	67.52
Bi	209	No Gas	12183501.90	12052546.07	101.09
Ge	72	No Gas	2330371.58	2410488.1	96.68
In	115	No Gas	16049932.53	15897428.5	100.96
Lu	175	No Gas	18645341.80	18289690.1433333	101.94
Rh	103	No Gas	15319027.73	15416412.72	99.37
Sc	45	No Gas	8852068.81	9412079.90333333	94.05
Tb	159	No Gas	19958948.45	19489435.1266667	102.41

Flag
<70% or >120%

Metals

Form III

Blanks

III
INITIAL CALIBRATION BLANK

Report No:	223060210	Blank ID:	1700
Instrument ID:	ICPMS2	Lab File ID:	2230605a_MS2.b\011_ICB.d
Analyst:	TDM	Analytical Batch:	767033
Analysis Date:	06/05/23 Time: 1119	Analytical Method:	EPA 6020B

ANALYTE	***RESULT***	***UNITS***	***Q***	***DL***	***LOD***	***LOQ***
Arsenic	0.50	ug/L	U	0.25	0.50	1.00
Iron	50.0	ug/L	U	25.0	50.0	100
Manganese	2.50	ug/L	U	1.25	2.50	5.00

FORM III - IN

Initial Calibration Blank (ICB) Report

Sample Name	1700	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	011_ICB.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 11:19:26	**Total Dilution**	1.0000
Sample Type	ICB	**Sample Pass/Fail**	Pass
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Pass

Units : ppb

QC Analyte Table

Name	Mass	ISTD	Mode	Conc	RSD	CPS	Limit	QC Flag
Li	7	6	No Gas	0.123	1.1	98138.11	2.5	
Be	9	6	No Gas	0.005	13.9	79.33	0.5	
B	11	6	No Gas	3.624	7.9	10784.16	5	
Sr	88	72	No Gas	0.003	5.1	1316.78	0.5	
Zr	90	72	No Gas	0.012	5.3	2263.54	0.5	
Mo	95	115	No Gas	0.143	2.7	3652.73	0.5	
Ag	107	115	No Gas	0.010	16.6	415.57	0.5	
Cd	111	115	No Gas	0.007	8.1	194.45	0.5	
Sb	121	115	No Gas	0.121	0.7	6556.04	1	
Ba	137	115	No Gas	0.003	15.3	113.33	0.5	
Tl	205	209	No Gas	-0.035	5.8	2920.42	0.5	
Pb	208	209	No Gas	0.010	18.2	1106.75	0.5	
Na	23	45	He	1.203	3.1	13319.65	50	
Mg	24	45	He	0.747	15.5	676.71	50	
Al	27	45	He	0.428	18.3	124.00	10	
Si	29	45	He	-58.602	36.4	3099.05	100	
K	39	45	He	-3.807	2.6	29760.60	50	
Ca	44	45	He	-14.996	7.2	553.35	250	
Ti	47	45	He	0.101	11.8	17.00	0.5	
V	51	72	He	0.018	20.2	387.79	0.5	
Cr	52	72	He	0.009	1.8	2115.73	0.5	
Mn	55	72	He	0.015	10.7	143.33	2.5	
Fe	57	72	He	0.693	20.1	373.35	50	
Co	59	72	He	0.005	25.3	130.00	0.5	
Ni	60	72	He	0.010	11.0	194.45	1	
Cu	63	45	He	0.008	5.9	556.68	0.5	
Zn	66	72	He	0.128	15.9	415.57	10	
As	75	72	He	-0.018	9.2	107.67	0.5	
Se	78	72	He	0.015	7.2	27.83	0.5	
Sn	120	115	He	0.011	22.0	467.79	0.5	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	6023688.24	6090049.07666667	98.91	
Ge	72	He	180833.42	183172.586666667	98.72	
In	115	He	1868611.88	1893460.06333333	98.69	
Lu	175	He	4378269.31	4453666.28666667	98.31	
Rh	103	He	5804624.07	5805949.63	99.98	
Sc	45	He	208302.53	210499.653333333	98.96	
Tb	159	He	7304954.89	7379981.76666667	98.98	

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
(Li)	6	No Gas	1481558.90	1489089.40333333	99.49	
Bi	209	No Gas	12701569.39	12052546.07	105.38	
Ge	72	No Gas	2428361.22	2410488.1	100.74	
In	115	No Gas	16614383.31	15897428.5	104.51	
Lu	175	No Gas	19179306.38	18289690.1433333	104.86	
Rh	103	No Gas	16057729.36	15416412.72	104.16	
Sc	45	No Gas	9635713.51	9412079.90333333	102.38	
Tb	159	No Gas	20430343.86	19489435.1266667	104.83	

III

METHOD BLANK

Report No:	223060210	Blank ID:	MB2489334
Instrument ID:	ICPMS2	Lab File ID:	2230605a_MS2.b\1216SMPL.d
Analyst:	TDM	Analytical Batch:	767033
Analysis Date:	06/05/23 Time: 1147	Analytical Method:	EPA 6020B

ANALYTE	***RESULT***	***UNITS***	***Q***	***DL***	***LOD***	***LOQ***
Arsenic	0.50	ug/L	U	0.25	0.50	1.00
Iron	50.0	ug/L	U	25.0	50.0	100
Manganese	1.98	ug/L	J	1.25	2.50	5.00

FORM III - IN

Method Blank (MB) Report

Sample Name	2489334	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1216SMPL.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 11:47:19	**Total Dilution**	1.0000
Sample Type	MBWATER	**Sample Pass/Fail**	Fail
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Pass

Units : ppb

QC Analyte Table

Name	Mass	ISTD	Mode	Conc	RSD	CPS	Limit	QC Flag
Li	7	6	No Gas	0.327	1.7	101541.15	2.5	
Be	9	6	No Gas	0.021	14.6	123.33	0.5	
B	11	6	No Gas	1.223	5.3	5537.87	5	
Sr	88	72	No Gas	0.052	1.6	4981.07	0.5	
Zr	90	72	No Gas	0.012	4.4	2288.01	0.5	
Mo	95	115	No Gas	0.450	2.1	8142.40	0.5	
Ag	107	115	No Gas	0.003	14.8	187.78	0.5	
Cd	111	115	No Gas	0.036	10.7	426.68	0.5	
Sb	121	115	No Gas	0.530	3.0	21019.14	1	
Ba	137	115	No Gas	0.191	6.6	2484.70	0.5	
Tl	205	209	No Gas	-0.017	8.7	4294.19	0.5	
Pb	208	209	No Gas	0.054	7.5	3263.88	0.5	
Na	23	45	He	32.549	2.8	28354.07	50	
Mg	24	45	He	6.925	1.9	1993.54	50	
Al	27	45	He	3.486	10.1	316.67	10	
Si	29	45	He	-107.137	38.0	2850.32	100	
K	39	45	He	-4.784	0.6	28334.24	50	
Ca	44	45	He	23.912	4.5	951.71	250	
Ti	47	45	He	0.243	27.5	30.33	0.5	
V	51	72	He	0.007	10.2	331.12	0.5	
Cr	52	72	He	0.005	6.5	2026.83	0.5	
Mn	55	72	He	1.979	1.5	4205.09	2.5	
Fe	57	72	He	2.110	1.2	496.69	50	
Co	59	72	He	0.021	19.7	272.23	0.5	
Ni	60	72	He	0.306	4.7	957.82	1	
Cu	63	45	He	0.114	1.7	1308.96	0.5	
Zn	66	72	He	16.361	2.3	15918.78	10	> 1/2 LOQ
As	75	72	He	0.105	6.0	225.00	0.5	
Se	78	72	He	-0.018	2.0	25.24	0.5	
Sn	120	115	He	0.636	1.7	3466.02	0.5	> 1/2 LOQ

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	5823897.00	6090049.07666667	95.63	
Ge	72	He	174780.51	183172.586666667	95.42	
In	115	He	1769932.69	1893460.06333333	93.48	
Lu	175	He	4150418.69	4453666.28666667	93.19	
Rh	103	He	5672832.83	5805949.63	97.71	
Sc	45	He	199925.91	210499.653333333	94.98	
Tb	159	He	7029190.73	7379981.76666667	95.25	

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
(Li)	6	No Gas	1512895.88	1489089.40333333	101.6	
Bi	209	No Gas	13293111.05	12052546.07	110.29	
Ge	72	No Gas	2440777.26	2410488.1	101.26	
In	115	No Gas	16926812.82	15897428.5	106.48	
Lu	175	No Gas	20124897.62	18289690.1433333	110.03	
Rh	103	No Gas	16325681.02	15416412.72	105.9	
Sc	45	No Gas	9602602.95	9412079.90333333	102.02	
Tb	159	No Gas	21368842.18	19489435.1266667	109.64	

III

CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	1900
Instrument ID:	ICPMS2	Lab File ID:	2230605a_MS2.b\1229_CCB.d
Analyst:	TDM	Analytical Batch:	767033
Analysis Date:	06/05/23 Time: 1232	Analytical Method:	EPA 6020B

ANALYTE	***RESULT***	***UNITS***	***Q***	***DL***	***LOD***	***LOQ***
Arsenic	0.50	ug/L	U	0.25	0.50	1.00
Iron	50.0	ug/L	U	25.0	50.0	100
Manganese	2.50	ug/L	U	1.25	2.50	5.00

FORM III - IN

Continuing Calibration Blank (CCB) Report

Sample Name	1900	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1229_CCB.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 12:32:39	**Total Dilution**	1.0000
Sample Type	CCB	**Sample Pass/Fail**	Pass
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Fail

Units : ppb

QC Analyte Table

Name	Mass	ISTD	Mode	Conc	RSD	CPS	Limit	QC Flag
Li	7	6	No Gas	0.949	1.2	69085.61	2.5	
Be	9	6	No Gas	0.001	31.8	44.00	0.5	
B	11	6	No Gas	2.701	7.1	5467.85	5	
Sr	88	72	No Gas	0.007	9.9	1583.49	0.5	
Zr	90	72	No Gas	0.005	3.1	1825.70	0.5	
Mo	95	115	No Gas	-0.008	5.5	1440.09	0.5	
Ag	107	115	No Gas	0.004	15.4	206.67	0.5	
Cd	111	115	No Gas	-0.005	6.0	96.66	0.5	
Sb	121	115	No Gas	0.135	2.4	6716.13	1	
Ba	137	115	No Gas	0.007	9.1	165.56	0.5	
Tl	205	209	No Gas	-0.067	8.6	853.39	0.5	
Pb	208	209	No Gas	0.002	6.3	730.05	0.5	
Na	23	45	He	16.532	1.8	19746.86	50	
Mg	24	45	He	1.335	17.3	753.37	50	
Al	27	45	He	0.464	8.6	117.33	10	
Si	29	45	He	52.390	37.2	3158.39	100	
K	39	45	He	9.082	0.5	30498.68	50	
Ca	44	45	He	7.835	1.7	753.37	250	
Ti	47	45	He	0.079	30.5	13.67	0.5	
V	51	72	He	0.036	11.5	427.79	0.5	
Cr	52	72	He	0.073	1.8	2290.21	0.5	
Mn	55	72	He	0.054	10.1	212.23	2.5	
Fe	57	72	He	0.570	6.9	336.68	50	
Co	59	72	He	0.002	6.0	96.66	0.5	
Ni	60	72	He	0.092	12.4	387.79	1	
Cu	63	45	He	0.014	9.1	561.13	0.5	
Zn	66	72	He	0.115	6.7	375.56	10	
As	75	72	He	-0.060	21.4	60.33	0.5	
Se	78	72	He	0.035	7.6	26.99	0.5	
Sn	120	115	He	-0.014	5.9	291.12	0.5	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	4942337.84	6090049.07666667	81.15	
Ge	72	He	168857.34	183172.586666667	92.18	
In	115	He	1597050.86	1893460.06333333	84.35	
Lu	175	He	3533196.82	4453666.28666667	79.33	
Rh	103	He	5193892.57	5805949.63	89.46	
Sc	45	He	193540.72	210499.653333333	91.94	
Tb	159	He	6065415.74	7379981.76666667	82.19	

Name	Mass	Mode	CPS	Ref CPS	Rec
(Li)	6	No Gas	980176.65	1489089.40333333	65.82
Bi	209	No Gas	12008728.15	12052546.07	99.64
Ge	72	No Gas	2317607.69	2410488.1	96.15
In	115	No Gas	15918404.08	15897428.5	100.13
Lu	175	No Gas	18404940.98	18289690.1433333	100.63
Rh	103	No Gas	15363304.39	15416412.72	99.66
Sc	45	No Gas	8805627.15	9412079.90333333	93.56
Tb	159	No Gas	19624168.46	19489435.1266667	100.69

Flag
<70% or >120%

Metals

Form IV

Interference Checks

IV

ICPMS INTERFERENCE CHECKS

Report No:	223060210	ICSA \ AB ID:	2000 \ 2100
Instrument ID:	ICPMS2	Analytical Batch:	767033
Analyst:	TDM	Analytical Method:	EPA 6020B
Lab File ID ICSA1:	2230605a_MS2.b\1212ICSA.d	Lab File ID ICSAB1:	2230605a_MS2.b\1213ICSB.d
Lab File ID ICSA2:		Lab File ID ICSAB2:	

Concentration Units: ug/L

		Analyzed (A/AB):	06/05/23 1133	06/05/23 1136				
ANALYTE	***TRUE A***	***TRUE AB***	***ICSA1***	***ICSAB1***	***%R***	***ICSA2***	***ICSAB2***	***%R***
Aluminum	1000	1000	980	981	98			
Antimony	0	0	0.0030	0.016				
Arsenic	0	10.0	-0.062	9.64	96			
Barium	0	0	0.0060	0.013				
Beryllium	0	0	-0.0040	-0.0070				
Boron	0	20.0	0.62	24.0	120			
Cadmium	0	10.0	0.0070	10.9	109			
Calcium	3000	3000	2840	2960	99			
Chromium	0	20.0	-0.0070	19.5	98			
Cobalt	0	20.0	0.028	19.8	99			
Copper	0	20.0	0.0090	20.9	104			
Iron	2500	2500	2440	2490	100			
Lead	0	0	0.0	0.0010				
Lithium	0	20.0	0.039	21.7	108			
Magnesium	1000	1000	1000	1020	102			
Manganese	0	20.0	0.053	20.2	101			
Molybdenum	20.0	20.0	18.0	18.3	92			
Nickel	0	20.0	0.014	20.1	100			
Potassium	1000	1000	989	994	99			
Selenium	0	10.0	0.0070	9.60	96			
Silicon	0	1000	-61	960	96			
Silver	0	5.00	0.0030	4.95	99			
Sodium	2500	2500	2440	2460	98			
Strontium	0	10.0	0.020	9.65	96			
Thallium	0	0	-0.066	-0.066				
Tin	0	10.0	0.020	9.60	96			
Titanium	20.0	20.0	19.1	19.6	98			
Vanadium	0	20.0	0.0050	19.2	96			
Zinc	0	20.0	0.70	20.7	104			
Zirconium	0	20.0	0.018	18.7	94			

FORM IV - IN

Interference Check Solution A (ICS-A) Report

Sample Name	2000	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1212ICSA.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 11:33:22	**Total Dilution**	1.0000
Sample Type	ICSA	**Sample Pass/Fail**	Pass
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Pass

Units : ppb

QC Analyte Table

Spiked Element Recovery: 80-120%

Name	Mass	ISTD	Mode	Conc	RSD	CPS	ExpValue	QC Flag
Li	7	6	No Gas	0.039	1.2	97448.01	2.5	
Be	9	6	No Gas	-0.004	34.6	52.00	0.5	
B	11	6	No Gas	0.618	3.8	4397.46	5	
Sr	88	72	No Gas	0.020	5.1	2580.33	0.5	
Zr	90	72	No Gas	0.018	3.0	2554.70	0.5	
Mo	95	115	No Gas	18.000	0.7	257451.84	20	
Ag	107	115	No Gas	0.003	12.3	184.45	0.5	
Cd	111	115	No Gas	0.007	13.9	192.22	0.5	
Sb	121	115	No Gas	0.003	3.3	2471.36	1	
Ba	137	115	No Gas	0.006	9.3	161.11	0.5	
Tl	205	209	No Gas	-0.066	5.7	980.07	0.5	
Pb	208	209	No Gas	0.000	12.6	673.38	0.5	
Na	23	45	He	2444.034	0.9	1295232.38	2500	
Mg	24	45	He	1000.614	0.9	230345.28	1000	
Al	27	45	He	979.716	1.4	67022.92	1000	
Si	29	45	He	-61.333	35.1	3134.38	100	
K	39	45	He	988.994	1.4	270228.63	1000	
Ca	44	45	He	2844.574	1.5	33214.08	3000	
Ti	47	45	He	19.124	3.3	1994.14	20	
V	51	72	He	0.005	13.6	342.23	0.5	
Cr	52	72	He	-0.007	5.9	2075.73	0.5	
Mn	55	72	He	0.053	8.7	230.00	2.5	
Fe	57	72	He	2439.169	0.8	248568.10	2500	
Co	59	72	He	0.028	14.9	352.23	0.5	
Ni	60	72	He	0.014	13.8	210.00	1	
Cu	63	45	He	0.009	4.2	576.69	0.5	
Zn	66	72	He	0.700	3.1	1002.27	10	
As	75	72	He	-0.062	24.8	64.33	0.5	
Se	78	72	He	0.007	1.3	28.05	0.5	
Sn	120	115	He	0.020	2.0	518.91	0.5	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	6048086.58	6090049.07666667	99.31	
Ge	72	He	184897.92	183172.586666667	100.94	
In	115	He	1878465.36	1893460.06333333	99.21	
Lu	175	He	4437861.91	4453666.28666667	99.65	
Rh	103	He	5765142.00	5805949.63	99.3	
Sc	45	He	211196.70	210499.653333333	100.33	
Tb	159	He	7382304.68	7379981.76666667	100.03	

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
(Li)	6	No Gas	1481981.17	1489089.40333333	99.52	
Bi	209	No Gas	12671065.64	12052546.07	105.13	
Ge	72	No Gas	2452987.61	2410488.1	101.76	
In	115	No Gas	16650082.32	15897428.5	104.73	
Lu	175	No Gas	19439427.21	18289690.1433333	106.29	
Rh	103	No Gas	15892008.81	15416412.72	103.08	
Sc	45	No Gas	9653911.84	9412079.90333333	102.57	
Tb	159	No Gas	20678451.36	19489435.1266667	106.1	

Interference Check Solution AB (ICS-AB) Report

Sample Name	2100	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1213ICSB.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 11:36:51	**Total Dilution**	1.0000
Sample Type	ICSB	**Sample Pass/Fail**	Pass
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Pass

Units : ppb

QC Analyte Table

Spiked Element Recovery: 80-120%

Name	Mass	ISTD	Mode	Conc	RSD	CPS	ExpValue	QC Flag
Li	7	6	No Gas	21.702	0.7	283658.48	20	
Be	9	6	No Gas	-0.007	12.6	42.00	0.5	
B	11	6	No Gas	23.977	1.9	52980.14	20	
Sr	88	72	No Gas	9.652	1.4	716070.72	10	
Zr	90	72	No Gas	18.688	1.8	879711.09	20	
Mo	95	115	No Gas	18.264	0.7	259963.88	20	
Ag	107	115	No Gas	4.954	1.4	180389.57	5	
Cd	111	115	No Gas	10.870	1.1	84414.68	10	
Sb	121	115	No Gas	0.016	1.2	2894.77	1	
Ba	137	115	No Gas	0.013	6.4	235.56	0.5	
Tl	205	209	No Gas	-0.066	16.6	976.74	0.5	
Pb	208	209	No Gas	0.001	3.2	716.71	0.5	
Na	23	45	He	2459.957	0.9	1289093.18	2500	
Mg	24	45	He	1016.473	0.3	231363.53	1000	
Al	27	45	He	981.462	1.2	66396.13	1000	
Si	29	45	He	959.528	21.8	5882.56	1000	
K	39	45	He	993.777	0.5	268325.68	1000	
Ca	44	45	He	2960.936	1.3	34154.46	3000	
Ti	47	45	He	19.605	4.8	2022.81	20	
V	51	72	He	19.227	1.6	78212.19	20	
Cr	52	72	He	19.540	0.8	105646.16	20	
Mn	55	72	He	20.220	1.3	43793.84	20	
Fe	57	72	He	2486.010	1.3	249907.13	2500	
Co	59	72	He	19.781	0.9	188982.65	20	
Ni	60	72	He	20.131	1.7	54646.96	20	
Cu	63	45	He	20.932	1.3	159506.82	20	
Zn	66	72	He	20.679	1.2	20918.37	20	
As	75	72	He	9.635	1.4	9999.44	10	
Se	78	72	He	9.595	2.9	536.06	10	
Sn	120	115	He	9.595	2.3	49292.31	10	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	5994626.78	6090049.07666667	98.43	
Ge	72	He	182415.83	183172.586666667	99.59	
In	115	He	1859307.88	1893460.06333333	98.2	
Lu	175	He	4387391.60	4453666.28666667	98.51	
Rh	103	He	5775027.27	5805949.63	99.47	
Sc	45	He	208836.14	210499.653333333	99.21	
Tb	159	He	7327412.18	7379981.76666667	99.29	

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
(Li)	6	No Gas	1454642.92	1489089.40333333	97.69	
Bi	209	No Gas	12567240.64	12052546.07	104.27	
Ge	72	No Gas	2453600.53	2410488.1	101.79	
In	115	No Gas	16570450.81	15897428.5	104.23	
Lu	175	No Gas	19392455.96	18289690.1433333	106.03	
Rh	103	No Gas	16005641.59	15416412.72	103.82	
Sc	45	No Gas	9668227.12	9412079.90333333	102.72	
Tb	159	No Gas	20524363.86	19489435.1266667	105.31	

Metals

Form V1

Matrix Spikes

V1

MS/MSD RECOVERY

Report No:	223060210	Parent Sample ID:	OT07-MW04-060123
Prep Method:	3010A	Parent GCAL ID:	22306021003
Prep Date:	06/05/23 Time: 0600	Prep Batch:	766941
Analytical Method:	EPA 6020B	Analytical Batch:	767033

GCAL QC ID:	**22306021006 MS**	Instrument ID:	ICPMS2
Analyst:	TDM	Lab File ID:	2230605a_MS2.b\1223SMPL.d
Analysis Date:	06/05/23 1211	Dilution:	1

ANALYTE	***UNITS***	***SPIKE ADDED***	***SAMPLE RESULT***	***MS RESULT***	***MS % REC***	***#***	***QC LIMITS***
Arsenic	ug/L	50	.93	52	102		84 - 116
Iron	ug/L	5000	151	5270	102		87 - 118
Manganese	ug/L	50	66.5	113	94		87 - 115

GCAL QC ID:	**22306021007 MSD**	Instrument ID:	ICPMS2
Analyst:	TDM	Lab File ID:	2230605a_MS2.b\1224SMPL.d
Analysis Date:	06/05/23 1215	Dilution:	1

ANALYTE	***UNITS***	***SPIKE ADDED***	***MSD RESULT***	***MSD % REC***	***#***	***% RPD***	***#***	***QC LIMITS %REC***	***QC LIMITS RPD***
Arsenic	ug/L	50	51.8	102		1		84 - 116	0 - 20
Iron	ug/L	5000	5210	101		1		87 - 118	0 - 20
Manganese	ug/L	50	115	98		2		87 - 115	0 - 20

RPD : 0 out of 3 outside limits

Spike Recovery: 0 out of 6 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM V (PART 1) - IN

Matrix Spike Sample (MS) Report

Sample Name 22306021006

File Name 1223SMPL.d

Data Path Name C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b

Acq Time 6/5/2023 12:11:45

Sample Type MS

Total Dilution 1.0000

Comment ICPMS-2,TDM

ISTD Ref FileName 004CALB.d

Sample QC Pass/Fial Fail

ISTD QC Pass/Fail Pass

QC Ref File Name 1222SMPL.d

QC Analyte Table

Name	Mass	Tune	Conc.	Conc. RSD	CPS	Reference Conc	Spk Amt	% Rec	%Low	%High	Flag
Li	7	No Gas	299.320	1.1	2037027.89	1.81187143512997	250	119	80	120	
Be	9	No Gas	58.280	1.4	129641.22	0.0316349780495815	50	116.5	80	120	
B	11	No Gas	480.628	0.6	758711.78	73.9645540902783	250	162.67	80	120	> +/- 20%
Sr	88	No Gas	412.694	0.5	29743108.72	162.828467037716	50	499.73	80	120	> +/- 20%
Zr	90	No Gas	10.516	0.8	482351.80	0.582540159720245	10	99.34	80	120	
Mo	95	No Gas	52.488	0.8	709082.02	0.554598515148509	50	103.87	80	120	
Ag	107	No Gas	51.264	1.0	1778606.53	0.00699813446788482	50	102.51	80	120	
Cd	111	No Gas	50.879	1.0	376094.02	0.0513613853824138	50	101.66	80	120	
Sb	121	No Gas	106.915	0.5	3515257.50	0.223371263913561	100	106.69	80	120	
Ba	137	No Gas	109.766	0.2	1287789.57	52.8577925988333	50	113.82	80	120	
Tl	205	No Gas	50.637	1.0	3059777.14	-0.0528443679417813	50	101.38	80	120	
Pb	208	No Gas	50.739	1.2	2210173.09	0.519339892532968	50	100.44	80	120	
Na	23	He	45242.166	0.2	21730464.68	20670.9858945964	5000	491.42	80	120	> +/- 20%
Mg	24	He	46698.123	1.5	9813901.94	13507.4529866846	5000	663.81	80	120	> +/- 20%
Al	27	He	1214.449	1.5	76000.63	163.056829490888	1000	105.14	80	120	
Si	29	He	10585.682	4.2	29754.78	8260.50489149299	5000	46.5	80	120	> +/- 20%
K	39	He	7740.176	1.5	1740477.58	5049.66117579662	5000	53.81	80	120	> +/- 20%
Ca	44	He	133181.906	0.7	1391904.35	36914.779350816	25000	385.07	80	120	> +/- 20%
Ti	47	He	55.866	2.7	5320.39	7.61707827383935	50	96.5	80	120	
V	51	He	54.023	0.6	205237.16	2.42900783792457	50	103.19	80	120	
Cr	52	He	53.124	0.1	265648.89	1.11280393902529	50	104.02	80	120	
Mn	55	He	113.313	0.6	229334.80	860.711422393556	50	-1494.8	80	120	> +/- 20%
Fe	57	He	5265.622	0.9	495372.58	5254.84502002217	5000	0.22	80	120	> +/- 20%
Co	59	He	51.842	0.8	463683.66	1.27574605730146	50	101.13	80	120	
Ni	60	He	104.039	0.5	263800.24	6.44769558018653	100	97.59	80	120	
Cu	63	He	53.827	1.3	378760.14	1.86460390848303	50	103.92	80	120	
Zn	66	He	1019.398	0.4	952498.09	22.9944155358434	1000	99.64	80	120	
As	75	He	52.039	0.8	50050.59	4.3583260798696	50	95.36	80	120	
Se	78	He	10.662	2.7	555.12	0.348258735316304	10	103.14	80	120	
Sn	120	He	54.197	0.5	237024.87	0.067008715233113	50	108.26	80	120	

QC ISTD Table

Matrix Spike Sample (MS) Report

Name	Mass	Tune Mode	CPS	CPS RSD	Ref CPS	% Rec	%QC Low	%QC High	QC Flag
(Li)	6	No Gas	1214899.26	1.8	1489089.40333333	81.59	70	120	
Sc	45	No Gas	9123909.64	0.8	9412079.90333333	96.94	70	120	
Ge	72	No Gas	2387291.09	1.2	2410488.1	99.04	70	120	
Rh	103	No Gas	14838813.30	0.5	15416412.72	96.25	70	120	
In	115	No Gas	15826855.57	0.7	15897428.5	99.56	70	120	
Tb	159	No Gas	20593555.11	0.3	19489435.1266667	105.67	70	120	
Lu	175	No Gas	19332672.21	0.3	18289690.1433333	105.7	70	120	
Bi	209	No Gas	12114553.15	1.0	12052546.07	100.51	70	120	
Sc	45	He	193205.20	0.7	210499.653333333	91.78	70	120	
Ge	72	He	170795.76	0.7	183172.586666667	93.24	70	120	
Rh	103	He	5103259.24	1.1	5805949.63	87.9	70	120	
In	115	He	1598579.15	1.6	1893460.06333333	84.43	70	120	
Tb	159	He	6582842.40	1.8	7379981.76666667	89.2	70	120	
Lu	175	He	3873628.48	0.9	4453666.28666667	86.98	70	120	
Bi	209	He	5148095.55	0.8	6090049.07666667	84.53	70	120	

Matrix Spike Duplicate (MSD) Sample Report

Sample Name	22306021007
File Name	1224SMPL.d
Data Path Name	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
Acq Time	6/5/2023 12:15:15
Sample Type	MSD
Total Dilution	1.0000
Comment	ICPMS-2,TDM
ISTD Ref FileName	004CALB.d
Sample QC Pass/Fial	Pass
ISTD Ref FileName	Pass
QC Ref File Name	1223SMPL.d

QC Analyte Table

Name	Mass	Tune Mode	Conc.	Conc. RSD	CPS	Reference Conc	RPD	RPD Max	Flag
Li	7	No Gas	295.958	1.7	1909738.83	299.319922482179	1.13	20	
Be	9	No Gas	57.947	1.4	122172.33	58.2796096354296	0.57	20	
B	11	No Gas	492.141	2.2	736134.39	480.628231636701	2.37	20	
Sr	88	No Gas	416.174	0.9	29763677.06	412.694315278725	0.84	20	
Zr	90	No Gas	10.430	0.7	474775.95	10.5164054823397	0.82	20	
Mo	95	No Gas	51.976	0.4	701476.71	52.4884684545942	0.98	20	
Ag	107	No Gas	50.419	0.9	1747371.04	51.2636197635508	1.66	20	
Cd	111	No Gas	50.323	1.5	371570.23	50.8792860553532	1.1	20	
Sb	121	No Gas	105.703	1.1	3471937.57	106.914552096389	1.14	20	
Ba	137	No Gas	111.902	0.5	1311495.36	109.765872843894	1.93	20	
Tl	205	No Gas	50.246	0.4	3012315.89	50.6374775461998	0.78	20	
Pb	208	No Gas	50.402	1.2	2178304.29	50.7389278767018	0.67	20	
Na	23	He	45412.513	1.1	21800374.68	45242.1655477931	0.38	20	
Mg	24	He	47357.033	1.5	9946696.93	46698.1233087174	1.4	20	
Al	27	He	1166.414	1.1	72959.05	1214.44868163217	4.04	20	
Si	29	He	10377.844	3.9	29224.39	10585.6820501386	1.98	20	
K	39	He	7750.628	0.8	1741983.15	7740.17606420886	0.13	20	
Ca	44	He	134062.403	0.8	1400398.83	133181.905583705	0.66	20	
Ti	47	He	53.714	2.8	5111.89	55.8663230615002	3.93	20	
V	51	He	53.465	0.3	202475.01	54.0232565471463	1.04	20	
Cr	52	He	52.680	1.0	262621.13	53.1235827525945	0.84	20	
Mn	55	He	115.315	0.5	232640.06	113.312735208285	1.75	20	
Fe	57	He	5205.790	0.1	488179.59	5265.62218587004	1.14	20	
Co	59	He	51.746	0.9	461322.60	51.8424844206753	0.19	20	
Ni	60	He	103.387	0.3	261312.09	104.039240219527	0.63	20	
Cu	63	He	53.291	2.2	374758.70	53.8268649792443	1	20	
Zn	66	He	1018.378	0.3	948513.37	1019.39753133315	0.1	20	
As	75	He	51.790	0.5	49651.23	52.0386757561472	0.48	20	
Se	78	He	9.866	1.0	513.95	10.6622589854251	7.76	20	
Sn	120	He	54.046	0.2	233343.49	54.1968893786676	0.28	20	

QC ISTD Table

Matrix Spike Duplicate (MSD) Sample Report

Name	Mass	Tune Mode	CPS	CPS RSD	Ref CPS	% Rec	%QC Low	%QC High	QC Flag
(Li)	6	No Gas	1151389.41	1.0	1489089.40333333	77.32	70	120	
Sc	45	No Gas	8971499.09	1.1	9412079.90333333	95.32	70	120	
Ge	72	No Gas	2368979.98	1.1	2410488.1	98.28	70	120	
Rh	103	No Gas	14771102.19	0.9	15416412.72	95.81	70	120	
In	115	No Gas	15810525.84	0.9	15897428.5	99.45	70	120	
Tb	159	No Gas	20531535.94	0.9	19489435.1266667	105.35	70	120	
Lu	175	No Gas	19396716.79	1.1	18289690.1433333	106.05	70	120	
Bi	209	No Gas	12019575.65	1.3	12052546.07	99.73	70	120	
Sc	45	He	193110.59	0.8	210499.653333333	91.74	70	120	
Ge	72	He	170254.72	0.8	183172.586666667	92.95	70	120	
Rh	103	He	5050750.49	1.7	5805949.63	86.99	70	120	
In	115	He	1578071.53	0.8	1893460.06333333	83.34	70	120	
Tb	159	He	6448322.19	1.1	7379981.76666667	87.38	70	120	
Lu	175	He	3802790.36	0.9	4453666.28666667	85.39	70	120	
Bi	209	He	5096241.49	1.0	6090049.07666667	83.68	70	120	

Metals

Form V2

Post Digestion Spikes

V2

POST DIGEST SPIKE SAMPLE RECOVERY

Report No:	223060210	GCAL PDS ID:	2489833
Matrix:	Water	Parent Sample ID:	OT07-MW06-060123 (22306021005)
Analyst:	TDM	Instrument ID:	ICPMS2
Analysis Date:	06/05/23 Time: 1218	Lab File ID:	2230605a_MS2.b\1225SMPL.d
Analytical Method:	EPA 6020B	Analytical Batch:	767033

ANALYTE	*UNITS*	*SPIKED SAMPLE RESULT*	*C*	*SAMPLE RESULT*	*C*	*SPIKE ADDED*	*% R*	*Q*	*LCL*	*UCL*
Arsenic	ug/L	54.3		4.36		50	100		80	120
Iron	ug/L	10100		5250		5000	97		80	120
Manganese	ug/L	888		861		50	54	*	80	120

FORM V (PART 2) - IN

Post Digestion Spike (PDS) Report

Sample Name	2489833
File Name	1225SMPL.d
Data Path Name	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
Acq Time	6/5/2023 12:18:44
Sample Type	PDS
Total Dilution	1.0000
Comment	ICPMS-2,TDM
ISTD Ref FileName	004CALB.d
Sample QC Pass/Fial	Fail
ISTD QC Pass/Fail	Pass
QC Ref File Name	1222SMPL.d

QC Analyte Table

Name	Mass	Tune	Conc.	Conc. RSD	CPS	Reference Conc	Spk Amt	% Rec	%Low	%High	Flag
Li	7	No Gas	289.392	1.3	1709956.43	1.81187143512997	250	114.92	75	125	
Be	9	No Gas	60.504	0.9	116711.24	0.0316349780495815	50	120.93	75	125	
B	11	No Gas	380.195	1.3	520802.45	73.9645540902783	250	117.36	75	125	
Sr	88	No Gas	214.353	0.2	15050048.94	162.828467037716	50	100.72	75	125	
Zr	90	No Gas	50.706	1.0	2259428.80	0.582540159720245	10	479.15	75	125	> +/- 25%
Mo	95	No Gas	51.459	1.0	681614.25	0.554598515148509	50	101.79	75	125	
Ag	107	No Gas	54.777	0.6	1863240.41	0.00699813446788482	50	109.54	75	125	
Cd	111	No Gas	51.245	0.5	371368.03	0.0513613853824138	50	102.38	75	125	
Sb	121	No Gas	98.336	1.7	3170172.58	0.223371263913561	100	98.12	75	125	
Ba	137	No Gas	104.571	0.7	1202856.06	52.8577925988333	50	101.67	75	125	
Tl	205	No Gas	50.588	1.9	2997630.79	-0.0528443679417813	50	101.28	75	125	
Pb	208	No Gas	51.441	1.5	2197472.31	0.519339892532968	50	101.82	75	125	
Na	23	He	25511.957	1.1	12185287.32	20670.9858945964	5000	99.38	75	125	
Mg	24	He	18303.587	0.5	3823821.19	13507.4529866846	5000	98.9	75	125	
Al	27	He	1139.855	1.0	70912.60	163.056829490888	1000	98.01	75	125	
Si	29	He	8124.588	5.8	23403.61	8260.50489149299	5000	61.27	75	125	> +/- 25%
K	39	He	10117.380	1.2	2252890.69	5049.66117579662	5000	100.67	75	125	
Ca	44	He	61075.323	0.8	634867.07	36914.779350816	25000	98.64	75	125	
Ti	47	He	57.687	1.7	5460.01	7.61707827383935	50	100.12	75	125	
V	51	He	53.873	2.1	204884.48	2.42900783792457	50	102.75	75	125	
Cr	52	He	52.861	2.2	264634.50	1.11280393902529	50	103.42	75	125	
Mn	55	He	887.673	1.3	1797791.11	860.711422393556	50	97.47	75	125	
Fe	57	He	10086.837	1.1	949693.63	5254.84502002217	5000	98.36	75	125	
Co	59	He	52.903	1.3	473683.42	1.27574605730146	50	103.17	75	125	
Ni	60	He	109.361	2.2	277577.22	6.44769558018653	100	102.74	75	125	
Cu	63	He	54.405	2.2	380517.37	1.86460390848303	50	104.9	75	125	
Zn	66	He	1020.586	1.0	954660.76	22.9944155358434	1000	99.76	75	125	
As	75	He	54.334	2.1	52308.11	4.3583260798696	50	99.96	75	125	
Se	78	He	10.115	2.4	528.56	0.348258735316304	10	97.75	75	125	
Sn	120	He	52.895	0.9	229327.53	0.067008715233113	50	105.65	75	125	

QC ISTD Table

Post Digestion Spike (PDS) Report

Name	Mass	Tune Mode	CPS	CPS RSD	Ref CPS	% Rec	%QC Low	%QC High	QC Flag
(Li)	6	No Gas	1053336.63	0.8	1489089.40333333	70.74	70	120	
Sc	45	No Gas	8674292.43	0.8	9412079.90333333	92.16	70	120	
Ge	72	No Gas	2325529.21	0.5	2410488.1	96.48	70	120	
Rh	103	No Gas	14643812.20	0.4	15416412.72	94.99	70	120	
In	115	No Gas	15517301.90	0.2	15897428.5	97.61	70	120	
Tb	159	No Gas	20041812.62	0.9	19489435.1266667	102.83	70	120	
Lu	175	No Gas	19024911.80	0.5	18289690.1433333	104.02	70	120	
Bi	209	No Gas	11882029.41	1.1	12052546.07	98.59	70	120	
Sc	45	He	192047.25	0.4	210499.653333333	91.23	70	120	
Ge	72	He	170990.61	0.6	183172.586666667	93.35	70	120	
Rh	103	He	5109184.10	0.9	5805949.63	88	70	120	
In	115	He	1584499.93	1.1	1893460.06333333	83.68	70	120	
Tb	159	He	6360645.95	2.0	7379981.76666667	86.19	70	120	
Lu	175	He	3733455.26	1.5	4453666.28666667	83.83	70	120	
Bi	209	He	5076700.03	1.9	6090049.07666667	83.36	70	120	

Metals

Form VII

Lab Control Spikes

VII

LABORATORY CONTROL SAMPLE

Report No:	223060210	GCAL ID:	2489335 (LCS)
Matrix:	Water	Instrument ID:	ICPMS2
Analyst:	TDM	Lab File ID:	2230605a_MS2.b\1217SMPL.d
Prep Date:	06/05/23 Time: 0600	Analysis Date:	06/05/23 Time: 1150
Prep Batch:	766941	Analytical Batch:	767033
Prep Method:	3010A	Analytical Method:	EPA 6020B

ANALYTE	UNITS	TRUE	FOUND	% R	Q	LCL	UCL
Arsenic	ug/L	50.0	49.7	99		84	116
Iron	ug/L	5000	5010	100		87	118
Manganese	ug/L	50.0	50.6	101		87	115

FORM VII - IN

Laboratory Control Sample (LCS) Report

Sample Name	2489335	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1217SMPL.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 11:50:48	**Total Dilution**	1.0000
Sample Type	LCS6020	**Sample Pass/Fail**	Pass
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Pass

Units : ppb

QC Analyte Table

Recovery Limits: 80-120% 6020B / 85-115% 200.8

Name	Mass	ISTD	Mode	Conc	RSD	CPS	SpkAmt	Rec	QC Flag
Li	7	6	No Gas	286.784	0.5	2399156.84	250	114.71	
Be	9	6	No Gas	55.131	0.6	150508.78	50	110.26	
B	11	6	No Gas	275.622	0.1	535195.79	250	110.25	
Sr	88	72	No Gas	50.368	0.7	3671714.74	50	100.74	
Zr	90	72	No Gas	9.937	0.6	461015.45	10	99.37	
Mo	95	115	No Gas	48.187	1.2	675356.68	50	96.37	
Ag	107	115	No Gas	49.815	0.8	1792644.24	50	99.63	
Cd	111	115	No Gas	49.076	0.8	376262.50	50	98.15	
Sb	121	115	No Gas	100.471	0.6	3426544.03	100	100.47	
Ba	137	115	No Gas	48.791	1.1	593777.51	50	97.58	
Tl	205	209	No Gas	48.834	0.7	3071564.95	50	97.67	
Pb	208	209	No Gas	48.996	1.6	2221530.02	50	97.99	
Na	23	45	He	5150.446	1.4	2556643.14	5000	103.01	
Mg	24	45	He	5140.660	0.3	1112116.16	5000	102.81	
Al	27	45	He	994.884	0.4	64082.89	1000	99.49	
Si	29	45	He	4843.968	7.0	15696.94	5000	96.88	
K	39	45	He	4970.558	1.1	1160752.48	5000	99.41	
Ca	44	45	He	24843.398	0.4	267749.73	25000	99.37	
Ti	47	45	He	49.070	0.9	4808.79	50	98.14	
V	51	72	He	50.279	0.7	195963.45	50	100.56	
Cr	52	72	He	50.279	0.7	258030.99	50	100.56	
Mn	55	72	He	50.631	1.8	105184.78	50	101.26	
Fe	57	72	He	5013.967	2.2	483984.06	5000	100.28	
Co	59	72	He	51.176	1.1	469569.56	50	102.35	
Ni	60	72	He	102.923	1.3	267726.39	100	102.92	
Cu	63	45	He	52.734	0.3	381853.45	50	105.47	
Zn	66	72	He	1018.885	1.0	976651.32	1000	101.89	
As	75	72	He	49.712	1.4	49056.61	50	99.42	
Se	78	72	He	9.600	0.7	515.35	10	96	
Sn	120	115	He	50.278	1.7	240087.32	50	100.56	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	5683547.41	6090049.07666667	93.33	
Ge	72	He	175239.57	183172.586666667	95.67	
In	115	He	1745137.29	1893460.06333333	92.17	
Lu	175	He	4188452.54	4453666.28666667	94.05	
Rh	103	He	5501753.39	5805949.63	94.76	
Sc	45	He	198828.46	210499.653333333	94.46	
Tb	159	He	7003254.48	7379981.76666667	94.9	

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
(Li)	6	No Gas	1490737.51	1489089.40333333	100.11	
Bi	209	No Gas	12610447.31	12052546.07	104.63	
Ge	72	No Gas	2414051.64	2410488.1	100.15	
In	115	No Gas	16416424.23	15897428.5	103.26	
Lu	175	No Gas	19734270.12	18289690.1433333	107.9	
Rh	103	No Gas	15562692.72	15416412.72	100.95	
Sc	45	No Gas	9430394.90	9412079.90333333	100.19	
Tb	159	No Gas	20903710.94	19489435.1266667	107.26	

US EPA Tune Check Sample Report

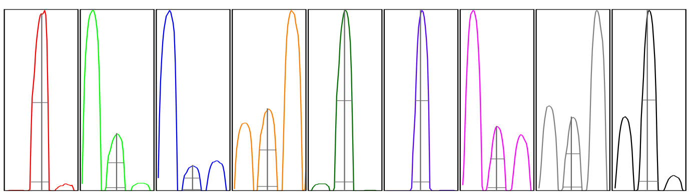
Batch Folder C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
Report Comment
Instrument Name G8403A JP14170244

[No Gas]

Mass	Count (Mean)	RSD% (Actual)	RSD% (Required)	RSD% (Flag)
9	7325	1.37	5.00	
24	59418	0.20	5.00	
25	8255	0.89	5.00	
26	10093	0.95	5.00	
59	272272	0.50	5.00	
115	251097	0.65	5.00	
206	59386	0.75	5.00	
207	52009	0.59	5.00	
208	126051	1.00	5.00	

Mass	Replicate 1 Count	Replicate 2 Count	Replicate 3 Count	Replicate 4 Count	Replicate 5 Count
9	7466	7309	7301	7359	7190
24	59458	59463	59332	59270	59565
25	8347	8211	8165	8244	8309
26	10008	10169	10117	10196	9979
59	269975	272032	273258	273105	272992
115	248731	251160	250424	252956	252215
206	58892	59117	59212	59931	59776
207	51826	51717	51899	52112	52493
208	124382	125883	126911	125467	127611

Integration Time [sec] = 0.1



Mass	Peak Height	Axis (Actual)	Axis (Required)	Axis (Flag)	Width-X% (Actual)	Width-X% (Required)	Width-X% (Flag)
9	1160	9.05	8.9 - 9.1		0.788	0.849	
24	9531	24.00	23.9 - 24.1		0.808	0.849	
25	1320	25.00	24.9 - 25.1		0.791	0.849	
26	1588	25.95	25.9 - 26.1		0.832	0.849	
59	45377	59.00	58.9 - 59.1		0.787	0.849	
115	45536	115.00	114.9 - 115.1		0.738	0.849	
206	10585	206.00	205.9 - 206.1		0.786	0.849	
207	9257	206.95	206.9 - 207.1		0.784	0.849	
208	22593	207.95	207.9 - 208.1		0.799	0.849	

X% = 5 Integration Time [sec] = 0.1 Acquisition Time [sec] = 235 Y Axis = Linear

Tune Parameters

Plasma Paramters

ParameterName	Value	Unit	ParameterName	Value	Unit	ParameterName	Value	Unit
RF Power	1550	W	Carrier Gas	1.00	L/min	S/C Temp	2	℃
RF Matching	1.60	V	Option Gas	0.0	%	Gas Switch	Dilution Gas	
Smpl Depth	8.0	mm	Nebulizer Pump	0.10	rps	Makeup/Dilution Gas	0.20	L/min

Lenses Parameters

ParameterName	Value	Unit	ParameterName	Value	Unit	ParameterName	Value	Unit
Extract 1	0.0	V	Omega Lens	12.0	V	Deflect	20.0	V
Extract 2	-250.0	V	Cell Entrance	-30	V	Plate Bias	-35	V
Omega Bias	-95	V	Cell Exit	-50	V			

Cell Parameters

ParameterName	Value	Unit	ParameterName	Value	Unit	ParameterName	Value	Unit
Use Gas	false		3rd Gas Flow	0	%	Energy Discrimination	4.0	V
He Flow	0.0	mL/min	OctP Bias	-8.0	V			
H2 Flow	0.0	mL/min	OctP RF	180	V			

Metals

Form IX

Serial Dilutions

IX
SERIAL DILUTIONS

Report No:	223060210	GCAL SD ID:	2489834
Matrix:	Water	Parent Sample ID:	OT07-MW06-060123 (22306021005)
Analyst:	TDM	Instrument ID:	ICPMS2
Analysis Date:	06/05/23 Time: 1222	Lab File ID:	2230605a_MS2.b\1226SMPL.d
Analytical Method:	EPA 6020B	Analytical Batch:	767033

ANALYTE	UNITS	PARENT SAMPLE RESULT	C	SERIAL DILUTION RESULT	C	% DIFF	Q	LCL	UCL
Arsenic	ug/L	4.36		4.26	J				
Iron	ug/L	5250		5410		3		0	10
Manganese	ug/L	861		888		3.1		0	10

FORM IX - IN

Sample Report

Sample Name	2489834	**Data Path Name**	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b
File Name	1226SMPL.d	**Comment**	ICPMS-2,TDM
Acq Time	6/5/2023 12:22:13	**Total Dilution**	5.0000
Sample Type	Sample	**Sample Pass/Fail**	Pass
ISTD Ref FileName	004CALB.d	**ISTD Pass/Fail**	Fail

Units : ppb

QC Analyte Table

Name	Mass	ISTD	Mode	MeasValue	Final Conc	RSD	CPS	LDR	QC Flag
Li	7	6	No Gas	1.212	6.059	1.1	73382.94	500	
Be	9	6	No Gas	0.015	0.075	21.8	74.67	1000	
B	11	6	No Gas	19.215	96.077	1.5	29860.14	500	
Sr	88	72	No Gas	33.252	166.262	1.6	2367491.84	1000	
Zr	90	72	No Gas	0.193	0.964	7.8	10344.98	100	
Mo	95	115	No Gas	0.228	1.138	1.1	4765.28	1000	
Ag	107	115	No Gas	0.009	0.047	14.6	400.01	100	
Cd	111	115	No Gas	0.011	0.057	20.2	224.45	1000	
Sb	121	115	No Gas	0.703	3.516	6.8	26213.39	1000	
Ba	137	115	No Gas	10.535	52.676	0.7	127846.22	1000	
Tl	205	209	No Gas	-0.054	-0.271	0.3	1673.50	1000	
Pb	208	209	No Gas	0.114	0.570	1.4	5644.75	1000	
Na	23	45	He	4259.432	21297.160	1.5	2101723.25	100000	
Mg	24	45	He	2742.63	13713.152	0.4	589449.69	100000	
Al	27	45	He	39.419	197.097	27.1	2610.98	20000	
Si	29	45	He	1692.748	8463.739	13.5	7458.58	10000	
K	39	45	He	1054.161	5270.803	1.2	267366.95	100000	
Ca	44	45	He	7555.367	37776.833	2.1	81346.97	500000	
Ti	47	45	He	1.453	7.266	16.9	147.67	1000	
V	51	72	He	0.492	2.458	0.9	2206.86	1000	
Cr	52	72	He	0.246	1.230	1.6	3242.62	1000	
Mn	55	72	He	177.644	888.219	1.1	367119.11	5000	
Fe	57	72	He	1081.673	5408.365	1.0	104155.06	100000	
Co	59	72	He	0.255	1.274	2.0	2409.12	1000	
Ni	60	72	He	1.381	6.907	1.8	3738.29	2000	
Cu	63	45	He	0.414	2.068	1.4	3442.66	1000	
Zn	66	72	He	5.395	26.973	3.0	5424.40	20000	
As	75	72	He	0.852	4.261	5.0	956.37	1000	
Se	78	72	He	0.138	0.692	11.6	33.11	50	
Sn	120	115	He	0.034	0.168	6.8	516.68	1000	

QC ISTD Table

Recovery Limits: 30 - 120% DOD 6020B / 70-130% 6020B / 60-125% 200.8

Name	Mass	Mode	CPS	Ref CPS	Rec	Flag
Bi	209	He	5131005.13	6090049.07666667	84.25	
Ge	72	He	174431.86	183172.586666667	95.23	
In	115	He	1652397.65	1893460.06333333	87.27	
Lu	175	He	3687765.78	4453666.28666667	82.8	
Rh	103	He	5253172.85	5805949.63	90.48	
Sc	45	He	197442.28	210499.653333333	93.8	
Tb	159	He	6311603.03	7379981.76666667	85.52	

Name	Mass	Mode	CPS	Ref CPS	Rec
(Li)	6	No Gas	1008762.99	1489089.40333333	67.74
Bi	209	No Gas	12207640.65	12052546.07	101.29
Ge	72	No Gas	2357192.20	2410488.1	97.79
In	115	No Gas	16325892.44	15897428.5	102.7
Lu	175	No Gas	19010062.22	18289690.1433333	103.94
Rh	103	No Gas	15402739.39	15416412.72	99.91
Sc	45	No Gas	8939106.31	9412079.90333333	94.97
Tb	159	No Gas	20298498.87	19489435.1266667	104.15

Flag
<70% or >120%

Metals

Form XIII

Preparation Logs

XIII

PREPARATION LOG

Report No: 223060210

Prep Method: EPA 3010A

Prep Batch: 766941

CLIENT SAMPLE ID	GCAL SAMPLE ID	PREP DATE	WEIGHT	UNITS	VOLUME	UNITS
LCS2489335	2489335	06/05/23			50	mL
MB2489334	2489334	06/05/23			50	mL
OT07-EB-060123	22306021001	06/05/23			50	mL
OT07-MW-806-060123	22306021008	06/05/23			50	mL
OT07-MW02-060123	22306021002	06/05/23			50	mL
OT07-MW04-060123	22306021003	06/05/23			50	mL
OT07-MW04-060123MS	22306021006	06/05/23			50	mL
OT07-MW04-060123MSD	22306021007	06/05/23			50	mL
OT07-MW05-060123	22306021004	06/05/23			50	mL
OT07-MW06-060123	22306021005	06/05/23			50	mL

FORM XIII - IN

Metals

Form XIV

Run Logs

XIV
ANALYSIS RUN LOG

Report No: 223060210
Instrument ID: ICPMS2
Analytical Batch: 767033
Analytical Method: EPA 6020B
Start Date: 06/05/23
End Date: 06/05/23

					Analyte Symbols																														
CLIENT SAMPLE ID	***GCAL SAMPLE ID***	***PF***	***D/F***	***TIME***	Al	Sb	As	Ba	Be	B	Cd	Ca	Cr	Co	Cu	Fe	Pb	Li	Mg	Mn	Hg	Mo	Ni	K	Se	Si	Ag	Na	Sr	Tl	Sn	Ti	V	Zn	Zr
IICALB	1300	*	1	1053			X									X				X															
IICAL2	1302	*	1	1056			X									X				X															
IICAL4	1304	*	1	1101			X									X				X															
IICAL5	1305	*	1	1108			X									X				X															
IICAL6	1306	*	1	1112			X									X				X															
ICV	1600	*	1	1115			X									X				X															
ICB	1700	*	1	1119			X									X				X															
LLCCV	1803	*	1	1129			X									X				X															
ICSA	2000	*	1	1133			X									X				X															
ICSAB	2100	*	1	1136			X									X				X															
LDR	2500	*	1	1140			X									X				X															
MB2489334	2489334	*	1	1147			X									X				X															
LCS2489335	2489335	*	1	1150			X									X				X															
OT07-EB-060123	22306021001	*	1	1154			X									X				X															
OT07-MW02-060123	22306021002	*	1	1157			X									X				X															
OT07-MW04-060123	22306021003	*	1	1201			X									X				X															
OT07-MW05-060123	22306021004	*	1	1204			X									X				X															
OT07-MW06-060123	22306021005	*	1	1208			X									X				X															
OT07-MW04-060123MS	22306021006	*	1	1211			X									X				X															
OT07-MW04-060123MSD	22306021007	*	1	1215			X									X				X															
OT07-MW06-060123PDS	2489833	*	1	1218			X									X				X															
OT07-MW06-060123SD	2489834	*	5	1222			X									X				X															
OT07-MW-806-060123	22306021008	*	1	1225			X									X				X															
CCV	1800	*	1	1229			X									X				X															
CCB	1900	*	1	1232			X									X				X															

FORM XIV - IN

Metals

Form XV

Internal Standards

XV (He)

ICP-MS INTERNAL STANDARDS RELATIVE INTENSITY SUMMARY

Report No:	223060210	Start Date:	06/05/23
Instrument ID:	ICPMS2	End Date:	06/05/23
Analytical Method	EPA 6020B	Analytical Batch:	767033

			Internal Standards %RI For:													
CLIENT SAMPLE ID	***GCAL SAMPLE ID***	***TIME***	***ISTD1***	***Q***	***ISTD2***	***Q***	***ISTD3***	***Q***	***ISTD4***	***Q***	***ISTD5***	***Q***	***ISTD6***	***Q***	***ISTD7***	***Q***
MB2489334	2489334	1147	96		95		93		93		98		95		95	
LCS2489335	2489335	1150	93		96		92		94		95		94		95	
OT07-EB-060123	22306021001	1154	94		96		92		92		97		94		94	
OT07-MW02-060123	22306021002	1157	91		96		90		92		92		94		93	
OT07-MW04-060123	22306021003	1201	87		94		88		89		89		93		91	
OT07-MW05-060123	22306021004	1204	88		94		88		87		90		93		89	
OT07-MW06-060123	22306021005	1208	86		94		87		87		90		93		89	
OT07-MW04-060123MS	22306021006	1211	85		93		84		87		88		92		89	
OT07-MW04-060123MSD	22306021007	1215	84		93		83		85		87		92		87	
OT07-MW06-060123PDS	2489833	1218	83		93		84		84		88		91		86	
OT07-MW06-060123SD	2489834	1222	84		95		87		83		90		94		86	
OT07-MW-806-060123	22306021008	1225	80		93		82		81		87		90		84	

ISTD 1:	Bismuth (He)	ISTD 4:	Lutetium (He)	ISTD 7:	Terbium (He)
ISTD 2:	Germanium (He)	ISTD 5:	Rhodium (He)		
ISTD 3:	Indium (He)	ISTD 6:	Scandium (He)		

FORM XV - IN

XV (No Gas)
ICP-MS INTERNAL STANDARDS RELATIVE INTENSITY SUMMARY

Report No: 223060210
Instrument ID: ICPMS2
Analytical Method EPA 6020B
Start Date: 06/05/23
End Date: 06/05/23
Analytical Batch: 767033

CLIENT SAMPLE ID	GCAL SAMPLE ID	TIME	ISTD8	Q	ISTD9	Q	ISTD10	Q	ISTD11	Q	ISTD12	Q	ISTD13	Q	ISTD14	Q
			Internal Standards %RI For:													
MB2489334	2489334	1147	110		101		106		110		106		102		110	
LCS2489335	2489335	1150	105		100		103		108		101		100		107	
OT07-EB-060123	22306021001	1154	110		101		108		111		107		103		110	
OT07-MW02-060123	22306021002	1157	105		101		104		109		101		102		108	
OT07-MW04-060123	22306021003	1201	101		100		101		106		99		100		106	
OT07-MW05-060123	22306021004	1204	106		100		104		108		101		100		108	
OT07-MW06-060123	22306021005	1208	102		99		101		106		98		97		105	
OT07-MW04-060123MS	22306021006	1211	101		99		100		106		96		97		106	
OT07-MW04-060123MSD	22306021007	1215	100		98		99		106		96		95		105	
OT07-MW06-060123PDS	2489833	1218	99		96		98		104		95		92		103	
OT07-MW06-060123SD	2489834	1222	101		98		103		104		100		95		104	
OT07-MW-806-060123	22306021008	1225	98		96		99		103		96		93		103	

ISTD 8: Bismuth (No Gas)
ISTD 9: Germanium (No Gas)
ISTD 10 Indium (No Gas)
ISTD 11 Lutetium (No Gas)
ISTD 12 Rhodium (No Gas)
ISTD 13 Scandium (No Gas)
ISTD 14 Terbium (No Gas)

FORM XV - IN

Metals

ICPMS ICAL

Pace Analytical

BATCH COVERSHEET

ANALYST TDM

DATE 06/05/23

ICP/MS Metals Analysis

BATCH **767033**

STANDARDS

ICAL Standard 100ppb	MET 360-2-3
ICV Standard	MET 360-3-4
ICSA Standard	MET 360-3-5
ICSAB Standard	MET 360-3-6
Internal Standard	met 359-15-1
Tune	met 355-31-4
P/A	met 355-45-5

Instrument ID	ICPMS-2
ICPMS Data File	2230605A_MS2

ADDITIONAL STANDARDS

LDR Standard	MET 360-1-7
ICAL Standard 10ppb	MET 359-18-1
ICAL Standard 1ppb	MET 359-18-2
ICAL Standard 0.5 ppb	MET 359-18-3
ICAL Standard 0.1ppb	MET 359-18-4

ACID MATRIX

2% HNO3 \ 0.5% HCL Solution	met 359-16-1
5% HNO3 \ 2% HCL Solution	met 359-15-6

LAB QC LIMITS

200.8 Correlation Coefficient (R) =0.998

6020B Correlation Coefficient (R) =0.995

ICV Recovery 90-110%

LLCCV Recovery 80-120%

ICSA \ ICSAB Recovery 80-120%

CCV Recovery 90-110%

EQUIPMENT

Pipette 1	met 112
Pipette 2	met 107
Pipette 3	met 109
Pipette 4	met 105
Pipette 5	met 111

Sample					
Data File	Acq. Date-Time	Type	Level	Sample Name	Total Dil.
001SMPL.d	6/5/2023 10:42	Sample		Blank	1
002SMPL.d	6/5/2023 10:46	Sample		Blank	1
003SMPL.d	6/5/2023 10:49	Sample		Blank	1
004CALB.d	6/5/2023 10:53	CalBlk	1	1300	1
005CALS.d	6/5/2023 10:56	CalStd	3	1302	1
006CALS.d	6/5/2023 11:01	CalStd	5	1304	1
007CALS.d	6/5/2023 11:04	CalStd		5 PPB	1
008CALS.d	6/5/2023 11:08	CalStd	6	1305	1
009CALS.d	6/5/2023 11:12	CalStd	7	1306	1
010_ICV.d	6/5/2023 11:15	ICV		1600	1
011_ICB.d	6/5/2023 11:19	ICB		1700	1
0120.1.d	6/5/2023 11:22	LLCCV0.1		1804	1
1210.5.d	6/5/2023 11:26	LLCCV0.5		1804	1
1211CCV1.d	6/5/2023 11:29	LLCCV1		1803	1
1212ICSA.d	6/5/2023 11:33	ICSA		2000	1
1213ICSB.d	6/5/2023 11:36	ICSB		2100	1
1214_QC1.d	6/5/2023 11:40	QC1		LDR	1
1215SMPL.d	6/5/2023 11:43	Sample		2500	1
1216SMPL.d	6/5/2023 11:47	MBWATER		2489334	1
1217SMPL.d	6/5/2023 11:50	LCS6020		2489335	1
1218SMPL.d	6/5/2023 11:54	Sample		22306021001	1
1219SMPL.d	6/5/2023 11:57	Sample		22306021002	1
1220SMPL.d	6/5/2023 12:01	Sample		22306021003	1
1221SMPL.d	6/5/2023 12:04	Sample		22306021004	1
1222SMPL.d	6/5/2023 12:08	AllRef		22306021005	1
1223SMPL.d	6/5/2023 12:11	MS		22306021006	1
1224SMPL.d	6/5/2023 12:15	MSD		22306021007	1
1225SMPL.d	6/5/2023 12:18	PDS		2489833	1
1226SMPL.d	6/5/2023 12:22	Sample		2489834	5
1227SMPL.d	6/5/2023 12:25	Sample		22306021008	1
1228_CCV.d	6/5/2023 12:29	CCV		1800	1
1229_CCB.d	6/5/2023 12:32	CCB		1900	1
1230SMPL.d	6/5/2023 12:36	MBWATER		2489600	1
1231SMPL.d	6/5/2023 12:39	LCS6020		2489601	1
1232SMPL.d	6/5/2023 12:43	Sample		22306010901	100
1233SMPL.d	6/5/2023 12:46	AllRef		22306010902	100
1234SMPL.d	6/5/2023 12:50	MS		2489602	100
1235SMPL.d	6/5/2023 12:53	MSD		2489603	100
1236SMPL.d	6/5/2023 12:57	PDS		PDS	100
1237SMPL.d	6/5/2023 13:00	Sample		SD	100
1238SMPL.d	6/5/2023 13:04	Sample		22306010903	100
1239SMPL.d	6/5/2023 13:07	Sample		22305312801	100
1240SMPL.d	6/5/2023 13:10	Sample		22306012701	100
1241SMPL.d	6/5/2023 13:14	Sample		22306015401	100
1242SMPL.d	6/5/2023 13:17	Sample		2489604	100
1243SMPL.d	6/5/2023 13:21	Sample		22306015402	100
1244SMPL.d	6/5/2023 13:24	Sample		F1	100
1245SMPL.d	6/5/2023 13:28	Sample		F2	100

1246SMPL.d	6/5/2023 13:31	Sample		22306014101	1
1247_CCV.d	6/5/2023 13:35	CCV		1800	1
1248_CCB.d	6/5/2023 13:38	CCB		1900	1
1249SMPL.d	6/5/2023 13:42	MBWATER		2489610	1
1250SMPL.d	6/5/2023 13:45	LCS200.8		2489611	1
1251SMPL.d	6/5/2023 13:49	AllRef		22306022401	10
1252SMPL.d	6/5/2023 13:52	MS		2489612	10
1253SMPL.d	6/5/2023 13:56	MSD		2489613	10
1254_CCV.d	6/5/2023 13:59	CCV		1800	1
1255_CCB.d	6/5/2023 14:03	CCB		1900	1
1256SMPL.d	6/5/2023 14:06	MBSOIL		2489359	40
1257SMPL.d	6/5/2023 14:10	LCS6020		2489360	40
1258SMPL.d	6/5/2023 14:13	AllRef		22306021301	400
1259SMPL.d	6/5/2023 14:17	MSSOIL		2489365	400
1260SMPL.d	6/5/2023 14:20	MSDSOIL		2489366	400
1261SMPL.d	6/5/2023 14:24	PDS		2489838	400
1262SMPL.d	6/5/2023 14:27	Sample		2489839	400
1263SMPL.d	6/5/2023 14:31	Sample		S2230621401	370.3704
1264_CCV.d	6/5/2023 14:34	CCV		1800	1
1265_CCB.d	6/5/2023 14:38	CCB		1900	1
1266SMPL.d	6/5/2023 14:41	Sample		Blank	1
1267SMPL.d	6/5/2023 14:45	Sample		Blank	1
1268SMPL.d	6/5/2023 14:48	Sample		Blank	1

Vial Number
1101
1102
1103
1104
1105
1106
1312
1107
1108
1109
1111
1112
1201
1202
1203
1204
1205
1206
2206
2207
2208
2209
2210
2211
2212
2301
2302
2303
2304
2305
1401
1501
2101
2102
2103
2104
2105
2106
2107
2108
2109
2110
2111
2112
2201
2202
2203
2204

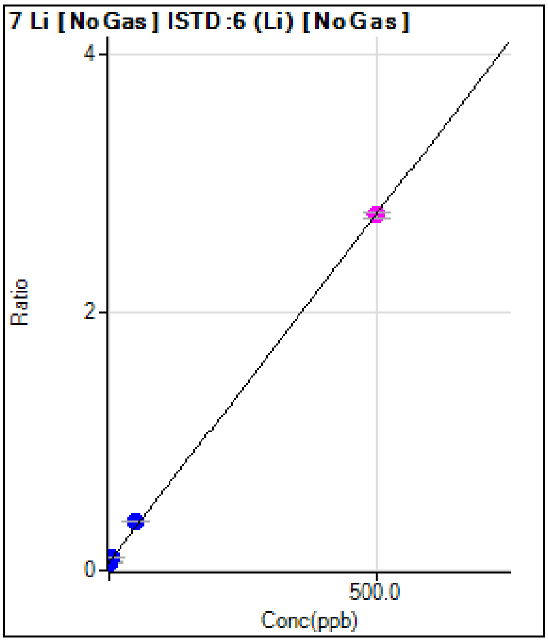
2205
1402
1502
2306
2307
2308
2309
2310
1403
1503
2311
2312
2401
2402
2403
2404
2405
2406
1404
1504
5
5
5

Tune Mode	Mass	Name	ISTD	R	a	b (blank)	DL	BEC	Units
No Gas	7	Li	6 (Li) [No Gas]	0.999892658	0.00538397	0.065376632	1.15469454	12.14283001	ppb
No Gas	9	Be	6 (Li) [No Gas]	0.999887771	0.001830558	4.34E-05	0.009404694	0.023685104	ppb
No Gas	11	B	6 (Li) [No Gas]	0.999715936	0.001295047	0.00207908	0.34318242	1.605409337	ppb
He	23	Na	45 Sc [He]	0.99999996	0.0024847	0.060982602	1.385168637	24.54324352	ppb
He	24	Mg	45 Sc [He]	0.999999743	0.00108769	0.002440271	1.061470405	2.243534806	ppb
He	27	Al	45 Sc [He]	0.999999666	0.000323535	0.000455923	0.225382889	1.409192357	ppb
He	29	Si	45 Sc [He]	0.999686471	1.31E-05	0.015692828	1370.578949	1200.90215	ppb
He	39	K	45 Sc [He]	0.999997313	0.001144909	0.147215268	4.820059249	128.5824655	ppb
He	44	Ca	45 Sc [He]	0.999999852	5.41E-05	0.003469118	15.68508559	64.16081708	ppb
He	47	Ti	45 Sc [He]	0.999998925	0.000492308	3.17E-05	0.016085887	0.064321334	ppb
He	51	V	72 Ge [He]	0.99999833	0.022211024	0.001734553	0.016937182	0.078094219	ppb
He	52	Cr	72 Ge [He]	0.999997883	0.029062807	0.011440387	0.014918862	0.393643558	ppb
He	55	Mn	72 Ge [He]	0.999999932	0.011844127	0.000618638	0.027847653	0.052231663	ppb
He	57	Fe	72 Ge [He]	0.999999857	0.000550473	0.001680522	3.983329046	3.052869174	ppb
He	59	Co	72 Ge [He]	0.999999947	0.052356498	0.000466644	0.00281484	0.008912818	ppb
He	60	Ni	72 Ge [He]	0.999999935	0.014836393	0.000934468	0.01407925	0.062984834	ppb
He	63	Cu	45 Sc [He]	0.999997907	0.036375683	0.00239233	0.01142454	0.065767292	ppb
He	66	Zn	72 Ge [He]	0.999999709	0.005469032	0.001594914	0.107649417	0.291626426	ppb
He	75	As	72 Ge [He]	0.999999477	0.00561772	0.000695292	0.039491229	0.123767691	ppb
He	78	Se	72 Ge [He]	0.999996019	0.000290801	0.000149608	0.189199988	0.514469731	ppb
No Gas	88	Sr	72 Ge [No Gas]	0.999998086	0.030189329	0.000460493	0.00481894	0.015253514	ppb
No Gas	90	Zr	72 Ge [No Gas]	0.999999271	0.019147496	0.000699249	0.005528673	0.036519082	ppb
No Gas	95	Mo	115 In [No Gas]	0.999988002	0.000851731	9.76E-05	0.013393807	0.114634325	ppb
No Gas	107	Ag	115 In [No Gas]	0.999999802	0.002192029	3.92E-06	0.002413091	0.001786598	ppb
No Gas	111	Cd	115 In [No Gas]	0.999999996	0.00046686	8.45E-06	0.007895153	0.018107535	ppb
No Gas	118	(Sn)	115 In [No Gas]	0.99999781	0.00139396	0.000110447	0.008634929	0.079232845	ppb
He	118	(Sn)	115 In [He]	0.99999066	0.00187764	0.000151522	0.053465712	0.080697976	ppb
He	120	Sn	115 In [He]	0.999994671	0.002731835	0.000220897	0.024779267	0.080860407	ppb
No Gas	121	Sb	115 In [No Gas]	0.999993519	0.002076137	0.000142049	0.007363711	0.068419779	ppb
No Gas	137	Ba	115 In [No Gas]	0.999996255	0.000741242	4.89E-06	0.001400291	0.006601863	ppb
He	156	[Se]	115 In [He]						ppb
No Gas	201	Hg							
No Gas	205	Tl	209 Bi [No Gas]	0.999996734	0.004979549	0.000406484	0.014950842	0.081630766	ppb
No Gas	206	(Pb)	209 Bi [No Gas]	0.999999657	0.001692184	2.13E-05	0.004930121	0.012597088	ppb
No Gas	207	(Pb)	209 Bi [No Gas]	0.999997148	0.001459813	2.13E-05	0.009968251	0.01461439	ppb
No Gas	208	Pb	209 Bi [No Gas]	0.999998245	0.003594467	5.23E-05	0.005137913	0.014551094	ppb
No Gas	6	(Li)							ppb
No Gas	45	Sc							ppb
He	45	Sc							ppb
No Gas	72	Ge							ppb
He	72	Ge							ppb
No Gas	103	Rh							ppb
He	103	Rh							ppb
No Gas	115	In							ppb
He	115	In							ppb
No Gas	159	Tb							ppb
He	159	Tb							ppb
No Gas	175	Lu							ppb
He	175	Lu							ppb
No Gas	209	Bi							ppb
He	209	Bi							ppb

Calibration for 1251SMPL.d

Batch Folder:	C:\Agilent\ICPMH\1\DATA\2230605a_MS2.b\
Analysis File:	2230605a_MS2.batch.bin
DA Date-Time:	6/6/2023 09:02:55
Calibration Title:	EPA6020
Calibration Method:	External Calibration
VIS Interpolation Fit:	

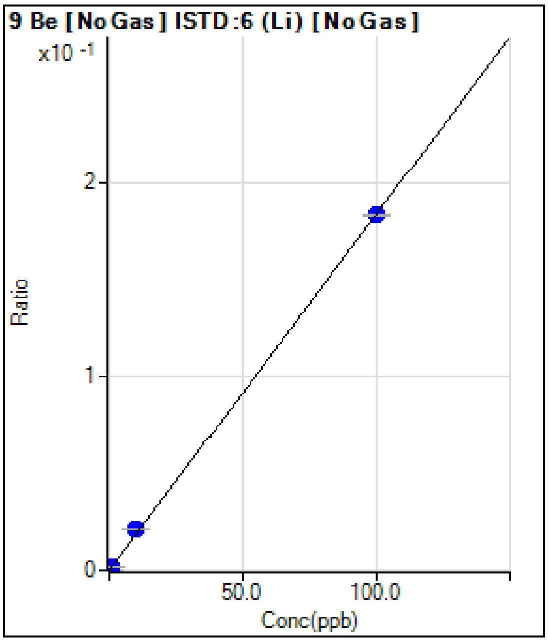
Level	Standard Data File	Sample Name	Acq. Date-Time
1	004CALB.d	1300	6/5/2023 10:53:25
2			
3	005CALS.d	1302	6/5/2023 10:56:54
4			
5	006CALS.d	1304	6/5/2023 11:01:18
6	008CALS.d	1305	6/5/2023 11:08:37
7	009CALS.d	1306	6/5/2023 11:12:17
8			



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1		0.000	0.000	97561.05	0.0654	P	3.2
2		0.000					
3		0.500	0.837	101238.69	0.0699	P	1.6
4		2.500					
5		5.000	5.889	144957.70	0.0971	P	1.9
6		50.000	57.470	569560.18	0.3748	P	0.7
7		500.000	499.244	4810077.95	2.7533	M	1.7
8							

y = 0.0054 * x + 0.0654
R = 0.9999
DL = 1.155
BEC = 12.14

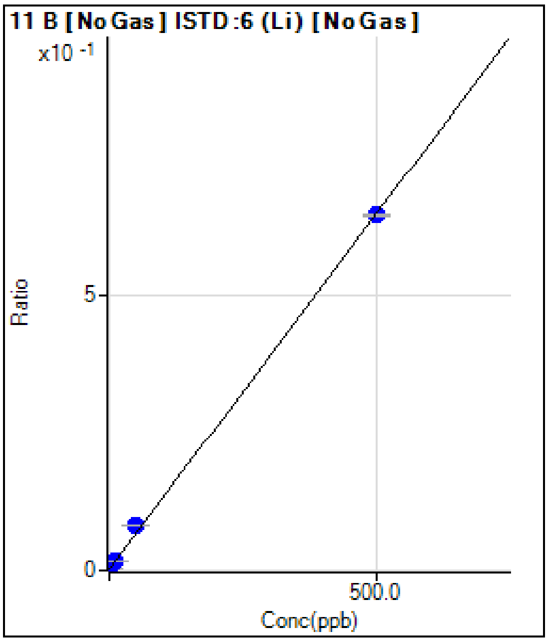
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1		0.000	0.000	64.67	0.0000	P	13.2
2		0.000					
3		0.100	0.121	384.68	0.0003	P	5.3
4		0.500					
5		1.000	1.163	3245.03	0.0022	P	1.4
6		10.000	11.509	32079.68	0.0211	P	0.6
7		100.000	99.847	319388.93	0.1828	P	0.8
8							

y = 0.0018 * x + 4.3357E-005
R = 0.9999
DL = 0.009405
BEC = 0.02369

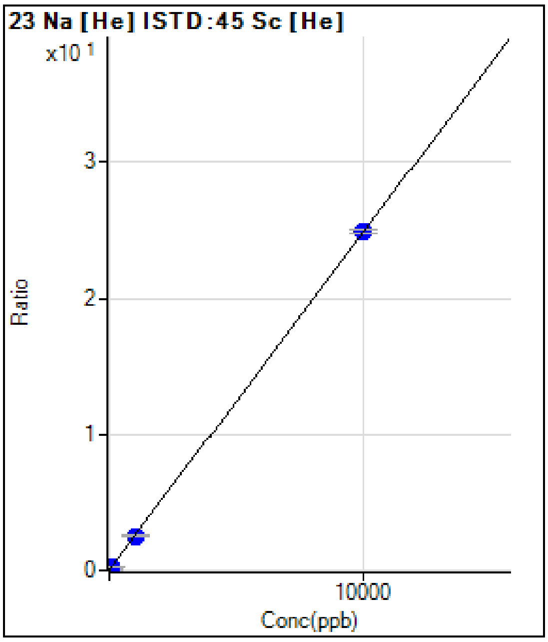
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	3103.76	0.0021	P	7.1
2	☐	0.000					
3	☐	1.000	0.988	4864.28	0.0034	P	10.9
4	☐	5.000					
5	☐	10.000	11.173	24707.60	0.0165	P	2.4
6	☐	50.000	61.858	124900.29	0.0822	P	1.4
7	☐	500.000	498.791	1132150.50	0.6480	P	0.4
8	☐						

y = 0.0013 * x + 0.0021
R = 0.9997
DL = 0.3432
BEC = 1.605

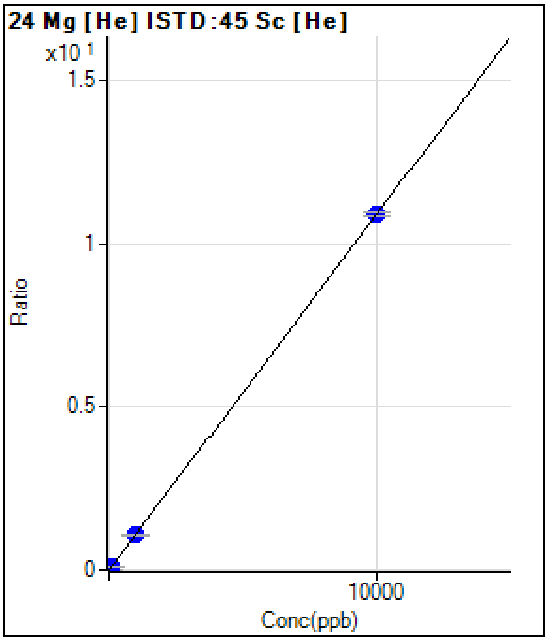
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	12835.86	0.0610	P	1.9
2	☐	0.000					
3	☐	10.000	6.562	16332.70	0.0773	P	2.5
4	☐	50.000					
5	☐	100.000	98.205	64385.01	0.3050	P	3.7
6	☐	1000.000	998.833	539667.53	2.5428	P	2.8
7	☐	10000.00	10000.13	5235069.09	24.908	P	1.3
8	☐						

y = 0.0025 * x + 0.0610
R = 1.0000
DL = 1.385
BEC = 24.54

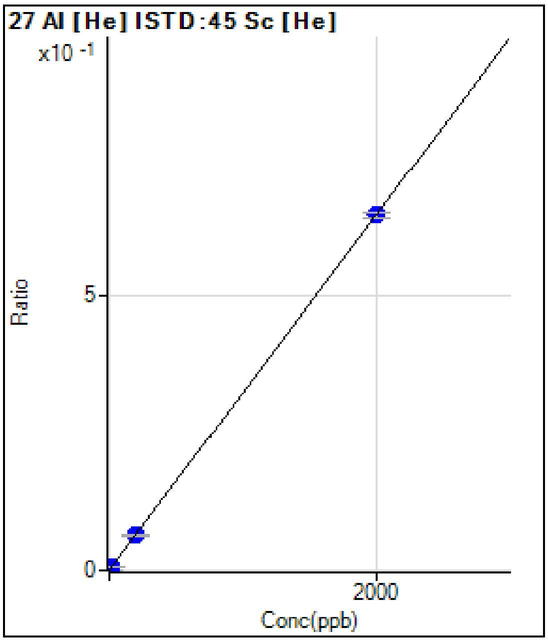
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	513.36	0.0024	P	15.8
2	☐	0.000					
3	☐	10.000	6.607	2033.54	0.0096	P	12.1
4	☐	50.000					
5	☐	100.000	100.257	23525.76	0.1115	P	7.5
6	☐	1000.000	992.816	229645.17	1.0823	P	7.3
7	☐	10000.00	10000.71	2286736.74	10.880	P	1.2
8	☐						

y = 0.0011 * x + 0.0024
R = 1.0000
DL = 1.061
BEC = 2.244

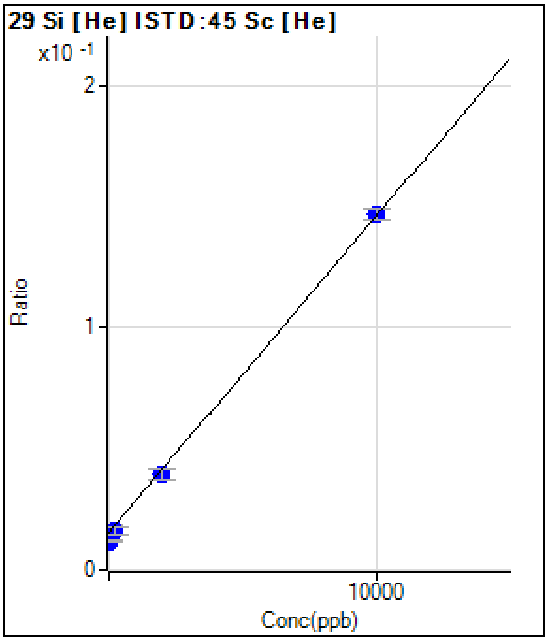
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	96.00	0.0005	P	5.3
2	☐	0.000					
3	☐	2.000	1.438	194.67	0.0009	P	10.4
4	☐	10.000					
5	☐	20.000	20.062	1465.42	0.0069	P	8.3
6	☐	200.000	198.344	13713.64	0.0646	P	5.8
7	☐	2000.000	2000.166	136100.92	0.6476	P	1.6
8	☐						

y = 3.2353E-004 * x + 4.5592E-004
R = 1.0000
DL = 0.2254
BEC = 1.409

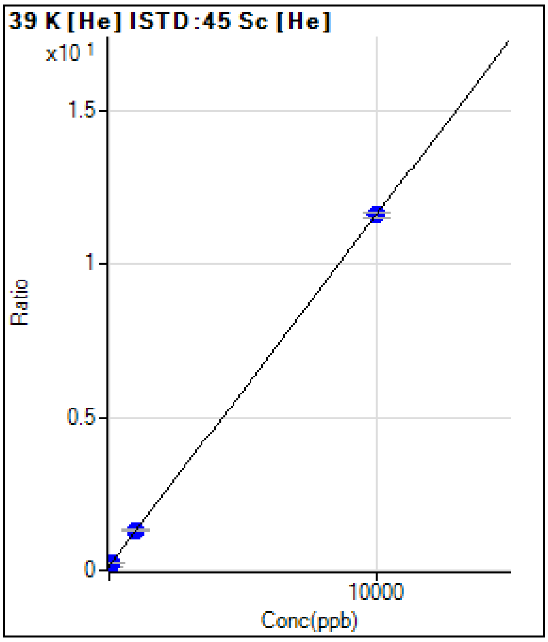
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	3295.76	0.0157	P	38.0
2	☐	0.000					
3	☐	20.000	-287.563	2521.56	0.0119	P	8.0
4	☐	100.000					
5	☐	200.000	36.639	3406.42	0.0162	P	19.7
6	☐	2000.000	1848.044	8452.44	0.0398	P	11.8
7	☐	10000.00	10034.27	30853.62	0.1468	P	3.1
8	☐						

y = 1.3068E-005 * x + 0.0157
R = 0.9997
DL = 1371
BEC = 1201

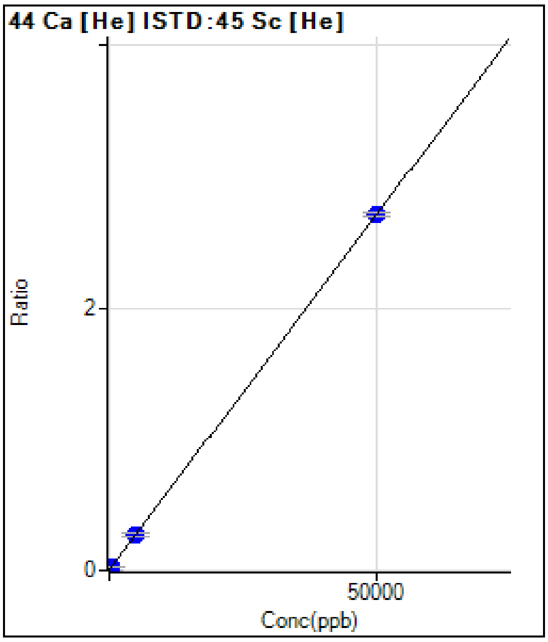
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	30986.26	0.1472	P	1.2
2	☐	0.000					
3	☐	10.000	4.585	32222.16	0.1525	P	0.4
4	☐	50.000					
5	☐	100.000	98.401	54868.39	0.2599	P	2.7
6	☐	1000.000	1020.174	279132.37	1.3152	P	2.7
7	☐	10000.00	9998.004	2436834.39	11.594	P	1.4
8	☐						

y = 0.0011 * x + 0.1472
R = 1.0000
DL = 4.82
BEC = 128.6

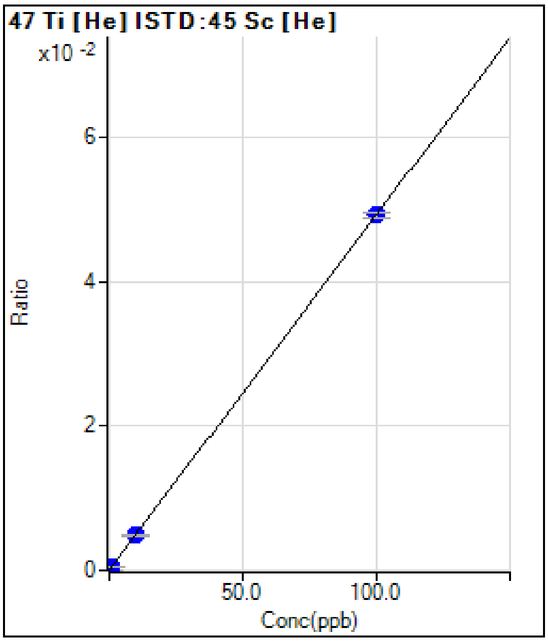
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	730.03	0.0035	P	8.1
2	☐	0.000					
3	☐	50.000	25.978	1030.05	0.0049	P	7.5
4	☐	250.000					
5	☐	500.000	501.496	6456.50	0.0306	P	3.1
6	☐	5000.000	4979.222	57862.80	0.2727	P	6.3
7	☐	50000.00	50002.08	568964.45	2.7070	P	0.9
8	☐						

y = 5.4069E-005 * x + 0.0035
R = 1.0000
DL = 15.69
BEC = 64.16

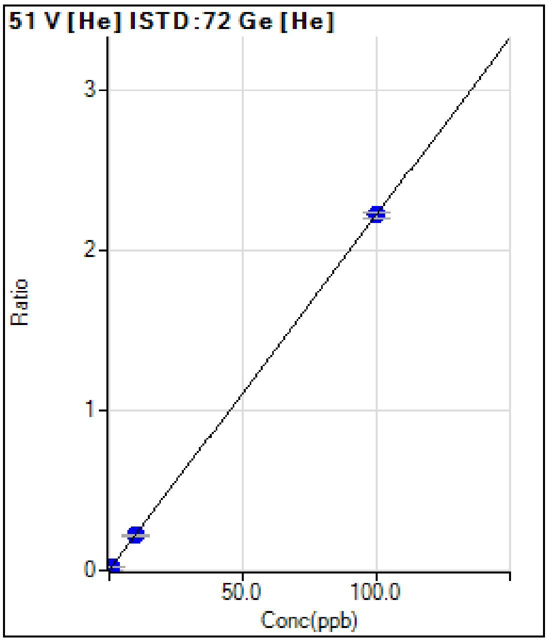
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	6.67	0.0000	P	8.3
2	☐	0.000					
3	☐	0.100	0.077	14.67	0.0001	P	34.0
4	☐	0.500					
5	☐	1.000	1.012	111.67	0.0005	P	16.4
6	☐	10.000	9.856	1036.37	0.0049	P	4.5
7	☐	100.000	100.014	10355.28	0.0493	P	1.6
8	☐						

y = 4.9231E-004 * x + 3.1666E-005
R = 1.0000
DL = 0.01609
BEC = 0.06432

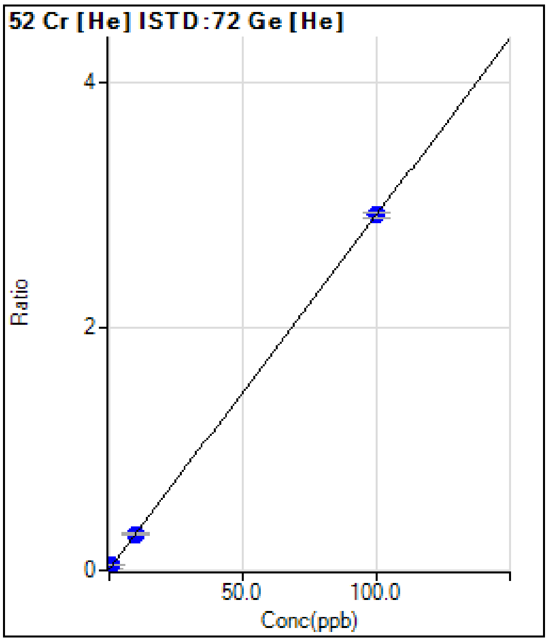
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	317.78	0.0017	P	7.2
2	☐	0.000					
3	☐	0.100	0.080	653.35	0.0035	P	6.7
4	☐	0.500					
5	☐	1.000	0.979	4339.57	0.0235	P	6.1
6	☐	10.000	9.810	40657.06	0.2196	P	6.3
7	☐	100.000	100.019	408152.98	2.2233	P	1.6
8	☐						

y = 0.0222 * x + 0.0017
R = 1.0000
DL = 0.01694
BEC = 0.07809

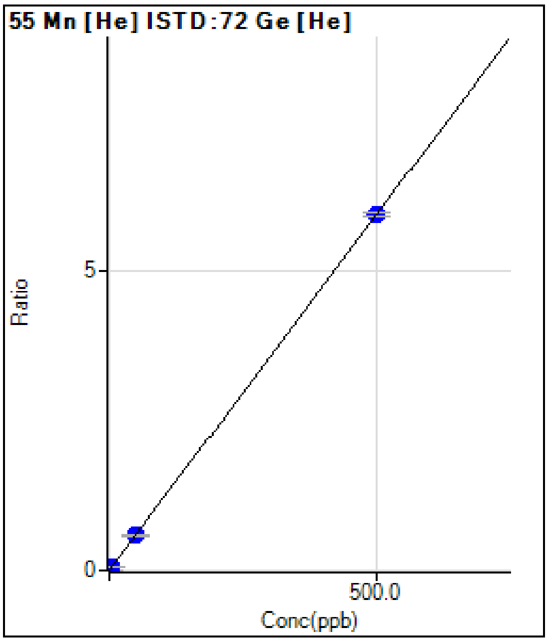
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	2095.73	0.0114	P	1.3
2	☐	0.000					
3	☐	0.100	0.046	2386.89	0.0128	P	4.4
4	☐	0.500					
5	☐	1.000	0.962	7285.20	0.0394	P	2.9
6	☐	10.000	9.776	54715.59	0.2956	P	6.2
7	☐	100.000	100.023	535765.81	2.9184	P	1.9
8	☐						

y = 0.0291 * x + 0.0114
R = 1.0000
DL = 0.01492
BEC = 0.3936

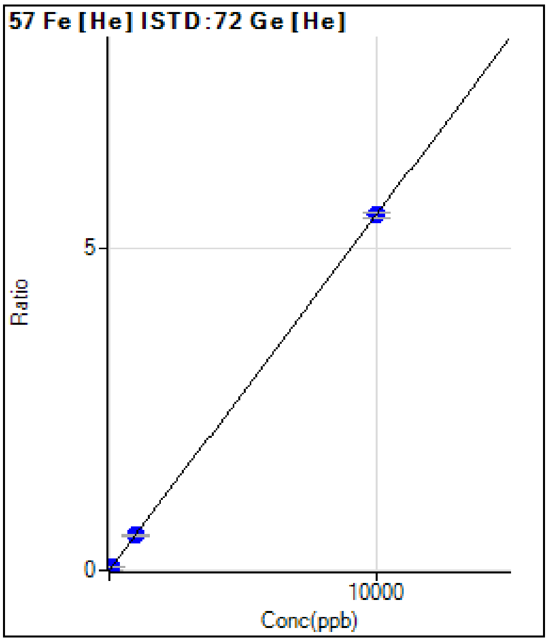
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	113.34	0.0006	P	17.8
2	☐	0.000					
3	☐	0.500	0.344	874.48	0.0047	P	14.2
4	☐	2.500					
5	☐	5.000	4.966	10987.54	0.0594	P	7.9
6	☐	50.000	49.817	109356.58	0.5907	P	4.8
7	☐	500.000	500.019	1087407.35	5.9229	P	1.0
8	☐						

y = 0.0118 * x + 6.1864E-004
R = 1.0000
DL = 0.02785
BEC = 0.05223

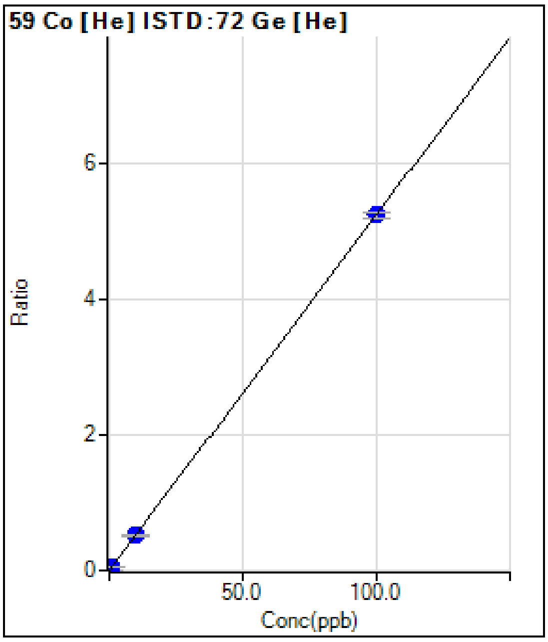
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	306.68	0.0017	P	43.5
2	☐	0.000					
3	☐	10.000	6.522	983.40	0.0053	P	7.1
4	☐	50.000					
5	☐	100.000	97.208	10200.58	0.0552	P	9.1
6	☐	1000.000	993.802	101587.46	0.5487	P	6.5
7	☐	10000.00	10000.65	1010946.26	5.5068	P	1.8
8	☐						

y = 5.5047E-004 * x + 0.0017
R = 1.0000
DL = 3.983
BEC = 3.053

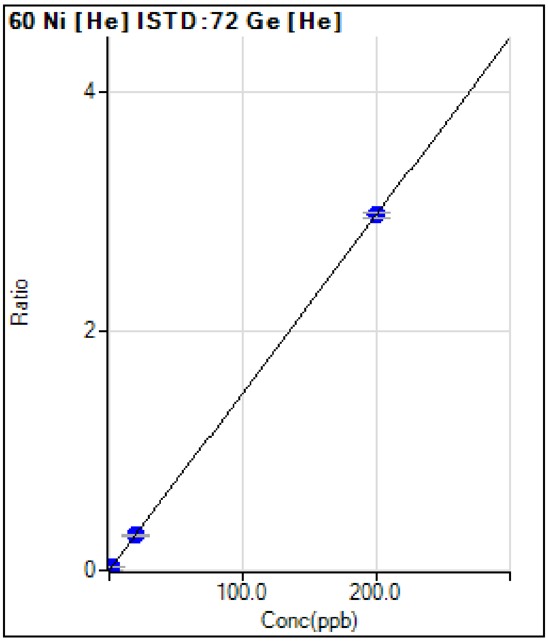
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	85.56	0.0005	P	10.5
2	☐	0.000					
3	☐	0.100	0.067	742.25	0.0040	P	14.5
4	☐	0.500					
5	☐	1.000	0.965	9423.12	0.0510	P	10.2
6	☐	10.000	9.974	96765.95	0.5227	P	5.1
7	☐	100.000	100.003	961272.26	5.2363	P	1.8
8	☐						

y = 0.0524 * x + 4.6664E-004
R = 1.0000
DL = 0.002815
BEC = 0.008913

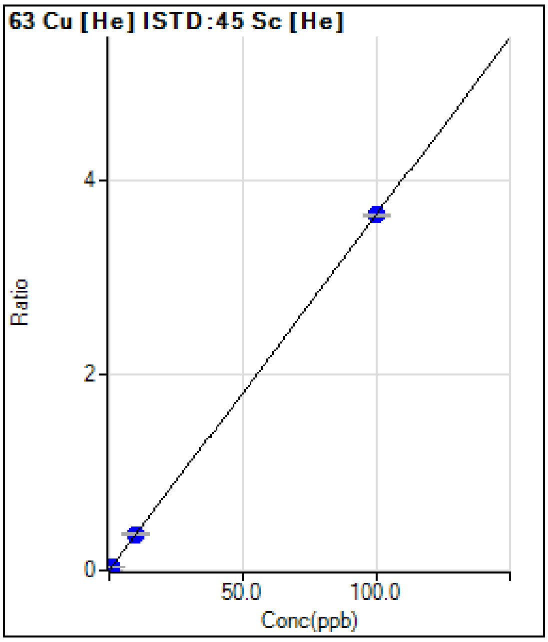
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	171.11	0.0009	P	7.5
2	☐	0.000					
3	☐	0.200	0.153	596.69	0.0032	P	14.7
4	☐	1.000					
5	☐	2.000	1.956	5537.77	0.0299	P	7.3
6	☐	20.000	20.029	55187.73	0.2981	P	5.6
7	☐	200.000	199.998	544906.73	2.9682	P	1.6
8	☐						

y = 0.0148 * x + 9.3447E-004
R = 1.0000
DL = 0.01408
BEC = 0.06298

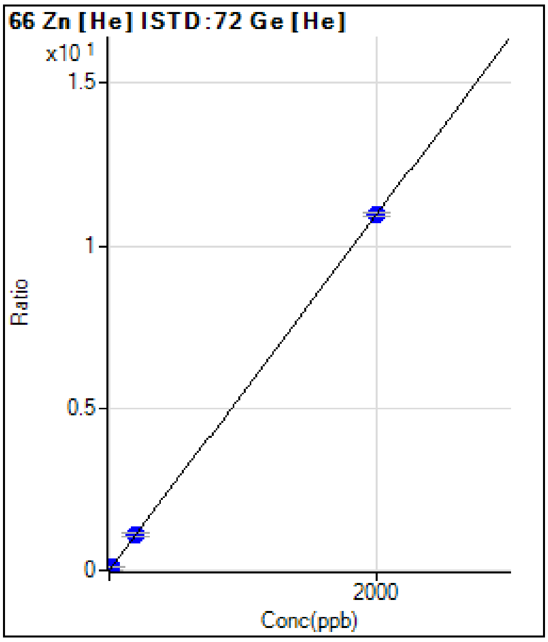
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	503.35	0.0024	P	5.8
2	☐	0.000					
3	☐	0.100	0.077	1098.94	0.0052	P	14.6
4	☐	0.500					
5	☐	1.000	1.020	8336.91	0.0395	P	7.2
6	☐	10.000	10.197	79218.36	0.3733	P	4.8
7	☐	100.000	99.980	764904.31	3.6392	P	0.8
8	☐						

y = 0.0364 * x + 0.0024
R = 1.0000
DL = 0.01142
BEC = 0.06577

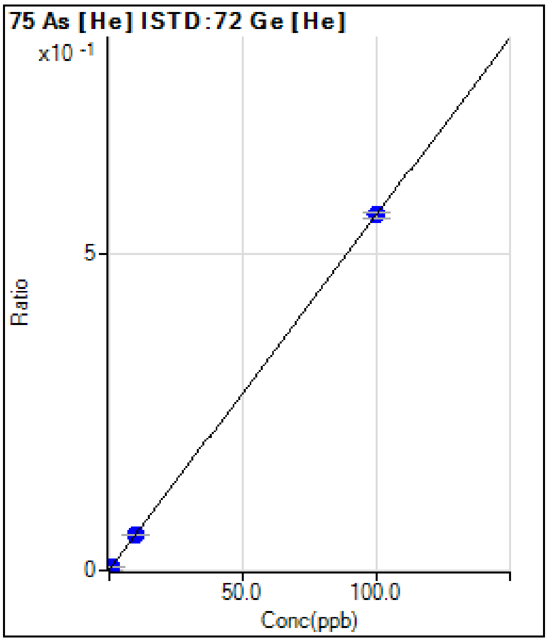
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	292.23	0.0016	P	12.3
2	☐	0.000					
3	☐	2.000	1.876	2211.31	0.0119	P	10.8
4	☐	10.000					
5	☐	20.000	20.183	20701.42	0.1120	P	6.4
6	☐	200.000	201.492	204307.99	1.1036	P	5.7
7	☐	2000.000	1999.849	2008250.40	10.938	P	1.2
8	☐						

y = 0.0055 * x + 0.0016
R = 1.0000
DL = 0.1076
BEC = 0.2916

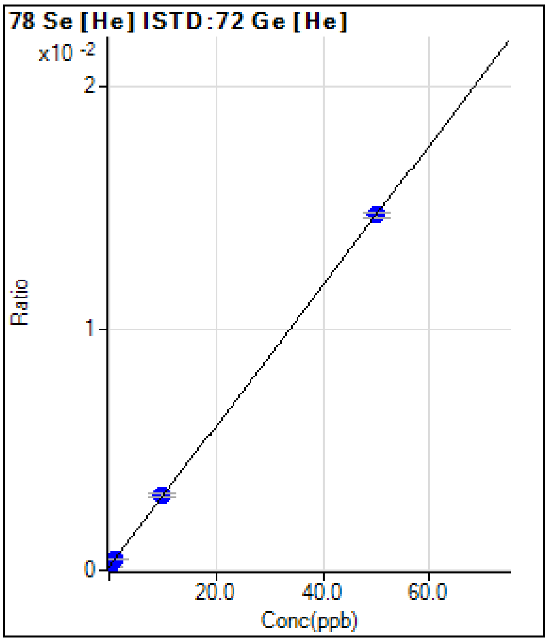
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	127.33	0.0007	P	10.6
2	☐	0.000					
3	☐	0.100	0.052	184.33	0.0010	P	1.8
4	☐	0.500					
5	☐	1.000	0.886	1049.04	0.0057	P	5.2
6	☐	10.000	9.911	10437.41	0.0564	P	4.0
7	☐	100.000	100.010	103271.03	0.5625	P	1.4
8	☐						

y = 0.0056 * x + 6.9529E-004
R = 1.0000
DL = 0.03949
BEC = 0.1238

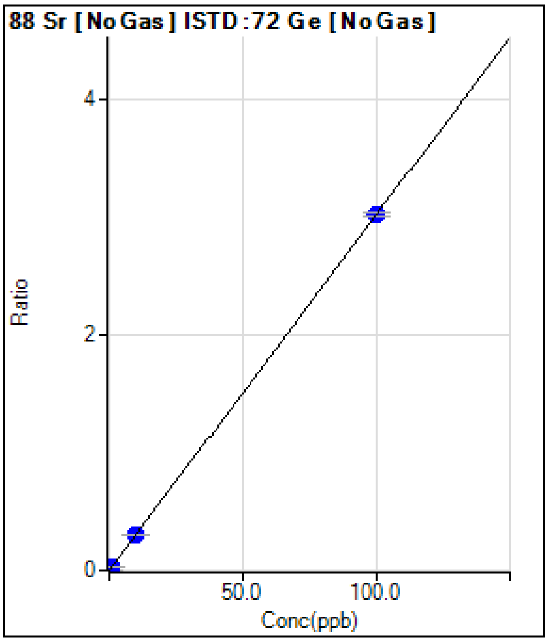
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	27.42	0.0001	P	12.3
2	☐	0.000					
3	☐	0.100	0.034	29.75	0.0002	P	9.1
4	☐	0.500					
5	☐	1.000	1.047	83.92	0.0005	P	7.4
6	☐	10.000	10.093	571.09	0.0031	P	4.9
7	☐	50.000	49.981	2695.83	0.0147	P	1.3
8	☐						

y = 2.9080E-004 * x + 1.4961E-004
R = 1.0000
DL = 0.1892
BEC = 0.5145

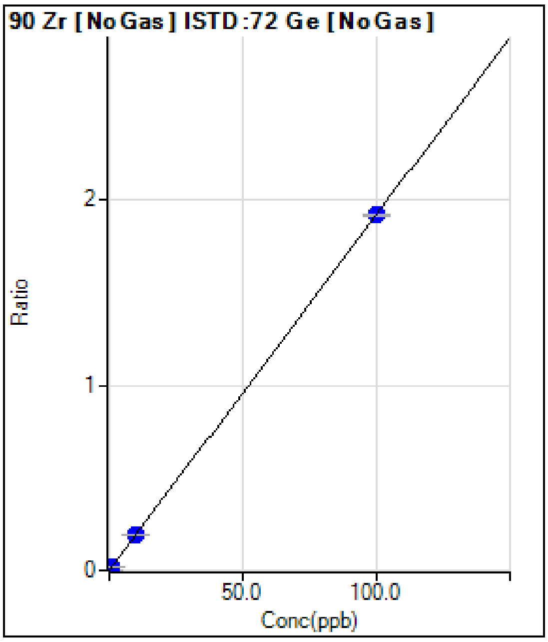
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	1110.09	0.0005	P	10.5
2	☐	0.000					
3	☐	0.100	0.097	8155.95	0.0034	P	2.2
4	☐	0.500					
5	☐	1.000	1.018	74714.84	0.0312	P	1.2
6	☐	10.000	10.195	741762.67	0.3082	P	0.6
7	☐	100.000	99.980	7289786.98	3.0188	P	1.0
8	☐						

y = 0.0302 * x + 4.6049E-004
R = 1.0000
DL = 0.004819
BEC = 0.01525

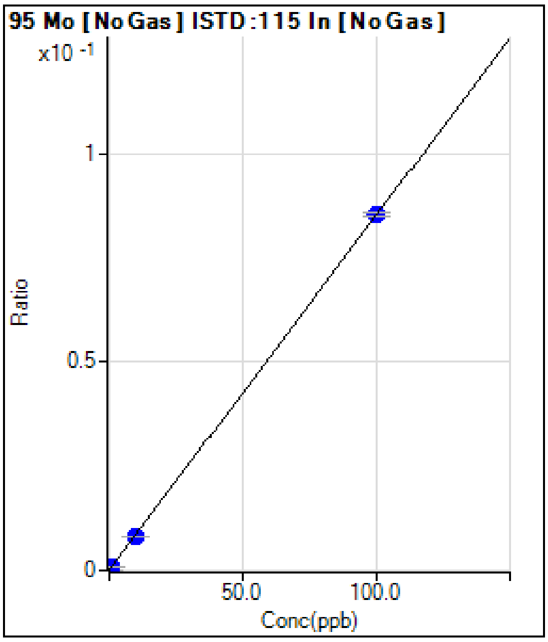
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	1685.67	0.0007	P	5.0
2	☐	0.000					
3	☐	0.100	0.082	5462.21	0.0023	P	0.6
4	☐	0.500					
5	☐	1.000	0.887	42351.10	0.0177	P	1.7
6	☐	10.000	10.029	463776.28	0.1927	P	0.7
7	☐	100.000	99.998	4625335.79	1.9154	P	0.5
8	☐						

y = 0.0191 * x + 6.9925E-004
R = 1.0000
DL = 0.005529
BEC = 0.03652

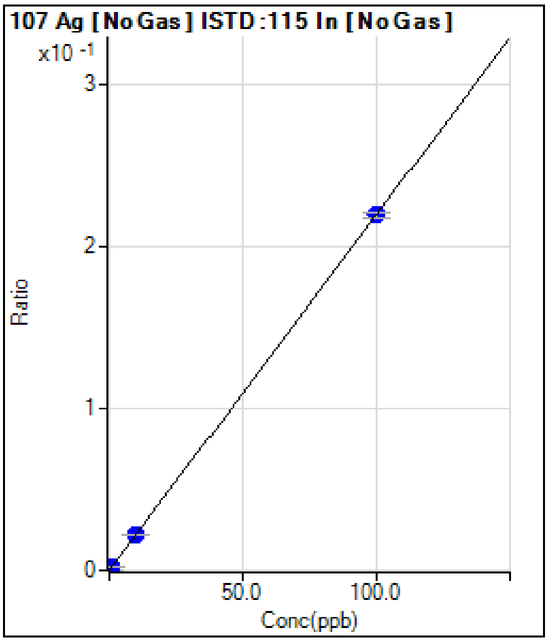
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	1552.32	0.0001	P	3.9
2	☐	0.000					
3	☐	0.100	0.066	2466.91	0.0002	P	5.5
4	☐	0.500					
5	☐	1.000	0.904	13964.69	0.0009	P	1.5
6	☐	10.000	9.487	134010.37	0.0082	P	0.5
7	☐	100.000	100.052	1354048.63	0.0853	P	1.0
8	☐						

y = 8.5173E-004 * x + 9.7638E-005
R = 1.0000
DL = 0.01339
BEC = 0.1146

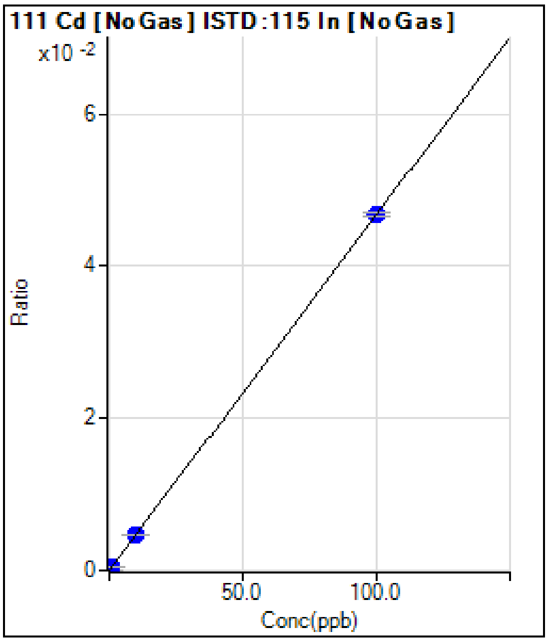
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	62.22	0.0000	P	45.0
2	☐	0.000					
3	☐	0.100	0.103	3668.29	0.0002	P	2.1
4	☐	0.500					
5	☐	1.000	1.005	35508.70	0.0022	P	0.2
6	☐	10.000	9.941	357124.54	0.0218	P	0.4
7	☐	100.000	100.006	3479258.33	0.2192	P	1.3
8	☐						

y = 0.0022 * x + 3.9163E-006
R = 1.0000
DL = 0.002413
BEC = 0.001787

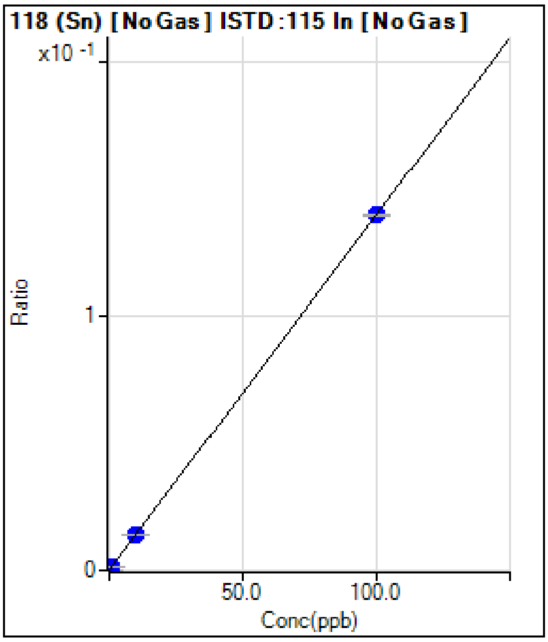
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1		0.000	0.000	134.45	0.0000	P	14.5
2		0.000					
3		0.100	0.094	838.92	0.0001	P	7.1
4		0.500					
5		1.000	0.996	7616.55	0.0005	P	2.1
6		10.000	10.003	76665.38	0.0047	P	0.4
7		100.000	100.000	741087.00	0.0467	P	1.0
8							

y = 4.6686E-004 * x + 8.4537E-006
R = 1.0000
DL = 0.007895
BEC = 0.01811

Weight: <None>
Min Conc: <None>

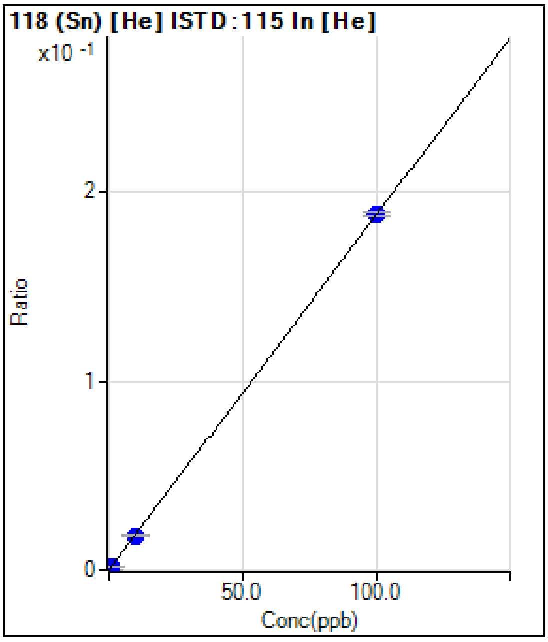


	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1		0.000	0.000	1755.68	0.0001	P	3.6
2		0.000					
3		0.100	0.134	4765.31	0.0003	P	3.4
4		0.500					
5		1.000	0.989	23954.89	0.0015	P	1.5
6		10.000	9.805	225777.75	0.0138	P	0.7
7		100.000	100.020	2214559.43	0.1395	P	0.8
8							

y = 0.0014 * x + 1.1045E-004
R = 1.0000
DL = 0.008635
BEC = 0.07923

Weight: <None>
Min Conc: <None>

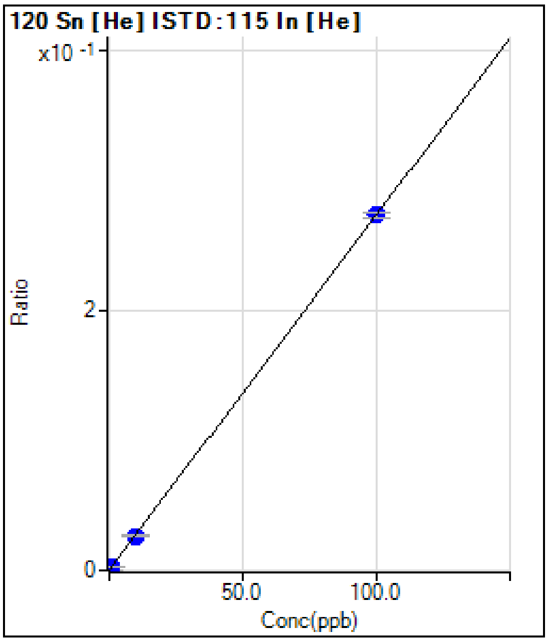
Calibration for 1251SMPL.d



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	286.67	0.0002	P	22.1
2	☐	0.000					
3	☐	0.100	0.114	698.91	0.0004	P	10.2
4	☐	0.500					
5	☐	1.000	0.932	3611.61	0.0019	P	6.9
6	☐	10.000	9.566	34522.25	0.0181	P	6.2
7	☐	100.000	100.044	350921.30	0.1880	P	1.2
8	☐						

y = 0.0019 * x + 1.5152E-004
R = 1.0000
DL = 0.05347
BEC = 0.0807

Weight: <None>
Min Conc: <None>

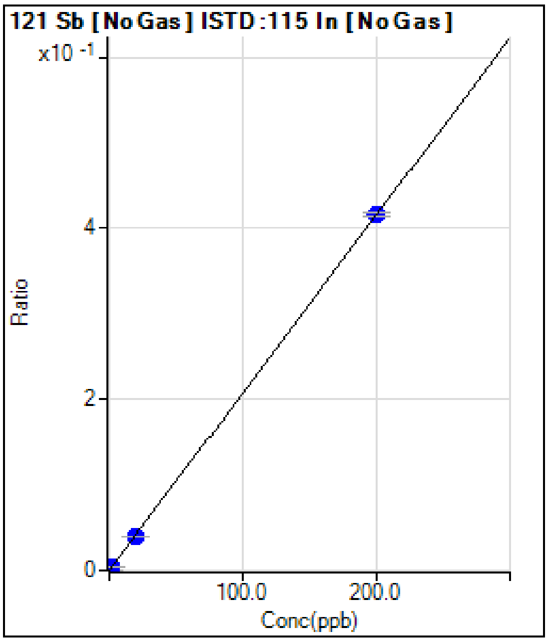


	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	417.79	0.0002	P	10.2
2	☐	0.000					
3	☐	0.100	0.083	855.59	0.0004	P	5.7
4	☐	0.500					
5	☐	1.000	0.922	5204.37	0.0027	P	10.1
6	☐	10.000	9.658	50715.37	0.0266	P	5.3
7	☐	100.000	100.035	510493.94	0.2735	P	1.6
8	☐						

y = 0.0027 * x + 2.2090E-004
R = 1.0000
DL = 0.02478
BEC = 0.08086

Weight: <None>
Min Conc: <None>

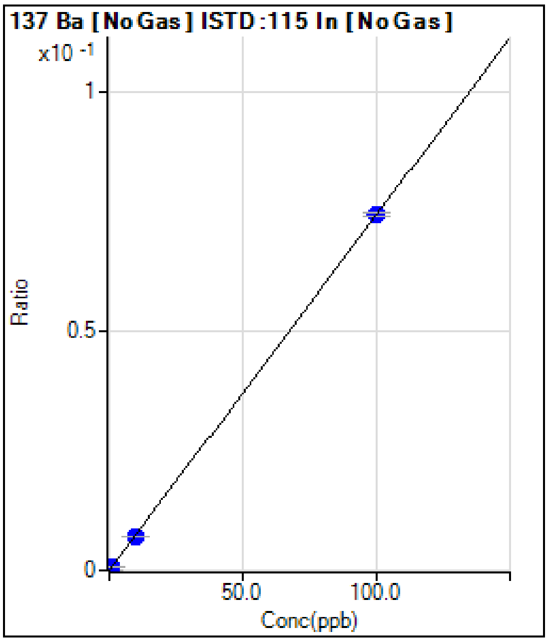
Calibration for 1251SMPL.d



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1		0.000	0.000	2257.99	0.0001	P	3.6
2		0.000					
3		0.200	0.202	9000.74	0.0006	P	1.9
4		1.000					
5		2.000	2.021	69803.97	0.0043	P	1.3
6		20.000	19.301	658953.90	0.0402	P	1.0
7		200.000	200.070	6594652.10	0.4155	P	0.9
8							

y = 0.0021 * x + 1.4205E-004
R = 1.0000
DL = 0.007364
BEC = 0.06842

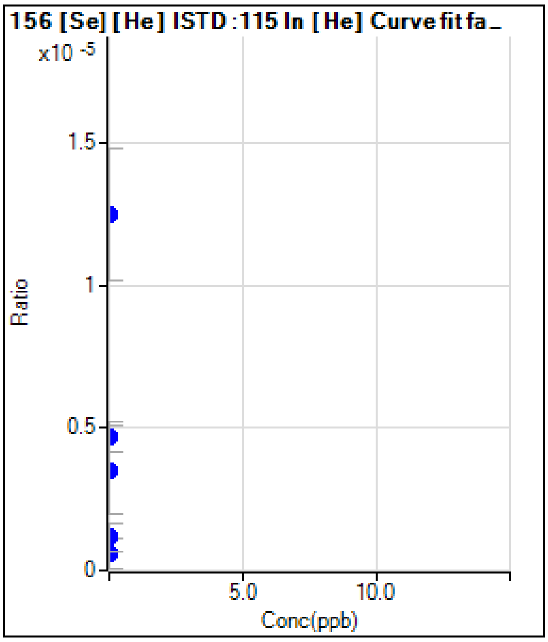
Weight: <None>
Min Conc: <None>



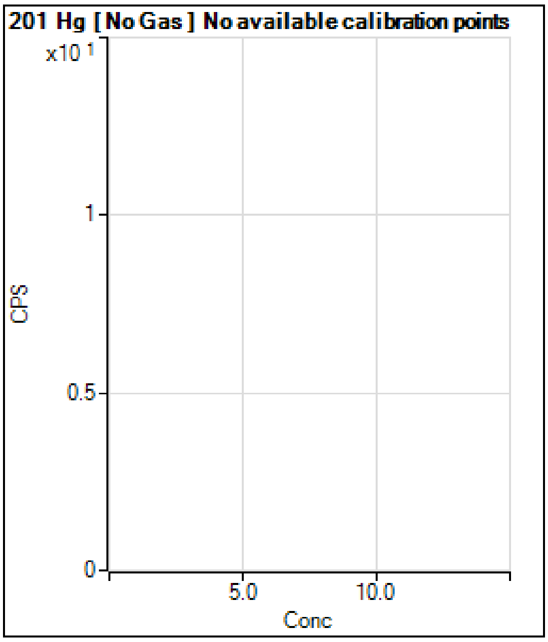
	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1		0.000	0.000	77.78	0.0000	P	7.1
2		0.000					
3		0.100	0.104	1312.30	0.0001	P	6.7
4		0.500					
5		1.000	0.982	11795.09	0.0007	P	2.1
6		10.000	9.728	118241.50	0.0072	P	0.4
7		100.000	100.027	1176837.11	0.0741	P	1.0
8		1000.000					

y = 7.4124E-004 * x + 4.8936E-006
R = 1.0000
DL = 0.0014
BEC = 0.006602

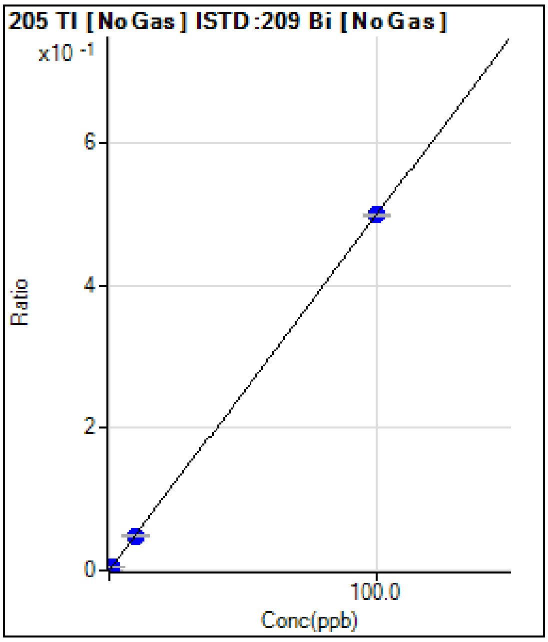
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000		1.11	0.0000	P	173.
2	☐	0.000					
3	☐	0.000		2.22	0.0000	P	86.6
4	☐	0.000					
5	☐	0.000		8.89	0.0000	P	23.1
6	☐	0.000		6.66	0.0000	P	87.9
7	☐	0.000		23.33	0.0000	P	37.1
8	☐						



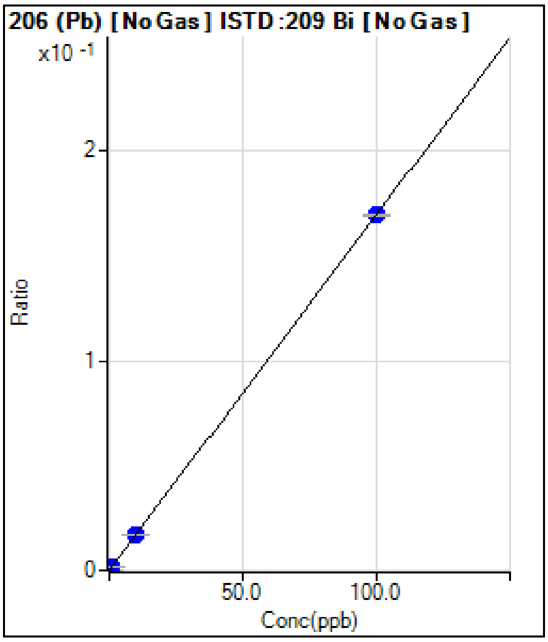
	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐			15.67		P	13.3
2	☐						
3	☐			18.83		P	18.8
4	☐						
5	☐			16.33		P	17.4
6	☐			123.33		P	7.1
7	☐			37.83		P	13.6
8	☐						



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	4901.10	0.0004	P	6.1
2	☐	0.000					
3	☐	0.100	0.075	9403.64	0.0008	P	1.9
4	☐	0.500					
5	☐	1.000	0.931	61362.12	0.0050	P	0.3
6	☐	10.000	9.728	602751.38	0.0488	P	1.5
7	☐	100.000	100.028	6010116.79	0.4985	P	0.8
8	☐						

y = 0.0050 * x + 4.0648E-004
R = 1.0000
DL = 0.01495
BEC = 0.08163

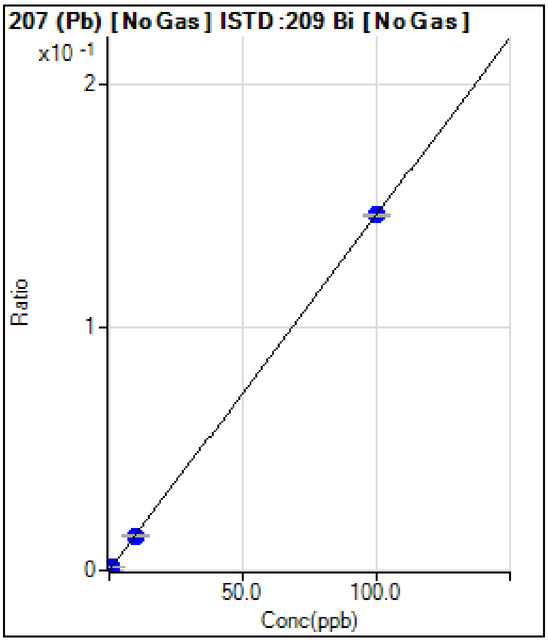
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	256.68	0.0000	P	13.0
2	☐	0.000					
3	☐	0.100	0.105	2396.96	0.0002	P	3.9
4	☐	0.500					
5	☐	1.000	0.989	20633.28	0.0017	P	3.7
6	☐	10.000	9.918	207352.00	0.0168	P	2.4
7	☐	100.000	100.008	2040764.09	0.1693	P	0.8
8	☐						

y = 0.0017 * x + 2.1317E-005
R = 1.0000
DL = 0.00493
BEC = 0.0126

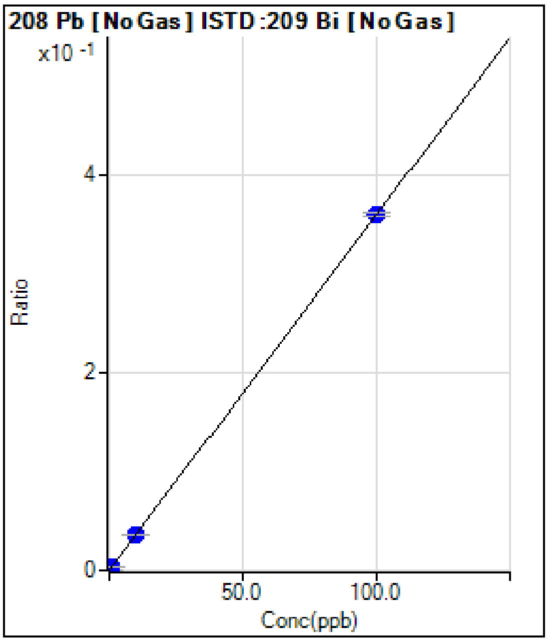
Weight: <None>
Min Conc: <None>



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	256.68	0.0000	P	22.7
2	☐	0.000					
3	☐	0.100	0.098	1980.23	0.0002	P	4.7
4	☐	0.500					
5	☐	1.000	0.965	17395.40	0.0014	P	2.8
6	☐	10.000	9.757	176009.17	0.0143	P	2.2
7	☐	100.000	100.025	1760722.11	0.1460	P	0.2
8	☐						

y = 0.0015 * x + 2.1334E-005
R = 1.0000
DL = 0.009968
BEC = 0.01461

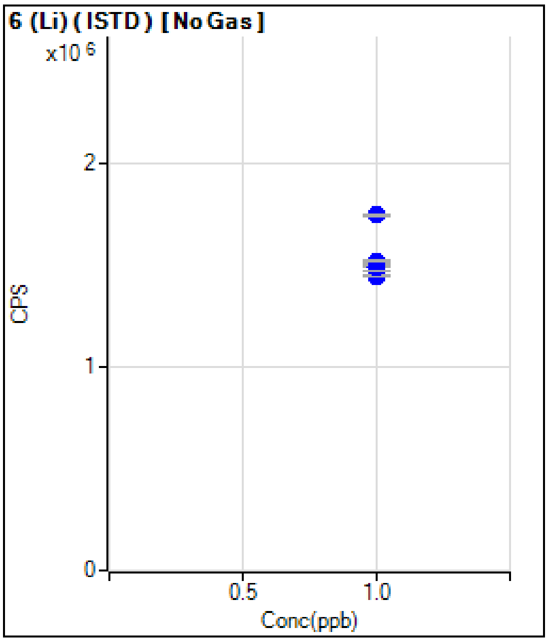
Weight: <None>
Min Conc: <None>



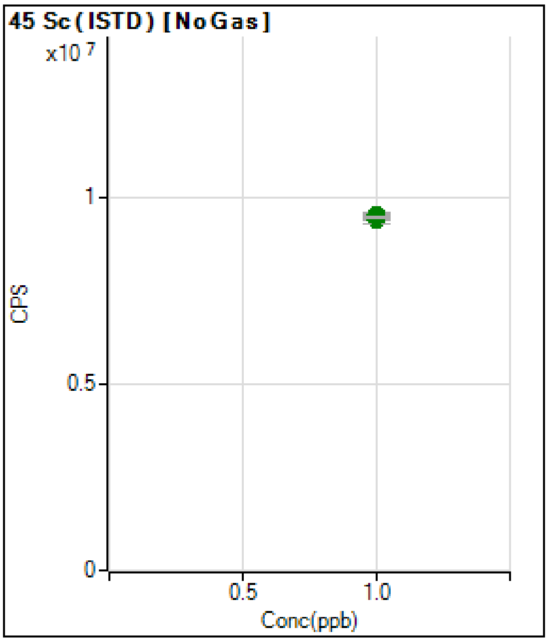
	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	0.000	0.000	630.04	0.0001	P	11.8
2	☐	0.000					
3	☐	0.100	0.099	4907.74	0.0004	P	1.6
4	☐	0.500					
5	☐	1.000	0.986	43759.74	0.0036	P	1.6
6	☐	10.000	9.812	435920.42	0.0353	P	0.2
7	☐	100.000	100.019	4335353.69	0.3596	P	1.2
8	☐						

y = 0.0036 * x + 5.2303E-005
R = 1.0000
DL = 0.005138
BEC = 0.01455

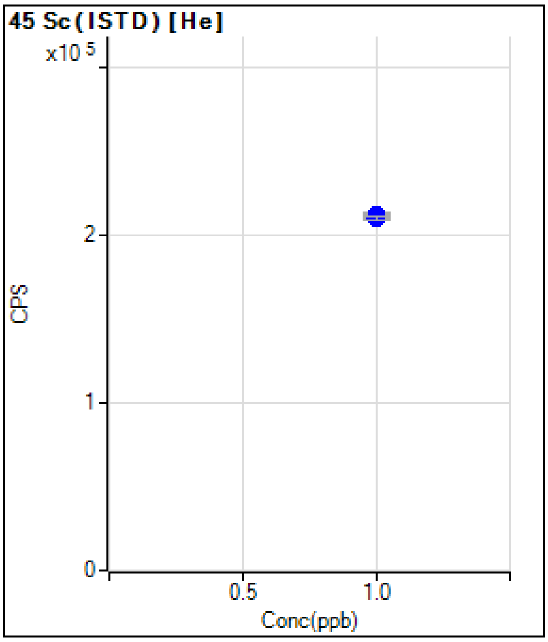
Weight: <None>
Min Conc: <None>



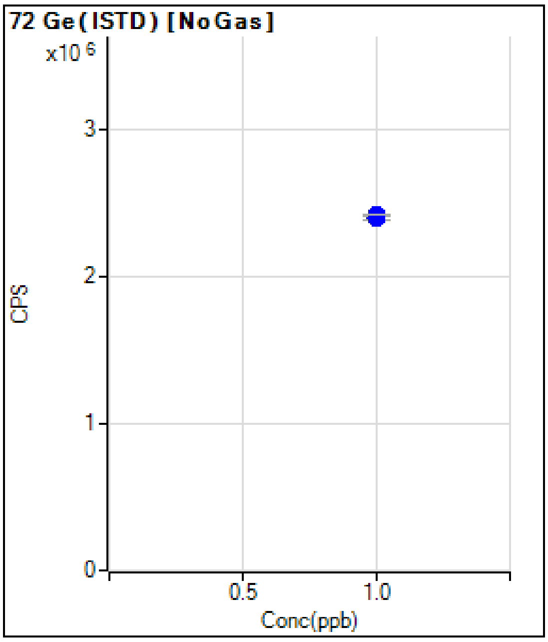
	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	1.000		1492504.04		P	0.8
2	☐	1.000					
3	☐	1.000		1448561.91		P	1.0
4	☐	1.000					
5	☐	1.000		1493531.64		P	2.3
6	☐	1.000		1519625.08		P	0.6
7	☐	1.000		1747076.75		P	0.7
8	☐	1.000					



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	1.000		9412079.90		A	1.2
2	☐	1.000					
3	☐	1.000		9352272.13		A	1.3
4	☐	1.000					
5	☐	1.000		9472370.18		A	0.9
6	☐	1.000		9506284.90		A	1.0
7	☐	1.000		9460583.79		A	0.6
8	☐	1.000					

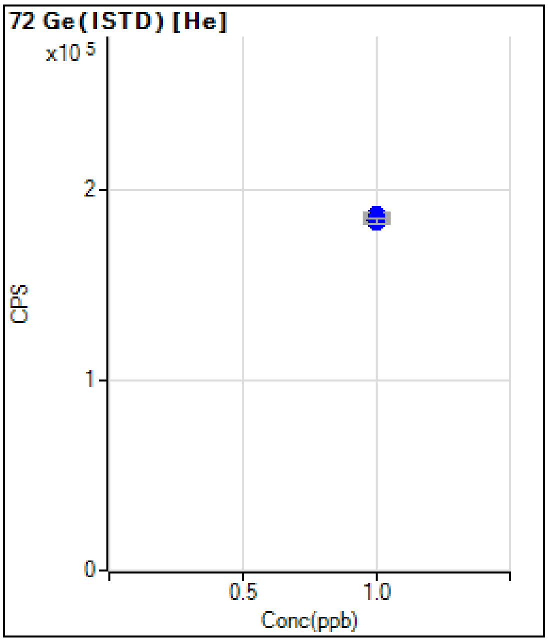


	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	1.000		210499.65		P	1.2
2	☐	1.000					
3	☐	1.000		211340.52		P	0.7
4	☐	1.000					
5	☐	1.000		211179.72		P	2.0
6	☐	1.000		212262.56		P	0.8
7	☐	1.000		210193.85		P	1.1
8	☐	1.000					

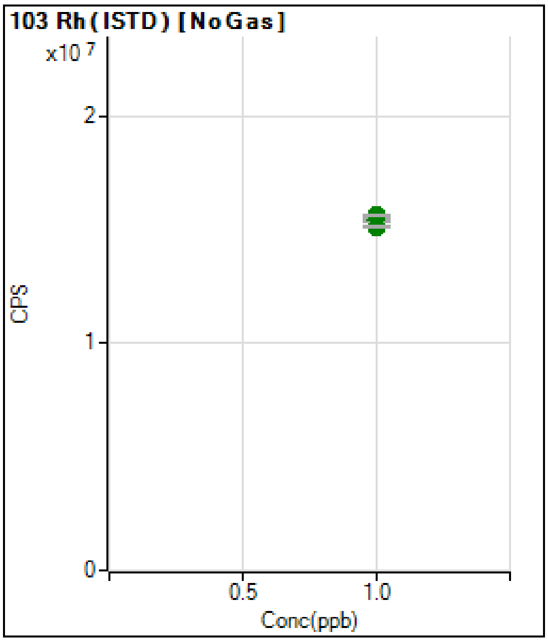


	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	1.000		2410488.10		P	0.5
2	☐	1.000					
3	☐	1.000		2396894.07		P	1.9
4	☐	1.000					
5	☐	1.000		2395685.95		P	0.7
6	☐	1.000		2406413.17		P	0.5
7	☐	1.000		2414784.14		P	0.1
8	☐	1.000					

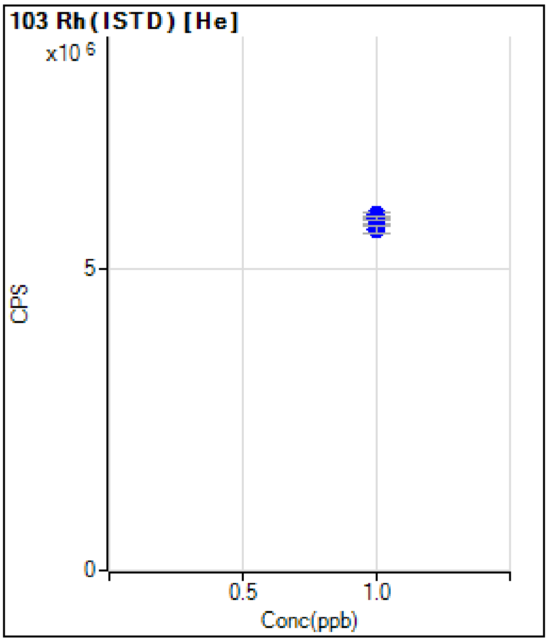
Calibration for 1251SMPL.d



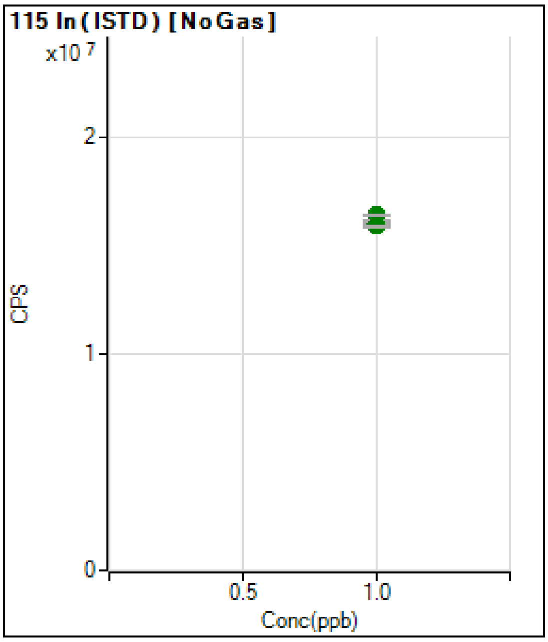
	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	1.000		183172.59		P	1.3
2	☐	1.000					
3	☐	1.000		186657.53		P	1.0
4	☐	1.000					
5	☐	1.000		185001.76		P	1.9
6	☐	1.000		185182.02		P	0.7
7	☐	1.000		183614.47		P	1.7
8	☐	1.000					



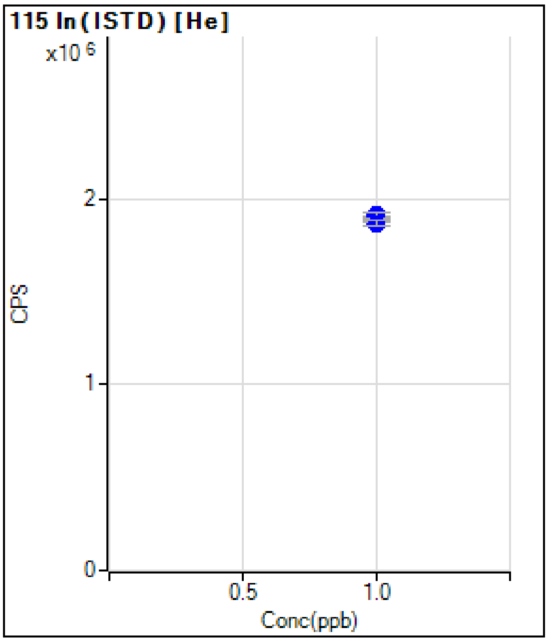
	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	1.000		15416412.7		A	0.5
2	☐	1.000					
3	☐	1.000		15455612.7		A	1.1
4	☐	1.000					
5	☐	1.000		15570582.7		A	1.1
6	☐	1.000		15640181.6		A	0.4
7	☐	1.000		15117514.4		A	0.7
8	☐	1.000					



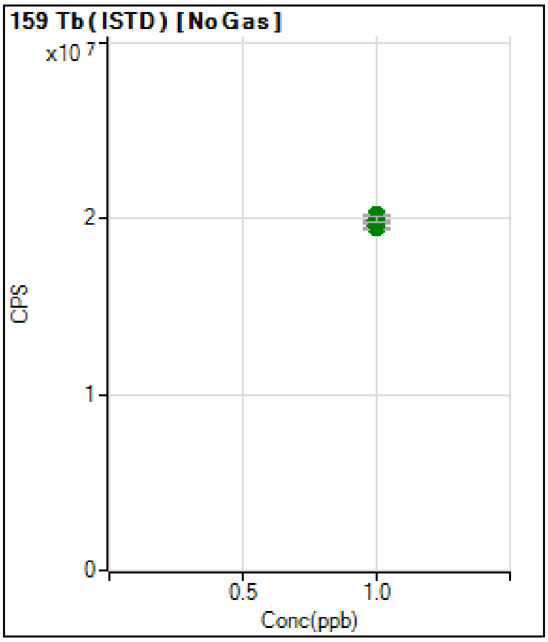
	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	1.000		5805949.63		P	1.7
2	☐	1.000					
3	☐	1.000		5904698.65		P	1.6
4	☐	1.000					
5	☐	1.000		5810527.41		P	2.6
6	☐	1.000		5848691.43		P	1.2
7	☐	1.000		5662777.97		P	2.0
8	☐	1.000					



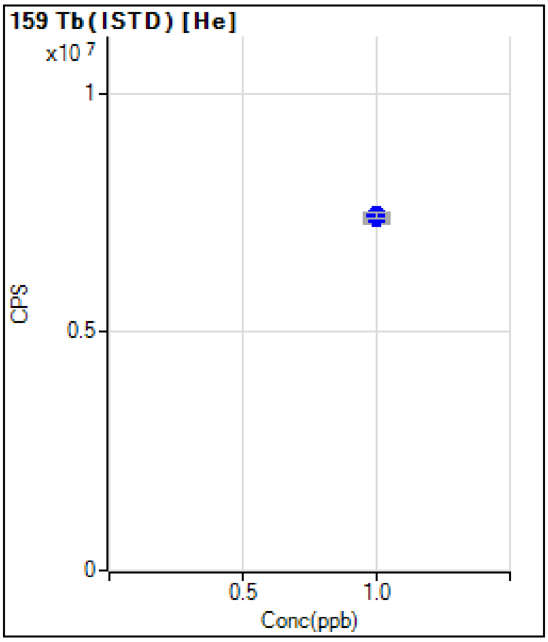
	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	1.000		15897456.5		A	0.5
2	☐	1.000					
3	☐	1.000		16024229.9		A	1.6
4	☐	1.000					
5	☐	1.000		16091374.9		A	0.8
6	☐	1.000		16386130.4		A	0.2
7	☐	1.000		15871288.8		A	0.6
8	☐	1.000					



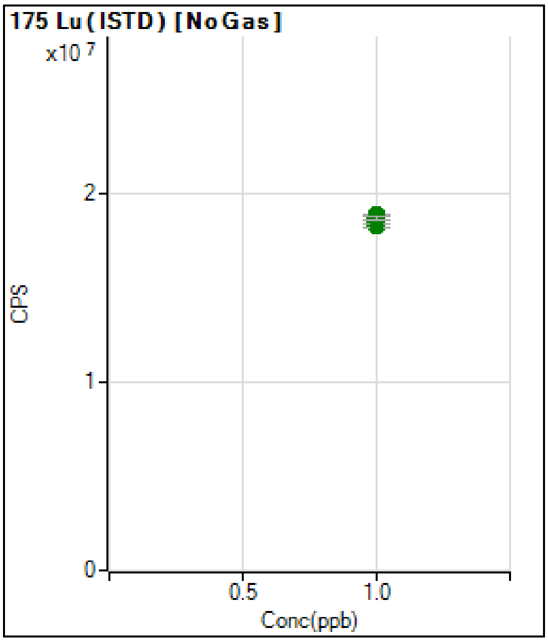
	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	1.000		1893464.65		P	1.7
2	☐	1.000					
3	☐	1.000		1913195.62		P	1.1
4	☐	1.000					
5	☐	1.000		1901675.41		P	2.5
6	☐	1.000		1907082.98		P	1.5
7	☐	1.000		1866856.59		P	1.7
8	☐	1.000					



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	1.000		19489435.1		A	0.3
2	☐	1.000					
3	☐	1.000		19619082.2		A	2.7
4	☐	1.000					
5	☐	1.000		19725032.2		A	0.7
6	☐	1.000		20198681.3		A	0.6
7	☐	1.000		19936810.1		A	1.7
8	☐	1.000					

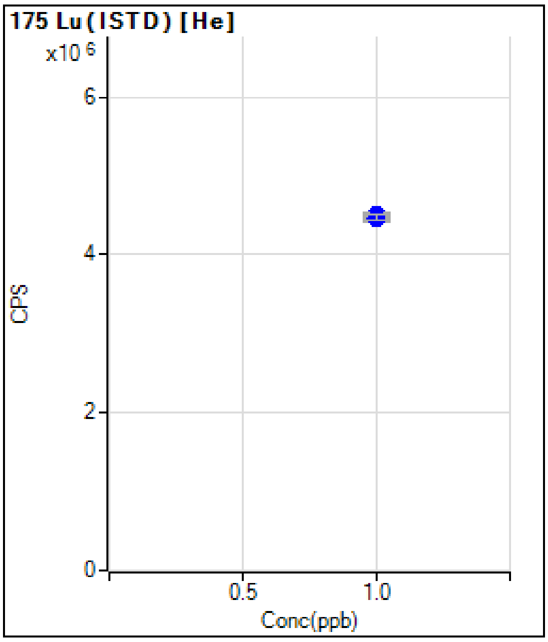


	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	1.000		7379981.77		P	2.4
2	☐	1.000					
3	☐	1.000		7394938.85		P	0.9
4	☐	1.000					
5	☐	1.000		7395911.76		P	3.1
6	☐	1.000		7444628.43		P	1.6
7	☐	1.000		7451801.14		P	1.8
8	☐	1.000					

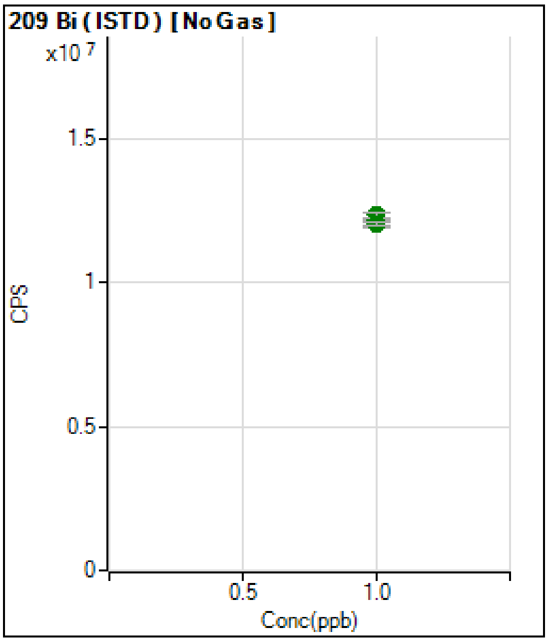


	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	1.000		18289690.1		A	1.2
2	☐	1.000					
3	☐	1.000		18407021.3		A	2.2
4	☐	1.000					
5	☐	1.000		18491520.5		A	1.2
6	☐	1.000		18854939.3		A	0.4
7	☐	1.000		18677033.4		A	1.2
8	☐	1.000					

Calibration for 1251SMPL.d

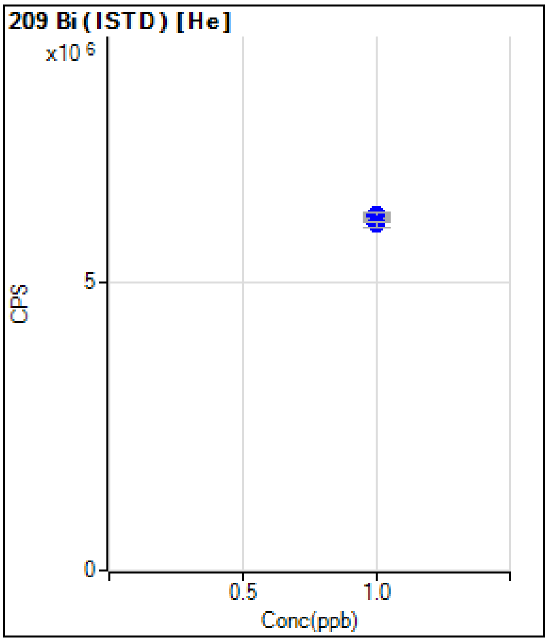


	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1		1.000		4453666.29		P	1.7
2		1.000					
3		1.000		4496257.54		P	0.8
4		1.000					
5		1.000		4471526.29		P	3.1
6		1.000		4501044.52		P	1.9
7		1.000		4471576.18		P	1.5
8		1.000					



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1		1.000		12052546.0		A	1.2
2		1.000					
3		1.000		12031302.3		A	2.4
4		1.000					
5		1.000		12171581.9		A	1.1
6		1.000		12341015.2		A	1.4
7		1.000		12056727.7		A	1.2
8		1.000					

Calibration for 1251SMPL.d



	Rjct	Conc.	Calc Conc.	CPS	Ratio	Det.	RSD
1	☐	1.000		6090049.08		P	1.0
2	☐	1.000					
3	☐	1.000		6146764.28		P	0.6
4	☐	1.000					
5	☐	1.000		6117190.74		P	2.7
6	☐	1.000		6154453.45		P	1.7
7	☐	1.000		5989814.70		P	1.4
8	☐	1.000					

Metals

Prep Sheets

3010A Metals Water Preparation

ANALYST/ TECH	JEL	START DATE/TIME	6/5/2023 06:00	END DATE/TIME	6/5/2023 10:00	BATCH	766941

#	CLIENT	TYPE	CLIENT ID	LAB ID	INITIAL VOL (mL)	FINAL VOL (mL)	COMMENT	STANDARDS\ REAGENTS
1	QC	MB	MB 2489334	2489334	50	50		GCAL - 8 - 250uL
2	QC	LCS	LCS 2489335	2489335	50	50		2132617
3	4887	SAMP	OT07-EB-060123	22306021001	50	50		Sb,Ag,Se SPIKE - 250uL
4	4887	SAMP	OT07-MW02-060123	22306021002	50	50		360-8-1
5	4887	SAMP	OT07-MW04-060123	22306021003	50	50		Li,B,Zr SPIKE - 250uL
6	4887	SAMP	OT07-MW05-060123	22306021004	50	50		360-2-1
7	4887	SAMP	OT07-MW06-060123	22306021005	50	50		Si SPIKE - 250uL
8	4887	SAMP	OT07-MW04-060123MS	22306021006	50	50		2131076
9	4887	SAMP	OT07-MW04-060123MSD	22306021007	50	50		1:1 HNO3
10	4887	SAMP	OT07-MW-806-060123	22306021008	50	50		226-42-16
11								1:1 HCL
12								226-42-13
13								
14								
15								
16								
17								
18								
19								
20								
21								
22								
23								
24								
25								
26								
27								Digestion Vessel Lot#
28								0260000251
29								
30								

EQUIPMENT\CONDITIONS

DIGESTION BLOCK\THERMOMETER ID	TEMPERATURE	PIPETTE 1
B1	**94**	**117**
PIPETTE 2	PIPETTE 3	
115		

NOTES

TCLPs are reduced in volume due to sample matrix. Matrix-Water. 6020_W_EX

SM 2320 B-2011

Form I

Sample Results

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW02-060123
Collect Date:	06/01/23 1230	LAB Sample ID:	22306021002
Matrix:	Water	Instrument ID:	TITR1
% Solids:	NA	Analyst:	MMP
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/06/23 1038
Prep Batch:	NA	Analytical Batch:	767075
Prep Method:	NA	Analytical Method:	SM 2320B

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Alkalinity	175	mg/L CaCO3		0.26	0.80	1.0

FORM I - GENCHEM

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123
Collect Date:	06/01/23 0935	LAB Sample ID:	22306021003
Matrix:	Water	Instrument ID:	TITR1
% Solids:	NA	Analyst:	MMP
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/06/23 1038
Prep Batch:	NA	Analytical Batch:	767075
Prep Method:	NA	Analytical Method:	SM 2320B

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Alkalinity	279	mg/L CaCO3		0.26	0.80	1.0

FORM I - GENCHEM

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW05-060123
Collect Date:	06/01/23 0910	LAB Sample ID:	22306021004
Matrix:	Water	Instrument ID:	TITR1
% Solids:	NA	Analyst:	MMP
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/06/23 1038
Prep Batch:	NA	Analytical Batch:	767075
Prep Method:	NA	Analytical Method:	SM 2320B

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Alkalinity	145	mg/L CaCO3		0.26	0.80	1.0

FORM I - GENCHEM

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW06-060123
Collect Date:	06/01/23 1100	LAB Sample ID:	22306021005
Matrix:	Water	Instrument ID:	TITR1
% Solids:	NA	Analyst:	MMP
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/06/23 1038
Prep Batch:	NA	Analytical Batch:	767075
Prep Method:	NA	Analytical Method:	SM 2320B

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Alkalinity	146	mg/L CaCO3		0.26	0.80	1.0

FORM I - GENCHEM

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW-806-060123
Collect Date:	06/01/23 1100	LAB Sample ID:	22306021008
Matrix:	Water	Instrument ID:	TITR1
% Solids:	NA	Analyst:	MMP
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/06/23 1038
Prep Batch:	NA	Analytical Batch:	767075
Prep Method:	NA	Analytical Method:	SM 2320B

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Alkalinity	168	mg/L CaCO3		0.26	0.80	1.0

FORM I - GENCHEM

SM 2320 B-2011

Form III

Blanks

III - METHOD BLANK

Report No:	223060210	Blank ID:	MB2489949
Matrix:	Water	Instrument ID:	TITR1
Analysis Date:	06/06/23 1038	Lab File ID:	NA
Analytical Method:	SM 2320B	Analytical Batch:	767075

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Alkalinity	0.80	mg/L CaCO3	U	0.26	0.80	1.0

FORM III - GENCHEM

SM 2320 B-2011

Form VI

Duplicates

VI - DUPLICATES

Report No:	223060210	Parent Sample ID:	OT07-MW02-060123
Prep Method:	NA	Parent LAB ID:	22306021002
Prep Date:	NA	Prep Batch:	NA
Analytical Method:	SM 2320 B-2011	Analytical Batch:	767075

LAB QC ID:	**2489952 DUP**	Instrument ID:	TITR1
Analyst:	MMP	Lab File ID:	NA
Analysis Date:	06/06/23 1038	Dilution:	1

ANALYTE	*UNITS*	*SAMPLE RESULT*	*Q*	*DUP RESULT*	*Q*	*RPD*	*#*	*RPD LIMITS*
Total Alkalinity	mg/L CaCO3	175		174		1		0 - 10.50

FORM VI - GENCHEM

SM 2320 B-2011

Form VII

Lab Control Spikes

VII - LABORATORY CONTROL SPIKE

Report No:	223060210	LAB ID:	LCS2489950
Matrix:	Water	Instrument ID:	TITR1
Analyst:	MMP	Lab File ID:	NA
Prep Date:	NA	Analysis Date:	06/06/23 1038
Prep Batch:	NA	Analytical Batch:	767075
Prep Method:	NA	Analytical Method:	SM 2320 B-2011

ANALYTE	UNITS	TRUE	FOUND	%REC	Q	% REC LIMITS
Total Alkalinity	mg/L CaCO3	200	199	99		90 - 110

FORM VII - GENCHEM

VII - LABORATORY CONTROL SPIKE DUPLICATE

Report No:	223060210	LAB ID:	LCSD2489951
Matrix:	Water	Instrument ID:	TITR1
Analyst:	MMP	Lab File ID:	NA
Prep Date:	NA	Analysis Date:	06/06/23 1038
Prep Batch:	NA	Analytical Batch:	767075
Prep Method:	NA	Analytical Method:	SM 2320 B-2011

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*%REC*	*Q*	*%REC LIMITS*	*RPD*	*Q*	*RPD LIMITS*
Total Alkalinity	mg/L CaCO3	200	200	100		90 - 110	1		0 - 11

FORM VII - GENCHEM

SM 2320 B-2011

Form XIV

Run Logs

XIV - ANALYSIS RUN LOG

Report No:	223060210	Analytical Batch:	767075	Start Date:	06/06/23
Instrument ID:	TITRATOR	Analytical Method:	SM 2320 B-2011	End Date:	06/06/23

CLIENT SAMPLE ID	*LAB SAMPLE ID*	*DILUTION*	*TIME*	*ANALYTES* Total Alkalinity
OT07-MW-806-060123	22306021008	1	1038	X
OT07-MW06-060123	22306021005	1	1038	X
OT07-MW05-060123	22306021004	1	1038	X
OT07-MW04-060123	22306021003	1	1038	X
OT07-MW02-060123DUP	2489952	1	1038	X
MB2489949	2489949	1	1038	X
LCSD2489951	2489951	1	1038	X
LCS2489950	2489950	1	1038	X
OT07-MW02-060123	22306021002	1	1038	X

FORM XIV - GENCHEM

SM 2320 B-2011

Raw Data

Pace Analytical

2320 B Alkalinity

767075

ANALYST/ TECH	MMP	START DATE/TIME	6/6/2023 10:38	END DATE/TIME	6/6/2023 13:18	BATCH	767075

#	CLIENT	TYPE	LAB ID	ALKT	ALKP	ALKB	ALKC	ALKOH	STANDARDS\ REAGENTS
1	QC	MB	2489949	0.00					0.02 N H2SO4 Lot
2	QC	LCS	2489950	198.72					2132573
3	QC	LCSD	2489951	199.72					0.02 N H2SO4 Exp
4	4887	SAMP	22306021002	175.31					04/30/27
5	QC	DUP	2489952	174.15					LCS Lot
6	4887	SAMP	22306021003	278.69					2133129
7	4887	SAMP	22306021004	144.68					LCS Exp
8	4887	SAMP	22306021005	146.18					05/31/2024
9	4887	SAMP	22306021008	168.33					
10	0080	CCV	1800	199.15					
11	0080	CCB	1900	0.00					
12									
13									
14									
15									
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29									
30									

EQUIPMENT\CONDITIONS

Instrument ID: TITR1		

NOTES

Method 1 Calibration pH electrode 05/03/2010 9:04
Measured 06/06/2023 10:36
User

ALL RESULTS

No.	ID	Sample size and results
1		50.0 mL pH,pM,pX 4.000 R1 = 167.108 mV
2		50.0 mL pH,pM,pX 7.000 R1 = 0.775 mV
3		50.0 mL pH,pM,pX 10.000 R1 = -166.914 mV

CALIBRATION

Sensor	DG111		
Buffer type	pH, pM, pX		
Zero point		7.006	pH0
Slope		-55.67	mV/pH
Calibration temperature		25.0	°C

Last page

Method	11 Alkalinity		10/04/2012 9:25
Measured	06/06/2023 10:38		
User			

RAW RESULTS

SAMPLE

No.		1			
Identification		2489949			
Titration stand		Stand 1			
Volume		50.0 mL			
Correction factor f		1.0			
Mol.mass M		100.00 g/mol			
Equivalent number z		1			
Temperature sensor		Manual			

DISPENSE

Titrant	H2SO4	0.02 mol/L t = 1.0		
Drive	No. 2	10 mL		
Volume		VDISP = 0.0	mL	
		QDISP = 0.0	mmol	

EP TITRATION

Titrant	H2SO4	0.02 mol/L t = 1.0		
Drive	No. 2	10 mL		
Sensor	DG111			
Excess		VEX = 0.0000	mL	
		QEX = 0.00000	mmol	

CALCULATION

Result	R1 = 3.61		Initial pH

CALCULATION

Result	R2 = 0.0000	mg/L	ALKP

DISPENSE

Titrant	H2SO4	0.02 mol/L t = 1.0		
Drive	No. 2	10 mL		
Volume		VDISP = 0.0	mL	
		QDISP = 0.0	mmol	

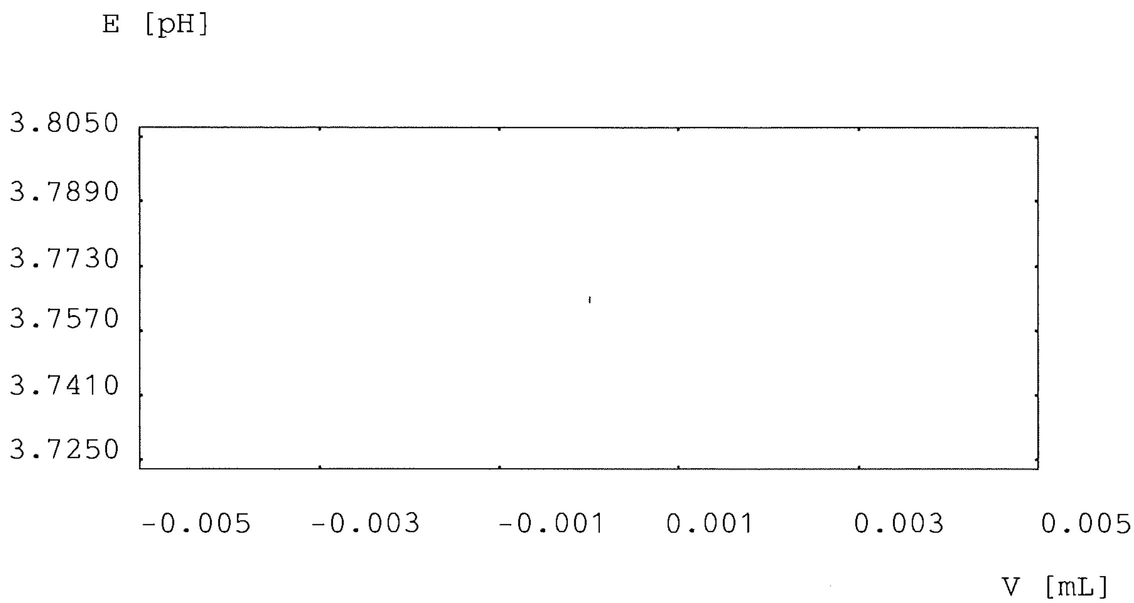
EP TITRATION

Titrant	H2SO4	0.02 mol/L t = 1.0		
Drive	No. 2	10 mL		
Sensor	DG111			
Excess		VEX = 0.0000	mL	
		QEX = 0.00000	mmol	

CALCULATION

Result	R3 = 0.00	mg/L	ALKT

E - V curve


E [pH]
3.8050
3.7890
3.7730
3.7570
3.7410
3.7250
-0.005
-0.003
-0.001
0.001
0.003
0.005
V [mL]

Method	11 Alkalinity		10/04/2012 9:25
Measured	06/06/2023 10:50		
User			

Sample aborted: No. 3

RAW RESULTS

SAMPLE

No.		4		
Identification		2489950		
Titration stand		Stand 1		
Volume		50.0 mL		
Correction factor f		1.0		
Mol.mass M		100.00 g/mol		
Equivalent number z		1		
Temperature sensor		Manual		

DISPENSE

Titrant	H2SO4	0.02 mol/L t = 1.0		
Drive	No. 2	10 mL		
Volume		VDISP = 0.0	mL	
		QDISP = 0.0	mmol	

EP TITRATION

Titrant	H2SO4	0.02 mol/L t = 1.0		
Drive	No. 2	10 mL		
Sensor	DG111			
Consumption	EP	VEQ1 = 5.9647	mL	
		Q1 = 0.11929	mmol	
		EPOT1 = 8.300	pH	
Excess		VEX = 0.0733	mL	
		QEX = 0.00147	mmol	

CALCULATION

Result		R1 = 10.45		Initial pH

CALCULATION

Result		R2 = 120.760	mg/L	ALKP

DISPENSE

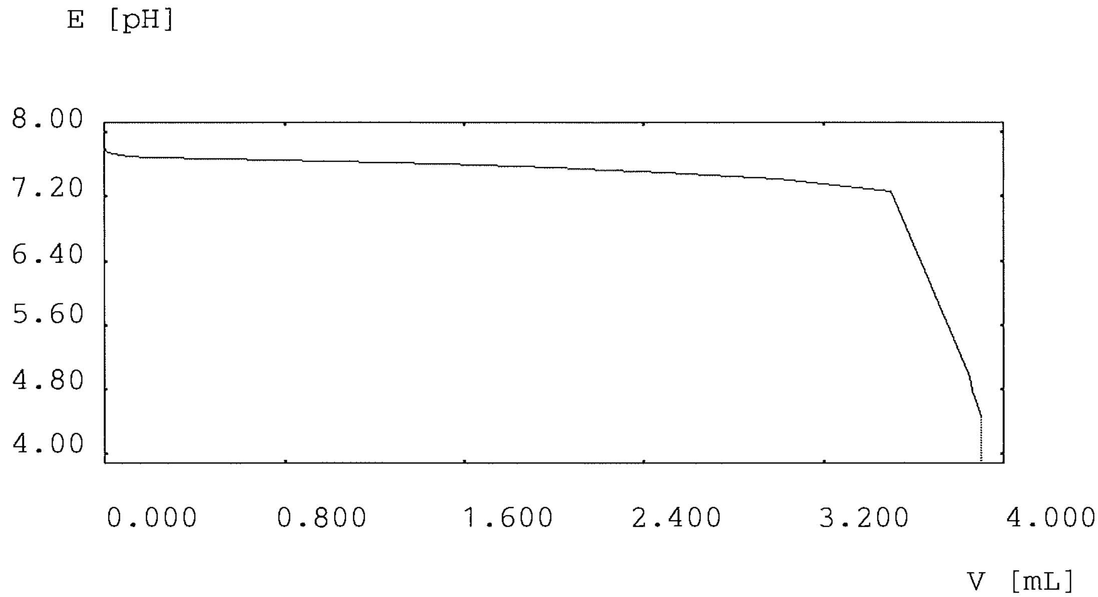
Titrant	H2SO4	0.02 mol/L t = 1.0		
Drive	No. 2	10 mL		
Volume		VDISP = 0.0	mL	
		QDISP = 0.0	mmol	

EP TITRATION

Titrant	H2SO4	0.02 mol/L t = 1.0		
Drive	No. 2	10 mL		
Sensor	DG111			
Consumption	EP	VEQ1 = 3.8978	mL	
		Q1 = 0.07796	mmol	
		EPOT1 = 4.500	pH	
Excess		VEX = 0.0042	mL	
		QEX = 0.00008	mmol	

CALCULATION					
Result	R3	=	198.72	mg/L	ALKT

E - V curve



Method	11 Alkalinity		10/04/2012 9:25
Measured	06/06/2023 10:54		
User			

Sample aborted: No. 3

RAW RESULTS

SAMPLE

No.		5
Identification		2489951
Titration stand		Stand 1
Volume		50.0 mL
Correction factor f		1.0
Mol.mass M		100.00 g/mol
Equivalent number z		1
Temperature sensor		Manual

DISPENSE

Titrant	H2SO4	0.02 mol/L t = 1.0		
Drive	No. 2	10 mL		
Volume		VDISP = 0.0	mL	
		QDISP = 0.0	mmol	

EP TITRATION

Titrant	H2SO4	0.02 mol/L t = 1.0		
Drive	No. 2	10 mL		
Sensor	DG111			
Consumption	EP	VEQ1 =	6.5274	mL
		Q1 =	0.13055	mmol
		EPOT1 =	8.300	pH
Excess		VEX =	0.1016	mL
		QEX =	0.00203	mmol

CALCULATION

Result	R1 =	10.35	Initial pH

CALCULATION

Result	R2 =	132.580 mg/L	ALKP

DISPENSE

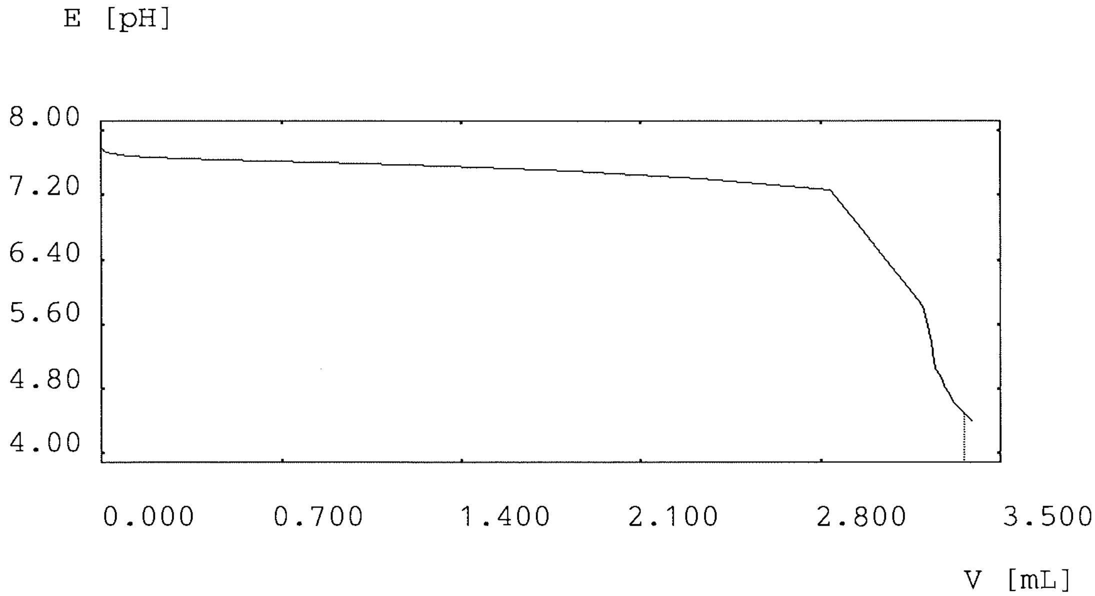
Titrant	H2SO4	0.02 mol/L t = 1.0		
Drive	No. 2	10 mL		
Volume		VDISP = 0.0	mL	
		QDISP = 0.0	mmol	

EP TITRATION

Titrant	H2SO4	0.02 mol/L t = 1.0		
Drive	No. 2	10 mL		
Sensor	DG111			
Consumption	EP	VEQ1 =	3.3572	mL
		Q1 =	0.06714	mmol
		EPOT1 =	4.500	pH
Excess		VEX =	0.0318	mL
		QEX =	0.00064	mmol

CALCULATION					
Result	R3	=	199.72	mg/L	ALKT

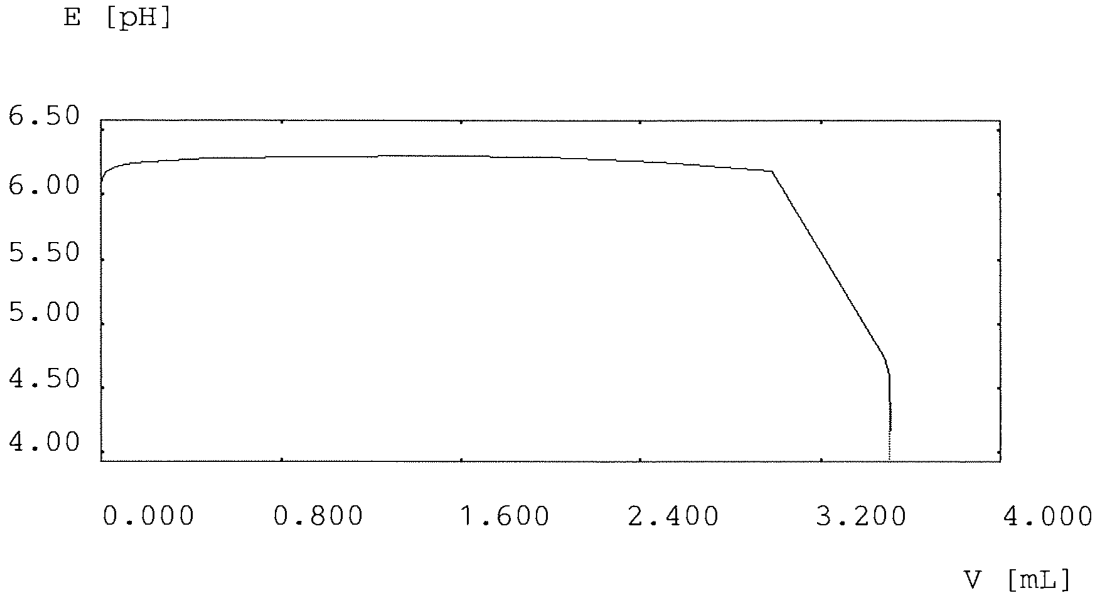
E - V curve



```
Method        11    Alkalinity                    10/04/2012   9:25
Measured      06/06/2023  11:11
User

Sample aborted: No. 3

Sample aborted: No. 6

Sample aborted: No. 8

RAW RESULTS

SAMPLE
  No.                          11
  Identification               2306021002
  Titration stand              Stand 1
  Volume                       20.0 mL
  Correction factor f          1.0
  Mol.mass M                   100.00 g/mol
  Equivalent number z          1
  Temperature sensor           Manual

DISPENSE
  Titrant          H2SO4  0.02 mol/L  t = 1.0
  Drive            No. 2       10 mL
  Volume                   VDISP = 0.0       mL
                           QDISP = 0.0       mmol

EP TITRATION
  Titrant          H2SO4  0.02 mol/L  t = 1.0
  Drive            No. 2       10 mL
  Sensor           DG111
  Excess                     VEX   =  0.0000  mL
                             QEX   =  0.00000 mmol

CALCULATION
  Result                      R1   =  5.92            Initial pH

CALCULATION
  Result                      R2   =  0.0000  mg/L    ALKP

DISPENSE
  Titrant          H2SO4  0.02 mol/L  t = 1.0
  Drive            No. 2       10 mL
  Volume                   VDISP = 0.0       mL
                           QDISP = 0.0       mmol

EP TITRATION
  Titrant          H2SO4  0.02 mol/L  t = 1.0
  Drive            No. 2       10 mL
  Sensor           DG111
  Consumption      EP        VEQ1  =  3.5062  mL
                               Q1  =  0.07012 mmol
                           EPOT1   =  4.500   pH
  Excess                     VEX   =  0.0038  mL
                             QEX   =  0.00008 mmol
```

CALCULATION					
Result	R3	=	175.31	mg/L	ALKT

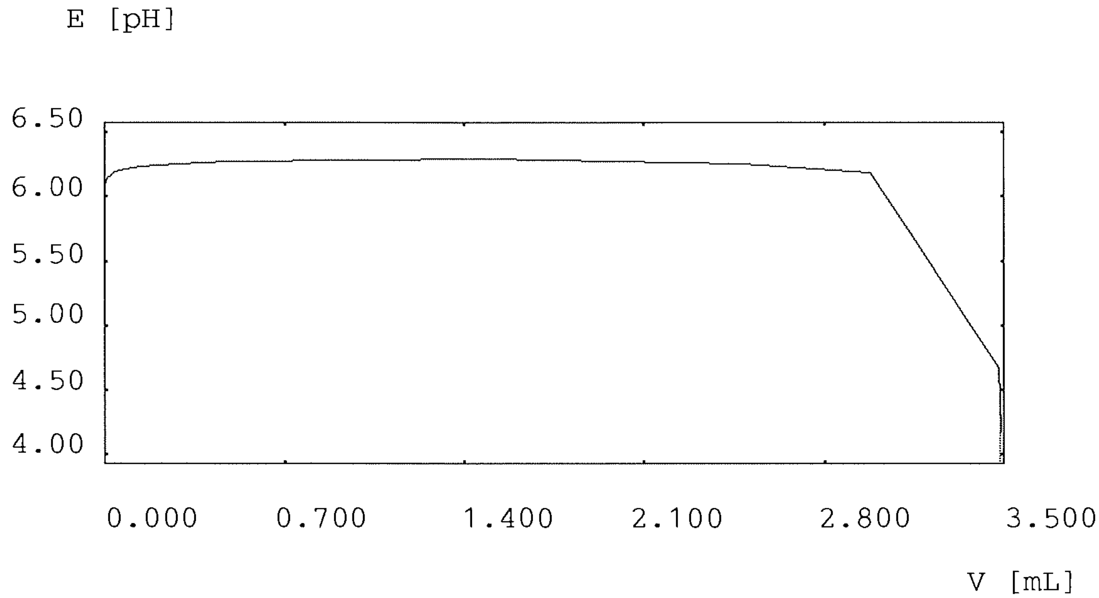
E - V curve



```
Method        11    Alkalinity                    10/04/2012   9:25
Measured      06/06/2023  11:15
User

Sample aborted: No. 3

Sample aborted: No. 6

Sample aborted: No. 8

RAW RESULTS

SAMPLE
  No.                          12
  Identification               2306021002
  Titration stand              Stand 1
  Volume                       20.0 mL
  Correction factor f          1.0
  Mol.mass M                   100.00 g/mol
  Equivalent number z          1
  Temperature sensor           Manual

DISPENSE
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Volume                       VDISP = 0.0       mL
                               QDISP = 0.0       mmol

EP TITRATION
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Sensor             DG111
  Excess                        VEX   =  0.0000  mL
                                QEX   =  0.00000 mmol

CALCULATION
  Result                        R1    =  5.94           Initial pH

CALCULATION
  Result                        R2    =  0.0000  mg/L   ALKP

DISPENSE
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Volume                       VDISP = 0.0       mL
                               QDISP = 0.0       mmol

EP TITRATION
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Sensor             DG111
  Consumption        EP         VEQ1  =  3.4830  mL
                                  Q1  =  0.06966 mmol
                               EPOT1  =  4.500   pH
  Excess                        VEX   =  0.0060  mL
                                QEX   =  0.00012 mmol
```

CALCULATION					
Result	R3	=	174.15	mg/L	ALKT

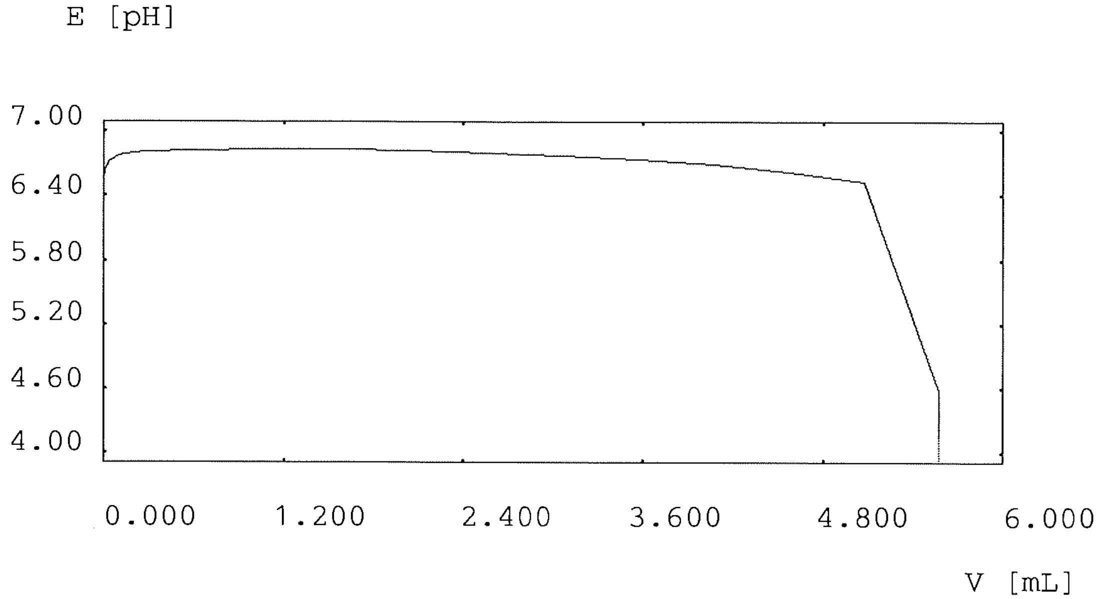
E - V curve



```
Method         11    Alkalinity                10/04/2012   9:25
Measured       06/06/2023  11:18
User

Sample aborted: No. 3

Sample aborted: No. 6

Sample aborted: No. 8

RAW RESULTS

SAMPLE
  No.                          13
  Identification               2306021003
  Titration stand              Stand 1
  Volume                       20.0 mL
  Correction factor f          1.0
  Mol.mass M                   100.00 g/mol
  Equivalent number z          1
  Temperature sensor           Manual

DISPENSE
  Titrant             H2SO4  0.02 mol/L  t = 1.0
  Drive               No. 2      10 mL
  Volume                       VDISP = 0.0       mL
                               QDISP = 0.0       mmol

EP TITRATION
  Titrant             H2SO4  0.02 mol/L  t = 1.0
  Drive               No. 2      10 mL
  Sensor              DG111
  Excess                        VEX   =  0.0000  mL
                                QEX   =  0.00000 mmol

CALCULATION
  Result                        R1    =  6.39            Initial pH

CALCULATION
  Result                        R2    =  0.0000  mg/L    ALKP

DISPENSE
  Titrant             H2SO4  0.02 mol/L  t = 1.0
  Drive               No. 2      10 mL
  Volume                       VDISP = 0.0       mL
                               QDISP = 0.0       mmol

EP TITRATION
  Titrant             H2SO4  0.02 mol/L  t = 1.0
  Drive               No. 2      10 mL
  Sensor              DG111
  Consumption         EP        VEQ1  =  5.5738  mL
                                  Q1  =  0.11148 mmol
                               EPOT1  =  4.500   pH
  Excess                        VEX   =  0.0042  mL
                                QEX   =  0.00008 mmol
```

CALCULATION					
Result	R3	=	278.69	mg/L	ALKT

E - V curve



Method	11 Alkalinity	10/04/2012	9:25
Measured	06/06/2023 12:45		
User			

Sample aborted: No. 3

Sample aborted: No. 6

Sample aborted: No. 8

RAW RESULTS

SAMPLE

No.	14
Identification	2306021004
Titration stand	Stand 1
Volume	20.0 mL
Correction factor f	1.0
Mol.mass M	100.00 g/mol
Equivalent number z	1
Temperature sensor	Manual

DISPENSE

Titrant	H2SO4	0.02 mol/L	t = 1.0	
Drive	No. 2	10 mL		
Volume		VDISP = 0.0		mL
		QDISP = 0.0		mmol

EP TITRATION

Titrant	H2SO4	0.02 mol/L	t = 1.0	
Drive	No. 2	10 mL		
Sensor	DG111			
Excess		VEX =	0.0000	mL
		QEX =	0.00000	mmol

CALCULATION

Result	R1 =	6.09		Initial pH

CALCULATION

Result	R2 =	0.0000	mg/L	ALKP

DISPENSE

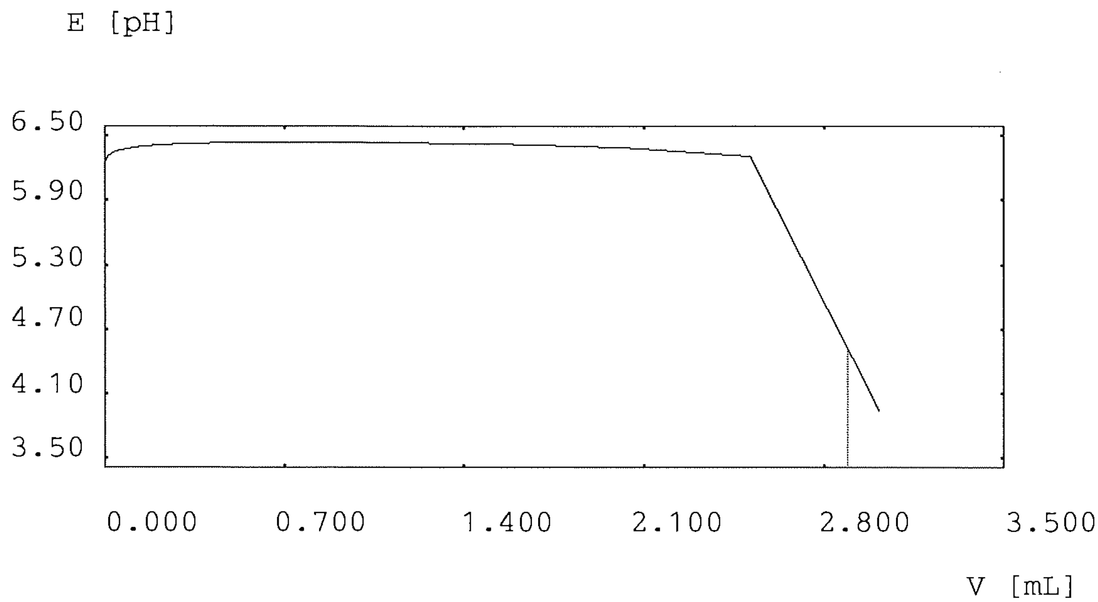
Titrant	H2SO4	0.02 mol/L	t = 1.0	
Drive	No. 2	10 mL		
Volume		VDISP = 0.0		mL
		QDISP = 0.0		mmol

EP TITRATION

Titrant	H2SO4	0.02 mol/L	t = 1.0	
Drive	No. 2	10 mL		
Sensor	DG111			
Consumption	EP	VEQ1 =	2.8937	mL
		Q1 =	0.05787	mmol
		EPOT1 =	4.500	pH
Excess		VEX =	0.1183	mL
		QEX =	0.00237	mmol

CALCULATION
Result R3 = 144.68 mg/L ALKT

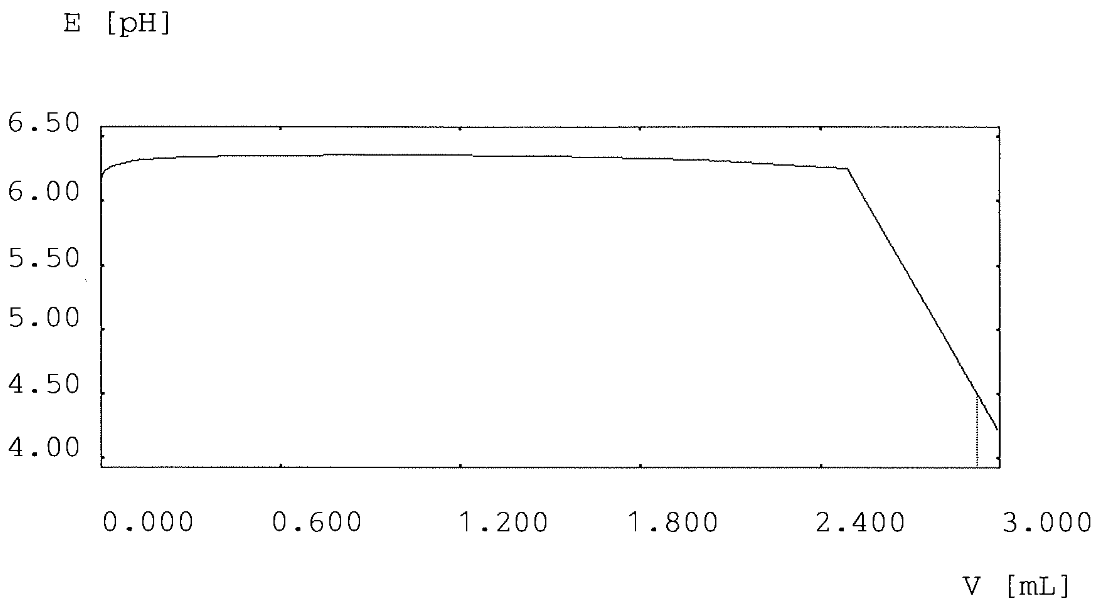
E - V curve



```
Method        11    Alkalinity                10/04/2012   9:25
Measured      06/06/2023  13:01
User

Sample aborted: No. 3

Sample aborted: No. 6

Sample aborted: No. 8

Sample aborted: No. 15

RAW RESULTS

SAMPLE
  No.                         16
  Identification              2306021005
  Titration stand             Stand 1
  Volume                      20.0 mL
  Correction factor f         1.0
  Mol.mass M                  100.00 g/mol
  Equivalent number z         1
  Temperature sensor          Manual

DISPENSE
  Titrant             H2SO4  0.02 mol/L  t = 1.0
  Drive               No. 2       10 mL
  Volume                      VDISP = 0.0      mL
                              QDISP = 0.0      mmol

EP TITRATION
  Titrant             H2SO4  0.02 mol/L  t = 1.0
  Drive               No. 2       10 mL
  Sensor              DG111
  Excess                        VEX  =  0.0000  mL
                                QEX  =  0.00000 mmol

CALCULATION
  Result                         R1  =  6.01           Initial pH

CALCULATION
  Result                         R2  =  0.0000  mg/L   ALKP

DISPENSE
  Titrant             H2SO4  0.02 mol/L  t = 1.0
  Drive               No. 2       10 mL
  Volume                      VDISP = 0.0      mL
                              QDISP = 0.0      mmol

EP TITRATION
  Titrant             H2SO4  0.02 mol/L  t = 1.0
  Drive               No. 2       10 mL
  Sensor              DG111
  Consumption         EP        VEQ1 =  2.9237  mL
                                  Q1 =  0.05847 mmol
                               EPOT1 =  4.500   pH
  Excess                        VEX  =  0.0694  mL
```

	QEX	=	0.00139 mmol	
CALCULATION				
Result	R3	=	146.18 mg/L	ALKT

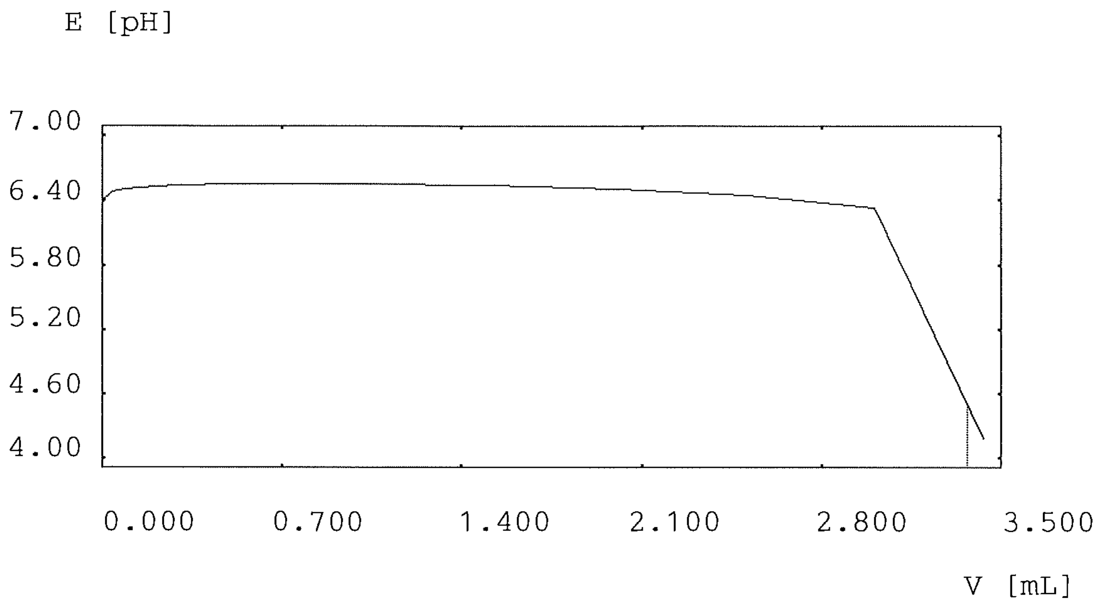
E - V curve



```
Method        11    Alkalinity                    10/04/2012   9:25
Measured      06/06/2023  13:10
User

Sample aborted: No. 3

Sample aborted: No. 6

Sample aborted: No. 8

Sample aborted: No. 15

RAW RESULTS

SAMPLE
  No.                          19
  Identification               2306021008
  Titration stand              Stand 1
  Volume                       20.0 mL
  Correction factor f          1.0
  Mol.mass M                   100.00 g/mol
  Equivalent number z          1
  Temperature sensor           Manual

DISPENSE
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Volume                       VDISP = 0.0       mL
                               QDISP = 0.0       mmol

EP TITRATION
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Sensor             DG111
  Excess                        VEX   =  0.0000  mL
                                QEX   =  0.00000 mmol

CALCULATION
  Result                         R1   =  6.21             Initial pH

CALCULATION
  Result                         R2   =  0.0000  mg/L   ALKP

DISPENSE
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Volume                       VDISP = 0.0       mL
                               QDISP = 0.0       mmol

EP TITRATION
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Sensor             DG111
  Consumption        EP         VEQ1  =  3.3667  mL
                                  Q1  =  0.06733 mmol
                               EPOT1  =  4.500   pH
  Excess                        VEX   =  0.0633  mL
```

	QEX	=	0.00127 mmol	
CALCULATION Result	R3	=	168.33 mg/L	ALKT

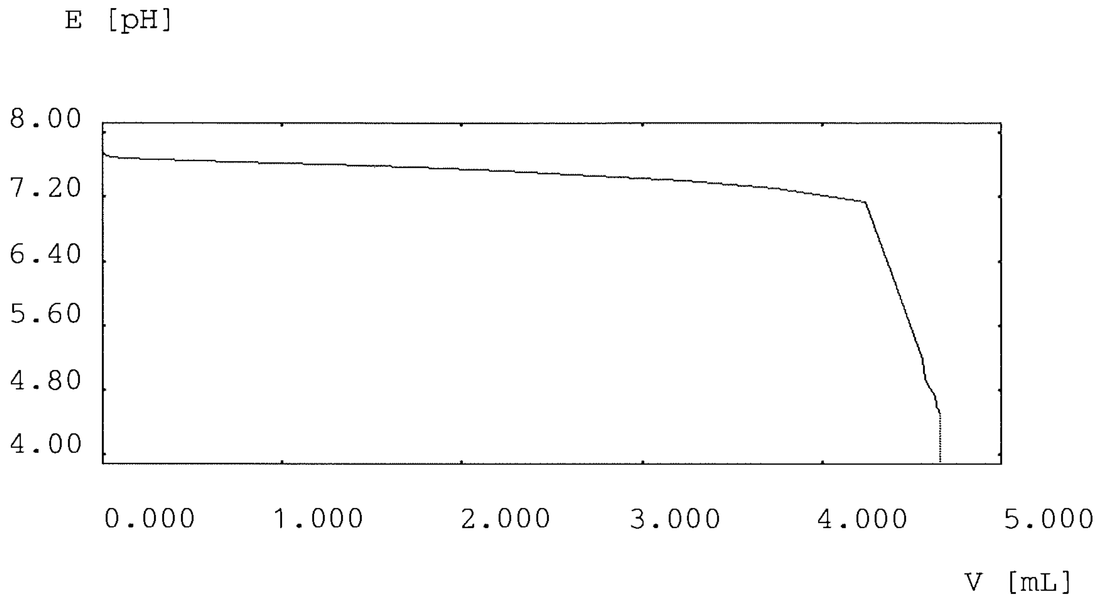
E - V curve



```
Method       11    Alkalinity                 10/04/2012    9:25
Measured     06/06/2023  13:13
User

Sample aborted: No. 3

Sample aborted: No. 6

Sample aborted: No. 8

Sample aborted: No. 15

RAW RESULTS

SAMPLE
  No.                          20
  Identification
  Titration stand              Stand 1
  Volume                       50.0 mL
  Correction factor f          1.0
  Mol.mass M                   100.00 g/mol
  Equivalent number z          1
  Temperature sensor           Manual

DISPENSE
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Volume                     VDISP = 0.0      mL
                             QDISP = 0.0      mmol

EP TITRATION
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Sensor             DG111
  Consumption        EP        VEQ1  =  5.2813  mL
                                 Q1  =  0.10563 mmol
                              EPOT1  =  8.300   pH
  Excess                       VEX   =  0.0167  mL
                               QEX   = -0.00033 mmol

CALCULATION
  Result                        R1   =  10.28          Initial pH

CALCULATION
  Result                        R2   =  105.960 mg/L   ALKP

DISPENSE
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Volume                     VDISP = 0.0      mL
                             QDISP = 0.0      mmol

EP TITRATION
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Sensor             DG111
```

Consumption	EP	VEQ1	=	4.6594	mL
		Q1	=	0.09319	mmol
		EPOT1	=	4.500	pH
Excess		VEX	=	0.0006	mL
		QEX	=	0.00001	mmol
CALCULATION					
Result		R3	=	199.15	mg/L ALKT

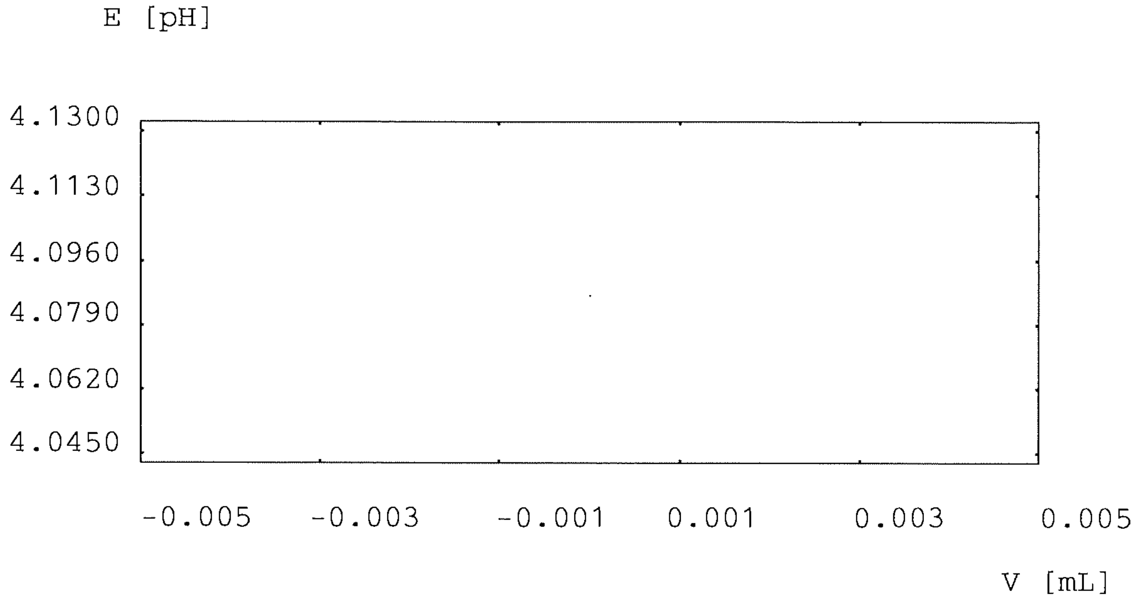
E - V curve



```
Method        11    Alkalinity              10/04/2012   9:25
Measured      06/06/2023  13:18
User

Sample aborted: No. 3

Sample aborted: No. 6

Sample aborted: No. 8

Sample aborted: No. 15

RAW RESULTS

SAMPLE
  No.                          21
  Identification
  Titration stand              Stand 1
  Volume                       50.0 mL
  Correction factor f          1.0
  Mol.mass M                   100.00 g/mol
  Equivalent number z          1
  Temperature sensor           Manual

DISPENSE
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Volume                     VDISP = 0.0      mL
                             QDISP = 0.0      mmol

EP TITRATION
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Sensor             DG111
  Excess                      VEX   =  0.0000  mL
                              QEX   =  0.00000 mmol

CALCULATION
  Result                       R1   =  3.98            Initial pH

CALCULATION
  Result                       R2   =  0.0000  mg/L    ALKP

DISPENSE
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Volume                     VDISP = 0.0      mL
                             QDISP = 0.0      mmol

EP TITRATION
  Titrant            H2SO4  0.02 mol/L  t = 1.0
  Drive              No. 2       10 mL
  Sensor             DG111
  Excess                      VEX   =  0.0000  mL
                              QEX   =  0.00000 mmol
```

CALCULATION					
Result	R3	=	0.00	mg/L	ALKT

E - V curve



EPA 9060A

Form I

Sample Results

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-EB-060123
Collect Date:	06/01/23 1214	LAB Sample ID:	22306021001
Matrix:	Water	Instrument ID:	TOC6
% Solids:	NA	Analyst:	CCL
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/06/23 1336
Prep Batch:	NA	Analytical Batch:	767077
Prep Method:	NA	Analytical Method:	EPA 9060A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	1.00	mg/L	U	0.300	1.00	2.00

FORM I - GENCHEM

Sample

Sample Name: 22306021001
Sample ID: 10210,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

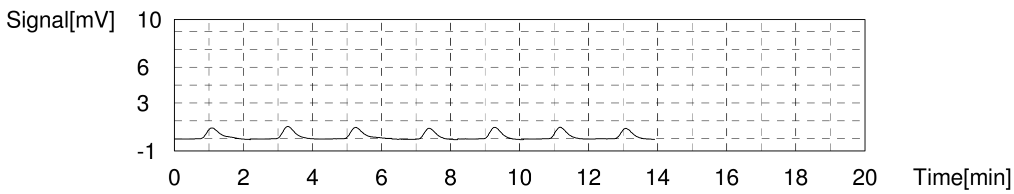
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:0.02803mg/L TC:0.5077mg/L IC:0.4796mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	2.631	0.6595mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 1:36:40 PM
2	2.487	0.6123mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 1:38:49 PM
3	2.452	0.6008mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 1:41:08 PM
4	2.151	0.5019mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 1:43:13 PM
5	2.246	0.5331mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 1:45:18 PM
6	2.258	0.5371mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 1:47:23 PM
7	2.019	0.4586mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 1:49:50 PM

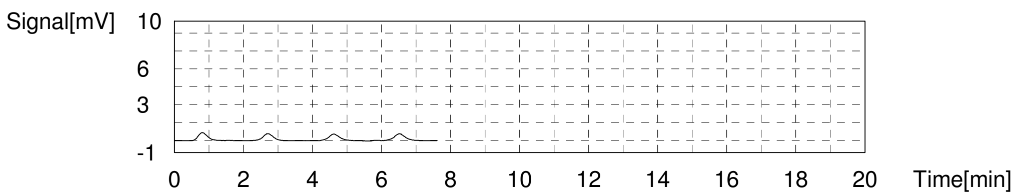
Mean Area 2.168
Mean Conc. 0.5077mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.255	0.4912mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 1:53:16 PM
2	1.177	0.4670mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 1:55:21 PM
3	1.191	0.4713mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 1:57:26 PM
4	1.248	0.4890mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 1:59:31 PM

Mean Area 1.218
Mean Conc. 0.4796mg/L



I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW02-060123
Collect Date:	06/01/23 1230	LAB Sample ID:	22306021002
Matrix:	Water	Instrument ID:	TOC6
% Solids:	NA	Analyst:	CCL
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/06/23 1412
Prep Batch:	NA	Analytical Batch:	767077
Prep Method:	NA	Analytical Method:	EPA 9060A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	5.95	mg/L		0.300	1.00	2.00

FORM I - GENCHEM

Sample

Sample Name:	22306021002
Sample ID:	10210,CCL,TOC6
Origin:	TOC4.met
Status	Completed
Chk. Result	

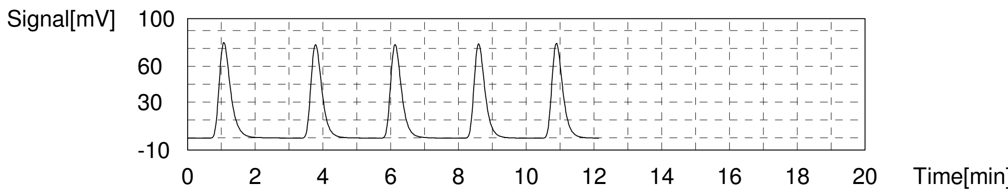
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:5.951mg/L TC:58.45mg/L IC:52.49mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	189.0	61.85mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 2:12:43 PM
2	178.7	58.47mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 2:15:15 PM
3	179.4	58.70mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 2:17:54 PM
4	177.0	57.91mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 2:20:23 PM
5	179.4	58.70mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 2:22:52 PM

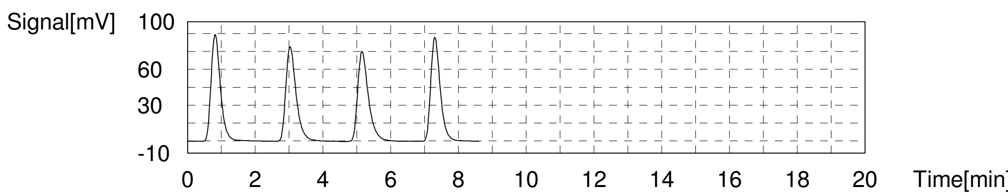
Mean Area 178.6
Mean Conc. 58.45mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	171.7	53.39mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 2:26:42 PM
2	167.6	52.11mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 2:29:00 PM
3	166.6	51.80mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 2:31:22 PM
4	169.4	52.67mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 2:33:40 PM

Mean Area 168.8
Mean Conc. 52.49mg/L



I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123
Collect Date:	06/01/23 0935	LAB Sample ID:	22306021003
Matrix:	Water	Instrument ID:	TOC6
% Solids:	NA	Analyst:	CCL
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/08/23 0053
Prep Batch:	NA	Analytical Batch:	767167
Prep Method:	NA	Analytical Method:	EPA 9060A

ANALYTE	RESULT	UNITS	Q	DL	LOD	LOQ
Total Organic Carbon	3.53	mg/L		0.300	1.00	2.00

FORM I - GENCHEM

Sample

Sample Name: 22306021003
Sample ID: 10212,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

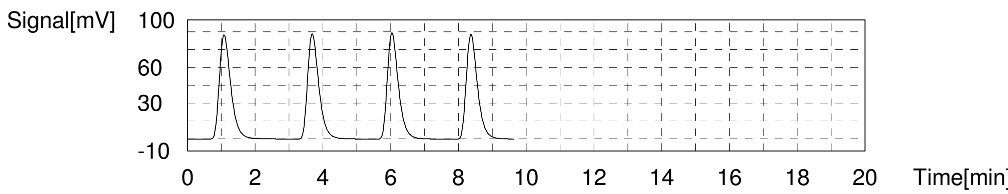
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:3.528mg/L TC:66.88mg/L IC:63.36mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	206.0	67.43mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 12:53:41 AM
2	203.7	66.68mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 12:56:13 AM
3	204.9	67.07mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 12:58:44 AM
4	202.7	66.35mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 1:01:16 AM

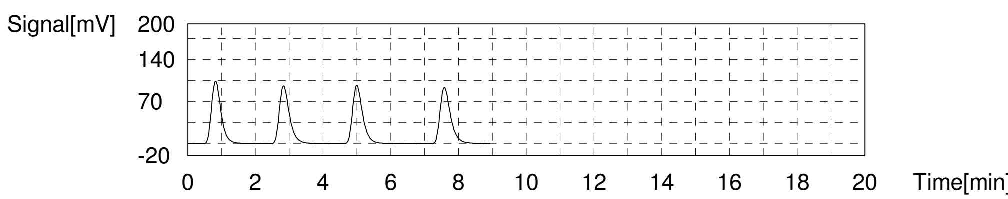
Mean Area 204.3
Mean Conc. 66.88mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	205.2	63.78mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 1:04:55 AM
2	202.5	62.94mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 1:07:16 AM
3	204.4	63.53mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 1:10:02 AM
4	203.2	63.16mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 1:12:23 AM

Mean Area 203.8
Mean Conc. 63.36mg/L



I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW05-060123
Collect Date:	06/01/23 0910	LAB Sample ID:	22306021004
Matrix:	Water	Instrument ID:	TOC6
% Solids:	NA	Analyst:	CCL
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/06/23 1440
Prep Batch:	NA	Analytical Batch:	767077
Prep Method:	NA	Analytical Method:	EPA 9060A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	2.70	mg/L		0.300	1.00	2.00

FORM I - GENCHEM

Sample

Sample Name: 22306021004
Sample ID: 10210,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

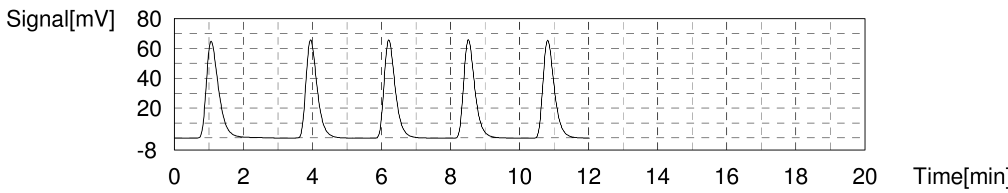
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:2.695mg/L TC:48.72mg/L IC:46.02mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	160.7	52.56mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 2:40:35 PM
2	149.2	48.78mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 2:43:02 PM
3	148.2	48.46mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 2:45:31 PM
4	148.7	48.62mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 2:48:00 PM
5	149.9	49.01mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 2:50:27 PM

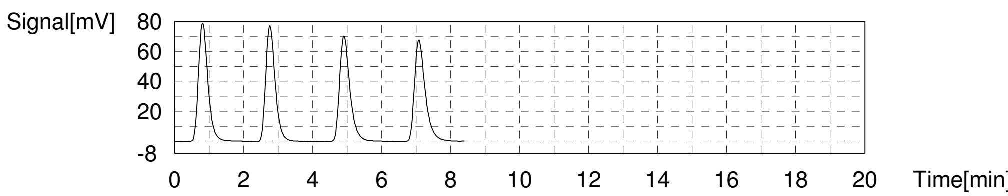
Mean Area 149.0
Mean Conc. 48.72mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	150.0	46.65mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 2:54:02 PM
2	148.1	46.06mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 2:56:21 PM
3	147.0	45.72mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 2:58:42 PM
4	146.8	45.66mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 3:01:02 PM

Mean Area 148.0
Mean Conc. 46.02mg/L



I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW06-060123
Collect Date:	06/01/23 1100	LAB Sample ID:	22306021005
Matrix:	Water	Instrument ID:	TOC6
% Solids:	NA	Analyst:	CCL
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/06/23 1507
Prep Batch:	NA	Analytical Batch:	767077
Prep Method:	NA	Analytical Method:	EPA 9060A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	10.7	mg/L		0.300	1.00	2.00

FORM I - GENCHEM

Sample

Sample Name: 22306021005
Sample ID: 10210,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

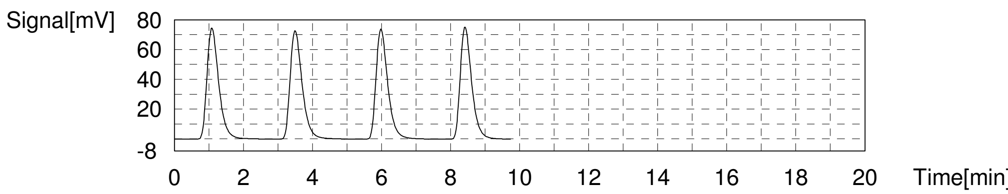
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:10.65mg/L TC:57.16mg/L IC:46.51mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	176.1	57.62mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 3:07:30 PM
2	175.4	57.39mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 3:10:10 PM
3	174.1	56.96mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 3:12:47 PM
4	173.3	56.70mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 3:15:21 PM

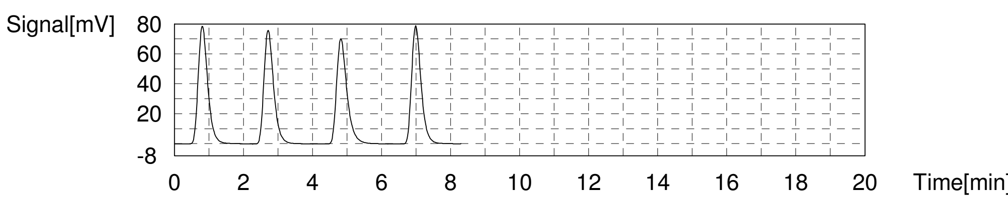
Mean Area 174.7
Mean Conc. 57.16mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	150.7	46.87mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 3:18:54 PM
2	148.4	46.16mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 3:21:11 PM
3	147.9	46.00mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 3:23:33 PM
4	151.2	47.02mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 3:25:51 PM

Mean Area 149.6
Mean Conc. 46.51mg/L



I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123MS
Collect Date:	06/01/23 0955	LAB Sample ID:	22306021006
Matrix:	Water	Instrument ID:	TOC6
% Solids:	NA	Analyst:	CCL
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/08/23 0119
Prep Batch:	NA	Analytical Batch:	767167
Prep Method:	NA	Analytical Method:	EPA 9060A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	53.9	mg/L		0.300	1.00	2.00

FORM I - GENCHEM

Sample

Sample Name: 22306021006
Sample ID: 10212,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

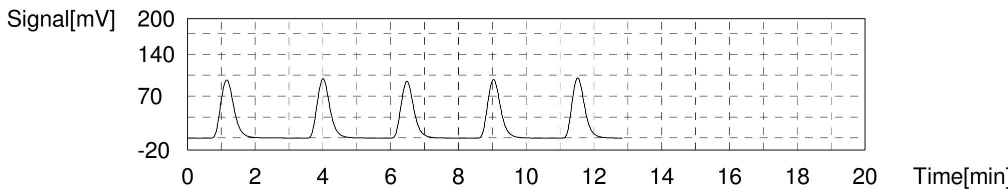
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:53.85mg/L TC:85.66mg/L IC:31.82mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	264.5	86.64mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 1:19:16 AM
2	259.7	85.07mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 1:21:56 AM
3	249.7	81.78mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 1:24:40 AM
4	260.2	85.23mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 1:27:21 AM
5	261.7	85.72mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 1:30:00 AM

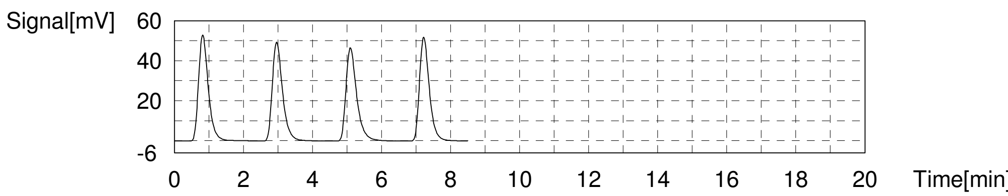
Mean Area 261.5
Mean Conc. 85.66mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	103.7	32.28mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 1:33:46 AM
2	101.9	31.72mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 1:36:05 AM
3	100.6	31.32mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 1:38:24 AM
4	102.6	31.94mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 1:40:40 AM

Mean Area 102.2
Mean Conc. 31.82mg/L



I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW-806-060123
Collect Date:	06/01/23 1100	LAB Sample ID:	22306021008
Matrix:	Water	Instrument ID:	TOC6
% Solids:	NA	Analyst:	CCL
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/06/23 1651
Prep Batch:	NA	Analytical Batch:	767077
Prep Method:	NA	Analytical Method:	EPA 9060A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	8.93	mg/L		0.300	1.00	2.00

FORM I - GENCHEM

Sample

Sample Name: 22306021008
Sample ID: 10210,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

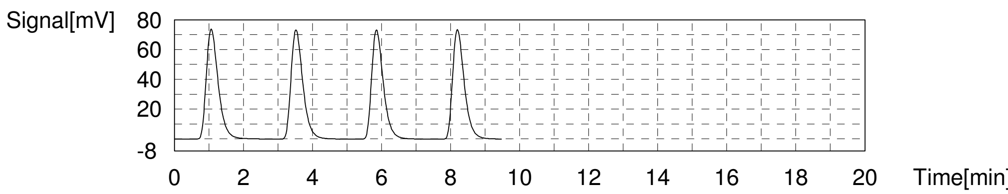
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:8.928mg/L TC:56.99mg/L IC:48.06mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	177.0	57.91mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 4:51:19 PM
2	174.4	57.06mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 4:53:50 PM
3	172.1	56.30mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 4:56:22 PM
4	173.3	56.70mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 4:58:53 PM

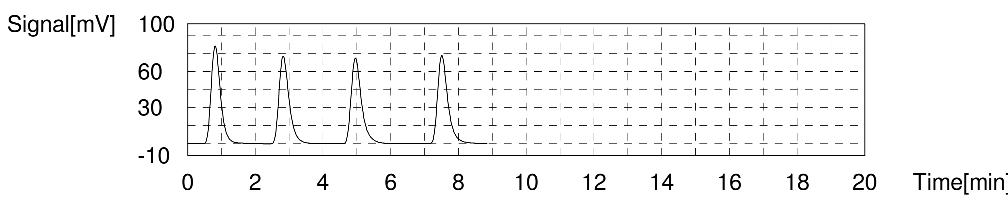
Mean Area 174.2
Mean Conc. 56.99mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	156.6	48.70mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 5:02:32 PM
2	152.9	47.55mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 5:04:51 PM
3	154.5	48.05mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 5:07:35 PM
4	154.2	47.96mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 5:09:55 PM

Mean Area 154.6
Mean Conc. 48.06mg/L



EPA 9060A

Form II

Calibration Verifications

II - CONTINUING CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	TOC6
Analysis Date:	06/06/23 0358	Lab File ID:	10210
Analytical Method:	EPA 9060A	Analytical Batch:	767077

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*% REC*	*LCL*	*UCL*	*Q*
Total Organic Carbon	mg/L	50.0	50.8	102	90	110	

FORM II - GENCHEM

Sample

Sample Name:	1800
Sample ID:	10210,CCL,TOC6
Origin:	TOC4.met
Status	Completed
Chk. Result	

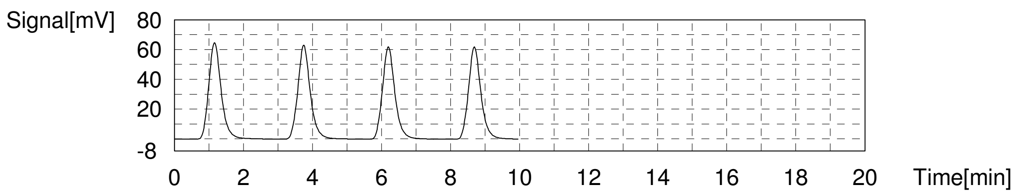
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:50.78mg/L TC:51.16mg/L IC:0.3782mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	160.7	52.56mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 3:58:13 AM
2	156.8	51.28mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 4:00:52 AM
3	154.3	50.46mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 4:03:32 AM
4	153.9	50.33mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 4:06:10 AM

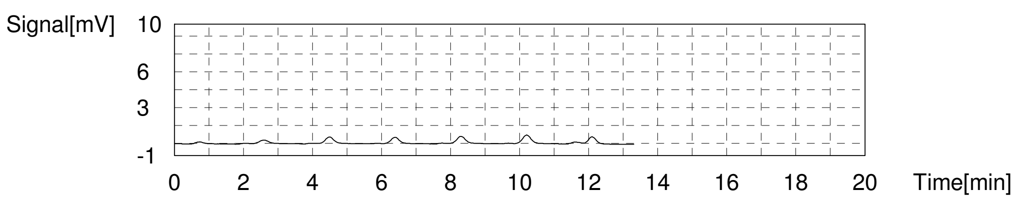
Mean Area 156.4
Mean Conc. 51.16mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.2610	0.1827mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 4:09:43 AM
2	0.5245	0.2645mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 4:11:48 AM
3	0.8910	0.3782mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 4:13:53 AM
4	0.8748	0.3732mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 4:15:58 AM
5	1.033	0.4223mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 4:18:03 AM
6	1.217	0.4794mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 4:20:08 AM
7	0.7643	0.3389mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 4:22:37 AM

Mean Area 0.8908
Mean Conc. 0.3782mg/L



II - CONTINUING CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	TOC6
Analysis Date:	06/06/23 1717	Lab File ID:	10210
Analytical Method:	EPA 9060A	Analytical Batch:	767077

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*% REC*	*LCL*	*UCL*	*Q*
Total Organic Carbon	mg/L	50.0	50.5	101	90	110	

FORM II - GENCHEM

Sample

Sample Name: 1800
Sample ID: 10210,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

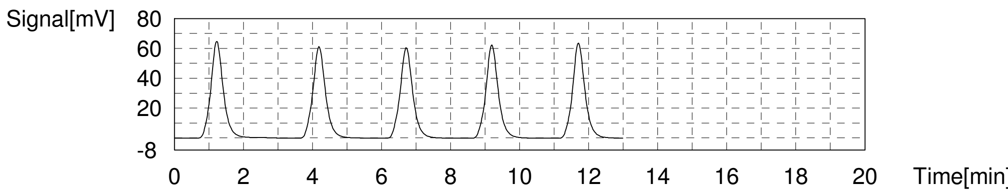
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:50.53mg/L TC:50.99mg/L IC:0.4612mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	162.5	53.15mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 5:17:08 PM
2	155.5	50.85mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 5:19:50 PM
3	153.6	50.23mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 5:22:29 PM
4	156.4	51.15mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 5:25:11 PM
5	158.2	51.74mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 5:27:53 PM

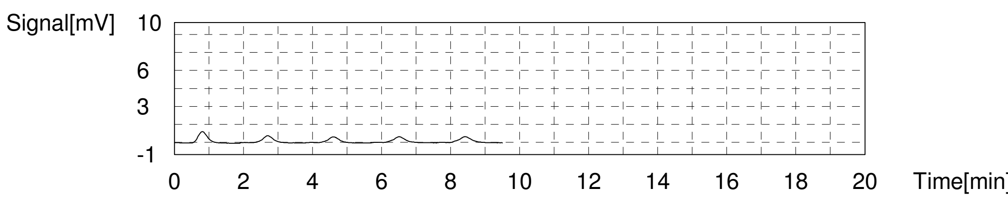
Mean Area 155.9
Mean Conc. 50.99mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.752	0.6454mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 5:31:25 PM
2	1.239	0.4862mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 5:33:30 PM
3	1.083	0.4378mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 5:35:35 PM
4	1.121	0.4496mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 5:37:40 PM
5	1.190	0.4710mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 5:39:45 PM

Mean Area 1.158
Mean Conc. 0.4612mg/L



II - CONTINUING CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	TOC6
Analysis Date:	06/06/23 2313	Lab File ID:	10212
Analytical Method:	EPA 9060A	Analytical Batch:	767167

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*% REC*	*LCL*	*UCL*	*Q*
Total Organic Carbon	mg/L	50.0	51.2	102	90	110	

FORM II - GENCHEM

Sample

Sample Name: 1800
Sample ID: 10212,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

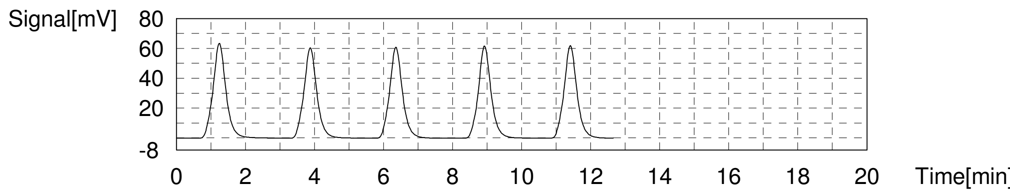
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:51.17mg/L TC:51.64mg/L IC:0.4706mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	164.3	53.74mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:13:42 PM
2	156.9	51.31mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:16:22 PM
3	156.7	51.25mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:19:06 PM
4	157.1	51.38mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:21:46 PM
5	160.9	52.63mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:24:27 PM

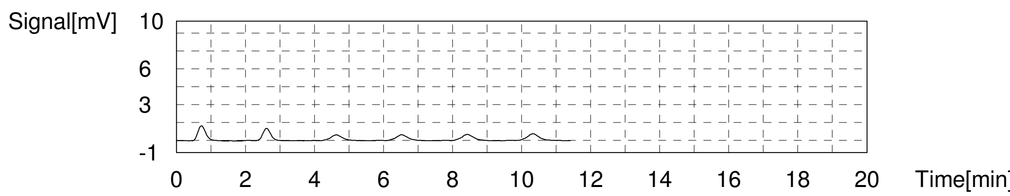
Mean Area 157.9
Mean Conc. 51.64mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.991	0.7196mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 11:27:59 PM
2	1.591	0.5955mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 11:30:04 PM
3	1.107	0.4453mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 11:32:09 PM
4	1.130	0.4524mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 11:34:14 PM
5	1.202	0.4748mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 11:36:19 PM
6	1.316	0.5101mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 11:38:24 PM

Mean Area 1.189
Mean Conc. 0.4706mg/L



II - CONTINUING CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	TOC6
Analysis Date:	06/07/23 1737	Lab File ID:	10212
Analytical Method:	EPA 9060A	Analytical Batch:	767167

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*% REC*	*LCL*	*UCL*	*Q*
Total Organic Carbon	mg/L	50.0	49.0	98	90	110	

FORM II - GENCHEM

Sample

Sample Name: 1800
Sample ID: 10212,CCL,TOC6
Origin: TOC3.met
Status Completed
Chk. Result

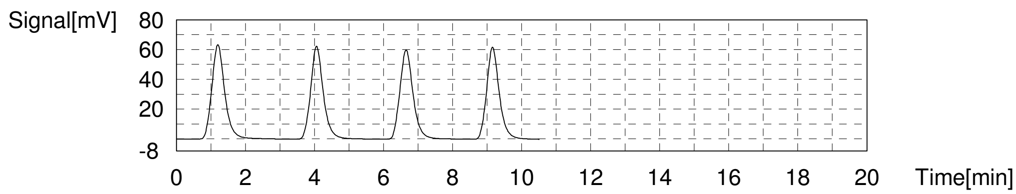
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:48.98mg/L TC:51.36mg/L IC:2.374mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	164.4	53.77mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 5:37:30 PM
2	157.9	51.64mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 5:40:17 PM
3	157.0	51.34mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 5:42:58 PM
4	156.2	51.08mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 5:45:42 PM

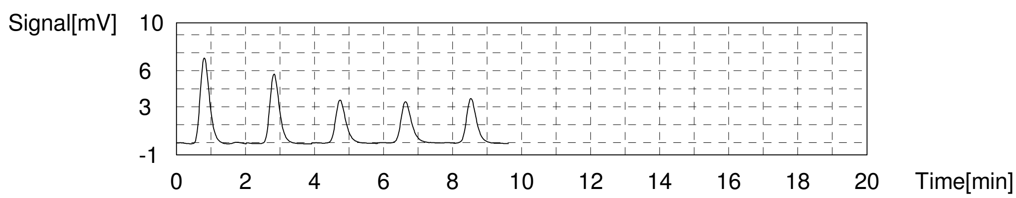
Mean Area 157.0
Mean Conc. 51.36mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	13.72	4.360mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 5:48:19 PM
2	11.08	3.540mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 5:50:24 PM
3	7.355	2.384mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 5:52:29 PM
4	7.178	2.329mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 5:54:34 PM
5	7.437	2.410mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 5:56:39 PM

Mean Area 7.323
Mean Conc. 2.374mg/L



II - CONTINUING CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	TOC6
Analysis Date:	06/07/23 2312	Lab File ID:	10212
Analytical Method:	EPA 9060A	Analytical Batch:	767167

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*% REC*	*LCL*	*UCL*	*Q*
Total Organic Carbon	mg/L	50.0	50.9	102	90	110	

FORM II - GENCHEM

Sample

Sample Name: 1800
Sample ID: 10212,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

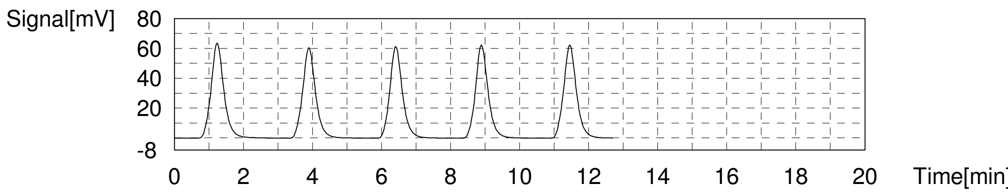
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:50.92mg/L TC:51.40mg/L IC:0.4864mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	163.1	53.35mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 11:12:29 PM
2	156.3	51.12mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 11:15:12 PM
3	154.6	50.56mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 11:17:51 PM
4	157.7	51.57mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 11:20:36 PM
5	160.1	52.36mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 11:23:15 PM

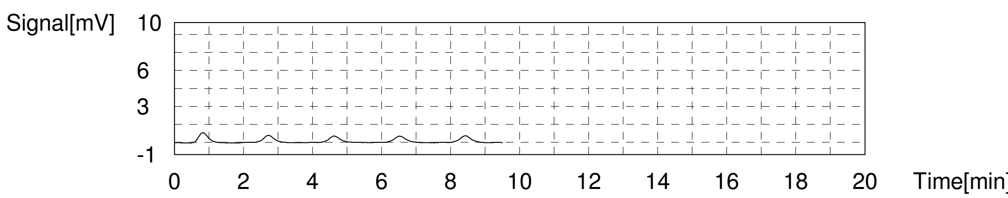
Mean Area 157.2
Mean Conc. 51.40mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.650	0.6138mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 11:26:47 PM
2	1.263	0.4937mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 11:28:52 PM
3	1.170	0.4648mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 11:30:57 PM
4	1.253	0.4906mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 11:33:02 PM
5	1.272	0.4965mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 11:35:07 PM

Mean Area 1.240
Mean Conc. 0.4864mg/L



II - CONTINUING CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	TOC6
Analysis Date:	06/08/23 0213	Lab File ID:	10212
Analytical Method:	EPA 9060A	Analytical Batch:	767167

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*% REC*	*LCL*	*UCL*	*Q*
Total Organic Carbon	mg/L	50.0	51.3	102	90	110	

FORM II - GENCHEM

Sample

Sample Name: 1800
Sample ID: 10212,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

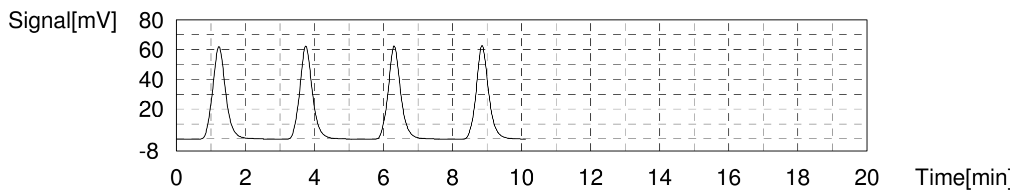
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:51.25mg/L TC:51.75mg/L IC:0.4962mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	161.4	52.79mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 2:13:44 AM
2	157.9	51.64mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 2:16:28 AM
3	156.9	51.31mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 2:19:11 AM
4	156.7	51.25mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 2:21:52 AM

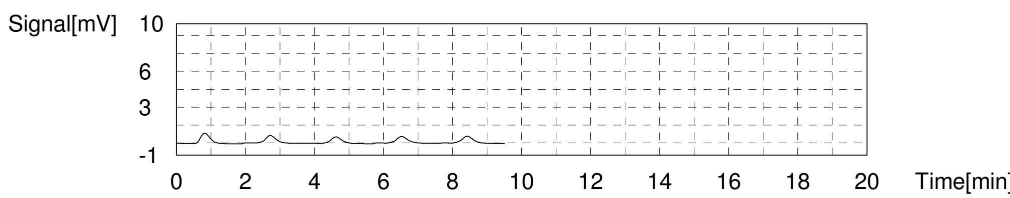
Mean Area 158.2
Mean Conc. 51.75mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.687	0.6253mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 2:25:25 AM
2	1.334	0.5157mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 2:27:30 AM
3	1.207	0.4763mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 2:29:35 AM
4	1.232	0.4841mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 2:31:40 AM
5	1.312	0.5089mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 2:33:45 AM

Mean Area 1.271
Mean Conc. 0.4962mg/L



EPA 9060A

Form III

Blanks

III - CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	CCB for HBN 767077
Matrix:	Water	Instrument ID:	TOC6
Analysis Date:	06/06/23 0459	Lab File ID:	10210
Analytical Method:	EPA 9060A	Analytical Batch:	767077

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	1.00	mg/L	U	0.300	1.00	2.00

FORM III - GENCHEM

Sample

Sample Name:	1900
Sample ID:	10210,CCL,TOC6
Origin:	TOC4.met
Status	Completed
Chk. Result	

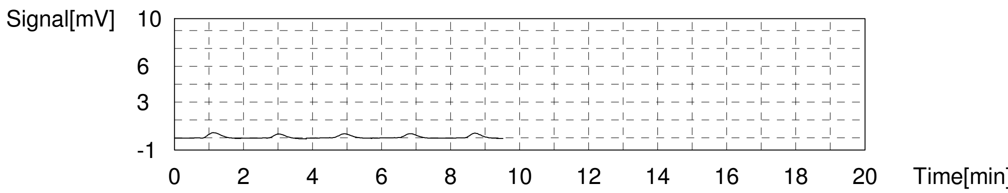
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:-0.1647mg/L TC:0.07622mg/L IC:0.2410mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.188	0.1857mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 4:59:16 AM
2	0.8565	0.07689mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 5:01:21 AM
3	0.8130	0.06261mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 5:03:26 AM
4	0.8268	0.06714mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 5:05:31 AM
5	0.9215	0.09824mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 5:07:36 AM

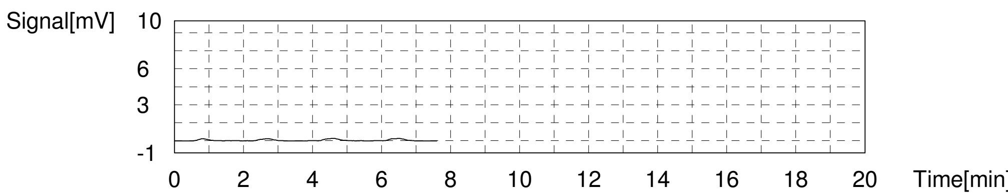
Mean Area 0.8545
Mean Conc. 0.07622mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.3205	0.2012mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 5:11:08 AM
2	0.4475	0.2406mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 5:13:13 AM
3	0.4798	0.2506mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 5:15:18 AM
4	0.5467	0.2714mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 5:17:23 AM

Mean Area 0.4486
Mean Conc. 0.2410mg/L



III - METHOD BLANK

Report No:	223060210	Blank ID:	MB2489958
Matrix:	Water	Instrument ID:	TOC6
Analysis Date:	06/06/23 1214	Lab File ID:	10210
Analytical Method:	EPA 9060A	Analytical Batch:	767077

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	1.00	mg/L	U	0.300	1.00	2.00

FORM III - GENCHEM

Sample

Sample Name: 2489958 MB
Sample ID: 10210,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

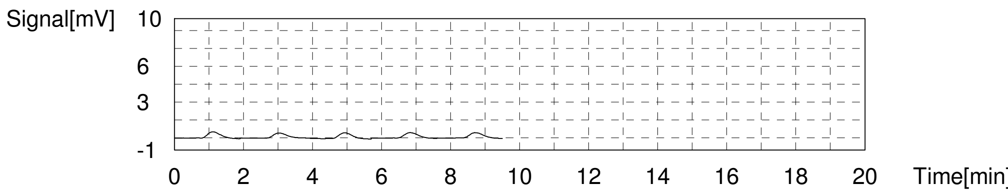
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:-0.01317mg/L TC:0.2171mg/L IC:0.2303mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.362	0.2429mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 12:14:13 PM
2	1.091	0.1539mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 12:16:18 PM
3	1.268	0.2120mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 12:18:23 PM
4	1.254	0.2074mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 12:20:28 PM
5	1.250	0.2061mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 12:22:33 PM

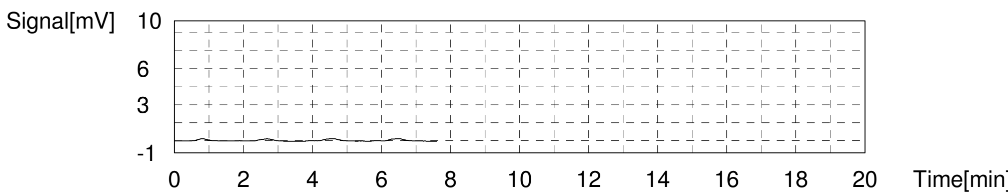
Mean Area 1.284
Mean Conc. 0.2171mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.2968	0.1938mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 12:26:05 PM
2	0.4273	0.2343mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 12:28:10 PM
3	0.4561	0.2433mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 12:30:15 PM
4	0.4765	0.2496mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 12:32:20 PM

Mean Area 0.4142
Mean Conc. 0.2303mg/L



III - CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	CCB for HBN 767077
Matrix:	Water	Instrument ID:	TOC6
Analysis Date:	06/06/23 1811	Lab File ID:	10210
Analytical Method:	EPA 9060A	Analytical Batch:	767077

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	1.00	mg/L	U	0.300	1.00	2.00

FORM III - GENCHEM

Sample

Sample Name:	1900
Sample ID:	10210,CCL,TOC6
Origin:	TOC4.met
Status	Completed
Chk. Result	

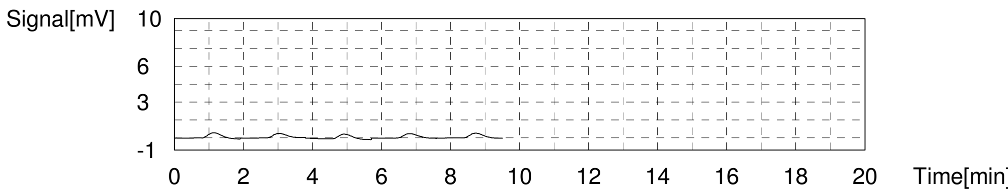
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:-0.1681mg/L TC:0.1312mg/L IC:0.2992mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.220	0.1962mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 6:11:34 PM
2	0.9444	0.1058mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 6:13:39 PM
3	1.072	0.1477mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 6:15:44 PM
4	1.028	0.1332mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 6:17:49 PM
5	1.043	0.1381mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 6:19:54 PM

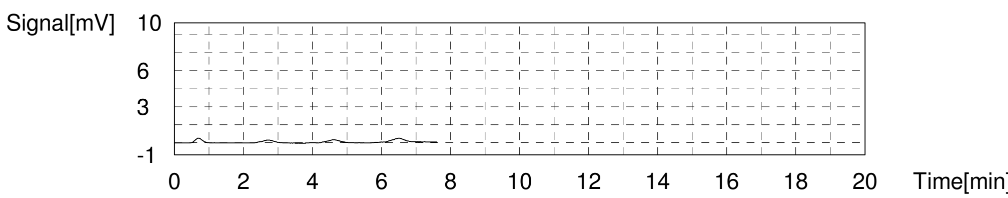
Mean Area 1.022
Mean Conc. 0.1312mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.5452	0.2709mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 6:23:26 PM
2	0.5828	0.2826mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 6:25:31 PM
3	0.6564	0.3054mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 6:27:36 PM
4	0.7613	0.3380mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 6:29:41 PM

Mean Area 0.6364
Mean Conc. 0.2992mg/L



III - CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	CCB for HBN 767167
Matrix:	Water	Instrument ID:	TOC6
Analysis Date:	06/07/23 0015	Lab File ID:	10212
Analytical Method:	EPA 9060A	Analytical Batch:	767167

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	1.0	mg/L	U	0.30	1.0	2.0

FORM III - GENCHEM

Sample

Sample Name:	1900
Sample ID:	10212,CCL,TOC6
Origin:	TOC4.met
Status	Completed
Chk. Result	

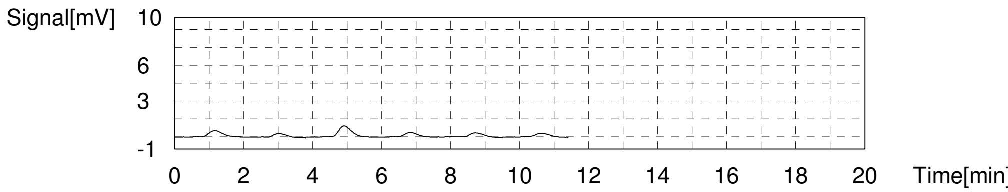
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:-0.1713mg/L TC:0.07024mg/L IC:0.2415mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.350	0.2389mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 12:15:49 AM
2	0.7985	0.05785mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 12:17:54 AM
3	2.264	0.5390mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 12:19:59 AM
4	0.8814	0.08507mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 12:22:04 AM
5	0.8931	0.08891mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 12:24:09 AM
6	0.7719	0.04912mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 12:26:14 AM

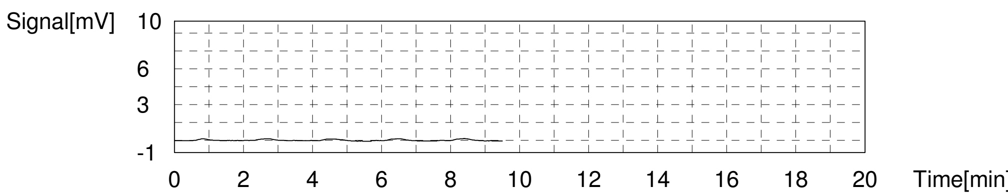
Mean Area 0.8362
Mean Conc. 0.07024mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.2430	0.1771mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 12:29:40 AM
2	0.3706	0.2167mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 12:31:45 AM
3	0.5076	0.2593mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 12:33:50 AM
4	0.4424	0.2390mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 12:35:55 AM
5	0.4809	0.2510mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 12:38:00 AM

Mean Area 0.4504
Mean Conc. 0.2415mg/L



III - METHOD BLANK

Report No:	223060210	Blank ID:	MB2490524
Matrix:	Water	Instrument ID:	TOC6
Analysis Date:	06/07/23 1325	Lab File ID:	10212
Analytical Method:	EPA 9060A	Analytical Batch:	767167

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	1.00	mg/L	U	0.300	1.00	2.00

FORM III - GENCHEM

Sample

Sample Name: 2490524 mb
Sample ID: 10212,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

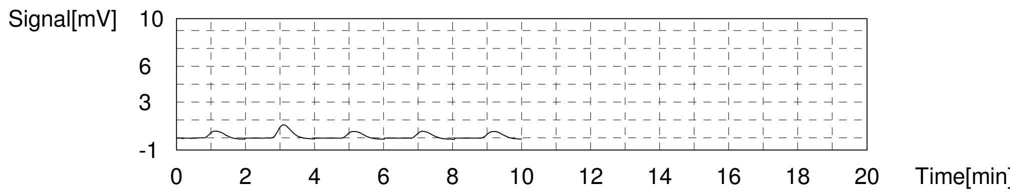
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:-0.03612mg/L TC:0.3798mg/L IC:0.4159mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.753	0.3713mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 1:25:45 PM
2	3.005	0.7823mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 1:27:57 PM
3	1.828	0.3959mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 1:30:09 PM
4	1.768	0.3762mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 1:32:22 PM
5	1.767	0.3758mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 1:34:30 PM

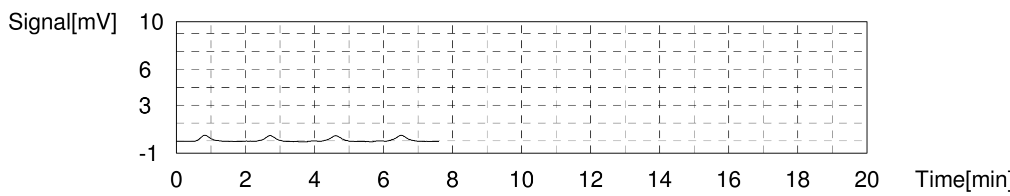
Mean Area 1.779
Mean Conc. 0.3798mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.9500	0.3966mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 1:38:02 PM
2	0.9985	0.4116mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 1:40:07 PM
3	1.001	0.4124mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 1:42:12 PM
4	1.100	0.4431mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 1:44:17 PM

Mean Area 1.012
Mean Conc. 0.4159mg/L



III - CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	CCB for HBN 767167
Matrix:	Water	Instrument ID:	TOC6
Analysis Date:	06/07/23 1826	Lab File ID:	10212
Analytical Method:	EPA 9060A	Analytical Batch:	767167

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	1.0	mg/L	U	0.30	1.0	2.0

FORM III - GENCHEM

Sample

Sample Name: 1900
Sample ID: 10212,CCL,TOC6
Origin: TOC3.met
Status Completed
Chk. Result

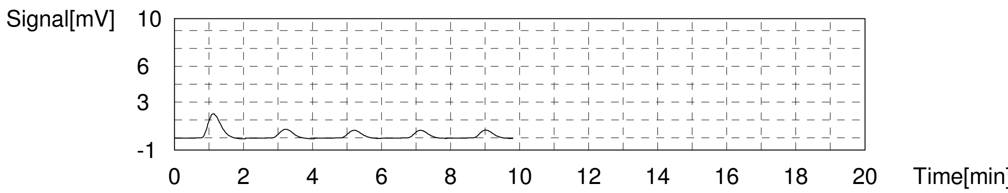
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:-0.07624mg/L TC:0.3686mg/L IC:0.4449mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	5.330	1.546mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 6:26:41 PM
2	2.073	0.4763mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 6:28:52 PM
3	1.744	0.3683mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 6:30:59 PM
4	1.733	0.3647mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 6:33:04 PM
5	1.758	0.3729mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 6:35:10 PM

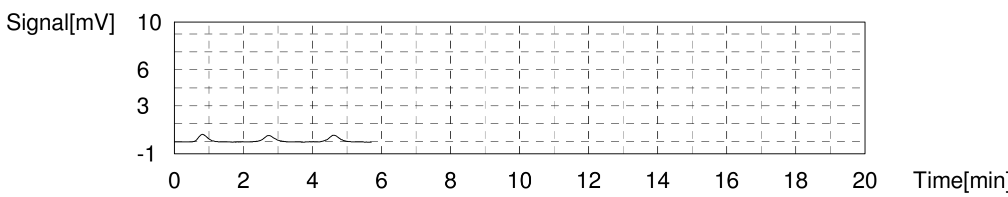
Mean Area 1.745
Mean Conc. 0.3686mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.129	0.4521mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 6:37:39 PM
2	1.051	0.4279mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 6:39:44 PM
3	1.137	0.4546mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 6:41:49 PM

Mean Area 1.106
Mean Conc. 0.4449mg/L



III - CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	CCB for HBN 767167
Matrix:	Water	Instrument ID:	TOC6
Analysis Date:	06/08/23 0010	Lab File ID:	10212
Analytical Method:	EPA 9060A	Analytical Batch:	767167

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	1.00	mg/L	U	0.300	1.00	2.00

FORM III - GENCHEM

Sample

Sample Name: 1900
Sample ID: 10212,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

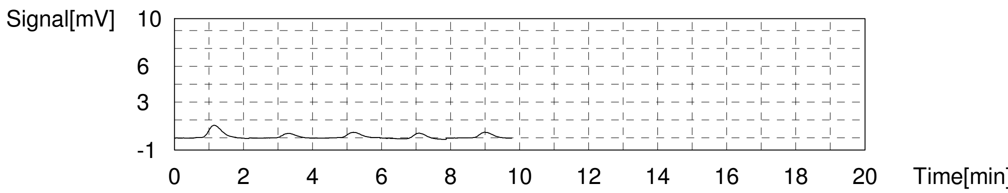
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:-0.1188mg/L TC:0.1710mg/L IC:0.2898mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	2.754	0.6999mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 12:10:28 AM
2	1.013	0.1283mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 12:12:33 AM
3	1.147	0.1723mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 12:14:38 AM
4	1.200	0.1897mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 12:16:44 AM
5	1.213	0.1939mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 12:18:49 AM

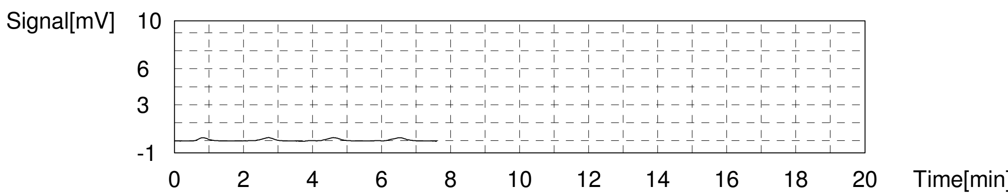
Mean Area 1.143
Mean Conc. 0.1710mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.5015	0.2574mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 12:22:21 AM
2	0.5942	0.2861mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 12:24:26 AM
3	0.6379	0.2997mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 12:26:31 AM
4	0.6905	0.3160mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 12:28:36 AM

Mean Area 0.6060
Mean Conc. 0.2898mg/L



III - CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	CCB for HBN 767167
Matrix:	Water	Instrument ID:	TOC6
Analysis Date:	06/08/23 0305	Lab File ID:	10212
Analytical Method:	EPA 9060A	Analytical Batch:	767167

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Total Organic Carbon	1.00	mg/L	U	0.300	1.00	2.00

FORM III - GENCHEM

Sample

Sample Name: 1900
Sample ID: 10212,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

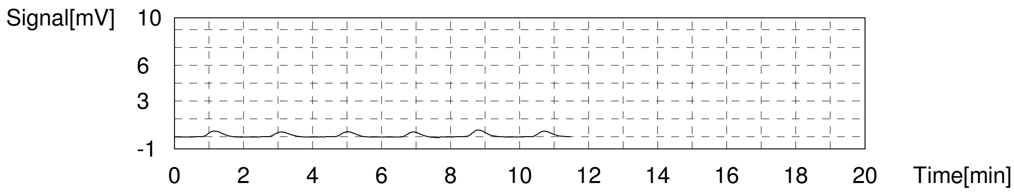
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:-0.1283mg/L TC:0.1762mg/L IC:0.3046mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.372	0.2462mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 3:05:38 AM
2	1.145	0.1716mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 3:07:45 AM
3	1.135	0.1683mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 3:09:50 AM
4	1.132	0.1674mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 3:11:55 AM
5	1.466	0.2770mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 3:14:00 AM
6	1.224	0.1976mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 3:16:05 AM

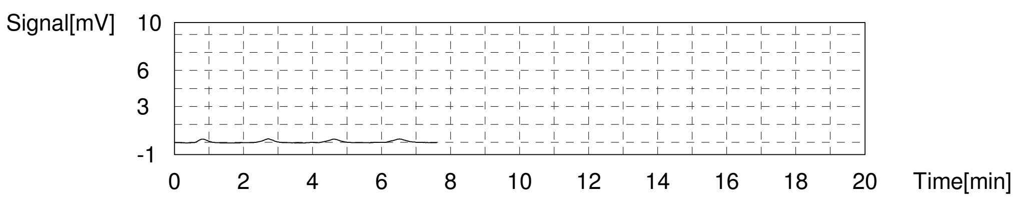
Mean Area 1.159
Mean Conc. 0.1762mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.5615	0.2760mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 3:19:31 AM
2	0.6406	0.3005mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 3:21:36 AM
3	0.6952	0.3175mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 3:23:41 AM
4	0.7170	0.3242mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 3:25:46 AM

Mean Area 0.6536
Mean Conc. 0.3046mg/L



EPA 9060A

Form V

Matrix Spikes

V - MS/MSD RECOVERY

Report No:	223060210	Parent Sample ID:	OT07-MW04-060123
Prep Date:	NA	Parent LAB ID:	22306021003
Prep Batch:	NA	Analytical Batch:	767167
Prep Method:	NA	Analytical Method:	EPA 9060A

LAB QC ID:	**22306021006 MS**	Instrument ID:	TOC6
Analyst:	CCL	Lab File ID:	10212
Analysis Date:	06/08/23 0119	Dilution:	1

ANALYTE	*UNITS*	*SPIKE ADDED*	*SAMPLE RESULT*	*MS RESULT*	*MS % REC*	*#*	*QC LIMITS*
Total Organic Carbon	mg/L	50	3.53	53.9	101		80 - 120

FORM V - GENCHEM

Sample

Sample Name: 22306021006
Sample ID: 10212,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

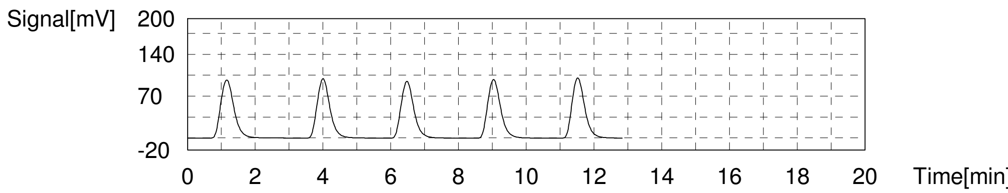
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:53.85mg/L TC:85.66mg/L IC:31.82mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	264.5	86.64mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 1:19:16 AM
2	259.7	85.07mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 1:21:56 AM
3	249.7	81.78mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 1:24:40 AM
4	260.2	85.23mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 1:27:21 AM
5	261.7	85.72mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/8/2023 1:30:00 AM

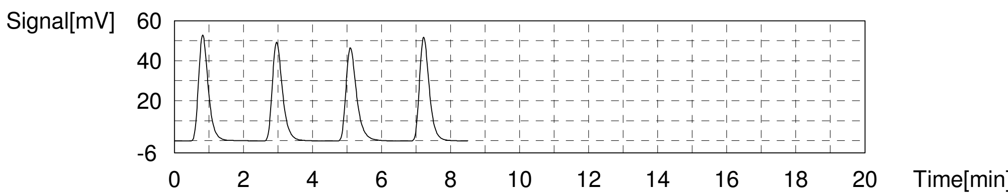
Mean Area 261.5
Mean Conc. 85.66mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	103.7	32.28mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 1:33:46 AM
2	101.9	31.72mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 1:36:05 AM
3	100.6	31.32mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 1:38:24 AM
4	102.6	31.94mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/8/2023 1:40:40 AM

Mean Area 102.2
Mean Conc. 31.82mg/L



EPA 9060A

Form VII

Lab Control Spikes

VII - LABORATORY CONTROL SPIKE

Report No:	223060210	LAB ID:	LCS2489959
Matrix:	Water	Instrument ID:	TOC6
Analyst:	CCL	Lab File ID:	10210
Prep Date:	NA	Analysis Date:	06/06/23 1124
Prep Batch:	NA	Analytical Batch:	767077
Prep Method:	NA	Analytical Method:	EPA 9060A

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*%REC*	*Q*	*% REC LIMITS*
Total Organic Carbon	mg/L	50.0	51.1	102		90 - 110

FORM VII - GENCHEM

Sample

Sample Name: 2489959 LCS
Sample ID: 10210,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

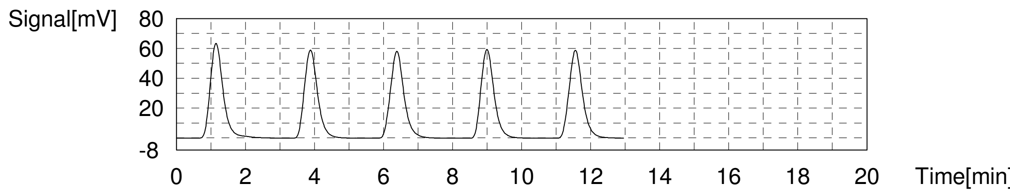
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:51.07mg/L TC:51.36mg/L IC:0.2904mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	164.4	53.77mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:24:03 AM
2	155.8	50.95mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:26:42 AM
3	157.9	51.64mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:29:32 AM
4	155.8	50.95mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:32:14 AM
5	158.7	51.90mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:35:09 AM

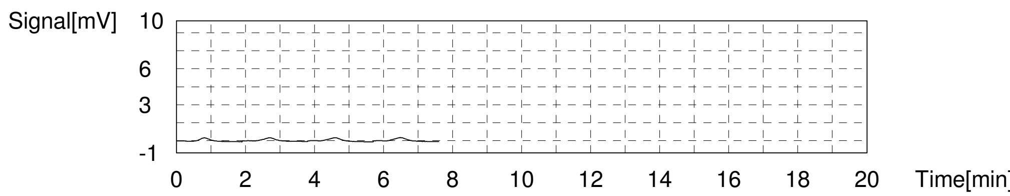
Mean Area 157.1
Mean Conc. 51.36mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.5244	0.2645mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 11:38:41 AM
2	0.6051	0.2895mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 11:40:46 AM
3	0.6410	0.3007mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 11:42:51 AM
4	0.6617	0.3071mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 11:44:56 AM

Mean Area 0.6081
Mean Conc. 0.2904mg/L



VII - LABORATORY CONTROL SPIKE DUPLICATE

Report No:	223060210	LAB ID:	LCSD2489960
Matrix:	Water	Instrument ID:	TOC6
Analyst:	CCL	Lab File ID:	10210
Prep Date:	NA	Analysis Date:	06/06/23 1149
Prep Batch:	NA	Analytical Batch:	767077
Prep Method:	NA	Analytical Method:	EPA 9060A

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*%REC*	*Q*	*%REC LIMITS*	*RPD*	*Q*	*RPD LIMITS*
Total Organic Carbon	mg/L	50.0	51.4	103		90 - 110	1		0 - 20

FORM VII - GENCHEM

Sample

Sample Name: 2489960
Sample ID: 10210,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

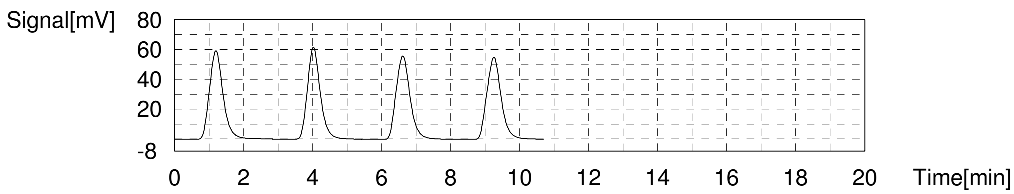
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:51.41mg/L TC:51.73mg/L IC:0.3255mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	161.8	52.92mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:49:52 AM
2	158.7	51.90mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:52:38 AM
3	155.7	50.92mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:55:26 AM
4	156.5	51.18mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/6/2023 11:58:16 AM

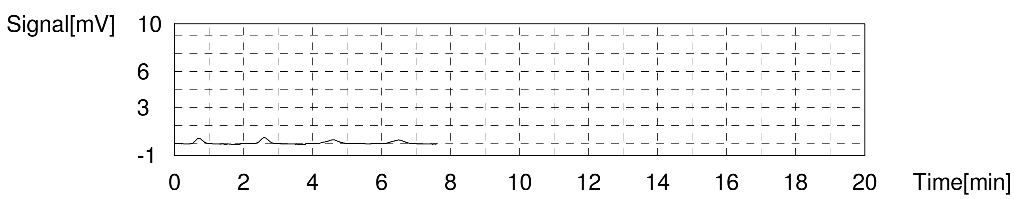
Mean Area 158.2
Mean Conc. 51.73mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.6980	0.3183mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 12:01:49 PM
2	0.8166	0.3552mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 12:03:54 PM
3	0.6516	0.3039mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 12:05:59 PM
4	0.7182	0.3246mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/6/2023 12:08:04 PM

Mean Area 0.7211
Mean Conc. 0.3255mg/L



VII - LABORATORY CONTROL SPIKE

Report No:	223060210	LAB ID:	LCS2490525
Matrix:	Water	Instrument ID:	TOC6
Analyst:	CCL	Lab File ID:	10212
Prep Date:	NA	Analysis Date:	06/07/23 1229
Prep Batch:	NA	Analytical Batch:	767167
Prep Method:	NA	Analytical Method:	EPA 9060A

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*%REC*	*Q*	*% REC LIMITS*
Total Organic Carbon	mg/L	50.0	52.2	104		90 - 110

FORM VII - GENCHEM

Sample

Sample Name: 2490525 lcs
Sample ID: 10212,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

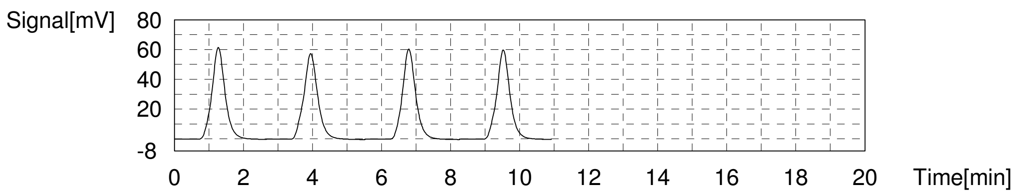
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:52.19mg/L TC:52.62mg/L IC:0.4298mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	163.1	53.35mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 12:29:38 PM
2	161.3	52.76mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 12:32:40 PM
3	159.9	52.30mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 12:35:35 PM
4	159.2	52.07mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 12:38:33 PM

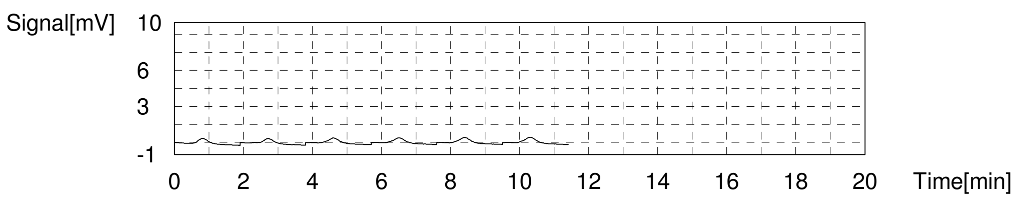
Mean Area 160.9
Mean Conc. 52.62mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.8692	0.3715mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 12:42:06 PM
2	0.6564	0.3054mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 12:44:18 PM
3	0.9620	0.4003mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 12:46:23 PM
4	1.028	0.4208mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 12:48:28 PM
5	1.114	0.4474mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 12:50:33 PM
6	1.125	0.4509mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 12:52:38 PM

Mean Area 1.057
Mean Conc. 0.4298mg/L



VII - LABORATORY CONTROL SPIKE DUPLICATE

Report No:	223060210	LAB ID:	LCSD2490526
Matrix:	Water	Instrument ID:	TOC6
Analyst:	CCL	Lab File ID:	10212
Prep Date:	NA	Analysis Date:	06/07/23 1257
Prep Batch:	NA	Analytical Batch:	767167
Prep Method:	NA	Analytical Method:	EPA 9060A

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*%REC*	*Q*	*%REC LIMITS*	*RPD*	*Q*	*RPD LIMITS*
Total Organic Carbon	mg/L	50.0	51.6	103		90 - 110	1		0 - 20

FORM VII - GENCHEM

Sample

Sample Name: 2490526
Sample ID: 10212,CCL,TOC6
Origin: TOC4.met
Status Completed
Chk. Result

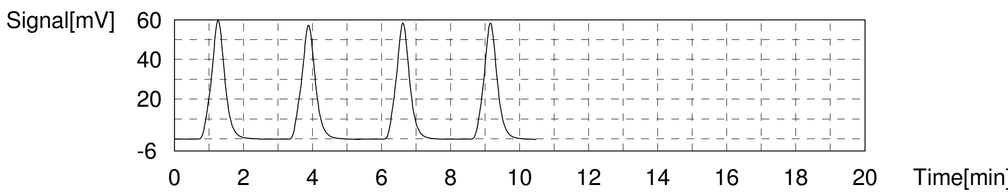
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:51.55mg/L TC:51.97mg/L IC:0.4181mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	162.1	53.02mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 12:57:15 PM
2	159.7	52.23mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 1:00:11 PM
3	156.9	51.31mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 1:02:54 PM
4	156.9	51.31mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	6/7/2023 1:05:39 PM

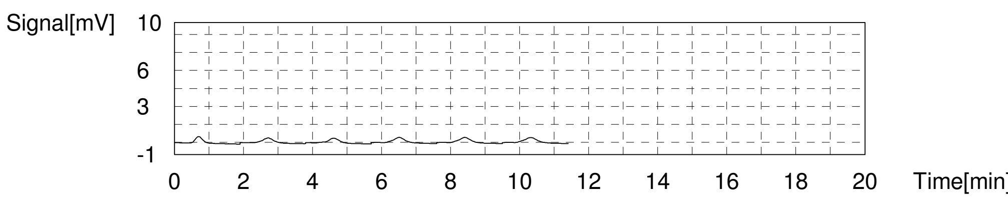
Mean Area 158.9
Mean Conc. 51.97mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.7780	0.3432mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 1:09:12 PM
2	0.8939	0.3791mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 1:11:17 PM
3	0.6236	0.2953mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 1:13:29 PM
4	1.030	0.4214mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 1:15:34 PM
5	1.050	0.4276mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 1:17:39 PM
6	1.104	0.4443mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	6/7/2023 1:19:44 PM

Mean Area 1.019
Mean Conc. 0.4181mg/L



EPA 9060A

Form XIV

Run Logs

XIV - ANALYSIS RUN LOG

Report No:	223060210	Analytical Batch:	767077	Start Date:	06/06/23
Instrument ID:	TOC6	Analytical Method:	EPA 9060A	End Date:	06/07/23

CLIENT SAMPLE ID	*LAB SAMPLE ID*	*DILUTION*	*TIME*	*ANALYTES* TOC
CCV	1800	1	0358	X
CCB	1900	1	0459	X
LCS2489959	2489959	1	1124	X
LCSD2489960	2489960	1	1149	X
MB2489958	2489958	1	1214	X
OT07-EB-060123	22306021001	1	1336	X
OT07-MW02-060123	22306021002	1	1412	X
OT07-MW05-060123	22306021004	1	1440	X
OT07-MW06-060123	22306021005	1	1507	X
OT07-MW-806-060123	22306021008	1	1651	X
CCV	1800	1	1717	X
CCB	1900	1	1811	X

FORM XIV - GENCHEM

XIV - ANALYSIS RUN LOG

Report No: 223060210 Analytical Batch: 767167 Start Date: 06/06/23

Instrument ID: TOC6 Analytical Method: EPA 9060A End Date: 06/08/23

CLIENT SAMPLE ID	*LAB SAMPLE ID*	*DILUTION*	*TIME*	*ANALYTES* TOC
CCV	1800	1	2313	X
CCB	1900	1	0015	X
LCS2490525	2490525	1	1229	X
LCSD2490526	2490526	1	1257	X
MB2490524	2490524	1	1325	X
CCV	1800	1	1737	X
CCB	1900	1	1826	X
CCV	1800	1	2312	X
CCB	1900	1	0010	X
OT07-MW04-060123	22306021003	1	0053	X
OT07-MW04-060123MS	22306021006	1	0119	X
OT07-MW04-060123MSD	22306021007	1	0148	X
CCV	1800	1	0213	X
CCB	1900	1	0305	X

FORM XIV - GENCHEM

EPA 9060A

Standards

BATCH COVERSHEET

ANALYST CCL

DATE 06/06/23

SM 5310 B-2011/9060A TOC

BATCH **767077**

TOTAL CARBON (TC) CALIBRATION

Calibration Standard Lot	2131595
Calibration Standard Exp	10/31/23
Calibration Standard ID	WLG-1245-81-3
Calibration Date	05/19/23
Calibration Exp	10/31/23

INORGANIC CARBON (IC) CALIBRATION

Calibration Standard Lot	2133029
Calibration Standard Exp	02/28/25
Calibration Standard ID	WLG-1245-81-5
Calibration Date	05/19/23
Calibration Exp	05/19/24

TOTAL CARBON (TC) QC

CCV/LCS Standard ID	WLG-1255-6-2
CCV/LCS Standard Exp	06/07/23
ICV Standard ID	WLG-1245-81-4
ICV Standard Exp	05/20/23
LLC 2 mg/L - Recovery %	89
ICV 25 mg/L - Recovery %	97.5
ICV 75 mg/L - Recovery %	97

INORGANIC CARBON (IC) QC

CCV/LCS Standard ID	WLG-1255-6-2
CCV/LCS Standard Exp	06/07/23
ICV Standard ID	WLG-1245-81-6
ICV Standard Exp	05/20/23
LLC 2 mg/L - Recovery %	108
ICV 25 mg/L - Recovery %	99
ICV 75 mg/L - Recovery %	99.7

CCV/ICV Control Limits 90-110%

LLC Control Limits 50-150%

Instrument ID TOC6

Pace Analytical®

BATCH COVERSHEET

ANALYST	CCL
DATE	06/07/23

SM 5310 B-2011/9060A TOC

BATCH 767167

TOTAL CARBON (TC) CALIBRATION

Calibration Standard Lot	2131595
Calibration Standard Exp	10/31/23
Calibration Standard ID	WLG-1245-81-3
Calibration Date	05/19/23
Calibration Exp	10/31/23

INORGANIC CARBON (IC) CALIBRATION

Calibration Standard Lot	2133029
Calibration Standard Exp	02/28/25
Calibration Standard ID	WLG-1245-81-5
Calibration Date	05/19/23
Calibration Exp	05/19/24

TOTAL CARBON (TC) QC

CCV/LCS Standard ID	WLG-1255-6-3
CCV/LCS Standard Exp	06/08/23
ICV Standard ID	WLG-1245-81-4
ICV Standard Exp	05/20/23
LLC 2 mg/L - Recovery %	89
ICV 25 mg/L - Recovery %	97.5
ICV 75 mg/L - Recovery %	97

INORGANIC CARBON (IC) QC

CCV/LCS Standard ID	WLG-1255-6-3
CCV/LCS Standard Exp	06/08/23
ICV Standard ID	WLG-1245-81-6
ICV Standard Exp	05/20/23
LLC 2 mg/L - Recovery %	108
ICV 25 mg/L - Recovery %	99
ICV 75 mg/L - Recovery %	99.7

CCV/ICV Control Limits 90-110%

LLC Control Limits 50-150%

Instrument ID: TOC6

EPA 9060A

Initial Calibration

Instr.Information

System	TOC 6
Instrument Options	TOC/ASI/IC Unit/
Catalyst	Regular Sensitivity

Cal. Curve

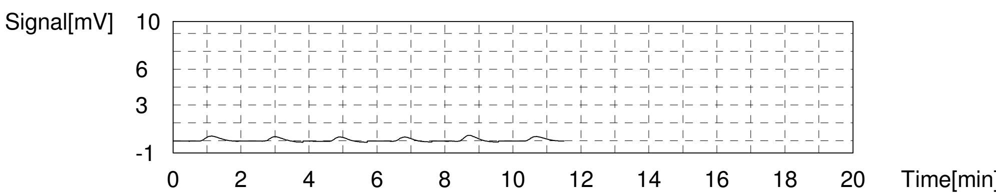
Sample Name:	TC ICAL
Sample ID:	CCL/ICAL
Cal. Curve:	TC ICAL.2023_05_19_10_37_44.cal
Status	Completed

Type	Anal.
Standard	TC

Conc: 0.000mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	1.245	50uL	1	*******		5/19/2023 10:42:51 AM
2	1.097	50uL	1	*******		5/19/2023 10:44:56 AM
3	1.046	50uL	1	*******		5/19/2023 10:47:01 AM
4	0.9251	50uL	1	*******	E	5/19/2023 10:49:06 AM
5	1.398	50uL	1	*******	E	5/19/2023 10:51:14 AM
6	1.142	50uL	1	*******		5/19/2023 10:53:21 AM

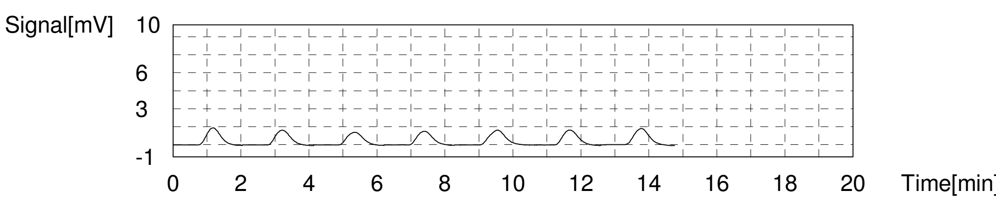
Acid Add. 0.000%
Mean Area 1.133



Conc: 1.000mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	4.097	50uL	1	*******		5/19/2023 10:58:48 AM
2	3.843	50uL	1	*******		5/19/2023 11:01:05 AM
3	3.439	50uL	1	*******	E	5/19/2023 11:03:20 AM
4	3.584	50uL	1	*******	E	5/19/2023 11:05:44 AM
5	4.103	50uL	1	*******		5/19/2023 11:08:07 AM
6	3.942	50uL	1	*******		5/19/2023 11:10:23 AM
7	4.258	50uL	1	*******	E	5/19/2023 11:12:56 AM

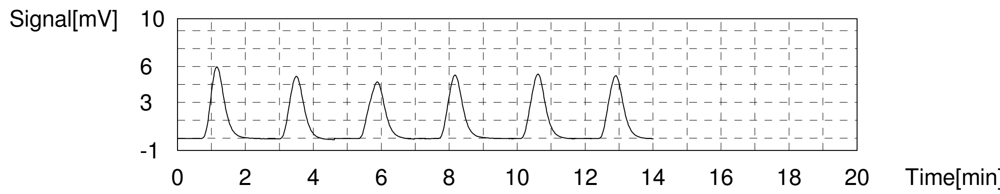
Acid Add. 0.000%
Mean Area 3.996



Conc: 5.000mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	16.53	50uL	1	*******	E	5/19/2023 11:18:39 AM
2	15.41	50uL	1	*******		5/19/2023 11:21:09 AM
3	15.27	50uL	1	*******		5/19/2023 11:23:41 AM
4	15.71	50uL	1	*******		5/19/2023 11:26:17 AM
5	16.10	50uL	1	*******	E	5/19/2023 11:28:46 AM
6	15.88	50uL	1	*******		5/19/2023 11:31:16 AM

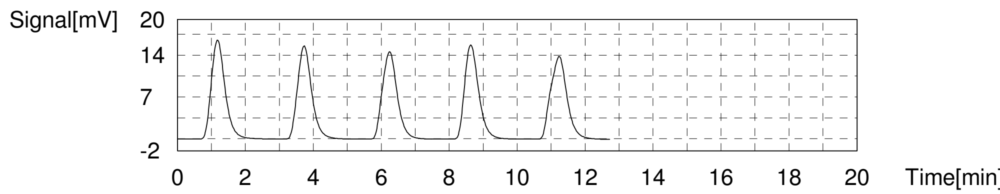
Acid Add. 0.000%
Mean Area 15.57



Conc: 15.00mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	46.65	50uL	1	*******		5/19/2023 11:37:12 AM
2	44.99	50uL	1	*******		5/19/2023 11:39:54 AM
3	44.28	50uL	1	*******	E	5/19/2023 11:42:32 AM
4	45.15	50uL	1	*******		5/19/2023 11:45:12 AM
5	46.30	50uL	1	*******		5/19/2023 11:48:09 AM

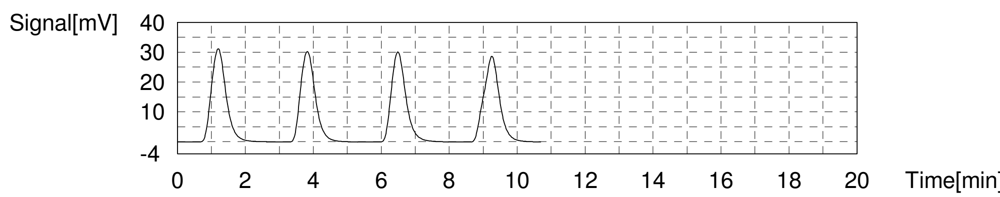
Acid Add. 0.000%
Mean Area 45.77



Conc: 30.00mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	93.35	50uL	1	*******		5/19/2023 11:54:17 AM
2	90.90	50uL	1	*******		5/19/2023 11:57:08 AM
3	89.24	50uL	1	*******		5/19/2023 11:59:59 AM
4	91.57	50uL	1	*******		5/19/2023 12:02:53 PM

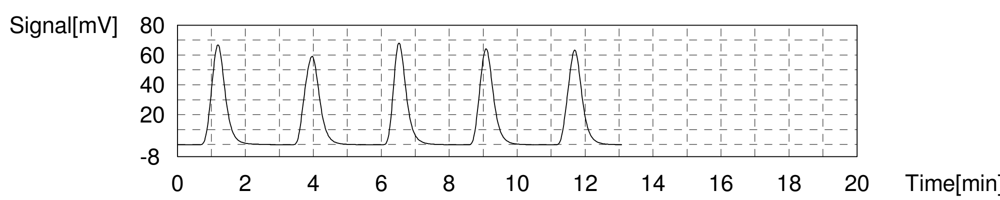
Acid Add. 0.000%
Mean Area 91.27



Conc: 60.00mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	187.8	50uL	1	*******		5/19/2023 12:09:08 PM
2	183.3	50uL	1	*******		5/19/2023 12:11:58 PM
3	178.1	50uL	1	*******	E	5/19/2023 12:14:40 PM
4	182.1	50uL	1	*******		5/19/2023 12:17:25 PM
5	184.2	50uL	1	*******		5/19/2023 12:20:12 PM

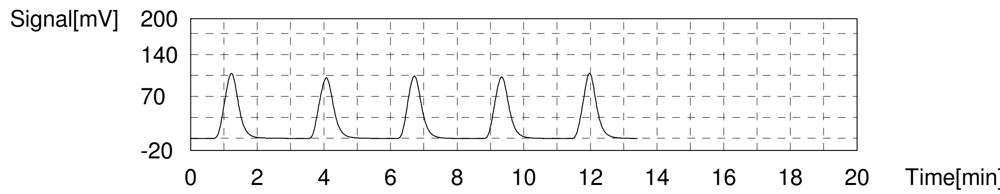
Acid Add. 0.000%
Mean Area 184.4



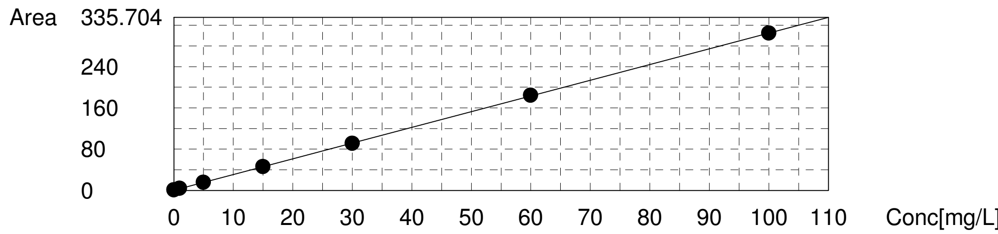
Conc: 100.0mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	317.6	50uL	1	*******	E	5/19/2023 12:26:33 PM
2	304.8	50uL	1	*******		5/19/2023 12:29:24 PM
3	301.5	50uL	1	*******		5/19/2023 12:32:11 PM
4	304.7	50uL	1	*******		5/19/2023 12:35:00 PM
5	308.6	50uL	1	*******		5/19/2023 12:37:50 PM

Acid Add. 0.000%
Mean Area 304.9



Slope: 3.046
Intercept 0.6223
r^2 1.0000
r 1.0000
Zero Shift No



Cal. Curve

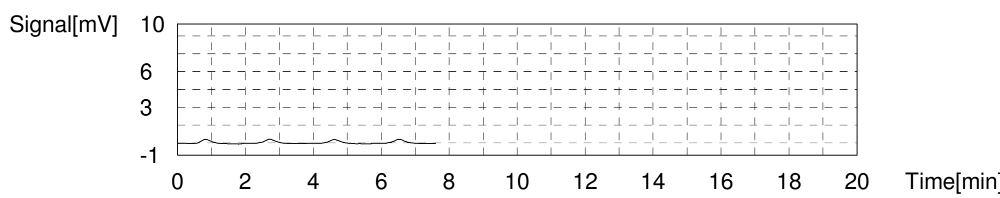
Sample Name: IC ICAL
Sample ID: CCL/ICAL
Cal. Curve: IC ICAL.2023_05_19_12_52_15.cal
Status Completed

Type	Anal.
Standard	IC

Conc: 0.000mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	0.6192	50uL	1	*******		5/19/2023 12:57:23 PM
2	0.6568	50uL	1	*******		5/19/2023 12:59:28 PM
3	0.6057	50uL	1	*******		5/19/2023 1:01:33 PM
4	0.6410	50uL	1	*******		5/19/2023 1:03:38 PM

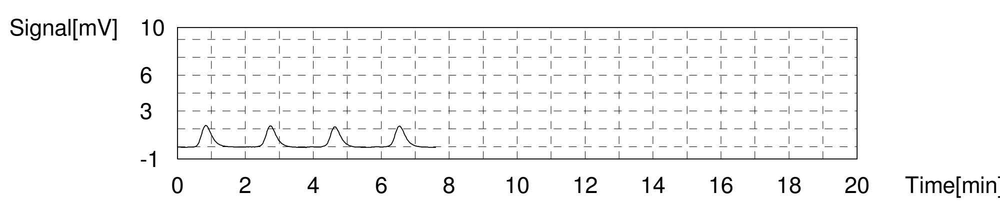
Acid Add. 0.000%
Mean Area 0.6307



Conc: 1.000mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	3.820	50uL	1	*******		5/19/2023 1:09:13 PM
2	3.759	50uL	1	*******		5/19/2023 1:11:18 PM
3	3.628	50uL	1	*******		5/19/2023 1:13:23 PM
4	3.712	50uL	1	*******		5/19/2023 1:15:28 PM

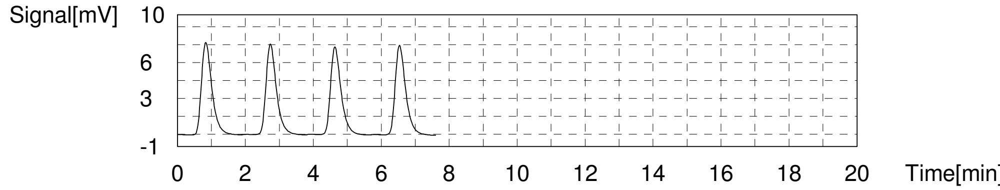
Acid Add. 0.000%
Mean Area 3.730



Conc: 5.000mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	16.35	50uL	1	*******		5/19/2023 1:20:59 PM
2	16.03	50uL	1	*******		5/19/2023 1:23:04 PM
3	15.61	50uL	1	*******		5/19/2023 1:25:09 PM
4	15.92	50uL	1	*******		5/19/2023 1:27:14 PM

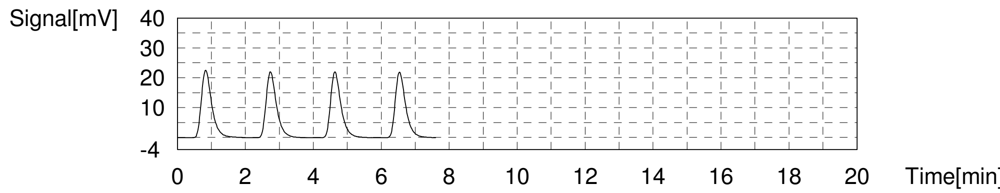
Acid Add. 0.000%
Mean Area 15.98



Conc: 15.00mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	47.65	50uL	1	*******		5/19/2023 1:32:45 PM
2	46.79	50uL	1	*******		5/19/2023 1:34:50 PM
3	46.36	50uL	1	*******		5/19/2023 1:36:55 PM
4	46.30	50uL	1	*******		5/19/2023 1:39:00 PM

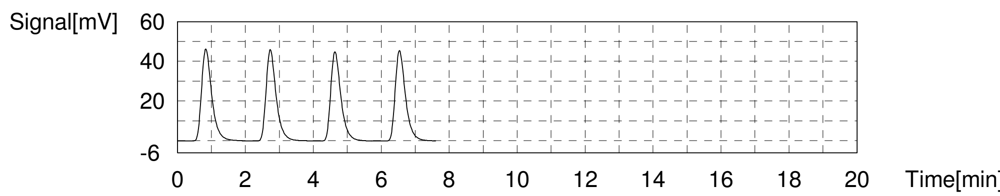
Acid Add. 0.000%
Mean Area 46.78



Conc: 30.00mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	96.48	50uL	1	*******		5/19/2023 1:44:31 PM
2	94.69	50uL	1	*******		5/19/2023 1:46:36 PM
3	93.79	50uL	1	*******		5/19/2023 1:48:41 PM
4	95.45	50uL	1	*******		5/19/2023 1:50:46 PM

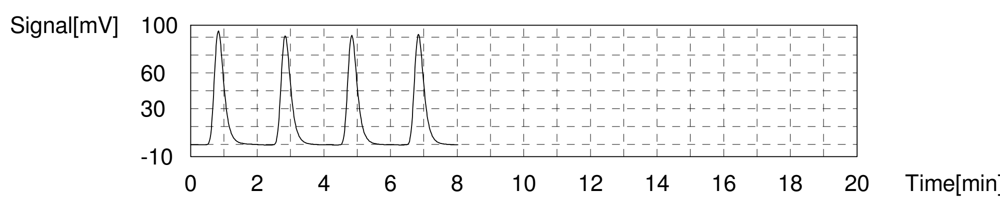
Acid Add. 0.000%
Mean Area 95.10



Conc: 60.00mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	197.4	50uL	1	*******		5/19/2023 1:56:23 PM
2	190.9	50uL	1	*******		5/19/2023 1:58:34 PM
3	190.9	50uL	1	*******		5/19/2023 2:00:45 PM
4	192.3	50uL	1	*******		5/19/2023 2:02:57 PM

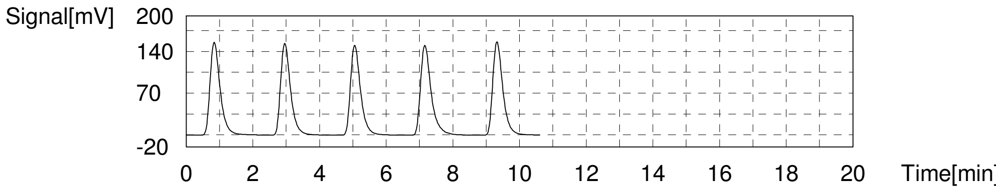
Acid Add. 0.000%
Mean Area 192.9



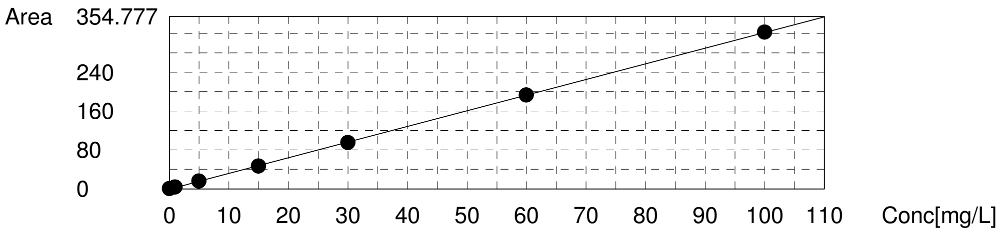
Conc: 100.0mg/L

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	326.5	50uL	1	*******		5/19/2023 2:08:41 PM
2	319.7	50uL	1	*******		5/19/2023 2:10:58 PM
3	316.0	50uL	1	*******		5/19/2023 2:13:15 PM
4	335.3	50uL	1	*******	E	5/19/2023 2:15:36 PM
5	327.9	50uL	1	*******		5/19/2023 2:17:55 PM

Acid Add. 0.000%
Mean Area 322.5



Slope: 3.222
Intercept -0.3278
r^2 0.9999
r 1.0000
Zero Shift No



Sample

Sample Name: CCB
Sample ID: TOC 4
Origin: TOC4.met
Status Completed
Chk. Result

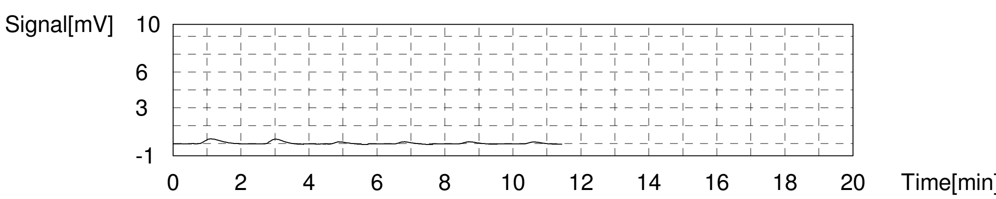
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:-0.2258mg/L TC:-0.05476mg/L IC:0.1710mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.273	0.2136mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 2:41:34 PM
2	1.105	0.1585mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 2:43:39 PM
3	0.5014	-0.03970mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 2:45:44 PM
4	0.4595	-0.05346mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 2:47:49 PM
5	0.4585	-0.05379mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 2:49:54 PM
6	0.4027	-0.07211mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 2:51:59 PM

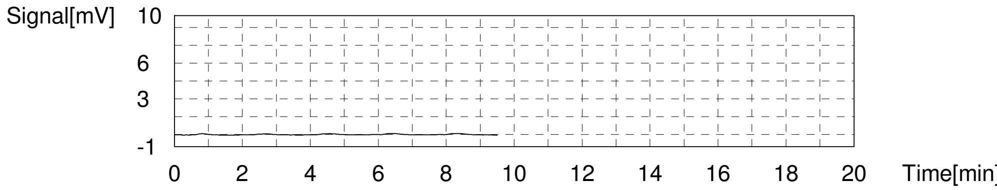
Mean Area 0.4555
Mean Conc. -0.05476mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.1656	0.1531mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 2:55:25 PM
2	0.000	0.1017mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 2:57:30 PM
3	0.2241	0.1713mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 2:59:35 PM
4	0.2353	0.1748mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 3:01:40 PM
5	0.2677	0.1848mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 3:03:45 PM

Mean Area 0.2232
Mean Conc. 0.1710mg/L



Sample

Sample Name: CCV TC
Sample ID: TOC 4
Origin: TOC4.met
Status Completed
Chk. Result

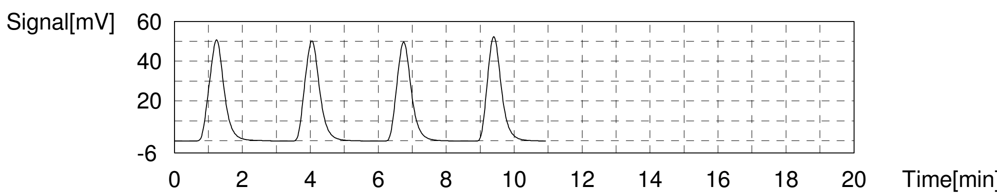
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:49.26mg/L TC:49.48mg/L IC:0.2232mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	154.4	50.49mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 3:14:21 PM
2	152.8	49.97mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 3:17:14 PM
3	147.6	48.26mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 3:20:06 PM
4	150.5	49.21mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 3:22:59 PM

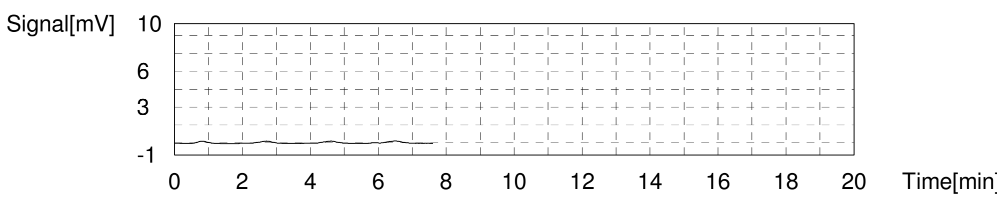
Mean Area 151.3
Mean Conc. 49.48mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.3482	0.2098mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 3:26:32 PM
2	0.3770	0.2187mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 3:28:37 PM
3	0.4148	0.2305mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 3:30:42 PM
4	0.4253	0.2337mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 3:32:47 PM

Mean Area 0.3913
Mean Conc. 0.2232mg/L



Sample

Sample Name: LLOQ TC
Sample ID: TOC 4
Origin: TOC4.met
Status Completed
Chk. Result

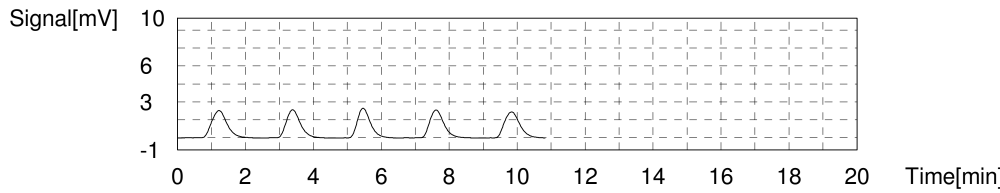
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:1.789mg/L TC:1.988mg/L IC:0.1996mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	7.040	2.107mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 3:40:00 PM
2	6.661	1.983mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 3:42:17 PM
3	6.602	1.963mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 3:44:36 PM
4	6.642	1.977mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 3:46:58 PM
5	6.805	2.030mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 3:49:20 PM

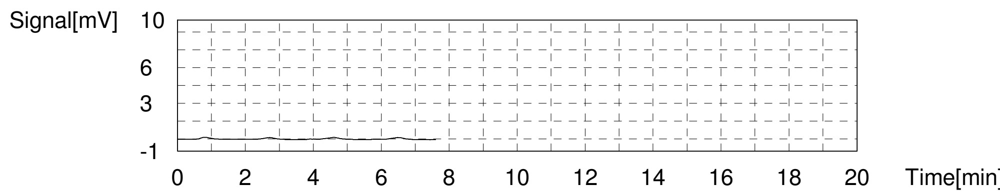
Mean Area 6.678
Mean Conc. 1.988mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.2722	0.1862mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 3:52:52 PM
2	0.3226	0.2018mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 3:54:57 PM
3	0.3367	0.2062mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 3:57:02 PM
4	0.3305	0.2043mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 3:59:07 PM

Mean Area 0.3155
Mean Conc. 0.1996mg/L



Sample

Sample Name: ICV 25 TC
Sample ID: TOC 4
Origin: TOC4.met
Status Completed
Chk. Result

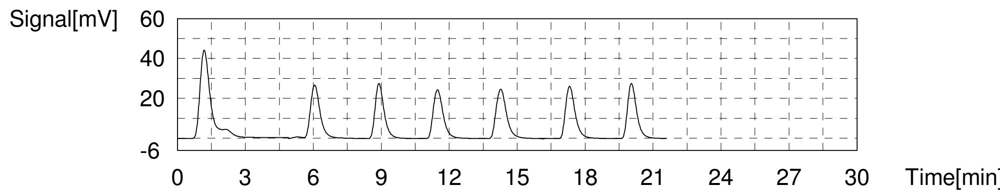
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:24.38mg/L TC:24.71mg/L IC:0.3282mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	151.3	49.47mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 4:09:01 PM
2	80.27	26.15mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 4:12:03 PM
3	75.43	24.56mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 4:14:48 PM
4	73.99	24.09mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 4:17:45 PM
5	79.67	25.95mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 4:21:01 PM
6	77.04	25.09mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 4:23:56 PM
7	77.06	25.10mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 4:27:10 PM

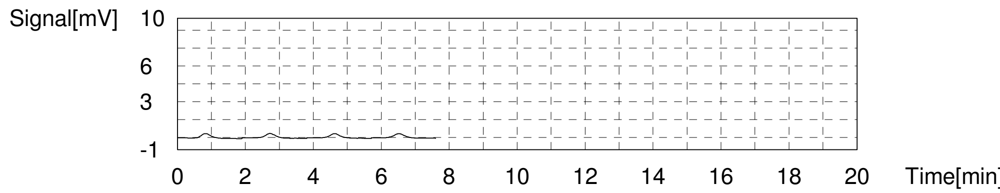
Mean Area 75.88
Mean Conc. 24.71mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.7177	0.3245mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 4:30:36 PM
2	0.7328	0.3291mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 4:32:41 PM
3	0.6931	0.3168mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 4:34:46 PM
4	0.7760	0.3426mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 4:36:51 PM

Mean Area 0.7299
Mean Conc. 0.3282mg/L



Sample

Sample Name: ICV 75 TC
Sample ID: TOC 4
Origin: TOC4.met
Status Completed
Chk. Result

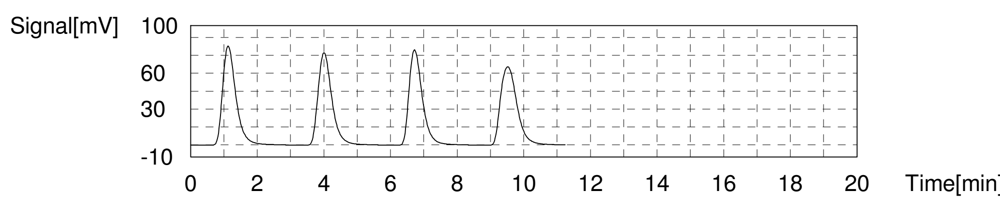
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:72.79mg/L TC:73.38mg/L IC:0.5889mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	229.2	75.05mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 5:02:11 PM
2	222.1	72.72mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 5:05:08 PM
3	220.7	72.26mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 5:08:02 PM
4	224.4	73.48mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 5:11:07 PM

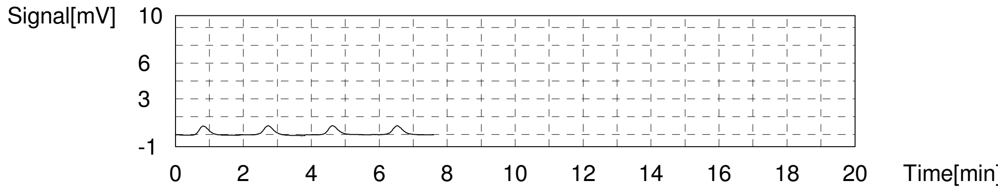
Mean Area 224.1
Mean Conc. 73.38mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	1.511	0.5707mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 5:14:40 PM
2	1.613	0.6023mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 5:16:45 PM
3	1.563	0.5868mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 5:18:50 PM
4	1.592	0.5958mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 5:20:55 PM

Mean Area 1.570
Mean Conc. 0.5889mg/L



Sample

Sample Name: CCB
Sample ID: TOC 4
Origin: TOC4.met
Status Completed
Chk. Result

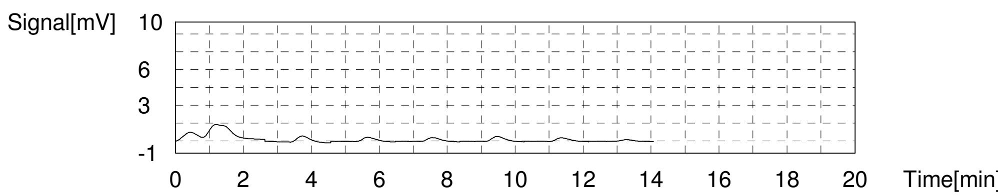
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:-0.1296mg/L TC:0.1038mg/L IC:0.2334mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	7.615	2.296mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 6:02:23 PM
2	1.373	0.2465mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 6:04:30 PM
3	0.9670	0.1132mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 6:06:34 PM
4	0.8352	0.06990mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 6:08:39 PM
5	1.111	0.1605mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 6:10:44 PM
6	0.8403	0.07157mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 6:12:49 PM
7	0.2654	-0.1172mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 6:15:16 PM

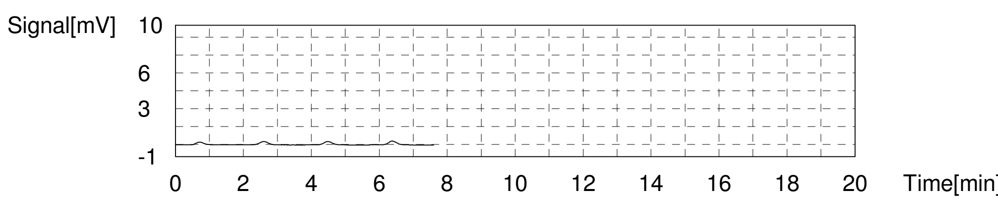
Mean Area 0.9384
Mean Conc. 0.1038mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.3257	0.2028mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 6:18:42 PM
2	0.4343	0.2365mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 6:20:47 PM
3	0.4569	0.2435mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 6:22:52 PM
4	0.4801	0.2507mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 6:24:57 PM

Mean Area 0.4243
Mean Conc. 0.2334mg/L



Sample

Sample Name: CCV IC
Sample ID: TOC 4
Origin: TOC4.met
Status Completed
Chk. Result

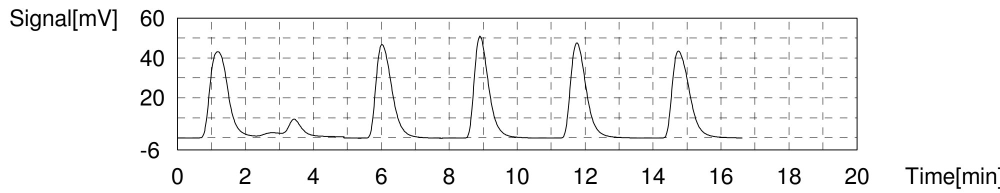
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:0.3491mg/L TC:49.66mg/L IC:49.31mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	192.5	63.00mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 6:34:52 PM
2	152.7	49.93mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 6:37:57 PM
3	150.2	49.11mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 6:40:57 PM
4	152.0	49.70mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 6:44:08 PM
5	152.6	49.90mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 6:47:19 PM

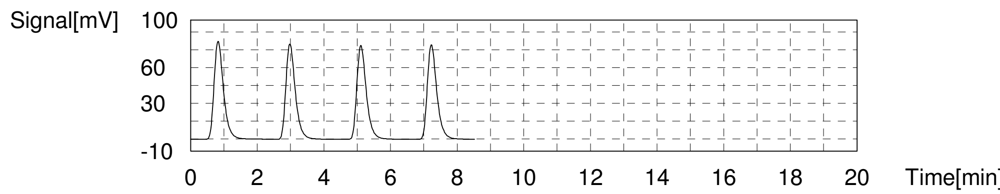
Mean Area 151.9
Mean Conc. 49.66mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	162.0	50.38mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 6:51:06 PM
2	157.9	49.10mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 6:53:25 PM
3	156.4	48.64mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 6:55:43 PM
4	158.0	49.13mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 6:58:01 PM

Mean Area 158.6
Mean Conc. 49.31mg/L



Sample

Sample Name: LLOQ IC
Sample ID: TOC 4
Origin: TOC4.met
Status Completed
Chk. Result

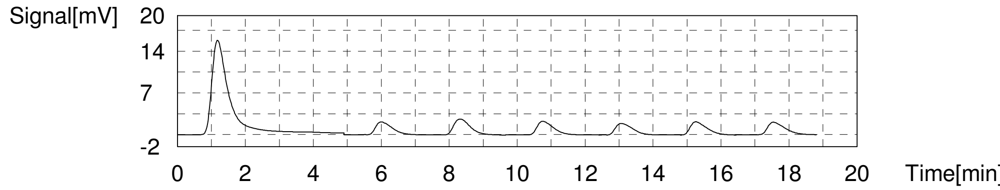
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:0.06123mg/L TC:2.225mg/L IC:2.164mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	57.17	18.57mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 7:07:55 PM
2	7.224	2.168mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 7:10:23 PM
3	9.005	2.752mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 7:13:02 PM
4	7.547	2.274mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 7:15:31 PM
5	6.637	1.975mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 7:17:54 PM
6	7.474	2.250mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 7:20:22 PM
7	7.352	2.210mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 7:23:17 PM

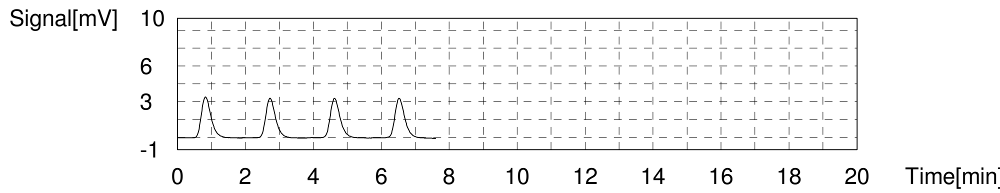
Mean Area 7.399
Mean Conc. 2.225mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	6.773	2.204mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 7:26:43 PM
2	6.555	2.136mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 7:28:48 PM
3	6.607	2.152mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 7:30:53 PM
4	6.645	2.164mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 7:32:58 PM

Mean Area 6.645
Mean Conc. 2.164mg/L



Sample

Sample Name: ICV 25 IC
Sample ID: TOC 4
Origin: TOC4.met
Status Completed
Chk. Result

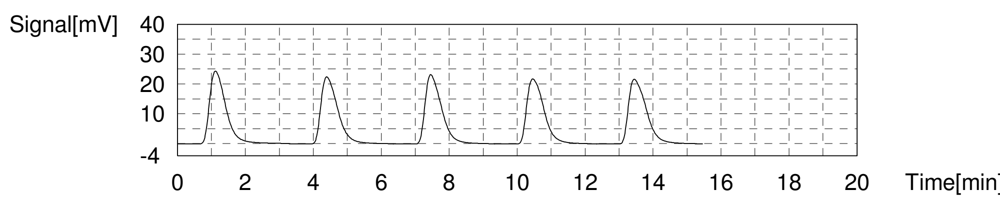
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:0.3904mg/L TC:25.16mg/L IC:24.77mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	80.88	26.35mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 7:41:19 PM
2	76.63	24.96mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 7:44:34 PM
3	77.49	25.24mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 7:47:44 PM
4	76.29	24.84mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 7:50:55 PM
5	78.60	25.60mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 7:54:13 PM

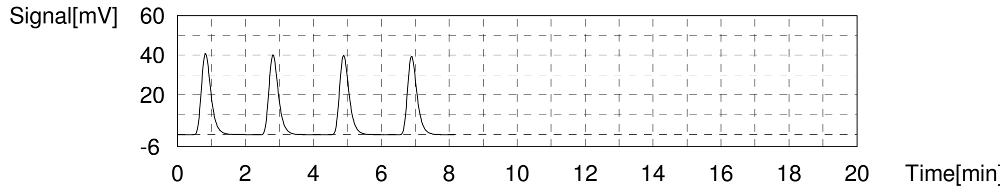
Mean Area 77.25
Mean Conc. 25.16mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	80.74	25.16mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 7:57:50 PM
2	79.77	24.86mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 8:00:06 PM
3	79.23	24.69mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 8:02:17 PM
4	78.22	24.38mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 8:04:33 PM

Mean Area 79.49
Mean Conc. 24.77mg/L



Sample

Sample Name: ICV 75 IC
Sample ID: TOC 4
Origin: TOC4.met
Status Completed
Chk. Result

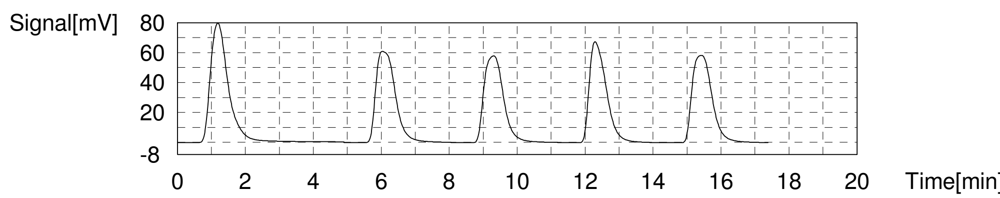
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:2.109mg/L TC:76.95mg/L IC:74.84mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	292.8	95.93mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 8:14:27 PM
2	240.9	78.89mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 8:17:47 PM
3	230.2	75.38mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 8:21:06 PM
4	232.5	76.13mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 8:24:17 PM
5	236.3	77.38mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 8:27:40 PM

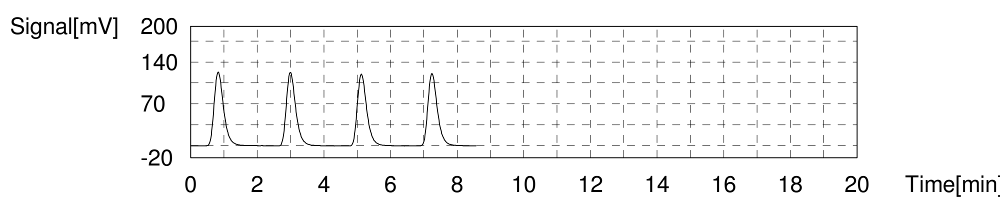
Mean Area 235.0
Mean Conc. 76.95mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	244.6	76.01mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 8:31:28 PM
2	242.7	75.42mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 8:33:47 PM
3	236.7	73.56mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 8:36:05 PM
4	239.3	74.36mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 8:38:26 PM

Mean Area 240.8
Mean Conc. 74.84mg/L



Sample

Sample Name: CCB
Sample ID: TOC 4
Origin: TOC4.met
Status Completed
Chk. Result

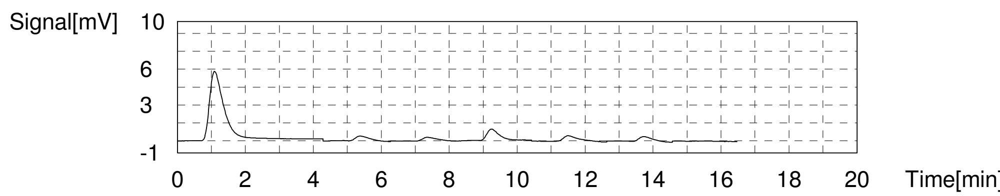
Type	Anal.	Manual Dilution	Result
Unknown	TOC	1.000	TOC:-0.05647mg/L TC:0.1907mg/L IC:0.2471mg/L

1. Det

Anal.: TC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	17.74	5.620mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 9:16:08 PM
2	1.208	0.1923mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 9:18:18 PM
3	0.8910	0.08822mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 9:20:23 PM
4	2.389	0.5801mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 9:22:49 PM
5	1.531	0.2984mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 9:25:13 PM
6	1.182	0.1838mg/L	50uL	1		TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 9:27:20 PM
7	0.000	-0.2043mg/L	50uL	1	E	TC ICAL.2023_05_19_10_37_44.cal	5/19/2023 9:29:47 PM

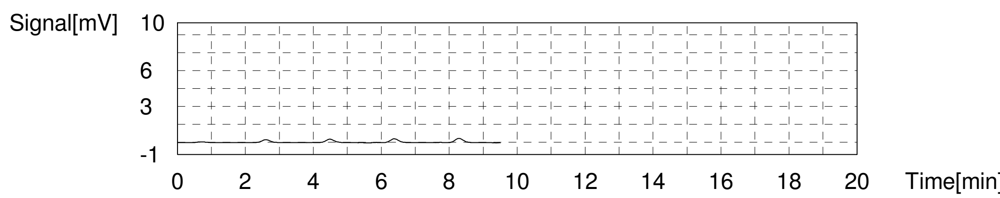
Mean Area 1.203
Mean Conc. 0.1907mg/L



Anal.: IC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Ex.	Cal. Curve	Date / Time
1	0.000	0.1017mg/L	50uL	1	E	IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 9:33:13 PM
2	0.3555	0.2121mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 9:35:18 PM
3	0.4374	0.2375mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 9:37:23 PM
4	0.5068	0.2590mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 9:39:28 PM
5	0.5744	0.2800mg/L	50uL	1		IC ICAL.2023_05_19_12_52_15.cal	5/19/2023 9:41:33 PM

Mean Area 0.4685
Mean Conc. 0.2471mg/L



EPA 9056A

Form I

Sample Results

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-EB-060123
Collect Date:	06/01/23 1214	LAB Sample ID:	22306021001
Matrix:	Water	Instrument ID:	BRWA13
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/02/23 1334
Prep Batch:	NA	Analytical Batch:	767085
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Chloride	0.050	mg/L	J	0.050	0.100	0.200
Nitrate	0.150	mg/L	U	0.100	0.150	0.200
Nitrite	0.150	mg/L	U	0.100	0.150	0.200
Sulfate	0.150	mg/L	U	0.100	0.150	0.200

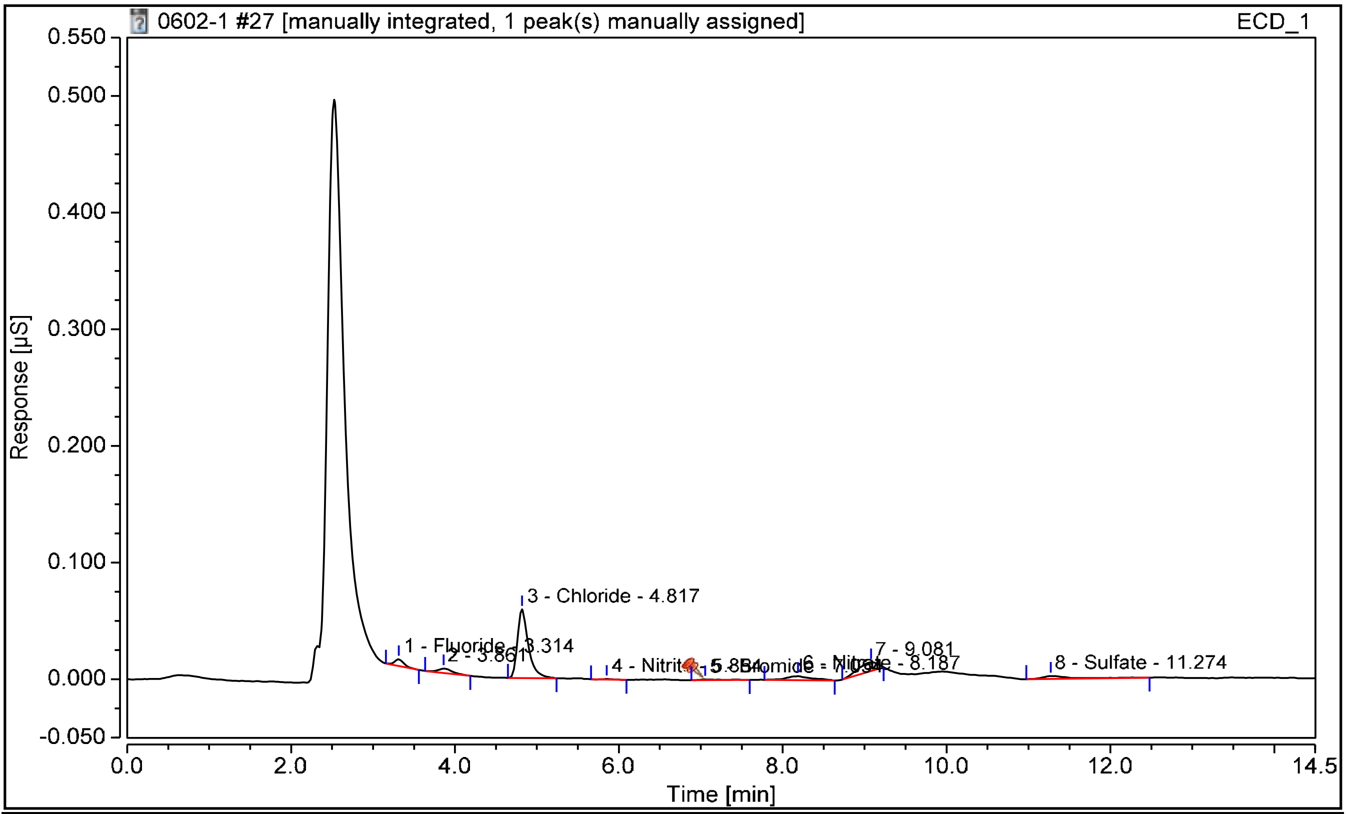
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021001**	*Run Time (min):*	**14.50**
Vial Number:	**13**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 13:34**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.314	0.001	0.0105	n.a.	n.a.	n.a.
3	Chloride	4.817	0.009	0.0496	n.a.	n.a.	n.a.
4	Nitrite	5.854	0.000	0.0091	n.a.	n.a.	n.a.
5	Bromide	7.054	0.000	0.0046	n.a.	n.a.	n.a.
6	Nitrate	8.187	0.001	0.0128	n.a.	n.a.	n.a.
8	Sulfate	11.274	0.001	0.0055	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW02-060123
Collect Date:	06/01/23 1230	LAB Sample ID:	22306021002
Matrix:	Water	Instrument ID:	BRWA13
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/02/23 1351
Prep Batch:	NA	Analytical Batch:	767085
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Chloride	4.08	mg/L		0.050	0.100	0.200
Nitrate	0.150	mg/L	U	0.100	0.150	0.200
Nitrite	0.150	mg/L	U	0.100	0.150	0.200

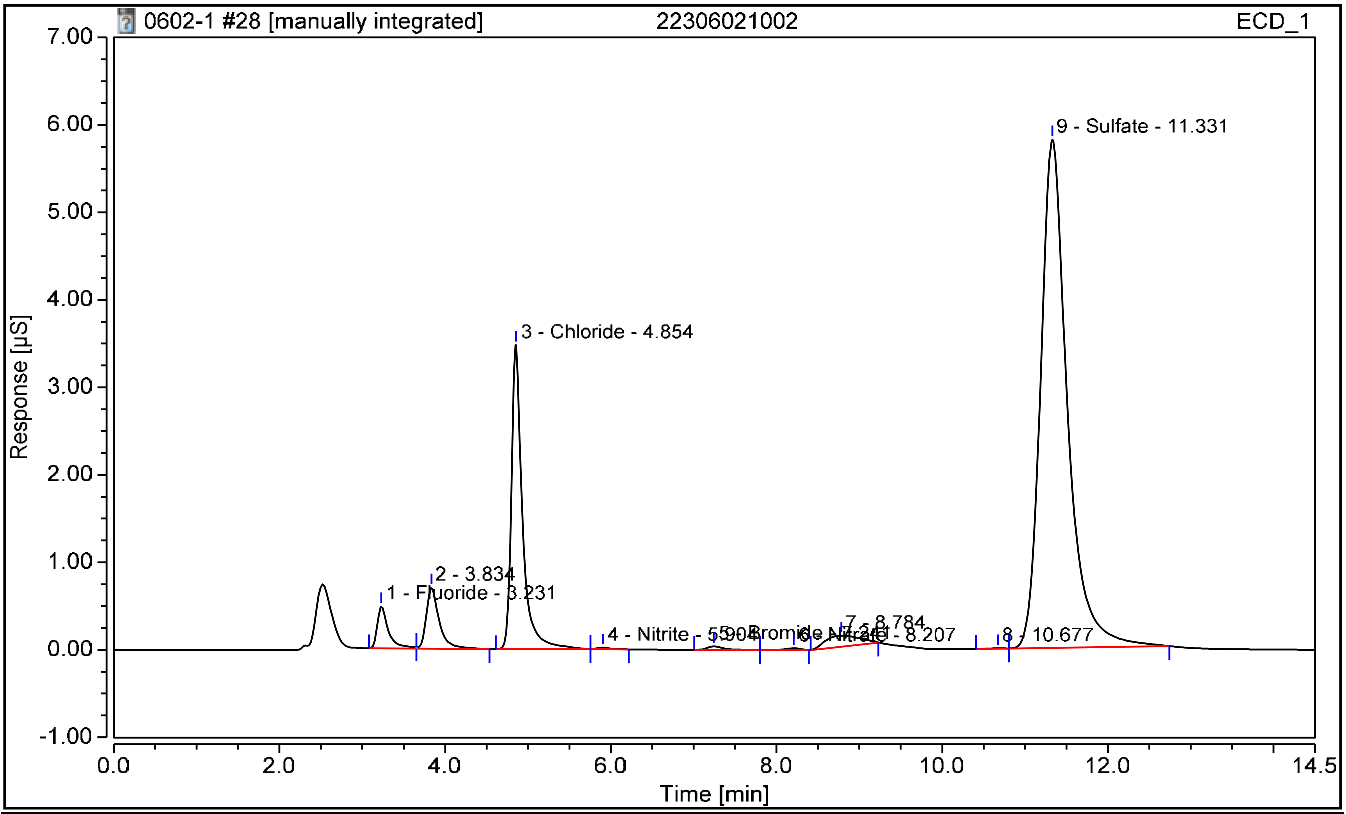
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021002**	*Run Time (min):*	**14.50**
Vial Number:	**14**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 13:51**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.231	0.075	0.3812	n.a.	n.a.	n.a.
3	Chloride	4.854	0.538	4.0812	n.a.	n.a.	n.a.
4	Nitrite	5.904	0.003	0.0183	n.a.	n.a.	n.a.
5	Bromide	7.241	0.009	0.1728	n.a.	n.a.	n.a.
6	Nitrate	8.207	0.004	0.0232	n.a.	n.a.	n.a.
9	Sulfate	11.331	2.179	21.7873	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW02-060123 DL
Collect Date:	06/01/23 1230	LAB Sample ID:	22306021002 DL
Matrix:	Water	Instrument ID:	IC4
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	5
Prep Date:	NA	Analysis Date:	06/07/23 1445
Prep Batch:	NA	Analytical Batch:	767223
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Sulfate	23.0	mg/L		0.500	0.750	1.00

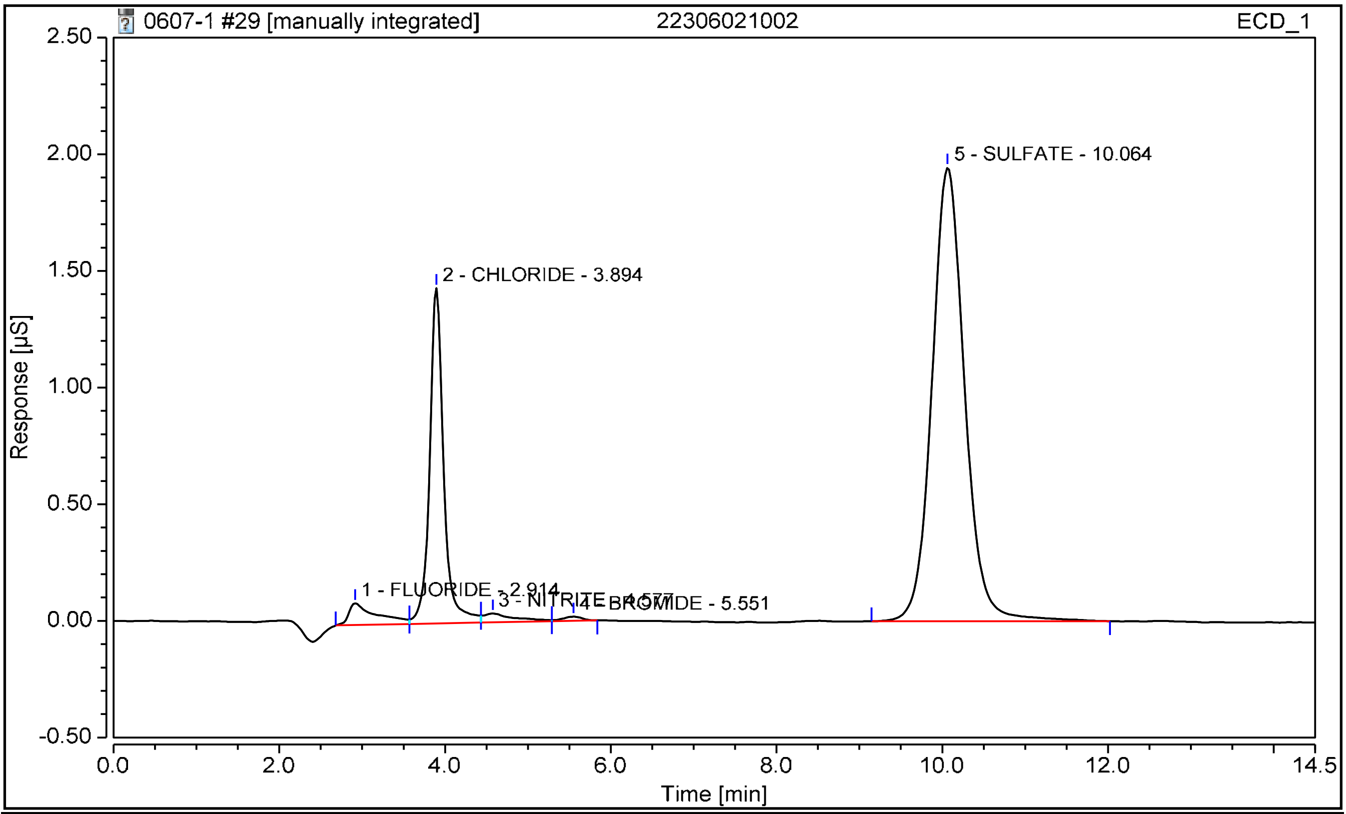
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021002**	*Operator:*	**CS**
Vial Number:	**15**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**5**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/07/2023 14:45**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	FLUORIDE	2.914	0.037	0.7785	n.a.	n.a.	n.a.
2	CHLORIDE	3.894	0.282	4.8321	n.a.	n.a.	n.a.
3	NITRITE	4.577	0.017	0.1851	n.a.	n.a.	n.a.
4	BROMIDE	5.551	0.005	0.2444	n.a.	n.a.	n.a.
n.a.	NITRATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
5	SULFATE	10.064	0.889	23.0430	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123
Collect Date:	06/01/23 0935	LAB Sample ID:	22306021003
Matrix:	Water	Instrument ID:	BRWA13
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/02/23 1207
Prep Batch:	NA	Analytical Batch:	767085
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	RESULT	UNITS	Q	DL	LOD	LOQ
Chloride	2.37	mg/L		0.050	0.100	0.200
Nitrate	0.158	mg/L	J	0.100	0.150	0.200
Nitrite	0.150	mg/L	U	0.100	0.150	0.200

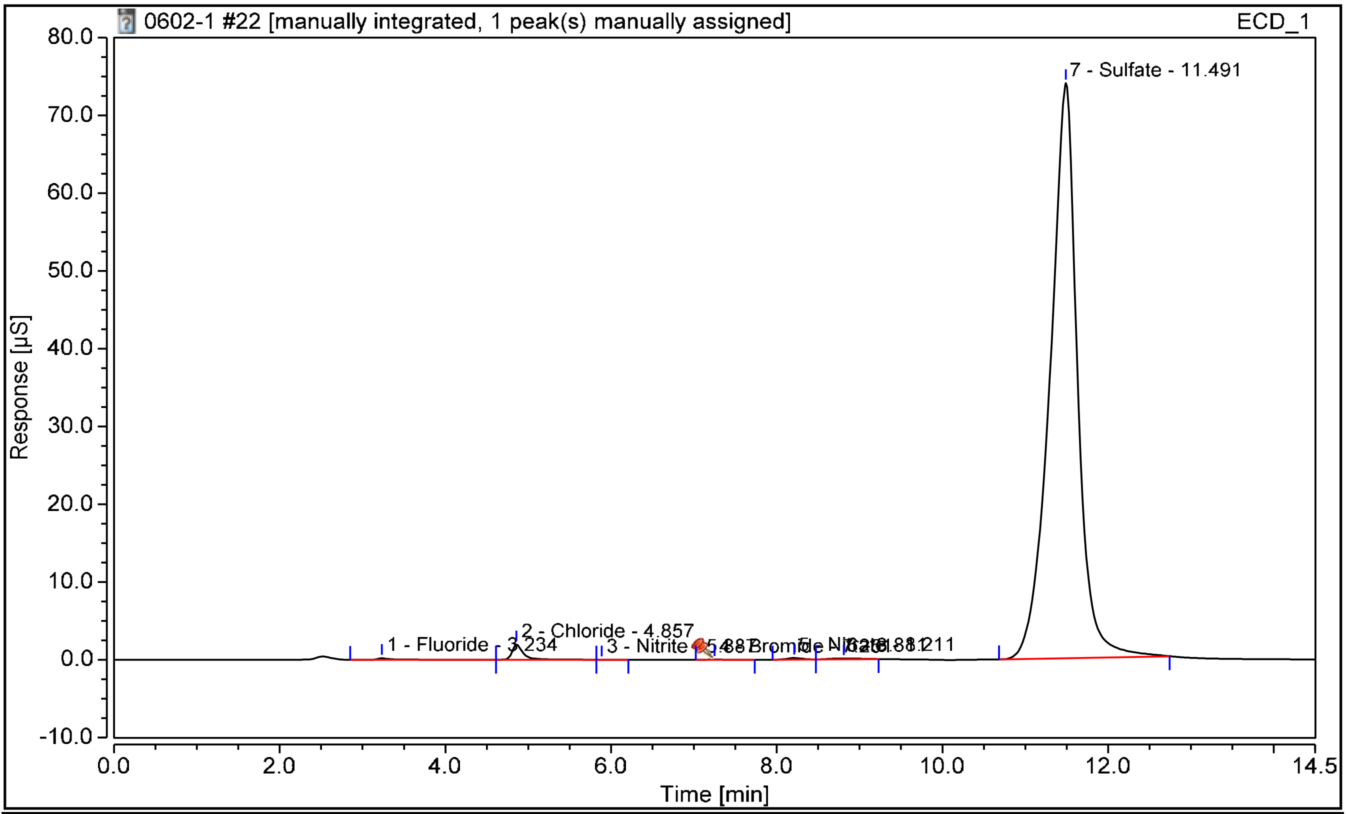
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021003**	*Run Time (min):*	**14.50**
Vial Number:	**8**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 12:07**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.234	0.051	0.2588	n.a.	n.a.	n.a.
2	Chloride	4.857	0.309	2.3712	n.a.	n.a.	n.a.
3	Nitrite	5.887	0.000	0.0094	n.a.	n.a.	n.a.
4	Bromide	7.251	0.005	0.1070	n.a.	n.a.	n.a.
5	Nitrate	8.211	0.045	0.1575	n.a.	n.a.	n.a.
7	Sulfate	11.491	28.368	190.1039	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123 DL
Collect Date:	06/01/23 0935	LAB Sample ID:	22306021003 DL
Matrix:	Water	Instrument ID:	IC4
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	50
Prep Date:	NA	Analysis Date:	06/08/23 1100
Prep Batch:	NA	Analytical Batch:	767281
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Sulfate	225	mg/L		5.00	7.50	10.0

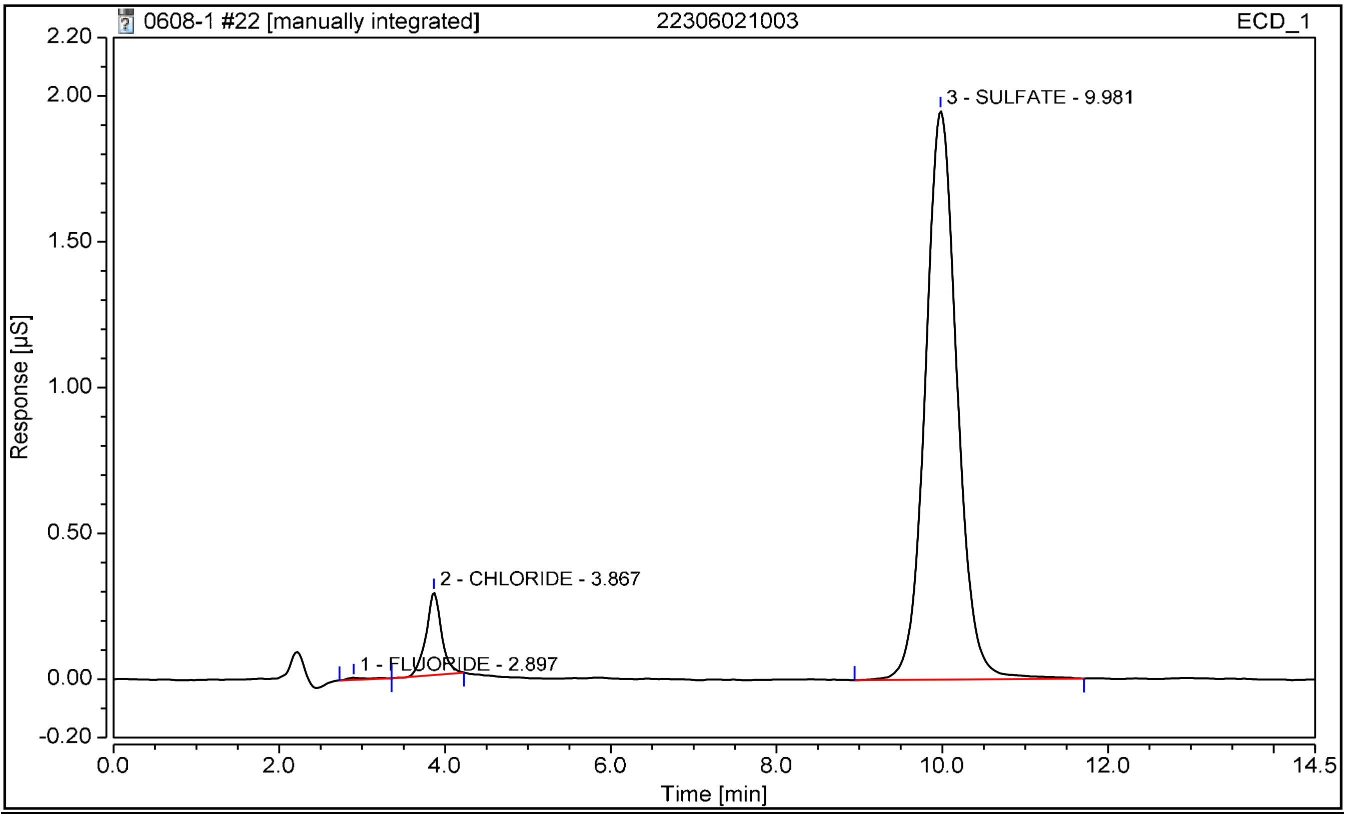
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021003**	*Operator:*	**CS**
Vial Number:	**8**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**50**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/08/2023 11:00**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	FLUORIDE	2.897	0.002	1.9995	n.a.	n.a.	n.a.
2	CHLORIDE	3.867	0.061	5.3827	n.a.	n.a.	n.a.
n.a.	NITRITE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	BROMIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	NITRATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
3	SULFATE	9.981	0.868	225.3484	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW05-060123
Collect Date:	06/01/23 0910	LAB Sample ID:	22306021004
Matrix:	Water	Instrument ID:	BRWA13
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/02/23 1150
Prep Batch:	NA	Analytical Batch:	767085
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Chloride	3.50	mg/L		0.050	0.100	0.200
Nitrate	0.186	mg/L	J	0.100	0.150	0.200
Nitrite	0.150	mg/L	U	0.100	0.150	0.200

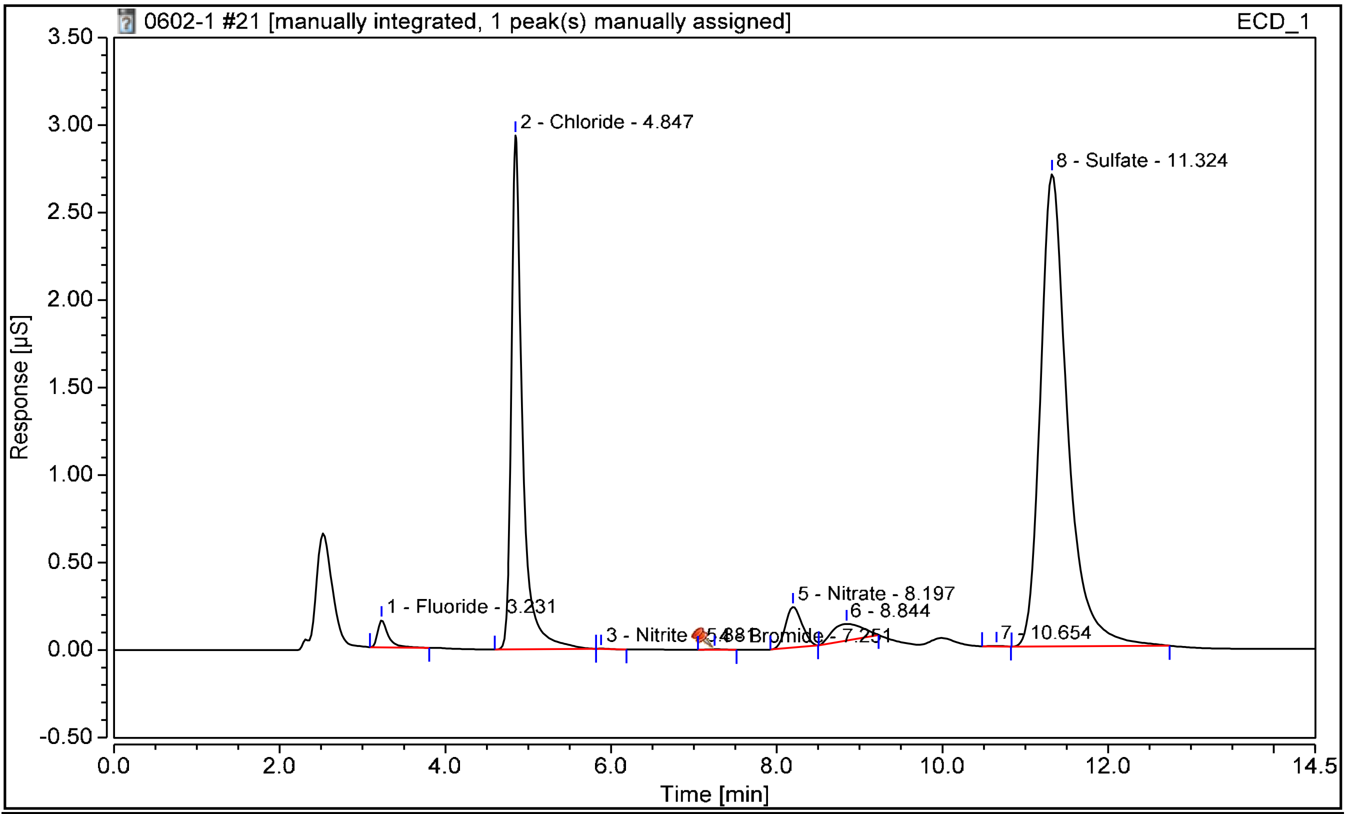
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021004**	*Run Time (min):*	**14.50**
Vial Number:	**7**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 11:50**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.231	0.023	0.1219	n.a.	n.a.	n.a.
2	Chloride	4.847	0.459	3.4993	n.a.	n.a.	n.a.
3	Nitrite	5.881	0.000	0.0091	n.a.	n.a.	n.a.
4	Bromide	7.251	0.001	0.0166	n.a.	n.a.	n.a.
5	Nitrate	8.197	0.054	0.1861	n.a.	n.a.	n.a.
8	Sulfate	11.324	1.029	10.6340	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW05-060123 DL
Collect Date:	06/01/23 0910	LAB Sample ID:	22306021004 DL
Matrix:	Water	Instrument ID:	IC4
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	2
Prep Date:	NA	Analysis Date:	06/07/23 1503
Prep Batch:	NA	Analytical Batch:	767223
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Sulfate	10.9	mg/L		0.200	0.300	0.400

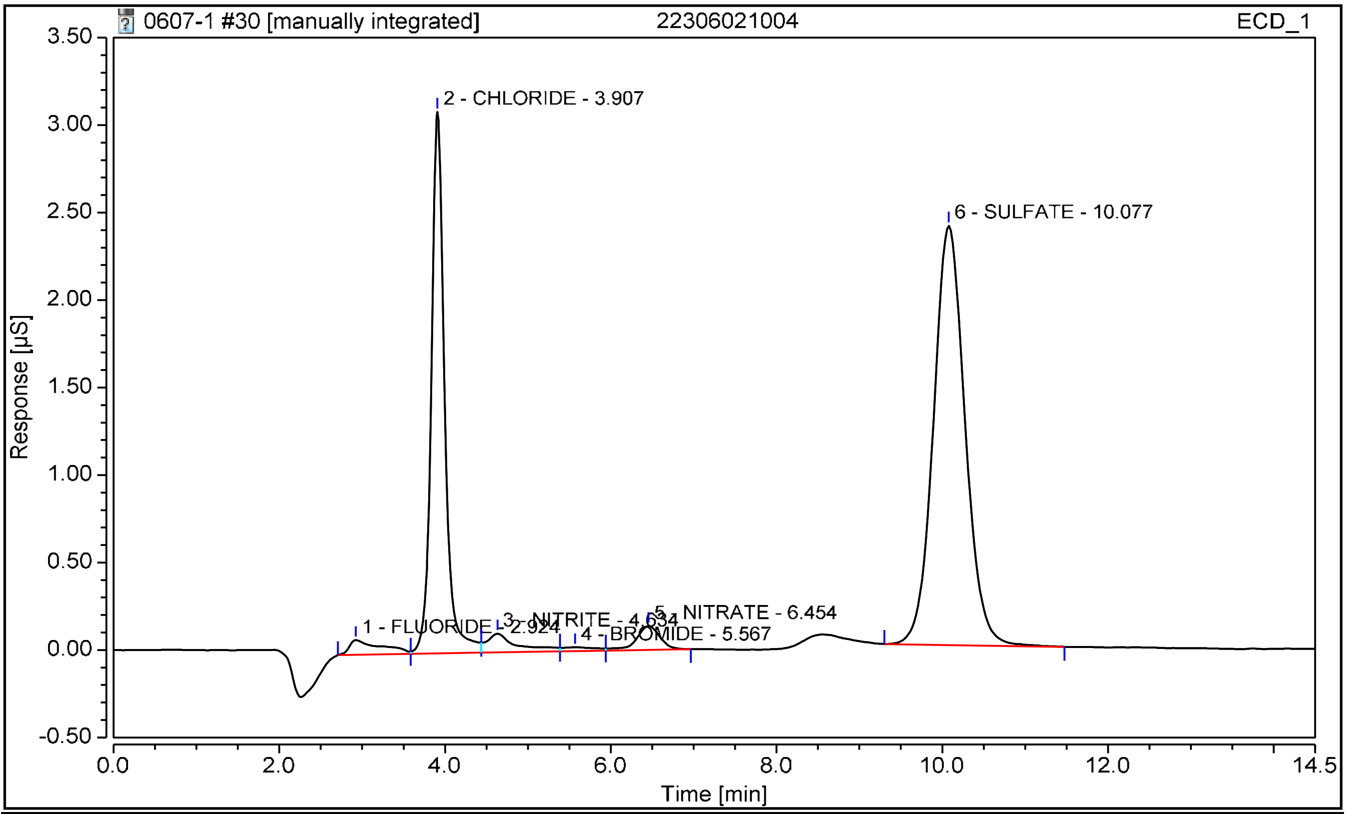
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021004**	*Operator:*	**CS**
Vial Number:	**16**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**2**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/07/2023 15:03**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	FLUORIDE	2.924	0.038	0.3212	n.a.	n.a.	n.a.
2	CHLORIDE	3.907	0.577	4.1571	n.a.	n.a.	n.a.
3	NITRITE	4.634	0.046	0.1688	n.a.	n.a.	n.a.
4	BROMIDE	5.567	0.010	0.1903	n.a.	n.a.	n.a.
5	NITRATE	6.454	0.045	0.2994	n.a.	n.a.	n.a.
6	SULFATE	10.077	1.058	10.9299	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW06-060123
Collect Date:	06/01/23 1100	LAB Sample ID:	22306021005
Matrix:	Water	Instrument ID:	BRWA13
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/02/23 1259
Prep Batch:	NA	Analytical Batch:	767085
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Chloride	1.50	mg/L		0.050	0.100	0.200
Nitrate	0.135	mg/L	J	0.100	0.150	0.200
Nitrite	0.150	mg/L	U	0.100	0.150	0.200

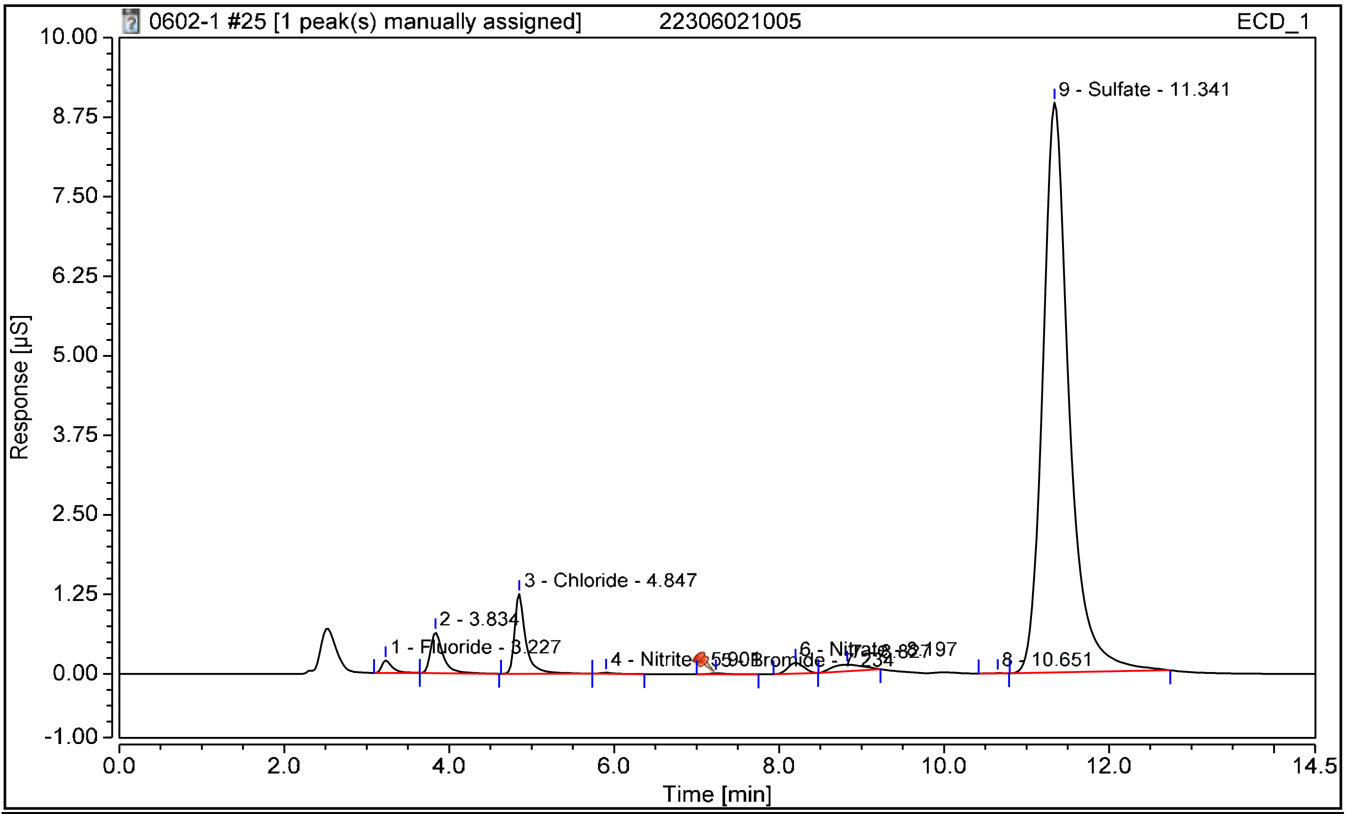
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021005**	*Run Time (min):*	**14.50**
Vial Number:	**11**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 12:59**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.227	0.029	0.1524	n.a.	n.a.	n.a.
3	Chloride	4.847	0.195	1.5024	n.a.	n.a.	n.a.
4	Nitrite	5.901	0.003	0.0199	n.a.	n.a.	n.a.
5	Bromide	7.234	0.004	0.0697	n.a.	n.a.	n.a.
6	Nitrate	8.197	0.038	0.1349	n.a.	n.a.	n.a.
9	Sulfate	11.341	3.320	32.2087	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW06-060123 DL
Collect Date:	06/01/23 1100	LAB Sample ID:	22306021005 DL
Matrix:	Water	Instrument ID:	IC4
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	5
Prep Date:	NA	Analysis Date:	06/07/23 1612
Prep Batch:	NA	Analytical Batch:	767223
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Sulfate	33.5	mg/L		0.500	0.750	1.00

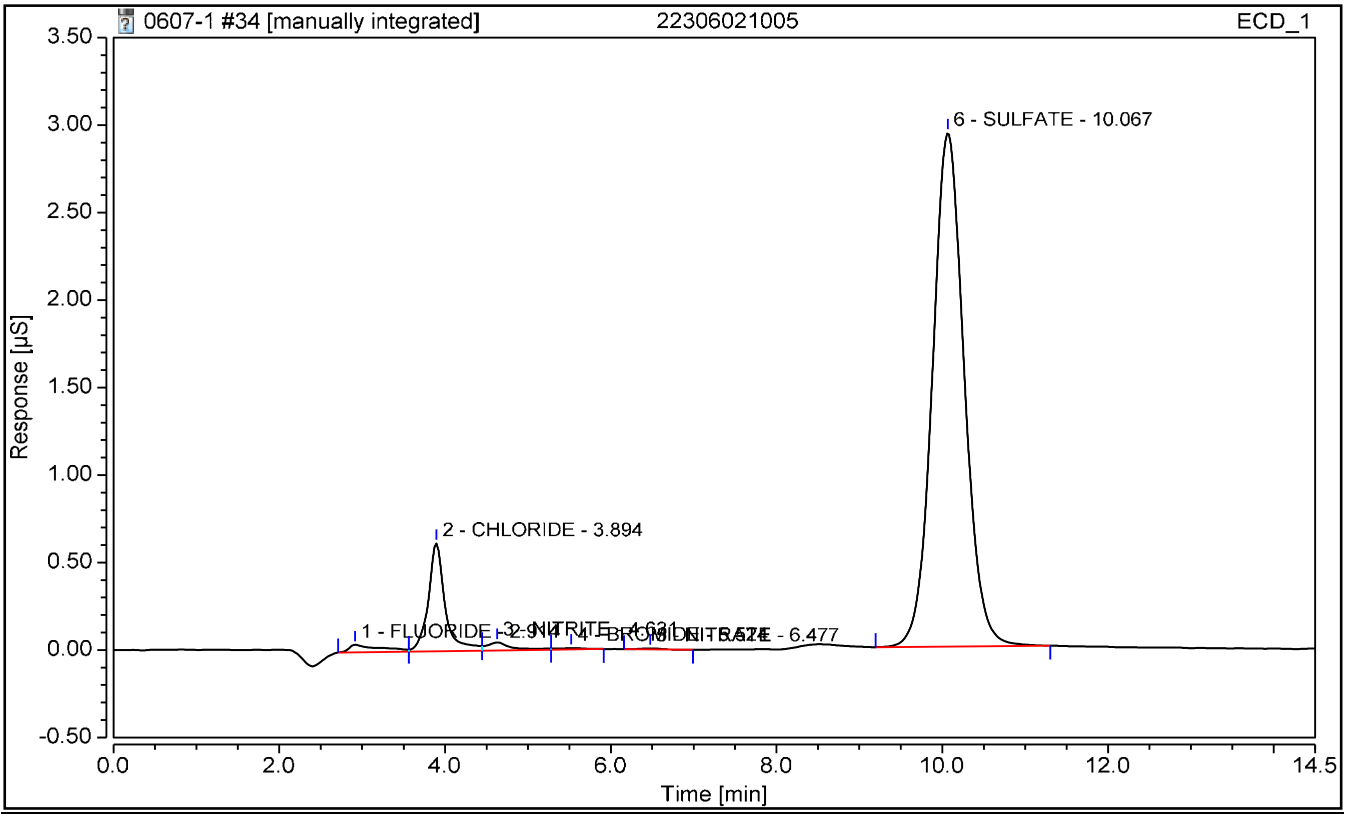
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021005**	*Operator:*	**CS**
Vial Number:	**20**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**5**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/07/2023 16:12**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	FLUORIDE	2.914	0.019	0.4800	n.a.	n.a.	n.a.
2	CHLORIDE	3.894	0.145	2.1934	n.a.	n.a.	n.a.
3	NITRITE	4.631	0.016	0.1765	n.a.	n.a.	n.a.
4	BROMIDE	5.524	0.003	0.1459	n.a.	n.a.	n.a.
5	NITRATE	6.477	0.001	0.4122	n.a.	n.a.	n.a.
6	SULFATE	10.067	1.305	33.4774	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123MS
Collect Date:	06/01/23 0955	LAB Sample ID:	22306021006
Matrix:	Water	Instrument ID:	BRWA13
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/02/23 1225
Prep Batch:	NA	Analytical Batch:	767085
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Chloride	4.79	mg/L		0.050	0.100	0.200
Nitrate	2.35	mg/L		0.100	0.150	0.200
Nitrite	2.44	mg/L		0.100	0.150	0.200

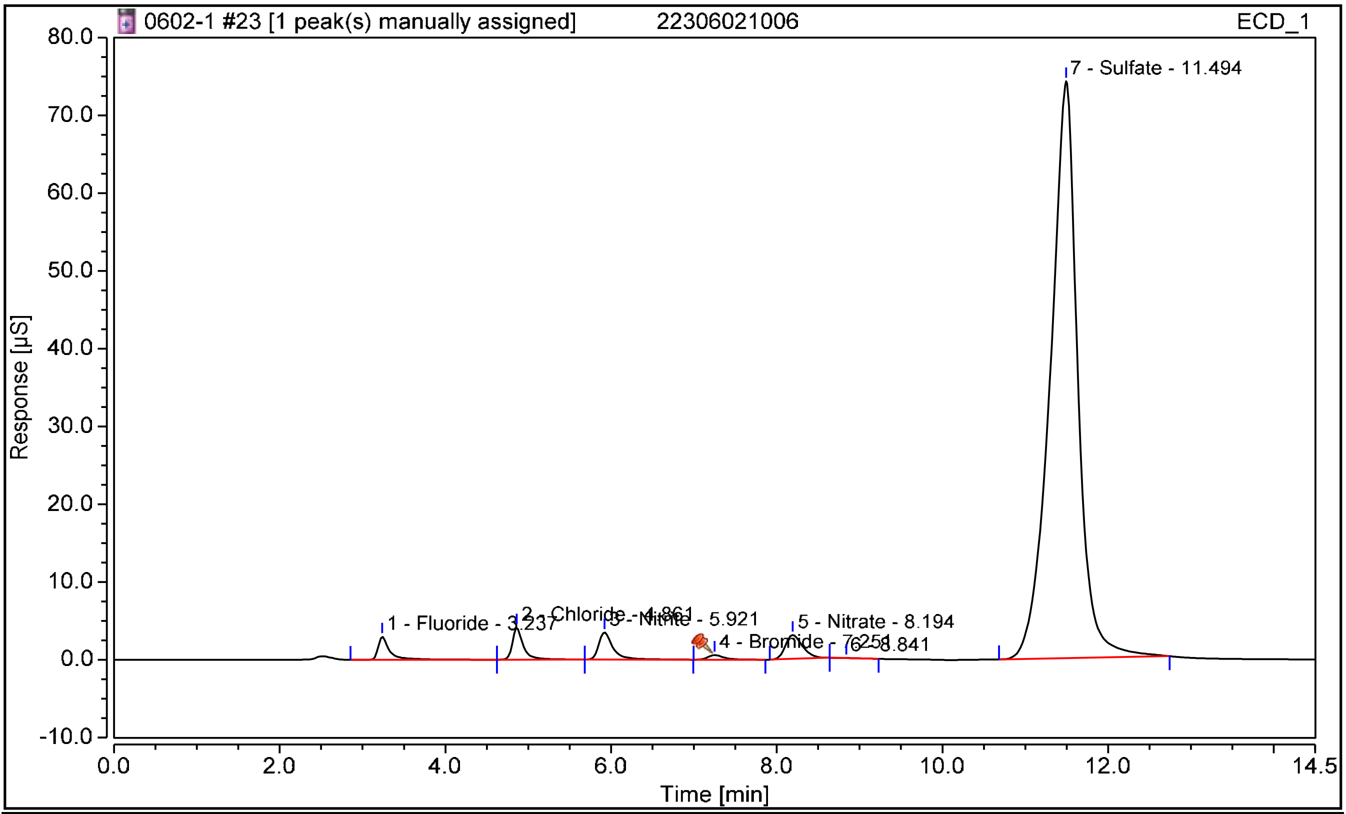
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021006**	*Run Time (min):*	**14.50**
Vial Number:	**9**	*Injection Volume:*	**5000.00**
Injection Type:	**Spiked**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 12:25**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.237	0.498	2.4483	n.a.	n.a.	n.a.
2	Chloride	4.861	0.634	4.7857	n.a.	n.a.	n.a.
3	Nitrite	5.921	0.708	2.4377	n.a.	n.a.	n.a.
4	Bromide	7.251	0.132	2.5472	n.a.	n.a.	n.a.
5	Nitrate	8.194	0.739	2.3540	n.a.	n.a.	n.a.
7	Sulfate	11.494	28.496	190.7269	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123MS DL
Collect Date:	06/01/23 0955	LAB Sample ID:	22306021006 DL
Matrix:	Water	Instrument ID:	IC4
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	50
Prep Date:	NA	Analysis Date:	06/08/23 1117
Prep Batch:	NA	Analytical Batch:	767281
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Sulfate	347	mg/L		5.00	7.50	10.0

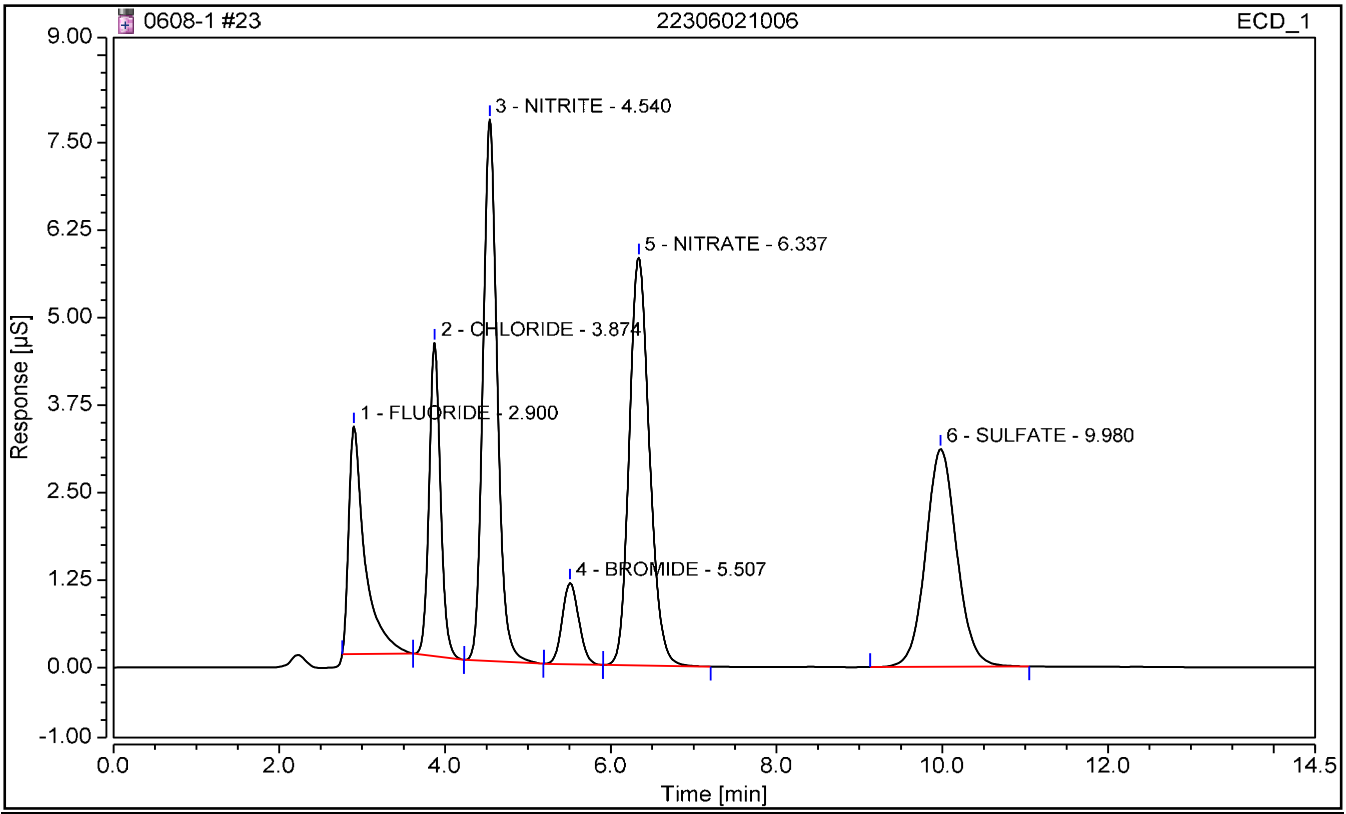
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021006**	*Operator:*	**CS**
Vial Number:	**9**	*Injection Volume:*	**5000.00**
Injection Type:	**Spiked**	*Dilution Factor:*	**50**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/08/2023 11:17**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	FLUORIDE	2.900	0.732	118.7787	n.a.	n.a.	n.a.
2	CHLORIDE	3.874	0.712	128.6810	n.a.	n.a.	n.a.
3	NITRITE	4.540	1.552	124.1540	n.a.	n.a.	n.a.
4	BROMIDE	5.507	0.271	124.7287	n.a.	n.a.	n.a.
5	NITRATE	6.337	1.646	124.5049	n.a.	n.a.	n.a.
6	SULFATE	9.980	1.354	346.8124	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123MSD
Collect Date:	06/01/23 1015	LAB Sample ID:	22306021007
Matrix:	Water	Instrument ID:	BRWA13
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/02/23 1242
Prep Batch:	NA	Analytical Batch:	767085
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	RESULT	UNITS	Q	DL	LOD	LOQ
Chloride	4.78	mg/L		0.050	0.100	0.200
Nitrate	2.36	mg/L		0.100	0.150	0.200
Nitrite	2.44	mg/L		0.100	0.150	0.200

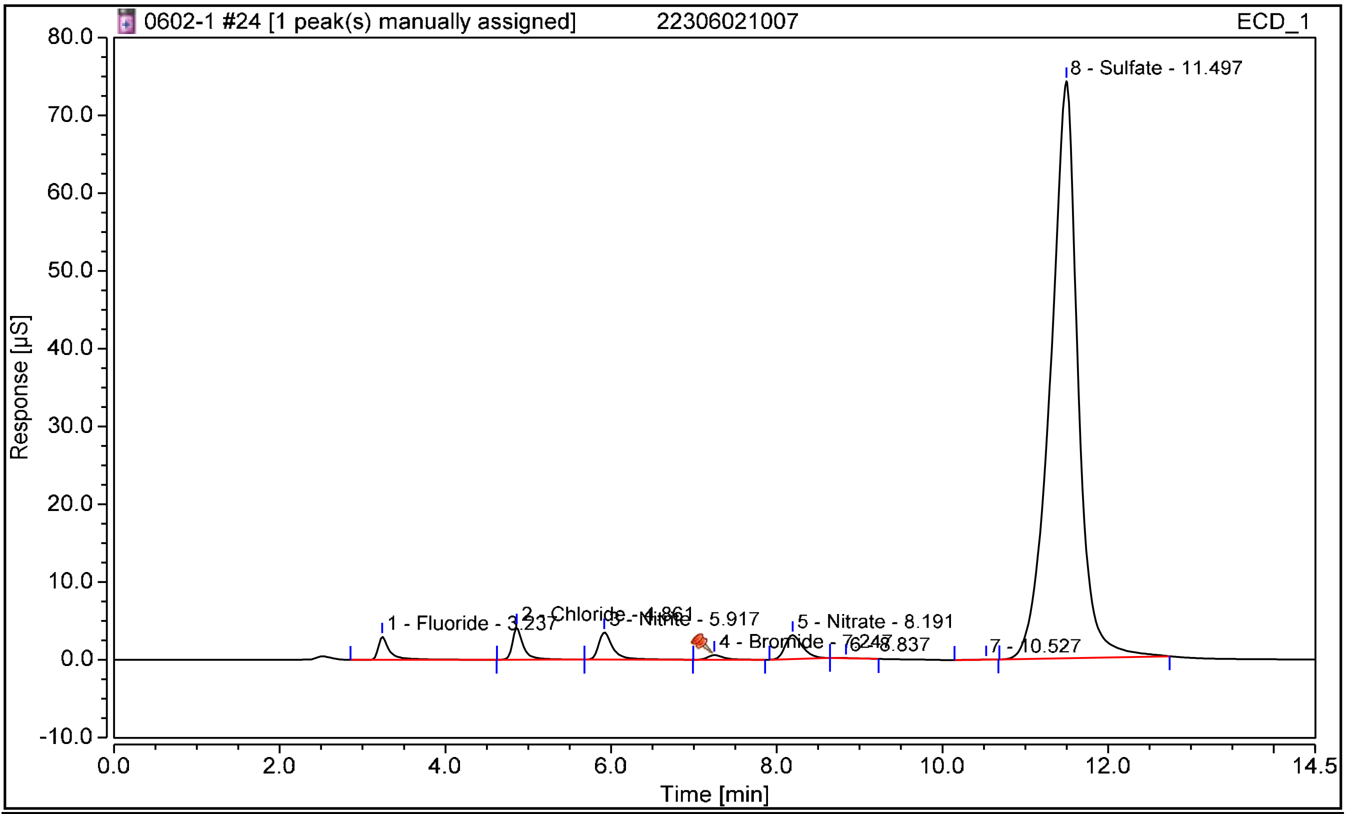
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021007**	*Run Time (min):*	**14.50**
Vial Number:	**10**	*Injection Volume:*	**5000.00**
Injection Type:	**Spiked**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 12:42**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.237	0.496	2.4362	n.a.	n.a.	n.a.
2	Chloride	4.861	0.633	4.7814	n.a.	n.a.	n.a.
3	Nitrite	5.917	0.709	2.4405	n.a.	n.a.	n.a.
4	Bromide	7.247	0.132	2.5466	n.a.	n.a.	n.a.
5	Nitrate	8.191	0.741	2.3594	n.a.	n.a.	n.a.
8	Sulfate	11.497	28.484	190.6701	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123MSD DL
Collect Date:	06/01/23 1015	LAB Sample ID:	22306021007 DL
Matrix:	Water	Instrument ID:	IC4
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	50
Prep Date:	NA	Analysis Date:	06/08/23 1135
Prep Batch:	NA	Analytical Batch:	767281
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Sulfate	348	mg/L		5.00	7.50	10.0

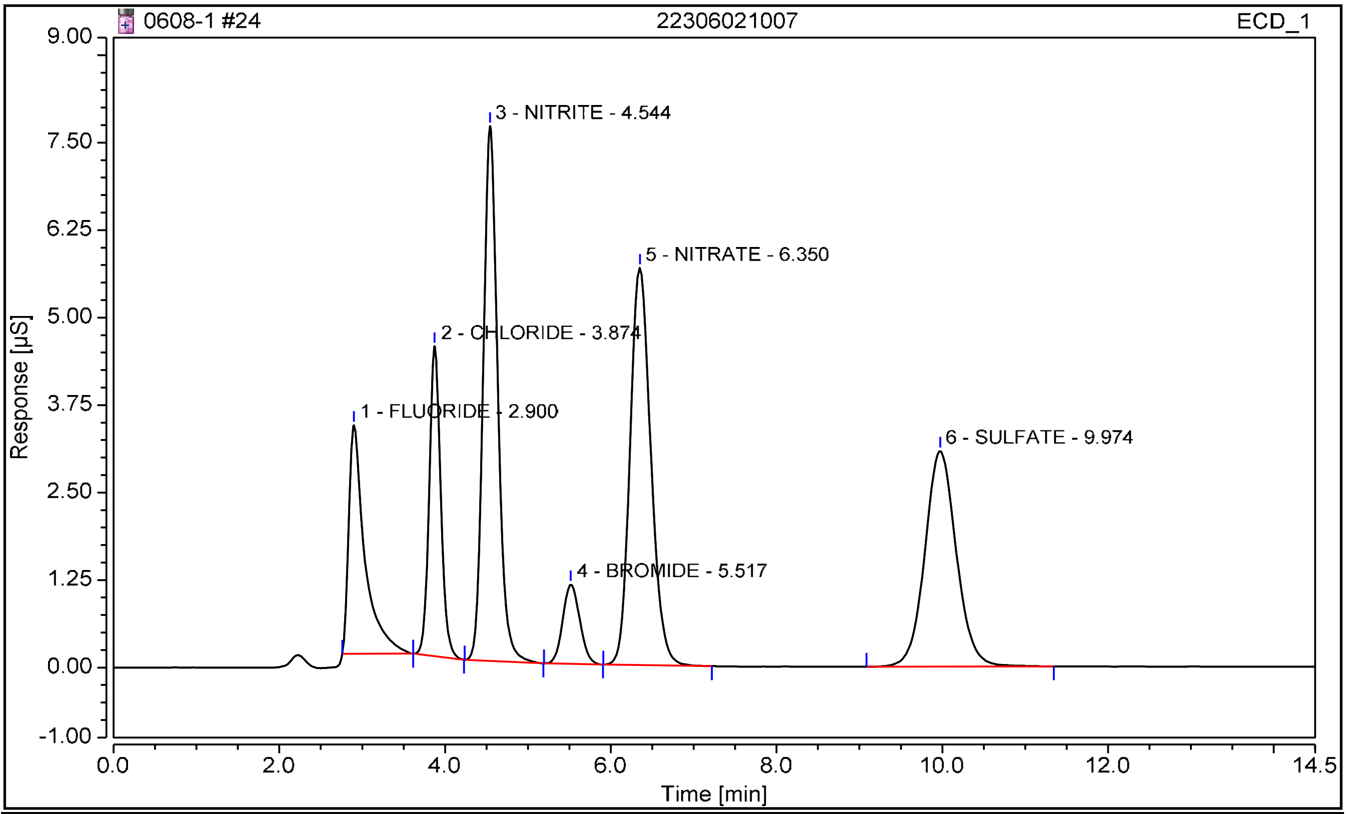
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021007**	*Operator:*	**CS**
Vial Number:	**10**	*Injection Volume:*	**5000.00**
Injection Type:	**Spiked**	*Dilution Factor:*	**50**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/08/2023 11:35**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	FLUORIDE	2.900	0.734	119.0742	n.a.	n.a.	n.a.
2	CHLORIDE	3.874	0.712	128.6777	n.a.	n.a.	n.a.
3	NITRITE	4.544	1.551	124.1067	n.a.	n.a.	n.a.
4	BROMIDE	5.517	0.271	124.6340	n.a.	n.a.	n.a.
5	NITRATE	6.350	1.646	124.5090	n.a.	n.a.	n.a.
6	SULFATE	9.974	1.358	347.8637	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW-806-060123
Collect Date:	06/01/23 1100	LAB Sample ID:	22306021008
Matrix:	Water	Instrument ID:	BRWA13
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	1
Prep Date:	NA	Analysis Date:	06/02/23 1317
Prep Batch:	NA	Analytical Batch:	767085
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Chloride	2.11	mg/L		0.050	0.100	0.200
Nitrate	0.150	mg/L	U	0.100	0.150	0.200
Nitrite	0.150	mg/L	U	0.100	0.150	0.200

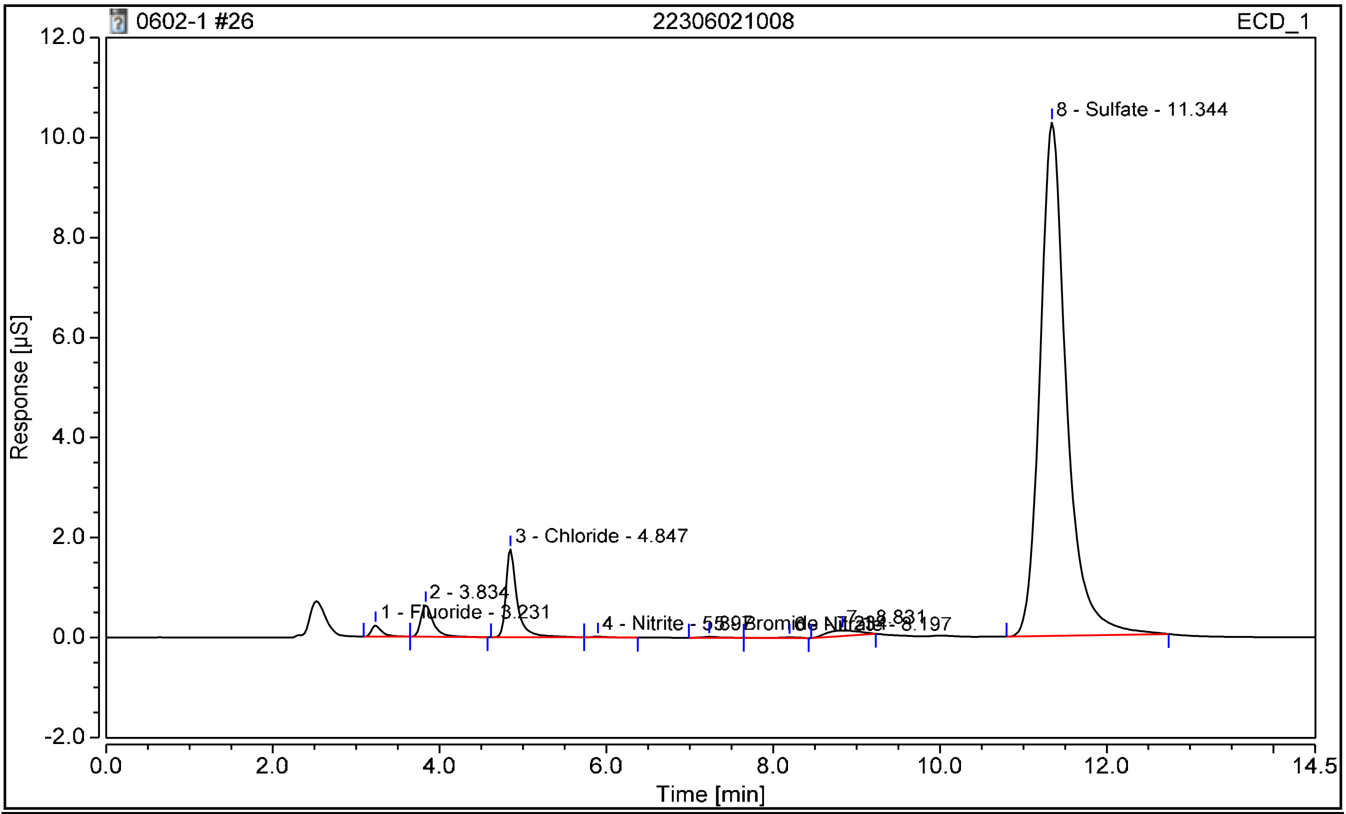
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021008**	*Run Time (min):*	**14.50**
Vial Number:	**12**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 13:17**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.231	0.033	0.1700	n.a.	n.a.	n.a.
3	Chloride	4.847	0.274	2.1082	n.a.	n.a.	n.a.
4	Nitrite	5.897	0.003	0.0185	n.a.	n.a.	n.a.
5	Bromide	7.234	0.004	0.0861	n.a.	n.a.	n.a.
6	Nitrate	8.197	0.004	0.0225	n.a.	n.a.	n.a.
8	Sulfate	11.344	3.787	36.3212	n.a.	n.a.	n.a.

I - GENERAL CHEMISTRY ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW-806-060123 DL
Collect Date:	06/01/23 1100	LAB Sample ID:	22306021008 DL
Matrix:	Water	Instrument ID:	IC4
% Solids:	NA	Analyst:	CS
Sample Amt:	NA	Lab File ID:	NA
Prep Vol.:	NA	Dilution Factor:	5
Prep Date:	NA	Analysis Date:	06/07/23 1520
Prep Batch:	NA	Analytical Batch:	767223
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Sulfate	37.9	mg/L		0.500	0.750	1.00

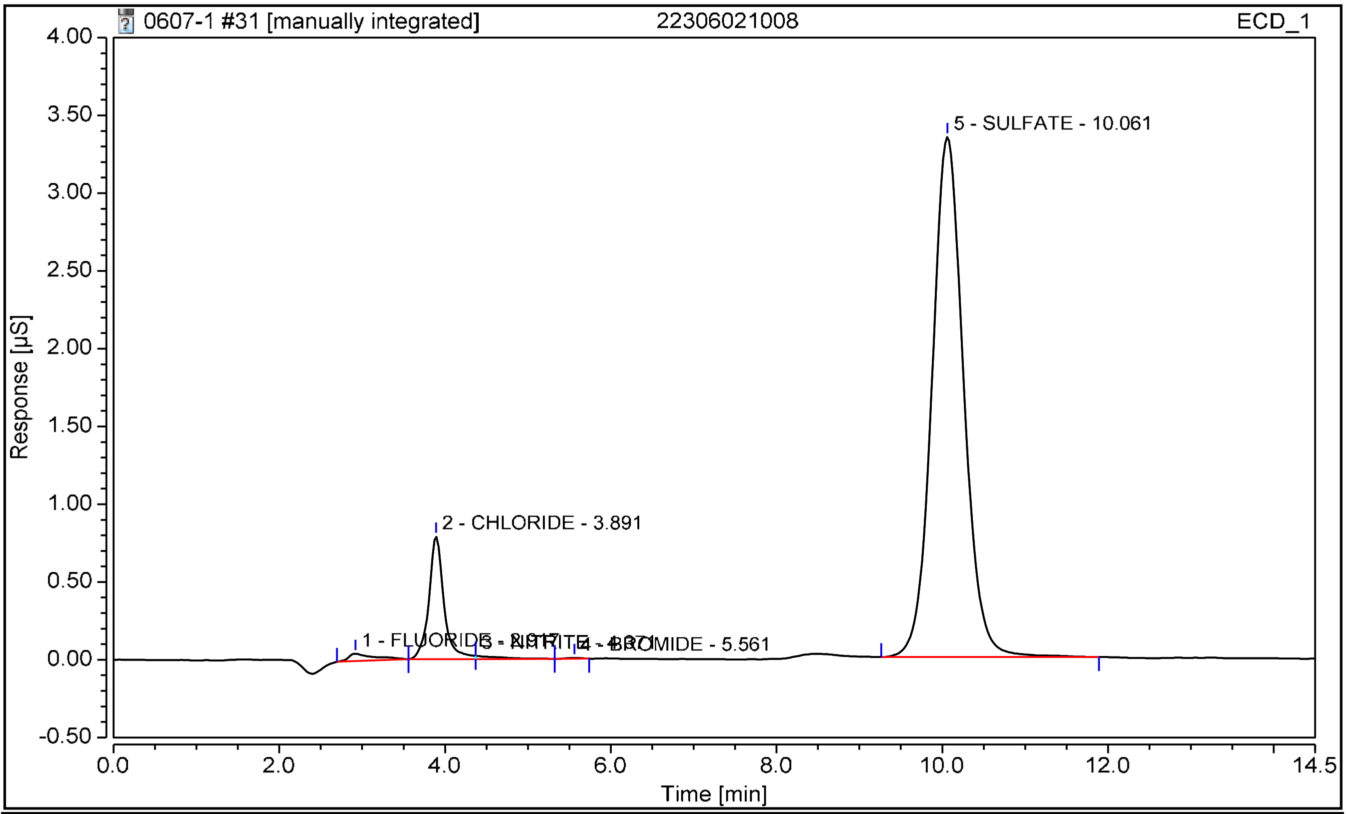
FORM I - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021008**	*Operator:*	**CS**
Vial Number:	**17**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**5**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/07/2023 15:20**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	FLUORIDE	2.917	0.017	0.4597	n.a.	n.a.	n.a.
2	CHLORIDE	3.891	0.166	2.5992	n.a.	n.a.	n.a.
3	NITRITE	4.371	0.007	0.1046	n.a.	n.a.	n.a.
4	BROMIDE	5.561	0.001	0.0676	n.a.	n.a.	n.a.
n.a.	NITRATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
5	SULFATE	10.061	1.483	37.8591	n.a.	n.a.	n.a.

EPA 9056A

Form II

Calibration Verifications

II - CONTINUING CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	BRWA13
Analysis Date:	06/02/23 1006	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767085

ANALYTE	UNITS	TRUE	FOUND	% REC	LCL	UCL	Q
Chloride	mg/L	2.50	2.47	98.8	90	110	
Nitrate	mg/L	2.50	2.54	102	90	110	
Nitrite	mg/L	2.50	2.50	99.9	90	110	
Sulfate	mg/L	2.50	2.50	100	90	110	

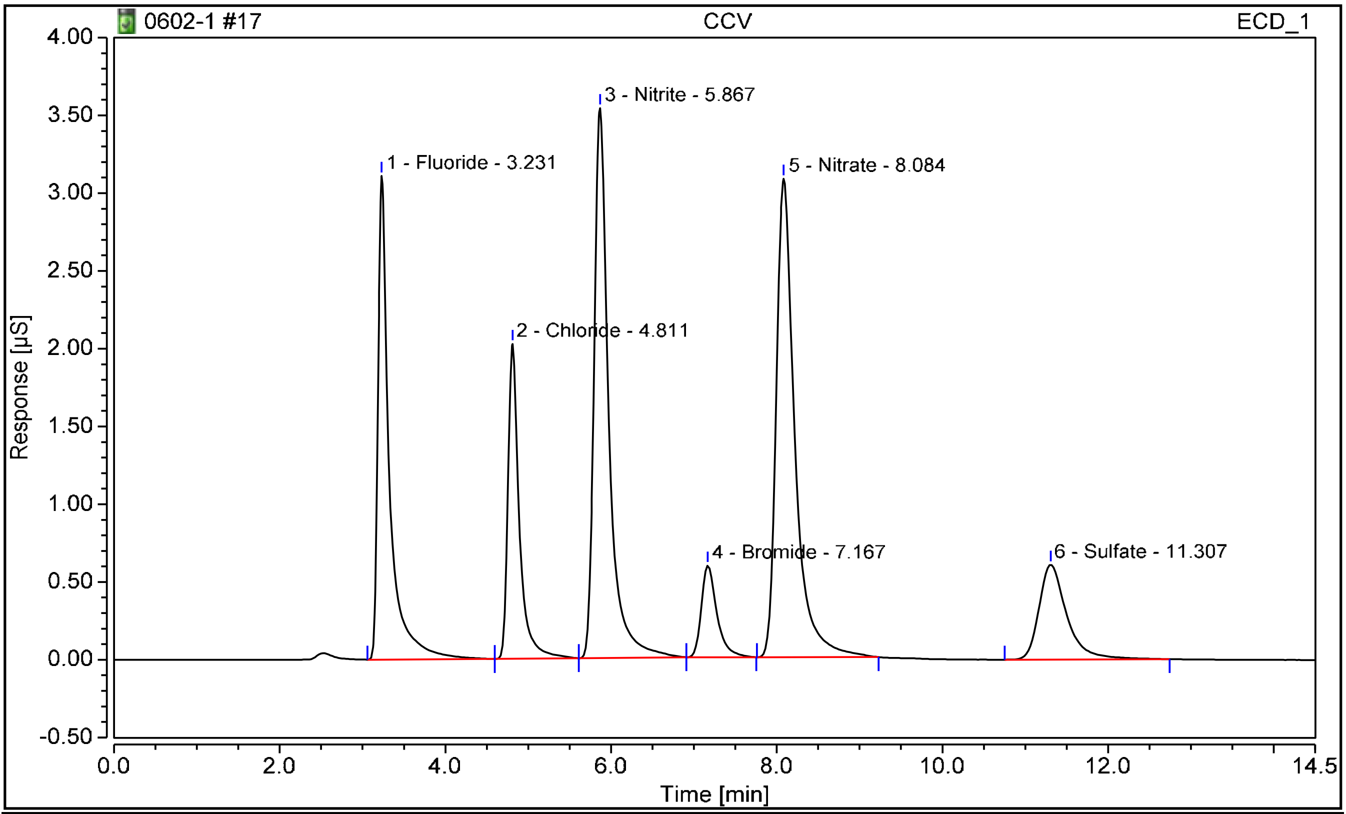
FORM II - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**CCV**	*Run Time (min):*	**14.50**
Vial Number:	**3**	*Injection Volume:*	**5000.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:	**CCV**	*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 10:06**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.231	0.501	2.4604	-247.5396	250.0000	0.98
2	Chloride	4.811	0.322	2.4687	-247.5313	250.0000	0.99
3	Nitrite	5.867	0.726	2.4979	-247.5021	250.0000	1.00
4	Bromide	7.167	0.129	2.4832	-247.5168	250.0000	0.99
5	Nitrate	8.084	0.800	2.5396	-247.4604	250.0000	1.02
6	Sulfate	11.307	0.237	2.5040	-247.4960	250.0000	1.00

II - CONTINUING CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	BRWA13
Analysis Date:	06/02/23 1711	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767085

ANALYTE	UNITS	TRUE	FOUND	% REC	LCL	UCL	Q
Chloride	mg/L	2.50	2.48	99.3	90	110	
Nitrate	mg/L	2.50	2.53	101	90	110	
Nitrite	mg/L	2.50	2.51	100	90	110	
Sulfate	mg/L	2.50	2.51	100	90	110	

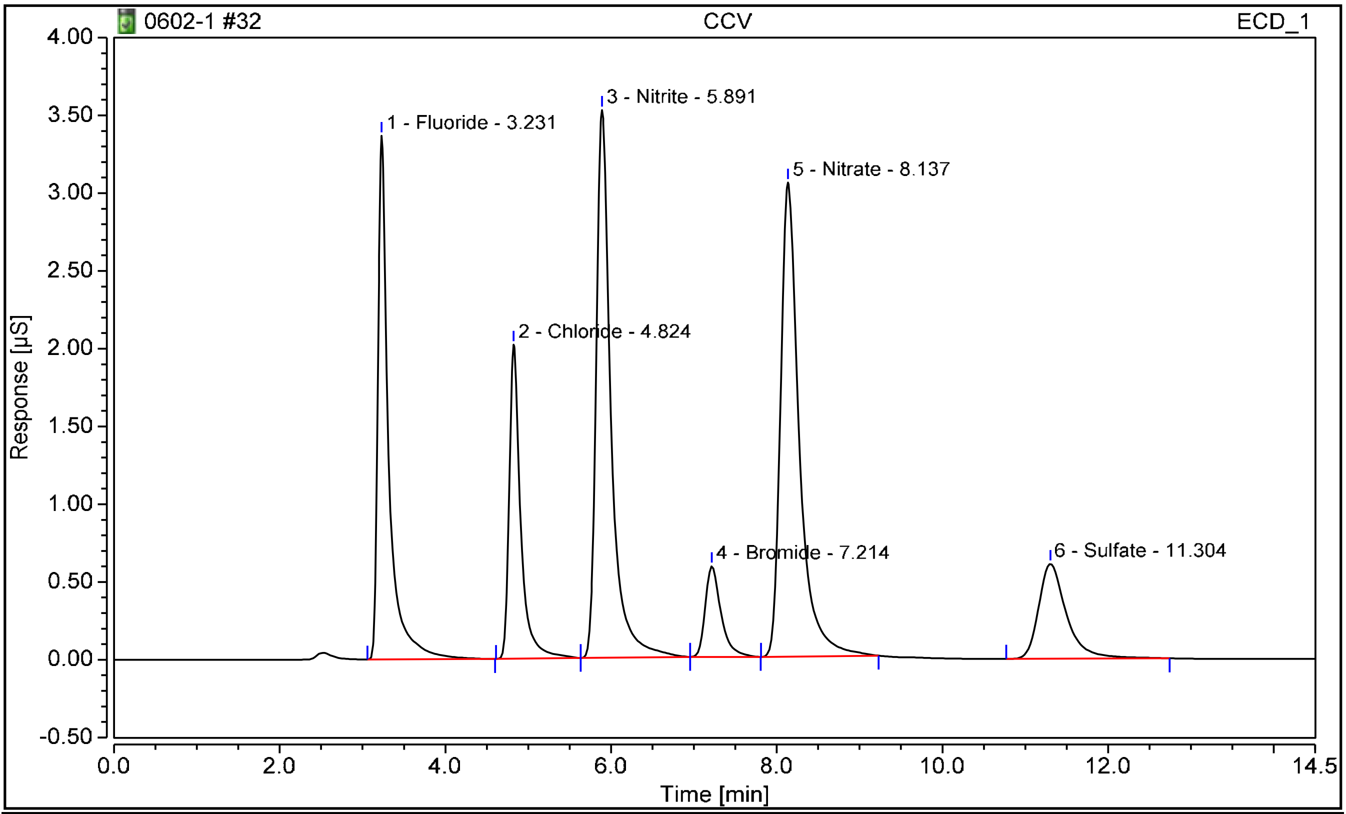
FORM II - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**CCV**	*Run Time (min):*	**14.50**
Vial Number:	**18**	*Injection Volume:*	**5000.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:	**CCV**	*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 17:11**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.231	0.512	2.5129	-247.4871	250.0000	1.01
2	Chloride	4.824	0.324	2.4827	-247.5173	250.0000	0.99
3	Nitrite	5.891	0.730	2.5106	-247.4894	250.0000	1.00
4	Bromide	7.214	0.129	2.4914	-247.5086	250.0000	1.00
5	Nitrate	8.137	0.798	2.5327	-247.4673	250.0000	1.01
6	Sulfate	11.304	0.237	2.5049	-247.4951	250.0000	1.00

II - CONTINUING CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	IC4
Analysis Date:	06/07/23 1018	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767223

ANALYTE	UNITS	TRUE	FOUND	% REC	LCL	UCL	Q
Sulfate	mg/L	2.50	2.47	98.7	90	110	

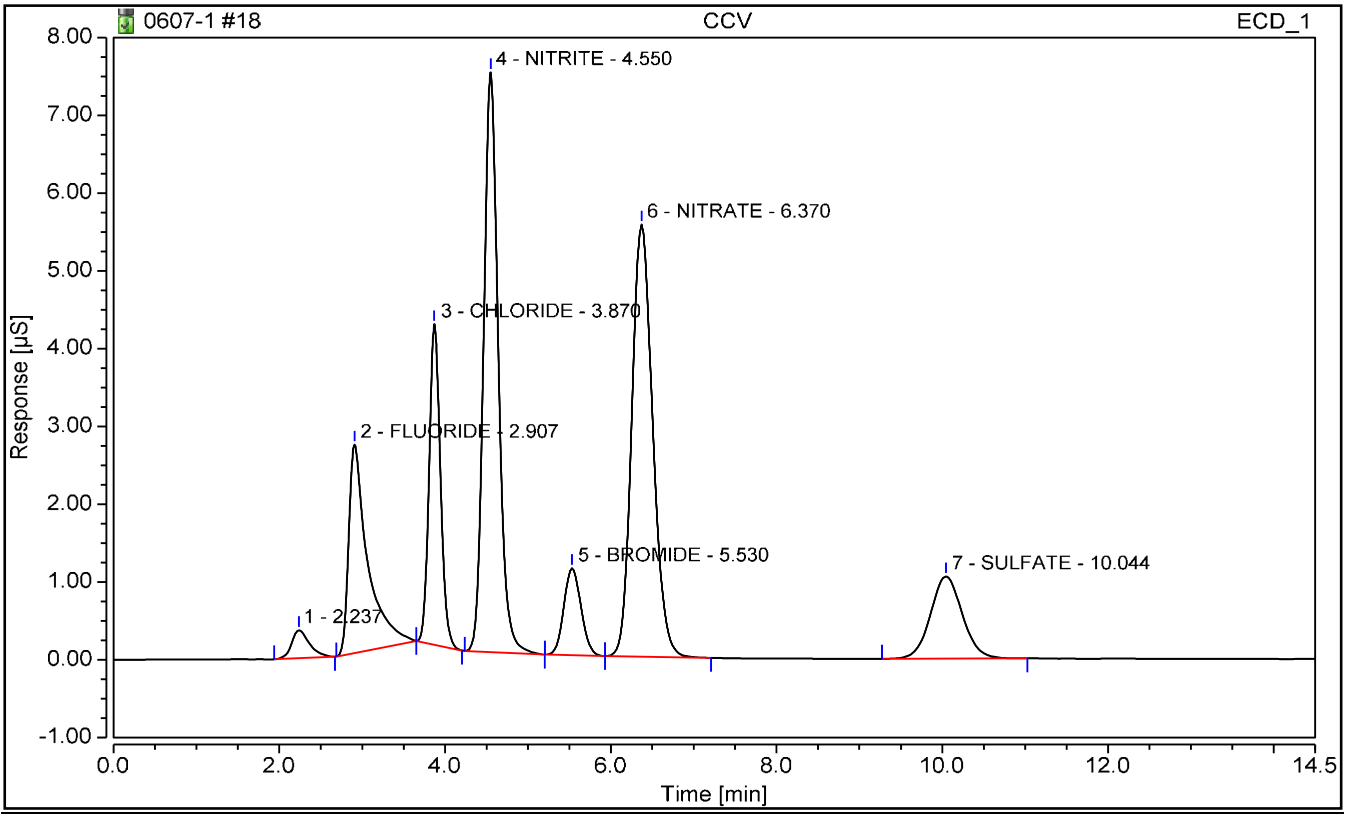
FORM II - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**CCV**	*Operator:*	**CS**
Vial Number:	**4**	*Injection Volume:*	**5000.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:	**CCV**	*Instrument Method:*	**TEST 1**
Injection Time:	**06/07/2023 10:18**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
2	FLUORIDE	2.907	0.712	2.3156	-0.1844	2.5000	92.62
3	CHLORIDE	3.870	0.650	2.3463	-0.1537	2.5000	93.85
4	NITRITE	4.550	1.525	2.4409	-0.0591	2.5000	97.64
5	BROMIDE	5.530	0.269	2.4753	-0.0247	2.5000	99.01
6	NITRATE	6.370	1.625	2.4603	-0.0397	2.5000	98.41
7	SULFATE	10.044	0.471	2.4673	-0.0327	2.5000	98.69

II - CONTINUING CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	IC4
Analysis Date:	06/07/23 1555	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767223

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*% REC*	*LCL*	*UCL*	*Q*
Sulfate	mg/L	2.50	2.50	100	90	110	

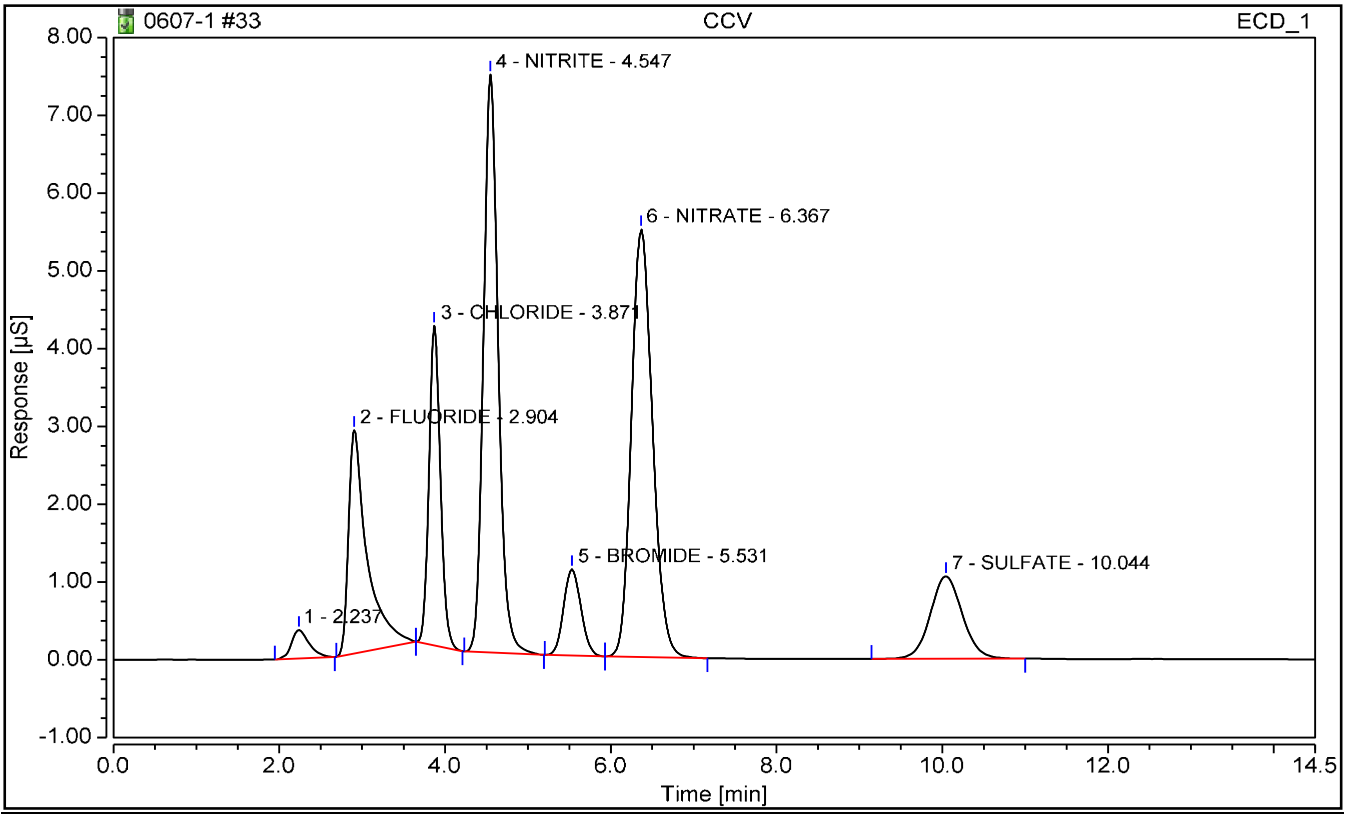
FORM II - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**CCV**	*Operator:*	**CS**
Vial Number:	**19**	*Injection Volume:*	**5000.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:	**CCV**	*Instrument Method:*	**TEST 1**
Injection Time:	**06/07/2023 15:55**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
2	FLUORIDE	2.904	0.746	2.4186	-0.0814	2.5000	96.74
3	CHLORIDE	3.871	0.653	2.3568	-0.1432	2.5000	94.27
4	NITRITE	4.547	1.535	2.4579	-0.0421	2.5000	98.32
5	BROMIDE	5.531	0.270	2.4882	-0.0118	2.5000	99.53
6	NITRATE	6.367	1.629	2.4657	-0.0343	2.5000	98.63
7	SULFATE	10.044	0.477	2.4991	-0.0009	2.5000	99.96

II - CONTINUING CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	IC4
Analysis Date:	06/07/23 1923	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767223

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*% REC*	*LCL*	*UCL*	*Q*
Sulfate	mg/L	2.50	2.50	99.9	90	110	

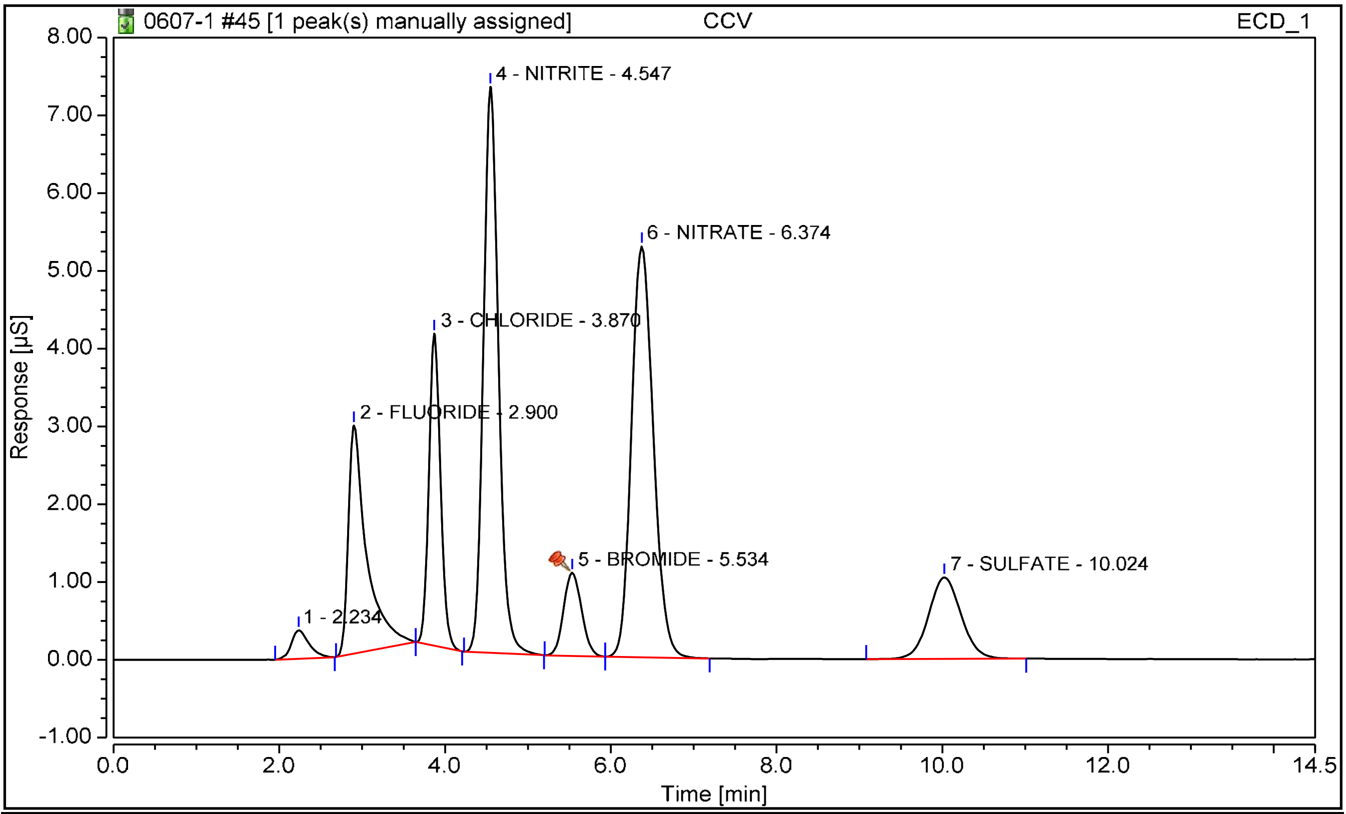
FORM II - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**CCV**	*Operator:*	**CS**
Vial Number:	**31**	*Injection Volume:*	**5000.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:	**CCV**	*Instrument Method:*	**TEST 1**
Injection Time:	**06/07/2023 19:23**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
2	FLUORIDE	2.900	0.753	2.4413	-0.0587	2.5000	97.65
3	CHLORIDE	3.870	0.652	2.3523	-0.1477	2.5000	94.09
4	NITRITE	4.547	1.538	2.4610	-0.0390	2.5000	98.44
5	BROMIDE	5.534	0.269	2.4756	-0.0244	2.5000	99.03
6	NITRATE	6.374	1.619	2.4518	-0.0482	2.5000	98.07
7	SULFATE	10.024	0.477	2.4969	-0.0031	2.5000	99.88

II - CONTINUING CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	IC4
Analysis Date:	06/08/23 0935	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767281

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*% REC*	*LCL*	*UCL*	*Q*
Sulfate	mg/L	2.50	2.48	99	90	110	

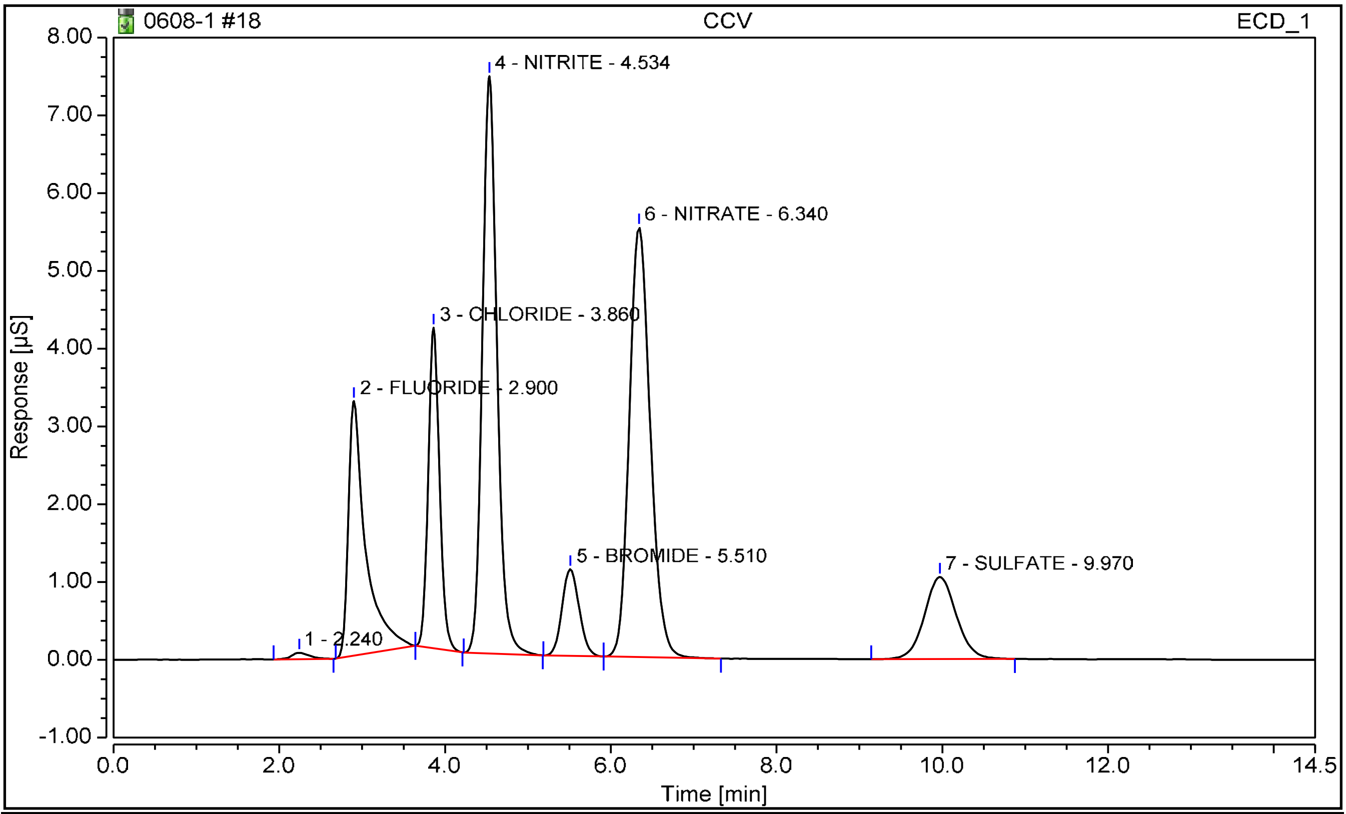
FORM II - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**CCV**	*Operator:*	**CS**
Vial Number:	**4**	*Injection Volume:*	**5000.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:	**CCV**	*Instrument Method:*	**TEST 1**
Injection Time:	**06/08/2023 09:35**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
2	FLUORIDE	2.900	0.792	2.5617	0.0617	2.5000	102.47
3	CHLORIDE	3.860	0.652	2.3556	-0.1444	2.5000	94.22
4	NITRITE	4.534	1.523	2.4378	-0.0622	2.5000	97.51
5	BROMIDE	5.510	0.268	2.4670	-0.0330	2.5000	98.68
6	NITRATE	6.340	1.621	2.4551	-0.0449	2.5000	98.20
7	SULFATE	9.970	0.473	2.4761	-0.0239	2.5000	99.05

II - CONTINUING CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	IC4
Analysis Date:	06/08/23 1652	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767281

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*% REC*	*LCL*	*UCL*	*Q*
Sulfate	mg/L	2.50	2.48	99.4	90	110	

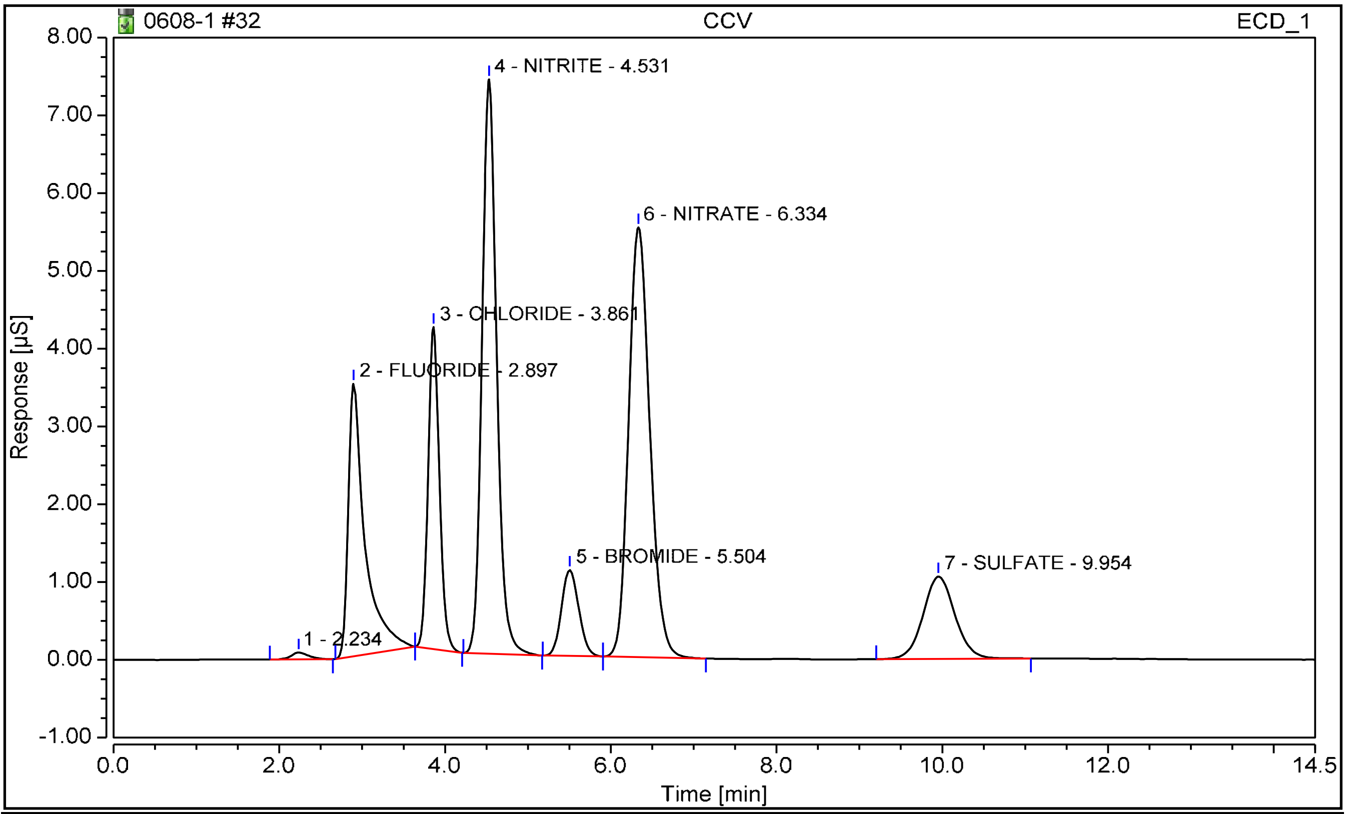
FORM II - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**CCV**	*Operator:*	**CS**
Vial Number:	**18**	*Injection Volume:*	**5000.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:	**CCV**	*Instrument Method:*	**TEST 1**
Injection Time:	**06/08/2023 16:52**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
2	FLUORIDE	2.897	0.818	2.6407	0.1407	2.5000	105.63
3	CHLORIDE	3.861	0.656	2.3684	-0.1316	2.5000	94.74
4	NITRITE	4.531	1.513	2.4231	-0.0769	2.5000	96.92
5	BROMIDE	5.504	0.268	2.4663	-0.0337	2.5000	98.65
6	NITRATE	6.334	1.636	2.4753	-0.0247	2.5000	99.01
7	SULFATE	9.954	0.474	2.4837	-0.0163	2.5000	99.35

EPA 9056A

Form III

Blanks

III - CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	CCB for HBN 767085
Matrix:	Water	Instrument ID:	BRWA13
Analysis Date:	06/02/23 0948	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767085

ANALYTE	RESULT	UNITS	Q	DL	LOD	LOQ
Chloride	0.100	mg/L	U	0.050	0.100	0.200
Nitrate	0.150	mg/L	U	0.100	0.150	0.200
Nitrite	0.150	mg/L	U	0.100	0.150	0.200
Sulfate	0.150	mg/L	U	0.100	0.150	0.200

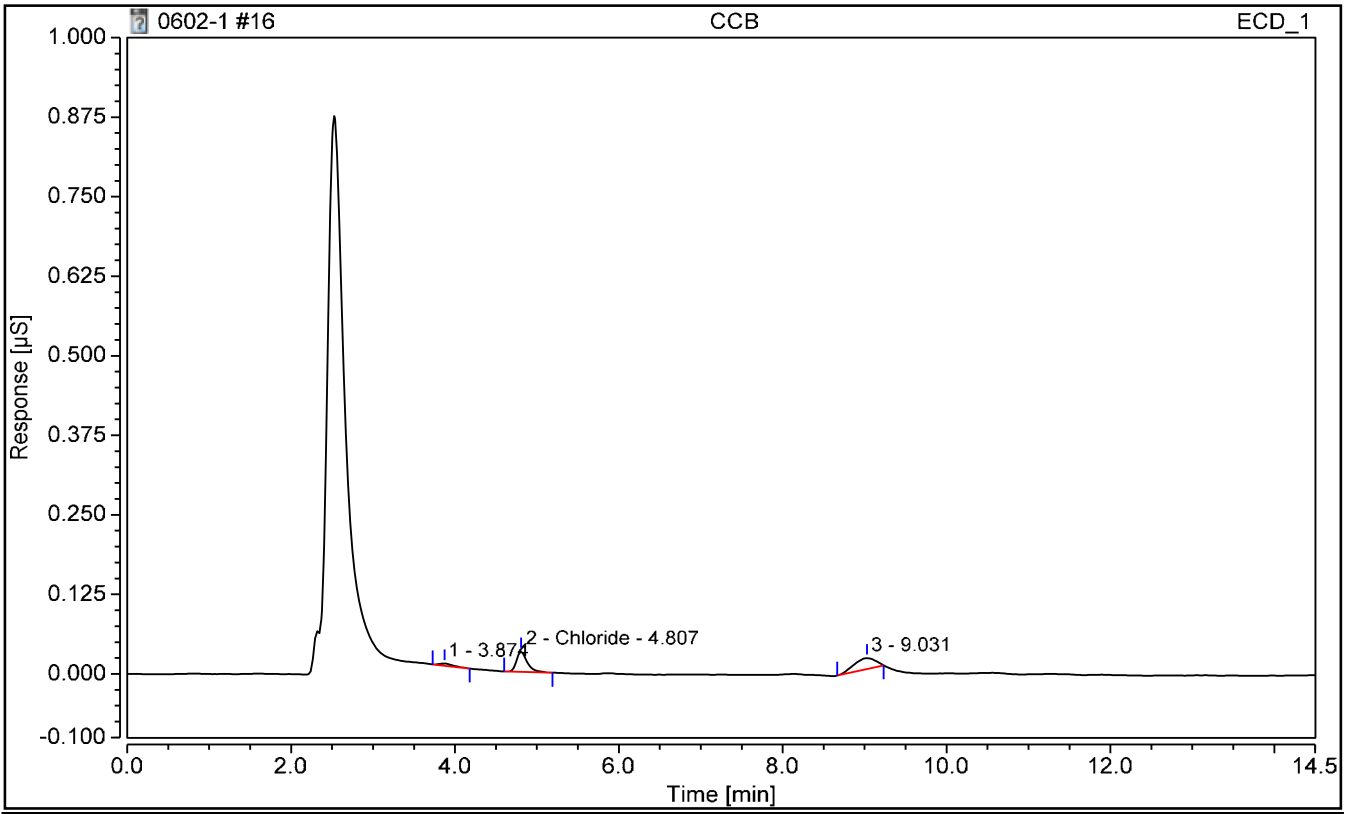
FORM III - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**CCB**	*Run Time (min):*	**14.50**
Vial Number:	**2**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 09:48**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
n.a.	Fluoride	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
2	Chloride	4.807	0.005	0.0163	n.a.	n.a.	n.a.
n.a.	Nitrite	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	Bromide	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	Nitrate	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	Sulfate	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

III - METHOD BLANK

Report No:	223060210	Blank ID:	MB2489994
Matrix:	Water	Instrument ID:	BRWA13
Analysis Date:	06/02/23 1023	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767085

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Chloride	0.100	mg/L	U	0.050	0.100	0.200
Nitrate	0.150	mg/L	U	0.100	0.150	0.200
Nitrite	0.150	mg/L	U	0.100	0.150	0.200
Sulfate	0.150	mg/L	U	0.100	0.150	0.200

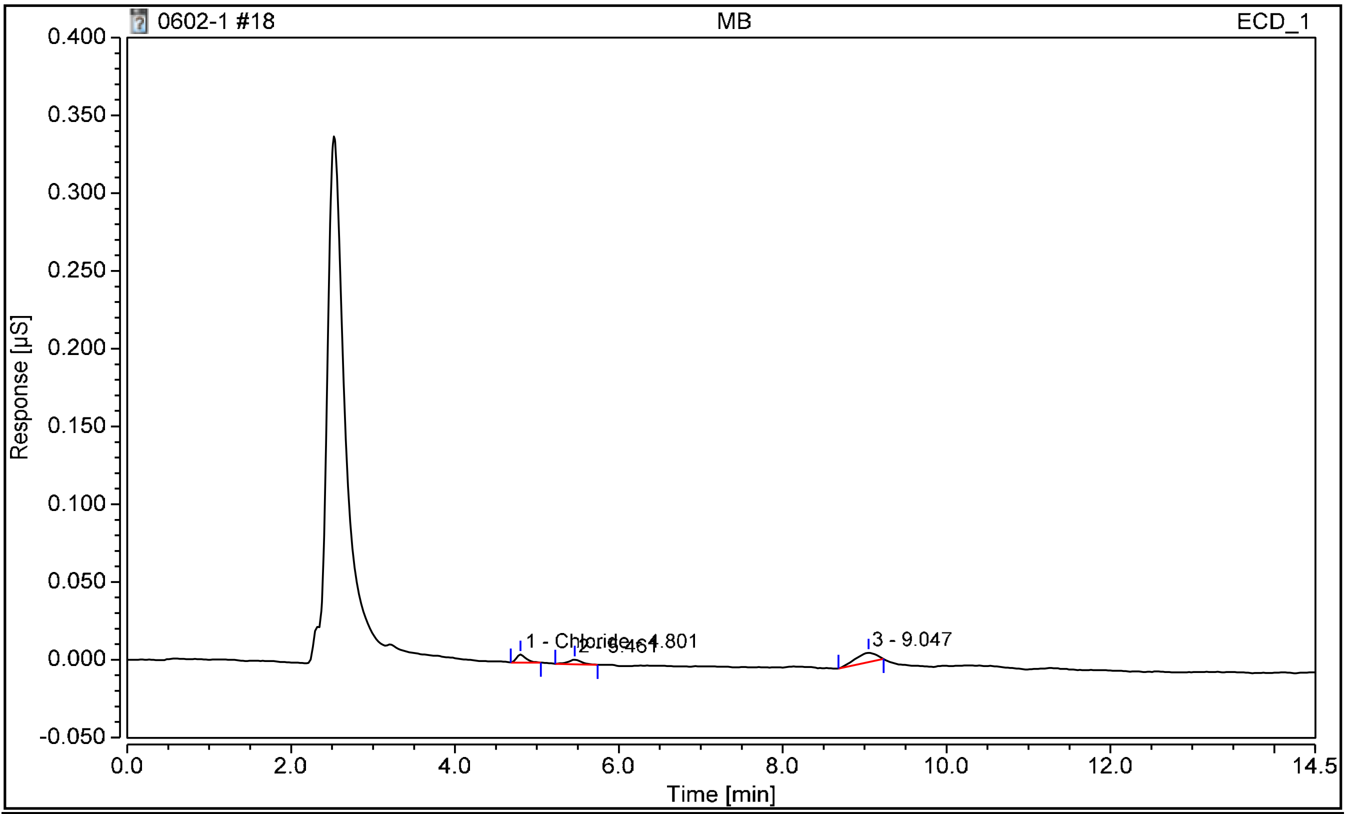
FORM III - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**MB**	*Run Time (min):*	**14.50**
Vial Number:	**4**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 10:23**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
n.a.	Fluoride	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
1	Chloride	4.801	0.001	n.a.	n.a.	n.a.	n.a.
n.a.	Nitrite	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	Bromide	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	Nitrate	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	Sulfate	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

III - CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	CCB for HBN 767085
Matrix:	Water	Instrument ID:	BRWA13
Analysis Date:	06/02/23 1654	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767085

ANALYTE	RESULT	UNITS	Q	DL	LOD	LOQ
Chloride	0.100	mg/L	U	0.050	0.100	0.200
Nitrate	0.150	mg/L	U	0.100	0.150	0.200
Nitrite	0.150	mg/L	U	0.100	0.150	0.200
Sulfate	0.150	mg/L	U	0.100	0.150	0.200

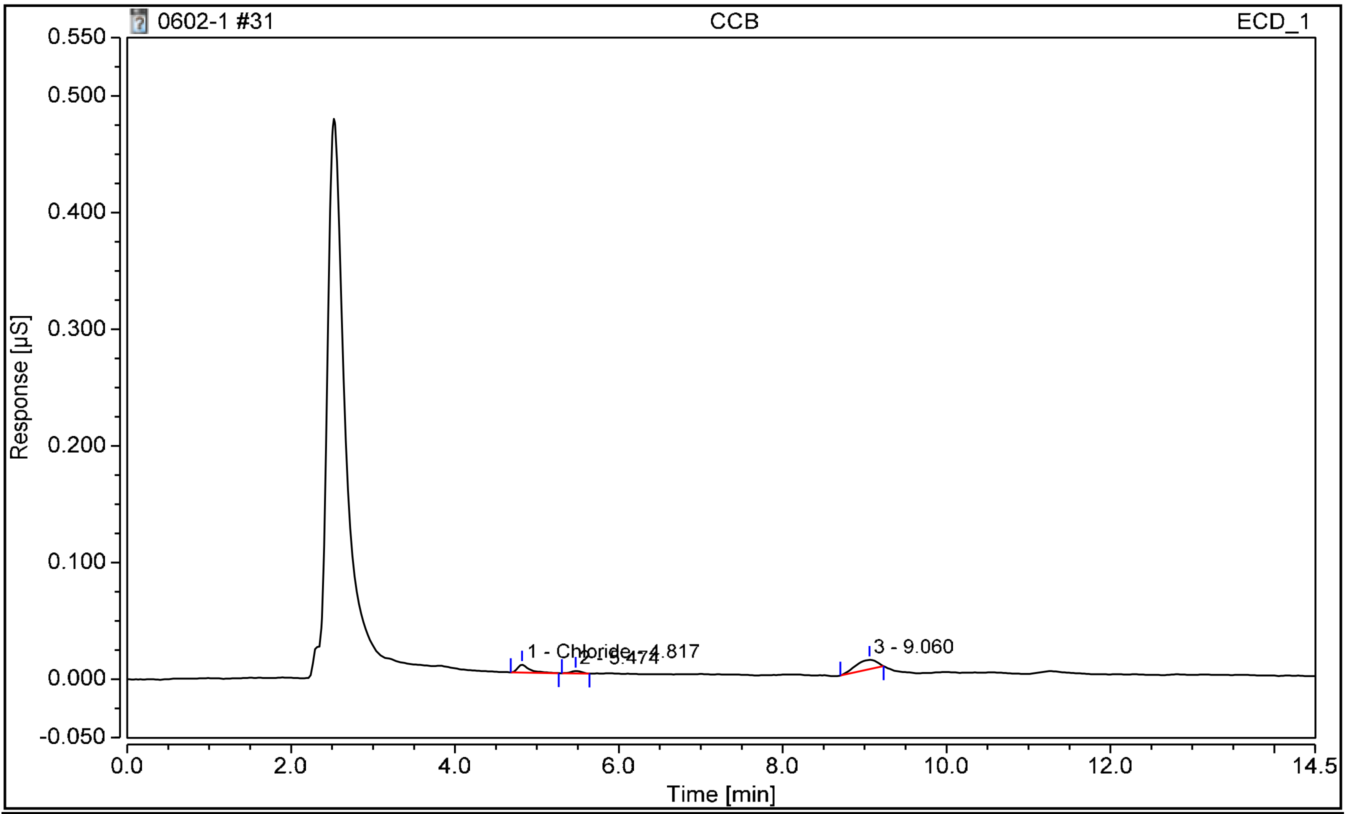
FORM III - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**CCB**	*Run Time (min):*	**14.50**
Vial Number:	**17**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 16:54**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
n.a.	Fluoride	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
1	Chloride	4.817	0.001	n.a.	n.a.	n.a.	n.a.
n.a.	Nitrite	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	Bromide	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	Nitrate	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	Sulfate	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

III - CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	CCB for HBN 767223
Matrix:	Water	Instrument ID:	IC4
Analysis Date:	06/07/23 1001	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767223

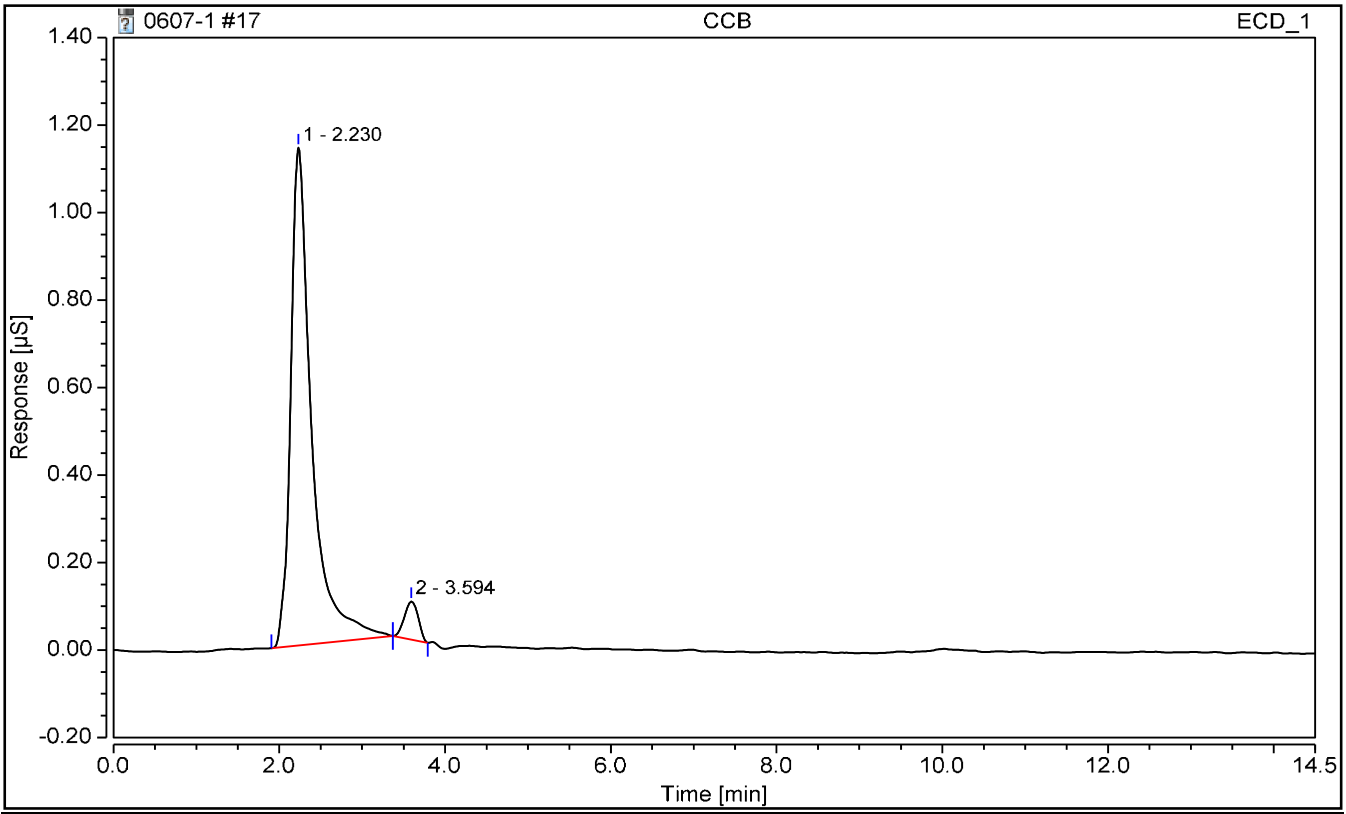
ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Sulfate	0.150	mg/L	U	0.100	0.150	0.200

Chromatogram and Results

Injection Details

Injection Name:	**CCB**	*Operator:*	**CS**
Vial Number:	**3**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/07/2023 10:01**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
n.a.	FLUORIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	CHLORIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	NITRITE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	BROMIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	NITRATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	SULFATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

III - METHOD BLANK

Report No:	223060210	Blank ID:	MB2490889
Matrix:	Water	Instrument ID:	IC4
Analysis Date:	06/07/23 1036	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767223

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Sulfate	0.150	mg/L	U	0.100	0.150	0.200

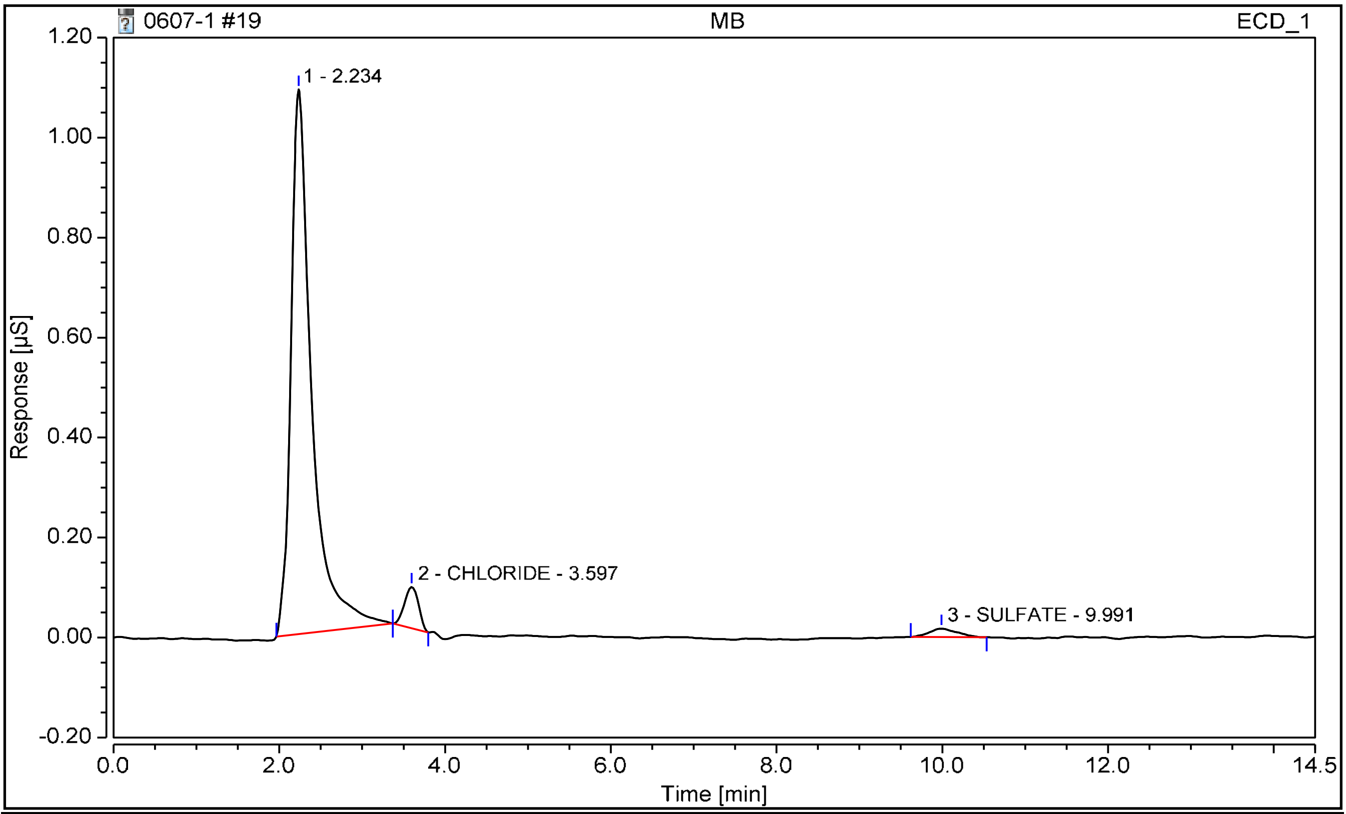
FORM III - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**MB**	*Operator:*	**CS**
Vial Number:	**5**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/07/2023 10:36**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
n.a.	FLUORIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
2	CHLORIDE	3.597	0.016	n.a.	n.a.	n.a.	n.a.
n.a.	NITRITE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	BROMIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	NITRATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
3	SULFATE	9.991	0.007	0.0189	n.a.	n.a.	n.a.

III - CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	CCB for HBN 767223
Matrix:	Water	Instrument ID:	IC4
Analysis Date:	06/07/23 1537	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767223

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Sulfate	0.150	mg/L	U	0.100	0.150	0.200

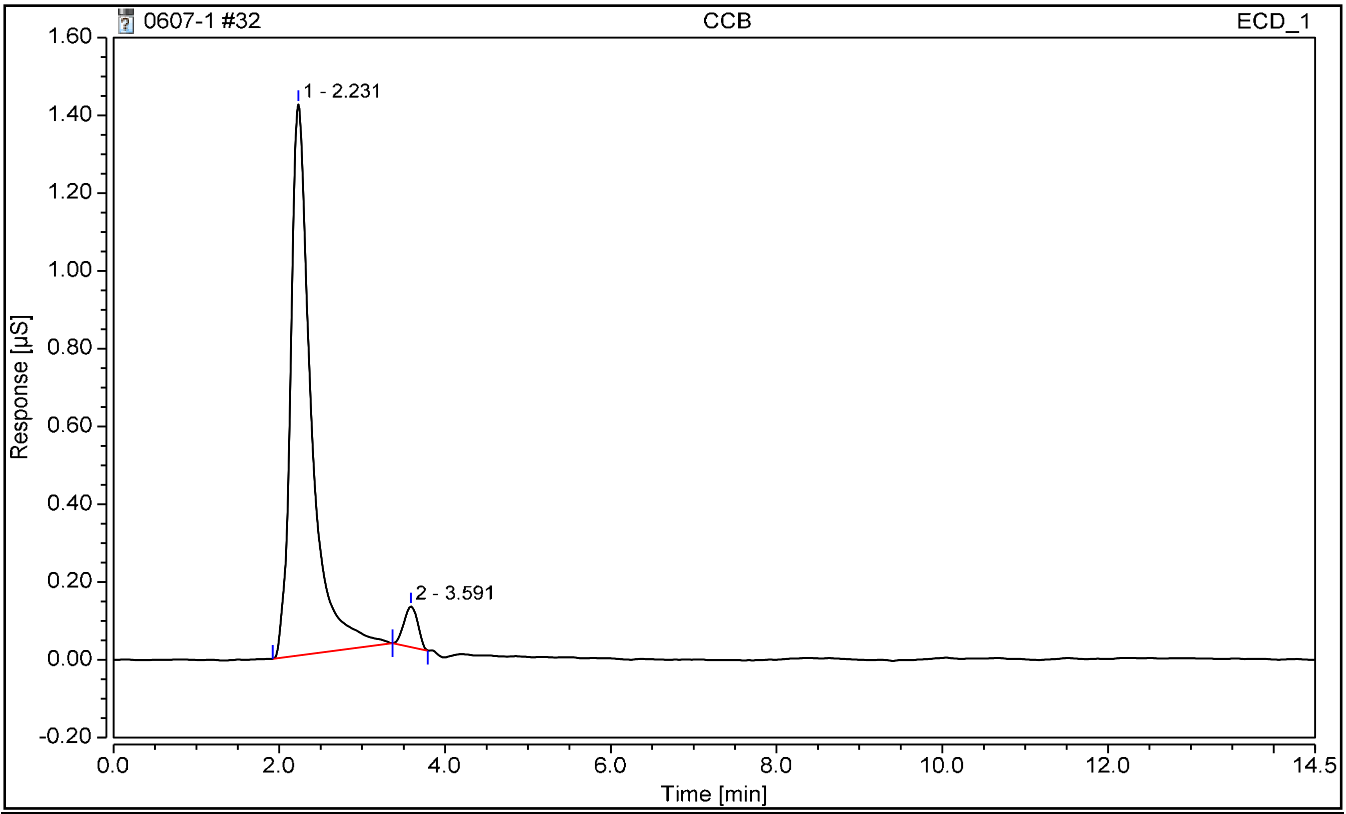
FORM III - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**CCB**	*Operator:*	**CS**
Vial Number:	**18**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/07/2023 15:37**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
n.a.	FLUORIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	CHLORIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	NITRITE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	BROMIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	NITRATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	SULFATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

III - CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	CCB for HBN 767223
Matrix:	Water	Instrument ID:	IC4
Analysis Date:	06/07/23 1906	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767223

ANALYTE	RESULT	UNITS	Q	DL	LOD	LOQ
Sulfate	0.150	mg/L	U	0.100	0.150	0.200

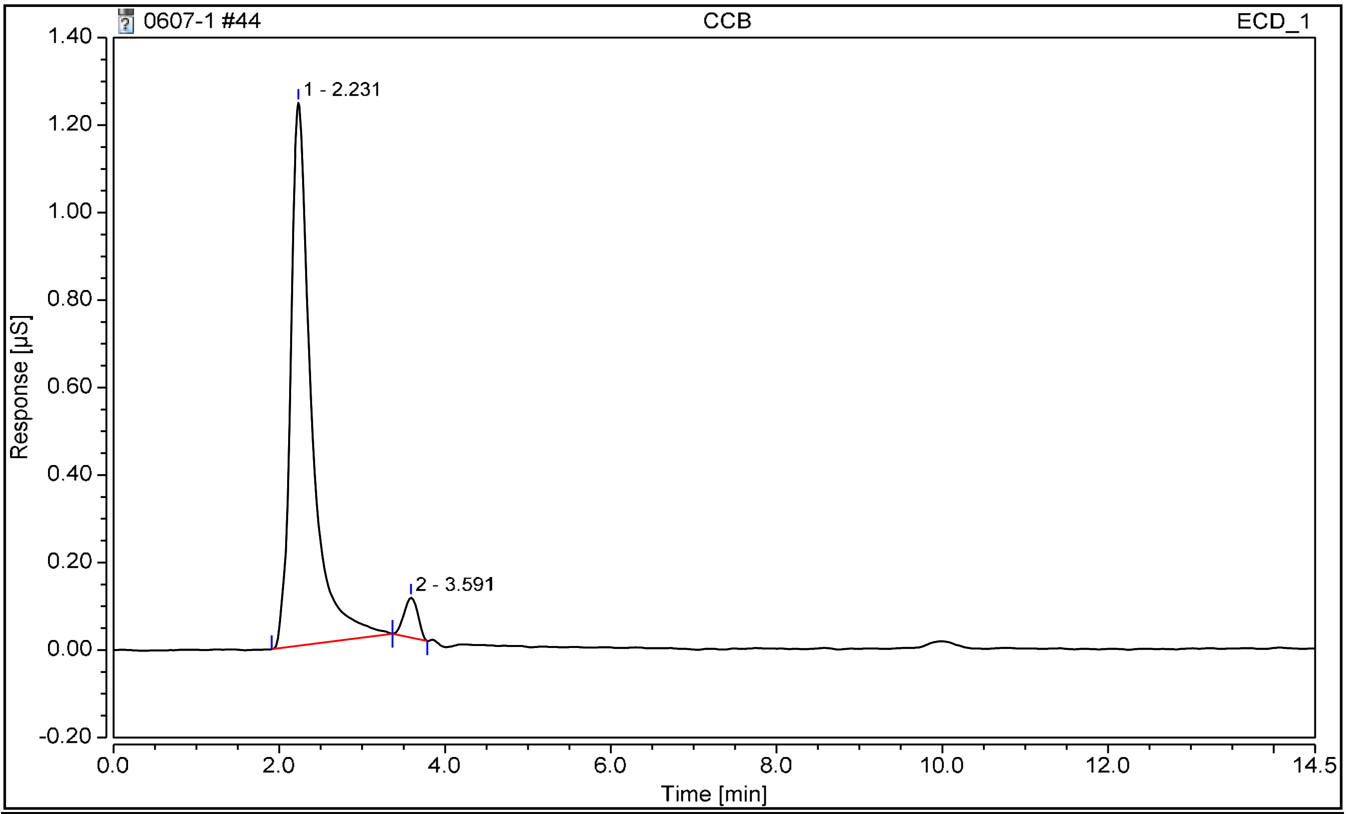
FORM III - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**CCB**	*Operator:*	**CS**
Vial Number:	**30**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/07/2023 19:06**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
n.a.	FLUORIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	CHLORIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	NITRITE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	BROMIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	NITRATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	SULFATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

III - CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	CCB for HBN 767281
Matrix:	Water	Instrument ID:	IC4
Analysis Date:	06/08/23 0917	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767281

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Sulfate	0.150	mg/L	U	0.100	0.150	0.200

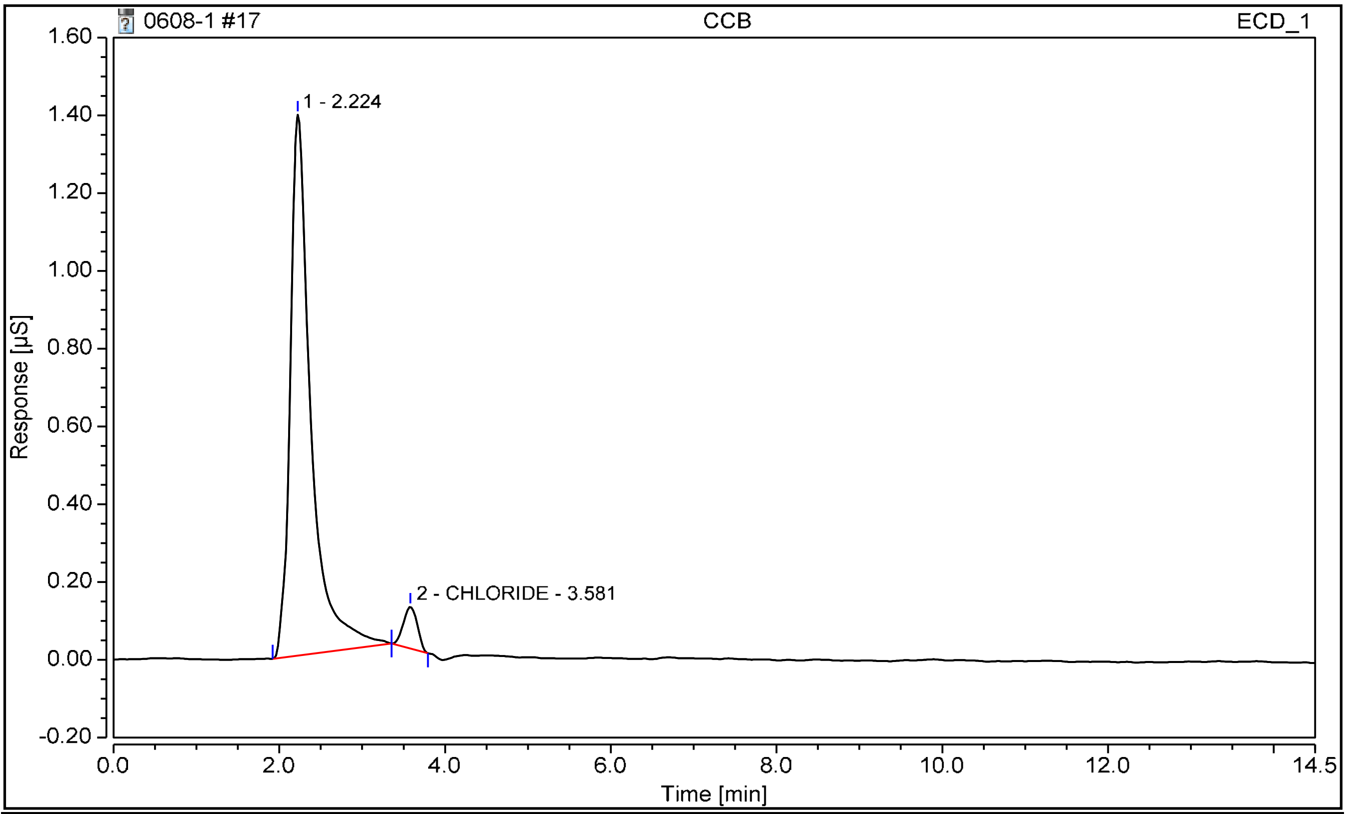
FORM III - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**CCB**	*Operator:*	**CS**
Vial Number:	**3**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/08/2023 09:17**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
n.a.	FLUORIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
2	CHLORIDE	3.581	0.021	n.a.	n.a.	n.a.	n.a.
n.a.	NITRITE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	BROMIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	NITRATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	SULFATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

III - METHOD BLANK

Report No:	223060210	Blank ID:	MB2491179
Matrix:	Water	Instrument ID:	IC4
Analysis Date:	06/08/23 0952	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767281

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Sulfate	0.150	mg/L	U	0.100	0.150	0.200

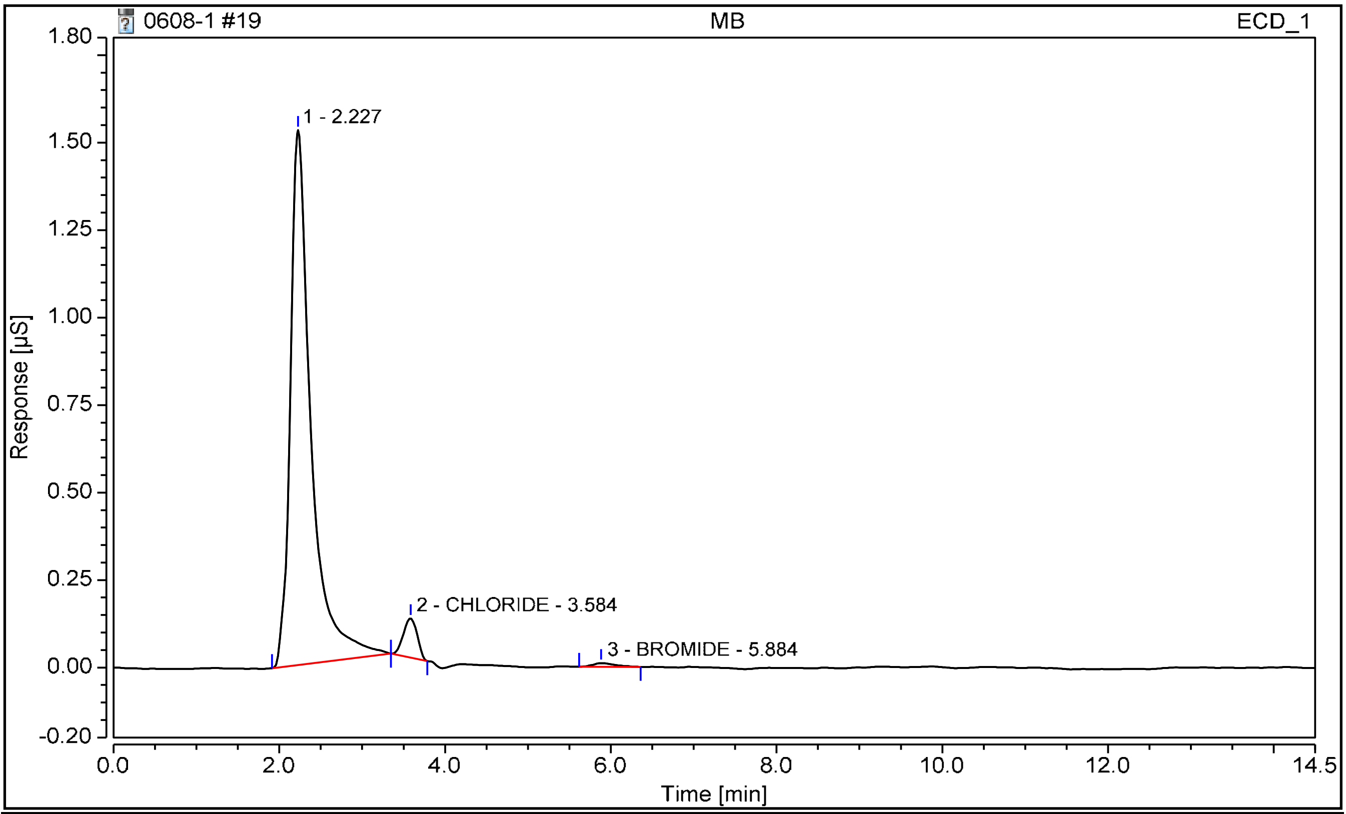
FORM III - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**MB**	*Operator:*	**CS**
Vial Number:	**5**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/08/2023 09:52**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
n.a.	FLUORIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
2	CHLORIDE	3.584	0.022	n.a.	n.a.	n.a.	n.a.
n.a.	NITRITE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
3	BROMIDE	5.884	0.003	0.0327	n.a.	n.a.	n.a.
n.a.	NITRATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	SULFATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

III - CONTINUING CALIBRATION BLANK

Report No:	223060210	Blank ID:	CCB for HBN 767281
Matrix:	Water	Instrument ID:	IC4
Analysis Date:	06/08/23 1635	Lab File ID:	NA
Analytical Method:	EPA 9056A	Analytical Batch:	767281

ANALYTE	*RESULT*	*UNITS*	*Q*	*DL*	*LOD*	*LOQ*
Sulfate	0.150	mg/L	U	0.100	0.150	0.200

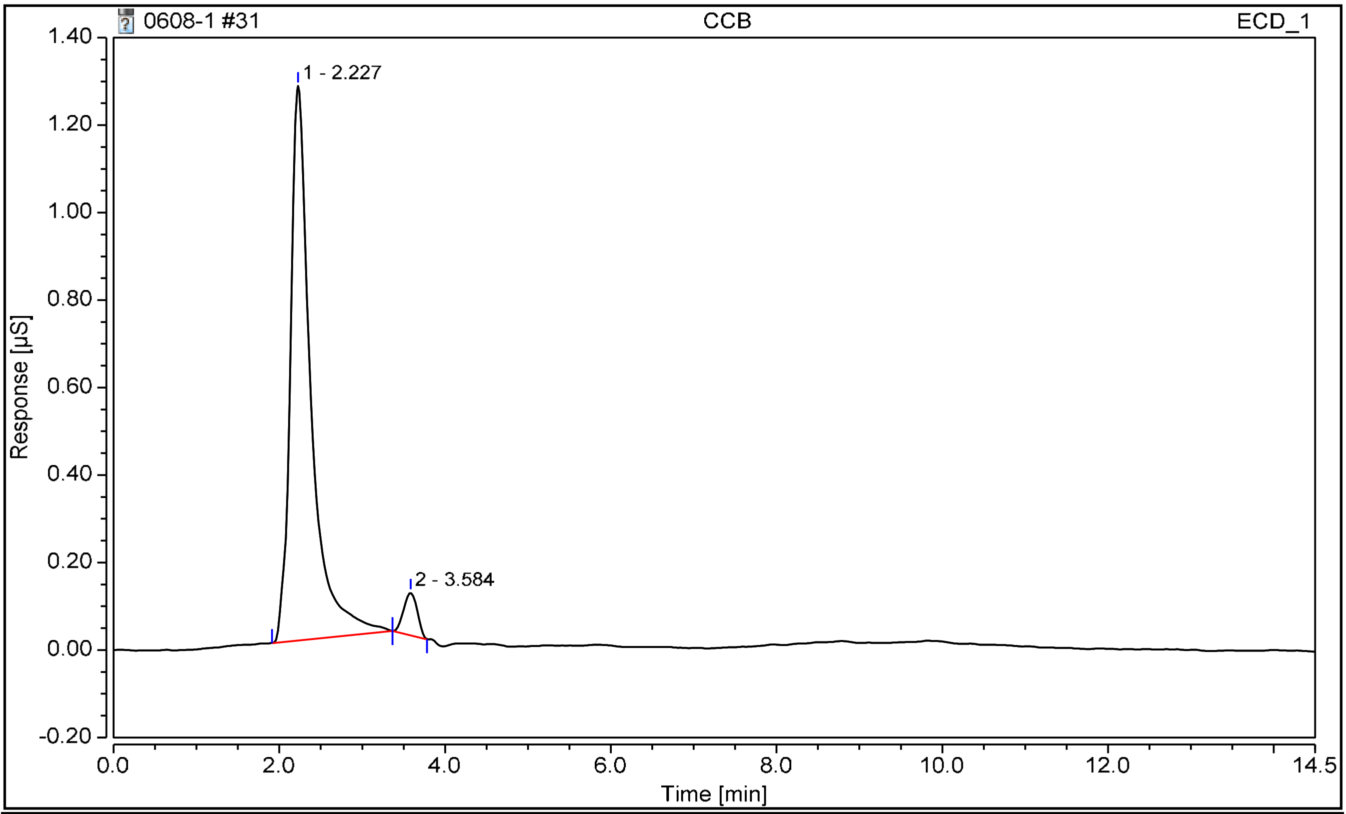
FORM III - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**CCB**	*Operator:*	**CS**
Vial Number:	**17**	*Injection Volume:*	**5000.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/08/2023 16:35**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
n.a.	FLUORIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	CHLORIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	NITRITE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	BROMIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	NITRATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	SULFATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

EPA 9056A

Form V

Matrix Spikes

V - MS/MSD RECOVERY

Report No:	223060210	Parent Sample ID:	OT07-MW04-060123
Prep Date:	NA	Parent LAB ID:	22306021003
Prep Batch:	NA	Analytical Batch:	767085
Prep Method:	NA	Analytical Method:	EPA 9056A

LAB QC ID:	**22306021006 MS**	Instrument ID:	BRWA13
Analyst:	CS	Lab File ID:	
Analysis Date:	06/02/23 1225	Dilution:	1

ANALYTE	*UNITS*	*SPIKE ADDED*	*SAMPLE RESULT*	*MS RESULT*	*MS % REC*	*#*	*QC LIMITS*
Chloride	mg/L	2.5	2.37	4.79	97		87 - 111
Nitrate	mg/L	2.5	.158	2.35	88		88 - 111
Nitrite	mg/L	2.5	0	2.44	97		87 - 111

LAB QC ID:	**22306021007 MSD**	Instrument ID:	BRWA13
Analyst:	CS	Lab File ID:	
Analysis Date:	06/02/23 1242	Dilution:	1

ANALYTE	*UNITS*	*SPIKE ADDED*	*MSD RESULT*	*MSD % REC*	*#*	*% RPD*	*#*	*%REC LIMITS*	*RPD LIMITS*
Chloride	mg/L	2.5	4.78	96		0		87 - 111	0 - 15
Nitrate	mg/L	2.5	2.36	88		0		88 - 111	0 - 15
Nitrite	mg/L	2.5	2.44	97		0		87 - 111	0 - 15

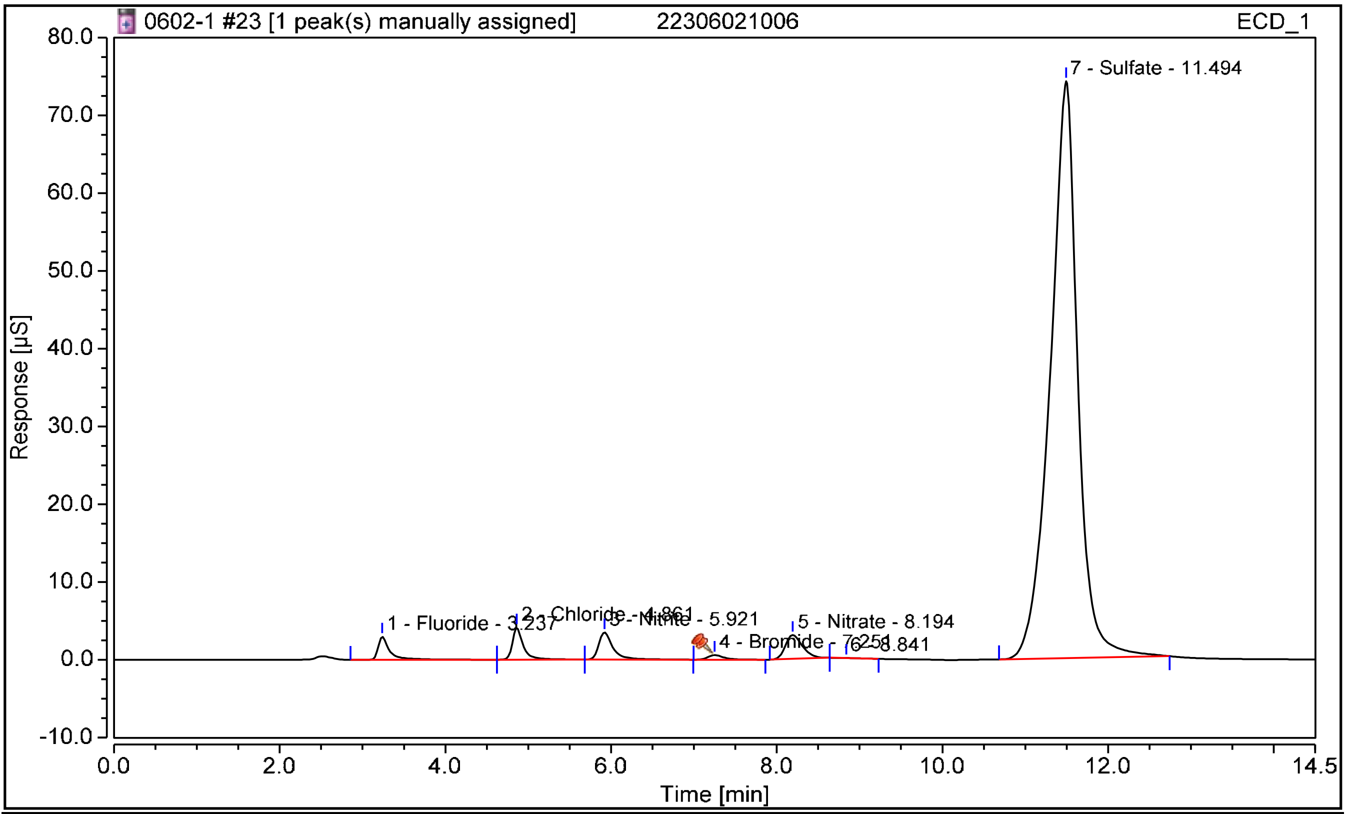
FORM V - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021006**	*Run Time (min):*	**14.50**
Vial Number:	**9**	*Injection Volume:*	**5000.00**
Injection Type:	**Spiked**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 12:25**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

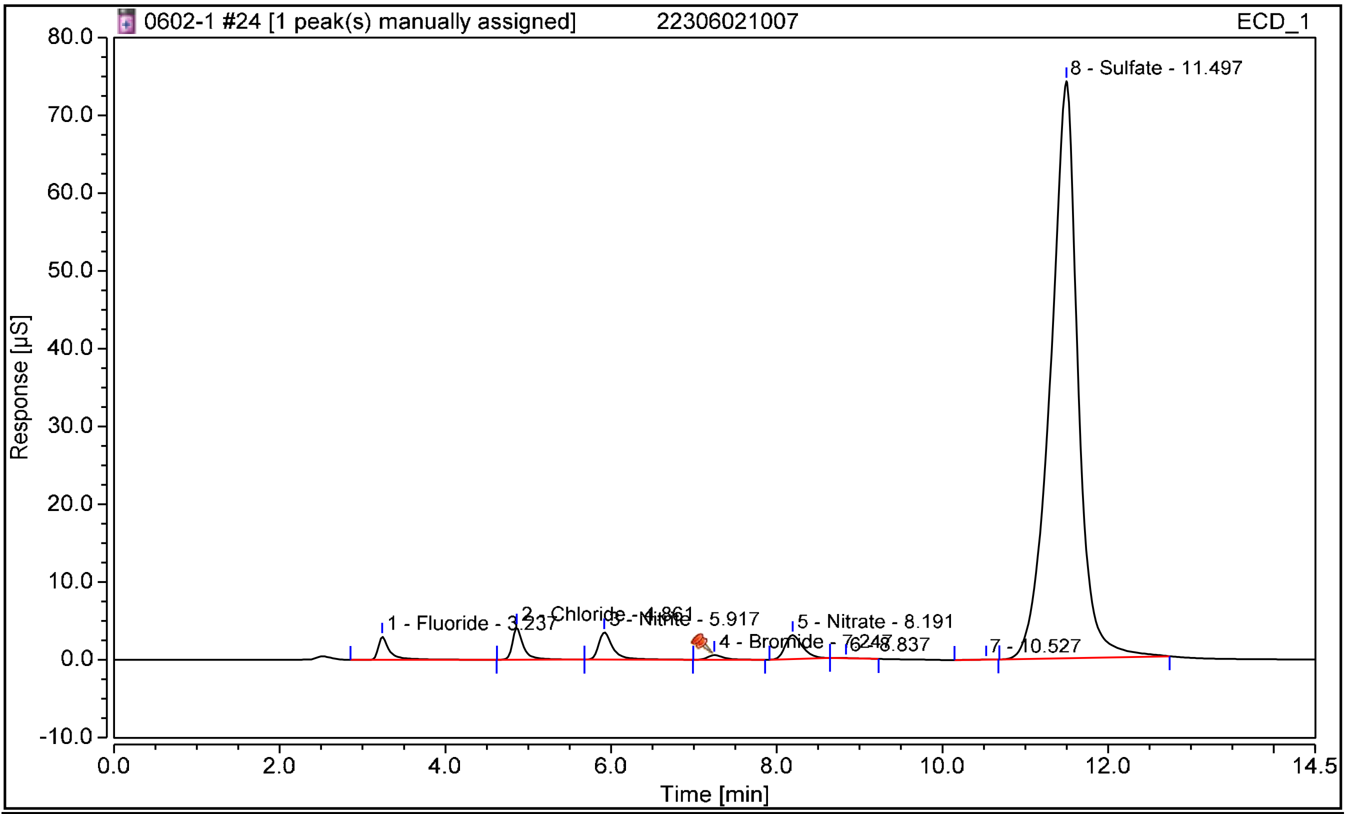
No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.237	0.498	2.4483	n.a.	n.a.	n.a.
2	Chloride	4.861	0.634	4.7857	n.a.	n.a.	n.a.
3	Nitrite	5.921	0.708	2.4377	n.a.	n.a.	n.a.
4	Bromide	7.251	0.132	2.5472	n.a.	n.a.	n.a.
5	Nitrate	8.194	0.739	2.3540	n.a.	n.a.	n.a.
7	Sulfate	11.494	28.496	190.7269	n.a.	n.a.	n.a.

Chromatogram and Results

Injection Details

Injection Name:	**22306021007**	*Run Time (min):*	**14.50**
Vial Number:	**10**	*Injection Volume:*	**5000.00**
Injection Type:	**Spiked**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 12:42**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.237	0.496	2.4362	n.a.	n.a.	n.a.
2	Chloride	4.861	0.633	4.7814	n.a.	n.a.	n.a.
3	Nitrite	5.917	0.709	2.4405	n.a.	n.a.	n.a.
4	Bromide	7.247	0.132	2.5466	n.a.	n.a.	n.a.
5	Nitrate	8.191	0.741	2.3594	n.a.	n.a.	n.a.
8	Sulfate	11.497	28.484	190.6701	n.a.	n.a.	n.a.

V - MS/MSD RECOVERY

Report No:	223060210	Parent Sample ID:	OT07-MW04-060123 DL
Prep Date:	NA	Parent LAB ID:	22306021003
Prep Batch:	NA	Analytical Batch:	767281
Prep Method:	NA	Analytical Method:	EPA 9056A

LAB QC ID:	**22306021006 MS**	Instrument ID:	IC4
Analyst:	CS	Lab File ID:	
Analysis Date:	06/08/23 1117	Dilution:	50

ANALYTE	*UNITS*	*SPIKE ADDED*	*SAMPLE RESULT*	*MS RESULT*	*MS % REC*	*#*	*QC LIMITS*
Sulfate	mg/L	125	225	347	97		87 - 112

LAB QC ID:	**22306021007 MSD**	Instrument ID:	IC4
Analyst:	CS	Lab File ID:	
Analysis Date:	06/08/23 1135	Dilution:	50

ANALYTE	*UNITS*	*SPIKE ADDED*	*MSD RESULT*	*MSD % REC*	*#*	*% RPD*	*#*	*%REC LIMITS*	*RPD LIMITS*
Sulfate	mg/L	125	348	98		0		87 - 112	0 - 15

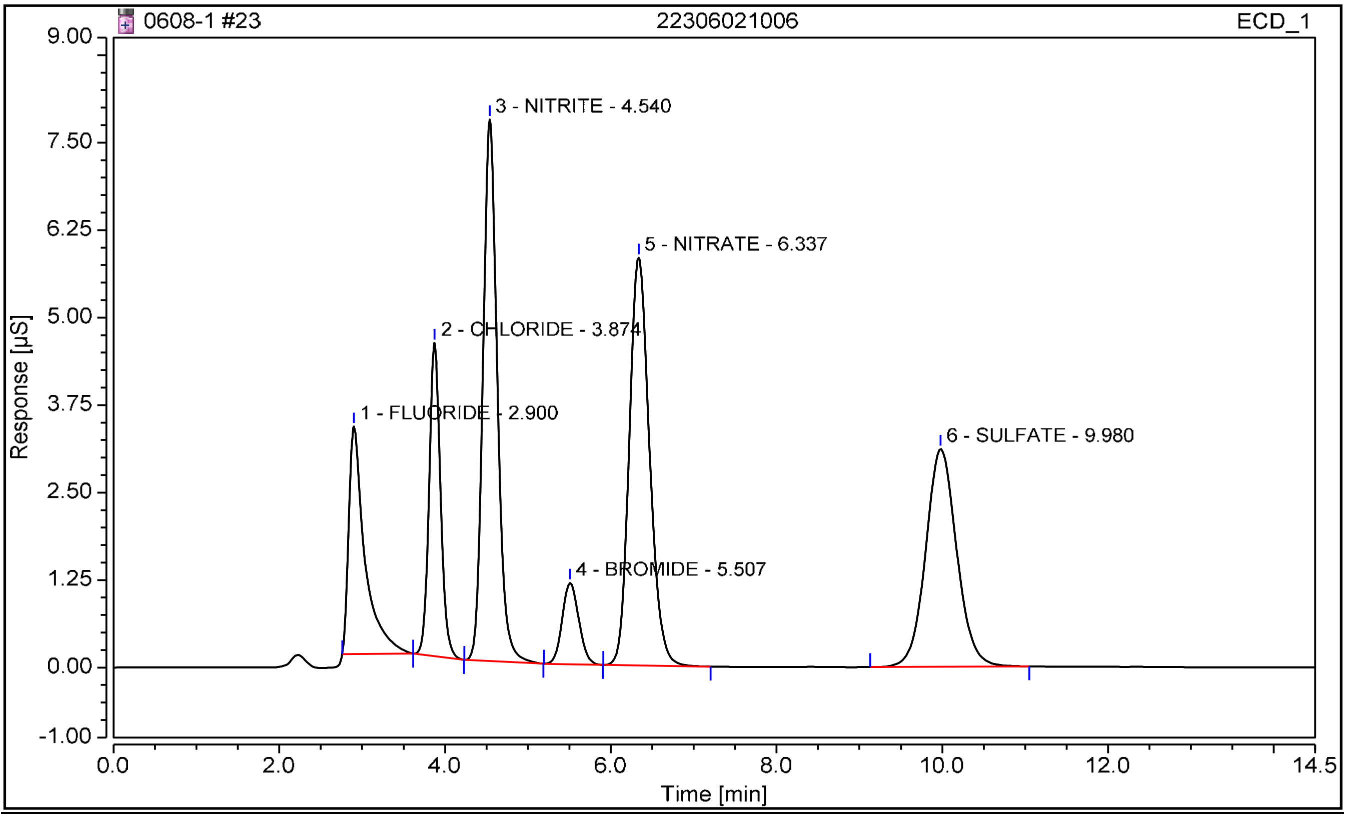
FORM V - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**22306021006**	*Operator:*	**CS**
Vial Number:	**9**	*Injection Volume:*	**5000.00**
Injection Type:	**Spiked**	*Dilution Factor:*	**50**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/08/2023 11:17**	*Processing Method:*	**9056**

Chromatogram



Integration Results

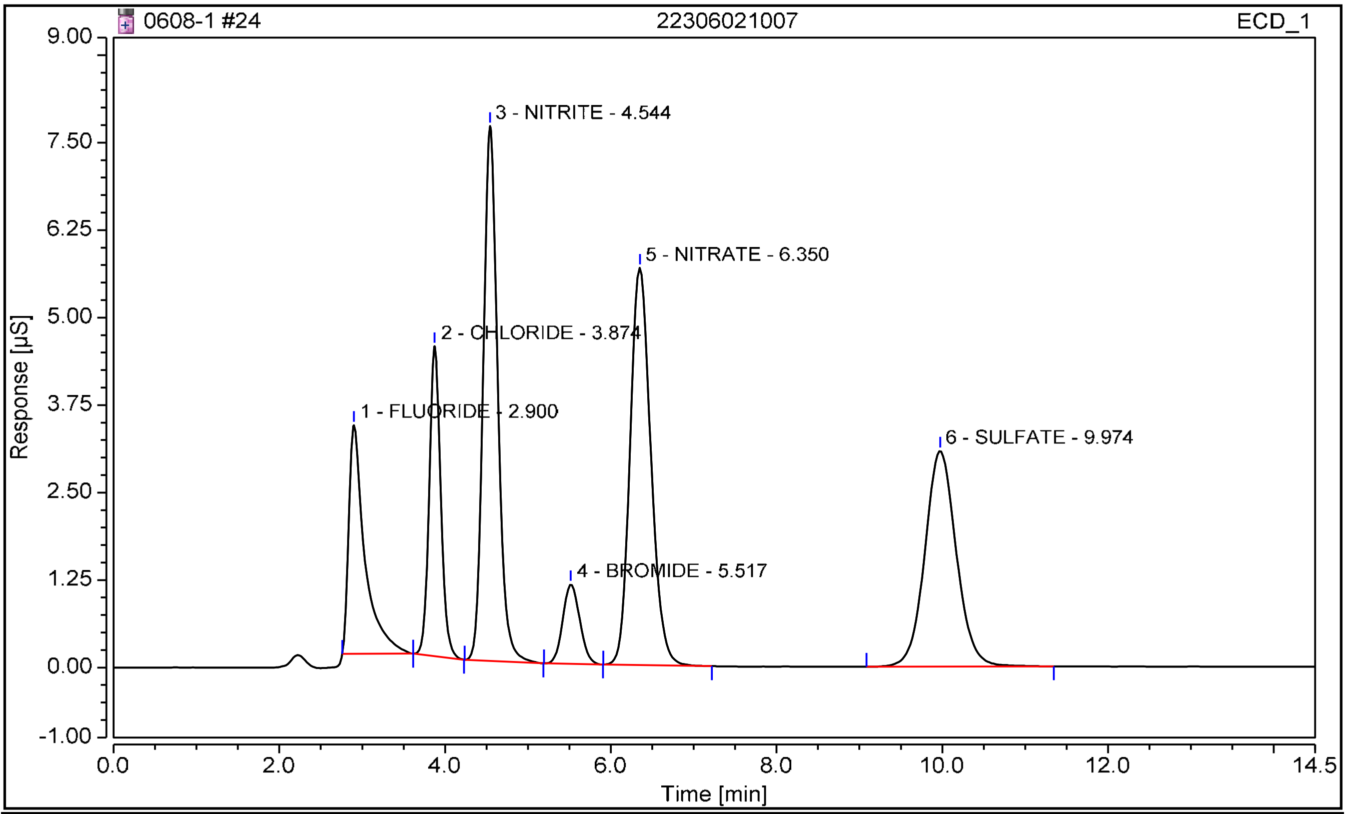
No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	FLUORIDE	2.900	0.732	118.7787	n.a.	n.a.	n.a.
2	CHLORIDE	3.874	0.712	128.6810	n.a.	n.a.	n.a.
3	NITRITE	4.540	1.552	124.1540	n.a.	n.a.	n.a.
4	BROMIDE	5.507	0.271	124.7287	n.a.	n.a.	n.a.
5	NITRATE	6.337	1.646	124.5049	n.a.	n.a.	n.a.
6	SULFATE	9.980	1.354	346.8124	n.a.	n.a.	n.a.

Chromatogram and Results

Injection Details

Injection Name:	**22306021007**	*Operator:*	**CS**
Vial Number:	**10**	*Injection Volume:*	**5000.00**
Injection Type:	**Spiked**	*Dilution Factor:*	**50**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/08/2023 11:35**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	FLUORIDE	2.900	0.734	119.0742	n.a.	n.a.	n.a.
2	CHLORIDE	3.874	0.712	128.6777	n.a.	n.a.	n.a.
3	NITRITE	4.544	1.551	124.1067	n.a.	n.a.	n.a.
4	BROMIDE	5.517	0.271	124.6340	n.a.	n.a.	n.a.
5	NITRATE	6.350	1.646	124.5090	n.a.	n.a.	n.a.
6	SULFATE	9.974	1.358	347.8637	n.a.	n.a.	n.a.

EPA 9056A

Form VII

Lab Control Spikes

VII - LABORATORY CONTROL SPIKE

Report No:	223060210	LAB ID:	LCS2489995
Matrix:	Water	Instrument ID:	BRWA13
Analyst:	CS	Lab File ID:	NA
Prep Date:	NA	Analysis Date:	06/02/23 1040
Prep Batch:	NA	Analytical Batch:	767085
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*%REC*	*Q*	*% REC LIMITS*
Chloride	mg/L	2.50	2.48	99		87 - 111
Nitrate	mg/L	2.50	2.54	102		88 - 111
Nitrite	mg/L	2.50	2.51	100		87 - 111
Sulfate	mg/L	2.50	2.51	100		87 - 112

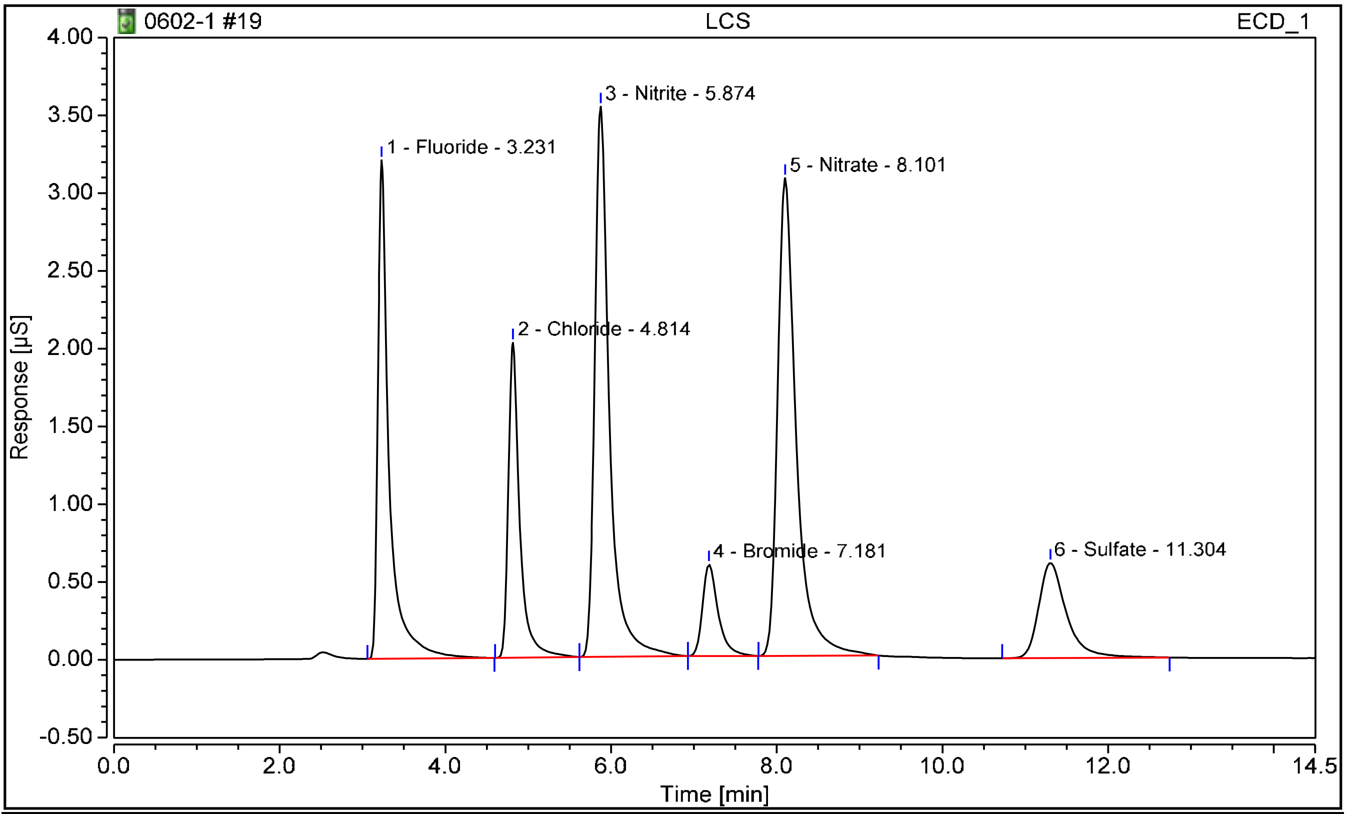
FORM VII - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**LCS**	*Run Time (min):*	**14.50**
Vial Number:	**5**	*Injection Volume:*	**5000.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 10:40**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.231	0.506	2.4840	n.a.	n.a.	n.a.
2	Chloride	4.814	0.323	2.4797	n.a.	n.a.	n.a.
3	Nitrite	5.874	0.728	2.5047	n.a.	n.a.	n.a.
4	Bromide	7.181	0.129	2.4853	n.a.	n.a.	n.a.
5	Nitrate	8.101	0.801	2.5410	n.a.	n.a.	n.a.
6	Sulfate	11.304	0.237	2.5101	n.a.	n.a.	n.a.

VII - LABORATORY CONTROL SPIKE DUPLICATE

Report No:	223060210	LAB ID:	LCSD2489996
Matrix:	Water	Instrument ID:	BRWA13
Analyst:	CS	Lab File ID:	NA
Prep Date:	NA	Analysis Date:	06/02/23 1133
Prep Batch:	NA	Analytical Batch:	767085
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*%REC*	*Q*	*%REC LIMITS*	*RPD*	*Q*	*RPD LIMITS*
Chloride	mg/L	2.50	2.48	99		87 - 111	0		0 - 15
Nitrate	mg/L	2.50	2.53	101		88 - 111	0		0 - 15
Nitrite	mg/L	2.50	2.51	100		87 - 111	0		0 - 15
Sulfate	mg/L	2.50	2.49	100		87 - 112	1		0 - 15

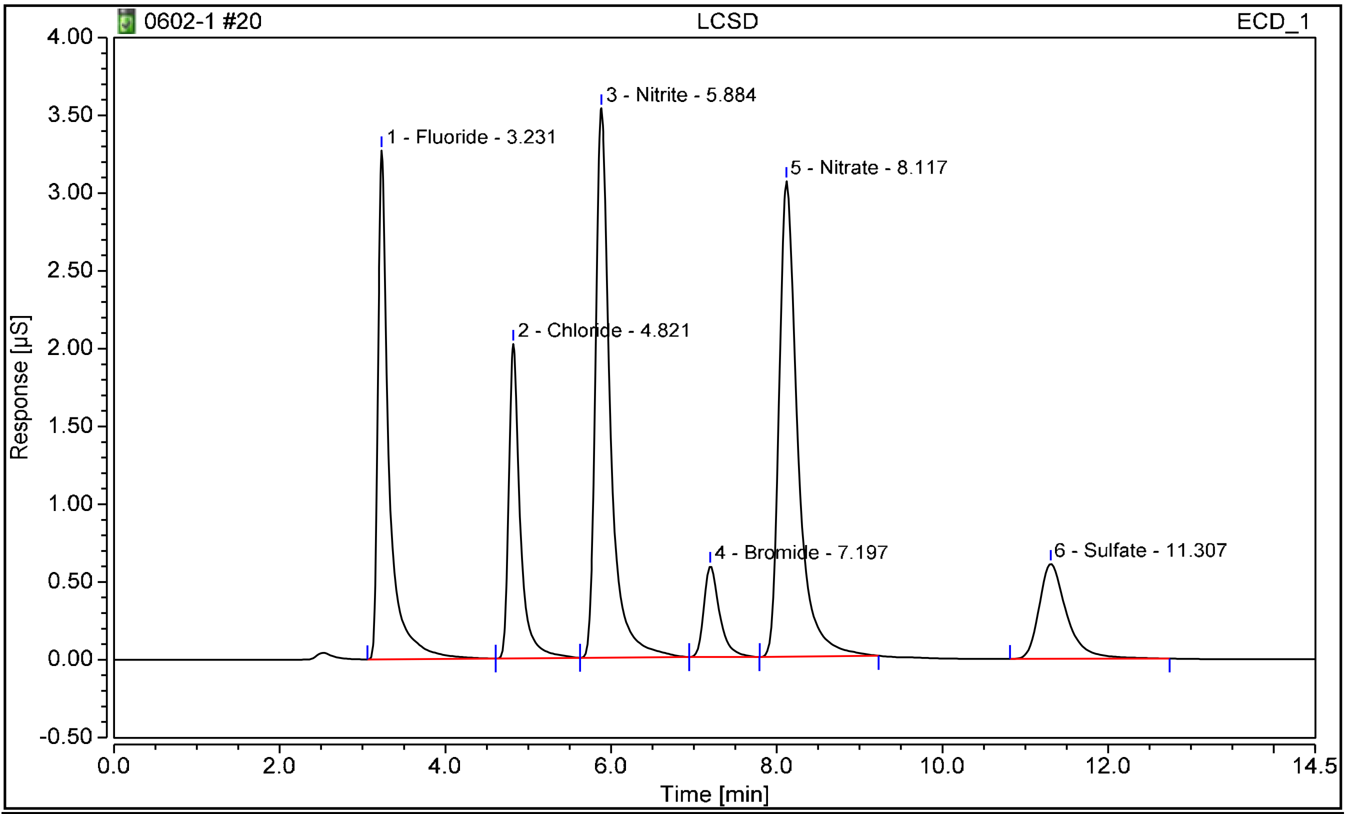
FORM VII - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**LCSD**	*Run Time (min):*	**14.50**
Vial Number:	**6**	*Injection Volume:*	**5000.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**06/02/2023 11:33**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.231	0.509	2.4977	n.a.	n.a.	n.a.
2	Chloride	4.821	0.323	2.4795	n.a.	n.a.	n.a.
3	Nitrite	5.884	0.730	2.5099	n.a.	n.a.	n.a.
4	Bromide	7.197	0.129	2.4833	n.a.	n.a.	n.a.
5	Nitrate	8.117	0.798	2.5332	n.a.	n.a.	n.a.
6	Sulfate	11.307	0.236	2.4928	n.a.	n.a.	n.a.

VII - LABORATORY CONTROL SPIKE

Report No:	223060210	LAB ID:	LCS2490890
Matrix:	Water	Instrument ID:	IC4
Analyst:	CS	Lab File ID:	NA
Prep Date:	NA	Analysis Date:	06/07/23 1053
Prep Batch:	NA	Analytical Batch:	767223
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*%REC*	*Q*	*% REC LIMITS*
Sulfate	mg/L	2.50	2.49	100		87 - 112

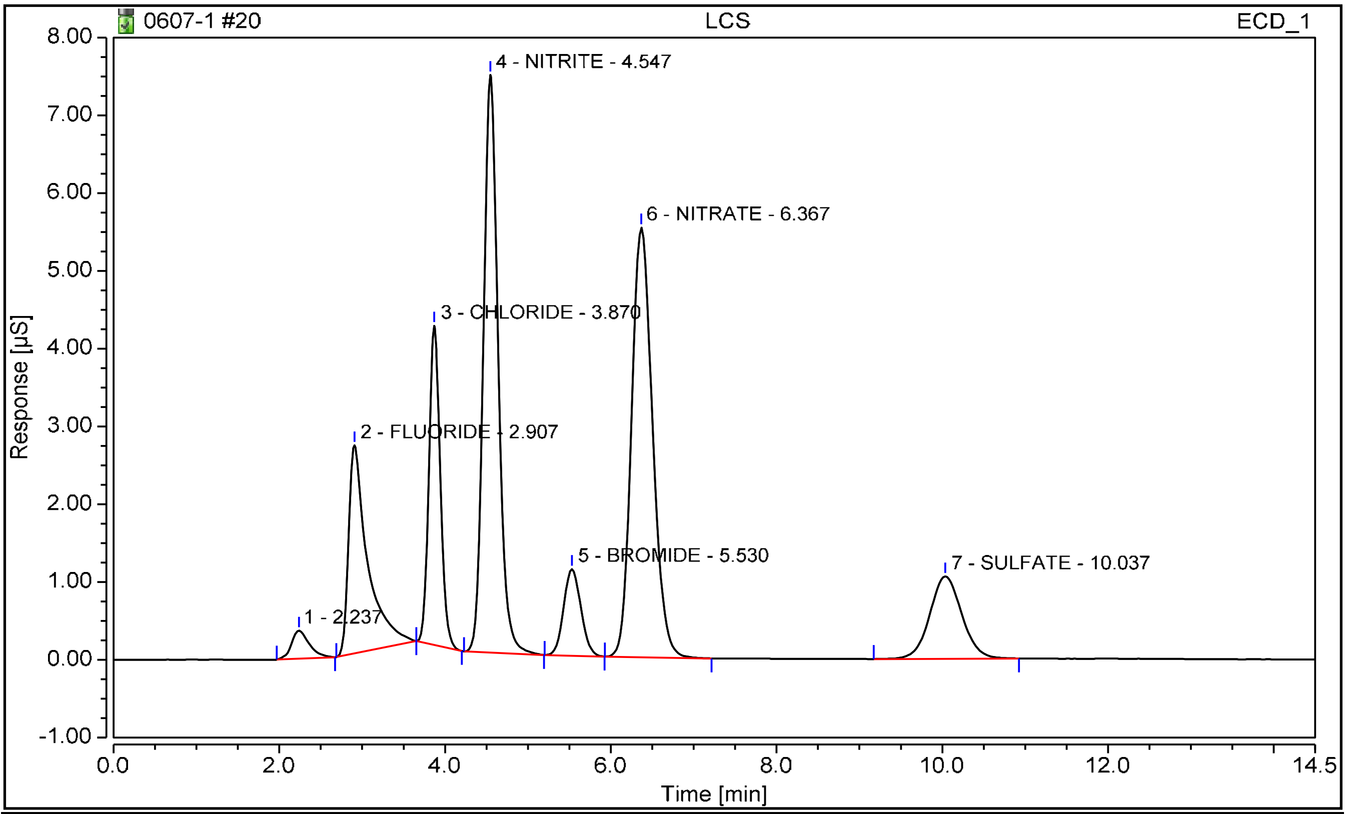
FORM VII - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**LCS**	*Operator:*	**CS**
Vial Number:	**6**	*Injection Volume:*	**5000.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/07/2023 10:53**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
2	FLUORIDE	2.907	0.715	2.3234	n.a.	n.a.	n.a.
3	CHLORIDE	3.870	0.650	2.3468	n.a.	n.a.	n.a.
4	NITRITE	4.547	1.527	2.4450	n.a.	n.a.	n.a.
5	BROMIDE	5.530	0.269	2.4756	n.a.	n.a.	n.a.
6	NITRATE	6.367	1.627	2.4629	n.a.	n.a.	n.a.
7	SULFATE	10.037	0.475	2.4878	n.a.	n.a.	n.a.

VII - LABORATORY CONTROL SPIKE DUPLICATE

Report No:	223060210	LAB ID:	LCSD2490891
Matrix:	Water	Instrument ID:	IC4
Analyst:	CS	Lab File ID:	NA
Prep Date:	NA	Analysis Date:	06/07/23 1110
Prep Batch:	NA	Analytical Batch:	767223
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*%REC*	*Q*	*%REC LIMITS*	*RPD*	*Q*	*RPD LIMITS*
Sulfate	mg/L	2.50	2.48	99		87 - 112	0		0 - 15

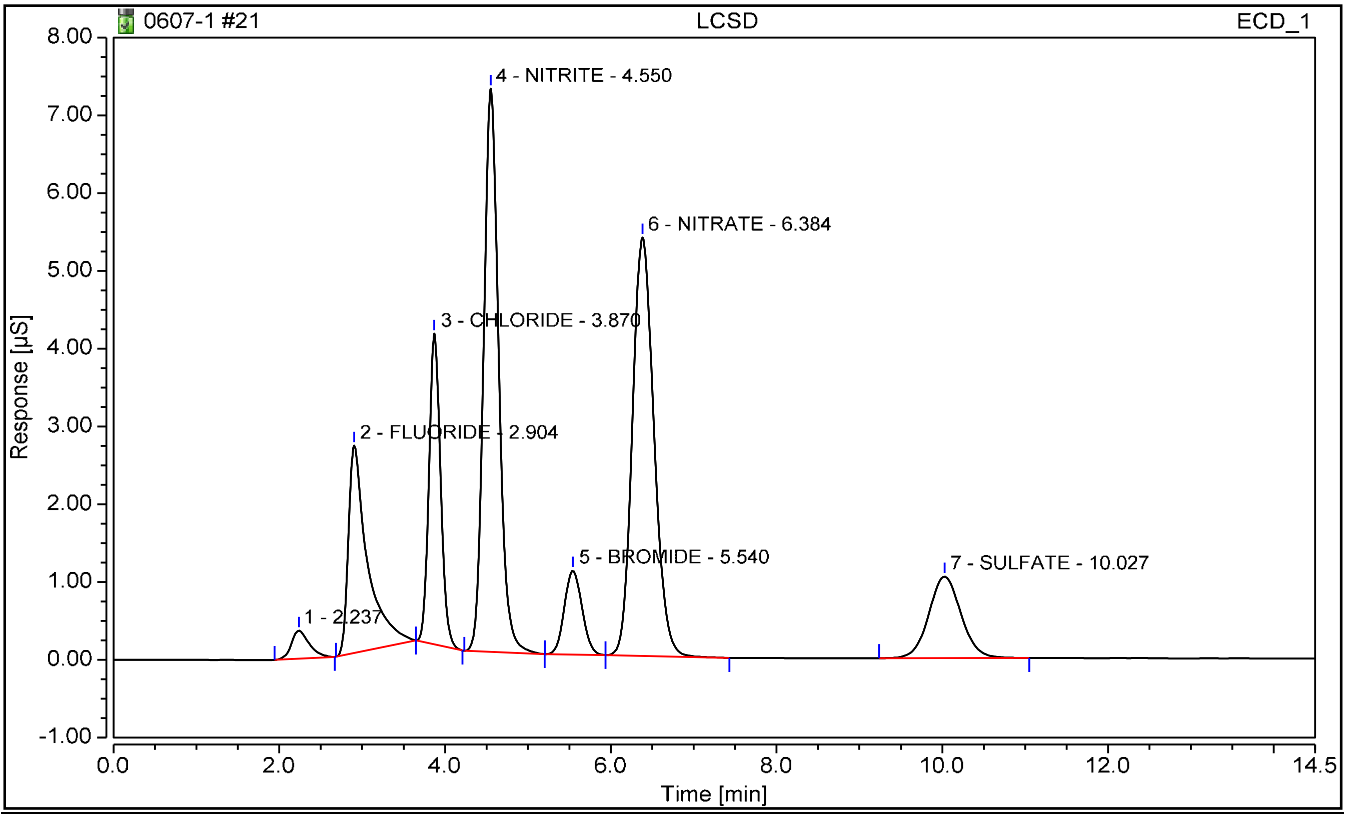
FORM VII - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**LCSD**	*Operator:*	**CS**
Vial Number:	**7**	*Injection Volume:*	**5000.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/07/2023 11:10**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
2	FLUORIDE	2.904	0.715	2.3238	n.a.	n.a.	n.a.
3	CHLORIDE	3.870	0.649	2.3444	n.a.	n.a.	n.a.
4	NITRITE	4.550	1.525	2.4415	n.a.	n.a.	n.a.
5	BROMIDE	5.540	0.268	2.4658	n.a.	n.a.	n.a.
6	NITRATE	6.384	1.625	2.4610	n.a.	n.a.	n.a.
7	SULFATE	10.027	0.473	2.4776	n.a.	n.a.	n.a.

VII - LABORATORY CONTROL SPIKE

Report No:	223060210	LAB ID:	LCS2491180
Matrix:	Water	Instrument ID:	IC4
Analyst:	CS	Lab File ID:	NA
Prep Date:	NA	Analysis Date:	06/08/23 1010
Prep Batch:	NA	Analytical Batch:	767281
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*%REC*	*Q*	*% REC LIMITS*
Sulfate	mg/L	2.50	2.49	100		87 - 112

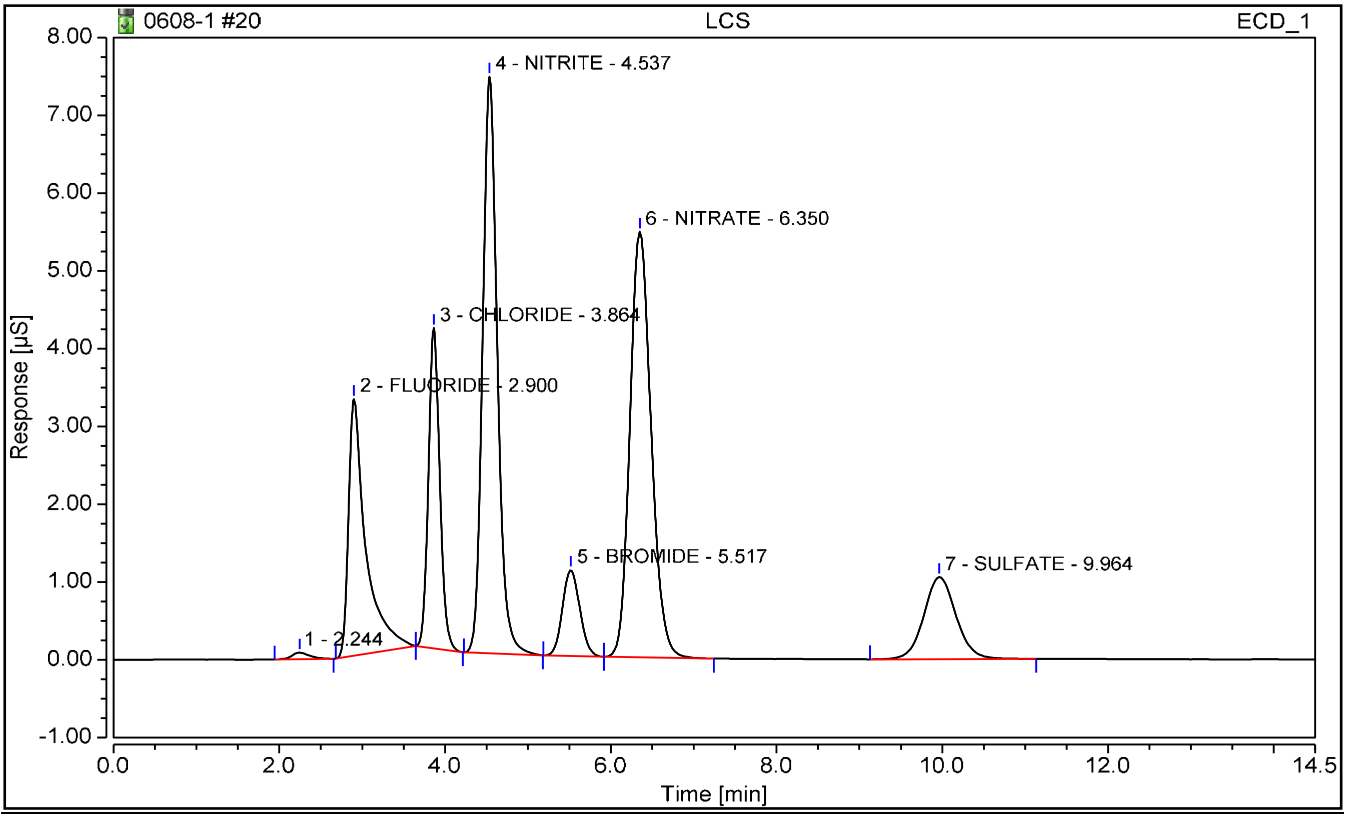
FORM VII - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**LCS**	*Operator:*	**CS**
Vial Number:	**6**	*Injection Volume:*	**5000.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/08/2023 10:10**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
2	FLUORIDE	2.900	0.796	2.5739	n.a.	n.a.	n.a.
3	CHLORIDE	3.864	0.653	2.3591	n.a.	n.a.	n.a.
4	NITRITE	4.537	1.525	2.4419	n.a.	n.a.	n.a.
5	BROMIDE	5.517	0.269	2.4757	n.a.	n.a.	n.a.
6	NITRATE	6.350	1.621	2.4546	n.a.	n.a.	n.a.
7	SULFATE	9.964	0.475	2.4882	n.a.	n.a.	n.a.

VII - LABORATORY CONTROL SPIKE DUPLICATE

Report No:	223060210	LAB ID:	LCSD2491181
Matrix:	Water	Instrument ID:	IC4
Analyst:	CS	Lab File ID:	NA
Prep Date:	NA	Analysis Date:	06/08/23 1043
Prep Batch:	NA	Analytical Batch:	767281
Prep Method:	NA	Analytical Method:	EPA 9056A

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*%REC*	*Q*	*%REC LIMITS*	*RPD*	*Q*	*RPD LIMITS*
Sulfate	mg/L	2.50	2.48	99		87 - 112	0		0 - 15

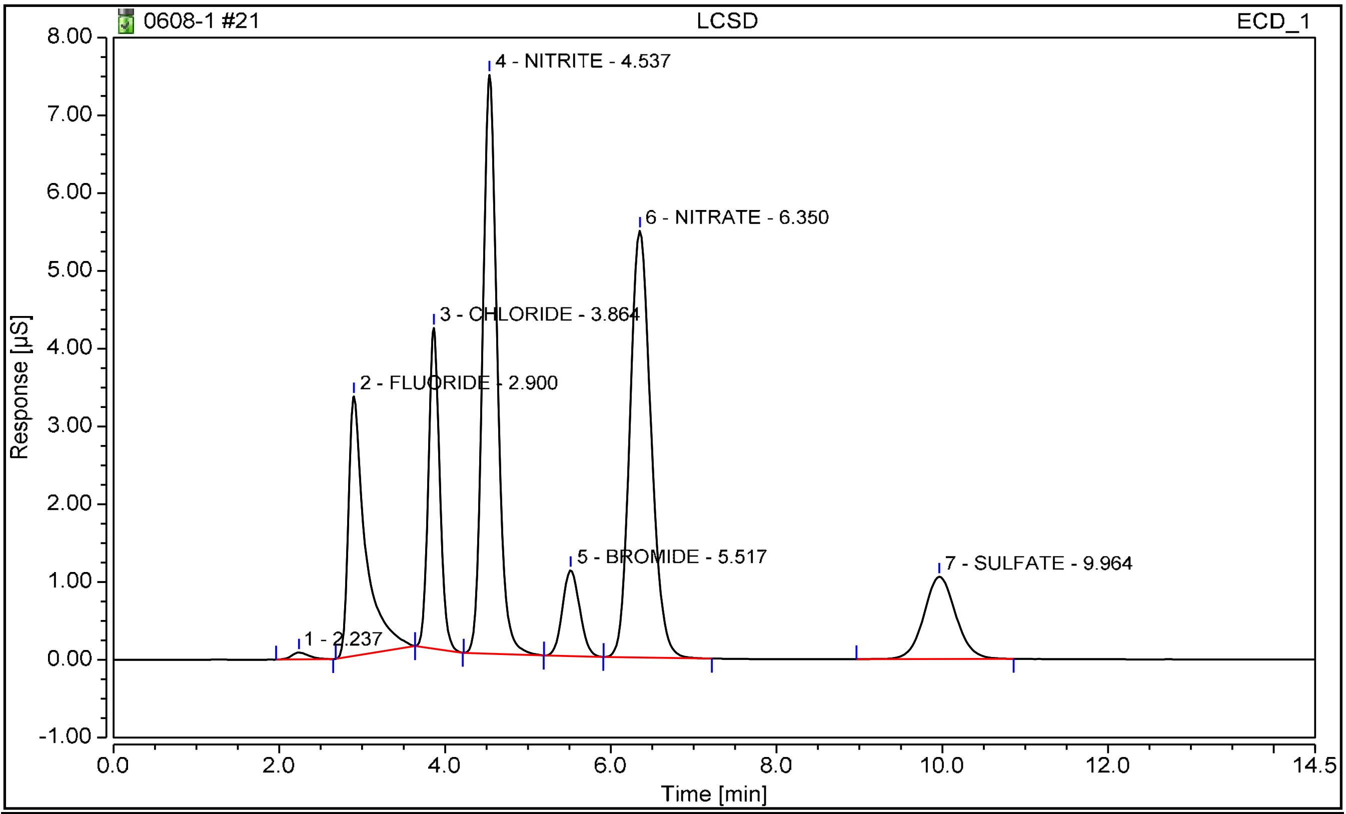
FORM VII - GENCHEM

Chromatogram and Results

Injection Details

Injection Name:	**LCSD**	*Operator:*	**CS**
Vial Number:	**7**	*Injection Volume:*	**5000.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**06/08/2023 10:43**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
2	FLUORIDE	2.900	0.800	2.5855	n.a.	n.a.	n.a.
3	CHLORIDE	3.864	0.654	2.3621	n.a.	n.a.	n.a.
4	NITRITE	4.537	1.529	2.4470	n.a.	n.a.	n.a.
5	BROMIDE	5.517	0.267	2.4624	n.a.	n.a.	n.a.
6	NITRATE	6.350	1.623	2.4577	n.a.	n.a.	n.a.
7	SULFATE	9.964	0.474	2.4821	n.a.	n.a.	n.a.

EPA 9056A

Form XIV

Run Logs

XIV - ANALYSIS RUN LOG

Report No:	223060210	Analytical Batch:	767085	Start Date:	06/02/23
Instrument ID:	BRWA13	Analytical Method:	EPA 9056A	End Date:	06/02/23

CLIENT SAMPLE ID	LAB SAMPLE ID	DILUTION	TIME	ANALYTES Chloride	Nitrate	Nitrite	Sulfate
CCB	1900	1	0948	X	X	X	X
CCV	1800	1	1006	X	X	X	X
MB2489994	2489994	1	1023	X	X	X	X
LCS2489995	2489995	1	1040	X	X	X	X
LCSD2489996	2489996	1	1133	X	X	X	X
OT07-MW05-060123	22306021004	1	1150	X	X	X	
OT07-MW04-060123	22306021003	1	1207	X	X	X	
OT07-MW04-060123MS	22306021006	1	1225	X	X	X	X
OT07-MW04-060123MSD	22306021007	1	1242	X	X	X	X
OT07-MW06-060123	22306021005	1	1259	X	X	X	
OT07-MW-806-060123	22306021008	1	1317	X	X	X	
OT07-EB-060123	22306021001	1	1334	X	X	X	X
OT07-MW02-060123	22306021002	1	1351	X	X	X	
CCB	1900	1	1654	X	X	X	X
CCV	1800	1	1711	X	X	X	X

FORM XIV - GENCHEM

XIV - ANALYSIS RUN LOG

Report No:	223060210	Analytical Batch:	767223	Start Date:	06/07/23
Instrument ID:	IC4	Analytical Method:	EPA 9056A	End Date:	06/07/23

CLIENT SAMPLE ID	*LAB SAMPLE ID*	*DILUTION*	*TIME*	*ANALYTES* Chloride	Nitrate	Nitrite	Sulfate
CCB	1900	1	1001	X			X
CCV	1800	1	1018	X			X
MB2490889	2490889	1	1036	X			X
LCS2490890	2490890	1	1053	X			X
LCSD2490891	2490891	1	1110	X			X
OT07-MW02-060123 DL	22306021002	5	1445				X
OT07-MW05-060123 DL	22306021004	2	1503				X
OT07-MW-806-060123 DL	22306021008	5	1520				X
CCB	1900	1	1537	X			X
CCV	1800	1	1555	X			X
OT07-MW06-060123 DL	22306021005	5	1612				X
CCB	1900	1	1906				X
CCV	1800	1	1923				X

FORM XIV - GENCHEM

XIV - ANALYSIS RUN LOG

Report No:	223060210	Analytical Batch:	767281	Start Date:	06/08/23
Instrument ID:	IC4	Analytical Method:	EPA 9056A	End Date:	06/08/23

CLIENT SAMPLE ID	*LAB SAMPLE ID*	*DILUTION*	*TIME*	*ANALYTES* Chloride	Nitrate	Nitrite	Sulfate
CCB	1900	1	0917				X
CCV	1800	1	0935				X
MB2491179	2491179	1	0952				X
LCS2491180	2491180	1	1010				X
LCSD2491181	2491181	1	1043				X
OT07-MW04-060123 DL	22306021003	50	1100				X
OT07-MW04-060123MS DL	22306021006	50	1117				X
OT07-MW04-060123MSD DL	22306021007	50	1135				X
CCB	1900	1	1635				X
CCV	1800	1	1652				X

FORM XIV - GENCHEM

EPA 9056A

Standards

Pace Analytical®

BATCH COVERSHEET

ANALYST	CS
DATE	06/06/23

Ion Chromatography 9056/ 300.0

BATCH **767085**

CALIBRATION STOCK INFORMATION

Calibration Standard ID	WLG1245-91-3
Calibration Standard Exp	9/30/23
Calibration Date	05/31/23
Calibration Exp	9/30/23

Instrument ID	BRWA13
Sequence Name	0602-1

VERIFICATION STANDARDS

CCV/LCS Standard ID	WLG1245-96-3
CCV/LCS Standard Exp Date	06/03/23
ICV Standard ID	WLG1245-90-2
ICV Standard Exp Date	06/01/23

REAGENTS

Eluent Lot	2133049
Eluent Exp	11/30/23

COMMON MANUAL INTEGRATION ABBREVIATIONS

NI: Peak was not identified by the data system

LT: peak integration was less area than it should be

GT: peak integration was more area than it should

BA: Baseline adjustment required

WP: Wrong peak was selected by data system

CO: Needed to split co-eluting peaks

RT: Retention time shift

INT: Interference correction

LAB QC LIMITS

300.0/9056A Correlation Coefficient (R) =0.995

Calibration Low Point %RE = 50%

Calibration Mid Point %RE = 10%

ICV Recovery 90-110%

LLCK/LLOQ Recovery 50-150%

CCV Recovery 90-110%

Pace Analytical®

BATCH COVERSHEET

ANALYST CS

DATE 06/08/23

Ion Chromatography 9056/ 300.0

BATCH 767223

CALIBRATION STOCK INFORMATION

Calibration Standard ID WLG1245-89-2

Calibration Standard Exp 9/30/23

Calibration Date 05/31/23

Calibration Exp 9/30/23

Instrument ID IC4

Sequence Name 0607-1

VERIFICATION STANDARDS

CCV/LCS Standard ID WLG1255-1-5

CCV/LCS Standard Exp Date 06/08/23

ICV Standard ID WLG1245-90-1

ICV Standard Exp Date 06/01/23

REAGENTS

Eluent Lot 2132501

Eluent Exp 10/13/23

COMMON MANUAL INTEGRATION ABBREVIATIONS

NI: Peak was not identified by the data system

LT: peak integration was less area than it should be

GT: peak integration was more area than it should

BA: Baseline adjustment required

WP: Wrong peak was selected by data system

CO: Needed to split co-eluting peaks

RT: Retention time shift

INT: Interference correction

LAB QC LIMITS

300.0/9056A Correlation Coefficient (R) =0.995

Calibration Low Point %RE = 50%

Calibration Mid Point %RE = 10%

ICV Recovery 90-110%

LLCK/LLOQ Recovery 50-150%

CCV Recovery 90-110%

BATCH COVERSHEET

ANALYST	CS
DATE	06/09/23

Ion Chromatography 9056/ 300.0

BATCH 767281

CALIBRATION STOCK INFORMATION

Calibration Standard ID	WLG1245-89-2
Calibration Standard Exp	9/30/23
Calibration Date	05/31/23
Calibration Exp	9/30/23

Instrument ID	IC4
Sequence Name	0608-1

VERIFICATION STANDARDS

CCV/LCS Standard ID	WLG1255-1-7
CCV/LCS Standard Exp Date	06/09/23
ICV Standard ID	WLG1245-90-1
ICV Standard Exp Date	06/01/23

REAGENTS

Eluent Lot	2132501
Eluent Exp	10/13/23

COMMON MANUAL INTEGRATION ABBREVIATIONS

NI: Peak was not identified by the data system

LT: peak integration was less area than it should be

GT: peak integration was more area than it should

BA: Baseline adjustment required

WP: Wrong peak was selected by data system

CO: Needed to split co-eluting peaks

RT: Retention time shift

INT: Interference correction

LAB QC LIMITS

300.0/9056A Correlation Coefficient (R) =0.995

Calibration Low Point %RE = 50%

Calibration Mid Point %RE = 10%

ICV Recovery 90-110%

LLCK/LLOQ Recovery 50-150%

CCV Recovery 90-110%

EPA 9056A

Initial Calibration

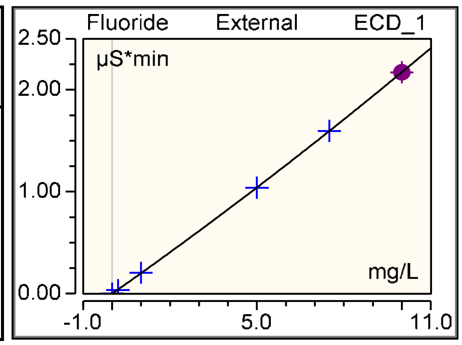
Calibration Batch Report

Sequence:	**0602-1**	*Injection Volume:*	**50.00**
Instrument Method:	**IC8 method 11Nov2022**	*Operator:*	**wet**
Injection Time:	**05/31/2023 14:28**	*Run Time:*	**14.5005**

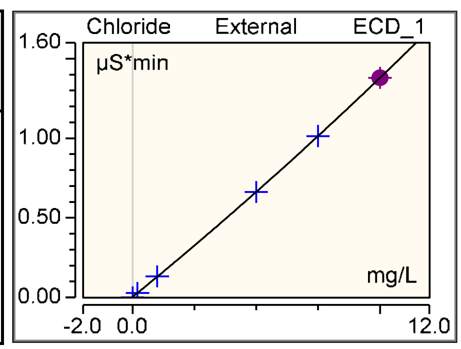
Calibration Summary

Peak Name	Eval.Type	Cal.Type	Points	Offset (C0)	Slope (C1)	Curve (C2)	Coeff.Det. %
Fluoride	Area	uad, WithOffs	5.000	-0.001	0.200	0.002	99.9989
Chloride	Area	uad, WithOffs	6.000	0.003	0.126	0.001	99.9992
Nitrite	Area	uad, WithOffs	5.000	-0.003	0.285	0.003	99.9998
Bromide	Area	uad, WithOffs	5.000	0.000	0.051	0.000	100.0000
Nitrate	Area	uad, WithOffs	5.000	-0.003	0.303	0.005	99.9971
Sulfate	Area	uad, WithOffs	5.000	0.000	0.094	0.000	99.9996
		AVERAGE:		-0.0005	0.1766	0.0019	99.9991

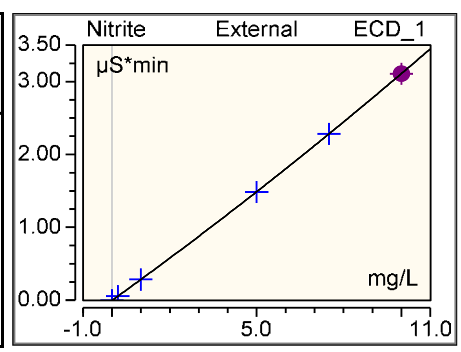
Injection Name	Ret.Time min ECD_1	Area µS*min ECD_1	Height µS ECD_1	Amount mg/L ECD_1
	Fluoride	Fluoride	Fluoride	Fluoride
1	n.a.	n.a.	n.a.	n.a.
2	3.234	0.0354	0.213	0.184
3	3.234	0.2047	1.122	1.022
4	3.237	1.0394	6.300	4.991
5	3.241	1.5964	9.935	7.504
6	3.244	2.1716	13.722	10.000
Average	3.238			
Rel. Std. Dev.	0.135 %			



Injection Name	Ret.Time min ECD_1	Area µS*min ECD_1	Height µS ECD_1	Amount mg/L ECD_1
	Chloride	Chloride	Chloride	Chloride
1	4.794	0.0005	0.004	n.a.
2	4.810	0.0286	0.176	0.203
3	4.807	0.1330	0.832	1.020
4	4.807	0.6624	4.209	4.995
5	4.804	1.0139	6.505	7.498
6	4.807	1.3797	8.896	10.002
Average	4.805			
Rel. Std. Dev.	0.120 %			

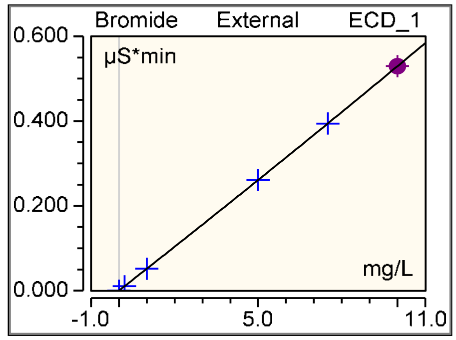


Injection Name	Ret.Time min ECD_1	Area µS*min ECD_1	Height µS ECD_1	Amount mg/L ECD_1
	Nitrite	Nitrite	Nitrite	Nitrite
1	n.a.	n.a.	n.a.	n.a.
2	5.857	0.0559	0.270	0.205
3	5.854	0.2841	1.376	0.995
4	5.861	1.4876	7.270	4.996
5	5.864	2.2852	11.166	7.507
6	5.871	3.1084	15.081	9.997
Average	5.861			

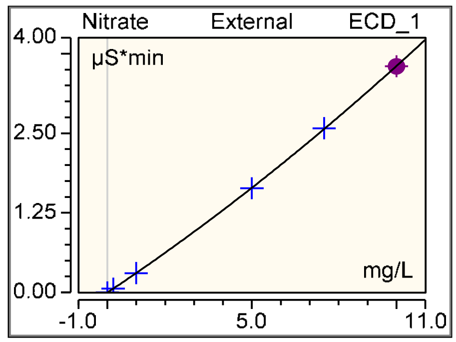


Rel. Std. Dev.	0.110 %

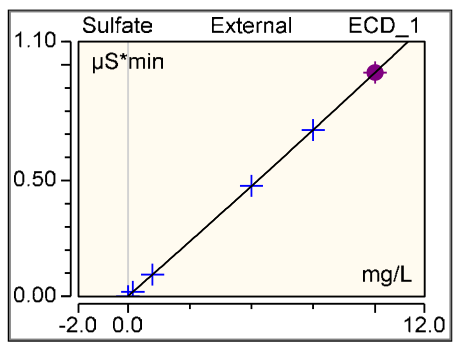
Injection Name	Ret.Time min ECD_1	Area µS*min ECD_1	Height µS ECD_1	Amount mg/L ECD_1
	Bromide	Bromide	Bromide	Bromide
1	n.a.	n.a.	n.a.	n.a.
2	7.154	0.0102	0.046	0.199
3	7.147	0.0516	0.234	1.000
4	7.141	0.2610	1.196	5.001
5	7.131	0.3942	1.817	7.499
6	7.134	0.5295	2.444	10.000
Average	7.141			
Rel. Std. Dev.	0.133 %			



Injection Name	Ret.Time min ECD_1	Area µS*min ECD_1	Height µS ECD_1	Amount mg/L ECD_1
	Nitrate	Nitrate	Nitrate	Nitrate
1	n.a.	n.a.	n.a.	n.a.
2	8.087	0.0608	0.233	0.209
3	8.074	0.3041	1.187	0.996
4	8.047	1.6347	6.442	4.975
5	8.034	2.5785	10.075	7.532
6	8.027	3.5505	13.857	9.989
Average	8.054			
Rel. Std. Dev.	0.320 %			



Injection Name	Ret.Time min ECD_1	Area µS*min ECD_1	Height µS ECD_1	Amount mg/L ECD_1
	Sulfate	Sulfate	Sulfate	Sulfate
1	n.a.	n.a.	n.a.	n.a.
2	11.337	0.0197	0.053	0.206
3	11.334	0.0934	0.244	0.991
4	11.337	0.4768	1.232	5.010
5	11.341	0.7183	1.865	7.491
6	11.344	0.9664	2.511	10.003
Average	11.338			
Rel. Std. Dev.	0.034 %			

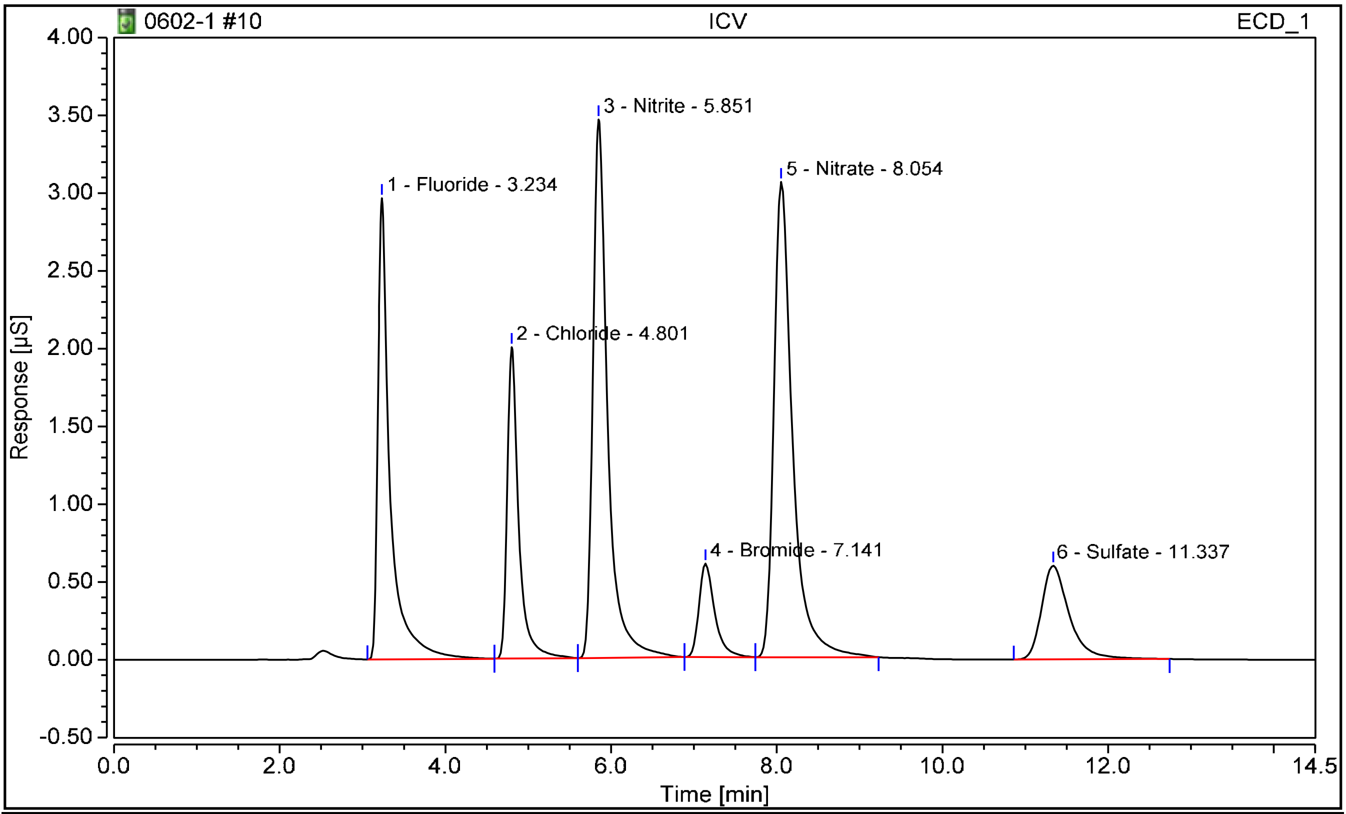


Chromatogram and Results

Injection Details

Injection Name:	**ICV**	*Run Time (min):*	**14.50**
Vial Number:	**10**	*Injection Volume:*	**50.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:	**ICV**	*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**05/31/2023 14:44**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

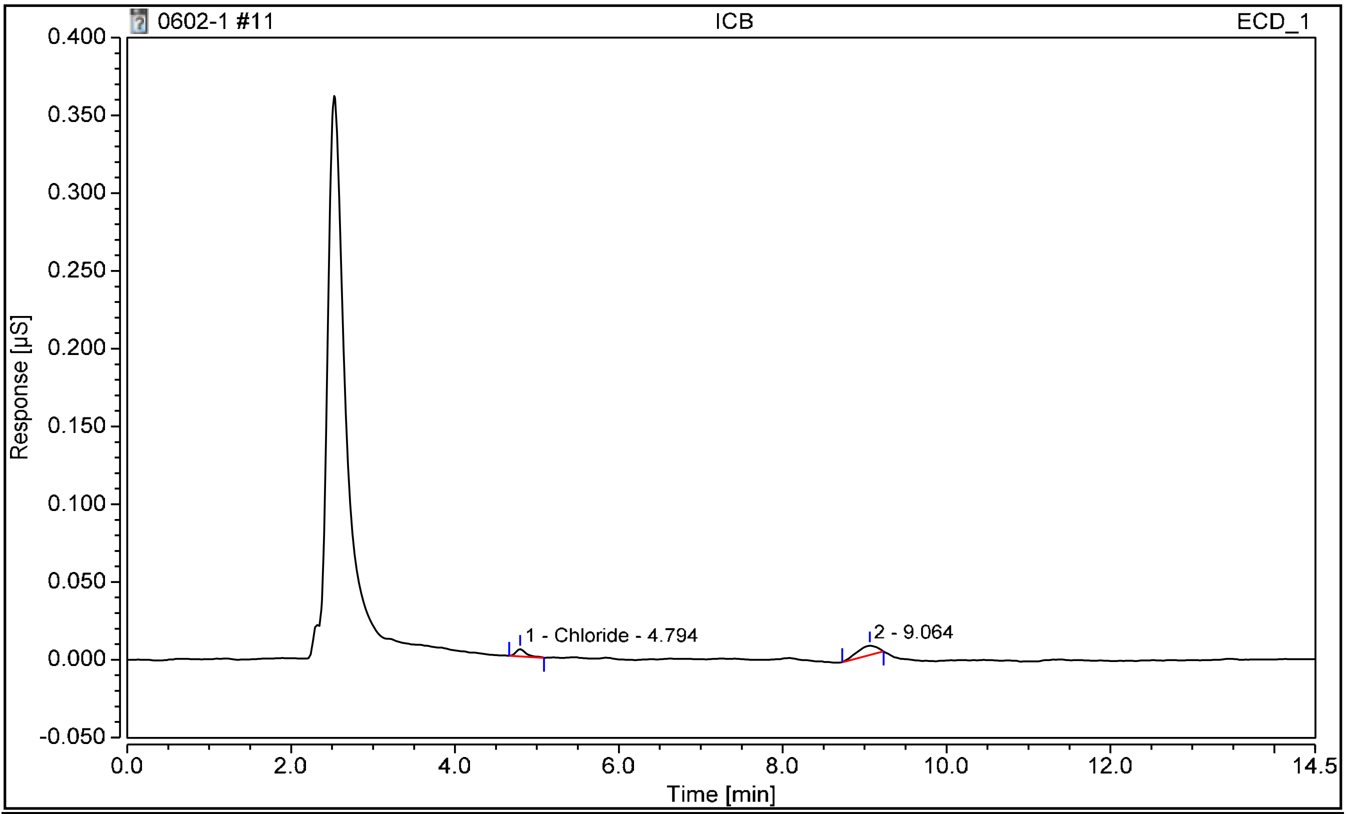
No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
1	Fluoride	3.234	0.503	2.4702	-0.0298	2.5000	98.81
2	Chloride	4.801	0.321	2.4616	-0.0384	2.5000	98.46
3	Nitrite	5.851	0.714	2.4548	-0.0452	2.5000	98.19
4	Bromide	7.141	0.133	2.5571	0.0571	2.5000	102.28
5	Nitrate	8.054	0.799	2.5343	0.0343	2.5000	101.37
6	Sulfate	11.337	0.235	2.4826	-0.0174	2.5000	99.30

Chromatogram and Results

Injection Details

Injection Name:	**ICB**	*Run Time (min):*	**14.50**
Vial Number:	**11**	*Injection Volume:*	**50.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**IC8 method 11Nov2022**
Injection Time:	**05/31/2023 15:01**	*Processing Method:*	**TMO**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
n.a.	Fluoride	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
1	Chloride	4.794	0.001	n.a.	n.a.	n.a.	n.a.
n.a.	Nitrite	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	Bromide	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	Nitrate	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	Sulfate	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

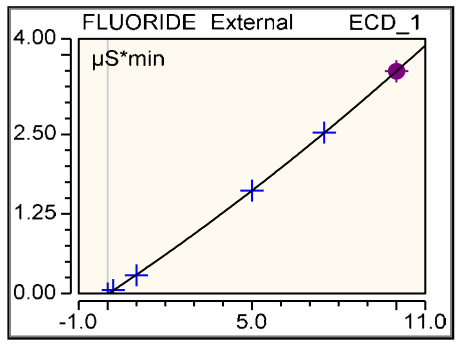
Calibration Batch Report

Sequence:	**0607-1**	*Injection Volume:*	**50.00**
Instrument Method:	**TEST 1**	*Operator:*	**LHM**
Injection Time:	**05/31/2023 13:59**	*Run Time:*	**14.500333**

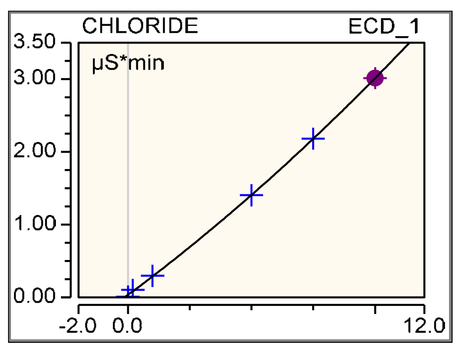
Calibration Summary

Peak Name	Eval.Type	Cal.Type	Points	Offset (C0)	Slope (C1)	Curve (C2)	Coeff.Det. %
FLUORIDE	Area	uad, WithOffs	5.000	-0.010	0.300	0.005	99.9985
CHLORIDE	Area	uad, WithOffs	6.000	0.034	0.252	0.005	99.9878
NITRITE	Area	uad, WithOffs	5.000	-0.006	0.612	0.006	99.9985
BROMIDE	Area	uad, WithOffs	5.000	0.000	0.107	0.001	99.9996
NITRATE	Area	uad, WithOffs	5.000	-0.052	0.645	0.015	99.9989
SULFATE	Area	uad, WithOffs	5.000	0.003	0.187	0.001	99.9999
		AVERAGE:		-0.0052	0.3504	0.0054	99.9972

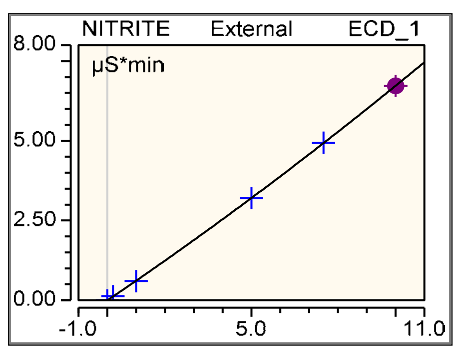
Injection Name	Ret.Time min ECD_1	Area μS*min ECD_1	Height μS ECD_1	Amount ECD_1
	FLUORIDE	FLUORIDE	FLUORIDE	FLUORIDE
1	n.a.	n.a.	n.a.	n.a.
2	2.930	0.0570	0.207	0.223
3	2.930	0.2868	1.090	0.973
4	2.937	1.6168	6.288	5.000
5	2.944	2.5267	9.911	7.508
6	2.947	3.4909	13.781	9.996
Average	2.938			
Rel. Std. Dev.	0.261 %			



Injection Name	Ret.Time min ECD_1	Area μS*min ECD_1	Height μS ECD_1	Amount ECD_1
	CHLORIDE	CHLORIDE	CHLORIDE	CHLORIDE
1	3.630	0.0114	0.058	n.a.
2	3.937	0.1039	0.474	0.275
3	3.940	0.2970	1.731	1.025
4	3.930	1.4040	8.991	4.988
5	3.927	2.1806	14.239	7.499
6	3.924	3.0132	19.849	10.003
Average	3.881			
Rel. Std. Dev.	3.177 %			

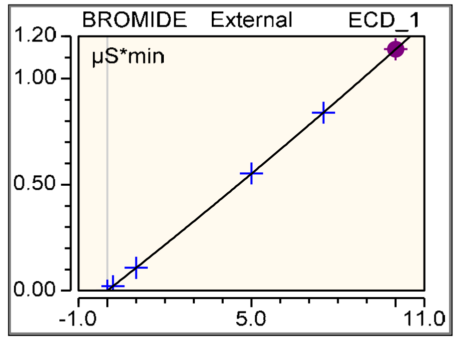


Injection Name	Ret.Time min ECD_1	Area μS*min ECD_1	Height μS ECD_1	Amount ECD_1
	NITRITE	NITRITE	NITRITE	NITRITE
1	n.a.	n.a.	n.a.	n.a.
2	4.640	0.1278	0.616	0.218
3	4.630	0.5997	2.932	0.980
4	4.630	3.2026	15.697	4.992
5	4.634	4.9410	24.019	7.517
6	4.640	6.7224	32.050	9.993
Average	4.635			

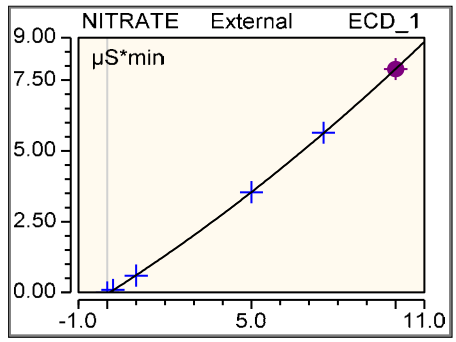


Rel. Std. Dev. 0.110 %

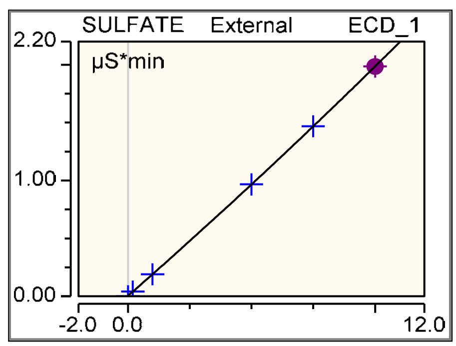
Injection Name	Ret.Time min ECD_1	Area µS*min ECD_1	Height µS ECD_1	Amount ECD_1
	BROMIDE	BROMIDE	BROMIDE	BROMIDE
1	n.a.	n.a.	n.a.	n.a.
2	5.650	0.0209	0.087	0.198
3	5.644	0.1075	0.451	1.001
4	5.623	0.5528	2.361	5.009
5	5.617	0.8393	3.637	7.488
6	5.610	1.1387	4.952	10.004
Average	5.629			
Rel. Std. Dev.	0.306 %			



Injection Name	Ret.Time min ECD_1	Area µS*min ECD_1	Height µS ECD_1	Amount ECD_1
	NITRATE	NITRATE	NITRATE	NITRATE
1	n.a.	n.a.	n.a.	n.a.
2	6.554	0.0888	0.303	0.217
3	6.527	0.5953	2.046	0.982
4	6.463	3.5413	12.550	4.993
5	6.441	5.6402	20.144	7.514
6	6.420	7.8922	28.125	9.994
Average	6.481			
Rel. Std. Dev.	0.880 %			



Injection Name	Ret.Time min ECD_1	Area µS*min ECD_1	Height µS ECD_1	Amount ECD_1
	SULFATE	SULFATE	SULFATE	SULFATE
1	n.a.	n.a.	n.a.	n.a.
2	10.407	0.0411	0.088	0.203
3	10.407	0.1903	0.424	0.995
4	10.387	0.9666	2.176	5.003
5	10.374	1.4679	3.318	7.498
6	10.364	1.9851	4.515	10.001
Average	10.388			
Rel. Std. Dev.	0.187 %			

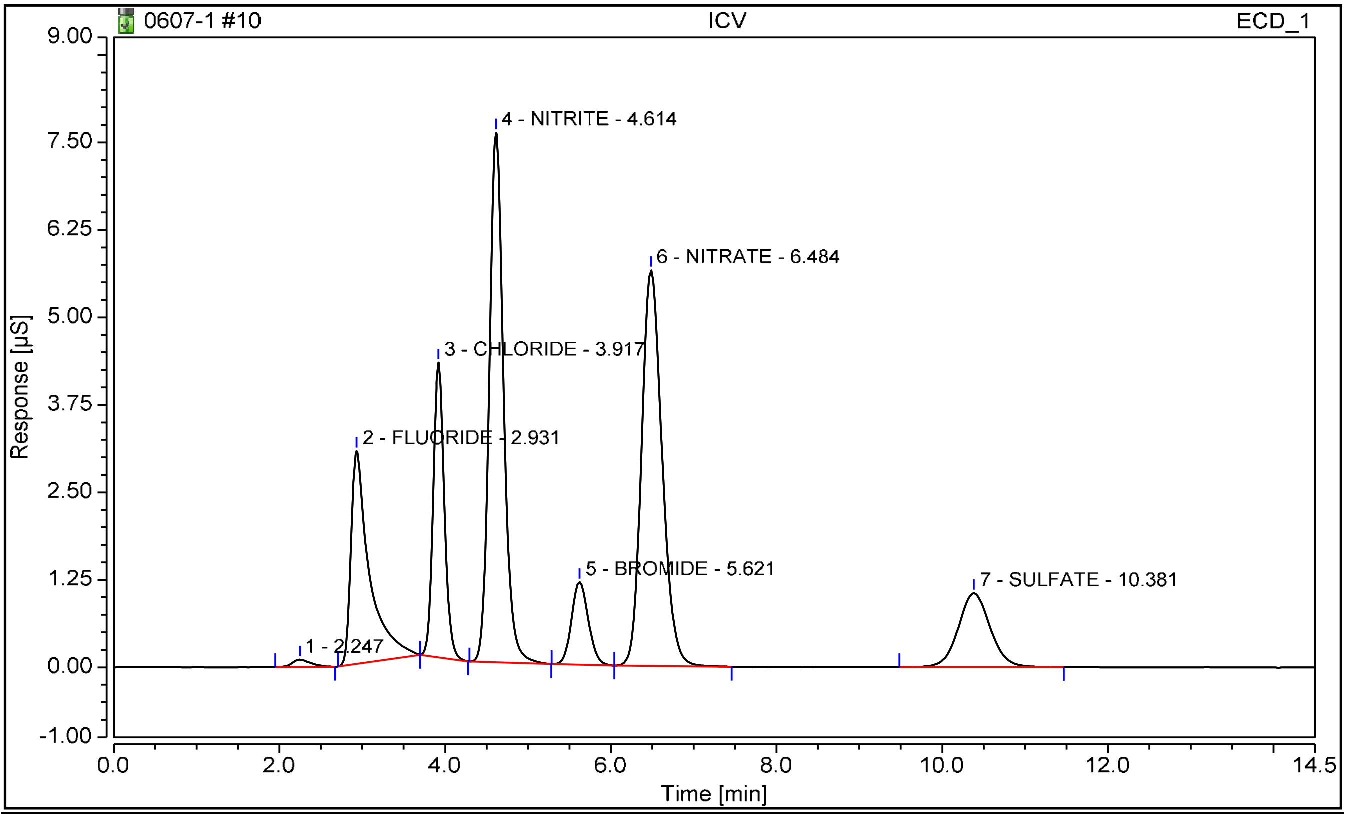


Chromatogram and Results

Injection Details

Injection Name:	**ICV**	*Operator:*	**LHM**
Vial Number:	**10**	*Injection Volume:*	**50.00**
Injection Type:	**Check Standard**	*Dilution Factor:*	**1**
Calibration Level:	**ICV**	*Instrument Method:*	**TEST 1**
Injection Time:	**05/31/2023 14:15**	*Processing Method:*	**9056**

Chromatogram



Integration Results

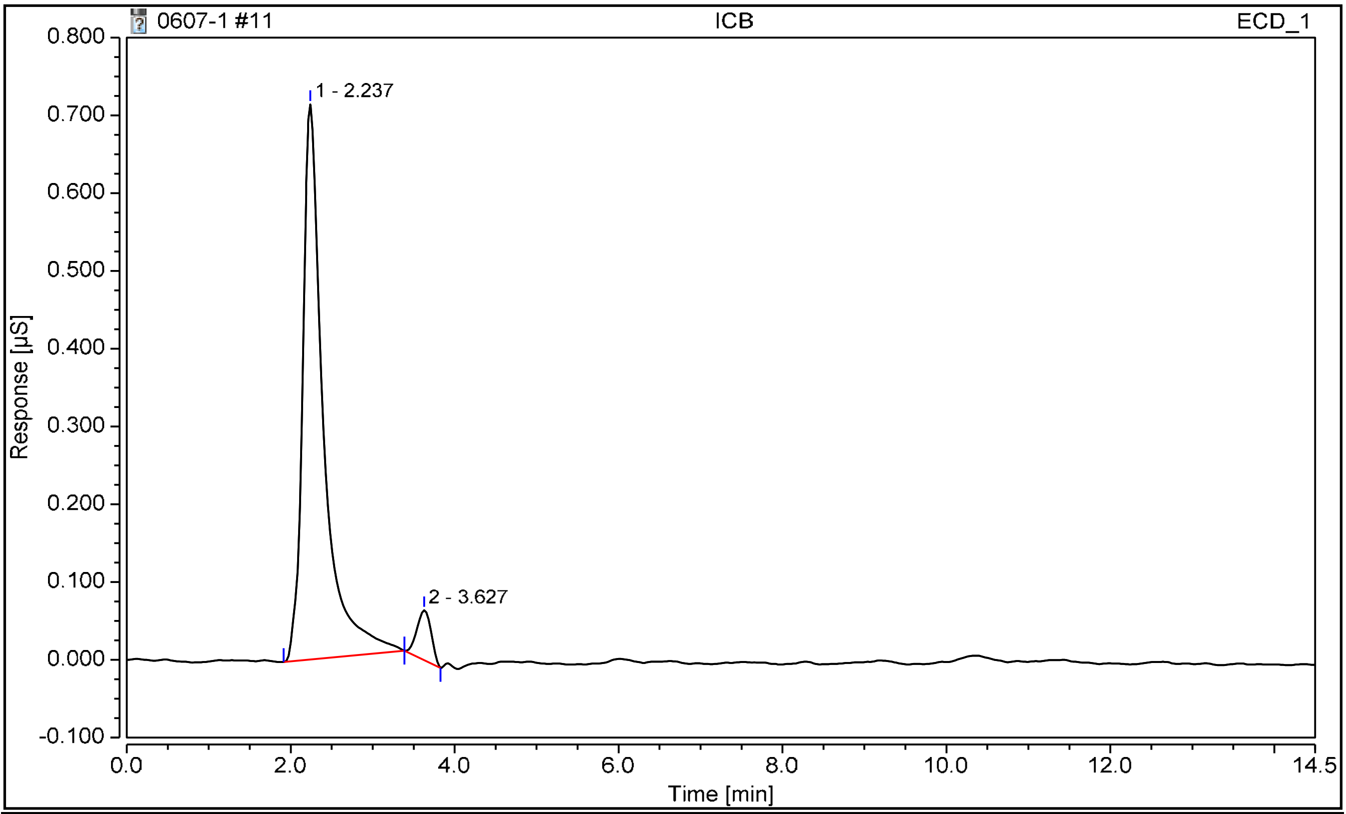
No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
2	FLUORIDE	2.931	0.785	2.5405	0.0405	2.5000	101.62
3	CHLORIDE	3.917	0.652	2.3557	-0.1443	2.5000	94.23
4	NITRITE	4.614	1.522	2.4368	-0.0632	2.5000	97.47
5	BROMIDE	5.621	0.278	2.5633	0.0633	2.5000	102.53
6	NITRATE	6.484	1.625	2.4607	-0.0393	2.5000	98.43
7	SULFATE	10.381	0.472	2.4717	-0.0283	2.5000	98.87

Chromatogram and Results

Injection Details

Injection Name:	**ICB**	*Operator:*	**LHM**
Vial Number:	**11**	*Injection Volume:*	**50.00**
Injection Type:	**Unknown**	*Dilution Factor:*	**1**
Calibration Level:		*Instrument Method:*	**TEST 1**
Injection Time:	**05/31/2023 14:31**	*Processing Method:*	**9056**

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area µS*min	Amount mg/L	Amnt.Dev. mg/L	True Value mg/L	Recovery %
n.a.	FLUORIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	CHLORIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	NITRITE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	BROMIDE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	NITRATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
n.a.	SULFATE	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

EPA 8260B DOD

Form 1A

Results

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-EB-060123		
Collect Date:	06/01/23	Time:	1214	GCAL Sample ID:	22306021001		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11		
Sample Amt:	5	mL		Lab File ID:	230602/j1695		
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942		
Analysis Date:	06/02/23	Time:	1446	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	0.500	U	0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	0.500	U	0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	0.500	U	0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	0.500	U	0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	0.500	U	0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	0.500	U	0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	0.500	U	0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	0.500	U	0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	0.500	U	0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	0.500	U	0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	0.500	U	0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	0.500	U	0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	0.500	U	0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	1.00	U	0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	0.500	U	0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	0.500	U	0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	0.500	U	0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	0.500	U	0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	0.500	U	0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	0.500	U	0.200	0.500	1.00
78-93-3	2-Butanone	0.500	U	0.200	0.500	5.00
95-49-8	2-Chlorotoluene	0.500	U	0.200	0.500	1.00
591-78-6	2-Hexanone	1.00	U	0.500	1.00	5.00
106-43-4	4-Chlorotoluene	0.500	U	0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	0.500	U	0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	0.500	U	0.200	0.500	5.00
67-64-1	Acetone	0.749	J	0.500	1.00	5.00
71-43-2	Benzene	0.500	U	0.200	0.500	1.00
108-86-1	Bromobenzene	0.500	U	0.200	0.500	1.00
74-97-5	Bromochloromethane	0.500	U	0.200	0.500	1.00
75-27-4	Bromodichloromethane	0.500	U	0.200	0.500	1.00
75-25-2	Bromoform	0.500	U	0.250	0.500	1.00
74-83-9	Bromomethane	1.00	U	0.500	1.00	2.00
75-15-0	Carbon disulfide	0.500	U	0.200	0.500	1.00
56-23-5	Carbon tetrachloride	0.500	U	0.250	0.500	1.00

FORM I VOA

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-EB-060123		
Collect Date:	06/01/23	Time:	1214	GCAL Sample ID:	22306021001		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11		
Sample Amt:	5	mL		Lab File ID:	230602/j1695		
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942		
Analysis Date:	06/02/23	Time:	1446	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	0.500	U	0.200	0.500	1.00
75-00-3	Chloroethane	0.500	U	0.250	0.500	1.00
67-66-3	Chloroform	0.500	U	0.200	0.500	1.00
74-87-3	Chloromethane	0.500	U	0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
124-48-1	Dibromochloromethane	0.500	U	0.200	0.500	1.00
74-95-3	Dibromomethane	0.500	U	0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	0.500	U	0.200	0.500	1.00
100-41-4	Ethylbenzene	0.500	U	0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	1.00	U	0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	0.500	U	0.200	0.500	1.00
136777-61-2	m,p-Xylene	0.500	U	0.200	0.500	2.00
75-09-2	Methylene chloride	0.825	J	0.200	0.500	5.00
91-20-3	Naphthalene	0.500	U	0.200	0.500	5.00
104-51-8	n-Butylbenzene	0.500	U	0.200	0.500	1.00
103-65-1	n-Propylbenzene	2.00	U	1.00	2.00	5.00
95-47-6	o-Xylene	0.500	U	0.200	0.500	1.00
135-98-8	sec-Butylbenzene	0.500	U	0.200	0.500	1.00
100-42-5	Styrene	0.500	U	0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	0.500	U	0.200	0.500	1.00
98-06-6	tert-Butylbenzene	0.500	U	0.200	0.500	1.00
127-18-4	Tetrachloroethene	0.500	U	0.200	0.500	1.00
108-88-3	Toluene	0.500	U	0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
79-01-6	Trichloroethene	0.500	U	0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	0.500	U	0.200	0.500	1.00
75-01-4	Vinyl chloride	0.500	U	0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv11.i/230602.s.b/j1695.d
Lab Smp Id: 22306021001
Inj Date : 02-JUN-2023 14:46
Operator : KN1 Inst ID: msv11.i
Smp Info : 22306021001*
Misc Info : MSV~52914~*1*KN1
Comment :
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Meth Date : 05-Jun-2023 08:22 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 15:02 Cal File: j1306.d
Als bottle: 1
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260bDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

						CONCENTRATIONS		
	QUANT SIG					ON-COLUMN	FINAL	
Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ug/L)	SIMILARITY
18 Methylene Chloride	49	3.760	3.760	(0.578)	2523	0.82522	0.825	
19 Acetone	43	3.807	3.812	(0.585)	1374	0.74901	0.749	
$ 41 Dibromofluoromethane	111	5.710	5.715	(0.878)	80362	58.2605	58.3	5441
$ 51 1,2-Dichloroethane-d4	65	6.229	6.229	(0.957)	105627	57.3456	57.3	
* 56 FLUOROBENZENE	96	6.507	6.507	(1.000)	276659	50.0000		
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	267434	51.2934	51.3	
* 90 CHLOROBENZENE-d5	82	9.238	9.239	(1.000)	108920	50.0000		
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	78609	50.0128	50.0	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	104291	50.0000		

Data File: /var/chem/msv11.i/230602.s.b/j1695.d

Page 1

Date : 02-JUN-2023 14:46

Client ID:

Instrument: msv11.i

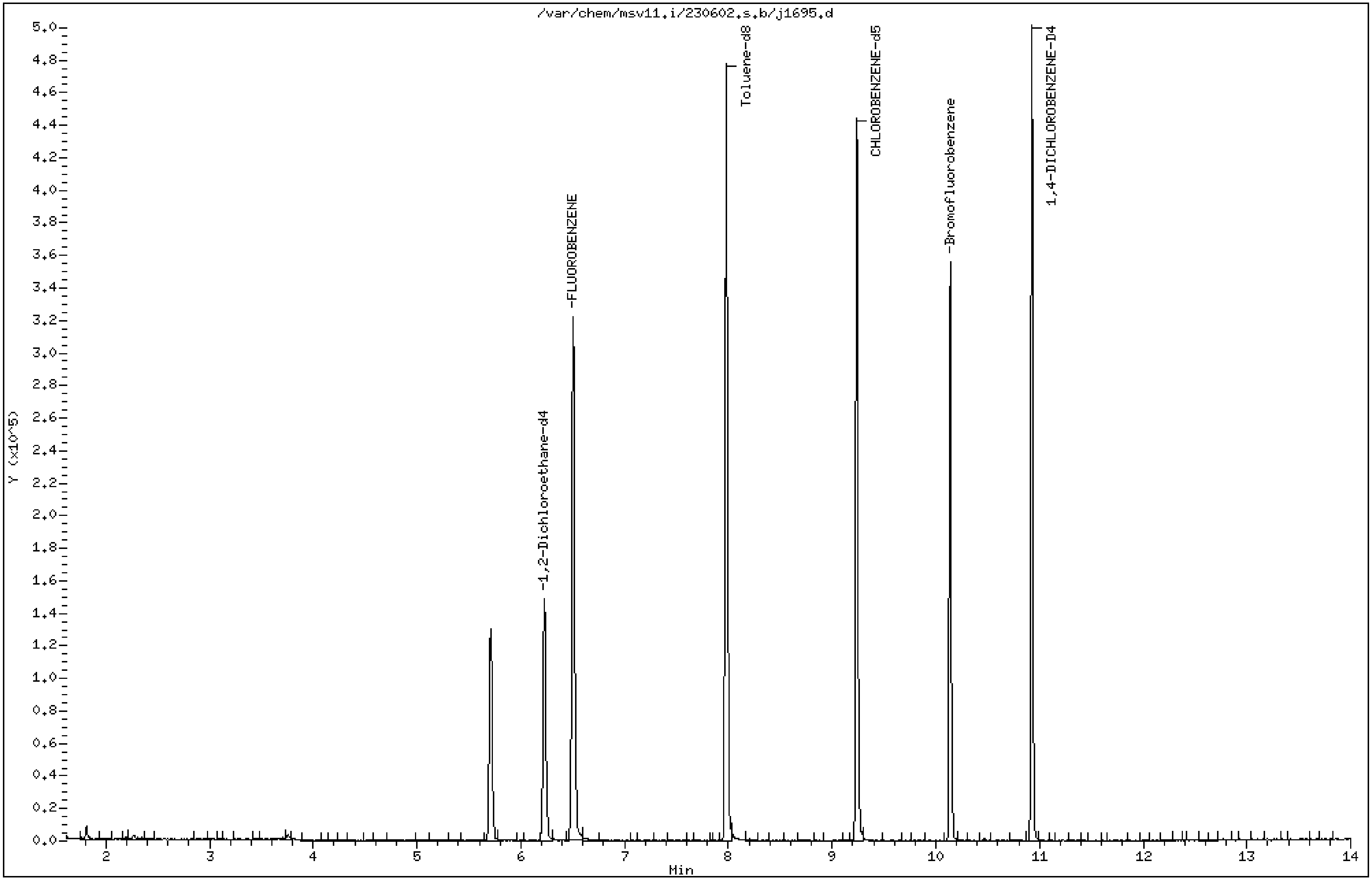
Sample Info: 22306021001*

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

Column diameter: 0.25



Data File: /var/chem/msv11.i/230602.s.b/j1695.d

Date : 02-JUN-2023 14:46

Client ID:

Instrument: msv11.i

Sample Info: 22306021001*

Purge Volume: 5.0

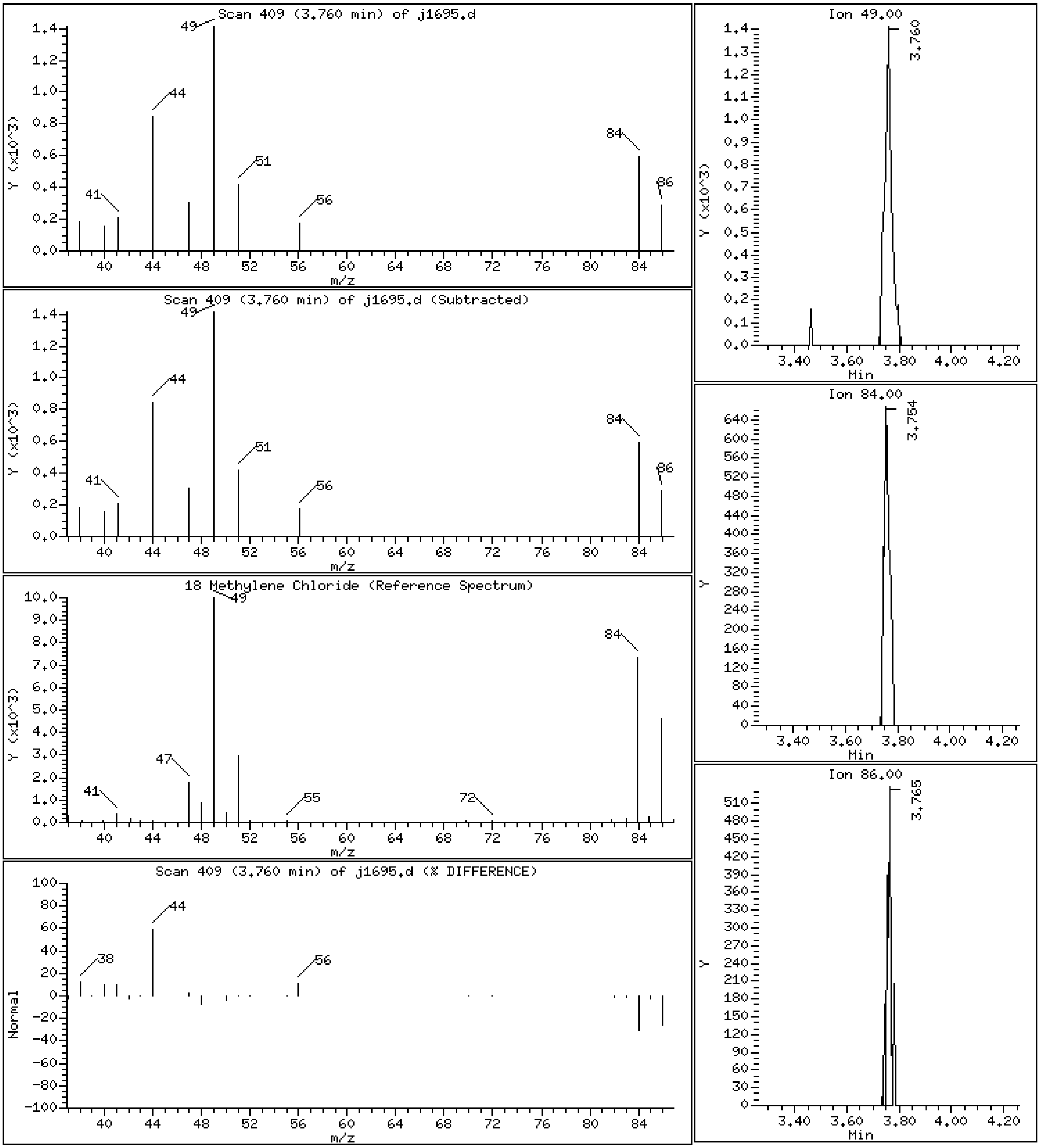
Operator: KN1

Column phase: RTX-VMS-30M

Column diameter: 0.25

18 Methylene Chloride

Concentration: 0.825 ug/L



Data File: /var/chem/msv11.i/230602.s.b/j1695.d

Date : 02-JUN-2023 14:46

Client ID:

Instrument: msv11.i

Sample Info: 22306021001*

Purge Volume: 5.0

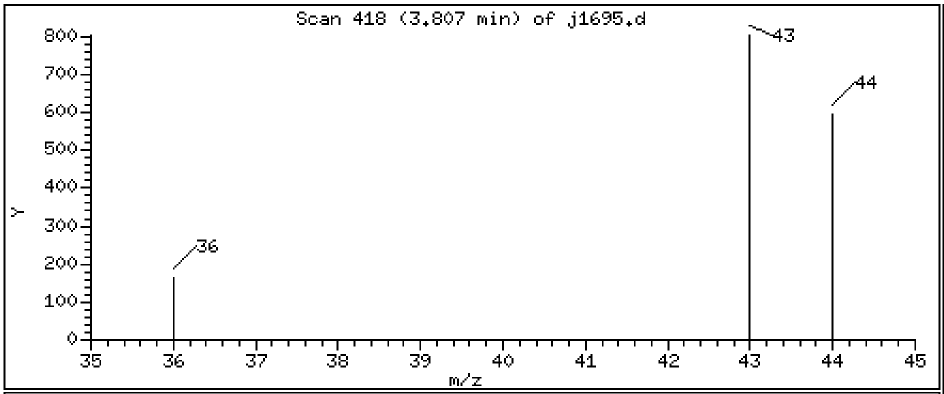
Operator: KN1

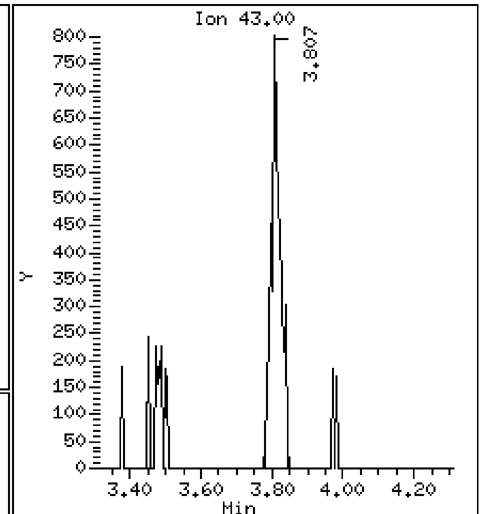
Column phase: RTX-VMS-30M

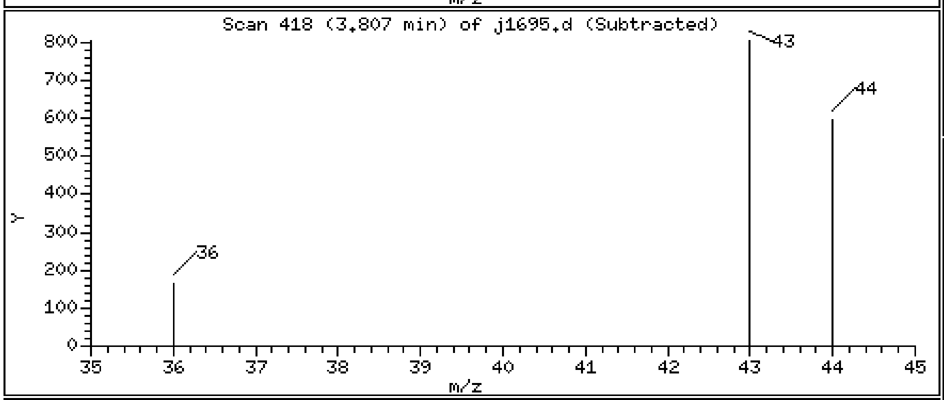
Column diameter: 0.25

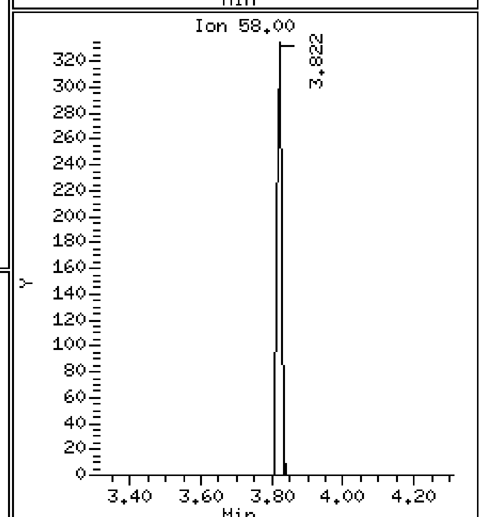
19 Acetone

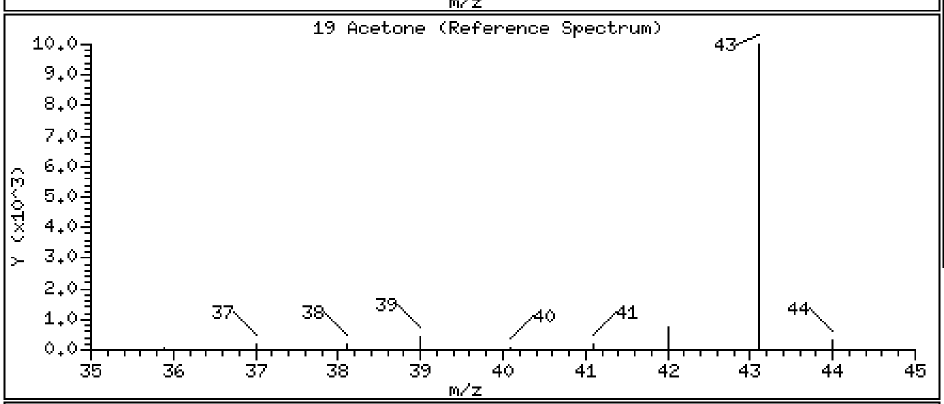
Concentration: 0.749 ug/L

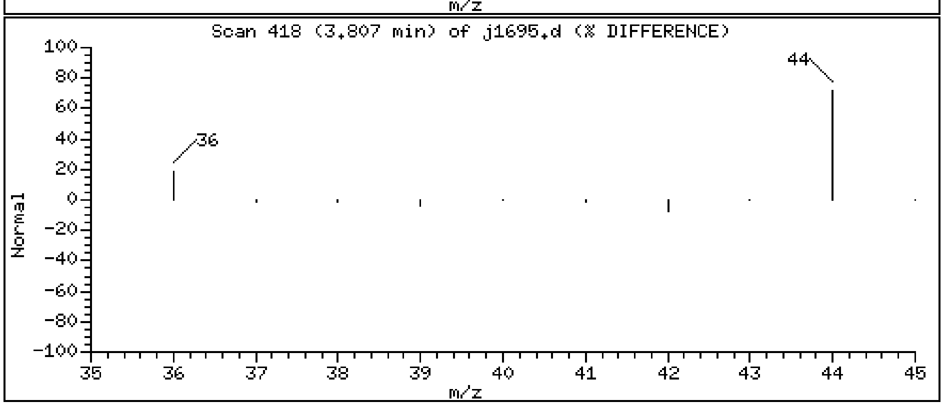








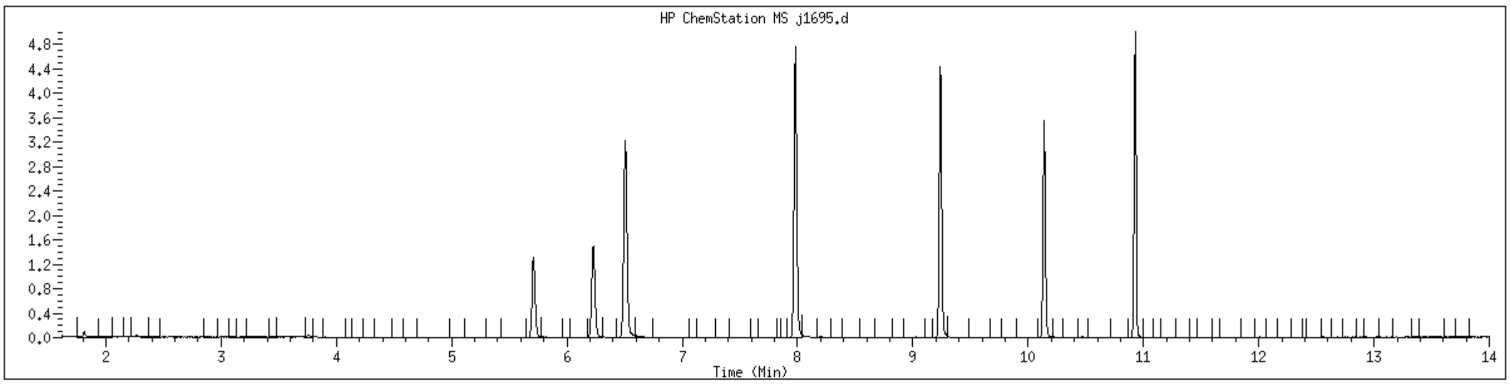




Report Date: 06/05/2023 09:16

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 22306021001 SampleType : SAMPLE
Injection Date: 06/02/2023 14:46 Instrument : msv11.i
Operator : KN1
Sample Info : 22306021001*
Misc Info : MSV~52914~*1*KN1
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260bDOD



NO MANUAL INTEGRATIONS

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-MW02-060123
Collect Date:	06/01/23	Time:	1230	GCAL Sample ID:	22306021002
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11
Sample Amt:	5	mL		Lab File ID:	230602/j1697
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942
Analysis Date:	06/02/23	Time:	1532	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	0.500	U	0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	0.500	U	0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	0.500	U	0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	0.500	U	0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	0.500	U	0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	0.500	U	0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	0.500	U	0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	0.500	U	0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	0.500	U	0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	0.500	U	0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	0.500	U	0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	0.500	U	0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	0.500	U	0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	1.00	U	0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	0.500	U	0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	0.500	U	0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	0.500	U	0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	0.500	U	0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	0.500	U	0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	0.500	U	0.200	0.500	1.00
78-93-3	2-Butanone	0.500	U	0.200	0.500	5.00
95-49-8	2-Chlorotoluene	0.500	U	0.200	0.500	1.00
591-78-6	2-Hexanone	1.00	U	0.500	1.00	5.00
106-43-4	4-Chlorotoluene	0.500	U	0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	0.500	U	0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	0.500	U	0.200	0.500	5.00
67-64-1	Acetone	1.00	U	0.500	1.00	5.00
71-43-2	Benzene	0.500	U	0.200	0.500	1.00
108-86-1	Bromobenzene	0.500	U	0.200	0.500	1.00
74-97-5	Bromochloromethane	0.500	U	0.200	0.500	1.00
75-27-4	Bromodichloromethane	0.500	U	0.200	0.500	1.00
75-25-2	Bromoform	0.500	U	0.250	0.500	1.00
74-83-9	Bromomethane	1.00	U	0.500	1.00	2.00
75-15-0	Carbon disulfide	0.500	U	0.200	0.500	1.00
56-23-5	Carbon tetrachloride	0.500	U	0.250	0.500	1.00

FORM I VOA

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-MW02-060123		
Collect Date:	06/01/23	Time:	1230	GCAL Sample ID:	22306021002		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11		
Sample Amt:	5	mL		Lab File ID:	230602/j1697		
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942		
Analysis Date:	06/02/23	Time:	1532	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	0.500	U	0.200	0.500	1.00
75-00-3	Chloroethane	0.500	U	0.250	0.500	1.00
67-66-3	Chloroform	0.500	U	0.200	0.500	1.00
74-87-3	Chloromethane	0.500	U	0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
124-48-1	Dibromochloromethane	0.500	U	0.200	0.500	1.00
74-95-3	Dibromomethane	0.500	U	0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	0.500	U	0.200	0.500	1.00
100-41-4	Ethylbenzene	0.500	U	0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	1.00	U	0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	0.500	U	0.200	0.500	1.00
136777-61-2	m,p-Xylene	0.500	U	0.200	0.500	2.00
75-09-2	Methylene chloride	0.500	U	0.200	0.500	5.00
91-20-3	Naphthalene	0.500	U	0.200	0.500	5.00
104-51-8	n-Butylbenzene	0.500	U	0.200	0.500	1.00
103-65-1	n-Propylbenzene	2.00	U	1.00	2.00	5.00
95-47-6	o-Xylene	0.500	U	0.200	0.500	1.00
135-98-8	sec-Butylbenzene	0.500	U	0.200	0.500	1.00
100-42-5	Styrene	0.500	U	0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	0.500	U	0.200	0.500	1.00
98-06-6	tert-Butylbenzene	0.500	U	0.200	0.500	1.00
127-18-4	Tetrachloroethene	0.500	U	0.200	0.500	1.00
108-88-3	Toluene	0.500	U	0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
79-01-6	Trichloroethene	0.500	U	0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	0.500	U	0.200	0.500	1.00
75-01-4	Vinyl chloride	0.500	U	0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv11.i/230602.s.b/j1697.d
Lab Smp Id: 22306021002
Inj Date : 02-JUN-2023 15:32
Operator : KN1 Inst ID: msv11.i
Smp Info : 22306021002*
Misc Info : MSV~52914~*1*KN1
Comment :
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Meth Date : 05-Jun-2023 08:22 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 15:02 Cal File: j1306.d
Als bottle: 1
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260bDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	CONCENTRATIONS FINAL (ug/L)	SIMILARITY
$ 41 Dibromofluoromethane	111	5.710	5.715	(0.877)	77430	58.9252	58.9	5308
$ 51 1,2-Dichloroethane-d4	65	6.229	6.229	(0.957)	103428	58.9429	58.9	
* 56 FLUOROBENZENE	96	6.512	6.507	(1.000)	263558	50.0000		
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	262188	52.0555	52.1	
* 90 CHLOROBENZENE-d5	82	9.238	9.239	(1.000)	105220	50.0000		
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	78813	51.9059	51.9	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	99785	50.0000		

Data File: /var/chem/msv11.i/230602.s.b/j1697.d

Date : 02-JUN-2023 15:32

Client ID:

Sample Info: 22306021002*

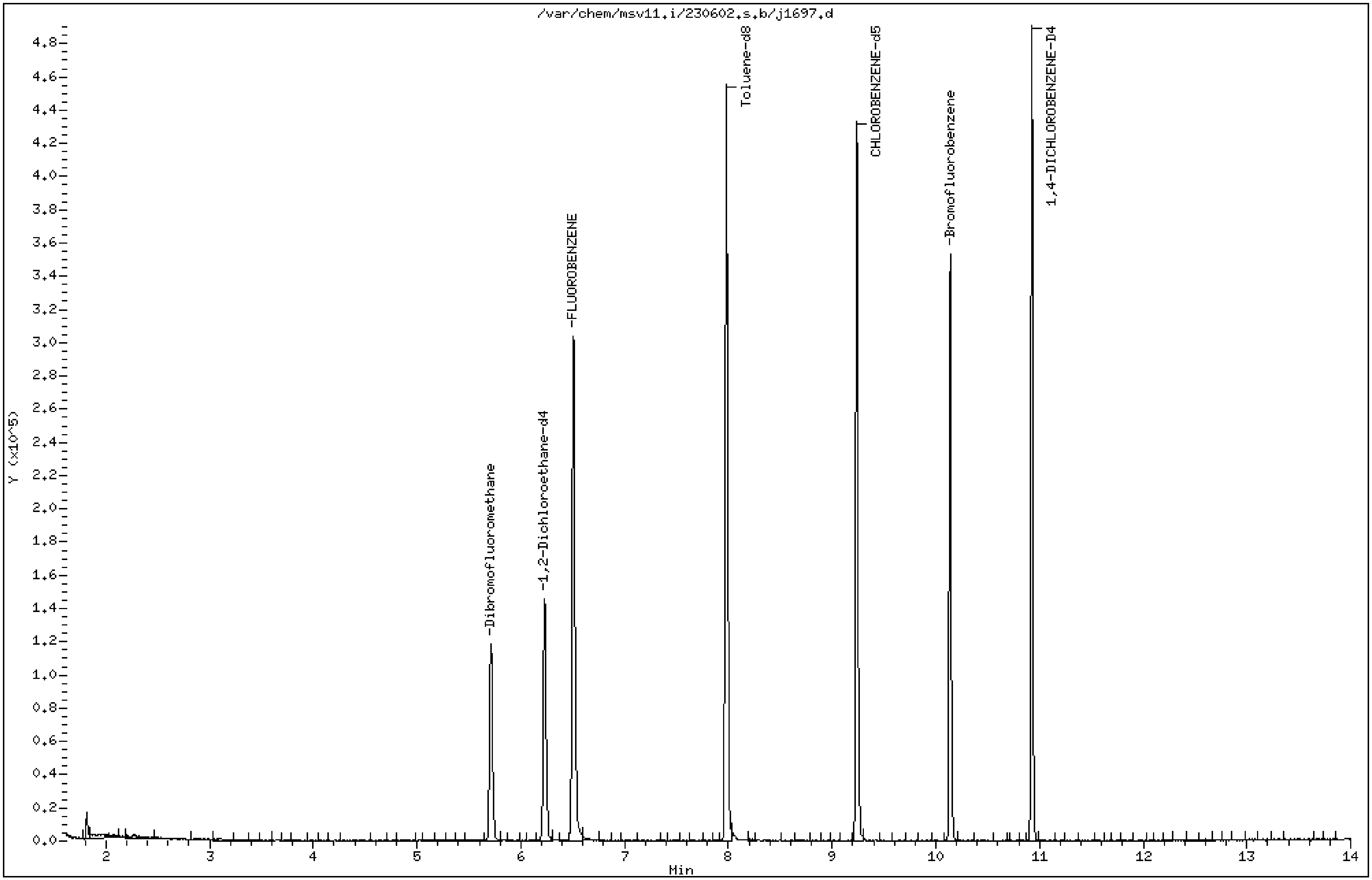
Purge Volume: 5.0

Column phase: RTX-VMS-30M

Instrument: msv11.i

Operator: KN1

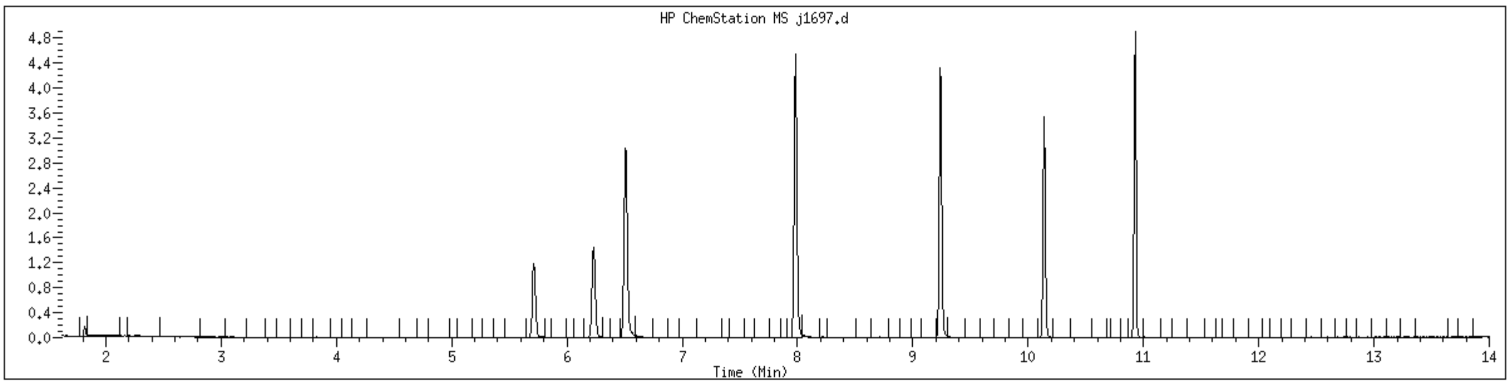
Column diameter: 0.25



Data file : /var/chem/msv11.i/230602.s.b/j1697.d Page: 1
Report Date: 06/05/2023 09:17

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 22306021002 SampleType : SAMPLE
Injection Date: 06/02/2023 15:32 Instrument : msv11.i
Operator : KN1
Sample Info : 22306021002*
Misc Info : MSV~52914~*1*KN1
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260bDOD



NO MANUAL INTEGRATIONS

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-MW04-060123
Collect Date:	06/01/23	Time:	0935	GCAL Sample ID:	22306021003
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11
Sample Amt:	5	mL		Lab File ID:	230602/j1698
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942
Analysis Date:	06/02/23	Time:	1556	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	0.500	U	0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	0.500	U	0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	0.500	U	0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	0.500	U	0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	0.500	U	0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	0.500	U	0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	0.500	U	0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	0.500	U	0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	0.500	U	0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	0.500	U	0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	0.500	U	0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	0.500	U	0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	0.500	U	0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	1.00	U	0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	0.500	U	0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	0.500	U	0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	0.500	U	0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	0.500	U	0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	0.500	U	0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	0.500	U	0.200	0.500	1.00
78-93-3	2-Butanone	0.500	U	0.200	0.500	5.00
95-49-8	2-Chlorotoluene	0.500	U	0.200	0.500	1.00
591-78-6	2-Hexanone	1.00	U	0.500	1.00	5.00
106-43-4	4-Chlorotoluene	0.500	U	0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	0.500	U	0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	0.500	U	0.200	0.500	5.00
67-64-1	Acetone	1.00	U	0.500	1.00	5.00
71-43-2	Benzene	0.500	U	0.200	0.500	1.00
108-86-1	Bromobenzene	0.500	U	0.200	0.500	1.00
74-97-5	Bromochloromethane	0.500	U	0.200	0.500	1.00
75-27-4	Bromodichloromethane	0.500	U	0.200	0.500	1.00
75-25-2	Bromoform	0.500	U	0.250	0.500	1.00
74-83-9	Bromomethane	1.00	U	0.500	1.00	2.00
75-15-0	Carbon disulfide	0.500	U	0.200	0.500	1.00
56-23-5	Carbon tetrachloride	0.500	U	0.250	0.500	1.00

FORM I VOA

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-MW04-060123		
Collect Date:	06/01/23	Time:	0935	GCAL Sample ID:	22306021003		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11		
Sample Amt:	5	mL		Lab File ID:	230602/j1698		
Injection Vol.:	1		(µL)	GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942		
Analysis Date:	06/02/23	Time:	1556	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	0.500	U	0.200	0.500	1.00
75-00-3	Chloroethane	0.500	U	0.250	0.500	1.00
67-66-3	Chloroform	0.500	U	0.200	0.500	1.00
74-87-3	Chloromethane	0.500	U	0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
124-48-1	Dibromochloromethane	0.500	U	0.200	0.500	1.00
74-95-3	Dibromomethane	0.500	U	0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	0.500	UJ	0.200	0.500	1.00
100-41-4	Ethylbenzene	0.500	U	0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	1.00	U	0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	0.500	U	0.200	0.500	1.00
136777-61-2	m,p-Xylene	0.500	U	0.200	0.500	2.00
75-09-2	Methylene chloride	0.500	U	0.200	0.500	5.00
91-20-3	Naphthalene	0.500	U	0.200	0.500	5.00
104-51-8	n-Butylbenzene	0.500	U	0.200	0.500	1.00
103-65-1	n-Propylbenzene	2.00	U	1.00	2.00	5.00
95-47-6	o-Xylene	0.500	U	0.200	0.500	1.00
135-98-8	sec-Butylbenzene	0.500	U	0.200	0.500	1.00
100-42-5	Styrene	0.500	U	0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	0.500	U	0.200	0.500	1.00
98-06-6	tert-Butylbenzene	0.500	U	0.200	0.500	1.00
127-18-4	Tetrachloroethene	0.500	U	0.200	0.500	1.00
108-88-3	Toluene	0.500	U	0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
79-01-6	Trichloroethene	0.500	U	0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	0.500	U	0.200	0.500	1.00
75-01-4	Vinyl chloride	0.500	U	0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv11.i/230602.s.b/j1698.d
Lab Smp Id: 22306021003
Inj Date : 02-JUN-2023 15:56
Operator : KN1 Inst ID: msv11.i
Smp Info : 22306021003*
Misc Info : MSV~52914~*1*KN1
Comment :
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Meth Date : 05-Jun-2023 08:22 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 15:02 Cal File: j1306.d
Als bottle: 1
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260bDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
$ 41 Dibromofluoromethane	111	5.715	5.715	(0.878)	77563	59.9426	59.9	5308(R)
$ 51 1,2-Dichloroethane-d4	65	6.234	6.229	(0.957)	101865	58.9532	59.0	
* 56 FLUOROBENZENE	96	6.512	6.507	(1.000)	259530	50.0000		
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	258442	50.4144	50.4	
* 90 CHLOROBENZENE-d5	82	9.238	9.239	(1.000)	107093	50.0000		
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	76478	49.4871	49.5	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	102497	50.0000		

QC Flag Legend

R - Spike/Surrogate failed recovery limits.

Data File: /var/chem/msv11.i/230602.s.b/j1698.d

Page 1

Date : 02-JUN-2023 15:56

Client ID:

Instrument: msv11.i

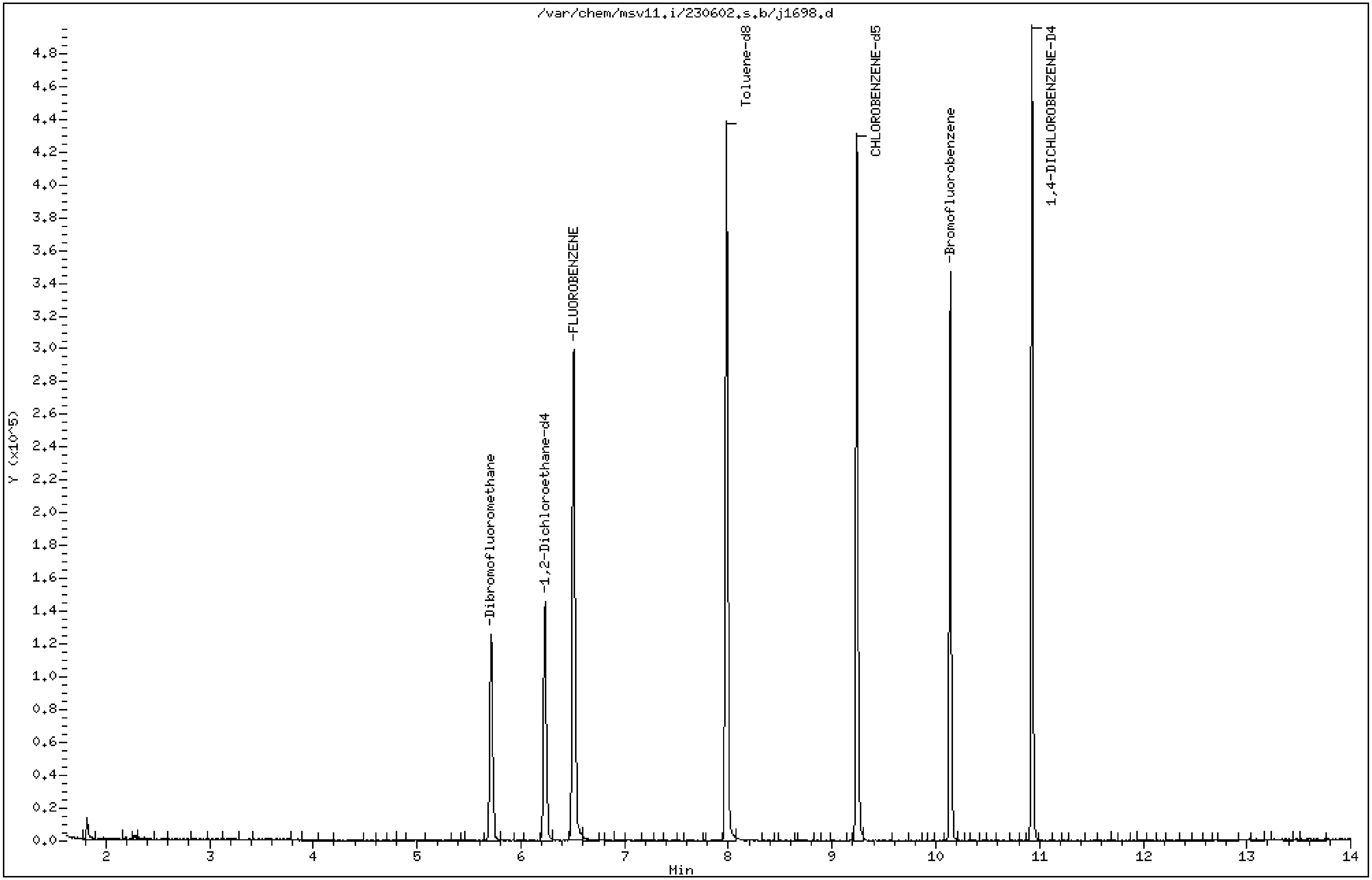
Sample Info: 22306021003*

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

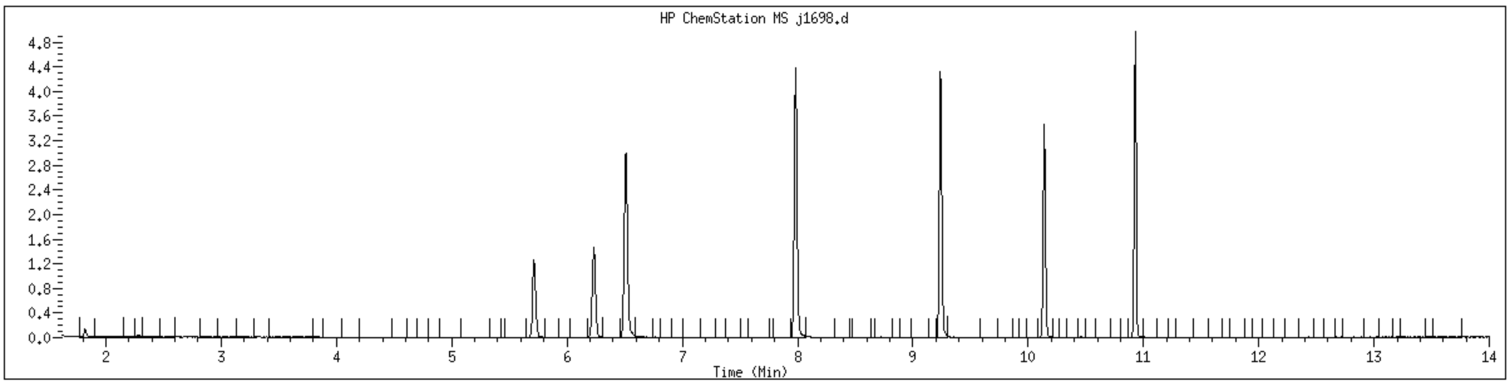
Column diameter: 0.25



Data file : /var/chem/msv11.i/230602.s.b/j1698.d Page: 1
Report Date: 06/05/2023 09:18

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID	: 22306021003	SampleType	: SAMPLE
Injection Date	: 06/02/2023 15:56	Instrument	: msv11.i
Operator	: KN1		
Sample Info	: 22306021003*		
Misc Info	: MSV~52914~*1*KN1		
Method	: /var/chem/msv11.i/230602.s.b/8260bDODw11.m		
Dilution	: 1.00		
Matrix	: WATER		
Integrator	: HP RTE	Compound Sublist	: 8260bDOD



NO MANUAL INTEGRATIONS

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-MW05-060123		
Collect Date:	06/01/23	Time:	0910	GCAL Sample ID:	22306021004		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11		
Sample Amt:	5	mL		Lab File ID:	230602/j1699		
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942		
Analysis Date:	06/02/23	Time:	1619	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	***ANALYTE***	***RESULT***	***Q***	***DL***	***LOD***	***LOQ***
630-20-6	1,1,1,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	0.500	U	0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	0.500	U	0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	0.500	U	0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	0.500	U	0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	0.500	U	0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	0.500	U	0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	0.500	U	0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	0.500	U	0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	0.500	U	0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	0.500	U	0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	0.500	U	0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	0.500	U	0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	0.500	U	0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	1.00	U	0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	0.500	U	0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	0.500	U	0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	0.500	U	0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	0.500	U	0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	0.500	U	0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	0.500	U	0.200	0.500	1.00
78-93-3	2-Butanone	0.500	U	0.200	0.500	5.00
95-49-8	2-Chlorotoluene	0.500	U	0.200	0.500	1.00
591-78-6	2-Hexanone	1.00	U	0.500	1.00	5.00
106-43-4	4-Chlorotoluene	0.500	U	0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	0.500	U	0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	0.500	U	0.200	0.500	5.00
67-64-1	Acetone	1.00	U	0.500	1.00	5.00
71-43-2	Benzene	0.500	U	0.200	0.500	1.00
108-86-1	Bromobenzene	0.500	U	0.200	0.500	1.00
74-97-5	Bromochloromethane	0.500	U	0.200	0.500	1.00
75-27-4	Bromodichloromethane	0.500	U	0.200	0.500	1.00
75-25-2	Bromoform	0.500	U	0.250	0.500	1.00
74-83-9	Bromomethane	1.00	U	0.500	1.00	2.00
75-15-0	Carbon disulfide	0.500	U	0.200	0.500	1.00
56-23-5	Carbon tetrachloride	0.500	U	0.250	0.500	1.00

FORM I VOA

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW05-060123
Collect Date:	06/01/23 Time: 0910	GCAL Sample ID:	22306021004
Matrix:	Water % Moisture: NA	Instrument ID:	MSV11
Sample Amt:	5 mL	Lab File ID:	230602/j1699
Injection Vol.:	1 (µL)	GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1 Analyst: KN1	Analytical Batch:	766942
Analysis Date:	06/02/23 Time: 1619	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	0.500	U	0.200	0.500	1.00
75-00-3	Chloroethane	0.500	U	0.250	0.500	1.00
67-66-3	Chloroform	0.500	U	0.200	0.500	1.00
74-87-3	Chloromethane	0.500	U	0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
124-48-1	Dibromochloromethane	0.500	U	0.200	0.500	1.00
74-95-3	Dibromomethane	0.500	U	0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	0.500	U	0.200	0.500	1.00
100-41-4	Ethylbenzene	0.500	U	0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	1.00	U	0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	0.500	U	0.200	0.500	1.00
136777-61-2	m,p-Xylene	0.500	U	0.200	0.500	2.00
75-09-2	Methylene chloride	0.500	U	0.200	0.500	5.00
91-20-3	Naphthalene	0.500	U	0.200	0.500	5.00
104-51-8	n-Butylbenzene	0.500	U	0.200	0.500	1.00
103-65-1	n-Propylbenzene	2.00	U	1.00	2.00	5.00
95-47-6	o-Xylene	0.500	U	0.200	0.500	1.00
135-98-8	sec-Butylbenzene	0.500	U	0.200	0.500	1.00
100-42-5	Styrene	0.500	U	0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	0.500	U	0.200	0.500	1.00
98-06-6	tert-Butylbenzene	0.500	U	0.200	0.500	1.00
127-18-4	Tetrachloroethene	0.500	U	0.200	0.500	1.00
108-88-3	Toluene	0.500	U	0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
79-01-6	Trichloroethene	0.500	U	0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	0.500	U	0.200	0.500	1.00
75-01-4	Vinyl chloride	0.500	U	0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv11.i/230602.s.b/j1699.d
Lab Smp Id: 22306021004
Inj Date : 02-JUN-2023 16:19
Operator : KN1 Inst ID: msv11.i
Smp Info : 22306021004*
Misc Info : MSV~52914~*1*KN1
Comment :
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Meth Date : 05-Jun-2023 08:22 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 15:02 Cal File: j1306.d
Als bottle: 1
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260bDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
$ 41 Dibromofluoromethane	111	5.710	5.715	(0.878)	78063	58.7192	58.7	5308
$ 51 1,2-Dichloroethane-d4	65	6.224	6.229	(0.956)	104293	58.7478	58.7	
* 56 FLUOROBENZENE	96	6.507	6.507	(1.000)	266645	50.0000		
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	263983	51.8331	51.8	
* 90 CHLOROBENZENE-d5	82	9.238	9.239	(1.000)	106395	50.0000		
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	80831	52.6470	52.6	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	104730	50.0000		

Data File: /var/chem/msv11.i/230602.s.b/j1699.d

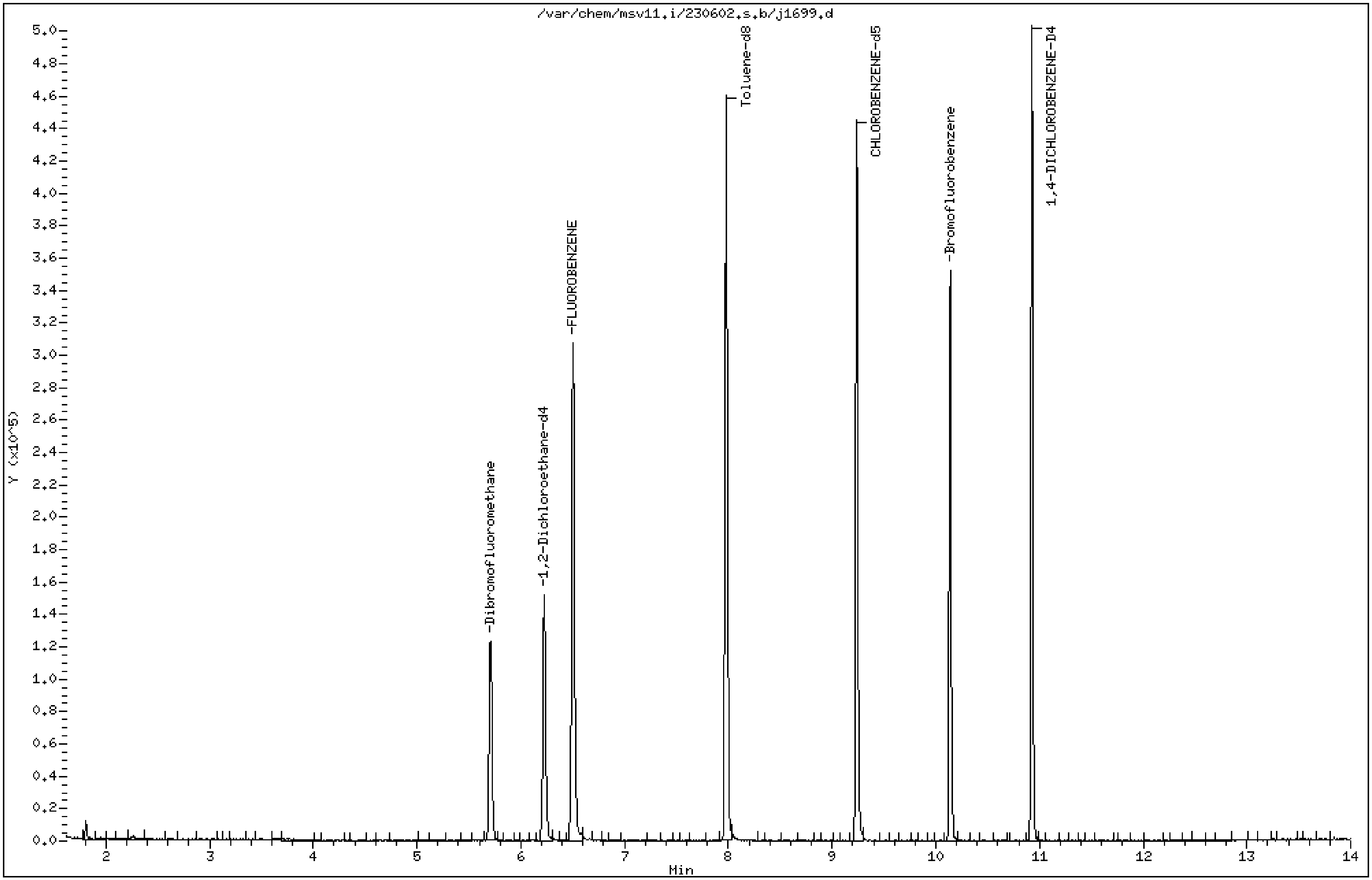
Date : 02-JUN-2023 16:19

Client ID: | Instrument: msv11.i

Sample Info: 22306021004*

Purge Volume: 5.0 | Operator: KN1

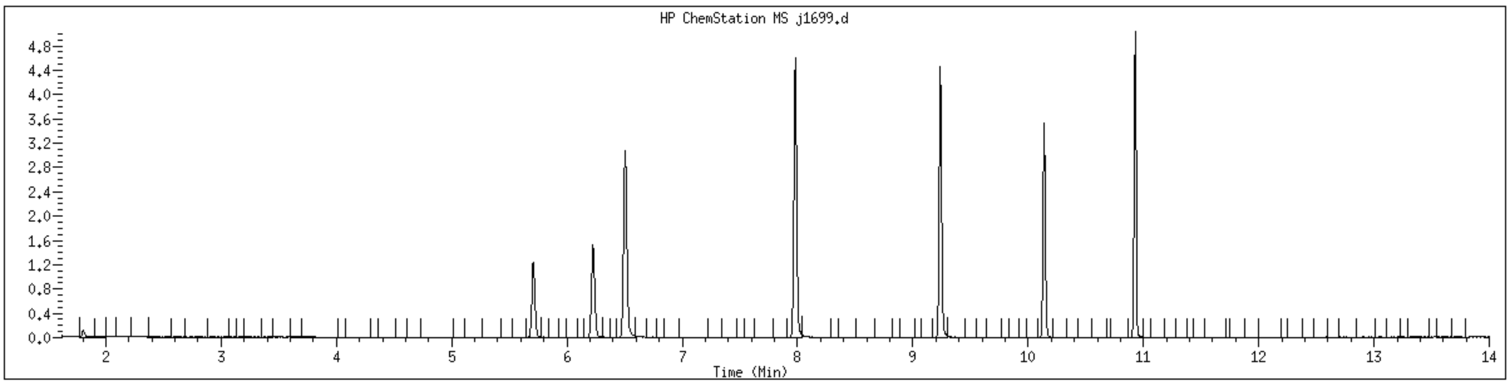
Column phase: RTX-VMS-30M | Column diameter: 0.25



Report Date: 06/05/2023 09:18

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 22306021004   SampleType   : SAMPLE
Injection Date: 06/02/2023 16:19   Instrument   : msv11.i
Operator   : KN1
Sample Info   : 22306021004*
Misc Info   : MSV~52914~*1*KN1
Method   : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260bDOD



NO MANUAL INTEGRATIONS

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-MW06-060123
Collect Date:	06/01/23	Time:	1100	GCAL Sample ID:	22306021005
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11
Sample Amt:	5	mL		Lab File ID:	230602/j1700
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942
Analysis Date:	06/02/23	Time:	1642	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	0.500	U	0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	0.500	U	0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	0.500	U	0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	0.500	U	0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	0.500	U	0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	0.500	U	0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	0.500	U	0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	0.500	U	0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	0.500	U	0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	0.500	U	0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	0.500	U	0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	0.500	U	0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	0.500	U	0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	14.3		0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	0.500	U	0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	0.500	U	0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	0.500	U	0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	0.500	U	0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	0.500	U	0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	0.500	U	0.200	0.500	1.00
78-93-3	2-Butanone	0.500	U	0.200	0.500	5.00
95-49-8	2-Chlorotoluene	0.500	U	0.200	0.500	1.00
591-78-6	2-Hexanone	1.00	U	0.500	1.00	5.00
106-43-4	4-Chlorotoluene	0.500	U	0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	0.500	U	0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	0.500	U	0.200	0.500	5.00
67-64-1	Acetone	1.00	U	0.500	1.00	5.00
71-43-2	Benzene	1.71		0.200	0.500	1.00
108-86-1	Bromobenzene	0.500	U	0.200	0.500	1.00
74-97-5	Bromochloromethane	0.500	U	0.200	0.500	1.00
75-27-4	Bromodichloromethane	0.500	U	0.200	0.500	1.00
75-25-2	Bromoform	0.500	U	0.250	0.500	1.00
74-83-9	Bromomethane	1.00	U	0.500	1.00	2.00
75-15-0	Carbon disulfide	0.500	U	0.200	0.500	1.00
56-23-5	Carbon tetrachloride	0.500	U	0.250	0.500	1.00

FORM I VOA

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-MW06-060123		
Collect Date:	06/01/23	Time:	1100	GCAL Sample ID:	22306021005		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11		
Sample Amt:	5	mL		Lab File ID:	230602/j1700		
Injection Vol.:	1		(µL)	GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942		
Analysis Date:	06/02/23	Time:	1642	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	0.500	U	0.200	0.500	1.00
75-00-3	Chloroethane	0.500	U	0.250	0.500	1.00
67-66-3	Chloroform	0.500	U	0.200	0.500	1.00
74-87-3	Chloromethane	0.500	U	0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	14.3		0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
124-48-1	Dibromochloromethane	0.500	U	0.200	0.500	1.00
74-95-3	Dibromomethane	0.500	U	0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	0.500	U	0.200	0.500	1.00
100-41-4	Ethylbenzene	0.500	U	0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	1.00	U	0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	0.500	U	0.200	0.500	1.00
136777-61-2	m,p-Xylene	0.500	U	0.200	0.500	2.00
75-09-2	Methylene chloride	0.500	U	0.200	0.500	5.00
91-20-3	Naphthalene	0.500	U	0.200	0.500	5.00
104-51-8	n-Butylbenzene	0.500	U	0.200	0.500	1.00
103-65-1	n-Propylbenzene	2.00	U	1.00	2.00	5.00
95-47-6	o-Xylene	0.500	U	0.200	0.500	1.00
135-98-8	sec-Butylbenzene	0.500	U	0.200	0.500	1.00
100-42-5	Styrene	0.500	U	0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	0.500	U	0.200	0.500	1.00
98-06-6	tert-Butylbenzene	0.500	U	0.200	0.500	1.00
127-18-4	Tetrachloroethene	0.500	U	0.200	0.500	1.00
108-88-3	Toluene	0.500	U	0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
79-01-6	Trichloroethene	0.500	U	0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	0.500	U	0.200	0.500	1.00
75-01-4	Vinyl chloride	94.6		0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv11.i/230602.s.b/j1700.d
Lab Smp Id: 22306021005
Inj Date : 02-JUN-2023 16:42
Operator : KN1 Inst ID: msv11.i
Smp Info : 22306021005*
Misc Info : MSV~52914~*1*KN1
Comment :
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Meth Date : 05-Jun-2023 08:22 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 15:02 Cal File: j1306.d
Als bottle: 1
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260bDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

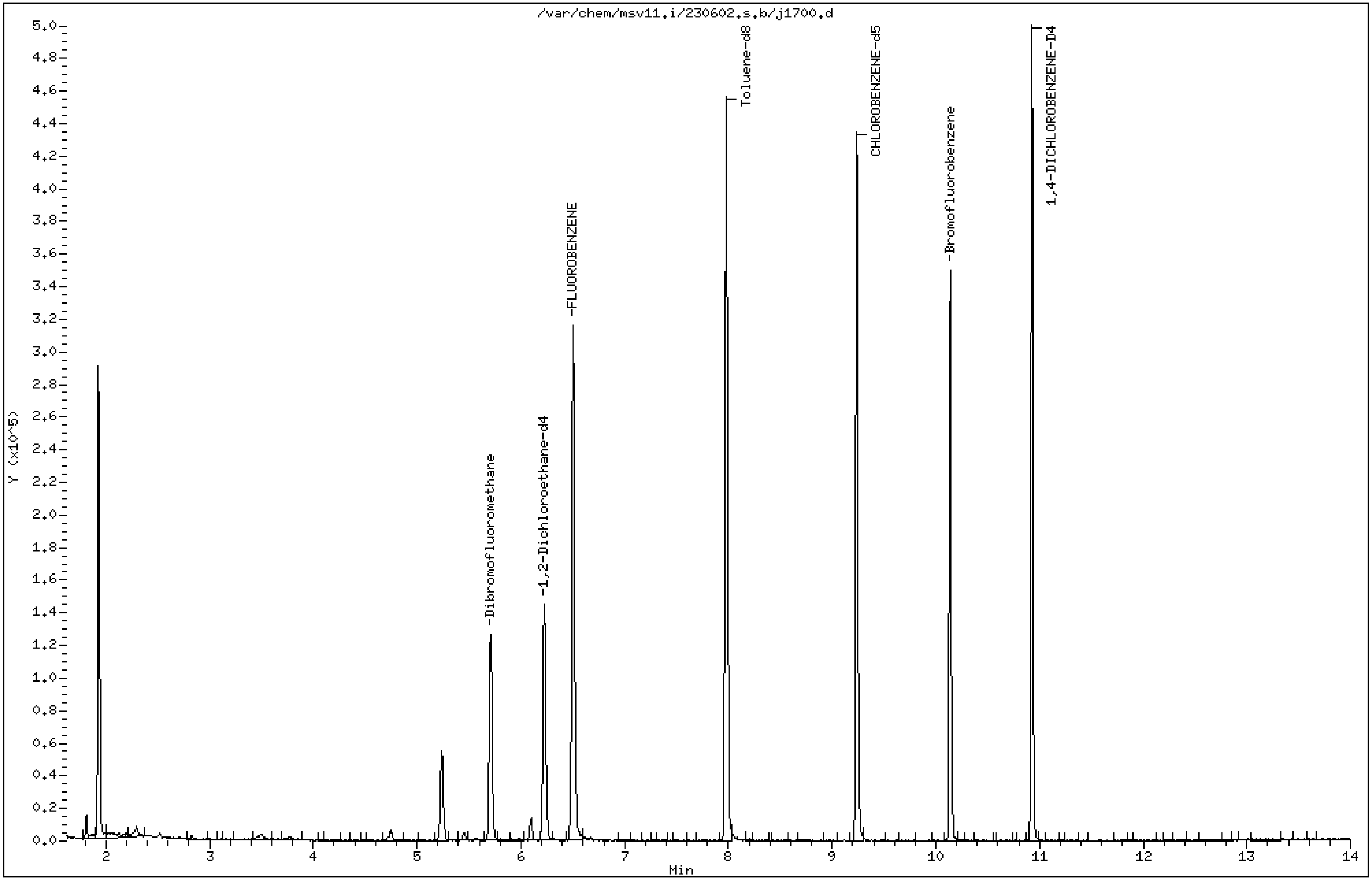
Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	CONCENTRATIONS FINAL (ug/L)	SIMILARITY
3 Vinyl Chloride +	62	1.929	1.940	(0.297)	196976	94.5710	94.6	
32 cis-1,2-Dichloroethene	61	5.238	5.238	(0.805)	35802	14.3057	14.3	
M 33 Total 1,2-Dichloroethene	61				35802	14.3057	14.3	
36 Cyclohexane	56	5.448	5.453	(0.837)	2500	0.77620	0.776	6572
$ 41 Dibromofluoromethane	111	5.710	5.715	(0.878)	79387	59.3735	59.4	5308
48 Benzene	78	6.103	6.098	(0.938)	9666	1.70837	1.71	
$ 51 1,2-Dichloroethane-d4	65	6.229	6.229	(0.957)	105223	58.9326	58.9	
* 56 FLUOROBENZENE	96	6.507	6.507	(1.000)	268179	50.0000		
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	265112	52.3837	52.4	
* 90 CHLOROBENZENE-d5	82	9.238	9.239	(1.000)	105727	50.0000		
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	77081	50.5217	50.5	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	103869	50.0000		

Data File: /var/chem/msv11.i/230602.s.b/j1700.d
Date : 02-JUN-2023 16:42
Client ID:
Sample Info: 22306021005*
Purge Volume: 5.0
Column phase: RTX-VMS-30M

Page 1

Instrument: msv11.i

Operator: KN1
Column diameter: 0.25



Data File: /var/chem/msv11.i/230602.s.b/j1700.d

Date : 02-JUN-2023 16:42

Client ID:

Instrument: msv11.i

Sample Info: 22306021005*

Purge Volume: 5.0

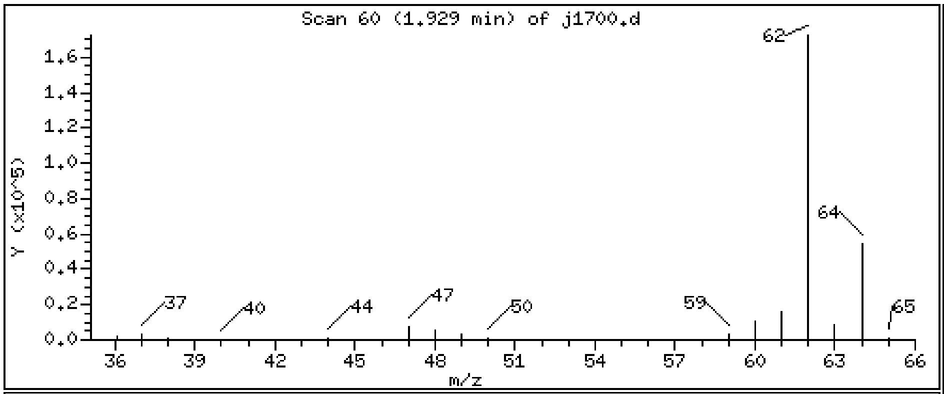
Operator: KN1

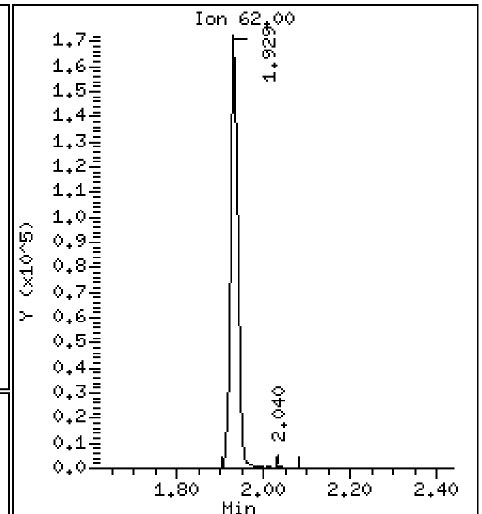
Column phase: RTX-VMS-30M

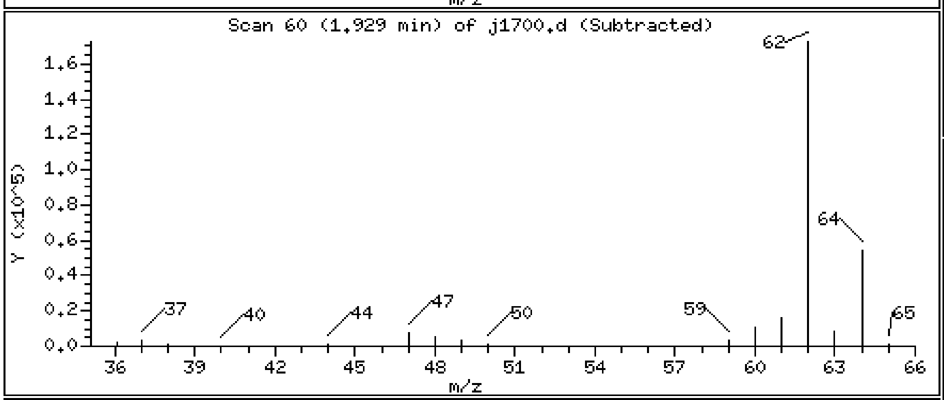
Column diameter: 0.25

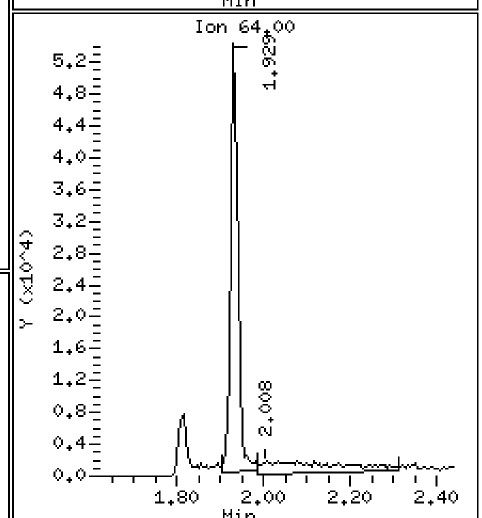
3 Vinyl Chloride +

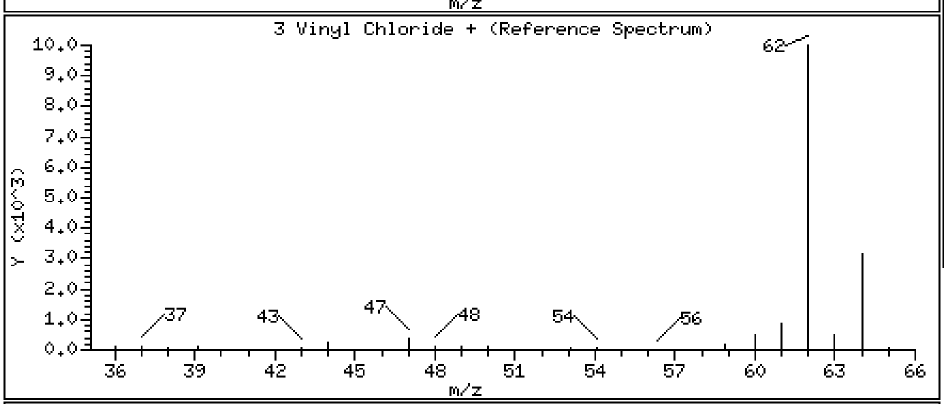
Concentration: 94.6 ug/L

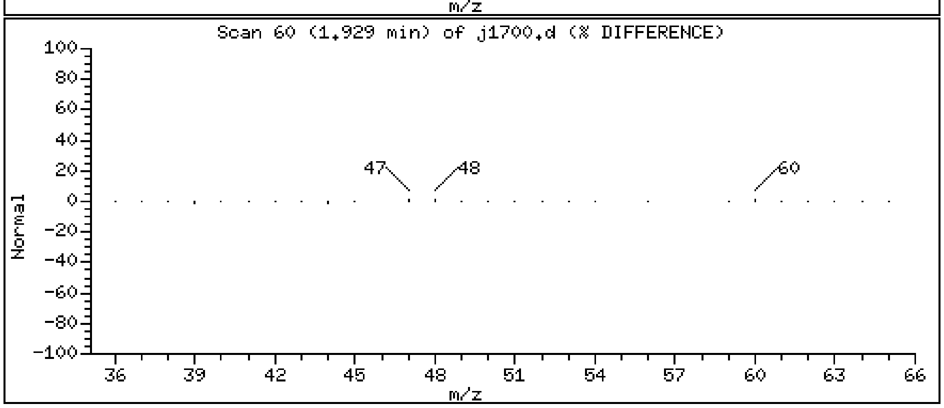












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Date : 02-JUN-2023 16:42

Client ID:

Instrument: msv11.i

Sample Info: 22306021005*

Purge Volume: 5.0

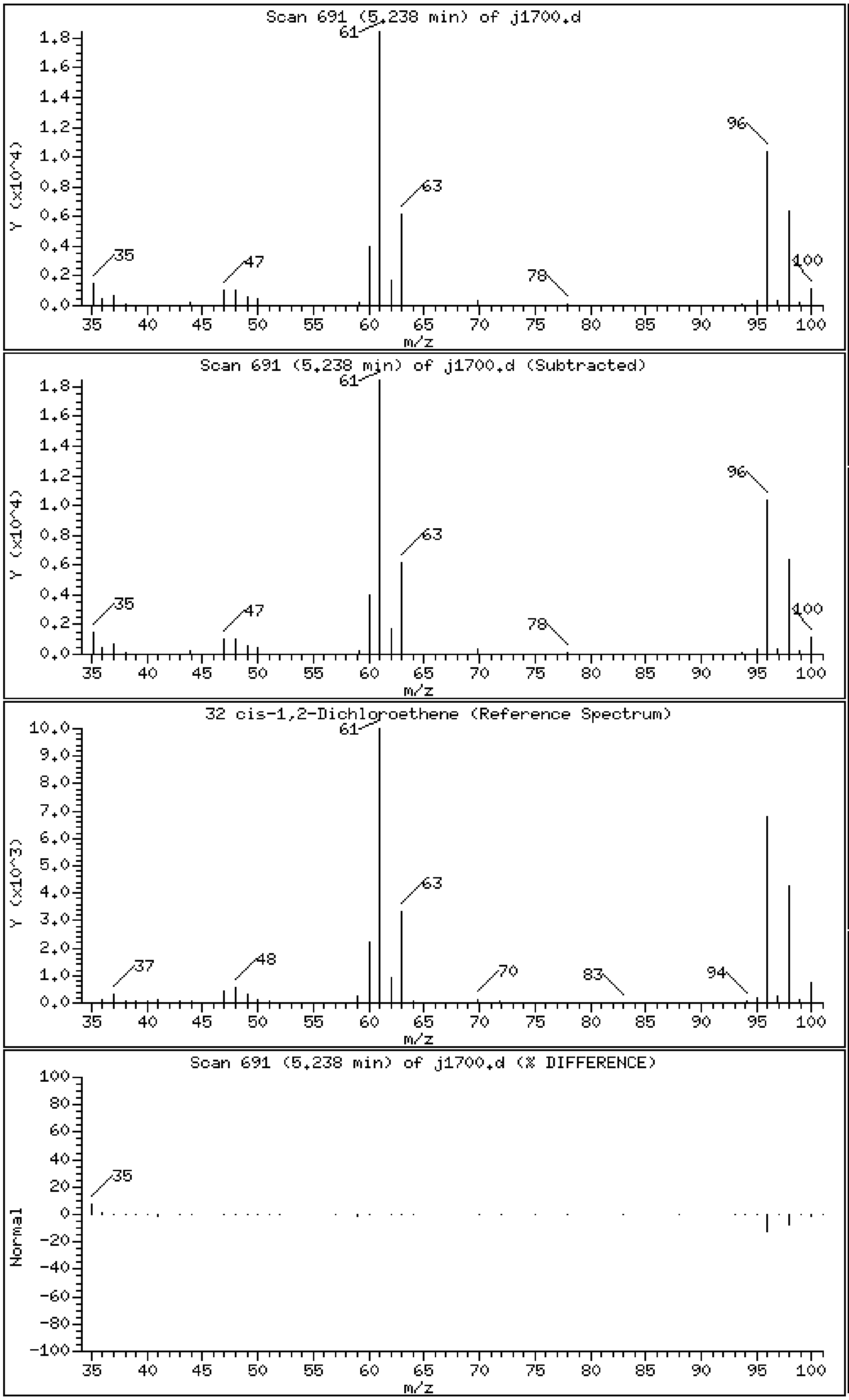
Operator: KN1

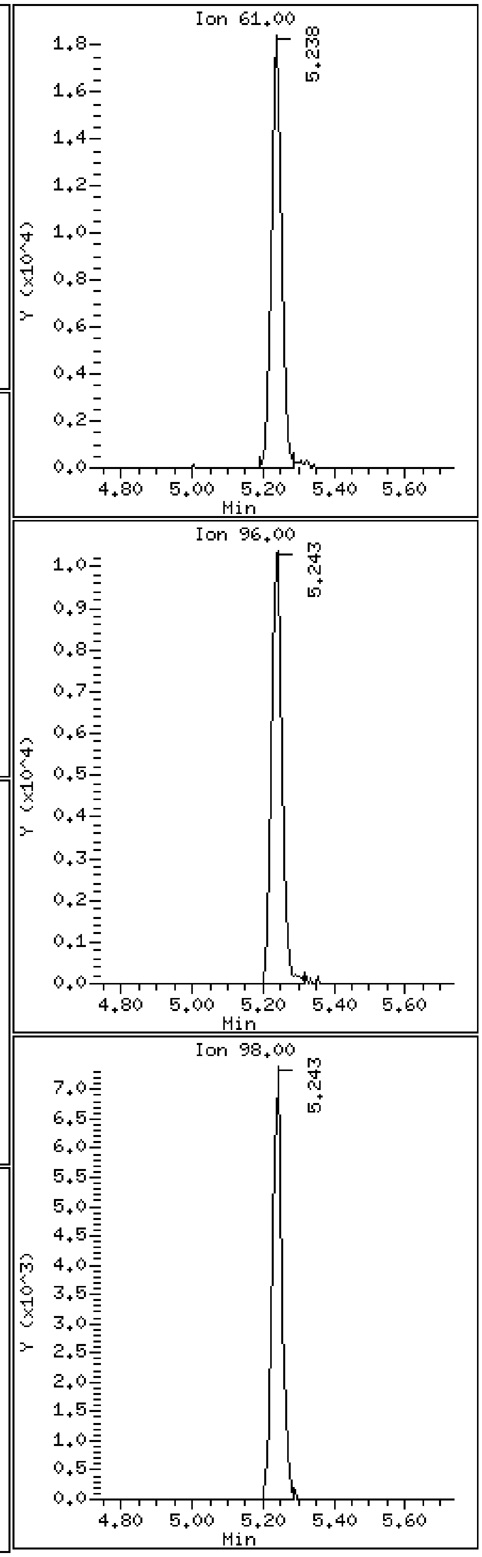
Column phase: RTX-VMS-30M

Column diameter: 0.25

32 cis-1,2-Dichloroethene

Concentration: 14.3 ug/L





Data File: /var/chem/msv11.i/230602.s.b/j1700.d

Date : 02-JUN-2023 16:42

Client ID:

Instrument: msv11.i

Sample Info: 22306021005*

Purge Volume: 5.0

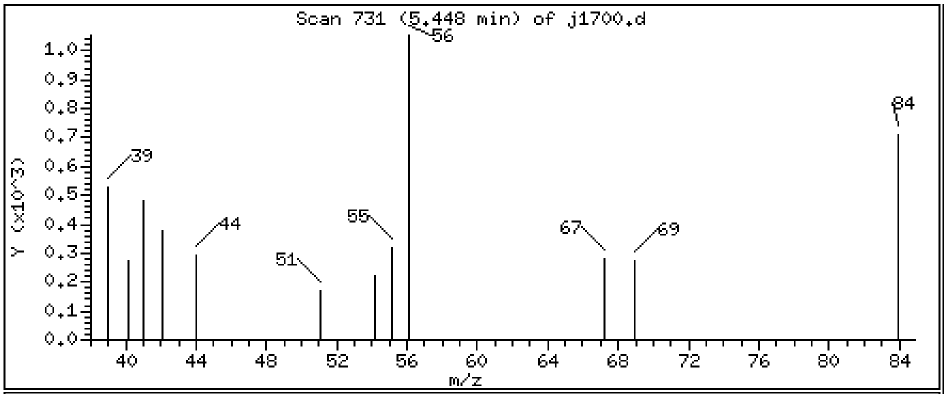
Operator: KN1

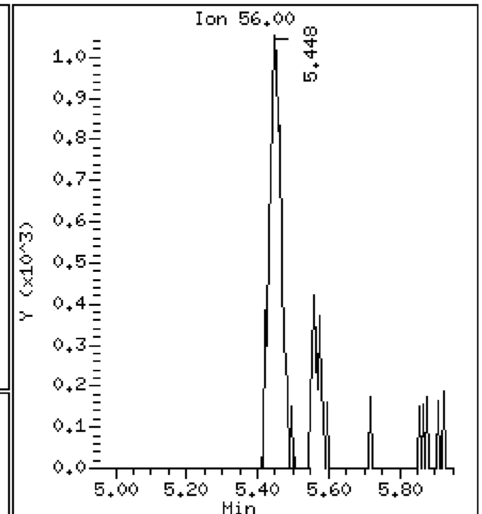
Column phase: RTX-VMS-30M

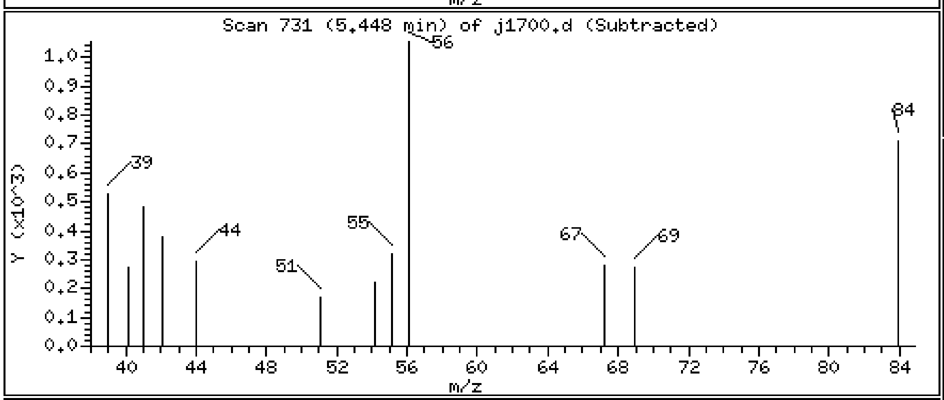
Column diameter: 0.25

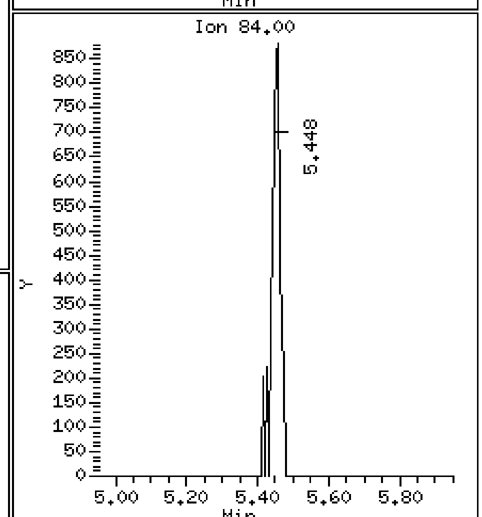
36 Cyclohexane

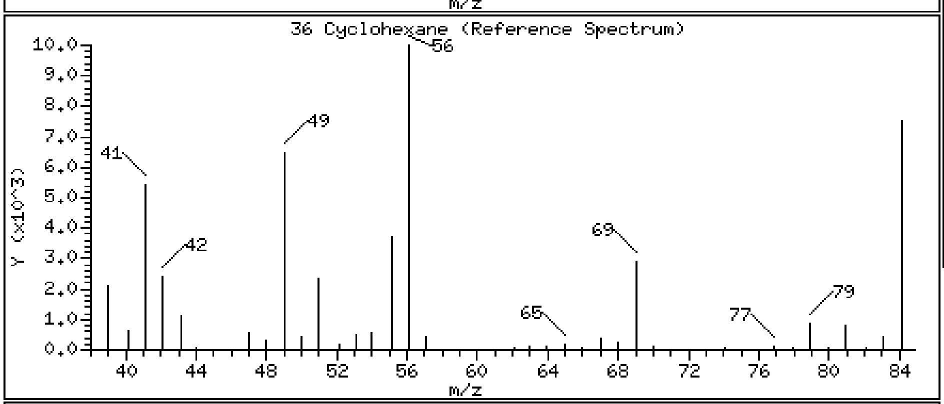
Concentration: 0.776 ug/L

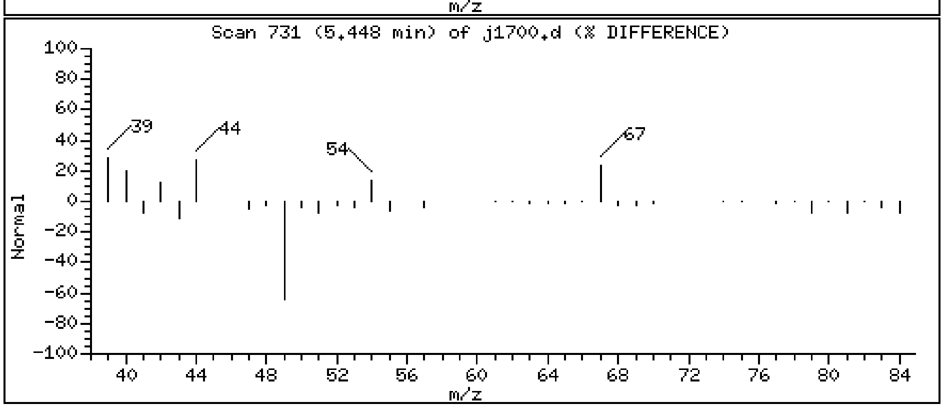












Data File: /var/chem/msv11.i/230602.s.b/j1700.d

Date : 02-JUN-2023 16:42

Client ID:

Instrument: msv11.i

Sample Info: 22306021005*

Purge Volume: 5.0

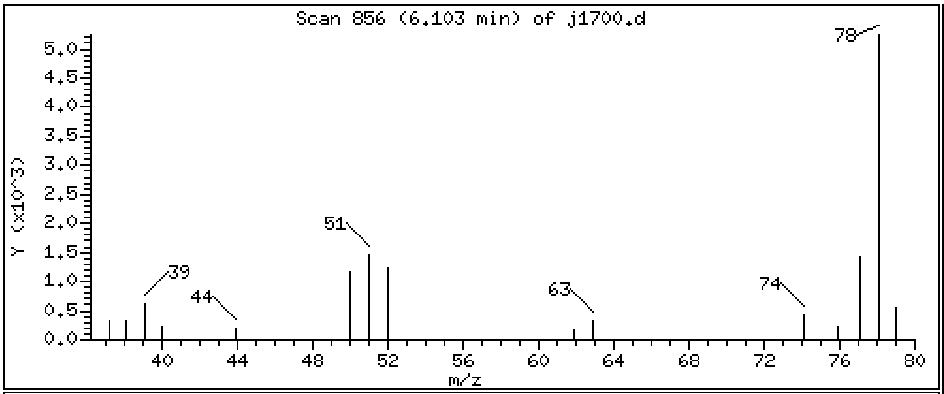
Operator: KN1

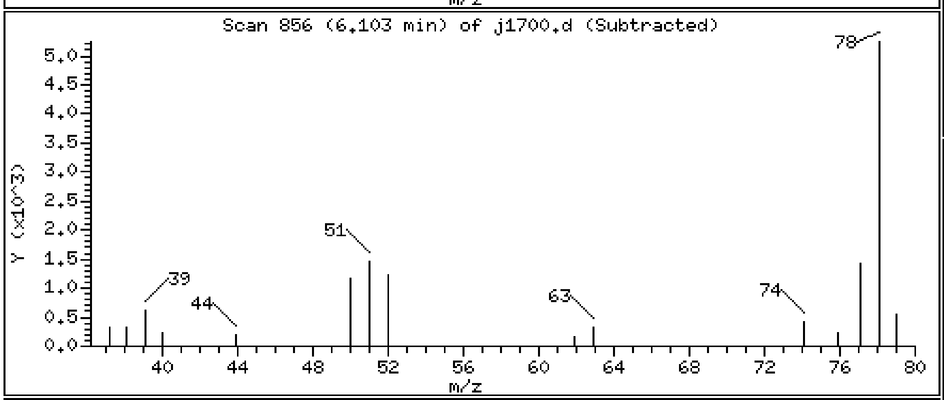
Column phase: RTX-VMS-30M

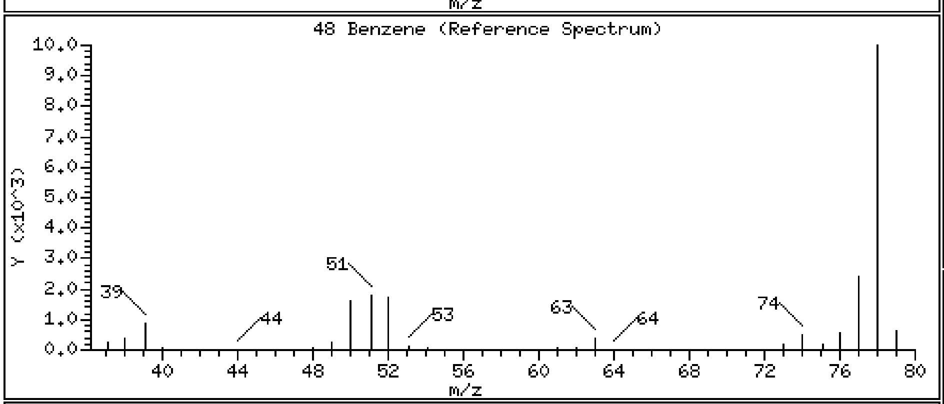
Column diameter: 0.25

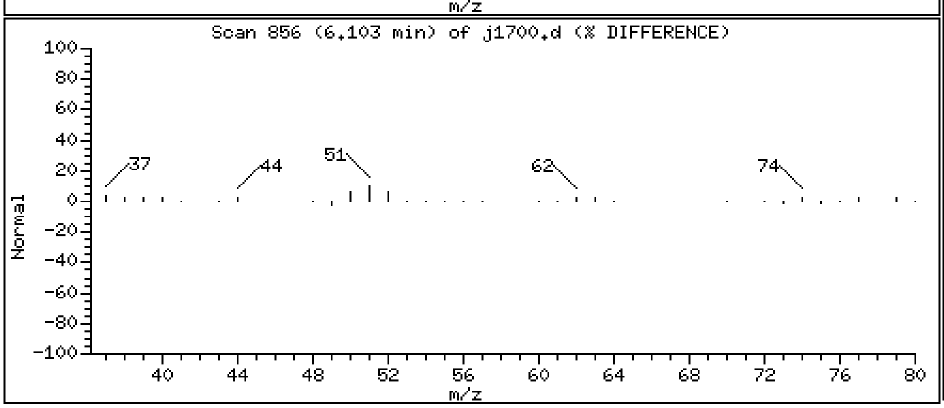
48 Benzene

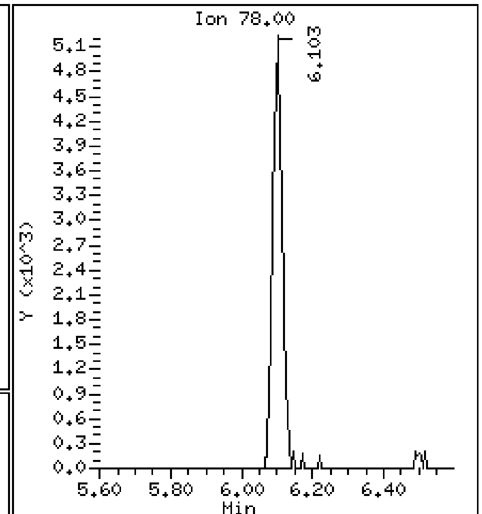
Concentration: 1.71 ug/L

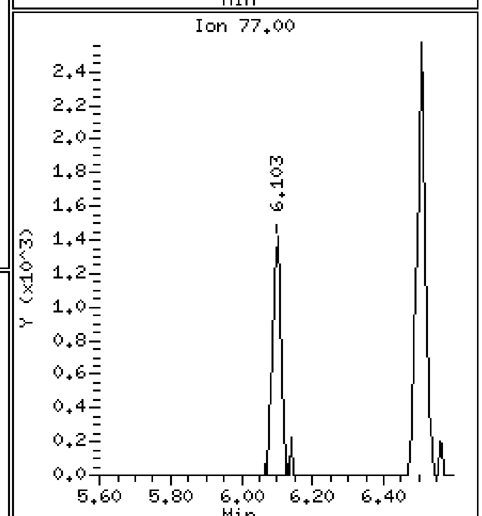


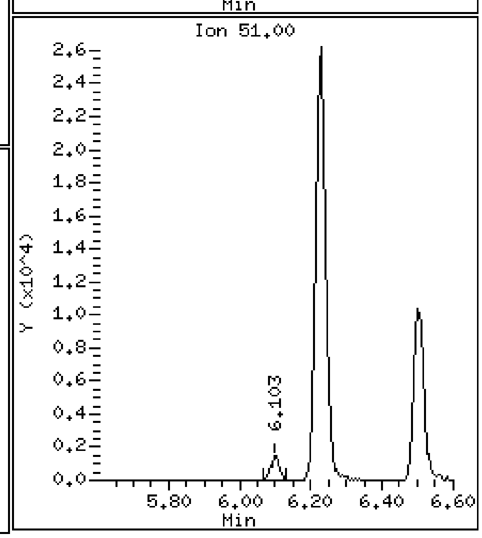








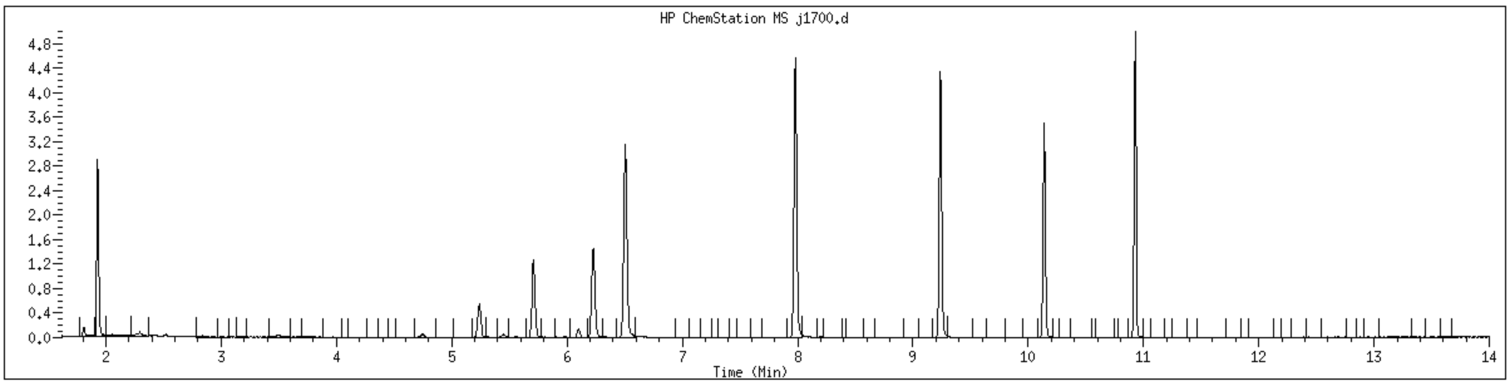




Data file : /var/chem/msv11.i/230602.s.b/j1700.d Page: 1
Report Date: 06/05/2023 09:19

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 22306021005 SampleType : SAMPLE
Injection Date: 06/02/2023 16:42 Instrument : msv11.i
Operator : KN1
Sample Info : 22306021005*
Misc Info : MSV~52914~*1*KN1
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260bDOD



NO MANUAL INTEGRATIONS

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123MS
Collect Date:	06/01/23 Time: 0955	GCAL Sample ID:	22306021006
Matrix:	Water % Moisture: NA	Instrument ID:	MSV11
Sample Amt:	5 mL	Lab File ID:	230602/j1705
Injection Vol.:	1 (µL)	GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1 Analyst: KN1	Analytical Batch:	766942
Analysis Date:	06/02/23 Time: 1838	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	47.3		0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	54.6		0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	41.5		0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	44.8		0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	50.1		0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	53.8		0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	51.8		0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	39.8		0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	38.6		0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	39.4		0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	44.9		0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	41.9		0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	46.1		0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	41.7		0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	54.4		0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	100		0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	50.3		0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	43.5		0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	41.5		0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	45.9		0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	40.8		0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	51.1		0.200	0.500	1.00
78-93-3	2-Butanone	183		0.200	0.500	5.00
95-49-8	2-Chlorotoluene	43.4		0.200	0.500	1.00
591-78-6	2-Hexanone	189		0.500	1.00	5.00
106-43-4	4-Chlorotoluene	41.6		0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	43.9		0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	233		0.200	0.500	5.00
67-64-1	Acetone	146		0.500	1.00	5.00
71-43-2	Benzene	50.7		0.200	0.500	1.00
108-86-1	Bromobenzene	42.3		0.200	0.500	1.00
74-97-5	Bromochloromethane	52.3		0.200	0.500	1.00
75-27-4	Bromodichloromethane	51.9		0.200	0.500	1.00
75-25-2	Bromoform	48.7		0.250	0.500	1.00
74-83-9	Bromomethane	47.8		0.500	1.00	2.00
75-15-0	Carbon disulfide	49.7		0.200	0.500	1.00
56-23-5	Carbon tetrachloride	55.0		0.250	0.500	1.00

FORM I VOA

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-MW04-060123MS		
Collect Date:	06/01/23	Time:	0955	GCAL Sample ID:	22306021006		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11		
Sample Amt:	5	mL		Lab File ID:	230602/j1705		
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942		
Analysis Date:	06/02/23	Time:	1838	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	47.2		0.200	0.500	1.00
75-00-3	Chloroethane	46.7		0.250	0.500	1.00
67-66-3	Chloroform	53.2		0.200	0.500	1.00
74-87-3	Chloromethane	43.0		0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	51.1		0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	49.9		0.200	0.500	1.00
124-48-1	Dibromochloromethane	48.1		0.200	0.500	1.00
74-95-3	Dibromomethane	51.6		0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	39.3		0.200	0.500	1.00
100-41-4	Ethylbenzene	47.3		0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	44.5		0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	50.1		0.200	0.500	1.00
136777-61-2	m,p-Xylene	95.3		0.200	0.500	2.00
75-09-2	Methylene chloride	50.5		0.200	0.500	5.00
91-20-3	Naphthalene	35.5		0.200	0.500	5.00
104-51-8	n-Butylbenzene	39.6		0.200	0.500	1.00
103-65-1	n-Propylbenzene	42.9		1.00	2.00	5.00
95-47-6	o-Xylene	49.0		0.200	0.500	1.00
135-98-8	sec-Butylbenzene	44.3		0.200	0.500	1.00
100-42-5	Styrene	49.1		0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	49.7		0.200	0.500	1.00
98-06-6	tert-Butylbenzene	44.0		0.200	0.500	1.00
127-18-4	Tetrachloroethene	48.3		0.200	0.500	1.00
108-88-3	Toluene	46.9		0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	49.2		0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	50.2		0.200	0.500	1.00
79-01-6	Trichloroethene	50.8		0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	52.0		0.200	0.500	1.00
75-01-4	Vinyl chloride	46.2		0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv11.i/230602.s.b/j1705.d
Lab Smp Id: 22306021006 Client Smp ID: MS
Inj Date : 02-JUN-2023 18:38
Operator : KN1 Inst ID: msv11.i
Smp Info : 22306021006*MS
Misc Info : MSV~52914~*1*KN1
Comment :
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Meth Date : 05-Jun-2023 08:22 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 15:02 Cal File: j1306.d
Als bottle: 1 QC Sample: MS
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260bDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

							CONCENTRATIONS		
	QUANT SIG					ON-COLUMN	FINAL		
Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ug/L)	SIMILARITY	
1 Dichlorodifluoromethane	85	1.673	1.673	(0.257)	73621	39.2765	39.3		
2 Chloromethane ++	50	1.856	1.856	(0.285)	141648	43.0362	43.0		
3 Vinyl Chloride +	62	1.935	1.940	(0.297)	112465	46.1660	46.2		
4 1-3 Butadiene	39	1.956	1.956	(0.301)	147339	55.1586	55.2	9815	
5 Bromomethane	94	2.249	2.250	(0.346)	47879	47.8377	47.8		
7 Chloroethane	64	2.370	2.370	(0.364)	63505	46.7480	46.7		
8 Trichlorofluoromethane	101	2.522	2.522	(0.388)	139446	52.0120	52.0		
11 1,1-Dichloroethene +	96	3.067	3.067	(0.471)	74552	53.8263	53.8		
12 Carbon Disulfide	76	3.104	3.099	(0.477)	209584	49.7269	49.7		
13 1,1,2Trichlotrifluoroethane	101	3.120	3.120	(0.479)	75863	52.5566	52.6		
15 Methyl Iodide	142	3.230	3.225	(0.496)	49079	29.6488	29.6		
17 Allyl chloride	39	3.633	3.628	(0.558)	134281	55.7198	55.7	9424	
18 Methylene Chloride	49	3.759	3.760	(0.578)	175144	50.5273	50.5		
19 Acetone	43	3.812	3.812	(0.586)	304519	146.418	146		

							CONCENTRATIONS		
		QUANT SIG					ON-COLUMN	FINAL	
	Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ug/L)	SIMILARITY
	20 trans-1,2-Dichloroethene	61	3.959	3.959	(0.608)	152494	49.1619	49.2	(R)
	21 Methyl Acetate	43	3.974	3.974	(0.611)	585353	259.756	260	9519(R)
	22 Hexane	57	4.069	4.069	(0.625)	222083	48.4679	48.5	9328
	23 MTBE	73	4.100	4.106	(0.630)	243569	49.6962	49.7	8806
	24 tert-Butyl Alcohol	59	4.226	4.226	(0.649)	91463	299.383	299	8810
	25 Acetonitrile	41	4.341	4.342	(0.667)	99869	248.770	249	9855
	26 Isopropyl Ether	45	4.551	4.551	(0.699)	415115	49.9774	50.0	9684
	28 1,1-Dichloroethane ++	63	4.656	4.656	(0.716)	196655	50.0699	50.1	
	29 Acrylonitrile	53	4.698	4.698	(0.722)	303983	283.714	284	
	31 Vinyl Acetate	43	4.949	4.944	(0.761)	36713	7.75765	7.76	(R)
	30 Ethyl Tert-butyl Ether	59	4.949	4.950	(0.761)	322484	47.3712	47.4	8814
	32 cis-1,2-Dichloroethene	61	5.238	5.238	(0.805)	149488	51.0704	51.1	
M	33 Total 1,2-Dichloroethene	61				301982	100.232	100	
	34 2,2-Dichloropropane	77	5.348	5.348	(0.822)	126530	51.0793	51.1	
	35 Bromochloromethane	128	5.448	5.448	(0.837)	45128	52.3021	52.3	
	36 Cyclohexane	56	5.458	5.453	(0.839)	197867	52.5250	52.5	8640
	37 Chloroform +	83	5.531	5.526	(0.850)	157054	53.2137	53.2	
	38 Carbon Tetrachloride	117	5.668	5.668	(0.871)	129458	54.9793	55.0	
$	41 Dibromofluoromethane	111	5.710	5.715	(0.878)	88272	56.4451	56.4	8350
	42 1,1,1-Trichloroethane	97	5.731	5.731	(0.881)	146242	54.5942	54.6	
	44 2-Butanone	43	5.825	5.825	(0.895)	334643	182.699	183	
	45 1,1-Dichloropropene	75	5.857	5.857	(0.900)	116383	51.7864	51.8	
	47 2,2,4 Trimethylpentane	57	5.977	5.977	(0.919)	309140	32.8601	32.9	9486
	48 Benzene	78	6.103	6.098	(0.938)	335311	50.6691	50.7	
$	51 1,2-Dichloroethane-d4	65	6.229	6.229	(0.957)	115229	55.1781	55.2	
	54 tert-amyl Methyl Ether	73	6.229	6.229	(0.957)	221350	46.4350	46.4	7343
	52 1,2-Dichloroethane	62	6.297	6.292	(0.968)	153592	54.4233	54.4	
	53 Isobutyl Alcohol	43	6.339	6.339	(0.974)	22986	190.183	190	9071
*	56 FLUOROBENZENE	96	6.507	6.507	(1.000)	313664	50.0000		
	57 Methyl Cyclohexane	83	6.674	6.669	(1.026)	124406	50.6572	50.7	9280
	58 Trichloroethene	130	6.674	6.675	(1.026)	96539	50.8182	50.8	
	60 Dibromomethane	93	7.073	7.073	(1.087)	56331	51.6464	51.6	
	62 1,2-Dichloropropane +	63	7.162	7.157	(1.101)	115900	50.2569	50.3	
	63 Bromodichloromethane	83	7.230	7.230	(1.111)	115569	51.8621	51.9	
	65 1,4- Dioxane	58	7.419	7.414	(1.140)	24750	1319.70	1320	8809
	66 1-Bromo-2-chloroethane	63	7.671	7.671	(1.179)	150218	49.7476	49.7	9760
	68 cis-1,3-Dichloropropene	75	7.812	7.812	(1.201)	128879	49.9256	49.9	
$	69 Toluene-d8	98	7.980	7.980	(0.864)	316356	50.2264	50.2	
	70 Toluene +	91	8.027	8.022	(0.869)	354715	46.9262	46.9	
M	72 1-3 Dichloropropene total	100				250266	100.119	100	0
	74 4-methyl-2-pentanone	43	8.336	8.337	(0.902)	685665	232.765	233	
	73 Tetrachloroethene	164	8.357	8.358	(0.905)	73918	48.2653	48.3	
	75 trans-1,3-Dichloropropene	75	8.378	8.373	(1.288)	121387	50.1934	50.2	
	78 1,1,2-Trichloroethane	97	8.509	8.504	(0.921)	76568	44.7675	44.8	
	79 Dibromochloromethane	129	8.656	8.657	(0.937)	88950	48.1242	48.1	
	80 1,3-Dichloropropane	76	8.724	8.725	(0.944)	132410	45.8630	45.9	
	82 1,2-Dibromoethane(EDB)	107	8.845	8.840	(0.957)	81182	46.1169	46.1	

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
83 2-Hexanone	43	9.008	9.008	(0.975)	465128	189.438	189	
89 1-Chlorohexane	91	9.233	9.233	(0.999)	99167	46.1886	46.2	9453
* 90 CHLOROBENZENE-d5	82	9.238	9.239	(1.000)	131582	50.0000		
91 Chlorobenzene ++	112	9.254	9.254	(1.002)	231438	47.2301	47.2	
92 Ethylbenzene +	106	9.270	9.270	(1.003)	122451	47.3110	47.3	
93 1,1,1,2-Tetrachloroethane	133	9.301	9.301	(1.007)	79148	47.2709	47.3	
94 p,m-Xylene	106	9.380	9.380	(1.015)	296277	95.2861	95.3	
M 95 TOTAL XYLENE	106				437757	144.307	144	
96 o-Xylene	106	9.700	9.700	(1.050)	141480	49.0206	49.0	
97 Styrene	104	9.736	9.737	(1.054)	201777	49.1134	49.1	
98 Bromoform ++	173	9.768	9.768	(1.057)	59237	48.7159	48.7	
99 Isopropylbenzene	105	9.930	9.931	(1.075)	360233	50.0994	50.1	
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	103771	54.6508	54.7	
102 cis-1,4-dichloro-2-butene	53	10.177	10.172	(0.931)	40382	33.7100	33.7	8969
103 Bromobenzene	77	10.213	10.214	(0.935)	162813	42.2637	42.3	
104 n-Propylbenzene	91	10.229	10.224	(0.936)	411337	42.9182	42.9	
105 1,1,2,2-Tetrachloroethane++	83	10.276	10.277	(0.940)	103159	41.5490	41.5	
106 2-Chlorotoluene	91	10.345	10.345	(0.947)	293158	43.4240	43.4	
107 1,3,5-Trimethylbenzene	105	10.360	10.361	(0.948)	312672	43.5489	43.5	
108 1,2,3-Trichloropropane	75	10.376	10.376	(0.950)	124786	38.6484	38.6	
109 trans-1,4-Dichloro-2-Butene	53	10.397	10.397	(0.952)	34234	29.6046	29.6	(RM1)
100 Cyclohexanone	55	10.402	10.402	(0.952)	21930	25.9541	26.0	8736(R)
110 4-Chlorotoluene	91	10.460	10.460	(0.957)	267040	41.5783	41.6	
111 tert-butylbenzene	91	10.591	10.586	(0.969)	172500	44.0306	44.0	
112 1,2,4-Trimethylbenzene	105	10.638	10.638	(0.974)	323714	44.8587	44.9	
113 sec-Butylbenzene	105	10.712	10.712	(0.980)	368731	44.3129	44.3	
115 p-Isopropyltoluene	119	10.811	10.806	(0.989)	338335	43.8553	43.9	
116 1,3-Dichlorobenzene	146	10.879	10.874	(0.996)	183714	41.5470	41.5	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	139318	50.0000		
118 1,4-Dichlorobenzene	146	10.937	10.937	(1.001)	181849	40.8218	40.8	
119 n-Butylbenzene	91	11.100	11.100	(1.016)	440929	39.5924	39.6	
120 1,2-Dichlorobenzene	146	11.231	11.226	(1.028)	178469	41.6911	41.7	
122 1,2-Dibromo-3-Chloropropane	157	11.776	11.771	(1.078)	24828	41.9297	41.9	
123 Hexachlorobutadiene	225	12.201	12.201	(1.117)	52927	44.5406	44.5	
124 1,2,4-Trichlorobenzene	180	12.227	12.227	(1.119)	103895	39.4364	39.4	
125 Naphthalene	128	12.447	12.447	(1.139)	277743	35.5155	35.5	
126 1,2,3-Trichlorobenzene	180	12.573	12.573	(1.151)	101902	39.8113	39.8	

QC Flag Legend

R - Spike/Surrogate failed recovery limits.
M1- Compound response manually integrated because
Target system did not integrate.

Data File: /var/chem/msv11.i/230602.s.b/j1705.d

Page 1

Date : 02-JUN-2023 18:38

Client ID: MS

Instrument: msv11.i

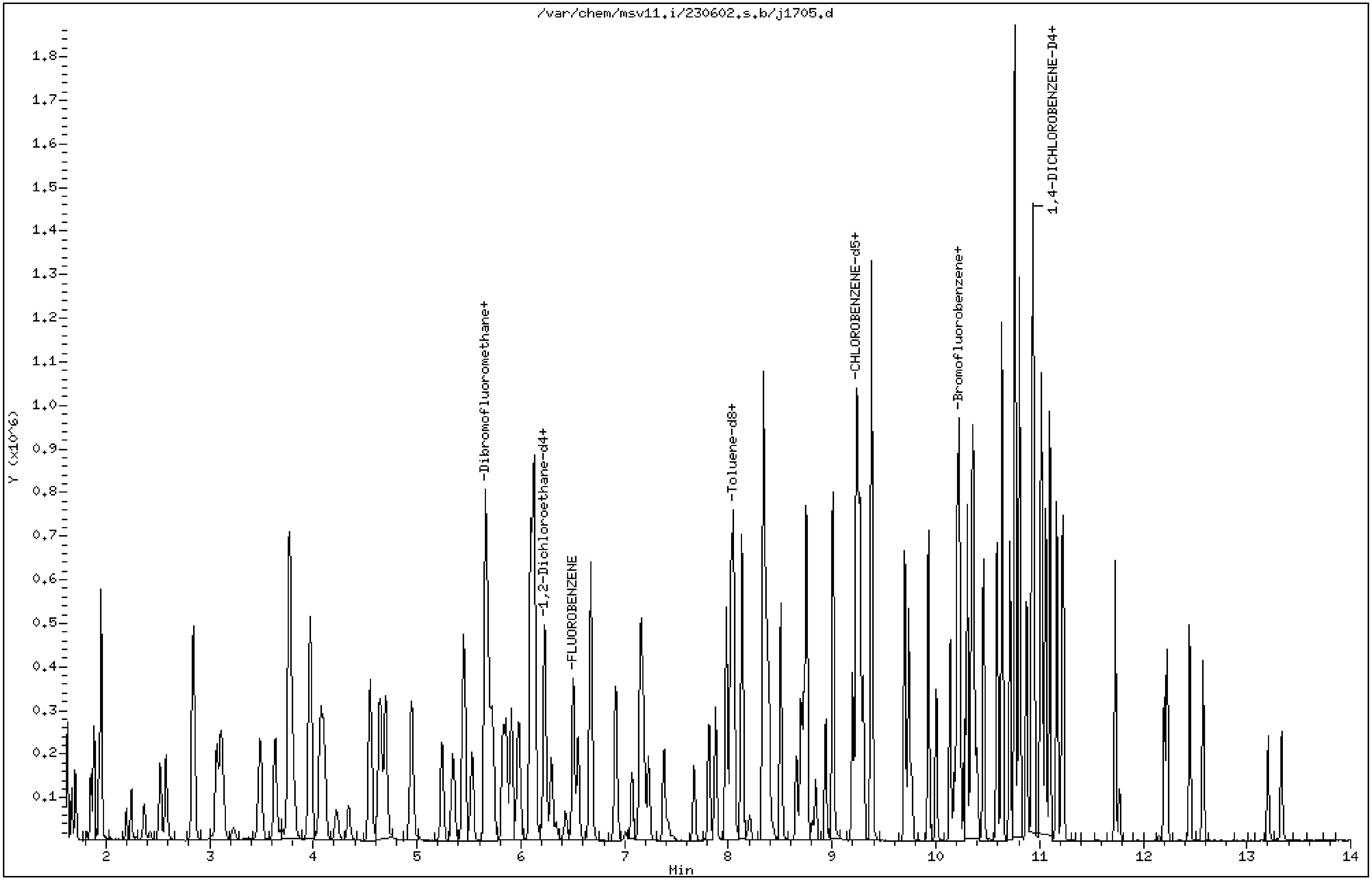
Sample Info: 22306021006*MS

Purge Volume: 5.0

Operator: KN1

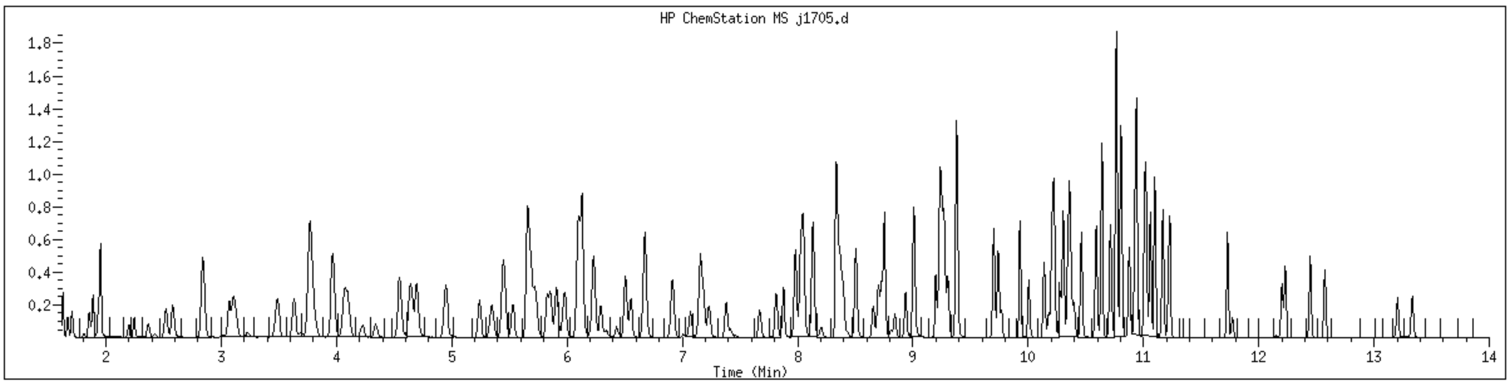
Column phase: RTX-VMS-30M

Column diameter: 0.25



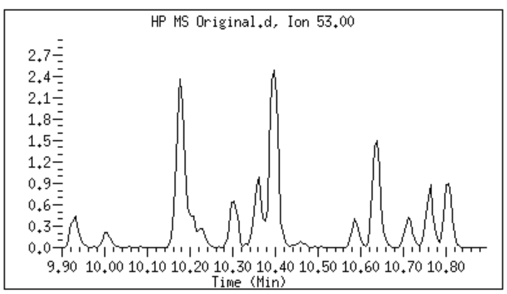
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID	: 22306021006	SampleType	: MS
Injection Date	: 06/02/2023 18:38	Instrument	: msv11.i
Operator	: KN1		
Sample Info	: 22306021006*MS		
Misc Info	: MSV~52914~*1*KN1		
Method	: /var/chem/msv11.i/230602.s.b/8260bDODw11.m		
Dilution	: 1.00		
Matrix	: WATER		
Integrator	: HP RTE	Compound Sublist	: 8260bDOD



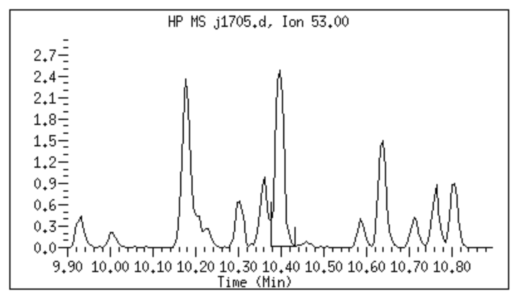
Original   Final

109 trans-1,4-Dichloro-2-Butene   CAS#: 110-57-6   Reason: M1



Electronic Signature Applied

User: kmn
Date: 06/05/2023 09:21



M1 - Target system did not integrate

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-MW04-060123MSD		
Collect Date:	06/01/23	Time:	1015	GCAL Sample ID:	22306021007		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11		
Sample Amt:	5	mL		Lab File ID:	230602/j1706		
Injection Vol.:	1		(µL)	GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942		
Analysis Date:	06/02/23	Time:	1902	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	47.8		0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	54.1		0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	41.1		0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	45.4		0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	50.1		0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	54.1		0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	52.5		0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	40.0		0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	38.5		0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	39.8		0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	44.9		0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	41.1		0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	46.2		0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	40.9		0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	53.4		0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	100		0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	49.8		0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	43.8		0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	41.0		0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	47.7		0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	41.1		0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	50.7		0.200	0.500	1.00
78-93-3	2-Butanone	187		0.200	0.500	5.00
95-49-8	2-Chlorotoluene	42.5		0.200	0.500	1.00
591-78-6	2-Hexanone	192		0.500	1.00	5.00
106-43-4	4-Chlorotoluene	42.2		0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	44.2		0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	238		0.200	0.500	5.00
67-64-1	Acetone	149		0.500	1.00	5.00
71-43-2	Benzene	49.8		0.200	0.500	1.00
108-86-1	Bromobenzene	41.3		0.200	0.500	1.00
74-97-5	Bromochloromethane	51.8		0.200	0.500	1.00
75-27-4	Bromodichloromethane	52.2		0.200	0.500	1.00
75-25-2	Bromoform	49.5		0.250	0.500	1.00
74-83-9	Bromomethane	50.4		0.500	1.00	2.00
75-15-0	Carbon disulfide	49.5		0.200	0.500	1.00
56-23-5	Carbon tetrachloride	54.9		0.250	0.500	1.00

FORM I VOA

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	OT07-MW04-060123MSD
Collect Date:	06/01/23 Time: 1015	GCAL Sample ID:	22306021007
Matrix:	Water % Moisture: NA	Instrument ID:	MSV11
Sample Amt:	5 mL	Lab File ID:	230602/j1706
Injection Vol.:	1 (µL)	GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1 Analyst: KN1	Analytical Batch:	766942
Analysis Date:	06/02/23 Time: 1902	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	47.5		0.200	0.500	1.00
75-00-3	Chloroethane	53.3		0.250	0.500	1.00
67-66-3	Chloroform	52.4		0.200	0.500	1.00
74-87-3	Chloromethane	43.5		0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	50.4		0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	50.0		0.200	0.500	1.00
124-48-1	Dibromochloromethane	49.8		0.200	0.500	1.00
74-95-3	Dibromomethane	52.1		0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	50.2		0.200	0.500	1.00
100-41-4	Ethylbenzene	48.2		0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	44.6		0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	51.2		0.200	0.500	1.00
136777-61-2	m,p-Xylene	97.5		0.200	0.500	2.00
75-09-2	Methylene chloride	49.1		0.200	0.500	5.00
91-20-3	Naphthalene	36.0		0.200	0.500	5.00
104-51-8	n-Butylbenzene	39.5		0.200	0.500	1.00
103-65-1	n-Propylbenzene	42.8		1.00	2.00	5.00
95-47-6	o-Xylene	50.3		0.200	0.500	1.00
135-98-8	sec-Butylbenzene	43.5		0.200	0.500	1.00
100-42-5	Styrene	50.6		0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	50.3		0.200	0.500	1.00
98-06-6	tert-Butylbenzene	43.7		0.200	0.500	1.00
127-18-4	Tetrachloroethene	51.0		0.200	0.500	1.00
108-88-3	Toluene	48.4		0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	49.6		0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	50.5		0.200	0.500	1.00
79-01-6	Trichloroethene	50.7		0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	56.8		0.200	0.500	1.00
75-01-4	Vinyl chloride	51.2		0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv11.i/230602.s.b/j1706.d
Lab Smp Id: 22306021007 Client Smp ID: MSD
Inj Date : 02-JUN-2023 19:02
Operator : KN1 Inst ID: msv11.i
Smp Info : 22306021007*MSD
Misc Info : MSV~52914~*1*KN1
Comment :
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Meth Date : 05-Jun-2023 08:22 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 15:02 Cal File: j1306.d
Als bottle: 1 QC Sample: MSD
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260bDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

						CONCENTRATIONS		
	QUANT SIG					ON-COLUMN	FINAL	
Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ug/L)	SIMILARITY
1 Dichlorodifluoromethane	85	1.668	1.673	(0.256)	98742	50.1710	50.2	
2 Chloromethane ++	50	1.851	1.856	(0.284)	150162	43.4514	43.5	
3 Vinyl Chloride +	62	1.930	1.940	(0.297)	130895	51.1739	51.2	
4 1-3 Butadiene	39	1.951	1.956	(0.300)	154432	55.0621	55.1	9743
5 Bromomethane	94	2.244	2.250	(0.345)	52888	50.3650	50.4	
7 Chloroethane	64	2.365	2.370	(0.363)	76044	53.3139	53.3	
8 Trichlorofluoromethane	101	2.517	2.522	(0.387)	159977	56.8297	56.8	
11 1,1-Dichloroethene +	96	3.067	3.067	(0.471)	78712	54.1248	54.1	
12 Carbon Disulfide	76	3.099	3.099	(0.476)	219266	49.5464	49.5	
13 1,1,2Trichlotrifluoroethane	101	3.115	3.120	(0.479)	80359	53.0215	53.0	
15 Methyl Iodide	142	3.225	3.225	(0.496)	66257	37.1298	37.1	
17 Allyl chloride	39	3.628	3.628	(0.558)	142015	56.1241	56.1	9359
18 Methylene Chloride	49	3.754	3.760	(0.577)	178675	49.0924	49.1	
19 Acetone	43	3.807	3.812	(0.585)	324592	148.641	149	

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
	20 trans-1,2-Dichloroethene	61	3.954	3.959	(0.608)	161656	49.6350	49.6	(R)
	21 Methyl Acetate	43	3.969	3.974	(0.610)	598591	252.987	253	9507(R)
	22 Hexane	57	4.069	4.069	(0.625)	236418	49.1405	49.1	9075
	23 MTBE	73	4.095	4.106	(0.629)	258989	50.3272	50.3	8831
	24 tert-Butyl Alcohol	59	4.221	4.226	(0.649)	101673	316.962	317	8814
	25 Acetonitrile	41	4.336	4.342	(0.666)	104565	248.070	248	9674
	26 Isopropyl Ether	45	4.546	4.551	(0.699)	437538	50.1697	50.2	9591
	28 1,1-Dichloroethane ++	63	4.651	4.656	(0.715)	206703	50.1232	50.1	
	29 Acrylonitrile	53	4.693	4.698	(0.721)	304948	271.068	271	
	31 Vinyl Acetate	43	4.944	4.944	(0.760)	37351	7.51679	7.52	(R)
	30 Ethyl Tert-butyl Ether	59	4.944	4.950	(0.760)	349534	48.9008	48.9	8711
	32 cis-1,2-Dichloroethene	61	5.238	5.238	(0.805)	154926	50.4089	50.4	
M	33 Total 1,2-Dichloroethene	61				316582	100.044	100	
	34 2,2-Dichloropropane	77	5.348	5.348	(0.822)	131804	50.6758	50.7	
	35 Bromochloromethane	128	5.443	5.448	(0.836)	46905	51.7741	51.8	
	36 Cyclohexane	56	5.453	5.453	(0.838)	207706	52.5124	52.5	8871
	37 Chloroform +	83	5.526	5.526	(0.849)	162395	52.4043	52.4	
	38 Carbon Tetrachloride	117	5.668	5.668	(0.871)	135650	54.8669	54.9	
$	41 Dibromofluoromethane	111	5.710	5.715	(0.878)	91513	55.7323	55.7	8810
	42 1,1,1-Trichloroethane	97	5.726	5.731	(0.880)	152095	54.0766	54.1	
	44 2-Butanone	43	5.825	5.825	(0.895)	359344	186.847	187	
	45 1,1-Dichloropropene	75	5.857	5.857	(0.900)	124001	52.5498	52.5	
	47 2,2,4 Trimethylpentane	57	5.977	5.977	(0.919)	328882	33.2946	33.3	9578
	48 Benzene	78	6.098	6.098	(0.937)	345697	49.7521	49.8	
$	51 1,2-Dichloroethane-d4	65	6.224	6.229	(0.956)	119361	54.4362	54.4	
	54 tert-amyl Methyl Ether	73	6.229	6.229	(0.957)	236515	47.2546	47.3	7225
	52 1,2-Dichloroethane	62	6.292	6.292	(0.967)	158181	53.3815	53.4	
	53 Isobutyl Alcohol	43	6.339	6.339	(0.974)	27206	214.385	214	9298
*	56 FLUOROBENZENE	96	6.507	6.507	(1.000)	329340	50.0000		
	57 Methyl Cyclohexane	83	6.669	6.669	(1.025)	130038	50.4302	50.4	9262
	58 Trichloroethene	130	6.675	6.675	(1.026)	101036	50.6539	50.7	
	60 Dibromomethane	93	7.068	7.073	(1.086)	59717	52.1447	52.1	
	62 1,2-Dichloropropane +	63	7.162	7.157	(1.101)	120642	49.8231	49.8	
	63 Bromodichloromethane	83	7.230	7.230	(1.111)	122213	52.2331	52.2	
	65 1,4- Dioxane	58	7.414	7.414	(1.139)	27536	1398.37	1400	8879
	66 1-Bromo-2-chloroethane	63	7.671	7.671	(1.179)	153044	48.2711	48.3	9705
	68 cis-1,3-Dichloropropene	75	7.812	7.812	(1.201)	135542	50.0075	50.0	
$	69 Toluene-d8	98	7.980	7.980	(0.864)	328566	51.1583	51.2	
	70 Toluene +	91	8.022	8.022	(0.868)	373085	48.4040	48.4	
M	72 1-3 Dichloropropene total	100				263804	100.519	101	0
	74 4-methyl-2-pentanone	43	8.337	8.337	(0.902)	714934	238.017	238	
	73 Tetrachloroethene	164	8.358	8.358	(0.905)	79648	51.0032	51.0	
	75 trans-1,3-Dichloropropene	75	8.373	8.373	(1.287)	128262	50.5117	50.5	
	78 1,1,2-Trichloroethane	97	8.510	8.504	(0.921)	79244	45.4381	45.4	
	79 Dibromochloromethane	129	8.657	8.657	(0.937)	93798	49.7678	49.8	
	80 1,3-Dichloropropane	76	8.725	8.725	(0.944)	140456	47.7112	47.7	
	82 1,2-Dibromoethane(EDB)	107	8.840	8.840	(0.957)	82912	46.1908	46.2	

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
83 2-Hexanone	43	9.008	9.008	(0.975)	480946	192.101	192	
89 1-Chlorohexane	91	9.233	9.233	(0.999)	105751	48.3047	48.3	9501
* 90 CHLOROBENZENE-d5	82	9.239	9.239	(1.000)	134171	50.0000		
91 Chlorobenzene ++	112	9.254	9.254	(1.002)	237229	47.4777	47.5	
92 Ethylbenzene +	106	9.270	9.270	(1.003)	127121	48.1676	48.2	
93 1,1,1,2-Tetrachloroethane	133	9.301	9.301	(1.007)	81663	47.8318	47.8	
94 p,m-Xylene	106	9.380	9.380	(1.015)	309212	97.5272	97.5	
M 95 TOTAL XYLENE	106				457241	147.827	148	
96 o-Xylene	106	9.700	9.700	(1.050)	148029	50.3000	50.3	
97 Styrene	104	9.737	9.737	(1.054)	211784	50.5544	50.6	
98 Bromoform ++	173	9.768	9.768	(1.057)	61341	49.4727	49.5	
99 Isopropylbenzene	105	9.931	9.931	(1.075)	375052	51.1538	51.2	
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	104655	54.0528	54.1	
102 cis-1,4-dichloro-2-butene	53	10.177	10.172	(0.931)	40872	32.6622	32.7	8945
103 Bromobenzene	77	10.214	10.214	(0.935)	166085	41.2722	41.3	
104 n-Propylbenzene	91	10.229	10.224	(0.936)	428862	42.8361	42.8	
105 1,1,2,2-Tetrachloroethane++	83	10.277	10.277	(0.940)	106654	41.1225	41.1	
106 2-Chlorotoluene	91	10.345	10.345	(0.947)	300009	42.5413	42.5	
107 1,3,5-Trimethylbenzene	105	10.361	10.361	(0.948)	328132	43.7507	43.8	
108 1,2,3-Trichloropropane	75	10.371	10.376	(0.949)	129698	38.4546	38.5	
109 trans-1,4-Dichloro-2-Butene	53	10.397	10.397	(0.952)	35403	29.3085	29.3	(RM1)
100 Cyclohexanone	55	10.408	10.402	(0.952)	26220	29.7063	29.7	8012(R)
110 4-Chlorotoluene	91	10.460	10.460	(0.957)	283195	42.2109	42.2	
111 tert-butylbenzene	91	10.586	10.586	(0.969)	178873	43.7078	43.7	
112 1,2,4-Trimethylbenzene	105	10.638	10.638	(0.974)	338247	44.8713	44.9	
113 sec-Butylbenzene	105	10.712	10.712	(0.980)	378329	43.5250	43.5	
115 p-Isopropyltoluene	119	10.811	10.806	(0.989)	356376	44.2214	44.2	
116 1,3-Dichlorobenzene	146	10.874	10.874	(0.995)	189602	41.0477	41.0	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	145532	50.0000		
118 1,4-Dichlorobenzene	146	10.937	10.937	(1.001)	191328	41.1158	41.1	
119 n-Butylbenzene	91	11.100	11.100	(1.016)	459326	39.4833	39.5	
120 1,2-Dichlorobenzene	146	11.231	11.226	(1.028)	182884	40.8983	40.9	
122 1,2-Dibromo-3-Chloropropane	157	11.771	11.771	(1.077)	25450	41.1450	41.1	
123 Hexachlorobutadiene	225	12.201	12.201	(1.117)	55387	44.6206	44.6	
124 1,2,4-Trichlorobenzene	180	12.227	12.227	(1.119)	109454	39.7725	39.8	
125 Naphthalene	128	12.447	12.447	(1.139)	294305	36.0168	36.0	
126 1,2,3-Trichlorobenzene	180	12.573	12.573	(1.151)	106985	40.0125	40.0	

QC Flag Legend

R - Spike/Surrogate failed recovery limits.
M1- Compound response manually integrated because
Target system did not integrate.

Data File: /var/chem/msv11.i/230602.s.b/j1706.d

Date : 02-JUN-2023 19:02

Client ID: MSD

Instrument: msv11.i

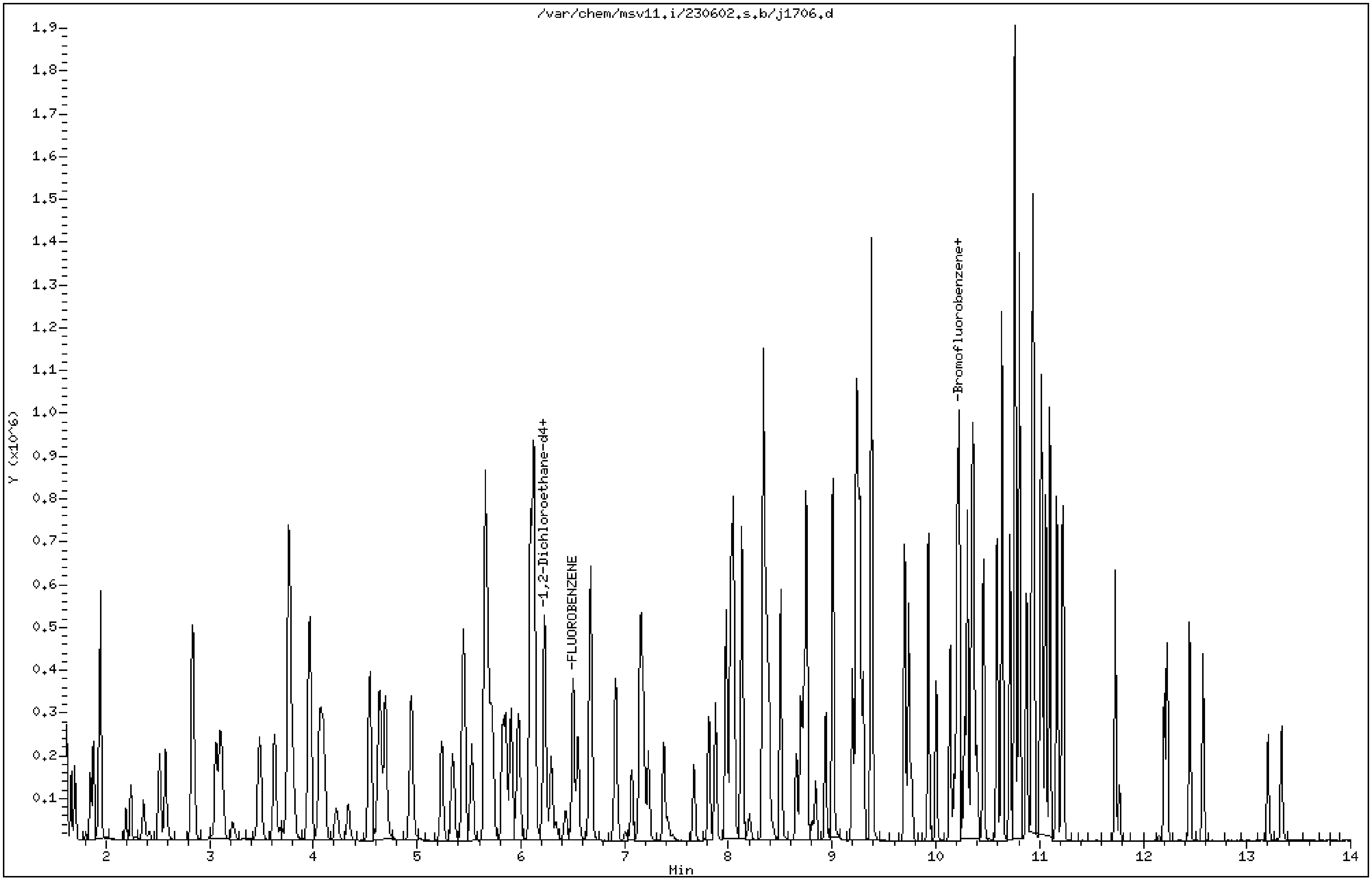
Sample Info: 22306021007*MSD

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

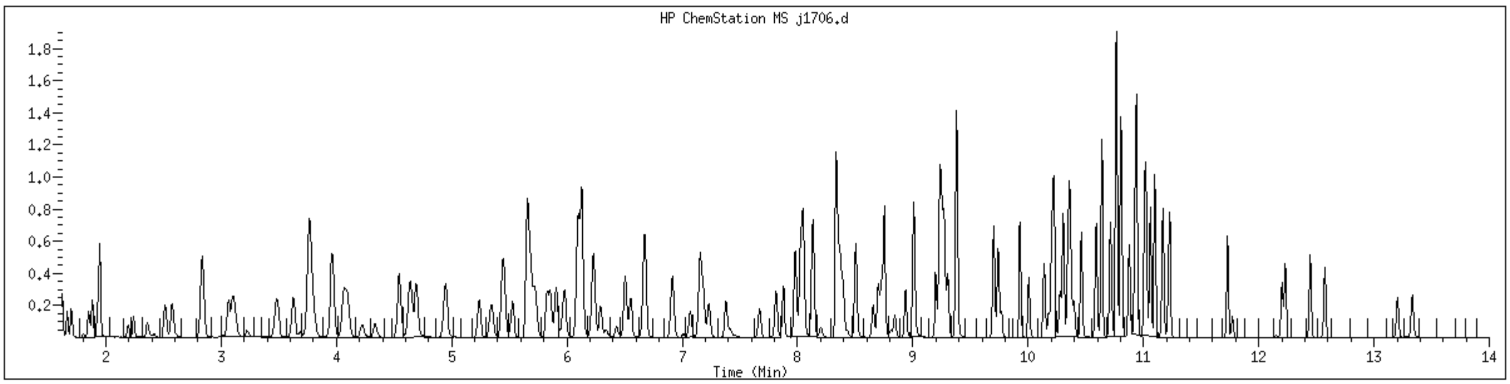
Column diameter: 0.25



Data file : /var/chem/msv11.i/230602.s.b/j1706.d   Page: 1
Report Date: 06/05/2023 09:22

MANUAL INTEGRATION GRAPHIC REPORT

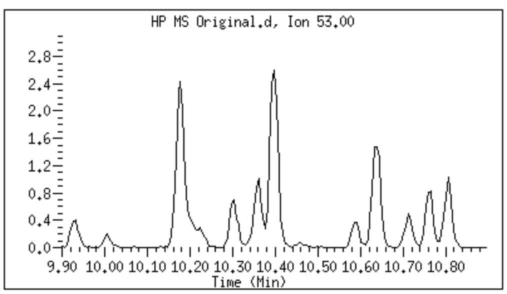
Lab ID   : 22306021007   SampleType   : MSD
Injection Date: 06/02/2023 19:02   Instrument   : msv11.i
Operator   : KN1
Sample Info   : 22306021007*MSD
Misc Info   : MSV~52914~*1*KN1
Method   : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260bDOD



Original   Final

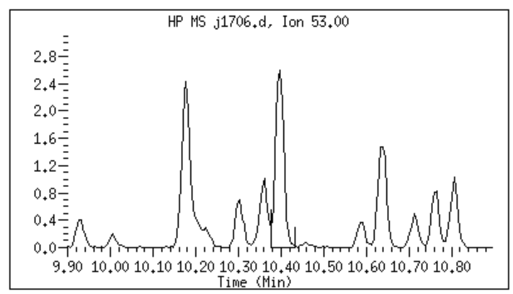
===

109 trans-1,4-Dichloro-2-Butene   CAS#: 110-57-6   Reason: M1



Electronic Signature
Applied

User: kmn
Date: 06/05/2023 09:22



M1 - Target system did not integrate

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-MW-806-060123
Collect Date:	06/01/23	Time:	1100	GCAL Sample ID:	22306021008
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV15
Sample Amt:	5	mL		Lab File ID:	230605/c0742
Injection Vol.:	1	(μL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	767028
Analysis Date:	06/05/23	Time:	1430	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	0.500	U	0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	0.500	U	0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	0.500	U	0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	0.500	U	0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	0.500	U	0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	0.500	U	0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	0.500	U	0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	0.500	U	0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	0.500	U	0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	0.500	U	0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	0.500	U	0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	0.500	U	0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	0.500	U	0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	13.5		0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	0.500	U	0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	0.500	U	0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	0.500	U	0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	0.500	U	0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	0.500	U	0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	0.500	U	0.200	0.500	1.00
78-93-3	2-Butanone	0.500	U	0.200	0.500	5.00
95-49-8	2-Chlorotoluene	0.500	U	0.200	0.500	1.00
591-78-6	2-Hexanone	1.00	U	0.500	1.00	5.00
106-43-4	4-Chlorotoluene	0.500	U	0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	0.500	U	0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	0.500	U	0.200	0.500	5.00
67-64-1	Acetone	1.00	U	0.500	1.00	5.00
71-43-2	Benzene	1.57		0.200	0.500	1.00
108-86-1	Bromobenzene	0.500	U	0.200	0.500	1.00
74-97-5	Bromochloromethane	0.500	U	0.200	0.500	1.00
75-27-4	Bromodichloromethane	0.500	U	0.200	0.500	1.00
75-25-2	Bromoform	0.500	U	0.250	0.500	1.00
74-83-9	Bromomethane	1.00	UQ	0.500	1.00	2.00
75-15-0	Carbon disulfide	0.500	U	0.200	0.500	1.00
56-23-5	Carbon tetrachloride	0.500	U	0.250	0.500	1.00

FORM I VOA

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-MW-806-060123
Collect Date:	06/01/23	Time:	1100	GCAL Sample ID:	22306021008
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV15
Sample Amt:	5	mL		Lab File ID:	230605/c0742
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	767028
Analysis Date:	06/05/23	Time:	1430	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	*ANALYTE*	*RESULT*	*Q*	*DL*	*LOD*	*LOQ*
108-90-7	Chlorobenzene	0.500	U	0.200	0.500	1.00
75-00-3	Chloroethane	0.500	U	0.250	0.500	1.00
67-66-3	Chloroform	0.500	U	0.200	0.500	1.00
74-87-3	Chloromethane	0.500	U	0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	13.5		0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
124-48-1	Dibromochloromethane	0.500	U	0.200	0.500	1.00
74-95-3	Dibromomethane	0.500	U	0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	0.500	UQ	0.200	0.500	1.00
100-41-4	Ethylbenzene	0.500	U	0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	1.00	U	0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	0.500	U	0.200	0.500	1.00
136777-61-2	m,p-Xylene	0.500	U	0.200	0.500	2.00
75-09-2	Methylene chloride	0.500	U	0.200	0.500	5.00
91-20-3	Naphthalene	0.500	U	0.200	0.500	5.00
104-51-8	n-Butylbenzene	0.500	U	0.200	0.500	1.00
103-65-1	n-Propylbenzene	2.00	U	1.00	2.00	5.00
95-47-6	o-Xylene	0.500	U	0.200	0.500	1.00
135-98-8	sec-Butylbenzene	0.500	U	0.200	0.500	1.00
100-42-5	Styrene	0.500	U	0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	0.500	U	0.200	0.500	1.00
98-06-6	tert-Butylbenzene	0.500	U	0.200	0.500	1.00
127-18-4	Tetrachloroethene	0.500	U	0.200	0.500	1.00
108-88-3	Toluene	0.500	U	0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
79-01-6	Trichloroethene	0.500	U	0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	0.500	U	0.200	0.500	1.00
75-01-4	Vinyl chloride	88.5		0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230605.s.b/c0742.d
Lab Smp Id: 22306021008 Client Smp ID: 22306021008
Inj Date : 05-JUN-2023 14:30
Operator : KN1 Inst ID: msv15.i
Smp Info : 22306021008*
Misc Info : MSV~529~*1*SMR
Comment :
Method : /var/chem/msv15.i/230605.s.b/8260bw15DOD.m
Meth Date : 06-Jun-2023 12:20 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

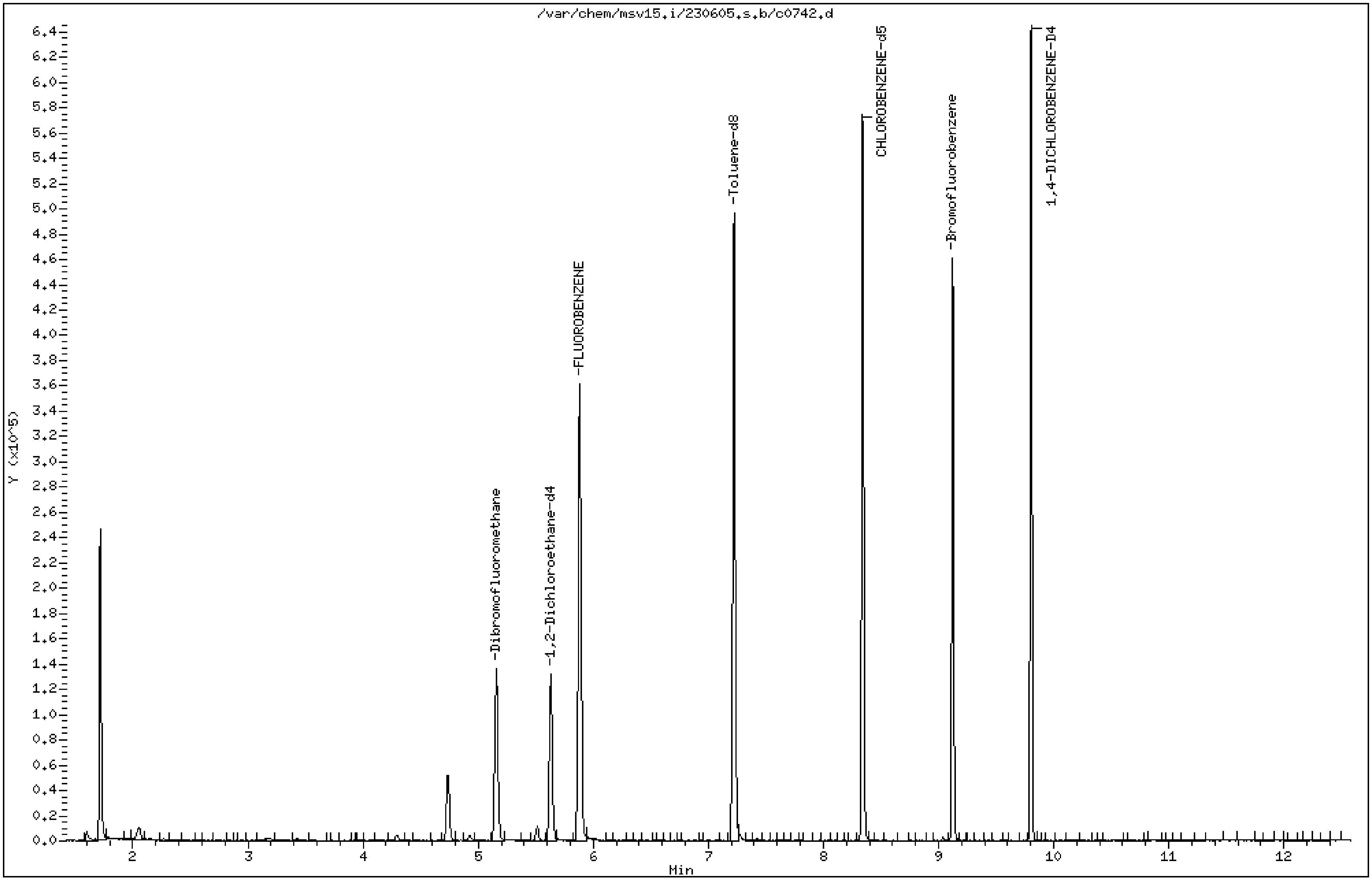
Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	CONCENTRATIONS FINAL (ug/L)	SIMILARITY
3 Vinyl Chloride +	62	1.716	1.719	(0.292)	167906	88.5246	88.5	9800
32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	28209	13.4742	13.5	9747
M 33 Total 1,2-Dichloroethene	61				28209	13.4742	13.5	0
36 Cyclohexane	56	4.918	4.928	(0.836)	1966	0.74035	0.740	6894
$ 42 Dibromofluoromethane	111	5.160	5.160	(0.877)	81679	51.3359	51.3	6890
48 Benzene	78	5.516	5.513	(0.938)	10436	1.57422	1.57	9044
$ 50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	86428	52.1699	52.2	9683
* 55 FLUOROBENZENE	96	5.883	5.883	(1.000)	322181	50.0000		9870
$ 74 Toluene-d8	98	7.224	7.224	(0.866)	286449	46.3958	46.4	9838
* 91 CHLOROBENZENE-d5	82	8.339	8.339	(1.000)	131738	50.0000		9855
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	93529	46.6455	46.6	9356
* 124 1,4-DICHLOROBENZENE-D4	152	9.806	9.806	(1.000)	121370	50.0000		9659

Data File: /var/chem/msv15.i/230605.s.b/c0742.d
Date : 05-JUN-2023 14:30
Client ID: 22306021008
Sample Info: 22306021008*
Purge Volume: 5.0
Column phase: RTX-VMS-30M

Instrument: msv15.i

Operator: KN1
Column diameter: 0.25



Date : 05-JUN-2023 14:30

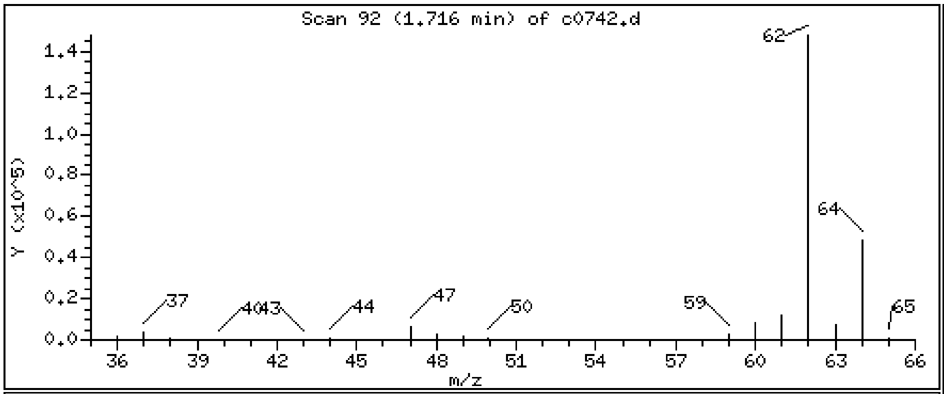
Client ID: 22306021008 Instrument: msv15.i

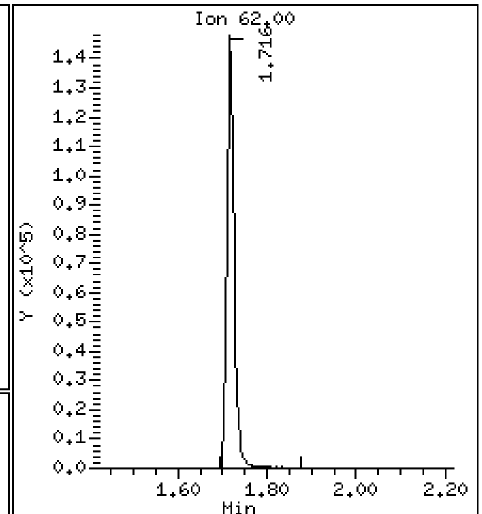
Sample Info: 22306021008*

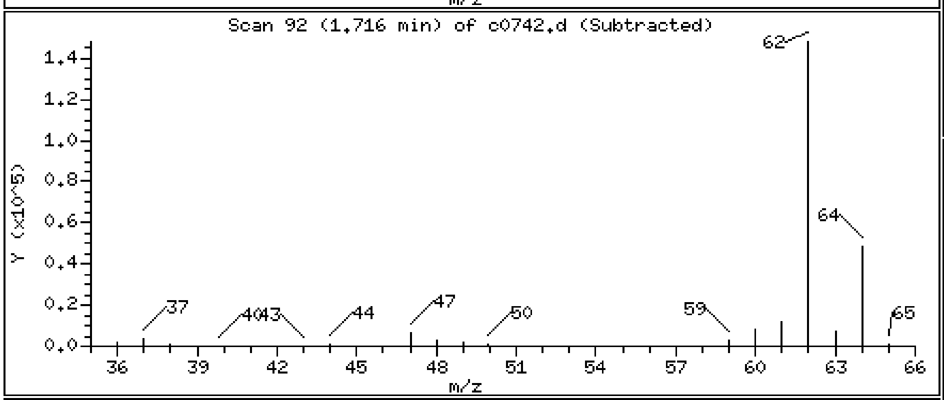
Purge Volume: 5.0 Operator: KN1

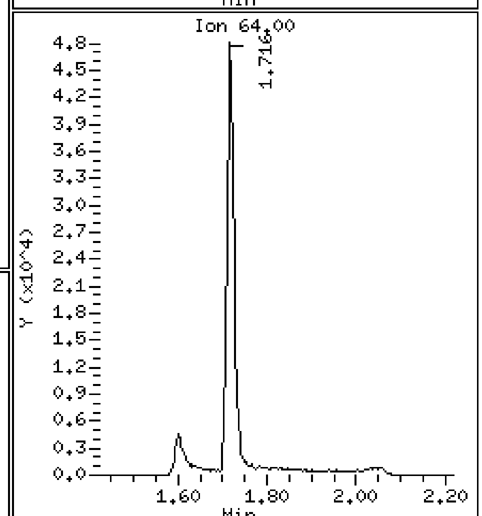
Column phase: RTX-VMS-30M Column diameter: 0.25

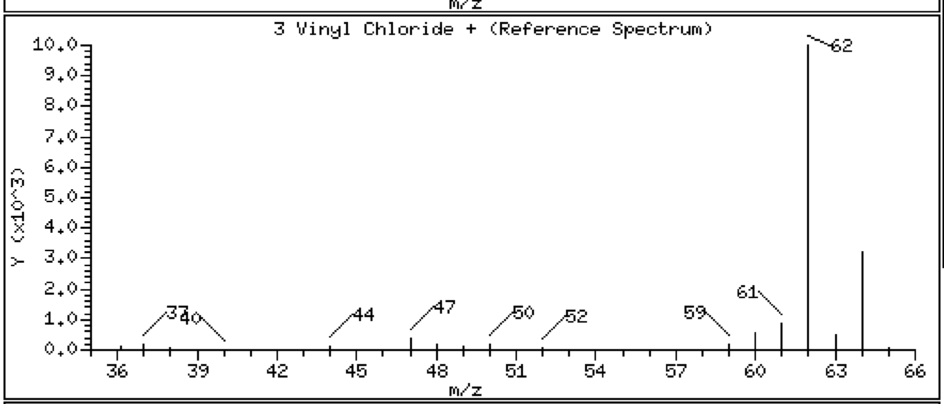
3 Vinyl Chloride + Concentration: 88.5 ug/L

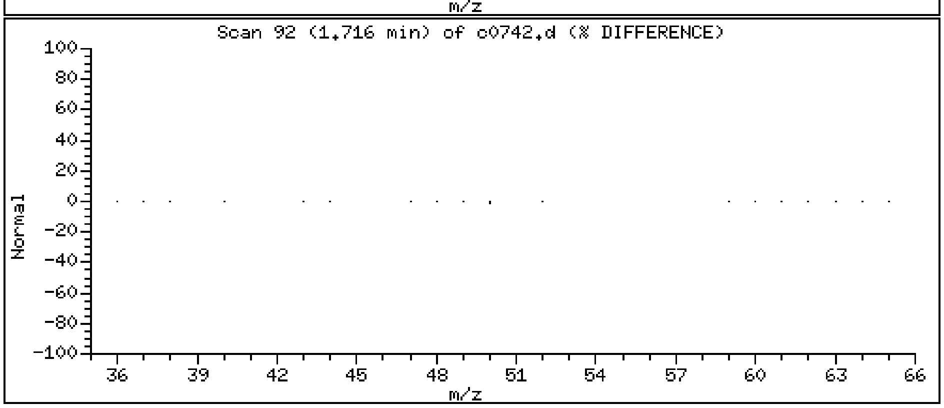












Data File: /var/chem/msv15.i/230605.s.b/c0742.d

Date : 05-JUN-2023 14:30

Client ID: 22306021008

Instrument: msv15.i

Sample Info: 22306021008*

Purge Volume: 5.0

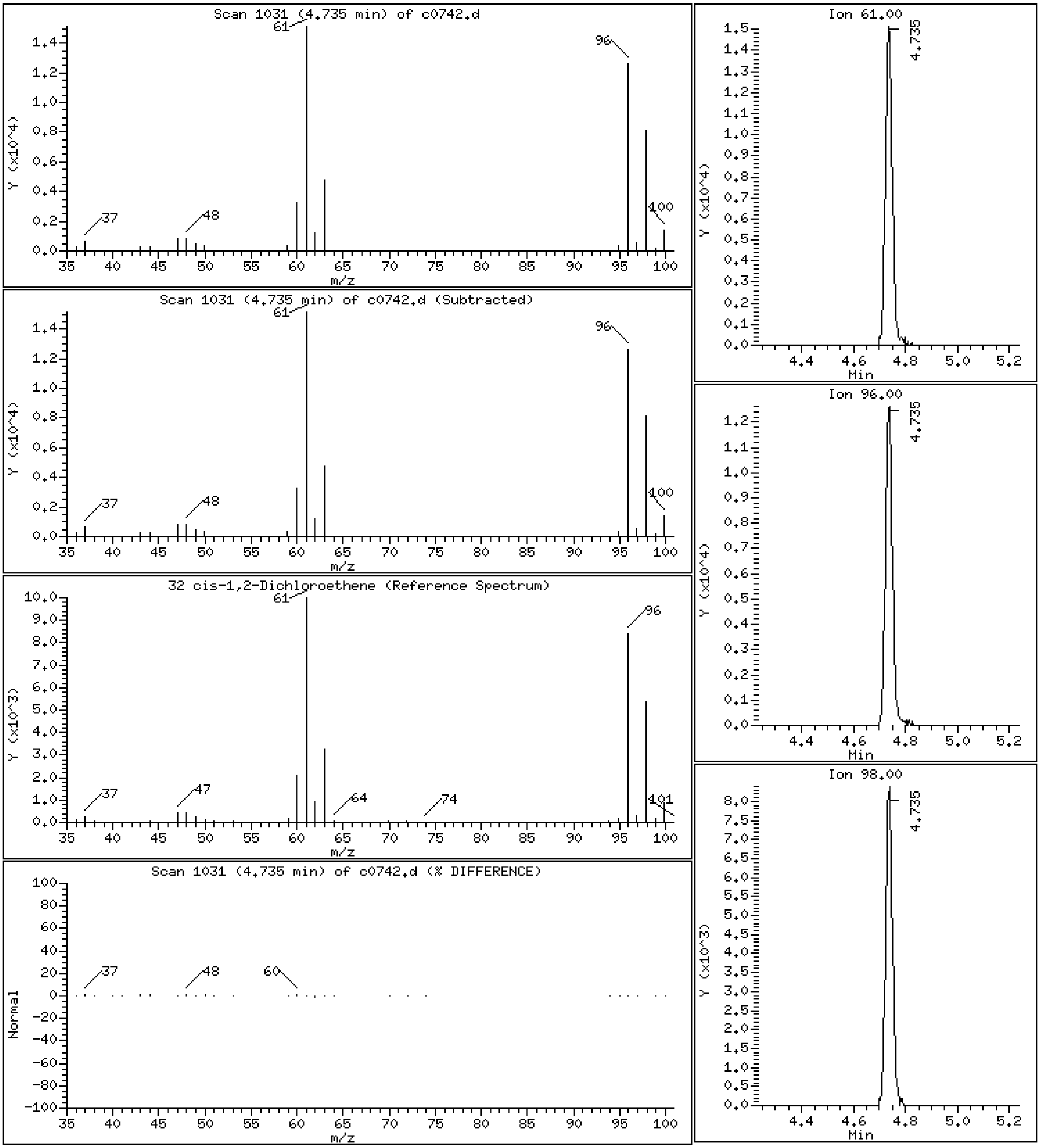
Operator: KN1

Column phase: RTX-VMS-30M

Column diameter: 0.25

32 cis-1,2-Dichloroethene

Concentration: 13.5 ug/L



Data File: /var/chem/msv15.i/230605.s.b/c0742.d

Date : 05-JUN-2023 14:30

Client ID: 22306021008

Sample Info: 22306021008*

Purge Volume: 5.0

Column phase: RTX-VMS-30M

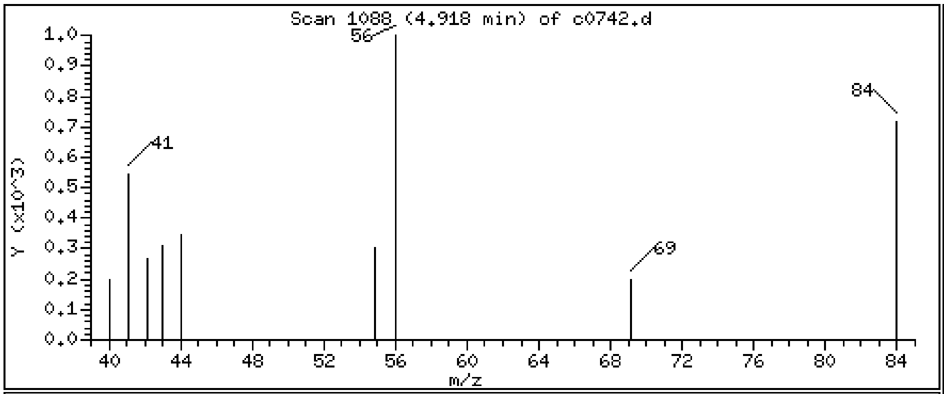
36 Cyclohexane

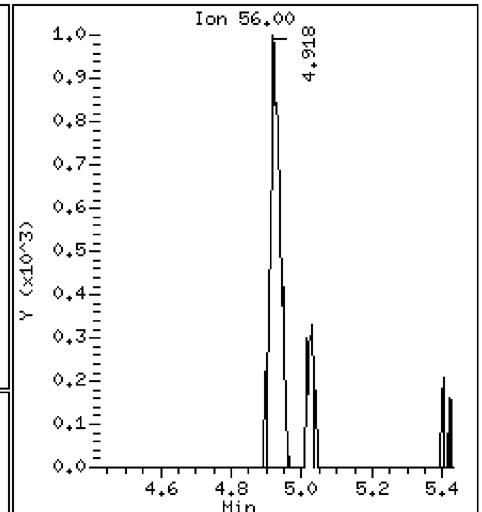
Instrument: msv15.i

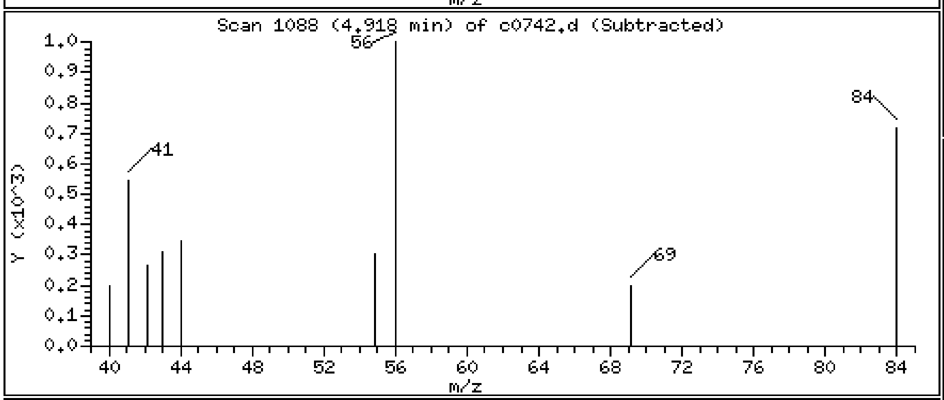
Operator: KN1

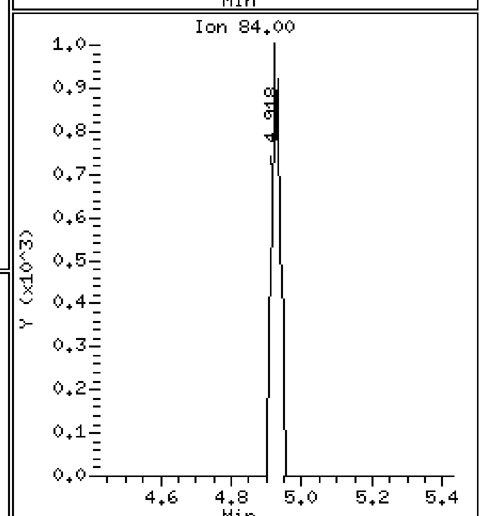
Column diameter: 0.25

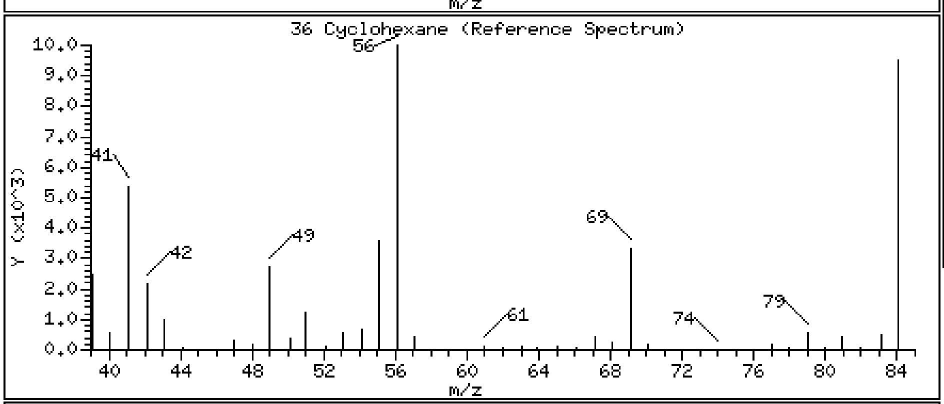
Concentration: 0.740 ug/L

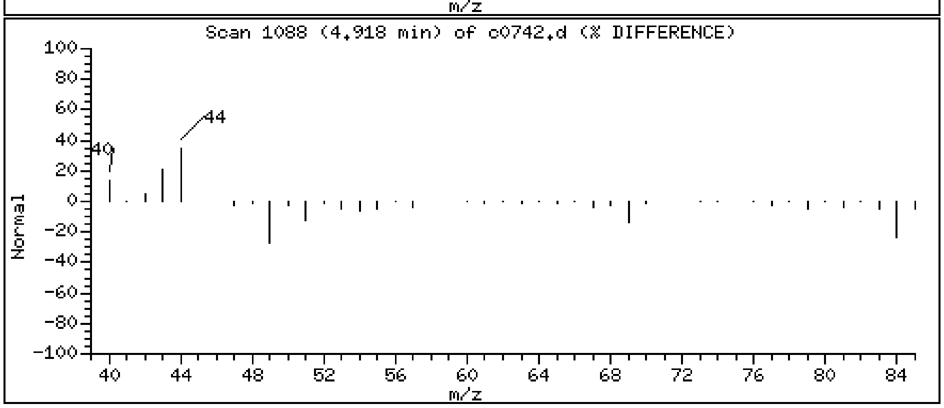












Data File: /var/chem/msv15.i/230605.s.b/c0742.d

Date : 05-JUN-2023 14:30

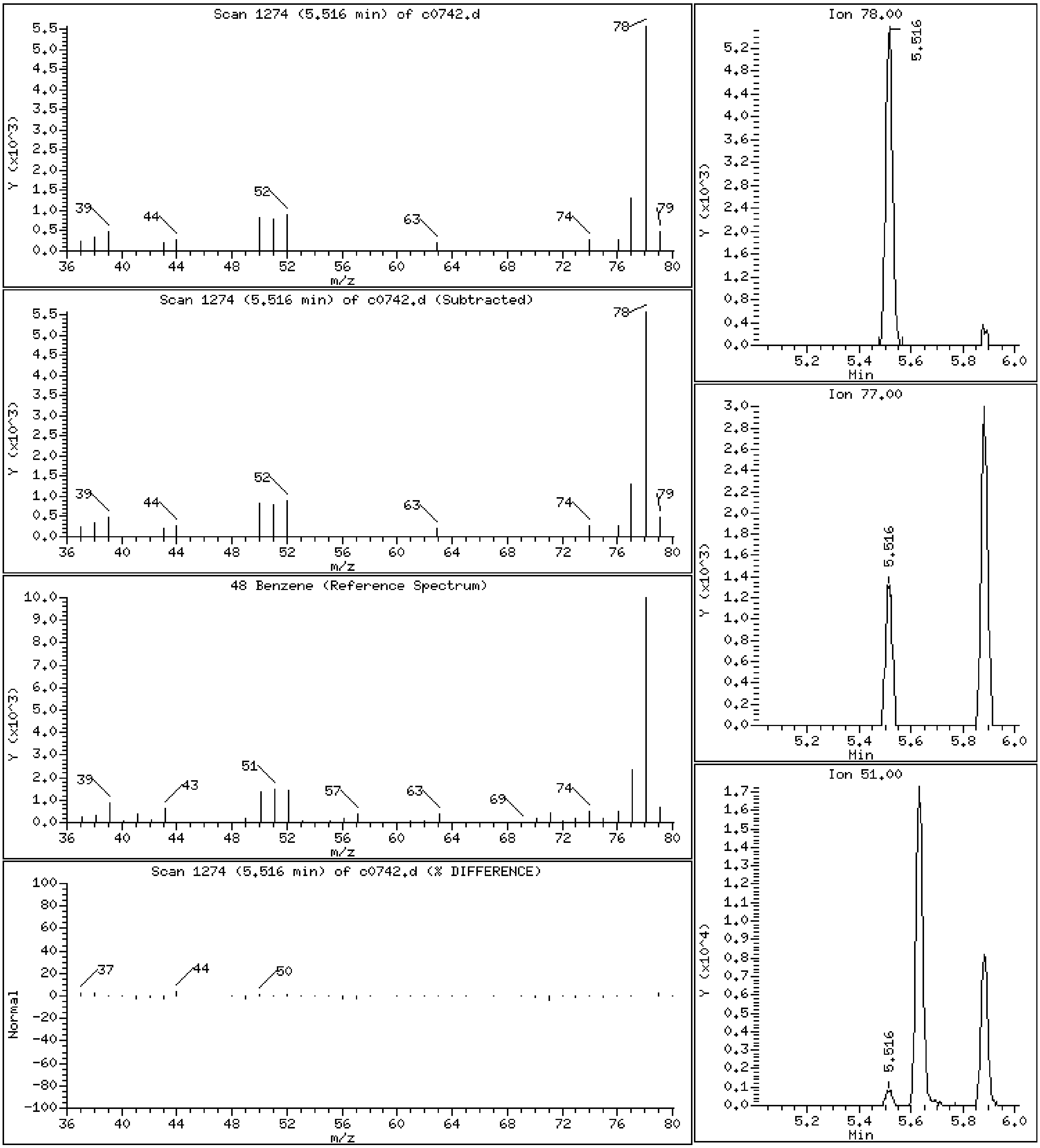
Client ID: 22306021008 Instrument: msv15.i

Sample Info: 22306021008*

Purge Volume: 5.0 Operator: KN1

Column phase: RTX-VMS-30M Column diameter: 0.25

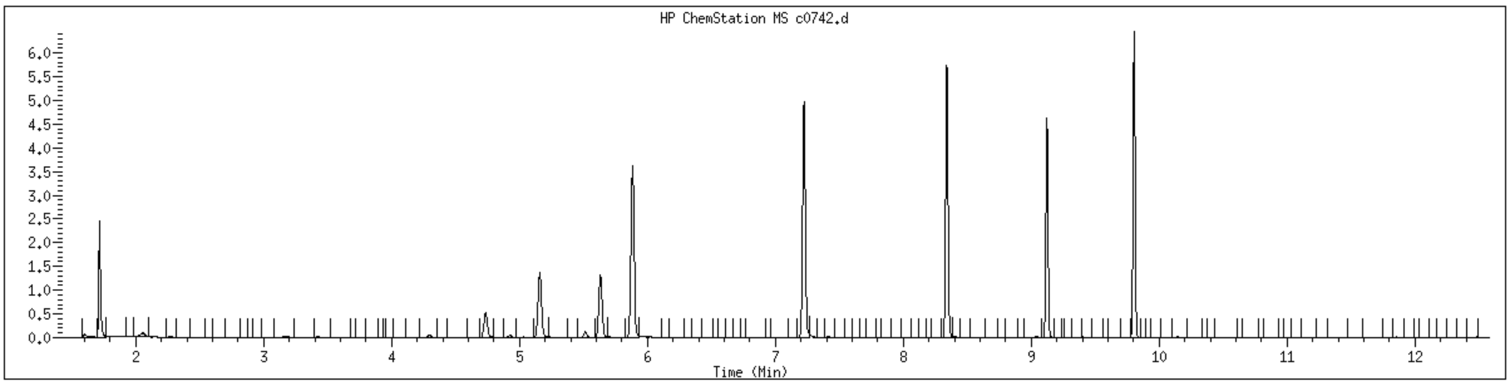
48 Benzene Concentration: 1.57 ug/L



Data file : /var/chem/msv15.i/230605.s.b/c0742.d Page: 1
Report Date: 06/06/2023 12:32

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 22306021008 SampleType : SAMPLE
Injection Date: 06/05/2023 14:30 Instrument : msv15.i
Operator : KN1
Sample Info : 22306021008*
Misc Info : MSV~529~*1*SMR
Method : /var/chem/msv15.i/230605.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-TB-01-060124
Collect Date:	06/01/23	Time:	0730	GCAL Sample ID:	22306021009
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV15
Sample Amt:	5	mL		Lab File ID:	230606/c0777
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	767081
Analysis Date:	06/06/23	Time:	1342	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	0.500	U	0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	0.500	U	0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	0.500	U	0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	0.500	U	0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	0.500	U	0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	0.500	U	0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	0.500	U	0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	0.500	U	0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	0.500	U	0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	0.500	UQ	0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	0.500	U	0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	0.500	U	0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	0.500	U	0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	1.00	U	0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	0.500	U	0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	0.500	U	0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	0.500	U	0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	0.500	U	0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	0.500	U	0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	0.500	U	0.200	0.500	1.00
78-93-3	2-Butanone	0.500	UQ	0.200	0.500	5.00
95-49-8	2-Chlorotoluene	0.500	U	0.200	0.500	1.00
591-78-6	2-Hexanone	1.00	U	0.500	1.00	5.00
106-43-4	4-Chlorotoluene	0.500	U	0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	0.500	U	0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	0.500	UQ	0.200	0.500	5.00
67-64-1	Acetone	1.22	JQ	0.500	1.00	5.00
71-43-2	Benzene	0.500	U	0.200	0.500	1.00
108-86-1	Bromobenzene	0.500	U	0.200	0.500	1.00
74-97-5	Bromochloromethane	0.500	U	0.200	0.500	1.00
75-27-4	Bromodichloromethane	0.500	U	0.200	0.500	1.00
75-25-2	Bromoform	0.500	U	0.250	0.500	1.00
74-83-9	Bromomethane	1.00	U	0.500	1.00	2.00
75-15-0	Carbon disulfide	0.500	U	0.200	0.500	1.00
56-23-5	Carbon tetrachloride	0.500	U	0.250	0.500	1.00

FORM I VOA

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	OT07-TB-01-060124		
Collect Date:	06/01/23	Time:	0730	GCAL Sample ID:	22306021009		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV15		
Sample Amt:	5	mL		Lab File ID:	230606/c0777		
Injection Vol.:	1		(µL)	GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	767081		
Analysis Date:	06/06/23	Time:	1342	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	0.500	U	0.200	0.500	1.00
75-00-3	Chloroethane	0.500	U	0.250	0.500	1.00
67-66-3	Chloroform	0.500	U	0.200	0.500	1.00
74-87-3	Chloromethane	0.500	U	0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
124-48-1	Dibromochloromethane	0.500	U	0.200	0.500	1.00
74-95-3	Dibromomethane	0.500	U	0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	0.500	U	0.200	0.500	1.00
100-41-4	Ethylbenzene	0.500	U	0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	1.00	U	0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	0.500	U	0.200	0.500	1.00
136777-61-2	m,p-Xylene	0.500	U	0.200	0.500	2.00
75-09-2	Methylene chloride	0.500	U	0.200	0.500	5.00
91-20-3	Naphthalene	0.500	U	0.200	0.500	5.00
104-51-8	n-Butylbenzene	0.500	U	0.200	0.500	1.00
103-65-1	n-Propylbenzene	2.00	U	1.00	2.00	5.00
95-47-6	o-Xylene	0.500	U	0.200	0.500	1.00
135-98-8	sec-Butylbenzene	0.500	U	0.200	0.500	1.00
100-42-5	Styrene	0.500	U	0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	0.500	UQ	0.200	0.500	1.00
98-06-6	tert-Butylbenzene	0.500	U	0.200	0.500	1.00
127-18-4	Tetrachloroethene	0.500	U	0.200	0.500	1.00
108-88-3	Toluene	0.500	U	0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
79-01-6	Trichloroethene	0.500	U	0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	0.500	U	0.200	0.500	1.00
75-01-4	Vinyl chloride	0.500	U	0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230606.s.b/c0777.d
Lab Smp Id: 22306021009 Client Smp ID: 22306021009
Inj Date : 06-JUN-2023 13:42
Operator : KN1 Inst ID: msv15.i
Smp Info : 22306021009*
Misc Info : MSV~52923~*1*SMR
Comment :
Method : /var/chem/msv15.i/230606.s.b/8260bw15DOD.m
Meth Date : 06-Jun-2023 16:25 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

						CONCENTRATIONS		
	QUANT SIG					ON-COLUMN	FINAL	
Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ug/L)	SIMILARITY
19 Acetone	43	3.475	3.468	(0.591)	907	1.22455	1.22	3047
$ 42 Dibromofluoromethane	111	5.160	5.160	(0.877)	86451	52.4138	52.4	6902
$ 50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	90670	52.7952	52.8	9630
* 55 FLUOROBENZENE	96	5.883	5.883	(1.000)	333991	50.0000		9876
$ 74 Toluene-d8	98	7.221	7.221	(0.866)	301519	46.5269	46.5	9834
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	138278	50.0000		9800
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	99533	47.2921	47.3	9431
* 124 1,4-DICHLOROBENZENE-D4	152	9.803	9.806	(1.000)	127564	50.0000		9593

Data File: /var/chem/msv15.i/230606.s.b/c0777.d

Page 1

Date : 06-JUN-2023 13:42

Client ID: 22306021009

Instrument: msv15.i

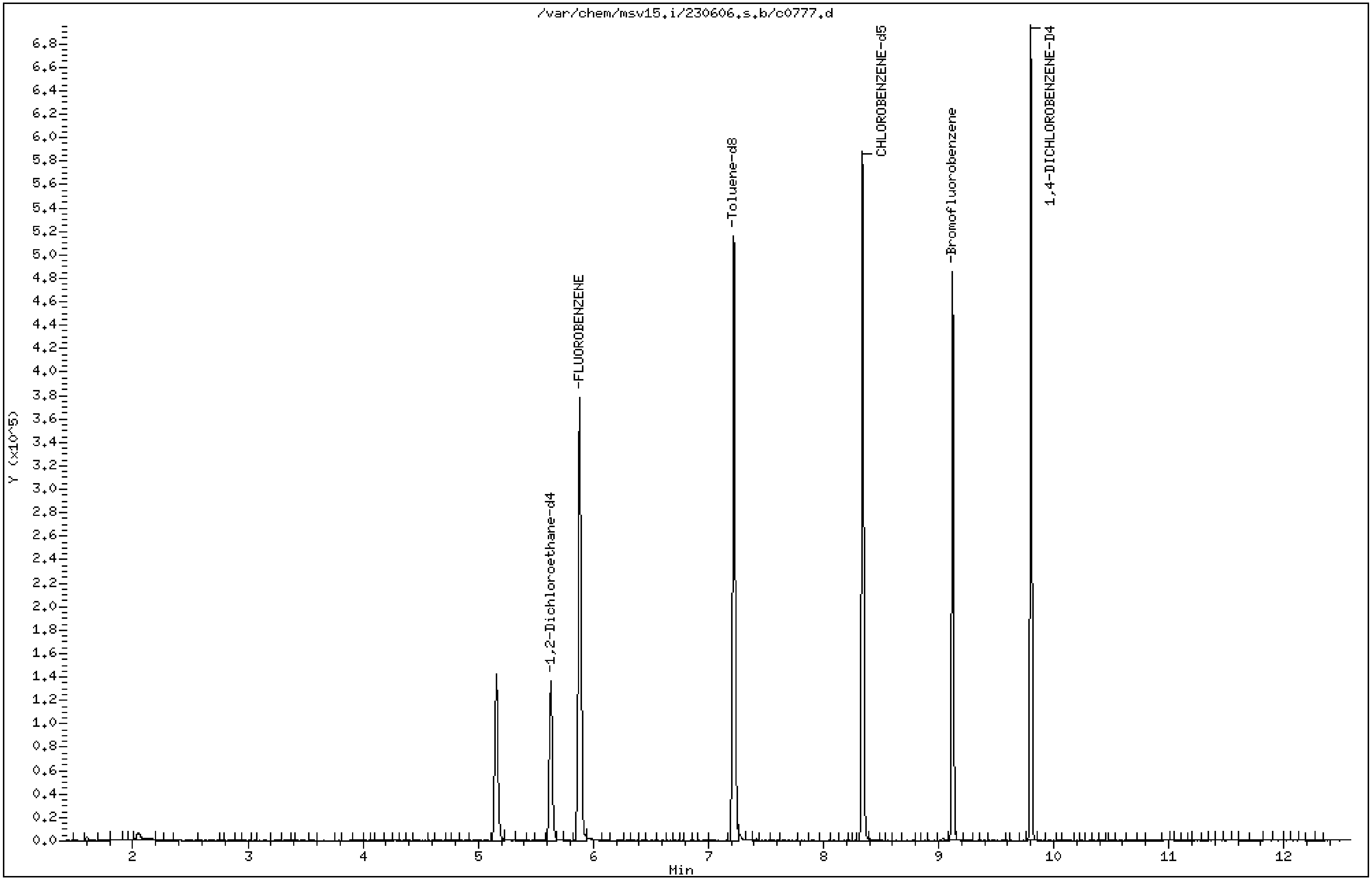
Sample Info: 22306021009*

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

Column diameter: 0.25



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Date : 06-JUN-2023 13:42

Client ID: 22306021009

Instrument: msv15.i

Sample Info: 22306021009*

Purge Volume: 5.0

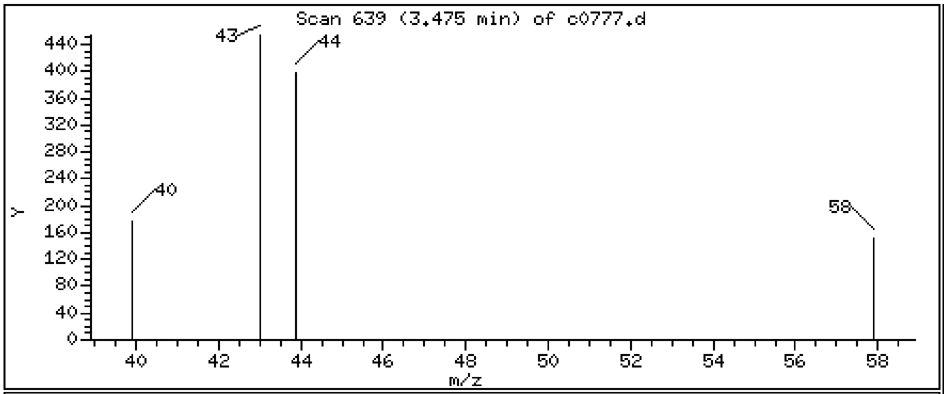
Operator: KN1

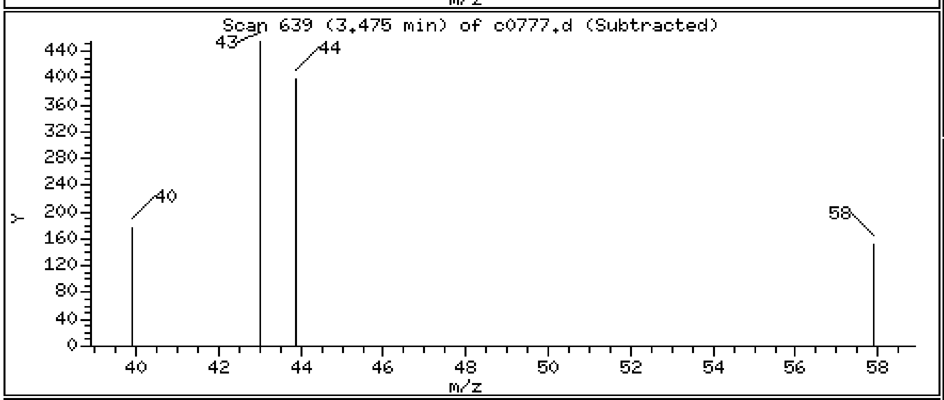
Column phase: RTX-VMS-30M

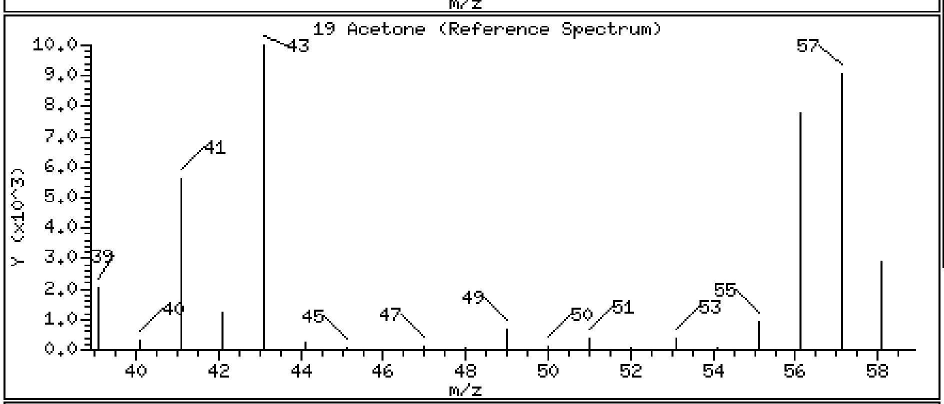
Column diameter: 0.25

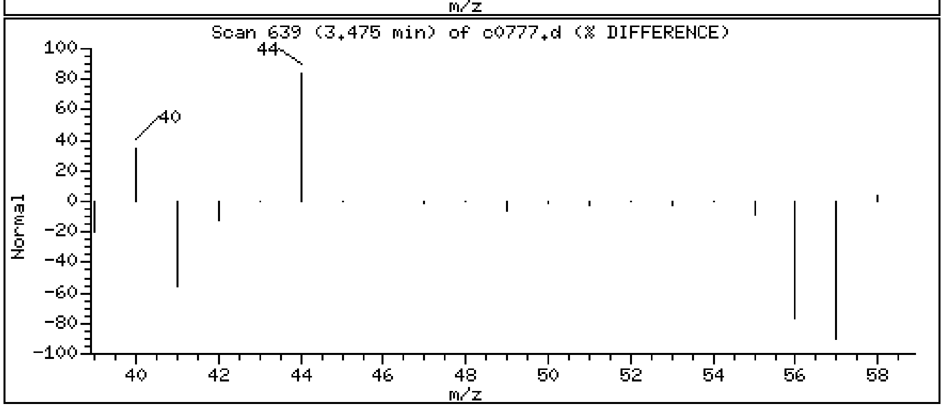
19 Acetone

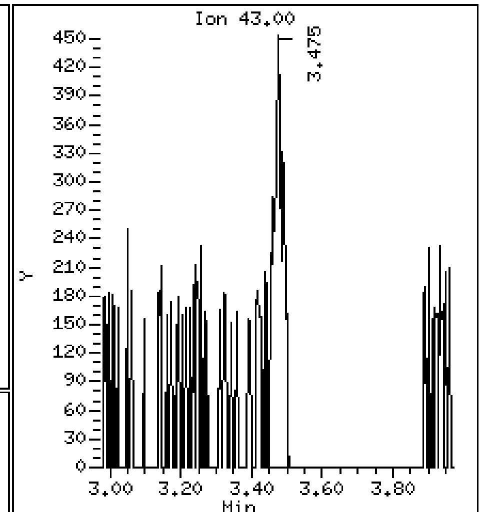
Concentration: 1.22 ug/L

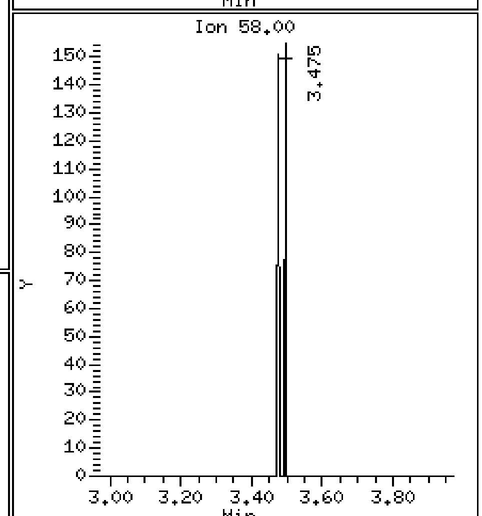








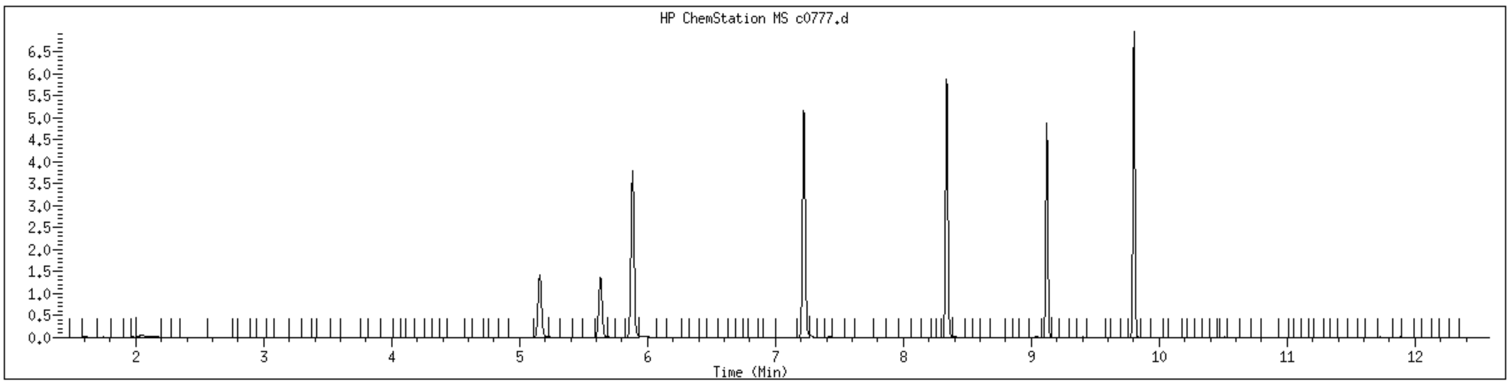




Report Date: 06/07/2023 11:30

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 22306021009 SampleType : SAMPLE
Injection Date: 06/06/2023 13:42 Instrument : msv15.i
Operator : KN1
Sample Info : 22306021009*
Misc Info : MSV~52923~*1*SMR
Method : /var/chem/msv15.i/230606.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	MB2489345
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2489345
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11
Sample Amt:	5	mL		Lab File ID:	230602/j1694
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942
Analysis Date:	06/02/23	Time:	1423	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	0.500	U	0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	0.500	U	0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	0.500	U	0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	0.500	U	0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	0.500	U	0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	0.500	U	0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	0.500	U	0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	0.500	U	0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	0.500	U	0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	0.500	U	0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	0.500	U	0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	0.500	U	0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	0.500	U	0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	1.00	U	0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	0.500	U	0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	0.500	U	0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	0.500	U	0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	0.500	U	0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	0.500	U	0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	0.500	U	0.200	0.500	1.00
78-93-3	2-Butanone	0.500	U	0.200	0.500	5.00
95-49-8	2-Chlorotoluene	0.500	U	0.200	0.500	1.00
591-78-6	2-Hexanone	1.00	U	0.500	1.00	5.00
106-43-4	4-Chlorotoluene	0.500	U	0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	0.500	U	0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	0.500	U	0.200	0.500	5.00
67-64-1	Acetone	1.00	U	0.500	1.00	5.00
71-43-2	Benzene	0.500	U	0.200	0.500	1.00
108-86-1	Bromobenzene	0.500	U	0.200	0.500	1.00
74-97-5	Bromochloromethane	0.500	U	0.200	0.500	1.00
75-27-4	Bromodichloromethane	0.500	U	0.200	0.500	1.00
75-25-2	Bromoform	0.500	U	0.250	0.500	1.00
74-83-9	Bromomethane	1.00	U	0.500	1.00	2.00
75-15-0	Carbon disulfide	0.500	U	0.200	0.500	1.00
56-23-5	Carbon tetrachloride	0.500	U	0.250	0.500	1.00

FORM I VOA

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	MB2489345
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2489345
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11
Sample Amt:	5	mL		Lab File ID:	230602/j1694
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942
Analysis Date:	06/02/23	Time:	1423	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	0.500	U	0.200	0.500	1.00
75-00-3	Chloroethane	0.500	U	0.250	0.500	1.00
67-66-3	Chloroform	0.500	U	0.200	0.500	1.00
74-87-3	Chloromethane	0.500	U	0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
124-48-1	Dibromochloromethane	0.500	U	0.200	0.500	1.00
74-95-3	Dibromomethane	0.500	U	0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	0.500	U	0.200	0.500	1.00
100-41-4	Ethylbenzene	0.500	U	0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	1.00	U	0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	0.500	U	0.200	0.500	1.00
136777-61-2	m,p-Xylene	0.500	U	0.200	0.500	2.00
75-09-2	Methylene chloride	0.500	U	0.200	0.500	5.00
91-20-3	Naphthalene	0.500	U	0.200	0.500	5.00
104-51-8	n-Butylbenzene	0.500	U	0.200	0.500	1.00
103-65-1	n-Propylbenzene	2.00	U	1.00	2.00	5.00
95-47-6	o-Xylene	0.500	U	0.200	0.500	1.00
135-98-8	sec-Butylbenzene	0.500	U	0.200	0.500	1.00
100-42-5	Styrene	0.500	U	0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	0.500	U	0.200	0.500	1.00
98-06-6	tert-Butylbenzene	0.500	U	0.200	0.500	1.00
127-18-4	Tetrachloroethene	0.500	U	0.200	0.500	1.00
108-88-3	Toluene	0.500	U	0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
79-01-6	Trichloroethene	0.500	U	0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	0.500	U	0.200	0.500	1.00
75-01-4	Vinyl chloride	0.500	U	0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv11.i/230602.s.b/j1694.d
Lab Smp Id: 2489345 Client Smp ID: MB
Inj Date : 02-JUN-2023 14:23
Operator : KN1 Inst ID: msv11.i
Smp Info : MB*MB
Misc Info : MSV~529~*1*KN1
Comment :
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Meth Date : 05-Jun-2023 08:22 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 15:02 Cal File: j1306.d
Als bottle: 1 QC Sample: BLANK
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260bDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
$ 41 Dibromofluoromethane	111	5.705	5.715	(0.877)	79932	59.4121	59.4	5353
$ 51 1,2-Dichloroethane-d4	65	6.224	6.229	(0.957)	104271	58.0389	58.0	
* 56 FLUOROBENZENE	96	6.502	6.507	(1.000)	269845	50.0000		
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	265518	52.2361	52.2	
* 90 CHLOROBENZENE-d5	82	9.238	9.239	(1.000)	106188	50.0000		
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	76701	50.0544	50.1	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	104077	50.0000		

Data File: /var/chem/msv11.i/230602.s.b/j1694.d

Page 1

Date : 02-JUN-2023 14:23

Client ID: MB

Instrument: msv11.i

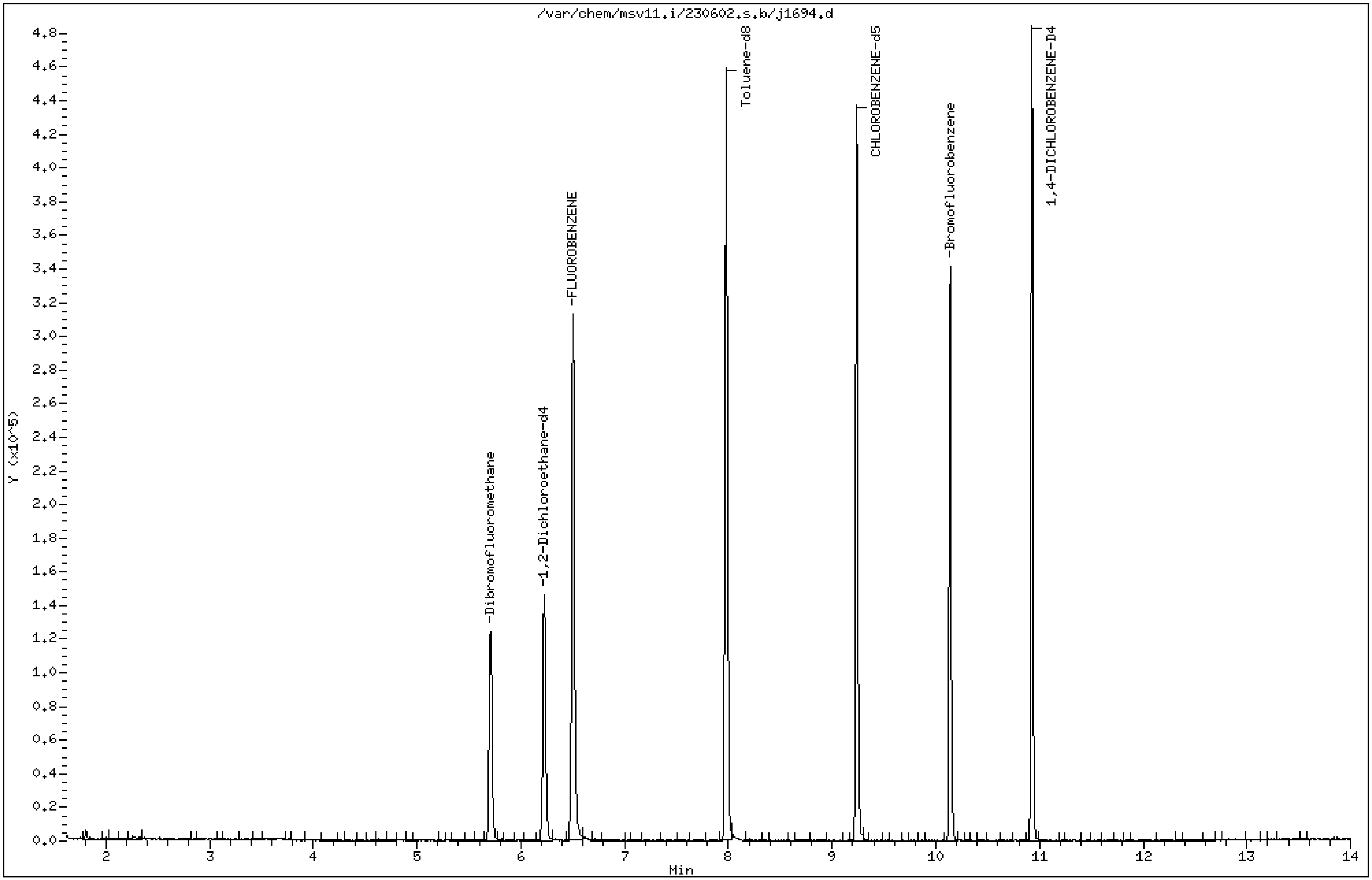
Sample Info: MB*MB

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

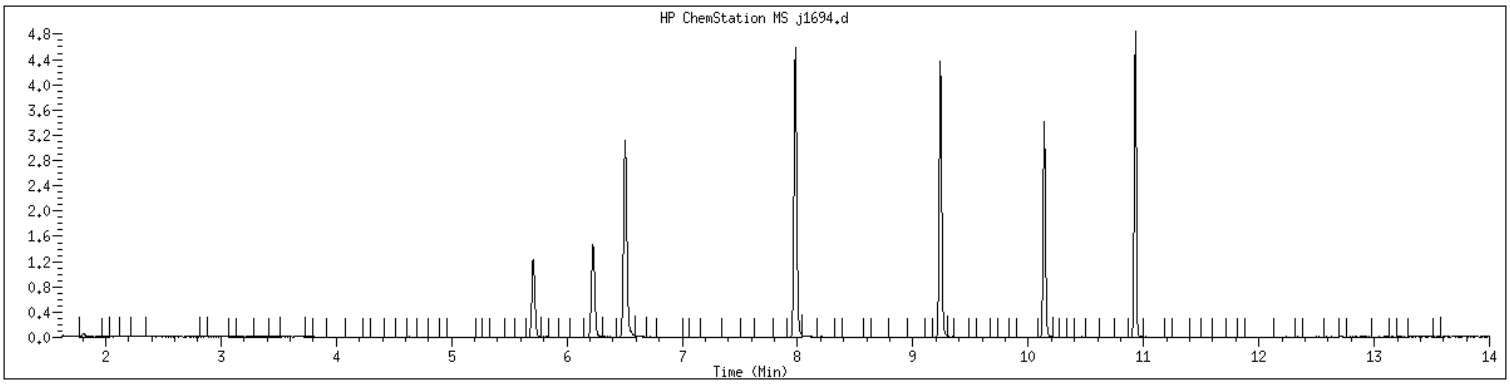
Column diameter: 0.25



Report Date: 06/05/2023 08:30

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID	: 2489345	SampleType	: BLANK
Injection Date	: 06/02/2023 14:23	Instrument	: msv11.i
Operator	: KN1		
Sample Info	: MB*MB		
Misc Info	: MSV~529~*1*KN1		
Method	: /var/chem/msv11.i/230602.s.b/8260bDODw11.m		
Dilution	: 1.00		
Matrix	: WATER		
Integrator	: HP RTE	Compound Sublist	: 8260bDOD



NO MANUAL INTEGRATIONS

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	LCS2489346		
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2489346		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11		
Sample Amt:	5	mL		Lab File ID:	230602/j1687L		
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942		
Analysis Date:	06/02/23	Time:	1123	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	50.2		0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	52.3		0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	45.6		0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	49.3		0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	50.1		0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	53.1		0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	52.0		0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	46.4		0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	43.4		0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	46.4		0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	50.4		0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	46.4		0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	49.6		0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	46.6		0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	53.1		0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	97.9		0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	49.7		0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	49.0		0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	46.3		0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	49.2		0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	46.3		0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	50.7		0.200	0.500	1.00
78-93-3	2-Butanone	217		0.200	0.500	5.00
95-49-8	2-Chlorotoluene	46.4		0.200	0.500	1.00
591-78-6	2-Hexanone	223		0.500	1.00	5.00
106-43-4	4-Chlorotoluene	47.4		0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	49.5		0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	254		0.200	0.500	5.00
67-64-1	Acetone	201		0.500	1.00	5.00
71-43-2	Benzene	49.3		0.200	0.500	1.00
108-86-1	Bromobenzene	45.8		0.200	0.500	1.00
74-97-5	Bromochloromethane	50.8		0.200	0.500	1.00
75-27-4	Bromodichloromethane	51.5		0.200	0.500	1.00
75-25-2	Bromoform	53.0		0.250	0.500	1.00
74-83-9	Bromomethane	40.2		0.500	1.00	2.00
75-15-0	Carbon disulfide	50.3		0.200	0.500	1.00
56-23-5	Carbon tetrachloride	52.6		0.250	0.500	1.00

FORM I VOA

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	LCS2489346
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2489346
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV11
Sample Amt:	5	mL		Lab File ID:	230602/j1687L
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	766942
Analysis Date:	06/02/23	Time:	1123	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	49.4		0.200	0.500	1.00
75-00-3	Chloroethane	47.4		0.250	0.500	1.00
67-66-3	Chloroform	51.8		0.200	0.500	1.00
74-87-3	Chloromethane	44.9		0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	49.6		0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	53.5		0.200	0.500	1.00
124-48-1	Dibromochloromethane	51.9		0.200	0.500	1.00
74-95-3	Dibromomethane	51.8		0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	41.9		0.200	0.500	1.00
100-41-4	Ethylbenzene	50.1		0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	49.2		0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	53.0		0.200	0.500	1.00
136777-61-2	m,p-Xylene	103		0.200	0.500	2.00
75-09-2	Methylene chloride	48.8		0.200	0.500	5.00
91-20-3	Naphthalene	43.0		0.200	0.500	5.00
104-51-8	n-Butylbenzene	46.2		0.200	0.500	1.00
103-65-1	n-Propylbenzene	48.1		1.00	2.00	5.00
95-47-6	o-Xylene	52.7		0.200	0.500	1.00
135-98-8	sec-Butylbenzene	49.6		0.200	0.500	1.00
100-42-5	Styrene	52.4		0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	50.5		0.200	0.500	1.00
98-06-6	tert-Butylbenzene	48.6		0.200	0.500	1.00
127-18-4	Tetrachloroethene	52.5		0.200	0.500	1.00
108-88-3	Toluene	50.3		0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	48.3		0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	51.3		0.200	0.500	1.00
79-01-6	Trichloroethene	52.1		0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	53.5		0.200	0.500	1.00
75-01-4	Vinyl chloride	48.5		0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv11.i/230602.s.b/j1687L.d
Lab Smp Id: 2489346 Client Smp ID: LCS
Inj Date : 02-JUN-2023 11:23
Operator : KN1 Inst ID: msv11.i
Smp Info : 1400*V11STD050*51364
Misc Info : MSV~529~*1*KN1
Comment :
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Meth Date : 05-Jun-2023 08:22 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 15:02 Cal File: j1306.d
Als bottle: 1 QC Sample: LCS
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260bDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	CONCENTRATIONS FINAL (ug/L)	SIMILARITY
1 Dichlorodifluoromethane	85	1.673	1.673	(0.257)	83739	41.8904	41.9	
2 Chloromethane ++	50	1.856	1.856	(0.285)	157538	44.8812	44.9	
3 Vinyl Chloride +	62	1.940	1.940	(0.298)	126068	48.5250	48.5	
4 1-3 Butadiene	39	1.956	1.956	(0.301)	151875	53.3135	53.3	9764
5 Bromomethane	94	2.250	2.250	(0.346)	43008	40.1783	40.2	
7 Chloroethane	64	2.370	2.370	(0.364)	68736	47.4455	47.4	
8 Trichlorofluoromethane	101	2.522	2.522	(0.388)	153108	53.5490	53.5	
11 1,1-Dichloroethene +	96	3.067	3.067	(0.471)	78495	53.1414	53.1	
12 Carbon Disulfide	76	3.099	3.099	(0.476)	226180	50.3252	50.3	
13 1,1,2Trichlotrifluoroethane	101	3.120	3.120	(0.479)	82330	53.4824	53.5	
15 Methyl Iodide	142	3.225	3.225	(0.496)	50963	28.9597	29.0	
16 Acrolein	56	3.450	3.450	(0.530)	126457	255.412	255	
17 Allyl chloride	39	3.628	3.628	(0.558)	136132	52.9676	53.0	9325
18 Methylene Chloride	49	3.760	3.760	(0.578)	180222	48.7522	48.8	

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
	19 Acetone	43	3.812	3.812	(0.586)	445972	201.068	201	
	20 trans-1,2-Dichloroethene	61	3.959	3.959	(0.608)	159712	48.2802	48.3	(R)
	21 Methyl Acetate	43	3.974	3.974	(0.611)	618261	257.262	257	9606(R)
	22 Hexane	57	4.069	4.069	(0.625)	250936	51.3520	51.4	9410
	23 MTBE	73	4.106	4.106	(0.631)	263951	50.4987	50.5	9140
	24 tert-Butyl Alcohol	59	4.226	4.226	(0.649)	81239	249.346	249	8844
	25 Acetonitrile	41	4.342	4.342	(0.667)	99344	232.041	232	9806
	26 Isopropyl Ether	45	4.551	4.551	(0.699)	442237	49.9248	49.9	9647
	28 1,1-Dichloroethane ++	63	4.656	4.656	(0.716)	209728	50.0707	50.1	
	29 Acrylonitrile	53	4.698	4.698	(0.722)	295617	258.712	259	
	31 Vinyl Acetate	43	4.944	4.944	(0.760)	260670	51.6484	51.6	
	30 Ethyl Tert-butyl Ether	59	4.950	4.950	(0.761)	352245	48.5184	48.5	7839
	32 cis-1,2-Dichloroethene	61	5.238	5.238	(0.805)	154742	49.5709	49.6	
M	33 Total 1,2-Dichloroethene	61				314454	97.8511	97.9	
	34 2,2-Dichloropropane	77	5.348	5.348	(0.822)	133827	50.6583	50.7	
	35 Bromochloromethane	128	5.448	5.448	(0.837)	46715	50.7675	50.8	
	36 Cyclohexane	56	5.453	5.453	(0.838)	213927	53.2493	53.2	9019
	37 Chloroform +	83	5.526	5.526	(0.849)	163142	51.8317	51.8	
	38 Carbon Tetrachloride	117	5.668	5.668	(0.871)	132179	52.6367	52.6	
$	41 Dibromofluoromethane	111	5.715	5.715	(0.878)	94035	56.3831	56.4	9339
	42 1,1,1-Trichloroethane	97	5.731	5.731	(0.881)	149463	52.3195	52.3	
	44 2-Butanone	43	5.825	5.825	(0.895)	424040	217.079	217	
	45 1,1-Dichloropropene	75	5.857	5.857	(0.900)	124609	51.9913	52.0	
	47 2,2,4 Trimethylpentane	57	5.977	5.977	(0.919)	405991	40.4656	40.5	8899
	48 Benzene	78	6.098	6.098	(0.937)	347695	49.2662	49.3	
$	51 1,2-Dichloroethane-d4	65	6.229	6.229	(0.957)	122522	55.0142	55.0	
	54 tert-amyl Methyl Ether	73	6.229	6.229	(0.957)	244719	48.1381	48.1	7299
	52 1,2-Dichloroethane	62	6.292	6.292	(0.967)	159754	53.0791	53.1	
	53 Isobutyl Alcohol	43	6.339	6.339	(0.974)	25297	196.261	196	9240
*	56 FLUOROBENZENE	96	6.507	6.507	(1.000)	334510	50.0000		
	57 Methyl Cyclohexane	83	6.669	6.669	(1.025)	136021	51.9352	51.9	9372
	58 Trichloroethene	130	6.675	6.675	(1.026)	105566	52.1070	52.1	
	60 Dibromomethane	93	7.073	7.073	(1.087)	60206	51.7592	51.8	
	62 1,2-Dichloropropane +	63	7.157	7.157	(1.100)	122214	49.6922	49.7	
	63 Bromodichloromethane	83	7.230	7.230	(1.111)	122405	51.5066	51.5	
	65 1,4- Dioxane	58	7.414	7.414	(1.139)	24353	1217.61	1220	9140
	66 1-Bromo-2-chloroethane	63	7.671	7.671	(1.179)	162384	50.4254	50.4	9187
	67 2-Chloroethyl vinyl ether	63	7.760	7.760	(1.193)	30411	104.365	104	(R)
	68 cis-1,3-Dichloropropene	75	7.812	7.812	(1.201)	147219	53.4762	53.5	
$	69 Toluene-d8	98	7.980	7.980	(0.864)	323479	51.3323	51.3	
	70 Toluene +	91	8.022	8.022	(0.868)	380236	50.2780	50.3	
M	72 1-3 Dichloropropene total	100				279645	104.822	105	0
	74 4-methyl-2-pentanone	43	8.337	8.337	(0.902)	749280	254.236	254	
	73 Tetrachloroethene	164	8.358	8.358	(0.905)	80514	52.5467	52.5	
	75 trans-1,3-Dichloropropene	75	8.373	8.373	(1.287)	132426	51.3456	51.3	
	78 1,1,2-Trichloroethane	97	8.504	8.504	(0.921)	84321	49.2766	49.3	
	79 Dibromochloromethane	129	8.657	8.657	(0.937)	95941	51.8812	51.9	

		QUANT SIG					CONCENTRATIONS ON-COLUMN	FINAL	
	Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ug/L)	SIMILARITY
	80 1,3-Dichloropropane	76	8.725	8.725	(0.944)	142205	49.2318	49.2	
	82 1,2-Dibromoethane(EDB)	107	8.840	8.840	(0.957)	87285	49.5597	49.6	
	83 2-Hexanone	43	9.008	9.008	(0.975)	548043	223.100	223	
	89 1-Chlorohexane	91	9.233	9.233	(0.999)	111898	52.0929	52.1	9523
*	90 CHLOROBENZENE-d5	82	9.239	9.239	(1.000)	131646	50.0000		
	91 Chlorobenzene ++	112	9.254	9.254	(1.002)	242112	49.3843	49.4	
	92 Ethylbenzene +	106	9.270	9.270	(1.003)	129832	50.1384	50.1	
	93 1,1,1,2-Tetrachloroethane	133	9.301	9.301	(1.007)	84111	50.2106	50.2	
	94 p,m-Xylene	106	9.380	9.380	(1.015)	319624	102.745	103	
M	95 TOTAL XYLENE	106				471750	155.428	155	
	96 o-Xylene	106	9.700	9.700	(1.050)	152126	52.6836	52.7	
	97 Styrene	104	9.737	9.737	(1.054)	215482	52.4237	52.4	
	98 Bromoform ++	173	9.768	9.768	(1.057)	64430	52.9608	53.0	
	99 Isopropylbenzene	105	9.931	9.931	(1.075)	381095	52.9750	53.0	
$	101 Bromofluorobenzene	174	10.140	10.140	(1.098)	100853	53.0882	53.1	
	102 cis-1,4-dichloro-2-butene	53	10.172	10.172	(0.931)	46220	40.0732	40.1	8847
	103 Bromobenzene	77	10.214	10.214	(0.935)	170045	45.8452	45.8	
	104 n-Propylbenzene	91	10.224	10.224	(0.936)	443615	48.0731	48.1	
	105 1,1,2,2-Tetrachloroethane++	83	10.277	10.277	(0.941)	109061	45.6221	45.6	
	106 2-Chlorotoluene	91	10.345	10.345	(0.947)	301612	46.4011	46.4	
	107 1,3,5-Trimethylbenzene	105	10.361	10.361	(0.948)	338691	48.9941	49.0	
	108 1,2,3-Trichloropropane	75	10.376	10.376	(0.950)	134843	43.3757	43.4	
	109 trans-1,4-Dichloro-2-Butene	53	10.397	10.397	(0.952)	44298	39.7865	39.8	
	100 Cyclohexanone	55	10.402	10.402	(0.952)	127020	156.132	156	5733
	110 4-Chlorotoluene	91	10.460	10.460	(0.957)	293144	47.4049	47.4	
	111 tert-butylbenzene	91	10.586	10.586	(0.969)	183438	48.6303	48.6	
	112 1,2,4-Trimethylbenzene	105	10.638	10.638	(0.974)	350528	50.4499	50.4	
	113 sec-Butylbenzene	105	10.712	10.712	(0.980)	397324	49.5927	49.6	
	115 p-Isopropyltoluene	119	10.806	10.806	(0.989)	367593	49.4874	49.5	
	116 1,3-Dichlorobenzene	146	10.874	10.874	(0.995)	197183	46.3147	46.3	
*	117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	134139	50.0000		
	118 1,4-Dichlorobenzene	146	10.937	10.937	(1.001)	198644	46.3136	46.3	
	119 n-Butylbenzene	91	11.100	11.100	(1.016)	495299	46.1916	46.2	
	120 1,2-Dichlorobenzene	146	11.226	11.226	(1.027)	191970	46.5764	46.6	
	122 1,2-Dibromo-3-Chloropropane	157	11.771	11.771	(1.077)	26471	46.4304	46.4	
	123 Hexachlorobutadiene	225	12.201	12.201	(1.117)	56343	49.2460	49.2	
	124 1,2,4-Trichlorobenzene	180	12.227	12.227	(1.119)	117740	46.4172	46.4	
	125 Naphthalene	128	12.447	12.447	(1.139)	324673	42.9768	43.0	
	126 1,2,3-Trichlorobenzene	180	12.573	12.573	(1.151)	114282	46.3731	46.4	

QC Flag Legend

R - Spike/Surrogate failed recovery limits.

Data File: /var/chem/msv11.i/230602.s.b/j1687L.d

Page 1

Date : 02-JUN-2023 11:23

Client ID: LCS

Instrument: msv11.i

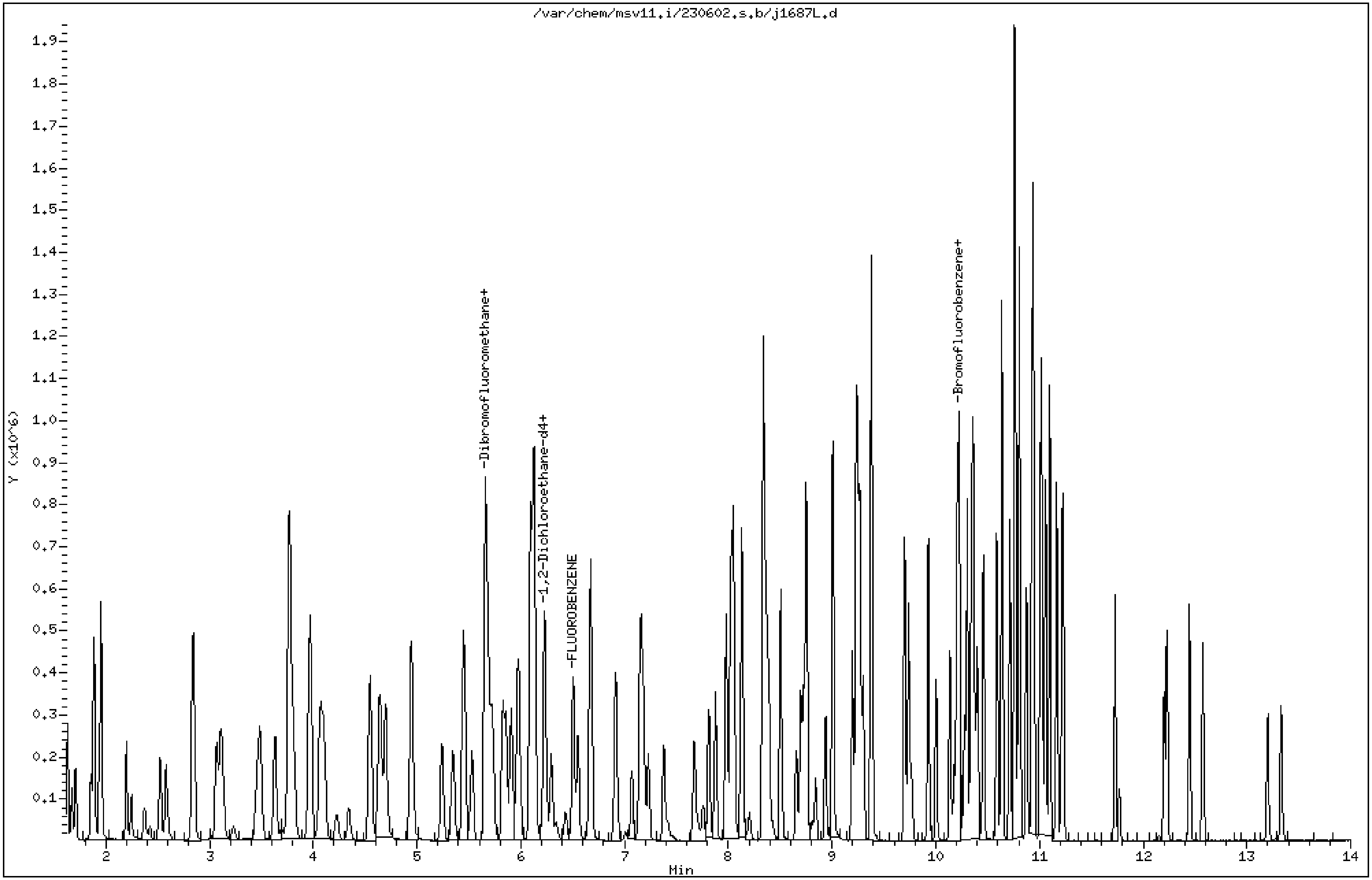
Sample Info: 1400*V11STD050*51364

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

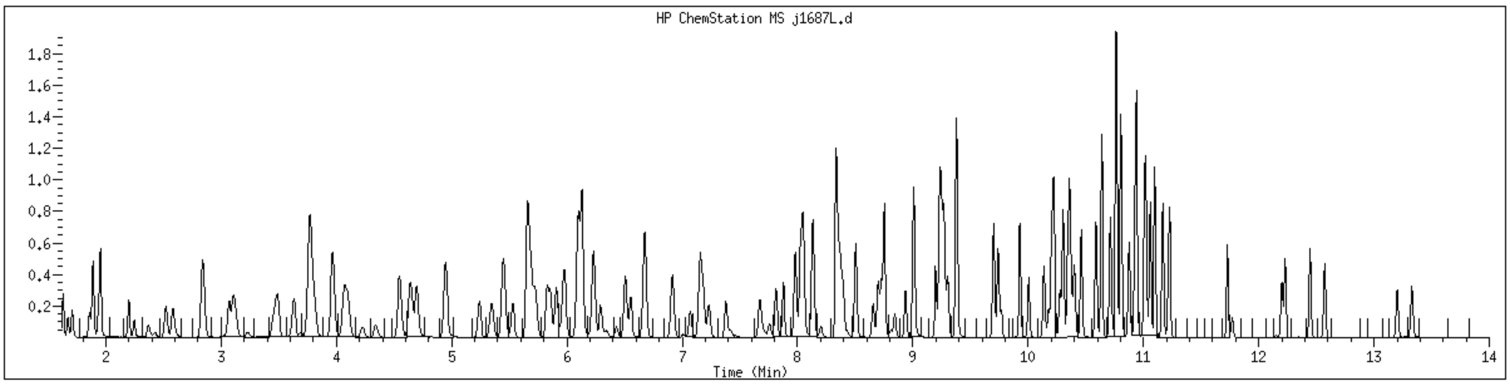
Column diameter: 0.25



Report Date: 06/05/2023 08:22

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 2489346 SampleType : LCS
Injection Date: 06/02/2023 11:23 Instrument : msv11.i
Operator : KN1
Sample Info : 1400*V11STD050*51364
Misc Info : MSV~529~*1*KN1
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260bDOD



NO MANUAL INTEGRATIONS

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	MB2489807
Collect Date:	NA Time: NA	GCAL Sample ID:	2489807
Matrix:	Water % Moisture: NA	Instrument ID:	MSV15
Sample Amt:	5 mL	Lab File ID:	230605/c0737
Injection Vol.:	1 (µL)	GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1 Analyst: KN1	Analytical Batch:	767028
Analysis Date:	06/05/23 Time: 1257	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	***ANALYTE***	***RESULT***	***Q***	***DL***	***LOD***	***LOQ***
630-20-6	1,1,1,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	0.500	U	0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	0.500	U	0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	0.500	U	0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	0.500	U	0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	0.500	U	0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	0.500	U	0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	0.500	U	0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	0.500	U	0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	0.500	U	0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	0.500	U	0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	0.500	U	0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	0.500	U	0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	0.500	U	0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	1.00	U	0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	0.500	U	0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	0.500	U	0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	0.500	U	0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	0.500	U	0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	0.500	U	0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	0.500	U	0.200	0.500	1.00
78-93-3	2-Butanone	0.500	U	0.200	0.500	5.00
95-49-8	2-Chlorotoluene	0.500	U	0.200	0.500	1.00
591-78-6	2-Hexanone	1.00	U	0.500	1.00	5.00
106-43-4	4-Chlorotoluene	0.500	U	0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	0.500	U	0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	0.500	U	0.200	0.500	5.00
67-64-1	Acetone	1.00	U	0.500	1.00	5.00
71-43-2	Benzene	0.500	U	0.200	0.500	1.00
108-86-1	Bromobenzene	0.500	U	0.200	0.500	1.00
74-97-5	Bromochloromethane	0.500	U	0.200	0.500	1.00
75-27-4	Bromodichloromethane	0.500	U	0.200	0.500	1.00
75-25-2	Bromoform	0.500	U	0.250	0.500	1.00
74-83-9	Bromomethane	1.00	U	0.500	1.00	2.00
75-15-0	Carbon disulfide	0.500	U	0.200	0.500	1.00
56-23-5	Carbon tetrachloride	0.500	U	0.250	0.500	1.00

FORM I VOA

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	MB2489807		
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2489807		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV15		
Sample Amt:	5	mL		Lab File ID:	230605/c0737		
Injection Vol.:	1		(µL)	GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	767028		
Analysis Date:	06/05/23	Time:	1257	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	0.500	U	0.200	0.500	1.00
75-00-3	Chloroethane	0.500	U	0.250	0.500	1.00
67-66-3	Chloroform	0.500	U	0.200	0.500	1.00
74-87-3	Chloromethane	0.500	U	0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
124-48-1	Dibromochloromethane	0.500	U	0.200	0.500	1.00
74-95-3	Dibromomethane	0.500	U	0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	0.500	U	0.200	0.500	1.00
100-41-4	Ethylbenzene	0.500	U	0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	1.00	U	0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	0.500	U	0.200	0.500	1.00
136777-61-2	m,p-Xylene	0.500	U	0.200	0.500	2.00
75-09-2	Methylene chloride	0.500	U	0.200	0.500	5.00
91-20-3	Naphthalene	0.500	U	0.200	0.500	5.00
104-51-8	n-Butylbenzene	0.500	U	0.200	0.500	1.00
103-65-1	n-Propylbenzene	2.00	U	1.00	2.00	5.00
95-47-6	o-Xylene	0.500	U	0.200	0.500	1.00
135-98-8	sec-Butylbenzene	0.500	U	0.200	0.500	1.00
100-42-5	Styrene	0.500	U	0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	0.500	U	0.200	0.500	1.00
98-06-6	tert-Butylbenzene	0.500	U	0.200	0.500	1.00
127-18-4	Tetrachloroethene	0.500	U	0.200	0.500	1.00
108-88-3	Toluene	0.500	U	0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
79-01-6	Trichloroethene	0.500	U	0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	0.500	U	0.200	0.500	1.00
75-01-4	Vinyl chloride	0.500	U	0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230605.s.b/c0737.d
Lab Smp Id: 2489807 Client Smp ID: MB
Inj Date : 05-JUN-2023 12:57
Operator : KN1 Inst ID: msv15.i
Smp Info : MB*MB
Misc Info : MSV~52919~*1*SMR
Comment :
Method : /var/chem/msv15.i/230605.s.b/8260bw15DOD.m
Meth Date : 06-Jun-2023 12:20 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 QC Sample: BLANK
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
$ 42 Dibromofluoromethane	111	5.159	5.160	(0.877)	86599	51.8594	51.9	6889
$ 50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	91538	52.6466	52.6	9742
* 55 FLUOROBENZENE	96	5.883	5.883	(1.000)	338140	50.0000		9837
$ 74 Toluene-d8	98	7.224	7.224	(0.866)	302516	46.9043	46.9	9799
* 91 CHLOROBENZENE-d5	82	8.339	8.339	(1.000)	137619	50.0000		9858
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	99980	47.7319	47.7	9391
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.806	(1.000)	129255	50.0000		9567

Data File: /var/chem/msv15.i/230605.s.b/c0737.d

Page 1

Date : 05-JUN-2023 12:57

Client ID: MB

Instrument: msv15.i

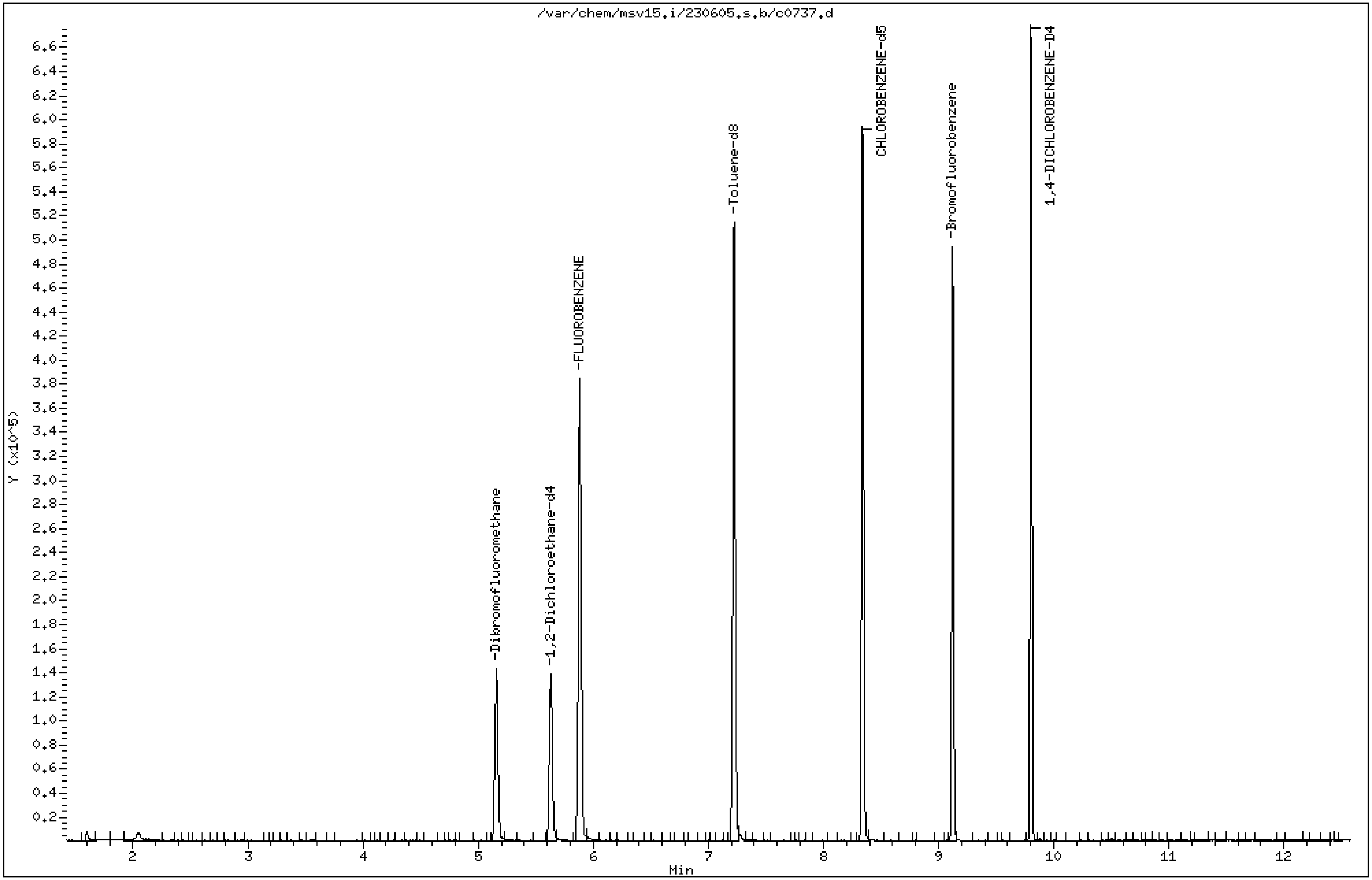
Sample Info: MB*MB

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

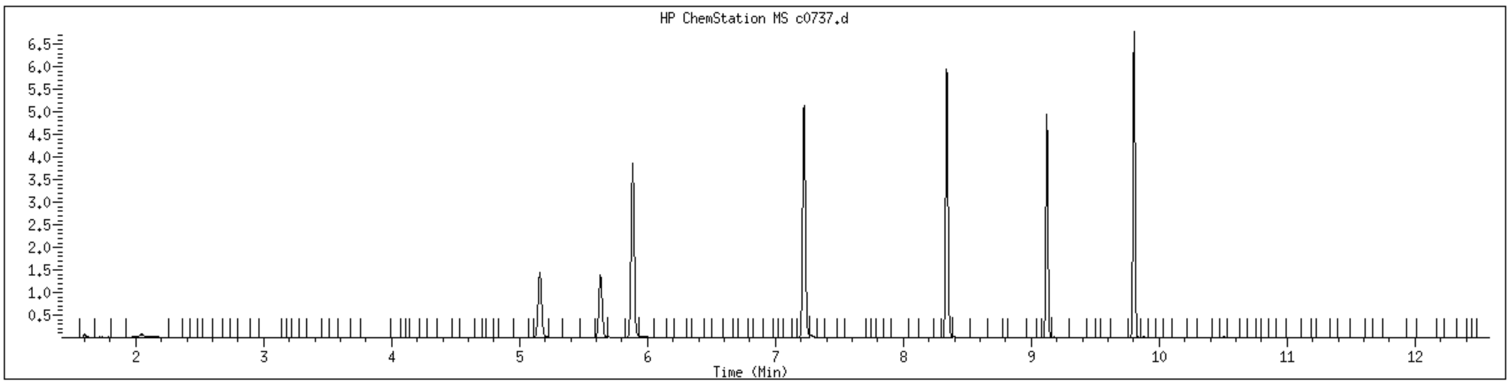
Column diameter: 0.25



Data file : /var/chem/msv15.i/230605.s.b/c0737.d Page: 1
Report Date: 06/06/2023 12:20

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 2489807 SampleType : BLANK
Injection Date: 06/05/2023 12:57 Instrument : msv15.i
Operator : KN1
Sample Info : MB*MB
Misc Info : MSV~52919~*1*SMR
Method : /var/chem/msv15.i/230605.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	LCS2489808
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2489808
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV15
Sample Amt:	5	mL		Lab File ID:	230605/c0730L
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	767028
Analysis Date:	06/05/23	Time:	1045	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	51.6		0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	51.7		0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	50.6		0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	51.9		0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	53.6		0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	53.6		0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	52.8		0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	50.7		0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	47.6		0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	51.4		0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	52.9		0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	41.8		0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	52.2		0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	51.7		0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	50.1		0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	107		0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	53.7		0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	53.1		0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	51.4		0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	51.4		0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	51.2		0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	48.2		0.200	0.500	1.00
78-93-3	2-Butanone	236		0.200	0.500	5.00
95-49-8	2-Chlorotoluene	52.1		0.200	0.500	1.00
591-78-6	2-Hexanone	227		0.500	1.00	5.00
106-43-4	4-Chlorotoluene	52.2		0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	53.1		0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	221		0.200	0.500	5.00
67-64-1	Acetone	234		0.500	1.00	5.00
71-43-2	Benzene	53.4		0.200	0.500	1.00
108-86-1	Bromobenzene	53.2		0.200	0.500	1.00
74-97-5	Bromochloromethane	52.5		0.200	0.500	1.00
75-27-4	Bromodichloromethane	52.8		0.200	0.500	1.00
75-25-2	Bromoform	45.5		0.250	0.500	1.00
74-83-9	Bromomethane	38.7		0.500	1.00	2.00
75-15-0	Carbon disulfide	53.3		0.200	0.500	1.00
56-23-5	Carbon tetrachloride	51.7		0.250	0.500	1.00

FORM I VOA

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210	Client Sample ID:	LCS2489808
Collect Date:	NA Time: NA	GCAL Sample ID:	2489808
Matrix:	Water % Moisture: NA	Instrument ID:	MSV15
Sample Amt:	5 mL	Lab File ID:	230605/c0730L
Injection Vol.:	1 (µL)	GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1 Analyst: KN1	Analytical Batch:	767028
Analysis Date:	06/05/23 Time: 1045	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	52.9		0.200	0.500	1.00
75-00-3	Chloroethane	52.8		0.250	0.500	1.00
67-66-3	Chloroform	51.6		0.200	0.500	1.00
74-87-3	Chloromethane	57.7		0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	52.7		0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	52.4		0.200	0.500	1.00
124-48-1	Dibromochloromethane	54.5		0.200	0.500	1.00
74-95-3	Dibromomethane	52.0		0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	60.8		0.200	0.500	1.00
100-41-4	Ethylbenzene	53.3		0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	51.8		0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	54.1		0.200	0.500	1.00
136777-61-2	m,p-Xylene	108		0.200	0.500	2.00
75-09-2	Methylene chloride	52.3		0.200	0.500	5.00
91-20-3	Naphthalene	48.2		0.200	0.500	5.00
104-51-8	n-Butylbenzene	52.5		0.200	0.500	1.00
103-65-1	n-Propylbenzene	52.3		1.00	2.00	5.00
95-47-6	o-Xylene	53.1		0.200	0.500	1.00
135-98-8	sec-Butylbenzene	53.2		0.200	0.500	1.00
100-42-5	Styrene	57.4		0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	45.6		0.200	0.500	1.00
98-06-6	tert-Butylbenzene	50.8		0.200	0.500	1.00
127-18-4	Tetrachloroethene	52.7		0.200	0.500	1.00
108-88-3	Toluene	53.1		0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	54.1		0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	51.0		0.200	0.500	1.00
79-01-6	Trichloroethene	52.8		0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	51.5		0.200	0.500	1.00
75-01-4	Vinyl chloride	54.7		0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230605.s.b/c0730L.d
Lab Smp Id: 2489808 Client Smp ID: LCS
Inj Date : 05-JUN-2023 10:45
Operator : KN1 Inst ID: msv15.i
Smp Info : 1400*V15STD050*51363
Misc Info : MSV~529~*1*SMR
Comment :
Method : /var/chem/msv15.i/230605.s.b/8260bw15DOD.m
Meth Date : 06-Jun-2023 12:20 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 QC Sample: LCS
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	91824	60.8001	60.8	9745
2 Chloromethane ++	50	1.652	1.652	(0.281)	78421	57.6564	57.7	9692
3 Vinyl Chloride +	62	1.719	1.719	(0.292)	104686	54.6592	54.7	9831
5 1-3 Butadiene	39	1.735	1.735	(0.295)	77137	54.0825	54.1	9671
6 Bromomethane	94	2.018	2.018	(0.343)	62158	38.7009	38.7	9792
7 Chloroethane	64	2.137	2.137	(0.363)	74395	52.8445	52.8	9242
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	143877	51.5336	51.5	9634
11 1,1-Dichloroethene +	96	2.783	2.783	(0.473)	85090	53.5681	53.6	8253
12 Carbon Disulfide	76	2.816	2.816	(0.479)	259083	53.3153	53.3	8014
13 1,1,2Trichlotrifluoroethane	101	2.835	2.835	(0.482)	93790	54.4104	54.4	9245
15 Methyl Iodide	142	2.938	2.938	(0.499)	69467	36.9988	37.0	9502
16 Acrolein	56	3.150	3.150	(0.535)	60475	250.017	250	5646
17 Allyl chloride	39	3.304	3.304	(0.562)	77882	55.7831	55.8	9774
18 Methylene Chloride	49	3.417	3.417	(0.581)	91875	52.2672	52.3	4916

							CONCENTRATIONS		
	QUANT SIG					ON-COLUMN	FINAL	
Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ug/L)	SIMILARITY
==========================	====	==	======	======	========	=======	=======	==========
19 Acetone	43	3.468	3.468	(0.590)	169081	234.355	234	5474(H)
20 trans-1,2-Dichloroethene	61	3.590	3.590	(0.610)	119061	54.0663	54.1	8707
21 Methyl Acetate	43	3.610	3.610	(0.614)	221306	246.394	246	9331(R)
22 Hexane	57	3.687	3.687	(0.627)	173341	54.6354	54.6	9243
23 MTBE	73	3.719	3.719	(0.632)	196003	45.6366	45.6	9656
24 tert-Butyl Alcohol	59	3.822	3.822	(0.650)	33518	176.917	177	8137
25 Acetonitrile	41	3.941	3.941	(0.670)	33700	263.830	264	9656
26 Isopropyl Ether	45	4.115	4.115	(0.699)	205589	53.6510	53.7	9666
27 Chloroprene	53	4.198	4.198	(0.714)	117325	54.3552	54.4	8460
28 1,1-Dichloroethane ++	63	4.214	4.214	(0.716)	157495	53.5596	53.6	9452
29 Acrylonitrile	53	4.259	4.259	(0.724)	127103	253.486	253	9870
31 Ethyl Tert-butyl Ether	59	4.471	4.471	(0.760)	208514	48.3487	48.3	8005
30 Vinyl Acetate	43	4.471	4.471	(0.760)	118793	50.0945	50.1	5687(R)
32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	111406	52.6989	52.7	9806
M 33 Total 1,2-Dichloroethene	61				230467	106.765	107	0
34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	125968	48.1958	48.2	9601
35 Bromochloromethane	128	4.922	4.922	(0.837)	43574	52.5014	52.5	9086
36 Cyclohexane	56	4.928	4.928	(0.838)	149114	55.6097	55.6	9427
37 Chloroform +	83	4.992	4.992	(0.849)	155217	51.6001	51.6	9808
39 Carbon Tetrachloride	117	5.118	5.118	(0.870)	125656	51.6743	51.7	8329
$ 42 Dibromofluoromethane	111	5.160	5.160	(0.877)	84111	52.3529	52.4	9018
43 1,1,1-Trichloroethane	97	5.179	5.179	(0.880)	143990	51.6907	51.7	9575
45 1,1-Dichloropropene	75	5.291	5.291	(0.899)	130836	52.8296	52.8	9557
44 2-Butanone	43	5.275	5.275	(0.897)	176440	235.616	236	6453
46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	291478	38.9714	39.0	9635
48 Benzene	78	5.513	5.513	(0.937)	357616	53.4227	53.4	9442
56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	212759	48.4131	48.4	7397
$ 50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	82274	49.1819	49.2	8607
52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	105474	50.0876	50.1	9779
51 Isobutyl Alcohol	43	5.722	5.722	(0.973)	9129	208.391	208	4969
* 55 FLUOROBENZENE	96	5.883	5.883	(1.000)	325329	50.0000		9863
58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	172487	54.8366	54.8	8332
57 Trichloroethene	130	6.031	6.031	(1.025)	108639	52.8419	52.8	8786
62 Dibromomethane	93	6.394	6.394	(1.087)	55972	51.9619	52.0	9463
64 1,2-Dichloropropane +	63	6.481	6.481	(1.102)	88852	53.7044	53.7	7912
65 Bromodichloromethane	83	6.539	6.539	(1.111)	118295	52.7859	52.8	9776
67 1,4- Dioxane	88	6.722	6.722	(1.143)	13697	1116.52	1120	9424
68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	113964	54.0679	54.1	9009
72 2-Chloroethyl vinyl ether	63	7.021	7.021	(1.193)	16128	38.0438	38.0	9586(R)
73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	139077	52.4171	52.4	9797
$ 74 Toluene-d8	98	7.224	7.224	(0.866)	314205	53.3186	53.3	9891
75 Toluene +	91	7.262	7.262	(0.871)	389395	53.0732	53.1	9615
77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	300603	220.847	221	8500
78 Tetrachloroethene	164	7.558	7.558	(0.906)	87386	52.6812	52.7	8464
79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	128191	50.9733	51.0	8720
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	77328	51.8904	51.9	9562
82 Dibromochloromethane	129	7.825	7.825	(0.938)	92418	54.5230	54.5	9592

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	104865	53.5152	53.5	9745
86 1,3-Dichloropropane	76	7.893	7.893	(0.946)	127652	51.4475	51.4	9604
M 83 1-3 Dichloropropene total	100				267268	103.390	103	0
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	83209	52.1819	52.2	9762
89 2-Hexanone	43	8.143	8.143	(0.976)	253816	226.764	227	9805
90 1-Chlorohexane	91	8.333	8.333	(0.999)	135078	51.5011	51.5	8017
* 91 CHLOROBENZENE-d5	82	8.339	8.339	(1.000)	125741	50.0000		9184
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	256874	52.9007	52.9	9333
93 Ethylbenzene +	106	8.368	8.368	(1.003)	140734	53.3034	53.3	9156
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	84143	51.5710	51.6	9238
95 p,m-Xylene	106	8.465	8.465	(1.015)	344380	107.887	108	9889
97 o-Xylene	106	8.745	8.745	(1.049)	157832	53.0988	53.1	9910
101 Styrene	104	8.777	8.777	(1.052)	219622	57.3824	57.4	9862
102 Bromoform ++	173	8.799	8.799	(1.055)	62190	45.5110	45.5	8706
104 Isopropylbenzene	105	8.944	8.944	(1.072)	420730	54.0756	54.1	9892
M 105 TOTAL XYLENE	106				502212	160.986	161	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	96349	50.3436	50.3	9465
107 cis-1,4-dichloro-2-butene	53	9.159	9.159	(0.934)	30588	49.8997	49.9	9592
109 n-Propylbenzene	91	9.201	9.201	(0.938)	492700	52.2698	52.3	8863
108 Bromobenzene	77	9.195	9.195	(0.938)	166908	53.2066	53.2	9656
110 1,1,2,2-Tetrachloroethane++	83	9.243	9.243	(0.943)	93250	50.6376	50.6	9743
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	313031	52.0983	52.1	8159
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	336932	53.1435	53.1	8424
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.951)	118843	47.5576	47.6	7276
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.953)	27971	46.0107	46.0	9472
116 Cyclohexanone	55	9.365	9.365	(0.955)	68311	224.077	224	9727
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	281322	52.1750	52.2	9778
118 tert-butylbenzene	91	9.516	9.516	(0.970)	180866	50.8101	50.8	9879
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.974)	329478	52.9476	52.9	9318
120 sec-Butylbenzene	105	9.622	9.622	(0.981)	435977	53.1595	53.2	9845
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	385365	53.0778	53.1	8874
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.995)	188620	51.4159	51.4	9685
125 1,4-Dichlorobenzene	146	9.815	9.815	(1.001)	193100	51.1776	51.2	9047
* 124 1,4-DICHLOROBENZENE-D4	152	9.806	9.806	(1.000)	120697	50.0000		9022
126 n-Butylbenzene	91	9.954	9.954	(1.015)	507004	52.5185	52.5	8983
127 1,2-Dichlorobenzene	146	10.066	10.066	(1.027)	171513	51.6808	51.7	9721
129 1,2-Dibromo-3-Chloropropane	157	10.558	10.558	(1.077)	19380	41.8172	41.8	9387
130 Hexachlorobutadiene	225	10.999	10.999	(1.122)	52502	51.7507	51.8	9233
131 1,2,4-Trichlorobenzene	180	11.031	11.031	(1.125)	104204	51.3671	51.4	9488
132 Naphthalene	128	11.291	11.291	(1.151)	229011	48.1759	48.2	9850
133 1,2,3-Trichlorobenzene	180	11.445	11.445	(1.167)	88675	50.6756	50.7	9473

QC Flag Legend

R - Spike/Surrogate failed recovery limits.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230605.s.b/c0730L.d

Page 1

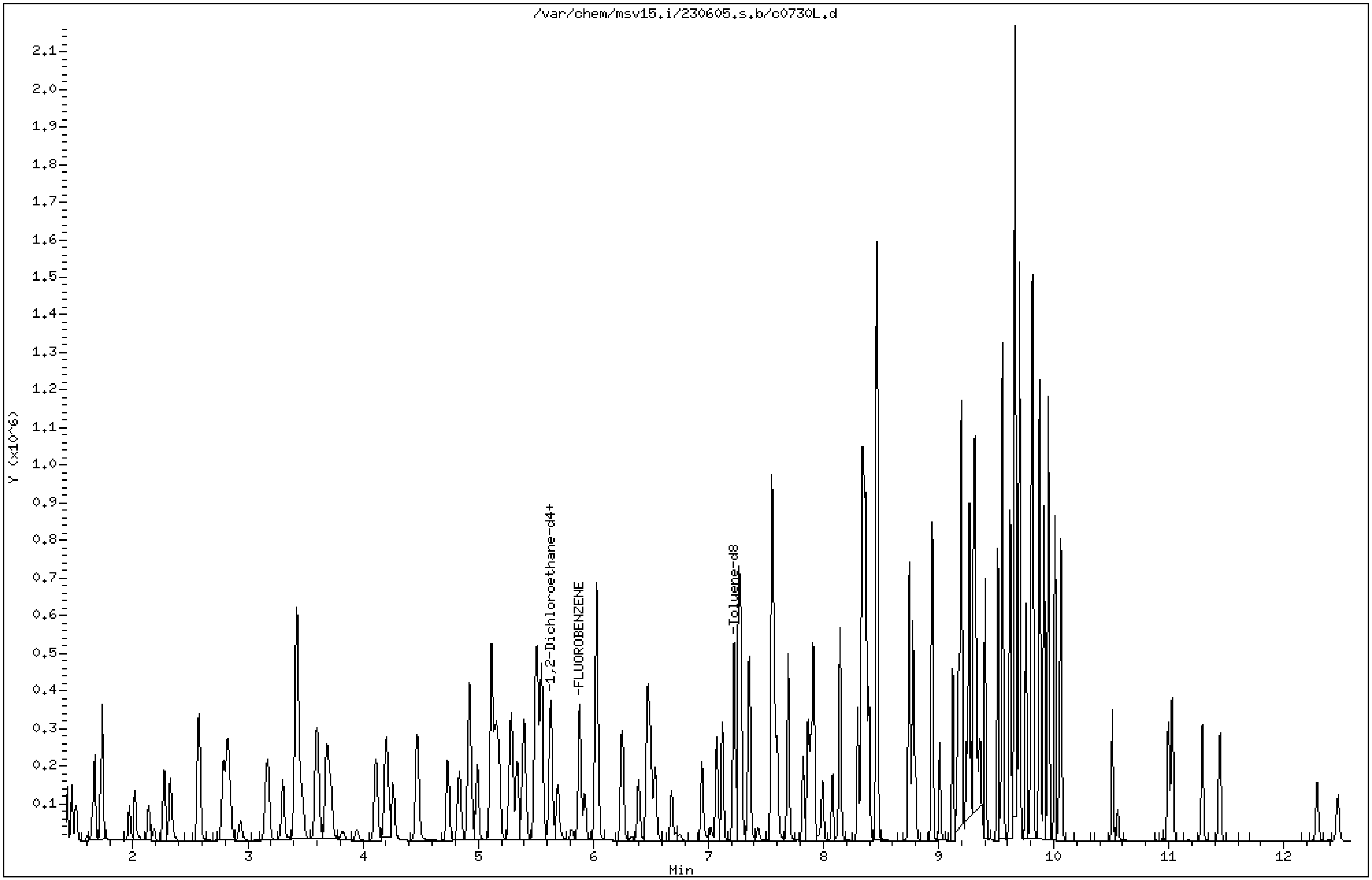
Date : 05-JUN-2023 10:45

Client ID: LCS | Instrument: msv15.i

Sample Info: 1400*V15STD050*51363

Purge Volume: 5.0 | Operator: KN1

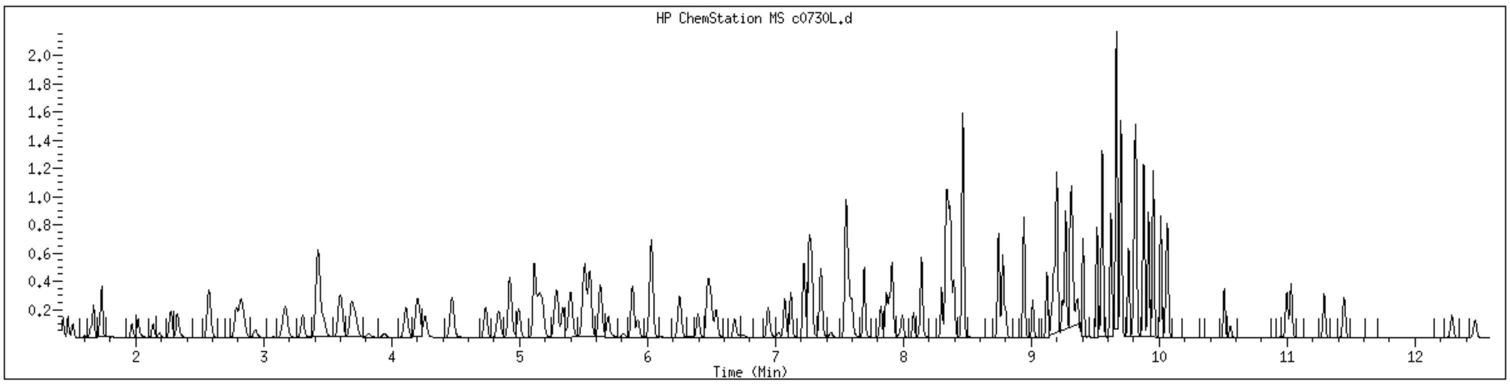
Column phase: RTX-VMS-30M | Column diameter: 0.25



Report Date: 06/06/2023 12:20

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 2489808 SampleType : LCS
Injection Date: 06/05/2023 10:45 Instrument : msv15.i
Operator : KN1
Sample Info : 1400*V15STD050*51363
Misc Info : MSV~529~*1*SMR
Method : /var/chem/msv15.i/230605.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	MB2489978		
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2489978		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV15		
Sample Amt:	5	mL		Lab File ID:	230606/c0774d		
Injection Vol.:	1		(µL)	GC Column:	RTX-VMS-30M	ID	.25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	767081		
Analysis Date:	06/06/23	Time:	1246	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	***ANALYTE***	***RESULT***	***Q***	***DL***	***LOD***	***LOQ***
630-20-6	1,1,1,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	0.500	U	0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	0.500	U	0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	0.500	U	0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	0.500	U	0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	0.500	U	0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	0.500	U	0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	0.500	U	0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	0.500	U	0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	0.500	U	0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	0.500	U	0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	0.500	U	0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	0.500	U	0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	0.500	U	0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	0.500	U	0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	1.00	U	0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	0.500	U	0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	0.500	U	0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	0.500	U	0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	0.500	U	0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	0.500	U	0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	0.500	U	0.200	0.500	1.00
78-93-3	2-Butanone	0.500	U	0.200	0.500	5.00
95-49-8	2-Chlorotoluene	0.500	U	0.200	0.500	1.00
591-78-6	2-Hexanone	1.00	U	0.500	1.00	5.00
106-43-4	4-Chlorotoluene	0.500	U	0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	0.500	U	0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	0.500	U	0.200	0.500	5.00
67-64-1	Acetone	1.00	U	0.500	1.00	5.00
71-43-2	Benzene	0.500	U	0.200	0.500	1.00
108-86-1	Bromobenzene	0.500	U	0.200	0.500	1.00
74-97-5	Bromochloromethane	0.500	U	0.200	0.500	1.00
75-27-4	Bromodichloromethane	0.500	U	0.200	0.500	1.00
75-25-2	Bromoform	0.500	U	0.250	0.500	1.00
74-83-9	Bromomethane	1.00	U	0.500	1.00	2.00
75-15-0	Carbon disulfide	0.500	U	0.200	0.500	1.00
56-23-5	Carbon tetrachloride	0.500	U	0.250	0.500	1.00

FORM I VOA

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	MB2489978
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2489978
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV15
Sample Amt:	5	mL		Lab File ID:	230606/c0774d
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	767081
Analysis Date:	06/06/23	Time:	1246	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	0.500	U	0.200	0.500	1.00
75-00-3	Chloroethane	0.500	U	0.250	0.500	1.00
67-66-3	Chloroform	0.500	U	0.200	0.500	1.00
74-87-3	Chloromethane	0.500	U	0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
124-48-1	Dibromochloromethane	0.500	U	0.200	0.500	1.00
74-95-3	Dibromomethane	0.500	U	0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	0.500	U	0.200	0.500	1.00
100-41-4	Ethylbenzene	0.500	U	0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	1.00	U	0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	0.500	U	0.200	0.500	1.00
136777-61-2	m,p-Xylene	0.500	U	0.200	0.500	2.00
75-09-2	Methylene chloride	0.500	U	0.200	0.500	5.00
91-20-3	Naphthalene	0.500	U	0.200	0.500	5.00
104-51-8	n-Butylbenzene	0.500	U	0.200	0.500	1.00
103-65-1	n-Propylbenzene	2.00	U	1.00	2.00	5.00
95-47-6	o-Xylene	0.500	U	0.200	0.500	1.00
135-98-8	sec-Butylbenzene	0.500	U	0.200	0.500	1.00
100-42-5	Styrene	0.500	U	0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	0.500	U	0.200	0.500	1.00
98-06-6	tert-Butylbenzene	0.500	U	0.200	0.500	1.00
127-18-4	Tetrachloroethene	0.500	U	0.200	0.500	1.00
108-88-3	Toluene	0.500	U	0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	0.500	U	0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	0.500	U	0.200	0.500	1.00
79-01-6	Trichloroethene	0.500	U	0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	0.500	U	0.200	0.500	1.00
75-01-4	Vinyl chloride	0.500	U	0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230606.s.b/c0774d.d
Lab Smp Id: 2489978 Client Smp ID: MB
Inj Date : 06-JUN-2023 12:46
Operator : KN1 Inst ID: msv15.i
Smp Info : MB*MB
Misc Info : MSV~52924~*1*SMR
Comment :
Method : /var/chem/msv15.i/230606.s.b/8260bw15DOD.m
Meth Date : 07-Jun-2023 11:45 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 QC Sample: BLANK
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	CONCENTRATIONS FINAL (ug/L)	SIMILARITY
$ 42 Dibromofluoromethane	111	5.159	5.160	(0.877)	87474	51.5175	51.5	6894
$ 50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	92680	52.4223	52.4	9650
* 55 FLUOROBENZENE	96	5.883	5.883	(1.000)	343823	50.0000		9839
$ 74 Toluene-d8	98	7.224	7.221	(0.866)	305669	46.0750	46.1	9863
* 91 CHLOROBENZENE-d5	82	8.339	8.340	(1.000)	141556	50.0000		9854
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	102283	47.4733	47.5	9396
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.806	(1.000)	132736	50.0000		9576

Data File: /var/chem/msv15.i/230606.s.b/c0774d.d

Page 1

Date : 06-JUN-2023 12:46

Client ID: MB

Instrument: msv15.i

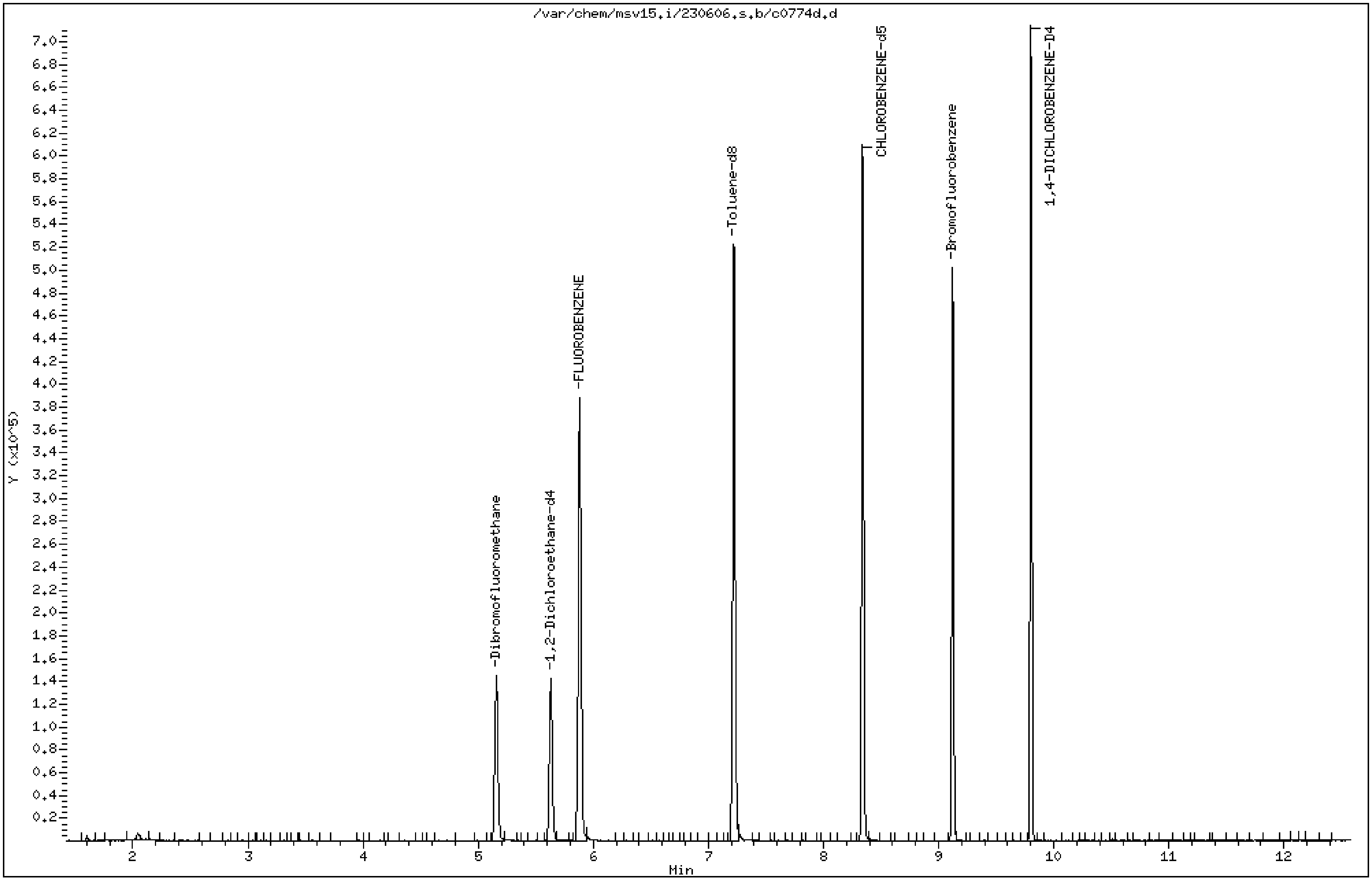
Sample Info: MB*MB

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

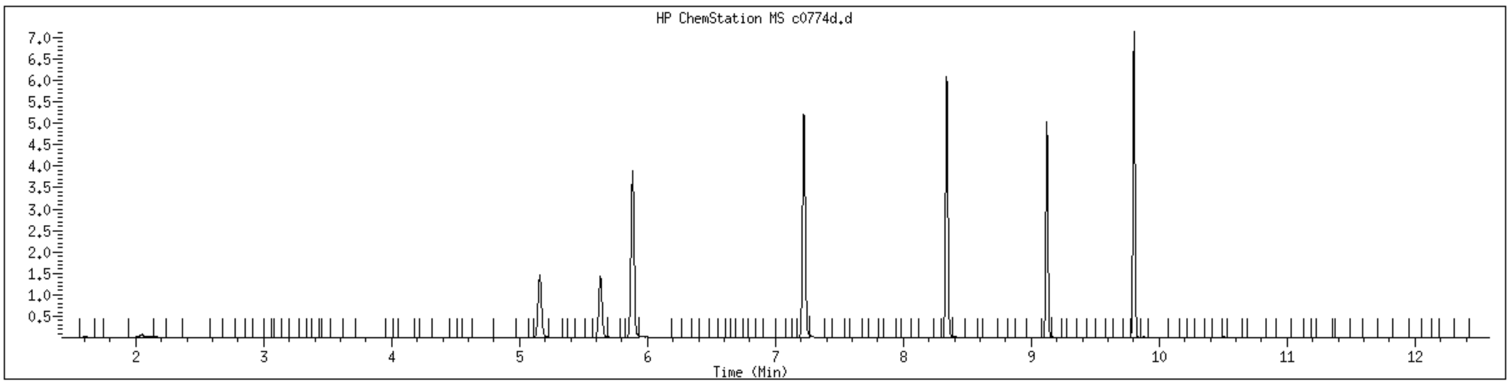
Column diameter: 0.25



Data file : /var/chem/msv15.i/230606.s.b/c0774d.d Page: 1
Report Date: 06/07/2023 11:45

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 2489978 SampleType : BLANK
Injection Date: 06/06/2023 12:46 Instrument : msv15.i
Operator : KN1
Sample Info : MB*MB
Misc Info : MSV~52924~*1*SMR
Method : /var/chem/msv15.i/230606.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	LCS2489979
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2489979
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV15
Sample Amt:	5	mL		Lab File ID:	230606/c0768Ld
Injection Vol.:	1	(μL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	767081
Analysis Date:	06/06/23	Time:	1053	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	51.3		0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	51.8		0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	52.4		0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	51.8		0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	53.7		0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	53.4		0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	52.7		0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	51.1		0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	48.1		0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	52.2		0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	53.1		0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	42.1		0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	52.3		0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	52.2		0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	50.5		0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	107		0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	55.3		0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	53.5		0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	52.3		0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	51.4		0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	51.9		0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	43.6		0.200	0.500	1.00
78-93-3	2-Butanone	246		0.200	0.500	5.00
95-49-8	2-Chlorotoluene	52.5		0.200	0.500	1.00
591-78-6	2-Hexanone	232		0.500	1.00	5.00
106-43-4	4-Chlorotoluene	53.1		0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	52.9		0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	226		0.200	0.500	5.00
67-64-1	Acetone	250		0.500	1.00	5.00
71-43-2	Benzene	53.7		0.200	0.500	1.00
108-86-1	Bromobenzene	51.6		0.200	0.500	1.00
74-97-5	Bromochloromethane	53.0		0.200	0.500	1.00
75-27-4	Bromodichloromethane	54.2		0.200	0.500	1.00
75-25-2	Bromoform	46.0		0.250	0.500	1.00
74-83-9	Bromomethane	42.6		0.500	1.00	2.00
75-15-0	Carbon disulfide	52.8		0.200	0.500	1.00
56-23-5	Carbon tetrachloride	51.3		0.250	0.500	1.00

FORM I VOA

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	LCS2489979		
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2489979		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV15		
Sample Amt:	5	mL		Lab File ID:	230606/c0768Ld		
Injection Vol.:	1		(µL)	GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	767081		
Analysis Date:	06/06/23	Time:	1053	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	52.6		0.200	0.500	1.00
75-00-3	Chloroethane	50.6		0.250	0.500	1.00
67-66-3	Chloroform	52.8		0.200	0.500	1.00
74-87-3	Chloromethane	58.4		0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	53.5		0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	52.9		0.200	0.500	1.00
124-48-1	Dibromochloromethane	54.8		0.200	0.500	1.00
74-95-3	Dibromomethane	53.4		0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	54.4		0.200	0.500	1.00
100-41-4	Ethylbenzene	52.8		0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	52.7		0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	53.3		0.200	0.500	1.00
136777-61-2	m,p-Xylene	107		0.200	0.500	2.00
75-09-2	Methylene chloride	53.2		0.200	0.500	5.00
91-20-3	Naphthalene	48.7		0.200	0.500	5.00
104-51-8	n-Butylbenzene	52.2		0.200	0.500	1.00
103-65-1	n-Propylbenzene	52.4		1.00	2.00	5.00
95-47-6	o-Xylene	53.0		0.200	0.500	1.00
135-98-8	sec-Butylbenzene	53.1		0.200	0.500	1.00
100-42-5	Styrene	56.3		0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	43.0		0.200	0.500	1.00
98-06-6	tert-Butylbenzene	51.0		0.200	0.500	1.00
127-18-4	Tetrachloroethene	50.9		0.200	0.500	1.00
108-88-3	Toluene	52.6		0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	53.5		0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	52.3		0.200	0.500	1.00
79-01-6	Trichloroethene	53.6		0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	49.8		0.200	0.500	1.00
75-01-4	Vinyl chloride	52.4		0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230606.s.b/c0768Ld.d
Lab Smp Id: 2489979 Client Smp ID: LCS
Inj Date : 06-JUN-2023 10:53
Operator : KN1 Inst ID: msv15.i
Smp Info : 1400*V15STD050*51363
Misc Info : MSV~529~*1*SMR
Comment :
Method : /var/chem/msv15.i/230606.s.b/8260bw15DOD.m
Meth Date : 07-Jun-2023 11:45 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 QC Sample: LCS
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	CONCENTRATIONS FINAL (ug/L)	SIMILARITY
1 Dichlorodifluoromethane	85	1.469	1.469	(0.250)	80445	54.3898	54.4	9734
2 Chloromethane ++	50	1.649	1.649	(0.280)	77812	58.4160	58.4	9771
3 Vinyl Chloride +	62	1.719	1.719	(0.292)	98210	52.3601	52.4	9798
5 1-3 Butadiene	39	1.735	1.735	(0.295)	71773	51.3836	51.4	9645
6 Bromomethane	94	2.018	2.018	(0.343)	67083	42.6488	42.6	9774
7 Chloroethane	64	2.137	2.137	(0.363)	69699	50.5537	50.6	9245
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	136042	49.7556	49.8	9667
11 1,1-Dichloroethene +	96	2.787	2.787	(0.474)	83085	53.4098	53.4	8429
12 Carbon Disulfide	76	2.816	2.816	(0.479)	251105	52.7641	52.8	7961
13 1,1,2Trichlotrifluoroethane	101	2.832	2.832	(0.481)	90035	53.3344	53.3	8782
15 Methyl Iodide	142	2.935	2.935	(0.499)	72665	39.4080	39.4	9532
16 Acrolein	56	3.150	3.150	(0.535)	58938	248.805	249	5377
17 Allyl chloride	39	3.304	3.304	(0.562)	72555	53.0642	53.1	9797
18 Methylene Chloride	49	3.420	3.420	(0.581)	91654	53.2419	53.2	4646

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
	19 Acetone	43	3.468	3.468	(0.590)	176697	250.080	250	5353(H)
	20 trans-1,2-Dichloroethene	61	3.591	3.591	(0.610)	115417	53.5177	53.5	8694
	21 Methyl Acetate	43	3.610	3.610	(0.614)	217268	247.003	247	9429(R)
	22 Hexane	57	3.687	3.687	(0.627)	165661	53.3167	53.3	9217
	23 MTBE	73	3.719	3.719	(0.632)	180671	42.9545	43.0	9734
	24 tert-Butyl Alcohol	59	3.822	3.822	(0.650)	22592	121.763	122	8219
	25 Acetonitrile	41	3.944	3.944	(0.670)	30669	245.168	245	9546
	26 Isopropyl Ether	45	4.112	4.112	(0.699)	207236	55.2222	55.2	9624
	27 Chloroprene	53	4.195	4.195	(0.713)	114782	54.2994	54.3	8866
	28 1,1-Dichloroethane ++	63	4.214	4.214	(0.716)	154640	53.6986	53.7	9472
	29 Acrylonitrile	53	4.259	4.259	(0.724)	124391	253.313	253	9800
	31 Ethyl Tert-butyl Ether	59	4.468	4.468	(0.760)	207414	49.1086	49.1	8113
	30 Vinyl Acetate	43	4.472	4.472	(0.760)	116251	50.0571	50.1	5591(R)
	32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	110691	53.4657	53.5	9801
M	33 Total 1,2-Dichloroethene	61				226108	106.983	107	0
	34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	111698	43.6380	43.6	9539
	35 Bromochloromethane	128	4.919	4.919	(0.836)	43088	53.0115	53.0	9385
	36 Cyclohexane	56	4.928	4.928	(0.838)	141619	53.9292	53.9	9440
	37 Chloroform +	83	4.992	4.992	(0.849)	155442	52.7655	52.8	9790
	39 Carbon Tetrachloride	117	5.118	5.118	(0.870)	122271	51.3435	51.3	8290
$	42 Dibromofluoromethane	111	5.160	5.160	(0.877)	82958	52.7250	52.7	9061
	43 1,1,1-Trichloroethane	97	5.179	5.179	(0.880)	141267	51.7834	51.8	9506
	45 1,1-Dichloropropene	75	5.292	5.292	(0.899)	127840	52.7092	52.7	9533
	44 2-Butanone	43	5.275	5.275	(0.897)	180439	246.041	246	6451
	46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	284971	38.9055	38.9	9769
	48 Benzene	78	5.513	5.513	(0.937)	352036	53.6990	53.7	9538
	56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	215120	49.9834	50.0	7355
$	50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	81357	49.6601	49.7	8590
	52 1,2-Dichloroethane	62	5.690	5.690	(0.967)	104140	50.4978	50.5	9817
	51 Isobutyl Alcohol	43	5.726	5.726	(0.973)	8392	195.610	196	3912
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	318605	50.0000		9837
	58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	167543	54.3889	54.4	8276
	57 Trichloroethene	130	6.034	6.034	(1.026)	107967	53.6233	53.6	9138
	62 Dibromomethane	93	6.398	6.398	(1.087)	56311	53.3799	53.4	9454
	64 1,2-Dichloropropane +	63	6.484	6.484	(1.102)	89550	55.2686	55.3	8467
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	119043	54.2408	54.2	9779
	67 1,4- Dioxane	88	6.716	6.716	(1.142)	13763	1145.57	1150	9354
	68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	112170	54.3399	54.3	8813
	72 2-Chloroethyl vinyl ether	63	7.021	7.021	(1.193)	21455	50.0881	50.1	9652(R)
	73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	137334	52.8525	52.9	9810
$	74 Toluene-d8	98	7.221	7.221	(0.866)	306041	51.5230	51.5	9901
	75 Toluene +	91	7.263	7.263	(0.871)	389066	52.6096	52.6	9625
	77 4-methyl-2-pentanone	43	7.549	7.549	(0.905)	309955	225.858	226	7919
	78 Tetrachloroethene	164	7.558	7.558	(0.906)	85105	50.9009	50.9	8425
	79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	128770	52.2841	52.3	8717
	81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	77770	51.7748	51.8	9561
	82 Dibromochloromethane	129	7.825	7.825	(0.938)	93680	54.8311	54.8	9629

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	104971	53.1463	53.1	9687
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	128505	51.3822	51.4	9576
M 83 1-3 Dichloropropene total	100				266104	105.137	105	0
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	84033	52.2824	52.3	9750
89 2-Hexanone	43	8.143	8.143	(0.976)	261397	231.635	232	9766
90 1-Chlorohexane	91	8.333	8.333	(0.999)	134371	50.8269	50.8	7798
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	126742	50.0000		9139
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	257255	52.5607	52.6	9437
93 Ethylbenzene +	106	8.369	8.369	(1.003)	140507	52.7971	52.8	9084
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	84374	51.3042	51.3	9205
95 p,m-Xylene	106	8.465	8.465	(1.015)	342660	106.500	107	9924
97 o-Xylene	106	8.745	8.745	(1.049)	158864	53.0239	53.0	9900
101 Styrene	104	8.777	8.777	(1.052)	217060	56.2651	56.3	9877
102 Bromoform ++	173	8.799	8.799	(1.055)	63321	45.9641	46.0	8672
104 Isopropylbenzene	105	8.944	8.944	(1.072)	417658	53.2568	53.3	9904
M 105 TOTAL XYLENE	106				501524	159.524	160	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	95976	49.7527	49.8	9553
107 cis-1,4-dichloro-2-butene	53	9.160	9.160	(0.934)	30327	50.4086	50.4	9546
109 n-Propylbenzene	91	9.201	9.201	(0.938)	484333	52.3529	52.4	8800
108 Bromobenzene	77	9.195	9.195	(0.938)	158960	51.6303	51.6	9707
110 1,1,2,2-Tetrachloroethane++	83	9.243	9.243	(0.943)	94635	52.3606	52.4	9812
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	309827	52.5393	52.5	8340
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	332891	53.4981	53.5	8231
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.951)	118062	48.1376	48.1	7494
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.953)	26724	44.7963	44.8	9350
116 Cyclohexanone	55	9.365	9.365	(0.955)	70414	235.274	235	9701
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	280949	53.0903	53.1	9817
118 tert-butylbenzene	91	9.516	9.516	(0.970)	178055	50.9655	51.0	9895
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.974)	324588	53.1472	53.1	9366
120 sec-Butylbenzene	105	9.623	9.623	(0.981)	427714	53.1372	53.1	9869
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	376821	52.8815	52.9	8920
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.995)	188425	52.3331	52.3	9693
125 1,4-Dichlorobenzene	146	9.815	9.815	(1.001)	192196	51.9003	51.9	9156
* 124 1,4-DICHLOROBENZENE-D4	152	9.806	9.806	(1.000)	118459	50.0000		8889
126 n-Butylbenzene	91	9.954	9.954	(1.015)	494437	52.1843	52.2	8940
127 1,2-Dichlorobenzene	146	10.066	10.066	(1.027)	169903	52.1629	52.2	9731
129 1,2-Dibromo-3-Chloropropane	157	10.558	10.558	(1.077)	19151	42.0981	42.1	9522
130 Hexachlorobutadiene	225	10.999	10.999	(1.122)	52472	52.6982	52.7	9283
131 1,2,4-Trichlorobenzene	180	11.031	11.031	(1.125)	103951	52.2105	52.2	9544
132 Naphthalene	128	11.291	11.291	(1.151)	227415	48.7440	48.7	9835
133 1,2,3-Trichlorobenzene	180	11.446	11.446	(1.167)	87720	51.0769	51.1	9485

QC Flag Legend

R - Spike/Surrogate failed recovery limits.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230606.s.b/c0768Ld.d

Date : 06-JUN-2023 10:53

Client ID: LCS

Sample Info: 1400*V15STD050*51363

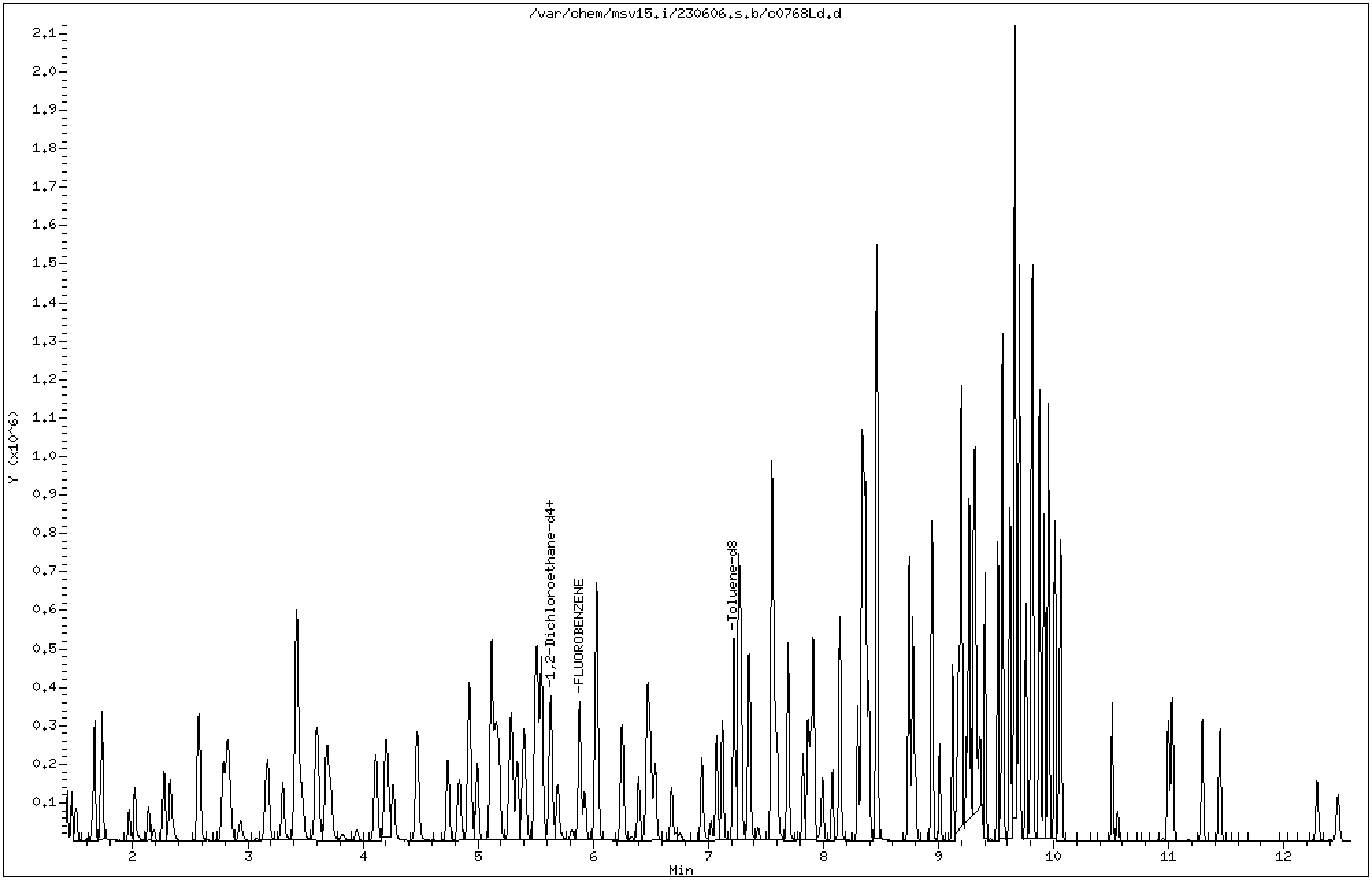
Purge Volume: 5.0

Column phase: RTX-VMS-30M

Instrument: msv15.i

Operator: KN1

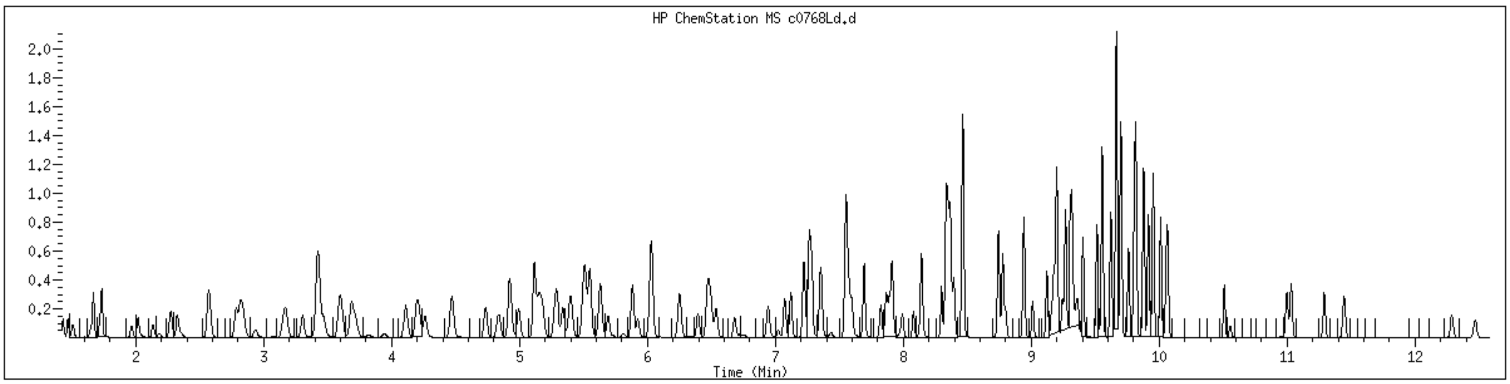
Column diameter: 0.25



Data file : /var/chem/msv15.i/230606.s.b/c0768Ld.d Page: 1
Report Date: 06/07/2023 11:45

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 2489979 SampleType : LCS
Injection Date: 06/06/2023 10:53 Instrument : msv15.i
Operator : KN1
Sample Info : 1400*V15STD050*51363
Misc Info : MSV~529~*1*SMR
Method : /var/chem/msv15.i/230606.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	LCSD2489980
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2489980
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV15
Sample Amt:	5	mL		Lab File ID:	230606/c0769
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	767081
Analysis Date:	06/06/23	Time:	1112	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	52.0		0.200	0.500	1.00
71-55-6	1,1,1-Trichloroethane	52.3		0.200	0.500	1.00
79-34-5	1,1,2,2-Tetrachloroethane	57.2		0.200	0.500	1.00
79-00-5	1,1,2-Trichloroethane	54.6		0.200	0.500	1.00
75-34-3	1,1-Dichloroethane	54.2		0.200	0.500	1.00
75-35-4	1,1-Dichloroethene	53.7		0.200	0.500	1.00
563-58-6	1,1-Dichloropropene	53.1		0.200	0.500	1.00
87-61-6	1,2,3-Trichlorobenzene	55.9		0.200	0.500	1.00
96-18-4	1,2,3-Trichloropropane	54.1		0.200	0.500	1.00
120-82-1	1,2,4-Trichlorobenzene	54.7		0.200	0.500	1.00
95-63-6	1,2,4-Trimethylbenzene	53.6		0.200	0.500	1.00
96-12-8	1,2-Dibromo-3-chloropropane	53.5		0.200	0.500	1.00
106-93-4	1,2-Dibromoethane	55.7		0.200	0.500	1.00
95-50-1	1,2-Dichlorobenzene	53.0		0.200	0.500	1.00
107-06-2	1,2-Dichloroethane	51.2		0.200	0.500	1.00
540-59-0	1,2-Dichloroethene (total)	107		0.400	1.00	2.00
78-87-5	1,2-Dichloropropane	54.8		0.200	0.500	1.00
108-67-8	1,3,5-Trimethylbenzene	53.1		0.200	0.500	1.00
541-73-1	1,3-Dichlorobenzene	52.5		0.200	0.500	1.00
142-28-9	1,3-Dichloropropane	52.9		0.200	0.500	1.00
106-46-7	1,4-Dichlorobenzene	51.7		0.200	0.500	1.00
594-20-7	2,2-Dichloropropane	50.4		0.200	0.500	1.00
78-93-3	2-Butanone	315		0.200	0.500	5.00
95-49-8	2-Chlorotoluene	52.1		0.200	0.500	1.00
591-78-6	2-Hexanone	284		0.500	1.00	5.00
106-43-4	4-Chlorotoluene	53.5		0.200	0.500	1.00
99-87-6	4-Isopropyltoluene	53.1		0.200	0.500	1.00
108-10-1	4-Methyl-2-pentanone	278		0.200	0.500	5.00
67-64-1	Acetone	301		0.500	1.00	5.00
71-43-2	Benzene	53.7		0.200	0.500	1.00
108-86-1	Bromobenzene	51.1		0.200	0.500	1.00
74-97-5	Bromochloromethane	51.7		0.200	0.500	1.00
75-27-4	Bromodichloromethane	53.9		0.200	0.500	1.00
75-25-2	Bromoform	50.2		0.250	0.500	1.00
74-83-9	Bromomethane	45.6		0.500	1.00	2.00
75-15-0	Carbon disulfide	52.8		0.200	0.500	1.00
56-23-5	Carbon tetrachloride	52.1		0.250	0.500	1.00

FORM I VOA

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060210			Client Sample ID:	LCSD2489980		
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2489980		
Matrix:	Water	% Moisture:	NA	Instrument ID:	MSV15		
Sample Amt:	5	mL		Lab File ID:	230606/c0769		
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	1	Analyst:	KN1	Analytical Batch:	767081		
Analysis Date:	06/06/23	Time:	1112	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/L

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	52.9		0.200	0.500	1.00
75-00-3	Chloroethane	52.5		0.250	0.500	1.00
67-66-3	Chloroform	52.6		0.200	0.500	1.00
74-87-3	Chloromethane	53.2		0.200	0.500	1.00
156-59-2	cis-1,2-Dichloroethene	53.1		0.200	0.500	1.00
10061-01-5	cis-1,3-Dichloropropene	53.8		0.200	0.500	1.00
124-48-1	Dibromochloromethane	56.3		0.200	0.500	1.00
74-95-3	Dibromomethane	54.4		0.250	0.500	1.00
75-71-8	Dichlorodifluoromethane	56.5		0.200	0.500	1.00
100-41-4	Ethylbenzene	53.9		0.200	0.500	1.00
87-68-3	Hexachlorobutadiene	54.7		0.500	1.00	5.00
98-82-8	Isopropylbenzene (Cumene)	54.2		0.200	0.500	1.00
136777-61-2	m,p-Xylene	108		0.200	0.500	2.00
75-09-2	Methylene chloride	53.0		0.200	0.500	5.00
91-20-3	Naphthalene	57.5		0.200	0.500	5.00
104-51-8	n-Butylbenzene	54.8		0.200	0.500	1.00
103-65-1	n-Propylbenzene	52.7		1.00	2.00	5.00
95-47-6	o-Xylene	54.4		0.200	0.500	1.00
135-98-8	sec-Butylbenzene	53.5		0.200	0.500	1.00
100-42-5	Styrene	56.1		0.200	0.500	1.00
1634-04-4	tert-Butyl methyl ether (MTBE)	53.8		0.200	0.500	1.00
98-06-6	tert-Butylbenzene	52.2		0.200	0.500	1.00
127-18-4	Tetrachloroethene	52.7		0.200	0.500	1.00
108-88-3	Toluene	53.4		0.200	0.500	1.00
156-60-5	trans-1,2-Dichloroethene	53.5		0.200	0.500	1.00
10061-02-6	trans-1,3-Dichloropropene	52.8		0.200	0.500	1.00
79-01-6	Trichloroethene	53.9		0.200	0.500	1.00
75-69-4	Trichlorofluoromethane	52.8		0.200	0.500	1.00
75-01-4	Vinyl chloride	54.2		0.200	0.500	1.00

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230606.s.b/c0769.d
Lab Smp Id: 2489980 Client Smp ID: LCSD
Inj Date : 06-JUN-2023 11:12
Operator : KN1 Inst ID: msv15.i
Smp Info : 1400*V15STD050*51363
Misc Info : MSV~529~*1*SMR
Comment :
Method : /var/chem/msv15.i/230606.s.b/8260bw15DOD.m
Meth Date : 06-Jun-2023 16:25 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 QC Sample: LCSD
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.469	(0.250)	86993	56.5377	56.5	9776
2 Chloromethane ++	50	1.652	1.649	(0.281)	73765	53.2318	53.2	9739
3 Vinyl Chloride +	62	1.716	1.719	(0.292)	105800	54.2209	54.2	9848
5 1-3 Butadiene	39	1.735	1.735	(0.295)	83563	57.5061	57.5	9689
6 Bromomethane	94	2.015	2.018	(0.343)	74599	45.5893	45.6	9837
7 Chloroethane	64	2.134	2.137	(0.363)	75302	52.5011	52.5	9259
8 Trichlorofluoromethane	101	2.269	2.272	(0.386)	150267	52.8286	52.8	9714
11 1,1-Dichloroethene +	96	2.787	2.787	(0.474)	86894	53.6938	53.7	8493
12 Carbon Disulfide	76	2.816	2.816	(0.479)	261606	52.8405	52.8	7972
13 1,1,2Trichlotrifluoroethane	101	2.835	2.832	(0.482)	95474	54.3647	54.4	9309
15 Methyl Iodide	142	2.938	2.935	(0.499)	78755	40.9876	41.0	9540
16 Acrolein	56	3.147	3.150	(0.535)	71527	290.249	290	6491
17 Allyl chloride	39	3.301	3.304	(0.561)	78496	55.1847	55.2	9830
18 Methylene Chloride	49	3.417	3.420	(0.581)	94966	53.0281	53.0	4929

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
	19 Acetone	43	3.468	3.468	(0.590)	221256	301.010	301	5307(H)
	20 trans-1,2-Dichloroethene	61	3.590	3.591	(0.610)	119933	53.4567	53.5	8584
	21 Methyl Acetate	43	3.610	3.610	(0.614)	274561	300.042	300	9563(R)
	22 Hexane	57	3.687	3.687	(0.627)	182404	56.4304	56.4	9261
	23 MTBE	73	3.719	3.719	(0.632)	235524	53.8259	53.8	9699
	24 tert-Butyl Alcohol	59	3.822	3.822	(0.650)	58835	304.813	305	8032
	25 Acetonitrile	41	3.944	3.944	(0.670)	40847	313.877	314	9554
	26 Isopropyl Ether	45	4.114	4.112	(0.699)	218738	56.0284	56.0	9610
	27 Chloroprene	53	4.198	4.195	(0.714)	116267	52.8705	52.9	8513
	28 1,1-Dichloroethane ++	63	4.217	4.214	(0.717)	162404	54.2093	54.2	9561
	29 Acrylonitrile	53	4.259	4.259	(0.724)	153604	300.682	301	9855
	31 Ethyl Tert-butyl Ether	59	4.471	4.468	(0.760)	231287	52.6389	52.6	8023
	30 Vinyl Acetate	43	4.471	4.472	(0.760)	129507	53.6042	53.6	5667(R)
	32 cis-1,2-Dichloroethene	61	4.738	4.735	(0.805)	114396	53.1141	53.1	9769
M	33 Total 1,2-Dichloroethene	61				234329	106.571	107	0
	34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	134273	50.4248	50.4	9610
	35 Bromochloromethane	128	4.925	4.919	(0.837)	43733	51.7200	51.7	8791
	36 Cyclohexane	56	4.928	4.928	(0.838)	151421	55.4274	55.4	9434
	37 Chloroform +	83	4.995	4.992	(0.849)	161249	52.6156	52.6	9827
	39 Carbon Tetrachloride	117	5.118	5.118	(0.870)	129181	52.1431	52.1	8116
$	42 Dibromofluoromethane	111	5.159	5.160	(0.877)	86400	52.7847	52.8	8361
	43 1,1,1-Trichloroethane	97	5.182	5.179	(0.881)	148427	52.2996	52.3	9251
	45 1,1-Dichloropropene	75	5.295	5.292	(0.900)	134014	53.1136	53.1	9516
	44 2-Butanone	43	5.275	5.275	(0.897)	240159	314.784	315	5553
	46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	345006	45.2765	45.3	9306
	48 Benzene	78	5.516	5.513	(0.938)	366392	53.7231	53.7	9619
	56 tert-amyl Methyl Ether	73	5.635	5.632	(0.958)	238307	53.2252	53.2	7187
$	50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	85245	50.0170	50.0	8492
	52 1,2-Dichloroethane	62	5.693	5.690	(0.968)	109738	51.1503	51.2	9786
	51 Isobutyl Alcohol	43	5.725	5.726	(0.973)	11348	254.262	254	3819
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	331449	50.0000		9831
	58 Methyl Cyclohexane	83	6.031	6.028	(1.025)	177637	55.4311	55.4	8172
	57 Trichloroethene	130	6.034	6.034	(1.026)	112889	53.8952	53.9	9009
	62 Dibromomethane	93	6.394	6.398	(1.087)	59737	54.4332	54.4	9479
	64 1,2-Dichloropropane +	63	6.484	6.484	(1.102)	92319	54.7696	54.8	8194
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	123050	53.8939	53.9	9793
	67 1,4- Dioxane	88	6.719	6.716	(1.142)	18816	1505.47	1510	9394
	68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	118660	55.2563	55.3	9166
	72 2-Chloroethyl vinyl ether	63	7.024	7.021	(1.194)	62518	132.309	132	9744
	73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	145486	53.8202	53.8	9787
$	74 Toluene-d8	98	7.224	7.221	(0.866)	319781	52.5298	52.5	9909
	75 Toluene +	91	7.262	7.263	(0.871)	404963	53.4304	53.4	9710
	77 4-methyl-2-pentanone	43	7.552	7.549	(0.906)	392065	278.128	278	7974
	78 Tetrachloroethene	164	7.558	7.558	(0.906)	90220	52.6508	52.7	7803
	79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	135266	52.7934	52.8	8450
	81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	84128	54.6485	54.6	9486
	82 Dibromochloromethane	129	7.825	7.825	(0.938)	98524	56.2669	56.3	9616

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	110240	54.4595	54.5	9695
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	135464	52.8504	52.9	9577
M 83 1-3 Dichloropropene total	100				280752	106.614	107	0
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	91701	55.6688	55.7	9700
89 2-Hexanone	43	8.143	8.143	(0.976)	329653	284.426	284	9737
90 1-Chlorohexane	91	8.333	8.333	(0.999)	146759	54.1657	54.2	8005
* 91 CHLOROBENZENE-d5	82	8.339	8.340	(1.000)	129894	50.0000		9207
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	265425	52.9140	52.9	9294
93 Ethylbenzene +	106	8.368	8.369	(1.003)	146948	53.8775	53.9	9162
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	87716	52.0421	52.0	9246
95 p,m-Xylene	106	8.465	8.465	(1.015)	355999	107.961	108	9912
97 o-Xylene	106	8.745	8.745	(1.049)	166937	54.3663	54.4	9910
101 Styrene	104	8.777	8.777	(1.052)	221689	56.0705	56.1	9903
102 Bromoform ++	173	8.799	8.799	(1.055)	71007	50.2119	50.2	8753
104 Isopropylbenzene	105	8.944	8.944	(1.072)	435405	54.1725	54.2	9911
M 105 TOTAL XYLENE	106				522936	162.328	162	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	99572	50.3642	50.4	9544
107 cis-1,4-dichloro-2-butene	53	9.159	9.160	(0.934)	34779	55.8273	55.8	9582
109 n-Propylbenzene	91	9.201	9.201	(0.938)	505181	52.7349	52.7	8793
108 Bromobenzene	77	9.195	9.195	(0.938)	162997	51.1271	51.1	9738
110 1,1,2,2-Tetrachloroethane++	83	9.243	9.243	(0.943)	107125	57.2398	57.2	9796
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	317933	52.0661	52.1	8334
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	342146	53.1010	53.1	8222
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.951)	137389	54.0979	54.1	7715
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.953)	31779	51.4086	51.4	9444(M2)
116 Cyclohexanone	55	9.365	9.365	(0.955)	99704	321.240	321	9714
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	293370	53.5374	53.5	9811
118 tert-butylbenzene	91	9.516	9.516	(0.970)	188679	52.1555	52.2	9889
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.974)	339014	53.6069	53.6	9355
120 sec-Butylbenzene	105	9.622	9.623	(0.981)	446229	53.5375	53.5	9856
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	391606	53.0729	53.1	8953
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.995)	195603	52.4648	52.5	9716
125 1,4-Dichlorobenzene	146	9.812	9.815	(1.001)	198353	51.7272	51.7	8586(H)
* 124 1,4-DICHLOROBENZENE-D4	152	9.806	9.806	(1.000)	122663	50.0000		8923
126 n-Butylbenzene	91	9.954	9.954	(1.015)	537921	54.8280	54.8	8892
127 1,2-Dichlorobenzene	146	10.066	10.066	(1.027)	178794	53.0112	53.0	9771
129 1,2-Dibromo-3-Chloropropane	157	10.558	10.558	(1.077)	25305	53.4890	53.5	9525
130 Hexachlorobutadiene	225	10.998	10.999	(1.122)	56420	54.7213	54.7	9298
131 1,2,4-Trichlorobenzene	180	11.031	11.031	(1.125)	112822	54.7239	54.7	9513
132 Naphthalene	128	11.291	11.291	(1.151)	277904	57.5243	57.5	9856
133 1,2,3-Trichlorobenzene	180	11.445	11.446	(1.167)	99425	55.9083	55.9	9464

QC Flag Legend

R - Spike/Surrogate failed recovery limits.
M2- Compound response manually integrated because
Target system integrated incorrectly.

QC Flag Legend

H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230606.s.b/c0769.d

Page 1

Date : 06-JUN-2023 11:12

Client ID: LCSD

Instrument: msv15.i

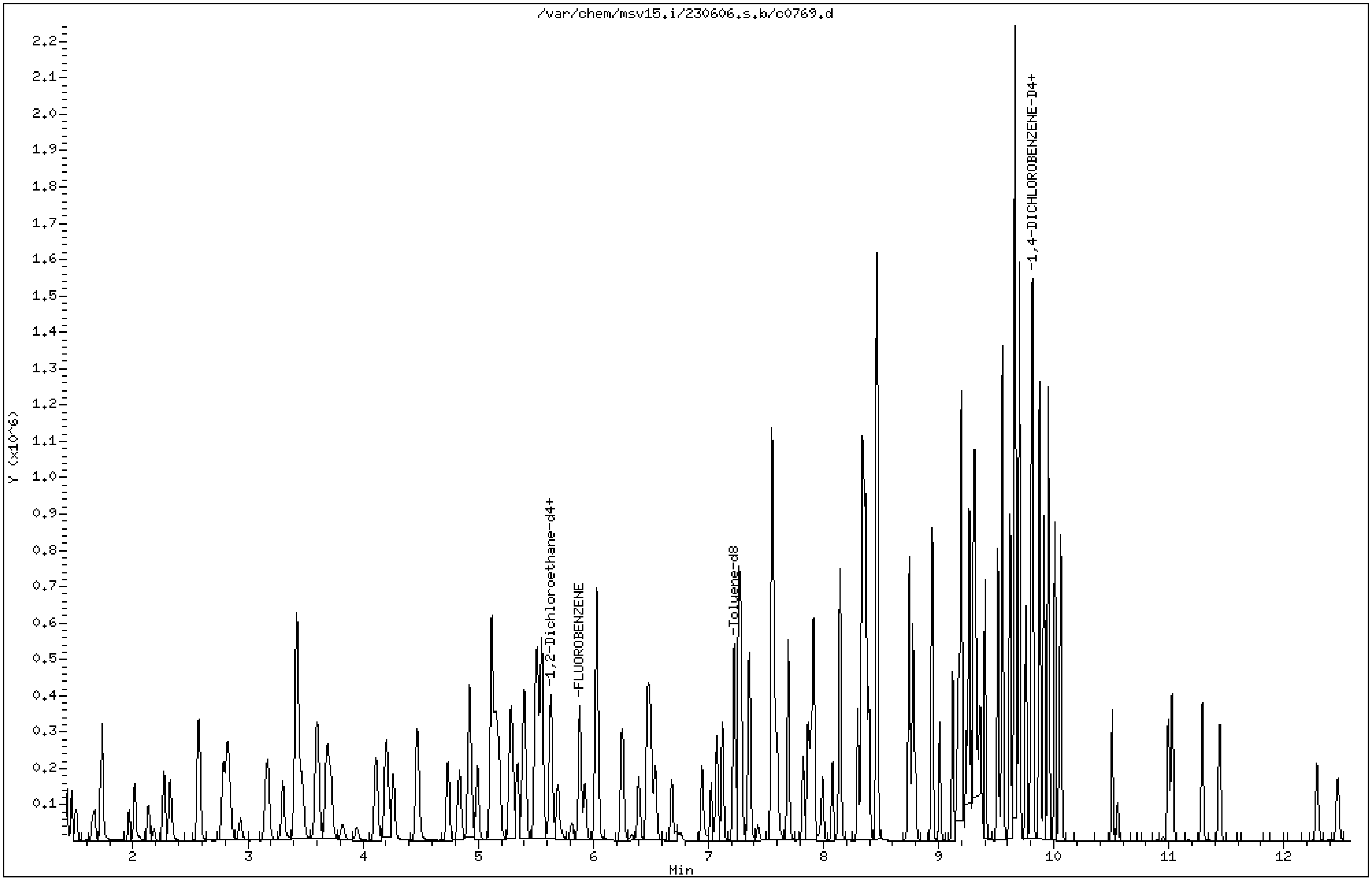
Sample Info: 1400*V15STD050*51363

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

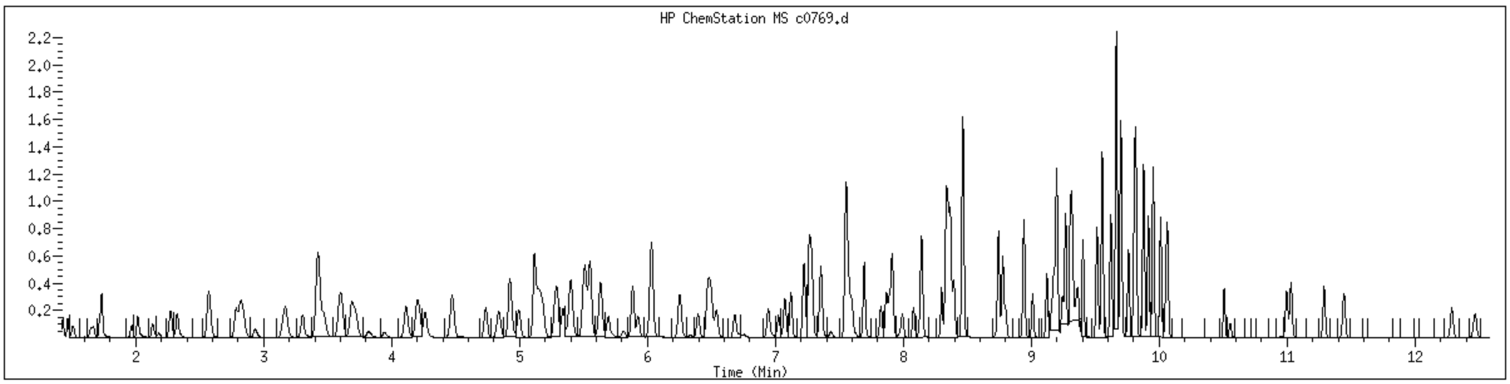
Column diameter: 0.25



Report Date: 06/06/2023 16:25

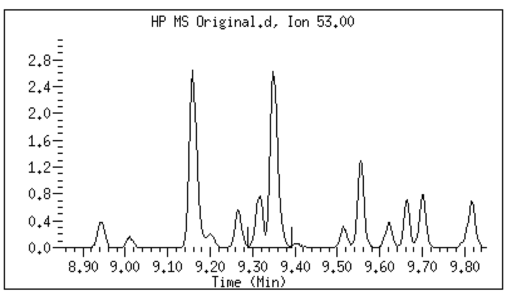
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 2489980   SampleType   : LCSD
Injection Date: 06/06/2023 11:12   Instrument   : msv15.i
Operator   : KN1
Sample Info   : 1400*V15STD050*51363
Misc Info   : MSV~529~*1*SMR
Method   : /var/chem/msv15.i/230606.s.b/8260bw15DOD.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260DOD



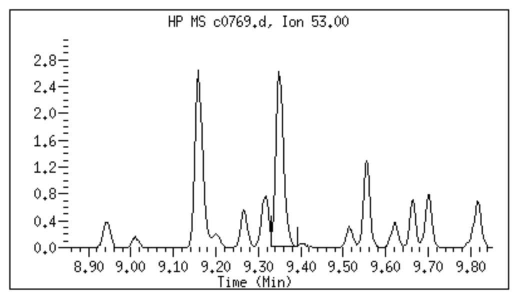
Original   Final

115 trans-1,4-Dichloro-2-Butene   CAS#: 110-57-6   Reason: M2



Electronic Signature Applied

User: kmn
Date: 06/06/2023 16:19



M2 - Target system integrated incorrectly

EPA 8260B DOD

Form 2A

Surrogates

Water

2A

WATER VOLATILE SYSTEM MONITORING COMPOUND RECOVERY

Report No: 223060210 Analytical Method: EPA 8260B

	Client Sample ID	GCALSample ID	SMC1	#	SMC2	#	SMC3	#	SMC4	#	TOT OUT
1 .	OT07-EB-060123	22306021001	115		100		117		103		0
2 .	OT07-MW02-060123	22306021002	118		104		118		104		0
3 .	OT07-MW04-060123	22306021003	118		99		120	*	101		1
4 .	OT07-MW05-060123	22306021004	117		105		117		104		0
5 .	OT07-MW06-060123	22306021005	118		101		119		105		0
6 .	OT07-MW04-060123MS	22306021006	110		109		113		100		0
7 .	OT07-MW04-060123MSD	22306021007	109		108		111		102		0
8 .	OT07-MW-806-060123	22306021008	104		93		103		93		0
9 .	OT07-TB-01-060124	22306021009	106		95		105		93		0
10 .	MB2489345	2489345	116		100		119		104		0
11 .	LCS2489346	2489346	110		106		113		103		0
12 .	MB2489807	2489807	105		95		104		94		0
13 .	LCS2489808	2489808	98		101		105		107		0
14 .	MB2489978	2489978	105		95		103		92		0
15 .	LCS2489979	2489979	99		100		105		103		0
16 .	LCSD2489980	2489980	100		101		106		105		0

		QC LIMITS
SMC 1	1,2-Dichloroethane-d4	81 - 118
SMC 2	4-Bromofluorobenzene	85 - 114
SMC 3	Dibromofluoromethane	80 - 119
SMC 4	Toluene-d8	89 - 112

Column to be used to flag recovery values

* Values outside of QC limits

FORM II VOA-1

EPA 8260B DOD

Form 3A

Spikes

Water

3A
WATER VOLATILE MS/MSD RECOVERY

Report No: 223060210

Parent Sample ID: OT07-MW04-060123

Analytical Method: EPA 8260B

Analytical Batch: 766942

GCAL QC ID: 22306021006

ANALYTE	UNITS	SPIKE ADDED	SAMPLE RESULT	MS RESULT	MS % REC	#	QC LIMITS
1,1,1,2-Tetrachloroethane	ug/L	50	0	47.3	95		78 - 124
1,1,1-Trichloroethane	ug/L	50	0	54.6	109		74 - 131
1,1,2,2-Tetrachloroethane	ug/L	50	0	41.5	83		71 - 121
1,1,2-Trichloroethane	ug/L	50	0	44.8	90		80 - 119
1,1-Dichloroethane	ug/L	50	0	50.1	100		77 - 125
1,1-Dichloroethene	ug/L	50	0	53.8	108		71 - 131
1,1-Dichloropropene	ug/L	50	0	51.8	104		79 - 125
1,2,3-Trichlorobenzene	ug/L	50	0	39.8	80		69 - 129
1,2,3-Trichloropropane	ug/L	50	0	38.6	77		73 - 122
1,2,4-Trichlorobenzene	ug/L	50	0	39.4	79		69 - 130
1,2,4-Trimethylbenzene	ug/L	50	0	44.9	90		76 - 124
1,2-Dibromo-3-chloropropane	ug/L	50	0	41.9	84		62 - 128
1,2-Dibromoethane	ug/L	50	0	46.1	92		77 - 121
1,2-Dichlorobenzene	ug/L	50	0	41.7	83		80 - 119
1,2-Dichloroethane	ug/L	50	0	54.4	109		73 - 128
1,2-Dichloroethene (total)	ug/L	100	0	100	100		79 - 121
1,2-Dichloropropane	ug/L	50	0	50.3	101		78 - 122
1,3,5-Trimethylbenzene	ug/L	50	0	43.5	87		75 - 124
1,3-Dichlorobenzene	ug/L	50	0	41.5	83		80 - 119
1,3-Dichloropropane	ug/L	50	0	45.9	92		80 - 119
1,4-Dichlorobenzene	ug/L	50	0	40.8	82		79 - 118
2,2-Dichloropropane	ug/L	50	0	51.1	102		60 - 139
2-Butanone	ug/L	250	0	183	73		56 - 143
2-Chlorotoluene	ug/L	50	0	43.4	87		79 - 122
2-Hexanone	ug/L	250	0	189	76		57 - 139
4-Chlorotoluene	ug/L	50	0	41.6	83		78 - 122
4-Isopropyltoluene	ug/L	50	0	43.9	88		77 - 127
4-Methyl-2-pentanone	ug/L	250	0	233	93		67 - 130
Acetone	ug/L	250	0	146	58		39 - 160
Benzene	ug/L	50	0	50.7	101		79 - 120
Bromobenzene	ug/L	50	0	42.3	85		80 - 120
Bromochloromethane	ug/L	50	0	52.3	105		78 - 123
Bromodichloromethane	ug/L	50	0	51.9	104		79 - 125
Bromoform	ug/L	50	0	48.7	97		66 - 130
Bromomethane	ug/L	50	0	47.8	96		53 - 141
Carbon disulfide	ug/L	50	0	49.7	99		64 - 133
Carbon tetrachloride	ug/L	50	0	55	110		72 - 136
Chlorobenzene	ug/L	50	0	47.2	94		82 - 118
Chloroethane	ug/L	50	0	46.7	93		60 - 138

RPD : 1 out of 66 outside limits

Spike Recovery: 0 out of 132 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III VOA-1

3A

WATER VOLATILE MS/MSD RECOVERY

Report No: 223060210

Parent Sample ID: OT07-MW04-060123

Analytical Method: EPA 8260B

Analytical Batch: 766942

Chloroform	ug/L	50	0	53.2	106		79	-	124
Chloromethane	ug/L	50	0	43	86		50	-	139
Dibromochloromethane	ug/L	50	0	48.1	96		74	-	126
Dibromomethane	ug/L	50	0	51.6	103		79	-	123
Dichlorodifluoromethane	ug/L	50	0	39.3	79		32	-	152
Ethylbenzene	ug/L	50	0	47.3	95		79	-	121
Hexachlorobutadiene	ug/L	50	0	44.5	89		66	-	134
Isopropylbenzene (Cumene)	ug/L	50	0	50.1	100		72	-	131
Methylene chloride	ug/L	50	0	50.5	101		74	-	124
Naphthalene	ug/L	50	0	35.5	71		61	-	128
Styrene	ug/L	50	0	49.1	98		78	-	123
Tetrachloroethene	ug/L	50	0	48.3	97		74	-	129
Toluene	ug/L	50	0	46.9	94		80	-	121
Trichloroethene	ug/L	50	0	50.8	102		79	-	123
Trichlorofluoromethane	ug/L	50	0	52	104		65	-	141
Vinyl chloride	ug/L	50	0	46.2	92		58	-	137
cis-1,2-Dichloroethene	ug/L	50	0	51.1	102		78	-	123
cis-1,3-Dichloropropene	ug/L	50	0	49.9	100		75	-	124
m,p-Xylene	ug/L	100	0	95.3	95		80	-	121
n-Butylbenzene	ug/L	50	0	39.6	79		75	-	128
n-Propylbenzene	ug/L	50	0	42.9	86		76	-	126
o-Xylene	ug/L	50	0	49	98		78	-	122
sec-Butylbenzene	ug/L	50	0	44.3	89		77	-	126
tert-Butyl methyl ether (MTBE)	ug/L	50	0	49.7	99		71	-	124
tert-Butylbenzene	ug/L	50	0	44	88		78	-	124
trans-1,2-Dichloroethene	ug/L	50	0	49.2	98		75	-	124
trans-1,3-Dichloropropene	ug/L	50	0	50.2	100		73	-	127

RPD : 1 out of 66 outside limits

Spike Recovery: 0 out of 132 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III VOA-1

3A
WATER VOLATILE MS/MSD RECOVERY

Report No: 223060210 Parent Sample ID: OT07-MW04-060123
Analytical Method: EPA 8260B Analytical Batch: 766942

GCAL QC ID: 22306021007

ANALYTE	UNITS	SPIKE ADDED	MSD RESULT	MSD % REC	#	% RPD	#	QC LIMITS REC	QC LIMITS RPD
1,1,1,2-Tetrachloroethane	ug/L	50	47.8	96		1		78 - 124	0 - 20
1,1,1-Trichloroethane	ug/L	50	54.1	108		.9		74 - 131	0 - 20
1,1,2,2-Tetrachloroethane	ug/L	50	41.1	82		1		71 - 121	0 - 20
1,1,2-Trichloroethane	ug/L	50	45.4	91		1		80 - 119	0 - 20
1,1-Dichloroethane	ug/L	50	50.1	100		0		77 - 125	0 - 20
1,1-Dichloroethene	ug/L	50	54.1	108		.6		71 - 131	0 - 20
1,1-Dichloropropene	ug/L	50	52.5	105		1		79 - 125	0 - 20
1,2,3-Trichlorobenzene	ug/L	50	40	80		.5		69 - 129	0 - 20
1,2,3-Trichloropropane	ug/L	50	38.5	77		.3		73 - 122	0 - 20
1,2,4-Trichlorobenzene	ug/L	50	39.8	80		1		69 - 130	0 - 20
1,2,4-Trimethylbenzene	ug/L	50	44.9	90		0		76 - 124	0 - 20
1,2-Dibromo-3-chloropropane	ug/L	50	41.1	82		2		62 - 128	0 - 20
1,2-Dibromoethane	ug/L	50	46.2	92		.2		77 - 121	0 - 20
1,2-Dichlorobenzene	ug/L	50	40.9	82		2		80 - 119	0 - 20
1,2-Dichloroethane	ug/L	50	53.4	107		2		73 - 128	0 - 20
1,2-Dichloroethene (total)	ug/L	100	100	100		0		79 - 121	0 - 20
1,2-Dichloropropane	ug/L	50	49.8	100		1		78 - 122	0 - 20
1,3,5-Trimethylbenzene	ug/L	50	43.8	88		.7		75 - 124	0 - 20
1,3-Dichlorobenzene	ug/L	50	41	82		1		80 - 119	0 - 20
1,3-Dichloropropane	ug/L	50	47.7	95		4		80 - 119	0 - 20
1,4-Dichlorobenzene	ug/L	50	41.1	82		.7		79 - 118	0 - 20
2,2-Dichloropropane	ug/L	50	50.7	101		.8		60 - 139	0 - 20
2-Butanone	ug/L	250	187	75		2		56 - 143	0 - 20
2-Chlorotoluene	ug/L	50	42.5	85		2		79 - 122	0 - 20
2-Hexanone	ug/L	250	192	77		2		57 - 139	0 - 20
4-Chlorotoluene	ug/L	50	42.2	84		1		78 - 122	0 - 20
4-Isopropyltoluene	ug/L	50	44.2	88		.7		77 - 127	0 - 20
4-Methyl-2-pentanone	ug/L	250	238	95		2		67 - 130	0 - 20
Acetone	ug/L	250	149	60		2		39 - 160	0 - 20
Benzene	ug/L	50	49.8	100		2		79 - 120	0 - 20
Bromobenzene	ug/L	50	41.3	83		2		80 - 120	0 - 20
Bromochloromethane	ug/L	50	51.8	104		1		78 - 123	0 - 20
Bromodichloromethane	ug/L	50	52.2	104		.6		79 - 125	0 - 20
Bromoform	ug/L	50	49.5	99		2		66 - 130	0 - 20
Bromomethane	ug/L	50	50.4	101		5		53 - 141	0 - 20
Carbon disulfide	ug/L	50	49.5	99		.4		64 - 133	0 - 20
Carbon tetrachloride	ug/L	50	54.9	110		.2		72 - 136	0 - 20
Chlorobenzene	ug/L	50	47.5	95		.6		82 - 118	0 - 20

RPD : 1 out of 66 outside limits

Spike Recovery: 0 out of 132 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III VOA-1

3A

WATER VOLATILE MS/MSD RECOVERY

Report No: 223060210

Parent Sample ID: OT07-MW04-060123

Analytical Method: EPA 8260B

Analytical Batch: 766942

Chloroethane	ug/L	50	53.3	107		13		60 - 138	0 - 20
Chloroform	ug/L	50	52.4	105		2		79 - 124	0 - 20
Chloromethane	ug/L	50	43.5	87		1		50 - 139	0 - 20
Dibromochloromethane	ug/L	50	49.8	100		3		74 - 126	0 - 20
Dibromomethane	ug/L	50	52.1	104		1		79 - 123	0 - 20
Dichlorodifluoromethane	ug/L	50	50.2	100		24	*	32 - 152	0 - 20
Ethylbenzene	ug/L	50	48.2	96		2		79 - 121	0 - 20
Hexachlorobutadiene	ug/L	50	44.6	89		.2		66 - 134	0 - 20
Isopropylbenzene (Cumene)	ug/L	50	51.2	102		2		72 - 131	0 - 20
Methylene chloride	ug/L	50	49.1	98		3		74 - 124	0 - 20
Naphthalene	ug/L	50	36	72		1		61 - 128	0 - 20
Styrene	ug/L	50	50.6	101		3		78 - 123	0 - 20
Tetrachloroethene	ug/L	50	51	102		5		74 - 129	0 - 20
Toluene	ug/L	50	48.4	97		3		80 - 121	0 - 20
Trichloroethene	ug/L	50	50.7	101		.2		79 - 123	0 - 20
Trichlorofluoromethane	ug/L	50	56.8	114		9		65 - 141	0 - 20
Vinyl chloride	ug/L	50	51.2	102		10		58 - 137	0 - 20
cis-1,2-Dichloroethene	ug/L	50	50.4	101		1		78 - 123	0 - 20
cis-1,3-Dichloropropene	ug/L	50	50	100		.2		75 - 124	0 - 20
m,p-Xylene	ug/L	100	97.5	98		2		80 - 121	0 - 20
n-Butylbenzene	ug/L	50	39.5	79		.3		75 - 128	0 - 20
n-Propylbenzene	ug/L	50	42.8	86		.2		76 - 126	0 - 20
o-Xylene	ug/L	50	50.3	101		3		78 - 122	0 - 20
sec-Butylbenzene	ug/L	50	43.5	87		2		77 - 126	0 - 20
tert-Butyl methyl ether (MTBE)	ug/L	50	50.3	101		1		71 - 124	0 - 20
tert-Butylbenzene	ug/L	50	43.7	87		.7		78 - 124	0 - 20
trans-1,2-Dichloroethene	ug/L	50	49.6	99		.8		75 - 124	0 - 20
trans-1,3-Dichloropropene	ug/L	50	50.5	101		.6		73 - 127	0 - 20

RPD : 1 out of 66 outside limits

Spike Recovery: 0 out of 132 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III VOA-1

3A

WATER VOLATILE LCS/LCSD RECOVERY

Report No: 223060210

Analytical Method: EPA 8260B

Analytical Batch: 767081

GCAL QC ID: 2489979

ANALYTE	UNITS	SPIKE ADDED	SAMPLE RESULT	LCS RESULT	LCS % REC	#	QC LIMITS
1,1,1,2-Tetrachloroethane	ug/L	50	0	51.3	103		78 - 124
1,1,1-Trichloroethane	ug/L	50	0	51.8	104		74 - 131
1,1,2,2-Tetrachloroethane	ug/L	50	0	52.4	105		71 - 121
1,1,2-Trichloroethane	ug/L	50	0	51.8	104		80 - 119
1,1-Dichloroethane	ug/L	50	0	53.7	107		77 - 125
1,1-Dichloroethene	ug/L	50	0	53.4	107		71 - 131
1,1-Dichloropropene	ug/L	50	0	52.7	105		79 - 125
1,2,3-Trichlorobenzene	ug/L	50	0	51.1	102		69 - 129
1,2,3-Trichloropropane	ug/L	50	0	48.1	96		73 - 122
1,2,4-Trichlorobenzene	ug/L	50	0	52.2	104		69 - 130
1,2,4-Trimethylbenzene	ug/L	50	0	53.1	106		76 - 124
1,2-Dibromo-3-chloropropane	ug/L	50	0	42.1	84		62 - 128
1,2-Dibromoethane	ug/L	50	0	52.3	105		77 - 121
1,2-Dichlorobenzene	ug/L	50	0	52.2	104		80 - 119
1,2-Dichloroethane	ug/L	50	0	50.5	101		73 - 128
1,2-Dichloroethene (total)	ug/L	100	0	107	107		79 - 121
1,2-Dichloropropane	ug/L	50	0	55.3	111		78 - 122
1,3,5-Trimethylbenzene	ug/L	50	0	53.5	107		75 - 124
1,3-Dichlorobenzene	ug/L	50	0	52.3	105		80 - 119
1,3-Dichloropropane	ug/L	50	0	51.4	103		80 - 119
1,4-Dichlorobenzene	ug/L	50	0	51.9	104		79 - 118
2,2-Dichloropropane	ug/L	50	0	43.6	87		60 - 139
2-Butanone	ug/L	250	0	246	98		56 - 143
2-Chlorotoluene	ug/L	50	0	52.5	105		79 - 122
2-Hexanone	ug/L	250	0	232	93		57 - 139
4-Chlorotoluene	ug/L	50	0	53.1	106		78 - 122
4-Isopropyltoluene	ug/L	50	0	52.9	106		77 - 127
4-Methyl-2-pentanone	ug/L	250	0	226	90		67 - 130
Acetone	ug/L	250	0	250	100		39 - 160
Benzene	ug/L	50	0	53.7	107		79 - 120
Bromobenzene	ug/L	50	0	51.6	103		80 - 120
Bromochloromethane	ug/L	50	0	53	106		78 - 123
Bromodichloromethane	ug/L	50	0	54.2	108		79 - 125
Bromoform	ug/L	50	0	46	92		66 - 130
Bromomethane	ug/L	50	0	42.6	85		53 - 141
Carbon disulfide	ug/L	50	0	52.8	106		64 - 133
Carbon tetrachloride	ug/L	50	0	51.3	103		72 - 136
Chlorobenzene	ug/L	50	0	52.6	105		82 - 118
Chloroethane	ug/L	50	0	50.6	101		60 - 138

RPD : 4 out of 66 outside limits

Spike Recovery: 0 out of 132 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III VOA-1

3A

WATER VOLATILE LCS/LCSD RECOVERY

Report No: 223060210

Analytical Method: EPA 8260B

Analytical Batch: 767081

Chloroform	ug/L	50	0	52.8	106		79	-	124
Chloromethane	ug/L	50	0	58.4	117		50	-	139
Dibromochloromethane	ug/L	50	0	54.8	110		74	-	126
Dibromomethane	ug/L	50	0	53.4	107		79	-	123
Dichlorodifluoromethane	ug/L	50	0	54.4	109		32	-	152
Ethylbenzene	ug/L	50	0	52.8	106		79	-	121
Hexachlorobutadiene	ug/L	50	0	52.7	105		66	-	134
Isopropylbenzene (Cumene)	ug/L	50	0	53.3	107		72	-	131
Methylene chloride	ug/L	50	0	53.2	106		74	-	124
Naphthalene	ug/L	50	0	48.7	97		61	-	128
Styrene	ug/L	50	0	56.3	113		78	-	123
Tetrachloroethene	ug/L	50	0	50.9	102		74	-	129
Toluene	ug/L	50	0	52.6	105		80	-	121
Trichloroethene	ug/L	50	0	53.6	107		79	-	123
Trichlorofluoromethane	ug/L	50	0	49.8	100		65	-	141
Vinyl chloride	ug/L	50	0	52.4	105		58	-	137
cis-1,2-Dichloroethene	ug/L	50	0	53.5	107		78	-	123
cis-1,3-Dichloropropene	ug/L	50	0	52.9	106		75	-	124
m,p-Xylene	ug/L	100	0	107	107		80	-	121
n-Butylbenzene	ug/L	50	0	52.2	104		75	-	128
n-Propylbenzene	ug/L	50	0	52.4	105		76	-	126
o-Xylene	ug/L	50	0	53	106		78	-	122
sec-Butylbenzene	ug/L	50	0	53.1	106		77	-	126
tert-Butyl methyl ether (MTBE)	ug/L	50	0	43	86		71	-	124
tert-Butylbenzene	ug/L	50	0	51	102		78	-	124
trans-1,2-Dichloroethene	ug/L	50	0	53.5	107		75	-	124
trans-1,3-Dichloropropene	ug/L	50	0	52.3	105		73	-	127

RPD : 4 out of 66 outside limits

Spike Recovery: 0 out of 132 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III VOA-1

3A

WATER VOLATILE LCS/LCSD RECOVERY

Report No: 223060210

Analytical Method: EPA 8260B

Analytical Batch: 767081

GCAL QC ID: 2489980

ANALYTE	UNITS	SPIKE ADDED	LCSD RESULT	LCSD % REC	#	% RPD	#	QC LIMITS REC	QC LIMITS RPD
1,1,1,2-Tetrachloroethane	ug/L	50	52	104		1		78 - 124	0 - 20
1,1,1-Trichloroethane	ug/L	50	52.3	105		1		74 - 131	0 - 20
1,1,2,2-Tetrachloroethane	ug/L	50	57.2	114		9		71 - 121	0 - 20
1,1,2-Trichloroethane	ug/L	50	54.6	109		5		80 - 119	0 - 20
1,1-Dichloroethane	ug/L	50	54.2	108		.9		77 - 125	0 - 20
1,1-Dichloroethene	ug/L	50	53.7	107		.6		71 - 131	0 - 20
1,1-Dichloropropene	ug/L	50	53.1	106		.8		79 - 125	0 - 20
1,2,3-Trichlorobenzene	ug/L	50	55.9	112		9		69 - 129	0 - 20
1,2,3-Trichloropropane	ug/L	50	54.1	108		12		73 - 122	0 - 20
1,2,4-Trichlorobenzene	ug/L	50	54.7	109		5		69 - 130	0 - 20
1,2,4-Trimethylbenzene	ug/L	50	53.6	107		.9		76 - 124	0 - 20
1,2-Dibromo-3-chloropropane	ug/L	50	53.5	107		24	*	62 - 128	0 - 20
1,2-Dibromoethane	ug/L	50	55.7	111		6		77 - 121	0 - 20
1,2-Dichlorobenzene	ug/L	50	53	106		2		80 - 119	0 - 20
1,2-Dichloroethane	ug/L	50	51.2	102		1		73 - 128	0 - 20
1,2-Dichloroethene (total)	ug/L	100	107	107		0		79 - 121	0 - 20
1,2-Dichloropropane	ug/L	50	54.8	110		.9		78 - 122	0 - 20
1,3,5-Trimethylbenzene	ug/L	50	53.1	106		.8		75 - 124	0 - 20
1,3-Dichlorobenzene	ug/L	50	52.5	105		.4		80 - 119	0 - 20
1,3-Dichloropropane	ug/L	50	52.9	106		3		80 - 119	0 - 20
1,4-Dichlorobenzene	ug/L	50	51.7	103		.4		79 - 118	0 - 20
2,2-Dichloropropane	ug/L	50	50.4	101		14		60 - 139	0 - 20
2-Butanone	ug/L	250	315	126		25	*	56 - 143	0 - 20
2-Chlorotoluene	ug/L	50	52.1	104		.8		79 - 122	0 - 20
2-Hexanone	ug/L	250	284	114		20		57 - 139	0 - 20
4-Chlorotoluene	ug/L	50	53.5	107		.8		78 - 122	0 - 20
4-Isopropyltoluene	ug/L	50	53.1	106		.4		77 - 127	0 - 20
4-Methyl-2-pentanone	ug/L	250	278	111		21	*	67 - 130	0 - 20
Acetone	ug/L	250	301	120		19		39 - 160	0 - 20
Benzene	ug/L	50	53.7	107		0		79 - 120	0 - 20
Bromobenzene	ug/L	50	51.1	102		1		80 - 120	0 - 20
Bromochloromethane	ug/L	50	51.7	103		2		78 - 123	0 - 20
Bromodichloromethane	ug/L	50	53.9	108		.6		79 - 125	0 - 20
Bromoform	ug/L	50	50.2	100		9		66 - 130	0 - 20
Bromomethane	ug/L	50	45.6	91		7		53 - 141	0 - 20
Carbon disulfide	ug/L	50	52.8	106		0		64 - 133	0 - 20
Carbon tetrachloride	ug/L	50	52.1	104		2		72 - 136	0 - 20
Chlorobenzene	ug/L	50	52.9	106		.6		82 - 118	0 - 20

RPD : 4 out of 66 outside limits

Spike Recovery: 0 out of 132 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III VOA-1

3A

WATER VOLATILE LCS/LCSD RECOVERY

Report No: 223060210

Analytical Method: EPA 8260B

Analytical Batch: 767081

Chloroethane	ug/L	50	52.5	105		4		60 - 138	0 - 20
Chloroform	ug/L	50	52.6	105		.4		79 - 124	0 - 20
Chloromethane	ug/L	50	53.2	106		9		50 - 139	0 - 20
Dibromochloromethane	ug/L	50	56.3	113		3		74 - 126	0 - 20
Dibromomethane	ug/L	50	54.4	109		2		79 - 123	0 - 20
Dichlorodifluoromethane	ug/L	50	56.5	113		4		32 - 152	0 - 20
Ethylbenzene	ug/L	50	53.9	108		2		79 - 121	0 - 20
Hexachlorobutadiene	ug/L	50	54.7	109		4		66 - 134	0 - 20
Isopropylbenzene (Cumene)	ug/L	50	54.2	108		2		72 - 131	0 - 20
Methylene chloride	ug/L	50	53	106		.4		74 - 124	0 - 20
Naphthalene	ug/L	50	57.5	115		17		61 - 128	0 - 20
Styrene	ug/L	50	56.1	112		.4		78 - 123	0 - 20
Tetrachloroethene	ug/L	50	52.7	105		3		74 - 129	0 - 20
Toluene	ug/L	50	53.4	107		2		80 - 121	0 - 20
Trichloroethene	ug/L	50	53.9	108		.6		79 - 123	0 - 20
Trichlorofluoromethane	ug/L	50	52.8	106		6		65 - 141	0 - 20
Vinyl chloride	ug/L	50	54.2	108		3		58 - 137	0 - 20
cis-1,2-Dichloroethene	ug/L	50	53.1	106		.8		78 - 123	0 - 20
cis-1,3-Dichloropropene	ug/L	50	53.8	108		2		75 - 124	0 - 20
m,p-Xylene	ug/L	100	108	108		.9		80 - 121	0 - 20
n-Butylbenzene	ug/L	50	54.8	110		5		75 - 128	0 - 20
n-Propylbenzene	ug/L	50	52.7	105		.6		76 - 126	0 - 20
o-Xylene	ug/L	50	54.4	109		3		78 - 122	0 - 20
sec-Butylbenzene	ug/L	50	53.5	107		.8		77 - 126	0 - 20
tert-Butyl methyl ether (MTBE)	ug/L	50	53.8	108		22	*	71 - 124	0 - 20
tert-Butylbenzene	ug/L	50	52.2	104		2		78 - 124	0 - 20
trans-1,2-Dichloroethene	ug/L	50	53.5	107		0		75 - 124	0 - 20
trans-1,3-Dichloropropene	ug/L	50	52.8	106		1		73 - 127	0 - 20

RPD : 4 out of 66 outside limits

Spike Recovery: 0 out of 132 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III VOA-1

3A
WATER VOLATILE LCS/LCSD RECOVERY

Report No: 223060210

Analytical Method: EPA 8260B

Analytical Batch: 767028

GCAL QC ID: 2489808

ANALYTE	UNITS	SPIKE ADDED	SAMPLE RESULT	LCS RESULT	LCS % REC	#	QC LIMITS
1,1,1,2-Tetrachloroethane	ug/L	50	0	51.6	103		78 - 124
1,1,1-Trichloroethane	ug/L	50	0	51.7	103		74 - 131
1,1,2,2-Tetrachloroethane	ug/L	50	0	50.6	101		71 - 121
1,1,2-Trichloroethane	ug/L	50	0	51.9	104		80 - 119
1,1-Dichloroethane	ug/L	50	0	53.6	107		77 - 125
1,1-Dichloroethene	ug/L	50	0	53.6	107		71 - 131
1,1-Dichloropropene	ug/L	50	0	52.8	106		79 - 125
1,2,3-Trichlorobenzene	ug/L	50	0	50.7	101		69 - 129
1,2,3-Trichloropropane	ug/L	50	0	47.6	95		73 - 122
1,2,4-Trichlorobenzene	ug/L	50	0	51.4	103		69 - 130
1,2,4-Trimethylbenzene	ug/L	50	0	52.9	106		76 - 124
1,2-Dibromo-3-chloropropane	ug/L	50	0	41.8	84		62 - 128
1,2-Dibromoethane	ug/L	50	0	52.2	104		77 - 121
1,2-Dichlorobenzene	ug/L	50	0	51.7	103		80 - 119
1,2-Dichloroethane	ug/L	50	0	50.1	100		73 - 128
1,2-Dichloroethene (total)	ug/L	100	0	107	107		79 - 121
1,2-Dichloropropane	ug/L	50	0	53.7	107		78 - 122
1,3,5-Trimethylbenzene	ug/L	50	0	53.1	106		75 - 124
1,3-Dichlorobenzene	ug/L	50	0	51.4	103		80 - 119
1,3-Dichloropropane	ug/L	50	0	51.4	103		80 - 119
1,4-Dichlorobenzene	ug/L	50	0	51.2	102		79 - 118
2,2-Dichloropropane	ug/L	50	0	48.2	96		60 - 139
2-Butanone	ug/L	250	0	236	94		56 - 143
2-Chlorotoluene	ug/L	50	0	52.1	104		79 - 122
2-Hexanone	ug/L	250	0	227	91		57 - 139
4-Chlorotoluene	ug/L	50	0	52.2	104		78 - 122
4-Isopropyltoluene	ug/L	50	0	53.1	106		77 - 127
4-Methyl-2-pentanone	ug/L	250	0	221	88		67 - 130
Acetone	ug/L	250	0	234	94		39 - 160
Benzene	ug/L	50	0	53.4	107		79 - 120
Bromobenzene	ug/L	50	0	53.2	106		80 - 120
Bromochloromethane	ug/L	50	0	52.5	105		78 - 123
Bromodichloromethane	ug/L	50	0	52.8	106		79 - 125
Bromoform	ug/L	50	0	45.5	91		66 - 130
Bromomethane	ug/L	50	0	38.7	77		53 - 141
Carbon disulfide	ug/L	50	0	53.3	107		64 - 133
Carbon tetrachloride	ug/L	50	0	51.7	103		72 - 136
Chlorobenzene	ug/L	50	0	52.9	106		82 - 118
Chloroethane	ug/L	50	0	52.8	106		60 - 138

RPD : 0 out of 0 outside limits

Spike Recovery: 0 out of 66 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III VOA-1

3A

WATER VOLATILE LCS/LCSD RECOVERY

Report No: 223060210

Analytical Method: EPA 8260B

Analytical Batch: 767028

Chloroform	ug/L	50	0	51.6	103		79	-	124
Chloromethane	ug/L	50	0	57.7	115		50	-	139
Dibromochloromethane	ug/L	50	0	54.5	109		74	-	126
Dibromomethane	ug/L	50	0	52	104		79	-	123
Dichlorodifluoromethane	ug/L	50	0	60.8	122		32	-	152
Ethylbenzene	ug/L	50	0	53.3	107		79	-	121
Hexachlorobutadiene	ug/L	50	0	51.8	104		66	-	134
Isopropylbenzene (Cumene)	ug/L	50	0	54.1	108		72	-	131
Methylene chloride	ug/L	50	0	52.3	105		74	-	124
Naphthalene	ug/L	50	0	48.2	96		61	-	128
Styrene	ug/L	50	0	57.4	115		78	-	123
Tetrachloroethene	ug/L	50	0	52.7	105		74	-	129
Toluene	ug/L	50	0	53.1	106		80	-	121
Trichloroethene	ug/L	50	0	52.8	106		79	-	123
Trichlorofluoromethane	ug/L	50	0	51.5	103		65	-	141
Vinyl chloride	ug/L	50	0	54.7	109		58	-	137
cis-1,2-Dichloroethene	ug/L	50	0	52.7	105		78	-	123
cis-1,3-Dichloropropene	ug/L	50	0	52.4	105		75	-	124
m,p-Xylene	ug/L	100	0	108	108		80	-	121
n-Butylbenzene	ug/L	50	0	52.5	105		75	-	128
n-Propylbenzene	ug/L	50	0	52.3	105		76	-	126
o-Xylene	ug/L	50	0	53.1	106		78	-	122
sec-Butylbenzene	ug/L	50	0	53.2	106		77	-	126
tert-Butyl methyl ether (MTBE)	ug/L	50	0	45.6	91		71	-	124
tert-Butylbenzene	ug/L	50	0	50.8	102		78	-	124
trans-1,2-Dichloroethene	ug/L	50	0	54.1	108		75	-	124
trans-1,3-Dichloropropene	ug/L	50	0	51	102		73	-	127

RPD : 0 out of 0 outside limits

Spike Recovery: 0 out of 66 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III VOA-1

3A
WATER VOLATILE LCS/LCSD RECOVERY

Report No: 223060210

Analytical Method: EPA 8260B

Analytical Batch: 766942

GCAL QC ID: 2489346

ANALYTE	UNITS	SPIKE ADDED	SAMPLE RESULT	LCS RESULT	LCS % REC	#	QC LIMITS
1,1,1,2-Tetrachloroethane	ug/L	50	0	50.2	100		78 - 124
1,1,1-Trichloroethane	ug/L	50	0	52.3	105		74 - 131
1,1,2,2-Tetrachloroethane	ug/L	50	0	45.6	91		71 - 121
1,1,2-Trichloroethane	ug/L	50	0	49.3	99		80 - 119
1,1-Dichloroethane	ug/L	50	0	50.1	100		77 - 125
1,1-Dichloroethene	ug/L	50	0	53.1	106		71 - 131
1,1-Dichloropropene	ug/L	50	0	52	104		79 - 125
1,2,3-Trichlorobenzene	ug/L	50	0	46.4	93		69 - 129
1,2,3-Trichloropropane	ug/L	50	0	43.4	87		73 - 122
1,2,4-Trichlorobenzene	ug/L	50	0	46.4	93		69 - 130
1,2,4-Trimethylbenzene	ug/L	50	0	50.4	101		76 - 124
1,2-Dibromo-3-chloropropane	ug/L	50	0	46.4	93		62 - 128
1,2-Dibromoethane	ug/L	50	0	49.6	99		77 - 121
1,2-Dichlorobenzene	ug/L	50	0	46.6	93		80 - 119
1,2-Dichloroethane	ug/L	50	0	53.1	106		73 - 128
1,2-Dichloroethene (total)	ug/L	100	0	97.9	98		79 - 121
1,2-Dichloropropane	ug/L	50	0	49.7	99		78 - 122
1,3,5-Trimethylbenzene	ug/L	50	0	49	98		75 - 124
1,3-Dichlorobenzene	ug/L	50	0	46.3	93		80 - 119
1,3-Dichloropropane	ug/L	50	0	49.2	98		80 - 119
1,4-Dichlorobenzene	ug/L	50	0	46.3	93		79 - 118
2,2-Dichloropropane	ug/L	50	0	50.7	101		60 - 139
2-Butanone	ug/L	250	0	217	87		56 - 143
2-Chlorotoluene	ug/L	50	0	46.4	93		79 - 122
2-Hexanone	ug/L	250	0	223	89		57 - 139
4-Chlorotoluene	ug/L	50	0	47.4	95		78 - 122
4-Isopropyltoluene	ug/L	50	0	49.5	99		77 - 127
4-Methyl-2-pentanone	ug/L	250	0	254	102		67 - 130
Acetone	ug/L	250	0	201	80		39 - 160
Benzene	ug/L	50	0	49.3	99		79 - 120
Bromobenzene	ug/L	50	0	45.8	92		80 - 120
Bromochloromethane	ug/L	50	0	50.8	102		78 - 123
Bromodichloromethane	ug/L	50	0	51.5	103		79 - 125
Bromoform	ug/L	50	0	53	106		66 - 130
Bromomethane	ug/L	50	0	40.2	80		53 - 141
Carbon disulfide	ug/L	50	0	50.3	101		64 - 133
Carbon tetrachloride	ug/L	50	0	52.6	105		72 - 136
Chlorobenzene	ug/L	50	0	49.4	99		82 - 118
Chloroethane	ug/L	50	0	47.4	95		60 - 138

RPD : 0 out of 0 outside limits

Spike Recovery: 0 out of 66 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III VOA-1

3A

WATER VOLATILE LCS/LCSD RECOVERY

Report No: 223060210

Analytical Method: EPA 8260B

Analytical Batch: 766942

Chloroform	ug/L	50	0	51.8	104		79	-	124
Chloromethane	ug/L	50	0	44.9	90		50	-	139
Dibromochloromethane	ug/L	50	0	51.9	104		74	-	126
Dibromomethane	ug/L	50	0	51.8	104		79	-	123
Dichlorodifluoromethane	ug/L	50	0	41.9	84		32	-	152
Ethylbenzene	ug/L	50	0	50.1	100		79	-	121
Hexachlorobutadiene	ug/L	50	0	49.2	98		66	-	134
Isopropylbenzene (Cumene)	ug/L	50	0	53	106		72	-	131
Methylene chloride	ug/L	50	0	48.8	98		74	-	124
Naphthalene	ug/L	50	0	43	86		61	-	128
Styrene	ug/L	50	0	52.4	105		78	-	123
Tetrachloroethene	ug/L	50	0	52.5	105		74	-	129
Toluene	ug/L	50	0	50.3	101		80	-	121
Trichloroethene	ug/L	50	0	52.1	104		79	-	123
Trichlorofluoromethane	ug/L	50	0	53.5	107		65	-	141
Vinyl chloride	ug/L	50	0	48.5	97		58	-	137
cis-1,2-Dichloroethene	ug/L	50	0	49.6	99		78	-	123
cis-1,3-Dichloropropene	ug/L	50	0	53.5	107		75	-	124
m,p-Xylene	ug/L	100	0	103	103		80	-	121
n-Butylbenzene	ug/L	50	0	46.2	92		75	-	128
n-Propylbenzene	ug/L	50	0	48.1	96		76	-	126
o-Xylene	ug/L	50	0	52.7	105		78	-	122
sec-Butylbenzene	ug/L	50	0	49.6	99		77	-	126
tert-Butyl methyl ether (MTBE)	ug/L	50	0	50.5	101		71	-	124
tert-Butylbenzene	ug/L	50	0	48.6	97		78	-	124
trans-1,2-Dichloroethene	ug/L	50	0	48.3	97		75	-	124
trans-1,3-Dichloropropene	ug/L	50	0	51.3	103		73	-	127

RPD : 0 out of 0 outside limits

Spike Recovery: 0 out of 66 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III VOA-1

EPA 8260B DOD

Form 4A

Method Blanks

4A
VOLATILE METHOD BLANK SUMMARY

Report No:	223060210	Method Blank ID:	2489345
Matrix:	Water	Instrument ID:	MSV11
Sample Amt:	5 mL	Lab File ID:	230602/j1694
Injection Vol.:	1 (µL)	GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1 Analyst: KN1	Analytical Batch:	766942
Analysis Date:	06/02/23 Time: 1423	Analytical Method:	EPA 8260B

THIS METHOD BLANK APPLIES TO THE FOLLOWING SAMPLES, MS AND MSD

	CLIENT SAMPLE ID	***GCAL SAMPLE ID***	***LAB FILE ID***	***DATE ANALYZED***	***TIME ANALYZED***
1 .	LCS2489346	2489346	230602/j1687L	06/02/23	1123
2 .	OT07-EB-060123	22306021001	230602/j1695	06/02/23	1446
3 .	OT07-MW02-060123	22306021002	230602/j1697	06/02/23	1532
4 .	OT07-MW04-060123	22306021003	230602/j1698	06/02/23	1556
5 .	OT07-MW05-060123	22306021004	230602/j1699	06/02/23	1619
6 .	OT07-MW06-060123	22306021005	230602/j1700	06/02/23	1642
7 .	OT07-MW04-060123MS	22306021006	230602/j1705	06/02/23	1838
8 .	OT07-MW04-060123MSD	22306021007	230602/j1706	06/02/23	1902

FORM IV VOA

4A

VOLATILE METHOD BLANK SUMMARY

Report No:	223060210	Method Blank ID:	2489807
Matrix:	Water	Instrument ID:	MSV15
Sample Amt:	5 mL	Lab File ID:	230605/c0737
Injection Vol.:	1 (µL)	GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1 Analyst: KN1	Analytical Batch:	767028
Analysis Date:	06/05/23 Time: 1257	Analytical Method:	EPA 8260B

THIS METHOD BLANK APPLIES TO THE FOLLOWING SAMPLES, MS AND MSD

	CLIENT SAMPLE ID	***GCAL SAMPLE ID***	***LAB FILE ID***	***DATE ANALYZED***	***TIME ANALYZED***
1 .	LCS2489808	2489808	230605/c0730L	06/05/23	1045
2 .	OT07-MW-806-060123	22306021008	230605/c0742	06/05/23	1430

FORM IV VOA

4A

VOLATILE METHOD BLANK SUMMARY

Report No:	223060210	Method Blank ID:	2489978
Matrix:	Water	Instrument ID:	MSV15
Sample Amt:	5 mL	Lab File ID:	230606/c0774d
Injection Vol.:	1 (µL)	GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	1 Analyst: KN1	Analytical Batch:	767081
Analysis Date:	06/06/23 Time: 1246	Analytical Method:	EPA 8260B

THIS METHOD BLANK APPLIES TO THE FOLLOWING SAMPLES, MS AND MSD

	CLIENT SAMPLE ID	***GCAL SAMPLE ID***	***LAB FILE ID***	***DATE ANALYZED***	***TIME ANALYZED***
1 .	LCS2489979	2489979	230606/c0768Ld	06/06/23	1053
2 .	LCSD2489980	2489980	230606/c0769	06/06/23	1112
3 .	OT07-TB-01-060124	22306021009	230606/c0777	06/06/23	1342

FORM IV VOA

EPA 8260B DOD

Form 5A

Tunes

5A

VOLATILE ORGANICS INSTRUMENT PERFORMANCE CHECK

BROMOFLUOROBENZENE (BFB)

Report No:	223060210	Tune ID:	1000
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV11
Injection Vol.:	1.0 (µL)	Lab File ID:	230517/j1293t
Analyst:	KN1	Analytical Batch:	766136
Analysis Date:	05/17/23 Time: 0939	Analytical Method:	EPA 8260B

m / e	ION ABUNDANCE CRITERIA	% Relative Abundance		
50	15.0 - 40.0% of mass 95	23.68	( )	
75	30.0 - 66.0% of mass 95	47.79	( )	
95	Base Peak, 100% relative abundance	100	( )	
96	5.0 -9.0% of mass 95	6.86	( )	
173	Less than 2.0% of mass 174	.87	(1.06)	1
174	50.0 - 120.0% of mass 95	82.43	( )	
175	4.0 - 9.0% of mass 174	5.91	(7.18)	1
176	95.0 - 101.0% of mass 174	80.01	(97.07)	1
177	5.0 - 9.0% of mass 176	4.81	(6.02)	2

1- Value is % mass 174 2- Value is % mass 176

THIS CHECK APPLIES TO THE FOLLOWING SAMPLES, MS, MSD, BLANKS, AND STANDARDS:

	CLIENT SAMPLE ID	GCAL SAMPLE ID	LAB FILE ID	DATE ANALYZED	TIME ANALYZED
1 .	V11STD020	1206	230517/j1299	05/17/23	1206
2 .	V11STD050	1207	230517/j1300	05/17/23	1229
3 .	V11STD100	1208	230517/j1301	05/17/23	1252
4 .	V11STD200	1209	230517/j1302	05/17/23	1315
5 .	V11STD001	1203	230517/j1304	05/17/23	1417
6 .	V11STD005	1204	230517/j1305	05/17/23	1440
7 .	V11STD010	1205	230517/j1306	05/17/23	1502
8 .	V11ICV051	1600	230517/j1309	05/17/23	1727

FORM V VOA

Data File: /var/chem/msv11.i/230517.s.b/j1293t.d

Date : 17-MAY-2023 09:39

Client ID: V11BFB

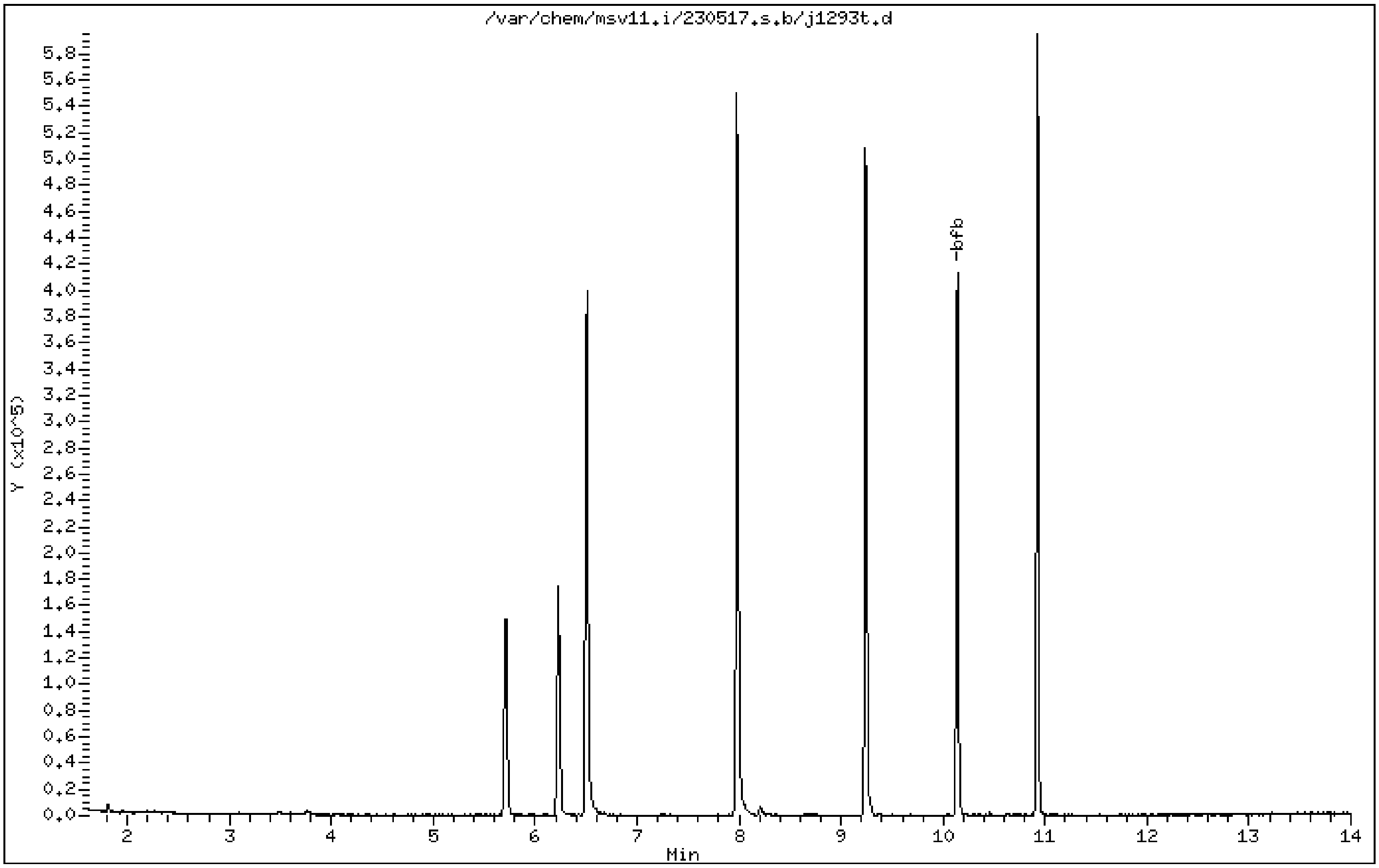
Instrument: msv11.i

Sample Info: BLANK*

Operator: KN1

Column phase: RTX-VMS-30M

Column diameter: 0.25



Data File: /var/chem/msv11.i/230517.s.b/j1293t.d

Date : 17-MAY-2023 09:39

Client ID: V11BFB

Instrument: msv11.i

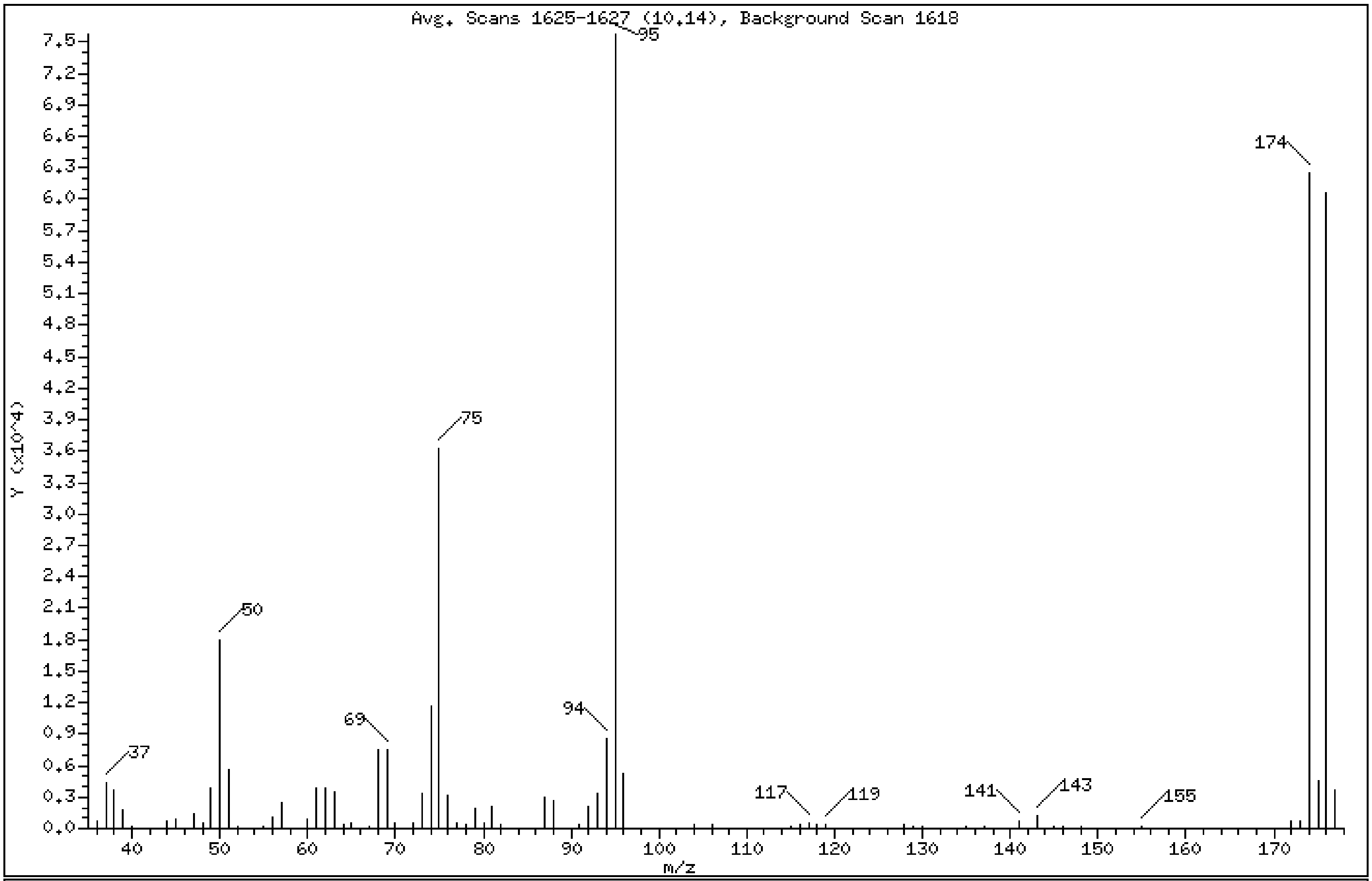
Sample Info: BLANK*

Operator: KN1

Column phase: RTX-VMS-30M

Column diameter: 0.25

1 bfb



m/e	ION ABUNDANCE CRITERIA	% RELATIVE ABUNDANCE
95	Base Peak, 100% relative abundance	100.00
50	15.00 - 40.00% of mass 95	23.68
75	30.00 - 60.00% of mass 95	47.79
96	5.00 - 9.00% of mass 95	6.86
173	Less than 2.00% of mass 174	0.88 (1.06)
174	50.00 - 120.00% of mass 95	82.43
175	4.00 - 9.00% of mass 174	5.92 (7.18)
176	95.00 - 101.00% of mass 174	80.02 (97.07)
177	5.00 - 9.00% of mass 176	4.82 (6.02)

Date : 17-MAY-2023 09:39

Client ID: V11BFB Instrument: msv11.i

Sample Info: BLANK*

Operator: KN1

Column phase: RTX-VMS-30M Column diameter: 0.25

Data File: j1293t.d

Spectrum: Avg. Scans 1625-1627 (10.14), Background Scan 1618

Location of Maximum: 95.00

Number of points: 76

m/z	Y	m/z	Y	m/z	Y	m/z	Y
36.00	699	63.00	3527	91.00	312	141.00	722
37.00	4272	64.00	427	92.00	2016	142.00	50
38.00	3693	65.00	486	93.00	3355	143.00	1144
39.00	1754	67.00	231	94.00	8466	145.00	259
40.00	143	68.00	7458	95.00	75760	146.00	123
44.00	640	69.00	7495	96.00	5195	148.00	135
45.00	827	70.00	549	97.00	51	149.00	67
46.00	80	72.00	453	104.00	264	150.00	63
47.00	1335	73.00	3268	106.00	333	155.00	166
48.00	472	74.00	11731	115.00	106	157.00	52
49.00	3786	75.00	36200	116.00	373	172.00	641
50.00	17936	76.00	3202	117.00	572	173.00	663
51.00	5528	77.00	558	118.00	320	174.00	62456
52.00	195	78.00	339	119.00	355	175.00	4486
55.00	245	79.00	1973	128.00	330	176.00	60624
56.00	1097	80.00	539	129.00	218	177.00	3650
57.00	2383	81.00	2040	130.00	175		
60.00	837	82.00	402	131.00	51		
61.00	3830	87.00	2933	135.00	164		
62.00	3819	88.00	2577	137.00	142		

5A

VOLATILE ORGANICS INSTRUMENT PERFORMANCE CHECK

BROMOFLUOROBENZENE (BFB)

Report No:	223060210		Tune ID:	1000
GC Column:	RTX-VMS-30M ID .25 (mm)		Instrument ID:	MSV11
Injection Vol.:	1.0 (µL)		Lab File ID:	230602/j1686t
Analyst:	KN1		Analytical Batch:	766942
Analysis Date:	06/02/23 Time: 1100		Analytical Method:	EPA 8260B

m / e	***ION ABUNDANCE CRITERIA***	***% Relative Abundance***		
50	15.0 - 40.0% of mass 95	25.23	( )	
75	30.0 - 66.0% of mass 95	49.85	( )	
95	Base Peak, 100% relative abundance	100	( )	
96	5.0 -9.0% of mass 95	6.75	( )	
173	Less than 2.0% of mass 174	.9	(1.12)	1
174	50.0 - 120.0% of mass 95	80.48	( )	
175	4.0 - 9.0% of mass 174	5.89	(7.33)	1
176	95.0 - 101.0% of mass 174	78.27	(97.26)	1
177	5.0 - 9.0% of mass 176	5.43	(6.94)	2

1- Value is % mass 174 2- Value is % mass 176

THIS CHECK APPLIES TO THE FOLLOWING SAMPLES, MS, MSD, BLANKS, AND STANDARDS:

	CLIENT SAMPLE ID	***GCAL SAMPLE ID***	***LAB FILE ID***	***DATE ANALYZED***	***TIME ANALYZED***
1 .	V11STD050	1400	230602/j1687	06/02/23	1123
2 .	LCS2489346	2489346	230602/j1687L	06/02/23	1123
3 .	MB2489345	2489345	230602/j1694	06/02/23	1423
4 .	OT07-EB-060123	22306021001	230602/j1695	06/02/23	1446
5 .	OT07-MW02-060123	22306021002	230602/j1697	06/02/23	1532
6 .	OT07-MW04-060123	22306021003	230602/j1698	06/02/23	1556
7 .	OT07-MW05-060123	22306021004	230602/j1699	06/02/23	1619
8 .	OT07-MW06-060123	22306021005	230602/j1700	06/02/23	1642
9 .	OT07-MW04-060123MS	22306021006	230602/j1705	06/02/23	1838
10 .	OT07-MW04-060123MSD	22306021007	230602/j1706	06/02/23	1902
11 .	V11STD050	1440	230602/j1707	06/02/23	1924

FORM V VOA

Data File: /var/chem/msv11.i/230602.s.b/j1686t.d

Page 1

Date : 02-JUN-2023 11:00

Client ID: V11BFB

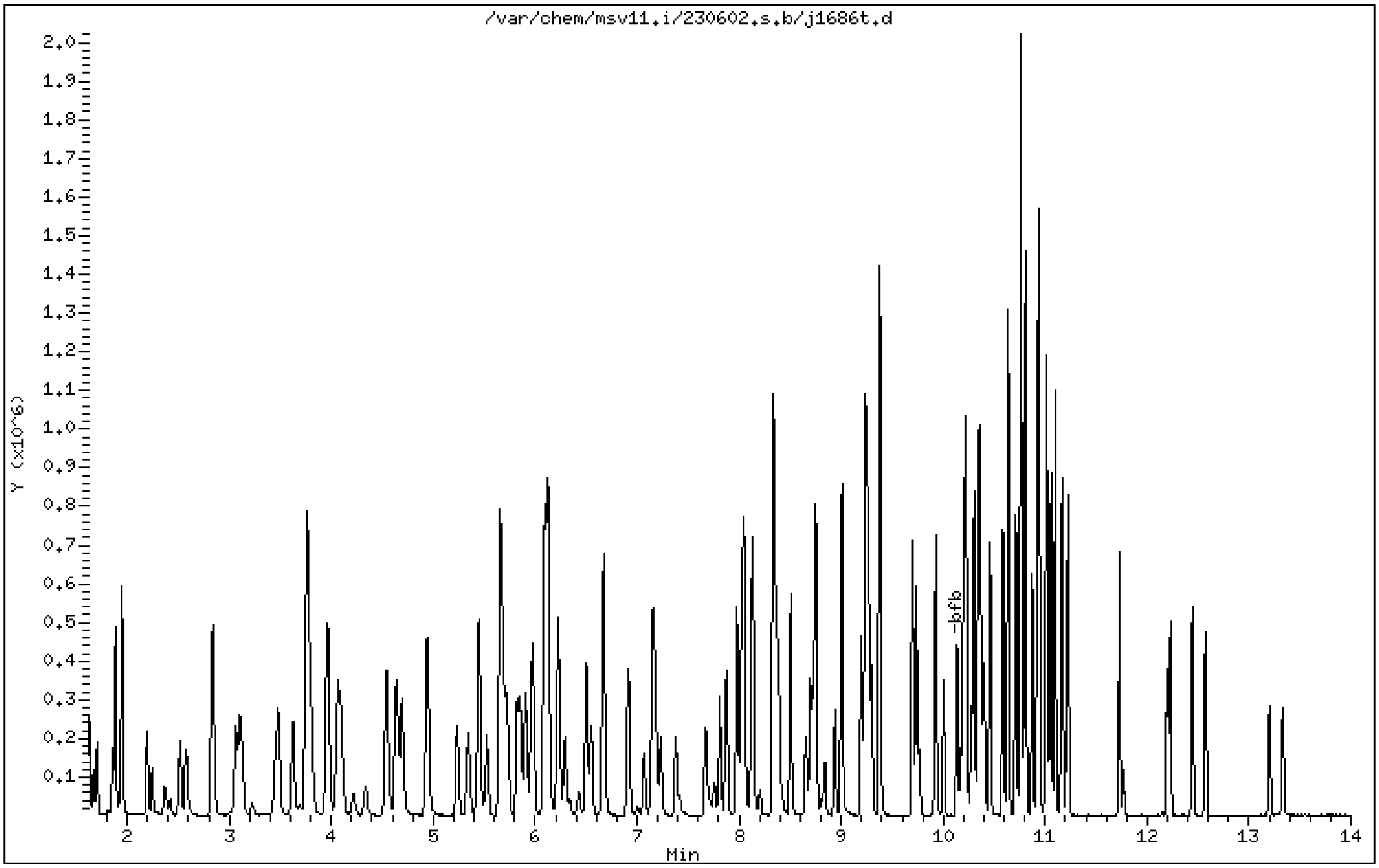
Instrument: msv11.i

Sample Info: 1400*V11STD050*51363

Operator: KN1

Column phase: RTX-VMS-30M

Column diameter: 0.25



Data File: /var/chem/msv11.i/230602.s.b/j1686t.d

Date : 02-JUN-2023 11:00

Client ID: V11BFB

Instrument: msv11.i

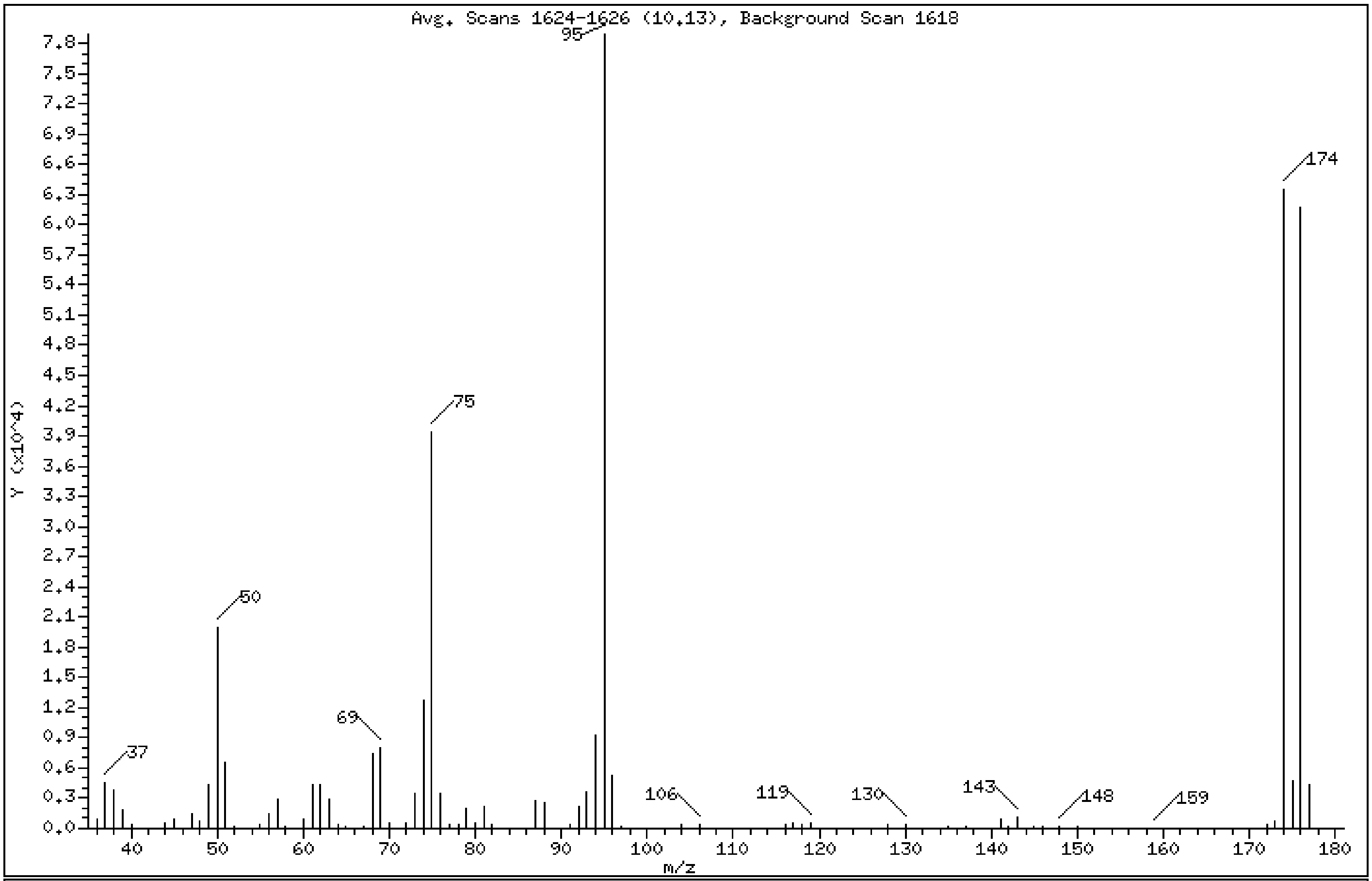
Sample Info: 1400*V11STD050*51363

Operator: KN1

Column phase: RTX-VMS-30M

Column diameter: 0.25

1 bfb



m/e	ION ABUNDANCE CRITERIA	% RELATIVE ABUNDANCE
95	Base Peak, 100% relative abundance	100.00
50	15.00 - 40.00% of mass 95	25.23
75	30.00 - 60.00% of mass 95	49.85
96	5.00 - 9.00% of mass 95	6.75
173	Less than 2.00% of mass 174	0.90 (1.12)
174	50.00 - 120.00% of mass 95	80.48
175	4.00 - 9.00% of mass 174	5.90 (7.33)
176	95.00 - 101.00% of mass 174	78.27 (97.26)
177	5.00 - 9.00% of mass 176	5.43 (6.94)

Date : 02-JUN-2023 11:00

Client ID: V11BFB Instrument: msv11.i

Sample Info: 1400*V11STD050*51363

Operator: KN1

Column phase: RTX-VMS-30M Column diameter: 0.25

Data File: j1686t.d

Spectrum: Avg. Scans 1624-1626 (10.13), Background Scan 1618

Location of Maximum: 95.00

Number of points: 84

m/z	Y	m/z	Y	m/z	Y	m/z	Y
36.00	903	63.00	2874	92.00	2200	143.00	1038
37.00	4454	64.00	296	93.00	3640	144.00	55
38.00	3886	65.00	180	94.00	9343	145.00	114
39.00	1751	67.00	144	95.00	78936	146.00	194
40.00	306	68.00	7432	96.00	5329	148.00	199
43.00	5	69.00	7986	97.00	111	149.00	51
44.00	507	70.00	586	104.00	310	150.00	112
45.00	987	72.00	516	105.00	72	155.00	59
46.00	71	73.00	3438	106.00	408	159.00	74
47.00	1486	74.00	12745	115.00	71	161.00	57
48.00	725	75.00	39352	116.00	352	172.00	414
49.00	4370	76.00	3407	117.00	524	173.00	712
50.00	19912	77.00	444	118.00	381	174.00	63528
51.00	6550	78.00	370	119.00	547	175.00	4657
52.00	223	79.00	2060	128.00	325	176.00	61784
55.00	364	80.00	494	130.00	385	177.00	4289
56.00	1412	81.00	2154	131.00	60	178.00	58
57.00	2844	82.00	447	135.00	94	180.00	56
58.00	148	86.00	51	136.00	50		
60.00	847	87.00	2693	137.00	142		
61.00	4330	88.00	2522	141.00	932		
62.00	4371	91.00	371	142.00	114		

5A
VOLATILE ORGANICS INSTRUMENT PERFORMANCE CHECK
BROMOFLUOROBENZENE (BFB)

Report No:	223060210		Tune ID:	1000
GC Column:	RTX-VMS-30M ID .25	(mm)	Instrument ID:	MSV15
Injection Vol.:	1.0	(µL)	Lab File ID:	230517/c0227t
Analyst:	CJR		Analytical Batch:	766139
Analysis Date:	05/17/23 Time: 1337		Analytical Method:	EPA 8260B

m / e	***ION ABUNDANCE CRITERIA***	***% Relative Abundance***		
50	15.0 - 40.0% of mass 95	17.86	( )	
75	30.0 - 66.0% of mass 95	50.49	( )	
95	Base Peak, 100% relative abundance	100	( )	
96	5.0 -9.0% of mass 95	6.84	( )	
173	Less than 2.0% of mass 174	.93	(1)	1
174	50.0 - 120.0% of mass 95	93.38	( )	
175	4.0 - 9.0% of mass 174	6.6	(7.07)	1
176	95.0 - 101.0% of mass 174	89.76	(96.13)	1
177	5.0 - 9.0% of mass 176	5.85	(6.52)	2

1- Value is % mass 174
2- Value is % mass 176

THIS CHECK APPLIES TO THE FOLLOWING SAMPLES, MS, MSD, BLANKS, AND STANDARDS:

	CLIENT SAMPLE ID	***GCAL SAMPLE ID***	***LAB FILE ID***	***DATE ANALYZED***	***TIME ANALYZED***
1 .	V15STD001	1203	230517/c0228	05/17/23	1428
2 .	V15STD005	1204	230517/c0229	05/17/23	1447
3 .	V15STD010	1205	230517/c0230	05/17/23	1505
4 .	V15STD020	1206	230517/c0231	05/17/23	1524
5 .	V15STD050	1207	230517/c0232	05/17/23	1543
6 .	V15STD100	1208	230517/c0233	05/17/23	1601
7 .	V15STD200	1209	230517/c0234	05/17/23	1620
8 .	V15ICV050	1600	230517/c0238	05/17/23	1735

FORM V VOA

Data File: /var/chem/msv15.i/230517.s.b/c0227t.d

Date : 17-MAY-2023 13:37

Client ID: V15BFB

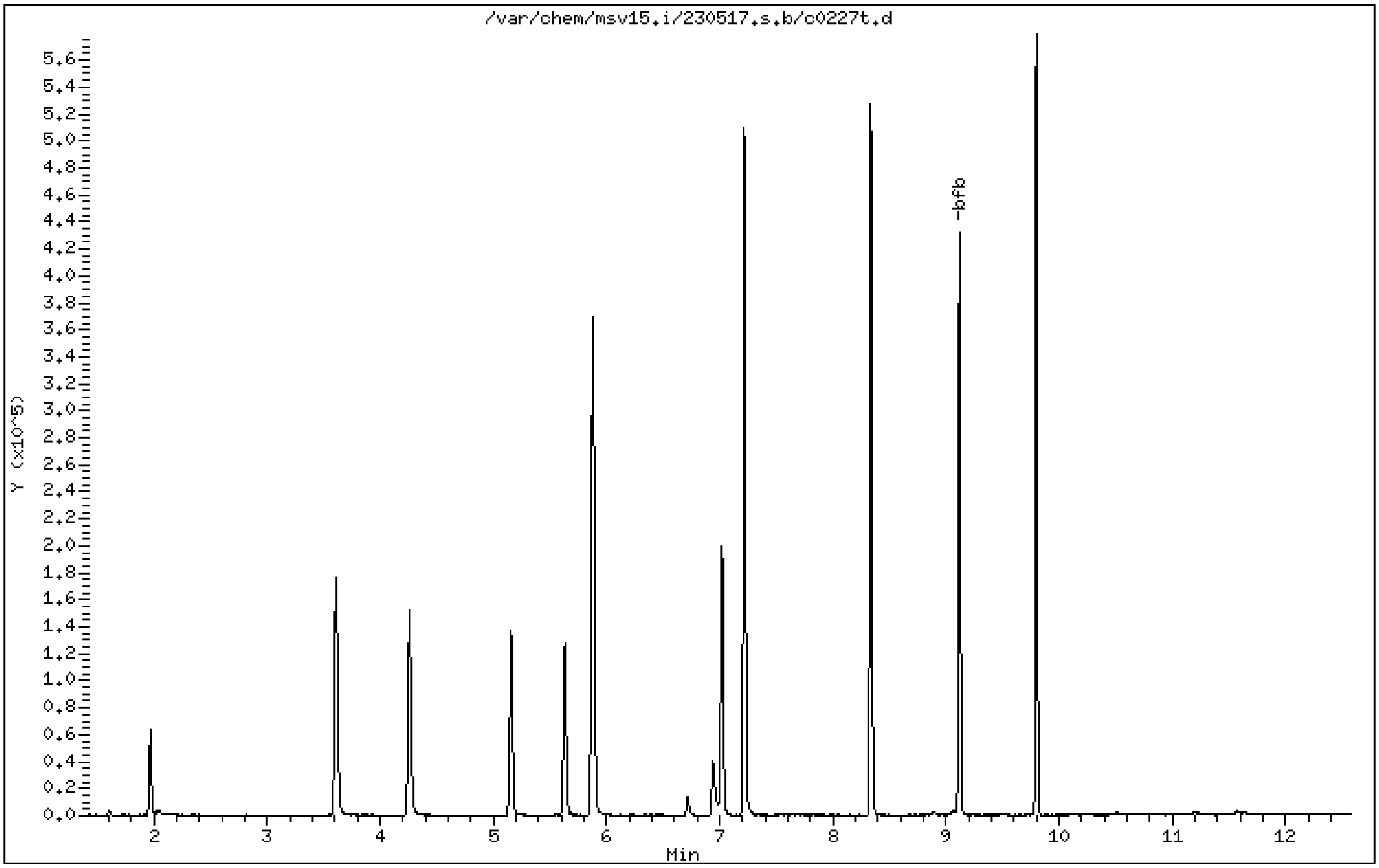
Instrument: msv15.i

Sample Info: HANDSPIKE*

Operator: CJR

Column phase: RTX-VMS-30M

Column diameter: 0.25



Data File: /var/chem/msv15.i/230517.s.b/c0227t.d

Date : 17-MAY-2023 13:37

Client ID: V15BFB

Instrument: msv15.i

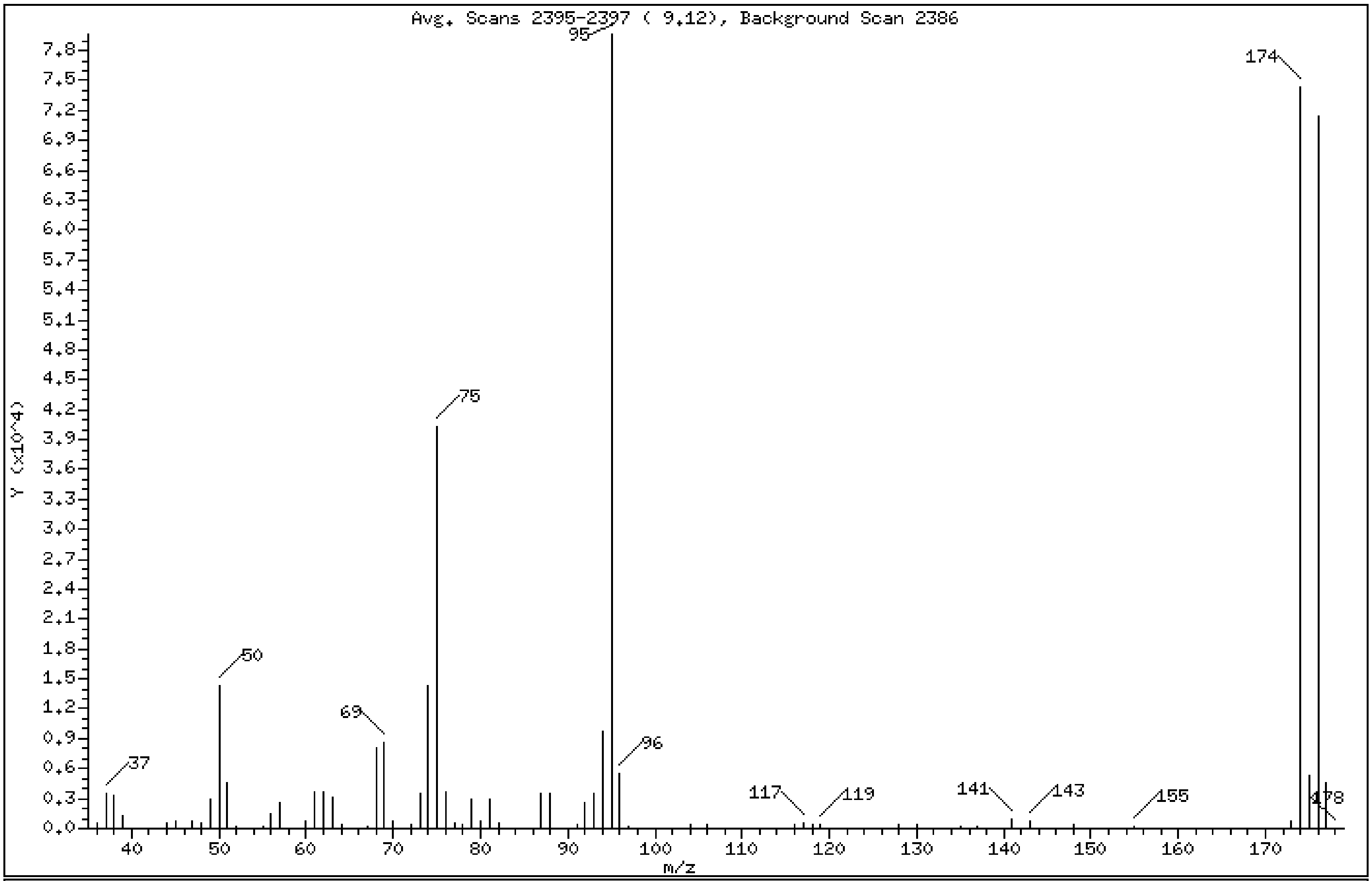
Sample Info: HANDSPIKE*

Operator: CJR

Column phase: RTX-VMS-30M

Column diameter: 0.25

1 bfb



m/e	ION ABUNDANCE CRITERIA	% RELATIVE ABUNDANCE
95	Base Peak, 100% relative abundance	100.00
50	15.00 - 40.00% of mass 95	17.86
75	30.00 - 60.00% of mass 95	50.49
96	5.00 - 9.00% of mass 95	6.84
173	Less than 2.00% of mass 174	0.94 (1.00)
174	50.00 - 120.00% of mass 95	93.38
175	4.00 - 9.00% of mass 174	6.60 (7.07)
176	95.00 - 101.00% of mass 174	89.76 (96.13)
177	5.00 - 9.00% of mass 176	5.86 (6.52)

Date : 17-MAY-2023 13:37

Client ID: V15BFB Instrument: msv15.i

Sample Info: HANDSPIKE*

Operator: CJR

Column phase: RTX-VMS-30M Column diameter: 0.25

Data File: c0227t.d
Spectrum: Avg. Scans 2395-2397 (9.12), Background Scan 2386
Location of Maximum: 95.00
Number of points: 71

m/z	Y	m/z	Y	m/z	Y	m/z	Y
36.00	583	62.00	3625	86.00	64	130.00	285
37.00	3521	63.00	3025	87.00	3493	131.00	53
38.00	3245	64.00	289	88.00	3484	135.00	111
39.00	1293	67.00	213	91.00	351	137.00	225
40.00	55	68.00	7996	92.00	2530	141.00	859
44.00	530	69.00	8577	93.00	3442	143.00	810
45.00	716	70.00	692	94.00	9736	146.00	76
47.00	769	72.00	403	95.00	79664	148.00	284
48.00	485	73.00	3476	96.00	5450	155.00	205
49.00	2961	74.00	14223	97.00	108	157.00	64
50.00	14227	75.00	40216	104.00	386	172.00	68
51.00	4506	76.00	3609	106.00	392	173.00	745
52.00	157	77.00	552	116.00	280	174.00	74392
55.00	251	78.00	386	117.00	554	175.00	5257
56.00	1394	79.00	2900	118.00	300	176.00	71512
57.00	2508	80.00	819	119.00	433	177.00	4665
60.00	808	81.00	2969	128.00	278	178.00	61
61.00	3669	82.00	602	129.00	58		

5A

VOLATILE ORGANICS INSTRUMENT PERFORMANCE CHECK

BROMOFLUOROBENZENE (BFB)

Report No:	223060210	Tune ID:	1000
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230605/c0728
Analyst:	CJR	Analytical Batch:	767028
Analysis Date:	06/05/23 Time: 0925	Analytical Method:	EPA 8260B

m / e	***ION ABUNDANCE CRITERIA***	***% Relative Abundance***		
50	15.0 - 40.0% of mass 95	19.45	( )	
75	30.0 - 66.0% of mass 95	52.83	( )	
95	Base Peak, 100% relative abundance	100	( )	
96	5.0 -9.0% of mass 95	6.8	( )	
173	Less than 2.0% of mass 174	1	(1.09)	1
174	50.0 - 120.0% of mass 95	91.86	( )	
175	4.0 - 9.0% of mass 174	6.7	(7.3)	1
176	95.0 - 101.0% of mass 174	89.52	(97.46)	1
177	5.0 - 9.0% of mass 176	5.71	(6.38)	2

1- Value is % mass 174   2- Value is % mass 176

THIS CHECK APPLIES TO THE FOLLOWING SAMPLES, MS, MSD, BLANKS, AND STANDARDS:

	CLIENT SAMPLE ID	***GCAL SAMPLE ID***	***LAB FILE ID***	***DATE ANALYZED***	***TIME ANALYZED***
1 .	V15STD050	1400	230605/c0730	06/05/23	1045
2 .	LCS2489808	2489808	230605/c0730L	06/05/23	1045
3 .	MB2489807	2489807	230605/c0737	06/05/23	1257
4 .	OT07-MW-806-060123	22306021008	230605/c0742	06/05/23	1430
5 .	V15STD050	1440	230605/c0761	06/05/23	2027

FORM V VOA

Data File: /var/chem/msv15.i/230605.s.b/c0728.d

Date : 05-JUN-2023 09:25

Client ID: V15BFB

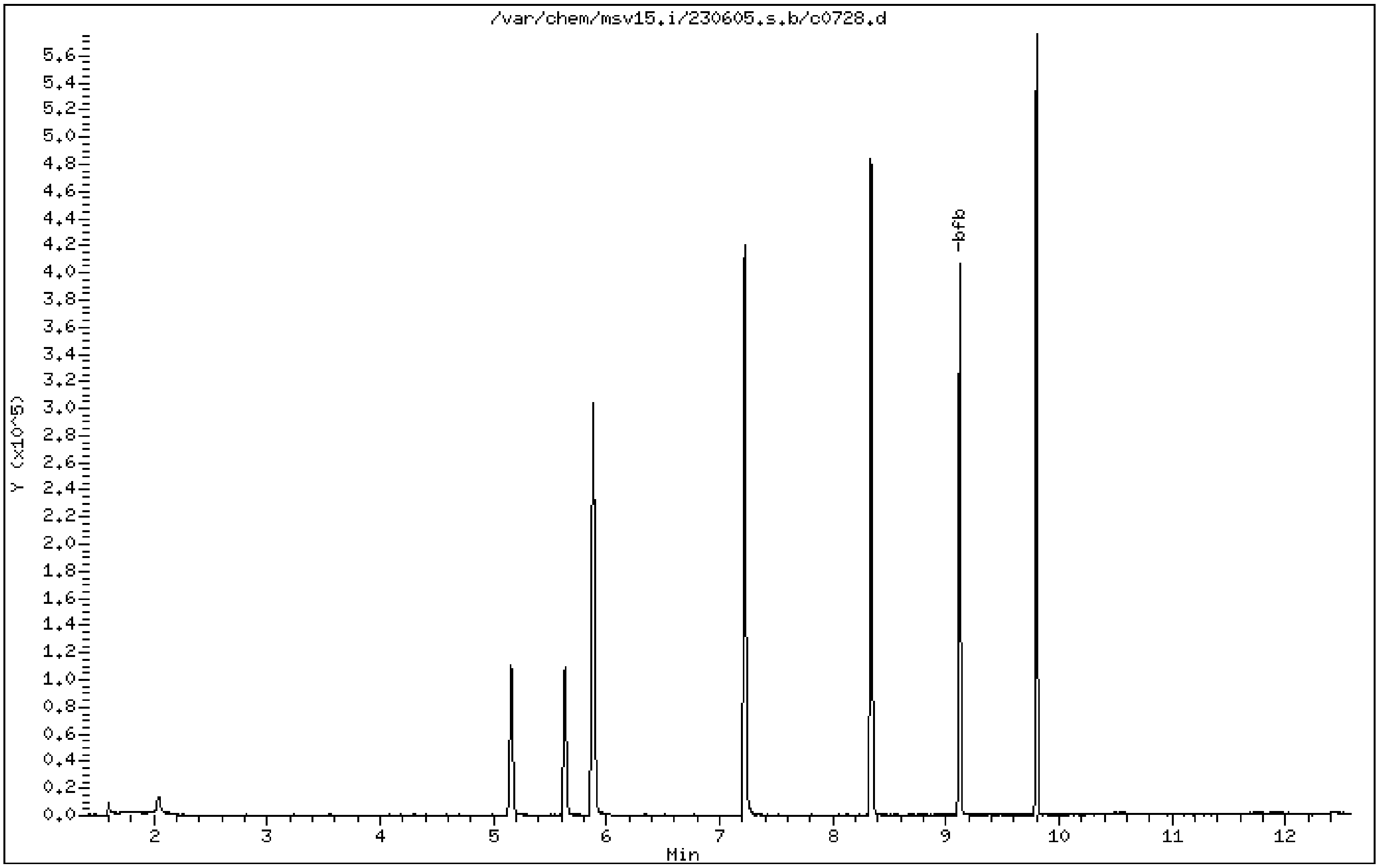
Instrument: msv15.i

Sample Info: 1000*V15BFB

Operator: CJR

Column phase: RTX-VMS-30M

Column diameter: 0.25



Data File: /var/chem/msv15.i/230605.s.b/c0728.d

Date : 05-JUN-2023 09:25

Client ID: V15BFB

Instrument: msv15.i

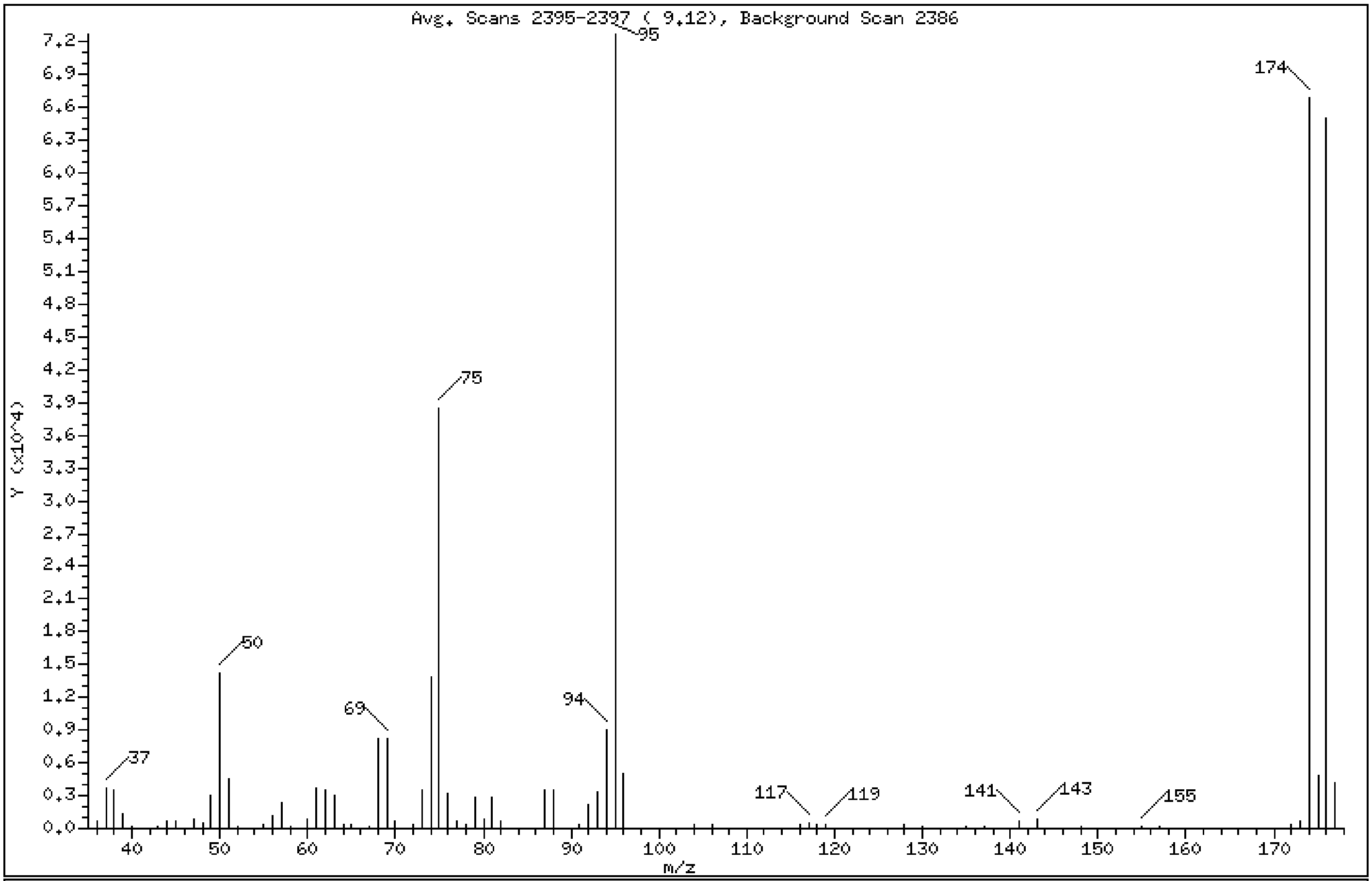
Sample Info: 1000*V15BFB

Operator: CJR

Column phase: RTX-VMS-30M

Column diameter: 0.25

1 bfb



m/e	ION ABUNDANCE CRITERIA	% RELATIVE ABUNDANCE
95	Base Peak, 100% relative abundance	100.00
50	15.00 - 40.00% of mass 95	19.45
75	30.00 - 60.00% of mass 95	52.83
96	5.00 - 9.00% of mass 95	6.80
173	Less than 2.00% of mass 174	1.00 (1.09)
174	50.00 - 120.00% of mass 95	91.86
175	4.00 - 9.00% of mass 174	6.71 (7.30)
176	95.00 - 101.00% of mass 174	89.53 (97.46)
177	5.00 - 9.00% of mass 176	5.71 (6.38)

Date : 05-JUN-2023 09:25

Client ID: V15BFB Instrument: msv15.i

Sample Info: 1000*V15BFB

Operator: CJR

Column phase: RTX-VMS-30M Column diameter: 0.25

Data File: c0728.d

Spectrum: Avg. Scans 2395-2397 (9.12), Background Scan 2386

Location of Maximum: 95.00

Number of points: 73

m/z	Y	m/z	Y	m/z	Y	m/z	Y
36.00	599	61.00	3691	82.00	647	130.00	232
37.00	3747	62.00	3538	87.00	3488	131.00	59
38.00	3515	63.00	2993	88.00	3509	135.00	121
39.00	1300	64.00	317	91.00	319	137.00	188
40.00	173	65.00	254	92.00	2125	141.00	740
43.00	110	67.00	212	93.00	3422	143.00	805
44.00	630	68.00	8204	94.00	8975	147.00	66
45.00	711	69.00	8256	95.00	72752	148.00	232
47.00	781	70.00	730	96.00	4944	155.00	204
48.00	494	72.00	411	97.00	71	157.00	120
49.00	3086	73.00	3537	104.00	356	172.00	415
50.00	14152	74.00	13818	105.00	66	173.00	729
51.00	4559	75.00	38432	106.00	350	174.00	66832
52.00	150	76.00	3254	116.00	292	175.00	4881
55.00	290	77.00	617	117.00	511	176.00	65136
56.00	1137	78.00	359	118.00	272	177.00	4155
57.00	2350	79.00	2768	119.00	389		
58.00	90	80.00	807	128.00	294		
60.00	777	81.00	2871	129.00	57		

5A

VOLATILE ORGANICS INSTRUMENT PERFORMANCE CHECK

BROMOFLUOROBENZENE (BFB)

Report No:	223060210	Tune ID:	1000
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230606/c0766d
Analyst:	CJR	Analytical Batch:	767081
Analysis Date:	06/06/23 Time: 0950	Analytical Method:	EPA 8260B

m / e	*ION ABUNDANCE CRITERIA*	*% Relative Abundance*		
50	15.0 - 40.0% of mass 95	17.75	( )	
75	30.0 - 66.0% of mass 95	50.35	( )	
95	Base Peak, 100% relative abundance	100	( )	
96	5.0 -9.0% of mass 95	6.73	( )	
173	Less than 2.0% of mass 174	.77	(.85)	1
174	50.0 - 120.0% of mass 95	90.99	( )	
175	4.0 - 9.0% of mass 174	6.65	(7.31)	1
176	95.0 - 101.0% of mass 174	87.12	(95.75)	1
177	5.0 - 9.0% of mass 176	5.95	(6.83)	2

1- Value is % mass 174 2- Value is % mass 176

THIS CHECK APPLIES TO THE FOLLOWING SAMPLES, MS, MSD, BLANKS, AND STANDARDS:

	CLIENT SAMPLE ID	*GCAL SAMPLE ID*	*LAB FILE ID*	*DATE ANALYZED*	*TIME ANALYZED*
1 .	V15STD050	1400	230606/c0768d	06/06/23	1053
2 .	LCS2489979	2489979	230606/c0768Ld	06/06/23	1053
3 .	LCSD2489980	2489980	230606/c0769	06/06/23	1112
4 .	MB2489978	2489978	230606/c0774d	06/06/23	1246
5 .	OT07-TB-01-060124	22306021009	230606/c0777	06/06/23	1342
6 .	V15STD050	1440	230606/c0796	06/06/23	1939

FORM V VOA

Data File: /var/chem/msv15.i/230606.s.b/c0766d.d

Page 1

Date : 06-JUN-2023 09:50

Client ID: V15BFB

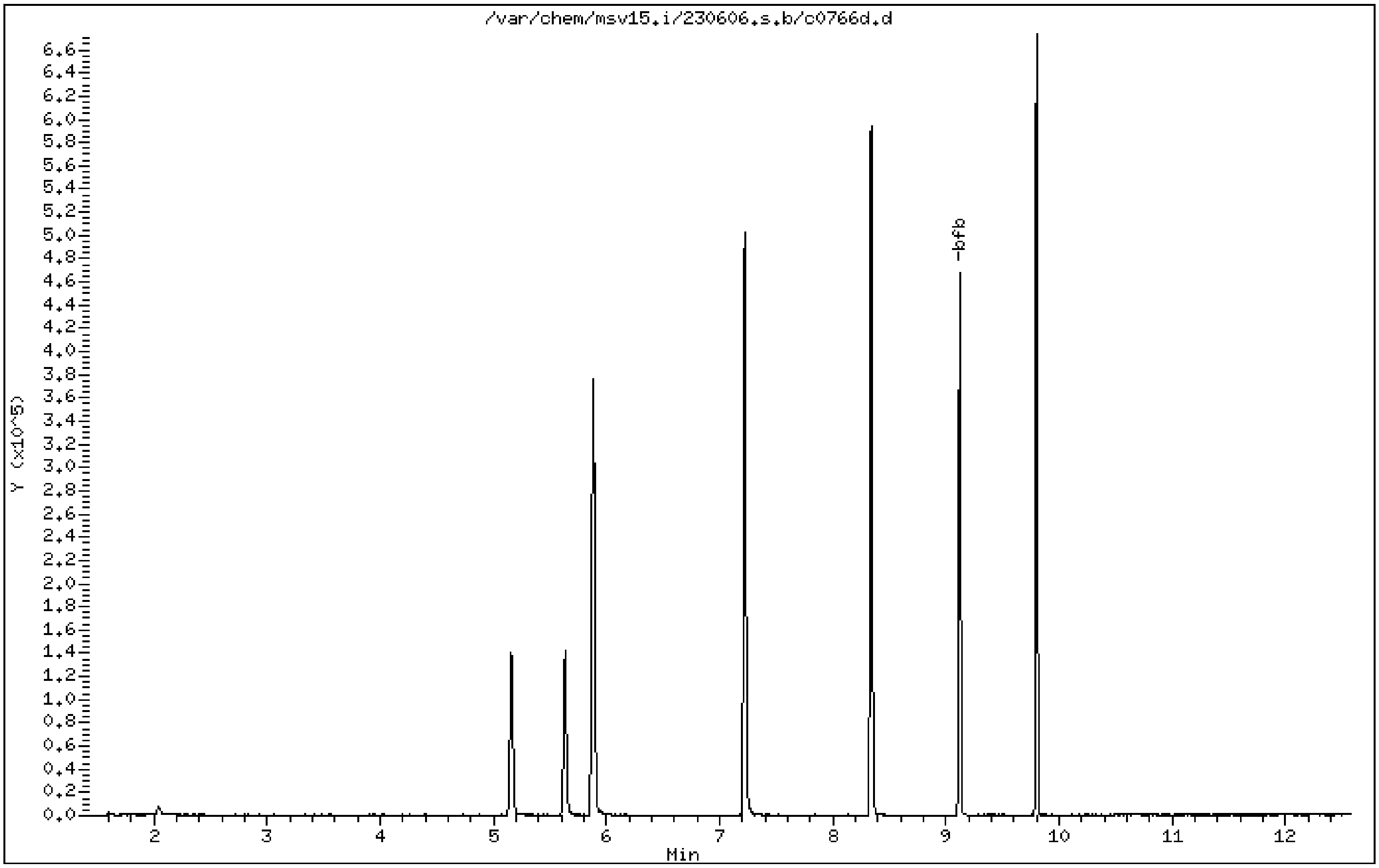
Instrument: msv15.i

Sample Info: 1000*V15BFB

Operator: CJR

Column phase: RTX-VMS-30M

Column diameter: 0.25



Data File: /var/chem/msv15.i/230606.s.b/c0766d.d

Date : 06-JUN-2023 09:50

Client ID: V15BFB

Instrument: msv15.i

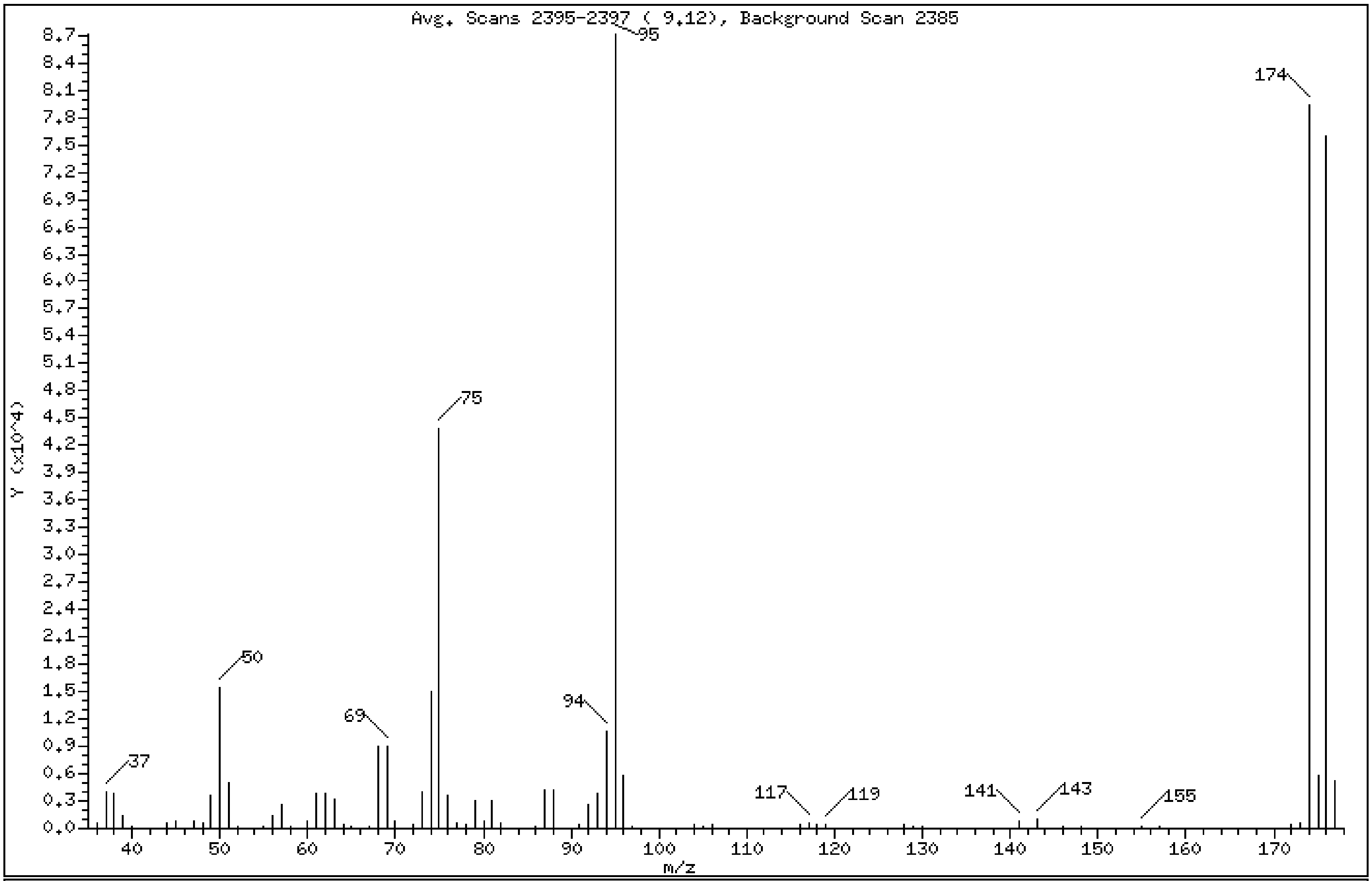
Sample Info: 1000*V15BFB

Operator: CJR

Column phase: RTX-VMS-30M

Column diameter: 0.25

1 bfb



m/e	ION ABUNDANCE CRITERIA	% RELATIVE ABUNDANCE
95	Base Peak, 100% relative abundance	100.00
50	15.00 - 40.00% of mass 95	17.75
75	30.00 - 60.00% of mass 95	50.35
96	5.00 - 9.00% of mass 95	6.73
173	Less than 2.00% of mass 174	0.77 (0.85)
174	50.00 - 120.00% of mass 95	90.99
175	4.00 - 9.00% of mass 174	6.65 (7.31)
176	95.00 - 101.00% of mass 174	87.13 (95.75)
177	5.00 - 9.00% of mass 176	5.95 (6.83)

Date : 06-JUN-2023 09:50

Client ID: V15BFB Instrument: msv15.i

Sample Info: 1000*V15BFB

Operator: CJR

Column phase: RTX-VMS-30M Column diameter: 0.25

Data File: c0766d.d

Spectrum: Avg. Scans 2395-2397 (9.12), Background Scan 2385

Location of Maximum: 95.00

Number of points: 74

m/z	Y	m/z	Y	m/z	Y	m/z	Y
36.00	697	61.00	3792	82.00	634	129.00	139
37.00	3923	62.00	3893	86.00	104	130.00	299
38.00	3768	63.00	3211	87.00	4248	135.00	56
39.00	1373	64.00	357	88.00	4147	137.00	87
40.00	132	65.00	208	91.00	342	141.00	859
43.00	5	67.00	261	92.00	2554	143.00	1014
44.00	567	68.00	9070	93.00	3846	146.00	175
45.00	790	69.00	9103	94.00	10562	148.00	226
47.00	870	70.00	780	95.00	87144	155.00	256
48.00	577	72.00	474	96.00	5869	157.00	116
49.00	3545	73.00	3974	97.00	207	159.00	50
50.00	15472	74.00	15080	104.00	457	172.00	436
51.00	5077	75.00	43872	105.00	115	173.00	672
52.00	190	76.00	3594	106.00	397	174.00	79288
55.00	243	77.00	529	116.00	323	175.00	5793
56.00	1450	78.00	425	117.00	546	176.00	75920
57.00	2609	79.00	3083	118.00	318	177.00	5189
58.00	136	80.00	847	119.00	452		
60.00	743	81.00	3054	128.00	324		

EPA 8260B DOD

Form 6A

Calibrations

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

Report No:	223060210			Instrument ID:	MSV11	**LabQC ID - FileID - Conc**	1203 ~ 230517/j1304 ~ 1
GC Column:	RTX-VMS-30M	ID	.25 (mm)	Analyst:	KN1	1204 ~ 230517/j1305 ~ 5	1205 ~ 230517/j1306 ~ 10
Calib. Date 1:	05/17/23	Time 1:	1206	Analytical Batch:	766136	1206 ~ 230517/j1299 ~ 20	1207 ~ 230517/j1300 ~ 50
Calib. Date 2:	05/17/23	Time 2:	1502	Analytical Method:	EPA 8260B	1208 ~ 230517/j1301 ~ 100	1209 ~ 230517/j1302 ~ 200

ANALYTE	*1201*	*1202*	*1203*	*1204*	*1205*	*1206*	*1207*	*1208*	*1209*	*RF / m / A*	*b / B*	*C*	*FIT*	*TYPE*	
1,1,1,2-Tetrachloroethane			0.534	0.627	0.620	0.641	0.678	0.665	0.688	0.636			8.123	A	
1,1,1-Trichloroethane			0.367	0.417	0.425	0.434	0.451	0.439	0.455	0.427			6.938	A	
1,1,2,2-Tetrachloroethane			0.835	0.925	0.901	0.927	0.891	0.842	0.916	0.891			4.271	A	
1,1,2-Trichloroethane			0.626	0.652	0.636	0.651	0.661	0.654	0.669	0.650			2.228	A	
1,1-Dichloroethane			0.571	0.640	0.638	0.639	0.629	0.617	0.649	0.626			4.192	A	
1,1-Dichloroethene			0.209	0.225	0.226	0.219	0.223	0.219	0.225	0.221			2.750	A	
1,1-Dichloropropene			0.303	0.343	0.364	0.368	0.376	0.369	0.385	0.358			7.674	A	
1,2,3-Trichlorobenzene (RSP			3451	10752	21139	44972	127711	274487	595849	0.918	0.000		0.997	W	
1,2,3-Trichlorobenzene			1.259	0.798	0.787	0.865	0.900	0.877	0.953						
1,2,3-Trichloropropane			1.027	1.091	1.139	1.174	1.183	1.191	1.306	1.159			7.542	A	
1,2,4-Trichlorobenzene			1.224	0.857	0.825	0.876	0.926	0.904	1.007	0.946			14.36	A	
1,2,4-Trimethylbenzene			2.165	2.517	2.604	2.624	2.708	2.644	2.867	2.590			8.345	A	
1,2-Dibromo-3-chloropropan			0.184	0.192	0.212	0.208	0.223	0.223	0.246	0.213			9.855	A	
1,2-Dibromoethane			0.630	0.639	0.639	0.673	0.698	0.693	0.711	0.669			4.919	A	
1,2-Dichlorobenzene			1.792	1.521	1.456	1.471	1.512	1.456	1.546	1.536			7.672	A	
1,2-Dichloroethane			0.417	0.430	0.440	0.470	0.471	0.456	0.466	0.450			4.728	A	
1,2-Dichloroethane-d4			0.316	0.321	0.322	0.348	0.347	0.335	0.340	0.333			3.932	A	
1,2-Dichloroethene (total)			0.487	0.468	0.484	0.476	0.482	0.471	0.497	0.481			2.066	A	
1,2-Dichloropropane			0.337	0.373	0.364	0.366	0.376	0.371	0.384	0.368			4.035	A	
1,3,5-Trimethylbenzene			2.182	2.578	2.506	2.661	2.635	2.607	2.868	2.577			8.038	A	
1,3-Butadiene				0.424	0.433	0.453	0.415	0.416	0.413	0.426			3.598	A	
1,3-Dichlorobenzene			1.922	1.525	1.515	1.530	1.534	1.486	1.596	1.587			9.550	A	
1,3-Dichloropropane			1.025	1.044	1.077	1.100	1.144	1.128	1.161	1.097			4.665	A	
1,3-Dichloropropylene			0.315	0.364	0.383	0.408	0.435	0.434	0.449	0.398			11.95	A	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines
to allow for evaluation against minimum RRF

RF = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

				LabQC ID - FileID - Conc	1203 ~ 230517/j1304 ~ 1
Report No:	223060210	Instrument ID:	MSV11	1204 ~ 230517/j1305 ~ 5	1205 ~ 230517/j1306 ~ 10
GC Column:	RTX-VMS-30M ID .25 (mm)	Analyst:	KN1	1206 ~ 230517/j1299 ~ 20	1207 ~ 230517/j1300 ~ 50
Calib. Date 1:	05/17/23 Time 1: 1206	Analytical Batch:	766136	1208 ~ 230517/j1301 ~ 100	1209 ~ 230517/j1302 ~ 200
Calib. Date 2:	05/17/23 Time 2: 1502	Analytical Method:	EPA 8260B		

ANALYTE	***1201***	***1202***	***1203***	***1204***	***1205***	***1206***	***1207***	***1208***	***1209***	***RF / m / A***	***b / B***	***C***	***FIT***	***TYPE***
1,4 Dioxane			0.002	0.003	0.003	0.003	0.003	0.003	0.004	0.003			13.74	A
1,4-Dichlorobenzene			1.892	1.616	1.524	1.546	1.532	1.496	1.584	1.599			8.466	A
1-Bromo-2-Chloroethane			0.437	0.460	0.473	0.491	0.497	0.498	0.512	0.481			5.389	A
1-Chlorohexane			0.910	0.672	0.749	0.792	0.856	0.851	0.881	0.816			10.20	A
1-Nitropropane (RSP)			278	2101	4015	9372	27878	54753	109437	0.177	0.013		0.996	W
1-Nitropropane			0.091	0.140	0.136	0.169	0.194	0.177	0.172					
2,2,4-Trimethylpentane				1.333	1.394	1.483	1.518	1.541	1.728	1.500			9.109	A
2,2-Dichloropropane			0.382	0.384	0.387	0.403	0.404	0.396	0.408	0.395			2.658	A
2,3-Dichloro-1-propene			0.397	0.412	0.423	0.438	0.462	0.461	0.480	0.439			6.902	A
2-Butanone			0.245	0.275	0.283	0.307	0.309	0.308	0.317	0.292			8.775	A
2-Chloroethylvinyl ether (RS			1057	5825	13618	26959	84008	177922	392563	0.045	0.086		0.997	W
2-Chloroethylvinyl ether			0.025	0.030	0.036	0.038	0.045	0.044	0.046					
2-Chlorotoluene			2.284	2.422	2.368	2.404	2.382	2.417	2.683	2.423			5.118	A
2-Hexanone			0.796	0.855	0.893	0.944	1.013	0.988	1.042	0.933			9.568	A
2-Nitropropane			0.249	0.327	0.285	0.308	0.352	0.358	0.373	0.322			13.72	A
4-Bromofluorobenzene				0.650	0.673	0.724	0.739	0.746	0.798	0.722			7.398	A
4-Chlorotoluene			2.285	2.276	2.248	2.292	2.301	2.241	2.491	2.305			3.688	A
4-Isopropyltoluene			2.574	2.775	2.763	2.807	2.793	2.737	2.932	2.769			3.833	A
4-Methyl-2-pentanone			0.934	1.031	1.121	1.156	1.219	1.203	1.172	1.119			9.186	A
Acetone			0.323	0.333	0.333	0.340	0.334	0.321	0.338	0.332			2.178	A
Acetonitrile			0.063	0.064	0.060	0.065	0.066	0.065	0.065	0.064			3.419	A
Acrolein			0.068	0.071	0.070	0.077	0.077	0.076	0.079	0.074			5.913	A
Acrylonitrile			0.147	0.152	0.166	0.184	0.182	0.178	0.186	0.171			9.429	A
Allyl chloride			0.376	0.377	0.402	0.414	0.418	0.385	0.318	0.384			8.789	A

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines
to allow for evaluation against minimum RRF

$\overline{RF}$ = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

				LabQC ID - FileID - Conc	1203 ~ 230517/j1304 ~ 1
Report No:	223060210	Instrument ID:	MSV11	1204 ~ 230517/j1305 ~ 5	1205 ~ 230517/j1306 ~ 10
GC Column:	RTX-VMS-30M ID .25 (mm)	Analyst:	KN1	1206 ~ 230517/j1299 ~ 20	1207 ~ 230517/j1300 ~ 50
Calib. Date 1:	05/17/23 Time 1: 1206	Analytical Batch:	766136	1208 ~ 230517/j1301 ~ 100	1209 ~ 230517/j1302 ~ 200
Calib. Date 2:	05/17/23 Time 2: 1502	Analytical Method:	EPA 8260B		

ANALYTE	*1201*	*1202*	*1203*	*1204*	*1205*	*1206*	*1207*	*1208*	*1209*	*RF / m / A*	*b / B*	*C*	*FIT*	*TYPE*	
Benzene			0.962	1.049	1.050	1.093	1.074	1.056	1.101	1.055			4.345	A	
Bromobenzene			1.364	1.407	1.347	1.356	1.361	1.341	1.501	1.383			4.096	A	
Bromochloromethane			0.124	0.147	0.143	0.156	0.138	0.125	0.129	0.138			8.611	A	
Bromodichloromethane			0.331	0.342	0.344	0.357	0.365	0.367	0.379	0.355			4.737	A	
Bromoform			0.356	0.447	0.447	0.475	0.516	0.531		0.462			13.55	A	
Bromomethane (RSP)			1908	7963	14337	24650	57857	121938	270172	0.157	-0.015		0.997	W	
Bromomethane			0.229	0.202	0.188	0.175	0.155	0.150	0.160						
Carbon disulfide (RSP)			8574	27220	51825	91911	246086	529394	1153996	0.666	-0.008		0.999	W	
Carbon disulfide			1.027	0.690	0.679	0.653	0.661	0.650	0.683						
Carbon tetrachloride			0.291	0.373	0.377	0.397	0.395	0.395	0.399	0.375			10.32	A	
Chlorobenzene			1.862	1.845	1.815	1.847	1.903	1.863	1.899	1.862			1.652	A	
Chloroethane			0.248	0.214	0.202	0.216	0.211	0.207		0.217			7.544	A	
Chloroform			0.410	0.474	0.486	0.479	0.484	0.469	0.491	0.470			5.845	A	
Chloromethane			0.518	0.535	0.515	0.551	0.527	0.504	0.523	0.525			2.873	A	
Chloroprene				0.510	0.552	0.562	0.578	0.589	0.613	0.567			6.185	A	
Cyclohexane			0.479	0.560	0.593	0.607	0.636	0.645	0.685	0.600			11.13	A	
Cyclohexanone			0.284	0.277	0.299	0.298	0.313	0.322	0.330	0.303			6.398	A	
Dibromochloromethane			0.579	0.640	0.670	0.711	0.759	0.758	0.800	0.702			11.03	A	
Dibromofluoromethane			0.199	0.242	0.251	0.269	0.265	0.256	0.263	0.249			9.683	A	
Dibromomethane			0.146	0.168	0.176	0.184	0.182	0.176	0.185	0.174			7.920	A	
Dichlorodifluoromethane			0.298	0.288	0.283	0.320	0.302	0.297	0.304	0.299			3.960	A	
Dicyclopentadiene			2.654	3.301	3.387	3.361	3.422	3.395	3.723	3.320			9.746	A	
Diethyl ether			0.273	0.297	0.310	0.298	0.304	0.303	0.317	0.300			4.682	A	
Ethyl acetate				0.450	0.483	0.504	0.512	0.506	0.525	0.497			5.358	A	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines
to allow for evaluation against minimum RRF

RF = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

Report No: 223060210	Instrument ID: MSV11	**LabQC ID - FileID - Conc**	1203 ~ 230517/j1304 ~ 1
GC Column: RTX-VMS-30M ID .25 (mm)	Analyst: KN1	1204 ~ 230517/j1305 ~ 5	1205 ~ 230517/j1306 ~ 10
Calib. Date 1: 05/17/23 Time 1: 1206	Analytical Batch: 766136	1206 ~ 230517/j1299 ~ 20	1207 ~ 230517/j1300 ~ 50
Calib. Date 2: 05/17/23 Time 2: 1502	Analytical Method: EPA 8260B	1208 ~ 230517/j1301 ~ 100	1209 ~ 230517/j1302 ~ 200

ANALYTE	*1201*	*1202*	*1203*	*1204*	*1205*	*1206*	*1207*	*1208*	*1209*	*$\overline{RF}$ / m / A*	*b / B*	*C*	*FIT*	*TYPE*	
Ethyl methacrylate			0.701	0.735	0.862	0.902	0.952	0.984	0.986	0.874			13.28	A	
Ethyl tert-butyl ether (ETBE)				0.977	1.046	1.074	1.107	1.113	1.194	1.085			6.704	A	
Ethylbenzene			0.865	0.963	0.979	0.976	1.044	1.010	1.049	0.984			6.323	A	
Hexachlorobutadiene				0.443	0.393	0.427	0.422	0.411	0.463	0.426			5.697	A	
Isobutyl alcohol			0.017	0.015	0.021	0.018	0.021	0.020	0.023	0.019			13.53	A	
Isopropylbenzene (Cumene)			2.202	2.398	2.616	2.743	2.975	2.981	3.211	2.732			12.98	A	
Methacrylonitrile			0.270	0.282	0.283	0.319	0.353	0.317	0.315	0.306			9.465	A	
Methyl Acetate			0.305	0.345	0.360	0.387	0.382	0.367	0.368	0.359			7.728	A	
Methyl iodide (RSP)				5821	13064	29806	101184	244432	492906	0.299	0.069		0.997	W	
Methyl iodide				0.148	0.171	0.212	0.272	0.300	0.292						
Methyl methacrylate			0.282	0.349	0.379	0.400	0.414	0.413	0.431	0.381			13.47	A	
Methylcyclohexane			0.330	0.371	0.388	0.396	0.408	0.411	0.435	0.391			8.581	A	
Methylene chloride			0.574	0.537	0.545	0.567	0.556	0.539	0.550	0.553			2.567	A	
Naphthalene (RSP)			7432	26020	56490	119049	374712	842819	1884818	2.860	0.013		0.992	W	
Naphthalene			2.711	1.932	2.104	2.291	2.639	2.691	3.014						
Propionitrile			0.071	0.061	0.068	0.071	0.072	0.070	0.071	0.069			5.349	A	
Styrene			1.203	1.425	1.445	1.562	1.711	1.719	1.863	1.561			14.29	A	
Tetrachloroethene			0.566	0.574	0.601	0.587	0.586	0.569	0.590	0.582			2.159	A	
Tetrahydrofuran				0.174	0.186	0.196	0.198	0.194	0.203	0.192			5.333	A	
Toluene			2.574	2.756	2.834	2.902	3.015	2.956	3.070	2.872			5.888	A	
Toluene-d8				2.269	2.278	2.436	2.465	2.412	2.501	2.393			4.074	A	
Trichloroethene			0.297	0.284	0.293	0.309	0.316	0.305	0.317	0.303			4.070	A	
Trichlorofluoromethane			0.399	0.429	0.417	0.454	0.438	0.427	0.427	0.427			3.978	A	
Trichlorotrifluoroethane			0.211	0.241	0.234	0.236	0.235	0.225	0.229	0.230			4.202	A	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines
to allow for evaluation against minimum RRF

$\overline{RF}$ = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

				LabQC ID - FileID - Conc	1203 ~ 230517/j1304 ~ 1
Report No:	223060210	Instrument ID:	MSV11	1204 ~ 230517/j1305 ~ 5	1205 ~ 230517/j1306 ~ 10
GC Column:	RTX-VMS-30M ID .25 (mm)	Analyst:	KN1	1206 ~ 230517/j1299 ~ 20	1207 ~ 230517/j1300 ~ 50
Calib. Date 1:	05/17/23 Time 1: 1206	Analytical Batch:	766136	1208 ~ 230517/j1301 ~ 100	1209 ~ 230517/j1302 ~ 200
Calib. Date 2:	05/17/23 Time 2: 1502	Analytical Method:	EPA 8260B		

ANALYTE	***1201***	***1202***	***1203***	***1204***	***1205***	***1206***	***1207***	***1208***	***1209***	***RF / m / A***	***b / B***	***C***	***FIT***	***TYPE***	
Vinyl acetate			0.654	0.671	0.691	0.823	0.835	0.811	0.796	0.754			10.46	A	
Vinyl chloride			0.398	0.386	0.376	0.384	0.384	0.398	0.394	0.388			2.166	A	
Xylene (total)			1.005	1.099	1.119	1.132	1.217	1.227	1.274	1.153			7.956	A	
cis-1,2-Dichloroethene			0.431	0.460	0.471	0.474	0.473	0.465	0.491	0.467			3.956	A	
cis-1,3-Dichloropropene			0.319	0.379	0.398	0.423	0.447	0.448	0.466	0.411			12.30	A	
diisopropyl Ether (DIPE)				1.246	1.241	1.308	1.364	1.351	1.434	1.324			5.603	A	
m,p-Xylene			1.085	1.122	1.145	1.170	1.235	1.238	1.276	1.182			5.913	A	
n-Butyl alcohol (RSP)				1431	2786	6305	17497	41129	90875	0.011	0.240		0.997	W	
n-Butyl alcohol				0.007	0.007	0.009	0.009	0.010	0.011						
n-Butylbenzene			4.012	3.865	3.716	3.950	4.074	4.031	4.330	3.997			4.757	A	
n-Hexane			0.643	0.733	0.717	0.740	0.743	0.743	0.794	0.730			6.191	A	
n-Propylbenzene			3.213	3.374	3.382	3.512	3.447	3.389	3.760	3.440			4.884	A	
o-Xylene			0.845	1.054	1.069	1.054	1.183	1.203	1.269	1.097			12.70	A	
sec-Butylbenzene			2.744	2.945	2.947	2.960	3.029	3.004	3.275	2.986			5.263	A	
t-Butanol (TBA)			0.040	0.043	0.047	0.050	0.052	0.054	0.055	0.049			11.47	A	
t-amyl methyl ether (TAME)				0.695	0.725	0.753	0.774	0.781	0.832	0.760			6.257	A	
tert-Butyl methyl ether (MTB			0.674	0.745	0.779	0.785	0.811	0.806	0.869	0.781			7.734	A	
tert-Butylbenzene			1.261	1.375	1.394	1.404	1.443	1.411	1.554	1.406			6.173	A	
trans-1,2-Dichloroethene			0.542	0.475	0.496	0.477	0.490	0.478	0.502	0.494			4.759	A	
trans-1,3-Dichloropropene			0.311	0.349	0.369	0.394	0.423	0.421	0.432	0.386			11.62	A	
trans-1,4-Dichloro-2-butene			0.355	0.358	0.390	0.421	0.445	0.444	0.491	0.415			12.06	A	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines
to allow for evaluation against minimum RRF

$\overline{RF}$ = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

GCAL, Inc.

Data file : /var/chem/msv11.i/230517.s.b/j1304.d
Lab Smp Id: 1203 Client Smp ID: V11STD001
Inj Date : 17-MAY-2023 14:17
Operator : KN1 Inst ID: msv11.i
Smp Info : 1203*V11STD001
Misc Info : MSV~528~*1*SMR
Comment :
Method : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Meth Date : 23-May-2023 17:14 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 14:17 Cal File: j1304.d
Als bottle: 1 Calibration Sample, Level: 3
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.662	1.662	(0.256)	2483	1.00000	0.996	
2 Chloromethane ++	50	1.851	1.851	(0.285)	4320	1.00000	0.987	
3 Vinyl Chloride +	62	1.930	1.930	(0.297)	3319	1.00000	1.02	
4 1-3 Butadiene	39	1.945	1.945	(0.299)	3796	1.00000	1.07	7617
5 Bromomethane	94	2.244	2.244	(0.345)	1908	1.00000	0.727	
7 Chloroethane	64	2.370	2.370	(0.365)	2073	1.00000	1.15	
8 Trichlorofluoromethane	101	2.512	2.512	(0.386)	3332	1.00000	0.934	
9 Isoprene	67	2.831	2.831	(0.435)	3088	1.00000	0.947	8579(a)
10 Ethyl Ether	59	2.847	2.847	(0.438)	2276	1.00000	0.908	5167
11 1,1-Dichloroethene +	96	3.062	3.062	(0.471)	1741	1.00000	0.945	
12 Carbon Disulfide	76	3.094	3.094	(0.476)	8574	1.00000	1.13	
13 1,1,2Trichlotrifluoroethane	101	3.120	3.120	(0.480)	1765	1.00000	0.919	
15 Methyl Iodide	142	3.225	3.225	(0.496)	1560	1.00000	4.09	
16 Acrolein	56	3.450	3.450	(0.531)	2834	5.00000	4.59	

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
17 Allyl chloride	39	3.623	3.623	(0.557)	3137	1.00000	0.978	6841
18 Methylene Chloride	49	3.754	3.754	(0.577)	4794	1.00000	1.04	
19 Acetone	43	3.807	3.807	(0.585)	13464	5.00000	4.87	
20 trans-1,2-Dichloroethene	61	3.948	3.948	(0.607)	4525	1.00000	1.10	
21 Methyl Acetate	43	3.974	3.974	(0.611)	12723	5.00000	4.24	8819
22 Hexane	57	4.053	4.053	(0.623)	5366	1.00000	0.880	8604
23 MTBE	73	4.106	4.106	(0.631)	5628	1.00000	0.863	8143
24 tert-Butyl Alcohol	59	4.226	4.226	(0.650)	1672	5.00000	4.11	6466
25 Acetonitrile	41	4.341	4.341	(0.668)	2620	5.00000	4.91	8759
26 Isopropyl Ether	45	4.546	4.546	(0.699)	8946	1.00000	0.810	9150
27 Chloroprene	53	4.630	4.630	(0.712)	3820	1.00000	0.807	8615
28 1,1-Dichloroethane ++	63	4.651	4.651	(0.715)	4766	1.00000	0.912	
29 Acrylonitrile	53	4.703	4.703	(0.723)	6119	5.00000	4.29	
31 Vinyl Acetate	43	4.950	4.950	(0.761)	5456	1.00000	0.867	(M1)
30 Ethyl Tert-butyl Ether	59	4.944	4.944	(0.760)	7172	1.00000	0.792	7690
M 33 Total 1,2-Dichloroethene	61				8123	2.00000	2.02	
32 cis-1,2-Dichloroethene	61	5.238	5.238	(0.806)	3598	1.00000	0.924	
34 2,2-Dichloropropane	77	5.338	5.338	(0.821)	3192	1.00000	0.969	
35 Bromochloromethane	128	5.442	5.442	(0.837)	1038	1.00000	0.904	
36 Cyclohexane	56	5.448	5.448	(0.838)	3996	1.00000	0.797	8897
37 Chloroform +	83	5.521	5.521	(0.849)	3425	1.00000	0.872	
39 Ethyl Acetate	43	5.663	5.663	(0.871)	16818	5.00000	4.06	6661
38 Carbon Tetrachloride	117	5.663	5.663	(0.871)	2425	1.00000	0.774	
40 Tetrahydrofuran	42	5.694	5.694	(0.876)	6947	5.00000	4.34	7232
$ 41 Dibromofluoromethane	111	5.710	5.710	(0.878)	1658	1.00000	0.797	6369
42 1,1,1-Trichloroethane	97	5.731	5.731	(0.881)	3065	1.00000	0.860	
44 2-Butanone	43	5.830	5.830	(0.897)	10218	5.00000	4.19	
45 1,1-Dichloropropene	75	5.867	5.867	(0.902)	2531	1.00000	0.846	
47 2,2,4 Trimethylpentane	57	5.967	5.967	(0.918)	9530	1.00000	0.761	7770
48 Benzene	78	6.098	6.098	(0.938)	8028	1.00000	0.912	
49 Propionitrile	54	6.103	6.103	(0.939)	2977	5.00000	5.17	7505
50 Methylacrylonitrile	41	6.135	6.135	(0.944)	11267	5.00000	4.42	9021
$ 51 1,2-Dichloroethane-d4	65	6.224	6.224	(0.957)	2638	1.00000	0.949	
54 tert-amyl Methyl Ether	73	6.234	6.234	(0.959)	4851	1.00000	0.765	3050(M3)
52 1,2-Dichloroethane	62	6.292	6.292	(0.968)	3479	1.00000	0.927	(M3)
53 Isobutyl Alcohol	43	6.344	6.344	(0.976)	690	5.00000	4.29	6808
* 56 FLUOROBENZENE	96	6.502	6.502	(1.000)	417309	50.0000		
57 Methyl Cyclohexane	83	6.664	6.664	(1.025)	2758	1.00000	0.844	8839
58 Trichloroethene	130	6.680	6.680	(1.027)	2476	1.00000	0.980	
59 n-Butanol	56	7.031	7.031	(1.081)	193	5.00000	14.2	5670(M3)
60 Dibromomethane	93	7.073	7.073	(1.088)	1216	1.00000	0.838	
61 2-3 Dichloro-1-Proprene	75	7.146	7.146	(1.099)	3310	1.00000	0.904	8648
62 1,2-Dichloropropane +	63	7.162	7.162	(1.102)	2816	1.00000	0.918	
63 Bromodichloromethane	83	7.230	7.230	(1.112)	2760	1.00000	0.931	
64 Methyl methacrylate	41	7.377	7.377	(1.135)	2353	1.00000	0.740	7719
65 1,4- Dioxane	58	7.430	7.430	(1.143)	487	25.0000	19.5	5769
66 1-Bromo-2-chloroethane	63	7.671	7.671	(1.180)	3649	1.00000	0.908	8256

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
67 2-Chloroethyl vinyl ether	63	7.760	7.760	(1.194)	1057	5.00000	7.07	(M1)
68 cis-1,3-Dichloropropene	75	7.812	7.812	(1.202)	2665	1.00000	0.776	
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	6675	1.00000	0.913	
70 Toluene +	91	8.022	8.022	(0.868)	7866	1.00000	0.896	(M1)
71 2-Nitropropane	43	8.206	8.206	(0.888)	761	1.00000	0.774	2799(H)
M 72 1-3 Dichloropropene total	100				5259	2.00000	1.58	0
74 4-methyl-2-pentanone	43	8.342	8.342	(0.903)	14269	5.00000	4.17	
73 Tetrachloroethene	164	8.358	8.358	(0.905)	1729	1.00000	0.972	
75 trans-1,3-Dichloropropene	75	8.379	8.379	(1.289)	2594	1.00000	0.806	
77 Ethyl Methacrylate	41	8.510	8.510	(0.921)	2141	1.00000	0.801	0(M1)
78 1,1,2-Trichloroethane	97	8.510	8.510	(0.921)	1914	1.00000	0.964	
79 Dibromochloromethane	129	8.651	8.651	(0.936)	1769	1.00000	0.824	
76 3,4-dichloro-1-butene	75	8.698	8.698	(0.942)	1743	1.00000	0.676	7974(a)
80 1,3-Dichloropropane	76	8.725	8.725	(0.944)	3131	1.00000	0.934	
81 1-Nitropropane	43	8.814	8.814	(0.954)	278	1.00000	1.17	5172(H)
82 1,2-Dibromoethane(EDB)	107	8.845	8.845	(0.957)	1924	1.00000	0.941	
83 2-Hexanone	43	9.008	9.008	(0.975)	12166	5.00000	4.27	
89 1-Chlorohexane	91	9.238	9.238	(1.000)	2780	1.00000	1.11	5032(H)
* 90 CHLOROBENZENE-d5	82	9.238	9.238	(1.000)	152805	50.0000		
91 Chlorobenzene ++	112	9.254	9.254	(1.002)	5690	1.00000	1.000	
92 Ethylbenzene +	106	9.275	9.275	(1.004)	2643	1.00000	0.879	
93 1,1,1,2-Tetrachloroethane	133	9.301	9.301	(1.007)	1633	1.00000	0.840	(M1)
94 p,m-Xylene	106	9.385	9.385	(1.016)	6632	2.00000	1.84	
M 95 TOTAL XYLENE	106				9213	3.00000	2.61	
96 o-Xylene	106	9.700	9.700	(1.050)	2581	1.00000	0.770	
97 Styrene	104	9.742	9.742	(1.054)	3677	1.00000	0.771	
98 Bromoform ++	173	9.763	9.763	(1.057)	1088	1.00000	0.770	
99 Isopropylbenzene	105	9.931	9.931	(1.075)	6731	1.00000	0.806	
$ 101 Bromofluorobenzene	174	10.146	10.146	(1.098)	2509	1.00000	1.14	
102 cis-1,4-dichloro-2-butene	53	10.177	10.177	(0.931)	877	1.00000	0.744	7507(a)
103 Bromobenzene	77	10.219	10.219	(0.935)	3740	1.00000	0.987	
104 n-Propylbenzene	91	10.229	10.229	(0.936)	8809	1.00000	0.934	
105 1,1,2,2-Tetrachloroethane++	83	10.277	10.277	(0.940)	2290	1.00000	0.937	(M1)
106 2-Chlorotoluene	91	10.345	10.345	(0.947)	6261	1.00000	0.943	
107 1,3,5-Trimethylbenzene	105	10.360	10.360	(0.948)	5981	1.00000	0.847	
108 1,2,3-Trichloropropane	75	10.376	10.376	(0.950)	2816	1.00000	0.887	
109 trans-1,4-Dichloro-2-Butene	53	10.402	10.402	(0.952)	974	1.00000	0.856	
100 Cyclohexanone	55	10.408	10.408	(0.952)	3898	5.00000	4.69	4252
110 4-Chlorotoluene	91	10.460	10.460	(0.957)	6264	1.00000	0.991	
111 tert-butylbenzene	91	10.591	10.591	(0.969)	3458	1.00000	0.897	
112 1,2,4-Trimethylbenzene	105	10.638	10.638	(0.974)	5936	1.00000	0.836	
113 sec-Butylbenzene	105	10.717	10.717	(0.981)	7522	1.00000	0.919	
114 Dicyclopentadiene	66	10.806	10.806	(0.989)	7275	1.00000	0.799	8193
115 p-Isopropyltoluene	119	10.811	10.811	(0.989)	7057	1.00000	0.930	
116 1,3-Dichlorobenzene	146	10.874	10.874	(0.995)	5270	1.00000	1.21	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	137065	50.0000		
118 1,4-Dichlorobenzene	146	10.937	10.937	(1.001)	5187	1.00000	1.18	

						AMOUNTS		
	QUANT SIG					CAL-AMT	ON-COL	
Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
119 n-Butylbenzene	91	11.100	11.100	(1.016)	10997	1.00000	1.00	
120 1,2-Dichlorobenzene	146	11.236	11.236	(1.028)	4912	1.00000	1.17	
122 1,2-Dibromo-3-Chloropropane	157	11.766	11.766	(1.077)	504	1.00000	0.865	
123 Hexachlorobutadiene	225	12.196	12.196	(1.116)	2379	1.00000	2.03	
124 1,2,4-Trichlorobenzene	180	12.232	12.232	(1.119)	3355	1.00000	1.29	
125 Naphthalene	128	12.452	12.452	(1.140)	7432	1.00000	1.61	
126 1,2,3-Trichlorobenzene	180	12.578	12.578	(1.151)	3451	1.00000	1.36	

QC Flag Legend

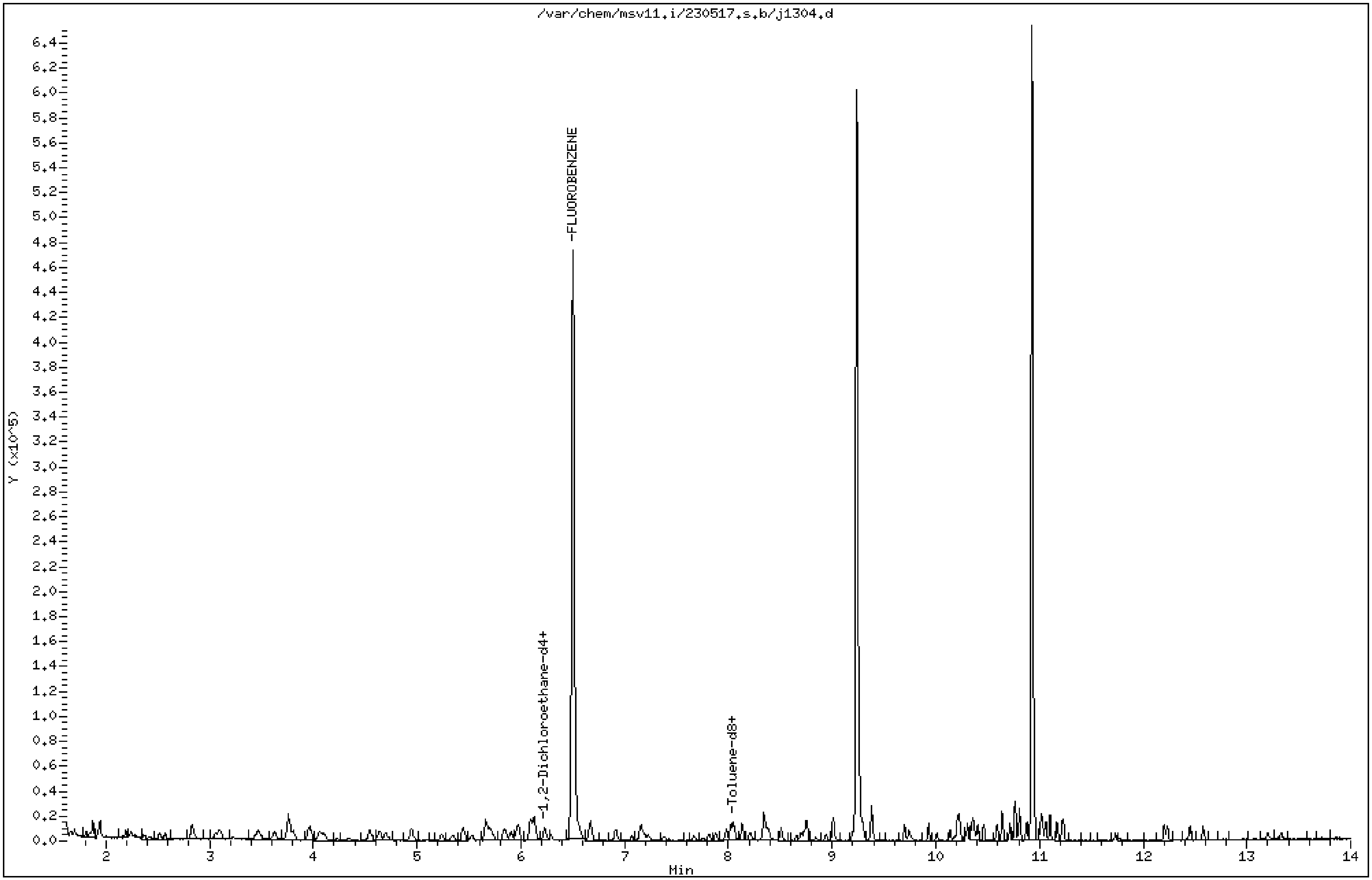
a - Target compound detected but, quantitated amount
    Below Limit Of Quantitation(BLOQ).
M1- Compound response manually integrated because
    Target system did not integrate.
M3- Compound response manually integrated because
    Target system integrated incorrect peak.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv11.i/230517.s.b/j1304.d
Date : 17-MAY-2023 14:17
Client ID: V11STD001
Sample Info: 1203*V11STD001
Purge Volume: 5.0
Column phase: RTX-VMS-30M

Page 1

Instrument: msv11.i

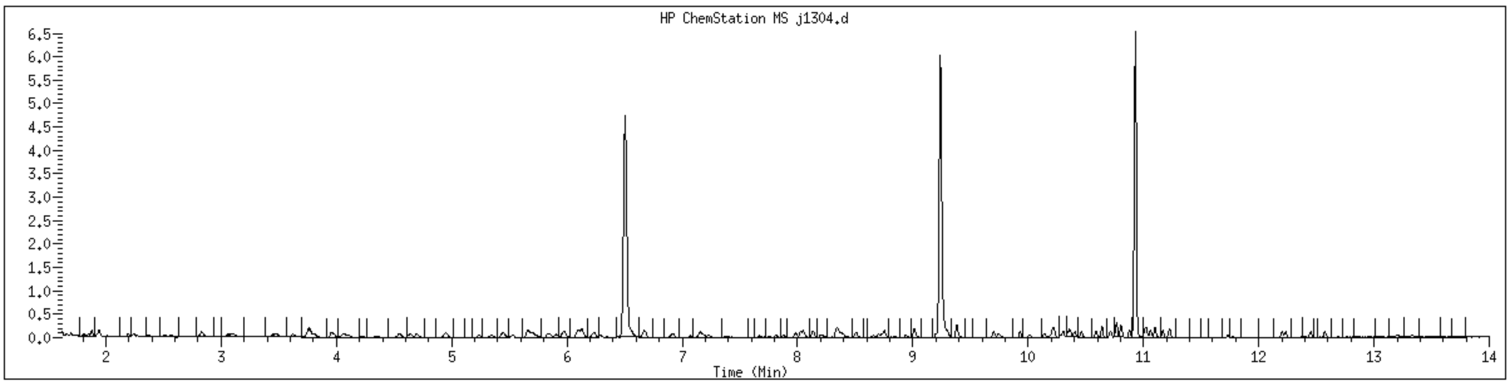
Operator: KN1
Column diameter: 0.25



Report Date: 05/23/2023 17:14

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID	: 1203	SampleType	: CALIB_3
Injection Date	: 05/17/2023 14:17	Instrument	: msv11.i
Operator	: KN1		
Sample Info	: 1203*V11STD001		
Misc Info	: MSV~528~*1*SMR		
Method	: /var/chem/msv11.i/230517.s.b/8260bDODw11.m		
Dilution	: 1.00		
Matrix	: WATER		
Integrator	: HP RTE	Compound Sublist:	8260b+AppIXDOD



Original   Final

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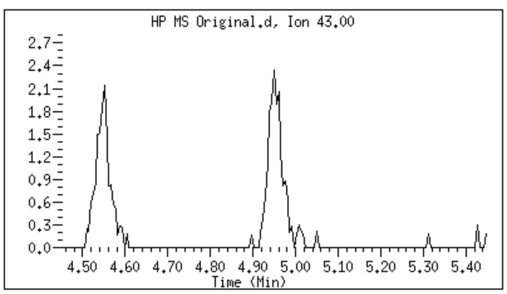
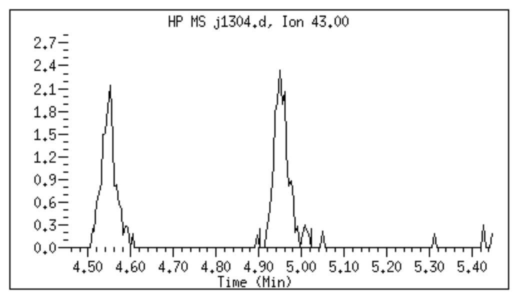
31 Vinyl Acetate   CAS#: 108-05-4   Reason: M1



Electronic Signature Applied

User: smr
Date: 05/17/2023 16:22



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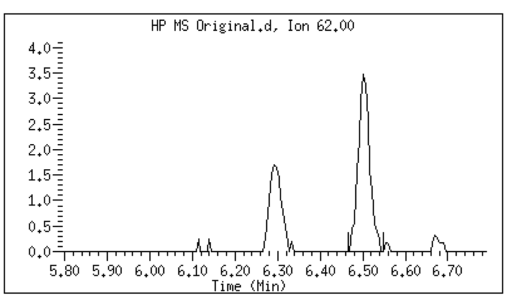
52 1,2-Dichloroethane   CAS#: 107-06-2   Reason: M3



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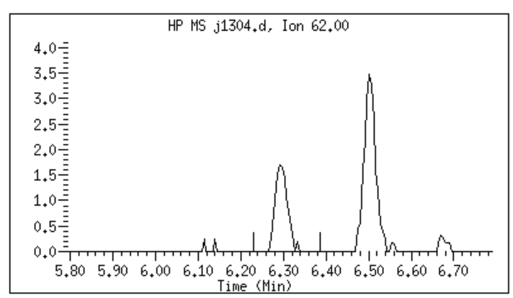
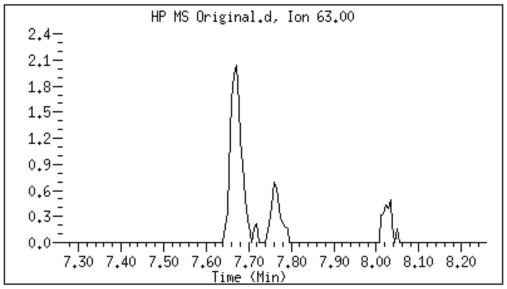


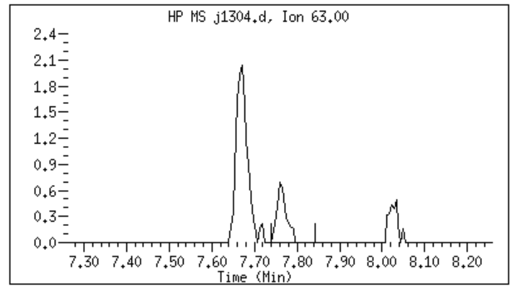
Original | Final

67 2-Chloroethyl vinyl ether   CAS#: 110-75-8   Reason: M1

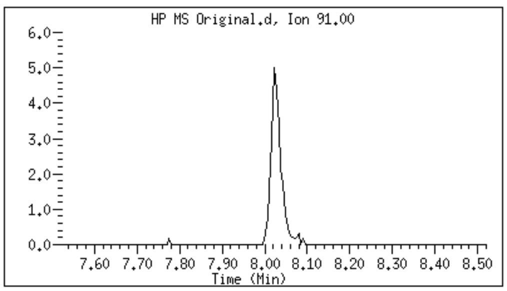


Electronic Signature Applied

User: smr
Date: 05/17/2023 16:05



70 Toluene +   CAS#: 108-88-3   Reason: M1



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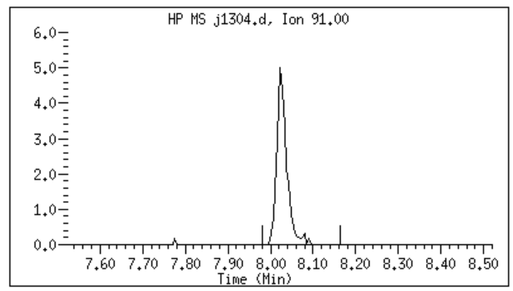
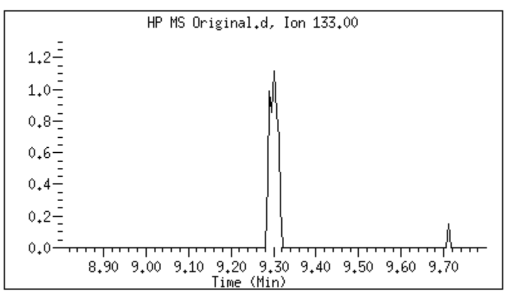
User: smr
Date: 05/17/2023 16:05

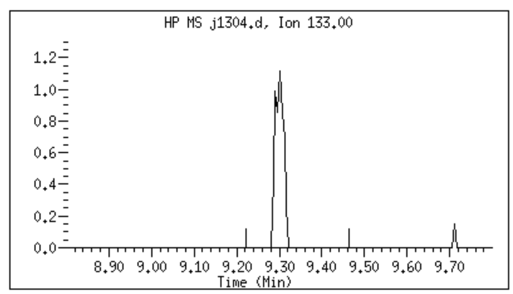


93 1,1,1,2-Tetrachloroethane   CAS#: 630-20-6   Reason: M1

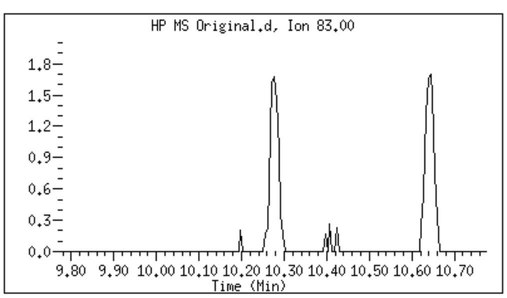


Electronic Signature Applied

User: smr
Date: 05/17/2023 16:09



105 1,1,2,2-Tetrachloroethane++   CAS#: 79-34-5   Reason: M1



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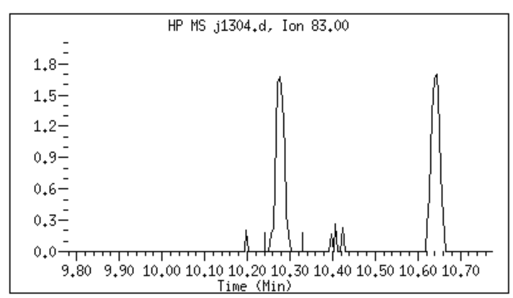
User: smr
Date: 05/17/2023 16:23



Original   Final

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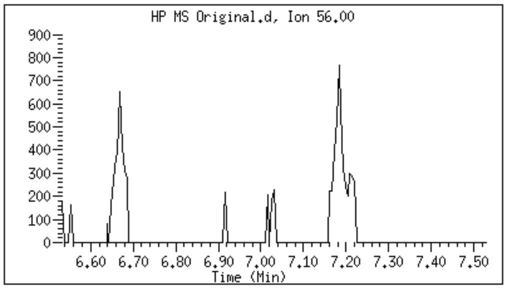
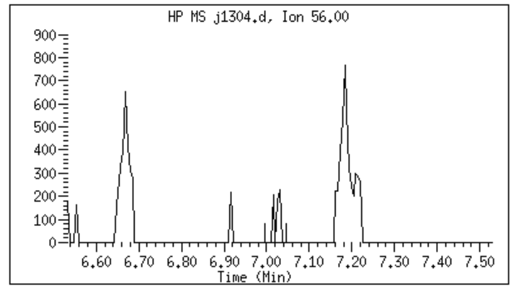
59 n-Butanol   CAS#: 71-36-3   Reason: M3



Electronic Signature Applied

User: smr
Date: 05/17/2023 16:04



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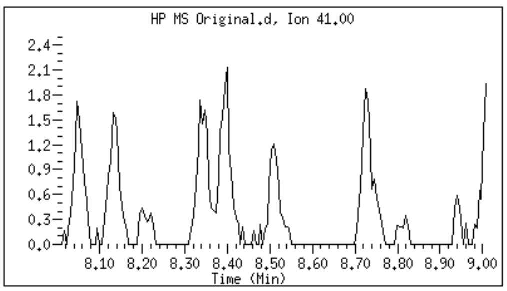
77 Ethyl Methacrylate   CAS#: 97-63-2   Reason: M1



Electronic Signature Applied

User: smr
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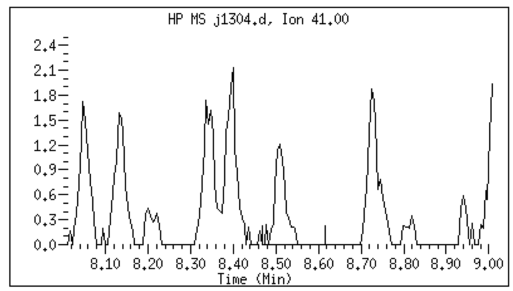

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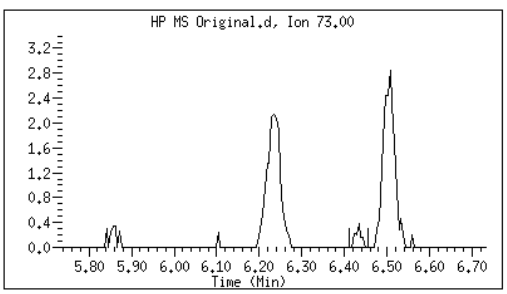
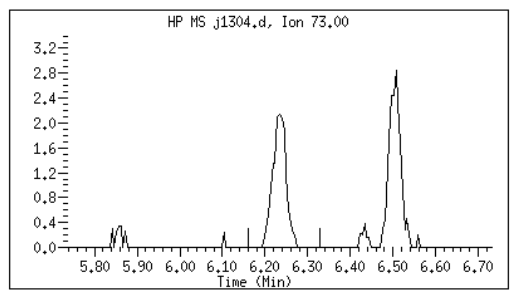
54 tert-amyl Methyl Ether   CAS#: 994-05-8   Reason: M3



Electronic Signature Applied

User: smr
Date: 05/17/2023 16:23



M1 - Target system did not integrate
M3 - Target system integrated incorrect peak

GCAL, Inc.

Data file : /var/chem/msv11.i/230517.s.b/j1305.d
Lab Smp Id: 1204 Client Smp ID: V11STD005
Inj Date : 17-MAY-2023 14:40
Operator : KN1 Inst ID: msv11.i
Smp Info : 1204*V11STD005
Misc Info : MSV~528~*1*SMR
Comment :
Method : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Meth Date : 23-May-2023 17:14 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 14:40 Cal File: j1305.d
Als bottle: 1 Calibration Sample, Level: 4
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.667	1.667	(0.256)	11358	5.00000	4.82	
2 Chloromethane ++	50	1.851	1.851	(0.284)	21105	5.00000	5.10	
3 Vinyl Chloride +	62	1.929	1.929	(0.297)	15204	5.00000	4.96	
4 1-3 Butadiene	39	1.945	1.945	(0.299)	16704	5.00000	4.97	9232
5 Bromomethane	94	2.244	2.244	(0.345)	7963	5.00000	5.70	
7 Chloroethane	64	2.375	2.375	(0.365)	8455	5.00000	4.95	
8 Trichlorofluoromethane	101	2.522	2.522	(0.388)	16916	5.00000	5.02	
9 Isoprene	67	2.831	2.831	(0.435)	15190	5.00000	4.93	9112(a)
10 Ethyl Ether	59	2.852	2.852	(0.438)	11713	5.00000	4.95	6498
11 1,1-Dichloroethene +	96	3.062	3.062	(0.471)	8866	5.00000	5.09	
12 Carbon Disulfide	76	3.099	3.099	(0.476)	27220	5.00000	4.77	
13 1,1,2Trichlotrifluoroethane	101	3.114	3.114	(0.479)	9501	5.00000	5.24	
15 Methyl Iodide	142	3.230	3.230	(0.496)	5821	5.00000	5.94	
16 Acrolein	56	3.450	3.450	(0.530)	14039	25.0000	24.1	

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
17 Allyl chloride	39	3.623	3.623	(0.557)	14847	5.00000	4.90	8728
18 Methylene Chloride	49	3.754	3.754	(0.577)	21162	5.00000	4.86	
19 Acetone	43	3.807	3.807	(0.585)	65658	25.0000	25.1	
20 trans-1,2-Dichloroethene	61	3.953	3.953	(0.608)	18728	5.00000	4.80	
21 Methyl Acetate	43	3.974	3.974	(0.611)	67961	25.0000	24.0	9576
22 Hexane	57	4.063	4.063	(0.625)	28896	5.00000	5.02	9370
23 MTBE	73	4.105	4.105	(0.631)	29370	5.00000	4.77	8928
24 tert-Butyl Alcohol	59	4.231	4.231	(0.650)	8478	25.0000	22.1	8229
25 Acetonitrile	41	4.341	4.341	(0.667)	12631	25.0000	25.0	9074
26 Isopropyl Ether	45	4.551	4.551	(0.699)	49133	5.00000	4.71	9410
27 Chloroprene	53	4.635	4.635	(0.712)	20124	5.00000	4.50	9214
28 1,1-Dichloroethane ++	63	4.656	4.656	(0.716)	25240	5.00000	5.11	
29 Acrylonitrile	53	4.698	4.698	(0.722)	29937	25.0000	22.2	
31 Vinyl Acetate	43	4.949	4.949	(0.761)	26443	5.00000	4.45	
30 Ethyl Tert-butyl Ether	59	4.949	4.949	(0.761)	38514	5.00000	4.50	7934
M 33 Total 1,2-Dichloroethene	61				36867	10.0000	9.73	
32 cis-1,2-Dichloroethene	61	5.238	5.238	(0.805)	18139	5.00000	4.93	
34 2,2-Dichloropropane	77	5.348	5.348	(0.822)	15125	5.00000	4.86	
35 Bromochloromethane	128	5.448	5.448	(0.837)	5792	5.00000	5.34	
36 Cyclohexane	56	5.453	5.453	(0.838)	22069	5.00000	4.66	9062
37 Chloroform +	83	5.526	5.526	(0.849)	18683	5.00000	5.04	
39 Ethyl Acetate	43	5.657	5.657	(0.869)	88702	25.0000	22.7	6549
38 Carbon Tetrachloride	117	5.663	5.663	(0.870)	14715	5.00000	4.97	
40 Tetrahydrofuran	42	5.689	5.689	(0.874)	34397	25.0000	22.7	8044
$ 41 Dibromofluoromethane	111	5.710	5.710	(0.878)	9554	5.00000	4.86	7411
42 1,1,1-Trichloroethane	97	5.725	5.725	(0.880)	16440	5.00000	4.88	
43 sec-Butanol	45	5.736	5.736	(0.882)	1510	5.00000	4.61	9264
44 2-Butanone	43	5.830	5.830	(0.896)	54306	25.0000	23.6	
45 1,1-Dichloropropene	75	5.857	5.857	(0.900)	13518	5.00000	4.79	
47 2,2,4 Trimethylpentane	57	5.972	5.972	(0.918)	52567	5.00000	4.45	7878
48 Benzene	78	6.098	6.098	(0.937)	41367	5.00000	4.97	
49 Propionitrile	54	6.108	6.108	(0.939)	12066	25.0000	22.2	8259
50 Methylacrylonitrile	41	6.134	6.134	(0.943)	55668	25.0000	23.1	9420
$ 51 1,2-Dichloroethane-d4	65	6.224	6.224	(0.956)	12660	5.00000	4.82	
54 tert-amyl Methyl Ether	73	6.234	6.234	(0.958)	27384	5.00000	4.57	6996
52 1,2-Dichloroethane	62	6.292	6.292	(0.967)	16946	5.00000	4.78	(H)
53 Isobutyl Alcohol	43	6.339	6.339	(0.974)	3023	25.0000	19.9	7941
* 56 FLUOROBENZENE	96	6.507	6.507	(1.000)	394285	50.0000		
57 Methyl Cyclohexane	83	6.669	6.669	(1.025)	14620	5.00000	4.74	9135
58 Trichloroethene	130	6.674	6.674	(1.026)	11194	5.00000	4.69	
59 n-Butanol	56	7.015	7.015	(1.078)	1431	25.0000	29.1	6099(M3)
60 Dibromomethane	93	7.073	7.073	(1.087)	6625	5.00000	4.83	
61 2-3 Dichloro-1-Proprene	75	7.152	7.152	(1.099)	16242	5.00000	4.69	9283
62 1,2-Dichloropropane +	63	7.162	7.162	(1.101)	14717	5.00000	5.08	
63 Bromodichloromethane	83	7.230	7.230	(1.111)	13495	5.00000	4.82	
64 Methyl methacrylate	41	7.382	7.382	(1.135)	13744	5.00000	4.58	9076
65 1,4- Dioxane	58	7.424	7.424	(1.141)	2575	125.000	109	8275

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
66 1-Bromo-2-chloroethane	63	7.676	7.676	(1.180)	18141	5.00000	4.78	9052
67 2-Chloroethyl vinyl ether	63	7.765	7.765	(1.193)	5825	25.0000	20.5	
68 cis-1,3-Dichloropropene	75	7.817	7.817	(1.201)	14962	5.00000	4.61	
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	34013	5.00000	4.74	
70 Toluene +	91	8.027	8.027	(0.869)	41303	5.00000	4.80	
71 2-Nitropropane	43	8.205	8.205	(0.888)	4902	5.00000	5.09	6430(H)
M 72 1-3 Dichloropropene total	100				28733	10.0000	9.14	0
74 4-methyl-2-pentanone	43	8.342	8.342	(0.903)	77236	25.0000	23.0	
73 Tetrachloroethene	164	8.363	8.363	(0.905)	8601	5.00000	4.93	
75 trans-1,3-Dichloropropene	75	8.373	8.373	(1.287)	13771	5.00000	4.53	
77 Ethyl Methacrylate	41	8.504	8.504	(0.921)	11011	5.00000	4.20	9010
78 1,1,2-Trichloroethane	97	8.504	8.504	(0.921)	9768	5.00000	5.01	
79 Dibromochloromethane	129	8.662	8.662	(0.938)	9592	5.00000	4.56	
76 3,4-dichloro-1-butene	75	8.698	8.698	(0.942)	10980	5.00000	4.34	9107(a)
80 1,3-Dichloropropane	76	8.724	8.724	(0.944)	15652	5.00000	4.76	
81 1-Nitropropane	43	8.808	8.808	(0.953)	2101	5.00000	4.62	6975
82 1,2-Dibromoethane(EDB)	107	8.845	8.845	(0.957)	9578	5.00000	4.78	
83 2-Hexanone	43	9.013	9.013	(0.976)	64080	25.0000	22.9	
89 1-Chlorohexane	91	9.233	9.233	(0.999)	10076	5.00000	4.12	5960
* 90 CHLOROBENZENE-d5	82	9.238	9.238	(1.000)	149890	50.0000		
91 Chlorobenzene ++	112	9.254	9.254	(1.002)	27661	5.00000	4.96	
92 Ethylbenzene +	106	9.270	9.270	(1.003)	14428	5.00000	4.89	
93 1,1,1,2-Tetrachloroethane	133	9.301	9.301	(1.007)	9391	5.00000	4.92	
94 p,m-Xylene	106	9.380	9.380	(1.015)	33631	10.0000	9.50	
M 95 TOTAL XYLENE	106				49432	15.0000	14.3	
96 o-Xylene	106	9.705	9.705	(1.050)	15801	5.00000	4.81	
97 Styrene	104	9.742	9.742	(1.054)	21362	5.00000	4.56	
98 Bromoform ++	173	9.768	9.768	(1.057)	6698	5.00000	4.84	
99 Isopropylbenzene	105	9.930	9.930	(1.075)	35940	5.00000	4.39	
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	9740	5.00000	4.50	
102 cis-1,4-dichloro-2-butene	53	10.177	10.177	(0.931)	4774	5.00000	4.12	8658(a)
103 Bromobenzene	77	10.219	10.219	(0.935)	18953	5.00000	5.09	
104 n-Propylbenzene	91	10.229	10.229	(0.936)	45439	5.00000	4.90	
105 1,1,2,2-Tetrachloroethane++	83	10.276	10.276	(0.940)	12456	5.00000	5.19	
106 2-Chlorotoluene	91	10.345	10.345	(0.947)	32622	5.00000	5.00	
107 1,3,5-Trimethylbenzene	105	10.360	10.360	(0.948)	34723	5.00000	5.00	
108 1,2,3-Trichloropropane	75	10.371	10.371	(0.949)	14698	5.00000	4.71	
109 trans-1,4-Dichloro-2-Butene	53	10.402	10.402	(0.952)	4821	5.00000	4.31	
100 Cyclohexanone	55	10.408	10.408	(0.952)	18636	25.0000	22.8	4642(H)
110 4-Chlorotoluene	91	10.460	10.460	(0.957)	30658	5.00000	4.94	
111 tert-butylbenzene	91	10.591	10.591	(0.969)	18520	5.00000	4.89	
112 1,2,4-Trimethylbenzene	105	10.638	10.638	(0.974)	33901	5.00000	4.86	
113 sec-Butylbenzene	105	10.712	10.712	(0.980)	39664	5.00000	4.93	
114 Dicyclopentadiene	66	10.806	10.806	(0.989)	44461	5.00000	4.97	8635
115 p-Isopropyltoluene	119	10.811	10.811	(0.989)	37374	5.00000	5.01	
116 1,3-Dichlorobenzene	146	10.879	10.879	(0.996)	20539	5.00000	4.81	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	134676	50.0000		

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
118 1,4-Dichlorobenzene	146	10.937	10.937	(1.001)	21768	5.00000	5.05	
119 n-Butylbenzene	91	11.100	11.100	(1.016)	52051	5.00000	4.83	
120 1,2-Dichlorobenzene	146	11.231	11.231	(1.028)	20485	5.00000	4.95	
122 1,2-Dibromo-3-Chloropropane	157	11.771	11.771	(1.077)	2582	5.00000	4.51	
123 Hexachlorobutadiene	225	12.201	12.201	(1.117)	5960	5.00000	5.19	
124 1,2,4-Trichlorobenzene	180	12.232	12.232	(1.119)	11540	5.00000	4.53	
125 Naphthalene	128	12.447	12.447	(1.139)	26020	5.00000	4.04	
126 1,2,3-Trichlorobenzene	180	12.573	12.573	(1.151)	10752	5.00000	4.34	

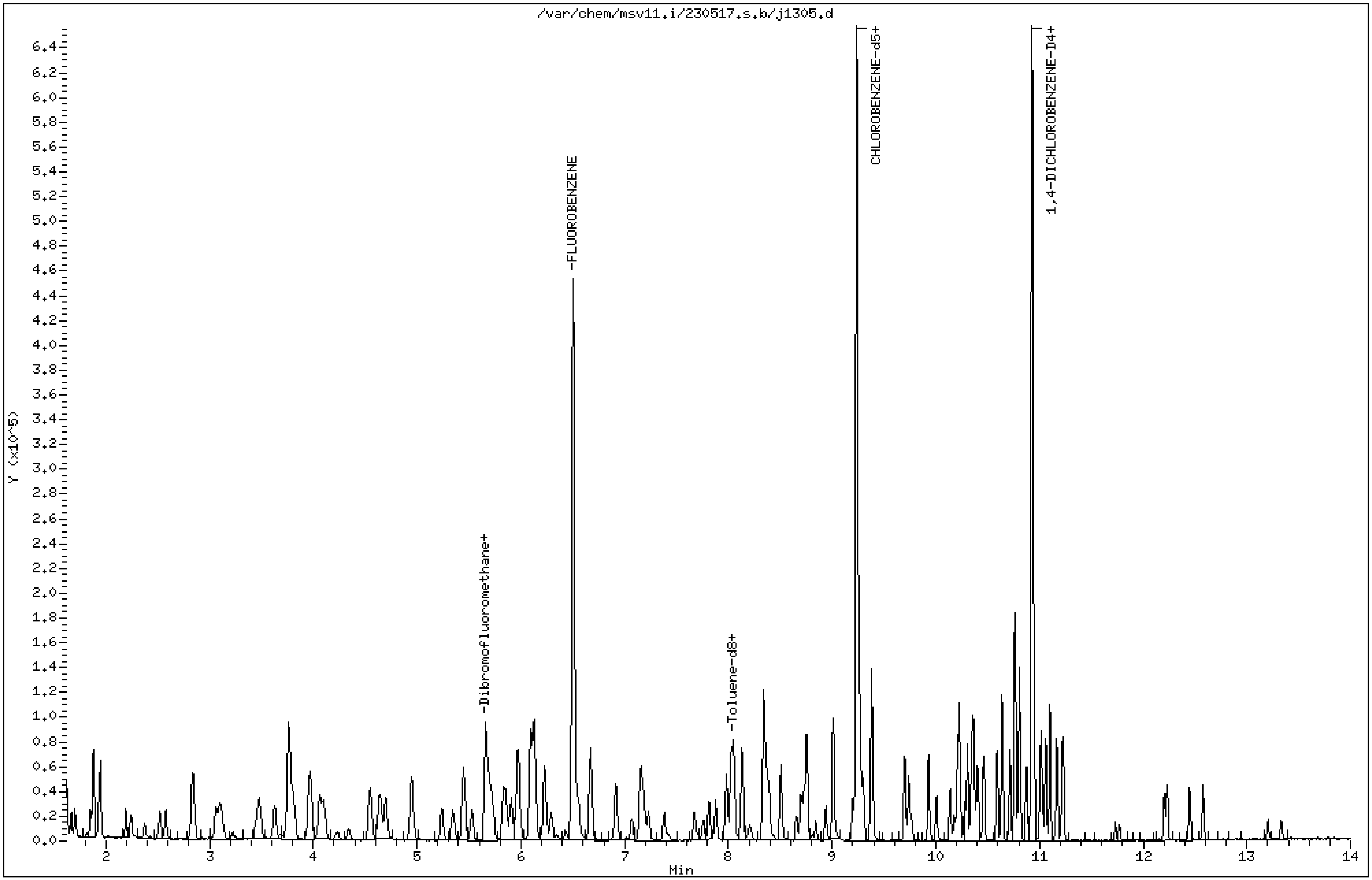
QC Flag Legend

a - Target compound detected but, quantitated amount
    Below Limit Of Quantitation(BLOQ).
M3- Compound response manually integrated because
    Target system integrated incorrect peak.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv11.i/230517.s.b/j1305.d
Date : 17-MAY-2023 14:40
Client ID: V11STD005
Sample Info: 1204*V11STD005
Purge Volume: 5.0
Column phase: RTX-VMS-30M

Page 1

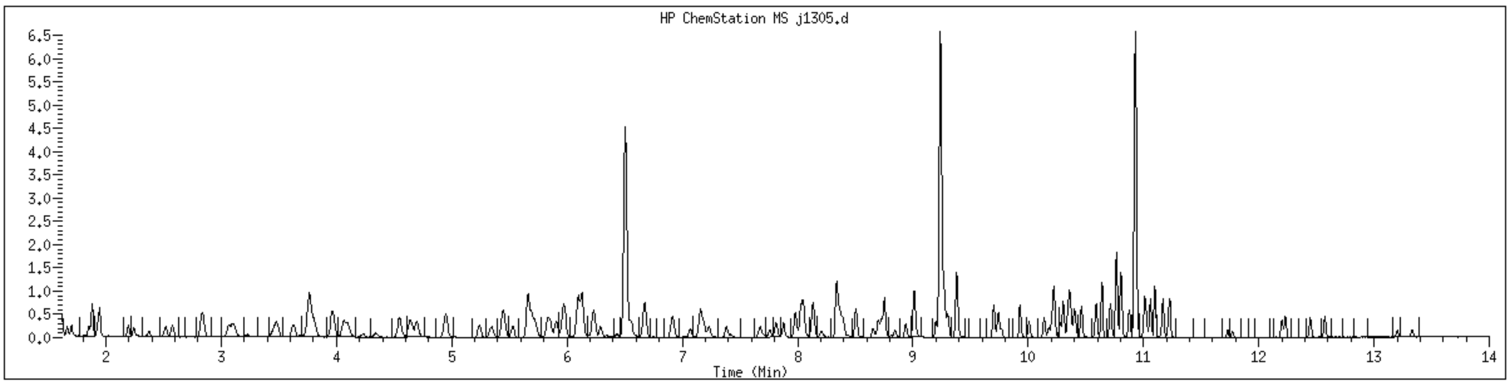
Instrument: msv11.i
Operator: KN1
Column diameter: 0.25



Data file : /var/chem/msv11.i/230517.s.b/j1305.d Page: 1
Report Date: 05/23/2023 17:14

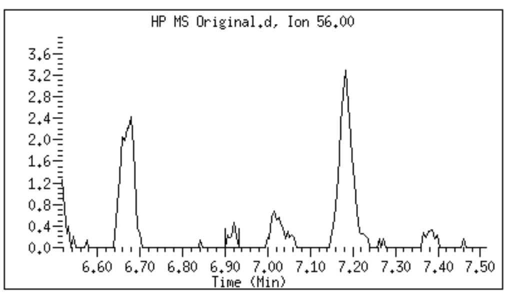
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1204 SampleType : CALIB_4
Injection Date: 05/17/2023 14:40 Instrument : msv11.i
Operator : KN1
Sample Info : 1204*V11STD005
Misc Info : MSV~528~*1*SMR
Method : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260b+AppIXDOD



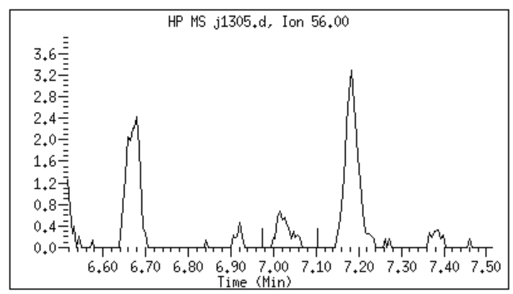
Original Final

59 n-Butanol CAS#: 71-36-3 Reason: M3



Electronic Signature
Applied

User: smr
Date: 05/17/2023 16:04



M3 - Target system integrated incorrect peak

GCAL, Inc.

Data file : /var/chem/msv11.i/230517.s.b/j1306.d
Lab Smp Id: 1205 Client Smp ID: V11STD010
Inj Date : 17-MAY-2023 15:02
Operator : KN1 Inst ID: msv11.i
Smp Info : 1205*V11STD010
Misc Info : MSV~528~*1*SMR
Comment :
Method : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Meth Date : 23-May-2023 17:14 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 15:02 Cal File: j1306.d
Als bottle: 1 Calibration Sample, Level: 5
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.667	1.667	(0.256)	21605	10.0000	9.47	
2 Chloromethane ++	50	1.856	1.856	(0.285)	39296	10.0000	9.81	
3 Vinyl Chloride +	62	1.935	1.935	(0.297)	28670	10.0000	9.67	
4 1-3 Butadiene	39	1.950	1.950	(0.300)	33074	10.0000	10.2	9578
5 Bromomethane	94	2.249	2.249	(0.346)	14337	10.0000	11.2	
7 Chloroethane	64	2.375	2.375	(0.365)	15457	10.0000	9.35	
8 Trichlorofluoromethane	101	2.522	2.522	(0.388)	31843	10.0000	9.76	
9 Isoprene	67	2.837	2.837	(0.436)	30609	10.0000	10.3	9183
10 Ethyl Ether	59	2.857	2.857	(0.439)	23670	10.0000	10.3	6880
11 1,1-Dichloroethene +	96	3.067	3.067	(0.471)	17237	10.0000	10.2	
12 Carbon Disulfide	76	3.099	3.099	(0.476)	51825	10.0000	9.78	
13 1,1,2Trichlotrifluoroethane	101	3.120	3.120	(0.479)	17866	10.0000	10.2	
15 Methyl Iodide	142	3.230	3.230	(0.496)	13064	10.0000	9.19	
16 Acrolein	56	3.455	3.455	(0.531)	26531	50.0000	47.0	

							AMOUNTS		
		QUANT SIG					CAL-AMT	ON-COL	
	Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
	==========================	====	==	======	======	========	=======	=======	==========
	17 Allyl chloride	39	3.628	3.628	(0.558)	30675	10.0000	10.5	9256
	18 Methylene Chloride	49	3.765	3.765	(0.579)	41595	10.0000	9.86	
	19 Acetone	43	3.817	3.817	(0.587)	126977	50.0000	50.2	
	20 trans-1,2-Dichloroethene	61	3.959	3.959	(0.608)	37908	10.0000	10.0	
	21 Methyl Acetate	43	3.980	3.980	(0.612)	137337	50.0000	50.1	9377
	22 Hexane	57	4.069	4.069	(0.625)	54727	10.0000	9.81	9308
	23 MTBE	73	4.105	4.105	(0.631)	59458	10.0000	9.97	8993
	24 tert-Butyl Alcohol	59	4.226	4.226	(0.649)	18017	50.0000	48.5	8899
	25 Acetonitrile	41	4.336	4.336	(0.666)	22746	50.0000	46.6	9535
	26 Isopropyl Ether	45	4.551	4.551	(0.699)	94773	10.0000	9.38	9382
	27 Chloroprene	53	4.635	4.635	(0.712)	42118	10.0000	9.72	9347
	28 1,1-Dichloroethane ++	63	4.656	4.656	(0.716)	48681	10.0000	10.2	
	29 Acrylonitrile	53	4.698	4.698	(0.722)	63563	50.0000	48.7	
	31 Vinyl Acetate	43	4.944	4.944	(0.760)	52752	10.0000	9.16	
	30 Ethyl Tert-butyl Ether	59	4.949	4.949	(0.761)	79865	10.0000	9.64	7843
M	33 Total 1,2-Dichloroethene	61				73834	20.0000	20.1	
	32 cis-1,2-Dichloroethene	61	5.243	5.243	(0.806)	35926	10.0000	10.1	
	34 2,2-Dichloropropane	77	5.348	5.348	(0.822)	29559	10.0000	9.80	
	35 Bromochloromethane	128	5.442	5.442	(0.836)	10946	10.0000	10.4	
	36 Cyclohexane	56	5.458	5.458	(0.839)	45285	10.0000	9.88	8577
	37 Chloroform +	83	5.531	5.531	(0.850)	37112	10.0000	10.3	
	39 Ethyl Acetate	43	5.657	5.657	(0.869)	184360	50.0000	48.6	6287
	38 Carbon Tetrachloride	117	5.668	5.668	(0.871)	28821	10.0000	10.1	
	40 Tetrahydrofuran	42	5.689	5.689	(0.874)	70853	50.0000	48.4	7939
$	41 Dibromofluoromethane	111	5.710	5.710	(0.878)	19131	10.0000	10.1	7234
	42 1,1,1-Trichloroethane	97	5.731	5.731	(0.881)	32426	10.0000	9.95	
	43 sec-Butanol	45	5.731	5.731	(0.881)	3647	10.0000	11.5	9476
	44 2-Butanone	43	5.830	5.830	(0.896)	108162	50.0000	48.5	
	45 1,1-Dichloropropene	75	5.857	5.857	(0.900)	27795	10.0000	10.2	
	47 2,2,4 Trimethylpentane	57	5.977	5.977	(0.919)	106444	10.0000	9.30	8157
	48 Benzene	78	6.103	6.103	(0.938)	80147	10.0000	9.95	
	49 Propionitrile	54	6.103	6.103	(0.938)	25816	50.0000	49.0	7421
	50 Methylacrylonitrile	41	6.129	6.129	(0.942)	107854	50.0000	46.2	9469
$	51 1,2-Dichloroethane-d4	65	6.229	6.229	(0.957)	24607	10.0000	9.68	
	54 tert-amyl Methyl Ether	73	6.234	6.234	(0.958)	55381	10.0000	9.55	6957
	52 1,2-Dichloroethane	62	6.297	6.297	(0.968)	33594	10.0000	9.78	(H)
	53 Isobutyl Alcohol	43	6.349	6.349	(0.976)	7954	50.0000	54.1	8799
*	56 FLUOROBENZENE	96	6.507	6.507	(1.000)	381749	50.0000		
	57 Methyl Cyclohexane	83	6.674	6.674	(1.026)	29652	10.0000	9.92	9306
	58 Trichloroethene	130	6.680	6.680	(1.027)	22338	10.0000	9.66	
	59 n-Butanol	56	7.020	7.020	(1.079)	2786	50.0000	46.5	8661
	60 Dibromomethane	93	7.073	7.073	(1.087)	13438	10.0000	10.1	
	61 2-3 Dichloro-1-Proprene	75	7.146	7.146	(1.098)	32272	10.0000	9.63	9293
	62 1,2-Dichloropropane +	63	7.162	7.162	(1.101)	27825	10.0000	9.91	
	63 Bromodichloromethane	83	7.230	7.230	(1.111)	26302	10.0000	9.70	
	64 Methyl methacrylate	41	7.382	7.382	(1.135)	28918	10.0000	9.94	9437
	65 1,4- Dioxane	58	7.419	7.419	(1.140)	5521	250.000	242	8984

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
66 1-Bromo-2-chloroethane	63	7.671	7.671	(1.179)	36145	10.0000	9.84	9499
67 2-Chloroethyl vinyl ether	63	7.760	7.760	(1.193)	13618	50.0000	43.6	
68 cis-1,3-Dichloropropene	75	7.817	7.817	(1.201)	30380	10.0000	9.67	
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	67278	10.0000	9.52	
70 Toluene +	91	8.027	8.027	(0.869)	83698	10.0000	9.87	
71 2-Nitropropane	43	8.205	8.205	(0.888)	8410	10.0000	8.86	8713(M3)
M 72 1-3 Dichloropropene total	100				58558	20.0000	19.2	0
74 4-methyl-2-pentanone	43	8.342	8.342	(0.903)	165487	50.0000	50.1	
73 Tetrachloroethene	164	8.363	8.363	(0.905)	17736	10.0000	10.3	
75 trans-1,3-Dichloropropene	75	8.378	8.378	(1.288)	28178	10.0000	9.57	
77 Ethyl Methacrylate	41	8.504	8.504	(0.921)	25470	10.0000	9.86	9314
78 1,1,2-Trichloroethane	97	8.509	8.509	(0.921)	18789	10.0000	9.79	
79 Dibromochloromethane	129	8.662	8.662	(0.938)	19791	10.0000	9.54	
76 3,4-dichloro-1-butene	75	8.698	8.698	(0.942)	22439	10.0000	9.01	9058
80 1,3-Dichloropropane	76	8.730	8.730	(0.945)	31814	10.0000	9.82	
81 1-Nitropropane	43	8.814	8.814	(0.954)	4015	10.0000	8.34	7282
82 1,2-Dibromoethane(EDB)	107	8.845	8.845	(0.957)	18884	10.0000	9.56	
83 2-Hexanone	43	9.008	9.008	(0.975)	131860	50.0000	47.9	
89 1-Chlorohexane	91	9.233	9.233	(0.999)	22119	10.0000	9.18	6806
* 90 CHLOROBENZENE-d5	82	9.238	9.238	(1.000)	147675	50.0000		
91 Chlorobenzene ++	112	9.254	9.254	(1.002)	53616	10.0000	9.75	
92 Ethylbenzene +	106	9.270	9.270	(1.003)	28904	10.0000	9.95	
93 1,1,1,2-Tetrachloroethane	133	9.301	9.301	(1.007)	18323	10.0000	9.75	
94 p,m-Xylene	106	9.380	9.380	(1.015)	67607	20.0000	19.4	
M 95 TOTAL XYLENE	106				99169	30.0000	29.1	
96 o-Xylene	106	9.705	9.705	(1.050)	31562	10.0000	9.74	
97 Styrene	104	9.742	9.742	(1.054)	42672	10.0000	9.25	
98 Bromoform ++	173	9.768	9.768	(1.057)	13203	10.0000	9.67	
99 Isopropylbenzene	105	9.930	9.930	(1.075)	77276	10.0000	9.58	
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	19876	10.0000	9.33	
102 cis-1,4-dichloro-2-butene	53	10.177	10.177	(0.931)	10712	10.0000	9.28	8475
103 Bromobenzene	77	10.219	10.219	(0.935)	36172	10.0000	9.75	
104 n-Propylbenzene	91	10.229	10.229	(0.936)	90808	10.0000	9.83	
105 1,1,2,2-Tetrachloroethane++	83	10.276	10.276	(0.940)	24195	10.0000	10.1	
106 2-Chlorotoluene	91	10.345	10.345	(0.947)	63576	10.0000	9.77	
107 1,3,5-Trimethylbenzene	105	10.360	10.360	(0.948)	67271	10.0000	9.72	
108 1,2,3-Trichloropropane	75	10.376	10.376	(0.950)	30579	10.0000	9.83	
109 trans-1,4-Dichloro-2-Butene	53	10.402	10.402	(0.952)	10480	10.0000	9.41	
100 Cyclohexanone	55	10.407	10.407	(0.952)	40081	50.0000	49.2	4808
110 4-Chlorotoluene	91	10.460	10.460	(0.957)	60363	10.0000	9.75	
111 tert-butylbenzene	91	10.591	10.591	(0.969)	37412	10.0000	9.91	
112 1,2,4-Trimethylbenzene	105	10.638	10.638	(0.974)	69910	10.0000	10.1	
113 sec-Butylbenzene	105	10.717	10.717	(0.981)	79114	10.0000	9.87	
114 Dicyclopentadiene	66	10.806	10.806	(0.989)	90920	10.0000	10.2	8790
115 p-Isopropyltoluene	119	10.811	10.811	(0.989)	74171	10.0000	9.98	
116 1,3-Dichlorobenzene	146	10.879	10.879	(0.996)	40662	10.0000	9.54	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	134234	50.0000		

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
118 1,4-Dichlorobenzene	146	10.937	10.937	(1.001)	40928	10.0000	9.54	
119 n-Butylbenzene	91	11.100	11.100	(1.016)	99760	10.0000	9.30	
120 1,2-Dichlorobenzene	146	11.231	11.231	(1.028)	39076	10.0000	9.47	
122 1,2-Dibromo-3-Chloropropane	157	11.771	11.771	(1.077)	5690	10.0000	9.97	
123 Hexachlorobutadiene	225	12.201	12.201	(1.117)	10563	10.0000	9.23	
124 1,2,4-Trichlorobenzene	180	12.232	12.232	(1.119)	22142	10.0000	8.72	
125 Naphthalene	128	12.447	12.447	(1.139)	56490	10.0000	8.02	
126 1,2,3-Trichlorobenzene	180	12.573	12.573	(1.151)	21139	10.0000	8.57	

QC Flag Legend

M3- Compound response manually integrated because
Target system integrated incorrect peak.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv11.i/230517.s.b/j1306.d

Page 1

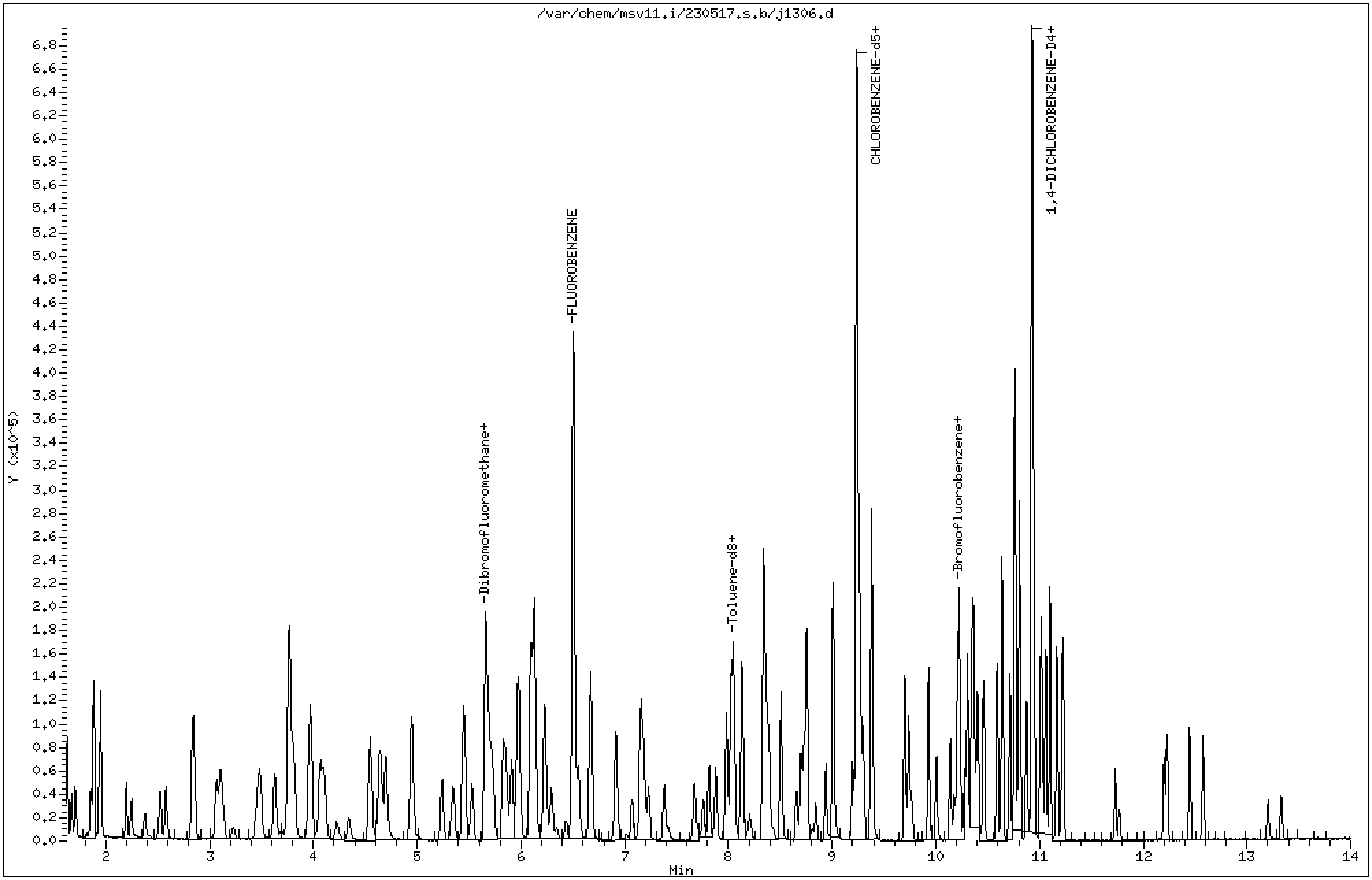
Date : 17-MAY-2023 15:02

Client ID: V11STD010 | Instrument: msv11.i

Sample Info: 1205*V11STD010

Purge Volume: 5.0 | Operator: KN1

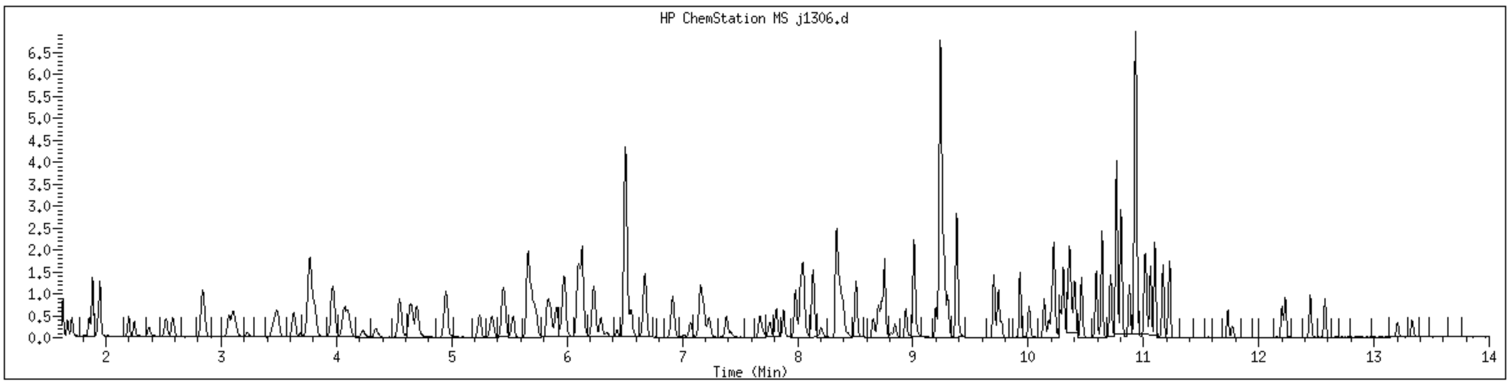
Column phase: RTX-VMS-30M | Column diameter: 0.25



Report Date: 05/23/2023 17:15

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1205 SampleType : CALIB_5
Injection Date: 05/17/2023 15:02 Instrument : msv11.i
Operator : KN1
Sample Info : 1205*V11STD010
Misc Info : MSV~528~*1*SMR
Method : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260b+AppIXDOD



Original Final

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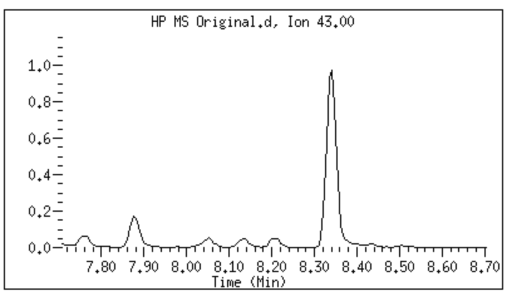
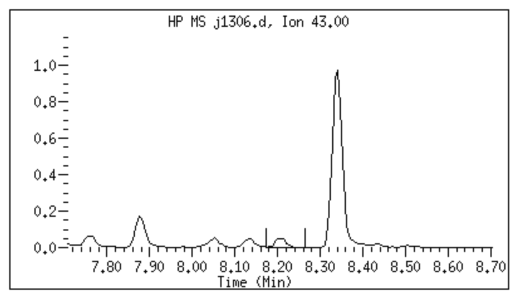
71 2-Nitropropane CAS#: 79-46-9 Reason: M3



Electronic Signature
Applied

User: smr
Date: 05/17/2023 16:08



M3 - Target system integrated incorrect peak

GCAL, Inc.

Data file : /var/chem/msv11.i/230517.s.b/j1299.d
Lab Smp Id: 1206 Client Smp ID: V11STD020
Inj Date : 17-MAY-2023 12:06
Operator : KN1 Inst ID: msv11.i
Smp Info : 1206*V11STD020
Misc Info : MSV~528~*1*SMR
Comment :
Method : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Meth Date : 23-May-2023 17:15 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 12:06 Cal File: j1299.d
Als bottle: 1 Calibration Sample, Level: 6
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.667	1.667	(0.256)	44998	20.0000	21.4	
2 Chloromethane ++	50	1.856	1.856	(0.285)	77541	20.0000	21.0	
3 Vinyl Chloride +	62	1.935	1.935	(0.297)	54031	20.0000	19.8	
4 1-3 Butadiene	39	1.951	1.951	(0.300)	63828	20.0000	21.3	9781
5 Bromomethane	94	2.249	2.249	(0.346)	24650	20.0000	21.5	
7 Chloroethane	64	2.375	2.375	(0.365)	30357	20.0000	19.9	
8 Trichlorofluoromethane	101	2.522	2.522	(0.388)	63955	20.0000	21.3	
9 Isoprene	67	2.837	2.837	(0.436)	54096	20.0000	19.7	9198
10 Ethyl Ether	59	2.852	2.852	(0.438)	41976	20.0000	19.9	5988
11 1,1-Dichloroethene +	96	3.073	3.073	(0.472)	30835	20.0000	19.8	
12 Carbon Disulfide	76	3.099	3.099	(0.476)	91911	20.0000	19.2	
13 1,1,2Trichlotrifluoroethane	101	3.120	3.120	(0.479)	33175	20.0000	20.5	
15 Methyl Iodide	142	3.230	3.230	(0.496)	29806	20.0000	17.6	
16 Acrolein	56	3.450	3.450	(0.530)	54279	100.000	104	

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
	17 Allyl chloride	39	3.628	3.628	(0.558)	58283	20.0000	21.5	9576
	18 Methylene Chloride	49	3.759	3.759	(0.578)	79826	20.0000	20.5	
	19 Acetone	43	3.812	3.812	(0.586)	239253	100.000	102	
	20 trans-1,2-Dichloroethene	61	3.959	3.959	(0.608)	67208	20.0000	19.3	
	21 Methyl Acetate	43	3.974	3.974	(0.611)	272794	100.000	108	9635
	22 Hexane	57	4.063	4.063	(0.625)	104264	20.0000	20.3	9482
	23 MTBE	73	4.105	4.105	(0.631)	110563	20.0000	20.1	8916
	24 tert-Butyl Alcohol	59	4.226	4.226	(0.649)	34916	100.000	102	8795
	25 Acetonitrile	41	4.341	4.341	(0.667)	46099	100.000	102	9742
	26 Isopropyl Ether	45	4.551	4.551	(0.699)	184262	20.0000	19.8	9631
	27 Chloroprene	53	4.635	4.635	(0.712)	79122	20.0000	19.8	9444
	28 1,1-Dichloroethane ++	63	4.656	4.656	(0.716)	89969	20.0000	20.4	
	29 Acrylonitrile	53	4.698	4.698	(0.722)	129294	100.000	108	
	31 Vinyl Acetate	43	4.944	4.944	(0.760)	115917	20.0000	21.8	
	30 Ethyl Tert-butyl Ether	59	4.950	4.950	(0.761)	151275	20.0000	19.8	7641
M	33 Total 1,2-Dichloroethene	61				134025	40.0000	39.6	
	32 cis-1,2-Dichloroethene	61	5.238	5.238	(0.805)	66817	20.0000	20.3	
	34 2,2-Dichloropropane	77	5.348	5.348	(0.822)	56729	20.0000	20.4	
	35 Bromochloromethane	128	5.442	5.442	(0.836)	21926	20.0000	22.6	
	36 Cyclohexane	56	5.453	5.453	(0.838)	85486	20.0000	20.2	9310
	37 Chloroform +	83	5.532	5.532	(0.850)	67411	20.0000	20.3	
	39 Ethyl Acetate	43	5.657	5.657	(0.869)	354653	100.000	101	6392
	38 Carbon Tetrachloride	117	5.668	5.668	(0.871)	55894	20.0000	21.1	
	40 Tetrahydrofuran	42	5.689	5.689	(0.874)	138351	100.000	102	7733
$	41 Dibromofluoromethane	111	5.715	5.715	(0.878)	37920	20.0000	21.6	8756
	42 1,1,1-Trichloroethane	97	5.736	5.736	(0.882)	61163	20.0000	20.3	
	43 sec-Butanol	45	5.736	5.736	(0.882)	5964	20.0000	20.4	9388
	44 2-Butanone	43	5.830	5.830	(0.896)	215864	100.000	105	
	45 1,1-Dichloropropene	75	5.862	5.862	(0.901)	51789	20.0000	20.5	
	47 2,2,4 Trimethylpentane	57	5.977	5.977	(0.919)	208907	20.0000	19.8	8247
	48 Benzene	78	6.103	6.103	(0.938)	153867	20.0000	20.7	
	49 Propionitrile	54	6.103	6.103	(0.938)	49854	100.000	103	7582
	50 Methylacrylonitrile	41	6.134	6.134	(0.943)	224700	100.000	104	9404
$	51 1,2-Dichloroethane-d4	65	6.229	6.229	(0.957)	48997	20.0000	20.9	
	54 tert-amyl Methyl Ether	73	6.234	6.234	(0.958)	106016	20.0000	19.8	6970
	52 1,2-Dichloroethane	62	6.297	6.297	(0.968)	66141	20.0000	20.9	
	53 Isobutyl Alcohol	43	6.344	6.344	(0.975)	12921	100.000	95.2	8994
*	56 FLUOROBENZENE	96	6.507	6.507	(1.000)	352069	50.0000		
	57 Methyl Cyclohexane	83	6.669	6.669	(1.025)	55799	20.0000	20.2	9260
	58 Trichloroethene	130	6.675	6.675	(1.026)	43450	20.0000	20.4	
	59 n-Butanol	56	7.010	7.010	(1.077)	6305	100.000	96.5	8775
	60 Dibromomethane	93	7.073	7.073	(1.087)	25924	20.0000	21.2	
	61 2-3 Dichloro-1-Proprene	75	7.146	7.146	(1.098)	61655	20.0000	19.9	9107
	62 1,2-Dichloropropane +	63	7.162	7.162	(1.101)	51598	20.0000	19.9	
	63 Bromodichloromethane	83	7.236	7.236	(1.112)	50332	20.0000	20.1	
	64 Methyl methacrylate	41	7.377	7.377	(1.134)	56269	20.0000	21.0	9230
	65 1,4- Dioxane	58	7.414	7.414	(1.139)	11230	500.000	533	9133

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
==========================	====	==	======	======	========	=======	=======	==========
66 1-Bromo-2-chloroethane	63	7.671	7.671	(1.179)	69205	20.0000	20.4	9295
67 2-Chloroethyl vinyl ether	63	7.760	7.760	(1.193)	26959	100.000	88.6	
68 cis-1,3-Dichloropropene	75	7.817	7.817	(1.201)	59572	20.0000	20.6	
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	135393	20.0000	20.4	
70 Toluene +	91	8.022	8.022	(0.868)	161292	20.0000	20.2	
71 2-Nitropropane	43	8.205	8.205	(0.888)	17091	20.0000	19.1	8335(H)
M 72 1-3 Dichloropropene total	100				114999	40.0000	41.0	0
74 4-methyl-2-pentanone	43	8.337	8.337	(0.902)	321181	100.000	103	
73 Tetrachloroethene	164	8.358	8.358	(0.905)	32639	20.0000	20.2	
75 trans-1,3-Dichloropropene	75	8.378	8.378	(1.288)	55427	20.0000	20.4	
77 Ethyl Methacrylate	41	8.504	8.504	(0.921)	50121	20.0000	20.6	9360
78 1,1,2-Trichloroethane	97	8.510	8.510	(0.921)	36206	20.0000	20.0	
79 Dibromochloromethane	129	8.656	8.656	(0.937)	39522	20.0000	20.2	
76 3,4-dichloro-1-butene	75	8.698	8.698	(0.942)	44208	20.0000	18.9	9111
80 1,3-Dichloropropane	76	8.730	8.730	(0.945)	61149	20.0000	20.1	
81 1-Nitropropane	43	8.814	8.814	(0.954)	9372	20.0000	19.7	7613
82 1,2-Dibromoethane(EDB)	107	8.840	8.840	(0.957)	37419	20.0000	20.1	
83 2-Hexanone	43	9.008	9.008	(0.975)	262311	100.000	101	
89 1-Chlorohexane	91	9.233	9.233	(0.999)	44028	20.0000	19.4	8087
* 90 CHLOROBENZENE-d5	82	9.238	9.238	(1.000)	138951	50.0000		
91 Chlorobenzene ++	112	9.254	9.254	(1.002)	102666	20.0000	19.8	
92 Ethylbenzene +	106	9.270	9.270	(1.003)	54234	20.0000	19.8	
93 1,1,1,2-Tetrachloroethane	133	9.301	9.301	(1.007)	35640	20.0000	20.2	
94 p,m-Xylene	106	9.380	9.380	(1.015)	130109	40.0000	39.6	
M 95 TOTAL XYLENE	106				188709	60.0000	58.9	
96 o-Xylene	106	9.700	9.700	(1.050)	58600	20.0000	19.2	
97 Styrene	104	9.736	9.736	(1.054)	86809	20.0000	20.0	
98 Bromoform ++	173	9.768	9.768	(1.057)	26381	20.0000	20.5	
99 Isopropylbenzene	105	9.930	9.930	(1.075)	152469	20.0000	20.1	
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	40215	20.0000	20.1	
102 cis-1,4-dichloro-2-butene	53	10.177	10.177	(0.931)	21867	20.0000	19.6	8936
103 Bromobenzene	77	10.219	10.219	(0.935)	70455	20.0000	19.6	
104 n-Propylbenzene	91	10.224	10.224	(0.936)	182518	20.0000	20.4	
105 1,1,2,2-Tetrachloroethane++	83	10.276	10.276	(0.940)	48182	20.0000	20.8	
106 2-Chlorotoluene	91	10.345	10.345	(0.947)	124914	20.0000	19.8	
107 1,3,5-Trimethylbenzene	105	10.360	10.360	(0.948)	138320	20.0000	20.7	
108 1,2,3-Trichloropropane	75	10.371	10.371	(0.949)	60992	20.0000	20.3	
109 trans-1,4-Dichloro-2-Butene	53	10.397	10.397	(0.952)	21870	20.0000	20.3	
100 Cyclohexanone	55	10.402	10.402	(0.952)	77437	100.000	98.3	5538
110 4-Chlorotoluene	91	10.460	10.460	(0.957)	119129	20.0000	19.9	
111 tert-butylbenzene	91	10.591	10.591	(0.969)	72971	20.0000	20.0	
112 1,2,4-Trimethylbenzene	105	10.638	10.638	(0.974)	136361	20.0000	20.3	
113 sec-Butylbenzene	105	10.712	10.712	(0.980)	153833	20.0000	19.8	
114 Dicyclopentadiene	66	10.806	10.806	(0.989)	174680	20.0000	20.2	8658
115 p-Isopropyltoluene	119	10.811	10.811	(0.989)	145902	20.0000	20.3	
116 1,3-Dichlorobenzene	146	10.874	10.874	(0.995)	79529	20.0000	19.3	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	129927	50.0000		

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
118 1,4-Dichlorobenzene	146	10.937	10.937	(1.001)	80359	20.0000	19.3	
119 n-Butylbenzene	91	11.100	11.100	(1.016)	205288	20.0000	19.8	
120 1,2-Dichlorobenzene	146	11.231	11.231	(1.028)	76470	20.0000	19.2	
122 1,2-Dibromo-3-Chloropropane	157	11.771	11.771	(1.077)	10801	20.0000	19.6	
123 Hexachlorobutadiene	225	12.201	12.201	(1.117)	22177	20.0000	20.0	
124 1,2,4-Trichlorobenzene	180	12.227	12.227	(1.119)	45533	20.0000	18.5	
125 Naphthalene	128	12.447	12.447	(1.139)	119049	20.0000	16.7	
126 1,2,3-Trichlorobenzene	180	12.578	12.578	(1.151)	44972	20.0000	18.8	

QC Flag Legend

H - Operator selected an alternate compound hit.

Data File: /var/chem/msv11.i/230517.s.b/j1299.d

Page 1

Date : 17-MAY-2023 12:06

Client ID: V11STD020

Instrument: msv11.i

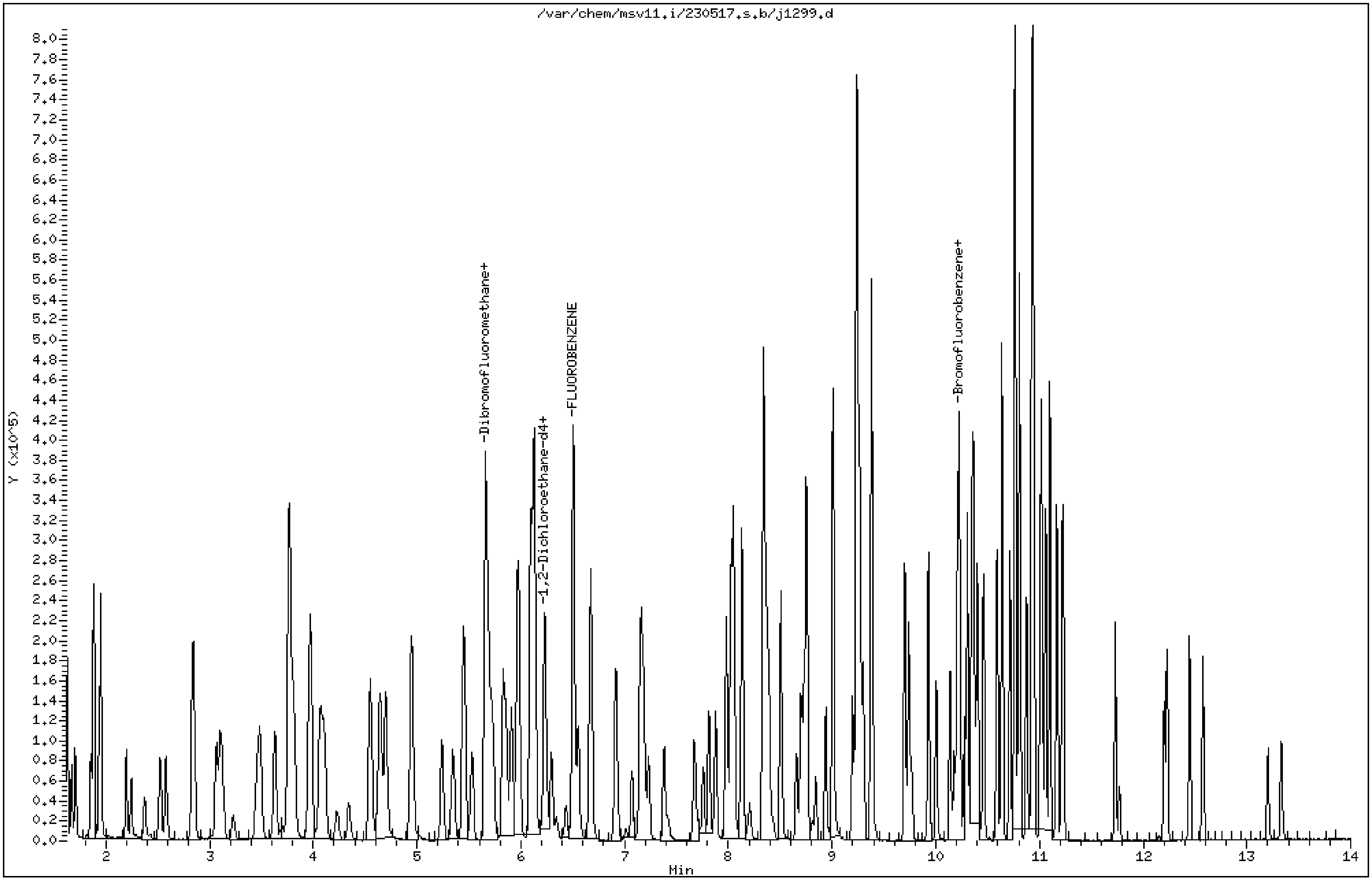
Sample Info: 1206*V11STD020

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

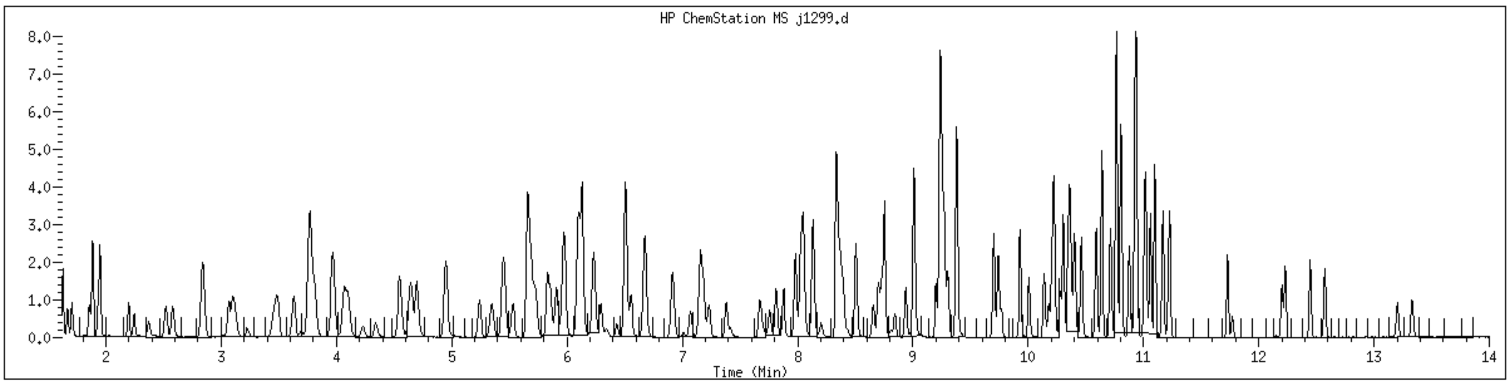
Column diameter: 0.25



Report Date: 05/23/2023 17:15

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID	: 1206	SampleType	: CALIB_6
Injection Date	: 05/17/2023 12:06	Instrument	: msv11.i
Operator	: KN1		
Sample Info	: 1206*V11STD020		
Misc Info	: MSV~528~*1*SMR		
Method	: /var/chem/msv11.i/230517.s.b/8260bDODw11.m		
Dilution	: 1.00		
Matrix	: WATER		
Integrator	: HP RTE	Compound Sublist	: 8260b+AppIXDOD



NO MANUAL INTEGRATIONS

GCAL, Inc.

Data file : /var/chem/msv11.i/230517.s.b/j1300.d
Lab Smp Id: 1207 Client Smp ID: V11STD050
Inj Date : 17-MAY-2023 12:29
Operator : KN1 Inst ID: msv11.i
Smp Info : 1207*V11STD050
Misc Info : MSV~528~*1*SMR
Comment :
Method : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Meth Date : 23-May-2023 17:15 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 12:29 Cal File: j1300.d
Als bottle: 1 Calibration Sample, Level: 7
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.662	1.662	(0.255)	112572	50.0000	50.6	
2 Chloromethane ++	50	1.851	1.851	(0.284)	196295	50.0000	50.2	
3 Vinyl Chloride +	62	1.930	1.930	(0.297)	142945	50.0000	49.4	
4 1-3 Butadiene	39	1.945	1.945	(0.299)	154680	50.0000	48.8	9690
5 Bromomethane	94	2.239	2.239	(0.344)	57857	50.0000	48.7	
7 Chloroethane	64	2.360	2.360	(0.363)	78749	50.0000	48.8	
8 Trichlorofluoromethane	101	2.512	2.512	(0.386)	163180	50.0000	51.3	
9 Isoprene	67	2.832	2.832	(0.435)	147238	50.0000	50.6	9196
10 Ethyl Ether	59	2.847	2.847	(0.438)	113334	50.0000	50.7	6612
11 1,1-Dichloroethene +	96	3.062	3.062	(0.471)	83070	50.0000	50.5	
12 Carbon Disulfide	76	3.094	3.094	(0.475)	246086	50.0000	49.2	
13 1,1,2Trichlotrifluoroethane	101	3.115	3.115	(0.479)	87358	50.0000	51.0	
15 Methyl Iodide	142	3.219	3.219	(0.495)	101184	50.0000	48.9	
16 Acrolein	56	3.445	3.445	(0.529)	143582	250.000	260	

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
17 Allyl chloride	39	3.623	3.623	(0.557)	155752	50.0000	54.4	9425
18 Methylene Chloride	49	3.754	3.754	(0.577)	207202	50.0000	50.3	
19 Acetone	43	3.807	3.807	(0.585)	622625	250.000	252	
20 trans-1,2-Dichloroethene	61	3.954	3.954	(0.608)	182611	50.0000	49.6	
21 Methyl Acetate	43	3.969	3.969	(0.610)	711116	250.000	266	9597
22 Hexane	57	4.058	4.058	(0.624)	276755	50.0000	50.9	9547
23 MTBE	73	4.095	4.095	(0.629)	301977	50.0000	51.9	8923
24 tert-Butyl Alcohol	59	4.221	4.221	(0.649)	97544	250.000	269	8967
25 Acetonitrile	41	4.331	4.331	(0.666)	122018	250.000	256	9797
26 Isopropyl Ether	45	4.546	4.546	(0.699)	507931	50.0000	51.5	9608
27 Chloroprene	53	4.630	4.630	(0.712)	215381	50.0000	51.0	9423
28 1,1-Dichloroethane ++	63	4.651	4.651	(0.715)	234234	50.0000	50.2	
29 Acrylonitrile	53	4.693	4.693	(0.721)	338885	250.000	266	
31 Vinyl Acetate	43	4.939	4.939	(0.759)	311050	50.0000	55.4	
30 Ethyl Tert-butyl Ether	59	4.944	4.944	(0.760)	412284	50.0000	51.0	7627
M 33 Total 1,2-Dichloroethene	61				358891	100.000	100	
32 cis-1,2-Dichloroethene	61	5.238	5.238	(0.805)	176280	50.0000	50.7	
34 2,2-Dichloropropane	77	5.343	5.343	(0.821)	150402	50.0000	51.1	
35 Bromochloromethane	128	5.443	5.443	(0.836)	51357	50.0000	50.1	
36 Cyclohexane	56	5.448	5.448	(0.837)	236677	50.0000	52.9	9270
37 Chloroform +	83	5.526	5.526	(0.849)	180149	50.0000	51.4	
39 Ethyl Acetate	43	5.652	5.652	(0.869)	952589	250.000	258	6272
38 Carbon Tetrachloride	117	5.663	5.663	(0.870)	146950	50.0000	52.6	
40 Tetrahydrofuran	42	5.684	5.684	(0.873)	368993	250.000	258	7636
$ 41 Dibromofluoromethane	111	5.710	5.710	(0.878)	98687	50.0000	53.2	8758
42 1,1,1-Trichloroethane	97	5.731	5.731	(0.881)	168016	50.0000	52.8	
43 sec-Butanol	45	5.731	5.731	(0.881)	15679	50.0000	50.6	9556
44 2-Butanone	43	5.825	5.825	(0.895)	575740	250.000	265	
45 1,1-Dichloropropene	75	5.857	5.857	(0.900)	140029	50.0000	52.5	
47 2,2,4 Trimethylpentane	57	5.972	5.972	(0.918)	565319	50.0000	50.6	8296
48 Benzene	78	6.098	6.098	(0.937)	400105	50.0000	50.9	
49 Propionitrile	54	6.103	6.103	(0.938)	133465	250.000	260	8027
50 Methylacrylonitrile	41	6.129	6.129	(0.942)	657846	250.000	289	9507
$ 51 1,2-Dichloroethane-d4	65	6.224	6.224	(0.956)	129327	50.0000	52.2	
54 tert-amyl Methyl Ether	73	6.229	6.229	(0.957)	288091	50.0000	50.9	7006
52 1,2-Dichloroethane	62	6.292	6.292	(0.967)	175265	50.0000	52.3	
53 Isobutyl Alcohol	43	6.339	6.339	(0.974)	38634	250.000	269	9145
* 56 FLUOROBENZENE	96	6.507	6.507	(1.000)	372401	50.0000		
57 Methyl Cyclohexane	83	6.669	6.669	(1.025)	151901	50.0000	52.1	9421
58 Trichloroethene	130	6.675	6.675	(1.026)	117545	50.0000	52.1	
59 n-Butanol	56	7.010	7.010	(1.077)	17497	250.000	234	9185
60 Dibromomethane	93	7.068	7.068	(1.086)	67707	50.0000	52.3	
61 2-3 Dichloro-1-Proprene	75	7.147	7.147	(1.098)	172082	50.0000	52.6	9299
62 1,2-Dichloropropane +	63	7.157	7.157	(1.100)	140106	50.0000	51.2	
63 Bromodichloromethane	83	7.230	7.230	(1.111)	136105	50.0000	51.4	
64 Methyl methacrylate	41	7.377	7.377	(1.134)	154085	50.0000	54.3	9296
65 1,4- Dioxane	58	7.414	7.414	(1.139)	29841	1250.00	1340	8705

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
66 1-Bromo-2-chloroethane	63	7.671	7.671	(1.179)	185014	50.0000	51.6	9225
67 2-Chloroethyl vinyl ether	63	7.760	7.760	(1.193)	84008	250.000	253	
68 cis-1,3-Dichloropropene	75	7.812	7.812	(1.201)	166464	50.0000	54.3	
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	353886	50.0000	51.5	
70 Toluene +	91	8.027	8.027	(0.869)	432902	50.0000	52.5	
71 2-Nitropropane	43	8.206	8.206	(0.888)	50609	50.0000	54.8	9131
M 72 1-3 Dichloropropene total	100				323899	100.000	109	0
74 4-methyl-2-pentanone	43	8.337	8.337	(0.902)	875291	250.000	272	
73 Tetrachloroethene	164	8.358	8.358	(0.905)	84196	50.0000	50.4	
75 trans-1,3-Dichloropropene	75	8.373	8.373	(1.287)	157435	50.0000	54.8	
77 Ethyl Methacrylate	41	8.504	8.504	(0.921)	136650	50.0000	54.4	9390
78 1,1,2-Trichloroethane	97	8.510	8.510	(0.921)	94928	50.0000	50.9	
79 Dibromochloromethane	129	8.657	8.657	(0.937)	108933	50.0000	54.0	
76 3,4-dichloro-1-butene	75	8.698	8.698	(0.942)	128660	50.0000	53.1	9220
80 1,3-Dichloropropane	76	8.725	8.725	(0.944)	164282	50.0000	52.1	
81 1-Nitropropane	43	8.809	8.809	(0.953)	27878	50.0000	55.6	7383(H)
82 1,2-Dibromoethane(EDB)	107	8.845	8.845	(0.957)	100158	50.0000	52.1	
83 2-Hexanone	43	9.008	9.008	(0.975)	726920	250.000	271	
89 1-Chlorohexane	91	9.233	9.233	(0.999)	122929	50.0000	52.5	9538
* 90 CHLOROBENZENE-d5	82	9.239	9.239	(1.000)	143574	50.0000		
91 Chlorobenzene ++	112	9.254	9.254	(1.002)	273183	50.0000	51.1	
92 Ethylbenzene +	106	9.270	9.270	(1.003)	149894	50.0000	53.1	
93 1,1,1,2-Tetrachloroethane	133	9.301	9.301	(1.007)	97327	50.0000	53.3	
94 p,m-Xylene	106	9.380	9.380	(1.015)	354550	100.000	105	
M 95 TOTAL XYLENE	106				524371	150.000	158	
96 o-Xylene	106	9.700	9.700	(1.050)	169821	50.0000	53.9	
97 Styrene	104	9.742	9.742	(1.054)	245637	50.0000	54.8	
98 Bromoform ++	173	9.768	9.768	(1.057)	74133	50.0000	55.9	
99 Isopropylbenzene	105	9.931	9.931	(1.075)	427099	50.0000	54.4	
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	106058	50.0000	51.2	
102 cis-1,4-dichloro-2-butene	53	10.177	10.177	(0.931)	63804	50.0000	52.3	9152
103 Bromobenzene	77	10.219	10.219	(0.935)	193183	50.0000	49.2	
104 n-Propylbenzene	91	10.224	10.224	(0.936)	489408	50.0000	50.1	
105 1,1,2,2-Tetrachloroethane++	83	10.277	10.277	(0.941)	126541	50.0000	50.0	
106 2-Chlorotoluene	91	10.345	10.345	(0.947)	338208	50.0000	49.2	
107 1,3,5-Trimethylbenzene	105	10.361	10.361	(0.948)	374037	50.0000	51.1	
108 1,2,3-Trichloropropane	75	10.371	10.371	(0.949)	167995	50.0000	51.1	
109 trans-1,4-Dichloro-2-Butene	53	10.397	10.397	(0.952)	63196	50.0000	53.6	
100 Cyclohexanone	55	10.402	10.402	(0.952)	222467	250.000	258	5396
110 4-Chlorotoluene	91	10.460	10.460	(0.957)	326684	50.0000	49.9	
111 tert-butylbenzene	91	10.591	10.591	(0.969)	204895	50.0000	51.3	
112 1,2,4-Trimethylbenzene	105	10.638	10.638	(0.974)	384424	50.0000	52.3	
113 sec-Butylbenzene	105	10.712	10.712	(0.980)	430030	50.0000	50.7	
114 Dicyclopentadiene	66	10.806	10.806	(0.989)	485759	50.0000	51.5	8618
115 p-Isopropyltoluene	119	10.811	10.811	(0.989)	396485	50.0000	50.4	
116 1,3-Dichlorobenzene	146	10.874	10.874	(0.995)	217851	50.0000	48.3	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	141971	50.0000		

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
118 1,4-Dichlorobenzene	146	10.937	10.937	(1.001)	217455	50.0000	47.9	
119 n-Butylbenzene	91	11.100	11.100	(1.016)	578453	50.0000	51.0	
120 1,2-Dichlorobenzene	146	11.231	11.231	(1.028)	214613	50.0000	49.2	
122 1,2-Dibromo-3-Chloropropane	157	11.771	11.771	(1.077)	31681	50.0000	52.5	
123 Hexachlorobutadiene	225	12.201	12.201	(1.117)	59895	50.0000	49.5	
124 1,2,4-Trichlorobenzene	180	12.227	12.227	(1.119)	131449	50.0000	49.0	
125 Naphthalene	128	12.447	12.447	(1.139)	374712	50.0000	46.8	
126 1,2,3-Trichlorobenzene	180	12.573	12.573	(1.151)	127711	50.0000	49.0	

QC Flag Legend

H - Operator selected an alternate compound hit.

Data File: /var/chem/msv11.i/230517.s.b/j1300.d

Page 1

Date : 17-MAY-2023 12:29

Client ID: V11STD050

Instrument: msv11.i

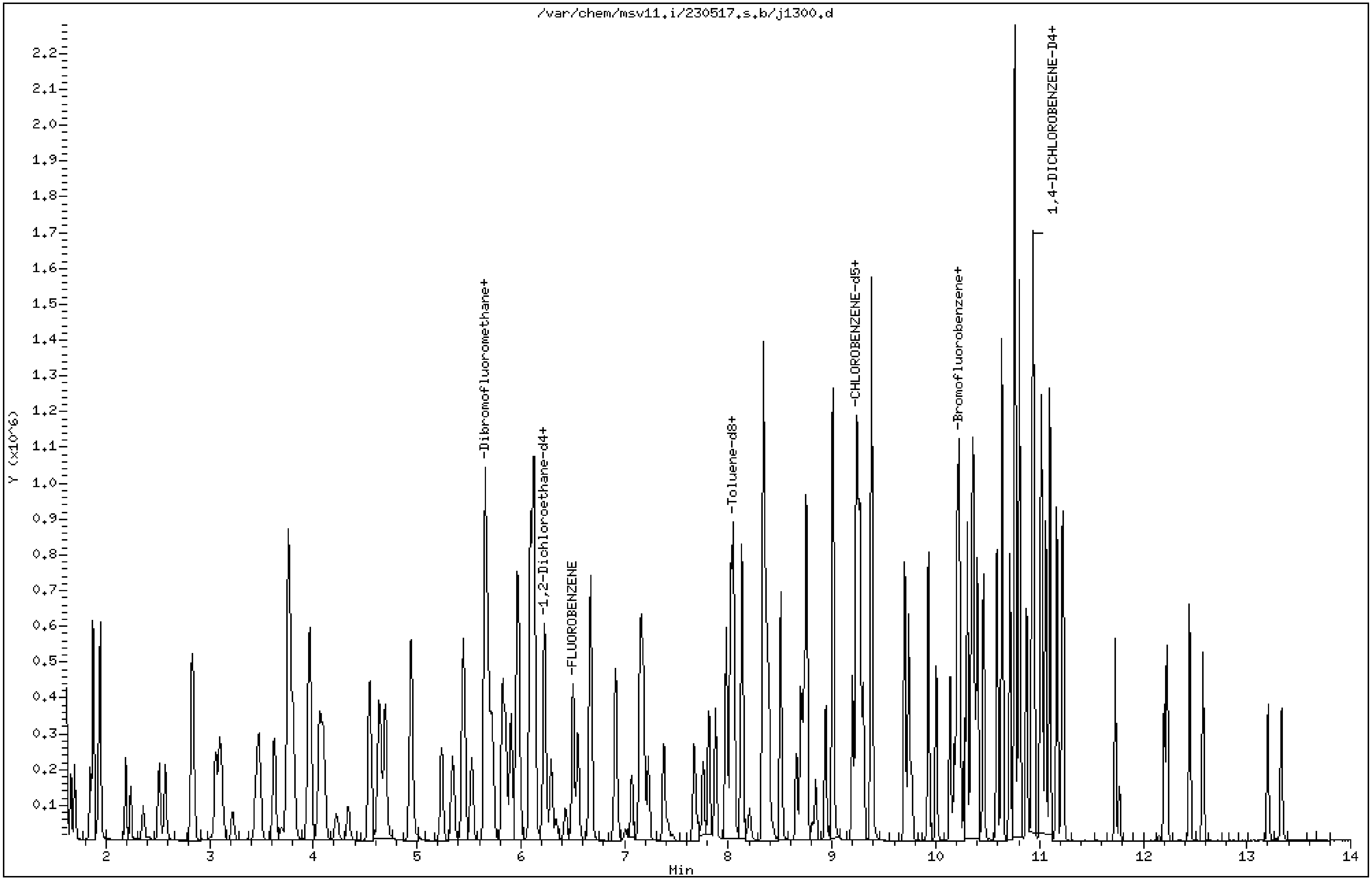
Sample Info: 1207*V11STD050

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

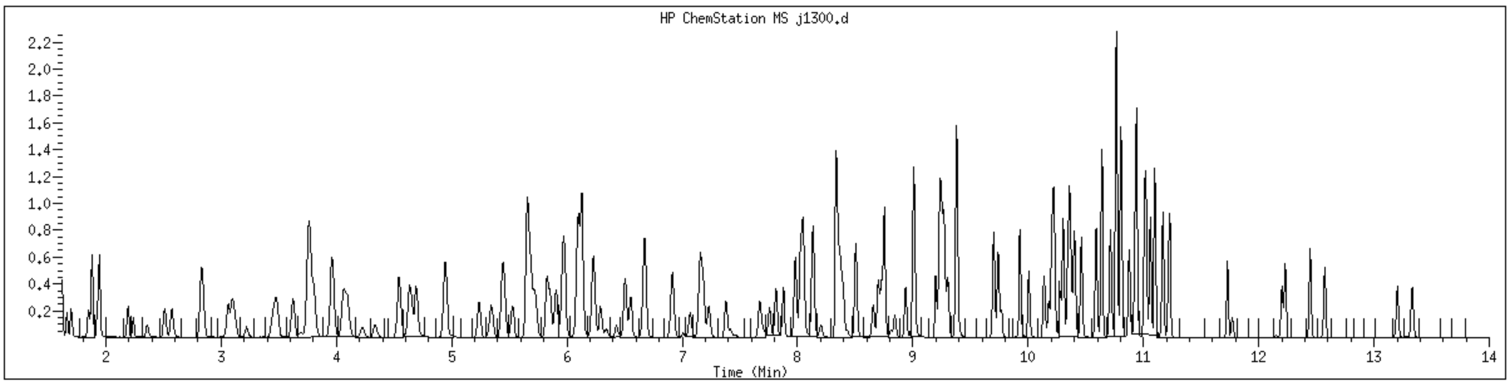
Column diameter: 0.25



Data file : /var/chem/msv11.i/230517.s.b/j1300.d   Page: 1
Report Date: 05/23/2023 17:15

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 1207   SampleType   : CALIB_7
Injection Date: 05/17/2023 12:29   Instrument   : msv11.i
Operator   : KN1
Sample Info   : 1207*V11STD050
Misc Info   : MSV~528~*1*SMR
Method   : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260b+AppIXDOD



NO MANUAL INTEGRATIONS

GCAL, Inc.

Data file : /var/chem/msv11.i/230517.s.b/j1301.d
Lab Smp Id: 1208 Client Smp ID: V11STD100
Inj Date : 17-MAY-2023 12:52
Operator : KN1 Inst ID: msv11.i
Smp Info : 1208*V11STD100
Misc Info : MSV~528~*1*SMR
Comment :
Method : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Meth Date : 23-May-2023 17:15 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 12:52 Cal File: j1301.d
Als bottle: 1 Calibration Sample, Level: 8
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.667	1.667	(0.256)	241948	100.000	99.4	
2 Chloromethane ++	50	1.851	1.851	(0.284)	410582	100.000	96.1	
3 Vinyl Chloride +	62	1.930	1.930	(0.297)	323923	100.000	102	
4 1-3 Butadiene	39	1.951	1.951	(0.300)	339193	100.000	97.8	9757
5 Bromomethane	94	2.239	2.239	(0.344)	121938	100.000	94.5	
7 Chloroethane	64	2.349	2.349	(0.361)	168566	100.000	95.6	
8 Trichlorofluoromethane	101	2.512	2.512	(0.386)	347834	100.000	99.9	
9 Isoprene	67	2.831	2.831	(0.435)	317733	100.000	99.8	9213
10 Ethyl Ether	59	2.847	2.847	(0.438)	246363	100.000	101	6639
11 1,1-Dichloroethene +	96	3.062	3.062	(0.471)	178373	100.000	99.2	
12 Carbon Disulfide	76	3.094	3.094	(0.475)	529394	100.000	97.1	
13 1,1,2Trichlotrifluoroethane	101	3.109	3.109	(0.478)	182904	100.000	97.6	
15 Methyl Iodide	142	3.225	3.225	(0.496)	244432	100.000	104	
16 Acrolein	56	3.450	3.450	(0.530)	309865	500.000	514	

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
17 Allyl chloride	39	3.623	3.623	(0.557)	313299	100.000	100	9380
18 Methylene Chloride	49	3.754	3.754	(0.577)	438622	100.000	97.5	
19 Acetone	43	3.807	3.807	(0.585)	1305790	500.000	484	
20 trans-1,2-Dichloroethene	61	3.953	3.953	(0.608)	388898	100.000	96.6	
21 Methyl Acetate	43	3.969	3.969	(0.610)	1496104	500.000	511	9607
22 Hexane	57	4.063	4.063	(0.625)	605122	100.000	102	9493
23 MTBE	73	4.095	4.095	(0.629)	656754	100.000	103	9017
24 tert-Butyl Alcohol	59	4.226	4.226	(0.649)	219280	500.000	553	8850
25 Acetonitrile	41	4.336	4.336	(0.666)	264960	500.000	508	9784
26 Isopropyl Ether	45	4.546	4.546	(0.699)	1099842	100.000	102	9696
27 Chloroprene	53	4.630	4.630	(0.712)	479321	100.000	104	9512
28 1,1-Dichloroethane ++	63	4.651	4.651	(0.715)	502523	100.000	98.6	
29 Acrylonitrile	53	4.698	4.698	(0.722)	726689	500.000	522	
31 Vinyl Acetate	43	4.939	4.939	(0.759)	660489	100.000	108	
30 Ethyl Tert-butyl Ether	59	4.944	4.944	(0.760)	906439	100.000	103	7626
M 33 Total 1,2-Dichloroethene	61				767785	200.000	196	
32 cis-1,2-Dichloroethene	61	5.238	5.238	(0.805)	378887	100.000	99.7	
34 2,2-Dichloropropane	77	5.343	5.343	(0.821)	322571	100.000	100	
35 Bromochloromethane	128	5.442	5.442	(0.836)	102199	100.000	91.2	
36 Cyclohexane	56	5.453	5.453	(0.838)	525073	100.000	107	8829
37 Chloroform +	83	5.526	5.526	(0.849)	382333	100.000	99.8	
39 Ethyl Acetate	43	5.652	5.652	(0.869)	2061040	500.000	510	6170
38 Carbon Tetrachloride	117	5.668	5.668	(0.871)	321983	100.000	105	
40 Tetrahydrofuran	42	5.684	5.684	(0.873)	789603	500.000	505	7568
$ 41 Dibromofluoromethane	111	5.710	5.710	(0.878)	208639	100.000	103	8775
42 1,1,1-Trichloroethane	97	5.731	5.731	(0.881)	357867	100.000	103	
43 sec-Butanol	45	5.731	5.731	(0.881)	31005	100.000	91.6	9650
44 2-Butanone	43	5.825	5.825	(0.895)	1252568	500.000	527	
45 1,1-Dichloropropene	75	5.857	5.857	(0.900)	300163	100.000	103	
47 2,2,4 Trimethylpentane	57	5.972	5.972	(0.918)	1255000	100.000	103	8337
48 Benzene	78	6.098	6.098	(0.937)	859867	100.000	100	
49 Propionitrile	54	6.103	6.103	(0.938)	283298	500.000	504	7905
50 Methylacrylonitrile	41	6.129	6.129	(0.942)	1291296	500.000	519	9468
$ 51 1,2-Dichloroethane-d4	65	6.229	6.229	(0.957)	273043	100.000	101	
54 tert-amyl Methyl Ether	73	6.229	6.229	(0.957)	636323	100.000	103	7031
52 1,2-Dichloroethane	62	6.292	6.292	(0.967)	371489	100.000	101	
53 Isobutyl Alcohol	43	6.344	6.344	(0.975)	83484	500.000	532	9156
* 56 FLUOROBENZENE	96	6.507	6.507	(1.000)	407194	50.0000		
57 Methyl Cyclohexane	83	6.669	6.669	(1.025)	334876	100.000	105	9376
58 Trichloroethene	130	6.675	6.675	(1.026)	248545	100.000	101	
59 n-Butanol	56	7.005	7.005	(1.077)	41129	500.000	489	9004(M3)
60 Dibromomethane	93	7.068	7.068	(1.086)	143345	100.000	101	
61 2-3 Dichloro-1-Proprene	75	7.146	7.146	(1.098)	375798	100.000	105	9326
62 1,2-Dichloropropane +	63	7.162	7.162	(1.101)	302441	100.000	101	
63 Bromodichloromethane	83	7.230	7.230	(1.111)	299205	100.000	103	
64 Methyl methacrylate	41	7.377	7.377	(1.134)	336400	100.000	108	9423
65 1,4- Dioxane	58	7.419	7.419	(1.140)	63819	2500.00	2620	8379

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
66 1-Bromo-2-chloroethane	63	7.671	7.671	(1.179)	405860	100.000	104	9346
67 2-Chloroethyl vinyl ether	63	7.760	7.760	(1.193)	177922	500.000	485	
68 cis-1,3-Dichloropropene	75	7.812	7.812	(1.201)	364770	100.000	109	
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	747971	100.000	101	
70 Toluene +	91	8.022	8.022	(0.868)	916637	100.000	103	
71 2-Nitropropane	43	8.205	8.205	(0.888)	110871	100.000	111	9426
M 72 1-3 Dichloropropene total	100				707343	200.000	218	0
74 4-methyl-2-pentanone	43	8.337	8.337	(0.902)	1865424	500.000	537	
73 Tetrachloroethene	164	8.358	8.358	(0.905)	176616	100.000	97.9	
75 trans-1,3-Dichloropropene	75	8.373	8.373	(1.287)	342573	100.000	109	
77 Ethyl Methacrylate	41	8.504	8.504	(0.921)	305033	100.000	112	9398
78 1,1,2-Trichloroethane	97	8.510	8.510	(0.921)	202722	100.000	101	
79 Dibromochloromethane	129	8.656	8.656	(0.937)	235110	100.000	108	
76 3,4-dichloro-1-butene	75	8.698	8.698	(0.942)	283272	100.000	108	9411
80 1,3-Dichloropropane	76	8.725	8.725	(0.944)	349949	100.000	103	
81 1-Nitropropane	43	8.808	8.808	(0.953)	54753	100.000	101	7356(H)
82 1,2-Dibromoethane(EDB)	107	8.840	8.840	(0.957)	214775	100.000	104	
83 2-Hexanone	43	9.008	9.008	(0.975)	1532348	500.000	530	
89 1-Chlorohexane	91	9.233	9.233	(0.999)	263859	100.000	104	8340
* 90 CHLOROBENZENE-d5	82	9.238	9.238	(1.000)	155064	50.0000		
91 Chlorobenzene ++	112	9.254	9.254	(1.002)	577800	100.000	100	
92 Ethylbenzene +	106	9.270	9.270	(1.003)	313253	100.000	103	
93 1,1,1,2-Tetrachloroethane	133	9.301	9.301	(1.007)	206224	100.000	105	
94 p,m-Xylene	106	9.380	9.380	(1.015)	768057	200.000	210	
M 95 TOTAL XYLENE	106				1141249	300.000	319	
96 o-Xylene	106	9.700	9.700	(1.050)	373192	100.000	110	
97 Styrene	104	9.742	9.742	(1.054)	533052	100.000	110	
98 Bromoform ++	173	9.768	9.768	(1.057)	164821	100.000	115	
99 Isopropylbenzene	105	9.930	9.930	(1.075)	924358	100.000	109	
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	231427	100.000	103	
102 cis-1,4-dichloro-2-butene	53	10.177	10.177	(0.931)	140625	100.000	104	9249
103 Bromobenzene	77	10.219	10.219	(0.935)	419980	100.000	97.0	
104 n-Propylbenzene	91	10.229	10.229	(0.936)	1061179	100.000	98.5	
105 1,1,2,2-Tetrachloroethane++	83	10.276	10.276	(0.941)	263606	100.000	94.5	
106 2-Chlorotoluene	91	10.345	10.345	(0.947)	756826	100.000	99.7	
107 1,3,5-Trimethylbenzene	105	10.360	10.360	(0.948)	816488	100.000	101	
108 1,2,3-Trichloropropane	75	10.376	10.376	(0.950)	372828	100.000	103	
109 trans-1,4-Dichloro-2-Butene	53	10.397	10.397	(0.952)	139116	100.000	107	
100 Cyclohexanone	55	10.402	10.402	(0.952)	503839	500.000	531	5553(A)
110 4-Chlorotoluene	91	10.460	10.460	(0.957)	701710	100.000	97.2	
111 tert-butylbenzene	91	10.591	10.591	(0.969)	441938	100.000	100	
112 1,2,4-Trimethylbenzene	105	10.638	10.638	(0.974)	828036	100.000	102	
113 sec-Butylbenzene	105	10.712	10.712	(0.980)	940864	100.000	101	
114 Dicyclopentadiene	66	10.806	10.806	(0.989)	1063064	100.000	102	8706
115 p-Isopropyltoluene	119	10.811	10.811	(0.989)	857166	100.000	98.9	
116 1,3-Dichlorobenzene	146	10.874	10.874	(0.995)	465417	100.000	93.7	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	156580	50.0000		

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
118 1,4-Dichlorobenzene	146	10.937	10.937	(1.001)	468493	100.000	93.6	
119 n-Butylbenzene	91	11.100	11.100	(1.016)	1262386	100.000	101	
120 1,2-Dichlorobenzene	146	11.231	11.231	(1.028)	456079	100.000	94.8	
122 1,2-Dibromo-3-Chloropropane	157	11.771	11.771	(1.077)	69796	100.000	105	
123 Hexachlorobutadiene	225	12.201	12.201	(1.117)	128726	100.000	96.4	
124 1,2,4-Trichlorobenzene	180	12.227	12.227	(1.119)	282947	100.000	95.6	
125 Naphthalene	128	12.447	12.447	(1.139)	842819	100.000	94.8	
126 1,2,3-Trichlorobenzene	180	12.578	12.578	(1.151)	274487	100.000	95.4	

QC Flag Legend

A - Target compound detected but, quantitated amount
exceeded maximum amount.
M3- Compound response manually integrated because
Target system integrated incorrect peak.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv11.i/230517.s.b/j1301.d

Date : 17-MAY-2023 12:52

Client ID: V11STD100

Sample Info: 1208*V11STD100

Purge Volume: 5.0

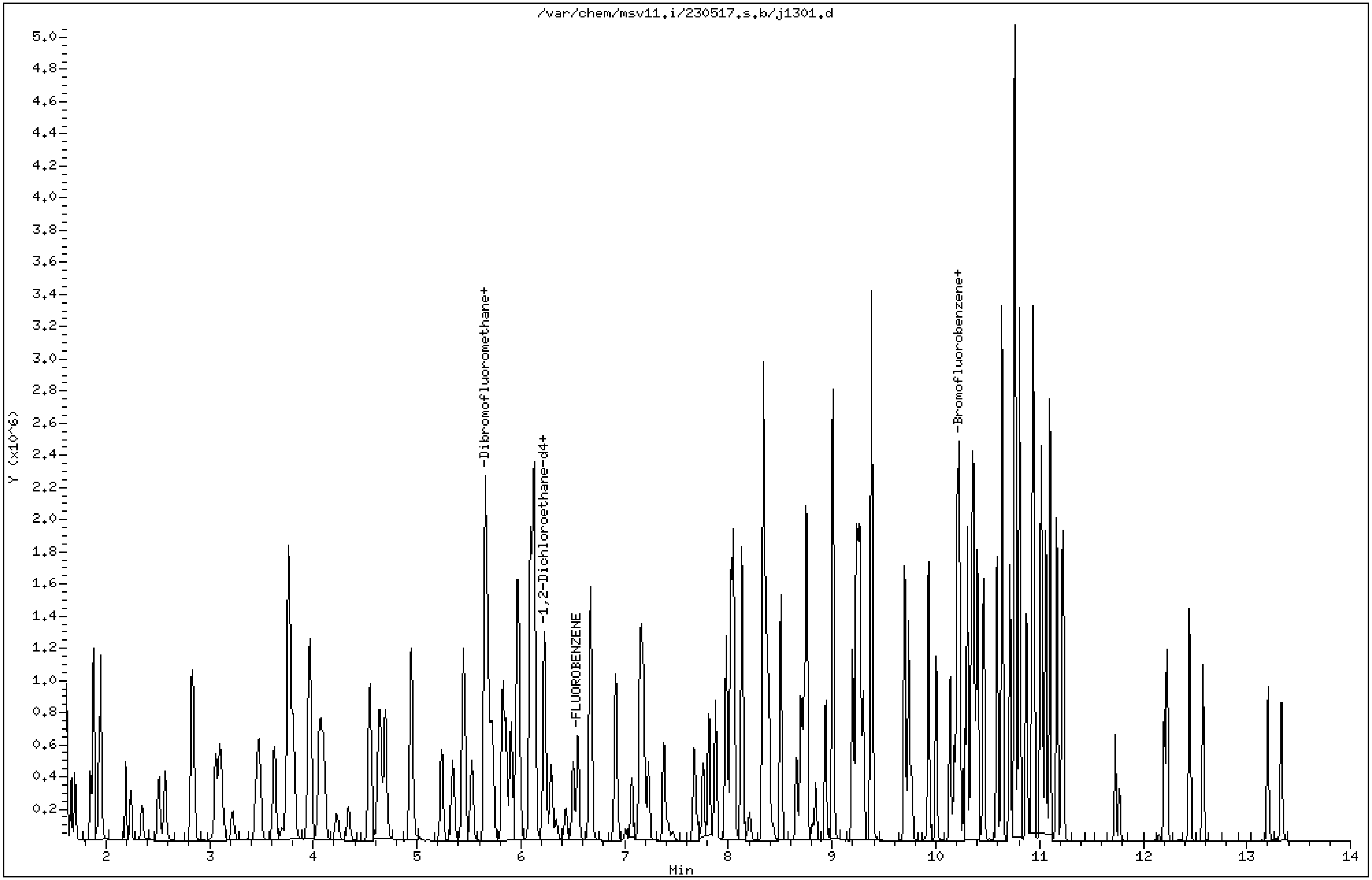
Column phase: RTX-VMS-30M

Page 1

Instrument: msv11.i

Operator: KN1

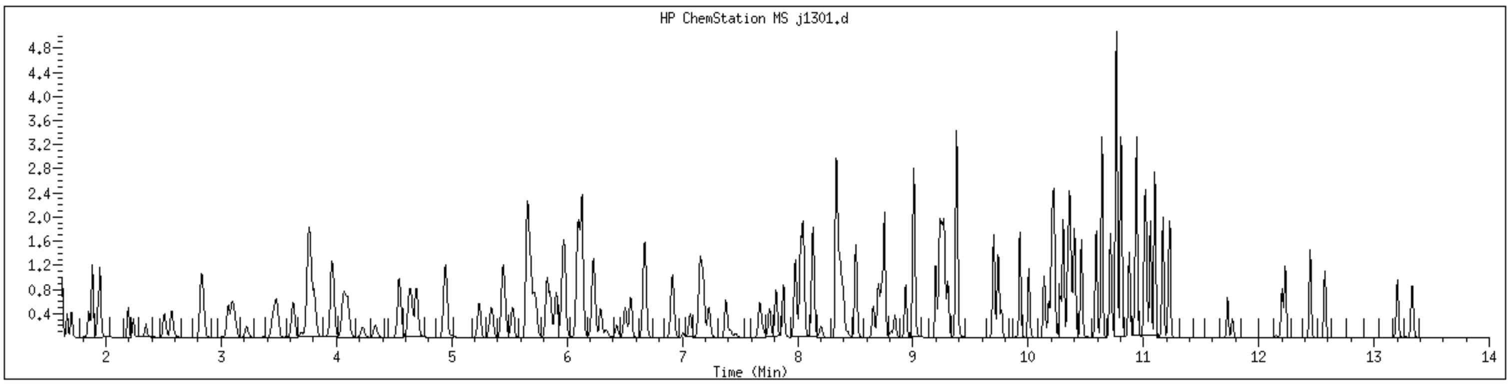
Column diameter: 0.25



Report Date: 05/23/2023 17:15

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1208 SampleType : CALIB_8
Injection Date: 05/17/2023 12:52 Instrument : msv11.i
Operator : KN1
Sample Info : 1208*V11STD100
Misc Info : MSV~528~*1*SMR
Method : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260b+AppIXDOD



Original Final

==

59 n-Butanol CAS#: 71-36-3 Reason: M3

Electronic Signature
Applied

GCAL, Inc.

Data file : /var/chem/msv11.i/230517.s.b/j1302.d
Lab Smp Id: 1209 Client Smp ID: V11STD200
Inj Date : 17-MAY-2023 13:15
Operator : KN1 Inst ID: msv11.i
Smp Info : 1209*V11STD200
Misc Info : MSV~528~*1*SMR
Comment :
Method : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Meth Date : 23-May-2023 17:15 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 13:15 Cal File: j1302.d
Als bottle: 1 Calibration Sample, Level: 9
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.667	1.667	(0.256)	513913	200.000	204	(A)
2 Chloromethane ++	50	1.856	1.856	(0.285)	884136	200.000	199	
3 Vinyl Chloride +	62	1.935	1.935	(0.297)	666220	200.000	203	(A)
4 1-3 Butadiene	39	1.956	1.956	(0.301)	697705	200.000	194	9748
5 Bromomethane	94	2.239	2.239	(0.344)	270172	200.000	203	(A)
7 Chloroethane	64	2.344	2.344	(0.360)	191164	200.000	104	
8 Trichlorofluoromethane	101	2.501	2.501	(0.384)	721315	200.000	200	
9 Isoprene	67	2.826	2.826	(0.434)	691894	200.000	210	8595(A)
10 Ethyl Ether	59	2.852	2.852	(0.438)	536395	200.000	211	8689
11 1,1-Dichloroethene +	96	3.062	3.062	(0.471)	380585	200.000	204	(A)
12 Carbon Disulfide	76	3.094	3.094	(0.475)	1153996	200.000	205	(A)
13 1,1,2Trichlotrifluoroethane	101	3.109	3.109	(0.478)	387781	200.000	199	
15 Methyl Iodide	142	3.219	3.219	(0.495)	492906	200.000	199	
16 Acrolein	56	3.450	3.450	(0.530)	668488	1000.00	1070	(A)

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
17 Allyl chloride	39	3.628	3.628	(0.558)	537569	200.000	166	9385
18 Methylene Chloride	49	3.754	3.754	(0.577)	929648	200.000	199	
19 Acetone	43	3.812	3.812	(0.586)	2852190	1000.00	1020	(AM2)
20 trans-1,2-Dichloroethene	61	3.953	3.953	(0.608)	848997	200.000	203	(A)
21 Methyl Acetate	43	3.974	3.974	(0.611)	3112722	1000.00	1030	9511(A)
22 Hexane	57	4.064	4.064	(0.625)	1341218	200.000	217	9544(A)
23 MTBE	73	4.100	4.100	(0.630)	1467649	200.000	222	9290(A)
24 tert-Butyl Alcohol	59	4.237	4.237	(0.651)	463046	1000.00	1130	8766(A)
25 Acetonitrile	41	4.341	4.341	(0.667)	552998	1000.00	1020	9789(A)
26 Isopropyl Ether	45	4.546	4.546	(0.699)	2423047	200.000	217	9675(A)
27 Chloroprene	53	4.630	4.630	(0.712)	1035473	200.000	216	9494(A)
28 1,1-Dichloroethane ++	63	4.651	4.651	(0.715)	1096532	200.000	207	(A)
29 Acrylonitrile	53	4.698	4.698	(0.722)	1575645	1000.00	1090	(A)
31 Vinyl Acetate	43	4.939	4.939	(0.759)	1345148	200.000	211	(A)
30 Ethyl Tert-butyl Ether	59	4.950	4.950	(0.761)	2017432	200.000	220	7902(A)
M 33 Total 1,2-Dichloroethene	61				1679427	400.000	414	
32 cis-1,2-Dichloroethene	61	5.238	5.238	(0.805)	830430	200.000	211	(A)
34 2,2-Dichloropropane	77	5.348	5.348	(0.822)	689598	200.000	207	(A)
35 Bromochloromethane	128	5.442	5.442	(0.836)	218086	200.000	188	
36 Cyclohexane	56	5.453	5.453	(0.838)	1156770	200.000	228	8896(A)
37 Chloroform +	83	5.526	5.526	(0.849)	829854	200.000	209	(A)
39 Ethyl Acetate	43	5.657	5.657	(0.869)	4436252	1000.00	1060	6562(A)
38 Carbon Tetrachloride	117	5.668	5.668	(0.871)	674789	200.000	213	(A)
40 Tetrahydrofuran	42	5.684	5.684	(0.873)	1712404	1000.00	1060	7750(A)
$ 41 Dibromofluoromethane	111	5.710	5.710	(0.878)	444459	200.000	211	8482(A)
42 1,1,1-Trichloroethane	97	5.731	5.731	(0.881)	769228	200.000	213	(A)
43 sec-Butanol	45	5.741	5.741	(0.882)	68947	200.000	196	9108
44 2-Butanone	43	5.825	5.825	(0.895)	2676808	1000.00	1090	(A)
45 1,1-Dichloropropene	75	5.857	5.857	(0.900)	650952	200.000	215	(A)
47 2,2,4 Trimethylpentane	57	5.972	5.972	(0.918)	2920089	200.000	230	8411(A)
48 Benzene	78	6.098	6.098	(0.937)	1859926	200.000	209	(A)
49 Propionitrile	54	6.108	6.108	(0.939)	597272	1000.00	1020	8426(A)
50 Methylacrylonitrile	41	6.135	6.135	(0.943)	2663377	1000.00	1030	9509(A)
$ 51 1,2-Dichloroethane-d4	65	6.229	6.229	(0.957)	575042	200.000	204	(A)
54 tert-amyl Methyl Ether	73	6.234	6.234	(0.958)	1405215	200.000	219	6813(A)
52 1,2-Dichloroethane	62	6.297	6.297	(0.968)	787491	200.000	207	(A)
53 Isobutyl Alcohol	43	6.350	6.350	(0.976)	190597	1000.00	1170	9084(A)
* 56 FLUOROBENZENE	96	6.507	6.507	(1.000)	422453	50.0000		
57 Methyl Cyclohexane	83	6.669	6.669	(1.025)	735738	200.000	222	9384(A)
58 Trichloroethene	130	6.680	6.680	(1.027)	536097	200.000	210	(A)
59 n-Butanol	56	7.015	7.015	(1.078)	90875	1000.00	1030	9119(A)
60 Dibromomethane	93	7.068	7.068	(1.086)	313324	200.000	213	(A)
61 2-3 Dichloro-1-Proprene	75	7.152	7.152	(1.099)	810908	200.000	219	9518(A)
62 1,2-Dichloropropane +	63	7.162	7.162	(1.101)	649261	200.000	209	(A)
63 Bromodichloromethane	83	7.230	7.230	(1.111)	640114	200.000	213	(A)
64 Methyl methacrylate	41	7.382	7.382	(1.135)	728293	200.000	226	9494(A)
65 1,4- Dioxane	58	7.419	7.419	(1.140)	150328	5000.00	5950	8140(A)

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
66 1-Bromo-2-chloroethane	63	7.671	7.671	(1.179)	865339	200.000	213	9558(A)
67 2-Chloroethyl vinyl ether	63	7.760	7.760	(1.193)	392563	1000.00	1030	
68 cis-1,3-Dichloropropene	75	7.812	7.812	(1.201)	787204	200.000	226	(A)
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	1593390	200.000	209	(A)
70 Toluene +	91	8.027	8.027	(0.869)	1956334	200.000	214	(A)
71 2-Nitropropane	43	8.206	8.206	(0.888)	237354	200.000	232	9727(A)
M 72 1-3 Dichloropropene total	100				1517920	400.000	451	0
74 4-methyl-2-pentanone	43	8.342	8.342	(0.903)	3735345	1000.00	1050	(A)
73 Tetrachloroethene	164	8.363	8.363	(0.905)	376205	200.000	203	(A)
75 trans-1,3-Dichloropropene	75	8.379	8.379	(1.288)	730716	200.000	224	(A)
77 Ethyl Methacrylate	41	8.504	8.504	(0.921)	628009	200.000	225	9461(A)
78 1,1,2-Trichloroethane	97	8.510	8.510	(0.921)	426280	200.000	206	(A)
79 Dibromochloromethane	129	8.656	8.656	(0.937)	509548	200.000	228	(A)
76 3,4-dichloro-1-butene	75	8.704	8.704	(0.942)	613530	200.000	228	9404(A)
80 1,3-Dichloropropane	76	8.730	8.730	(0.945)	739569	200.000	212	(A)
81 1-Nitropropane	43	8.814	8.814	(0.954)	109437	200.000	195	6068(M3H)
82 1,2-Dibromoethane(EDB)	107	8.845	8.845	(0.957)	453094	200.000	213	(A)
83 2-Hexanone	43	9.008	9.008	(0.975)	3320051	1000.00	1120	(A)
89 1-Chlorohexane	91	9.233	9.233	(0.999)	561308	200.000	216	7130(A)
* 90 CHLOROBENZENE-d5	82	9.238	9.238	(1.000)	159292	50.0000		
91 Chlorobenzene ++	112	9.254	9.254	(1.002)	1209789	200.000	204	(A)
92 Ethylbenzene +	106	9.270	9.270	(1.003)	668120	200.000	213	(A)
93 1,1,1,2-Tetrachloroethane	133	9.301	9.301	(1.007)	438584	200.000	216	(A)
94 p,m-Xylene	106	9.380	9.380	(1.015)	1625743	400.000	432	(A)
M 95 TOTAL XYLENE	106				2434379	600.000	663	
96 o-Xylene	106	9.700	9.700	(1.050)	808636	200.000	231	(A)
97 Styrene	104	9.742	9.742	(1.054)	1187286	200.000	239	(A)
98 Bromoform ++	173	9.768	9.768	(1.057)	371611	200.000	252	(A)
99 Isopropylbenzene	105	9.931	9.931	(1.075)	2045770	200.000	235	(A)
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	508408	200.000	221	(A)
102 cis-1,4-dichloro-2-butene	53	10.177	10.177	(0.931)	316911	200.000	236	9385(A)
103 Bromobenzene	77	10.214	10.214	(0.935)	938836	200.000	217	(A)
104 n-Propylbenzene	91	10.229	10.229	(0.936)	2351153	200.000	219	(A)
105 1,1,2,2-Tetrachloroethane++	83	10.277	10.277	(0.941)	572628	200.000	206	(A)
106 2-Chlorotoluene	91	10.345	10.345	(0.947)	1677897	200.000	222	(A)
107 1,3,5-Trimethylbenzene	105	10.360	10.360	(0.948)	1793409	200.000	223	(A)
108 1,2,3-Trichloropropane	75	10.376	10.376	(0.950)	816817	200.000	225	(A)
109 trans-1,4-Dichloro-2-Butene	53	10.397	10.397	(0.952)	307188	200.000	237	(A)
100 Cyclohexanone	55	10.408	10.408	(0.952)	1031115	1000.00	1090	4881(AH)
110 4-Chlorotoluene	91	10.460	10.460	(0.957)	1557641	200.000	216	(A)
111 tert-butylbenzene	91	10.591	10.591	(0.969)	971437	200.000	221	(A)
112 1,2,4-Trimethylbenzene	105	10.638	10.638	(0.974)	1792480	200.000	221	(A)
113 sec-Butylbenzene	105	10.717	10.717	(0.981)	2047868	200.000	219	(A)
114 Dicyclopentadiene	66	10.806	10.806	(0.989)	2327787	200.000	224	8773(A)
115 p-Isopropyltoluene	119	10.811	10.811	(0.989)	1833310	200.000	212	(A)
116 1,3-Dichlorobenzene	146	10.880	10.880	(0.996)	997729	200.000	201	(A)
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	156319	50.0000		

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
118 1,4-Dichlorobenzene	146	10.937	10.937	(1.001)	990651	200.000	198	
119 n-Butylbenzene	91	11.100	11.100	(1.016)	2707449	200.000	217	(A)
120 1,2-Dichlorobenzene	146	11.231	11.231	(1.028)	966911	200.000	201	(A)
122 1,2-Dibromo-3-Chloropropane	157	11.771	11.771	(1.077)	153948	200.000	232	(A)
123 Hexachlorobutadiene	225	12.201	12.201	(1.117)	289588	200.000	217	(A)
124 1,2,4-Trichlorobenzene	180	12.227	12.227	(1.119)	629935	200.000	213	(A)
125 Naphthalene	128	12.447	12.447	(1.139)	1884818	200.000	211	(A)
126 1,2,3-Trichlorobenzene	180	12.573	12.573	(1.151)	595849	200.000	208	(A)

QC Flag Legend

A - Target compound detected but, quantitated amount exceeded maximum amount.
M2- Compound response manually integrated because Target system integrated incorrectly.
M3- Compound response manually integrated because Target system integrated incorrect peak.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv11.i/230517.s.b/j1302.d

Page 1

Date : 17-MAY-2023 13:15

Client ID: V11STD200

Instrument: msv11.i

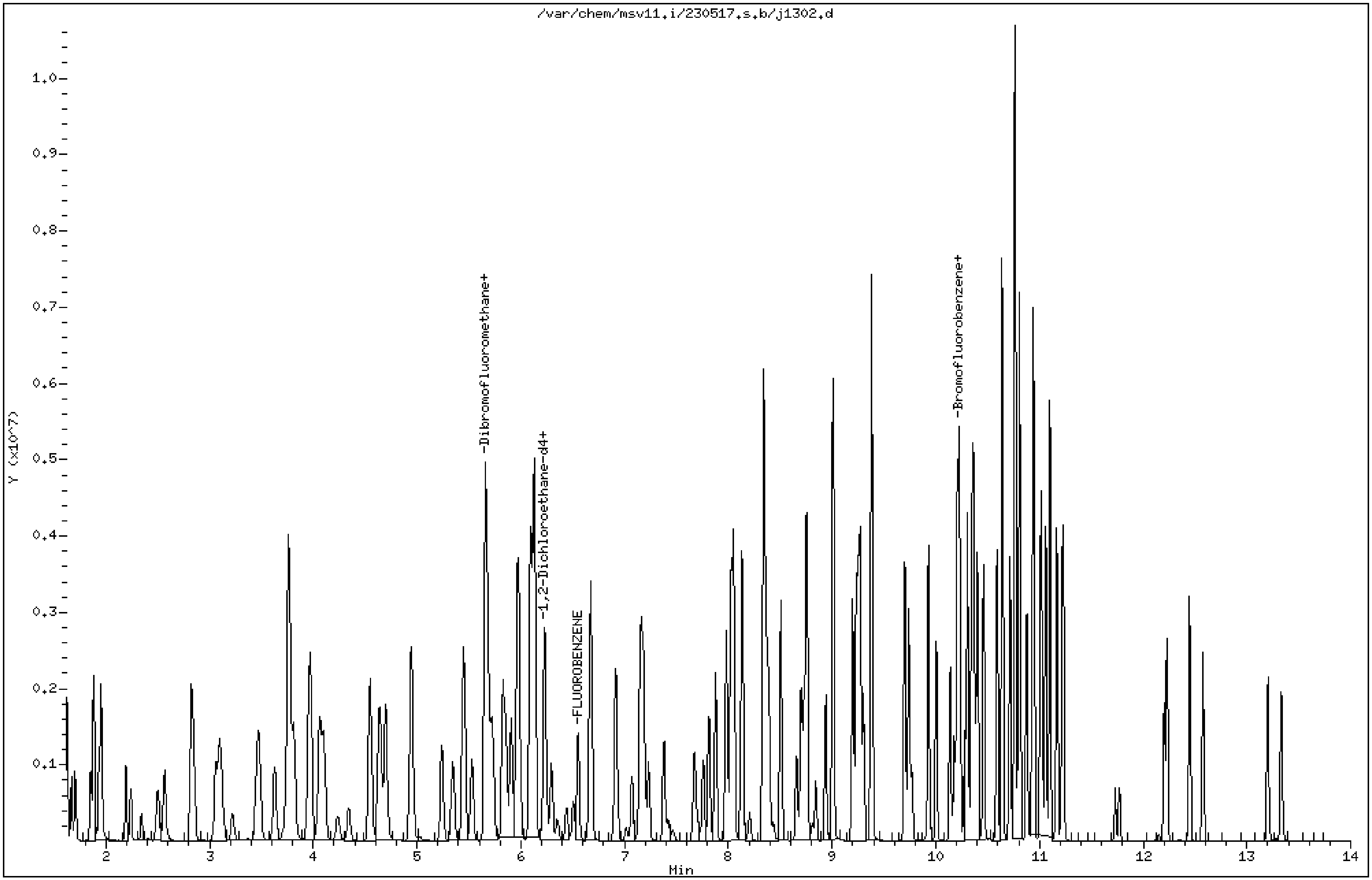
Sample Info: 1209*V11STD200

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

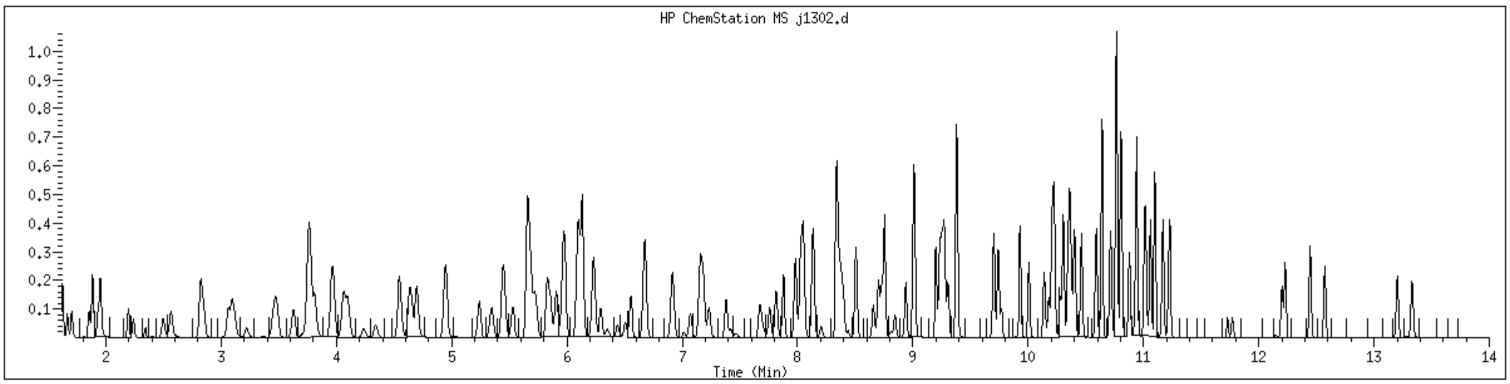
Column diameter: 0.25



Data file : /var/chem/msv11.i/230517.s.b/j1302.d   Page: 1
Report Date: 05/23/2023 17:15

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 1209   SampleType   : CALIB_9
Injection Date: 05/17/2023 13:15   Instrument   : msv11.i
Operator   : KN1
Sample Info   : 1209*V11STD200
Misc Info   : MSV~528~*1*SMR
Method   : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260b+AppIXDOD



Original   Final

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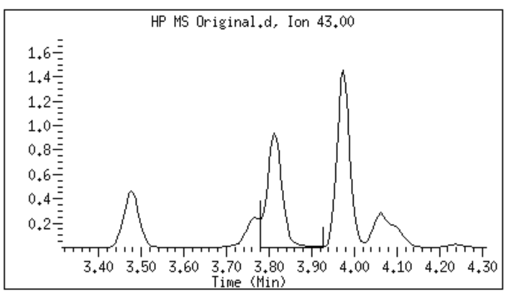
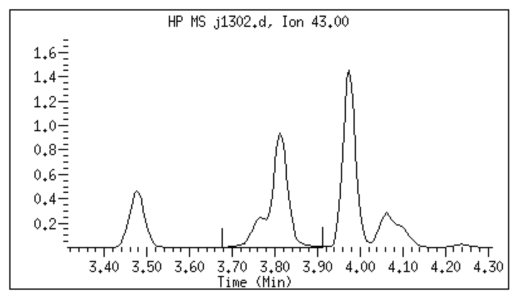
19 Acetone   CAS#: 67-64-1   Reason: M2



Electronic Signature Applied

User: kmn
Date: 05/17/2023 13:47



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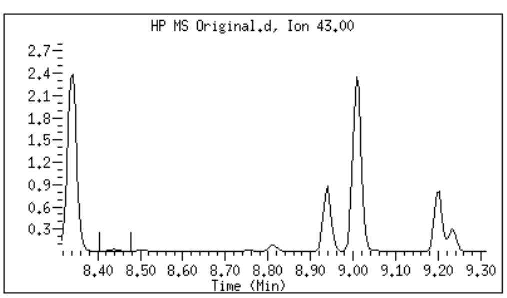
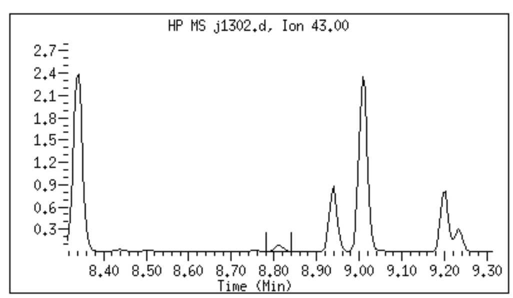
81 1-Nitropropane   CAS#: 108-03-2   Reason: M3



Electronic Signature Applied

User: kmn
Date: 05/17/2023 13:48



M2 - Target system integrated incorrectly
M3 - Target system integrated incorrect peak

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

		LabQC ID - FileID - Conc	1203 ~ 230517/c0228 ~ 1
Report No: 223060210	Instrument ID: MSV15	1204 ~ 230517/c0229 ~ 5	1205 ~ 230517/c0230 ~ 10
GC Column: RTX-VMS-30M ID .25 (mm)	Analyst: CJR	1206 ~ 230517/c0231 ~ 20	1207 ~ 230517/c0232 ~ 50
Calib. Date 1: 05/17/23 Time 1: 1428	Analytical Batch: 766139	1208 ~ 230517/c0233 ~ 100	1209 ~ 230517/c0234 ~ 200
Calib. Date 2: 05/17/23 Time 2: 1620	Analytical Method: EPA 8260B		

ANALYTE	*1201*	*1202*	*1203*	*1204*	*1205*	*1206*	*1207*	*1208*	*1209*	*RF / m / A*	*b / B*	*C*	*FIT*	*TYPE*	
1,1,1,2-Tetrachloroethane			0.689	0.580	0.587	0.639	0.660	0.688	0.699	0.649			7.532	A	
1,1,1-Trichloroethane			0.370	0.406	0.400	0.446	0.449	0.463	0.462	0.428			8.438	A	
1,1,2,2-Tetrachloroethane			0.623	0.775	0.805	0.826	0.806	0.767	0.739	0.763			8.956	A	
1,1,2-Trichloroethane			0.498	0.574	0.586	0.609	0.621	0.627	0.634	0.593			7.930	A	
1,1-Dichloroethane			0.387	0.452	0.436	0.467	0.473	0.473	0.474	0.452			7.012	A	
1,1-Dichloroethene			0.228	0.231	0.227	0.249	0.255	0.260	0.259	0.244			6.049	A	
1,1-Dichloropropene			0.351	0.366	0.359	0.395	0.400	0.395	0.399	0.381			5.599	A	
1,2,3-Trichlorobenzene			0.563	0.674	0.710	0.755	0.767	0.785	0.820	0.725			11.87	A	
1,2,3-Trichloropropane			1.037	0.998	1.019	1.101	1.072	1.027	0.993	1.035			3.774	A	
1,2,4-Trichlorobenzene			0.701	0.800	0.794	0.864	0.905	0.887	0.932	0.840			9.520	A	
1,2,4-Trimethylbenzene			2.030	2.501	2.556	2.795	2.762	2.704	2.698	2.578			10.24	A	
1,2-Dibromo-3-chloropropan			159	1562	3400	8227	22062	51173	114746	0.196	0.017		0.998	W	
1,2-Dibromo-3-chloropropan			0.079	0.151	0.157	0.183	0.183	0.192	0.200						
1,2-Dibromoethane			0.492	0.620	0.633	0.656	0.672	0.684	0.681	0.634			10.57	A	
1,2-Dichlorobenzene			1.211	1.328	1.349	1.412	1.437	1.437	1.449	1.375			6.254	A	
1,2-Dichloroethane			0.291	0.318	0.324	0.336	0.333	0.335	0.330	0.324			4.858	A	
1,2-Dichloroethane-d4			0.268	0.233	0.242	0.266	0.261	0.266	0.264	0.257			5.296	A	
1,2-Dichloroethene (total)			0.298	0.317	0.322	0.344	0.346	0.346	0.349	0.332			5.867	A	
1,2-Dichloropropane			0.223	0.241	0.245	0.263	0.267	0.268	0.273	0.254			7.163	A	
1,3,5-Trimethylbenzene			2.180	2.566	2.599	2.836	2.777	2.749	2.677	2.626			8.337	A	
1,3-Butadiene			0.205	0.216	0.210	0.241	0.216	0.220	0.226	0.219			5.335	A	
1,3-Dichlorobenzene			1.405	1.475	1.487	1.562	1.571	1.563	1.576	1.520			4.297	A	
1,3-Dichloropropane			0.855	0.979	0.977	1.040	1.030	1.008	1.017	0.987			6.346	A	
1,3-Dichloropropylene			0.353	0.368	0.376	0.410	0.418	0.427	0.428	0.397			7.748	A	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines
to allow for evaluation against minimum RRF

RF = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

				LabQC ID - FileID - Conc	1203 ~ 230517/c0228 ~ 1
Report No:	223060210	Instrument ID:	MSV15	1204 ~ 230517/c0229 ~ 5	1205 ~ 230517/c0230 ~ 10
GC Column:	RTX-VMS-30M ID .25 (mm)	Analyst:	CJR	1206 ~ 230517/c0231 ~ 20	1207 ~ 230517/c0232 ~ 50
Calib. Date 1:	05/17/23 Time 1: 1428	Analytical Batch:	766139	1208 ~ 230517/c0233 ~ 100	1209 ~ 230517/c0234 ~ 200
Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Method:	EPA 8260B		

ANALYTE	***1201***	***1202***	***1203***	***1204***	***1205***	***1206***	***1207***	***1208***	***1209***	***RF / m / A***	***b / B***	***C***	***FIT***	***TYPE***	
1,4 Dioxane				0.002	0.002	0.002	0.002	0.002	0.002	0.002			8.418	A	
1,4-Dichlorobenzene			1.456	1.496	1.520	1.622	1.618	1.602	1.628	1.563			4.521	A	
1-Bromo-2-Chloroethane			0.283	0.310	0.317	0.335	0.338	0.345	0.339	0.324			6.761	A	
1-Chlorohexane			0.935	0.993	1.053	1.001	1.141	1.067	1.110	1.043			6.876	A	
1-Nitropropane			0.000	0.072	0.071	0.070	0.073	0.082	0.075	0.074			6.121	A	
2,2,4-Trimethylpentane			0.856	1.088	1.095	1.189	1.242	1.279	1.297	1.150			13.35	A	
2,2-Dichloropropane			0.354	0.395	0.391	0.419	0.417	0.421	0.415	0.402			6.083	A	
2,3-Dichloro-1-propene			0.354	0.390	0.386	0.420	0.442	0.448	0.459	0.414			9.328	A	
2-Butanone			0.082	0.111	0.111	0.125	0.123	0.127	0.126	0.115			13.88	A	
2-Chloroethylvinyl ether (RS			1244	8287	19577	35793	105023	237890	569730	0.074	0.089		0.991	W	
2-Chloroethylvinyl ether			0.042	0.054	0.063	0.056	0.063	0.070	0.078						
2-Chlorotoluene			2.240	2.509	2.503	2.651	2.591	2.484	2.446	2.489			5.210	A	
2-Hexanone (RSP)			2993	22444	46436	109907	287469	616069	1298993	0.450	0.052		0.999	W	
2-Hexanone			0.260	0.389	0.388	0.441	0.445	0.453	0.448						
2-Nitropropane (RSP)			147	1643	3719	8113	19389	44203	94852	0.163	0.012		0.999	W	
2-Nitropropane			0.064	0.143	0.156	0.163	0.150	0.162	0.164						
4-Bromofluorobenzene			0.843	0.716	0.700	0.744	0.761	0.769	0.794	0.761			6.324	A	
4-Chlorotoluene			1.953	2.229	2.308	2.411	2.317	2.246	2.172	2.234			6.512	A	
4-Isopropyltoluene			2.454	2.856	2.926	3.216	3.265	3.156	3.180	3.008			9.564	A	
4-Methyl-2-pentanone (RSP)			3673	26975	57708	132162	343899	740988	1596464	0.548	0.054		0.999	W	
4-Methyl-2-pentanone			0.319	0.468	0.483	0.530	0.532	0.545	0.551						
Acetone			0.091	0.112	0.102	0.115	0.117	0.125	0.114	0.111			9.992	A	
Acetonitrile			0.015	0.019	0.020	0.022	0.021	0.021	0.020	0.020			11.00	A	
Acrolein			0.031	0.035	0.036	0.039	0.039	0.040	0.040	0.037			9.334	A	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines
to allow for evaluation against minimum RRF

RF = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

				LabQC ID - FileID - Conc	1203 ~ 230517/c0228 ~ 1
Report No:	223060210	Instrument ID:	MSV15	1204 ~ 230517/c0229 ~ 5	1205 ~ 230517/c0230 ~ 10
GC Column:	RTX-VMS-30M ID .25 (mm)	Analyst:	CJR	1206 ~ 230517/c0231 ~ 20	1207 ~ 230517/c0232 ~ 50
Calib. Date 1:	05/17/23 Time 1: 1428	Analytical Batch:	766139	1208 ~ 230517/c0233 ~ 100	1209 ~ 230517/c0234 ~ 200
Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Method:	EPA 8260B		

ANALYTE	1201	1202	1203	1204	1205	1206	1207	1208	1209	RF / m / A	b / B	C	FIT	TYPE	
Acrylonitrile			0.068	0.072	0.073	0.081	0.079	0.084	0.081	0.077			7.748	A	
Allyl chloride (RSP)			889	6819	14165	30710	79708	157006	283118	0.215	0.000		0.990	W	
Allyl chloride			0.150	0.224	0.227	0.239	0.239	0.230	0.194						
Benzene			0.892	0.998	0.985	1.066	1.065	1.095	1.101	1.029			7.292	A	
Bromobenzene			1.234	1.364	1.281	1.377	1.336	1.267	1.237	1.300			4.559	A	
Bromochloromethane			0.111	0.130	0.132	0.137	0.132	0.124	0.125	0.128			6.591	A	
Bromodichloromethane			0.282	0.328	0.331	0.353	0.365	0.373	0.379	0.344			9.852	A	
Bromoform (RSP)			720	4668	9910	23618	64825	146620	332459	0.554	0.017		0.996	W	
Bromoform			0.313	0.405	0.414	0.474	0.501	0.539	0.574						
Bromomethane			0.197	0.222	0.227	0.242	0.266	0.291	0.283	0.247			14.01	A	
Carbon disulfide			0.724	0.726	0.698	0.767	0.762	0.772	0.778	0.747			4.073	A	
Carbon tetrachloride			0.319	0.340	0.345	0.381	0.395	0.407	0.429	0.374			10.66	A	
Chlorobenzene			1.703	1.851	1.869	1.967	1.989	2.072	2.065	1.931			6.835	A	
Chloroethane			0.189	0.204	0.215	0.225	0.222	0.225	0.234	0.216			7.027	A	
Chloroform			0.421	0.447	0.439	0.475	0.483	0.481	0.491	0.462			5.720	A	
Chloromethane			0.189	0.206	0.202	0.218	0.210	0.216	0.223	0.209			5.502	A	
Chloroprene			0.285	0.316	0.313	0.355	0.346	0.352	0.355	0.332			8.187	A	
Cyclohexane			0.358	0.391	0.387	0.440	0.432	0.436	0.442	0.412			8.047	A	
Cyclohexanone (RSP)			762	6175	13437	31285	80908	170231	352267	0.127	0.026		0.998	W	
Cyclohexanone			0.076	0.119	0.124	0.139	0.134	0.127	0.123						
Dibromochloromethane			0.507	0.620	0.629	0.699	0.728	0.755	0.780	0.674			14.11	A	
Dibromofluoromethane			0.218	0.229	0.233	0.253	0.260	0.264	0.272	0.247			8.180	A	
Dibromomethane			0.131	0.161	0.162	0.172	0.176	0.179	0.178	0.166			10.37	A	
Dichlorodifluoromethane			0.189	0.226	0.228	0.238	0.236	0.252	0.257	0.232			9.641	A	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines to allow for evaluation against minimum RRF

$\overline{RF}$ = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

				LabQC ID - FileID - Conc	1203 ~ 230517/c0228 ~ 1
Report No:	223060210	Instrument ID:	MSV15	1204 ~ 230517/c0229 ~ 5	1205 ~ 230517/c0230 ~ 10
GC Column:	RTX-VMS-30M ID .25 (mm)	Analyst:	CJR	1206 ~ 230517/c0231 ~ 20	1207 ~ 230517/c0232 ~ 50
Calib. Date 1:	05/17/23 Time 1: 1428	Analytical Batch:	766139	1208 ~ 230517/c0233 ~ 100	1209 ~ 230517/c0234 ~ 200
Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Method:	EPA 8260B		

ANALYTE	***1201***	***1202***	***1203***	***1204***	***1205***	***1206***	***1207***	***1208***	***1209***	***RF / m / A***	***b / B***	***C***	***FIT***	***TYPE***	
Dicyclopentadiene			2.596	3.146	3.125	3.418	3.368	3.122	3.097	3.125			8.526	A	
Diethyl ether			0.132	0.161	0.163	0.174	0.177	0.183	0.187	0.168			10.97	A	
Ethyl acetate			0.152	0.193	0.199	0.212	0.213	0.223	0.223	0.202			12.28	A	
Ethyl methacrylate				0.367	0.376	0.425	0.423	0.425	0.429	0.408			6.879	A	
Ethyl tert-butyl ether (ETBE)			0.539	0.624	0.643	0.694	0.701	0.718	0.721	0.663			9.954	A	
Ethylbenzene			0.854	1.018	1.012	1.085	1.105	1.135	1.139	1.050			9.550	A	
Hexachlorobutadiene			0.325	0.409	0.395	0.436	0.441	0.446	0.489	0.420			12.25	A	
Isobutyl alcohol				0.006	0.007	0.007	0.007	0.007	0.007	0.007			6.979	A	
Isopropylbenzene (Cumene)			2.611	2.933	2.968	3.262	3.296	3.326	3.260	3.094			8.606	A	
Methacrylonitrile			0.087	0.113	0.116	0.122	0.122	0.129	0.129	0.117			12.30	A	
Methyl Acetate			0.125	0.128	0.135	0.144	0.145	0.149	0.141	0.138			6.660	A	
Methyl iodide (RSP)				5565	14998	37592	102973	212199	417991	0.302	0.033		0.997	W	
Methyl iodide				0.183	0.240	0.293	0.309	0.311	0.286						
Methyl methacrylate				0.145	0.160	0.179	0.183	0.196	0.193	0.176			11.25	A	
Methylcyclohexane			0.383	0.454	0.452	0.510	0.522	0.525	0.538	0.483			11.61	A	
Methylene chloride			0.258	0.265	0.259	0.278	0.275	0.276	0.279	0.270			3.396	A	
Naphthalene			1.450	1.776	1.925	2.079	2.122	2.176	2.257	1.969			14.21	A	
Propionitrile			0.019	0.028	0.029	0.031	0.030	0.031	0.031	0.028			14.77	A	
Styrene			1.151	1.366	1.439	1.574	1.668	1.718	1.737	1.522			14.12	A	
Tetrachloroethene			0.560	0.637	0.610	0.666	0.678	0.711	0.754	0.660			9.766	A	
Tetrahydrofuran				0.070	0.072	0.077	0.076	0.079	0.078	0.075			4.789	A	
Toluene			2.466	2.815	2.866	3.025	3.049	3.058	3.142	2.917			7.867	A	
Toluene-d8			2.108	2.223	2.238	2.436	2.449	2.464	2.486	2.343			6.411	A	
Trichloroethene			0.250	0.297	0.297	0.326	0.333	0.345	0.365	0.316			12.11	A	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines
to allow for evaluation against minimum RRF

RF = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

				LabQC ID - FileID - Conc	1203 ~ 230517/c0228 ~ 1
Report No:	223060210	Instrument ID:	MSV15	1204 ~ 230517/c0229 ~ 5	1205 ~ 230517/c0230 ~ 10
GC Column:	RTX-VMS-30M ID .25 (mm)	Analyst:	CJR	1206 ~ 230517/c0231 ~ 20	1207 ~ 230517/c0232 ~ 50
Calib. Date 1:	05/17/23 Time 1: 1428	Analytical Batch:	766139	1208 ~ 230517/c0233 ~ 100	1209 ~ 230517/c0234 ~ 200
Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Method:	EPA 8260B		

ANALYTE	***1201***	***1202***	***1203***	***1204***	***1205***	***1206***	***1207***	***1208***	***1209***	***RF / m / A***	***b / B***	***C***	***FIT***	***TYPE***	
Trichlorofluoromethane			0.391	0.406	0.407	0.447	0.446	0.456	0.450	0.429			6.255	A	
Trichlorotrifluoroethane			0.222	0.258	0.248	0.274	0.278	0.284	0.291	0.265			9.027	A	
Vinyl acetate			0.335	0.347	0.359	0.380	0.377	0.384	0.370	0.364			5.027	A	
Vinyl chloride			0.262	0.280	0.294	0.305	0.300	0.306	0.315	0.294			6.153	A	
Xylene (total)			1.018	1.173	1.202	1.284	1.309	1.351	1.345	1.240			9.615	A	
cis-1,2-Dichloroethene			0.301	0.312	0.318	0.336	0.336	0.335	0.336	0.325			4.524	A	
cis-1,3-Dichloropropene			0.376	0.381	0.383	0.422	0.428	0.430	0.435	0.408			6.502	A	
diisopropyl Ether (DIPE)			0.487	0.557	0.570	0.621	0.625	0.631	0.632	0.589			9.224	A	
m,p-Xylene			1.043	1.191	1.230	1.311	1.337	1.392	1.380	1.269			9.804	A	
n-Butyl alcohol (RSP)				406	1071	2315	6770	13877	29929	0.004	0.184		0.999	W	
n-Butyl alcohol				0.003	0.003	0.004	0.004	0.004	0.004						
n-Butylbenzene			3.111	3.881	3.920	4.292	4.350	4.230	4.209	3.999			10.77	A	
n-Hexane			0.395	0.468	0.475	0.519	0.519	0.518	0.518	0.488			9.511	A	
n-Propylbenzene			3.472	3.870	3.924	4.233	4.160	3.910	3.764	3.905			6.458	A	
o-Xylene			0.967	1.137	1.147	1.230	1.253	1.267	1.273	1.182			9.281	A	
sec-Butylbenzene			2.870	3.354	3.338	3.702	3.648	3.462	3.408	3.397			8.001	A	
t-Butanol (TBA)				0.025	0.025	0.030	0.030	0.032	0.033	0.029			11.96	A	
t-amyl methyl ether (TAME)			0.516	0.619	0.651	0.711	0.722	0.753	0.756	0.675			12.86	A	
tert-Butyl methyl ether (MTB			0.519	0.613	0.647	0.690	0.702	0.725	0.726	0.660			11.33	A	
tert-Butylbenzene			1.253	1.504	1.473	1.588	1.567	1.479	1.459	1.475			7.408	A	
tert-amyl alcohol (TAA)			0.081	0.091	0.094	0.104	0.108	0.111	0.112	0.100			11.52	A	
trans-1,2-Dichloroethene			0.296	0.323	0.326	0.352	0.355	0.356	0.361	0.338			7.187	A	
trans-1,3-Dichloropropene			0.331	0.356	0.368	0.398	0.408	0.424	0.421	0.387			9.208	A	
trans-1,4-Dichloro-2-butene (			295	2548	5589	12313	31798	67695	141123	0.253	0.005		0.998	W	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

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to allow for evaluation against minimum RRF

$\overline{RF}$ = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

Report No:	223060210	Instrument ID:	MSV15	**LabQC ID - FileID - Conc**	1203 ~ 230517/c0228 ~ 1
GC Column:	RTX-VMS-30M ID .25 (mm)	Analyst:	CJR	1204 ~ 230517/c0229 ~ 5	1205 ~ 230517/c0230 ~ 10
Calib. Date 1:	05/17/23 Time 1: 1428	Analytical Batch:	766139	1206 ~ 230517/c0231 ~ 20	1207 ~ 230517/c0232 ~ 50
Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Method:	EPA 8260B	1208 ~ 230517/c0233 ~ 100	1209 ~ 230517/c0234 ~ 200

ANALYTE	*1201*	*1202*	*1203*	*1204*	*1205*	*1206*	*1207*	*1208*	*1209*	*RF / m / A*	*b / B*	*C*	*FIT*	*TYPE*	
trans-1,4-Dichloro-2-butene			0.147	0.246	0.258	0.274	0.264	0.253	0.246						

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

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to allow for evaluation against minimum RRF

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m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0228.d
Lab Smp Id: 1203   Client Smp ID: V15STD001
Inj Date : 17-MAY-2023 14:28
Operator : CJR   Inst ID: msv15.i
Smp Info : 1203*V15STD001
Misc Info : MSV~527~*1*SMR
Comment   :
Method   : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr   Quant Type: ISTD
Cal Date : 17-MAY-2023 14:28   Cal File: c0228.d
Als bottle: 1   Calibration Sample, Level: 3
Dil Factor: 1.00000
Integrator: HP RTE   Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable   Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	1117	1.00000	0.813	7981
2 Chloromethane ++	50	1.649	1.649	(0.280)	1117	1.00000	0.902	8081
3 Vinyl Chloride +	62	1.719	1.719	(0.292)	1550	1.00000	0.889	7630
5 1-3 Butadiene	39	1.735	1.735	(0.295)	1215	1.00000	0.936	8374
6 Bromomethane	94	2.021	2.021	(0.344)	1167	1.00000	0.798	6338
7 Chloroethane	64	2.144	2.144	(0.364)	1121	1.00000	0.875	5692
8 Trichlorofluoromethane	101	2.276	2.276	(0.387)	2314	1.00000	0.911	8545
9 Isoprene	67	2.565	2.565	(0.436)	1820	1.00000	0.866	3894(a)
10 Ethyl Ether	59	2.584	2.584	(0.439)	783	1.00000	0.786	7309
11 1,1-Dichloroethene +	96	2.787	2.787	(0.474)	1352	1.00000	0.935	7941
12 Carbon Disulfide	76	2.812	2.812	(0.478)	4288	1.00000	0.969	6865
13 1,1,2Trichlotrifluoroethane	101	2.825	2.825	(0.480)	1314	1.00000	0.837	7000
15 Methyl Iodide	142	2.935	2.935	(0.499)	399	1.00000	1.85	7203
16 Acrolein	56	3.150	3.150	(0.535)	914	5.00000	4.15	4621

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
	17 Allyl chloride	39	3.311	3.311	(0.563)	889	1.00000	0.696	7439
	18 Methylene Chloride	49	3.420	3.420	(0.581)	1527	1.00000	0.954	4607
	19 Acetone	43	3.481	3.481	(0.592)	2691	5.00000	4.10	2977
	20 trans-1,2-Dichloroethene	61	3.587	3.587	(0.610)	1751	1.00000	0.874	8521
	21 Methyl Acetate	43	3.613	3.613	(0.614)	3689	5.00000	4.51	8424
	22 Hexane	57	3.674	3.674	(0.625)	2341	1.00000	0.811	8367
	23 MTBE	73	3.722	3.722	(0.633)	3072	1.00000	0.786	7791
	24 tert-Butyl Alcohol	59	3.825	3.825	(0.650)	141	5.00000	0.818	0(M1)
	25 Acetonitrile	41	3.941	3.941	(0.670)	449	5.00000	3.86	7055(M1)
	26 Isopropyl Ether	45	4.115	4.115	(0.699)	2883	1.00000	0.827	8475
	27 Chloroprene	53	4.198	4.198	(0.714)	1688	1.00000	0.859	0(M1)
	28 1,1-Dichloroethane ++	63	4.208	4.208	(0.715)	2295	1.00000	0.857	0(M1)
	29 Acrylonitrile	53	4.269	4.269	(0.726)	2016	5.00000	4.42	8501(M1)
	30 Vinyl Acetate	43	4.478	4.478	(0.761)	1983	1.00000	0.919	0(M1)
	31 Ethyl Tert-butyl Ether	59	4.475	4.475	(0.761)	3191	1.00000	0.813	7123
	32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	1781	1.00000	0.926	8626
M	33 Total 1,2-Dichloroethene	61				3532	2.00000	1.80	0
	34 2,2-Dichloropropane	77	4.832	4.832	(0.821)	2094	1.00000	0.880	8052
	35 Bromochloromethane	128	4.918	4.918	(0.836)	660	1.00000	0.874	8707
	36 Cyclohexane	56	4.922	4.922	(0.837)	2121	1.00000	0.869	8275
	37 Chloroform +	83	4.992	4.992	(0.849)	2493	1.00000	0.910	7638
	38 Ethyl Acetate	43	5.118	5.118	(0.870)	4501	5.00000	3.76	5435
	39 Carbon Tetrachloride	117	5.118	5.118	(0.870)	1892	1.00000	0.855	8652
	40 Tetrahydrofuran	42	5.166	5.166	(0.878)	1339	5.00000	3.00	7738
$	42 Dibromofluoromethane	111	5.160	5.160	(0.877)	1289	1.00000	0.881	6908
	43 1,1,1-Trichloroethane	97	5.176	5.176	(0.880)	2190	1.00000	0.864	7976
	44 2-Butanone	43	5.279	5.279	(0.897)	2441	5.00000	3.58	7013
	45 1,1-Dichloropropene	75	5.295	5.295	(0.900)	2076	1.00000	0.921	8493
	46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	5071	1.00000	0.745	8261(M1)
	47 Propionitrile	54	5.529	5.529	(0.940)	573	5.00000	3.40	0(M1)
	48 Benzene	78	5.510	5.510	(0.937)	5285	1.00000	0.867	8631
	49 Methylacrylonitrile	41	5.552	5.552	(0.944)	2580	5.00000	3.73	6827
	56 tert-amyl Methyl Ether	73	5.635	5.635	(0.958)	3055	1.00000	0.764	0(M1)
$	50 1,2-Dichloroethane-d4	65	5.629	5.629	(0.957)	1586	1.00000	1.04	7517
	52 1,2-Dichloroethane	62	5.690	5.690	(0.967)	1724	1.00000	0.899	0(M1)
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	296139	50.0000		9838
	57 Trichloroethene	130	6.037	6.037	(1.026)	1478	1.00000	0.790	8964
	58 Methyl Cyclohexane	83	6.021	6.021	(1.024)	2268	1.00000	0.792	8167
	59 Tert-amyl alcohol	59	6.246	6.246	(1.062)	2411	5.00000	4.06	6622
	60 n-Butanol	56	6.333	6.333	(1.077)	40	5.00000	10.8	0(M1)
	62 Dibromomethane	93	6.394	6.394	(1.087)	773	1.00000	0.788	8244
	63 2-3 Dichloro-1-Proprene	75	6.462	6.462	(1.098)	2096	1.00000	0.855	8662
	64 1,2-Dichloropropane +	63	6.478	6.478	(1.101)	1322	1.00000	0.878	7097
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	1669	1.00000	0.818	6990
	66 Methyl methacrylate	41	6.684	6.684	(1.136)	682	1.00000	0.654	7630
	67 1,4- Dioxane	88	6.725	6.725	(1.143)	1435	25.0000	129	7786
	68 1-Bromo-2-chloroethane	63	6.947	6.947	(1.181)	1677	1.00000	0.874	0(M1)

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	1244	5.00000	7.28	8497
73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	2225	1.00000	0.921	8076
$ 74 Toluene-d8	98	7.221	7.221	(0.866)	4852	1.00000	0.900	8965
75 Toluene +	91	7.262	7.262	(0.871)	5676	1.00000	0.845	9140
76 2-Nitropropane	43	7.436	7.436	(0.892)	147	1.00000	1.01	6280(M1)
77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	3673	5.00000	5.60	8940
78 Tetrachloroethene	164	7.552	7.552	(0.906)	1288	1.00000	0.848	7934
79 trans-1,3-Dichloropropene	75	7.571	7.571	(1.287)	1959	1.00000	0.856	7773
80 Ethyl Methacrylate	41	7.697	7.697	(0.923)	576	1.00000	0.614	0(M1)
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	1146	1.00000	0.840	7775
82 Dibromochloromethane	129	7.825	7.825	(0.938)	1166	1.00000	0.752	7123
M 83 1-3 Dichloropropene total	100				4184	2.00000	1.78	0
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	1374	1.00000	0.766	8212(a)
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	1969	1.00000	0.867	8499
88 1,2-Dibromoethane(EDB)	107	7.996	7.996	(0.959)	1133	1.00000	0.776	7873
89 2-Hexanone	43	8.147	8.147	(0.977)	2993	5.00000	5.51	8901
90 1-Chlorohexane	91	8.330	8.330	(0.999)	2151	1.00000	0.896	2514
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	115086	50.0000		9715
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	3920	1.00000	0.882	3030
93 Ethylbenzene +	106	8.368	8.368	(1.003)	1966	1.00000	0.814	8026
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	1585	1.00000	1.06	8215
95 p,m-Xylene	106	8.462	8.462	(1.015)	4800	2.00000	1.64	9540
97 o-Xylene	106	8.745	8.745	(1.049)	2226	1.00000	0.818	9362
101 Styrene	104	8.777	8.777	(1.052)	2650	1.00000	0.756	9109
102 Bromoform ++	173	8.799	8.799	(1.055)	720	1.00000	1.42	5747
104 Isopropylbenzene	105	8.944	8.944	(1.072)	6010	1.00000	0.844	9186
M 105 TOTAL XYLENE	106				7026	3.00000	2.46	0
$ 106 Bromofluorobenzene	174	9.121	9.121	(1.094)	1941	1.00000	1.11	8401
107 cis-1,4-dichloro-2-butene	53	9.156	9.156	(0.934)	319	1.00000	0.626	0(aM1)
109 n-Propylbenzene	91	9.198	9.198	(0.938)	6966	1.00000	0.889	9095
108 Bromobenzene	77	9.192	9.192	(0.937)	2475	1.00000	0.949	8775
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.942)	1249	1.00000	0.816	0(M1)
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	4494	1.00000	0.900	8094
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	4374	1.00000	0.830	7843
114 1,2,3-Trichloropropane	75	9.327	9.327	(0.951)	2080	1.00000	1.00	6425
115 trans-1,4-Dichloro-2-Butene	53	9.346	9.346	(0.953)	295	1.00000	0.820	6672
116 Cyclohexanone	55	9.368	9.368	(0.955)	762	5.00000	4.30	8114
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	3917	1.00000	0.874	9150
118 tert-butylbenzene	91	9.513	9.513	(0.970)	2514	1.00000	0.850	9368
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.974)	4072	1.00000	0.787	8689
120 sec-Butylbenzene	105	9.622	9.622	(0.981)	5758	1.00000	0.845	9251
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	4924	1.00000	0.816	8524
122 Dicyclopentadiene	66	9.700	9.700	(0.989)	5208	1.00000	0.831	7745
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.995)	2819	1.00000	0.925	8991
* 124 1,4-DICHLOROBENZENE-D4	152	9.806	9.806	(1.000)	100307	50.0000		9667
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	2920	1.00000	0.931	3199(H)
126 n-Butylbenzene	91	9.954	9.954	(1.015)	6242	1.00000	0.778	8690

						AMOUNTS		
	QUANT SIG					CAL-AMT	ON-COL	
Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
127 1,2-Dichlorobenzene	146	10.066	10.066	(1.027)	2430	1.00000	0.881	8254
129 1,2-Dibromo-3-Chloropropane	157	10.558	10.558	(1.077)	159	1.00000	1.24	0(M1)
130 Hexachlorobutadiene	225	10.999	10.999	(1.122)	652	1.00000	0.773	7245
131 1,2,4-Trichlorobenzene	180	11.031	11.031	(1.125)	1406	1.00000	0.834	8026
132 Naphthalene	128	11.288	11.288	(1.151)	2909	1.00000	0.736	8473
133 1,2,3-Trichlorobenzene	180	11.446	11.446	(1.167)	1129	1.00000	0.776	6978

QC Flag Legend

a - Target compound detected but, quantitated amount
Below Limit Of Quantitation(BLOQ).
M1- Compound response manually integrated because
Target system did not integrate.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0228.d

Page 1

Date : 17-MAY-2023 14:28

Client ID: V15STD001

Instrument: msv15.i

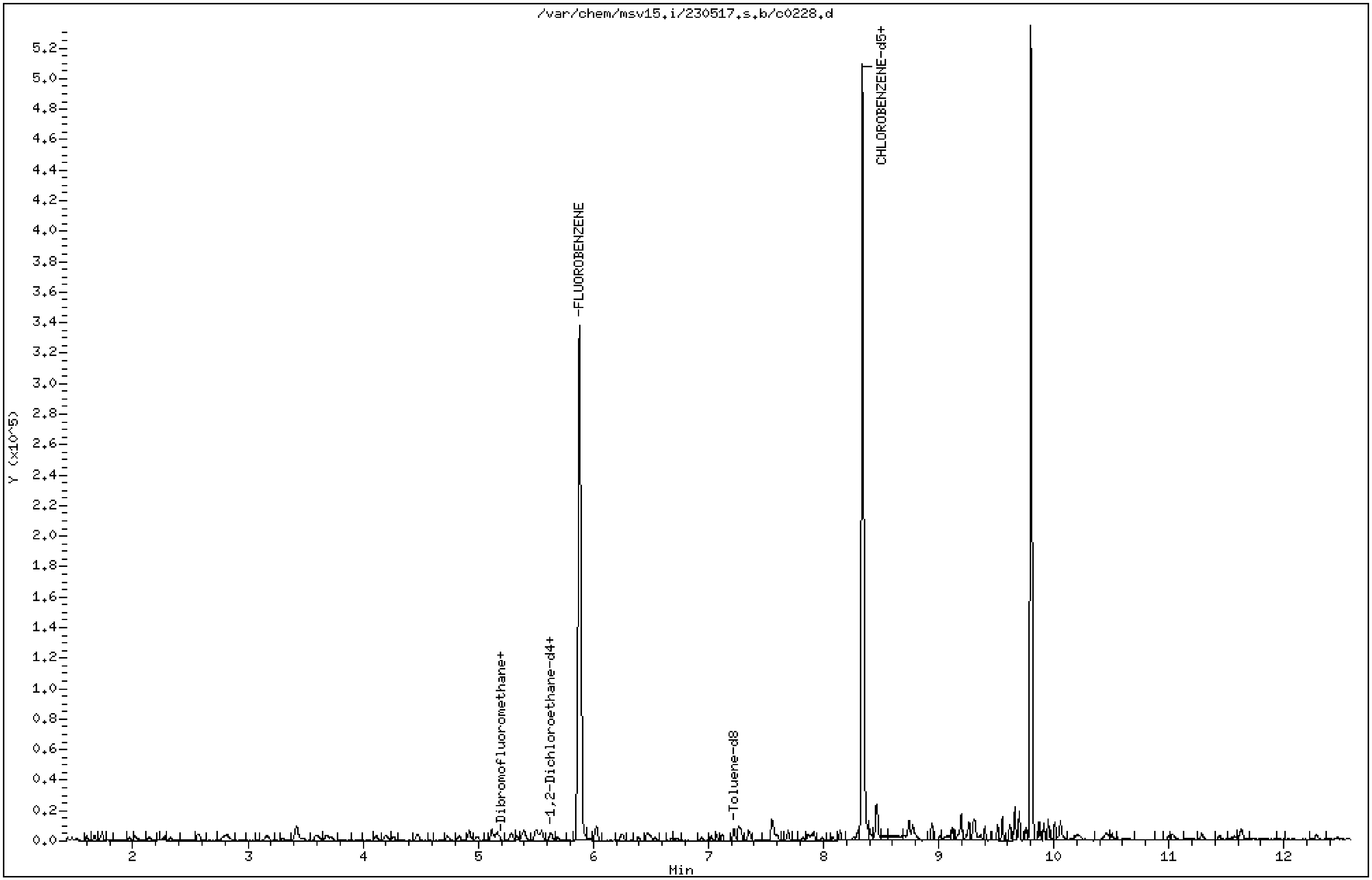
Sample Info: 1203*V15STD001

Purge Volume: 5.0

Operator: CJR

Column phase: RTX-VMS-30M

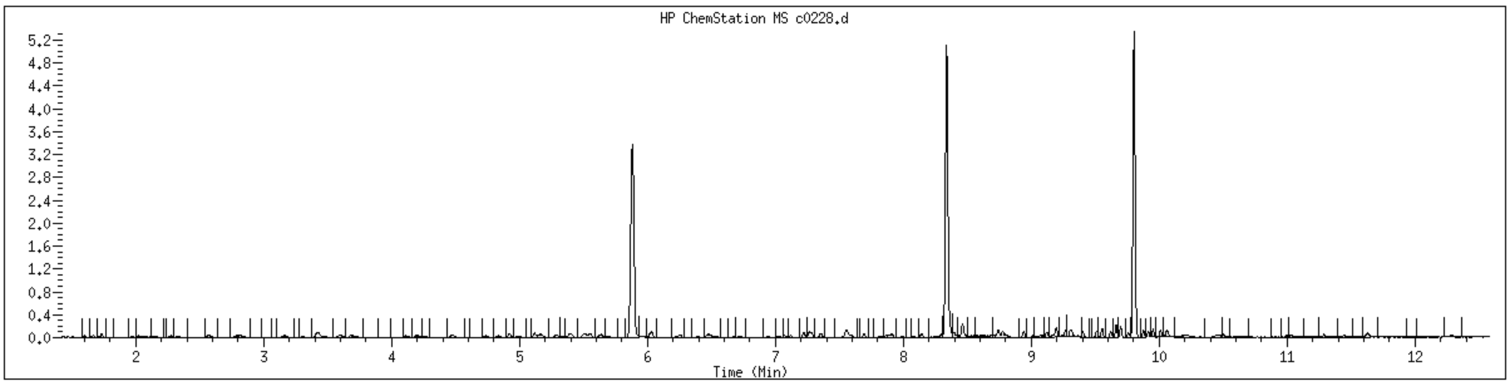
Column diameter: 0.25



Report Date: 05/23/2023 16:34

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 1203   SampleType   : CALIB_3
Injection Date: 05/17/2023 14:28   Instrument   : msv15.i
Operator   : CJR
Sample Info   : 1203*V15STD001
Misc Info   : MSV~527~*1*SMR
Method   : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260b+AppIXDOD



Original   Final

==

29 Acrylonitrile   CAS#: 107-13-1   Reason: M1

Electronic Signature
Applied

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0229.d
Lab Smp Id: 1204 Client Smp ID: V15STD005
Inj Date : 17-MAY-2023 14:47
Operator : CJR Inst ID: msv15.i
Smp Info : 1204*V15STD005
Misc Info : MSV~527~*1*SMR
Comment :
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr Quant Type: ISTD
Cal Date : 17-MAY-2023 14:47 Cal File: c0229.d
Als bottle: 1 Calibration Sample, Level: 4
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	6870	5.00000	4.86	9447
2 Chloromethane ++	50	1.652	1.652	(0.281)	6263	5.00000	4.92	9311
3 Vinyl Chloride +	62	1.719	1.719	(0.292)	8525	5.00000	4.75	9304
5 1-3 Butadiene	39	1.735	1.735	(0.295)	6564	5.00000	4.92	9510
6 Bromomethane	94	2.018	2.018	(0.343)	6749	5.00000	4.49	9197
7 Chloroethane	64	2.137	2.137	(0.363)	6205	5.00000	4.71	8861
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	12359	5.00000	4.73	9124
9 Isoprene	67	2.571	2.571	(0.437)	10068	5.00000	4.66	4501(a)
10 Ethyl Ether	59	2.587	2.587	(0.440)	4907	5.00000	4.79	8213
11 1,1-Dichloroethene +	96	2.787	2.787	(0.474)	7051	5.00000	4.74	8512
12 Carbon Disulfide	76	2.816	2.816	(0.479)	22125	5.00000	4.86	7846
13 1,1,2Trichlotrifluoroethane	101	2.841	2.841	(0.483)	7874	5.00000	4.88	8604
15 Methyl Iodide	142	2.935	2.935	(0.499)	5565	5.00000	4.65	8826
16 Acrolein	56	3.156	3.156	(0.537)	5341	25.0000	23.6	4526

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
17 Allyl chloride	39	3.308	3.308	(0.562)	6819	5.00000	5.21	9512
18 Methylene Chloride	49	3.420	3.420	(0.581)	8077	5.00000	4.91	4855
19 Acetone	43	3.475	3.475	(0.591)	17117	25.0000	25.3	4888(H)
20 trans-1,2-Dichloroethene	61	3.591	3.591	(0.610)	9832	5.00000	4.77	9142
21 Methyl Acetate	43	3.619	3.619	(0.615)	19451	25.0000	23.1	8637
22 Hexane	57	3.684	3.684	(0.626)	14269	5.00000	4.80	8905
23 MTBE	73	3.729	3.729	(0.634)	18657	5.00000	4.64	9421
24 tert-Butyl Alcohol	59	3.835	3.835	(0.652)	3739	25.0000	21.1	0(M1)
25 Acetonitrile	41	3.944	3.944	(0.670)	2904	25.0000	24.3	9271
26 Isopropyl Ether	45	4.115	4.115	(0.699)	16980	5.00000	4.73	9377
27 Chloroprene	53	4.198	4.198	(0.714)	9634	5.00000	4.77	8675
28 1,1-Dichloroethane ++	63	4.214	4.214	(0.716)	13766	5.00000	5.00	9244
29 Acrylonitrile	53	4.266	4.266	(0.725)	10944	25.0000	23.3	9481
30 Vinyl Acetate	43	4.475	4.475	(0.761)	10571	5.00000	4.76	5764
31 Ethyl Tert-butyl Ether	59	4.478	4.478	(0.761)	19016	5.00000	4.71	8017
32 cis-1,2-Dichloroethene	61	4.738	4.738	(0.805)	9499	5.00000	4.80	9504
M 33 Total 1,2-Dichloroethene	61				19331	10.0000	9.57	0
34 2,2-Dichloropropane	77	4.835	4.835	(0.822)	12035	5.00000	4.92	9292
35 Bromochloromethane	128	4.922	4.922	(0.837)	3963	5.00000	5.10	9439
36 Cyclohexane	56	4.928	4.928	(0.838)	11895	5.00000	4.74	9099
37 Chloroform +	83	4.996	4.996	(0.849)	13628	5.00000	4.84	9429
38 Ethyl Acetate	43	5.121	5.121	(0.870)	29353	25.0000	23.8	5908
41 sec-Butanol	45	5.179	5.179	(0.880)	218	5.00000	2.34	0(M1)
39 Carbon Tetrachloride	117	5.118	5.118	(0.870)	10352	5.00000	4.55	8107
40 Tetrahydrofuran	42	5.160	5.160	(0.877)	10647	25.0000	23.2	8163
$ 42 Dibromofluoromethane	111	5.160	5.160	(0.877)	6988	5.00000	4.65	7654
43 1,1,1-Trichloroethane	97	5.179	5.179	(0.880)	12365	5.00000	4.74	9516
44 2-Butanone	43	5.279	5.279	(0.897)	16865	25.0000	24.1	6896
45 1,1-Dichloropropene	75	5.295	5.295	(0.900)	11134	5.00000	4.80	9184
46 2,2,4 Trimethylpentane	57	5.404	5.404	(0.919)	33148	5.00000	4.73	8645
47 Propionitrile	54	5.536	5.536	(0.941)	4217	25.0000	24.3	8622
48 Benzene	78	5.517	5.517	(0.938)	30390	5.00000	4.85	9547
49 Methylacrylonitrile	41	5.555	5.555	(0.944)	17141	25.0000	24.1	7322
56 tert-amyl Methyl Ether	73	5.635	5.635	(0.958)	18841	5.00000	4.58	0(M1)
$ 50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	7109	5.00000	4.54	8209
52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	9684	5.00000	4.91	9460
51 Isobutyl Alcohol	43	5.732	5.732	(0.974)	886	25.0000	21.6	2591(H)
* 55 FLUOROBENZENE	96	5.883	5.883	(1.000)	304598	50.0000		9849
57 Trichloroethene	130	6.034	6.034	(1.026)	9041	5.00000	4.70	9137
58 Methyl Cyclohexane	83	6.025	6.025	(1.024)	13832	5.00000	4.70	8749
59 Tert-amyl alcohol	59	6.256	6.256	(1.063)	13852	25.0000	22.7	6786
60 n-Butanol	56	6.336	6.336	(1.077)	406	25.0000	25.3	0(M1)
62 Dibromomethane	93	6.394	6.394	(1.087)	4899	5.00000	4.86	9054
63 2-3 Dichloro-1-Proprene	75	6.468	6.468	(1.099)	11872	5.00000	4.71	8806
64 1,2-Dichloropropane +	63	6.484	6.484	(1.102)	7338	5.00000	4.74	8050
65 Bromodichloromethane	83	6.539	6.539	(1.111)	9982	5.00000	4.76	9358
66 Methyl methacrylate	41	6.677	6.677	(1.135)	4413	5.00000	4.12	9007

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
67 1,4- Dioxane	88	6.722	6.722	(1.143)	1658	125.000	144	7925
68 1-Bromo-2-chloroethane	63	6.947	6.947	(1.181)	9449	5.00000	4.79	8920
72 2-Chloroethyl vinyl ether	63	7.021	7.021	(1.193)	8287	25.0000	22.9	9327
73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	11598	5.00000	4.67	9450
$ 74 Toluene-d8	98	7.224	7.224	(0.866)	25628	5.00000	4.74	9708
75 Toluene +	91	7.262	7.262	(0.871)	32455	5.00000	4.83	9594
76 2-Nitropropane	43	7.436	7.436	(0.892)	1643	5.00000	4.99	6263(M3)
77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	26975	25.0000	24.0	8386
78 Tetrachloroethene	164	7.558	7.558	(0.906)	7348	5.00000	4.83	8044
79 trans-1,3-Dichloropropene	75	7.578	7.578	(1.288)	10829	5.00000	4.60	9230
80 Ethyl Methacrylate	41	7.697	7.697	(0.923)	4235	5.00000	4.51	7480
81 1,1,2-Trichloroethane	97	7.690	7.690	(0.922)	6622	5.00000	4.85	9421
82 Dibromochloromethane	129	7.825	7.825	(0.938)	7142	5.00000	4.60	9365
M 83 1-3 Dichloropropene total	100				22427	10.0000	9.27	0
84 3,4-dichloro-1-butene	75	7.864	7.864	(0.943)	8232	5.00000	4.58	9669(a)
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	11280	5.00000	4.96	9428
87 1-Nitropropane	43	7.976	7.976	(0.956)	825	5.00000	4.85	6201(M3)
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	7147	5.00000	4.89	9183
89 2-Hexanone	43	8.143	8.143	(0.976)	22444	25.0000	24.2	9600
90 1-Chlorohexane	91	8.333	8.333	(0.999)	11448	5.00000	4.76	3265
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	115277	50.0000		9742
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	21340	5.00000	4.79	4912
93 Ethylbenzene +	106	8.368	8.368	(1.003)	11735	5.00000	4.85	9125
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	6687	5.00000	4.47	9145
95 p,m-Xylene	106	8.465	8.465	(1.015)	27464	10.0000	9.38	9801
97 o-Xylene	106	8.745	8.745	(1.049)	13105	5.00000	4.81	9625
101 Styrene	104	8.774	8.774	(1.052)	15750	5.00000	4.49	9736
102 Bromoform ++	173	8.799	8.799	(1.055)	4668	5.00000	4.51	8307
104 Isopropylbenzene	105	8.941	8.941	(1.072)	33812	5.00000	4.74	9580
M 105 TOTAL XYLENE	106				40569	15.0000	14.2	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	8258	5.00000	4.71	9335
107 cis-1,4-dichloro-2-butene	53	9.156	9.156	(0.934)	2427	5.00000	4.61	9285(a)
109 n-Propylbenzene	91	9.201	9.201	(0.939)	40109	5.00000	4.96	8649
108 Bromobenzene	77	9.192	9.192	(0.938)	14139	5.00000	5.25	9052
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.943)	8032	5.00000	5.08	9527
112 2-Chlorotoluene	91	9.301	9.301	(0.949)	26003	5.00000	5.04	7697
113 1,3,5-Trimethylbenzene	105	9.314	9.314	(0.950)	26594	5.00000	4.89	9014
114 1,2,3-Trichloropropane	75	9.327	9.327	(0.951)	10340	5.00000	4.82	6905
115 trans-1,4-Dichloro-2-Butene	53	9.346	9.346	(0.953)	2548	5.00000	5.10	9136
116 Cyclohexanone	55	9.362	9.362	(0.955)	6175	25.0000	24.8	9506
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	23095	5.00000	4.99	9575
118 tert-butylbenzene	91	9.513	9.513	(0.970)	15583	5.00000	5.10	9653
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	25913	5.00000	4.85	9319
120 sec-Butylbenzene	105	9.622	9.622	(0.982)	34756	5.00000	4.94	9665
121 p-Isopropyltoluene	119	9.700	9.700	(0.990)	29602	5.00000	4.75	8812
122 Dicyclopentadiene	66	9.700	9.700	(0.990)	32601	5.00000	5.03	7854
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.996)	15286	5.00000	4.85	9445

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.803	9.803	(1.000)	103631	50.0000		9619
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	15506	5.00000	4.79	4773(H)
126 n-Butylbenzene	91	9.954	9.954	(1.015)	40223	5.00000	4.85	8824
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	13759	5.00000	4.83	9317
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	1562	5.00000	4.68	8525
130 Hexachlorobutadiene	225	10.995	10.995	(1.122)	4238	5.00000	4.87	9004
131 1,2,4-Trichlorobenzene	180	11.028	11.028	(1.125)	8295	5.00000	4.76	9147
132 Naphthalene	128	11.288	11.288	(1.152)	18400	5.00000	4.51	9663
133 1,2,3-Trichlorobenzene	180	11.439	11.439	(1.167)	6986	5.00000	4.65	9151

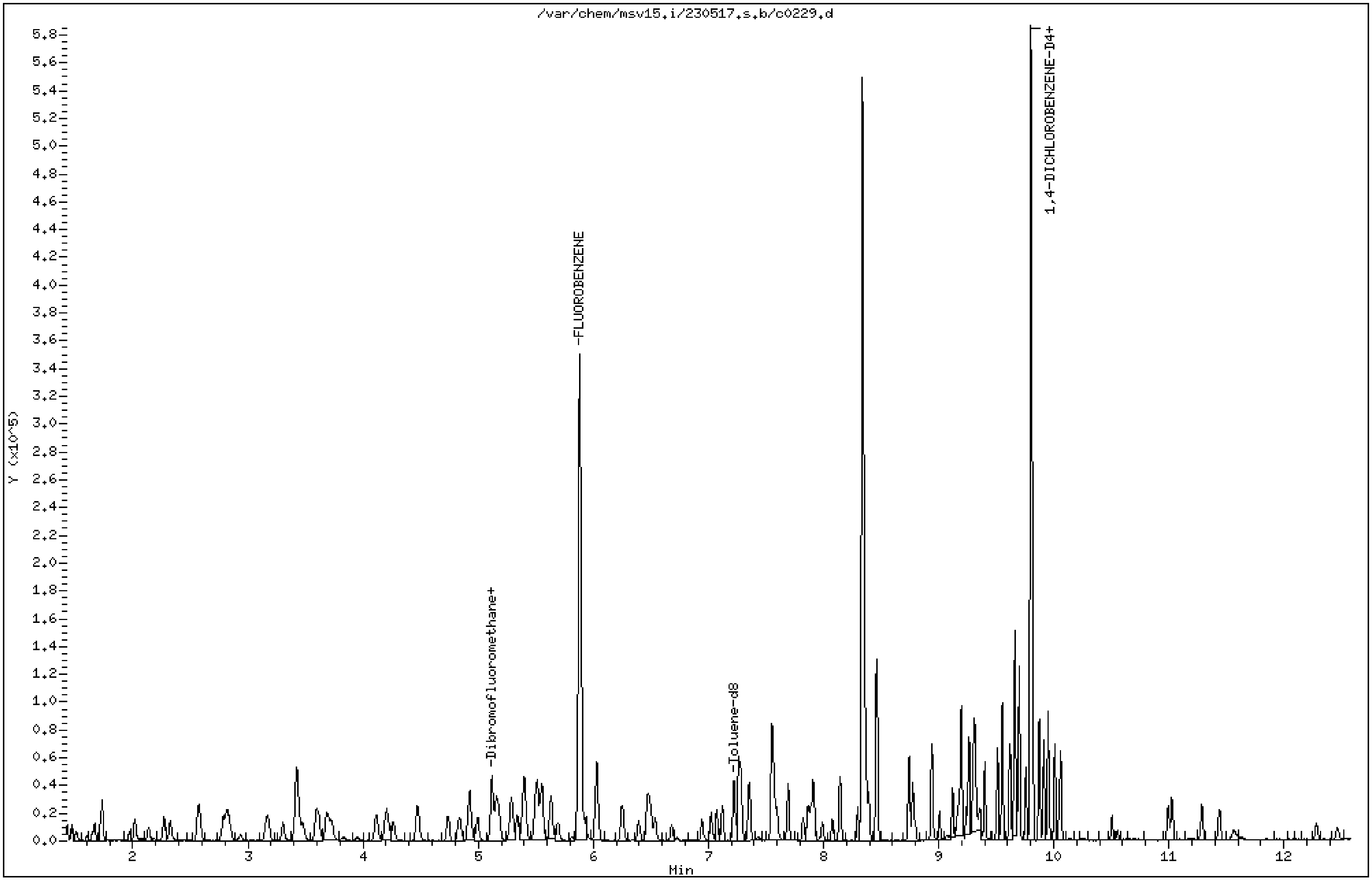
QC Flag Legend

a - Target compound detected but, quantitated amount Below Limit Of Quantitation(BLOQ).
M1- Compound response manually integrated because Target system did not integrate.
M3- Compound response manually integrated because Target system integrated incorrect peak.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0229.d
Date : 17-MAY-2023 14:47
Client ID: V15STD005
Sample Info: 1204*V15STD005
Purge Volume: 5.0
Column phase: RTX-VMS-30M

Page 1

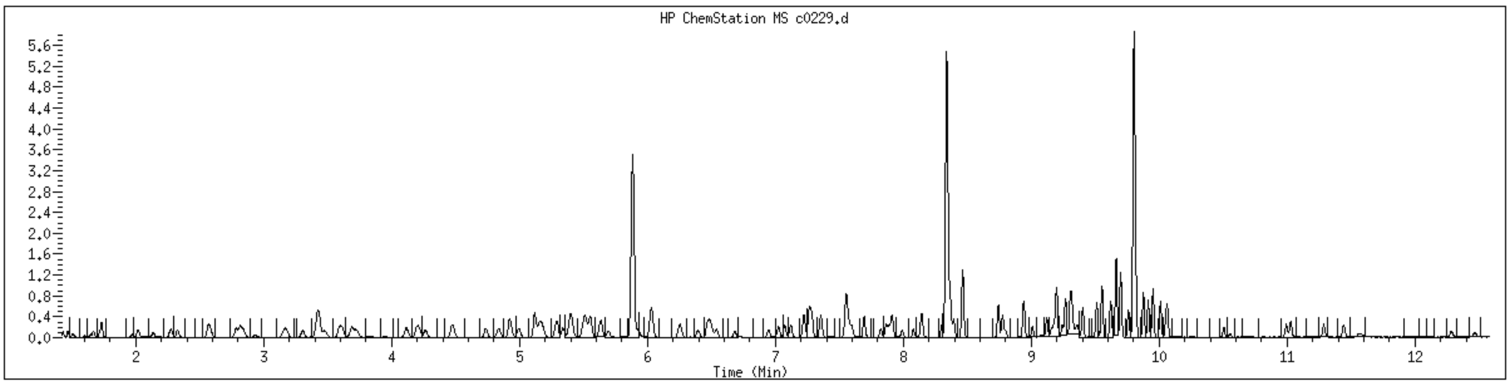
Instrument: msv15.i
Operator: CJR
Column diameter: 0.25



Data file : /var/chem/msv15.i/230517.s.b/c0229.d   Page: 1
Report Date: 05/23/2023 16:34

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID	: 1204	SampleType	: CALIB_4
Injection Date	: 05/17/2023 14:47	Instrument	: msv15.i
Operator	: CJR		
Sample Info	: 1204*V15STD005		
Misc Info	: MSV~527~*1*SMR		
Method	: /var/chem/msv15.i/230517.s.b/8260bw15DOD.m		
Dilution	: 1.00		
Matrix	: WATER		
Integrator	: HP RTE	Compound Sublist:	8260b+AppIXDOD



Original   Final

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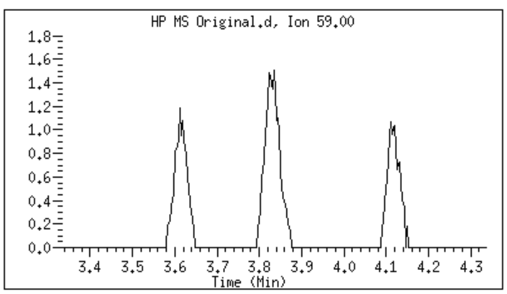
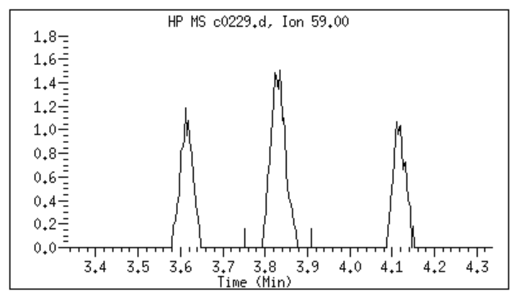
24 tert-Butyl Alcohol   CAS#: 75-65-0   Reason: M1



Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:01



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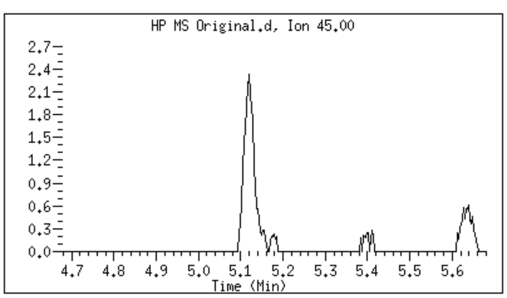
41 sec-Butanol   CAS#: 78-92-2   Reason: M1



Electronic Signature Applied

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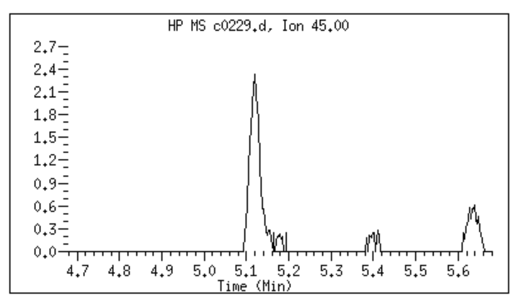
Date: 05/17/2023 16:10



Original   Final

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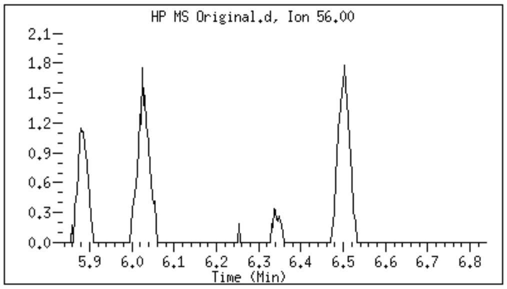
60 n-Butanol   CAS#: 71-36-3   Reason: M1



Electronic Signature
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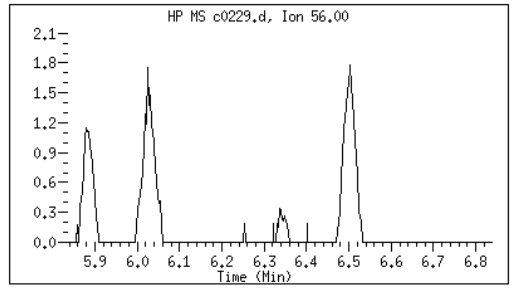
User: kmn
Date: 05/17/2023 16:11



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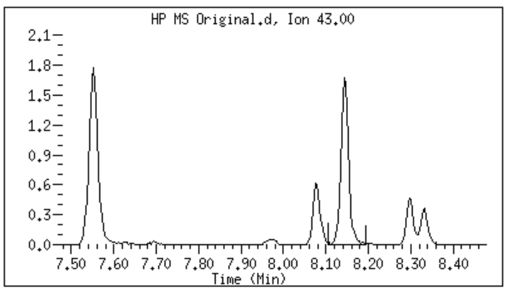
87 1-Nitropropane   CAS#: 108-03-2   Reason: M3



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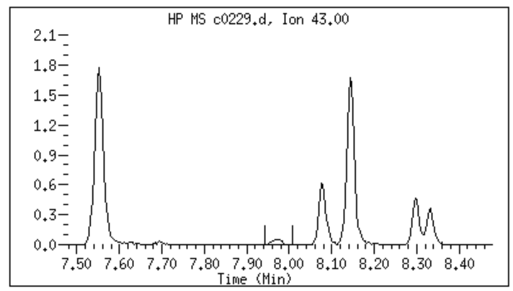

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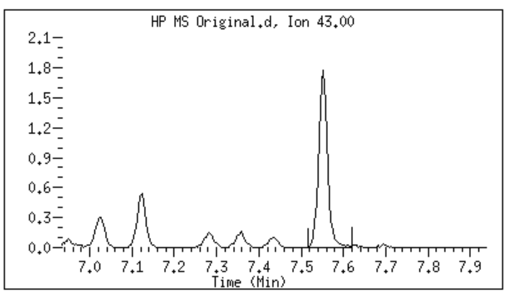
76 2-Nitropropane   CAS#: 79-46-9   Reason: M3



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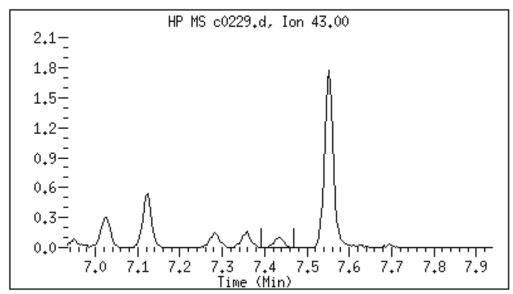

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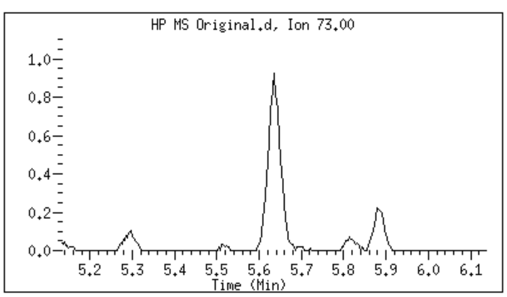
56 tert-amyl Methyl Ether   CAS#: 994-05-8   Reason: M1



Electronic Signature
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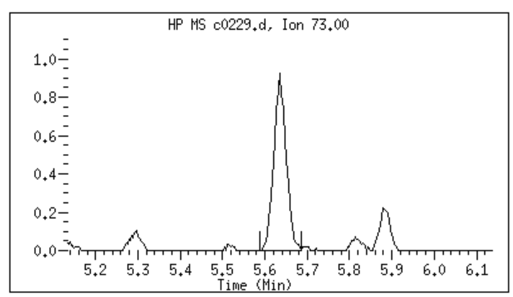
User: kmn
Date: 05/17/2023 16:11



M1 - Target system did not integrate
M3 - Target system integrated incorrect peak

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0230.d
Lab Smp Id: 1205 Client Smp ID: V15STD010
Inj Date : 17-MAY-2023 15:05
Operator : CJR Inst ID: msv15.i
Smp Info : 1205*V15STD010
Misc Info : MSV~527~*1*SMR
Comment :
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr Quant Type: ISTD
Cal Date : 17-MAY-2023 15:05 Cal File: c0230.d
Als bottle: 1 Calibration Sample, Level: 5
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	14263	10.0000	9.83	9700
2 Chloromethane ++	50	1.655	1.655	(0.281)	12647	10.0000	9.68	9525
3 Vinyl Chloride +	62	1.719	1.719	(0.292)	18349	10.0000	9.97	9540
5 1-3 Butadiene	39	1.735	1.735	(0.295)	13136	10.0000	9.59	9546
6 Bromomethane	94	2.018	2.018	(0.343)	14184	10.0000	9.19	9485
7 Chloroethane	64	2.140	2.140	(0.364)	13449	10.0000	9.94	9099
8 Trichlorofluoromethane	101	2.275	2.275	(0.387)	25447	10.0000	9.49	9463
9 Isoprene	67	2.571	2.571	(0.437)	20734	10.0000	9.35	4545
10 Ethyl Ether	59	2.584	2.584	(0.439)	10191	10.0000	9.70	7856
11 1,1-Dichloroethene +	96	2.787	2.787	(0.474)	14180	10.0000	9.29	8431
12 Carbon Disulfide	76	2.816	2.816	(0.479)	43652	10.0000	9.35	8143
13 1,1,2Trichlotrifluoroethane	101	2.835	2.835	(0.482)	15524	10.0000	9.37	9161
15 Methyl Iodide	142	2.938	2.938	(0.499)	14998	10.0000	9.58	9298
16 Acrolein	56	3.153	3.153	(0.536)	11173	50.0000	48.1	5597

		QUANT SIG					AMOUNTS		
							CAL-AMT	ON-COL	
	Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
	==========================	====	==	======	======	========	=======	=======	==========
	17 Allyl chloride	39	3.304	3.304	(0.562)	14165	10.0000	10.6	9531
	18 Methylene Chloride	49	3.420	3.420	(0.581)	16213	10.0000	9.60	4808
	19 Acetone	43	3.475	3.475	(0.591)	31942	50.0000	46.1	4801(M3H)
	20 trans-1,2-Dichloroethene	61	3.594	3.594	(0.611)	20360	10.0000	9.62	8792
	21 Methyl Acetate	43	3.613	3.613	(0.614)	42193	50.0000	48.9	9581
	22 Hexane	57	3.687	3.687	(0.627)	29696	10.0000	9.74	9271
	23 MTBE	73	3.722	3.722	(0.633)	40428	10.0000	9.80	9613
	24 tert-Butyl Alcohol	59	3.832	3.832	(0.651)	7929	50.0000	43.6	0(M1)
	25 Acetonitrile	41	3.951	3.951	(0.672)	6105	50.0000	49.7	9201
	26 Isopropyl Ether	45	4.115	4.115	(0.699)	35627	10.0000	9.68	9608
	27 Chloroprene	53	4.198	4.198	(0.714)	19593	10.0000	9.45	8523
	28 1,1-Dichloroethane ++	63	4.218	4.218	(0.717)	27282	10.0000	9.66	9184
	29 Acrylonitrile	53	4.266	4.266	(0.725)	22963	50.0000	47.7	9705
	30 Vinyl Acetate	43	4.478	4.478	(0.761)	22447	10.0000	9.85	5566
	31 Ethyl Tert-butyl Ether	59	4.475	4.475	(0.761)	40182	10.0000	9.70	8160
	32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	19871	10.0000	9.78	9665
M	33 Total 1,2-Dichloroethene	61				40231	20.0000	19.4	0
	34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	24426	10.0000	9.73	9446
	35 Bromochloromethane	128	4.915	4.915	(0.835)	8274	10.0000	10.4	9691
	36 Cyclohexane	56	4.928	4.928	(0.838)	24198	10.0000	9.39	8982
	37 Chloroform +	83	4.992	4.992	(0.849)	27444	10.0000	9.50	9653
	38 Ethyl Acetate	43	5.118	5.118	(0.870)	62159	50.0000	49.2	5949
	41 sec-Butanol	45	5.172	5.172	(0.879)	846	10.0000	8.84	5494
	39 Carbon Tetrachloride	117	5.121	5.121	(0.870)	21594	10.0000	9.24	8162
	40 Tetrahydrofuran	42	5.156	5.156	(0.876)	22473	50.0000	47.8	8495
$	42 Dibromofluoromethane	111	5.156	5.156	(0.876)	14586	10.0000	9.45	7353
	43 1,1,1-Trichloroethane	97	5.182	5.182	(0.881)	25029	10.0000	9.35	9255
	44 2-Butanone	43	5.282	5.282	(0.898)	34712	50.0000	48.2	7410
	45 1,1-Dichloropropene	75	5.295	5.295	(0.900)	22455	10.0000	9.44	9417
	46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	68436	10.0000	9.52	8674
	47 Propionitrile	54	5.533	5.533	(0.940)	9033	50.0000	50.8	8496
	48 Benzene	78	5.516	5.516	(0.938)	61559	10.0000	9.57	9746
	49 Methylacrylonitrile	41	5.552	5.552	(0.944)	36232	50.0000	49.6	7428
	56 tert-amyl Methyl Ether	73	5.635	5.635	(0.958)	40710	10.0000	9.64	7197
$	50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	15141	10.0000	9.42	8410
	52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	20228	10.0000	10.0	9571
	51 Isobutyl Alcohol	43	5.729	5.729	(0.974)	2126	50.0000	50.5	2571(H)
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	312572	50.0000		9889
	57 Trichloroethene	130	6.034	6.034	(1.026)	18554	10.0000	9.39	9173
	58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	28236	10.0000	9.34	8375
	59 Tert-amyl alcohol	59	6.253	6.253	(1.063)	29410	50.0000	47.0	6748
	60 n-Butanol	56	6.336	6.336	(1.077)	1071	50.0000	50.5	7961
	62 Dibromomethane	93	6.394	6.394	(1.087)	10112	10.0000	9.77	9161
	63 2-3 Dichloro-1-Proprene	75	6.465	6.465	(1.099)	24100	10.0000	9.31	9084
	64 1,2-Dichloropropane +	63	6.481	6.481	(1.102)	15321	10.0000	9.64	7653
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	20686	10.0000	9.61	9655
	66 Methyl methacrylate	41	6.680	6.680	(1.136)	9989	10.0000	9.08	9633

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
	67 1,4- Dioxane	88	6.725	6.725	(1.143)	2688	250.000	228	8111
	68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	19847	10.0000	9.80	9201
	72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	19577	50.0000	46.9	9712
	73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	23948	10.0000	9.39	9684
$	74 Toluene-d8	98	7.224	7.224	(0.866)	53506	10.0000	9.55	9711
	75 Toluene +	91	7.266	7.266	(0.871)	68537	10.0000	9.83	9427
	76 2-Nitropropane	43	7.436	7.436	(0.892)	3719	10.0000	10.2	6280(M3)
	77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	57708	50.0000	46.7	8332
	78 Tetrachloroethene	164	7.558	7.558	(0.906)	14595	10.0000	9.25	8090
	79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	23020	10.0000	9.53	8768
	80 Ethyl Methacrylate	41	7.693	7.693	(0.922)	8992	10.0000	9.23	7322
	81 1,1,2-Trichloroethane	97	7.696	7.696	(0.923)	14000	10.0000	9.88	9406
	82 Dibromochloromethane	129	7.822	7.822	(0.938)	15051	10.0000	9.34	9376
M	83 1-3 Dichloropropene total	100				46968	20.0000	18.9	0
	84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	17920	10.0000	9.62	9664
	86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	23361	10.0000	9.90	9471
	87 1-Nitropropane	43	7.963	7.963	(0.955)	1689	10.0000	9.58	6165(M3)
	88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	15124	10.0000	9.98	9582
	89 2-Hexanone	43	8.143	8.143	(0.976)	46436	50.0000	45.8	9665
	90 1-Chlorohexane	91	8.330	8.330	(0.999)	25187	10.0000	10.1	4639
*	91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	119551	50.0000		9744
	92 Chlorobenzene ++	112	8.352	8.352	(1.002)	44691	10.0000	9.68	6492
	93 Ethylbenzene +	106	8.365	8.365	(1.003)	24200	10.0000	9.64	9513
	94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	14046	10.0000	9.05	9178
	95 p,m-Xylene	106	8.462	8.462	(1.015)	58802	20.0000	19.4	9834
	97 o-Xylene	106	8.745	8.745	(1.049)	27430	10.0000	9.71	9818
	101 Styrene	104	8.777	8.777	(1.052)	34395	10.0000	9.45	9627
	102 Bromoform ++	173	8.799	8.799	(1.055)	9910	10.0000	8.34	8431
	104 Isopropylbenzene	105	8.941	8.941	(1.072)	70970	10.0000	9.59	9764
M	105 TOTAL XYLENE	106				86232	30.0000	29.1	0
$	106 Bromofluorobenzene	174	9.121	9.121	(1.094)	16738	10.0000	9.20	9267
	107 cis-1,4-dichloro-2-butene	53	9.159	9.159	(0.934)	5294	10.0000	9.64	9530
	109 n-Propylbenzene	91	9.198	9.198	(0.938)	84880	10.0000	10.0	9156
	108 Bromobenzene	77	9.195	9.195	(0.938)	27711	10.0000	9.86	9652
	110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.943)	17408	10.0000	10.5	9513
	112 2-Chlorotoluene	91	9.304	9.304	(0.949)	54141	10.0000	10.1	9036
	113 1,3,5-Trimethylbenzene	105	9.314	9.314	(0.950)	56216	10.0000	9.90	8672
	114 1,2,3-Trichloropropane	75	9.327	9.327	(0.951)	22041	10.0000	9.84	7091
	115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	5589	10.0000	10.4	9127
	116 Cyclohexanone	55	9.365	9.365	(0.955)	13437	50.0000	50.2	9649
	117 4-Chlorotoluene	91	9.401	9.401	(0.959)	49913	10.0000	10.3	9702
	118 tert-butylbenzene	91	9.513	9.513	(0.970)	31856	10.0000	9.99	9734
	119 1,2,4-Trimethylbenzene	105	9.552	9.552	(0.974)	55290	10.0000	9.92	9287
	120 sec-Butylbenzene	105	9.619	9.619	(0.981)	72201	10.0000	9.82	9736
	121 p-Isopropyltoluene	119	9.700	9.700	(0.990)	63300	10.0000	9.73	8748
	122 Dicyclopentadiene	66	9.700	9.700	(0.990)	67596	10.0000	10.0	7818
	123 1,3-Dichlorobenzene	146	9.757	9.757	(0.995)	32156	10.0000	9.78	9499

						AMOUNTS		
	QUANT SIG					CAL-AMT	ON-COL	
Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.802	(1.000)	108151	50.0000		9497
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	32888	10.0000	9.73	6212(H)
126 n-Butylbenzene	91	9.954	9.954	(1.015)	84800	10.0000	9.80	8788
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	29173	10.0000	9.81	9455
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	3400	10.0000	8.86	9015
130 Hexachlorobutadiene	225	10.992	10.992	(1.121)	8551	10.0000	9.41	9039
131 1,2,4-Trichlorobenzene	180	11.028	11.028	(1.125)	17174	10.0000	9.45	9283
132 Naphthalene	128	11.288	11.288	(1.152)	41641	10.0000	9.78	9784
133 1,2,3-Trichlorobenzene	180	11.442	11.442	(1.167)	15358	10.0000	9.79	9253

QC Flag Legend

M1- Compound response manually integrated because
Target system did not integrate.
M3- Compound response manually integrated because
Target system integrated incorrect peak.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0230.d

Page 1

Date : 17-MAY-2023 15:05

Client ID: V15STD010

Instrument: msv15.i

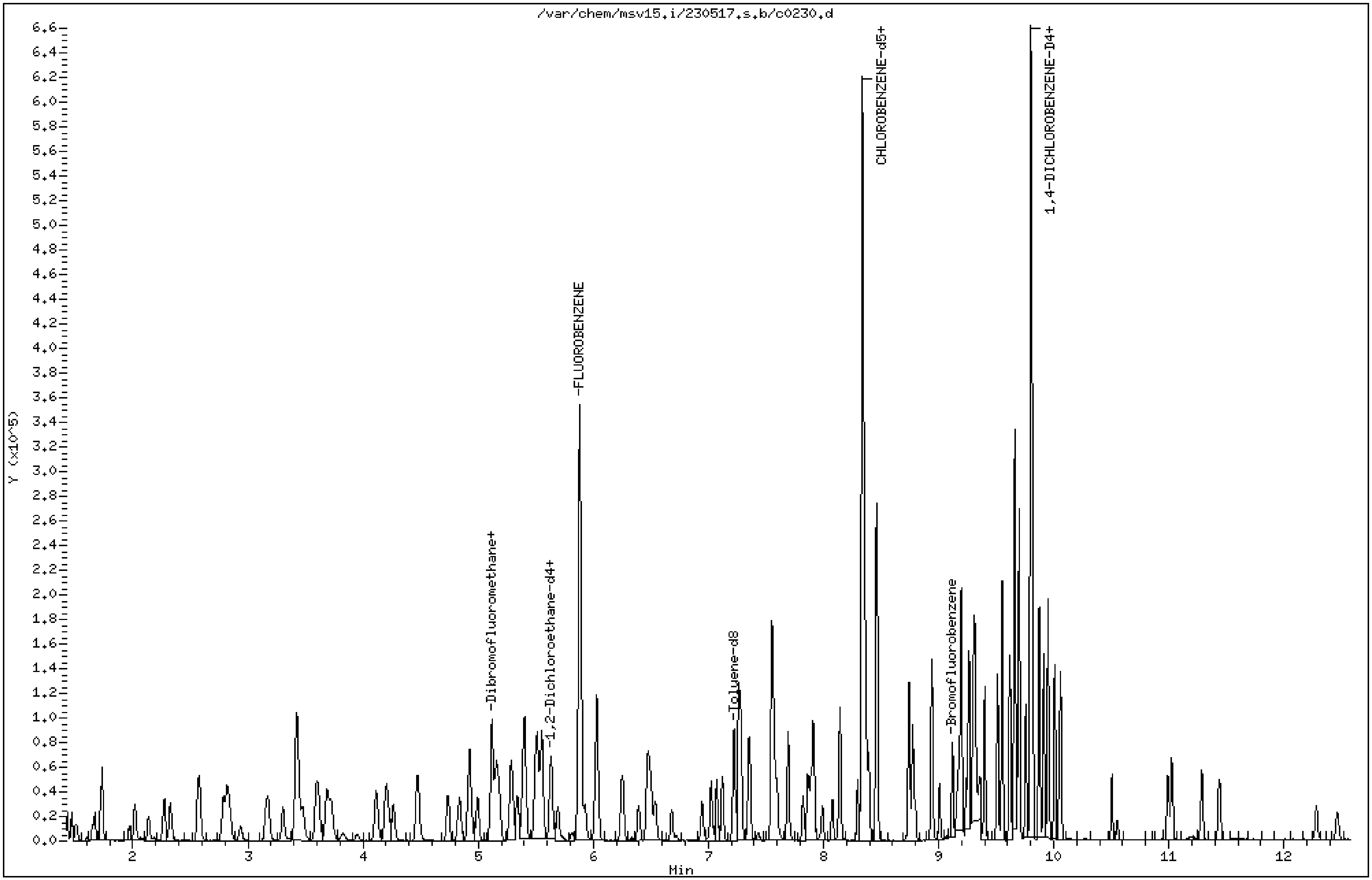
Sample Info: 1205*V15STD010

Purge Volume: 5.0

Operator: CJR

Column phase: RTX-VMS-30M

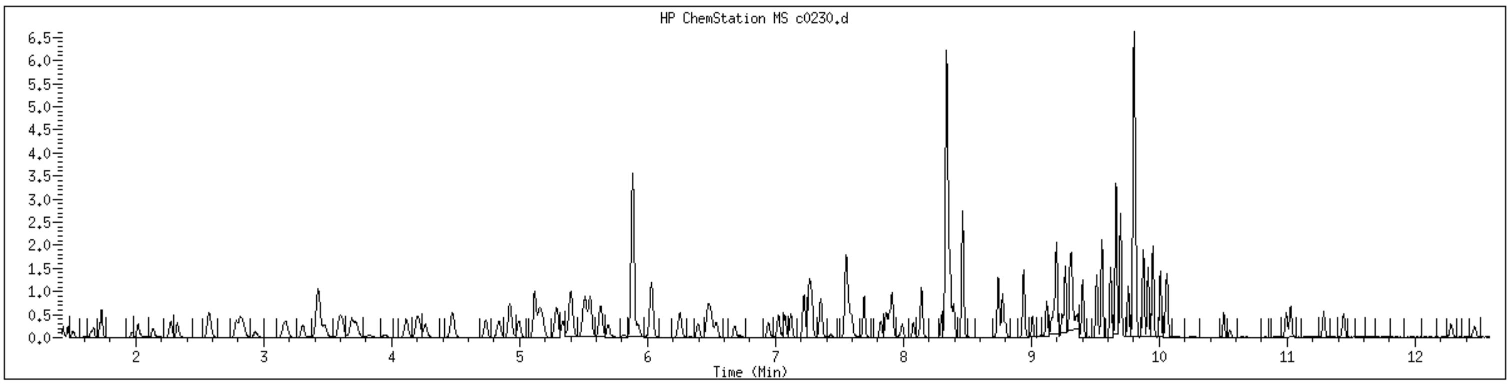
Column diameter: 0.25



Report Date: 05/23/2023 16:34

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 1205   SampleType   : CALIB_5
Injection Date: 05/17/2023 15:05   Instrument   : msv15.i
Operator   : CJR
Sample Info   : 1205*V15STD010
Misc Info   : MSV~527~*1*SMR
Method   : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260b+AppIXDOD



Original   Final

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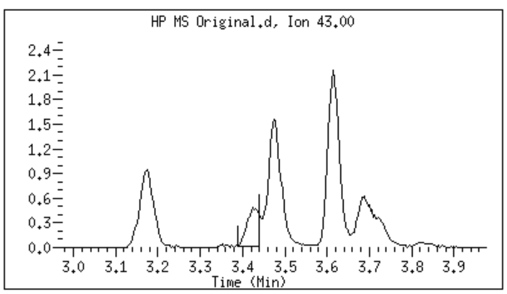
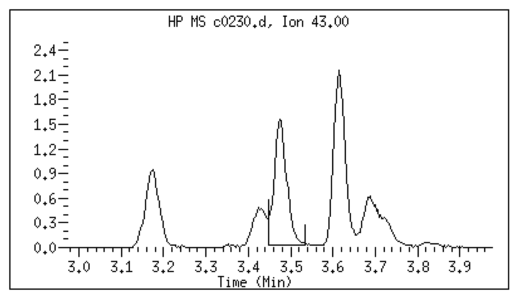
19 Acetone   CAS#: 67-64-1   Reason: M3



Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:09



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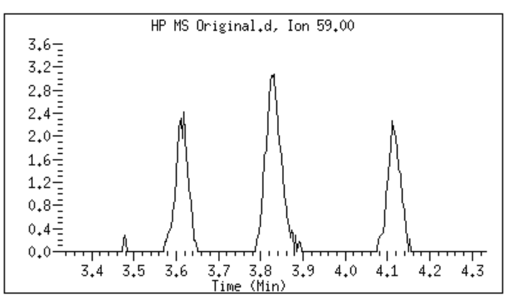
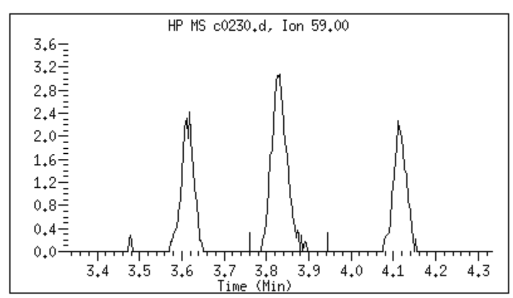
24 tert-Butyl Alcohol   CAS#: 75-65-0   Reason: M1



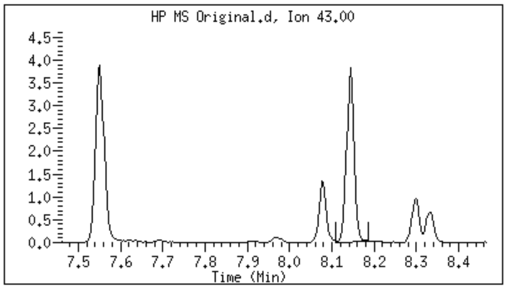
Electronic Signature Applied

User: smr
Date: 05/17/2023 16:54



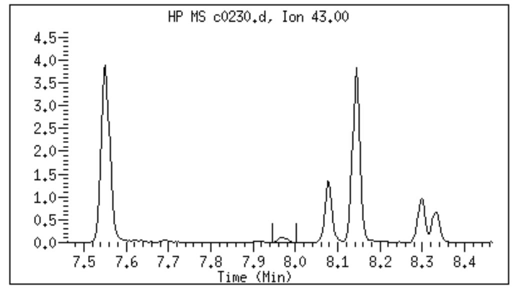
Original   Final

87 1-Nitropropane   CAS#: 108-03-2   Reason: M3

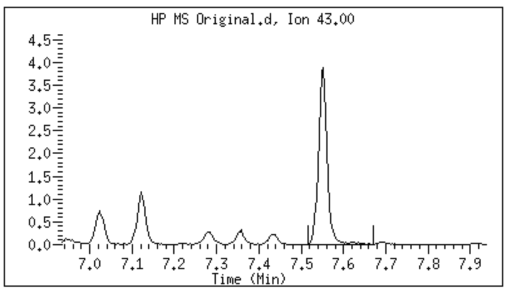


Electronic Signature
Applied

User: kmn
Date: 05/17/2023 16:09

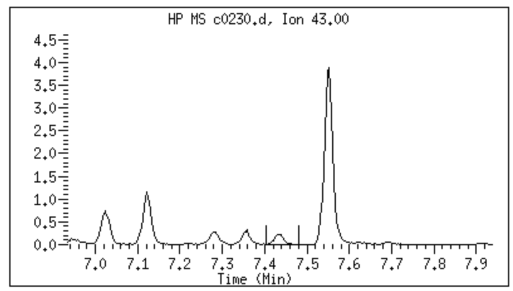


76 2-Nitropropane   CAS#: 79-46-9   Reason: M3



Electronic Signature
Applied

User: kmn
Date: 05/17/2023 16:09



M1 - Target system did not integrate
M3 - Target system integrated incorrect peak

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0231.d
Lab Smp Id: 1206 Client Smp ID: V15STD020
Inj Date : 17-MAY-2023 15:24
Operator : CJR Inst ID: msv15.i
Smp Info : 1206*V15STD020
Misc Info : MSV~527~*1*SMR
Comment :
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr Quant Type: ISTD
Cal Date : 17-MAY-2023 15:24 Cal File: c0231.d
Als bottle: 1 Calibration Sample, Level: 6
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	30602	20.0000	20.5	9579
2 Chloromethane ++	50	1.652	1.652	(0.281)	27979	20.0000	20.8	9764
3 Vinyl Chloride +	62	1.719	1.719	(0.292)	39220	20.0000	20.7	9633
5 1-3 Butadiene	39	1.735	1.735	(0.295)	30933	20.0000	22.0	9743
6 Bromomethane	94	2.018	2.018	(0.343)	31149	20.0000	19.6	9615
7 Chloroethane	64	2.137	2.137	(0.363)	28920	20.0000	20.8	9217
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	57432	20.0000	20.8	9556
9 Isoprene	67	2.571	2.571	(0.437)	46930	20.0000	20.6	4547
10 Ethyl Ether	59	2.587	2.587	(0.440)	22321	20.0000	20.7	8374
11 1,1-Dichloroethene +	96	2.787	2.787	(0.474)	31976	20.0000	20.4	8452
12 Carbon Disulfide	76	2.816	2.816	(0.479)	98564	20.0000	20.5	8107
13 1,1,2Trichlotrifluoroethane	101	2.835	2.835	(0.482)	35157	20.0000	20.7	9209
15 Methyl Iodide	142	2.938	2.938	(0.499)	37592	20.0000	21.0	9394
16 Acrolein	56	3.150	3.150	(0.535)	24838	100.000	104	6296

							AMOUNTS		
		QUANT SIG					CAL-AMT	ON-COL	
	Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
	==========================	====	==	======	======	========	=======	=======	==========
	17 Allyl chloride	39	3.304	3.304	(0.562)	30710	20.0000	22.3	9620
	18 Methylene Chloride	49	3.420	3.420	(0.581)	35774	20.0000	20.6	4893
	19 Acetone	43	3.472	3.472	(0.590)	73902	100.000	104	5274(H)
	20 trans-1,2-Dichloroethene	61	3.594	3.594	(0.611)	45226	20.0000	20.8	8649
	21 Methyl Acetate	43	3.616	3.616	(0.615)	92707	100.000	105	9190
	22 Hexane	57	3.690	3.690	(0.627)	66680	20.0000	21.3	9214
	23 MTBE	73	3.719	3.719	(0.632)	88622	20.0000	20.9	9656
	24 tert-Butyl Alcohol	59	3.829	3.829	(0.651)	19001	100.000	102	4488(M1)
	25 Acetonitrile	41	3.947	3.947	(0.671)	13943	100.000	111	9401
	26 Isopropyl Ether	45	4.118	4.118	(0.700)	79721	20.0000	21.1	9666
	27 Chloroprene	53	4.195	4.195	(0.713)	45592	20.0000	21.4	9048
	28 1,1-Dichloroethane ++	63	4.214	4.214	(0.716)	59997	20.0000	20.7	9338
	29 Acrylonitrile	53	4.263	4.263	(0.725)	52154	100.000	105	9726
	30 Vinyl Acetate	43	4.475	4.475	(0.761)	48754	20.0000	20.8	5654
	31 Ethyl Tert-butyl Ether	59	4.475	4.475	(0.761)	89151	20.0000	20.9	8078
	32 cis-1,2-Dichloroethene	61	4.738	4.738	(0.805)	43138	20.0000	20.7	9753
M	33 Total 1,2-Dichloroethene	61				88364	40.0000	41.5	0
	34 2,2-Dichloropropane	77	4.841	4.841	(0.823)	53854	20.0000	20.9	9473
	35 Bromochloromethane	128	4.922	4.922	(0.837)	17654	20.0000	21.5	9337
	36 Cyclohexane	56	4.928	4.928	(0.838)	56469	20.0000	21.3	9183
	37 Chloroform +	83	4.992	4.992	(0.849)	60968	20.0000	20.5	9692
	38 Ethyl Acetate	43	5.118	5.118	(0.870)	135862	100.000	105	5979
	41 sec-Butanol	45	5.176	5.176	(0.880)	1986	20.0000	20.2	3813(M2)
	39 Carbon Tetrachloride	117	5.121	5.121	(0.870)	48921	20.0000	20.4	8218
	40 Tetrahydrofuran	42	5.153	5.153	(0.876)	49466	100.000	102	8726
$	42 Dibromofluoromethane	111	5.156	5.156	(0.876)	32486	20.0000	20.5	7668
	43 1,1,1-Trichloroethane	97	5.182	5.182	(0.881)	57293	20.0000	20.8	9219
	44 2-Butanone	43	5.279	5.279	(0.897)	80056	100.000	108	6570
	45 1,1-Dichloropropene	75	5.291	5.291	(0.899)	50739	20.0000	20.8	9100
	46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	152803	20.0000	20.7	8849
	47 Propionitrile	54	5.529	5.529	(0.940)	19658	100.000	108	7888
	48 Benzene	78	5.517	5.517	(0.938)	136973	20.0000	20.7	9590
	49 Methylacrylonitrile	41	5.555	5.555	(0.944)	78410	100.000	105	7324
	56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	91354	20.0000	21.1	7198
$	50 1,2-Dichloroethane-d4	65	5.636	5.636	(0.958)	34198	20.0000	20.7	8488
	52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	43155	20.0000	20.8	9643
	51 Isobutyl Alcohol	43	5.726	5.726	(0.973)	4569	100.000	106	3872(H)
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	321153	50.0000		9858
	57 Trichloroethene	130	6.034	6.034	(1.026)	41878	20.0000	20.6	9179
	58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	65519	20.0000	21.1	8441
	59 Tert-amyl alcohol	59	6.250	6.250	(1.062)	66952	100.000	104	6639
	60 n-Butanol	56	6.336	6.336	(1.077)	2315	100.000	96.2	8107
	62 Dibromomethane	93	6.394	6.394	(1.087)	22119	20.0000	20.8	9312
	63 2-3 Dichloro-1-Proprene	75	6.465	6.465	(1.099)	53953	20.0000	20.3	9062
	64 1,2-Dichloropropane +	63	6.484	6.484	(1.102)	33848	20.0000	20.7	8182
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	45364	20.0000	20.5	9637
	66 Methyl methacrylate	41	6.684	6.684	(1.136)	23055	20.0000	20.4	9786

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
67 1,4- Dioxane	88	6.722	6.722	(1.143)	6140	500.000	507	8714
68 1-Bromo-2-chloroethane	63	6.947	6.947	(1.181)	43039	20.0000	20.7	9169
72 2-Chloroethyl vinyl ether	63	7.025	7.025	(1.194)	35793	100.000	80.0	9654
73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	54197	20.0000	20.7	9655
$ 74 Toluene-d8	98	7.221	7.221	(0.866)	121420	20.0000	20.8	9790
75 Toluene +	91	7.262	7.262	(0.871)	150767	20.0000	20.7	9583
76 2-Nitropropane	43	7.433	7.433	(0.891)	8113	20.0000	20.6	6241(M3)
77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	132162	100.000	99.5	8363
78 Tetrachloroethene	164	7.558	7.558	(0.906)	33221	20.0000	20.2	8128
79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	51089	20.0000	20.6	8725
80 Ethyl Methacrylate	41	7.693	7.693	(0.922)	21202	20.0000	20.9	7438
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	30378	20.0000	20.6	9445
82 Dibromochloromethane	129	7.825	7.825	(0.938)	34865	20.0000	20.8	9495
M 83 1-3 Dichloropropene total	100				105286	40.0000	41.3	0
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	39972	20.0000	20.6	9657
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	51849	20.0000	21.1	9495
87 1-Nitropropane	43	7.970	7.970	(0.956)	3514	20.0000	19.1	6124(M3)
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	32723	20.0000	20.7	9553
89 2-Hexanone	43	8.143	8.143	(0.976)	109907	100.000	101	9733
90 1-Chlorohexane	91	8.333	8.333	(0.999)	49908	20.0000	19.2	5571
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	124613	50.0000		9548
92 Chlorobenzene ++	112	8.349	8.349	(1.001)	98051	20.0000	20.4	6798
93 Ethylbenzene +	106	8.369	8.369	(1.003)	54089	20.0000	20.7	9039
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	31843	20.0000	19.7	9185
95 p,m-Xylene	106	8.462	8.462	(1.015)	130713	40.0000	41.3	9833
97 o-Xylene	106	8.741	8.741	(1.048)	61312	20.0000	20.8	9794
101 Styrene	104	8.777	8.777	(1.052)	78466	20.0000	20.7	9812
102 Bromoform ++	173	8.799	8.799	(1.055)	23618	20.0000	18.0	8549
104 Isopropylbenzene	105	8.941	8.941	(1.072)	162607	20.0000	21.1	9788
M 105 TOTAL XYLENE	106				192025	60.0000	62.1	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	37087	20.0000	19.6	9512
107 cis-1,4-dichloro-2-butene	53	9.156	9.156	(0.934)	12077	20.0000	21.1	9554
109 n-Propylbenzene	91	9.198	9.198	(0.938)	190472	20.0000	21.7	9186
108 Bromobenzene	77	9.192	9.192	(0.938)	61947	20.0000	21.2	9231
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.943)	37146	20.0000	21.6	9635
112 2-Chlorotoluene	91	9.301	9.301	(0.949)	119264	20.0000	21.3	7837
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	127607	20.0000	21.6	7926
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	49542	20.0000	21.3	8317
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	12313	20.0000	21.9	9166
116 Cyclohexanone	55	9.365	9.365	(0.955)	31285	100.000	111	9750
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	108488	20.0000	21.6	9724
118 tert-butylbenzene	91	9.513	9.513	(0.970)	71451	20.0000	21.5	9781
119 1,2,4-Trimethylbenzene	105	9.552	9.552	(0.974)	125754	20.0000	21.7	9318
120 sec-Butylbenzene	105	9.619	9.619	(0.981)	166584	20.0000	21.8	9815
121 p-Isopropyltoluene	119	9.700	9.700	(0.990)	144696	20.0000	21.4	8803
122 Dicyclopentadiene	66	9.700	9.700	(0.990)	153772	20.0000	21.9	7930
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.996)	70260	20.0000	20.5	9588

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.803	9.803	(1.000)	112488	50.0000		9420
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	72969	20.0000	20.8	7603(H)
126 n-Butylbenzene	91	9.950	9.950	(1.015)	193138	20.0000	21.5	8931
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	63555	20.0000	20.5	9518
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	8227	20.0000	19.5	9153
130 Hexachlorobutadiene	225	10.992	10.992	(1.121)	19628	20.0000	20.8	9018
131 1,2,4-Trichlorobenzene	180	11.028	11.028	(1.125)	38876	20.0000	20.6	9350
132 Naphthalene	128	11.288	11.288	(1.152)	93547	20.0000	21.1	9762
133 1,2,3-Trichlorobenzene	180	11.442	11.442	(1.167)	33962	20.0000	20.8	9324

QC Flag Legend

M1- Compound response manually integrated because Target system did not integrate.
M2- Compound response manually integrated because Target system integrated incorrectly.
M3- Compound response manually integrated because Target system integrated incorrect peak.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0231.d

Page 1

Date : 17-MAY-2023 15:24

Client ID: V15STD020

Instrument: msv15.i

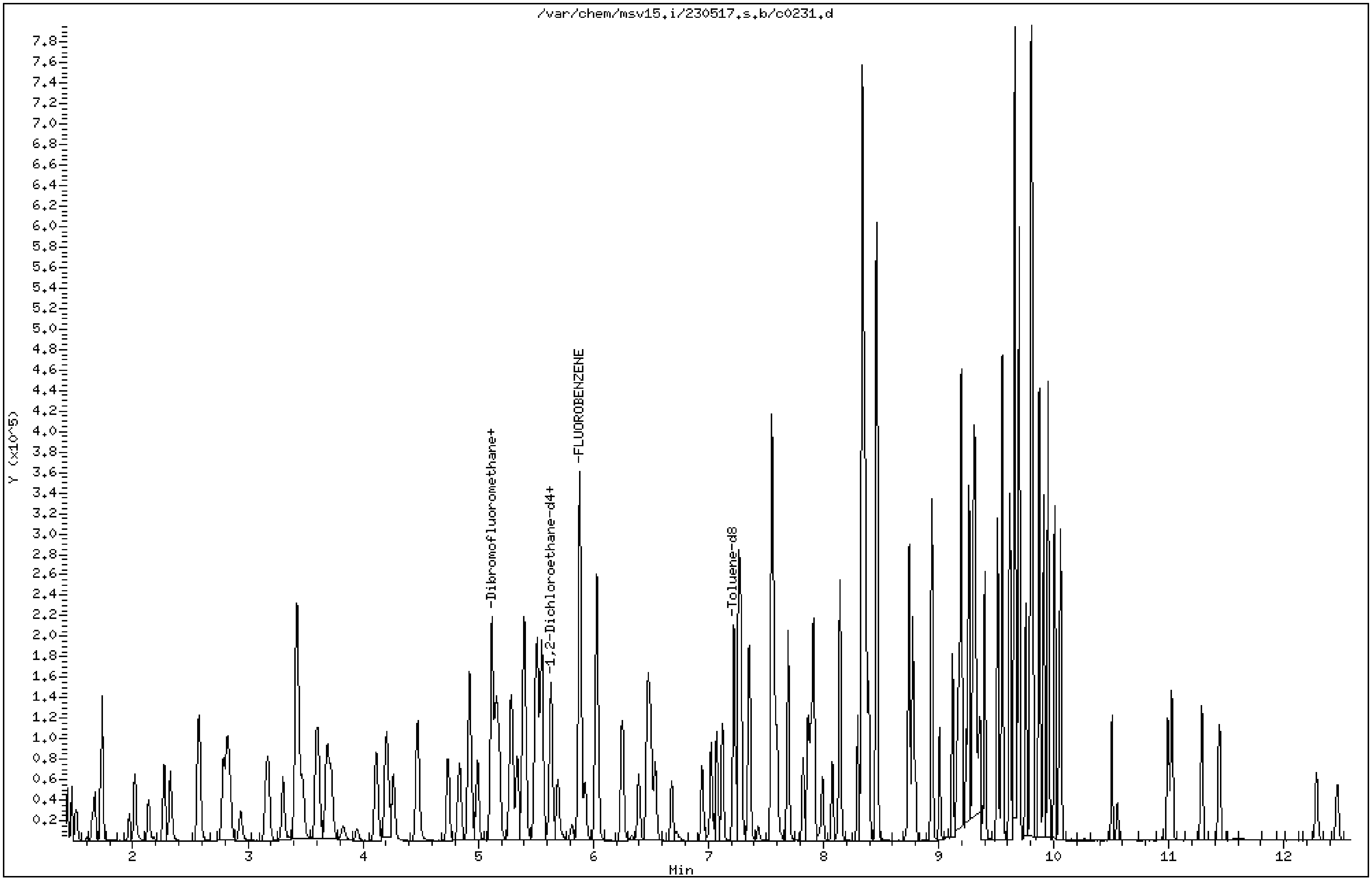
Sample Info: 1206*V15STD020

Purge Volume: 5.0

Operator: CJR

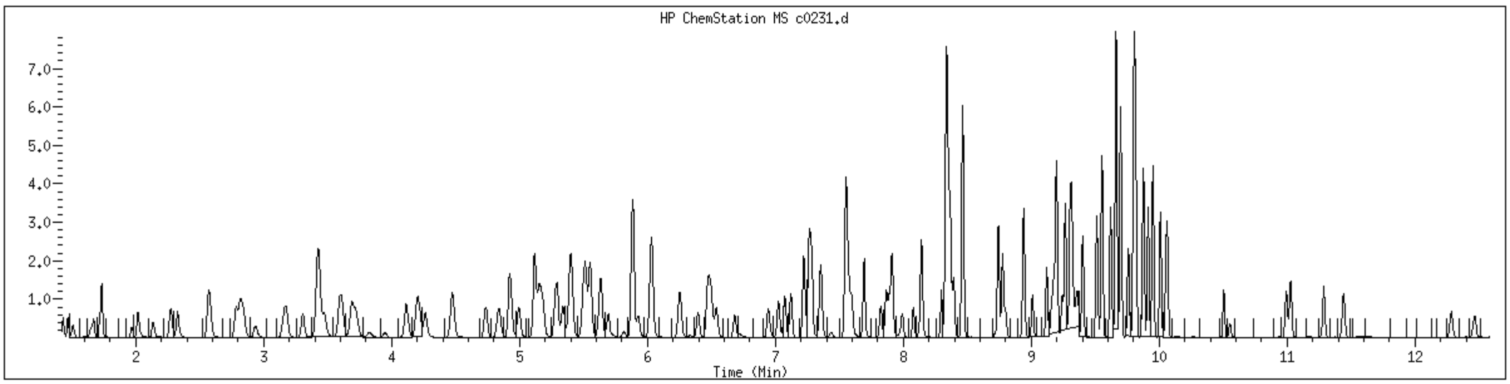
Column phase: RTX-VMS-30M

Column diameter: 0.25



MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 1206   SampleType   : CALIB_6
Injection Date: 05/17/2023 15:24   Instrument   : msv15.i
Operator   : CJR
Sample Info   : 1206*V15STD020
Misc Info   : MSV~527~*1*SMR
Method   : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260b+AppIXDOD



Original   Final

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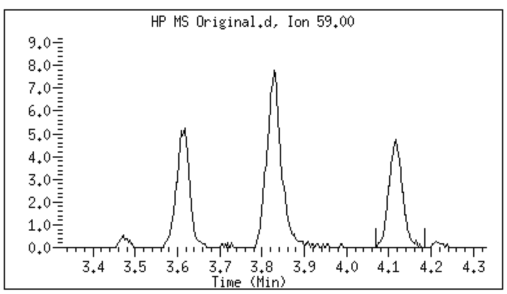
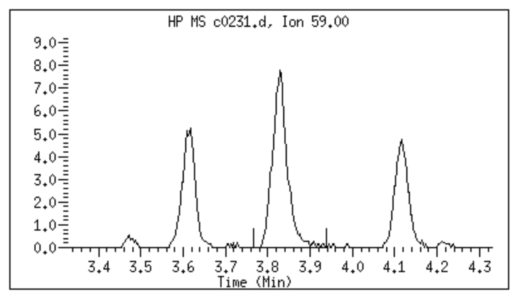
24 tert-Butyl Alcohol   CAS#: 75-65-0   Reason: M1



Electronic Signature
Applied

User: smr
Date: 05/17/2023 16:54



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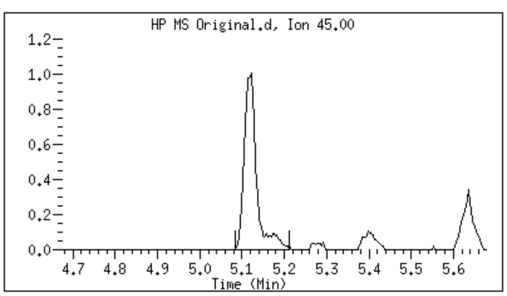
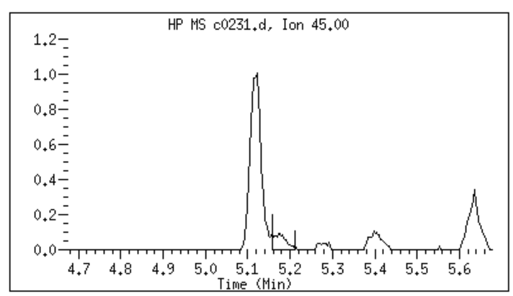
41 sec-Butanol   CAS#: 78-92-2   Reason: M2



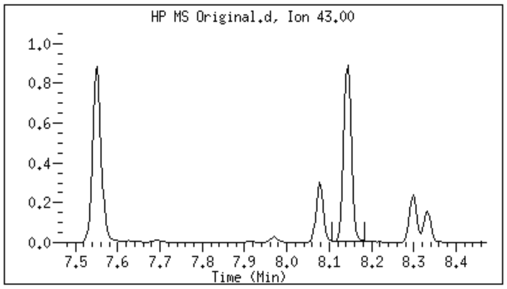
Electronic Signature
Applied

User: kmn
Date: 05/17/2023 16:07



Original Final

87 1-Nitropropane CAS#: 108-03-2 Reason: M3



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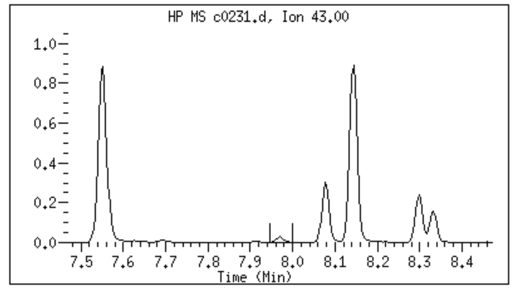
User: kmn
Date: 05/17/2023 16:08



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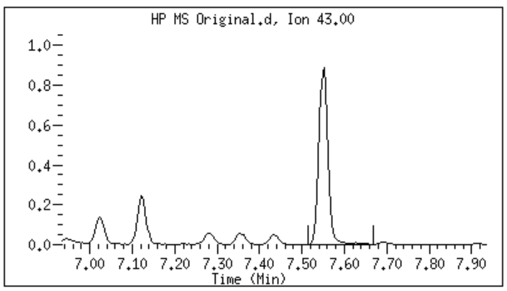
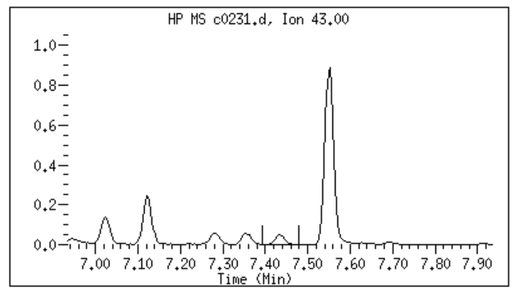
76 2-Nitropropane CAS#: 79-46-9 Reason: M3



Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:08



M1 - Target system did not integrate
M2 - Target system integrated incorrectly
M3 - Target system integrated incorrect peak

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0232.d
Lab Smp Id: 1207 Client Smp ID: V15STD050
Inj Date : 17-MAY-2023 15:43
Operator : CJR Inst ID: msv15.i
Smp Info : 1207*V15STD050
Misc Info : MSV~527~*1*SMR
Comment :
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr Quant Type: ISTD
Cal Date : 17-MAY-2023 15:43 Cal File: c0232.d
Als bottle: 1 Calibration Sample, Level: 7
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	78410	50.0000	50.7	9584
2 Chloromethane ++	50	1.649	1.649	(0.280)	69968	50.0000	50.3	9752
3 Vinyl Chloride +	62	1.716	1.716	(0.292)	99748	50.0000	50.9	9767
5 1-3 Butadiene	39	1.735	1.735	(0.295)	71997	50.0000	49.3	9711
6 Bromomethane	94	2.015	2.015	(0.343)	88397	50.0000	53.8	9715
7 Chloroethane	64	2.134	2.134	(0.363)	74021	50.0000	51.4	9207
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	148578	50.0000	52.0	9605
9 Isoprene	67	2.568	2.568	(0.437)	123576	50.0000	52.3	4543
10 Ethyl Ether	59	2.584	2.584	(0.439)	59003	50.0000	52.7	8054
11 1,1-Dichloroethene +	96	2.784	2.784	(0.473)	84852	50.0000	52.2	8264
12 Carbon Disulfide	76	2.816	2.816	(0.479)	253765	50.0000	51.0	7974
13 1,1,2Trichlotrifluoroethane	101	2.835	2.835	(0.482)	92398	50.0000	52.4	9297
15 Methyl Iodide	142	2.938	2.938	(0.499)	102973	50.0000	52.9	9533
16 Acrolein	56	3.150	3.150	(0.535)	65649	250.000	265	5940

							AMOUNTS		
		QUANT SIG					CAL-AMT	ON-COL	
	Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
	==========================	====	==	======	======	========	=======	=======	==========
	17 Allyl chloride	39	3.304	3.304	(0.562)	79708	50.0000	55.8	9726
	18 Methylene Chloride	49	3.420	3.420	(0.581)	91561	50.0000	50.9	4726
	19 Acetone	43	3.472	3.472	(0.590)	194463	250.000	263	5202(H)
	20 trans-1,2-Dichloroethene	61	3.591	3.591	(0.610)	118341	50.0000	52.5	8758
	21 Methyl Acetate	43	3.613	3.613	(0.614)	241456	250.000	263	9514
	22 Hexane	57	3.687	3.687	(0.627)	172742	50.0000	53.2	9240
	23 MTBE	73	3.722	3.722	(0.633)	233809	50.0000	53.2	9696
	24 tert-Butyl Alcohol	59	3.828	3.828	(0.651)	49824	250.000	257	4397(M1)
	25 Acetonitrile	41	3.944	3.944	(0.670)	34560	250.000	264	9591
	26 Isopropyl Ether	45	4.115	4.115	(0.699)	208071	50.0000	53.1	9642
	27 Chloroprene	53	4.198	4.198	(0.714)	115317	50.0000	52.2	8500
	28 1,1-Dichloroethane ++	63	4.218	4.218	(0.717)	157635	50.0000	52.4	9536
	29 Acrylonitrile	53	4.259	4.259	(0.724)	131821	250.000	257	9793
	30 Vinyl Acetate	43	4.475	4.475	(0.761)	125423	50.0000	51.7	5643
	31 Ethyl Tert-butyl Ether	59	4.475	4.475	(0.761)	233451	50.0000	52.9	8063
	32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	111972	50.0000	51.8	9712
M	33 Total 1,2-Dichloroethene	61				230313	100.000	104	0
	34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	138874	50.0000	51.9	9556
	35 Bromochloromethane	128	4.918	4.918	(0.836)	43994	50.0000	51.8	9323
	36 Cyclohexane	56	4.928	4.928	(0.838)	143772	50.0000	52.4	9392
	37 Chloroform +	83	4.996	4.996	(0.849)	160796	50.0000	52.2	9820
	38 Ethyl Acetate	43	5.115	5.115	(0.869)	355035	250.000	264	5963
	41 sec-Butanol	45	5.172	5.172	(0.879)	5762	50.0000	56.5	4882
	39 Carbon Tetrachloride	117	5.121	5.121	(0.870)	131370	50.0000	52.8	8278
	40 Tetrahydrofuran	42	5.144	5.144	(0.874)	125936	250.000	251	9684
$	42 Dibromofluoromethane	111	5.160	5.160	(0.877)	86430	50.0000	52.6	8697
	43 1,1,1-Trichloroethane	97	5.182	5.182	(0.881)	149626	50.0000	52.5	9183
	44 2-Butanone	43	5.275	5.275	(0.897)	205452	250.000	268	5939
	45 1,1-Dichloropropene	75	5.291	5.291	(0.899)	133125	50.0000	52.5	9458
	46 2,2,4 Trimethylpentane	57	5.398	5.398	(0.917)	413577	50.0000	54.0	8947
	47 Propionitrile	54	5.533	5.533	(0.940)	50566	250.000	267	8629
	48 Benzene	78	5.513	5.513	(0.937)	354755	50.0000	51.8	9352
	49 Methylacrylonitrile	41	5.552	5.552	(0.944)	203546	250.000	262	7394
	56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	240391	50.0000	53.4	7175
$	50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	86756	50.0000	50.7	8536
	52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	110787	50.0000	51.4	9778
	51 Isobutyl Alcohol	43	5.726	5.726	(0.973)	11454	250.000	255	3975(H)
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	332949	50.0000		9863
	57 Trichloroethene	130	6.031	6.031	(1.025)	110781	50.0000	52.7	8874
	58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	173750	50.0000	54.0	8337
	59 Tert-amyl alcohol	59	6.253	6.253	(1.063)	179846	250.000	270	6492
	60 n-Butanol	56	6.333	6.333	(1.077)	6770	250.000	254	8815
	62 Dibromomethane	93	6.394	6.394	(1.087)	58579	50.0000	53.1	9370
	63 2-3 Dichloro-1-Proprene	75	6.465	6.465	(1.099)	147136	50.0000	53.4	9187
	64 1,2-Dichloropropane +	63	6.484	6.484	(1.102)	88787	50.0000	52.4	8237
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	121677	50.0000	53.1	9706
	66 Methyl methacrylate	41	6.680	6.680	(1.136)	61021	50.0000	52.1	9763

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
67 1,4- Dioxane	88	6.722	6.722	(1.143)	14798	1250.00	1180	9142
68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	112416	50.0000	52.1	9201
72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	105023	250.000	218	9726
73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	142505	50.0000	52.5	9741
$ 74 Toluene-d8	98	7.221	7.221	(0.866)	316629	50.0000	52.2	9835
75 Toluene +	91	7.262	7.262	(0.871)	394310	50.0000	52.3	9567
76 2-Nitropropane	43	7.433	7.433	(0.891)	19389	50.0000	46.7	7539(M3)
77 4-methyl-2-pentanone	43	7.549	7.549	(0.905)	343899	250.000	245	7731
78 Tetrachloroethene	164	7.558	7.558	(0.906)	87675	50.0000	51.4	8245
79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	135976	50.0000	52.8	8844
80 Ethyl Methacrylate	41	7.693	7.693	(0.922)	54743	50.0000	51.9	7514
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	80250	50.0000	52.4	9466
82 Dibromochloromethane	129	7.825	7.825	(0.938)	94202	50.0000	54.0	9607
M 83 1-3 Dichloropropene total	100				278481	100.000	105	0
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	109440	50.0000	54.3	9652
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	133214	50.0000	52.2	9499
87 1-Nitropropane	43	7.970	7.970	(0.956)	9402	50.0000	49.3	6046(M3)
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	86960	50.0000	53.0	9621
89 2-Hexanone	43	8.143	8.143	(0.976)	287469	250.000	249	9714
90 1-Chlorohexane	91	8.330	8.330	(0.999)	147569	50.0000	54.7	8330
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	129315	50.0000		9070
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	257209	50.0000	51.5	9673
93 Ethylbenzene +	106	8.368	8.368	(1.003)	142956	50.0000	52.6	8968
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	85304	50.0000	50.8	9187
95 p,m-Xylene	106	8.462	8.462	(1.015)	345906	100.000	105	9859
97 o-Xylene	106	8.745	8.745	(1.049)	162009	50.0000	53.0	9868
101 Styrene	104	8.777	8.777	(1.052)	215667	50.0000	54.8	9851
102 Bromoform ++	173	8.799	8.799	(1.055)	64825	50.0000	46.1	8519
104 Isopropylbenzene	105	8.941	8.941	(1.072)	426261	50.0000	53.3	9857
M 105 TOTAL XYLENE	106				507915	150.000	158	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	98372	50.0000	50.0	9556
107 cis-1,4-dichloro-2-butene	53	9.156	9.156	(0.934)	32211	50.0000	52.7	9669
109 n-Propylbenzene	91	9.198	9.198	(0.938)	501136	50.0000	53.3	9127
108 Bromobenzene	77	9.192	9.192	(0.938)	160928	50.0000	51.4	9297
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.943)	97142	50.0000	52.9	9701
112 2-Chlorotoluene	91	9.301	9.301	(0.949)	312075	50.0000	52.0	7880
113 1,3,5-Trimethylbenzene	105	9.314	9.314	(0.950)	334535	50.0000	52.9	8628
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	129079	50.0000	51.8	8535
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	31798	50.0000	52.4	9147
116 Cyclohexanone	55	9.362	9.362	(0.955)	80908	250.000	266	9741
117 4-Chlorotoluene	91	9.401	9.401	(0.959)	279122	50.0000	51.9	9711
118 tert-butylbenzene	91	9.513	9.513	(0.970)	188790	50.0000	53.1	9779
119 1,2,4-Trimethylbenzene	105	9.552	9.552	(0.974)	332703	50.0000	53.6	9348
120 sec-Butylbenzene	105	9.619	9.619	(0.981)	439395	50.0000	53.7	9863
121 p-Isopropyltoluene	119	9.700	9.700	(0.990)	393228	50.0000	54.3	8839
122 Dicyclopentadiene	66	9.696	9.696	(0.989)	405727	50.0000	53.9	8017
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.996)	189196	50.0000	51.7	9736

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.803	9.803	(1.000)	120452	50.0000		9013
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	194889	50.0000	51.8	9052
126 n-Butylbenzene	91	9.950	9.950	(1.015)	523927	50.0000	54.4	8921
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	173109	50.0000	52.3	9590
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	22062	50.0000	47.6	9460
130 Hexachlorobutadiene	225	10.995	10.995	(1.122)	53170	50.0000	52.5	9129
131 1,2,4-Trichlorobenzene	180	11.028	11.028	(1.125)	108954	50.0000	53.8	9480
132 Naphthalene	128	11.285	11.285	(1.151)	255601	50.0000	53.9	9810
133 1,2,3-Trichlorobenzene	180	11.442	11.442	(1.167)	92445	50.0000	52.9	9351

QC Flag Legend

M1- Compound response manually integrated because Target system did not integrate.
M3- Compound response manually integrated because Target system integrated incorrect peak.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0232.d

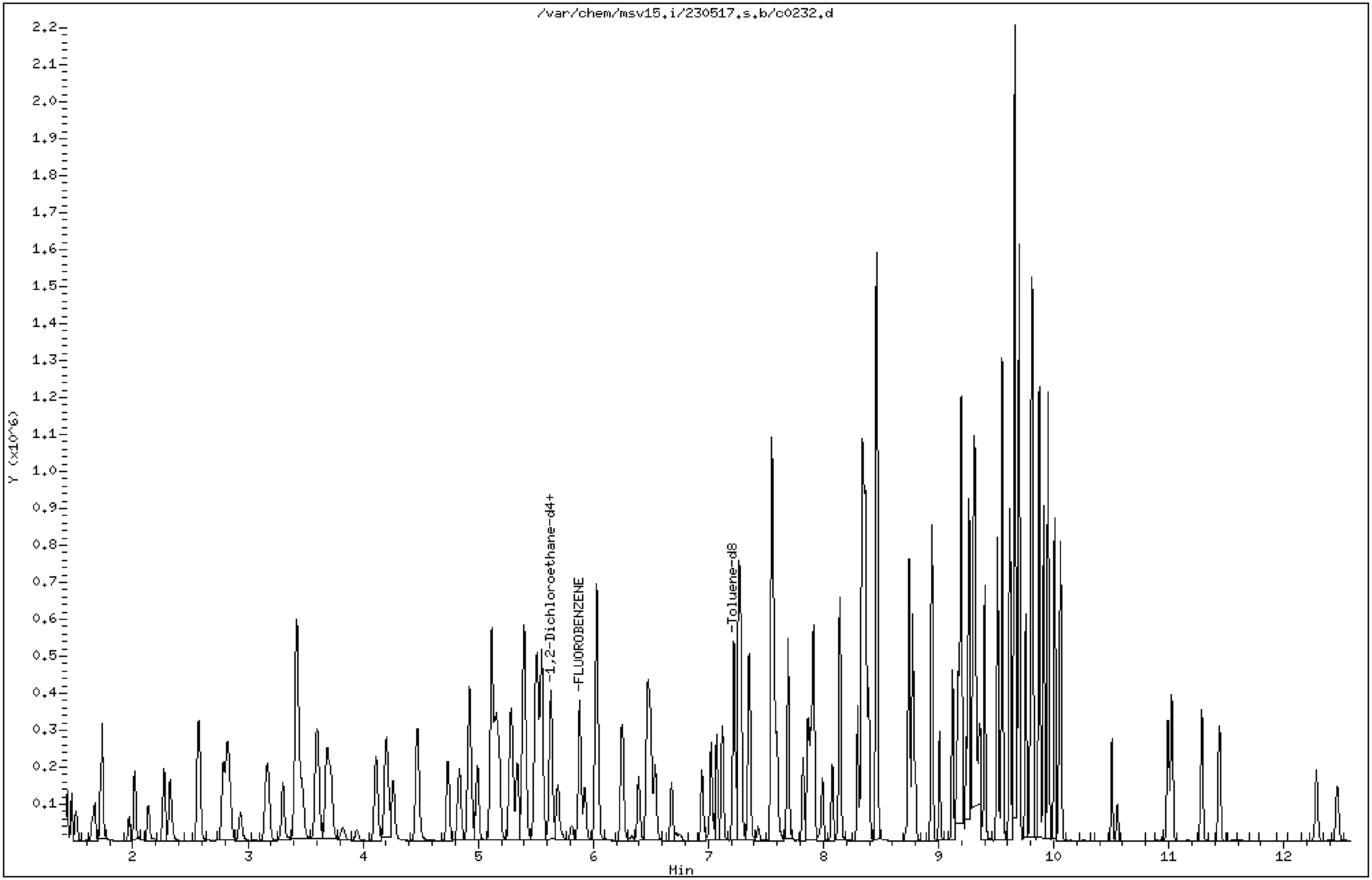
Date : 17-MAY-2023 15:43

Client ID: V15STD050 Instrument: msv15.i

Sample Info: 1207*V15STD050

Purge Volume: 5.0 Operator: CJR

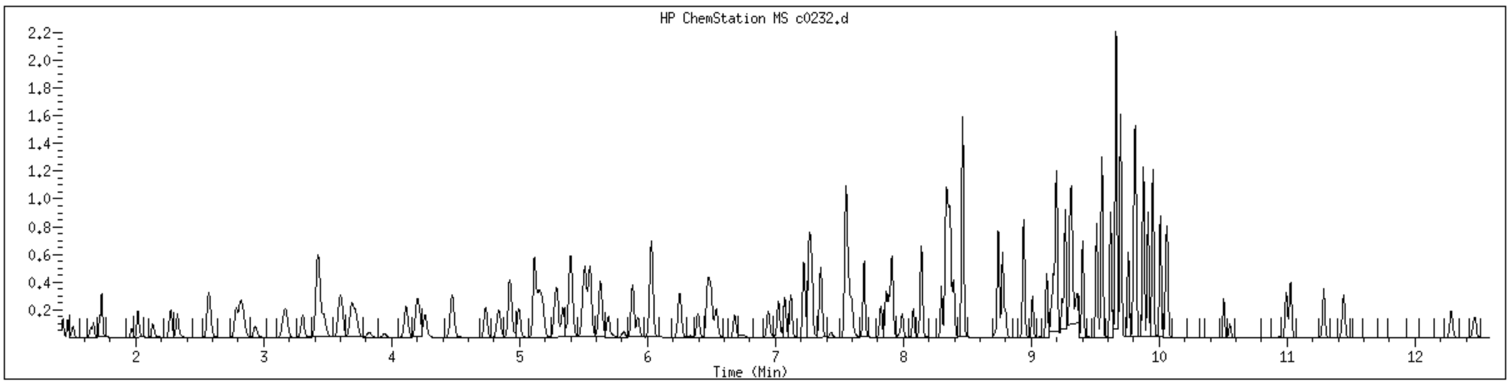
Column phase: RTX-VMS-30M Column diameter: 0.25



Report Date: 05/23/2023 16:34

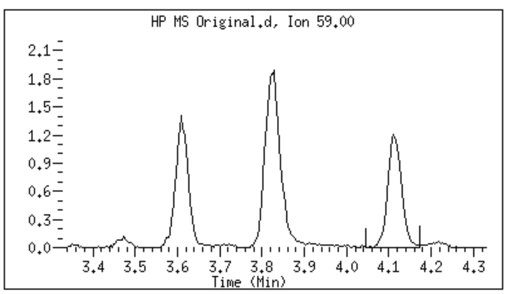
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID	: 1207	SampleType	: CALIB_7
Injection Date	: 05/17/2023 15:43	Instrument	: msv15.i
Operator	: CJR		
Sample Info	: 1207*V15STD050		
Misc Info	: MSV~527~*1*SMR		
Method	: /var/chem/msv15.i/230517.s.b/8260bw15DOD.m		
Dilution	: 1.00		
Matrix	: WATER		
Integrator	: HP RTE	Compound Sublist	: 8260b+AppIXDOD



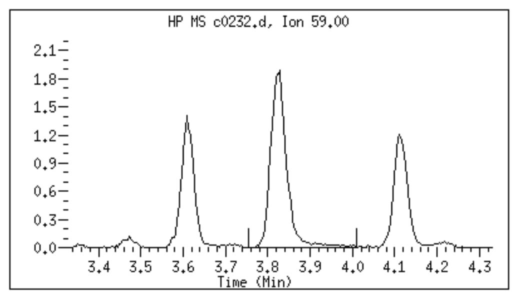
Original   Final

24 tert-Butyl Alcohol   CAS#: 75-65-0   Reason: M1

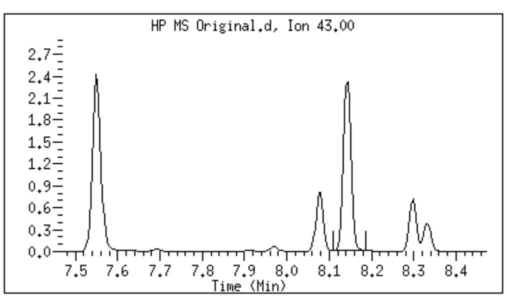


Electronic Signature Applied

User: smr
Date: 05/17/2023 16:54

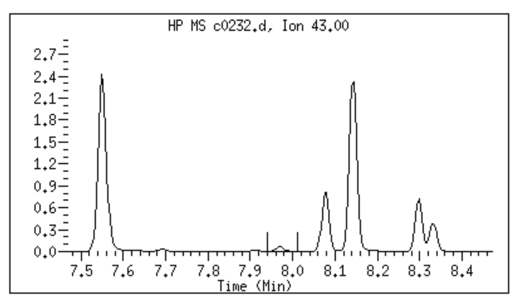


87 1-Nitropropane   CAS#: 108-03-2   Reason: M3



Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:06



Original Final

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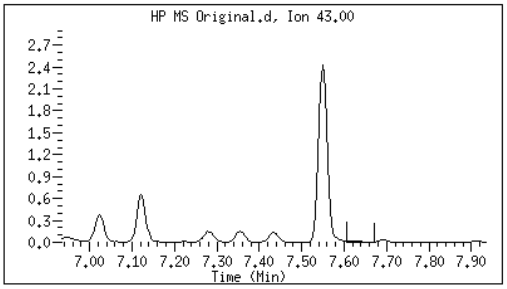
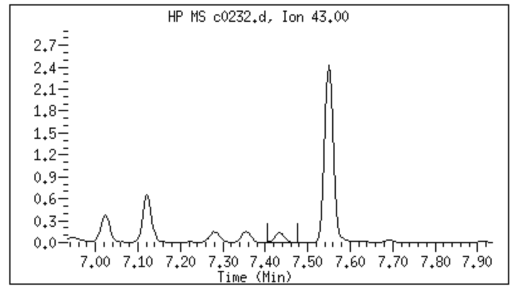
76 2-Nitropropane CAS#: 79-46-9 Reason: M3



Electronic Signature
Applied

User: kmn
Date: 05/17/2023 16:05



M1 - Target system did not integrate
M3 - Target system integrated incorrect peak

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0233.d
Lab Smp Id: 1208 Client Smp ID: V15STD100
Inj Date : 17-MAY-2023 16:01
Operator : CJR Inst ID: msv15.i
Smp Info : 1208*V15STD100
Misc Info : MSV~527~*1*SMR
Comment :
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr Quant Type: ISTD
Cal Date : 17-MAY-2023 16:01 Cal File: c0233.d
Als bottle: 1 Calibration Sample, Level: 8
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	172031	100.000	109	9731
2 Chloromethane ++	50	1.652	1.652	(0.281)	147309	100.000	103	9792
3 Vinyl Chloride +	62	1.716	1.716	(0.292)	208727	100.000	104	9708
5 1-3 Butadiene	39	1.735	1.735	(0.295)	150376	100.000	101	9651
6 Bromomethane	94	2.015	2.015	(0.342)	198723	100.000	118	9684
7 Chloroethane	64	2.131	2.131	(0.362)	153830	100.000	104	9179
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	311532	100.000	106	9663
9 Isoprene	67	2.568	2.568	(0.436)	262217	100.000	108	4561
10 Ethyl Ether	59	2.584	2.584	(0.439)	124721	100.000	109	8162
11 1,1-Dichloroethene +	96	2.787	2.787	(0.474)	177451	100.000	107	8490
12 Carbon Disulfide	76	2.815	2.815	(0.479)	526778	100.000	103	7993
13 1,1,2Trichlotrifluoroethane	101	2.835	2.835	(0.482)	193690	100.000	107	9285
15 Methyl Iodide	142	2.934	2.934	(0.499)	212199	100.000	105	9496
16 Acrolein	56	3.150	3.150	(0.535)	137707	500.000	543	5928

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
17 Allyl chloride	39	3.304	3.304	(0.562)	157006	100.000	107	9615
18 Methylene Chloride	49	3.417	3.417	(0.581)	188254	100.000	102	5029
19 Acetone	43	3.468	3.468	(0.590)	426163	500.000	563	5381(H)
20 trans-1,2-Dichloroethene	61	3.590	3.590	(0.610)	243290	100.000	105	8636
21 Methyl Acetate	43	3.613	3.613	(0.614)	507232	500.000	538	9509
22 Hexane	57	3.687	3.687	(0.627)	353751	100.000	106	9204
23 MTBE	73	3.716	3.716	(0.632)	494605	100.000	110	9663
24 tert-Butyl Alcohol	59	3.825	3.825	(0.650)	110597	500.000	557	4363(M1)
25 Acetonitrile	41	3.944	3.944	(0.670)	71541	500.000	534	9484
26 Isopropyl Ether	45	4.111	4.111	(0.699)	430561	100.000	107	9614
27 Chloroprene	53	4.195	4.195	(0.713)	239972	100.000	106	8963
28 1,1-Dichloroethane ++	63	4.217	4.217	(0.717)	323020	100.000	105	9477
29 Acrylonitrile	53	4.262	4.262	(0.725)	288059	500.000	548	9893
30 Vinyl Acetate	43	4.471	4.471	(0.760)	262383	100.000	105	5563
31 Ethyl Tert-butyl Ether	59	4.471	4.471	(0.760)	490115	100.000	108	8076
32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	228949	100.000	103	9760
M 33 Total 1,2-Dichloroethene	61				472239	200.000	209	0
34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	287486	100.000	105	9526
35 Bromochloromethane	128	4.921	4.921	(0.837)	84873	100.000	97.5	8900
36 Cyclohexane	56	4.928	4.928	(0.838)	297498	100.000	106	9469
37 Chloroform +	83	4.992	4.992	(0.849)	327993	100.000	104	9797
38 Ethyl Acetate	43	5.114	5.114	(0.869)	761216	500.000	552	5982
41 sec-Butanol	45	5.169	5.169	(0.879)	10464	100.000	100	4608(M2)
39 Carbon Tetrachloride	117	5.121	5.121	(0.870)	277815	100.000	109	8131
40 Tetrahydrofuran	42	5.147	5.147	(0.875)	269211	500.000	524	9634
$ 42 Dibromofluoromethane	111	5.156	5.156	(0.876)	180094	100.000	107	8340
43 1,1,1-Trichloroethane	97	5.179	5.179	(0.880)	316198	100.000	108	9662
44 2-Butanone	43	5.275	5.275	(0.897)	434935	500.000	554	5885
45 1,1-Dichloropropene	75	5.294	5.294	(0.900)	269553	100.000	104	9464
46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	872671	100.000	111	8987
47 Propionitrile	54	5.536	5.536	(0.941)	107416	500.000	554	8501
48 Benzene	78	5.516	5.516	(0.938)	747133	100.000	106	9713
49 Methylacrylonitrile	41	5.552	5.552	(0.944)	438552	500.000	550	7375
56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	514030	100.000	112	7143
$ 50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	181227	100.000	103	8566
52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	228401	100.000	103	9796
51 Isobutyl Alcohol	43	5.725	5.725	(0.973)	24027	500.000	523	4027(H)
* 55 FLUOROBENZENE	96	5.883	5.883	(1.000)	341246	50.0000		9856
57 Trichloroethene	130	6.034	6.034	(1.026)	235220	100.000	109	9160
58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	358295	100.000	109	8253
59 Tert-amyl alcohol	59	6.249	6.249	(1.062)	377263	500.000	552	6377
60 n-Butanol	56	6.336	6.336	(1.077)	13877	500.000	500	9175
62 Dibromomethane	93	6.394	6.394	(1.087)	122260	100.000	108	9423
63 2-3 Dichloro-1-Proprene	75	6.468	6.468	(1.099)	305492	100.000	108	8952
64 1,2-Dichloropropane +	63	6.481	6.481	(1.102)	182874	100.000	105	7750
65 Bromodichloromethane	83	6.539	6.539	(1.111)	254840	100.000	108	9785
66 Methyl methacrylate	41	6.680	6.680	(1.136)	133664	100.000	111	9798

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
67 1,4- Dioxane	88	6.719	6.719	(1.142)	31682	2500.00	2460	9397
68 1-Bromo-2-chloroethane	63	6.947	6.947	(1.181)	235395	100.000	106	9300
72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	237890	500.000	477	9722
73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	293449	100.000	105	9773
$ 74 Toluene-d8	98	7.224	7.224	(0.866)	670484	100.000	105	9863
75 Toluene +	91	7.262	7.262	(0.871)	832367	100.000	105	9644
76 2-Nitropropane	43	7.433	7.433	(0.891)	44203	100.000	100	9543
77 4-methyl-2-pentanone	43	7.548	7.548	(0.905)	740988	500.000	500	7634
78 Tetrachloroethene	164	7.558	7.558	(0.906)	193548	100.000	108	8247
79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	289421	100.000	110	8816
80 Ethyl Methacrylate	41	7.693	7.693	(0.922)	115542	100.000	104	7595
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	170515	100.000	106	9445
82 Dibromochloromethane	129	7.825	7.825	(0.938)	205410	100.000	112	9607
M 83 1-3 Dichloropropene total	100				582870	200.000	215	0
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	236353	100.000	111	9615
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	274252	100.000	102	9513
87 1-Nitropropane	43	7.966	7.966	(0.955)	22429	100.000	112	9565
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	186234	100.000	108	9676
89 2-Hexanone	43	8.143	8.143	(0.976)	616069	500.000	505	9678
90 1-Chlorohexane	91	8.333	8.333	(0.999)	290313	100.000	102	9047
* 91 CHLOROBENZENE-d5	82	8.339	8.339	(1.000)	136082	50.0000		8492
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	563797	100.000	107	9208
93 Ethylbenzene +	106	8.368	8.368	(1.003)	308926	100.000	108	8989
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	187313	100.000	106	9173
95 p,m-Xylene	106	8.465	8.465	(1.015)	757967	200.000	219	9923
97 o-Xylene	106	8.744	8.744	(1.049)	344807	100.000	107	9910
101 Styrene	104	8.777	8.777	(1.052)	467692	100.000	113	9871
102 Bromoform ++	173	8.799	8.799	(1.055)	146620	100.000	98.1	8533
104 Isopropylbenzene	105	8.944	8.944	(1.072)	905107	100.000	107	9904
M 105 TOTAL XYLENE	106				1102774	300.000	327	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	209299	100.000	101	9636
107 cis-1,4-dichloro-2-butene	53	9.156	9.156	(0.934)	69381	100.000	102	9591(M1)
109 n-Propylbenzene	91	9.201	9.201	(0.939)	1044472	100.000	100	8574
108 Bromobenzene	77	9.191	9.191	(0.938)	338565	100.000	97.5	9005
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.943)	204844	100.000	101	9750
112 2-Chlorotoluene	91	9.301	9.301	(0.949)	663599	100.000	99.8	7671
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	734421	100.000	105	8034
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	274378	100.000	99.2	8060
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	67695	100.000	100	9296
116 Cyclohexanone	55	9.365	9.365	(0.955)	170231	500.000	503	9679
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	600124	100.000	101	9858
118 tert-butylbenzene	91	9.513	9.513	(0.970)	395010	100.000	100	9883
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	722324	100.000	105	9387
120 sec-Butylbenzene	105	9.622	9.622	(0.982)	924935	100.000	102	9859
121 p-Isopropyltoluene	119	9.699	9.699	(0.990)	843229	100.000	105	8905
122 Dicyclopentadiene	66	9.699	9.699	(0.990)	834113	100.000	99.9	7872
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.996)	417578	100.000	103	9817

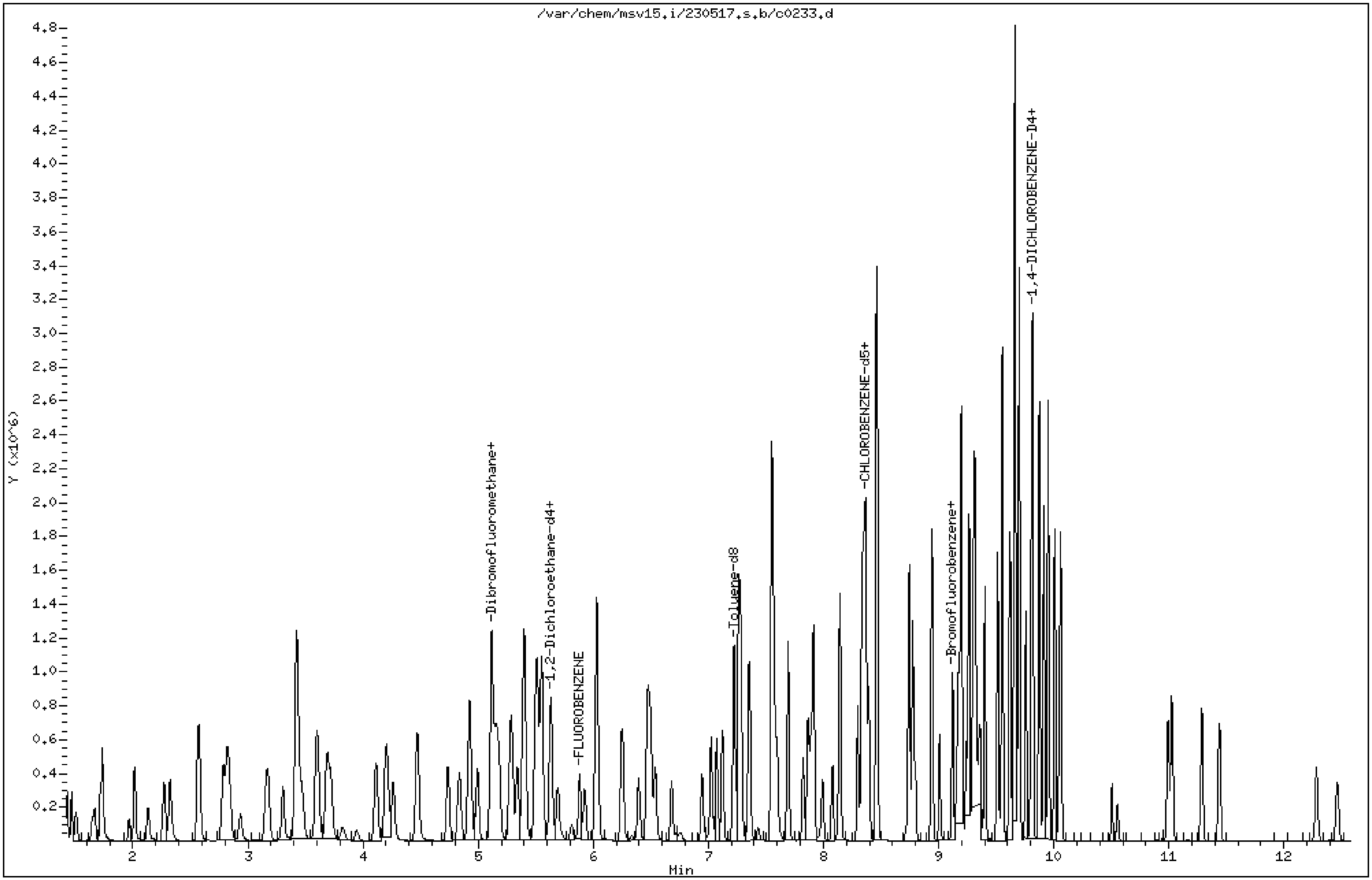
Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.802	(1.000)	133580	50.0000		8572
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	427974	100.000	102	9389
126 n-Butylbenzene	91	9.953	9.953	(1.015)	1130200	100.000	106	8876
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	384022	100.000	105	9684
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	51173	100.000	98.6	9553
130 Hexachlorobutadiene	225	10.995	10.995	(1.122)	119125	100.000	106	9234
131 1,2,4-Trichlorobenzene	180	11.027	11.027	(1.125)	237015	100.000	106	9503
132 Naphthalene	128	11.288	11.288	(1.152)	581316	100.000	110	9846
133 1,2,3-Trichlorobenzene	180	11.439	11.439	(1.167)	209737	100.000	108	9447

QC Flag Legend

M1- Compound response manually integrated because
Target system did not integrate.
M2- Compound response manually integrated because
Target system integrated incorrectly.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0233.d
Date : 17-MAY-2023 16:01
Client ID: V15STD100
Sample Info: 1208*V15STD100
Purge Volume: 5.0
Column phase: RTX-VMS-30M

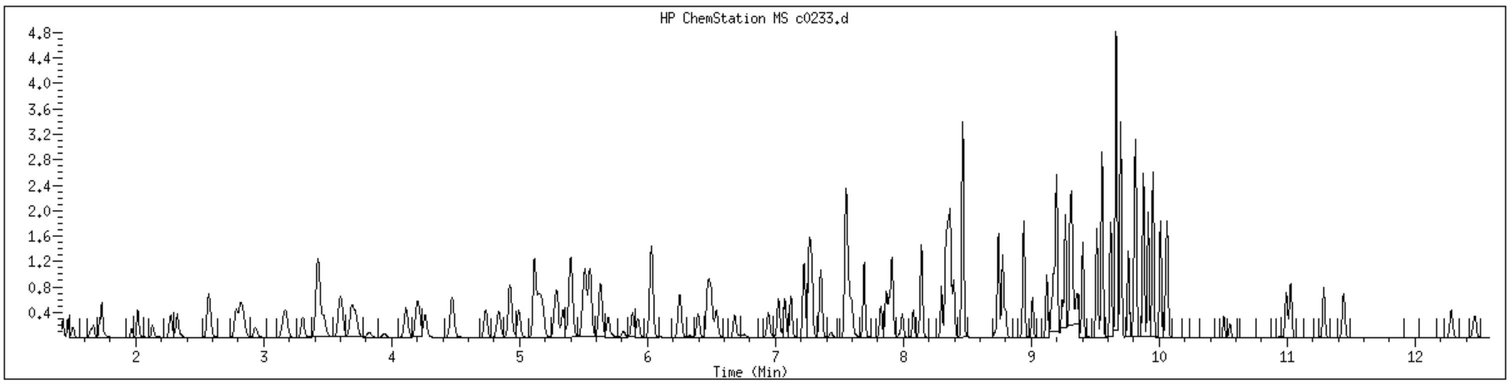
Instrument: msv15.i
Operator: CJR
Column diameter: 0.25



Data file : /var/chem/msv15.i/230517.s.b/c0233.d
Report Date: 05/23/2023 16:34

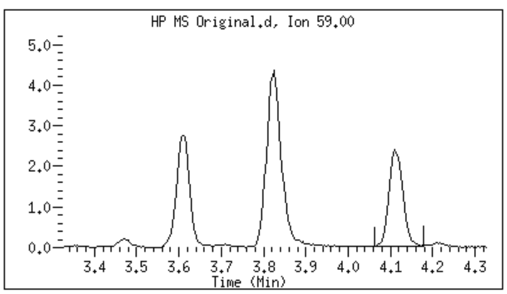
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1208   SampleType : CALIB_8
Injection Date: 05/17/2023 16:01   Instrument : msv15.i
Operator : CJR
Sample Info : 1208*V15STD100
Misc Info : MSV~527~*1*SMR
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE   Compound Sublist: 8260b+AppIXDOD



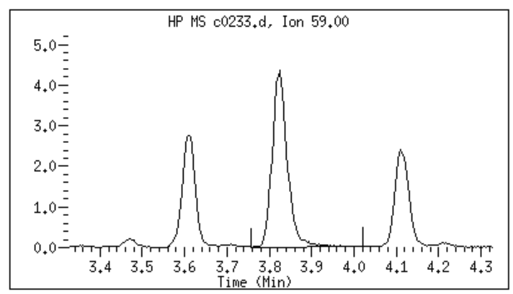
Original   Final

24 tert-Butyl Alcohol   CAS#: 75-65-0   Reason: M1

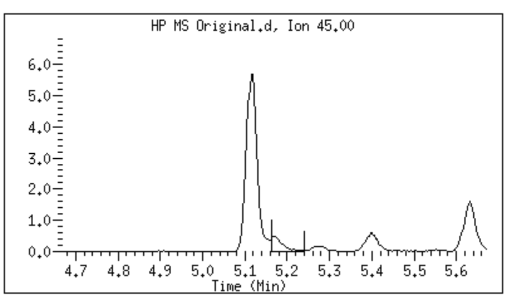


Electronic Signature Applied

User: smr
Date: 05/17/2023 16:54

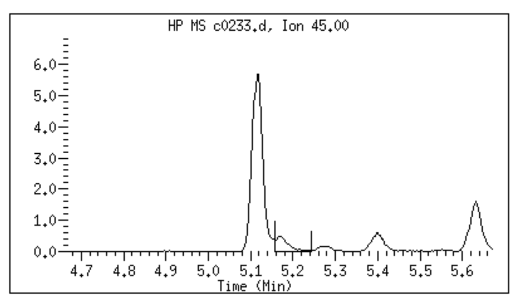


41 sec-Butanol   CAS#: 78-92-2   Reason: M2



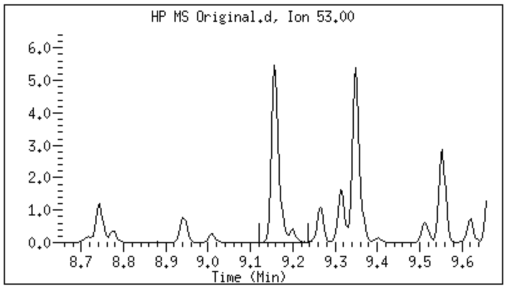
Electronic Signature Applied

User: smr
Date: 05/17/2023 16:56



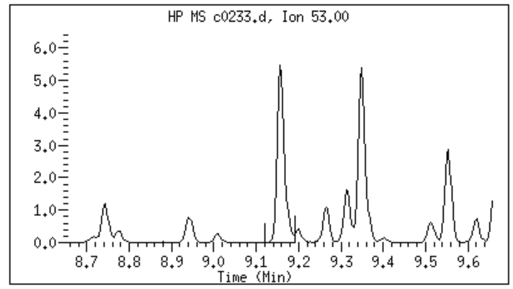
Original Final

107 cis-1,4-dichloro-2-butene CAS#: 1476-11-5 Reason: M1



Electronic Signature Applied

User: smr
Date: 05/17/2023 17:03



M1 - Target system did not integrate
M2 - Target system integrated incorrectly

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0234.d
Lab Smp Id: 1209 Client Smp ID: V15STD200
Inj Date : 17-MAY-2023 16:20
Operator : CJR Inst ID: msv15.i
Smp Info : 1209*V15STD200
Misc Info : MSV~527~*1*SMR
Comment :
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 Calibration Sample, Level: 9
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	374622	200.000	221	9697(A)
2 Chloromethane ++	50	1.651	1.651	(0.281)	325407	200.000	213	9715(A)
3 Vinyl Chloride +	62	1.716	1.716	(0.292)	459198	200.000	214	9835(A)
5 1-3 Butadiene	39	1.735	1.735	(0.295)	330256	200.000	206	9684(A)
6 Bromomethane	94	2.015	2.015	(0.342)	413368	200.000	230	9638(A)
7 Chloroethane	64	2.124	2.124	(0.361)	340914	200.000	216	9248(A)
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	657182	200.000	210	9737(A)
9 Isoprene	67	2.568	2.568	(0.436)	574359	200.000	222	4548(A)
10 Ethyl Ether	59	2.581	2.581	(0.439)	272599	200.000	222	7783
11 1,1-Dichloroethene +	96	2.786	2.786	(0.474)	377300	200.000	212	8635(A)
12 Carbon Disulfide	76	2.812	2.812	(0.478)	1135413	200.000	208	8986(A)
13 1,1,2Trichlotrifluoroethane	101	2.831	2.831	(0.481)	424355	200.000	220	9157(A)
15 Methyl Iodide	142	2.938	2.938	(0.499)	417991	200.000	191	0(M1)
16 Acrolein	56	3.150	3.150	(0.535)	292509	1000.00	1080	5736(A)

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
17 Allyl chloride	39	3.304	3.304	(0.562)	283118	200.000	181	9676
18 Methylene Chloride	49	3.417	3.417	(0.581)	407794	200.000	207	4968(A)
19 Acetone	43	3.471	3.471	(0.590)	831654	1000.00	1030	5046(AH)
20 trans-1,2-Dichloroethene	61	3.593	3.593	(0.611)	526882	200.000	213	8305(A)
21 Methyl Acetate	43	3.613	3.613	(0.614)	1028831	1000.00	1020	9597(A)
22 Hexane	57	3.687	3.687	(0.627)	756413	200.000	213	9320(A)
23 MTBE	73	3.716	3.716	(0.632)	1059252	200.000	220	9677(A)
24 tert-Butyl Alcohol	59	3.825	3.825	(0.650)	239814	1000.00	1130	4409(AM1)
25 Acetonitrile	41	3.947	3.947	(0.671)	147588	1000.00	1030	9518(A)
26 Isopropyl Ether	45	4.111	4.111	(0.699)	922362	200.000	215	9619(A)
27 Chloroprene	53	4.195	4.195	(0.713)	517468	200.000	214	9019(A)
28 1,1-Dichloroethane ++	63	4.217	4.217	(0.717)	691595	200.000	210	9485(A)
29 Acrylonitrile	53	4.262	4.262	(0.725)	592835	1000.00	1050	9903(A)
30 Vinyl Acetate	43	4.471	4.471	(0.760)	539371	200.000	203	5449(A)
31 Ethyl Tert-butyl Ether	59	4.471	4.471	(0.760)	1051655	200.000	217	8094(A)
32 cis-1,2-Dichloroethene	61	4.738	4.738	(0.805)	490815	200.000	207	9726(A)
M 33 Total 1,2-Dichloroethene	61				1017697	400.000	420	0
34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	605510	200.000	207	9535(A)
35 Bromochloromethane	128	4.921	4.921	(0.837)	182538	200.000	196	8720
36 Cyclohexane	56	4.928	4.928	(0.838)	644703	200.000	214	9502(A)
37 Chloroform +	83	4.995	4.995	(0.849)	716125	200.000	212	9803(A)
38 Ethyl Acetate	43	5.114	5.114	(0.869)	1626356	1000.00	1100	5994(A)
41 sec-Butanol	45	5.169	5.169	(0.879)	21713	200.000	194	4338(M2)
39 Carbon Tetrachloride	117	5.118	5.118	(0.870)	625925	200.000	230	8202(A)
40 Tetrahydrofuran	42	5.143	5.143	(0.874)	570538	1000.00	1040	9434(A)
$ 42 Dibromofluoromethane	111	5.159	5.159	(0.877)	396531	200.000	220	9082(A)
43 1,1,1-Trichloroethane	97	5.182	5.182	(0.881)	674327	200.000	216	9398(A)
44 2-Butanone	43	5.275	5.275	(0.897)	918709	1000.00	1090	5783(A)
45 1,1-Dichloropropene	75	5.291	5.291	(0.899)	582804	200.000	210	9452(A)
46 2,2,4 Trimethylpentane	57	5.400	5.400	(0.918)	1892720	200.000	226	9026(A)
47 Propionitrile	54	5.532	5.532	(0.940)	223502	1000.00	1080	8548(A)
48 Benzene	78	5.516	5.516	(0.938)	1605982	200.000	214	9719(A)
49 Methylacrylonitrile	41	5.552	5.552	(0.944)	939794	1000.00	1100	7432(A)
56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	1103297	200.000	224	7099(A)
$ 50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	385290	200.000	205	8615(A)
52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	480930	200.000	204	9772(A)
51 Isobutyl Alcohol	43	5.728	5.728	(0.974)	49197	1000.00	1000	3200(A)
* 55 FLUOROBENZENE	96	5.883	5.883	(1.000)	364826	50.0000		9873
57 Trichloroethene	130	6.034	6.034	(1.026)	533090	200.000	231	9215(A)
58 Methyl Cyclohexane	83	6.027	6.027	(1.025)	785744	200.000	223	8229(A)
59 Tert-amyl alcohol	59	6.253	6.253	(1.063)	817261	1000.00	1120	6299(A)
60 n-Butanol	56	6.333	6.333	(1.077)	29929	1000.00	999	9196
62 Dibromomethane	93	6.394	6.394	(1.087)	260479	200.000	216	9493(A)
63 2-3 Dichloro-1-Proprene	75	6.468	6.468	(1.099)	669179	200.000	222	9024(A)
64 1,2-Dichloropropane +	63	6.484	6.484	(1.102)	397850	200.000	214	8183(A)
65 Bromodichloromethane	83	6.539	6.539	(1.111)	552483	200.000	220	9781(A)
66 Methyl methacrylate	41	6.680	6.680	(1.136)	280924	200.000	219	9697(A)

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
67 1,4- Dioxane	88	6.719	6.719	(1.142)	68178	5000.00	4960	9417
68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	495032	200.000	209	9411(A)
72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	569730	1000.00	1060	9694(A)
73 cis-1,3-Dichloropropene	75	7.072	7.072	(1.202)	634947	200.000	213	9786(A)
$ 74 Toluene-d8	98	7.224	7.224	(0.866)	1440134	200.000	212	9897(A)
75 Toluene +	91	7.262	7.262	(0.871)	1820208	200.000	215	9702(A)
76 2-Nitropropane	43	7.436	7.436	(0.892)	94852	200.000	202	9563(A)
77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	1596464	1000.00	1010	8326(A)
78 Tetrachloroethene	164	7.558	7.558	(0.906)	436848	200.000	229	8108(A)
79 trans-1,3-Dichloropropene	75	7.577	7.577	(1.288)	614230	200.000	218	9168(A)
80 Ethyl Methacrylate	41	7.693	7.693	(0.922)	248686	200.000	211	7566(A)
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	367035	200.000	214	9419(A)
82 Dibromochloromethane	129	7.825	7.825	(0.938)	451726	200.000	231	9679(A)
M 83 1-3 Dichloropropene total	100				1249177	400.000	431	0
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	508123	200.000	225	9574(A)
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	589359	200.000	206	9555(A)
87 1-Nitropropane	43	7.970	7.970	(0.956)	43235	200.000	202	9621(A)
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	394224	200.000	215	9733(A)
89 2-Hexanone	43	8.143	8.143	(0.976)	1298993	1000.00	999	9648
90 1-Chlorohexane	91	8.333	8.333	(0.999)	643268	200.000	213	9631(A)
* 91 CHLOROBENZENE-d5	82	8.339	8.339	(1.000)	144815	50.0000		7544
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	1196185	200.000	214	8717(A)
93 Ethylbenzene +	106	8.368	8.368	(1.003)	659880	200.000	217	8987(A)
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	404707	200.000	215	9155(A)
95 p,m-Xylene	106	8.465	8.465	(1.015)	1599176	400.000	435	9904(A)
97 o-Xylene	106	8.744	8.744	(1.049)	737290	200.000	215	9912(A)
101 Styrene	104	8.777	8.777	(1.052)	1006134	200.000	228	9833(A)
102 Bromoform ++	173	8.799	8.799	(1.055)	332459	200.000	208	8477(A)
104 Isopropylbenzene	105	8.941	8.941	(1.072)	1888524	200.000	211	9879(A)
M 105 TOTAL XYLENE	106				2336466	600.000	650	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	459747	200.000	209	9703(A)
107 cis-1,4-dichloro-2-butene	53	9.159	9.159	(0.934)	142937	200.000	196	9533(M2)
109 n-Propylbenzene	91	9.201	9.201	(0.939)	2159345	200.000	193	8677
108 Bromobenzene	77	9.191	9.191	(0.938)	709893	200.000	190	8885
110 1,1,2,2-Tetrachloroethane++	83	9.243	9.243	(0.943)	423864	200.000	194	9821
112 2-Chlorotoluene	91	9.301	9.301	(0.949)	1403108	200.000	197	7581
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	1535851	200.000	204	8068(A)
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	569775	200.000	192	7911
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	141123	200.000	195	9319
116 Cyclohexanone	55	9.365	9.365	(0.955)	352267	1000.00	968	9703
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	1246147	200.000	194	9826
118 tert-butylbenzene	91	9.513	9.513	(0.970)	836925	200.000	198	9856
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	1547624	200.000	209	9326(A)
120 sec-Butylbenzene	105	9.619	9.619	(0.981)	1955174	200.000	201	9850(A)
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	1824169	200.000	211	9040(A)
122 Dicyclopentadiene	66	9.699	9.699	(0.990)	1776527	200.000	198	7839
123 1,3-Dichlorobenzene	146	9.760	9.760	(0.996)	904099	200.000	207	9847(A)

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.802	(1.000)	143421	50.0000		7717
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	933693	200.000	208	9349(A)
126 n-Butylbenzene	91	9.953	9.953	(1.015)	2414411	200.000	210	8894(A)
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	831202	200.000	211	9752(A)
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	114746	200.000	205	9757(A)
130 Hexachlorobutadiene	225	10.995	10.995	(1.122)	280592	200.000	233	9361(A)
131 1,2,4-Trichlorobenzene	180	11.027	11.027	(1.125)	534470	200.000	222	9588(A)
132 Naphthalene	128	11.288	11.288	(1.152)	1294831	200.000	229	9873(A)
133 1,2,3-Trichlorobenzene	180	11.442	11.442	(1.167)	470431	200.000	226	9599(A)

QC Flag Legend

A - Target compound detected but, quantitated amount exceeded maximum amount.
M1- Compound response manually integrated because Target system did not integrate.
M2- Compound response manually integrated because Target system integrated incorrectly.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0234.d

Page 1

Date : 17-MAY-2023 16:20

Client ID: V15STD200

Instrument: msv15.i

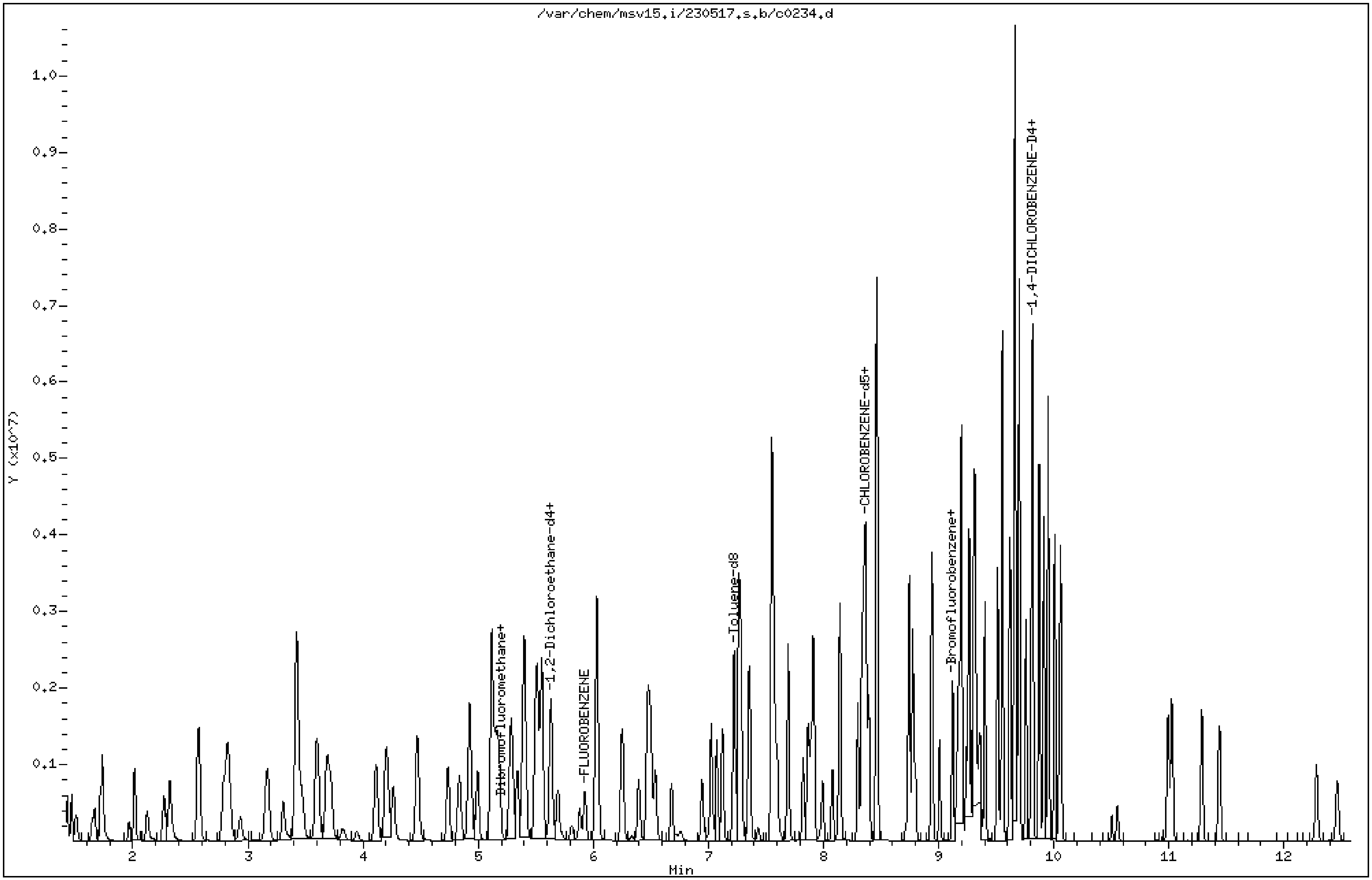
Sample Info: 1209*V15STD200

Purge Volume: 5.0

Operator: CJR

Column phase: RTX-VMS-30M

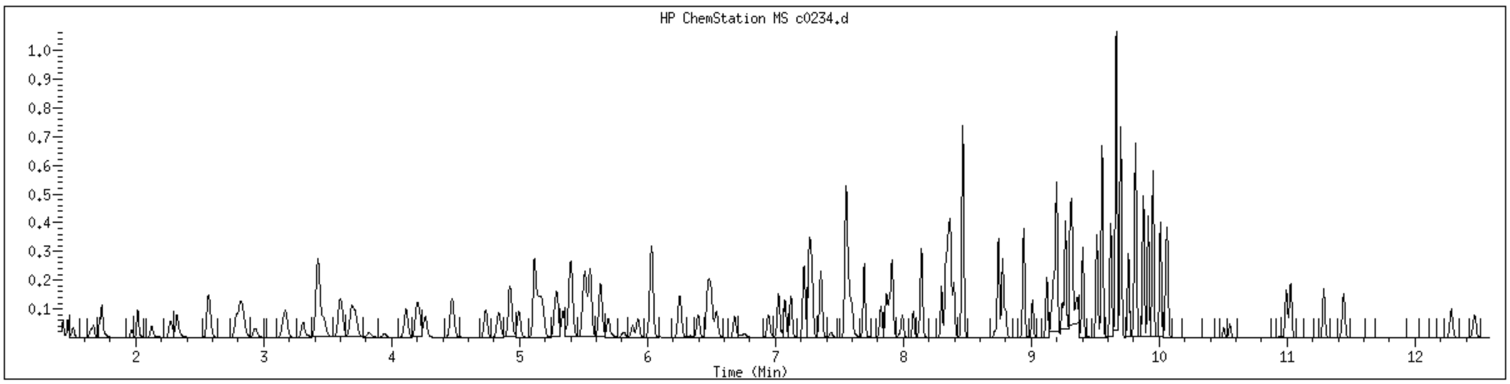
Column diameter: 0.25



Data file : /var/chem/msv15.i/230517.s.b/c0234.d   Page: 1
Report Date: 05/23/2023 16:34

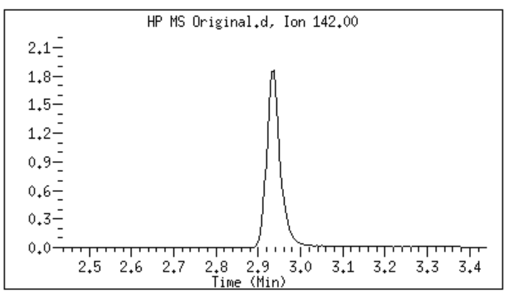
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID	: 1209	SampleType	: CALIB_9
Injection Date	: 05/17/2023 16:20	Instrument	: msv15.i
Operator	: CJR		
Sample Info	: 1209*V15STD200		
Misc Info	: MSV~527~*1*SMR		
Method	: /var/chem/msv15.i/230517.s.b/8260bw15DOD.m		
Dilution	: 1.00		
Matrix	: WATER		
Integrator	: HP RTE	Compound Sublist	: 8260b+AppIXDOD



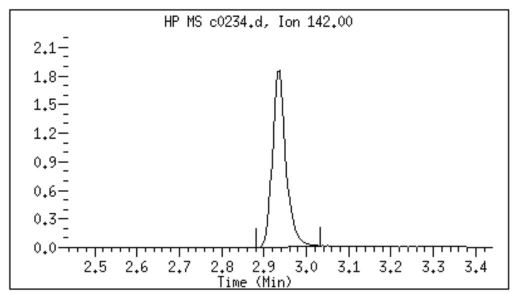
Original   Final

15 Methyl Iodide   CAS#: 74-88-4   Reason: M1

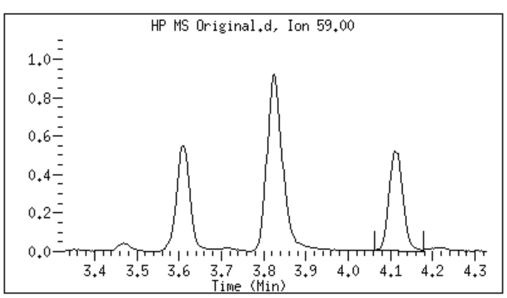


Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:34



24 tert-Butyl Alcohol   CAS#: 75-65-0   Reason: M1



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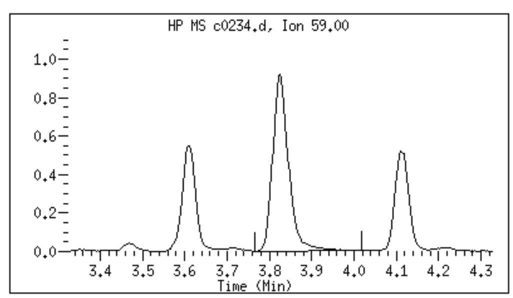
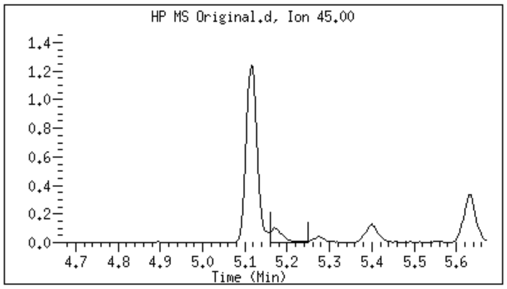
Date: 05/17/2023 16:55



Original Final

41 sec-Butanol CAS#: 78-92-2 Reason: M2



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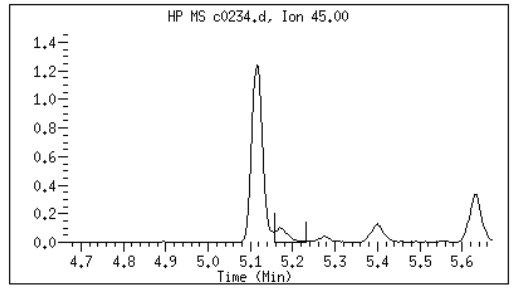
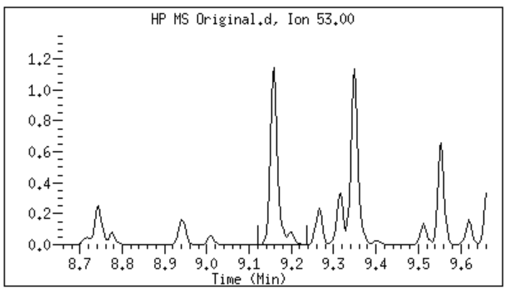
Date: 05/17/2023 16:58



107 cis-1,4-dichloro-2-butene CAS#: 1476-11-5 Reason: M2



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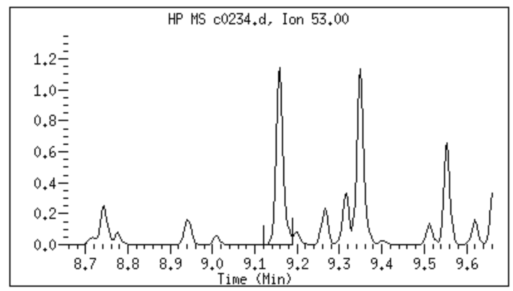
Date: 05/17/2023 17:03



M1 - Target system did not integrate
M2 - Target system integrated incorrectly

EPA 8260B DOD

Form 6I

ICAL Verifications

6I

ORGANICS INITIAL CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	MSV11
Analysis Date:	05/17/23 1727	Lab File ID:	230517/j1309
Analytical Method:	EPA 8260B DOD	Analytical Batch:	766136

ANALYTE	UNITS	TRUE	FOUND	%REC	LCL	UCL	Q
1,1,1,2-Tetrachloroethane	ug/L	50.0	49.1	98	80	120	
1,1,1-Trichloroethane	ug/L	50.0	47.7	95	80	120	
1,1,2,2-Tetrachloroethane	ug/L	50.0	46.0	92	80	120	
1,1,2-Trichloroethane	ug/L	50.0	47.0	94	80	120	
1,1-Dichloroethane	ug/L	50.0	47.5	95	80	120	
1,1-Dichloroethene	ug/L	50.0	47.1	94	80	120	
1,1-Dichloropropene	ug/L	50.0	50.1	100	80	120	
1,2,3-Trichlorobenzene	ug/L	50.0	46.3	93	80	120	
1,2,3-Trichloropropane	ug/L	50.0	45.3	91	80	120	
1,2,4-Trichlorobenzene	ug/L	50.0	48.1	96	80	120	
1,2,4-Trimethylbenzene	ug/L	50.0	50.1	100	80	120	
1,2-Dibromo-3-chloropropane	ug/L	50.0	46.8	94	80	120	
1,2-Dibromoethane	ug/L	50.0	47.1	94	80	120	
1,2-Dichlorobenzene	ug/L	50.0	45.8	92	80	120	
1,2-Dichloroethane	ug/L	50.0	46.8	94	80	120	
1,2-Dichloroethene(Total)	ug/L	100	94.9	95	80	120	
1,2-Dichloropropane	ug/L	50.0	49.6	99	80	120	
1,3,5-Trimethylbenzene	ug/L	50.0	49.7	99	80	120	
1,3-Dichlorobenzene	ug/L	50.0	45.8	92	80	120	
1,3-Dichloropropane	ug/L	50.0	47.2	94	80	120	
1,4-Dichlorobenzene	ug/L	50.0	46.0	92	80	120	
2,2-Dichloropropane	ug/L	50.0	49.2	98	80	120	
2-Butanone	ug/L	250	183	73	80	120	*
2-Chlorotoluene	ug/L	50.0	46.9	94	80	120	
2-Hexanone	ug/L	250	195	78	80	120	*
4-Chlorotoluene	ug/L	50.0	48.8	98	80	120	
4-Isopropyltoluene	ug/L	50.0	49.0	98	80	120	
4-Methyl-2-pentanone	ug/L	250	228	91	80	120	
Acetone	ug/L	250	178	71	80	120	*
Benzene	ug/L	50.0	48.2	96	80	120	
Bromobenzene	ug/L	50.0	46.8	94	80	120	
Bromochloromethane	ug/L	50.0	50.0	100	80	120	
Bromodichloromethane	ug/L	50.0	47.9	96	80	120	
Bromoform	ug/L	50.0	48.1	96	80	120	
Bromomethane	ug/L	50.0	47.8	96	80	120	
Carbon disulfide	ug/L	50.0	44.3	89	80	120	
Carbon tetrachloride	ug/L	50.0	48.7	97	80	120	
Chlorobenzene	ug/L	50.0	47.8	96	80	120	

FORM 6I - ORG

ORGANICS INITIAL CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	MSV11
Analysis Date:	05/17/23 1727	Lab File ID:	230517/j1309
Analytical Method:	EPA 8260B DOD	Analytical Batch:	766136

ANALYTE	UNITS	TRUE	FOUND	%REC	LCL	UCL	Q
Chloroethane	ug/L	50.0	45.5	91	80	120	
Chloroform	ug/L	50.0	48.9	98	80	120	
Chloromethane	ug/L	50.0	34.4	69	80	120	*
cis-1,2-Dichloroethene	ug/L	50.0	48.2	96	80	120	
cis-1,3-Dichloropropene	ug/L	50.0	50.9	102	80	120	
Dibromochloromethane	ug/L	50.0	48.7	97	80	120	
Dibromomethane	ug/L	50.0	48.9	98	80	120	
Dichlorodifluoromethane	ug/L	50.0	42.2	84	80	120	
Ethylbenzene	ug/L	50.0	49.2	98	80	120	
Hexachlorobutadiene	ug/L	50.0	49.8	100	80	120	
Isopropylbenzene (Cumene)	ug/L	50.0	50.9	102	80	120	
m,p-Xylene	ug/L	100	99.0	99	80	120	
Methylene chloride	ug/L	50.0	45.9	92	80	120	
Naphthalene	ug/L	50.0	42.0	84	80	120	
n-Butylbenzene	ug/L	50.0	49.0	98	80	120	
n-Propylbenzene	ug/L	50.0	48.9	98	80	120	
o-Xylene	ug/L	50.0	50.9	102	80	120	
sec-Butylbenzene	ug/L	50.0	49.1	98	80	120	
Styrene	ug/L	50.0	49.9	100	80	120	
tert-Butyl methyl ether (MTBE)	ug/L	50.0	49.0	98	80	120	
tert-Butylbenzene	ug/L	50.0	48.8	98	80	120	
Tetrachloroethene	ug/L	50.0	46.8	94	80	120	
Toluene	ug/L	50.0	49.0	98	80	120	
trans-1,2-Dichloroethene	ug/L	50.0	46.7	93	80	120	
trans-1,3-Dichloropropene	ug/L	50.0	50.3	101	80	120	
Trichloroethene	ug/L	50.0	47.2	94	80	120	
Trichlorofluoromethane	ug/L	50.0	44.9	90	80	120	
Vinyl chloride	ug/L	50.0	43.1	86	80	120	

GCAL, Inc.

Data file : /var/chem/msv11.i/230517.s.b/j1309.d
Lab Smp Id: 1600 Client Smp ID: V11ICV051
Inj Date : 17-MAY-2023 17:27
Operator : KN1 Inst ID: msv11.i
Smp Info : 1600*V11ICV051
Misc Info : MSV~528~*1*SMR
Comment :
Method : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Meth Date : 23-May-2023 17:15 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 13:15 Cal File: j1302.d
Als bottle: 1 QC Sample: LCS
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
1 Dichlorodifluoromethane	85	1.673	1.667	(0.257)	106779	42.2345	42.2	
2 Chloromethane ++	50	1.861	1.856	(0.286)	152700	34.3964	34.4	(R)
3 Vinyl Chloride +	62	1.940	1.935	(0.298)	141503	43.0647	43.1	
4 1-3 Butadiene	39	1.956	1.956	(0.301)	152045	42.2005	42.2	9641
5 Bromomethane	94	2.255	2.239	(0.346)	64575	47.8343	47.8	
7 Chloroethane	64	2.375	2.344	(0.365)	83398	45.5157	45.5	
8 Trichlorofluoromethane	101	2.527	2.501	(0.388)	162392	44.9069	44.9	
9 Isoprene	67	2.842	2.826	(0.437)	157551	47.6516	47.7	9324
10 Ethyl Ether	59	2.857	2.852	(0.439)	119637	47.0825	47.1	6179
11 1,1-Dichloroethene +	96	3.072	3.062	(0.472)	88002	47.1062	47.1	
12 Carbon Disulfide	76	3.109	3.094	(0.478)	251939	44.2733	44.3	
13 1,1,2Trichlotrifluoroethane	101	3.120	3.109	(0.479)	96136	49.3780	49.4	
15 Methyl Iodide	142	3.235	3.219	(0.497)	91395	39.6138	39.6	(R)
16 Acrolein	56	3.455	3.450	(0.531)	146692	234.261	234	

						CONCENTRATIONS		
	QUANT SIG					ON-COLUMN	FINAL	
Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ug/L)	SIMILARITY
17 Allyl chloride	39	3.633	3.628	(0.558)	152681	46.9710	47.0	9397
18 Methylene Chloride	49	3.765	3.754	(0.579)	214684	45.9178	45.9	
19 Acetone	43	3.817	3.812	(0.587)	499787	178.162	178	(R)
20 trans-1,2-Dichloroethene	61	3.959	3.953	(0.608)	195351	46.6919	46.7	
21 Methyl Acetate	43	3.979	3.974	(0.612)	769979	253.324	253	9617
22 Hexane	57	4.074	4.064	(0.626)	316293	51.1774	51.2	9315
23 MTBE	73	4.105	4.100	(0.631)	323790	48.9796	49.0	9062
24 tert-Butyl Alcohol	59	4.231	4.237	(0.650)	70086	170.084	170	8865(R)
25 Acetonitrile	41	4.341	4.341	(0.667)	119463	220.623	221	9764
26 Isopropyl Ether	45	4.546	4.546	(0.699)	544615	48.6122	48.6	9658
27 Chloroprene	53	4.640	4.630	(0.713)	226352	47.1581	47.2	9293
28 1,1-Dichloroethane ++	63	4.661	4.651	(0.716)	251533	47.4807	47.5	
29 Acrylonitrile	53	4.698	4.698	(0.722)	349253	241.670	242	
31 Vinyl Acetate	43	4.944	4.939	(0.760)	271812	42.5823	42.6	
30 Ethyl Tert-butyl Ether	59	4.949	4.950	(0.761)	444919	48.4549	48.5	7929
M 33 Total 1,2-Dichloroethene	61				385664	94.8958	94.9	
32 cis-1,2-Dichloroethene	61	5.243	5.238	(0.806)	190313	48.2038	48.2	
34 2,2-Dichloropropane	77	5.348	5.348	(0.822)	164377	49.1975	49.2	
35 Bromochloromethane	128	5.448	5.442	(0.837)	58161	49.9753	50.0	
36 Cyclohexane	56	5.458	5.453	(0.839)	262018	51.5672	51.6	8827
37 Chloroform +	83	5.531	5.526	(0.850)	194801	48.9346	48.9	
39 Ethyl Acetate	43	5.657	5.657	(0.869)	881394	209.776	210	5528
38 Carbon Tetrachloride	117	5.668	5.668	(0.871)	154799	48.7404	48.7	
40 Tetrahydrofuran	42	5.689	5.684	(0.874)	335413	206.581	207	7375
$ 41 Dibromofluoromethane	111	5.715	5.710	(0.878)	108523	51.4489	51.4	8943
42 1,1,1-Trichloroethane	97	5.736	5.731	(0.882)	172497	47.7426	47.7	
43 sec-Butanol	45	5.731	5.741	(0.881)	13076	37.1795	37.2	9310(R)
44 2-Butanone	43	5.830	5.825	(0.896)	452160	183.019	183	(R)
45 1,1-Dichloropropene	75	5.862	5.857	(0.901)	151887	50.1068	50.1	
47 2,2,4 Trimethylpentane	57	5.982	5.972	(0.919)	439829	34.6615	34.7	9788(R)
48 Benzene	78	6.103	6.098	(0.938)	429967	48.1705	48.2	
49 Propionitrile	54	6.103	6.108	(0.938)	129160	221.266	221	7481
50 Methylacrylonitrile	41	6.129	6.135	(0.942)	592836	229.216	229	9531
$ 51 1,2-Dichloroethane-d4	65	6.229	6.229	(0.957)	138339	49.1134	49.1	
54 tert-amyl Methyl Ether	73	6.229	6.234	(0.957)	310698	48.3231	48.3	7177
52 1,2-Dichloroethane	62	6.297	6.297	(0.968)	178173	46.8068	46.8	
53 Isobutyl Alcohol	43	6.339	6.350	(0.974)	35122	215.446	215	9235
* 56 FLUOROBENZENE	96	6.507	6.507	(1.000)	423072	50.0000		
57 Methyl Cyclohexane	83	6.674	6.669	(1.026)	171915	51.8996	51.9	9418
58 Trichloroethene	130	6.680	6.680	(1.027)	121047	47.2412	47.2	
59 n-Butanol	56	7.010	7.015	(1.077)	16643	197.677	198	9074(R)
60 Dibromomethane	93	7.073	7.068	(1.087)	72009	48.9474	48.9	
61 2-3 Dichloro-1-Proprene	75	7.152	7.152	(1.099)	182918	49.2525	49.3	9526
62 1,2-Dichloropropane +	63	7.162	7.162	(1.101)	154188	49.5693	49.6	(M1)
63 Bromodichloromethane	83	7.235	7.230	(1.112)	143935	47.8879	47.9	
64 Methyl methacrylate	41	7.382	7.382	(1.135)	158036	49.0281	49.0	9474
65 1,4- Dioxane	58	7.419	7.419	(1.140)	29063	1148.92	1150	8751

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
66 1-Bromo-2-chloroethane	63	7.671	7.671	(1.179)	195213	47.9302	47.9	9358
67 2-Chloroethyl vinyl ether	63	7.760	7.760	(1.193)	91977	243.623	244	
68 cis-1,3-Dichloropropene	75	7.817	7.812	(1.201)	177350	50.9358	50.9	
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	398291	50.3904	50.4	
70 Toluene +	91	8.027	8.027	(0.869)	465047	49.0258	49.0	
71 2-Nitropropane	43	8.205	8.206	(0.888)	47186	44.4363	44.4	9088
M 72 1-3 Dichloropropene total	100				341296	101.196	101	0
74 4-methyl-2-pentanone	43	8.336	8.342	(0.902)	841451	227.628	228	
73 Tetrachloroethene	164	8.363	8.363	(0.905)	89989	46.8237	46.8	
75 trans-1,3-Dichloropropene	75	8.378	8.379	(1.288)	163946	50.2603	50.3	
77 Ethyl Methacrylate	41	8.504	8.504	(0.921)	139651	48.3655	48.4	9398
78 1,1,2-Trichloroethane	97	8.509	8.510	(0.921)	100887	47.0048	47.0	
79 Dibromochloromethane	129	8.656	8.656	(0.937)	112941	48.6923	48.7	
76 3,4-dichloro-1-butene	75	8.698	8.704	(0.942)	139451	50.0702	50.1	9306
80 1,3-Dichloropropane	76	8.730	8.730	(0.945)	171109	47.2287	47.2	
81 1-Nitropropane	43	8.814	8.814	(0.954)	26897	46.7237	46.7	7527
82 1,2-Dibromoethane(EDB)	107	8.845	8.845	(0.957)	104017	47.0865	47.1	
83 2-Hexanone	43	9.008	9.008	(0.975)	601429	195.196	195	(R)
89 1-Chlorohexane	91	9.233	9.233	(0.999)	135475	50.2826	50.3	9429
* 90 CHLOROBENZENE-d5	82	9.238	9.238	(1.000)	165122	50.0000		
91 Chlorobenzene ++	112	9.254	9.254	(1.002)	293996	47.8098	47.8	
92 Ethylbenzene +	106	9.270	9.270	(1.003)	159879	49.2247	49.2	
93 1,1,1,2-Tetrachloroethane	133	9.301	9.301	(1.007)	103106	49.0715	49.1	
94 p,m-Xylene	106	9.380	9.380	(1.015)	386401	99.0288	99.0	
M 95 TOTAL XYLENE	106				570781	149.937	150	
96 o-Xylene	106	9.705	9.700	(1.050)	184380	50.9083	50.9	
97 Styrene	104	9.742	9.742	(1.054)	257259	49.8988	49.9	
98 Bromoform ++	173	9.768	9.768	(1.057)	73375	48.0858	48.1	
99 Isopropylbenzene	105	9.930	9.931	(1.075)	459513	50.9258	50.9	
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	120885	50.7323	50.7	
102 cis-1,4-dichloro-2-butene	53	10.177	10.177	(0.931)	59597	44.0600	44.1	9148
103 Bromobenzene	77	10.219	10.214	(0.935)	203578	46.8012	46.8	
104 n-Propylbenzene	91	10.229	10.229	(0.936)	528878	48.8706	48.9	
105 1,1,2,2-Tetrachloroethane++	83	10.276	10.277	(0.940)	128888	45.9742	46.0	
106 2-Chlorotoluene	91	10.345	10.345	(0.947)	357692	46.9229	46.9	
107 1,3,5-Trimethylbenzene	105	10.360	10.360	(0.948)	402659	49.6677	49.7	
108 1,2,3-Trichloropropane	75	10.376	10.376	(0.950)	165159	45.3019	45.3	
109 trans-1,4-Dichloro-2-Butene	53	10.397	10.397	(0.952)	56009	42.8949	42.9	
100 Cyclohexanone	55	10.402	10.408	(0.952)	179401	188.036	188	5978(R)
110 4-Chlorotoluene	91	10.460	10.460	(0.957)	353787	48.7843	48.8	
111 tert-butylbenzene	91	10.591	10.591	(0.969)	215996	48.8270	48.8	
112 1,2,4-Trimethylbenzene	105	10.638	10.638	(0.974)	407991	50.0708	50.1	
113 sec-Butylbenzene	105	10.717	10.717	(0.981)	461517	49.1198	49.1	
114 Dicyclopentadiene	66	10.806	10.806	(0.989)	521074	49.8812	49.9	8738
115 p-Isopropyltoluene	119	10.811	10.811	(0.989)	426520	48.9624	49.0	
116 1,3-Dichlorobenzene	146	10.879	10.880	(0.996)	228725	45.8099	45.8	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	157311	50.0000		

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
118 1,4-Dichlorobenzene	146	10.937	10.937	(1.001)	231594	46.0423	46.0	
119 n-Butylbenzene	91	11.100	11.100	(1.016)	616160	48.9988	49.0	
120 1,2-Dichlorobenzene	146	11.231	11.231	(1.028)	221421	45.8087	45.8	
122 1,2-Dibromo-3-Chloropropane	157	11.771	11.771	(1.077)	31288	46.7957	46.8	
123 Hexachlorobutadiene	225	12.201	12.201	(1.117)	66820	49.8005	49.8	
124 1,2,4-Trichlorobenzene	180	12.227	12.227	(1.119)	142941	48.0515	48.1	
125 Naphthalene	128	12.447	12.447	(1.139)	372332	42.0403	42.0	
126 1,2,3-Trichlorobenzene	180	12.573	12.573	(1.151)	133699	46.2607	46.3	

QC Flag Legend

R - Spike/Surrogate failed recovery limits.
M1- Compound response manually integrated because
Target system did not integrate.

Data File: /var/chem/msv11.i/230517.s.b/j1309.d

Page 1

Date : 17-MAY-2023 17:27

Client ID: V11ICV051

Instrument: msv11.i

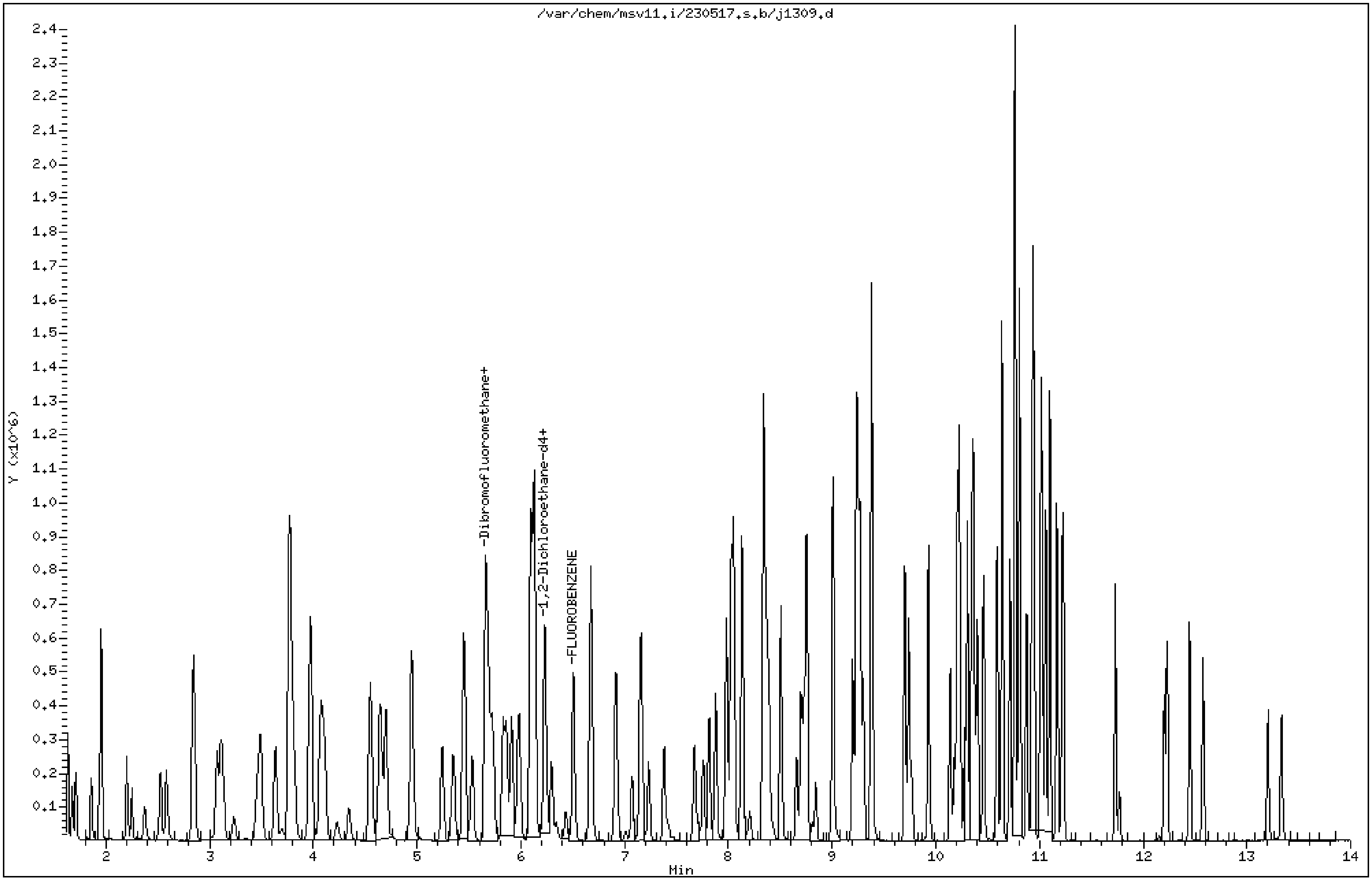
Sample Info: 1600*V11ICV051

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

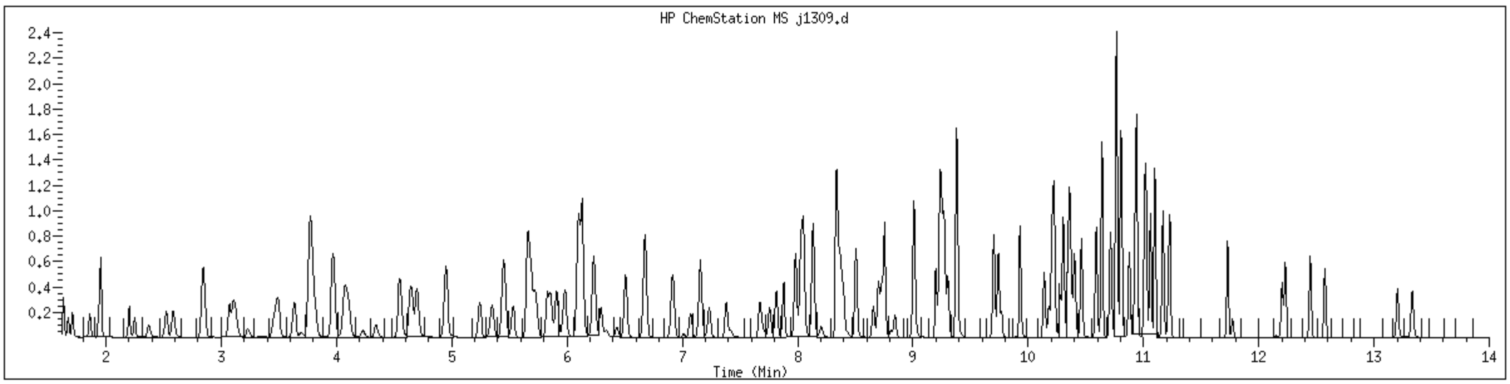
Column diameter: 0.25



Data file : /var/chem/msv11.i/230517.s.b/j1309.d Page: 1
Report Date: 05/23/2023 17:15

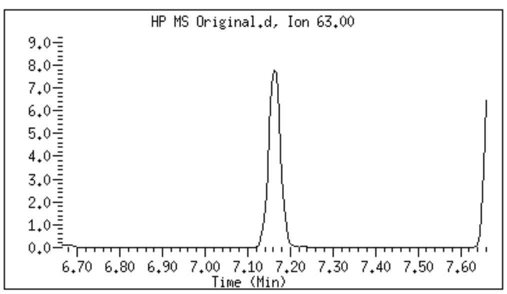
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1600 SampleType : LCS
Injection Date: 05/17/2023 17:27 Instrument : msv11.i
Operator : KN1
Sample Info : 1600*V11ICV051
Misc Info : MSV~528~*1*SMR
Method : /var/chem/msv11.i/230517.s.b/8260bDODw11.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260b+AppIXDOD



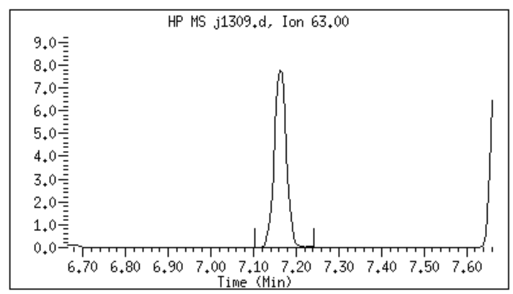
Original Final

62 1,2-Dichloropropane + CAS#: 78-87-5 Reason: M1



Electronic Signature Applied

User: kmn
Date: 05/17/2023 17:49



M1 - Target system did not integrate

6I

ORGANICS INITIAL CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	MSV15
Analysis Date:	05/17/23 1735	Lab File ID:	230517/c0238
Analytical Method:	EPA 8260B DOD	Analytical Batch:	766139

ANALYTE	UNITS	TRUE	FOUND	%REC	LCL	UCL	Q
1,1,1,2-Tetrachloroethane	ug/L	50.0	48.4	97	80	120	
1,1,1-Trichloroethane	ug/L	50.0	48.9	98	80	120	
1,1,2,2-Tetrachloroethane	ug/L	50.0	49.7	99	80	120	
1,1,2-Trichloroethane	ug/L	50.0	48.6	97	80	120	
1,1-Dichloroethane	ug/L	50.0	48.1	96	80	120	
1,1-Dichloroethene	ug/L	50.0	48.1	96	80	120	
1,1-Dichloropropene	ug/L	50.0	47.7	95	80	120	
1,2,3-Trichlorobenzene	ug/L	50.0	52.0	104	80	120	
1,2,3-Trichloropropane	ug/L	50.0	47.8	96	80	120	
1,2,4-Trichlorobenzene	ug/L	50.0	51.4	103	80	120	
1,2,4-Trimethylbenzene	ug/L	50.0	48.8	98	80	120	
1,2-Dibromo-3-chloropropane	ug/L	50.0	48.4	97	80	120	
1,2-Dibromoethane	ug/L	50.0	50.3	101	80	120	
1,2-Dichlorobenzene	ug/L	50.0	48.9	98	80	120	
1,2-Dichloroethane	ug/L	50.0	47.4	95	80	120	
1,2-Dichloroethene(Total)	ug/L	100	95.4	95	80	120	
1,2-Dichloropropane	ug/L	50.0	48.7	97	80	120	
1,3,5-Trimethylbenzene	ug/L	50.0	48.0	96	80	120	
1,3-Dichlorobenzene	ug/L	50.0	48.0	96	80	120	
1,3-Dichloropropane	ug/L	50.0	46.9	94	80	120	
1,4-Dichlorobenzene	ug/L	50.0	48.3	97	80	120	
2,2-Dichloropropane	ug/L	50.0	49.8	100	80	120	
2-Butanone	ug/L	250	220	88	80	120	
2-Chlorotoluene	ug/L	50.0	47.3	95	80	120	
2-Hexanone	ug/L	250	215	86	80	120	
4-Chlorotoluene	ug/L	50.0	47.5	95	80	120	
4-Isopropyltoluene	ug/L	50.0	49.3	99	80	120	
4-Methyl-2-pentanone	ug/L	250	233	93	80	120	
Acetone	ug/L	250	189	76	80	120	*
Benzene	ug/L	50.0	47.7	95	80	120	
Bromobenzene	ug/L	50.0	49.0	98	80	120	
Bromochloromethane	ug/L	50.0	48.3	97	80	120	
Bromodichloromethane	ug/L	50.0	49.5	99	80	120	
Bromoform	ug/L	50.0	45.3	91	80	120	
Bromomethane	ug/L	50.0	52.7	105	80	120	
Carbon disulfide	ug/L	50.0	44.2	88	80	120	
Carbon tetrachloride	ug/L	50.0	48.3	97	80	120	
Chlorobenzene	ug/L	50.0	49.2	98	80	120	

FORM 6I - ORG

6I

ORGANICS INITIAL CALIBRATION VERIFICATION

Report No:	223060210	Instrument ID:	MSV15
Analysis Date:	05/17/23 1735	Lab File ID:	230517/c0238
Analytical Method:	EPA 8260B DOD	Analytical Batch:	766139

ANALYTE	UNITS	TRUE	FOUND	%REC	LCL	UCL	Q
Chloroethane	ug/L	50.0	48.0	96	80	120	
Chloroform	ug/L	50.0	48.4	97	80	120	
Chloromethane	ug/L	50.0	39.6	79	80	120	*
cis-1,2-Dichloroethene	ug/L	50.0	47.6	95	80	120	
cis-1,3-Dichloropropene	ug/L	50.0	49.1	98	80	120	
Dibromochloromethane	ug/L	50.0	51.6	103	80	120	
Dibromomethane	ug/L	50.0	50.2	100	80	120	
Dichlorodifluoromethane	ug/L	50.0	45.2	90	80	120	
Ethylbenzene	ug/L	50.0	49.7	99	80	120	
Hexachlorobutadiene	ug/L	50.0	50.8	102	80	120	
Isopropylbenzene (Cumene)	ug/L	50.0	50.0	100	80	120	
m,p-Xylene	ug/L	100	99.2	99	80	120	
Methylene chloride	ug/L	50.0	47.3	95	80	120	
Naphthalene	ug/L	50.0	52.4	105	80	120	
n-Butylbenzene	ug/L	50.0	50.0	100	80	120	
n-Propylbenzene	ug/L	50.0	47.4	95	80	120	
o-Xylene	ug/L	50.0	49.9	100	80	120	
sec-Butylbenzene	ug/L	50.0	48.1	96	80	120	
Styrene	ug/L	50.0	51.1	102	80	120	
tert-Butyl methyl ether (MTBE)	ug/L	50.0	51.1	102	80	120	
tert-Butylbenzene	ug/L	50.0	48.0	96	80	120	
Tetrachloroethene	ug/L	50.0	48.9	98	80	120	
Toluene	ug/L	50.0	48.2	96	80	120	
trans-1,2-Dichloroethene	ug/L	50.0	47.7	95	80	120	
trans-1,3-Dichloropropene	ug/L	50.0	49.9	100	80	120	
Trichloroethene	ug/L	50.0	48.2	96	80	120	
Trichlorofluoromethane	ug/L	50.0	48.7	97	80	120	
Vinyl chloride	ug/L	50.0	44.7	89	80	120	

FORM 6I - ORG

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0238.d
Lab Smp Id: 1600   Client Smp ID: V15ICV050
Inj Date : 17-MAY-2023 17:35
Operator : CJR   Inst ID: msv15.i
Smp Info : 1600*V15ICV050
Misc Info : MSV~527~*1*SMR
Comment   :
Method   : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr   Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20   Cal File: c0234.d
Als bottle: 1   QC Sample: LCS
Dil Factor: 1.00000
Integrator: HP RTE   Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable   Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	CONCENTRATIONS FINAL (ug/L)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	77603	45.1510	45.2	9645
2 Chloromethane ++	50	1.651	1.651	(0.281)	61323	39.6168	39.6	9769(R)
3 Vinyl Chloride +	62	1.719	1.716	(0.292)	97449	44.7088	44.7	9748
5 1-3 Butadiene	39	1.735	1.735	(0.295)	73091	45.0296	45.0	9710
6 Bromomethane	94	2.018	2.015	(0.343)	96342	52.7085	52.7	9662
7 Chloroethane	64	2.134	2.124	(0.363)	76883	47.9874	48.0	9218
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	154711	48.6924	48.7	9629
9 Isoprene	67	2.568	2.568	(0.436)	127374	48.4794	48.5	4543
10 Ethyl Ether	59	2.584	2.581	(0.439)	62644	50.3212	50.3	8119
11 1,1-Dichloroethene +	96	2.783	2.786	(0.473)	87031	48.1441	48.1	8240
12 Carbon Disulfide	76	2.815	2.812	(0.479)	244627	44.2342	44.2	8010
13 1,1,2Trichlotrifluoroethane	101	2.835	2.831	(0.482)	94362	48.1020	48.1	9333
15 Methyl Iodide	142	2.934	2.938	(0.499)	92242	42.8982	42.9	9558
16 Acrolein	56	3.153	3.150	(0.536)	73042	265.343	265	5651

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
	17 Allyl chloride	39	3.304	3.304	(0.562)	78905	49.6600	49.7	9653
	18 Methylene Chloride	49	3.420	3.417	(0.581)	94560	47.2694	47.3	4846
	19 Acetone	43	3.471	3.471	(0.590)	154872	188.623	189	5308(RH)
	20 trans-1,2-Dichloroethene	61	3.590	3.593	(0.610)	119659	47.7467	47.7	8663
	21 Methyl Acetate	43	3.613	3.613	(0.614)	289070	282.801	283	9378
	22 Hexane	57	3.690	3.687	(0.627)	178006	49.3001	49.3	9241
	23 MTBE	73	3.719	3.716	(0.632)	249745	51.0961	51.1	9736
	24 tert-Butyl Alcohol	59	3.828	3.825	(0.651)	38701	179.496	179	7982(R)
	25 Acetonitrile	41	3.944	3.947	(0.670)	39150	269.318	269	9584
	26 Isopropyl Ether	45	4.111	4.111	(0.699)	212742	48.7834	48.8	9674
	27 Chloroprene	53	4.198	4.195	(0.714)	117752	47.9358	47.9	8544
	28 1,1-Dichloroethane ++	63	4.217	4.217	(0.717)	161086	48.1359	48.1	9450
	29 Acrylonitrile	53	4.262	4.262	(0.725)	156392	274.065	274	9847
	30 Vinyl Acetate	43	4.475	4.471	(0.761)	119847	44.4086	44.4	5352
	31 Ethyl Tert-butyl Ether	59	4.475	4.471	(0.761)	244877	49.8928	49.9	8154
	32 cis-1,2-Dichloroethene	61	4.738	4.738	(0.805)	114542	47.6100	47.6	9765
M	33 Total 1,2-Dichloroethene	61				234201	95.3567	95.4	0
	34 2,2-Dichloropropane	77	4.835	4.838	(0.822)	148076	49.7823	49.8	9494
	35 Bromochloromethane	128	4.921	4.921	(0.837)	45598	48.2758	48.3	9214
	36 Cyclohexane	56	4.928	4.928	(0.838)	145283	47.6088	47.6	9276
	37 Chloroform +	83	4.995	4.995	(0.849)	165566	48.3641	48.4	9818
	38 Ethyl Acetate	43	5.114	5.114	(0.869)	365177	244.082	244	5844
	41 sec-Butanol	45	5.169	5.169	(0.879)	6061	53.4915	53.5	4661
	39 Carbon Tetrachloride	117	5.121	5.118	(0.870)	133664	48.3000	48.3	8663
	40 Tetrahydrofuran	42	5.147	5.143	(0.875)	132874	238.436	238	9668
$	42 Dibromofluoromethane	111	5.159	5.159	(0.877)	95490	52.2260	52.2	8503
	43 1,1,1-Trichloroethane	97	5.179	5.182	(0.880)	154965	48.8826	48.9	9702
	44 2-Butanone	43	5.275	5.275	(0.897)	187809	220.376	220	6124
	45 1,1-Dichloropropene	75	5.291	5.291	(0.899)	134442	47.7008	47.7	9415
	46 2,2,4 Trimethylpentane	57	5.401	5.400	(0.918)	271077	31.8473	31.8	9752(R)
	47 Propionitrile	54	5.532	5.532	(0.940)	56466	268.203	268	8644
	48 Benzene	78	5.516	5.516	(0.938)	363251	47.6822	47.7	9659
	49 Methylacrylonitrile	41	5.552	5.552	(0.944)	218562	252.818	253	7411
	56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	253152	50.6170	50.6	7287
$	50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	95934	50.3913	50.4	8616
	52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	113628	47.4145	47.4	9758
	51 Isobutyl Alcohol	43	5.725	5.728	(0.973)	12910	258.954	259	3960
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	370239	50.0000		9869
	57 Trichloroethene	130	6.034	6.034	(1.026)	112727	48.1794	48.2	9113
	58 Methyl Cyclohexane	83	6.028	6.027	(1.025)	175444	49.0110	49.0	8381
	59 Tert-amyl alcohol	59	6.253	6.253	(1.063)	185786	250.439	250	6416
	60 n-Butanol	56	6.330	6.333	(1.076)	8397	282.755	283	9117
	62 Dibromomethane	93	6.397	6.394	(1.087)	61518	50.1831	50.2	9433
	63 2-3 Dichloro-1-Proprene	75	6.468	6.468	(1.099)	151981	49.5895	49.6	8925
	64 1,2-Dichloropropane +	63	6.484	6.484	(1.102)	91663	48.6830	48.7	8206
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	126290	49.5178	49.5	9738
	66 Methyl methacrylate	41	6.680	6.680	(1.136)	64193	49.2674	49.3	9691

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
67 1,4- Dioxane	88	6.722	6.719	(1.143)	19360	1386.71	1390	9562
68 1-Bromo-2-chloroethane	63	6.947	6.944	(1.181)	116609	48.6121	48.6	9204
72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	112521	210.472	210	9677
73 cis-1,3-Dichloropropene	75	7.072	7.072	(1.202)	148184	49.0749	49.1	9747
$ 74 Toluene-d8	98	7.224	7.224	(0.866)	350079	50.9102	50.9	9881
75 Toluene +	91	7.262	7.262	(0.871)	412375	48.1671	48.2	9649
76 2-Nitropropane	43	7.433	7.436	(0.891)	22691	48.1291	48.1	9503
77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	369543	232.524	233	8204
78 Tetrachloroethene	164	7.558	7.558	(0.906)	94581	48.8642	48.9	8070
79 trans-1,3-Dichloropropene	75	7.574	7.577	(1.287)	142850	49.9121	49.9	8713
80 Ethyl Methacrylate	41	7.693	7.693	(0.922)	56383	47.1315	47.1	7547
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	84497	48.5919	48.6	9434
82 Dibromochloromethane	129	7.822	7.825	(0.938)	102055	51.5977	51.6	9577
M 83 1-3 Dichloropropene total	100				291034	98.9870	99.0	0
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	115834	50.6589	50.7	9630
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	135904	46.9398	46.9	9532
87 1-Nitropropane	43	7.970	7.970	(0.956)	12569	58.0825	58.1	9477
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	93654	50.3325	50.3	9662
89 2-Hexanone	43	8.143	8.143	(0.976)	280409	214.833	215	9661
90 1-Chlorohexane	91	8.333	8.333	(0.999)	145965	47.6929	47.7	7375
* 91 CHLOROBENZENE-d5	82	8.339	8.339	(1.000)	146725	50.0000		9164
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	278963	49.2335	49.2	9515
93 Ethylbenzene +	106	8.368	8.368	(1.003)	153222	49.7336	49.7	9003
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	92203	48.4290	48.4	9195
95 p,m-Xylene	106	8.462	8.465	(1.015)	369416	99.1790	99.2	9876
97 o-Xylene	106	8.741	8.744	(1.048)	173049	49.8921	49.9	9904
101 Styrene	104	8.777	8.777	(1.052)	228037	51.0600	51.1	9848
102 Bromoform ++	173	8.799	8.799	(1.055)	72295	45.3428	45.3	8561
104 Isopropylbenzene	105	8.941	8.941	(1.072)	453954	50.0014	50.0	9864
M 105 TOTAL XYLENE	106				542465	149.071	149	0
$ 106 Bromofluorobenzene	174	9.121	9.124	(1.094)	112552	50.3992	50.4	9478
107 cis-1,4-dichloro-2-butene	53	9.156	9.159	(0.934)	35844	49.5760	49.6	9651
109 n-Propylbenzene	91	9.198	9.201	(0.938)	527111	47.4110	47.4	9138
108 Bromobenzene	77	9.191	9.191	(0.938)	181205	48.9742	49.0	9236
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.243	(0.943)	108001	49.7234	49.7	9716
112 2-Chlorotoluene	91	9.301	9.301	(0.949)	335416	47.3292	47.3	7811
113 1,3,5-Trimethylbenzene	105	9.314	9.317	(0.950)	359160	48.0291	48.0	8699
114 1,2,3-Trichloropropane	75	9.326	9.330	(0.951)	140772	47.7607	47.8	6907
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	34815	48.5408	48.5	9207
116 Cyclohexanone	55	9.365	9.365	(0.955)	84106	233.849	234	9735
117 4-Chlorotoluene	91	9.400	9.404	(0.959)	301767	47.4503	47.5	9763
118 tert-butylbenzene	91	9.513	9.513	(0.970)	201507	47.9946	48.0	9862
119 1,2,4-Trimethylbenzene	105	9.552	9.555	(0.974)	358167	48.7993	48.8	9381
120 sec-Butylbenzene	105	9.619	9.619	(0.981)	465691	48.1419	48.1	9891
121 p-Isopropyltoluene	119	9.699	9.703	(0.990)	422559	49.3442	49.3	8922
122 Dicyclopentadiene	66	9.699	9.699	(0.990)	421000	47.3238	47.3	7799
123 1,3-Dichlorobenzene	146	9.761	9.760	(0.996)	207739	48.0105	48.0	9805

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.802	(1.000)	142360	50.0000		9109
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	214779	48.2611	48.3	8952
126 n-Butylbenzene	91	9.953	9.953	(1.015)	569717	50.0344	50.0	8827
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	191245	48.8574	48.9	9694
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	26530	48.4000	48.4	9538
130 Hexachlorobutadiene	225	10.995	10.995	(1.122)	60750	50.7686	50.8	9257
131 1,2,4-Trichlorobenzene	180	11.027	11.027	(1.125)	122925	51.3747	51.4	9528
132 Naphthalene	128	11.285	11.288	(1.151)	293974	52.4313	52.4	9820
133 1,2,3-Trichlorobenzene	180	11.442	11.442	(1.167)	107422	52.0474	52.0	9471

QC Flag Legend

R - Spike/Surrogate failed recovery limits.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0238.d

Page 1

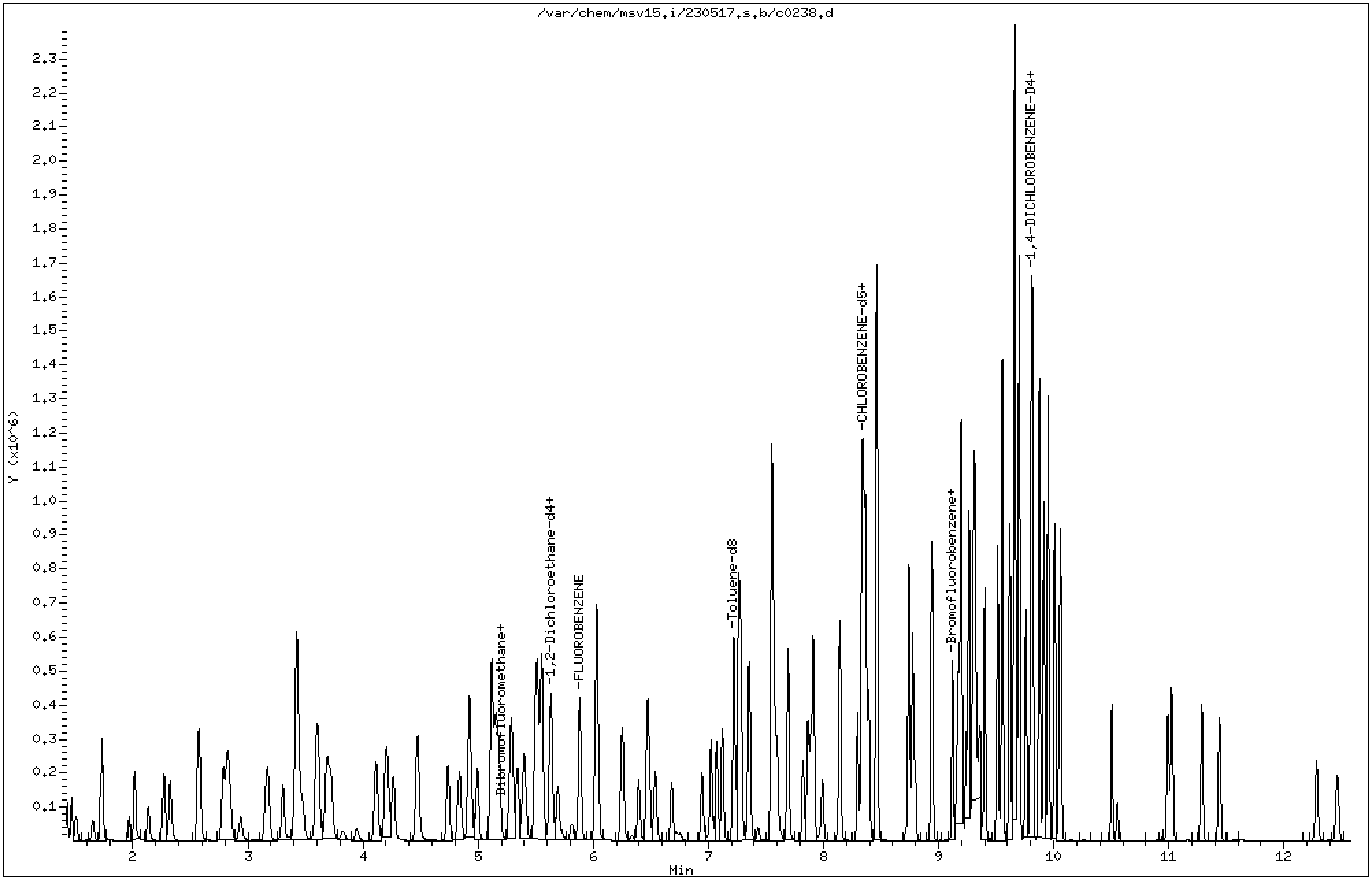
Date : 17-MAY-2023 17:35

Client ID: V15ICV050 | Instrument: msv15.i

Sample Info: 1600*V15ICV050

Purge Volume: 5.0 | Operator: CJR

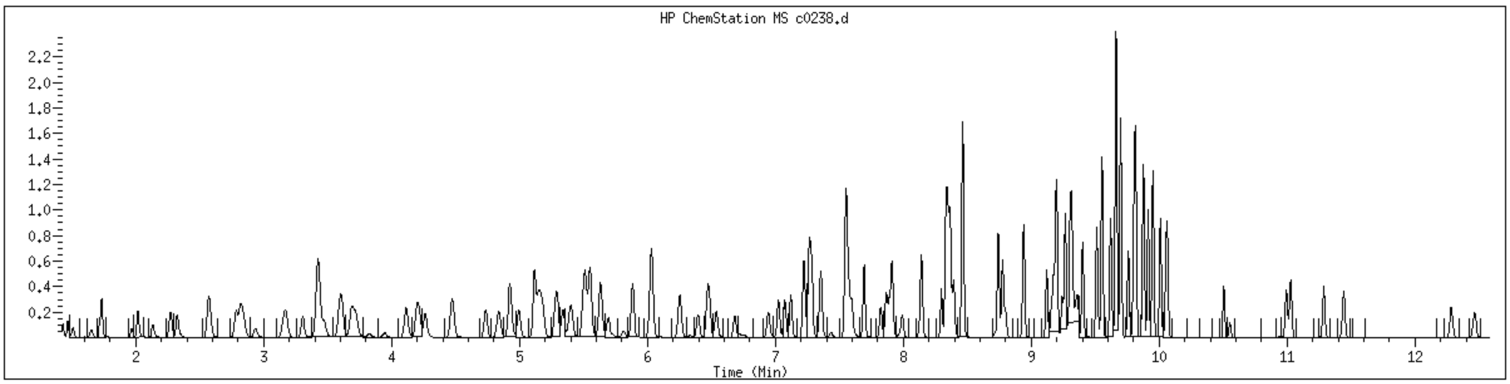
Column phase: RTX-VMS-30M | Column diameter: 0.25



Data file : /var/chem/msv15.i/230517.s.b/c0238.d   Page: 1
Report Date: 05/23/2023 16:34

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 1600   SampleType   : LCS
Injection Date: 05/17/2023 17:35   Instrument   : msv15.i
Operator   : CJR
Sample Info   : 1600*V15ICV050
Misc Info   : MSV~527~*1*SMR
Method   : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260b+AppIXDOD



NO MANUAL INTEGRATIONS

EPA 8260B DOD

Form 7A

CCAL Verifications

7A
VOLATILE CONTINUING CALIBRATION CHECK

Report No:	223060210			CCAL ID:	1400
GC Column:	RTX-VMS-30M	ID .25	(mm)	Instrument ID:	MSV11
Injection Vol.:	1.0		(µL)	Lab File ID:	230602/j1687
Init. Calib. Date 1:	05/17/23	Time 1:	1206	Analyst:	KN1
Init. Calib. Date 2:	05/17/23	Time 2:	1502	Analytical Batch:	766942
Analysis Date:	06/02/23	Time:	1123	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
1,1,1,2-Tetrachloroethane	50.0	50.2	0.636	0.639	.01	.42	20	A	
1,1,1-Trichloroethane	50.0	52.3	0.427	0.447	.01	4.64	20	A	
1,1,2,2-Tetrachloroethane	50.0	45.6	0.891	0.813	.3	-8.76	20	A	
1,1,2-Trichloroethane	50.0	49.3	0.650	0.641	.01	-1.45	20	A	
1,1-Dichloroethane	50.0	50.1	0.626	0.627	.1	.14	20	A	
1,1-Dichloroethene	50.0	53.1	0.221	0.235	.01	6.28	20	A	
1,1-Dichloropropene	50.0	52.0	0.358	0.373	.01	3.98	20	A	
1,2,3-Trichlorobenzene	50.0	46.4	0.918	0.852	.01	-7.2	20	W	
1,2,3-Trichloropropane	50.0	43.4	1.159	1.005	.01	-13.2	20	A	
1,2,4-Trichlorobenzene	50.0	46.4	0.946	0.878	.01	-7.17	20	A	
1,2,4-Trimethylbenzene	50.0	50.4	2.590	2.613	.01	.9	20	A	
1,2-Dibromo-3-chloropropane	50.0	46.4	0.213	0.197	.01	-7.14	20	A	
1,2-Dibromoethane	50.0	49.6	0.669	0.663	.01	-.88	20	A	
1,2-Dichlorobenzene	50.0	46.6	1.536	1.431	.01	-6.85	20	A	
1,2-Dichloroethane	50.0	53.1	0.450	0.478	.01	6.16	20	A	
1,2-Dichloroethane-d4	50.0	55.0	0.333	0.366	.01	10	20	A	
1,2-Dichloroethene (total)	100.0	97.9	0.481	0.470	.01	-2.19	20	A	
1,2-Dichloropropane	50.0	49.7	0.368	0.365	.01	-.62	20	A	
1,3,5-Trimethylbenzene	50.0	49.0	2.577	2.525	.01	-2.01	20	A	
1,3-Dichlorobenzene	50.0	46.3	1.587	1.470	.01	-7.37	20	A	
1,3-Dichloropropane	50.0	49.2	1.097	1.080	.01	-1.54	20	A	
1,4-Dichlorobenzene	50.0	46.3	1.599	1.481	.01	-7.37	20	A	
2,2-Dichloropropane	50.0	50.7	0.395	0.400	.01	1.32	20	A	
2-Butanone	250.0	217.0	0.292	0.254	.01	-13.1	20	A	
2-Chlorotoluene	50.0	46.4	2.423	2.248	.01	-7.2	20	A	
2-Hexanone	250.0	223.0	0.933	0.833	.01	-10.7	20	A	
4-Bromofluorobenzene	50.0	53.1	0.722	0.766	.01	6.18	20	A	
4-Chlorotoluene	50.0	47.4	2.305	2.185	.01	-5.19	20	A	
4-Isopropyltoluene	50.0	49.5	2.769	2.740	.01	-1.03	20	A	
4-Methyl-2-pentanone	250.0	254.0	1.119	1.138	.01	1.69	20	A	
Acetone	250.0	201.0	0.332	0.267	.01	-19.5	20	A	
Benzene	50.0	49.3	1.055	1.039	.01	-1.47	20	A	
Bromobenzene	50.0	45.8	1.383	1.268	.01	-8.31	20	A	
Bromochloromethane	50.0	50.8	0.138	0.140	.01	1.53	20	A	
Bromodichloromethane	50.0	51.5	0.355	0.366	.01	3.01	20	A	
Bromoform	50.0	53.0	0.462	0.489	.1	5.92	20	A	
Bromomethane	50.0	40.2	0.157	0.129	.01	-19.6	20	W	
Carbon disulfide	50.0	50.3	0.666	0.676	.01	.6	20	W	
Carbon tetrachloride	50.0	52.6	0.375	0.395	.01	5.27	20	A	

FORM V II VOA

7A
VOLATILE CONTINUING CALIBRATION CHECK

Report No:	223060210	CCAL ID:	1400
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV11
Injection Vol.:	1.0 (µL)	Lab File ID:	230602/j1687
Init. Calib. Date 1:	05/17/23 Time 1: 1206	Analyst:	KN1
Init. Calib. Date 2:	05/17/23 Time 2: 1502	Analytical Batch:	766942
Analysis Date:	06/02/23 Time: 1123	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
Chlorobenzene	50.0	49.4	1.862	1.839	.3	-1.23	20	A	
Chloroethane	50.0	47.4	0.217	0.205	.01	-5.11	20	A	
Chloroform	50.0	51.8	0.470	0.488	.01	3.66	20	A	
Chloromethane	50.0	44.9	0.525	0.471	.1	-10.2	20	A	
Dibromochloromethane	50.0	51.9	0.702	0.729	.01	3.76	20	A	
Dibromofluoromethane	50.0	56.4	0.249	0.281	.01	12.8	20	A	
Dibromomethane	50.0	51.8	0.174	0.180	.01	3.52	20	A	
Dichlorodifluoromethane	50.0	41.9	0.299	0.250	.01	-16.2	20	A	
Ethylbenzene	50.0	50.1	0.984	0.986	.01	.28	20	A	
Hexachlorobutadiene	50.0	49.2	0.426	0.420	.01	-1.51	20	A	
Isopropylbenzene (Cumene)	50.0	53.0	2.732	2.895	.01	5.95	20	A	
Methylene chloride	50.0	48.8	0.553	0.539	.01	-2.5	20	A	
Naphthalene	50.0	43.0	2.860	2.420	.01	-14	20	W	
Styrene	50.0	52.4	1.561	1.637	.01	4.85	20	A	
Tetrachloroethene	50.0	52.5	0.582	0.612	.01	5.09	20	A	
Toluene	50.0	50.3	2.872	2.888	.01	.56	20	A	
Toluene-d8	50.0	51.3	2.393	2.457	.01	2.66	20	A	
Trichloroethene	50.0	52.1	0.303	0.316	.01	4.21	20	A	
Trichlorofluoromethane	50.0	53.5	0.427	0.458	.01	7.1	20	A	
Vinyl chloride	50.0	48.5	0.388	0.377	.01	-2.95	20	A	
cis-1,2-Dichloroethene	50.0	49.6	0.467	0.463	.01	-.86	20	A	
cis-1,3-Dichloropropene	50.0	53.5	0.411	0.440	.01	6.95	20	A	
m,p-Xylene	100.0	103.0	1.182	1.214	.01	2.74	20	A	
n-Butylbenzene	50.0	46.2	3.997	3.692	.01	-7.62	20	A	
n-Propylbenzene	50.0	48.1	3.440	3.307	.01	-3.85	20	A	
o-Xylene	50.0	52.7	1.097	1.156	.01	5.37	20	A	
sec-Butylbenzene	50.0	49.6	2.986	2.962	.01	-.81	20	A	
tert-Butyl methyl ether (MTBE)	50.0	50.5	0.781	0.789	.01	1	20	A	
tert-Butylbenzene	50.0	48.6	1.406	1.368	.01	-2.74	20	A	
trans-1,2-Dichloroethene	50.0	48.3	0.494	0.477	.01	-3.44	20	A	
trans-1,3-Dichloropropene	50.0	51.3	0.386	0.396	.01	2.69	20	A	

FORM V II VOA

GCAL, Inc.

Data file : /var/chem/msv11.i/230602.s.b/j1687.d
Lab Smp Id: 1400 Client Smp ID: V11STD050
Inj Date : 02-JUN-2023 11:23
Operator : KN1 Inst ID: msv11.i
Smp Info : 1400*V11STD050*51364
Misc Info : MSV~529~*1*KN1
Comment :
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Meth Date : 05-Jun-2023 08:22 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 15:02 Cal File: j1306.d
Als bottle: 1 Continuing Calibration Sample
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260bDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.673	1.673	(0.257)	83739	50.0000	41.9	
2 Chloromethane ++	50	1.856	1.856	(0.285)	157538	50.0000	44.9	
3 Vinyl Chloride +	62	1.940	1.940	(0.298)	126068	50.0000	48.5	
4 1-3 Butadiene	39	1.956	1.956	(0.301)	151875	50.0000	53.3	9764
5 Bromomethane	94	2.250	2.250	(0.346)	43008	50.0000	40.2	
7 Chloroethane	64	2.370	2.370	(0.364)	68736	50.0000	47.4	
8 Trichlorofluoromethane	101	2.522	2.522	(0.388)	153108	50.0000	53.5	
11 1,1-Dichloroethene +	96	3.067	3.067	(0.471)	78495	50.0000	53.1	
12 Carbon Disulfide	76	3.099	3.099	(0.476)	226180	50.0000	50.3	
13 1,1,2Trichlotrifluoroethane	101	3.120	3.120	(0.479)	82330	50.0000	53.5	
15 Methyl Iodide	142	3.225	3.225	(0.496)	50963	50.0000	29.0	
16 Acrolein	56	3.450	3.450	(0.530)	126457	250.000	255	
17 Allyl chloride	39	3.628	3.628	(0.558)	136132	50.0000	53.0	9325
18 Methylene Chloride	49	3.760	3.760	(0.578)	180222	50.0000	48.8	

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
19 Acetone	43	3.812	3.812	(0.586)	445972	250.000	201	
20 trans-1,2-Dichloroethene	61	3.959	3.959	(0.608)	159712	50.0000	48.3	
21 Methyl Acetate	43	3.974	3.974	(0.611)	618261	250.000	257	9606
22 Hexane	57	4.069	4.069	(0.625)	250936	50.0000	51.4	9410
23 MTBE	73	4.106	4.106	(0.631)	263951	50.0000	50.5	9140
24 tert-Butyl Alcohol	59	4.226	4.226	(0.649)	81239	250.000	249	8844
25 Acetonitrile	41	4.342	4.342	(0.667)	99344	250.000	232	9806
26 Isopropyl Ether	45	4.551	4.551	(0.699)	442237	50.0000	49.9	9647
28 1,1-Dichloroethane ++	63	4.656	4.656	(0.716)	209728	50.0000	50.1	
29 Acrylonitrile	53	4.698	4.698	(0.722)	295617	250.000	259	
31 Vinyl Acetate	43	4.944	4.944	(0.760)	260670	50.0000	51.6	
30 Ethyl Tert-butyl Ether	59	4.950	4.950	(0.761)	352245	50.0000	48.5	7839
32 cis-1,2-Dichloroethene	61	5.238	5.238	(0.805)	154742	50.0000	49.6	
M 33 Total 1,2-Dichloroethene	61				314454	100.000	97.9	
34 2,2-Dichloropropane	77	5.348	5.348	(0.822)	133827	50.0000	50.7	
35 Bromochloromethane	128	5.448	5.448	(0.837)	46715	50.0000	50.8	
36 Cyclohexane	56	5.453	5.453	(0.838)	213927	50.0000	53.2	9019
37 Chloroform +	83	5.526	5.526	(0.849)	163142	50.0000	51.8	
38 Carbon Tetrachloride	117	5.668	5.668	(0.871)	132179	50.0000	52.6	
$ 41 Dibromofluoromethane	111	5.715	5.715	(0.878)	94035	50.0000	56.4	9339
42 1,1,1-Trichloroethane	97	5.731	5.731	(0.881)	149463	50.0000	52.3	
44 2-Butanone	43	5.825	5.825	(0.895)	424040	250.000	217	
45 1,1-Dichloropropene	75	5.857	5.857	(0.900)	124609	50.0000	52.0	
47 2,2,4 Trimethylpentane	57	5.977	5.977	(0.919)	405991	50.0000	40.5	8899
48 Benzene	78	6.098	6.098	(0.937)	347695	50.0000	49.3	
$ 51 1,2-Dichloroethane-d4	65	6.229	6.229	(0.957)	122522	50.0000	55.0	
54 tert-amyl Methyl Ether	73	6.229	6.229	(0.957)	244719	50.0000	48.1	7299
52 1,2-Dichloroethane	62	6.292	6.292	(0.967)	159754	50.0000	53.1	
53 Isobutyl Alcohol	43	6.339	6.339	(0.974)	25297	250.000	196	9240
* 56 FLUOROBENZENE	96	6.507	6.507	(1.000)	334510	50.0000		
57 Methyl Cyclohexane	83	6.669	6.669	(1.025)	136021	50.0000	51.9	9372
58 Trichloroethene	130	6.675	6.675	(1.026)	105566	50.0000	52.1	
60 Dibromomethane	93	7.073	7.073	(1.087)	60206	50.0000	51.8	
62 1,2-Dichloropropane +	63	7.157	7.157	(1.100)	122214	50.0000	49.7	
63 Bromodichloromethane	83	7.230	7.230	(1.111)	122405	50.0000	51.5	
65 1,4- Dioxane	58	7.414	7.414	(1.139)	24353	1250.00	1220	9140
66 1-Bromo-2-chloroethane	63	7.671	7.671	(1.179)	162384	50.0000	50.4	9187
67 2-Chloroethyl vinyl ether	63	7.760	7.760	(1.193)	30411	250.000	104	
68 cis-1,3-Dichloropropene	75	7.812	7.812	(1.201)	147219	50.0000	53.5	
$ 69 Toluene-d8	98	7.980	7.980	(0.864)	323479	50.0000	51.3	
70 Toluene +	91	8.022	8.022	(0.868)	380236	50.0000	50.3	
M 72 1-3 Dichloropropene total	100				279645	100.000	105	0
74 4-methyl-2-pentanone	43	8.337	8.337	(0.902)	749280	250.000	254	
73 Tetrachloroethene	164	8.358	8.358	(0.905)	80514	50.0000	52.5	
75 trans-1,3-Dichloropropene	75	8.373	8.373	(1.287)	132426	50.0000	51.3	
78 1,1,2-Trichloroethane	97	8.504	8.504	(0.921)	84321	50.0000	49.3	
79 Dibromochloromethane	129	8.657	8.657	(0.937)	95941	50.0000	51.9	

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
80 1,3-Dichloropropane	76	8.725	8.725	(0.944)	142205	50.0000	49.2	
82 1,2-Dibromoethane(EDB)	107	8.840	8.840	(0.957)	87285	50.0000	49.6	
83 2-Hexanone	43	9.008	9.008	(0.975)	548043	250.000	223	
89 1-Chlorohexane	91	9.233	9.233	(0.999)	111898	50.0000	52.1	9523
* 90 CHLOROBENZENE-d5	82	9.239	9.239	(1.000)	131646	50.0000		
91 Chlorobenzene ++	112	9.254	9.254	(1.002)	242112	50.0000	49.4	
92 Ethylbenzene +	106	9.270	9.270	(1.003)	129832	50.0000	50.1	
93 1,1,1,2-Tetrachloroethane	133	9.301	9.301	(1.007)	84111	50.0000	50.2	
94 p,m-Xylene	106	9.380	9.380	(1.015)	319624	100.000	103	
M 95 TOTAL XYLENE	106				471750	150.000	155	
96 o-Xylene	106	9.700	9.700	(1.050)	152126	50.0000	52.7	
97 Styrene	104	9.737	9.737	(1.054)	215482	50.0000	52.4	
98 Bromoform ++	173	9.768	9.768	(1.057)	64430	50.0000	53.0	
99 Isopropylbenzene	105	9.931	9.931	(1.075)	381095	50.0000	53.0	
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	100853	50.0000	53.1	
102 cis-1,4-dichloro-2-butene	53	10.172	10.172	(0.931)	46220	50.0000	40.1	8847
103 Bromobenzene	77	10.214	10.214	(0.935)	170045	50.0000	45.8	
104 n-Propylbenzene	91	10.224	10.224	(0.936)	443615	50.0000	48.1	
105 1,1,2,2-Tetrachloroethane++	83	10.277	10.277	(0.941)	109061	50.0000	45.6	
106 2-Chlorotoluene	91	10.345	10.345	(0.947)	301612	50.0000	46.4	
107 1,3,5-Trimethylbenzene	105	10.361	10.361	(0.948)	338691	50.0000	49.0	
108 1,2,3-Trichloropropane	75	10.376	10.376	(0.950)	134843	50.0000	43.4	
109 trans-1,4-Dichloro-2-Butene	53	10.397	10.397	(0.952)	44298	50.0000	39.8	
100 Cyclohexanone	55	10.402	10.402	(0.952)	127020	250.000	156	5733
110 4-Chlorotoluene	91	10.460	10.460	(0.957)	293144	50.0000	47.4	
111 tert-butylbenzene	91	10.586	10.586	(0.969)	183438	50.0000	48.6	
112 1,2,4-Trimethylbenzene	105	10.638	10.638	(0.974)	350528	50.0000	50.4	
113 sec-Butylbenzene	105	10.712	10.712	(0.980)	397324	50.0000	49.6	
115 p-Isopropyltoluene	119	10.806	10.806	(0.989)	367593	50.0000	49.5	
116 1,3-Dichlorobenzene	146	10.874	10.874	(0.995)	197183	50.0000	46.3	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	134139	50.0000		
118 1,4-Dichlorobenzene	146	10.937	10.937	(1.001)	198644	50.0000	46.3	
119 n-Butylbenzene	91	11.100	11.100	(1.016)	495299	50.0000	46.2	
120 1,2-Dichlorobenzene	146	11.226	11.226	(1.027)	191970	50.0000	46.6	
122 1,2-Dibromo-3-Chloropropane	157	11.771	11.771	(1.077)	26471	50.0000	46.4	
123 Hexachlorobutadiene	225	12.201	12.201	(1.117)	56343	50.0000	49.2	
124 1,2,4-Trichlorobenzene	180	12.227	12.227	(1.119)	117740	50.0000	46.4	
125 Naphthalene	128	12.447	12.447	(1.139)	324673	50.0000	43.0	
126 1,2,3-Trichlorobenzene	180	12.573	12.573	(1.151)	114282	50.0000	46.4	

Data File: /var/chem/msv11.i/230602.s.b/j1687.d

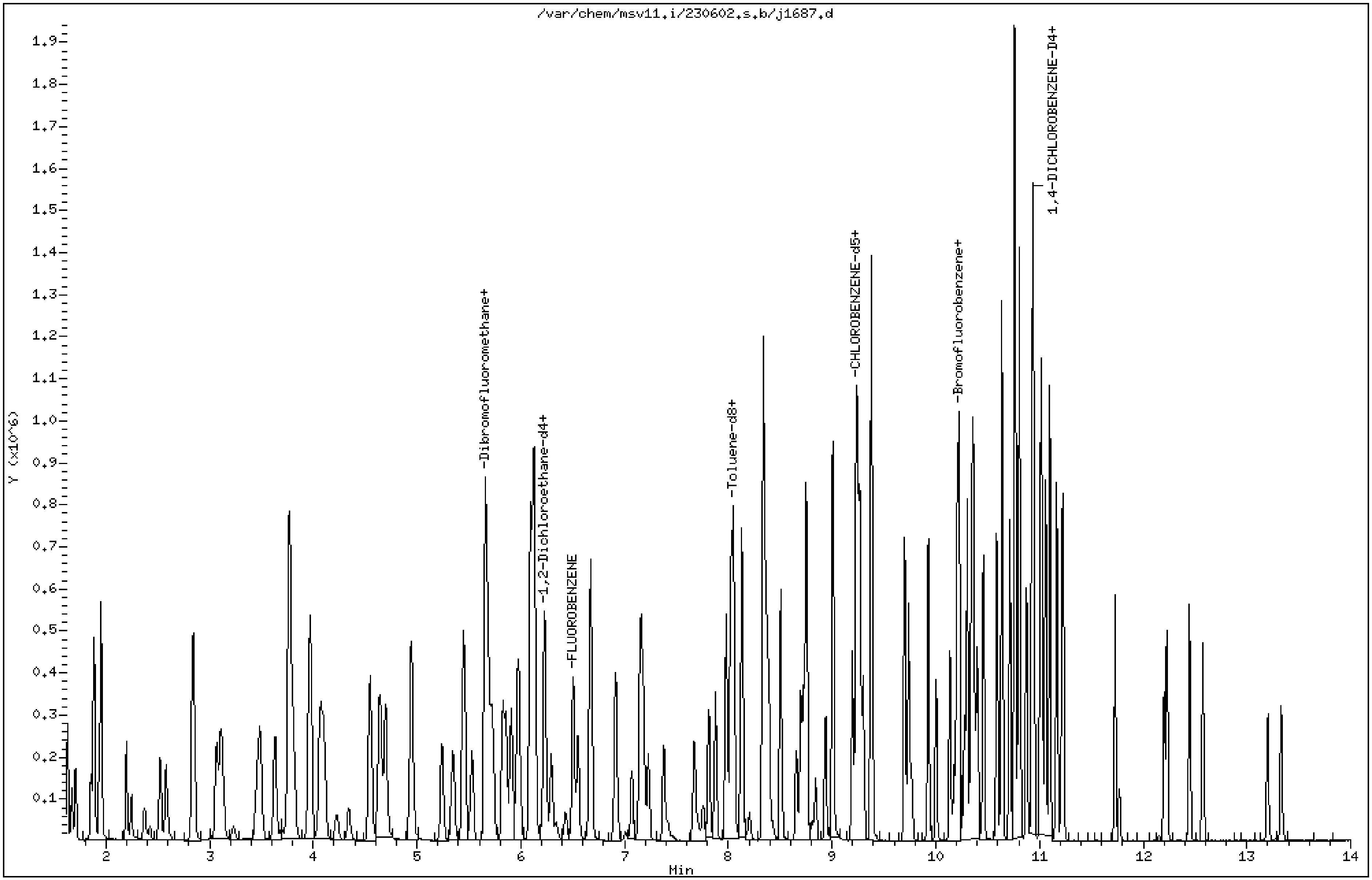
Date : 02-JUN-2023 11:23

Client ID: V11STD050 | Instrument: msv11.i

Sample Info: 1400*V11STD050*51364

Purge Volume: 5.0 | Operator: KN1

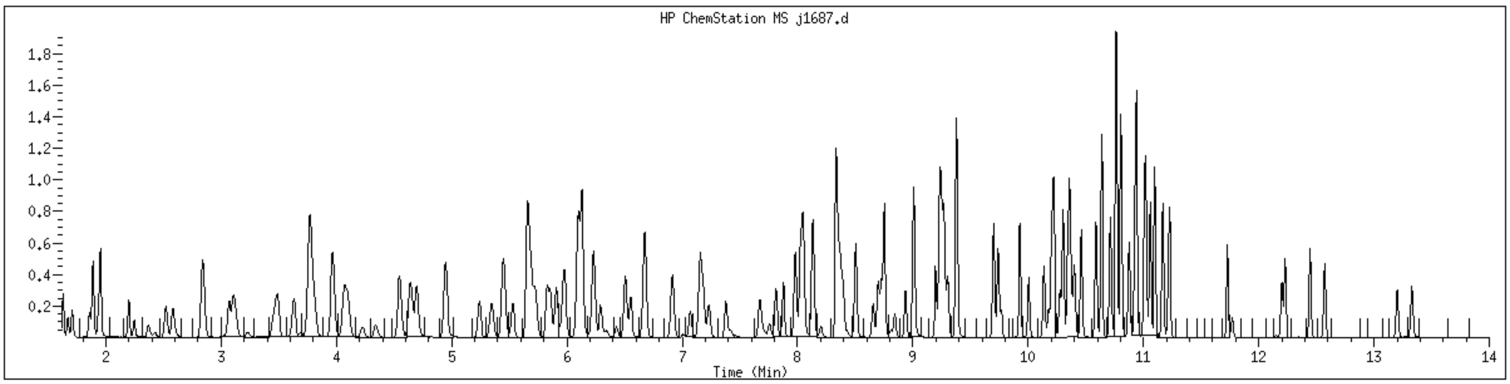
Column phase: RTX-VMS-30M | Column diameter: 0.25



Data file : /var/chem/msv11.i/230602.s.b/j1687.d Page: 1
Report Date: 06/05/2023 08:22

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1400 SampleType : CCALIB_7
Injection Date: 06/02/2023 11:23 Instrument : msv11.i
Operator : KN1
Sample Info : 1400*V11STD050*51364
Misc Info : MSV~529~*1*KN1
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260bDOD



NO MANUAL INTEGRATIONS

7A
VOLATILE CLOSING CALIBRATION CHECK

Report No:	223060210			CCAL ID:	1440
GC Column:	RTX-VMS-30M	ID	.25 (mm)	Instrument ID:	MSV11
Injection Vol.:	1.0		(µL)	Lab File ID:	230602/j1707
Init. Calib. Date 1:	05/17/23	Time 1:	1206	Analyst:	KN1
Init. Calib. Date 2:	05/17/23	Time 2:	1502	Analytical Batch:	766942
Analysis Date:	06/02/23	Time:	1924	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
1,1,1,2-Tetrachloroethane	50.0	50.7	0.636	0.645	.01	1.41	50	A	
1,1,1-Trichloroethane	50.0	54.4	0.427	0.464	.01	8.77	50	A	
1,1,2,2-Tetrachloroethane	50.0	46.6	0.891	0.830	.3	-6.87	50	A	
1,1,2-Trichloroethane	50.0	48.1	0.650	0.625	.01	-3.84	50	A	
1,1-Dichloroethane	50.0	50.1	0.626	0.628	.1	.24	50	A	
1,1-Dichloroethene	50.0	53.5	0.221	0.236	.01	6.94	50	A	
1,1-Dichloropropene	50.0	52.7	0.358	0.377	.01	5.37	50	A	
1,2,3-Trichlorobenzene	50.0	46.6	0.918	0.856	.01	-6.8	50	W	
1,2,3-Trichloropropane	50.0	44.7	1.159	1.035	.01	-10.6	50	A	
1,2,4-Trichlorobenzene	50.0	45.6	0.946	0.863	.01	-8.7	50	A	
1,2,4-Trimethylbenzene	50.0	50.9	2.590	2.638	.01	1.87	50	A	
1,2-Dibromo-3-chloropropane	50.0	49.1	0.213	0.208	.01	-1.9	50	A	
1,2-Dibromoethane	50.0	48.1	0.669	0.643	.01	-3.87	50	A	
1,2-Dichlorobenzene	50.0	48.0	1.536	1.475	.01	-3.99	50	A	
1,2-Dichloroethane	50.0	54.1	0.450	0.487	.01	8.15	50	A	
1,2-Dichloroethane-d4	50.0	57.0	0.333	0.379	.01	13.9	50	A	
1,2-Dichloroethene (total)	100.0	101.0	0.481	0.487	.01	1.29	50	A	
1,2-Dichloropropane	50.0	50.7	0.368	0.372	.01	1.3	50	A	
1,3,5-Trimethylbenzene	50.0	50.4	2.577	2.596	.01	.76	50	A	
1,3-Dichlorobenzene	50.0	46.9	1.587	1.490	.01	-6.11	50	A	
1,3-Dichloropropane	50.0	48.0	1.097	1.052	.01	-4.07	50	A	
1,4-Dichlorobenzene	50.0	47.6	1.599	1.523	.01	-4.76	50	A	
2,2-Dichloropropane	50.0	50.1	0.395	0.395	.01	.1	50	A	
2-Butanone	250.0	215.0	0.292	0.252	.01	-13.8	50	A	
2-Chlorotoluene	50.0	50.4	2.423	2.441	.01	.75	50	A	
2-Hexanone	250.0	218.0	0.933	0.815	.01	-12.6	50	A	
4-Bromofluorobenzene	50.0	52.9	0.722	0.763	.01	5.74	50	A	
4-Chlorotoluene	50.0	47.7	2.305	2.201	.01	-4.51	50	A	
4-Isopropyltoluene	50.0	49.4	2.769	2.738	.01	-1.11	50	A	
4-Methyl-2-pentanone	250.0	250.0	1.119	1.120	.01	.03	50	A	
Acetone	250.0	199.0	0.332	0.263	.01	-20.5	50	A	
Benzene	50.0	50.3	1.055	1.061	.01	.54	50	A	
Bromobenzene	50.0	47.1	1.383	1.303	.01	-5.78	50	A	
Bromochloromethane	50.0	50.5	0.138	0.139	.01	.91	50	A	
Bromodichloromethane	50.0	53.2	0.355	0.378	.01	6.44	50	A	
Bromoform	50.0	51.3	0.462	0.474	.1	2.69	50	A	
Bromomethane	50.0	46.0	0.157	0.147	.01	-8	50	W	
Carbon disulfide	50.0	49.5	0.666	0.665	.01	-1	50	W	
Carbon tetrachloride	50.0	53.5	0.375	0.402	.01	7	50	A	

FORM V II VOA

7A
VOLATILE CLOSING CALIBRATION CHECK

Report No:	223060210			CCAL ID:	1440
GC Column:	RTX-VMS-30M	ID .25	(mm)	Instrument ID:	MSV11
Injection Vol.:	1.0		(µL)	Lab File ID:	230602/j1707
Init. Calib. Date 1:	05/17/23	Time 1:	1206	Analyst:	KN1
Init. Calib. Date 2:	05/17/23	Time 2:	1502	Analytical Batch:	766942
Analysis Date:	06/02/23	Time:	1924	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
Chlorobenzene	50.0	48.8	1.862	1.818	.3	-2.35	50	A	
Chloroethane	50.0	49.5	0.217	0.215	.01	-.9	50	A	
Chloroform	50.0	52.8	0.470	0.497	.01	5.59	50	A	
Chloromethane	50.0	50.4	0.525	0.529	.1	.74	50	A	
Dibromochloromethane	50.0	51.2	0.702	0.719	.01	2.32	50	A	
Dibromofluoromethane	50.0	57.8	0.249	0.288	.01	15.6	50	A	
Dibromomethane	50.0	52.0	0.174	0.181	.01	3.99	50	A	
Dichlorodifluoromethane	50.0	47.1	0.299	0.281	.01	-5.88	50	A	
Ethylbenzene	50.0	49.7	0.984	0.978	.01	-.55	50	A	
Hexachlorobutadiene	50.0	49.5	0.426	0.422	.01	-1.08	50	A	
Isopropylbenzene (Cumene)	50.0	52.9	2.732	2.889	.01	5.73	50	A	
Methylene chloride	50.0	50.4	0.553	0.557	.01	.88	50	A	
Naphthalene	50.0	43.6	2.860	2.456	.01	-12.8	50	W	
Styrene	50.0	52.4	1.561	1.635	.01	4.72	50	A	
Tetrachloroethene	50.0	50.1	0.582	0.583	.01	.12	50	A	
Toluene	50.0	49.5	2.872	2.845	.01	-.95	50	A	
Toluene-d8	50.0	51.3	2.393	2.458	.01	2.69	50	A	
Trichloroethene	50.0	53.0	0.303	0.321	.01	6.07	50	A	
Trichlorofluoromethane	50.0	55.6	0.427	0.475	.01	11.2	50	A	
Vinyl chloride	50.0	50.2	0.388	0.390	.01	.31	50	A	
cis-1,2-Dichloroethene	50.0	51.5	0.467	0.481	.01	3.07	50	A	
cis-1,3-Dichloropropene	50.0	53.2	0.411	0.438	.01	6.4	50	A	
m,p-Xylene	100.0	102.0	1.182	1.205	.01	1.96	50	A	
n-Butylbenzene	50.0	44.8	3.997	3.582	.01	-10.3	50	A	
n-Propylbenzene	50.0	48.6	3.440	3.342	.01	-2.84	50	A	
o-Xylene	50.0	52.6	1.097	1.153	.01	5.13	50	A	
sec-Butylbenzene	50.0	49.4	2.986	2.950	.01	-1.22	50	A	
tert-Butyl methyl ether (MTBE)	50.0	51.8	0.781	0.810	.01	3.69	50	A	
tert-Butylbenzene	50.0	50.2	1.406	1.411	.01	.36	50	A	
trans-1,2-Dichloroethene	50.0	49.8	0.494	0.493	.01	-.38	50	A	
trans-1,3-Dichloropropene	50.0	51.4	0.386	0.396	.01	2.72	50	A	

FORM V II VOA

GCAL, Inc.

Data file : /var/chem/msv11.i/230602.s.b/j1707.d
Lab Smp Id: 1440 Client Smp ID: V11STD050
Inj Date : 02-JUN-2023 19:24
Operator : KN1 Inst ID: msv11.i
Smp Info : 1440*V11STD050*51364
Misc Info : MSV~52914~*1*KN1
Comment :
Method : /var/chem/msv11.i/230602.s.b/8260bDODw11clos.m
Meth Date : 05-Jun-2023 12:55 rprator Quant Type: ISTD
Cal Date : 17-MAY-2023 15:02 Cal File: j1306.d
Als bottle: 1 Continuing Calibration Sample
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260bDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.678	1.678	(0.258)	90918	50.0000	47.1	
2 Chloromethane ++	50	1.861	1.861	(0.286)	170872	50.0000	50.4	
3 Vinyl Chloride +	62	1.945	1.945	(0.299)	125934	50.0000	50.2	
4 1-3 Butadiene	39	1.961	1.961	(0.301)	166506	50.0000	60.5	9729
5 Bromomethane	94	2.255	2.255	(0.346)	47515	50.0000	46.0	
7 Chloroethane	64	2.375	2.375	(0.365)	69376	50.0000	49.5	
8 Trichlorofluoromethane	101	2.527	2.527	(0.388)	153633	50.0000	55.6	
11 1,1-Dichloroethene +	96	3.078	3.078	(0.473)	76335	50.0000	53.5	
12 Carbon Disulfide	76	3.114	3.114	(0.479)	214833	50.0000	49.5	
13 1,1,2Trichlotrifluoroethane	101	3.130	3.130	(0.481)	79675	50.0000	53.6	
15 Methyl Iodide	142	3.240	3.240	(0.498)	53207	50.0000	31.0	
16 Acrolein	56	3.460	3.460	(0.532)	117835	250.000	246	
17 Allyl chloride	39	3.639	3.639	(0.559)	138028	50.0000	55.6	9412
18 Methylene Chloride	49	3.770	3.770	(0.579)	180212	50.0000	50.4	

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
	19 Acetone	43	3.817	3.817	(0.587)	425607	250.000	199	
	20 trans-1,2-Dichloroethene	61	3.964	3.964	(0.609)	159242	50.0000	49.8	
	21 Methyl Acetate	43	3.980	3.980	(0.612)	639401	250.000	275	9620
	22 Hexane	57	4.074	4.074	(0.626)	236433	50.0000	50.1	9371
	23 MTBE	73	4.105	4.105	(0.631)	261904	50.0000	51.8	8873
	24 tert-Butyl Alcohol	59	4.231	4.231	(0.650)	89877	250.000	285	8883
	25 Acetonitrile	41	4.347	4.347	(0.668)	103400	250.000	250	9826
	26 Isopropyl Ether	45	4.551	4.551	(0.699)	438051	50.0000	51.2	9690
	28 1,1-Dichloroethane ++	63	4.661	4.661	(0.716)	202900	50.0000	50.1	
	29 Acrylonitrile	53	4.703	4.703	(0.723)	295080	250.000	267	
	31 Vinyl Acetate	43	4.944	4.944	(0.760)	234110	50.0000	48.0	
	30 Ethyl Tert-butyl Ether	59	4.955	4.955	(0.761)	352514	50.0000	50.2	7948
	32 cis-1,2-Dichloroethene	61	5.243	5.243	(0.806)	155486	50.0000	51.5	
M	33 Total 1,2-Dichloroethene	61				314728	100.000	101	
	34 2,2-Dichloropropane	77	5.353	5.353	(0.823)	127790	50.0000	50.1	
	35 Bromochloromethane	128	5.448	5.448	(0.837)	44870	50.0000	50.5	
	36 Cyclohexane	56	5.458	5.458	(0.839)	203833	50.0000	52.5	8972
	37 Chloroform +	83	5.537	5.537	(0.851)	160611	50.0000	52.8	
	38 Carbon Tetrachloride	117	5.668	5.668	(0.871)	129848	50.0000	53.5	
$	41 Dibromofluoromethane	111	5.715	5.715	(0.878)	93171	50.0000	57.8	8714
	42 1,1,1-Trichloroethane	97	5.736	5.736	(0.882)	150161	50.0000	54.4	
	44 2-Butanone	43	5.830	5.830	(0.896)	406606	250.000	215	
	45 1,1-Dichloropropene	75	5.862	5.862	(0.901)	122039	50.0000	52.7	
	47 2,2,4 Trimethylpentane	57	5.982	5.982	(0.919)	365231	50.0000	37.7	9043
	48 Benzene	78	6.103	6.103	(0.938)	342879	50.0000	50.3	
$	51 1,2-Dichloroethane-d4	65	6.229	6.229	(0.957)	122599	50.0000	57.0	
	54 tert-amyl Methyl Ether	73	6.234	6.234	(0.958)	240625	50.0000	49.0	7214
	52 1,2-Dichloroethane	62	6.297	6.297	(0.968)	157292	50.0000	54.1	
	53 Isobutyl Alcohol	43	6.344	6.344	(0.975)	27937	250.000	224	9148
*	56 FLUOROBENZENE	96	6.507	6.507	(1.000)	323297	50.0000		
	57 Methyl Cyclohexane	83	6.674	6.674	(1.026)	129770	50.0000	51.3	9326
	58 Trichloroethene	130	6.680	6.680	(1.027)	103849	50.0000	53.0	
	60 Dibromomethane	93	7.073	7.073	(1.087)	58451	50.0000	52.0	
	62 1,2-Dichloropropane +	63	7.162	7.162	(1.101)	120394	50.0000	50.7	
	63 Bromodichloromethane	83	7.230	7.230	(1.111)	122238	50.0000	53.2	
	65 1,4- Dioxane	58	7.419	7.419	(1.140)	26792	1250.00	1390	8841
	66 1-Bromo-2-chloroethane	63	7.671	7.671	(1.179)	159503	50.0000	51.2	9247
	67 2-Chloroethyl vinyl ether	63	7.765	7.765	(1.193)	16292	250.000	59.8	
	68 cis-1,3-Dichloropropene	75	7.817	7.817	(1.201)	141552	50.0000	53.2	
$	69 Toluene-d8	98	7.980	7.980	(0.864)	318784	50.0000	51.3	
	70 Toluene +	91	8.027	8.027	(0.869)	369006	50.0000	49.5	
M	72 1-3 Dichloropropene total	100				269574	100.000	105	0
	74 4-methyl-2-pentanone	43	8.336	8.336	(0.902)	726096	250.000	250	
	73 Tetrachloroethene	164	8.363	8.363	(0.905)	75569	50.0000	50.1	
	75 trans-1,3-Dichloropropene	75	8.378	8.378	(1.288)	128022	50.0000	51.4	
	78 1,1,2-Trichloroethane	97	8.509	8.509	(0.921)	81057	50.0000	48.1	
	79 Dibromochloromethane	129	8.662	8.662	(0.938)	93206	50.0000	51.2	

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
80 1,3-Dichloropropane	76	8.724	8.724	(0.944)	136496	50.0000	48.0	
82 1,2-Dibromoethane(EDB)	107	8.845	8.845	(0.957)	83401	50.0000	48.1	
83 2-Hexanone	43	9.008	9.008	(0.975)	528466	250.000	218	
89 1-Chlorohexane	91	9.233	9.233	(0.999)	101452	50.0000	47.9	9359
* 90 CHLOROBENZENE-d5	82	9.238	9.238	(1.000)	129699	50.0000		
91 Chlorobenzene ++	112	9.254	9.254	(1.002)	235832	50.0000	48.8	
92 Ethylbenzene +	106	9.270	9.270	(1.003)	126857	50.0000	49.7	
93 1,1,1,2-Tetrachloroethane	133	9.301	9.301	(1.007)	83680	50.0000	50.7	
94 p,m-Xylene	106	9.380	9.380	(1.015)	312499	100.000	102	
M 95 TOTAL XYLENE	106				462032	150.000	155	
96 o-Xylene	106	9.700	9.700	(1.050)	149533	50.0000	52.6	
97 Styrene	104	9.736	9.736	(1.054)	212031	50.0000	52.4	
98 Bromoform ++	173	9.768	9.768	(1.057)	61540	50.0000	51.3	
99 Isopropylbenzene	105	9.930	9.930	(1.075)	374663	50.0000	52.9	
$ 101 Bromofluorobenzene	174	10.140	10.140	(1.098)	98951	50.0000	52.9	
102 cis-1,4-dichloro-2-butene	53	10.177	10.177	(0.931)	41286	50.0000	37.6	9037
103 Bromobenzene	77	10.219	10.219	(0.935)	166280	50.0000	47.1	
104 n-Propylbenzene	91	10.229	10.229	(0.936)	426588	50.0000	48.6	
105 1,1,2,2-Tetrachloroethane++	83	10.276	10.276	(0.940)	105927	50.0000	46.6	
106 2-Chlorotoluene	91	10.345	10.345	(0.947)	311614	50.0000	50.4	
107 1,3,5-Trimethylbenzene	105	10.360	10.360	(0.948)	331413	50.0000	50.4	
108 1,2,3-Trichloropropane	75	10.376	10.376	(0.950)	132103	50.0000	44.7	
109 trans-1,4-Dichloro-2-Butene	53	10.397	10.397	(0.952)	40948	50.0000	38.6	
100 Cyclohexanone	55	10.402	10.402	(0.952)	115171	250.000	149	6099
110 4-Chlorotoluene	91	10.460	10.460	(0.957)	280949	50.0000	47.7	
111 tert-butylbenzene	91	10.591	10.591	(0.969)	180128	50.0000	50.2	
112 1,2,4-Trimethylbenzene	105	10.638	10.638	(0.974)	336776	50.0000	50.9	
113 sec-Butylbenzene	105	10.712	10.712	(0.980)	376565	50.0000	49.4	
115 p-Isopropyltoluene	119	10.811	10.811	(0.989)	349523	50.0000	49.4	
116 1,3-Dichlorobenzene	146	10.879	10.879	(0.996)	190188	50.0000	46.9	
* 117 1,4-DICHLOROBENZENE-D4	152	10.927	10.927	(1.000)	127649	50.0000		
118 1,4-Dichlorobenzene	146	10.937	10.937	(1.001)	194367	50.0000	47.6	
119 n-Butylbenzene	91	11.100	11.100	(1.016)	457270	50.0000	44.8	
120 1,2-Dichlorobenzene	146	11.231	11.231	(1.028)	188286	50.0000	48.0	
122 1,2-Dibromo-3-Chloropropane	157	11.771	11.771	(1.077)	26612	50.0000	49.1	
123 Hexachlorobutadiene	225	12.201	12.201	(1.117)	53848	50.0000	49.5	
124 1,2,4-Trichlorobenzene	180	12.227	12.227	(1.119)	110189	50.0000	45.6	
125 Naphthalene	128	12.447	12.447	(1.139)	313496	50.0000	43.6	
126 1,2,3-Trichlorobenzene	180	12.573	12.573	(1.151)	109255	50.0000	46.6	

Data File: /var/chem/msv11.i/230602.s.b/j1707.d

Date : 02-JUN-2023 19:24

Client ID: V11STD050

Sample Info: 1440*V11STD050*51364

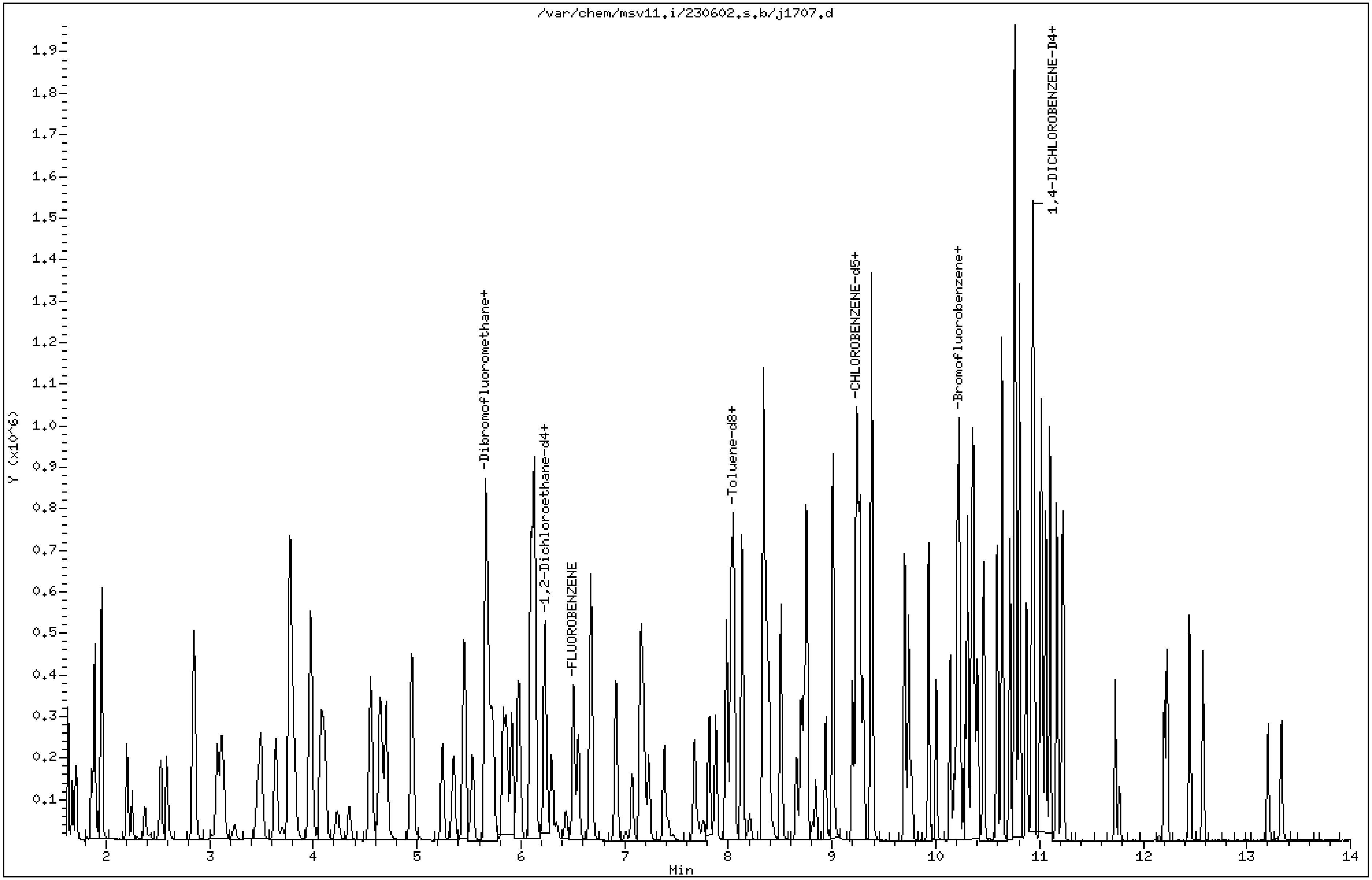
Purge Volume: 5.0

Column phase: RTX-VMS-30M

Instrument: msv11.i

Operator: KN1

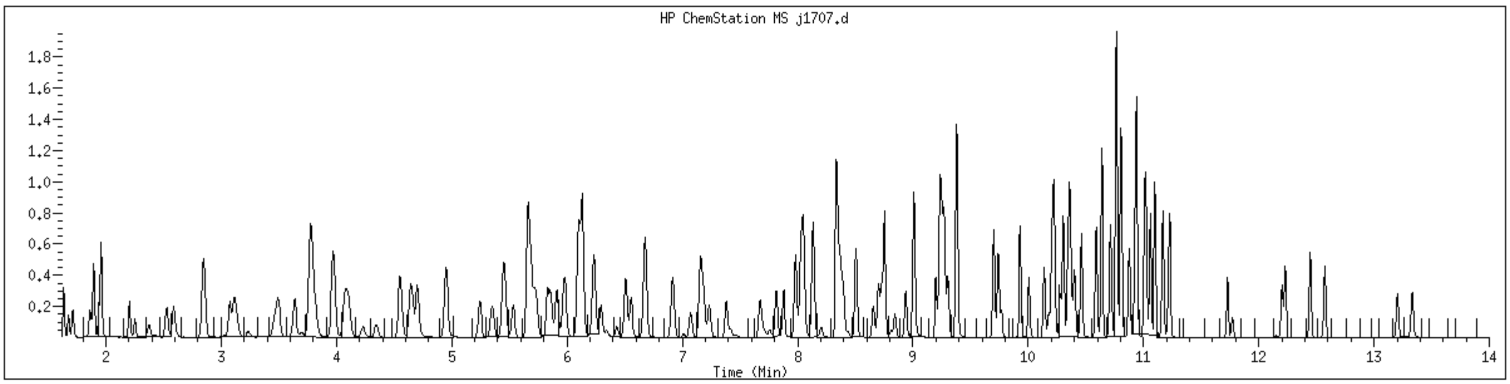
Column diameter: 0.25



Report Date: 06/05/2023 12:55

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 1440   SampleType   : CCALIB_7
Injection Date: 06/02/2023 19:24   Instrument   : msv11.i
Operator   : KN1
Sample Info   : 1440*V11STD050*51364
Misc Info   : MSV~52914~*1*KN1
Method   : /var/chem/msv11.i/230602.s.b/8260bDODw11clos.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260bDOD



NO MANUAL INTEGRATIONS

7A
VOLATILE CONTINUING CALIBRATION CHECK

Report No:	223060210			CCAL ID:	1400
GC Column:	RTX-VMS-30M	ID	.25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0		(µL)	Lab File ID:	230605/c0730
Init. Calib. Date 1:	05/17/23	Time 1:	1428	Analyst:	KN1
Init. Calib. Date 2:	05/17/23	Time 2:	1620	Analytical Batch:	767028
Analysis Date:	06/05/23	Time:	1045	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
1,1,1,2-Tetrachloroethane	50.0	51.6	0.649	0.669	.01	3.14	20	A	
1,1,1-Trichloroethane	50.0	51.7	0.428	0.443	.01	3.38	20	A	
1,1,2,2-Tetrachloroethane	50.0	50.6	0.763	0.773	.3	1.28	20	A	
1,1,2-Trichloroethane	50.0	51.9	0.593	0.615	.01	3.78	20	A	
1,1-Dichloroethane	50.0	53.6	0.452	0.484	.1	7.12	20	A	
1,1-Dichloroethene	50.0	53.6	0.244	0.262	.01	7.14	20	A	
1,1-Dichloropropene	50.0	52.8	0.381	0.402	.01	5.66	20	A	
1,2,3-Trichlorobenzene	50.0	50.7	0.725	0.735	.01	1.35	20	A	
1,2,3-Trichloropropane	50.0	47.6	1.035	0.985	.01	-4.88	20	A	
1,2,4-Trichlorobenzene	50.0	51.4	0.840	0.863	.01	2.73	20	A	
1,2,4-Trimethylbenzene	50.0	52.9	2.578	2.730	.01	5.9	20	A	
1,2-Dibromo-3-chloropropane	50.0	41.8	0.196	0.161	.01	-16.4	20	W	
1,2-Dibromoethane	50.0	52.2	0.634	0.662	.01	4.36	20	A	
1,2-Dichlorobenzene	50.0	51.7	1.375	1.421	.01	3.36	20	A	
1,2-Dichloroethane	50.0	50.1	0.324	0.324	.01	.18	20	A	
1,2-Dichloroethane-d4	50.0	49.2	0.257	0.253	.01	-1.64	20	A	
1,2-Dichloroethene (total)	100.0	107.0	0.332	0.354	.01	6.79	20	A	
1,2-Dichloropropane	50.0	53.7	0.254	0.273	.01	7.41	20	A	
1,3,5-Trimethylbenzene	50.0	53.1	2.626	2.792	.01	6.29	20	A	
1,3-Dichlorobenzene	50.0	51.4	1.520	1.563	.01	2.83	20	A	
1,3-Dichloropropane	50.0	51.4	0.987	1.015	.01	2.89	20	A	
1,4-Dichlorobenzene	50.0	51.2	1.563	1.600	.01	2.36	20	A	
2,2-Dichloropropane	50.0	48.2	0.402	0.387	.01	-3.61	20	A	
2-Butanone	250.0	236.0	0.115	0.108	.01	-5.75	20	A	
2-Chlorotoluene	50.0	52.1	2.489	2.594	.01	4.2	20	A	
2-Hexanone	250.0	227.0	0.450	0.404	.01	-9.2	20	W	
4-Bromofluorobenzene	50.0	50.3	0.761	0.766	.01	.69	20	A	
4-Chlorotoluene	50.0	52.2	2.234	2.331	.01	4.35	20	A	
4-Isopropyltoluene	50.0	53.1	3.008	3.193	.01	6.16	20	A	
4-Methyl-2-pentanone	250.0	221.0	0.548	0.478	.01	-11.6	20	W	
Acetone	250.0	234.0	0.111	0.104	.01	-6.26	20	A	
Benzene	50.0	53.4	1.029	1.099	.01	6.85	20	A	
Bromobenzene	50.0	53.2	1.300	1.383	.01	6.41	20	A	
Bromochloromethane	50.0	52.5	0.128	0.134	.01	5	20	A	
Bromodichloromethane	50.0	52.8	0.344	0.364	.01	5.57	20	A	
Bromoform	50.0	45.5	0.554	0.495	.1	-9	20	W	
Bromomethane	50.0	38.7	0.247	0.191	.01	-22.6	20	A	*
Carbon disulfide	50.0	53.3	0.747	0.796	.01	6.63	20	A	
Carbon tetrachloride	50.0	51.7	0.374	0.386	.01	3.35	20	A	

FORM V II VOA

7A
VOLATILE CONTINUING CALIBRATION CHECK

Report No:	223060210			CCAL ID:	1400
GC Column:	RTX-VMS-30M	ID	.25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0		(µL)	Lab File ID:	230605/c0730
Init. Calib. Date 1:	05/17/23	Time 1:	1428	Analyst:	KN1
Init. Calib. Date 2:	05/17/23	Time 2:	1620	Analytical Batch:	767028
Analysis Date:	06/05/23	Time:	1045	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
Chlorobenzene	50.0	52.9	1.931	2.043	.3	5.8	20	A	
Chloroethane	50.0	52.8	0.216	0.229	.01	5.69	20	A	
Chloroform	50.0	51.6	0.462	0.477	.01	3.2	20	A	
Chloromethane	50.0	57.7	0.209	0.241	.1	15.3	20	A	
Dibromochloromethane	50.0	54.5	0.674	0.735	.01	9.05	20	A	
Dibromofluoromethane	50.0	52.4	0.247	0.259	.01	4.71	20	A	
Dibromomethane	50.0	52.0	0.166	0.172	.01	3.92	20	A	
Dichlorodifluoromethane	50.0	60.8	0.232	0.282	.01	21.6	20	A	*
Ethylbenzene	50.0	53.3	1.050	1.119	.01	6.61	20	A	
Hexachlorobutadiene	50.0	51.8	0.420	0.435	.01	3.5	20	A	
Isopropylbenzene (Cumene)	50.0	54.1	3.094	3.346	.01	8.15	20	A	
Methylene chloride	50.0	52.3	0.270	0.282	.01	4.53	20	A	
Naphthalene	50.0	48.2	1.969	1.897	.01	-3.65	20	A	
Styrene	50.0	57.4	1.522	1.747	.01	14.8	20	A	
Tetrachloroethene	50.0	52.7	0.660	0.695	.01	5.36	20	A	
Toluene	50.0	53.1	2.917	3.097	.01	6.15	20	A	
Toluene-d8	50.0	53.3	2.343	2.499	.01	6.64	20	A	
Trichloroethene	50.0	52.8	0.316	0.334	.01	5.68	20	A	
Trichlorofluoromethane	50.0	51.5	0.429	0.442	.01	3.07	20	A	
Vinyl chloride	50.0	54.7	0.294	0.322	.01	9.32	20	A	
cis-1,2-Dichloroethene	50.0	52.7	0.325	0.342	.01	5.4	20	A	
cis-1,3-Dichloropropene	50.0	52.4	0.408	0.428	.01	4.83	20	A	
m,p-Xylene	100.0	108.0	1.269	1.369	.01	7.89	20	A	
n-Butylbenzene	50.0	52.5	3.999	4.201	.01	5.04	20	A	
n-Propylbenzene	50.0	52.3	3.905	4.082	.01	4.54	20	A	
o-Xylene	50.0	53.1	1.182	1.255	.01	6.2	20	A	
sec-Butylbenzene	50.0	53.2	3.397	3.612	.01	6.32	20	A	
tert-Butyl methyl ether (MTBE)	50.0	45.6	0.660	0.602	.01	-8.73	20	A	
tert-Butylbenzene	50.0	50.8	1.475	1.499	.01	1.62	20	A	
trans-1,2-Dichloroethene	50.0	54.1	0.338	0.366	.01	8.13	20	A	
trans-1,3-Dichloropropene	50.0	51.0	0.387	0.394	.01	1.95	20	A	

FORM V II VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230605.s.b/c0730.d
Lab Smp Id: 1400 Client Smp ID: V15STD050
Inj Date : 05-JUN-2023 10:45
Operator : KN1 Inst ID: msv15.i
Smp Info : 1400*V15STD050*51363
Misc Info : MSV~529~*1*SMR
Comment :
Method : /var/chem/msv15.i/230605.s.b/8260bw15DOD.m
Meth Date : 06-Jun-2023 12:20 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 Continuing Calibration Sample
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	91824	50.0000	60.8	9745
2 Chloromethane ++	50	1.652	1.652	(0.281)	78421	50.0000	57.7	9692
3 Vinyl Chloride +	62	1.719	1.719	(0.292)	104686	50.0000	54.7	9831
5 1-3 Butadiene	39	1.735	1.735	(0.295)	77137	50.0000	54.1	9671
6 Bromomethane	94	2.018	2.018	(0.343)	62158	50.0000	38.7	9792
7 Chloroethane	64	2.137	2.137	(0.363)	74395	50.0000	52.8	9242
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	143877	50.0000	51.5	9634
11 1,1-Dichloroethene +	96	2.783	2.783	(0.473)	85090	50.0000	53.6	8253
12 Carbon Disulfide	76	2.816	2.816	(0.479)	259083	50.0000	53.3	8014
13 1,1,2Trichlotrifluoroethane	101	2.835	2.835	(0.482)	93790	50.0000	54.4	9245
15 Methyl Iodide	142	2.938	2.938	(0.499)	69467	50.0000	37.0	9502
16 Acrolein	56	3.150	3.150	(0.535)	60475	250.000	250	5646
17 Allyl chloride	39	3.304	3.304	(0.562)	77882	50.0000	55.8	9774
18 Methylene Chloride	49	3.417	3.417	(0.581)	91875	50.0000	52.3	4916

							AMOUNTS		
		QUANT SIG					CAL-AMT	ON-COL	
	Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
	19 Acetone	43	3.468	3.468	(0.590)	169081	250.000	234	5474(H)
	20 trans-1,2-Dichloroethene	61	3.590	3.590	(0.610)	119061	50.0000	54.1	8707
	21 Methyl Acetate	43	3.610	3.610	(0.614)	221306	250.000	246	9331
	22 Hexane	57	3.687	3.687	(0.627)	173341	50.0000	54.6	9243
	23 MTBE	73	3.719	3.719	(0.632)	196003	50.0000	45.6	9656
	24 tert-Butyl Alcohol	59	3.822	3.822	(0.650)	33518	250.000	177	8137
	25 Acetonitrile	41	3.941	3.941	(0.670)	33700	250.000	264	9656
	26 Isopropyl Ether	45	4.115	4.115	(0.699)	205589	50.0000	53.7	9666
	27 Chloroprene	53	4.198	4.198	(0.714)	117325	50.0000	54.4	8460
	28 1,1-Dichloroethane ++	63	4.214	4.214	(0.716)	157495	50.0000	53.6	9452
	29 Acrylonitrile	53	4.259	4.259	(0.724)	127103	250.000	253	9870
	31 Ethyl Tert-butyl Ether	59	4.471	4.471	(0.760)	208514	50.0000	48.3	8005
	30 Vinyl Acetate	43	4.471	4.471	(0.760)	118793	50.0000	50.1	5687
	32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	111406	50.0000	52.7	9806
M	33 Total 1,2-Dichloroethene	61				230467	100.000	107	0
	34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	125968	50.0000	48.2	9601
	35 Bromochloromethane	128	4.922	4.922	(0.837)	43574	50.0000	52.5	9086
	36 Cyclohexane	56	4.928	4.928	(0.838)	149114	50.0000	55.6	9427
	37 Chloroform +	83	4.992	4.992	(0.849)	155217	50.0000	51.6	9808
	39 Carbon Tetrachloride	117	5.118	5.118	(0.870)	125656	50.0000	51.7	8329
$	42 Dibromofluoromethane	111	5.160	5.160	(0.877)	84111	50.0000	52.4	9018
	43 1,1,1-Trichloroethane	97	5.179	5.179	(0.880)	143990	50.0000	51.7	9575
	45 1,1-Dichloropropene	75	5.291	5.291	(0.899)	130836	50.0000	52.8	9557
	44 2-Butanone	43	5.275	5.275	(0.897)	176440	250.000	236	6453
	46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	291478	50.0000	39.0	9635
	48 Benzene	78	5.513	5.513	(0.937)	357616	50.0000	53.4	9442
	56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	212759	50.0000	48.4	7397
$	50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	82274	50.0000	49.2	8607
	52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	105474	50.0000	50.1	9779
	51 Isobutyl Alcohol	43	5.722	5.722	(0.973)	9129	250.000	208	4969
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	325329	50.0000		9863
	58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	172487	50.0000	54.8	8332
	57 Trichloroethene	130	6.031	6.031	(1.025)	108639	50.0000	52.8	8786
	62 Dibromomethane	93	6.394	6.394	(1.087)	55972	50.0000	52.0	9463
	64 1,2-Dichloropropane +	63	6.481	6.481	(1.102)	88852	50.0000	53.7	7912
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	118295	50.0000	52.8	9776
	67 1,4- Dioxane	88	6.722	6.722	(1.143)	13697	1250.00	1120	9424
	68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	113964	50.0000	54.1	9009
	72 2-Chloroethyl vinyl ether	63	7.021	7.021	(1.193)	16128	250.000	38.0	9586
	73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	139077	50.0000	52.4	9797
$	74 Toluene-d8	98	7.224	7.224	(0.866)	314205	50.0000	53.3	9891
	75 Toluene +	91	7.262	7.262	(0.871)	389395	50.0000	53.1	9615
	77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	300603	250.000	221	8500
	78 Tetrachloroethene	164	7.558	7.558	(0.906)	87386	50.0000	52.7	8464
	79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	128191	50.0000	51.0	8720
	81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	77328	50.0000	51.9	9562
	82 Dibromochloromethane	129	7.825	7.825	(0.938)	92418	50.0000	54.5	9592

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	104865	50.0000	53.5	9745
86 1,3-Dichloropropane	76	7.893	7.893	(0.946)	127652	50.0000	51.4	9604
M 83 1-3 Dichloropropene total	100				267268	100.000	103	0
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	83209	50.0000	52.2	9762
89 2-Hexanone	43	8.143	8.143	(0.976)	253816	250.000	227	9805
90 1-Chlorohexane	91	8.333	8.333	(0.999)	135078	50.0000	51.5	8017
* 91 CHLOROBENZENE-d5	82	8.339	8.339	(1.000)	125741	50.0000		9184
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	256874	50.0000	52.9	9333
93 Ethylbenzene +	106	8.368	8.368	(1.003)	140734	50.0000	53.3	9156
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	84143	50.0000	51.6	9238
95 p,m-Xylene	106	8.465	8.465	(1.015)	344380	100.000	108	9889
97 o-Xylene	106	8.745	8.745	(1.049)	157832	50.0000	53.1	9910
101 Styrene	104	8.777	8.777	(1.052)	219622	50.0000	57.4	9862
102 Bromoform ++	173	8.799	8.799	(1.055)	62190	50.0000	45.5	8706
104 Isopropylbenzene	105	8.944	8.944	(1.072)	420730	50.0000	54.1	9892
M 105 TOTAL XYLENE	106				502212	150.000	161	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	96349	50.0000	50.3	9465
107 cis-1,4-dichloro-2-butene	53	9.159	9.159	(0.934)	30588	50.0000	49.9	9592
109 n-Propylbenzene	91	9.201	9.201	(0.938)	492700	50.0000	52.3	8863
108 Bromobenzene	77	9.195	9.195	(0.938)	166908	50.0000	53.2	9656
110 1,1,2,2-Tetrachloroethane++	83	9.243	9.243	(0.943)	93250	50.0000	50.6	9743
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	313031	50.0000	52.1	8159
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	336932	50.0000	53.1	8424
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.951)	118843	50.0000	47.6	7276
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.953)	27971	50.0000	46.0	9472
116 Cyclohexanone	55	9.365	9.365	(0.955)	68311	250.000	224	9727
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	281322	50.0000	52.2	9778
118 tert-butylbenzene	91	9.516	9.516	(0.970)	180866	50.0000	50.8	9879
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.974)	329478	50.0000	52.9	9318
120 sec-Butylbenzene	105	9.622	9.622	(0.981)	435977	50.0000	53.2	9845
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	385365	50.0000	53.1	8874
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.995)	188620	50.0000	51.4	9685
125 1,4-Dichlorobenzene	146	9.815	9.815	(1.001)	193100	50.0000	51.2	9047
* 124 1,4-DICHLOROBENZENE-D4	152	9.806	9.806	(1.000)	120697	50.0000		9022
126 n-Butylbenzene	91	9.954	9.954	(1.015)	507004	50.0000	52.5	8983
127 1,2-Dichlorobenzene	146	10.066	10.066	(1.027)	171513	50.0000	51.7	9721
129 1,2-Dibromo-3-Chloropropane	157	10.558	10.558	(1.077)	19380	50.0000	41.8	9387
130 Hexachlorobutadiene	225	10.999	10.999	(1.122)	52502	50.0000	51.8	9233
131 1,2,4-Trichlorobenzene	180	11.031	11.031	(1.125)	104204	50.0000	51.4	9488
132 Naphthalene	128	11.291	11.291	(1.151)	229011	50.0000	48.2	9850
133 1,2,3-Trichlorobenzene	180	11.445	11.445	(1.167)	88675	50.0000	50.7	9473

QC Flag Legend

H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230605.s.b/c0730.d

Page 1

Date : 05-JUN-2023 10:45

Client ID: V15STD050

Instrument: msv15.i

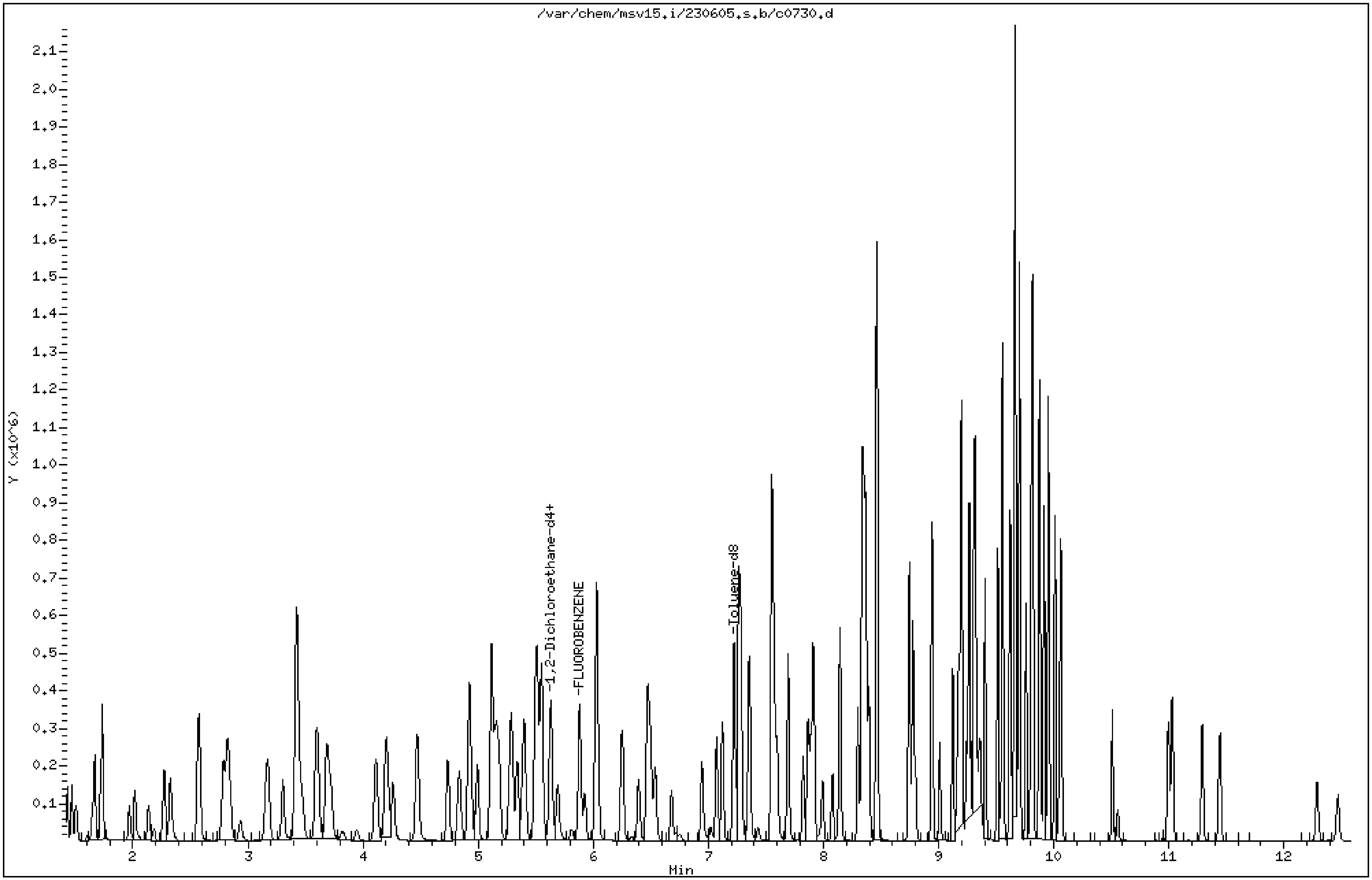
Sample Info: 1400*V15STD050*51363

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

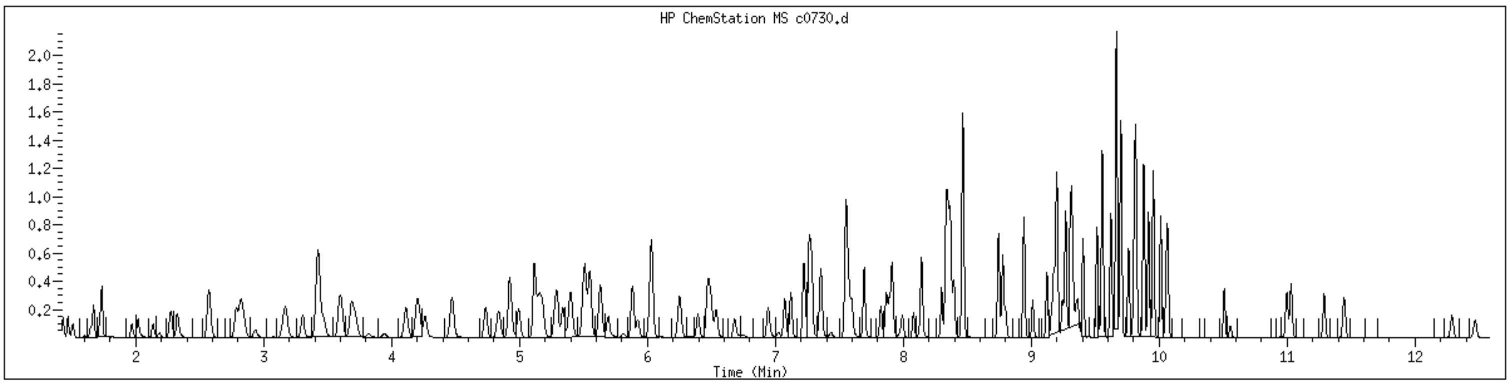
Column diameter: 0.25



Data file : /var/chem/msv15.i/230605.s.b/c0730.d Page: 1
Report Date: 06/06/2023 12:20

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1400 SampleType : CCALIB_7
Injection Date: 06/05/2023 10:45 Instrument : msv15.i
Operator : KN1
Sample Info : 1400*V15STD050*51363
Misc Info : MSV~529~*1*SMR
Method : /var/chem/msv15.i/230605.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

7A
VOLATILE CLOSING CALIBRATION CHECK

Report No:	223060210	CCAL ID:	1440
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230605/c0761
Init. Calib. Date 1:	05/17/23 Time 1: 1428	Analyst:	KN1
Init. Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Batch:	767028
Analysis Date:	06/05/23 Time: 2027	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
1,1,1,2-Tetrachloroethane	50.0	50.7	0.649	0.658	.01	1.47	50	A	
1,1,1-Trichloroethane	50.0	51.7	0.428	0.442	.01	3.3	50	A	
1,1,2,2-Tetrachloroethane	50.0	55.2	0.763	0.842	.3	10.4	50	A	
1,1,2-Trichloroethane	50.0	53.1	0.593	0.629	.01	6.15	50	A	
1,1-Dichloroethane	50.0	54.2	0.452	0.490	.1	8.42	50	A	
1,1-Dichloroethene	50.0	53.9	0.244	0.263	.01	7.82	50	A	
1,1-Dichloropropene	50.0	53.1	0.381	0.404	.01	6.18	50	A	
1,2,3-Trichlorobenzene	50.0	54.5	0.725	0.790	.01	8.95	50	A	
1,2,3-Trichloropropane	50.0	51.2	1.035	1.060	.01	2.38	50	A	
1,2,4-Trichlorobenzene	50.0	52.8	0.840	0.888	.01	5.68	50	A	
1,2,4-Trimethylbenzene	50.0	53.4	2.578	2.751	.01	6.71	50	A	
1,2-Dibromo-3-chloropropane	50.0	49.2	0.196	0.190	.01	-1.6	50	W	
1,2-Dibromoethane	50.0	54.2	0.634	0.687	.01	8.41	50	A	
1,2-Dichlorobenzene	50.0	52.5	1.375	1.444	.01	5.06	50	A	
1,2-Dichloroethane	50.0	50.8	0.324	0.329	.01	1.57	50	A	
1,2-Dichloroethane-d4	50.0	48.8	0.257	0.251	.01	-2.3	50	A	
1,2-Dichloroethene (total)	100.0	107.0	0.332	0.354	.01	6.87	50	A	
1,2-Dichloropropane	50.0	54.7	0.254	0.278	.01	9.49	50	A	
1,3,5-Trimethylbenzene	50.0	52.9	2.626	2.781	.01	5.87	50	A	
1,3-Dichlorobenzene	50.0	52.8	1.520	1.604	.01	5.53	50	A	
1,3-Dichloropropane	50.0	53.9	0.987	1.063	.01	7.72	50	A	
1,4-Dichlorobenzene	50.0	52.5	1.563	1.642	.01	5.05	50	A	
2,2-Dichloropropane	50.0	48.2	0.402	0.387	.01	-3.65	50	A	
2-Butanone	250.0	300.0	0.115	0.138	.01	20	50	A	
2-Chlorotoluene	50.0	52.0	2.489	2.586	.01	3.91	50	A	
2-Hexanone	250.0	267.0	0.450	0.477	.01	6.8	50	W	
4-Bromofluorobenzene	50.0	49.5	0.761	0.753	.01	-1.08	50	A	
4-Chlorotoluene	50.0	52.7	2.234	2.353	.01	5.33	50	A	
4-Isopropyltoluene	50.0	52.4	3.008	3.153	.01	4.82	50	A	
4-Methyl-2-pentanone	250.0	265.0	0.548	0.576	.01	6	50	W	
Acetone	250.0	276.0	0.111	0.122	.01	10.4	50	A	
Benzene	50.0	53.5	1.029	1.101	.01	7.05	50	A	
Bromobenzene	50.0	51.0	1.300	1.327	.01	2.08	50	A	
Bromochloromethane	50.0	52.6	0.128	0.134	.01	5.21	50	A	
Bromodichloromethane	50.0	52.6	0.344	0.363	.01	5.27	50	A	
Bromoform	50.0	47.9	0.554	0.521	.1	-4.2	50	W	
Bromomethane	50.0	46.3	0.247	0.228	.01	-7.47	50	A	
Carbon disulfide	50.0	53.6	0.747	0.800	.01	7.16	50	A	
Carbon tetrachloride	50.0	51.9	0.374	0.388	.01	3.83	50	A	

FORM V II VOA

7A
VOLATILE CLOSING CALIBRATION CHECK

Report No:	223060210	CCAL ID:	1440
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230605/c0761
Init. Calib. Date 1:	05/17/23 Time 1: 1428	Analyst:	KN1
Init. Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Batch:	767028
Analysis Date:	06/05/23 Time: 2027	Analytical Method:	EPA 8260B

ANALYTE	***TRUE***	***CONC***	***RRF***	***RRF CCV***	***Min RRF***	***%D/%Drift***	***Max %D/ %Drift***	***TYPE***	***Q***
Chlorobenzene	50.0	52.4	1.931	2.022	.3	4.72	50	A	
Chloroethane	50.0	54.6	0.216	0.236	.01	9.15	50	A	
Chloroform	50.0	51.9	0.462	0.480	.01	3.73	50	A	
Chloromethane	50.0	53.6	0.209	0.224	.1	7.21	50	A	
Dibromochloromethane	50.0	55.3	0.674	0.746	.01	10.7	50	A	
Dibromofluoromethane	50.0	51.3	0.247	0.253	.01	2.63	50	A	
Dibromomethane	50.0	53.2	0.166	0.176	.01	6.43	50	A	
Dichlorodifluoromethane	50.0	59.6	0.232	0.277	.01	19.2	50	A	
Ethylbenzene	50.0	52.8	1.050	1.108	.01	5.54	50	A	
Hexachlorobutadiene	50.0	53.0	0.420	0.445	.01	5.97	50	A	
Isopropylbenzene (Cumene)	50.0	53.0	3.094	3.282	.01	6.07	50	A	
Methylene chloride	50.0	53.3	0.270	0.288	.01	6.59	50	A	
Naphthalene	50.0	54.7	1.969	2.155	.01	9.44	50	A	
Styrene	50.0	55.0	1.522	1.674	.01	10	50	A	
Tetrachloroethene	50.0	52.3	0.660	0.689	.01	4.52	50	A	
Toluene	50.0	53.1	2.917	3.099	.01	6.21	50	A	
Toluene-d8	50.0	51.5	2.343	2.413	.01	2.97	50	A	
Trichloroethene	50.0	54.6	0.316	0.345	.01	9.22	50	A	
Trichlorofluoromethane	50.0	51.4	0.429	0.441	.01	2.86	50	A	
Vinyl chloride	50.0	55.5	0.294	0.327	.01	11	50	A	
cis-1,2-Dichloroethene	50.0	52.9	0.325	0.344	.01	5.72	50	A	
cis-1,3-Dichloropropene	50.0	51.8	0.408	0.422	.01	3.5	50	A	
m,p-Xylene	100.0	107.0	1.269	1.352	.01	6.51	50	A	
n-Butylbenzene	50.0	51.8	3.999	4.140	.01	3.52	50	A	
n-Propylbenzene	50.0	52.8	3.905	4.127	.01	5.7	50	A	
o-Xylene	50.0	52.9	1.182	1.251	.01	5.83	50	A	
sec-Butylbenzene	50.0	53.4	3.397	3.631	.01	6.89	50	A	
tert-Butyl methyl ether (MTBE)	50.0	52.6	0.660	0.695	.01	5.23	50	A	
tert-Butylbenzene	50.0	50.5	1.475	1.488	.01	.92	50	A	
trans-1,2-Dichloroethene	50.0	54.0	0.338	0.365	.01	7.97	50	A	
trans-1,3-Dichloropropene	50.0	51.9	0.387	0.401	.01	3.78	50	A	

FORM V II VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230605.s.b/c0761.d
Lab Smp Id: 1440 Client Smp ID: V15STD050
Inj Date : 05-JUN-2023 20:27
Operator : KN1 Inst ID: msv15.i
Smp Info : 1440*V15STD050
Misc Info : MSV~52919~*1*SMR
Comment :
Method : /var/chem/msv15.i/230605.s.b/8260bw15DODclos.m
Meth Date : 06-Jun-2023 12:21 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 Continuing Calibration Sample
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.465	1.465	(0.249)	91092	50.0000	59.6	9694
2 Chloromethane ++	50	1.648	1.648	(0.280)	73766	50.0000	53.6	9827
3 Vinyl Chloride +	62	1.716	1.716	(0.292)	107523	50.0000	55.5	9797
5 1-3 Butadiene	39	1.735	1.735	(0.295)	86057	50.0000	59.6	9666
6 Bromomethane	94	2.015	2.015	(0.343)	75178	50.0000	46.3	9740
7 Chloroethane	64	2.134	2.134	(0.363)	77736	50.0000	54.6	9250
8 Trichlorofluoromethane	101	2.269	2.269	(0.386)	145277	50.0000	51.4	9739
11 1,1-Dichloroethene +	96	2.783	2.783	(0.473)	86638	50.0000	53.9	8382
12 Carbon Disulfide	76	2.812	2.812	(0.478)	263427	50.0000	53.6	8157
13 1,1,2Trichlotrifluoroethane	101	2.835	2.835	(0.482)	93317	50.0000	53.5	9349
15 Methyl Iodide	142	2.934	2.934	(0.499)	80990	50.0000	42.4	9536
16 Acrolein	56	3.150	3.150	(0.536)	60791	250.000	248	5488
17 Allyl chloride	39	3.304	3.304	(0.562)	78889	50.0000	55.8	9792
18 Methylene Chloride	49	3.420	3.420	(0.582)	94781	50.0000	53.3	4676

							AMOUNTS	
		QUANT SIG				CAL-AMT	ON-COL	
	Compounds	MASS	RT	EXP RT REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
	19 Acetone	43	3.468	3.468 (0.590)	201439	250.000	276	5274(H)
	20 trans-1,2-Dichloroethene	61	3.590	3.590 (0.611)	120280	50.0000	54.0	8447
	21 Methyl Acetate	43	3.610	3.610 (0.614)	271145	250.000	298	9710
	22 Hexane	57	3.684	3.684 (0.626)	176960	50.0000	55.1	9326
	23 MTBE	73	3.716	3.716 (0.632)	228619	50.0000	52.6	9633
	24 tert-Butyl Alcohol	59	3.819	3.819 (0.649)	52071	250.000	272	8073
	25 Acetonitrile	41	3.944	3.944 (0.671)	37195	250.000	288	9613
	26 Isopropyl Ether	45	4.111	4.111 (0.699)	215983	50.0000	55.7	9688
	27 Chloroprene	53	4.198	4.198 (0.714)	115953	50.0000	53.1	8435
	28 1,1-Dichloroethane ++	63	4.217	4.217 (0.717)	161279	50.0000	54.2	9522
	29 Acrylonitrile	53	4.262	4.262 (0.725)	146932	250.000	290	9856
	31 Ethyl Tert-butyl Ether	59	4.471	4.471 (0.760)	227243	50.0000	52.1	8479
	30 Vinyl Acetate	43	4.471	4.471 (0.760)	84846	50.0000	35.4	4668
	32 cis-1,2-Dichloroethene	61	4.735	4.735 (0.805)	113063	50.0000	52.9	9829
M	33 Total 1,2-Dichloroethene	61			233343	100.000	107	0
	34 2,2-Dichloropropane	77	4.838	4.838 (0.823)	127397	50.0000	48.2	9668
	35 Bromochloromethane	128	4.921	4.921 (0.837)	44173	50.0000	52.6	9037
	36 Cyclohexane	56	4.925	4.925 (0.838)	149891	50.0000	55.3	9128
	37 Chloroform +	83	4.992	4.992 (0.849)	157846	50.0000	51.9	9795
	39 Carbon Tetrachloride	117	5.118	5.118 (0.870)	127719	50.0000	51.9	8157
$	42 Dibromofluoromethane	111	5.159	5.159 (0.877)	83414	50.0000	51.3	8587
	43 1,1,1-Trichloroethane	97	5.182	5.182 (0.881)	145571	50.0000	51.7	9162
	45 1,1-Dichloropropene	75	5.294	5.294 (0.900)	133032	50.0000	53.1	9512
	44 2-Butanone	43	5.275	5.275 (0.897)	227394	250.000	300	5879
	46 2,2,4 Trimethylpentane	57	5.401	5.401 (0.919)	324004	50.0000	42.8	9309
	48 Benzene	78	5.516	5.516 (0.938)	362518	50.0000	53.5	9721
	56 tert-amyl Methyl Ether	73	5.632	5.632 (0.958)	232785	50.0000	52.4	7215
$	50 1,2-Dichloroethane-d4	65	5.632	5.632 (0.958)	82678	50.0000	48.8	8475
	52 1,2-Dichloroethane	62	5.693	5.693 (0.968)	108202	50.0000	50.8	9767
	51 Isobutyl Alcohol	43	5.732	5.732 (0.975)	12063	250.000	272	3055
*	55 FLUOROBENZENE	96	5.880	5.880 (1.000)	329152	50.0000		9844
	58 Methyl Cyclohexane	83	6.028	6.028 (1.025)	172473	50.0000	54.2	8301
	57 Trichloroethene	130	6.031	6.031 (1.026)	113591	50.0000	54.6	8907
	62 Dibromomethane	93	6.394	6.394 (1.087)	57996	50.0000	53.2	9448
	64 1,2-Dichloropropane +	63	6.481	6.481 (1.102)	91635	50.0000	54.7	7846
	65 Bromodichloromethane	83	6.539	6.539 (1.112)	119341	50.0000	52.6	9772
	67 1,4- Dioxane	88	6.719	6.719 (1.143)	17432	1250.00	1400	9318
	68 1-Bromo-2-chloroethane	63	6.944	6.944 (1.181)	115974	50.0000	54.4	9392
	72 2-Chloroethyl vinyl ether	63	7.024	7.024 (1.195)	90914	250.000	192	9771
	73 cis-1,3-Dichloropropene	75	7.073	7.073 (1.203)	138923	50.0000	51.8	9813
$	74 Toluene-d8	98	7.224	7.224 (0.866)	311979	50.0000	51.5	9901
	75 Toluene +	91	7.262	7.262 (0.871)	400674	50.0000	53.1	9648
	77 4-methyl-2-pentanone	43	7.548	7.548 (0.905)	372119	250.000	265	7598
	78 Tetrachloroethene	164	7.558	7.558 (0.906)	89138	50.0000	52.3	7982
	79 trans-1,3-Dichloropropene	75	7.574	7.574 (1.288)	132029	50.0000	51.9	8662
	81 1,1,2-Trichloroethane	97	7.693	7.693 (0.922)	81332	50.0000	53.1	9472
	82 Dibromochloromethane	129	7.825	7.825 (0.938)	96474	50.0000	55.3	9630

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
84 3,4-dichloro-1-butene	75	7.864	7.864	(0.943)	106761	50.0000	53.0	9708
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	137425	50.0000	53.9	9570
M 83 1-3 Dichloropropene total	100				270952	100.000	104	0
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	88881	50.0000	54.2	9735
89 2-Hexanone	43	8.143	8.143	(0.976)	308161	250.000	267	9740
90 1-Chlorohexane	91	8.333	8.333	(0.999)	144197	50.0000	53.5	7743
* 91 CHLOROBENZENE-d5	82	8.339	8.339	(1.000)	129300	50.0000		9128
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	261455	50.0000	52.4	9518
93 Ethylbenzene +	106	8.368	8.368	(1.003)	143270	50.0000	52.8	9073
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	85119	50.0000	50.7	9214
95 p,m-Xylene	106	8.465	8.465	(1.015)	349591	100.000	107	9915
97 o-Xylene	106	8.744	8.744	(1.049)	161734	50.0000	52.9	9906
101 Styrene	104	8.777	8.777	(1.052)	216492	50.0000	55.0	9874
102 Bromoform ++	173	8.799	8.799	(1.055)	67332	50.0000	47.9	8640
104 Isopropylbenzene	105	8.941	8.941	(1.072)	424298	50.0000	53.0	9843
M 105 TOTAL XYLENE	106				511325	150.000	159	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	97340	50.0000	49.5	9573
107 cis-1,4-dichloro-2-butene	53	9.156	9.156	(0.934)	30919	50.0000	50.8	9558
109 n-Propylbenzene	91	9.201	9.201	(0.939)	494489	50.0000	52.8	8622
108 Bromobenzene	77	9.195	9.195	(0.938)	158936	50.0000	51.0	9828
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.943)	100911	50.0000	55.2	9791
112 2-Chlorotoluene	91	9.301	9.301	(0.949)	309865	50.0000	52.0	7676
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	333138	50.0000	52.9	8035
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	126979	50.0000	51.2	8256
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	29030	50.0000	48.1	9231
116 Cyclohexanone	55	9.365	9.365	(0.955)	89708	250.000	296	9688
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	281867	50.0000	52.7	9866
118 tert-butylbenzene	91	9.513	9.513	(0.970)	178294	50.0000	50.5	9873
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	329551	50.0000	53.4	9351
120 sec-Butylbenzene	105	9.619	9.619	(0.981)	435074	50.0000	53.4	9858
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	377715	50.0000	52.4	8942
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.996)	192147	50.0000	52.8	9793
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	196722	50.0000	52.5	8917
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.802	(1.000)	119806	50.0000		9105
126 n-Butylbenzene	91	9.953	9.953	(1.015)	495974	50.0000	51.8	8893
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	173038	50.0000	52.5	9689
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	22720	50.0000	49.2	9424
130 Hexachlorobutadiene	225	10.995	10.995	(1.122)	53356	50.0000	53.0	9319
131 1,2,4-Trichlorobenzene	180	11.027	11.027	(1.125)	106398	50.0000	52.8	9520
132 Naphthalene	128	11.288	11.288	(1.152)	258194	50.0000	54.7	9871
133 1,2,3-Trichlorobenzene	180	11.439	11.439	(1.167)	94622	50.0000	54.5	9537

QC Flag Legend

H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230605.s.b/c0761.d

Page 1

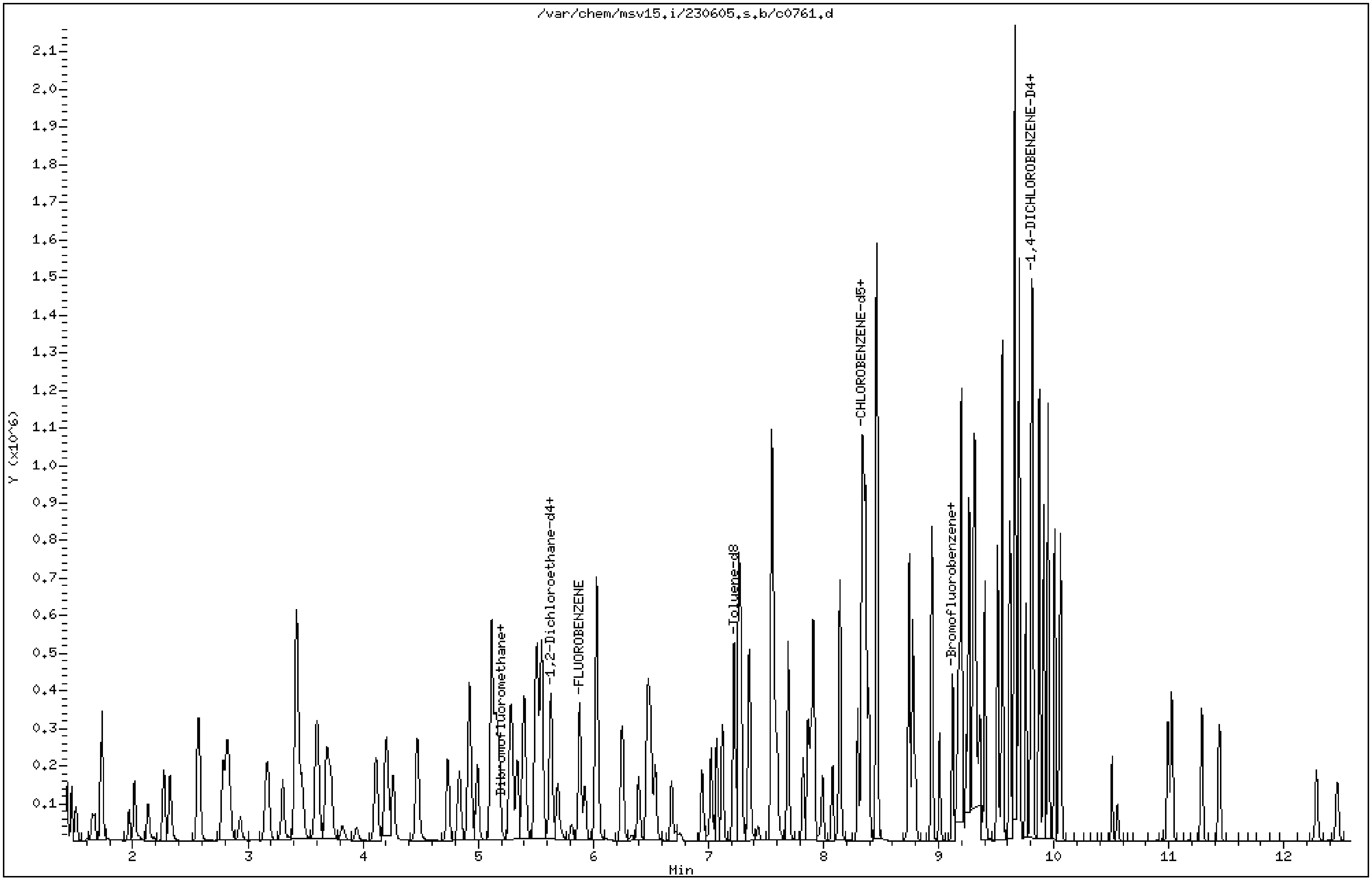
Date : 05-JUN-2023 20:27

Client ID: V15STD050 Instrument: msv15.i

Sample Info: 1440*V15STD050

Purge Volume: 5.0 Operator: KN1

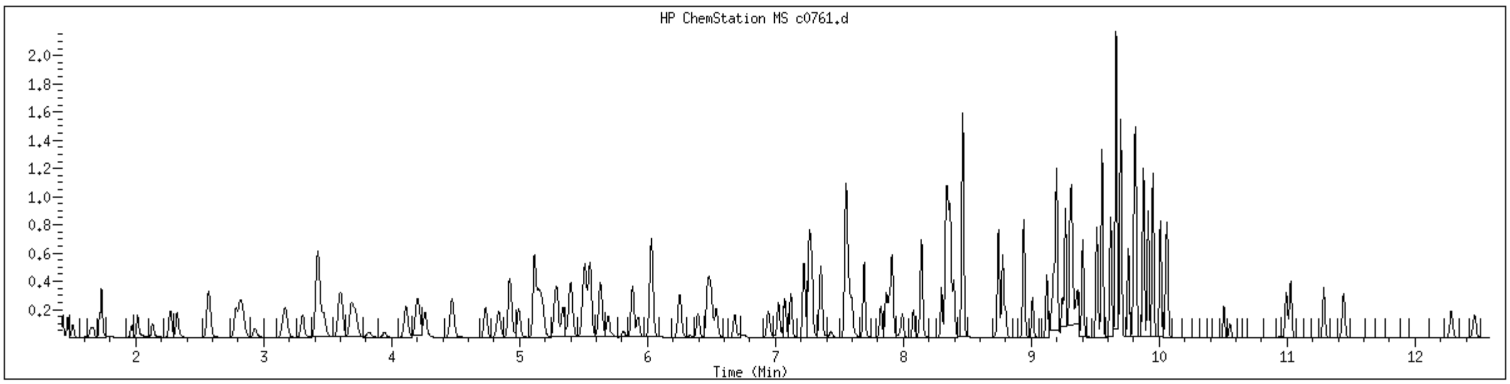
Column phase: RTX-VMS-30M Column diameter: 0.25



Data file : /var/chem/msv15.i/230605.s.b/c0761.d Page: 1
Report Date: 06/06/2023 12:21

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1440 SampleType : CCALIB_7
Injection Date: 06/05/2023 20:27 Instrument : msv15.i
Operator : KN1
Sample Info : 1440*V15STD050
Misc Info : MSV~52919~*1*SMR
Method : /var/chem/msv15.i/230605.s.b/8260bw15DODclos.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

7A
VOLATILE CONTINUING CALIBRATION CHECK

Report No:	223060210	CCAL ID:	1400
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230606/c0768d
Init. Calib. Date 1:	05/17/23 Time 1: 1428	Analyst:	KN1
Init. Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Batch:	767081
Analysis Date:	06/06/23 Time: 1053	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
1,1,1,2-Tetrachloroethane	50.0	51.3	0.649	0.666	.01	2.61	20	A	
1,1,1-Trichloroethane	50.0	51.8	0.428	0.443	.01	3.57	20	A	
1,1,2,2-Tetrachloroethane	50.0	52.4	0.763	0.799	.3	4.72	20	A	
1,1,2-Trichloroethane	50.0	51.8	0.593	0.614	.01	3.55	20	A	
1,1-Dichloroethane	50.0	53.7	0.452	0.485	.1	7.4	20	A	
1,1-Dichloroethene	50.0	53.4	0.244	0.261	.01	6.82	20	A	
1,1-Dichloropropene	50.0	52.7	0.381	0.401	.01	5.42	20	A	
1,2,3-Trichlorobenzene	50.0	51.1	0.725	0.741	.01	2.15	20	A	
1,2,3-Trichloropropane	50.0	48.1	1.035	0.997	.01	-3.72	20	A	
1,2,4-Trichlorobenzene	50.0	52.2	0.840	0.878	.01	4.42	20	A	
1,2,4-Trimethylbenzene	50.0	53.1	2.578	2.740	.01	6.29	20	A	
1,2-Dibromo-3-chloropropane	50.0	42.1	0.196	0.162	.01	-15.8	20	W	
1,2-Dibromoethane	50.0	52.3	0.634	0.663	.01	4.56	20	A	
1,2-Dichlorobenzene	50.0	52.2	1.375	1.434	.01	4.33	20	A	
1,2-Dichloroethane	50.0	50.5	0.324	0.327	.01	1	20	A	
1,2-Dichloroethane-d4	50.0	49.7	0.257	0.255	.01	-.68	20	A	
1,2-Dichloroethene (total)	100.0	107.0	0.332	0.355	.01	6.98	20	A	
1,2-Dichloropropane	50.0	55.3	0.254	0.281	.01	10.5	20	A	
1,3,5-Trimethylbenzene	50.0	53.5	2.626	2.810	.01	7	20	A	
1,3-Dichlorobenzene	50.0	52.3	1.520	1.591	.01	4.67	20	A	
1,3-Dichloropropane	50.0	51.4	0.987	1.014	.01	2.76	20	A	
1,4-Dichlorobenzene	50.0	51.9	1.563	1.622	.01	3.8	20	A	
2,2-Dichloropropane	50.0	43.6	0.402	0.351	.01	-12.7	20	A	
2-Butanone	250.0	246.0	0.115	0.113	.01	-1.58	20	A	
2-Chlorotoluene	50.0	52.5	2.489	2.615	.01	5.08	20	A	
2-Hexanone	250.0	232.0	0.450	0.412	.01	-7.2	20	W	
4-Bromofluorobenzene	50.0	49.8	0.761	0.757	.01	-.49	20	A	
4-Chlorotoluene	50.0	53.1	2.234	2.372	.01	6.18	20	A	
4-Isopropyltoluene	50.0	52.9	3.008	3.181	.01	5.76	20	A	
4-Methyl-2-pentanone	250.0	226.0	0.548	0.489	.01	-9.6	20	W	
Acetone	250.0	250.0	0.111	0.111	.01	.03	20	A	
Benzene	50.0	53.7	1.029	1.105	.01	7.4	20	A	
Bromobenzene	50.0	51.6	1.300	1.342	.01	3.26	20	A	
Bromochloromethane	50.0	53.0	0.128	0.135	.01	6.02	20	A	
Bromodichloromethane	50.0	54.2	0.344	0.374	.01	8.48	20	A	
Bromoform	50.0	46.0	0.554	0.500	.1	-8	20	W	
Bromomethane	50.0	42.6	0.247	0.211	.01	-14.7	20	A	
Carbon disulfide	50.0	52.8	0.747	0.788	.01	5.53	20	A	
Carbon tetrachloride	50.0	51.3	0.374	0.384	.01	2.69	20	A	

FORM V II VOA

7A
VOLATILE CONTINUING CALIBRATION CHECK

Report No:	223060210	CCAL ID:	1400
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230606/c0768d
Init. Calib. Date 1:	05/17/23 Time 1: 1428	Analyst:	KN1
Init. Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Batch:	767081
Analysis Date:	06/06/23 Time: 1053	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
Chlorobenzene	50.0	52.6	1.931	2.030	.3	5.12	20	A	
Chloroethane	50.0	50.6	0.216	0.219	.01	1.11	20	A	
Chloroform	50.0	52.8	0.462	0.488	.01	5.53	20	A	
Chloromethane	50.0	58.4	0.209	0.244	.1	16.8	20	A	
Dibromochloromethane	50.0	54.8	0.674	0.739	.01	9.66	20	A	
Dibromofluoromethane	50.0	52.7	0.247	0.260	.01	5.45	20	A	
Dibromomethane	50.0	53.4	0.166	0.177	.01	6.76	20	A	
Dichlorodifluoromethane	50.0	54.4	0.232	0.252	.01	8.78	20	A	
Ethylbenzene	50.0	52.8	1.050	1.109	.01	5.59	20	A	
Hexachlorobutadiene	50.0	52.7	0.420	0.443	.01	5.4	20	A	
Isopropylbenzene (Cumene)	50.0	53.3	3.094	3.295	.01	6.51	20	A	
Methylene chloride	50.0	53.2	0.270	0.288	.01	6.48	20	A	
Naphthalene	50.0	48.7	1.969	1.920	.01	-2.51	20	A	
Styrene	50.0	56.3	1.522	1.713	.01	12.5	20	A	
Tetrachloroethene	50.0	50.9	0.660	0.671	.01	1.8	20	A	
Toluene	50.0	52.6	2.917	3.070	.01	5.22	20	A	
Toluene-d8	50.0	51.5	2.343	2.415	.01	3.05	20	A	
Trichloroethene	50.0	53.6	0.316	0.339	.01	7.25	20	A	
Trichlorofluoromethane	50.0	49.8	0.429	0.427	.01	-.49	20	A	
Vinyl chloride	50.0	52.4	0.294	0.308	.01	4.72	20	A	
cis-1,2-Dichloroethene	50.0	53.5	0.325	0.347	.01	6.93	20	A	
cis-1,3-Dichloropropene	50.0	52.9	0.408	0.431	.01	5.71	20	A	
m,p-Xylene	100.0	107.0	1.269	1.352	.01	6.5	20	A	
n-Butylbenzene	50.0	52.2	3.999	4.174	.01	4.37	20	A	
n-Propylbenzene	50.0	52.4	3.905	4.089	.01	4.71	20	A	
o-Xylene	50.0	53.0	1.182	1.253	.01	6.05	20	A	
sec-Butylbenzene	50.0	53.1	3.397	3.611	.01	6.27	20	A	
tert-Butyl methyl ether (MTBE)	50.0	43.0	0.660	0.567	.01	-14.0	20	A	
tert-Butylbenzene	50.0	51.0	1.475	1.503	.01	1.93	20	A	
trans-1,2-Dichloroethene	50.0	53.5	0.338	0.362	.01	7.04	20	A	
trans-1,3-Dichloropropene	50.0	52.3	0.387	0.404	.01	4.57	20	A	

FORM V II VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230606.s.b/c0768d.d
Lab Smp Id: 1400 Client Smp ID: V15STD050
Inj Date : 06-JUN-2023 10:53
Operator : KN1 Inst ID: msv15.i
Smp Info : 1400*V15STD050*51363
Misc Info : MSV~529~*1*SMR
Comment :
Method : /var/chem/msv15.i/230606.s.b/8260bw15DOD.m
Meth Date : 07-Jun-2023 11:45 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 Continuing Calibration Sample
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

	QUANT SIG					AMOUNTS		
						CAL-AMT	ON-COL	
Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.469	1.469	(0.250)	80445	50.0000	54.4	9734
2 Chloromethane ++	50	1.649	1.649	(0.280)	77812	50.0000	58.4	9771
3 Vinyl Chloride +	62	1.719	1.719	(0.292)	98210	50.0000	52.4	9798
5 1-3 Butadiene	39	1.735	1.735	(0.295)	71773	50.0000	51.4	9645
6 Bromomethane	94	2.018	2.018	(0.343)	67083	50.0000	42.6	9774
7 Chloroethane	64	2.137	2.137	(0.363)	69699	50.0000	50.6	9245
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	136042	50.0000	49.8	9667
11 1,1-Dichloroethene +	96	2.787	2.787	(0.474)	83085	50.0000	53.4	8429
12 Carbon Disulfide	76	2.816	2.816	(0.479)	251105	50.0000	52.8	7961
13 1,1,2Trichlotrifluoroethane	101	2.832	2.832	(0.481)	90035	50.0000	53.3	8782
15 Methyl Iodide	142	2.935	2.935	(0.499)	72665	50.0000	39.4	9532
16 Acrolein	56	3.150	3.150	(0.535)	58938	250.000	249	5377
17 Allyl chloride	39	3.304	3.304	(0.562)	72555	50.0000	53.1	9797
18 Methylene Chloride	49	3.420	3.420	(0.581)	91654	50.0000	53.2	4646

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
	19 Acetone	43	3.468	3.468	(0.590)	176697	250.000	250	5353(H)
	20 trans-1,2-Dichloroethene	61	3.591	3.591	(0.610)	115417	50.0000	53.5	8694
	21 Methyl Acetate	43	3.610	3.610	(0.614)	217268	250.000	247	9429
	22 Hexane	57	3.687	3.687	(0.627)	165661	50.0000	53.3	9217
	23 MTBE	73	3.719	3.719	(0.632)	180671	50.0000	43.0	9734
	24 tert-Butyl Alcohol	59	3.822	3.822	(0.650)	22592	250.000	122	8219
	25 Acetonitrile	41	3.944	3.944	(0.670)	30669	250.000	245	9546
	26 Isopropyl Ether	45	4.112	4.112	(0.699)	207236	50.0000	55.2	9624
	27 Chloroprene	53	4.195	4.195	(0.713)	114782	50.0000	54.3	8866
	28 1,1-Dichloroethane ++	63	4.214	4.214	(0.716)	154640	50.0000	53.7	9472
	29 Acrylonitrile	53	4.259	4.259	(0.724)	124391	250.000	253	9800
	31 Ethyl Tert-butyl Ether	59	4.468	4.468	(0.760)	207414	50.0000	49.1	8113
	30 Vinyl Acetate	43	4.472	4.472	(0.760)	116251	50.0000	50.1	5591
	32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	110691	50.0000	53.5	9801
M	33 Total 1,2-Dichloroethene	61				226108	100.000	107	0
	34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	111698	50.0000	43.6	9539
	35 Bromochloromethane	128	4.919	4.919	(0.836)	43088	50.0000	53.0	9385
	36 Cyclohexane	56	4.928	4.928	(0.838)	141619	50.0000	53.9	9440
	37 Chloroform +	83	4.992	4.992	(0.849)	155442	50.0000	52.8	9790
	39 Carbon Tetrachloride	117	5.118	5.118	(0.870)	122271	50.0000	51.3	8290
$	42 Dibromofluoromethane	111	5.160	5.160	(0.877)	82958	50.0000	52.7	9061
	43 1,1,1-Trichloroethane	97	5.179	5.179	(0.880)	141267	50.0000	51.8	9506
	45 1,1-Dichloropropene	75	5.292	5.292	(0.899)	127840	50.0000	52.7	9533
	44 2-Butanone	43	5.275	5.275	(0.897)	180439	250.000	246	6451
	46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	284971	50.0000	38.9	9769
	48 Benzene	78	5.513	5.513	(0.937)	352036	50.0000	53.7	9538
	56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	215120	50.0000	50.0	7355
$	50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	81357	50.0000	49.7	8590
	52 1,2-Dichloroethane	62	5.690	5.690	(0.967)	104140	50.0000	50.5	9817
	51 Isobutyl Alcohol	43	5.726	5.726	(0.973)	8392	250.000	196	3912
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	318605	50.0000		9837
	58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	167543	50.0000	54.4	8276
	57 Trichloroethene	130	6.034	6.034	(1.026)	107967	50.0000	53.6	9138
	62 Dibromomethane	93	6.398	6.398	(1.087)	56311	50.0000	53.4	9454
	64 1,2-Dichloropropane +	63	6.484	6.484	(1.102)	89550	50.0000	55.3	8467
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	119043	50.0000	54.2	9779
	67 1,4- Dioxane	88	6.716	6.716	(1.142)	13763	1250.00	1150	9354
	68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	112170	50.0000	54.3	8813
	72 2-Chloroethyl vinyl ether	63	7.021	7.021	(1.193)	21455	250.000	50.1	9652
	73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	137334	50.0000	52.9	9810
$	74 Toluene-d8	98	7.221	7.221	(0.866)	306041	50.0000	51.5	9901
	75 Toluene +	91	7.263	7.263	(0.871)	389066	50.0000	52.6	9625
	77 4-methyl-2-pentanone	43	7.549	7.549	(0.905)	309955	250.000	226	7919
	78 Tetrachloroethene	164	7.558	7.558	(0.906)	85105	50.0000	50.9	8425
	79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	128770	50.0000	52.3	8717
	81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	77770	50.0000	51.8	9561
	82 Dibromochloromethane	129	7.825	7.825	(0.938)	93680	50.0000	54.8	9629

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	104971	50.0000	53.1	9687
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	128505	50.0000	51.4	9576
M 83 1-3 Dichloropropene total	100				266104	100.000	105	0
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	84033	50.0000	52.3	9750
89 2-Hexanone	43	8.143	8.143	(0.976)	261397	250.000	232	9766
90 1-Chlorohexane	91	8.333	8.333	(0.999)	134371	50.0000	50.8	7798
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	126742	50.0000		9139
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	257255	50.0000	52.6	9437
93 Ethylbenzene +	106	8.369	8.369	(1.003)	140507	50.0000	52.8	9084
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	84374	50.0000	51.3	9205
95 p,m-Xylene	106	8.465	8.465	(1.015)	342660	100.000	107	9924
97 o-Xylene	106	8.745	8.745	(1.049)	158864	50.0000	53.0	9900
101 Styrene	104	8.777	8.777	(1.052)	217060	50.0000	56.3	9877
102 Bromoform ++	173	8.799	8.799	(1.055)	63321	50.0000	46.0	8672
104 Isopropylbenzene	105	8.944	8.944	(1.072)	417658	50.0000	53.3	9904
M 105 TOTAL XYLENE	106				501524	150.000	160	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	95976	50.0000	49.8	9553
107 cis-1,4-dichloro-2-butene	53	9.160	9.160	(0.934)	30327	50.0000	50.4	9546
109 n-Propylbenzene	91	9.201	9.201	(0.938)	484333	50.0000	52.4	8800
108 Bromobenzene	77	9.195	9.195	(0.938)	158960	50.0000	51.6	9707
110 1,1,2,2-Tetrachloroethane++	83	9.243	9.243	(0.943)	94635	50.0000	52.4	9812
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	309827	50.0000	52.5	8340
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	332891	50.0000	53.5	8231
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.951)	118062	50.0000	48.1	7494
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.953)	26724	50.0000	44.8	9350
116 Cyclohexanone	55	9.365	9.365	(0.955)	70414	250.000	235	9701
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	280949	50.0000	53.1	9817
118 tert-butylbenzene	91	9.516	9.516	(0.970)	178055	50.0000	51.0	9895
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.974)	324588	50.0000	53.1	9366
120 sec-Butylbenzene	105	9.623	9.623	(0.981)	427714	50.0000	53.1	9869
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	376821	50.0000	52.9	8920
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.995)	188425	50.0000	52.3	9693
125 1,4-Dichlorobenzene	146	9.815	9.815	(1.001)	192196	50.0000	51.9	9156
* 124 1,4-DICHLOROBENZENE-D4	152	9.806	9.806	(1.000)	118459	50.0000		8889
126 n-Butylbenzene	91	9.954	9.954	(1.015)	494437	50.0000	52.2	8940
127 1,2-Dichlorobenzene	146	10.066	10.066	(1.027)	169903	50.0000	52.2	9731
129 1,2-Dibromo-3-Chloropropane	157	10.558	10.558	(1.077)	19151	50.0000	42.1	9522
130 Hexachlorobutadiene	225	10.999	10.999	(1.122)	52472	50.0000	52.7	9283
131 1,2,4-Trichlorobenzene	180	11.031	11.031	(1.125)	103951	50.0000	52.2	9544
132 Naphthalene	128	11.291	11.291	(1.151)	227415	50.0000	48.7	9835
133 1,2,3-Trichlorobenzene	180	11.446	11.446	(1.167)	87720	50.0000	51.1	9485

QC Flag Legend

H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230606.s.b/c0768d.d

Date : 06-JUN-2023 10:53

Client ID: V15STD050

Sample Info: 1400*V15STD050*51363

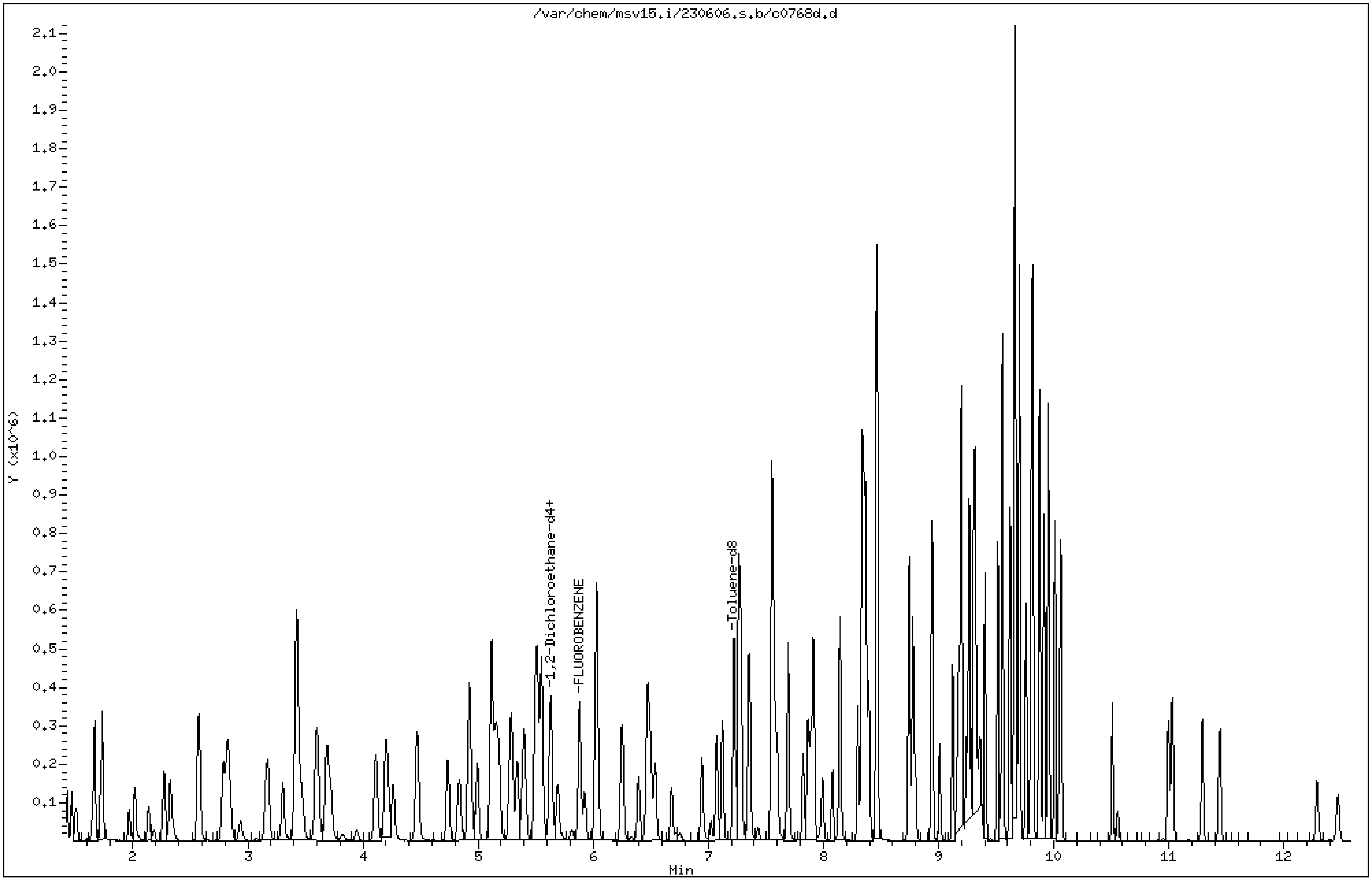
Purge Volume: 5.0

Column phase: RTX-VMS-30M

Instrument: msv15.i

Operator: KN1

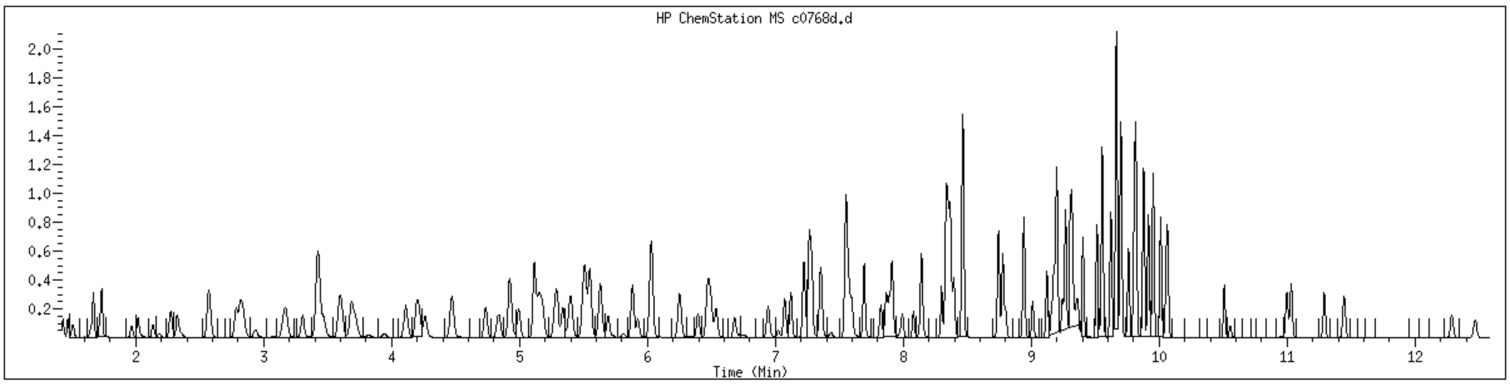
Column diameter: 0.25



Data file : /var/chem/msv15.i/230606.s.b/c0768d.d Page: 1
Report Date: 06/07/2023 11:45

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1400 SampleType : CCALIB_7
Injection Date: 06/06/2023 10:53 Instrument : msv15.i
Operator : KN1
Sample Info : 1400*V15STD050*51363
Misc Info : MSV~529~*1*SMR
Method : /var/chem/msv15.i/230606.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

7A

VOLATILE CLOSING CALIBRATION CHECK

Report No:	223060210	CCAL ID:	1440
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230606/c0796
Init. Calib. Date 1:	05/17/23 Time 1: 1428	Analyst:	KN1
Init. Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Batch:	767081
Analysis Date:	06/06/23 Time: 1939	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
1,1,1,2-Tetrachloroethane	50.0	49.2	0.649	0.638	.01	-1.69	50	A	
1,1,1-Trichloroethane	50.0	50.7	0.428	0.434	.01	1.35	50	A	
1,1,2,2-Tetrachloroethane	50.0	53.3	0.763	0.814	.3	6.67	50	A	
1,1,2-Trichloroethane	50.0	52.3	0.593	0.620	.01	4.61	50	A	
1,1-Dichloroethane	50.0	54.8	0.452	0.495	.1	9.63	50	A	
1,1-Dichloroethene	50.0	52.8	0.244	0.258	.01	5.57	50	A	
1,1-Dichloropropene	50.0	53.6	0.381	0.408	.01	7.26	50	A	
1,2,3-Trichlorobenzene	50.0	51.3	0.725	0.743	.01	2.51	50	A	
1,2,3-Trichloropropane	50.0	48.6	1.035	1.005	.01	-2.9	50	A	
1,2,4-Trichlorobenzene	50.0	51.5	0.840	0.866	.01	3.03	50	A	
1,2,4-Trimethylbenzene	50.0	53.1	2.578	2.740	.01	6.29	50	A	
1,2-Dibromo-3-chloropropane	50.0	45.2	0.196	0.174	.01	-9.6	50	W	
1,2-Dibromoethane	50.0	52.0	0.634	0.660	.01	4.1	50	A	
1,2-Dichlorobenzene	50.0	52.5	1.375	1.444	.01	5.02	50	A	
1,2-Dichloroethane	50.0	52.1	0.324	0.337	.01	4.13	50	A	
1,2-Dichloroethane-d4	50.0	50.7	0.257	0.261	.01	1.34	50	A	
1,2-Dichloroethene (total)	100.0	108.0	0.332	0.358	.01	8.05	50	A	
1,2-Dichloropropane	50.0	56.0	0.254	0.285	.01	12	50	A	
1,3,5-Trimethylbenzene	50.0	53.2	2.626	2.793	.01	6.36	50	A	
1,3-Dichlorobenzene	50.0	52.1	1.520	1.583	.01	4.18	50	A	
1,3-Dichloropropane	50.0	51.5	0.987	1.016	.01	3	50	A	
1,4-Dichlorobenzene	50.0	51.8	1.563	1.620	.01	3.64	50	A	
2,2-Dichloropropane	50.0	46.1	0.402	0.370	.01	-7.79	50	A	
2-Butanone	250.0	274.0	0.115	0.126	.01	9.57	50	A	
2-Chlorotoluene	50.0	51.8	2.489	2.578	.01	3.56	50	A	
2-Hexanone	250.0	243.0	0.450	0.434	.01	-2.8	50	W	
4-Bromofluorobenzene	50.0	49.1	0.761	0.748	.01	-1.76	50	A	
4-Chlorotoluene	50.0	53.0	2.234	2.368	.01	6.01	50	A	
4-Isopropyltoluene	50.0	52.4	3.008	3.152	.01	4.81	50	A	
4-Methyl-2-pentanone	250.0	244.0	0.548	0.530	.01	-2.4	50	W	
Acetone	250.0	256.0	0.111	0.041	.01	-63.0	50	A	*
Benzene	50.0	54.8	1.029	1.128	.01	9.6	50	A	
Bromobenzene	50.0	51.8	1.300	1.345	.01	3.52	50	A	
Bromochloromethane	50.0	54.0	0.128	0.138	.01	8.08	50	A	
Bromodichloromethane	50.0	54.0	0.344	0.372	.01	7.99	50	A	
Bromoform	50.0	45.2	0.554	0.491	.1	-9.6	50	W	
Bromomethane	50.0	37.0	0.247	0.183	.01	-25.9	50	A	
Carbon disulfide	50.0	53.4	0.747	0.798	.01	6.82	50	A	
Carbon tetrachloride	50.0	50.2	0.374	0.375	.01	.47	50	A	

FORM V II VOA

7A
VOLATILE CLOSING CALIBRATION CHECK

Report No:	223060210	CCAL ID:	1440
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230606/c0796
Init. Calib. Date 1:	05/17/23 Time 1: 1428	Analyst:	KN1
Init. Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Batch:	767081
Analysis Date:	06/06/23 Time: 1939	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
Chlorobenzene	50.0	51.4	1.931	1.984	.3	2.77	50	A	
Chloroethane	50.0	55.1	0.216	0.239	.01	10.3	50	A	
Chloroform	50.0	52.7	0.462	0.488	.01	5.49	50	A	
Chloromethane	50.0	55.6	0.209	0.232	.1	11.1	50	A	
Dibromochloromethane	50.0	53.3	0.674	0.719	.01	6.63	50	A	
Dibromofluoromethane	50.0	52.9	0.247	0.261	.01	5.8	50	A	
Dibromomethane	50.0	53.9	0.166	0.178	.01	7.82	50	A	
Dichlorodifluoromethane	50.0	58.9	0.232	0.273	.01	17.7	50	A	
Ethylbenzene	50.0	52.0	1.050	1.091	.01	3.92	50	A	
Hexachlorobutadiene	50.0	50.5	0.420	0.425	.01	1.08	50	A	
Isopropylbenzene (Cumene)	50.0	52.1	3.094	3.225	.01	4.23	50	A	
Methylene chloride	50.0	55.3	0.270	0.299	.01	10.6	50	A	
Naphthalene	50.0	51.4	1.969	2.026	.01	2.87	50	A	
Styrene	50.0	54.1	1.522	1.647	.01	8.22	50	A	
Tetrachloroethene	50.0	48.6	0.660	0.641	.01	-2.76	50	A	
Toluene	50.0	51.8	2.917	3.022	.01	3.58	50	A	
Toluene-d8	50.0	50.9	2.343	2.386	.01	1.82	50	A	
Trichloroethene	50.0	53.9	0.316	0.341	.01	7.8	50	A	
Trichlorofluoromethane	50.0	50.8	0.429	0.436	.01	1.65	50	A	
Vinyl chloride	50.0	56.7	0.294	0.334	.01	13.4	50	A	
cis-1,2-Dichloroethene	50.0	53.3	0.325	0.346	.01	6.63	50	A	
cis-1,3-Dichloropropene	50.0	52.3	0.408	0.426	.01	4.59	50	A	
m,p-Xylene	100.0	103.0	1.269	1.311	.01	3.32	50	A	
n-Butylbenzene	50.0	50.7	3.999	4.057	.01	1.45	50	A	
n-Propylbenzene	50.0	52.8	3.905	4.126	.01	5.65	50	A	
o-Xylene	50.0	52.0	1.182	1.229	.01	4.01	50	A	
sec-Butylbenzene	50.0	52.8	3.397	3.584	.01	5.5	50	A	
tert-Butyl methyl ether (MTBE)	50.0	51.6	0.660	0.681	.01	3.23	50	A	
tert-Butylbenzene	50.0	50.1	1.475	1.478	.01	.24	50	A	
trans-1,2-Dichloroethene	50.0	54.7	0.338	0.370	.01	9.41	50	A	
trans-1,3-Dichloropropene	50.0	51.2	0.387	0.396	.01	2.43	50	A	

FORM V II VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230606.s.b/c0796.d
Lab Smp Id: 1440 Client Smp ID: V15STD050
Inj Date : 06-JUN-2023 19:39
Operator : KN1 Inst ID: msv15.i
Smp Info : 1440*V15STD050
Misc Info : MSV~52923~*1*SMR
Comment :
Method : /var/chem/msv15.i/230606.s.b/8260bw15DODclos.m
Meth Date : 07-Jun-2023 11:19 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 Continuing Calibration Sample
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	83612	50.0000	58.9	9670
2 Chloromethane ++	50	1.648	1.648	(0.280)	71085	50.0000	55.6	9827
3 Vinyl Chloride +	62	1.716	1.716	(0.292)	102114	50.0000	56.7	9871
5 1-3 Butadiene	39	1.735	1.735	(0.295)	80373	50.0000	59.9	9695
6 Bromomethane	94	2.015	2.015	(0.343)	55899	50.0000	37.0	9852
7 Chloroethane	64	2.134	2.134	(0.363)	73005	50.0000	55.1	9167
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	133452	50.0000	50.8	9674
11 1,1-Dichloroethene +	96	2.783	2.783	(0.473)	78856	50.0000	52.8	8303
12 Carbon Disulfide	76	2.816	2.816	(0.479)	244107	50.0000	53.4	7983
13 1,1,2Trichlotrifluoroethane	101	2.832	2.832	(0.481)	85927	50.0000	53.0	8917
15 Methyl Iodide	142	2.931	2.931	(0.498)	81247	50.0000	45.6	9556
16 Acrolein	56	3.150	3.150	(0.535)	53739	250.000	236	5504
17 Allyl chloride	39	3.301	3.301	(0.561)	72379	50.0000	55.1	9796
18 Methylene Chloride	49	3.417	3.417	(0.581)	91399	50.0000	55.3	5070

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
19 Acetone	43	3.468	3.430	(0.590)	173676	250.000	256	5472(H)
20 trans-1,2-Dichloroethene	61	3.590	3.590	(0.610)	113304	50.0000	54.7	8567
21 Methyl Acetate	43	3.610	3.610	(0.614)	245747	250.000	291	9567
22 Hexane	57	3.687	3.687	(0.627)	163285	50.0000	54.7	9352
23 MTBE	73	3.716	3.716	(0.632)	208495	50.0000	51.6	9481
24 tert-Butyl Alcohol	59	3.822	3.822	(0.650)	40993	250.000	230	8052
25 Acetonitrile	41	3.944	3.944	(0.670)	35225	250.000	293	9586
26 Isopropyl Ether	45	4.111	4.111	(0.699)	204892	50.0000	56.9	9717
27 Chloroprene	53	4.195	4.195	(0.713)	107071	50.0000	52.7	8874
28 1,1-Dichloroethane ++	63	4.217	4.217	(0.717)	151595	50.0000	54.8	9524
29 Acrylonitrile	53	4.262	4.262	(0.725)	134998	250.000	286	9860
31 Ethyl Tert-butyl Ether	59	4.471	4.471	(0.760)	207649	50.0000	51.2	8562
30 Vinyl Acetate	43	4.475	4.475	(0.761)	71943	50.0000	32.3	4450
32 cis-1,2-Dichloroethene	61	4.738	4.738	(0.805)	106004	50.0000	53.3	9824
M 33 Total 1,2-Dichloroethene	61				219308	100.000	108	0
34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	113335	50.0000	46.1	9565
35 Bromochloromethane	128	4.922	4.922	(0.837)	42182	50.0000	54.0	9282
36 Cyclohexane	56	4.928	4.928	(0.838)	140964	50.0000	55.9	9288
37 Chloroform +	83	4.995	4.995	(0.849)	149220	50.0000	52.7	9825
39 Carbon Tetrachloride	117	5.121	5.121	(0.870)	114884	50.0000	50.2	8125
$ 42 Dibromofluoromethane	111	5.159	5.159	(0.877)	79935	50.0000	52.9	8638
43 1,1,1-Trichloroethane	97	5.182	5.182	(0.881)	132770	50.0000	50.7	9195
45 1,1-Dichloropropene	75	5.291	5.291	(0.899)	124923	50.0000	53.6	9432
44 2-Butanone	43	5.275	5.275	(0.897)	192926	250.000	274	5922
46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	297879	50.0000	42.3	9297
48 Benzene	78	5.513	5.513	(0.937)	345001	50.0000	54.8	9358
56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	212395	50.0000	51.4	7315
$ 50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	79718	50.0000	50.7	8485
52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	103115	50.0000	52.1	9734
51 Isobutyl Alcohol	43	5.729	5.729	(0.974)	9650	250.000	234	3558
* 55 FLUOROBENZENE	96	5.883	5.883	(1.000)	305977	50.0000		9845
58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	159012	50.0000	53.7	8293
57 Trichloroethene	130	6.031	6.031	(1.025)	104225	50.0000	53.9	8904
62 Dibromomethane	93	6.394	6.394	(1.087)	54618	50.0000	53.9	9407
64 1,2-Dichloropropane +	63	6.481	6.481	(1.102)	87118	50.0000	56.0	7850
65 Bromodichloromethane	83	6.539	6.539	(1.111)	113807	50.0000	54.0	9762
67 1,4- Dioxane	88	6.719	6.719	(1.142)	11965	1250.00	1040	9034
68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	109993	50.0000	55.5	9470
72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	104489	250.000	236	9751
73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	130498	50.0000	52.3	9787
$ 74 Toluene-d8	98	7.220	7.220	(0.866)	295410	50.0000	50.9	9868
75 Toluene +	91	7.262	7.262	(0.871)	374127	50.0000	51.8	9647
77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	327787	250.000	244	8189
78 Tetrachloroethene	164	7.558	7.558	(0.906)	79410	50.0000	48.6	8086
79 trans-1,3-Dichloropropene	75	7.577	7.577	(1.288)	121138	50.0000	51.2	9177
81 1,1,2-Trichloroethane	97	7.696	7.696	(0.923)	76750	50.0000	52.3	9576
82 Dibromochloromethane	129	7.822	7.822	(0.938)	88982	50.0000	53.3	9603

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	97230	50.0000	50.4	9738
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	125819	50.0000	51.5	9550
M 83 1-3 Dichloropropene total	100				251636	100.000	104	0
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	81720	50.0000	52.0	9720
89 2-Hexanone	43	8.143	8.143	(0.976)	268428	250.000	243	9784
90 1-Chlorohexane	91	8.333	8.333	(0.999)	132481	50.0000	51.3	7820
* 91 CHLOROBENZENE-d5	82	8.339	8.339	(1.000)	123809	50.0000		9136
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	245678	50.0000	51.4	9357
93 Ethylbenzene +	106	8.368	8.368	(1.003)	135078	50.0000	52.0	9116
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	78969	50.0000	49.2	9240
95 p,m-Xylene	106	8.465	8.465	(1.015)	324747	100.000	103	9914
97 o-Xylene	106	8.745	8.745	(1.049)	152207	50.0000	52.0	9896
101 Styrene	104	8.777	8.777	(1.052)	203920	50.0000	54.1	9863
102 Bromoform ++	173	8.799	8.799	(1.055)	60752	50.0000	45.2	8698
104 Isopropylbenzene	105	8.944	8.944	(1.072)	399229	50.0000	52.1	9879
M 105 TOTAL XYLENE	106				476954	150.000	155	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	92567	50.0000	49.1	9465
107 cis-1,4-dichloro-2-butene	53	9.159	9.159	(0.934)	26541	50.0000	46.0	9572
109 n-Propylbenzene	91	9.201	9.201	(0.939)	469137	50.0000	52.8	8748
108 Bromobenzene	77	9.195	9.195	(0.938)	152972	50.0000	51.8	9777
110 1,1,2,2-Tetrachloroethane++	83	9.243	9.243	(0.943)	92534	50.0000	53.3	9789
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	293119	50.0000	51.8	8352
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	317644	50.0000	53.2	8278
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	114306	50.0000	48.6	7665
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	26071	50.0000	45.5	9296
116 Cyclohexanone	55	9.365	9.365	(0.955)	73677	250.000	256	9767
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	269248	50.0000	53.0	9847
118 tert-butylbenzene	91	9.513	9.513	(0.970)	168091	50.0000	50.1	9861
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	311571	50.0000	53.1	9355
120 sec-Butylbenzene	105	9.622	9.622	(0.982)	407589	50.0000	52.8	9842
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	358472	50.0000	52.4	8925
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.996)	180026	50.0000	52.1	9700
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	184209	50.0000	51.8	8693
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.802	(1.000)	113712	50.0000		9156
126 n-Butylbenzene	91	9.953	9.953	(1.015)	461334	50.0000	50.7	9025
127 1,2-Dichlorobenzene	146	10.066	10.066	(1.027)	164183	50.0000	52.5	9758
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	19754	50.0000	45.2	9440
130 Hexachlorobutadiene	225	10.998	10.998	(1.122)	48305	50.0000	50.5	9340
131 1,2,4-Trichlorobenzene	180	11.031	11.031	(1.125)	98459	50.0000	51.5	9513
132 Naphthalene	128	11.288	11.288	(1.152)	230351	50.0000	51.4	9826
133 1,2,3-Trichlorobenzene	180	11.442	11.442	(1.167)	84495	50.0000	51.3	9494

QC Flag Legend

H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230606.s.b/c0796.d

Date : 06-JUN-2023 19:39

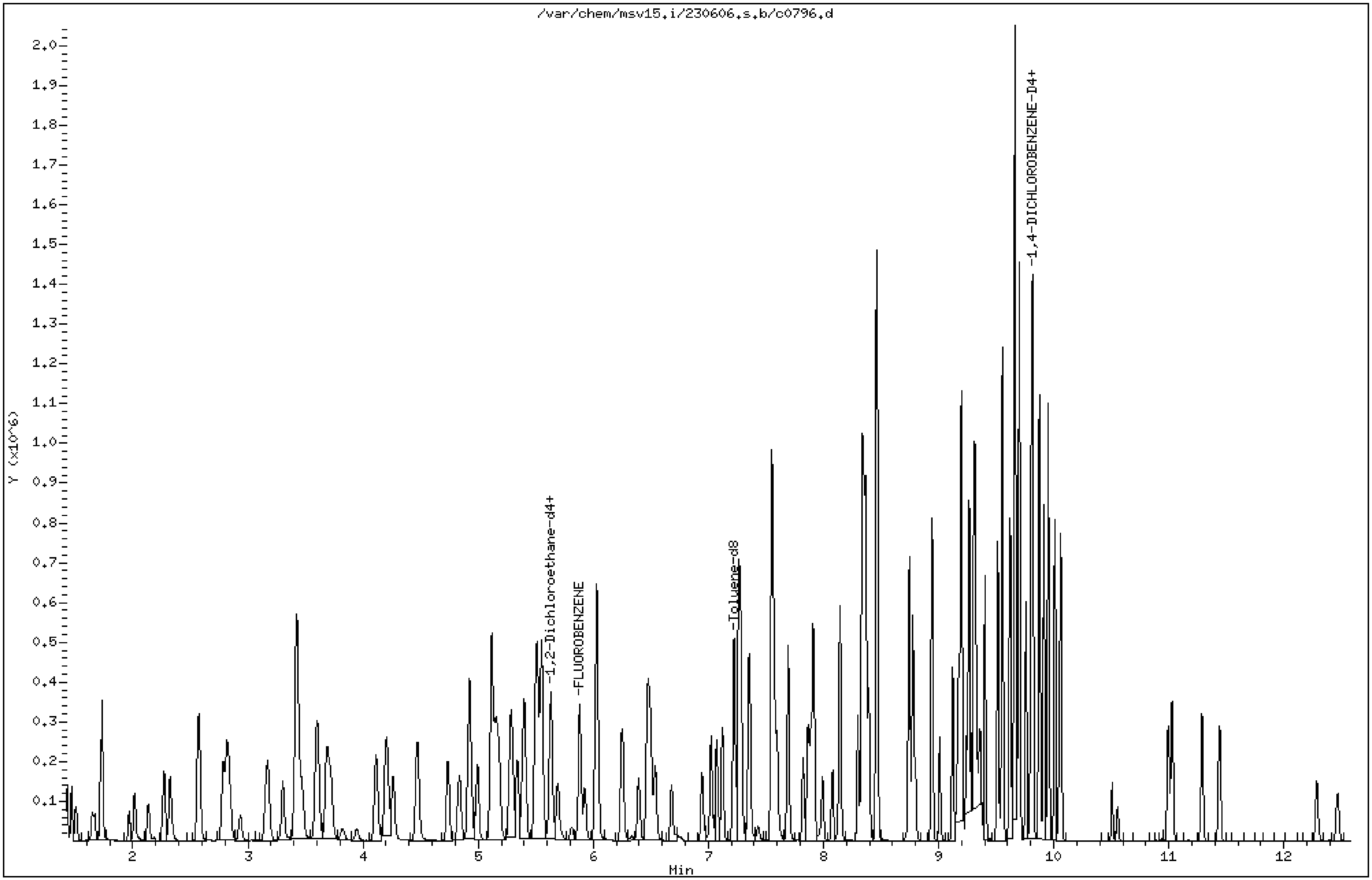
Client ID: V15STD050 | Instrument: msv15.i

Sample Info: 1440*V15STD050

Purge Volume: 5.0 | Operator: KN1

Column phase: RTX-VMS-30M | Column diameter: 0.25

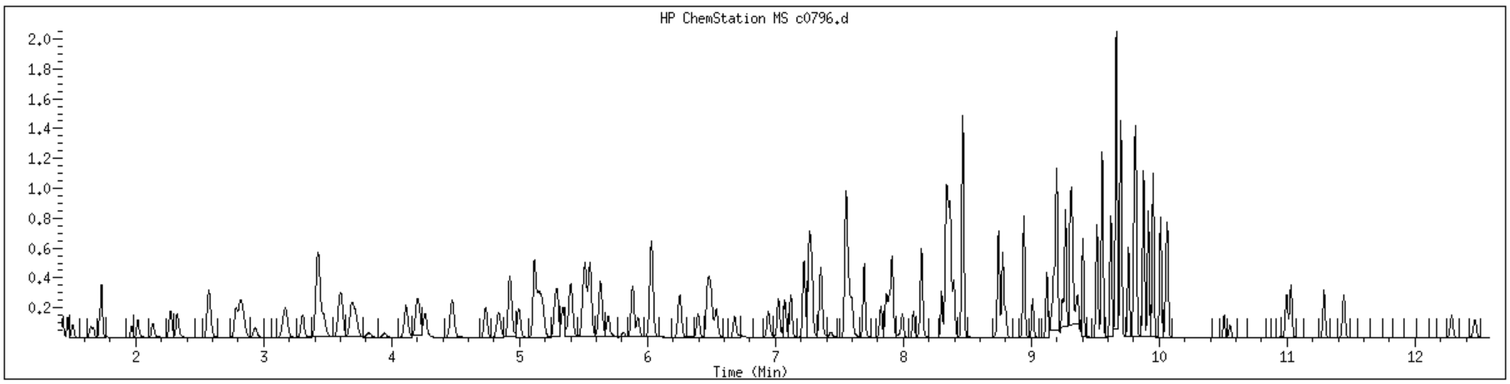
Page 1



Report Date: 06/07/2023 11:42

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 1440   SampleType   : CCALIB_7
Injection Date: 06/06/2023 19:39   Instrument   : msv15.i
Operator   : KN1
Sample Info   : 1440*V15STD050
Misc Info   : MSV~52923~*1*SMR
Method   : /var/chem/msv15.i/230606.s.b/8260bw15DODclos.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

EPA 8260B DOD

Form 8A

Internal Standards

8A
VOLATILE INTERNAL STANDARD AREA AND RT SUMMARY

Report No:	223060210	Standard ID:	1400
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV11
Injection Vol.:	1.0 (µL)	Lab File ID:	230602/j1687
Analyst:	KN1	Analytical Batch:	766942
Analysis Date:	06/02/23 Time: 1123	Analytical Method:	EPA 8260B

	IS 1 Area	RT	IS 2 Area	RT	IS 3 Area	RT
STANDARD	131646	9.24	134139	10.93	334510	6.51

CLIENT SAMPLE ID	GCAL SAMP ID	IS 1 Area	#	RT	#	IS 2 Area	#	RT	#	IS 3 Area	#	RT	#
LCS2489346	2489346	131646		9.24		134139		10.93		334510		6.51	
MB2489345	2489345	106188		9.24		104077		10.93		269845		6.5	
OT07-EB-060123	22306021001	108920		9.24		104291		10.93		276659		6.51	
OT07-MW02-060123	22306021002	105220		9.24		99785		10.93		263558		6.51	
OT07-MW04-060123	22306021003	107093		9.24		102497		10.93		259530		6.51	
OT07-MW05-060123	22306021004	106395		9.24		104730		10.93		266645		6.51	
OT07-MW06-060123	22306021005	105727		9.24		103869		10.93		268179		6.51	
OT07-MW04-060123MS	22306021006	131582		9.24		139318		10.93		313664		6.51	
OT07-MW04-060123MSD	22306021007	134171		9.24		145532		10.93		329340		6.51	

IS 1 ID : Chlorobenzene-d5
IS 2 ID : 1,4-Dichlorobenzene-d4
IS 3 ID : Fluorobenzene

Column used to flag values outside QC limits with an asterisk
* Value outside of QC limits

AREA UPPER LIMIT = +100% of internal standard area
AREA LOWER LIMIT = -50% of internal standard area

RT UPPER LIMIT = +0.17 minutes of internal standard RT
RT LOWER LIMIT = -0.17 minutes of internal standard RT

FORM V III VOA

8A

VOLATILE INTERNAL STANDARD AREA AND RT SUMMARY

Report No: 223060210

GC Column: RTX-VMS-30M ID .25 (mm)

Injection Vol.: 1.0 (µL)

Analyst: KN1

Analysis Date: 06/05/23 Time: 1045

Standard ID: 1400

Instrument ID: MSV15

Lab File ID: 230605/c0730

Analytical Batch: 767028

Analytical Method: EPA 8260B

		IS 1 Area	#	RT	#	IS 2 Area	#	RT	#	IS 3 Area	#	RT	#
STANDARD		125741		8.34		120697		9.81		325329		5.88	
CLIENT SAMPLE ID	***GCAL SAMP ID***												
LCS2489808	2489808	125741		8.34		120697		9.81		325329		5.88	
MB2489807	2489807	137619		8.34		129255		9.8		338140		5.88	
OT07-MW-806-060123	22306021008	131738		8.34		121370		9.81		322181		5.88	

IS 1 ID : Chlorobenzene-d5

IS 2 ID : 1,4-Dichlorobenzene-d4

IS 3 ID : Fluorobenzene

Column used to flag values outside QC limits with an asterisk

* Value outside of QC limits

AREA UPPER LIMIT = +100% of internal standard area

AREA LOWER LIMIT = -50% of internal standard area

RT UPPER LIMIT = +0.17 minutes of internal standard RT

RT LOWER LIMIT = -0.17 minutes of internal standard RT

FORM V III VOA

8A

VOLATILE INTERNAL STANDARD AREA AND RT SUMMARY

Report No:	223060210	Standard ID:	1400
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230606/c0768d
Analyst:	KN1	Analytical Batch:	767081
Analysis Date:	06/06/23 Time: 1053	Analytical Method:	EPA 8260B

		IS 1 Area	#	RT	#	IS 2 Area	#	RT	#	IS 3 Area	#	RT	#
STANDARD		126742		8.34		118459		9.81		318605		5.88	
CLIENT SAMPLE ID	***GCAL SAMP ID***												
LCS2489979	2489979	126742		8.34		118459		9.81		318605		5.88	
LCSD2489980	2489980	129894		8.34		122663		9.81		331449		5.88	
MB2489978	2489978	141556		8.34		132736		9.8		343823		5.88	
OT07-TB-01-060124	22306021009	138278		8.34		127564		9.8		333991		5.88	

IS 1 ID : Chlorobenzene-d5

IS 2 ID : 1,4-Dichlorobenzene-d4

IS 3 ID : Fluorobenzene

Column used to flag values outside QC limits with an asterisk

* Value outside of QC limits

AREA UPPER LIMIT = +100% of internal standard area

AREA LOWER LIMIT = -50% of internal standard area

RT UPPER LIMIT = +0.17 minutes of internal standard RT

RT LOWER LIMIT = -0.17 minutes of internal standard RT

FORM V III VOA

EPA 8260B DOD

RunLogs

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 17-MAY-2023
Instrument: msv11.i
Analyst(s): KN1

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
BLANK	8260b+App	j1291.d	5.00 ml	17-MAY-2023 08:52	1.0	KN1	
BLANK	all	j1291t.d	0.00 ml	17-MAY-2023 08:52	1.0	KN1	
BLANK	8260b+App	j1292.d	5.00 ml	17-MAY-2023 09:15	1.0	KN1	
BLANK	all	j1292t.d	0.00 ml	17-MAY-2023 09:15	1.0	KN1	
BLANK	8260b+App	j1293.d	5.00 ml	17-MAY-2023 09:39	1.0	KN1	
BLANK	all	j1293t.d	0.00 ml	17-MAY-2023 09:39	1.0	KN1	
HANDSPIKE	8260b+App	j1294.d	5.00 ml	17-MAY-2023 10:05	1.0	KN1	
BLANK	8260b+App	j1295.d	5.00 ml	17-MAY-2023 10:27	1.0	KN1	
1203	8260b+App	j1296.d	5.00 ml	17-MAY-2023 10:57	1.0	KN1	
1204	8260b+App	j1297.d	5.00 ml	17-MAY-2023 11:20	1.0	KN1	
1205	8260b+App	j1298.d	5.00 ml	17-MAY-2023 11:43	1.0	KN1	
1206	8260b+App	j1299.d	5.00 ml	17-MAY-2023 12:06	1.0	KN1	
1207	8260b+App	j1300.d	5.00 ml	17-MAY-2023 12:29	1.0	KN1	
1208	8260b+App	j1301.d	5.00 ml	17-MAY-2023 12:52	1.0	KN1	
1209	8260b+App	j1302.d	5.00 ml	17-MAY-2023 13:15	1.0	KN1	
BLANK	8260b+App	j1303.d	5.00 ml	17-MAY-2023 13:38	1.0	KN1	
1203	8260b+App	j1304.d	5.00 ml	17-MAY-2023 14:17	1.0	KN1	
1203	8260b+App	j1304llc.d	5.00 ml	17-MAY-2023 14:17	1.0	KN1	
1204	8260b+App	j1305.d	5.00 ml	17-MAY-2023 14:40	1.0	KN1	
1204	8260b+App	j1305llc.d	5.00 ml	17-MAY-2023 14:40	1.0	KN1	
1205	8260b+App	j1306.d	5.00 ml	17-MAY-2023 15:02	1.0	KN1	
1600	8260b+App	j1307.d	5.00 ml	17-MAY-2023 15:57	1.0	KN1	

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 17-MAY-2023
Instrument: msv11.i
Analyst(s): KN1

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
1600	all	j1307t.d	0.00 ml	17-MAY-2023 15:57	1.0	KN1	
1600	8260b+App	j1309.d	5.00 ml	17-MAY-2023 17:27	1.0	KN1	
1000	all	j1309t.d	0.00 ml	17-MAY-2023 17:27	1.0	KN1	

Legend:
- Peak was not identified (NI)
- Peak integration was less area than it should be (LT)
- Peak Integration was more area than it should be (GT)
- Baseline adjustment required (BA)
- Wrong peak identified (WP)
- Split co-eluting peaks (CO)
- Retention time shift (RT)
- Interference correction (INT)
- No spectral match (for deletion of analytes, where capabilities exist) (SM)

&l1O

LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 02-JUN-2023
Instrument: msv11.i
Analyst(s): KN1

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
1400	8260b+App	j1686.d	5.00 ml	02-JUN-2023 11:00	1.0	KN1	
1000	all	j1686t.d	0.00 ml	02-JUN-2023 11:00	1.0	KN1	
1400	8260bDOD	j1687.d	5.00 ml	02-JUN-2023 11:23	1.0	KN1	
2489346	8260bDOD	j1687L.d	5.00 ml	02-JUN-2023 11:23	1.0	KN1	
BLANK	8260bDOD	j1688.d	5.00 ml	02-JUN-2023 12:04	1.0	KN1	
BLANK	8260bDOD	j1689.d	5.00 ml	02-JUN-2023 12:27	1.0	KN1	
5PPB	8260b+App	j1690.d	5.00 ml	02-JUN-2023 12:50	1.0	KN1	
2PPB	8260b+App	j1691.d	5.00 ml	02-JUN-2023 13:13	1.0	KN1	
1PPB	8260b+App	j1692.d	5.00 ml	02-JUN-2023 13:36	1.0	KN1	
0.5PPB	8260b+App	j1693.d	5.00 ml	02-JUN-2023 13:59	1.0	KN1	
2489345	8260bDOD	j1694.d	5.00 ml	02-JUN-2023 14:23	1.0	KN1	
22306021001	8260bDOD	j1695.d	5.00 ml	02-JUN-2023 14:46	1.0	KN1	
22305316104	Short2DOD	j1696.d	5.00 ml	02-JUN-2023 15:09	1.0	KN1	
22306021002	8260bDOD	j1697.d	5.00 ml	02-JUN-2023 15:32	1.0	KN1	
22306021003	8260bDOD	j1698.d	5.00 ml	02-JUN-2023 15:56	1.0	KN1	
22306021004	8260bDOD	j1699.d	5.00 ml	02-JUN-2023 16:19	1.0	KN1	
22306021005	8260bDOD	j1700.d	5.00 ml	02-JUN-2023 16:42	1.0	KN1	
22306021008	8260bDOD	j1701.d	5.00 ml	02-JUN-2023 17:06	1.0	KN1	
22305316101	Short2DOD	j1702.d	5.00 ml	02-JUN-2023 17:29	1.0	KN1	
22305316102	Short2DOD	j1703.d	5.00 ml	02-JUN-2023 17:52	1.0	KN1	
22305316103	Short2DOD	j1704.d	5.00 ml	02-JUN-2023 18:15	1.0	KN1	
22306021006	8260bDOD	j1705.d	5.00 ml	02-JUN-2023 18:38	1.0	KN1	

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 02-JUN-2023
Instrument: msv11.i
Analyst(s): KN1

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
22306021007	8260bDOD	j1706.d	5.00 ml	02-JUN-2023 19:02	1.0	KN1	
1400	8260bDOD	j1707.d	5.00 ml	02-JUN-2023 19:24	1.0	KN1	
BLANK	8260bDOD	j1708.d	5.00 ml	02-JUN-2023 19:47	1.0	KN1	
BLANK	8260bDOD	j1709.d	5.00 ml	02-JUN-2023 20:11	1.0	KN1	
BLANK	8260bDOD	j1710.d	5.00 ml	02-JUN-2023 20:34	1.0	KN1	

Legend:
- Peak was not identified (NI)
- Peak integration was less area than it should be (LT)
- Peak Integration was more area than it should be (GT)
- Baseline adjustment required (BA)
- Wrong peak identified (WP)
- Split co-eluting peaks (CO)
- Retention time shift (RT)
- Interference correction (INT)
- No spectral match (for deletion of analytes, where capabilities exist) (SM)

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 17-MAY-2023
Instrument: msv15.i
Analyst(s): CJR

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
BLANK	8260b+App	c0221b.d	5.00 ml	17-MAY-2023 09:56	1.0	CJR	
1000	all	c0221.d	0.00 ml	17-MAY-2023 09:56	1.0	CJR	
BLANK	8260b+App	c0222.d	5.00 ml	17-MAY-2023 10:42	1.0	KN1	
BLANK	8260b+App	c0223.d	5.00 ml	17-MAY-2023 11:25	1.0	CJR	
BLANK	all	c0223t.d	0.00 ml	17-MAY-2023 11:25	1.0	CJR	
BLANK	8260b+App	c0224.d	5.00 ml	17-MAY-2023 11:44	1.0	CJR	
BLANK	all	c0224t.d	0.00 ml	17-MAY-2023 11:44	1.0	CJR	
BLANK	8260b+App	c0225.d	5.00 ml	17-MAY-2023 12:42	1.0	CJR	
BLANK	all	c0225t.d	0.00 ml	17-MAY-2023 12:42	1.0	CJR	
BLANK	8260b+App	c0226.d	5.00 ml	17-MAY-2023 13:15	1.0	CJR	
BLANK	all	c0226t.d	0.00 ml	17-MAY-2023 13:15	1.0	CJR	
HANDSPIKE	8260b+App	c0227.d	5.00 ml	17-MAY-2023 13:37	1.0	CJR	
HANDSPIKE	all	c0227t.d	0.00 ml	17-MAY-2023 13:37	1.0	CJR	
1203	8260b+App	c0228.d	5.00 ml	17-MAY-2023 14:28	1.0	CJR	
1203	8260b+App	c0228llc.d	5.00 ml	17-MAY-2023 14:28	1.0	CJR	
1204	8260b+App	c0229.d	5.00 ml	17-MAY-2023 14:47	1.0	CJR	
1204	8260b+App	c0229llc.d	5.00 ml	17-MAY-2023 14:47	1.0	CJR	
1205	8260b+App	c0230.d	5.00 ml	17-MAY-2023 15:05	1.0	CJR	
1206	8260b+App	c0231.d	5.00 ml	17-MAY-2023 15:24	1.0	CJR	
1207	8260b+App	c0232.d	5.00 ml	17-MAY-2023 15:43	1.0	CJR	
1208	8260b+App	c0233.d	5.00 ml	17-MAY-2023 16:01	1.0	CJR	
1209	8260b+App	c0234.d	5.00 ml	17-MAY-2023 16:20	1.0	CJR	

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 17-MAY-2023
Instrument: msv15.i
Analyst(s): CJR

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
BLANK	8260b+App	c0235.d	5.00 ml	17-MAY-2023 16:39	1.0	CJR	
1600	8260b+App	c0238.d	5.00 ml	17-MAY-2023 17:35	1.0	CJR	
1600	all	c0238t.d	0.00 ml	17-MAY-2023 17:35	1.0	CJR	

Legend:
- Peak was not identified (NI)
- Peak integration was less area than it should be (LT)
- Peak Integration was more area than it should be (GT)
- Baseline adjustment required (BA)
- Wrong peak identified (WP)
- Split co-eluting peaks (CO)
- Retention time shift (RT)
- Interference correction (INT)
- No spectral match (for deletion of analytes, where capabilities exist) (SM)

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 05-JUN-2023
Instrument: msv15.i
Analyst(s): KN1

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
BLANK	8260DOD	c0727.d	5.00 ml	05-JUN-2023 09:06	1.0	KN1	
1000	all	c0728.d	0.00 ml	05-JUN-2023 09:25	1.0	CJR	
1400	8260b+App	c0729.d	5.00 ml	05-JUN-2023 10:09	1.0	KN1	
1400	8260DOD	c0730.d	5.00 ml	05-JUN-2023 10:45	1.0	KN1	
2489808	8260DOD	c0730L.d	5.00 ml	05-JUN-2023 10:45	1.0	KN1	
1400	8260DOD	c0731.d	5.00 ml	05-JUN-2023 11:04	1.0	KN1	
BLANK	8260DOD	c0732.d	5.00 ml	05-JUN-2023 11:23	1.0	KN1	
5PPB	8260b+App	c0733.d	5.00 ml	05-JUN-2023 11:41	1.0	KN1	
2PPB	8260b+App	c0734.d	5.00 ml	05-JUN-2023 12:00	1.0	KN1	
1PPB	8260b+App	c0735.d	5.00 ml	05-JUN-2023 12:19	1.0	KN1	
0.5PPB	8260b+App	c0736.d	5.00 ml	05-JUN-2023 12:38	1.0	KN1	
2489807	8260DOD	c0737.d	5.00 ml	05-JUN-2023 12:57	1.0	KN1	
22306030901	8260DOD	c0738.d	5.00 ml	05-JUN-2023 13:15	1.0	KN1	
22306030908	8260DOD	c0739.d	5.00 ml	05-JUN-2023 13:34	1.0	KN1	
22306031001	8260DOD	c0740.d	5.00 ml	05-JUN-2023 13:53	1.0	KN1	
22306021003	8260DOD	c0741.d	5.00 ml	05-JUN-2023 14:12	1.0	KN1	
22306021008	8260DOD	c0742.d	5.00 ml	05-JUN-2023 14:30	1.0	KN1	
22306030902	8260DOD	c0743.d	5.00 ml	05-JUN-2023 14:49	1.0	KN1	
22306030904	8260DOD	c0744.d	5.00 ml	05-JUN-2023 15:08	1.0	KN1	
22306030905	8260DOD	c0745.d	5.00 ml	05-JUN-2023 15:27	1.0	KN1	
22306030906	8260DOD	c0746.d	5.00 ml	05-JUN-2023 15:46	1.0	KN1	
22306031002	8260DOD	c0747.d	5.00 ml	05-JUN-2023 16:04	1.0	KN1	

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 05-JUN-2023
Instrument: msv15.i
Analyst(s): KN1

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
22306031005	8260DOD	c0748.d	5.00 ml	05-JUN-2023 16:23	1.0	KN1	
22306031006	8260DOD	c0749.d	5.00 ml	05-JUN-2023 16:42	1.0	KN1	
22306031009	8260DOD	c0750.d	5.00 ml	05-JUN-2023 17:00	1.0	KN1	
22306031010	8260DOD	c0751.d	5.00 ml	05-JUN-2023 17:19	1.0	KN1	
22306031011	8260DOD	c0752.d	5.00 ml	05-JUN-2023 17:38	1.0	KN1	
22306031012	8260DOD	c0753.d	5.00 ml	05-JUN-2023 17:57	1.0	KN1	
22306031013	8260DOD	c0754.d	5.00 ml	05-JUN-2023 18:15	1.0	KN1	
22306030907	8260DOD	c0755.d	5.00 ml	05-JUN-2023 18:34	2.0	KN1	
22306031008	8260DOD	c0756.d	5.00 ml	05-JUN-2023 18:53	5.0	KN1	
22306031007	8260DOD	c0757.d	5.00 ml	05-JUN-2023 19:12	10.0	KN1	
22306030903	8260DOD	c0758.d	5.00 ml	05-JUN-2023 19:31	25.0	KN1	
22306031003	8260DOD	c0759.d	5.00 ml	05-JUN-2023 19:50	1.0	KN1	
22306031004	8260DOD	c0760.d	5.00 ml	05-JUN-2023 20:08	1.0	KN1	
1440	8260DOD	c0761.d	5.00 ml	05-JUN-2023 20:27	1.0	KN1	
BLANK	8260DOD	c0762.d	5.00 ml	05-JUN-2023 20:46	1.0	KN1	
BLANK	8260DOD	c0763.d	5.00 ml	05-JUN-2023 21:05	1.0	KN1	
BLANK	8260DOD	c0764.d	5.00 ml	05-JUN-2023 21:24	1.0	KN1	

Legend:
- Peak was not identified (NI)
- Peak integration was less area than it should be (LT)
- Peak Integration was more area than it should be (GT)
- Baseline adjustment required (BA)
- Wrong peak identified (WP)
- Split co-eluting peaks (CO)
- Retention time shift (RT)
- Interference correction (INT)
- No spectral match (for deletion of analytes, where capabilities exist) (SM)

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 06-JUN-2023
Instrument: msv15.i
Analyst(s): KN1

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
BLANK	8260DOD	c0765.d	5.00 ml	06-JUN-2023 09:31	1.0	KN1	
1000	all	c0766.d	0.00 ml	06-JUN-2023 09:50	1.0	CJR	
1000	all	c0766d.d	0.00 ml	06-JUN-2023 09:50	1.0	CJR	
1400	8260b+App	c0767.d	5.00 ml	06-JUN-2023 10:21	1.0	KN1	
1400	8260DOD	c0768.d	5.00 ml	06-JUN-2023 10:53	1.0	KN1	
1400	8260DOD	c0768d.d	5.00 ml	06-JUN-2023 10:53	1.0	KN1	
2489982	8260DOD	c0768L.d	5.00 ml	06-JUN-2023 10:53	1.0	KN1	
2489979	8260DOD	c0768Ld.d	5.00 ml	06-JUN-2023 10:53	1.0	KN1	
2489980	8260DOD	c0769.d	5.00 ml	06-JUN-2023 11:12	1.0	KN1	
BLANK	8260DOD	c0770.d	5.00 ml	06-JUN-2023 11:30	1.0	KN1	
5PPB	8260b+App	c0771.d	5.00 ml	06-JUN-2023 11:49	1.0	KN1	
2PPB	8260b+App	c0772.d	5.00 ml	06-JUN-2023 12:08	1.0	KN1	
1PPB	8260b+App	c0773.d	5.00 ml	06-JUN-2023 12:27	1.0	KN1	
2489981	8260DOD	c0774.d	5.00 ml	06-JUN-2023 12:46	1.0	KN1	
2489978	8260DOD	c0774d.d	5.00 ml	06-JUN-2023 12:46	1.0	KN1	
22306030105	8260DOD	c0775.d	5.00 ml	06-JUN-2023 13:05	1.0	KN1	
22306024702	8260DOD	c0776.d	5.00 ml	06-JUN-2023 13:24	1.0	KN1	
22306021009	8260DOD	c0777.d	5.00 ml	06-JUN-2023 13:42	1.0	KN1	
22306031014	8260DOD	c0778.d	5.00 ml	06-JUN-2023 14:01	1.0	KN1	
22306031101	8260DOD	c0779.d	5.00 ml	06-JUN-2023 14:20	1.0	KN1	
22306024701	8260DOD	c0780.d	5.00 ml	06-JUN-2023 14:39	1.0	KN1	
22306031102	8260DOD	c0781.d	5.00 ml	06-JUN-2023 14:58	5.0	KN1	

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 06-JUN-2023
Instrument: msv15.i
Analyst(s): KN1

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
22306053102	8260DOD	c0782.d	5.00 ml	06-JUN-2023 15:16	1.0	KN1	
22306052103	BTEX	c0783.d	5.00 ml	06-JUN-2023 15:35	1.0	KN1	
22306052104	BTEX	c0784.d	5.00 ml	06-JUN-2023 15:54	1.0	KN1	
22306052101	BTEX	c0785.d	5.00 ml	06-JUN-2023 16:13	10.0	KN1	
22306052102	BTEX	c0786.d	5.00 ml	06-JUN-2023 16:32	10.0	KN1	
22306030103	8260DOD	c0787.d RR@5X	5.00 ml	06-JUN-2023 16:50	100.0	KN1	OD
22306030104	8260DOD	c0788.d RR@25X	5.00 ml	06-JUN-2023 17:09	200.0	KN1	OD
22306030102	8260DOD	c0789.d	5.00 ml	06-JUN-2023 17:28	500.0	KN1	
22306030101	8260DOD	c0790.d RR@500X	5.00 ml	06-JUN-2023 17:47	2000.0	KN1	OD
BLANK	8260DOD	c0791.d	5.00 ml	06-JUN-2023 18:05	1.0	KN1	
22306015401	TCLP	c0792.d	5.00 ml	06-JUN-2023 18:24	50.0	KN1	
22306015402	TCLP	c0793.d	5.00 ml	06-JUN-2023 18:43	50.0	KN1	
22305312801	TCLP	c0794.d	5.00 ml	06-JUN-2023 19:02	100.0	KN1	
22305312701	TCLP	c0795.d	5.00 ml	06-JUN-2023 19:21	100.0	KN1	
1440	8260DOD	c0796.d	5.00 ml	06-JUN-2023 19:39	1.0	KN1	
BLANK	8260DOD	c0797.d	5.00 ml	06-JUN-2023 19:58	1.0	KN1	
BLANK	8260DOD	c0798.d	5.00 ml	06-JUN-2023 20:17	1.0	KN1	
BLANK	8260DOD	c0799.d	5.00 ml	06-JUN-2023 20:35	1.0	KN1	
BLANK	8260DOD	c0800.d	5.00 ml	06-JUN-2023 20:54	1.0	KN1	

Legend:
- Peak was not identified (NI)
- Peak integration was less area than it should be (LT)
- Peak Integration was more area than it should be (GT)
- Baseline adjustment required (BA)
- Wrong peak identified (WP)
- Split co-eluting peaks (CO)
- Retention time shift (RT)
- Interference correction (INT)
- No spectral match (for deletion of analytes, where capabilities exist) (SM)

EPA 8260B DOD

VOA pH Logs

Volatile pH Check

ANALYST/ TECH	DWR	START DATE/TIME	6/7/2023 10:45	END DATE/TIME	6/7/2023 11:00	BATCH	767081

#	CLIENT	TYPE	LAB ID	pH	Comment
1	4612	TB	22306024702	<2	
2	4612	SAMP	22306024701	<2	
3	4887	TB	22306021009	<2	
4	4878	EQBK	22306031014	<2	
5	4838	TB	22306031101	<2	
6	4838	SAMP	22306031102	<2	
7					
8					
9					
10					
11					
12					
13					
14					
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NOTES

LOT #228022

Volatile pH Check

ANALYST/ TECH	DWR	START DATE/TIME	6/5/2023 12:45	END DATE/TIME	6/5/2023 13:00	BATCH	766942

#	CLIENT	TYPE	LAB ID	pH	Comment
1	4887	EQBK	22306021001	<2	
2	4838	FD	22305316104	<2	
3	4887	SAMP	22306021002	<2	
4	4887	SAMP	22306021003	<2	
5	4887	SAMP	22306021004	<2	
6	4887	SAMP	22306021005	<2	
7	4838	SAMP	22305316101	<2	
8	4838	MS	22305316102	<2	
9	4838	MSD	22305316103	<2	
10	4887	MS	22306021006	<2	
11	4887	MSD	22306021007	<2	
12					
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14					
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LOT #228022

Pace Analytical®

Volatile pH Check

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ANALYST/ TECH	DWR	START DATE/TIME	6/6/2023 09:30	END DATE/TIME	6/6/2023 09:45	BATCH	767028

#	CLIENT	TYPE	LAB ID	pH	Comment
1	4887	SAMP	22306021008	<2	
2	4878	TB	22306030901	<2	
3	4878	SAMP	22306030902	<2	
4	4878	SAMP	22306030903	<2	
5	4878	SAMP	22306030904	<2	
6	4878	SAMP	22306030905	<2	
7	4878	SAMP	22306030906	<2	
8	4878	SAMP	22306030907	<2	
9	4878	EQBK	22306030908	<2	
10	4878	TB	22306031001	<2	
11	4878	SAMP	22306031002	<2	
12	4878	MS	22306031003	<2	
13	4878	MSD	22306031004	<2	
14	4878	SAMP	22306031005	<2	
15	4878	FD	22306031006	<2	
16	4878	SAMP	22306031007	<2	
17	4878	SAMP	22306031008	<2	
18	4878	SAMP	22306031009	<2	
19	4878	FD	22306031010	<2	
20	4878	SAMP	22306031011	<2	
21	4878	SAMP	22306031012	<2	
22	4878	SAMP	22306031013	<2	
23					
24					
25					
26					
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30					

NOTES

LOT #228022

Client ID: 4887 - Plexus Consultants
SDG: 223060210
PM: CAM

Pace

CHAIN-OF-CUSTODY / Analytical Request

The Chain-of-Custody is a LEGAL DOCUMENT. All relevant fields

Submitting a sample via this chain of custody constitutes acknowledgment and acceptance of the Pace Terms and Conditions found at https://info

Section A Required Client Information:	Section B Required Project Information:	Section C Invoice Information:
Company: PLEXUS CONSULTANTS	Report To: Patrick Reilley	Attention: Same as Section A
Address: 5510 Cherokee Ave.	Copy To:	Company Name:
Suite 350, Alexandria, VA 22312		Address:
Email: preilley@plexsci.com	Purchase Order #:	Pace Quote:
Phone: (703) 989-8405 Fax:	Project Name: Scott AFB - OT007	Pace Project Manager: carolyn.dicharry@pacelabs.com,
Requested Due Date:	Project #:	Pace Profile #: 271212

Regulatory Agency	State / Location
	IL

SAMPLE ID — One Character per box. (A-Z, 0-9 / , -) Sample Ids must be unique

MATRIX	CODE
Drinking Water	DW
Water	WT
Waste Water	WW
Product	P
Soil/Solid	SL
Oil	OL
Wipe	WP
Air	AR
Other	OT
Tissue	TS

Requested Analysis Filtered (Y/N)

ITEM #	SAMPLE ID	MATRIX CODE (see valid codes to left)	SAMPLE TYPE (G=GRAB C=COMP)	COLLECTED START DATE	START TIME	END DATE	END TIME	SAMPLE TEMP AT COLLECTION	# OF CONTAINERS	Unpreserved	H2SO4	HNO3	HCl	NaOH	Na2S2O3	Methanol	Other	Analyses Test Y/N	VOC by 8260	Metals by 6020 (As, Fe, Mn)	TOC	Alkalinity	Anions; IC (Cl, NO3, NO2, S)	Dissolved Gases, RSK-175	Trip Blank					Residual Chlorine (Y/N)	
1	OT07-EB-060123	WT		6/1/23	12:14	6/1/23	12:14		11	2		1	8						x	x	x	x	x	x							1
2	OT07-MW02-060123	WT		06/01/23	12:30	06/01/23	12:30		11	2		1	8						x	x	x	x	x	x							2
3	OT07-MW04-060123	WT		6/1/23	9:35	6/1/23	9:35		11	2		1	8						x	x	x	x	x	x							3
4	OT07-MW05-060123	WT		6/1/23	9:10	6/1/23	9:10		11	2		1	8						x	x	x	x	x	x							4
5	OT07-MW06-060123	WT		6/1/23	11:00	6/1/23	11:00		11	2		1	8						x	x	x	x	x	x							5
6	OT07-MW04-060123MS	WT		6/1/23	9:55	6/1/23	9:55		11	2		1	8						x	x	x	x	x	x							6
7	OT07-MW04-060123MSD	WT		6/1/23	10:15	6/1/23	10:15		11	2		1	8						x	x	x	x	x	x							7
8	OT07-MW-806-~~Dup~~060123	WT		6/1/23	11:00	6/1/23	11:00		11	2		1	8						x	x	x	x	x	x							8
9	OT07-TB-01-060123	WT		6/1/23	7:30	6/1/23	7:30		2				2																		9
10																															
11																															
12																															

ADDITIONAL COMMENTS	RELINQUISHED BY / AFFILIATION	DATE	TIME	ACCEPTED BY / AFFILIATION	DATE	TIME	SAMPLE CONDITIONS
	[signature] Plexus	6/1/23	18:00	UPS	6/1/23	18:00	
	UPS	6/2/23	10:17	[signature]	6/2/23	10:17	2.1+0.0=2.1

UPS 1Z9394770115241747

SAMPLER NAME AND SIGNATURE

PRINT Name of SAMPLER: John Filan

SIGNATURE of SAMPLER: [signature]

DATE Signed: 6/1/23

TEMP in C	Received on Ice (Y/N)	Custody Sealed Cooler (Y/N)	Samples Intact (Y/N)

SAMPLE RECEIVING CHECKLIST

223060210

SAMPLE DELIVERY GROUP 223060210	
Client **PM** CAM 4887 - Flexus Consultants	**Transport Method** UPS
Profile Number 271212	**Received By** Roberts, George S.
Line Item(s) 21 - W - OT007 Full VOA/TOC/RSK/Anions	**Receive Date(s)** 06/02/23

CHECKLIST	YES	NO
Samples received with proper thermal preservation?	☑	☐
Radioactivity is <1600 cpm? If no, record cpm value in notes section.	☐	☐
COC relinquished and complete (including sampleIDs, collect times, and sampler)?	☑	☐
All containers received in good condition and within hold time?	☑	☐
All sample labels and containers received match the chain of custody?	☑	☐
Preservative added to any containers?	☐	☑
If received, was headspace for VOC water containers < 6mm?	☑	☐
Samples collected in containers provided by Pace Gulf Coast?	☑	☐

COOLERS

Airbill Thermometer ID: E43	Temp °C
1Z93947Y0115241747	2.1

DISCREPANCIES

None

LAB PRESERVATIONS

None

NOTES

Revision 1.6

Page 1 of 1

ANALYTICAL REPORT

June 09, 2023

Pace Analytical - Baton Rouge, LA - DOD

Sample Delivery Group:	L1623041
Samples Received:	06/06/2023
Project Number:	223060210
Description:	Plexus Scott AFB
Site:	01
Report To:	Carolyn Dicharry 7979 Innovation Park Drive Baton Rouge, LA 70820

Entire Report Reviewed By: Nancy F. McLain

Nancy McLain
Project Manager

Pace Analytical National
12065 Lebanon Rd Mount Juliet, TN 37122 615-758-5858 800-767-5859 www.pacenational.com

TABLE OF CONTENTS

	Collected by	Collected date/time	Received date/time
OT07-EB-060123 L1623041-01 GW		06/01/23 12:14	06/06/23 09:00

Method	Batch	Dilution	Preparation date/time	Analysis date/time	Analyst	Location
Volatile Organic Compounds (GC) by Method RSK175	WG2073383	1	06/08/23 11:10	06/08/23 11:10	CCM	Mt. Juliet, TN

	Collected by	Collected date/time	Received date/time
OT07-MW02-060123 L1623041-02 GW		06/01/23 12:30	06/06/23 09:00

Method	Batch	Dilution	Preparation date/time	Analysis date/time	Analyst	Location
Volatile Organic Compounds (GC) by Method RSK175	WG2073383	1	06/08/23 11:13	06/08/23 11:13	CCM	Mt. Juliet, TN

	Collected by	Collected date/time	Received date/time
OT07-MW04-060123 L1623041-03 GW		06/01/23 09:35	06/06/23 09:00

Method	Batch	Dilution	Preparation date/time	Analysis date/time	Analyst	Location
Volatile Organic Compounds (GC) by Method RSK175	WG2074059	1	06/09/23 11:06	06/09/23 11:06	BAW	Mt. Juliet, TN

	Collected by	Collected date/time	Received date/time
OT07-MW05-060123 L1623041-04 GW		06/01/23 09:10	06/06/23 09:00

Method	Batch	Dilution	Preparation date/time	Analysis date/time	Analyst	Location
Volatile Organic Compounds (GC) by Method RSK175	WG2073383	1	06/08/23 11:22	06/08/23 11:22	CCM	Mt. Juliet, TN

	Collected by	Collected date/time	Received date/time
OT07-MW06-060123 L1623041-05 GW		06/01/23 11:00	06/06/23 09:00

Method	Batch	Dilution	Preparation date/time	Analysis date/time	Analyst	Location
Volatile Organic Compounds (GC) by Method RSK175	WG2073383	1	06/08/23 11:29	06/08/23 11:29	CCM	Mt. Juliet, TN

	Collected by	Collected date/time	Received date/time
OT07-MW04-060123MS L1623041-06 GW		06/01/23 09:55	06/06/23 09:00

Method	Batch	Dilution	Preparation date/time	Analysis date/time	Analyst	Location
Volatile Organic Compounds (GC) by Method RSK175	WG2074059	1	06/09/23 11:08	06/09/23 11:08	BAW	Mt. Juliet, TN

	Collected by	Collected date/time	Received date/time
OT07-MW04-060123MSD L1623041-07 GW		06/01/23 10:15	06/06/23 09:00

Method	Batch	Dilution	Preparation date/time	Analysis date/time	Analyst	Location
Volatile Organic Compounds (GC) by Method RSK175	WG2074059	1	06/09/23 11:11	06/09/23 11:11	BAW	Mt. Juliet, TN

	Collected by	Collected date/time	Received date/time
OT07-MW806-060123 L1623041-08 GW		06/01/23 11:00	06/06/23 09:00

Method	Batch	Dilution	Preparation date/time	Analysis date/time	Analyst	Location
Volatile Organic Compounds (GC) by Method RSK175	WG2074059	1	06/09/23 11:14	06/09/23 11:14	BAW	Mt. Juliet, TN

All sample aliquots were received at the correct temperature, in the proper containers, with the appropriate preservatives, and within method specified holding times, unless qualified or notated within the report. Where applicable, all MDL (LOD) and RDL (LOQ) values reported for environmental samples have been corrected for the dilution factor used in the analysis. All Method and Batch Quality Control are within established criteria except where addressed in this case narrative, a non-conformance form or properly qualified within the sample results. By my digital signature below, I affirm to the best of my knowledge, all problems/anomalies observed by the laboratory as having the potential to affect the quality of the data have been identified by the laboratory, and no information or data have been knowingly withheld that would affect the quality of the data.

Nancy F. McLain

Nancy McLain
Project Manager

Report Revision History

Level II Report - Version 1: 06/09/23 15:51

RSK175 Volatile Organic Compounds (GC)

3A-OR

MATRIX SPIKE / MATRIX SPIKE DUPLICATE RECOVERY

L1623041-03,06,07,08

SAMPLE NO.:
R3934861-5
R3934861-6

MS Sample / File ID:	R3934861-5 / 0609B_27	**SDG:**	L1623041
MSD Sample / File ID:	R3934861-6 / 0609B_28	**Analytical Batch:**	WG2074059
OS Sample / File ID:	L1623041-03 / 0609B_03	**Matrix:**	GW
Instrument ID:	AIRGC2		
Analytical Method:	RSK175		

Analyte	Spike Amount	OS Result	MS Result	MSD Result	MS Rec.	MSD Rec.	Dilution	Rec. Limits	RPD	RPD Limit
	ug/l	*ug/l*	*ug/l*	*ug/l*	*%*	*%*		*%*	*%*	*%*
Methane	67.8	32.8	99.1	103	97.8	104	1	73.0 - 125	3.86	30
Ethane	129	10.0	123	130	95.3	101	1	74.0 - 131	5.53	30
Ethene	127	10.0	122	129	96.1	102	1	72.0 - 133	5.58	30

*: Value outside the established quality control limits.

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

LABORATORY CONTROL SAMPLE
LABORATORY CONTROL SAMPLE DUPLICATE
RECOVERY
L1623041-01,02,04,05

SAMPLE NO.:
R3934254-1
R3934254-5

LCS Sample / File ID:	R3934254-1 / 0608_01LCS	**SDG:**	L1623041
LCSD Sample / File ID:	R3934254-5 / 0608_27	**Analytical Batch:**	WG2073383
Instrument ID:	AIRGC2	**Dilution Factor:**	1
Analytical Method:	RSK175	**Matrix:**	GW

Analyte	Spike Amount	LCS Result	LCSD Result	LCS Rec.	LCSD Rec.	Rec. Limits	RPD	RPD Limit
	ug/l	*ug/l*	*ug/l*	*%*	*%*	*%*	*%*	*%*
Methane	67.8	67.0	64.5	98.8	95.1	73.0 - 125	3.80	30
Ethane	129	118	119	91.5	92.2	74.0 - 131	0.844	30
Ethene	127	119	120	93.7	94.5	72.0 - 133	0.837	30

*: Value outside the established quality control limits.

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

LABORATORY CONTROL SAMPLE
LABORATORY CONTROL SAMPLE DUPLICATE
RECOVERY
L1623041-03,06,07,08

SAMPLE NO.:
R3934861-1
R3934861-7

LCS Sample / File ID:	R3934861-1 / 0609B_01LCS	**SDG:**	L1623041
LCSD Sample / File ID:	R3934861-7 / 0609B_29	**Analytical Batch:**	WG2074059
Instrument ID:	AIRGC2	**Dilution Factor:**	1
Analytical Method:	RSK175	**Matrix:**	GW

Analyte	Spike Amount	LCS Result	LCSD Result	LCS Rec.	LCSD Rec.	Rec. Limits	RPD	RPD Limit
	ug/l	*ug/l*	*ug/l*	*%*	*%*	*%*	*%*	*%*
Methane	67.8	70.1	69.3	103	102	73.0 - 125	1.15	30
Ethane	129	119	117	92.2	90.7	74.0 - 131	1.69	30
Ethene	127	119	118	93.7	92.9	72.0 - 133	0.844	30

*: Value outside the established quality control limits.

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

LABORATORY DUPLICATE

L1623041-01,02,04,05

SAMPLE NO.:
R3934254-3

DUP Sample / File ID:	R3934254-3 / 0608_14	**SDG:**	L1623041
OS Sample / File ID:	L1619855-03 / 0608_03	**Analytical Batch:**	WG2073383
Lab File ID:	0608_14	**Dilution Factor:**	1
Instrument ID:	AIRGC2	**Matrix:**	GW
Analytical Method:	RSK175		

Analyte	**OS Result**	**DUP Result**	**RPD**	**RPD Limits**
	ug/l	*ug/l*	*%*	*%*
Methane	6.00	6.00	0.000	30
Ethane	10.0	10.0	0.000	30
Ethene	10.0	10.0	0.000	30

*: Value outside the established quality control limits.

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

LABORATORY DUPLICATE
L1623041-01,02,04,05

SAMPLE NO.:
R3934254-4

DUP Sample / File ID:	R3934254-4 / 0608_26	**SDG:**	L1623041
OS Sample / File ID:	L1622529-13 / 0608_15	**Analytical Batch:**	WG2073383
Lab File ID:	0608_26	**Dilution Factor:**	1
Instrument ID:	AIRGC2	**Matrix:**	GW
Analytical Method:	RSK175		

Analyte	OS Result	DUP Result	RPD	RPD Limits
	ug/l	*ug/l*	*%*	*%*
Methane	6.00	6.00	0.000	30
Ethane	10.0	10.0	0.000	30
Ethene	10.0	10.0	0.000	30

*: Value outside the established quality control limits.

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

LABORATORY DUPLICATE

L1623041-03,06,07,08

SAMPLE NO.:
R3934861-3

DUP Sample / File ID:	R3934861-3 / 0609B_14	**SDG:**	L1623041
OS Sample / File ID:	L1623162-01 / 0609B_13	**Analytical Batch:**	WG2074059
Lab File ID:	0609B_14	**Dilution Factor:**	1
Instrument ID:	AIRGC2	**Matrix:**	GW
Analytical Method:	RSK175		

Analyte	OS Result	DUP Result	RPD	RPD Limits
	ug/l	*ug/l*	*%*	*%*
Methane	6.00	6.00	0.000	30
Ethane	10.0	10.0	0.000	30
Ethene	10.0	10.0	0.000	30

*: Value outside the established quality control limits.

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

LABORATORY DUPLICATE

L1623041-03,06,07,08

SAMPLE NO.:
R3934861-4

DUP Sample / File ID:	R3934861-4 / 0609B_26	**SDG:**	L1623041
OS Sample / File ID:	L1623669-02 / 0609B_24	**Analytical Batch:**	WG2074059
Lab File ID:	0609B_26	**Dilution Factor:**	1
Instrument ID:	AIRGC2	**Matrix:**	GW
Analytical Method:	RSK175		

Analyte	OS Result	DUP Result	RPD	RPD Limits
	ug/l	*ug/l*	*%*	*%*
Methane	687	687	0.000	30
Ethane	10.0	10.0	0.000	30
Ethene	10.0	10.0	0.000	30

*: Value outside the established quality control limits.

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

METHOD BLANK

SAMPLE NO.: R3934254-2

Lab Sample ID:	R3934254-2	**SDG:**	L1623041
Lab File ID:	0608_02	**Preparation Date/Time:**	06/08/23 09:48
Instrument ID:	AIRGC2	**Analysis Date/Time:**	06/08/23 09:48
Analytical Batch:	WG2073383	**Dilution Factor:**	1
Analytical Method:	RSK175	**Matrix:**	GW

Sample ID	**Lab Sample ID**	**Instrument**	**File ID**	**Analysis** *date/time*
LCS	R3934254-1	AIRGC2	0608_01LCS	06/08/23 09:45
OS	L1619855-03	AIRGC2	0608_03	06/08/23 09:54
DUP	R3934254-3	AIRGC2	0608_14	06/08/23 10:35
OS	L1622529-13	AIRGC2	0608_15	06/08/23 10:41
OT07-EB-060123	L1623041-01	AIRGC2	0608_21	06/08/23 11:10
OT07-MW02-060123	L1623041-02	AIRGC2	0608_22	06/08/23 11:13
OT07-MW05-060123	L1623041-04	AIRGC2	0608_24	06/08/23 11:22
OT07-MW06-060123	L1623041-05	AIRGC2	0608_25	06/08/23 11:29
DUP	R3934254-4	AIRGC2	0608_26	06/08/23 11:37
LCSD	R3934254-5	AIRGC2	0608_27	06/08/23 11:40

4A-OR

METHOD BLANK

SAMPLE NO.:
R3934861-2

Lab Sample ID:	R3934861-2	**SDG:**	L1623041
Lab File ID:	0609B_02	**Preparation Date/Time:**	06/09/23 11:03
Instrument ID:	AIRGC2	**Analysis Date/Time:**	06/09/23 11:03
Analytical Batch:	WG2074059	**Dilution Factor:**	1
Analytical Method:	RSK175	**Matrix:**	GW

Sample ID	**Lab Sample ID**	**Instrument**	**File ID**	**Analysis** *date/time*
LCS	R3934861-1	AIRGC2	0609B_01LCS	06/09/23 10:53
OT07-MW04-060123	L1623041-03	AIRGC2	0609B_03	06/09/23 11:06
OT07-MW04-060123MS	L1623041-06	AIRGC2	0609B_04	06/09/23 11:08
OT07-MW04-060123MSD	L1623041-07	AIRGC2	0609B_05	06/09/23 11:11
OT07-MW806-060123	L1623041-08	AIRGC2	0609B_06	06/09/23 11:14
OS	L1623162-01	AIRGC2	0609B_13	06/09/23 11:32
DUP	R3934861-3	AIRGC2	0609B_14	06/09/23 12:52
OS	L1623669-02	AIRGC2	0609B_24	06/09/23 13:27
DUP	R3934861-4	AIRGC2	0609B_26	06/09/23 13:33
MS	R3934861-5	AIRGC2	0609B_27	06/09/23 13:38
MSD	R3934861-6	AIRGC2	0609B_28	06/09/23 13:42
LCSD	R3934861-7	AIRGC2	0609B_29	06/09/23 13:46

SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
OT07-EB-060123

Lab Sample ID:	L1623041-01	**SDG:**	L1623041
Client Sample ID:	OT07-EB-060123	**Collected Date/Time:**	06/01/23 12:14
Lab File ID:	0608_21	**Received Date/Time:**	06/06/23 09:00
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/08/23 11:10
Analytical Batch:	WG2073383	**Analysis Date/Time:**	06/08/23 11:10
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

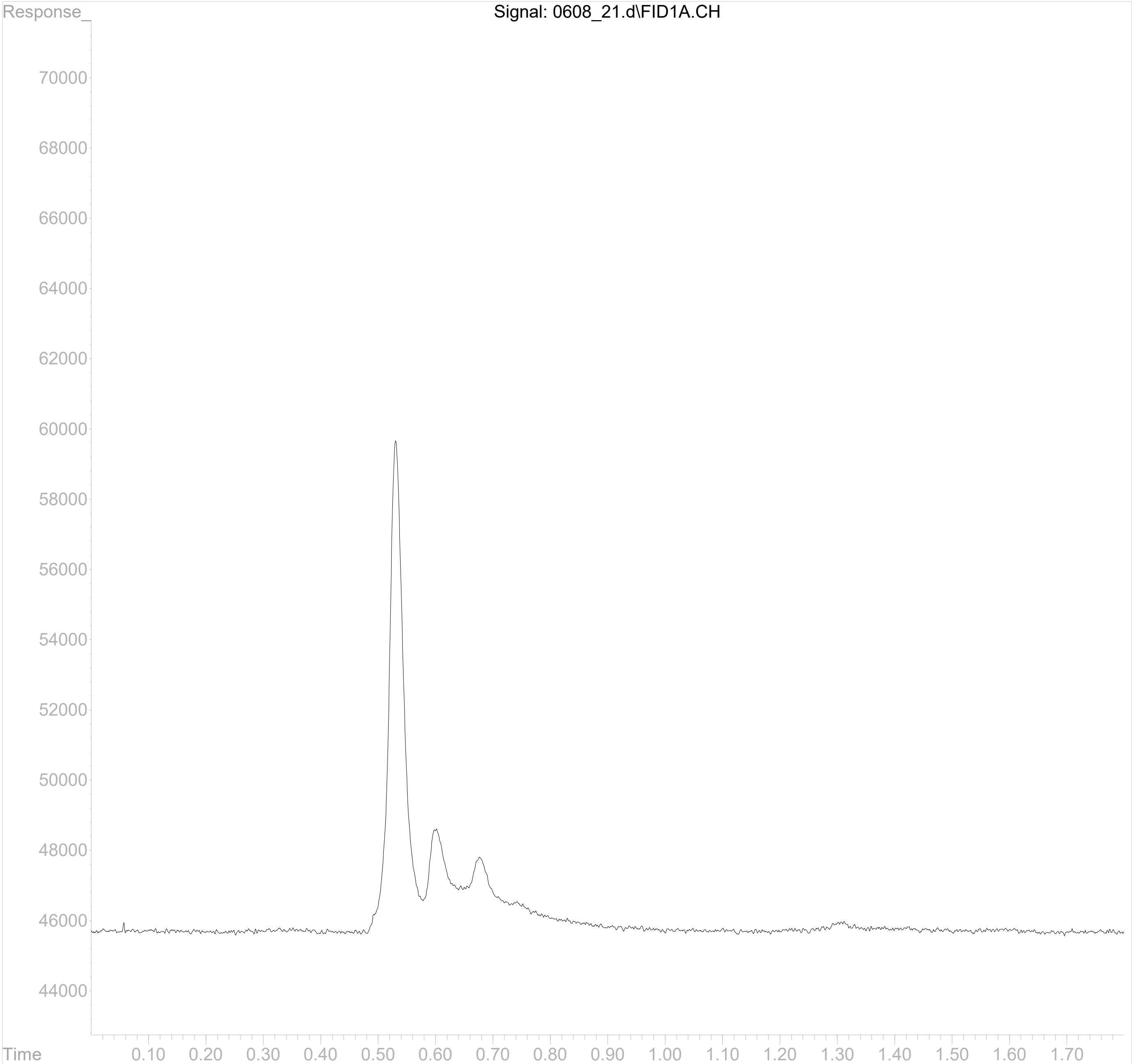
Analyte	CAS	RT	Result *ug/l*	Qualifier	DL *ug/l*	LOD *ug/l*	LOQ *ug/l*
Methane	74-82-8	0	6.00	U	2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060823\
Data File : 0608_21.d
Signal(s) : FID1A.CH
Acq On    : 08 Jun 2023  11:10 am
Operator  :
Sample    : L1623041-01 1x WG2073383
Misc      :
ALS Vial  : 21   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:13:33 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :

        Compound                     R.T.       Response    Conc Units
 ---------------------------------------------------------------------------------

 Target Compounds
 ----------------------------------------------------------------------------

(f)=RT Delta > 1/2 Window                                (m)=manual int.
```

Quantitation Report   (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060823\
Data File : 0608_21.d
Signal(s) : FID1A.CH
Acq On    : 08 Jun 2023  11:10 am
Operator  :
Sample    : L1623041-01 1x WG2073383
Misc      :
ALS Vial  : 21   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:13:33 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



1A-OR

SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
OT07-MW02-060123

Lab Sample ID:	L1623041-02	**SDG:**	L1623041
Client Sample ID:	OT07-MW02-060123	**Collected Date/Time:**	06/01/23 12:30
Lab File ID:	0608_22	**Received Date/Time:**	06/06/23 09:00
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/08/23 11:13
Analytical Batch:	WG2073383	**Analysis Date/Time:**	06/08/23 11:13
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

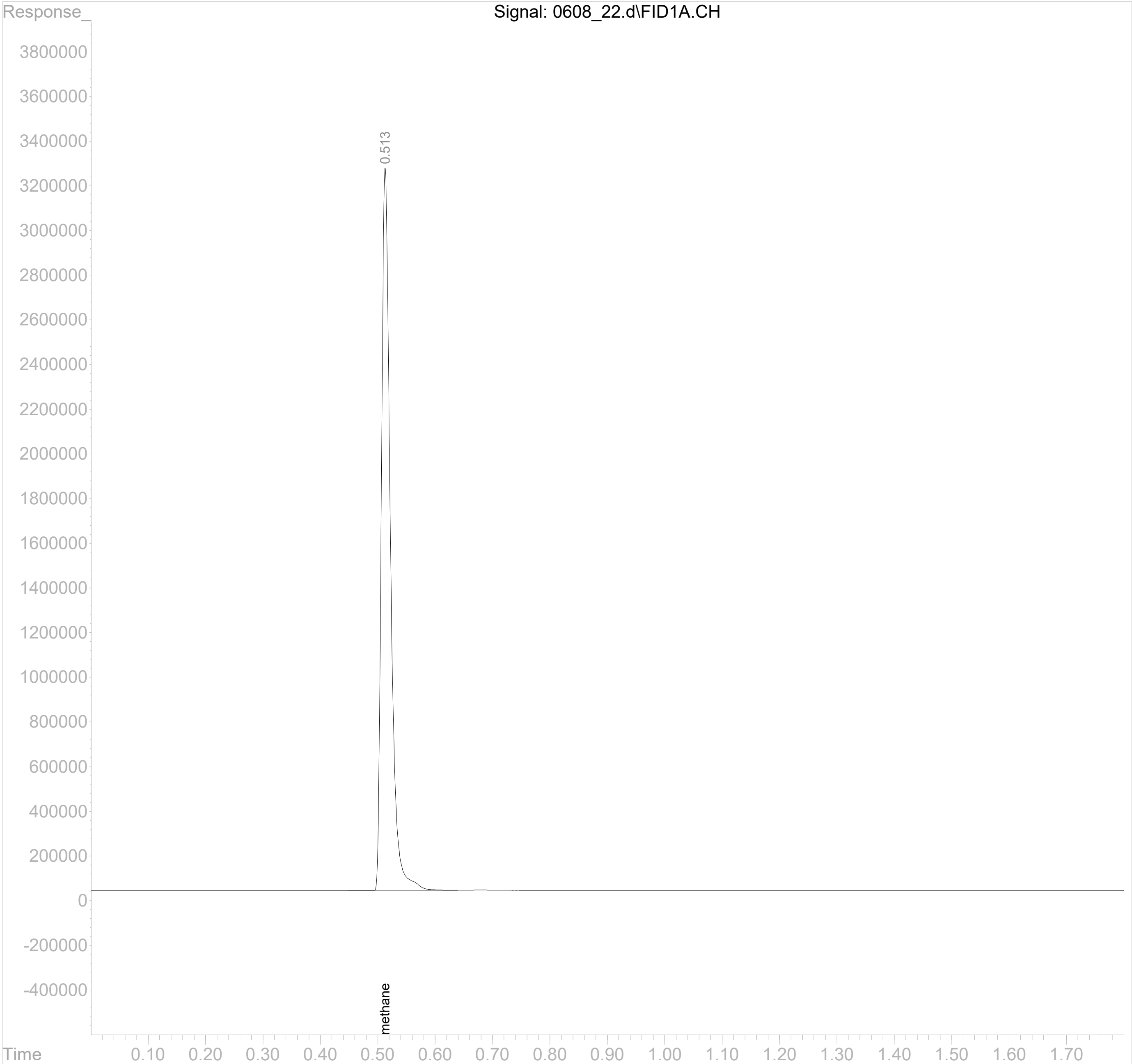
Analyte	CAS	RT	Result	Qualifier	DL	LOD	LOQ
			ug/l		*ug/l*	*ug/l*	*ug/l*
Methane	74-82-8	0.51	1480		2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060823\
Data File : 0608_22.d
Signal(s) : FID1A.CH
Acq On    : 08 Jun 2023  11:13 am
Operator  :
Sample    : L1623041-02 1x WG2073383
Misc      :
ALS Vial  : 22   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:16:38 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :

        Compound                     R.T.       Response    Conc Units
 -------------------------------------------------------------------------------

 Target Compounds
 1)     methane                    0.51       35508608    1.4769396 mg/l
 --------------------------------------------------------------------------

(f)=RT Delta > 1/2 Window                                (m)=manual int.
```

Quantitation Report   (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060823\
Data File : 0608_22.d
Signal(s) : FID1A.CH
Acq On    : 08 Jun 2023  11:13 am
Operator  :
Sample    : L1623041-02 1x WG2073383
Misc      :
ALS Vial  : 22   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:16:38 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
OT07-MW04-060123

Lab Sample ID:	L1623041-03	**SDG:**	L1623041
Client Sample ID:	OT07-MW04-060123	**Collected Date/Time:**	06/01/23 09:35
Lab File ID:	0609B_03	**Received Date/Time:**	06/06/23 09:00
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/09/23 11:06
Analytical Batch:	WG2074059	**Analysis Date/Time:**	06/09/23 11:06
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

Analyte	**CAS**	**RT**	**Result**	**Qualifier**	**DL**	**LOD**	**LOQ**
			ug/l		*ug/l*	*ug/l*	*ug/l*
Methane	74-82-8	0.52	32.8		2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_03.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 11:06 am
Operator :
Sample : L1623041-03 1x WG2074059
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:08:33 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

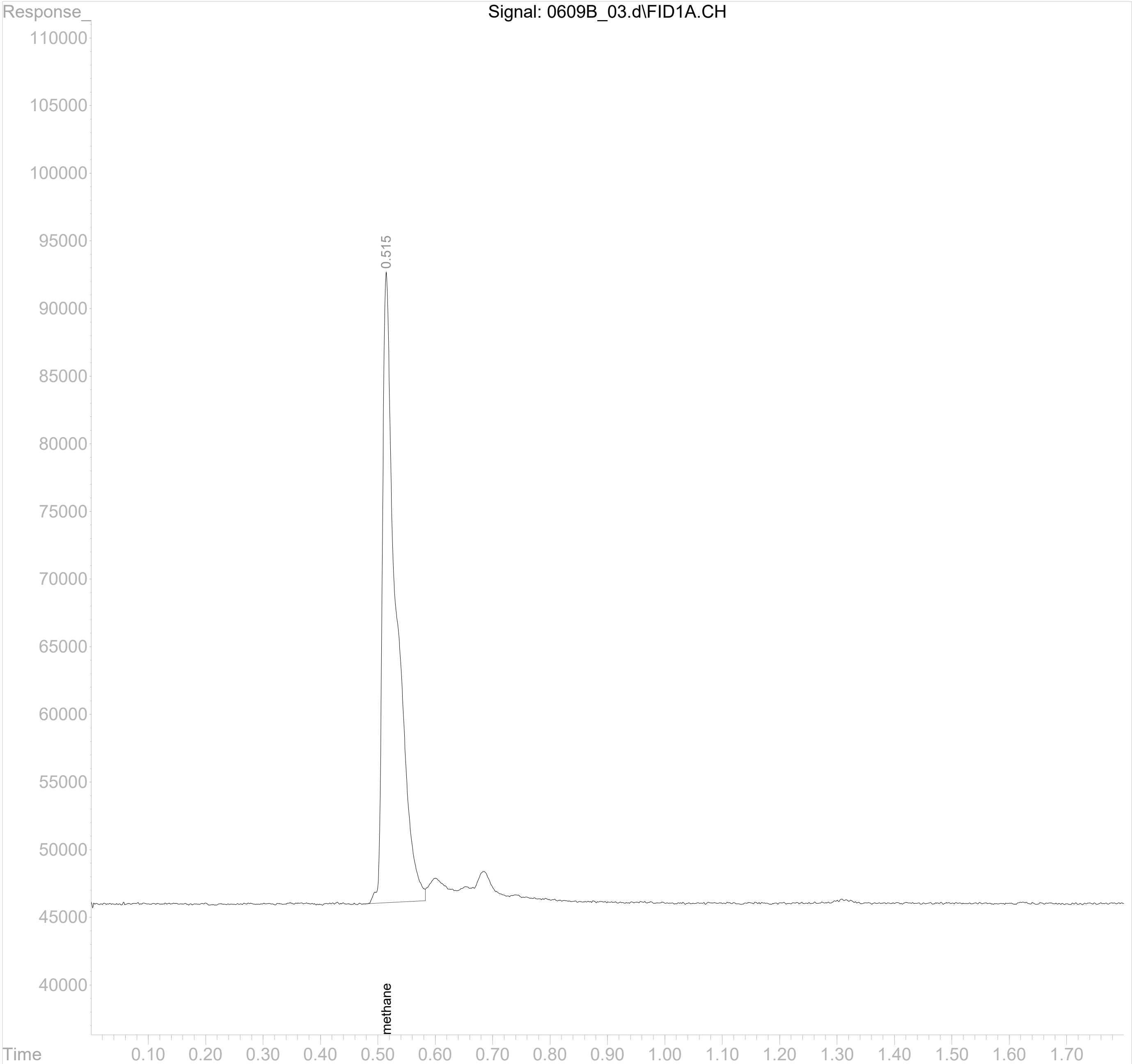
Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.52	788418	0.0327933 mg/l

(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_03.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  11:06 am
Operator  :
Sample    : L1623041-03 1x WG2074059
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:08:33 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
OT07-MW05-060123

Lab Sample ID:	L1623041-04	**SDG:**	L1623041
Client Sample ID:	OT07-MW05-060123	**Collected Date/Time:**	06/01/23 09:10
Lab File ID:	0608_24	**Received Date/Time:**	06/06/23 09:00
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/08/23 11:22
Analytical Batch:	WG2073383	**Analysis Date/Time:**	06/08/23 11:22
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	___________
Total Solids (%):	___________	**Final Wt/Vol:**	20 mL

Analyte	**CAS**	**RT**	**Result** *ug/l*	**Qualifier**	**DL** *ug/l*	**LOD** *ug/l*	**LOQ** *ug/l*
Methane	74-82-8	0	6.00	U	2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report (QT Reviewed)

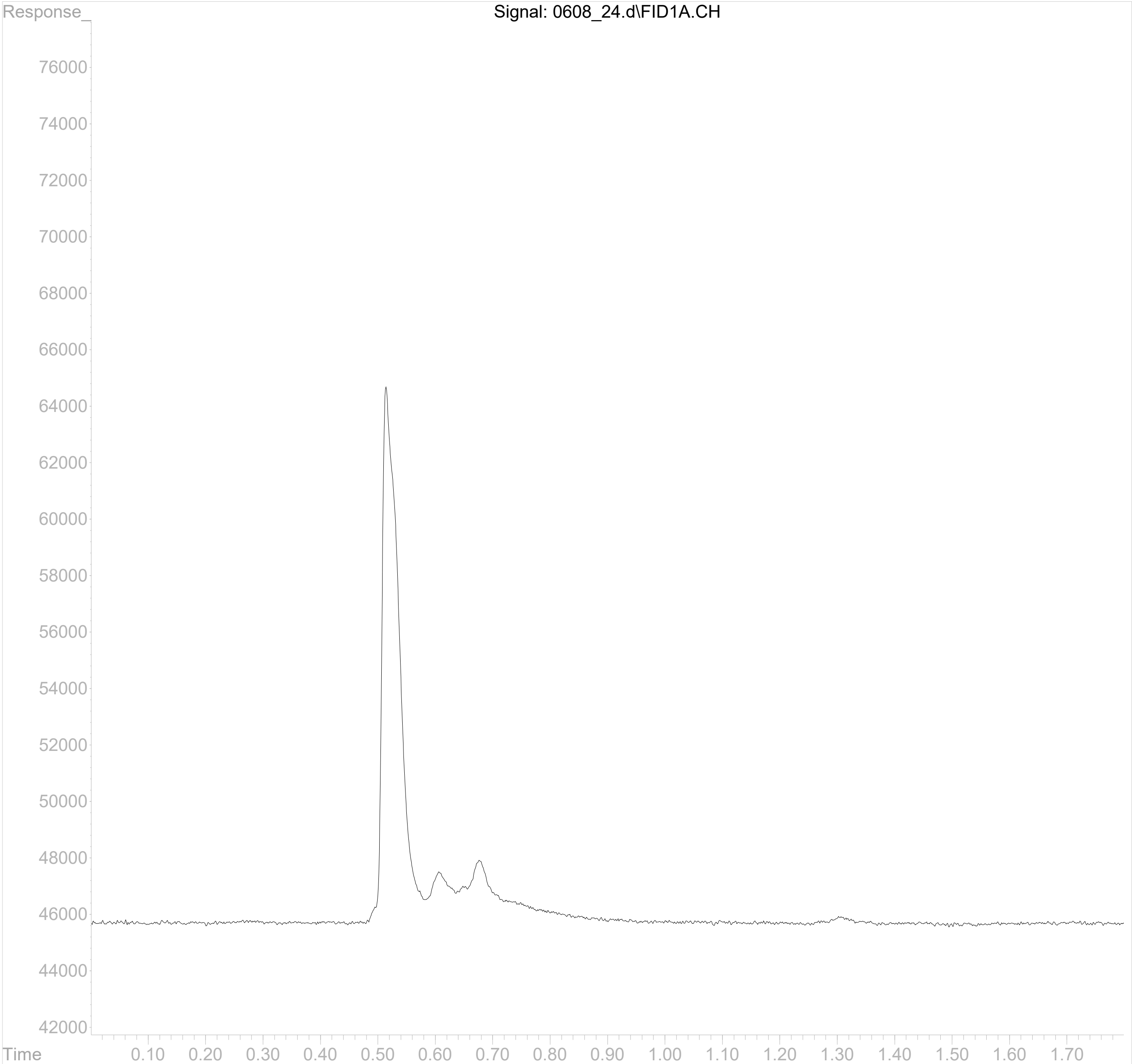
```
Data Path : C:\msdchem\1\data\060823\
Data File : 0608_24.d
Signal(s) : FID1A.CH
Acq On    : 08 Jun 2023  11:22 am
Operator  :
Sample    : L1623041-04 1x WG2073383
Misc      :
ALS Vial  : 24   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:28:26 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :

        Compound                     R.T.       Response    Conc Units
 -------------------------------------------------------------------------------

 Target Compounds
 ---------------------------------------------------------------------------

(f)=RT Delta > 1/2 Window                                (m)=manual int.
```

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_24.d
Signal(s) : FID1A.CH
Acq On : 08 Jun 2023 11:22 am
Operator :
Sample : L1623041-04 1x WG2073383
Misc :
ALS Vial : 24 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:28:26 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



1A-OR

SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
OT07-MW06-060123

Lab Sample ID:	L1623041-05	**SDG:**	L1623041
Client Sample ID:	OT07-MW06-060123	**Collected Date/Time:**	06/01/23 11:00
Lab File ID:	0608_25	**Received Date/Time:**	06/06/23 09:00
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/08/23 11:29
Analytical Batch:	WG2073383	**Analysis Date/Time:**	06/08/23 11:29
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

Analyte	**CAS**	**RT**	**Result** *ug/l*	**Qualifier**	**DL** *ug/l*	**LOD** *ug/l*	**LOQ** *ug/l*
Methane	74-82-8	0.51	327		2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060823\
Data File : 0608_25.d
Signal(s) : FID1A.CH
Acq On    : 08 Jun 2023  11:29 am
Operator  :
Sample    : L1623041-05 1x WG2073383
Misc      :
ALS Vial  : 25   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:36:36 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```

Compound	R.T.	Response	Conc Units
Target Compounds			
1) methane	0.51	7871738	0.3274159 mg/l

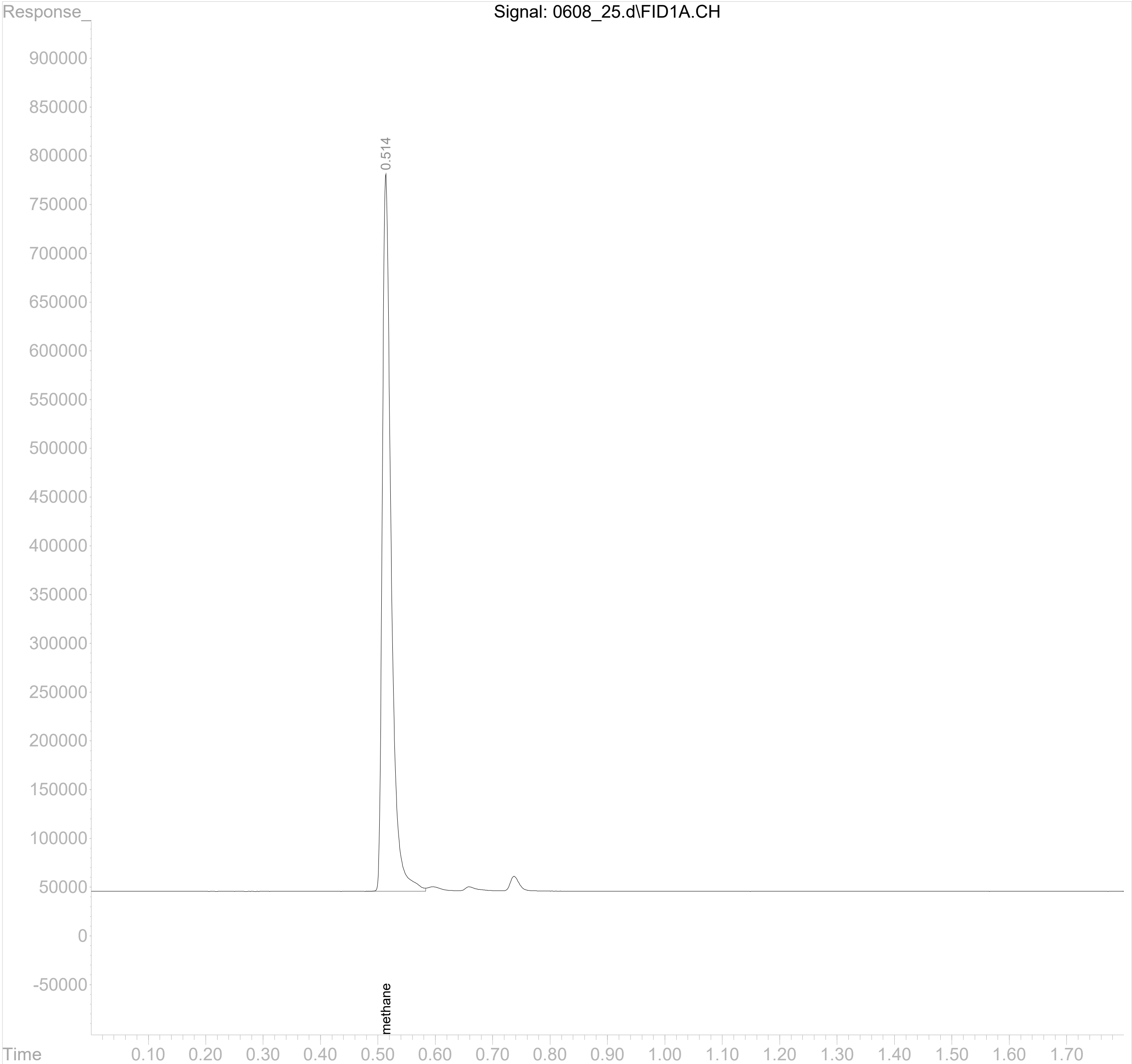
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_25.d
Signal(s) : FID1A.CH
Acq On : 08 Jun 2023 11:29 am
Operator :
Sample : L1623041-05 1x WG2073383
Misc :
ALS Vial : 25 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:36:36 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
OT07-MW04-060123MS

Lab Sample ID:	L1623041-06	**SDG:**	L1623041
Client Sample ID:	OT07-MW04-060123MS	**Collected Date/Time:**	06/01/23 09:55
Lab File ID:	0609B_04	**Received Date/Time:**	06/06/23 09:00
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/09/23 11:08
Analytical Batch:	WG2074059	**Analysis Date/Time:**	06/09/23 11:08
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

Analyte	CAS	RT	Result	Qualifier	DL	LOD	LOQ
			ug/l		*ug/l*	*ug/l*	*ug/l*
Methane	74-82-8	0.52	42.8		2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_04.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  11:08 am
Operator  :
Sample    : L1623041-06 1x WG2074059
Misc      :
ALS Vial  : 4   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:12:25 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```

Compound	R.T.	Response	Conc Units
Target Compounds			
1) methane	0.52	1028311	0.0427714 mg/l

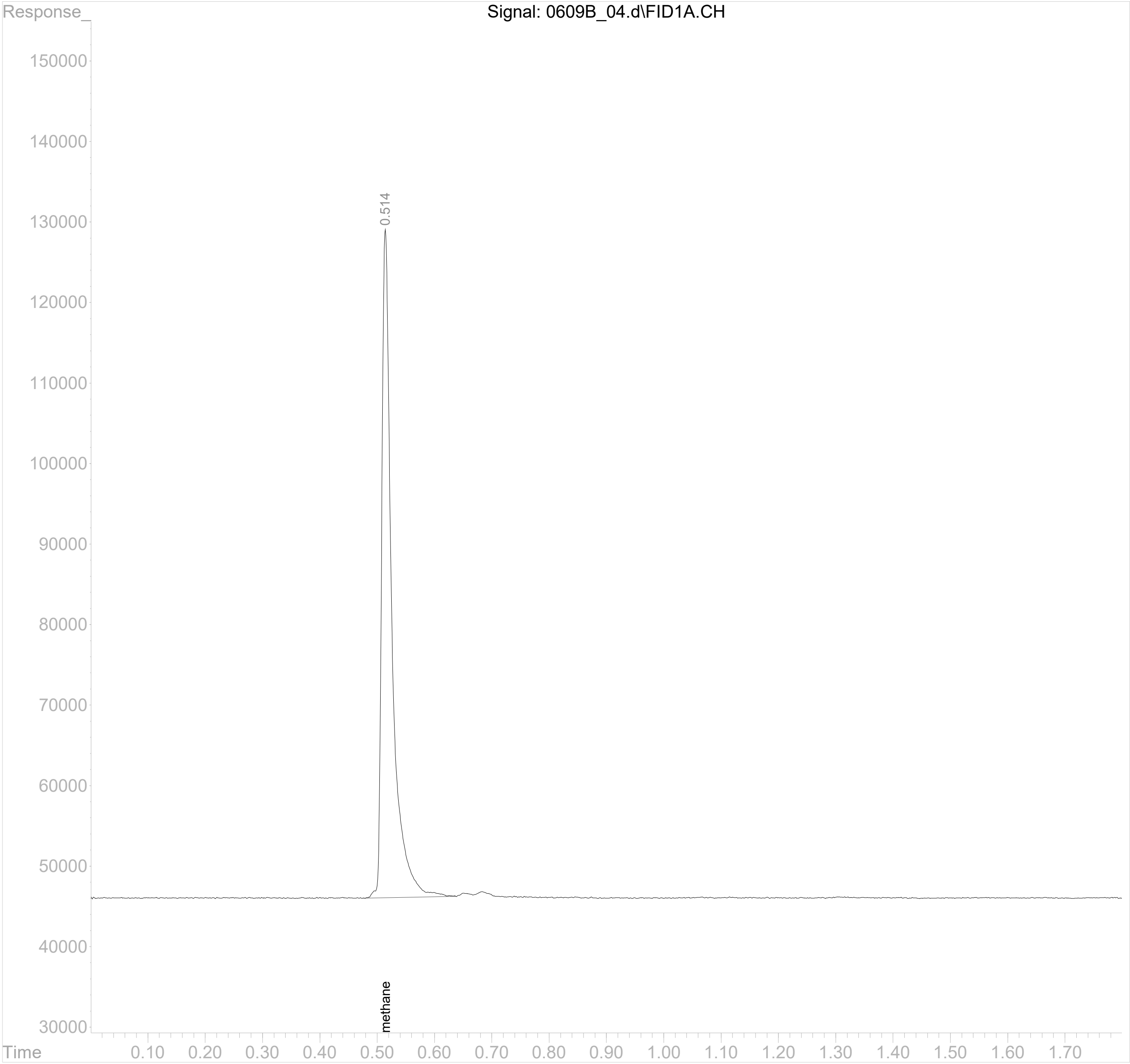
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report   (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_04.d
Signal(s) : FID1A.CH
Acq On   : 09 Jun 2023 11:08 am
Operator :
Sample   : L1623041-06 1x WG2074059
Misc   :
ALS Vial : 4   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:12:25 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



1A-OR

SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
OT07-MW04-060123MSD

Lab Sample ID:	L1623041-07	**SDG:**	L1623041
Client Sample ID:	OT07-MW04-060123MSD	**Collected Date/Time:**	06/01/23 10:15
Lab File ID:	0609B_05	**Received Date/Time:**	06/06/23 09:00
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/09/23 11:11
Analytical Batch:	WG2074059	**Analysis Date/Time:**	06/09/23 11:11
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

Analyte	**CAS**	**RT**	**Result** *ug/l*	**Qualifier**	**DL** *ug/l*	**LOD** *ug/l*	**LOQ** *ug/l*
Methane	74-82-8	0	6.00	U	2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report (QT Reviewed)

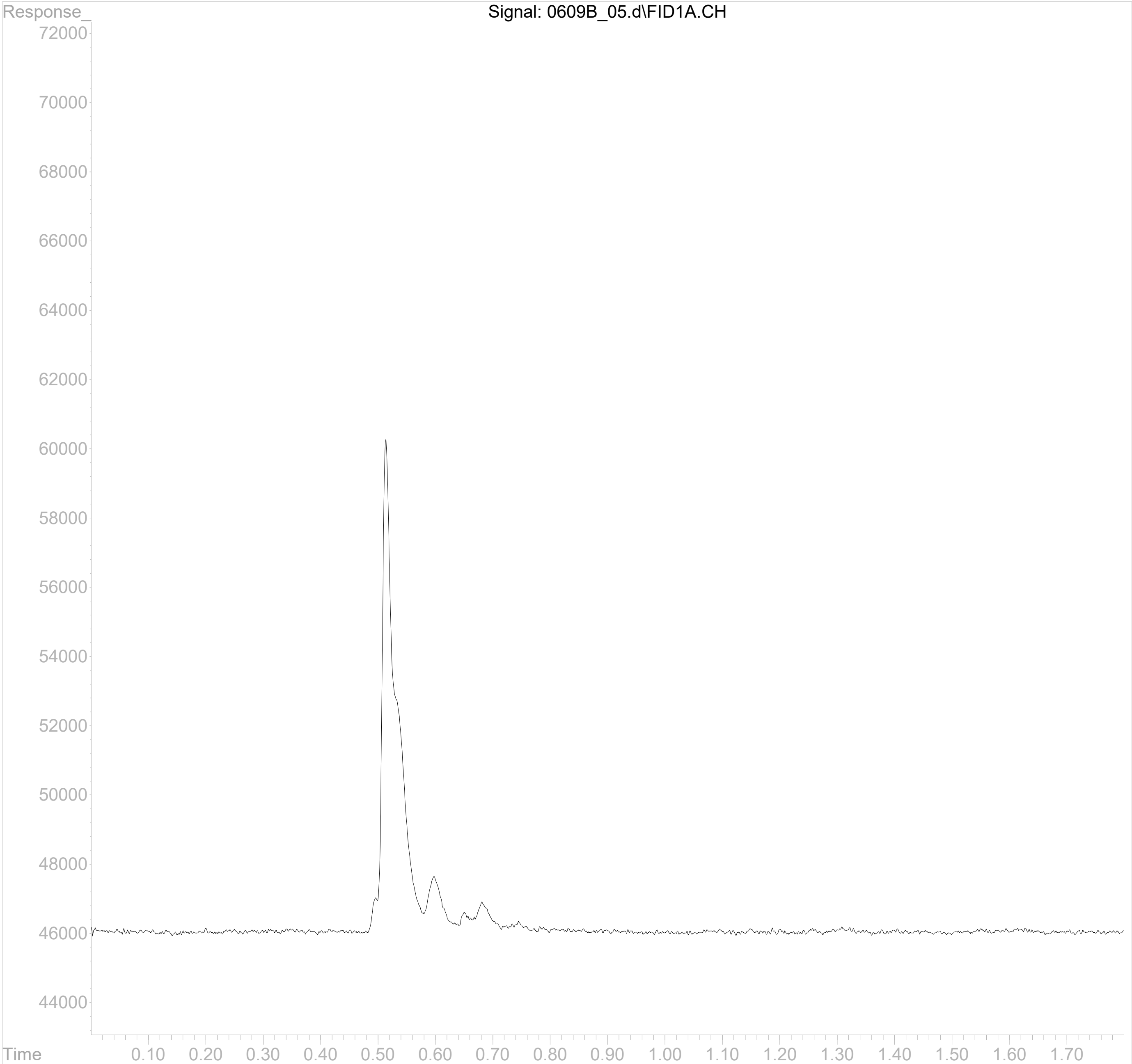
```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_05.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  11:11 am
Operator  :
Sample    : L1623041-07 1x WG2074059
Misc      :
ALS Vial  : 5   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:17:05 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :

        Compound                     R.T.       Response    Conc Units
 --------------------------------------------------------------------------------

 Target Compounds
 ---------------------------------------------------------------------------

(f)=RT Delta > 1/2 Window                                (m)=manual int.
```

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_05.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 11:11 am
Operator :
Sample : L1623041-07 1x WG2074059
Misc :
ALS Vial : 5 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:17:05 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



1A-OR

SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
OT07-MW806-060123

Lab Sample ID:	L1623041-08	**SDG:**	L1623041
Client Sample ID:	OT07-MW806-060123	**Collected Date/Time:**	06/01/23 11:00
Lab File ID:	0609B_06	**Received Date/Time:**	06/06/23 09:00
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/09/23 11:14
Analytical Batch:	WG2074059	**Analysis Date/Time:**	06/09/23 11:14
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

Analyte	CAS	RT	Result	Qualifier	DL	LOD	LOQ
			ug/l		*ug/l*	*ug/l*	*ug/l*
Methane	74-82-8	0.51	501		2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report (QT Reviewed)

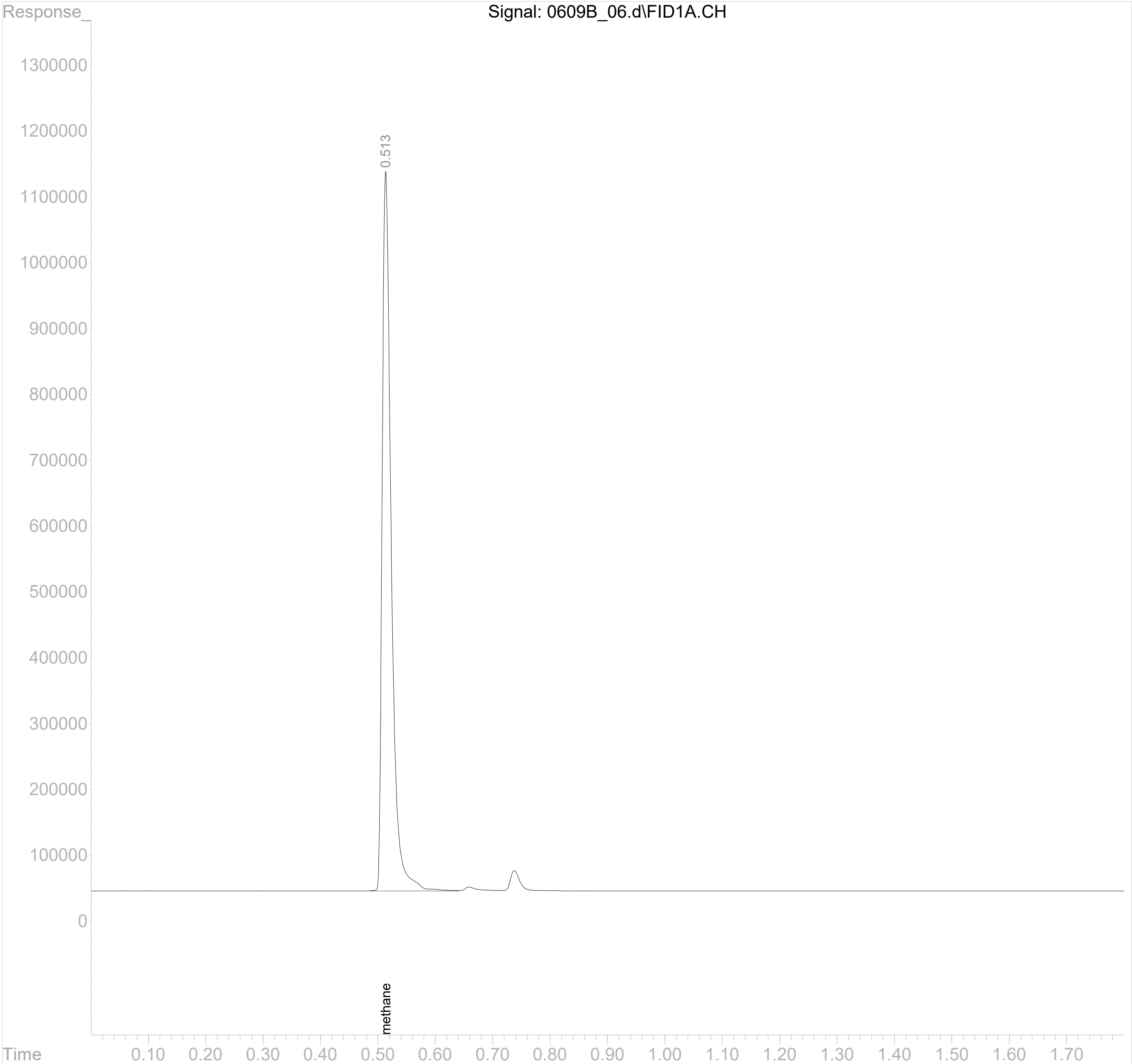
```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_06.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  11:14 am
Operator  :
Sample    : L1623041-08 1x WG2074059
Misc      :
ALS Vial  : 6   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:20:36 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	12047532	0.5011032 mg/l

(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_06.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  11:14 am
Operator  :
Sample    : L1623041-08 1x WG2074059
Misc      :
ALS Vial  : 6   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:20:36 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



6B-OR

RETENTION TIME
INITIAL CALIBRATION OF
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Analytical Method:	RSK175		

Analyte	CS: 1	CS: 2	CS: 3.0	CS: 4	CS: 5	CS: 6	CS: 7
METHANE	0.51	0.51	0.51	0.51	0.51	0.51	0.51
ETHANE	0.66	0.66	0.66	0.66	0.66	0.66	0.66
ETHENE	0.74	0.74	0.74	0.74	0.74	0.74	0.74
File ID:	1123A_01	1123A_02	1123A_03	1123A_04	1123A_05	1123A_06	1123A_07

6C-OR

CALIBRATION FACTOR
INITIAL CALIBRATION OF
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Analytical Method:	RSK175		

Analyte	RRF: 1	RRF: 2	RRF: 3.0	RRF: 4	RRF: 5	RRF: 6	RRF: 7
METHANE	23620000	25250000	24960000	25220000	23130000	22800000	23320000
ETHANE	24710000	25180000	25750000	27040000	24030000	22820000	24320000
ETHENE	27510000	25760000	26540000	26520000	23600000	22530000	24260000
File ID:	1123A_01	1123A_02	1123A_03	1123A_04	1123A_05	1123A_06	1123A_07

6C-OR

CALIBRATION FACTOR
INITIAL CALIBRATION OF
SINGLE COMPONENT ANALYTES

SDG: L1623041
Instrument ID: AIRGC2
Analytical Method: RSK175

Calibration (begin) date/time: 11/23/22 12:44
Calibration (end) date/time: 11/23/22 13:32

Analyte	RRF. Avg	%RSD	COD
METHANE	24042020	4.42	
ETHANE	24835840	5.41	
ETHENE	25246600	7.18	

Response Factor Report AIRGC2

Method Path : C:\msdchem\1\methods\
Method File : GC2RSKK23V.M
Title :
Last Update : Wed Nov 23 13:35:39 2022
Response Via : Initial Calibration

Calibration Files

1	=1123A_01.d	2	=1123A_02.d	3.0	=1123A_03.d
4	=1123A_04.d	5	=1123A_05.d	6	=1123A_06.d

	Compound	1	2	3.0	4	5	6	Avg	%RSD
1)	methane	2.362	2.525	2.496	2.522	2.313	2.280	2.404 E7	4.42
2)	ethane	2.471	2.518	2.575	2.704	2.403	2.282	2.484 E7	5.41
3)	ethene	2.751	2.576	2.654	2.652	2.360	2.253	2.525 E7	7.18
4)	acetylene	1.773	1.839	1.907	1.872	1.682	1.657	1.781 E7	5.37
5)	propane	2.410	2.549	2.618	2.648	2.361	2.228	2.448 E7	6.52

(#) = Out of Range ### Number of calibration levels exceeded format ###

Quantitation Report (QT Reviewed)

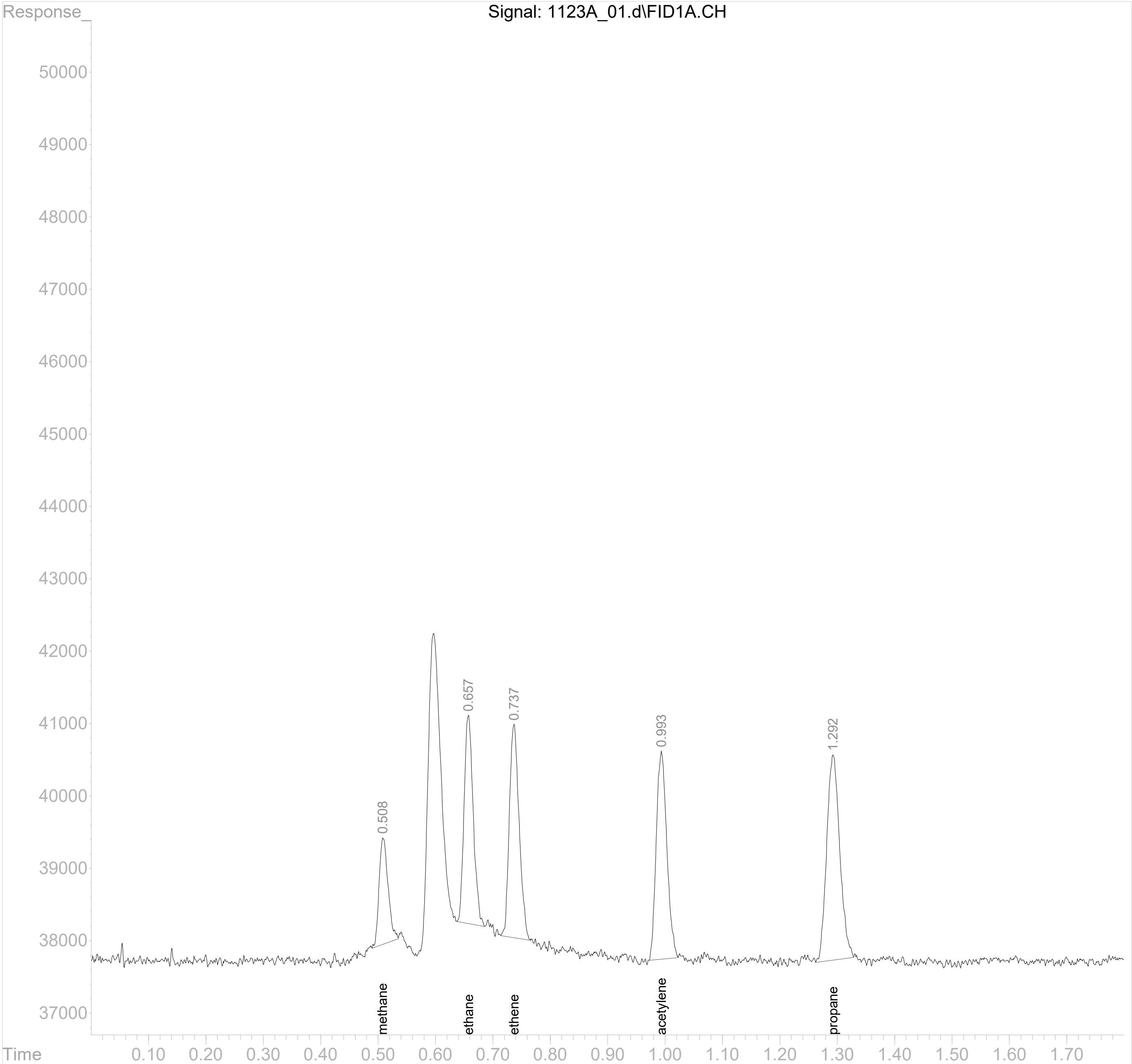
```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:18:38 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:16:49 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	16011	0.0006378 mg/l m
2)	ethane	0.66	31872	0.0012516 mg/l m
3)	ethene	0.74	34939	0.0013360 mg/l m
4)	acetylene	0.99	36887	0.0019694 mg/l m
5)	propane	1.29	44819	0.0017348 mg/l m

(f)=RT Delta > 1/2 Window (m)=manual int.

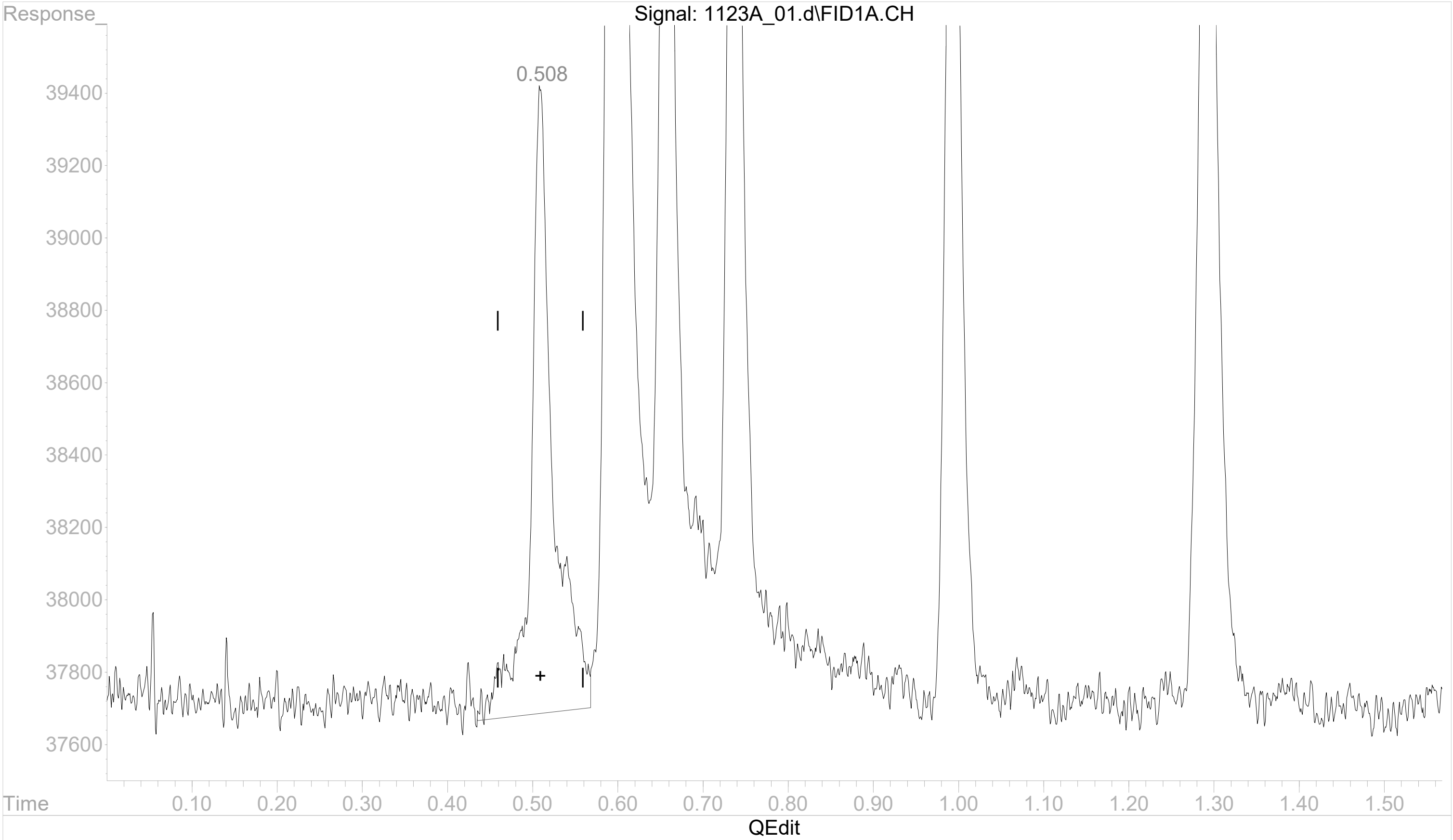
Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:18:38 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:16:49 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:58:10 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:58:00 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(1) methane

0.509min   0.0012196 mg/l

response   32099

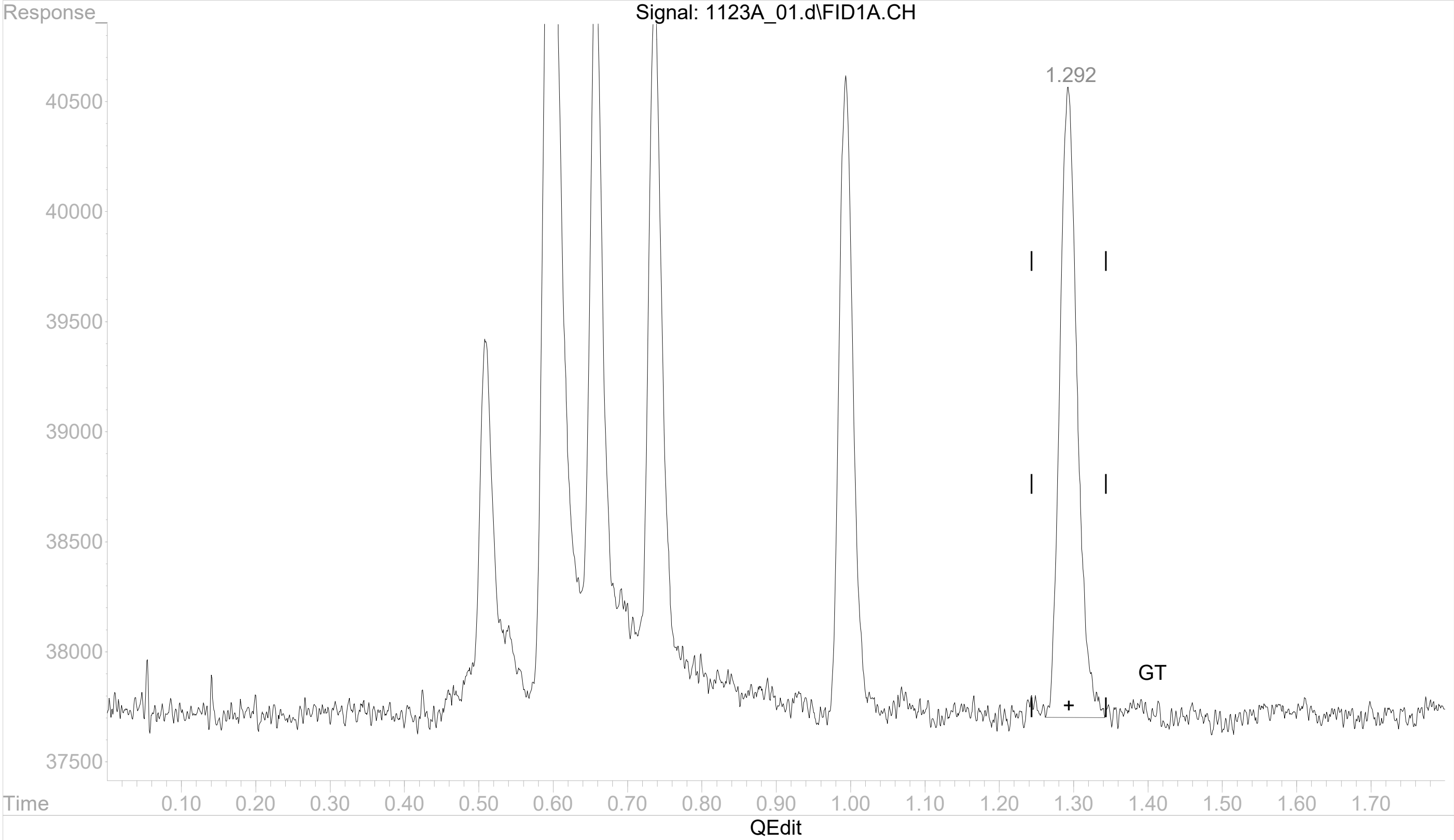
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:44 pm
Operator :
Sample : STD AGC 1 RSK175
Misc :
ALS Vial : 1 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:58:10 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:58:00 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(5) propane

1.292min 0.0018188 mg/l m

response 46676

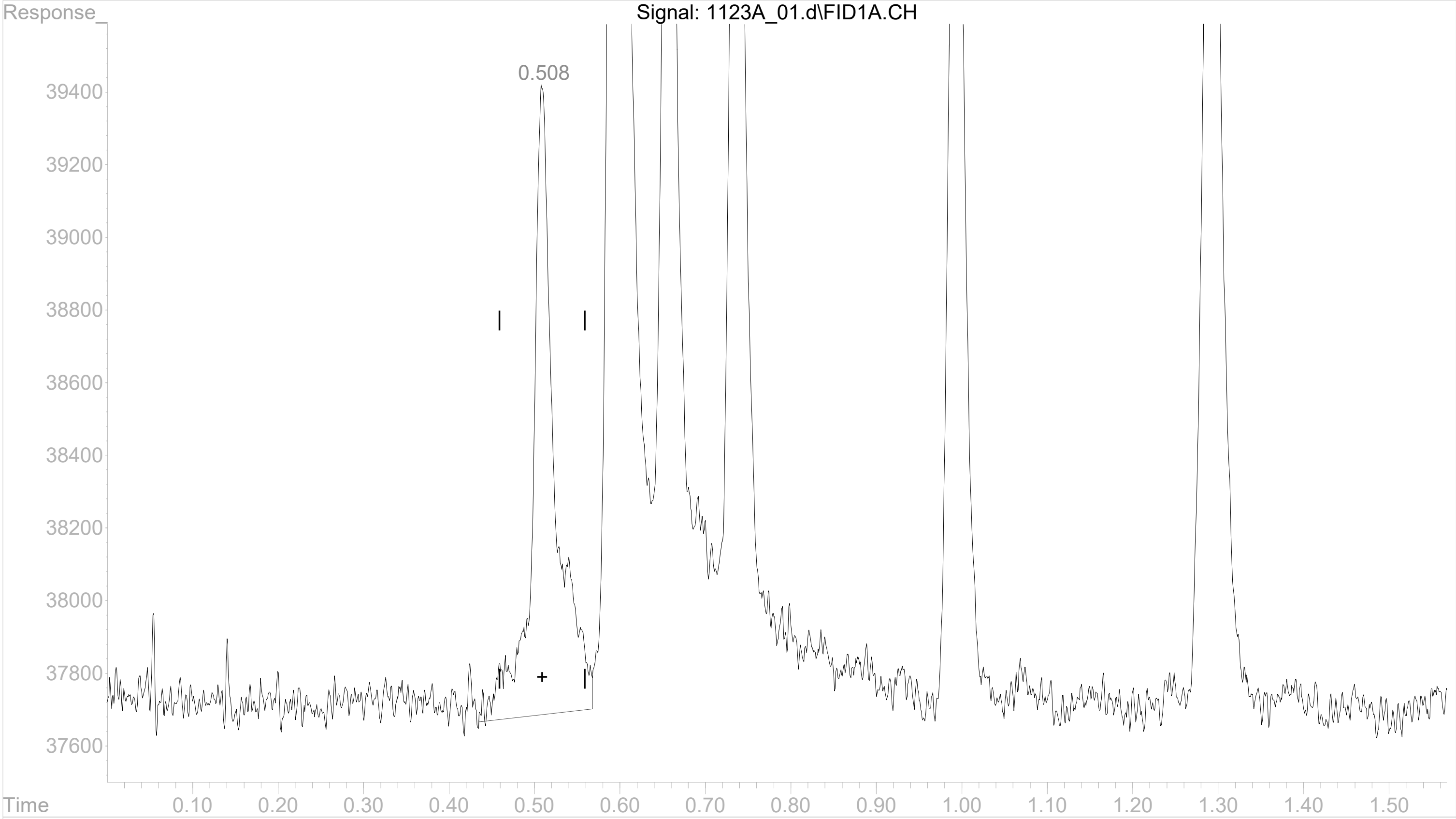
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:44 pm
Operator :
Sample : STD AGC 1 RSK175
Misc :
ALS Vial : 1 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:17:03 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:16:49 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(1) methane

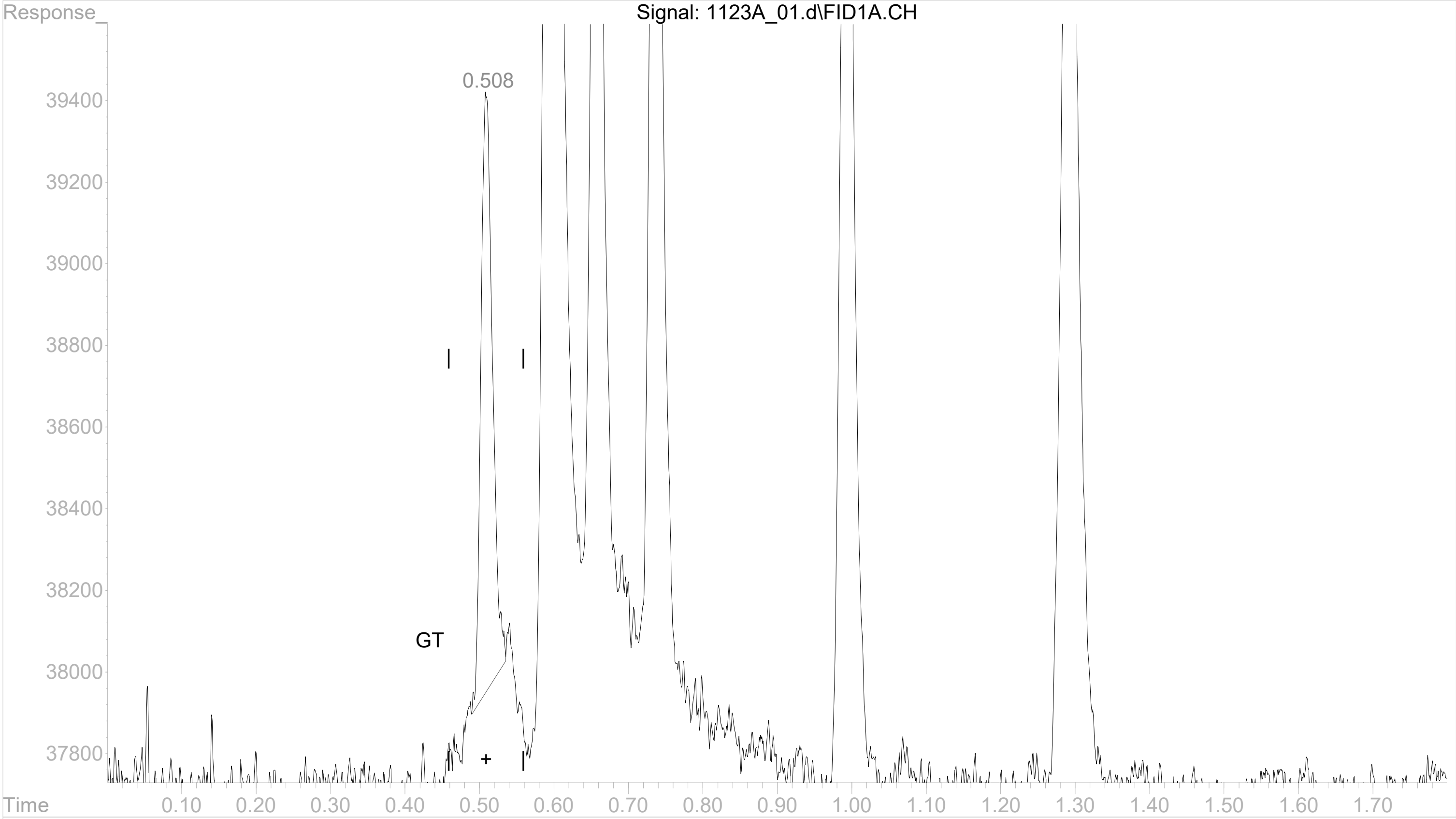
0.509min 0.0012787 mg/l

response 32099

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:17:03 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:16:49 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(1) methane

0.508min 0.0006378 mg/l m

response 16011

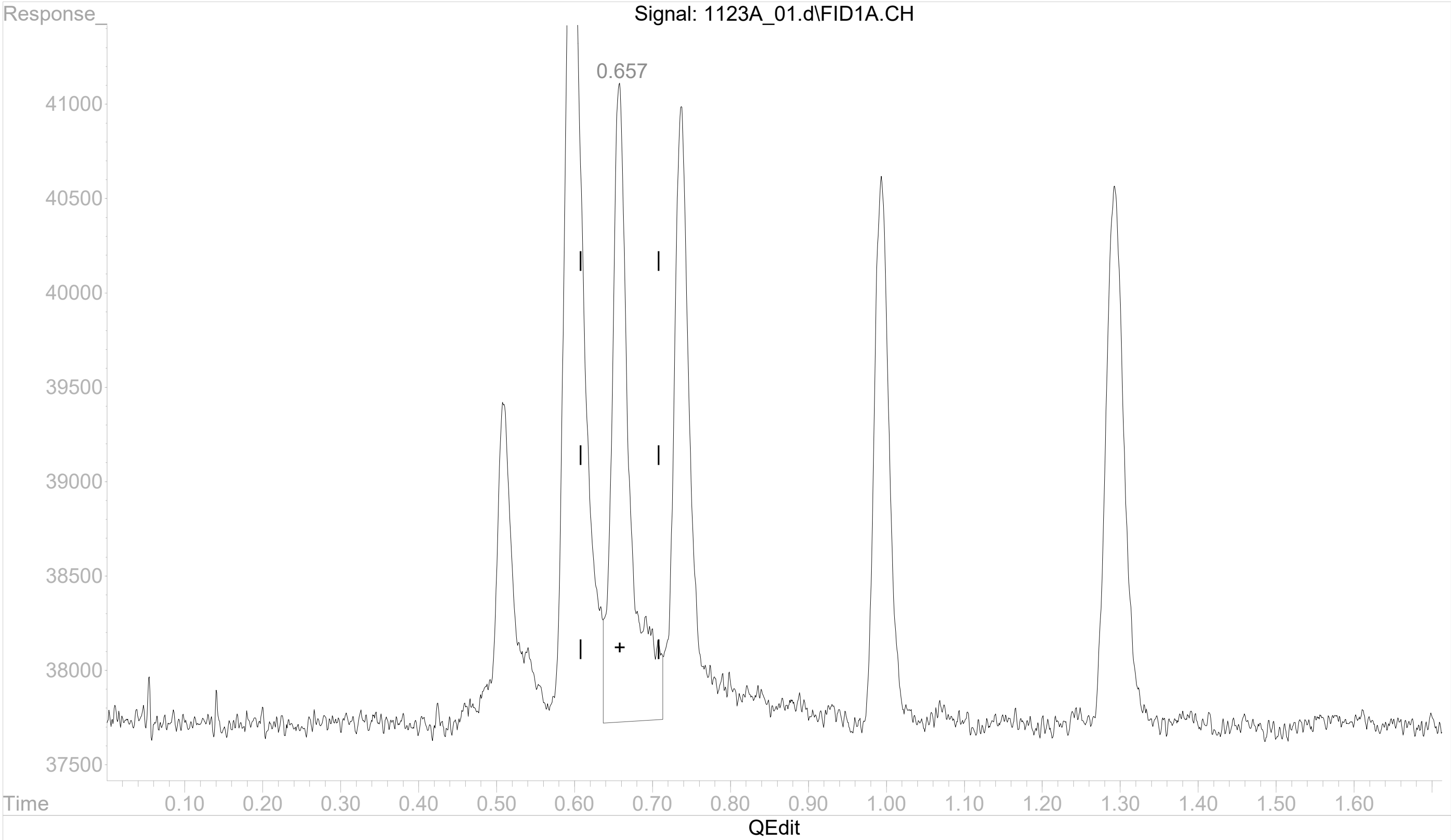
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:44 pm
Operator :
Sample : STD AGC 1 RSK175
Misc :
ALS Vial : 1 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:17:03 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:16:49 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(2) ethane

0.658min 0.0021169 mg/l

response 53907

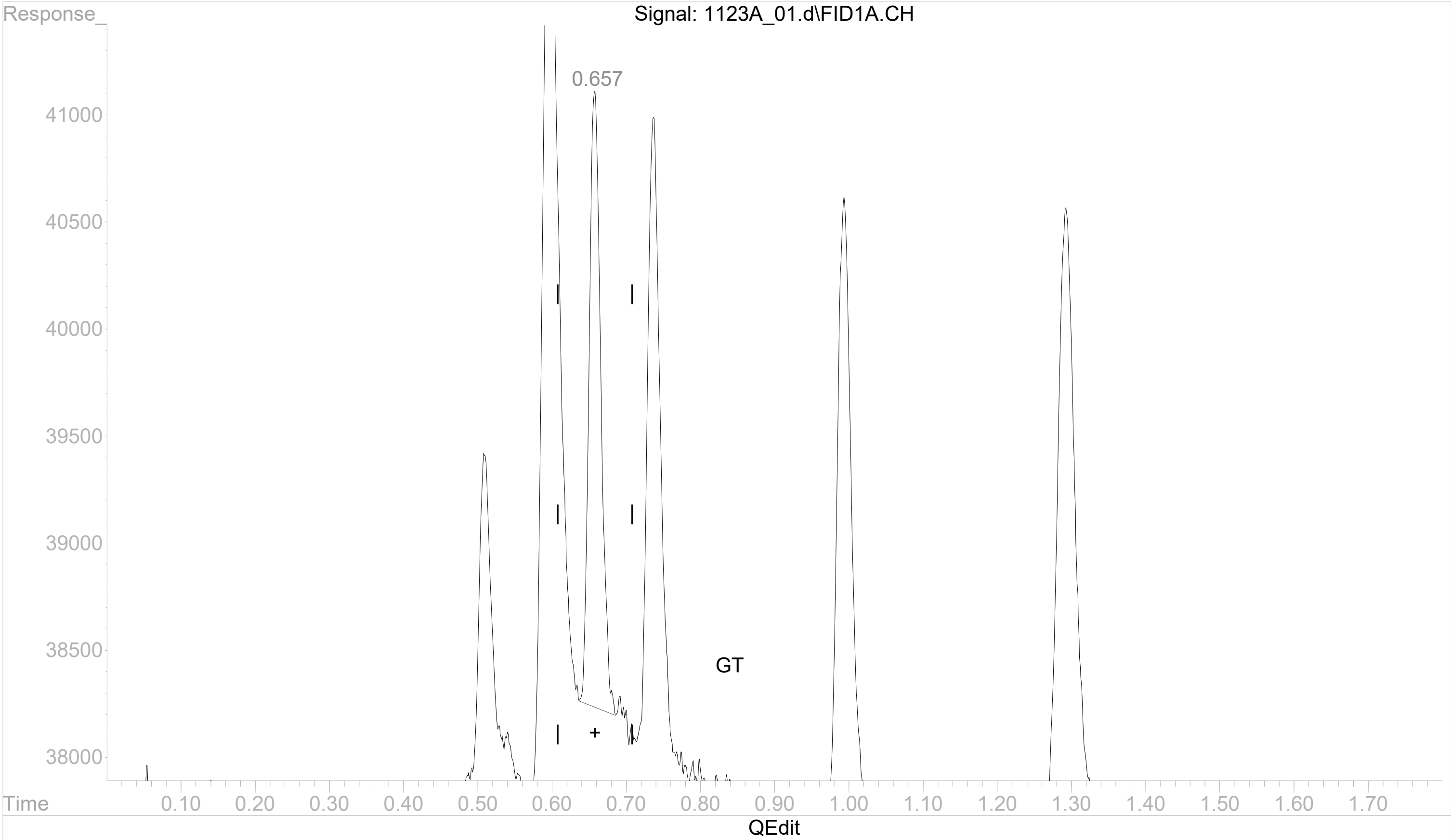
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On   : 23 Nov 2022 12:44 pm
Operator :
Sample   : STD AGC 1 RSK175
Misc   :
ALS Vial : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:17:03 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:16:49 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(2) ethane

0.657min 0.0012516 mg/l m

response 31872

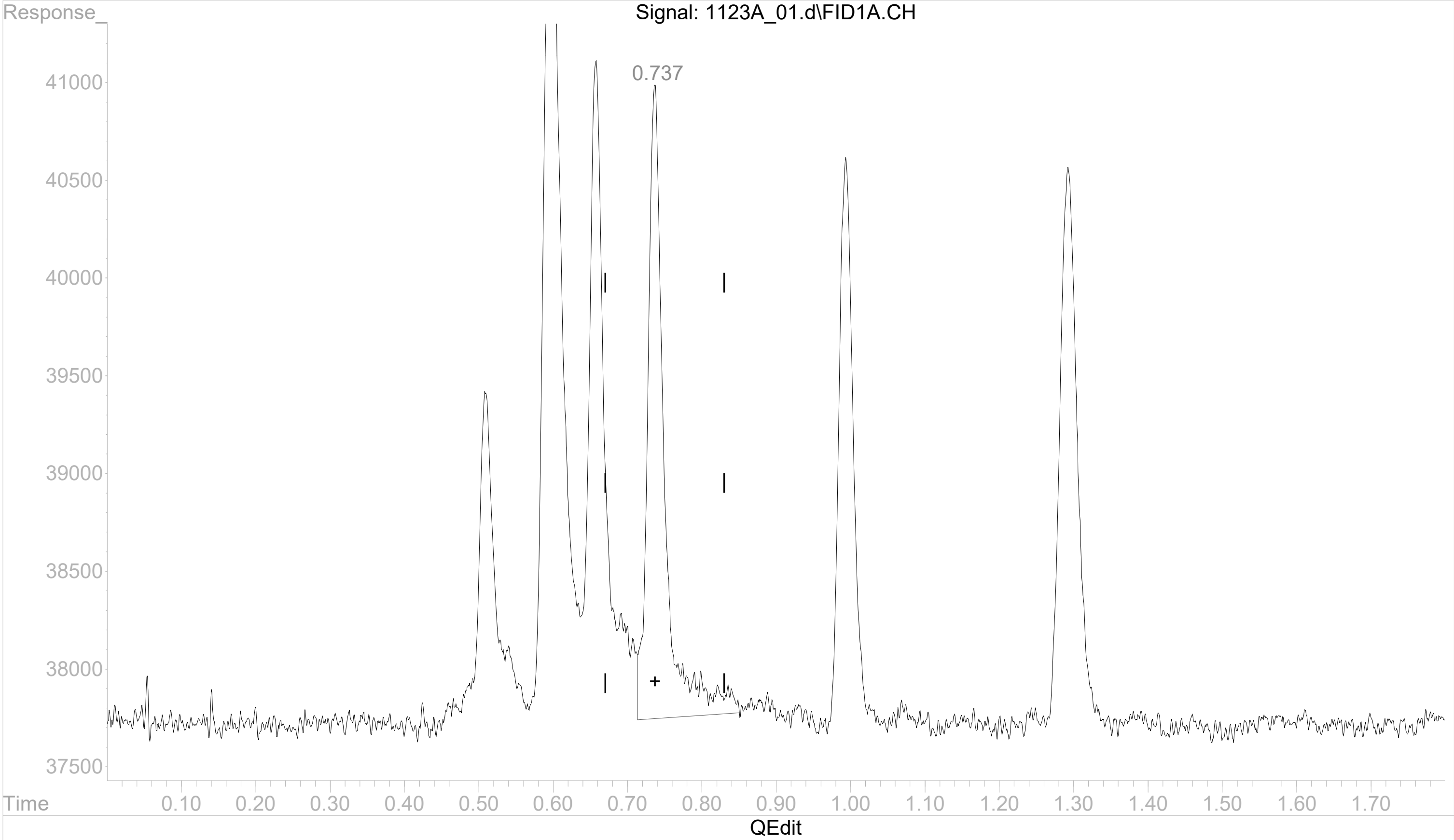
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:44 pm
Operator :
Sample : STD AGC 1 RSK175
Misc :
ALS Vial : 1 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:17:03 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:16:49 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(3) ethene

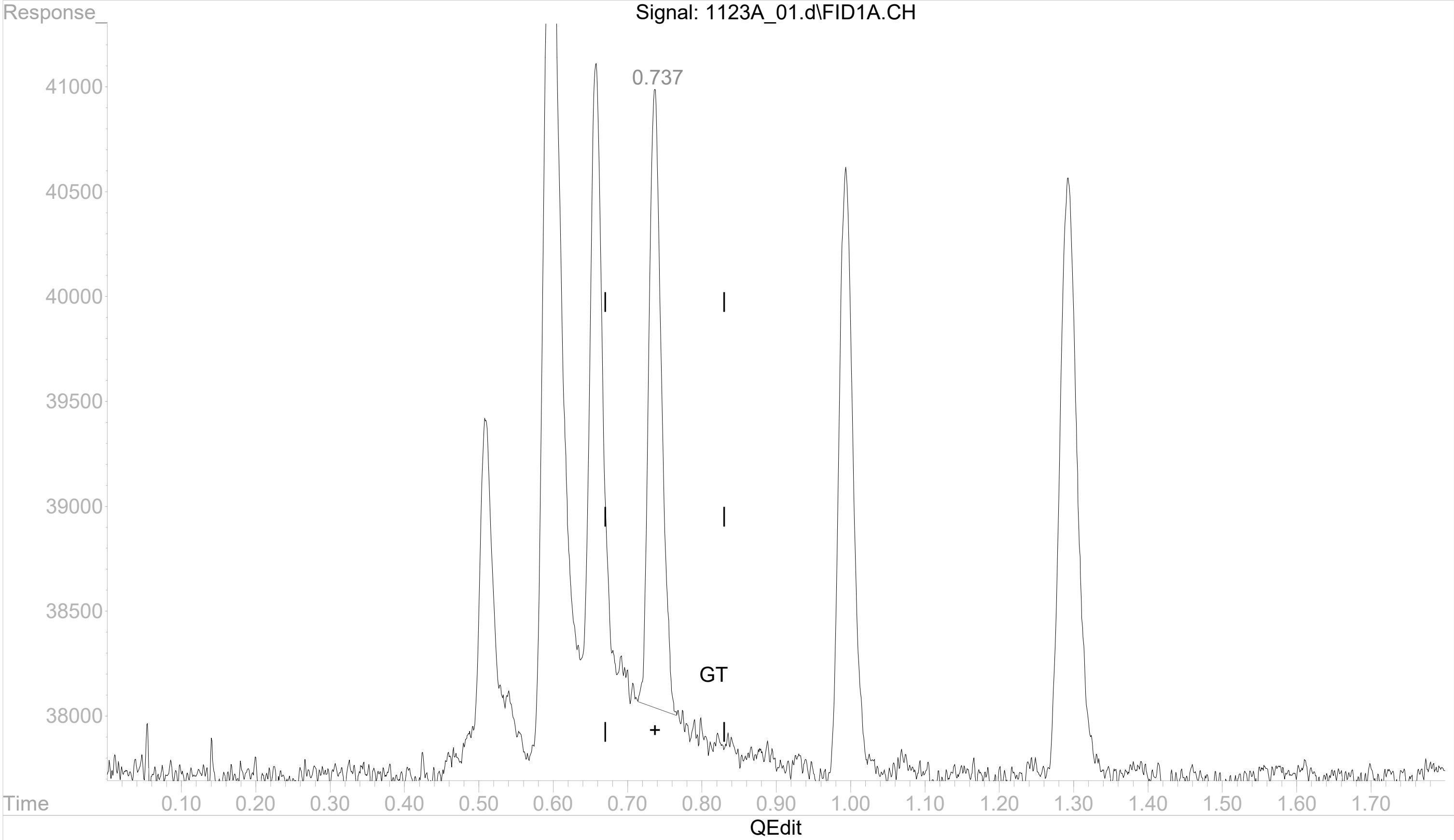
0.737min 0.0019480 mg/l

response 50943

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:17:03 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:16:49 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(3) ethene

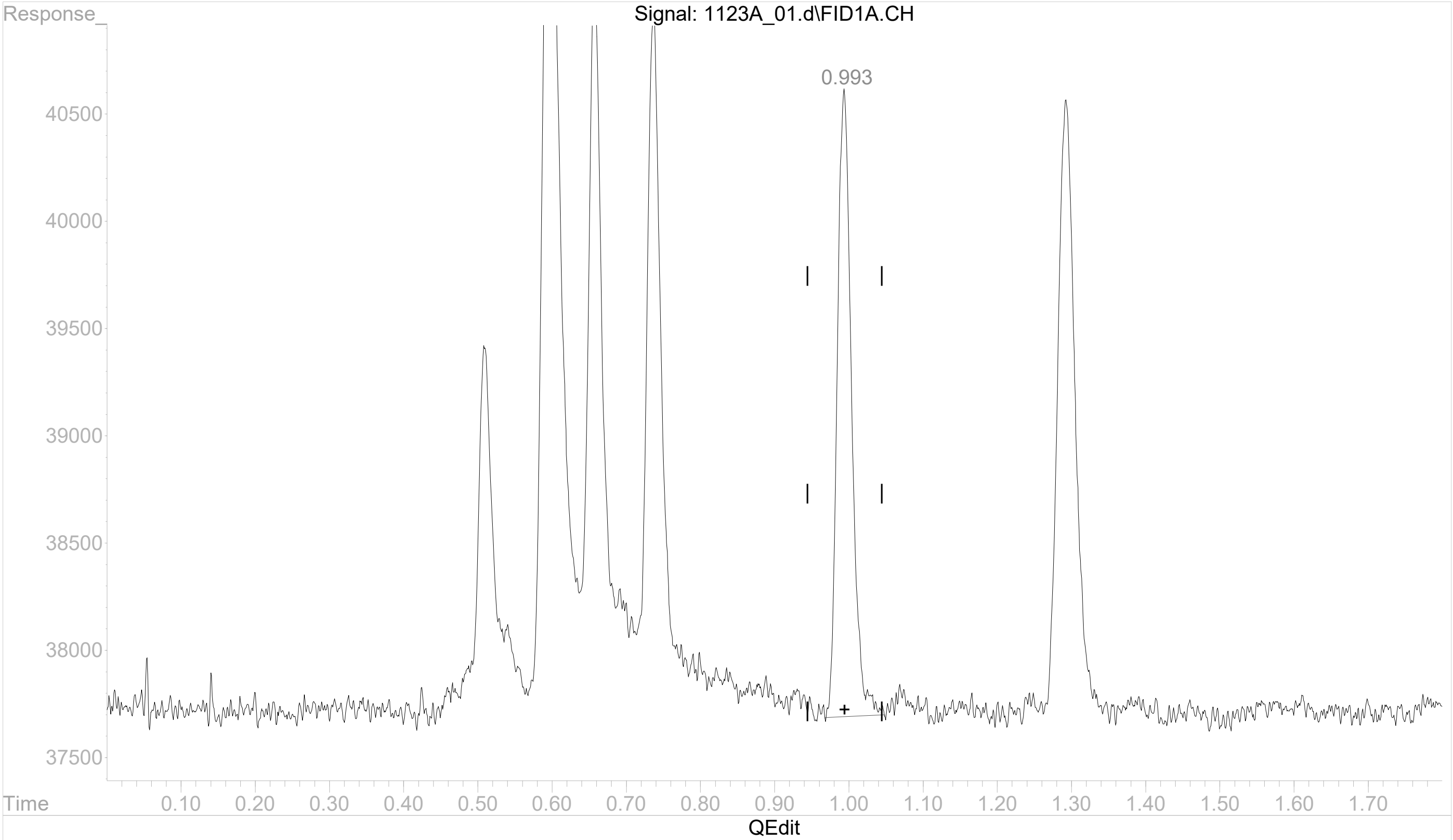
0.737min 0.0013360 mg/l m

response 34939

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:17:03 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:16:49 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(4) acetylene

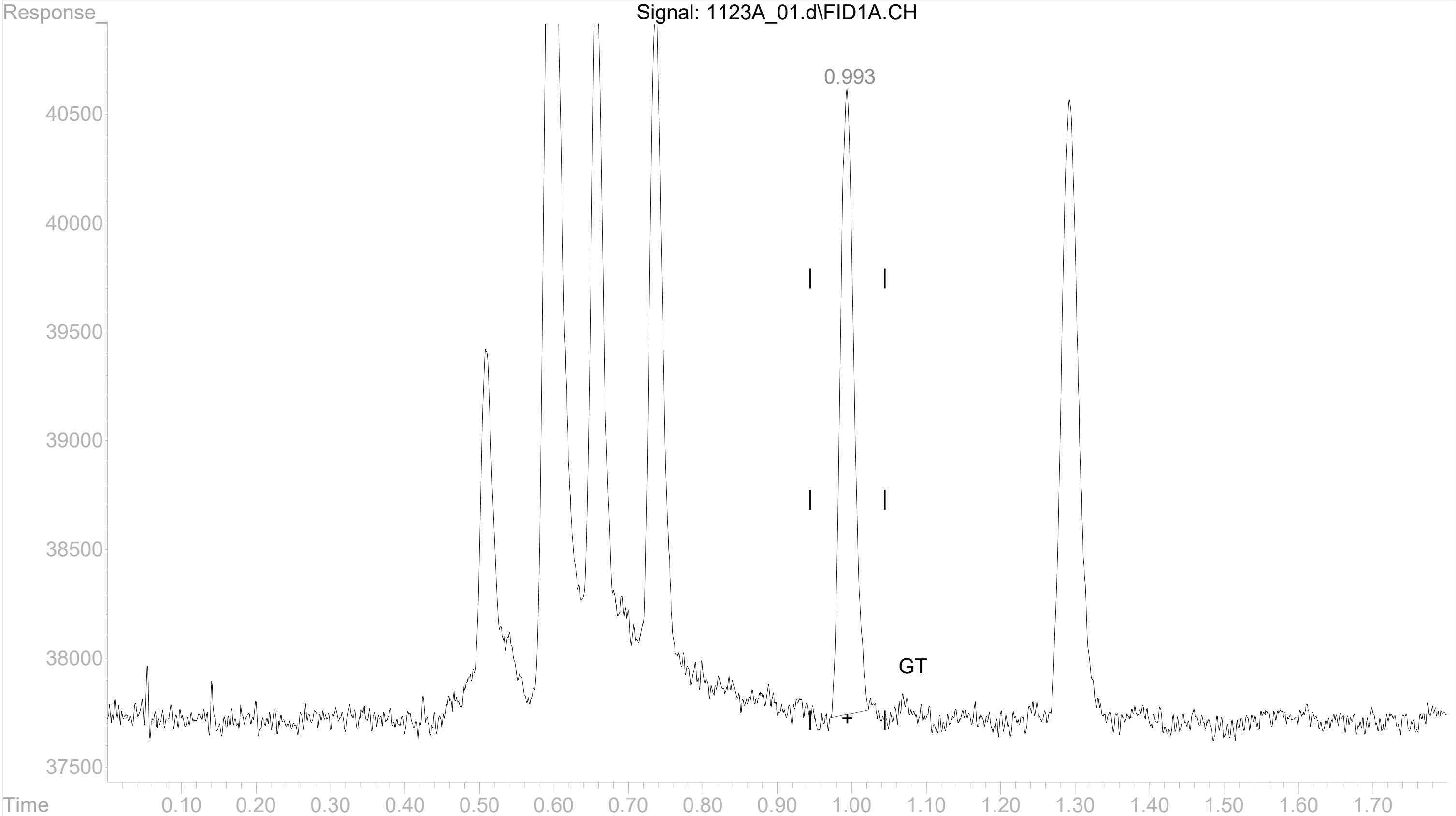
0.994min 0.0021035 mg/l

response 39399

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:17:03 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:16:49 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(4) acetylene

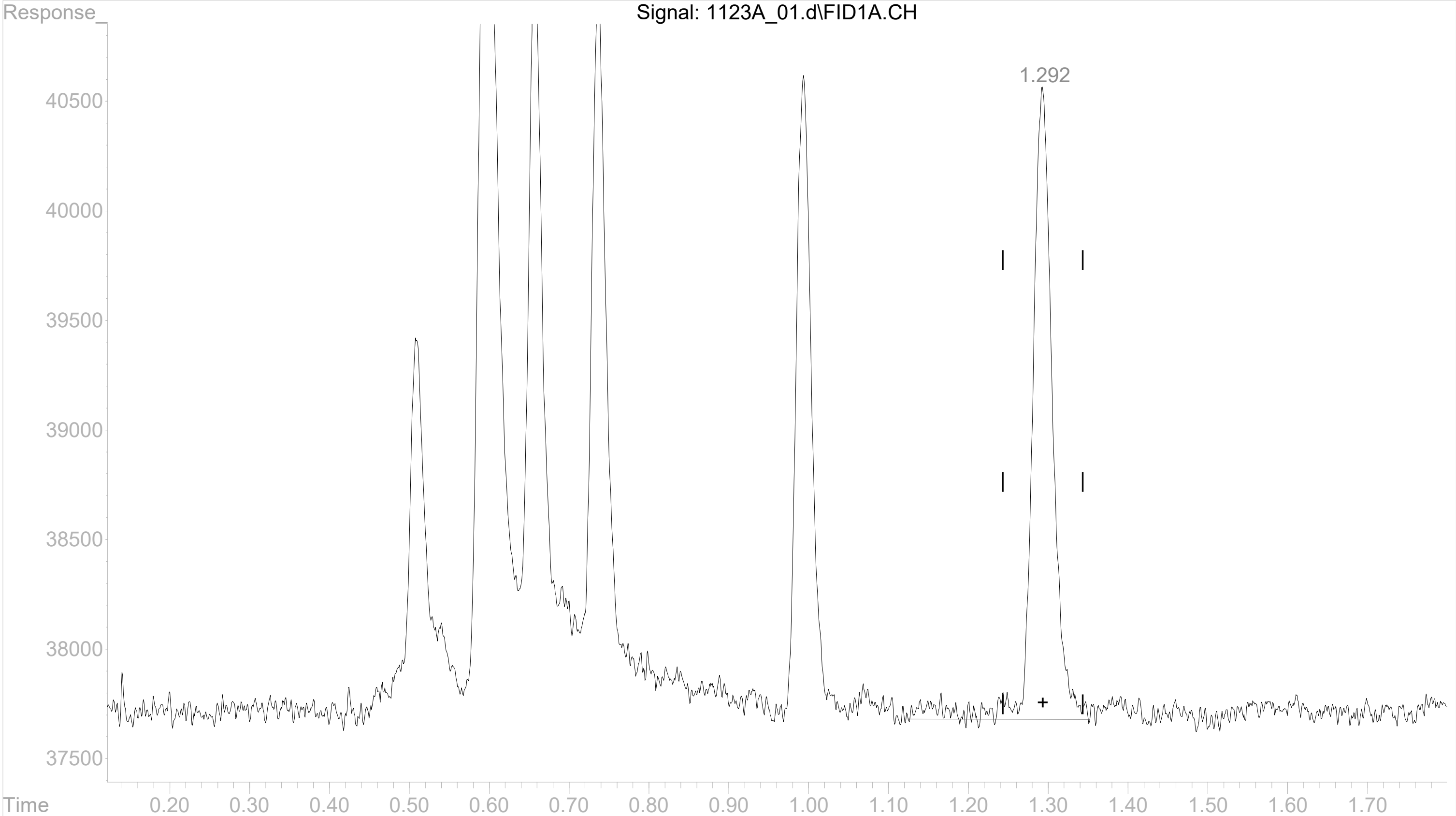
0.993min 0.0019694 mg/l m

response 36887

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:17:03 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:16:49 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(5) propane

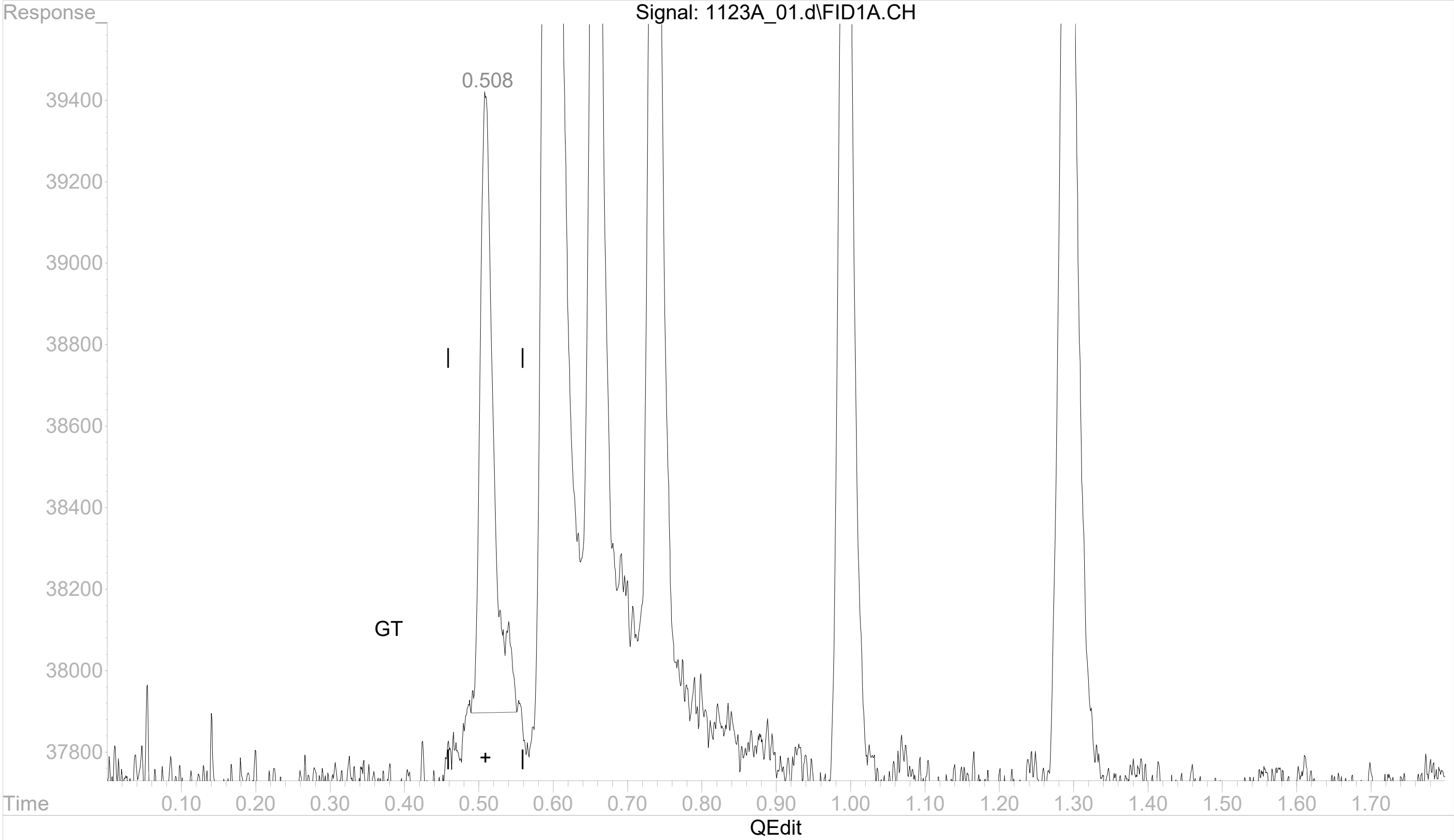
1.293min 0.0019928 mg/l

response 51484

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:58:10 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:58:00 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(1) methane

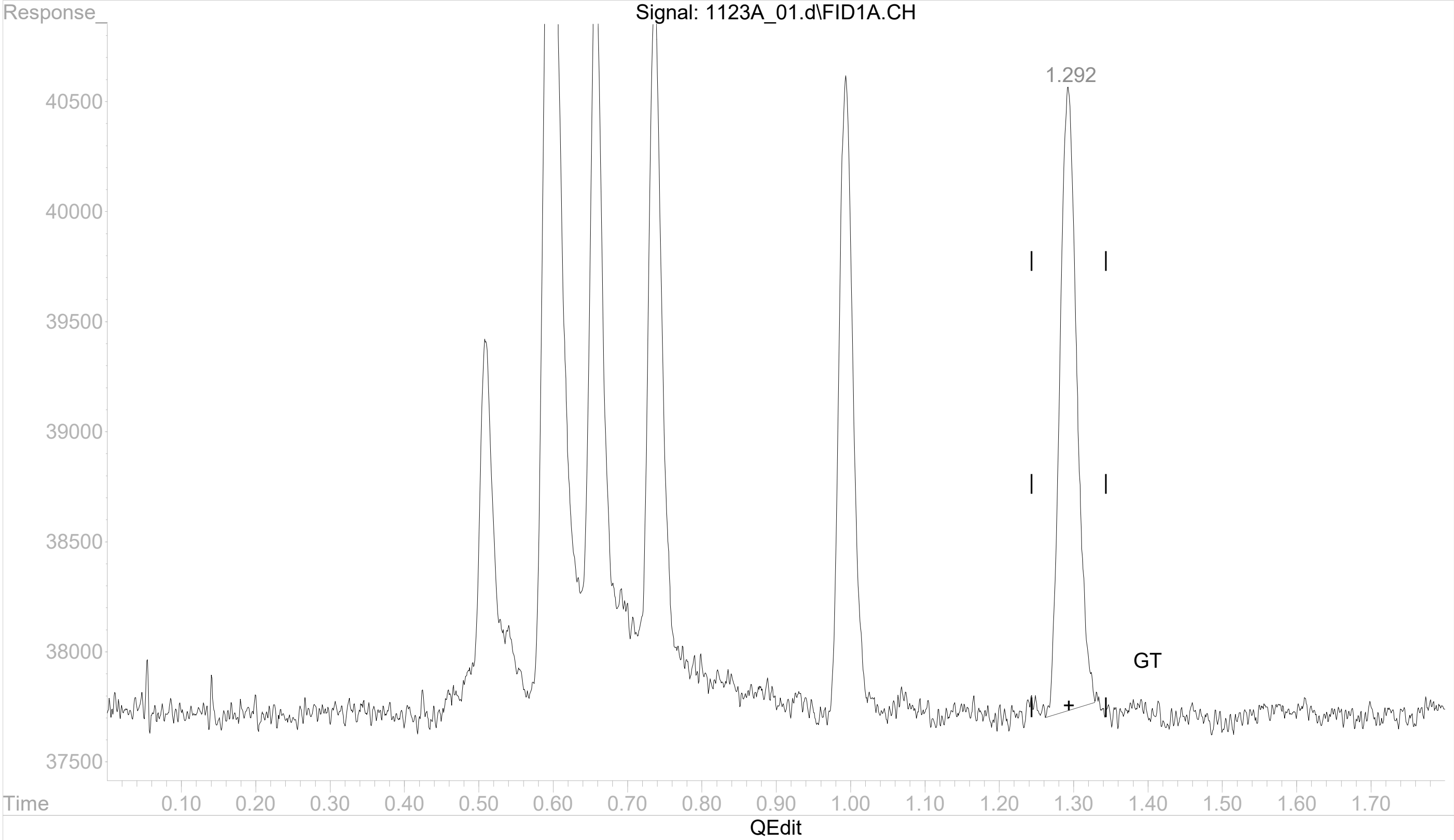
0.508min   0.0007253 mg/l m

response   19090

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:17:03 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:16:49 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(5) propane

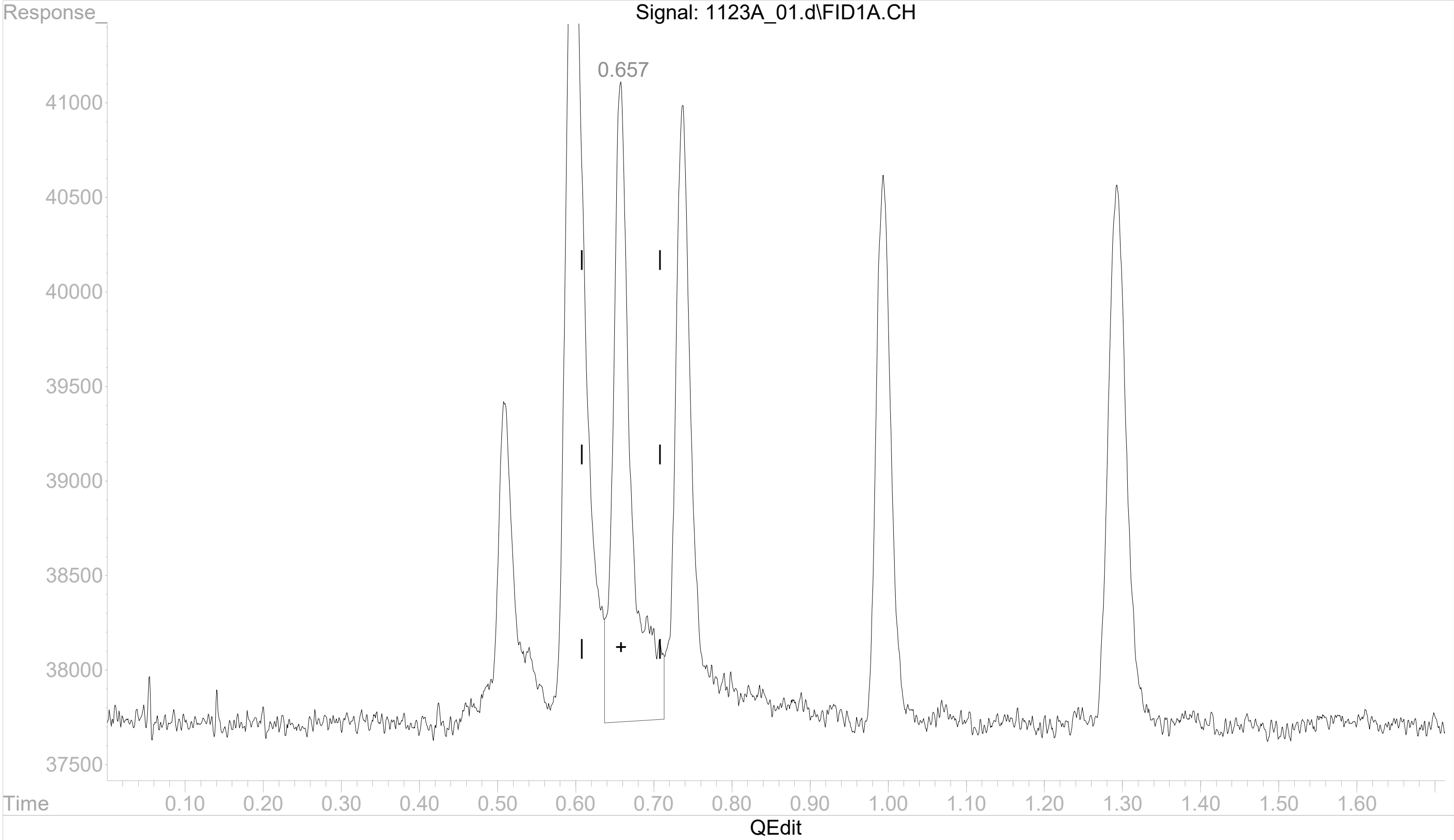
1.292min 0.0017348 mg/l m

response 44819

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:58:10 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:58:00 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(2) ethane

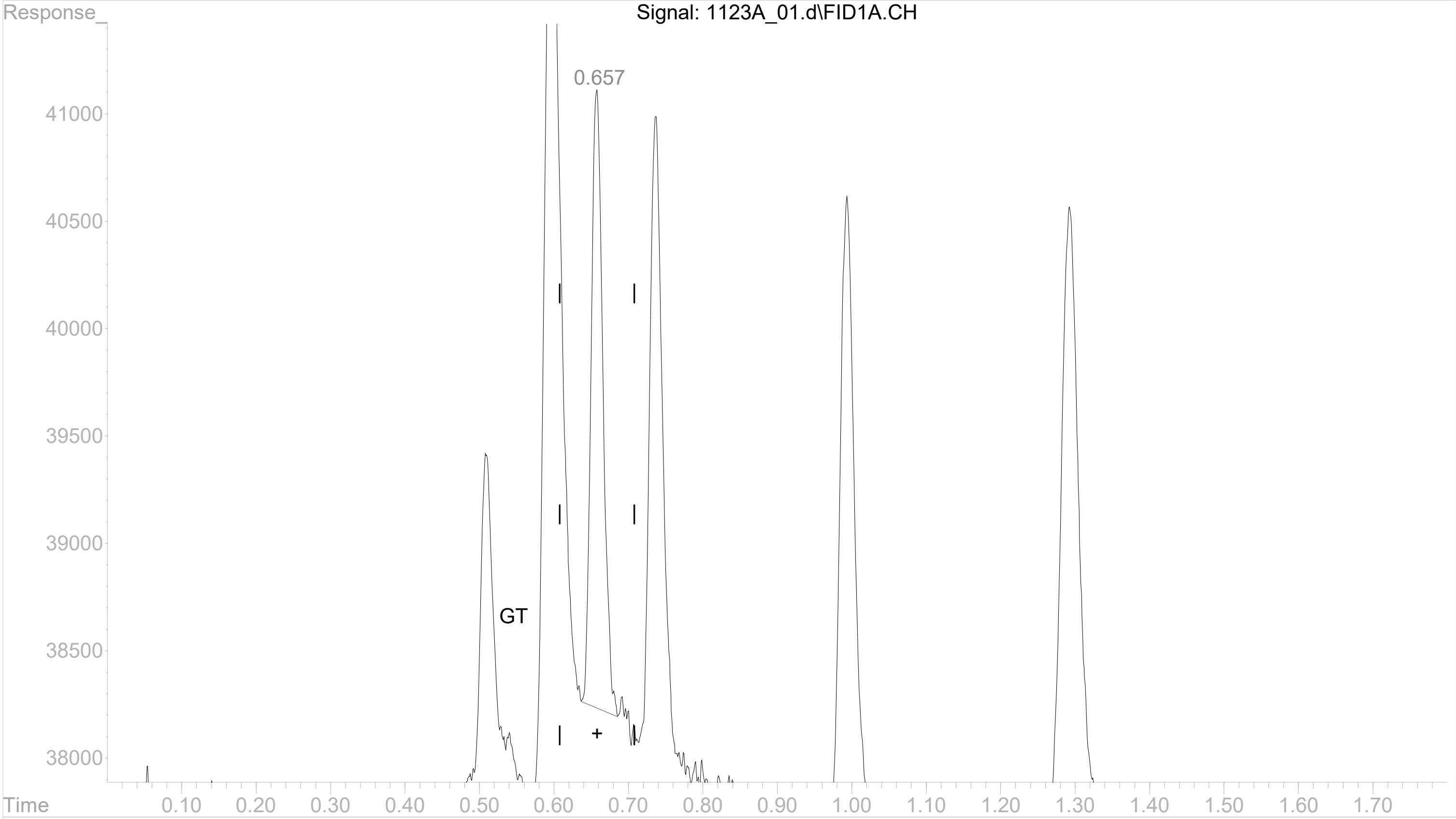
0.658min 0.0021171 mg/l

response 53907

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:58:10 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:58:00 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(2) ethane

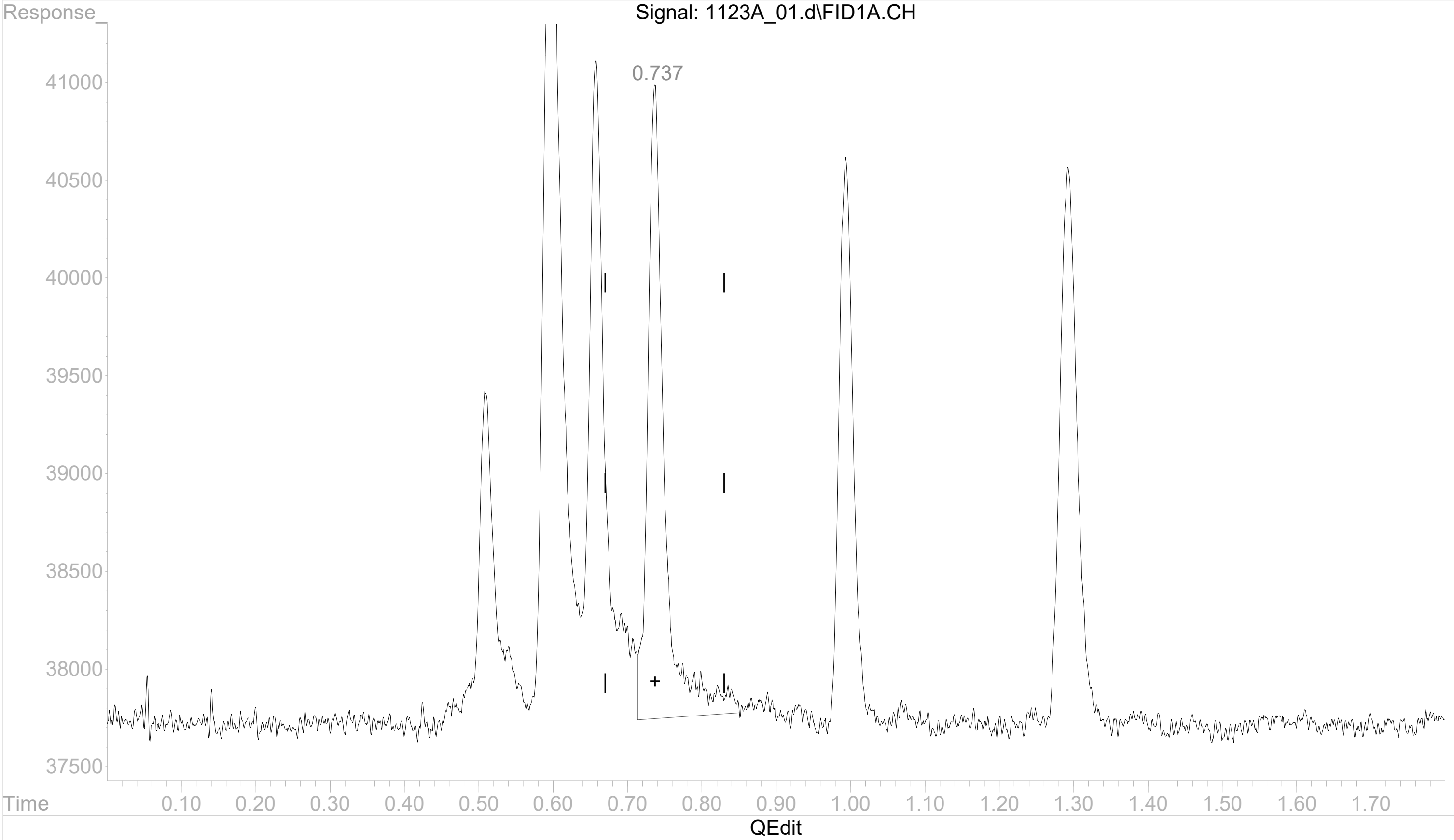
0.657min   0.0012535 mg/l m

response   31917

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:58:10 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:58:00 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(3) ethene

0.737min 0.0019486 mg/l

response 50943

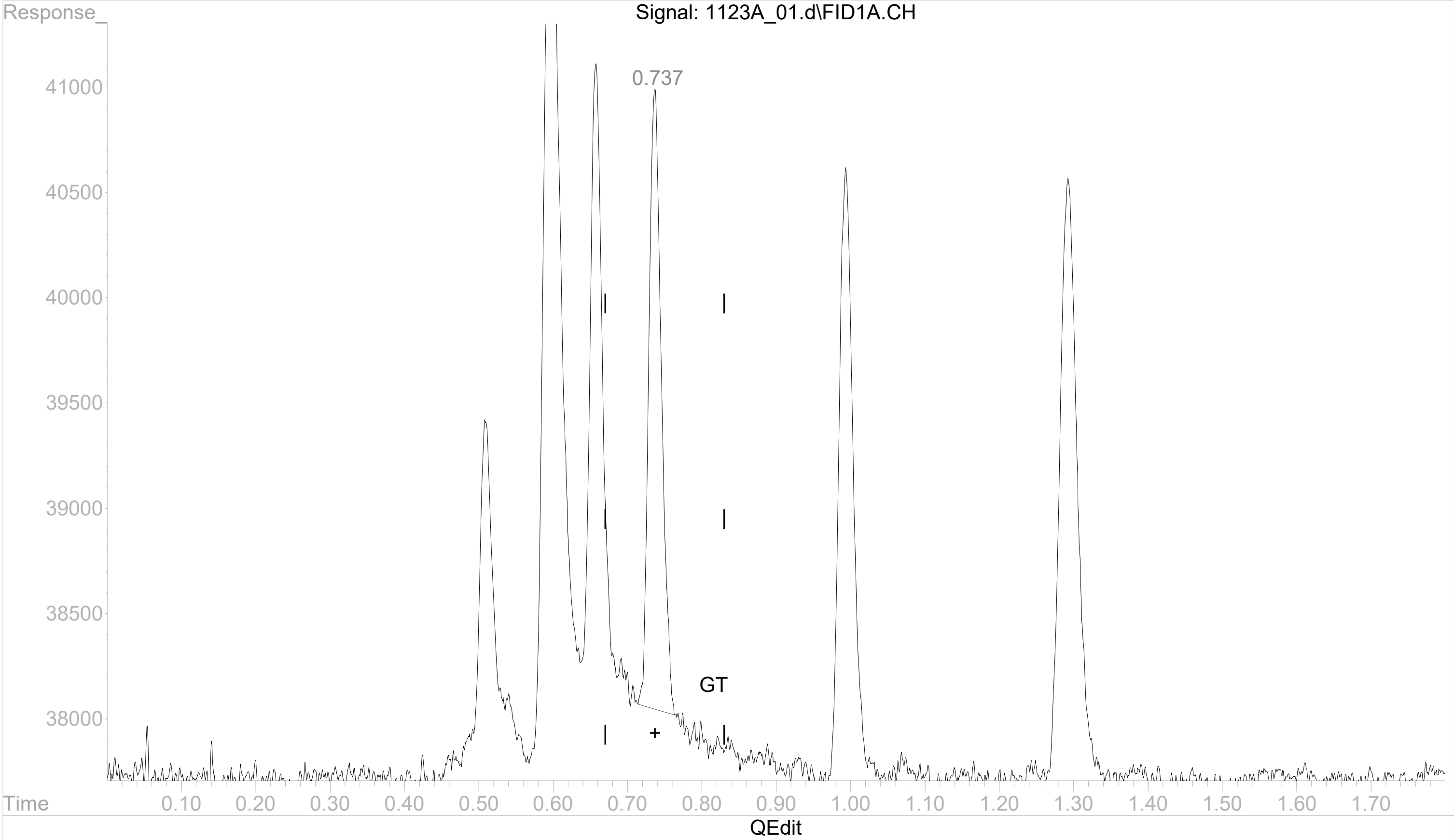
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On   : 23 Nov 2022 12:44 pm
Operator :
Sample   : STD AGC 1 RSK175
Misc   :
ALS Vial : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:58:10 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:58:00 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(3) ethene

0.737min   0.0013277 mg/l m

response   34710

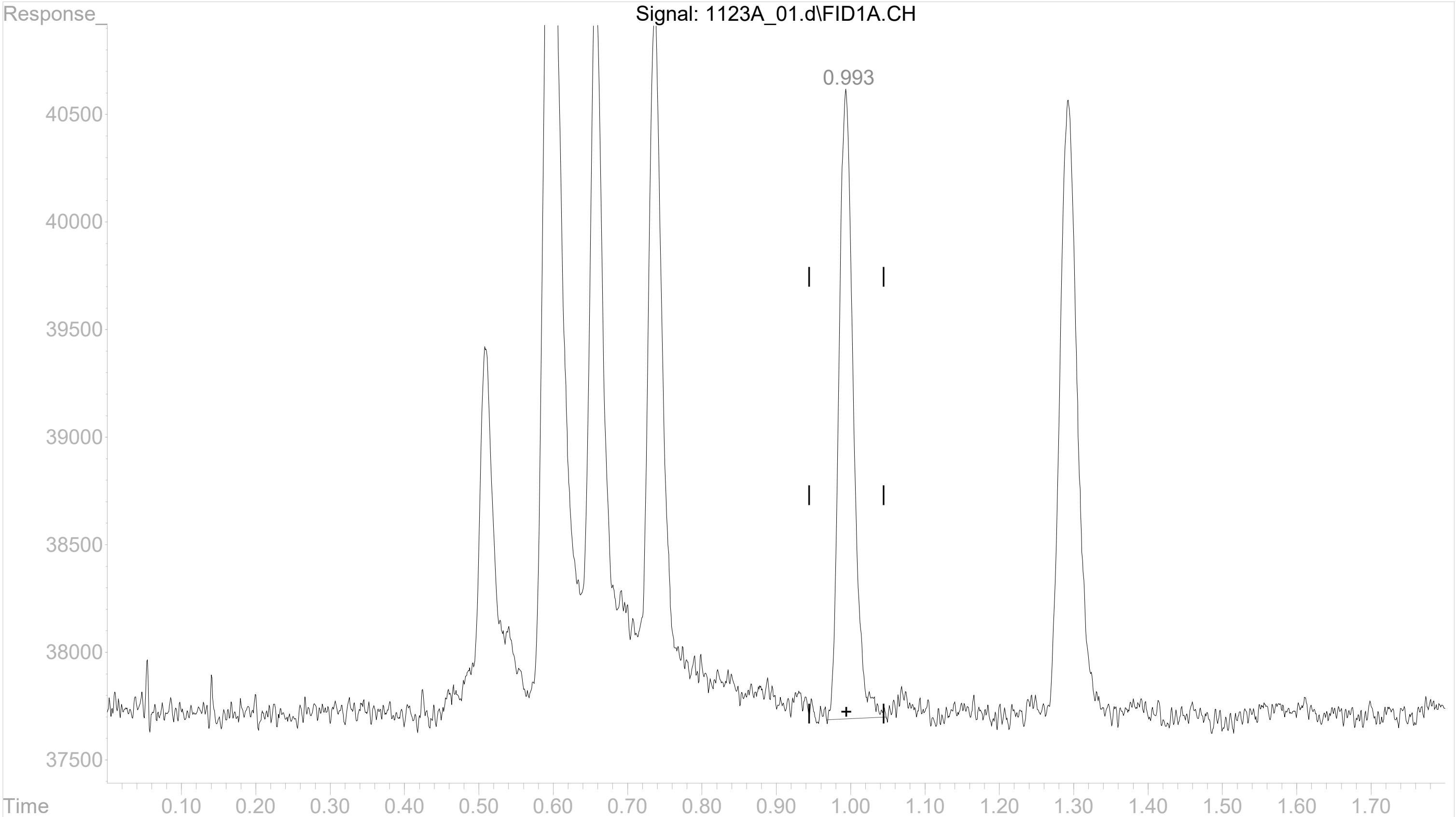
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:44 pm
Operator :
Sample : STD AGC 1 RSK175
Misc :
ALS Vial : 1 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:58:10 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:58:00 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(4) acetylene

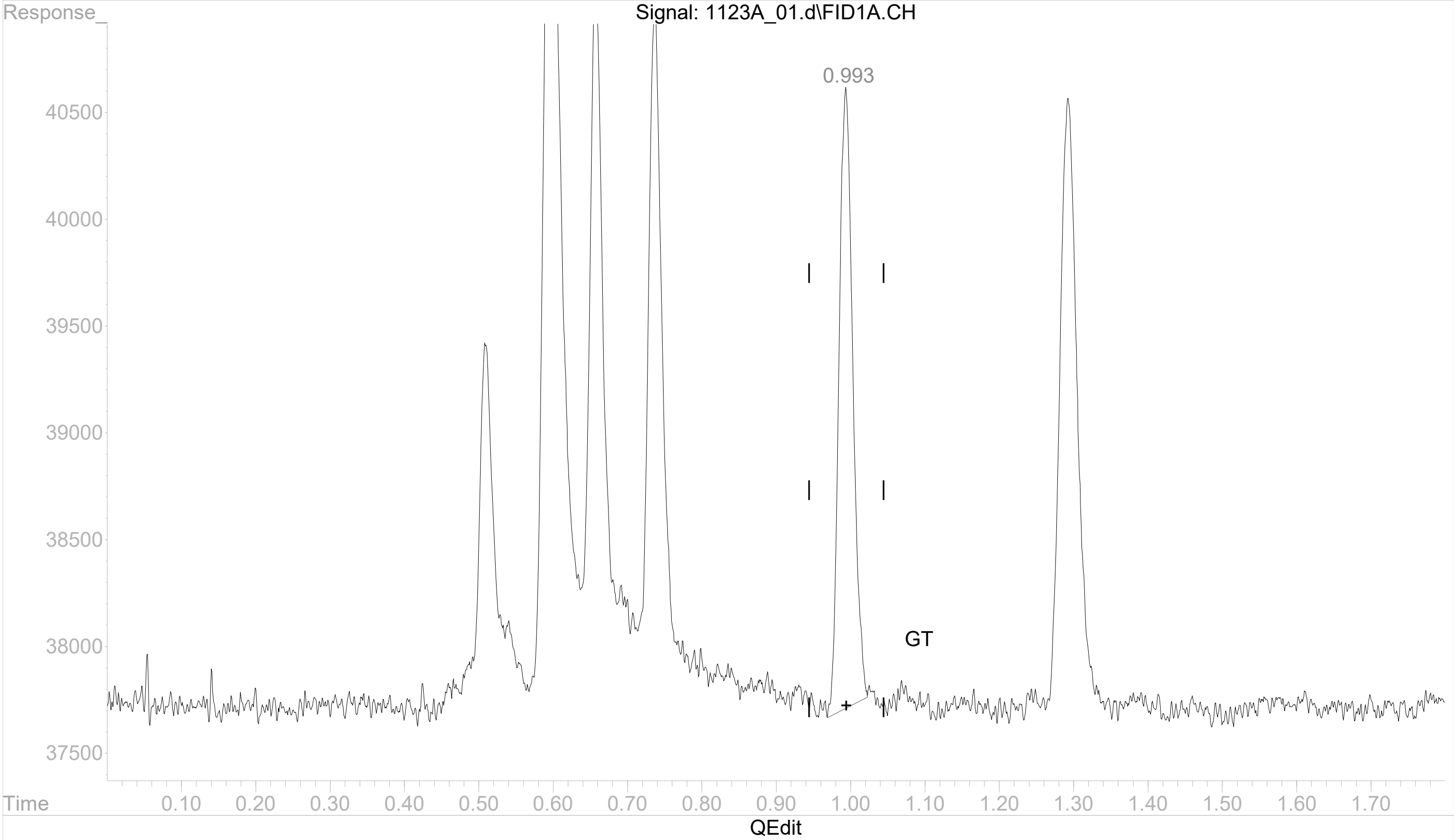
0.994min 0.0021068 mg/l

response 39399

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:58:10 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:58:00 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(4) acetylene

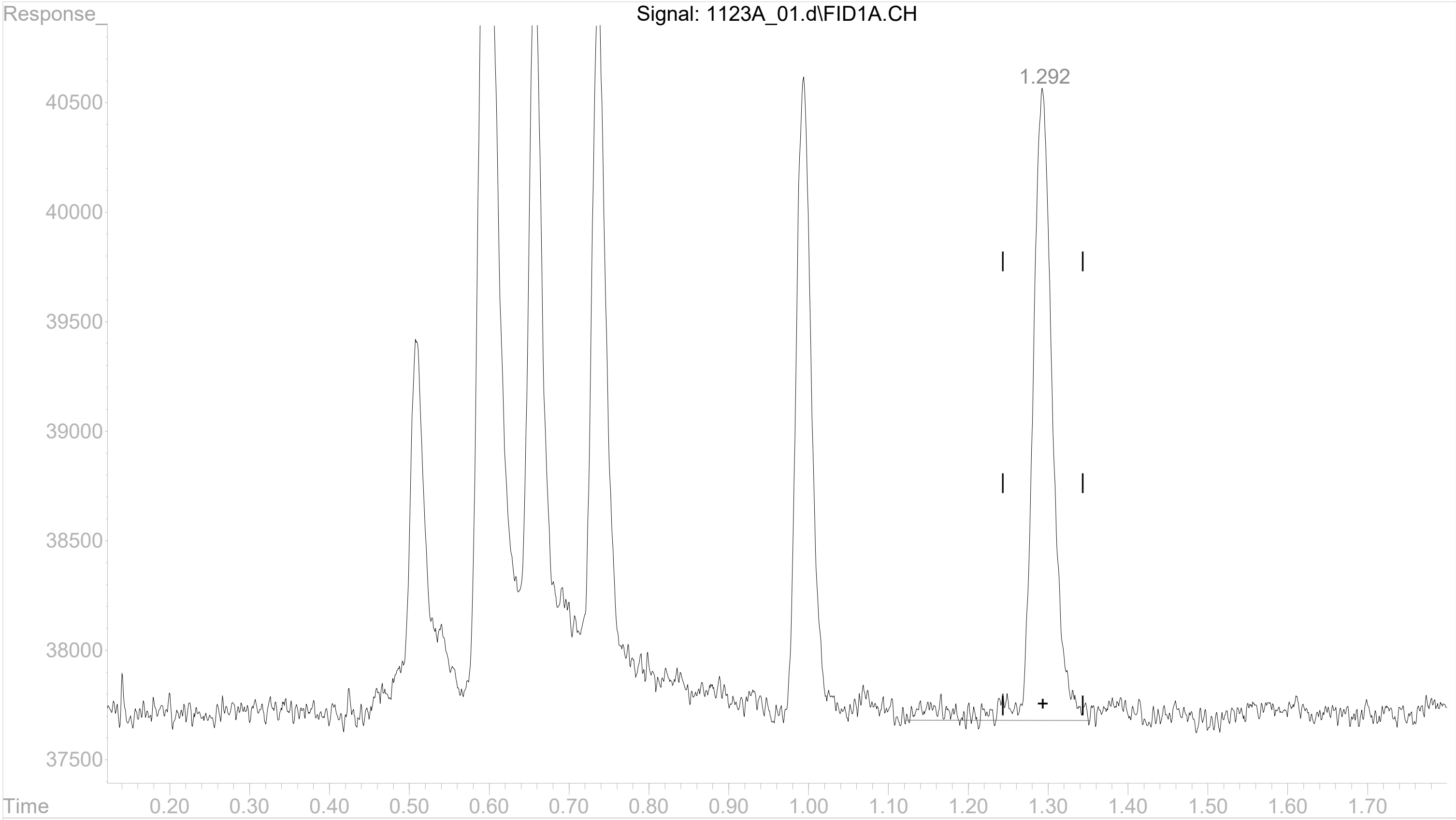
0.993min 0.0020211 mg/l m

response 37797

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_01.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:44 pm
Operator  :
Sample    : STD AGC 1 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:58:10 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:58:00 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(5) propane

1.293min 0.0020061 mg/l

response 51484

(+) = Expected Retention Time

Quantitation Report (QT Reviewed)

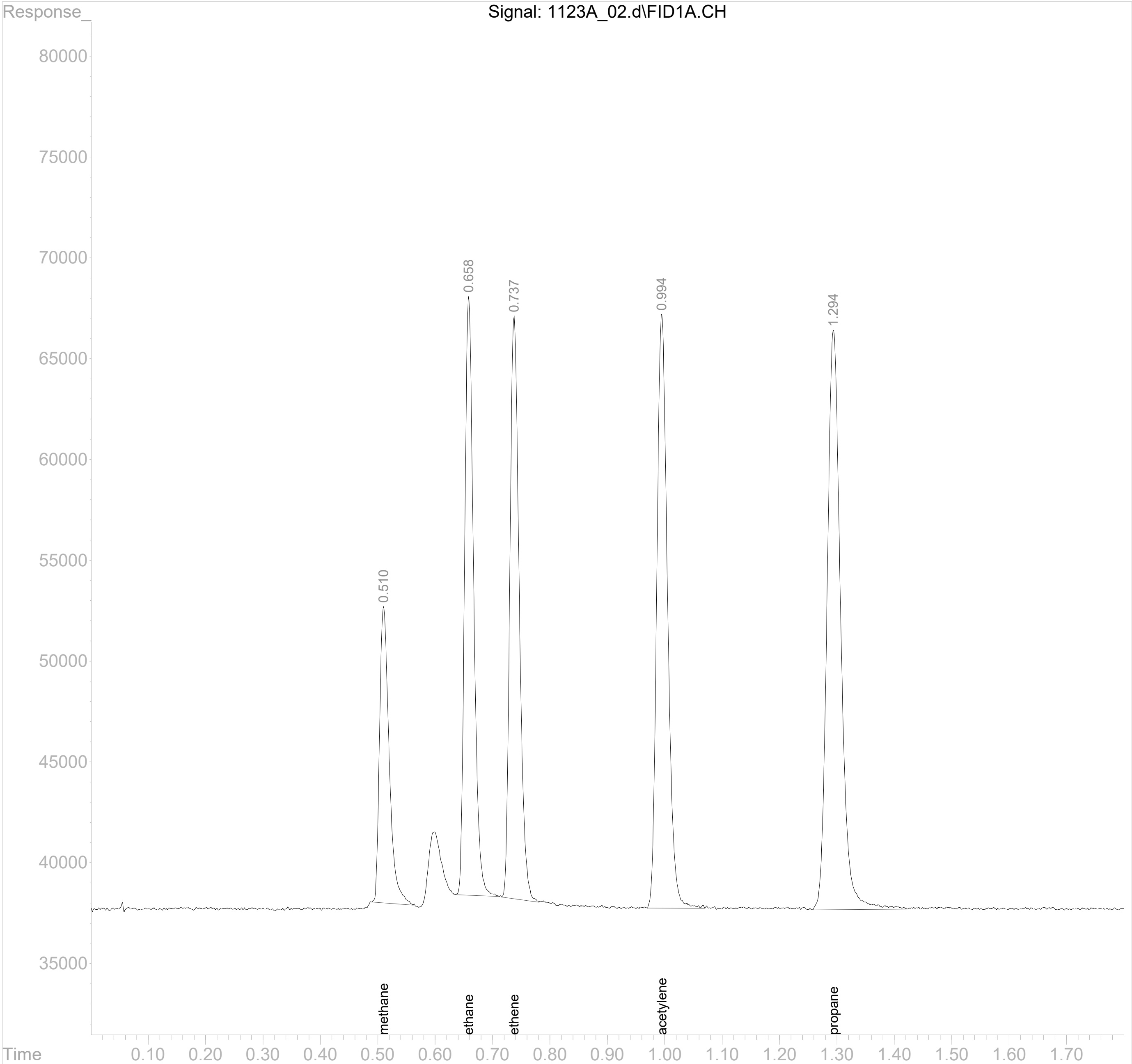
```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:48 pm
Operator  :
Sample    : STD AGC 2 RSK175
Misc      :
ALS Vial  : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:16:38 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:15:14 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	171200	0.0068600 mg/l m
2)	ethane	0.66	324777	0.0126107 mg/l m
3)	ethene	0.74	327153	0.0123255 mg/l m
4)	acetylene	0.99	382572	0.0200636 mg/l m
5)	propane	1.29	474128	0.0181100 mg/l m

(f)=RT Delta > 1/2 Window (m)=manual int.

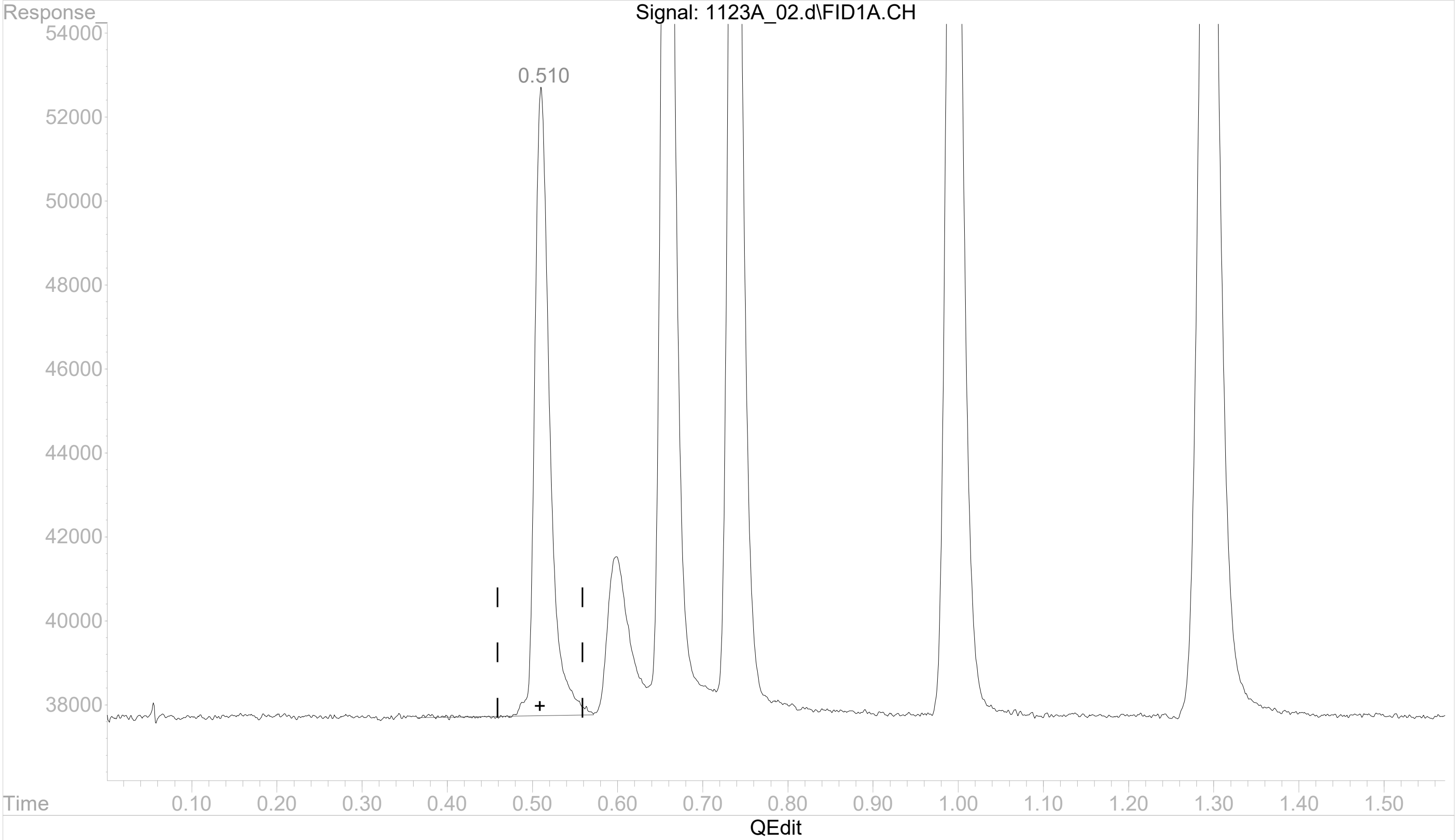
Quantitation Report   (QT Reviewed)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:48 pm
Operator  :
Sample    : STD AGC 2 RSK175
Misc      :
ALS Vial  : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:16:38 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:15:14 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:48 pm
Operator  :
Sample    : STD AGC 2 RSK175
Misc      :
ALS Vial  : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:56:18 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:56:06 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(1) methane

0.511min   0.0070668 mg/l

response   182804

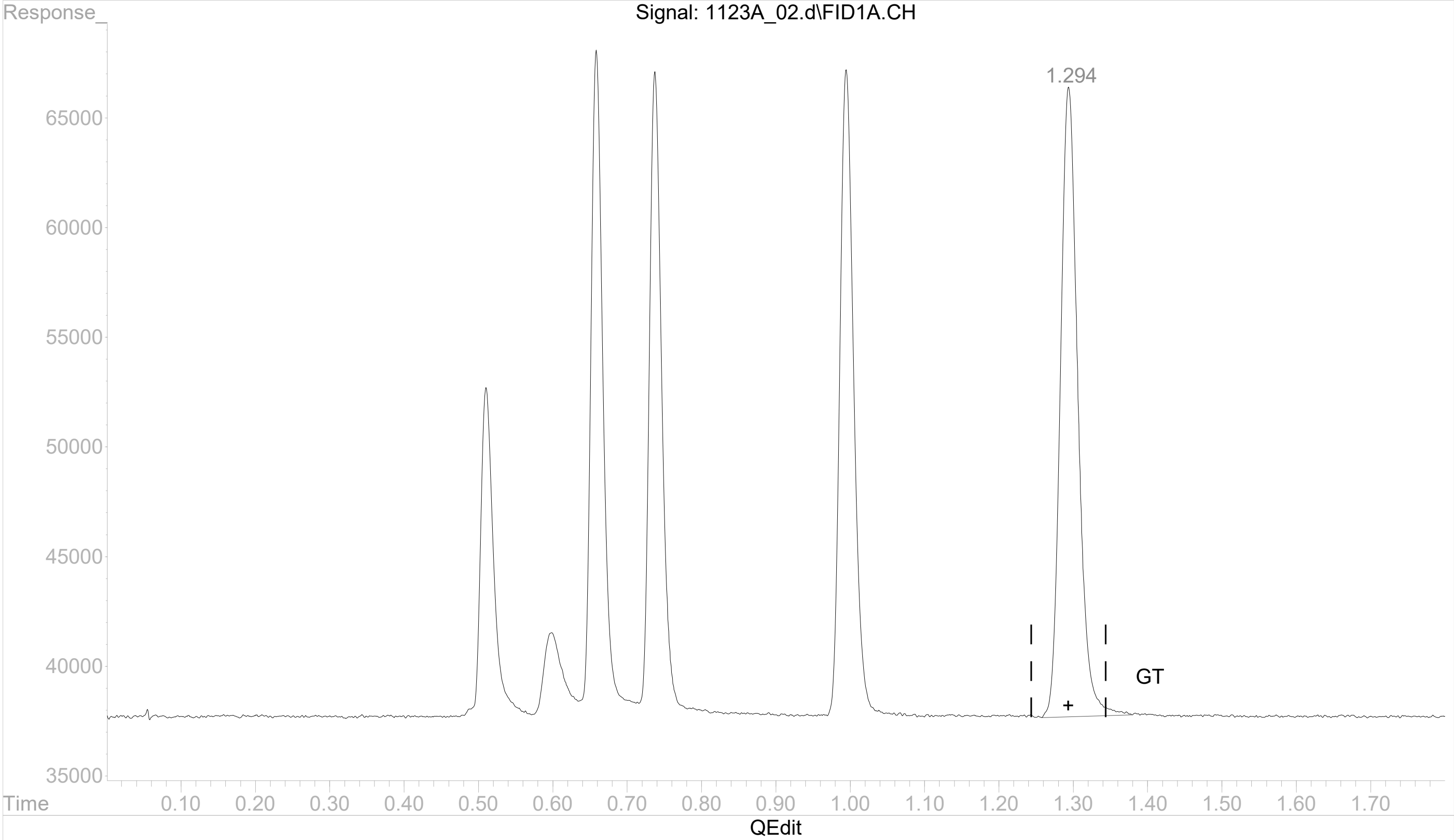
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On   : 23 Nov 2022 12:48 pm
Operator :
Sample   : STD AGC 2 RSK175
Misc   :
ALS Vial : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:56:18 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:56:06 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(5) propane

1.294min   0.0178938 mg/l m

response   468102

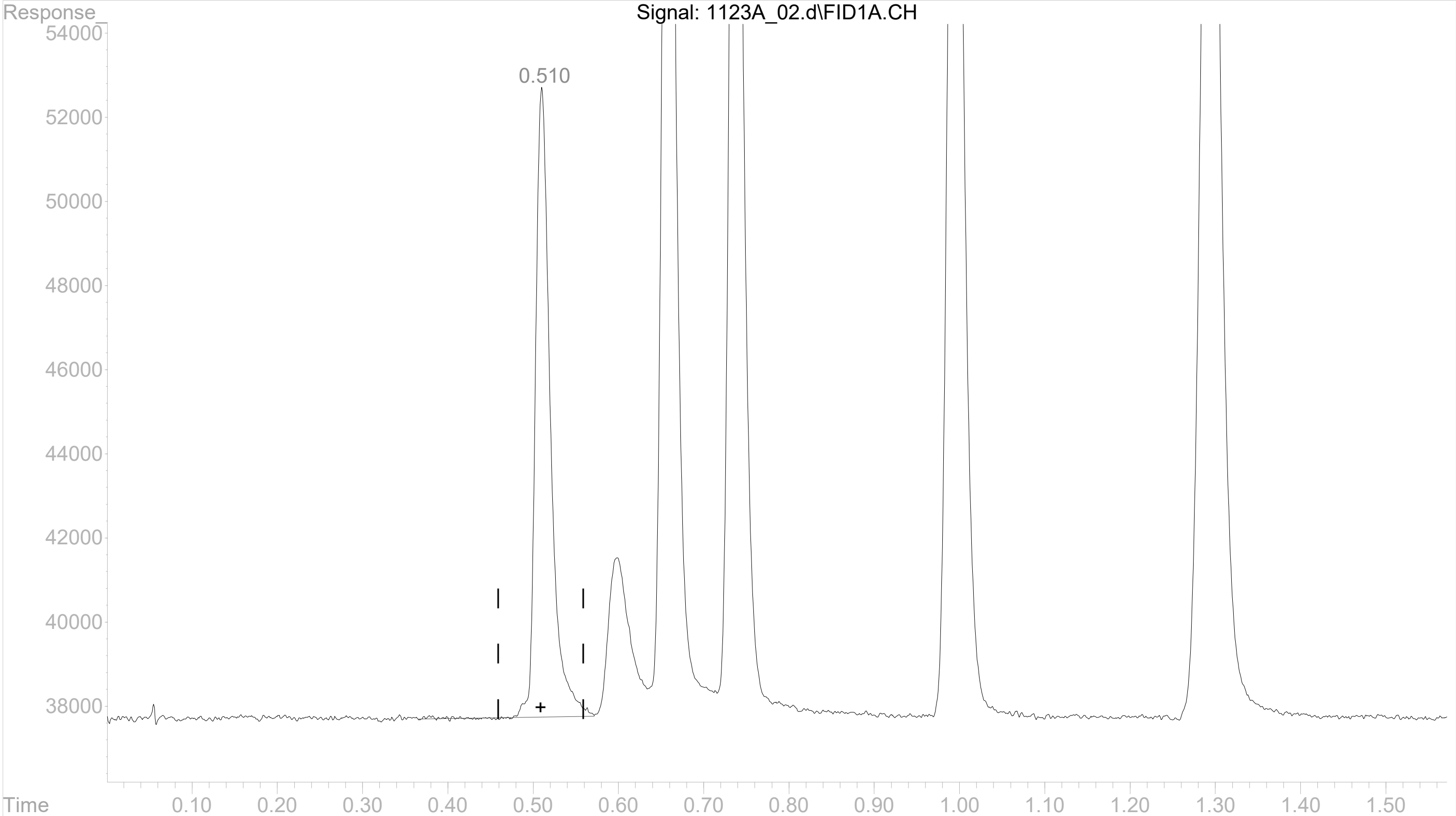
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:48 pm
Operator :
Sample : STD AGC 2 RSK175
Misc :
ALS Vial : 2 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:15:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:15:14 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(1) methane

0.511min 0.0073250 mg/l

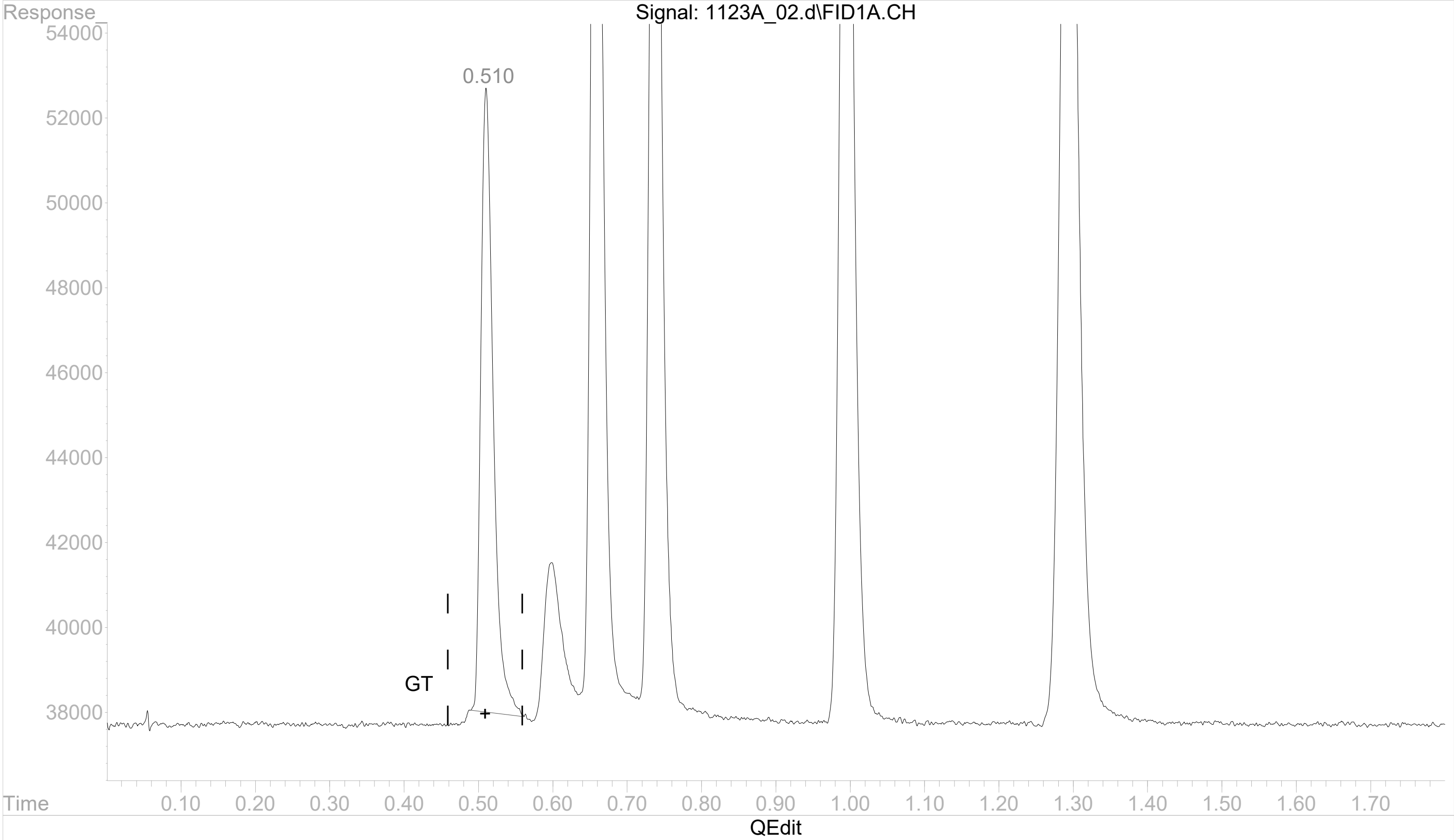
response 182804

(+) = Expected Retention Time

GC2RSKK23V.M Wed Nov 23 13:15:35 2022 Page: 1

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:48 pm
Operator  :
Sample    : STD AGC 2 RSK175
Misc      :
ALS Vial  : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:15:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:15:14 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(1) methane

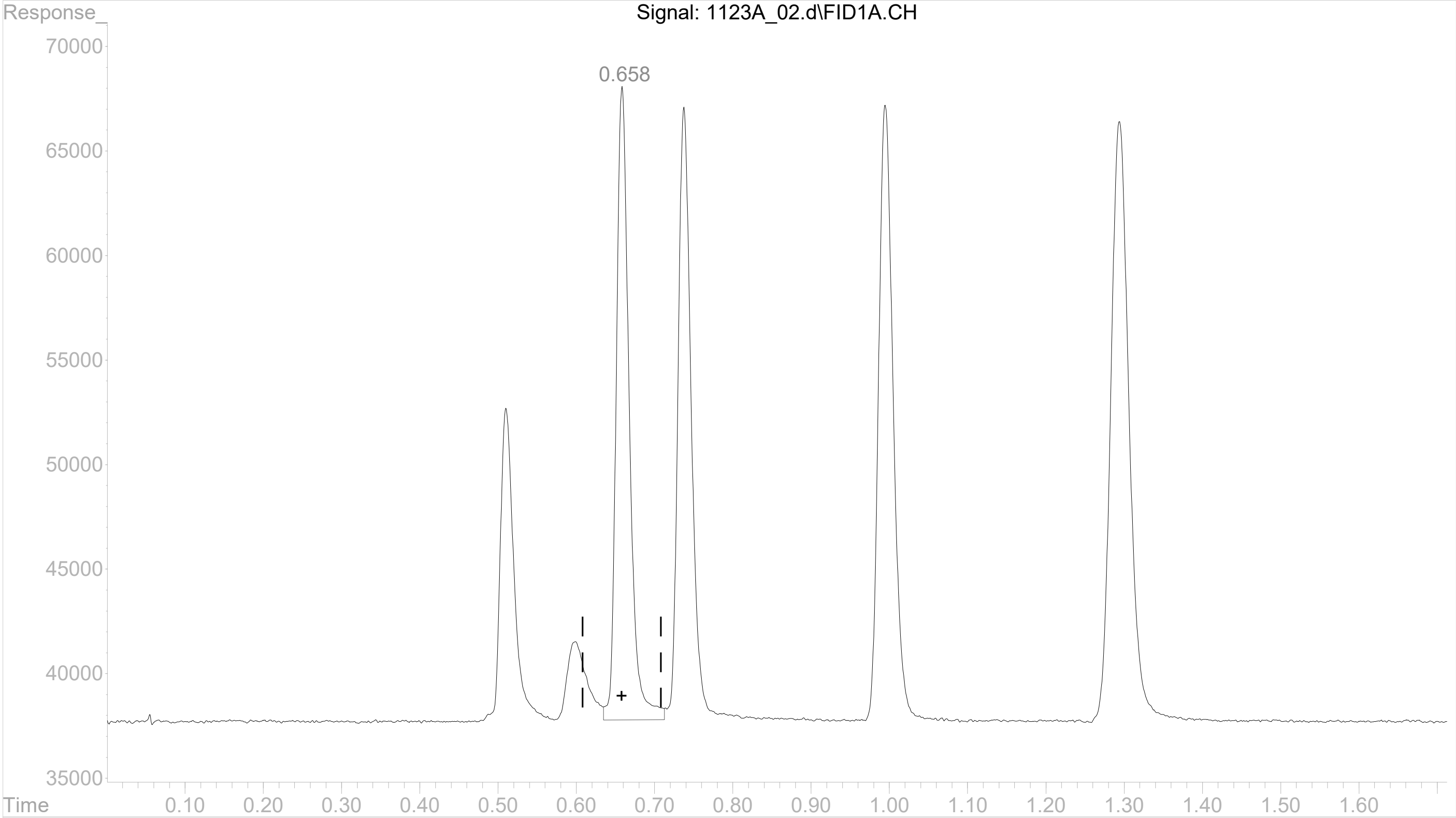
0.510min 0.0068600 mg/l m

response 171200

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:48 pm
Operator  :
Sample    : STD AGC 2 RSK175
Misc      :
ALS Vial  : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:15:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:15:14 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(2) ethane

0.659min 0.0136455 mg/l

response 351427

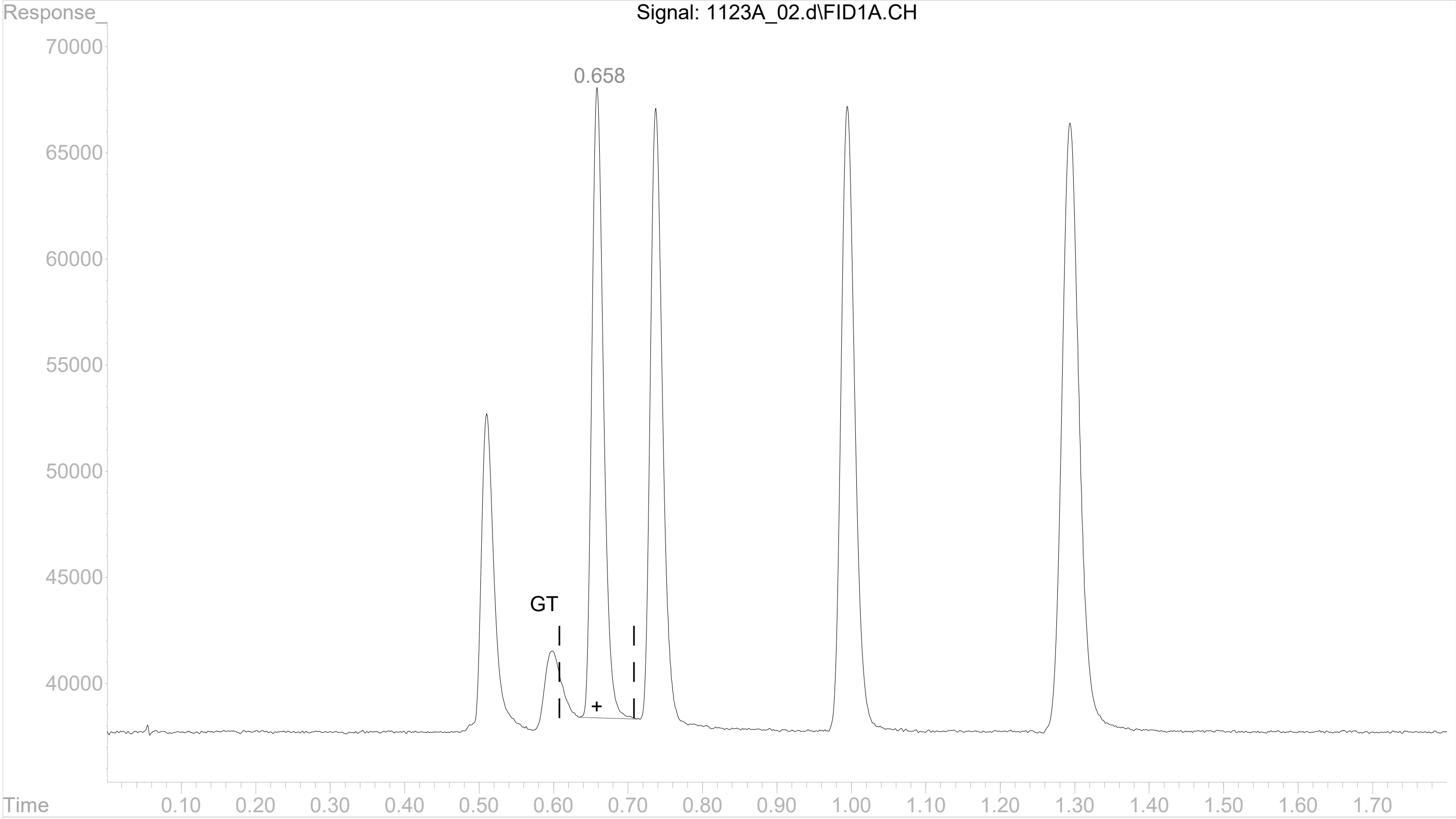
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:48 pm
Operator :
Sample : STD AGC 2 RSK175
Misc :
ALS Vial : 2 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:15:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:15:14 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(2) ethane

0.658min 0.0126107 mg/l m

response 324777

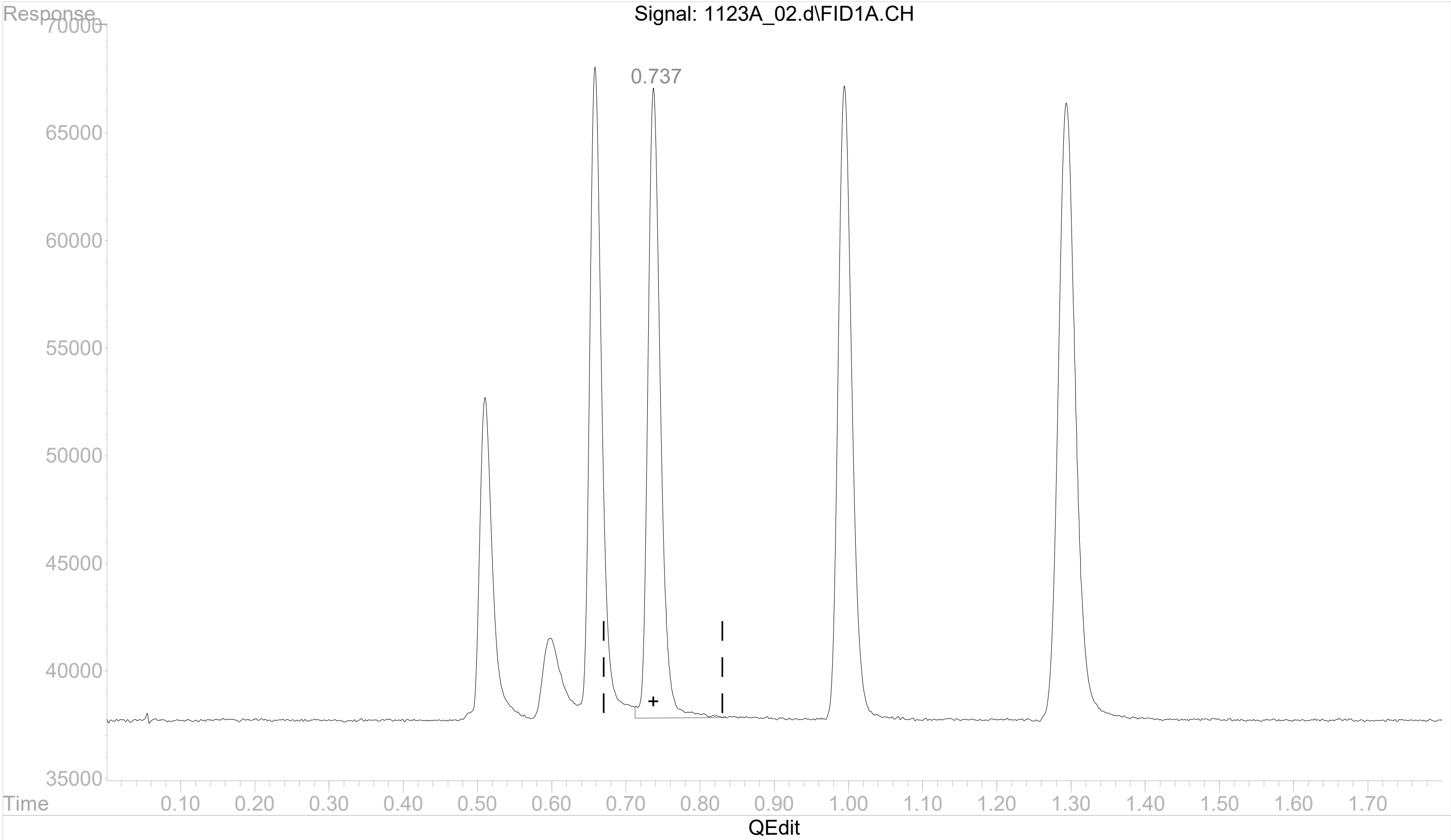
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:48 pm
Operator :
Sample : STD AGC 2 RSK175
Misc :
ALS Vial : 2 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:15:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:15:14 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(3) ethene

0.738min 0.0130432 mg/l

response 346201

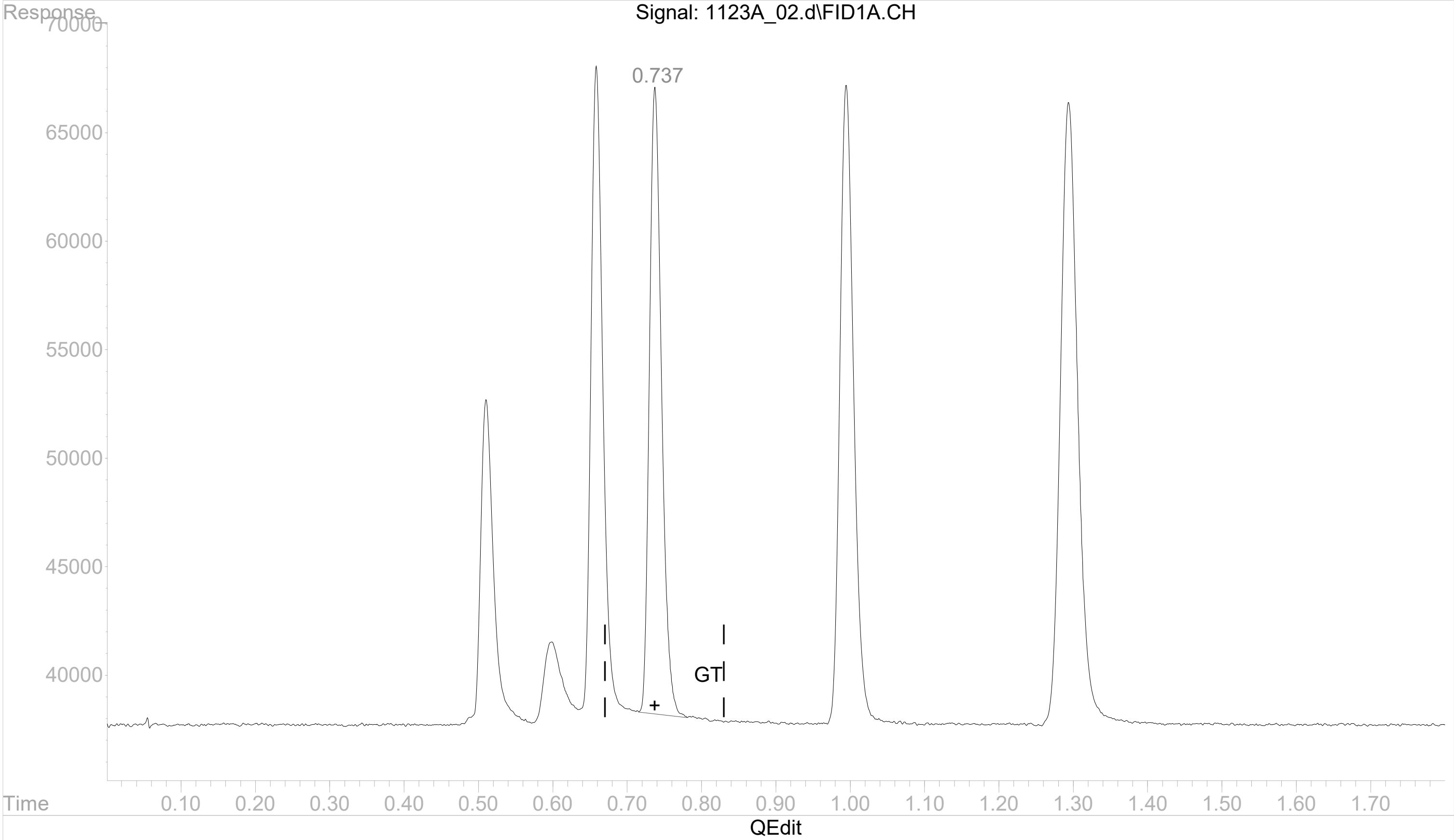
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:48 pm
Operator :
Sample : STD AGC 2 RSK175
Misc :
ALS Vial : 2 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:15:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:15:14 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(3) ethene

0.737min 0.0123255 mg/l m

response 327153

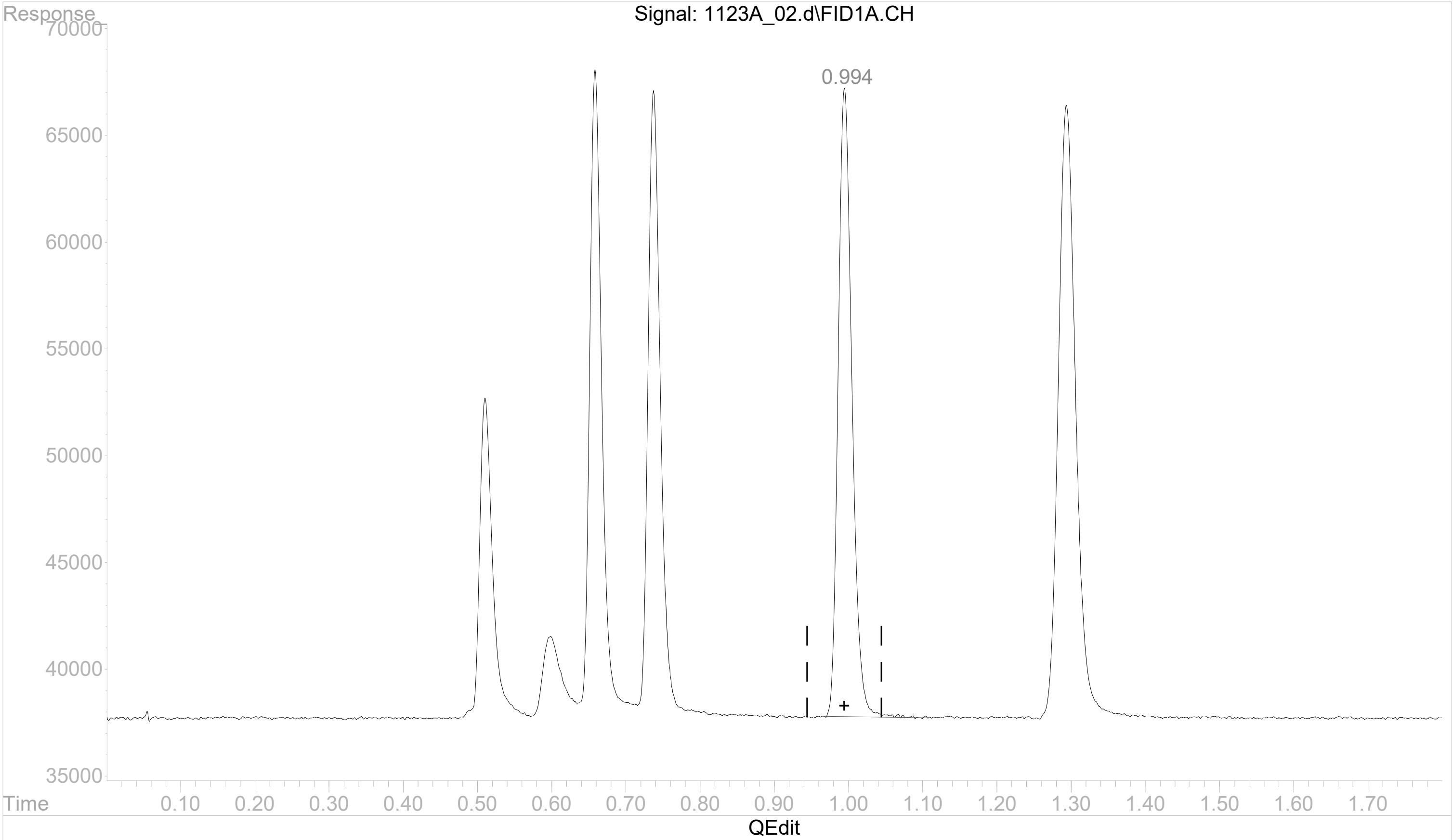
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:48 pm
Operator :
Sample : STD AGC 2 RSK175
Misc :
ALS Vial : 2 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:15:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:15:14 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(4) acetylene

0.995min 0.0199965 mg/l

response 381292

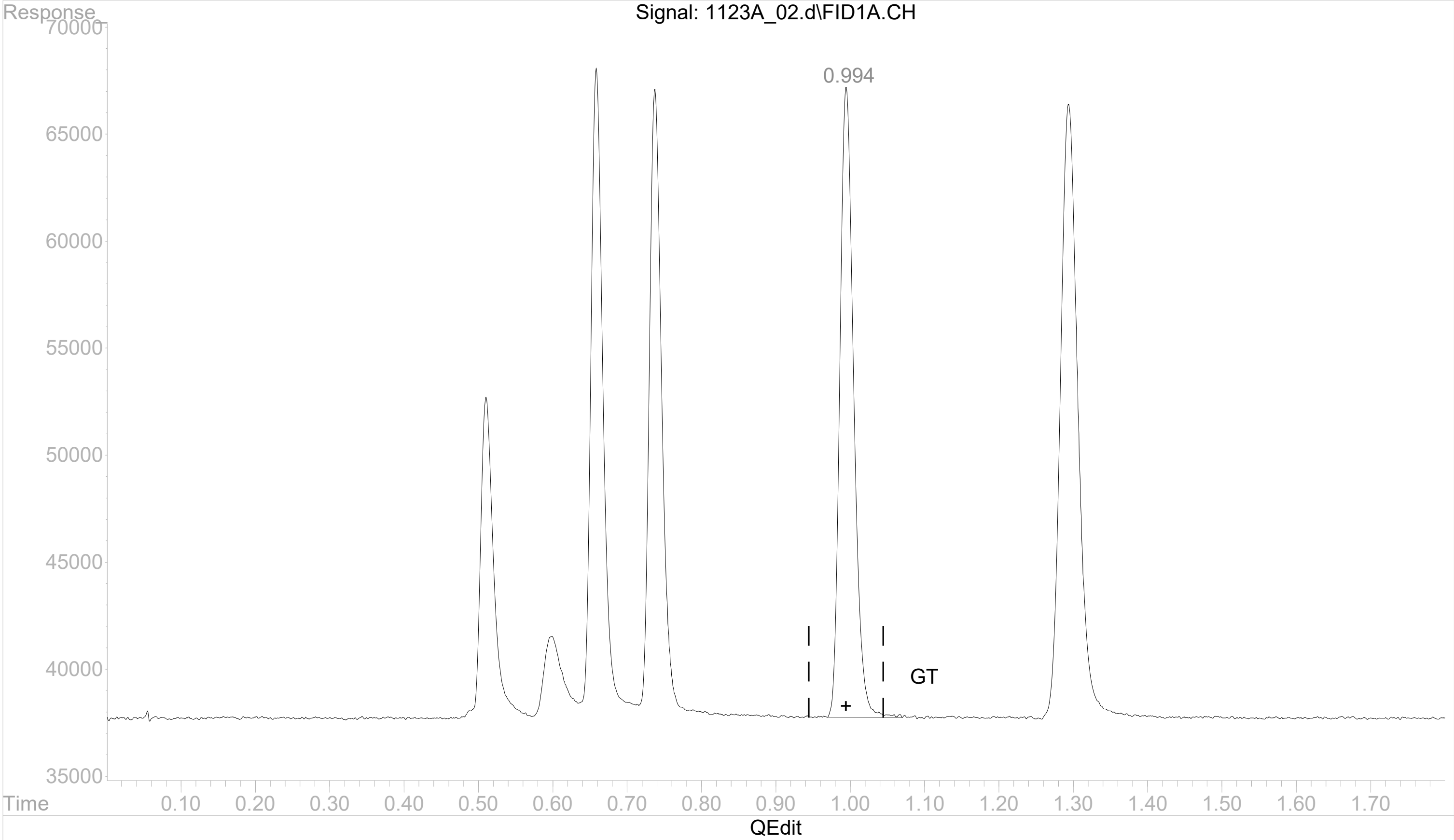
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:48 pm
Operator :
Sample : STD AGC 2 RSK175
Misc :
ALS Vial : 2 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:15:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:15:14 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(4) acetylene

0.994min 0.0200636 mg/l m

response 382572

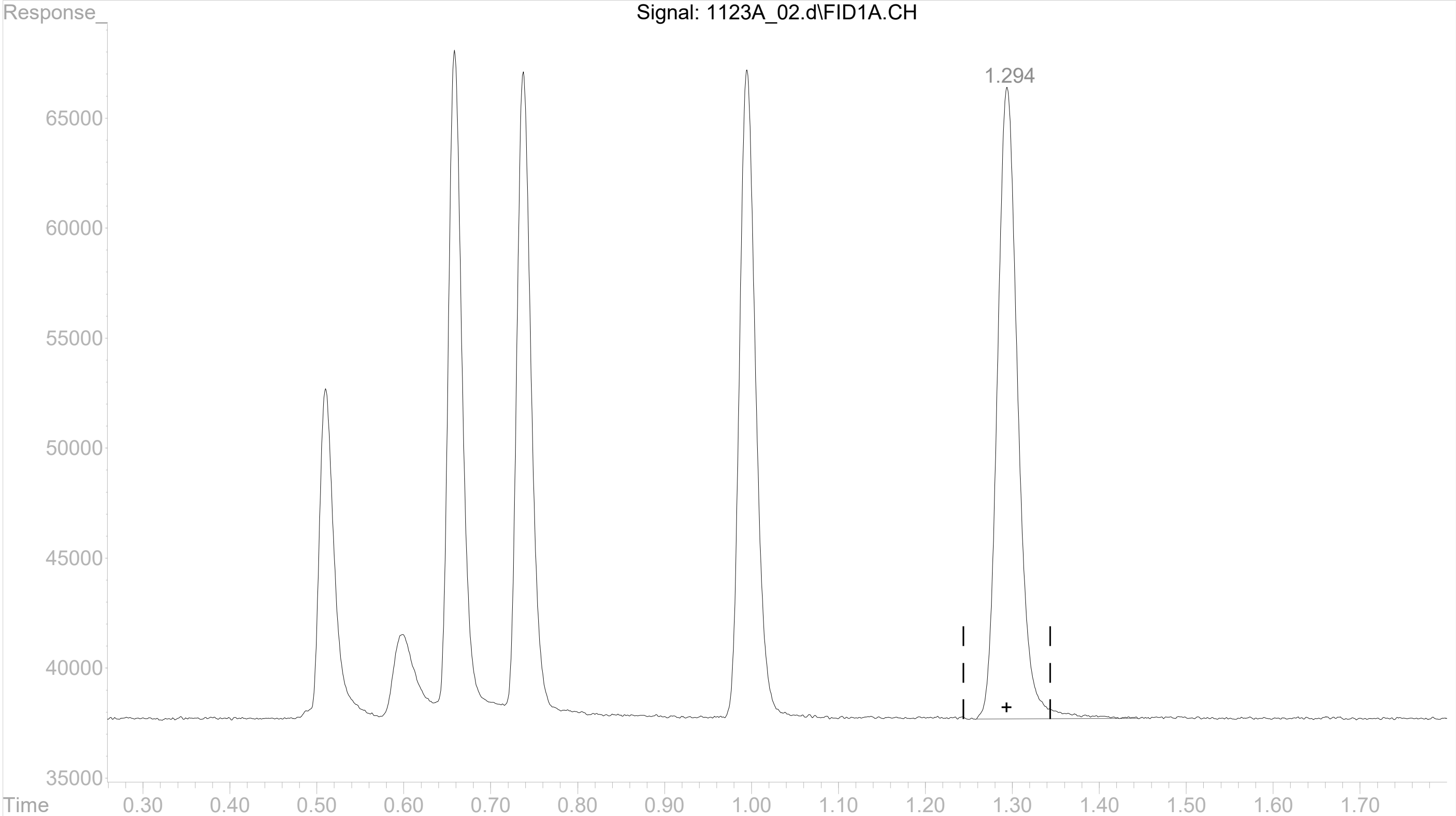
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:48 pm
Operator :
Sample : STD AGC 2 RSK175
Misc :
ALS Vial : 2 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:15:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:15:14 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(5) propane

1.294min 0.0180519 mg/l

response 472606

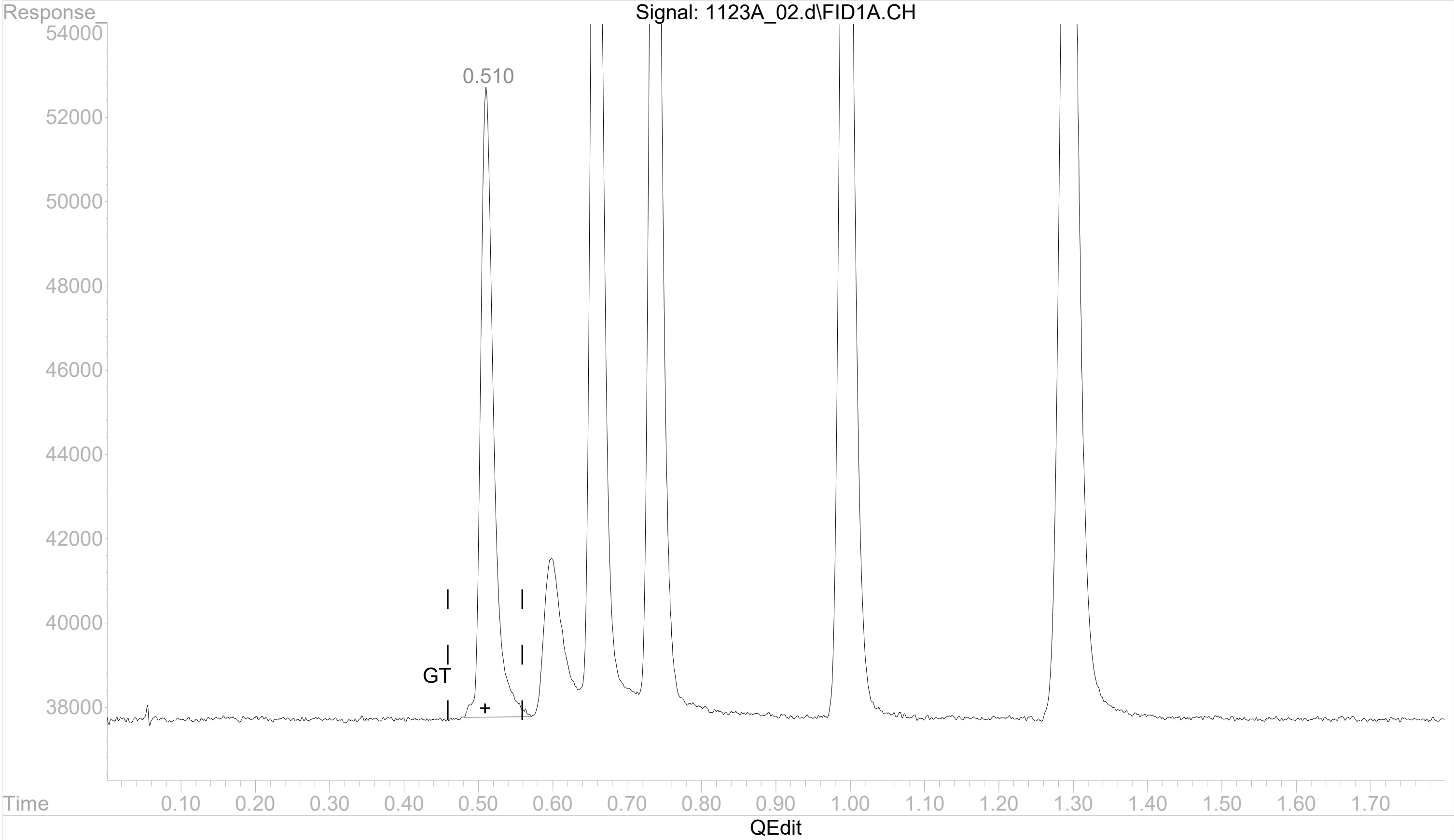
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:48 pm
Operator :
Sample : STD AGC 2 RSK175
Misc :
ALS Vial : 2 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:56:18 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:56:06 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(1) methane

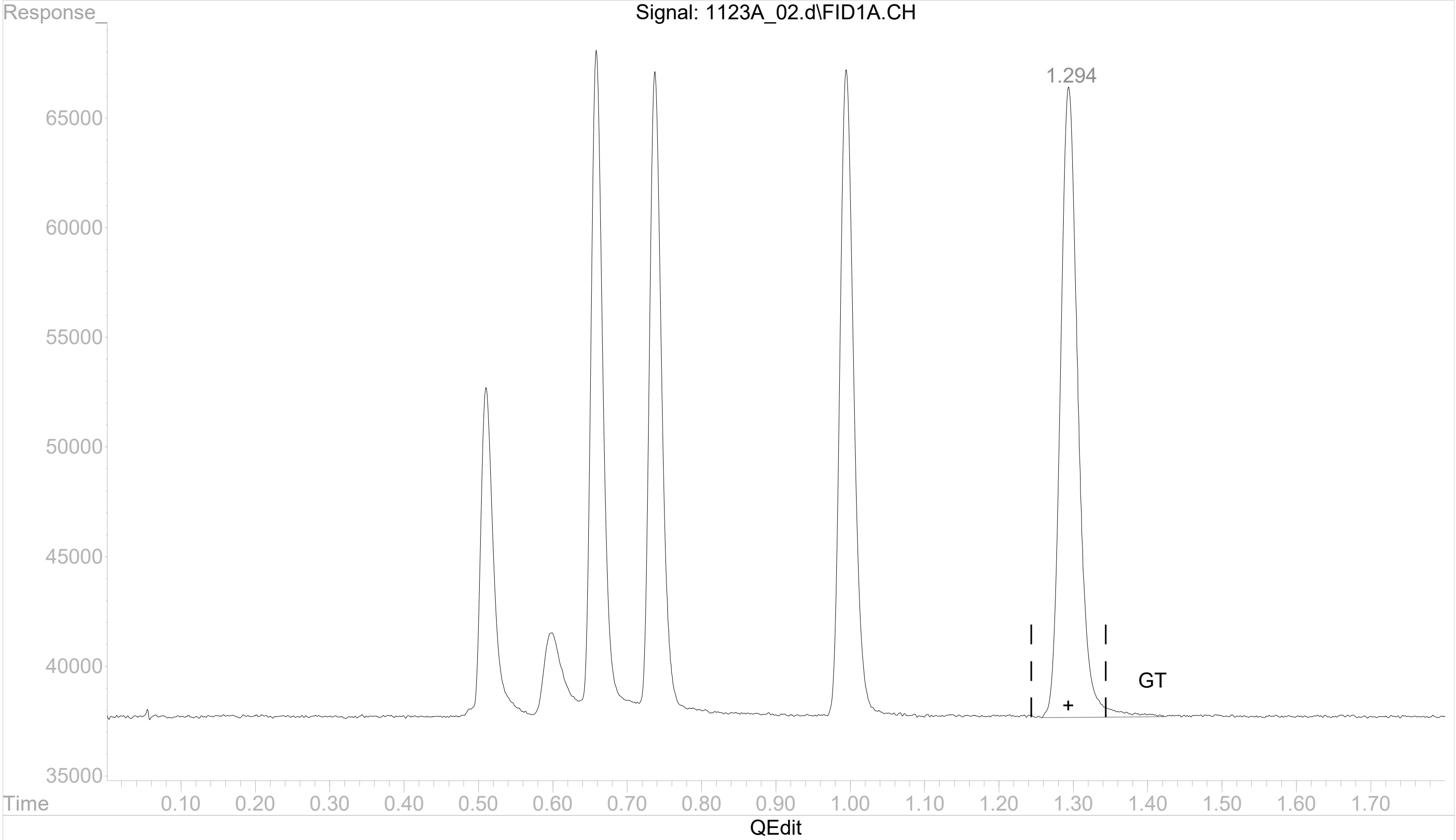
0.510min 0.0070165 mg/l m

response 181503

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:48 pm
Operator  :
Sample    : STD AGC 2 RSK175
Misc      :
ALS Vial  : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:15:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:15:14 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(5) propane

1.294min 0.0181100 mg/l m

response 474128

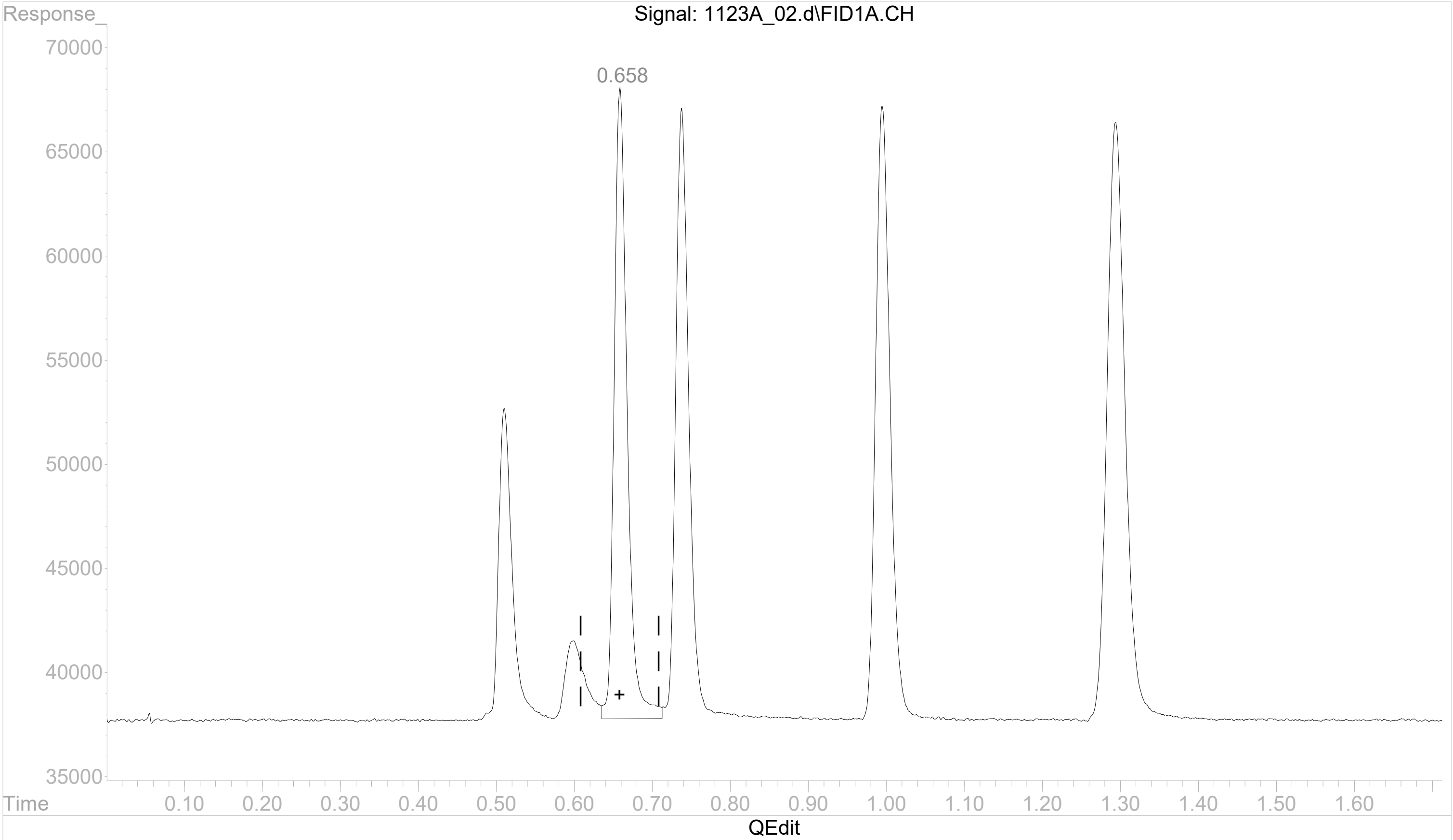
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:48 pm
Operator :
Sample : STD AGC 2 RSK175
Misc :
ALS Vial : 2 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:56:18 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:56:06 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(2) ethane

0.659min 0.0136426 mg/l

response 351427

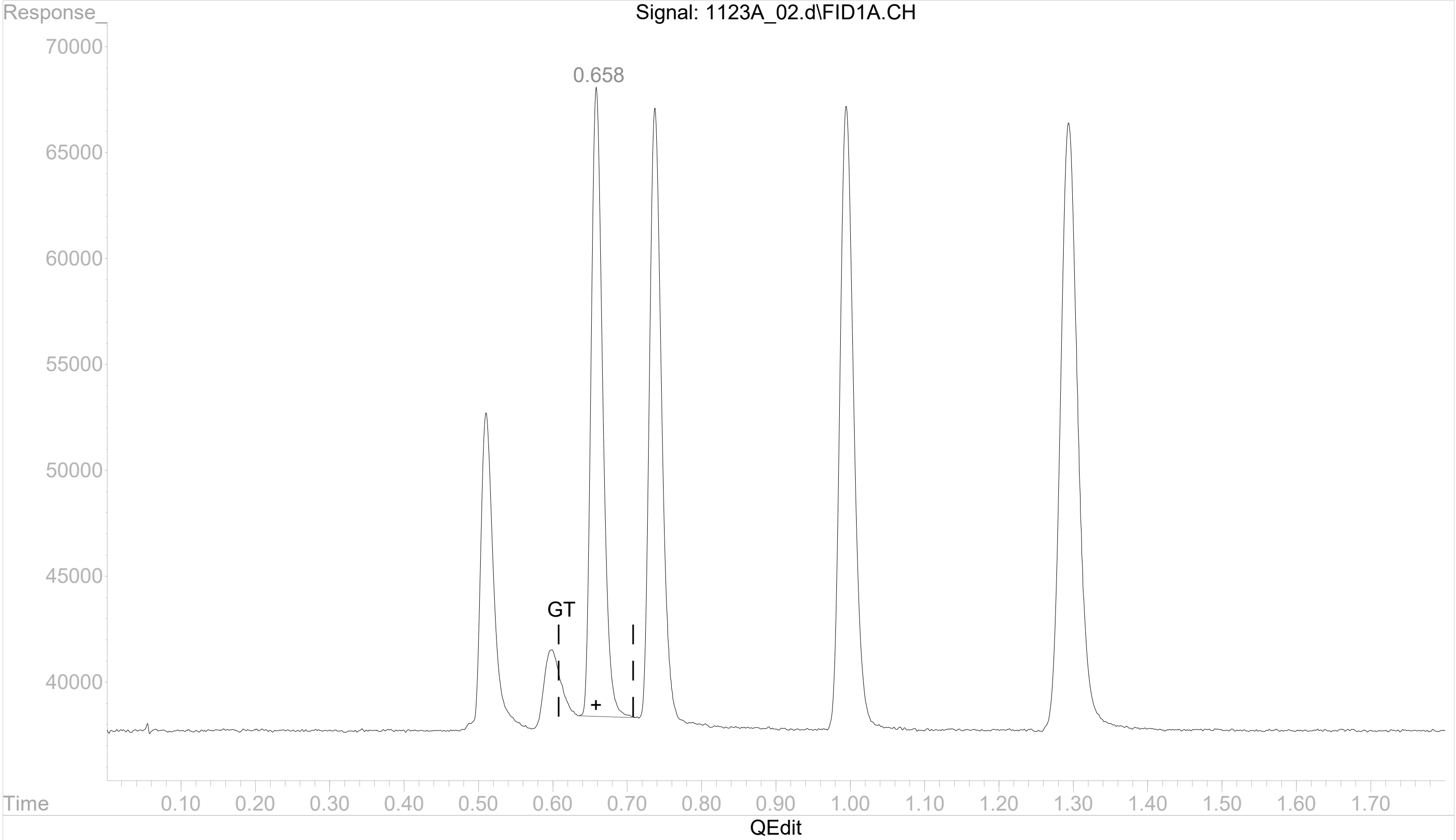
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On   : 23 Nov 2022 12:48 pm
Operator :
Sample   : STD AGC 2 RSK175
Misc   :
ALS Vial : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:56:18 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:56:06 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(2) ethane

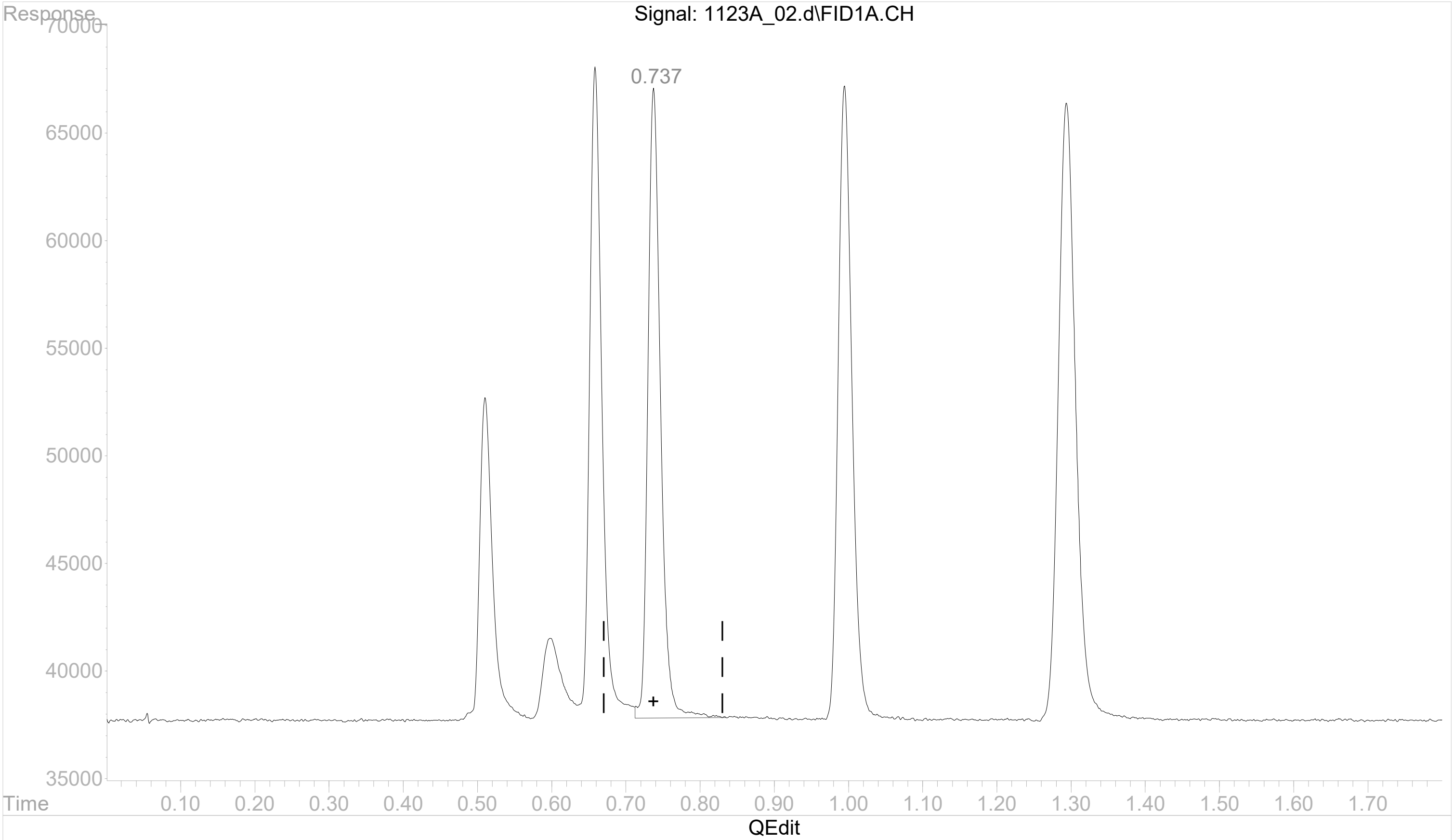
0.658min   0.0126025 mg/l m

response   324633

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:48 pm
Operator  :
Sample    : STD AGC 2 RSK175
Misc      :
ALS Vial  : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:56:18 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:56:06 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(3) ethene

0.738min   0.0130426 mg/l

response   346201

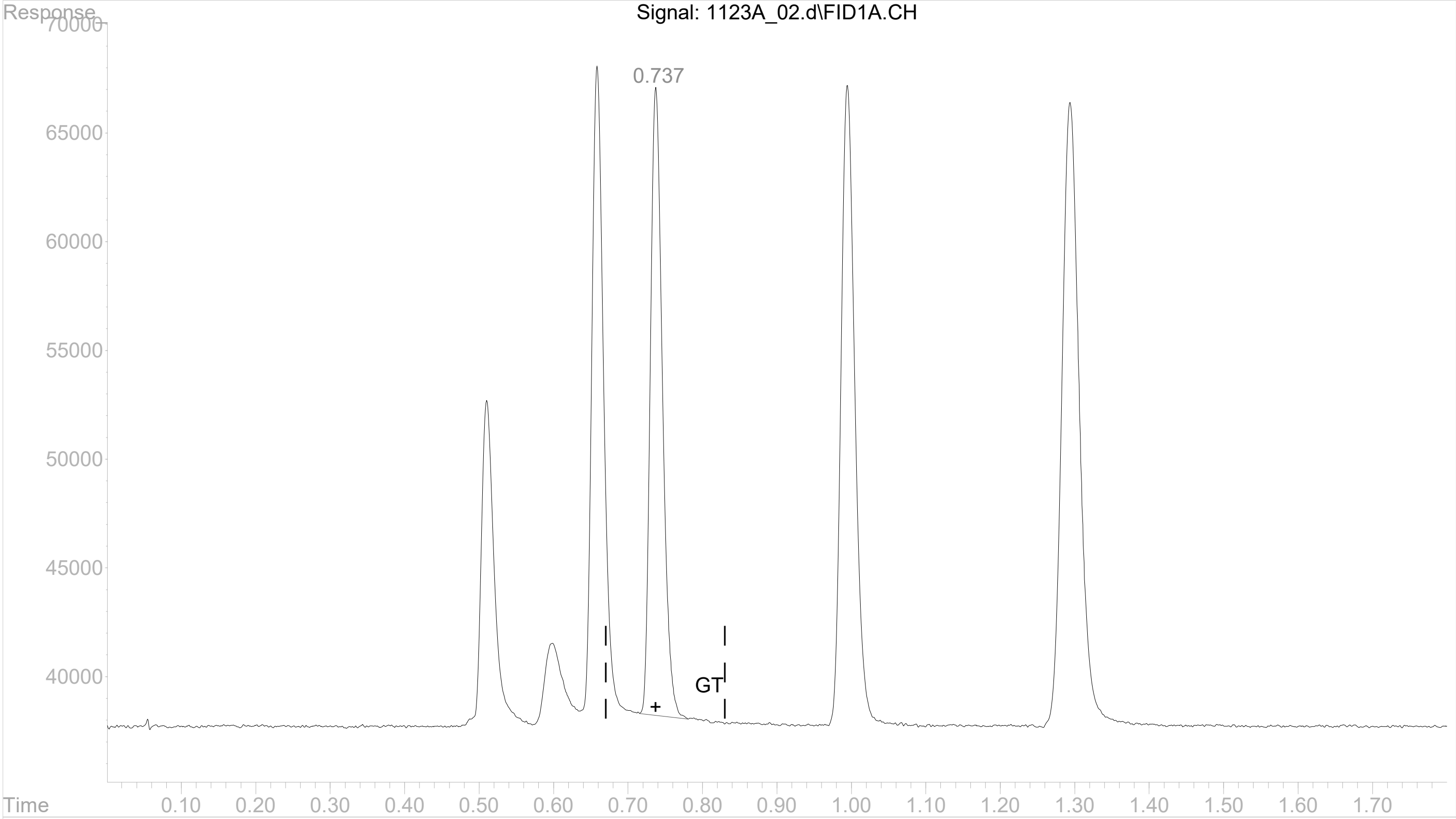
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:48 pm
Operator :
Sample : STD AGC 2 RSK175
Misc :
ALS Vial : 2 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:56:18 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:56:06 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(3) ethene

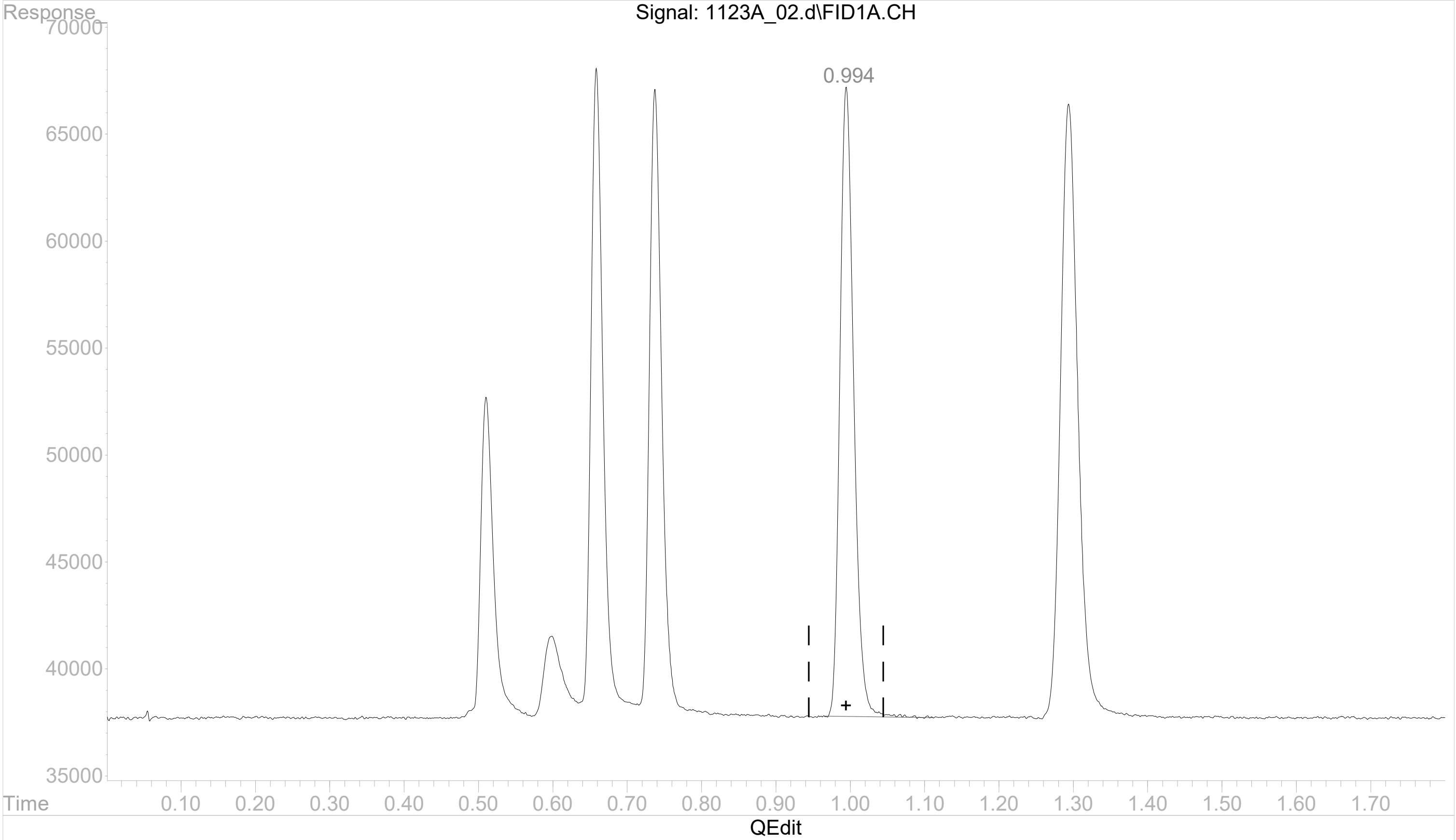
0.737min 0.0123164 mg/l m

response 326924

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:48 pm
Operator  :
Sample    : STD AGC 2 RSK175
Misc      :
ALS Vial  : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:56:18 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:56:06 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(4) acetylene

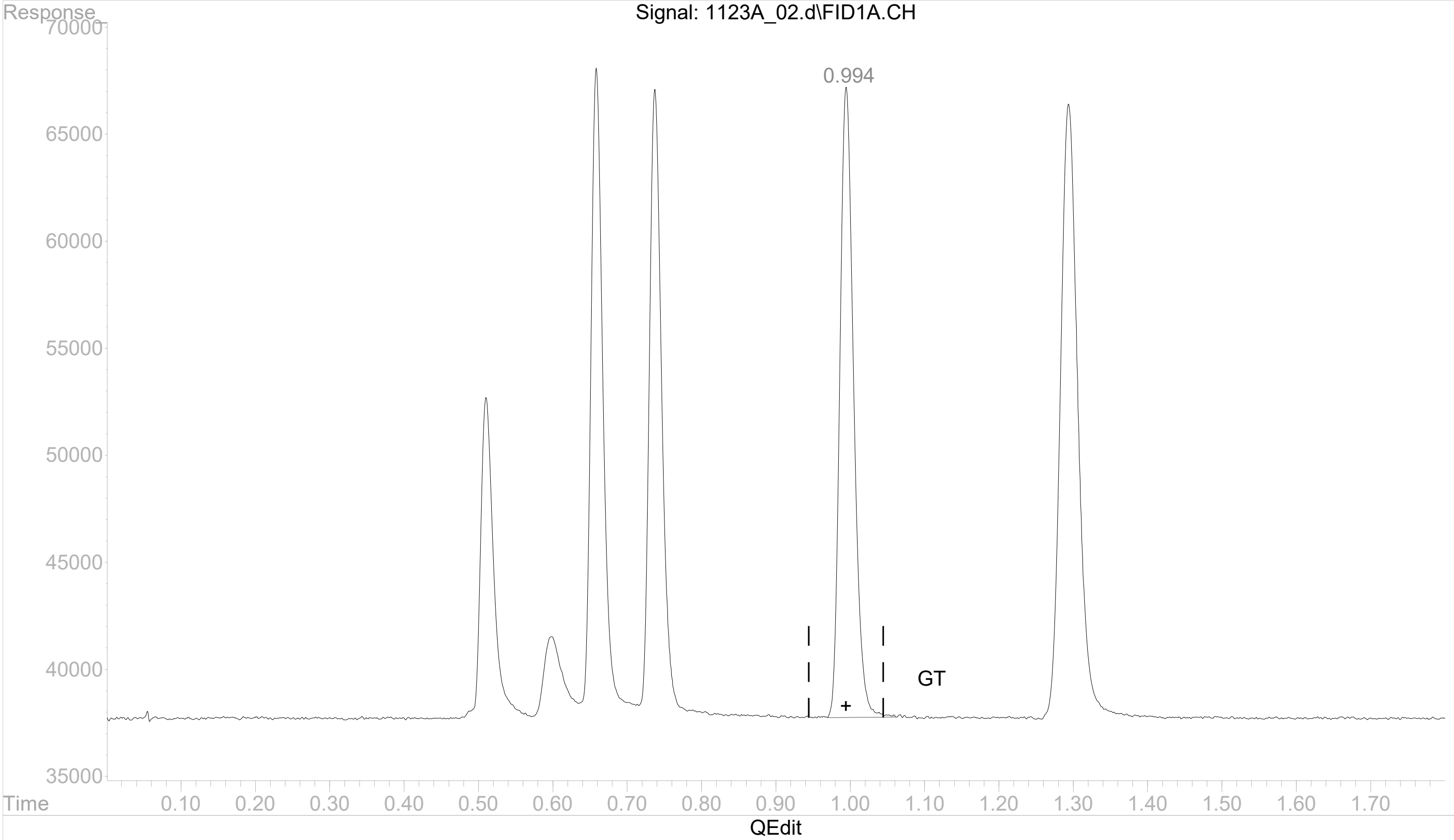
0.995min   0.0199971 mg/l

response   381292

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:48 pm
Operator  :
Sample    : STD AGC 2 RSK175
Misc      :
ALS Vial  : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:56:18 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:56:06 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(4) acetylene

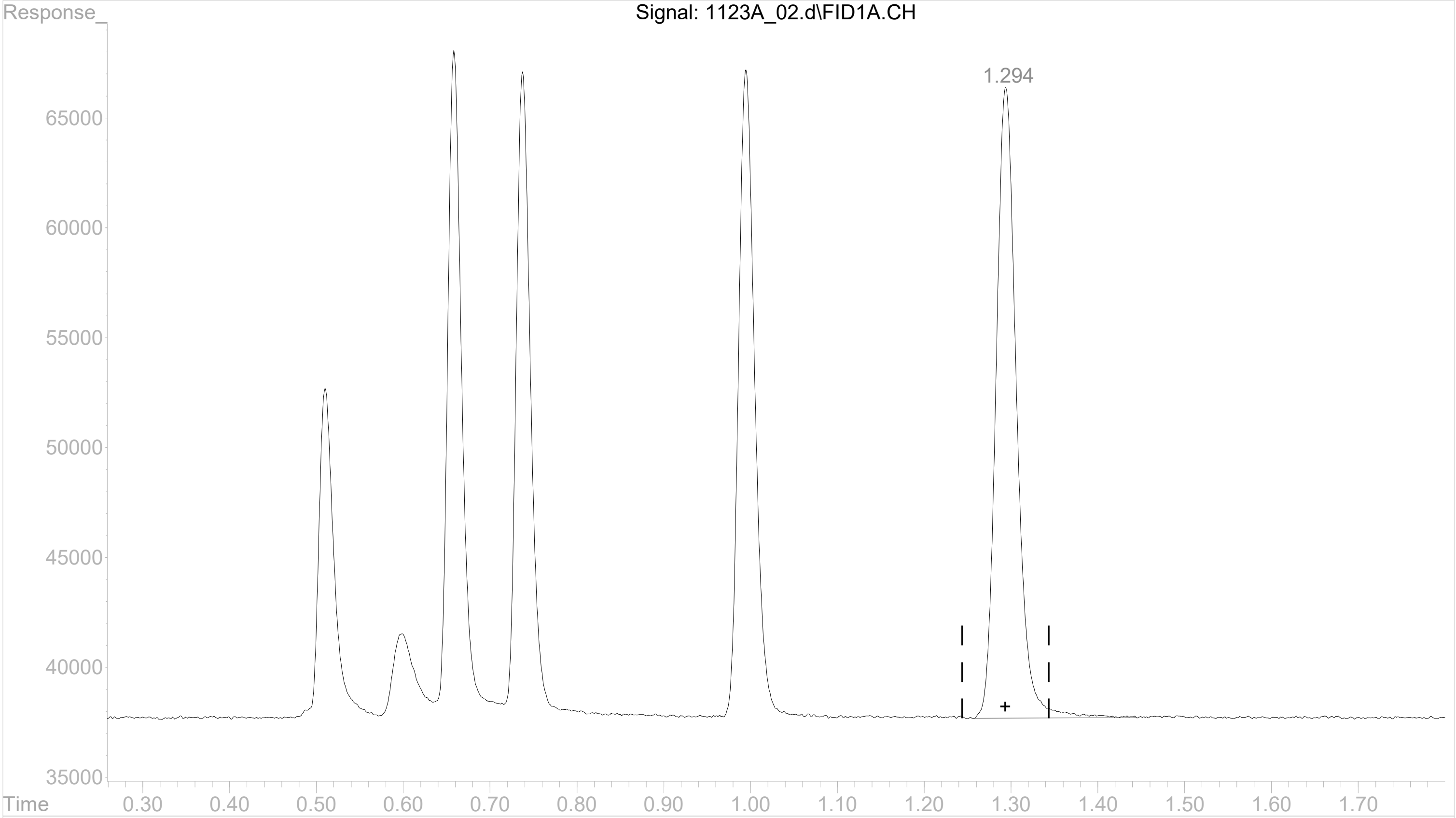
0.994min   0.0200007 mg/l m

response   381361

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_02.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:48 pm
Operator  :
Sample    : STD AGC 2 RSK175
Misc      :
ALS Vial  : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:56:18 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:56:06 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(5) propane

1.294min 0.0180660 mg/l

response 472606

(+) = Expected Retention Time

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:49:37 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	1691961	0.0703752 mg/l m
2)	ethane	0.66	3322107	0.1337626 mg/l m
3)	ethene	0.74	3372607	0.1335866 mg/l m
4)	acetylene	0.99	3966364	0.2227357 mg/l m
5)	propane	1.29	4869358	0.1989238 mg/l m

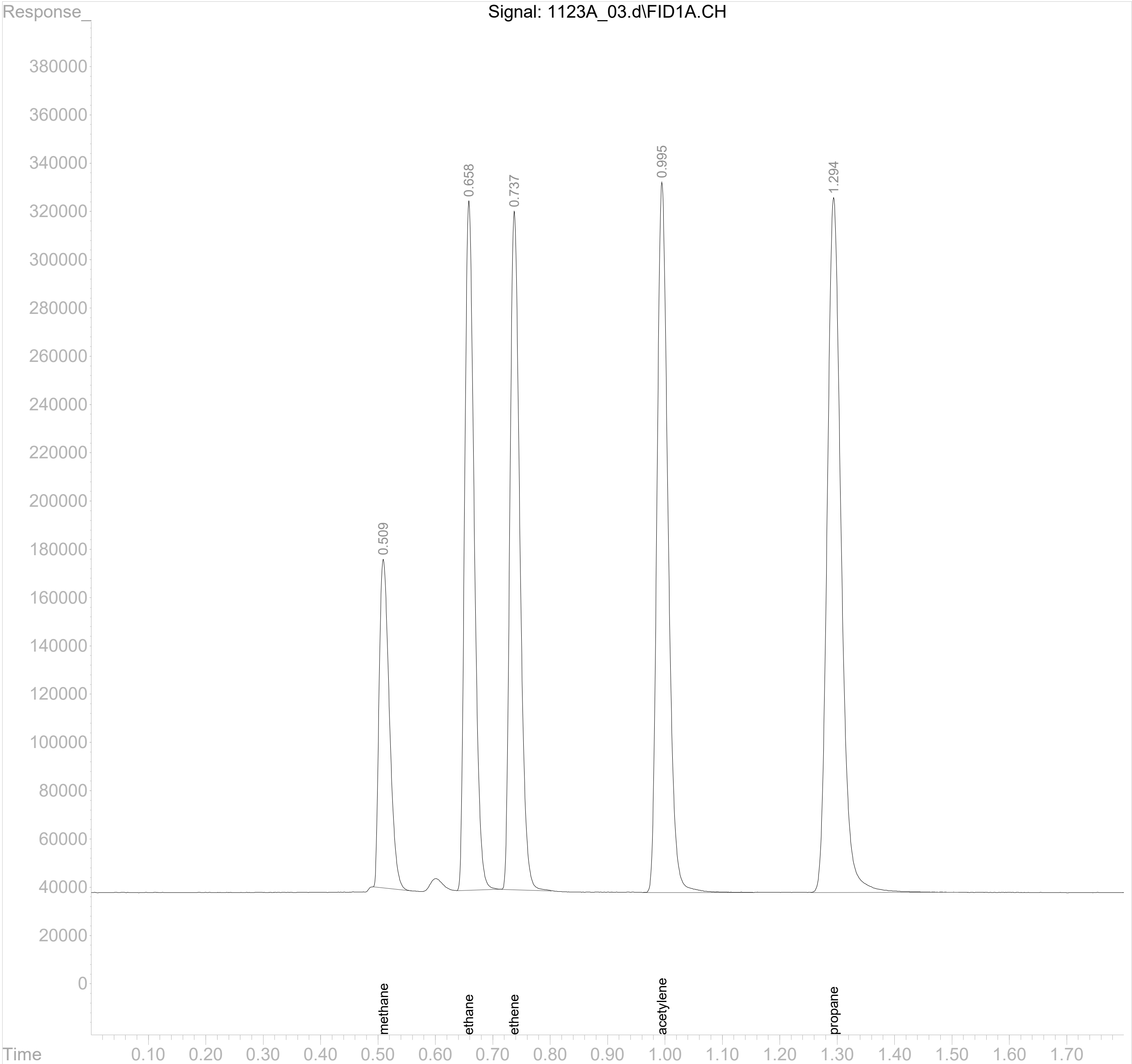
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:49:37 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

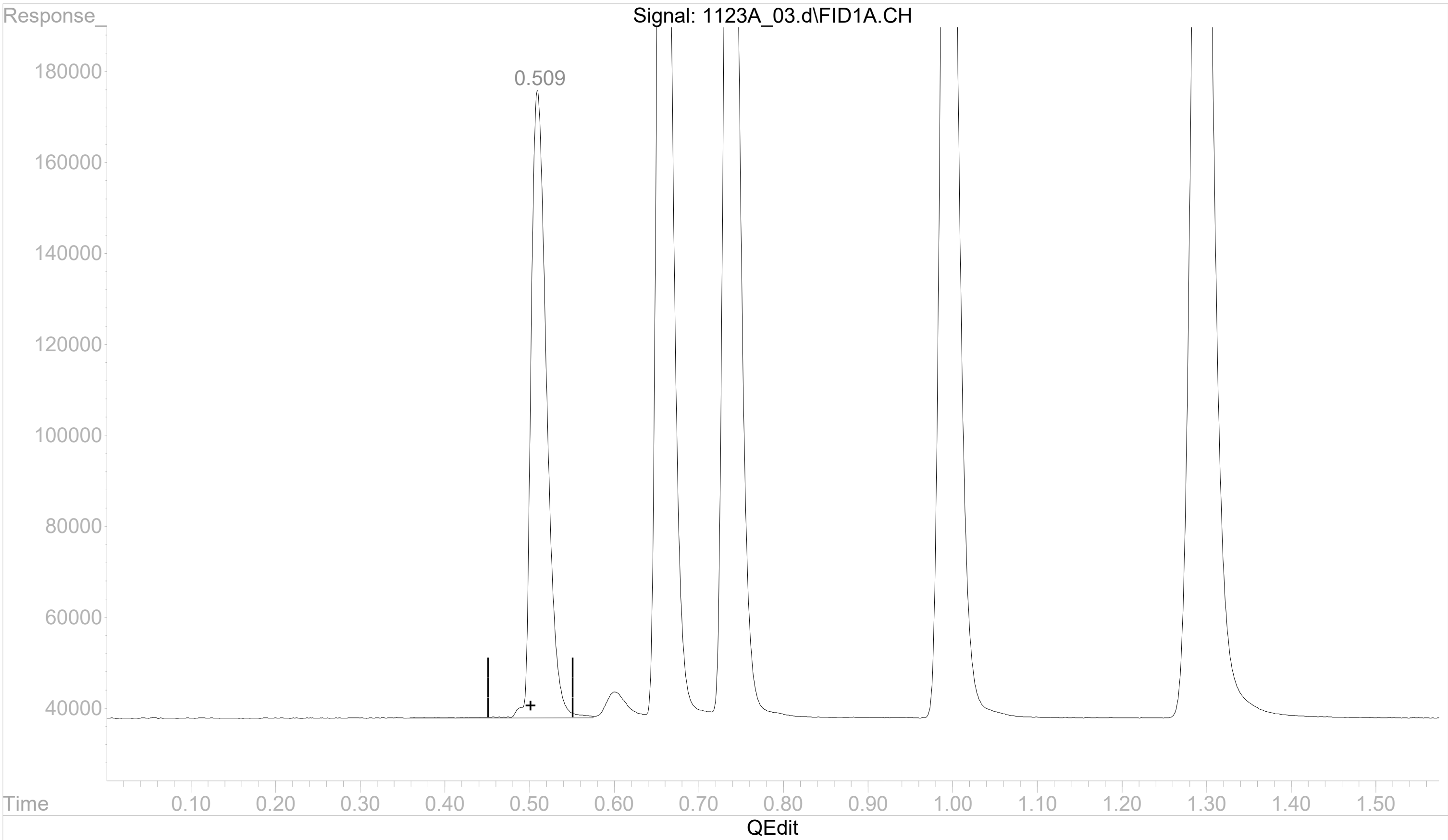


Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:54:17 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:37:57 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(1) methane

0.510min 0.0000000 mg/l

response 1775634

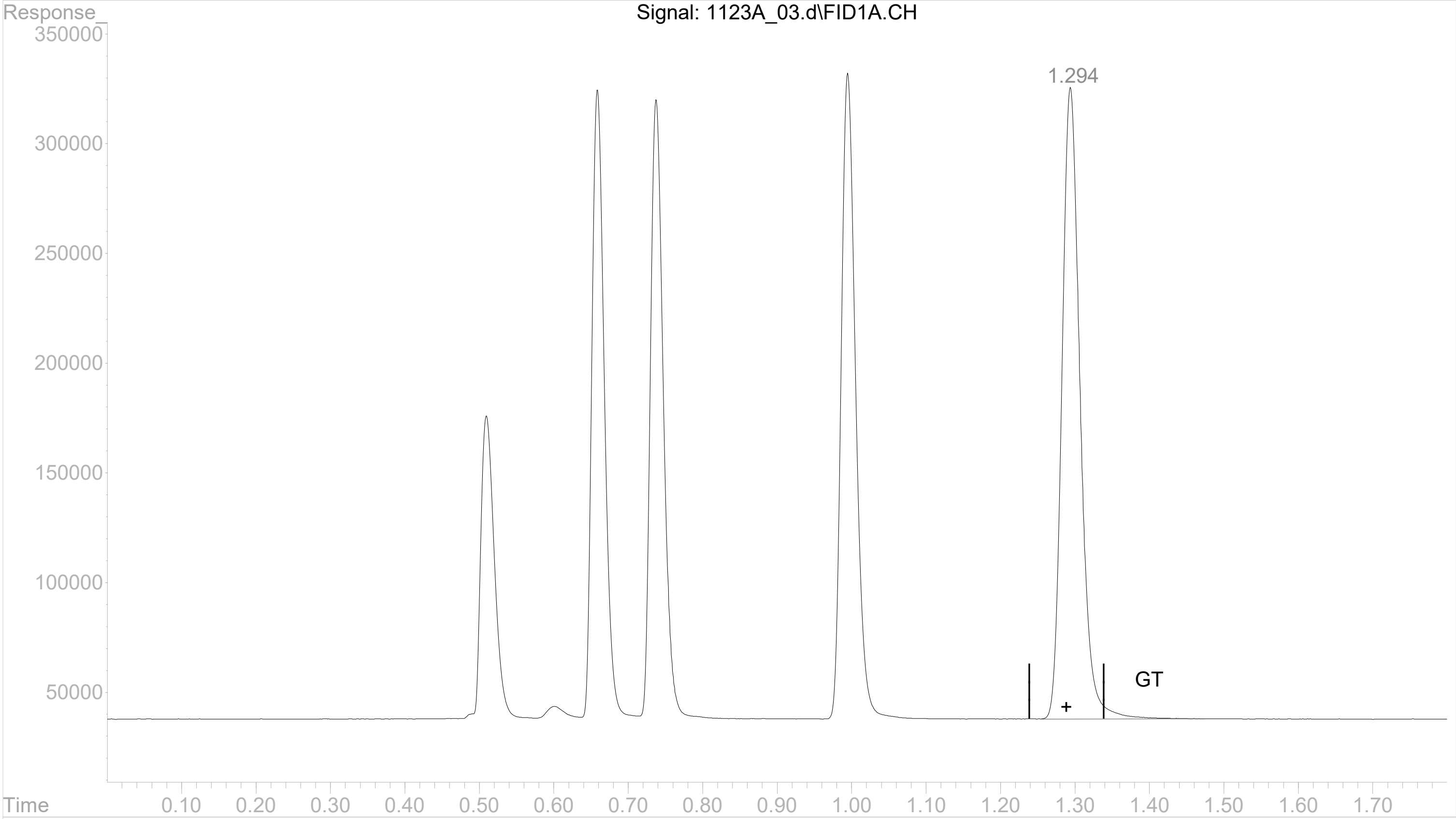
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:54:17 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:37:57 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(5) propane

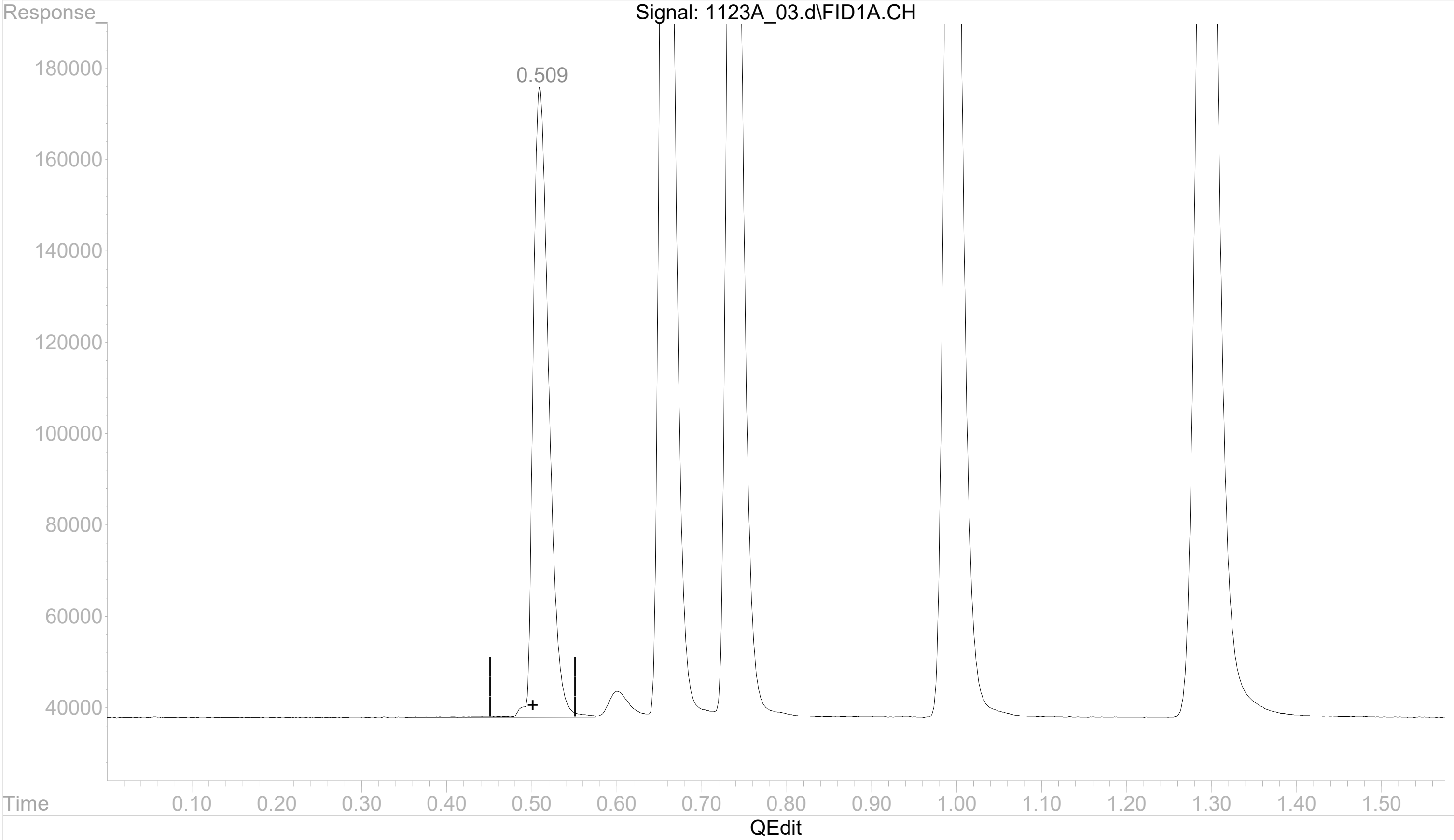
1.294min 0.0000000 mg/l m

response 4865751

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:13:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:07:03 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(1) methane

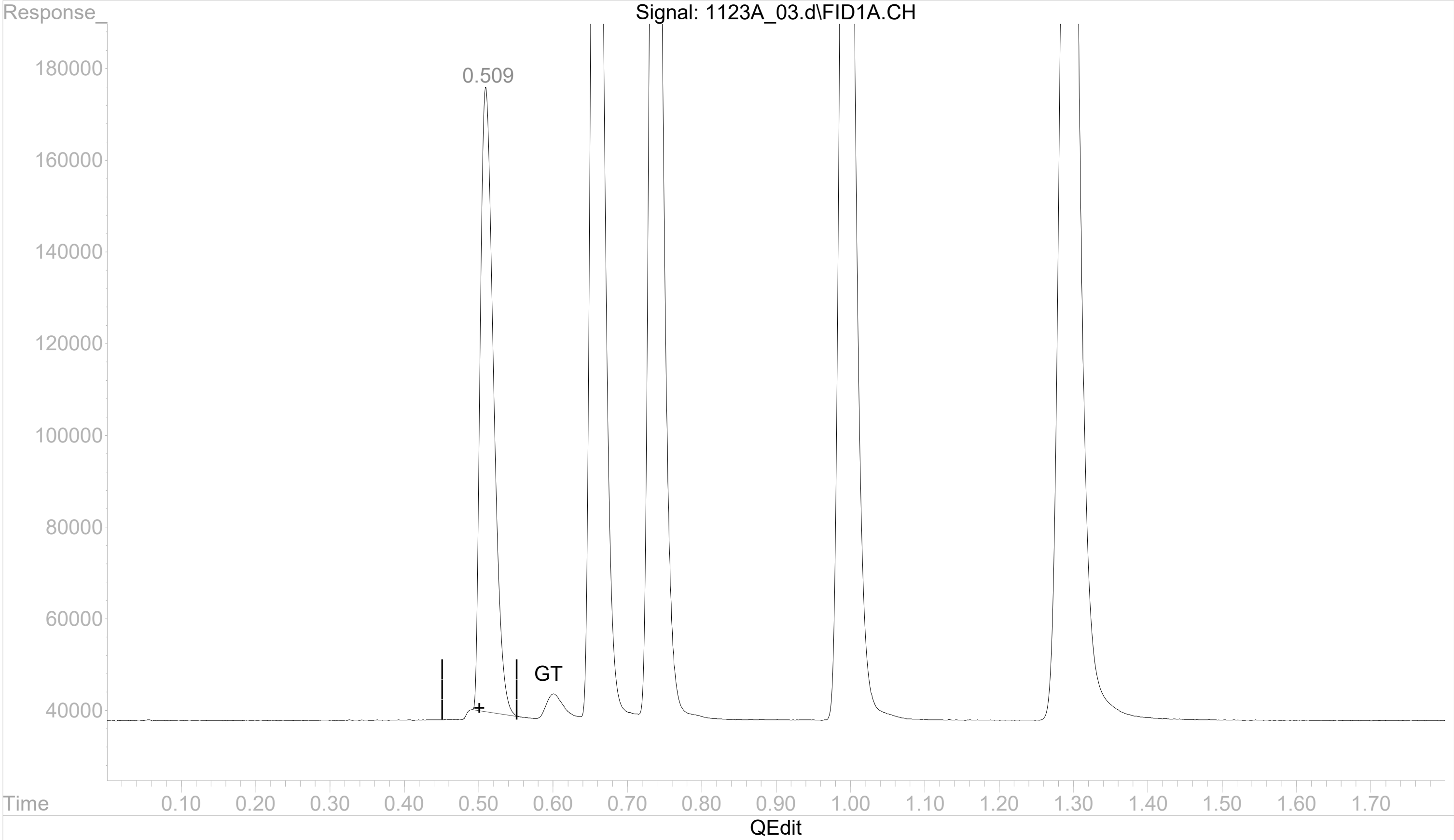
0.510min 0.0000000 mg/l

response 1775634

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:13:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:07:03 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(1) methane

0.509min 0.0000000 mg/l m

response 1692034

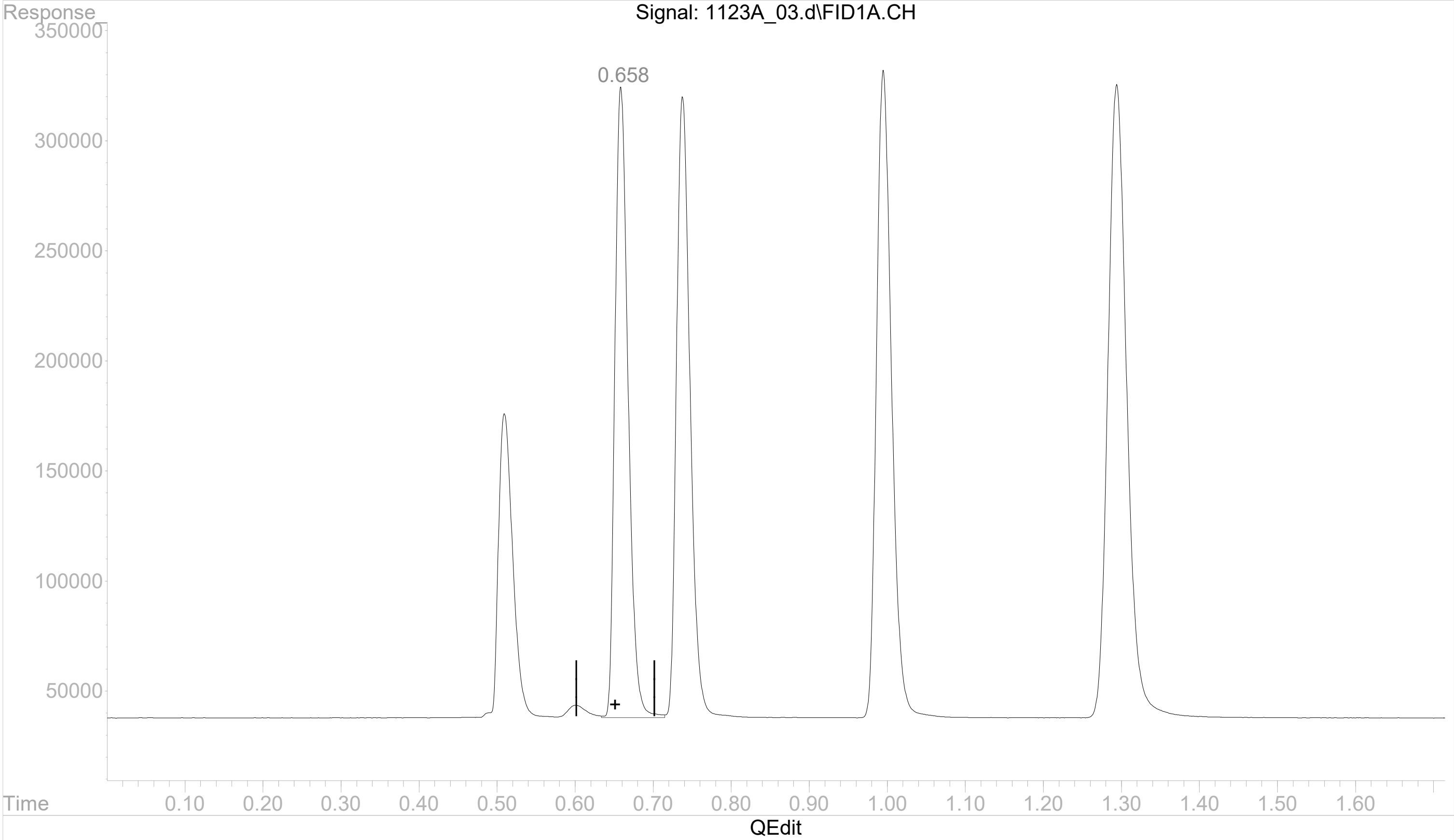
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:13:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:07:03 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(2) ethane

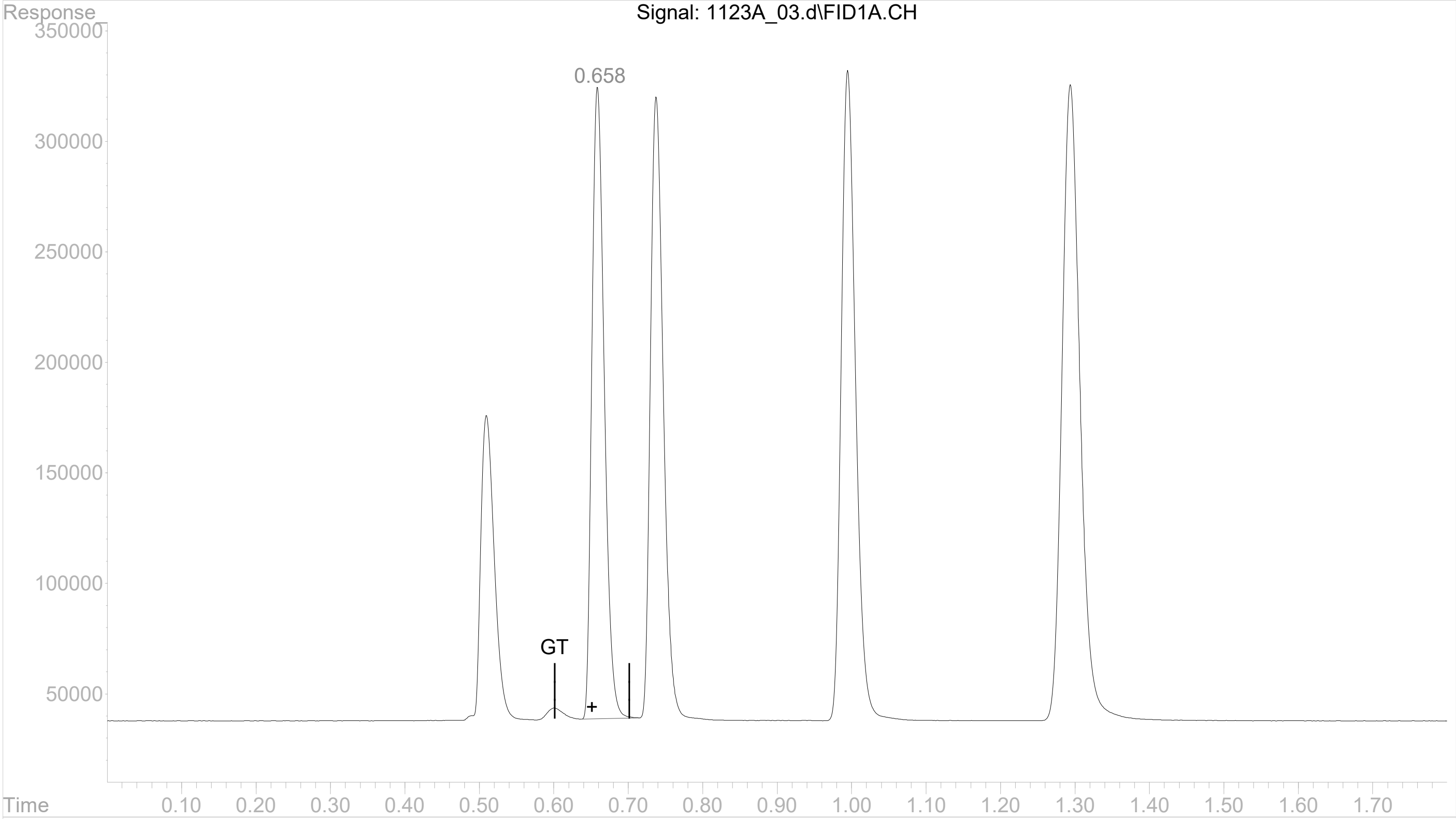
0.659min 0.0000000 mg/l

response 3369902

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:13:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:07:03 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(2) ethane

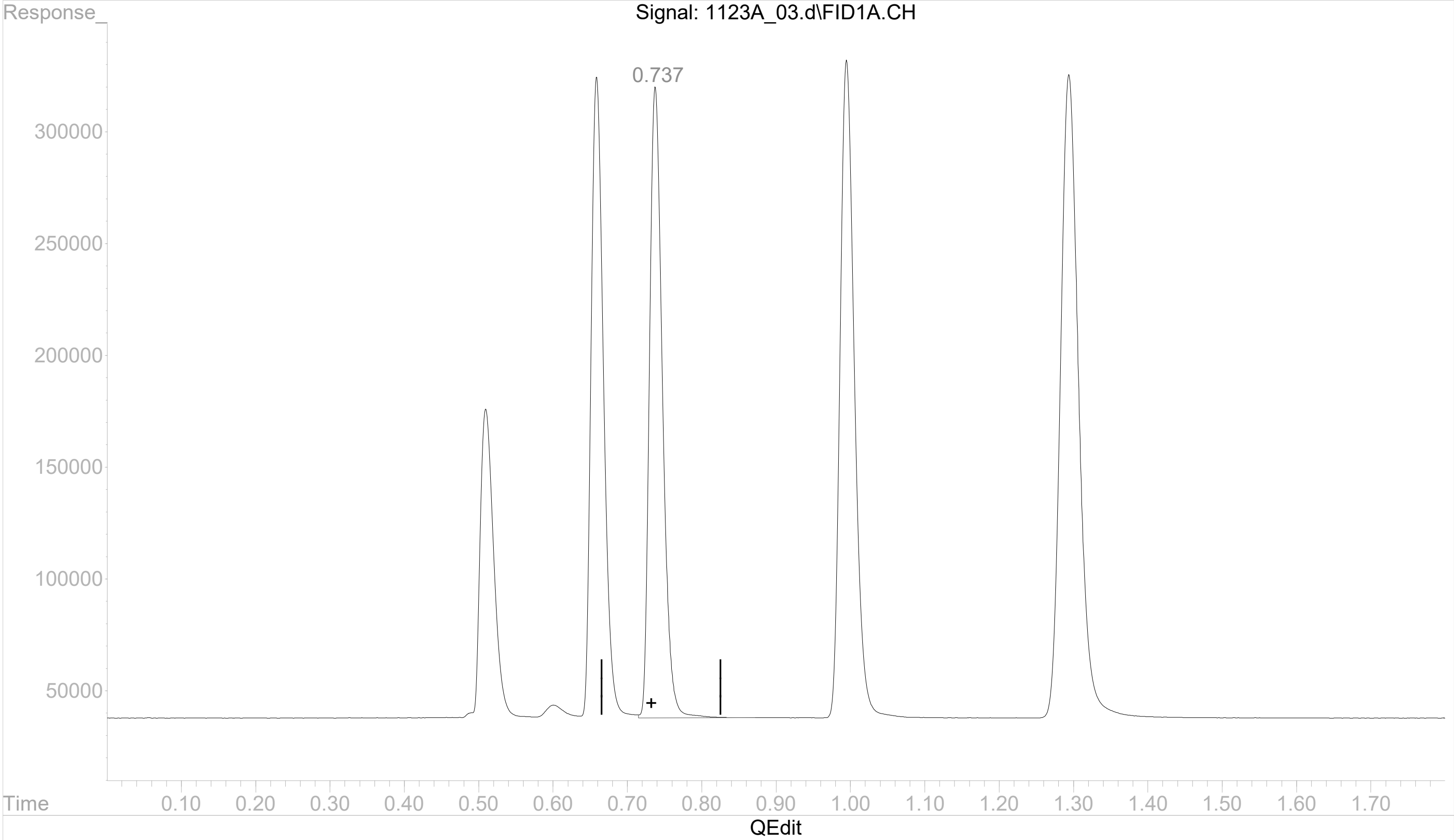
0.658min 0.0000000 mg/l m

response 3322272

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:13:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:07:03 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(3) ethene

0.738min   0.0000000 mg/l

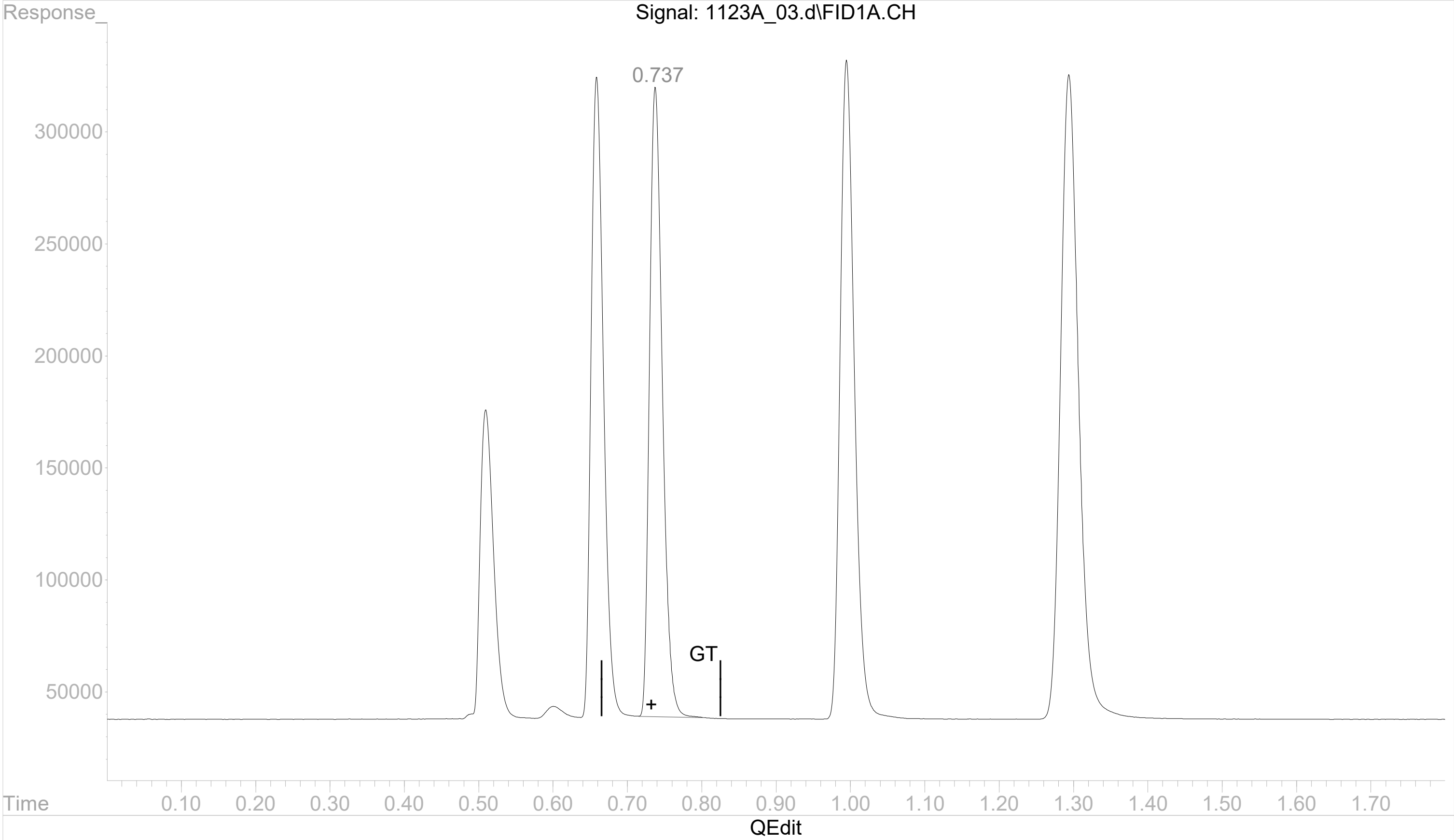
response   3426893

(+) = Expected Retention Time

GC2RSKK23V.M Wed Nov 23 13:14:09 2022 Page: 1

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:13:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:07:03 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(3) ethene

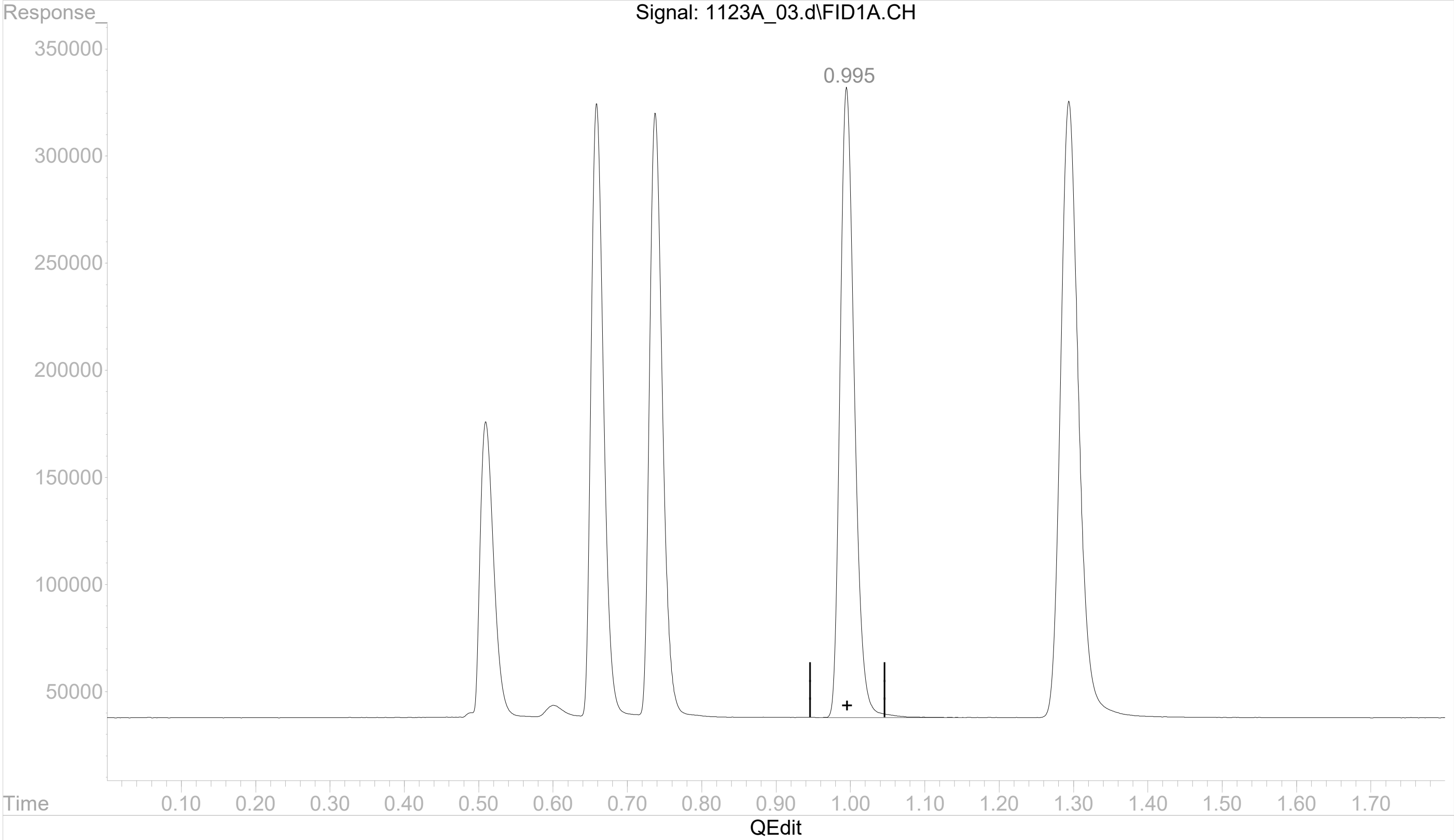
0.737min 0.0000000 mg/l m

response 3370926

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:13:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:07:03 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(4) acetylene

0.995min   0.0000000 mg/l

response   3961909

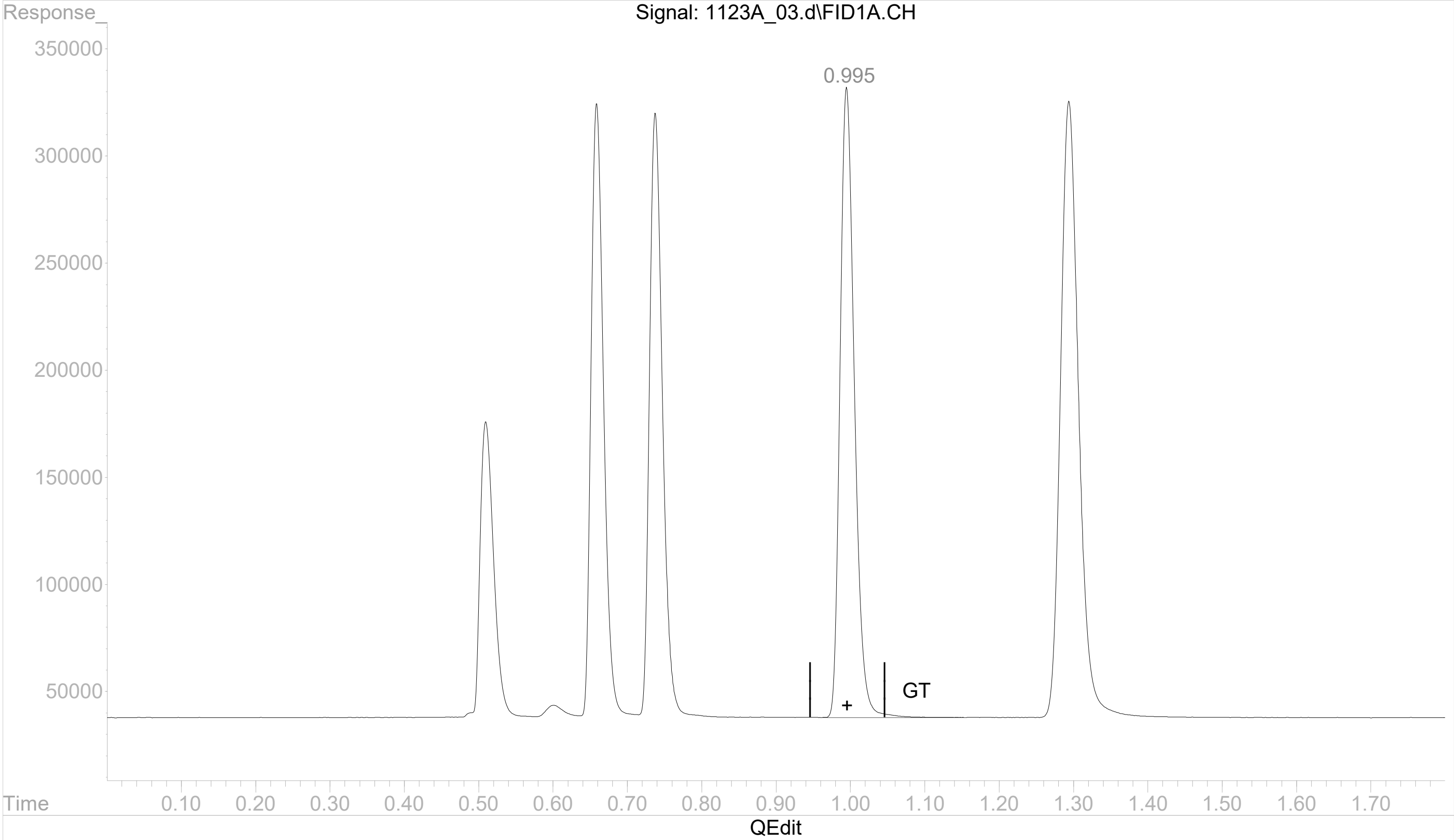
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:13:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:07:03 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(4) acetylene

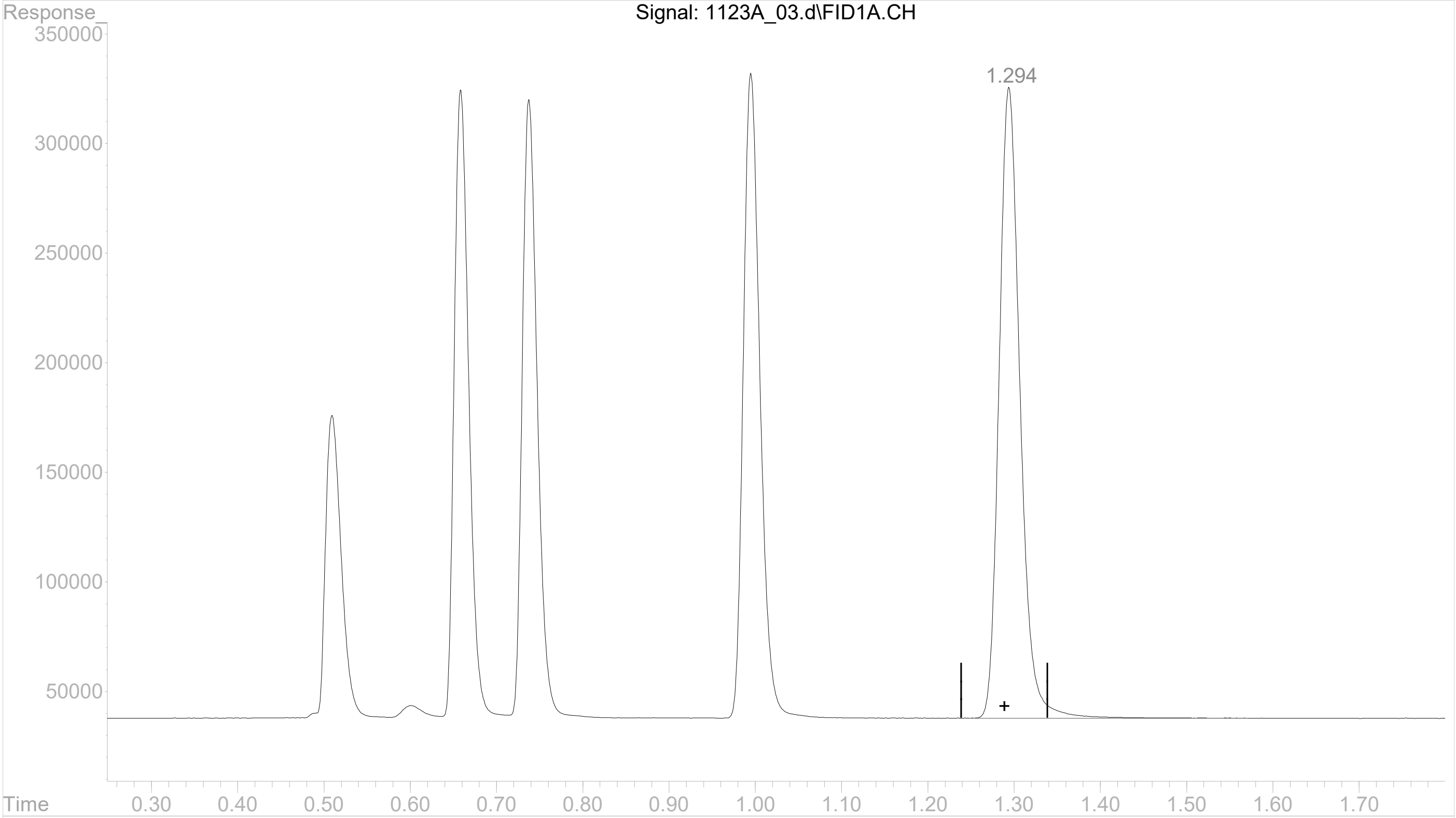
0.995min 0.0000000 mg/l m

response 3966140

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:13:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:07:03 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(5) propane

1.294min 0.0000000 mg/l

response 4872647

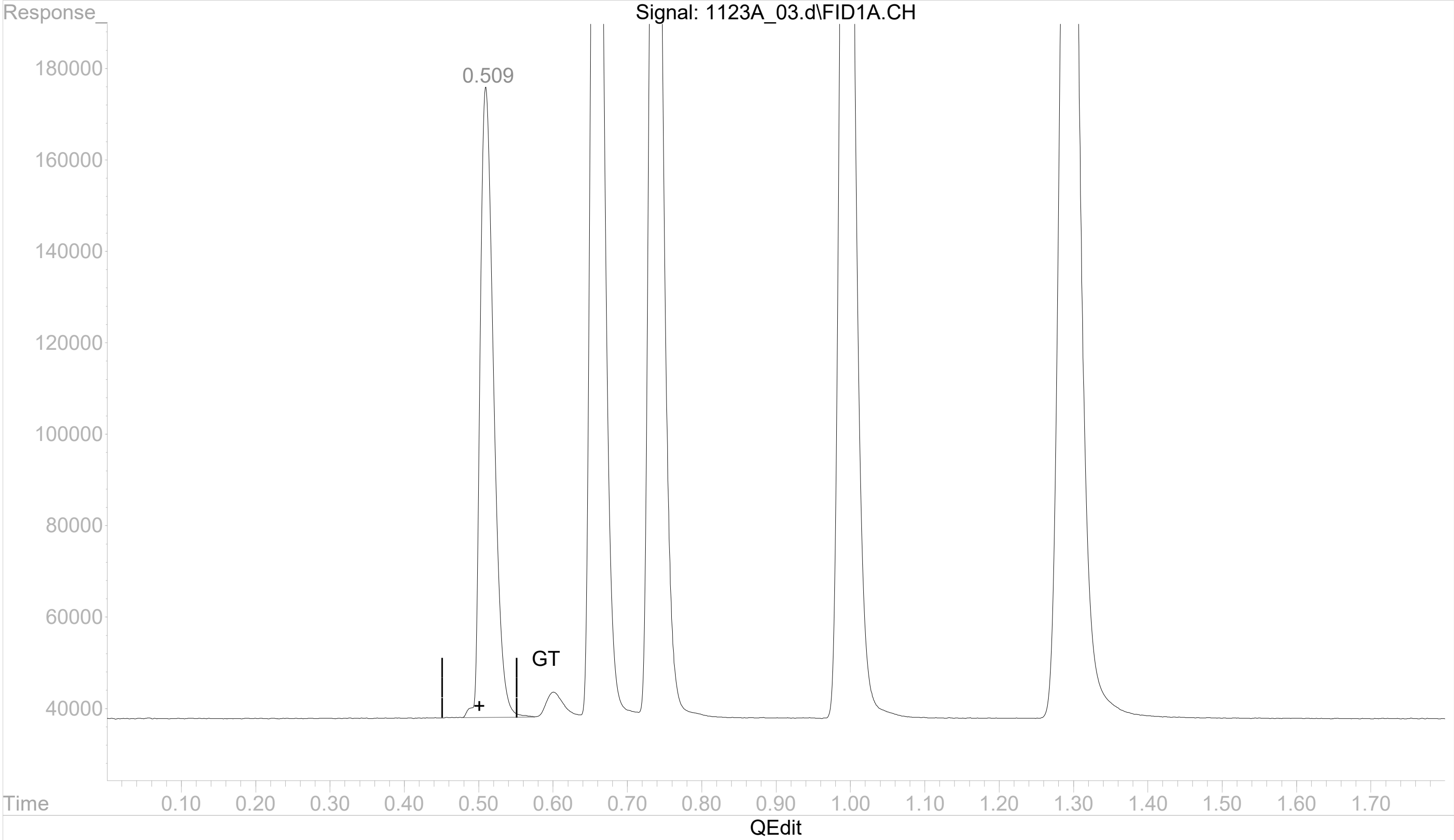
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:54:17 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:37:57 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(1) methane

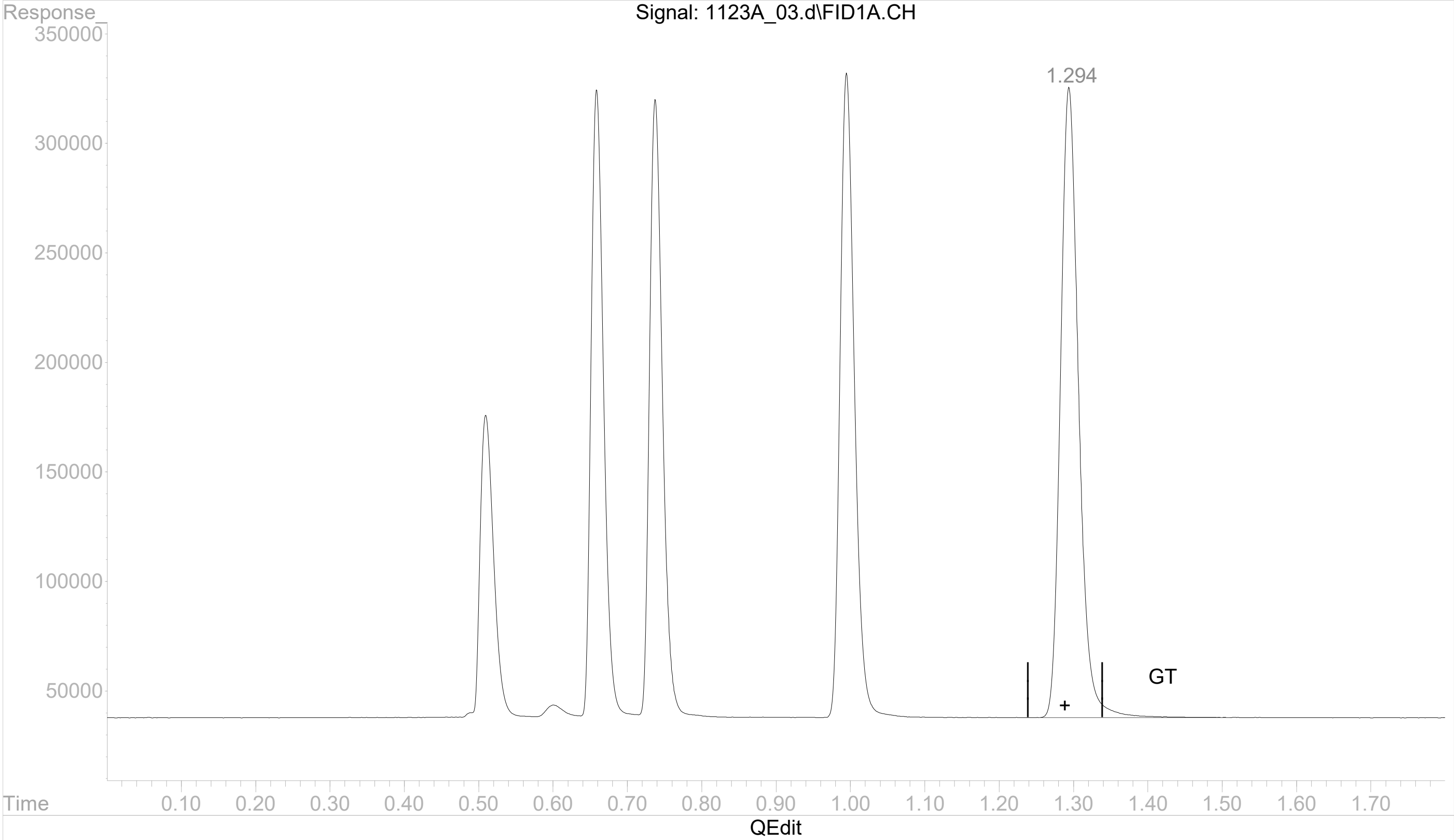
0.509min 0.0000000 mg/l m

response 1753854

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:13:28 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:07:03 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(5) propane

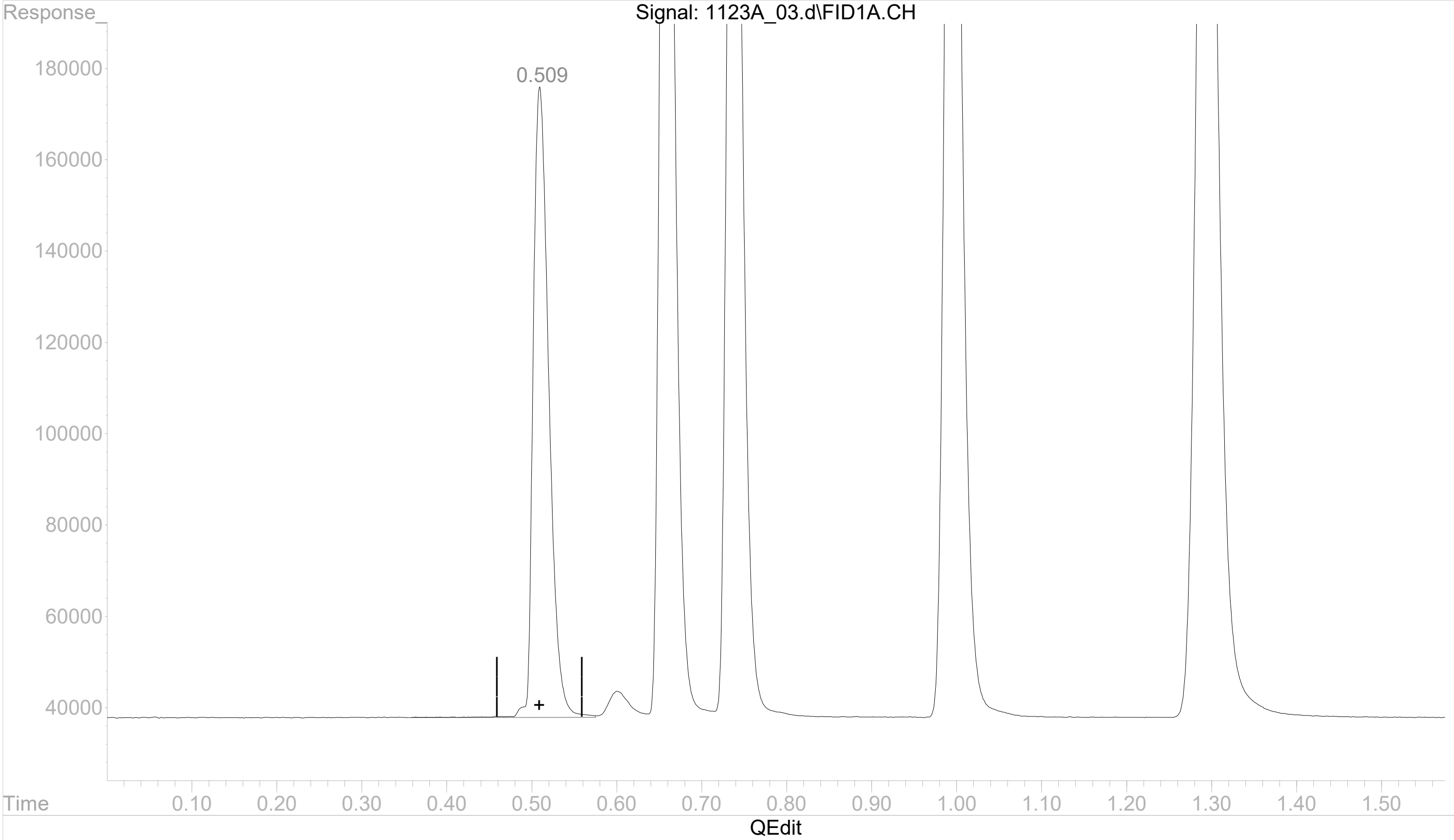
1.294min 0.0000000 mg/l m

response 4869571

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:48:16 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(1) methane

0.510min 0.0738555 mg/l

response 1775634

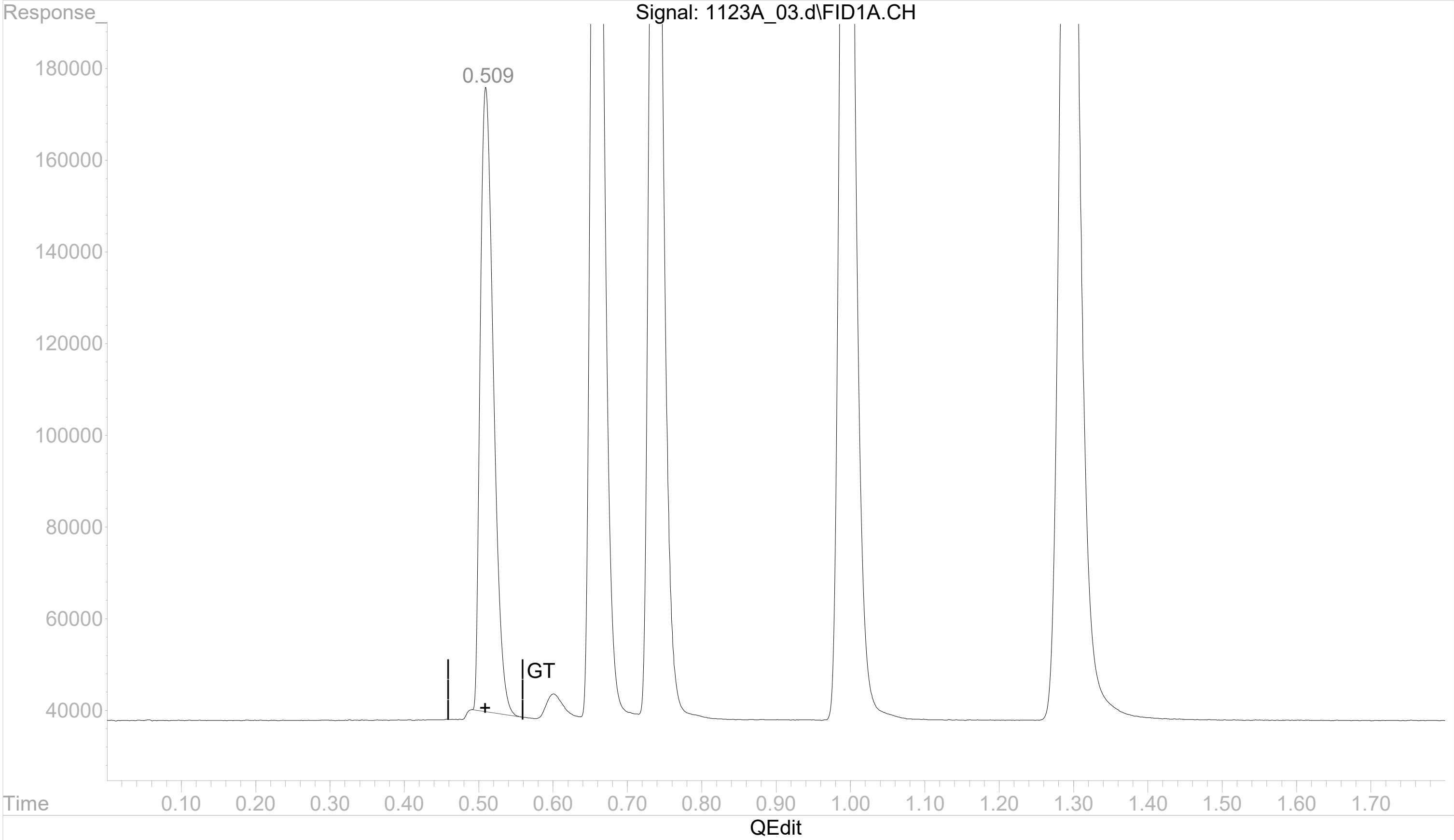
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:48:16 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(1) methane

0.509min 0.0703752 mg/l m

response 1691961

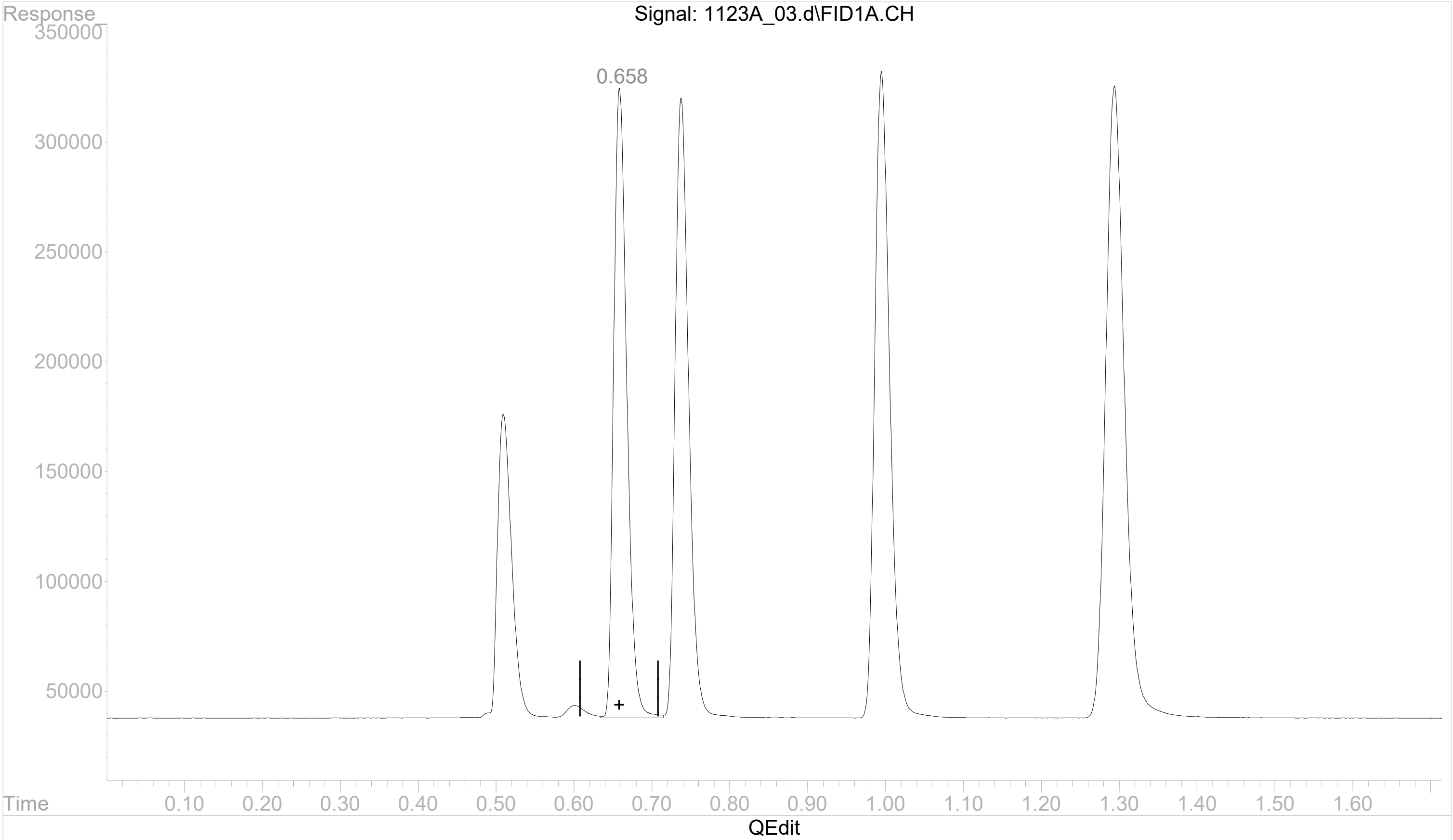
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On   : 23 Nov 2022 12:51 pm
Operator :
Sample   : STD AGC 3.0 RSK175
Misc   :
ALS Vial : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:48:16 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(2) ethane

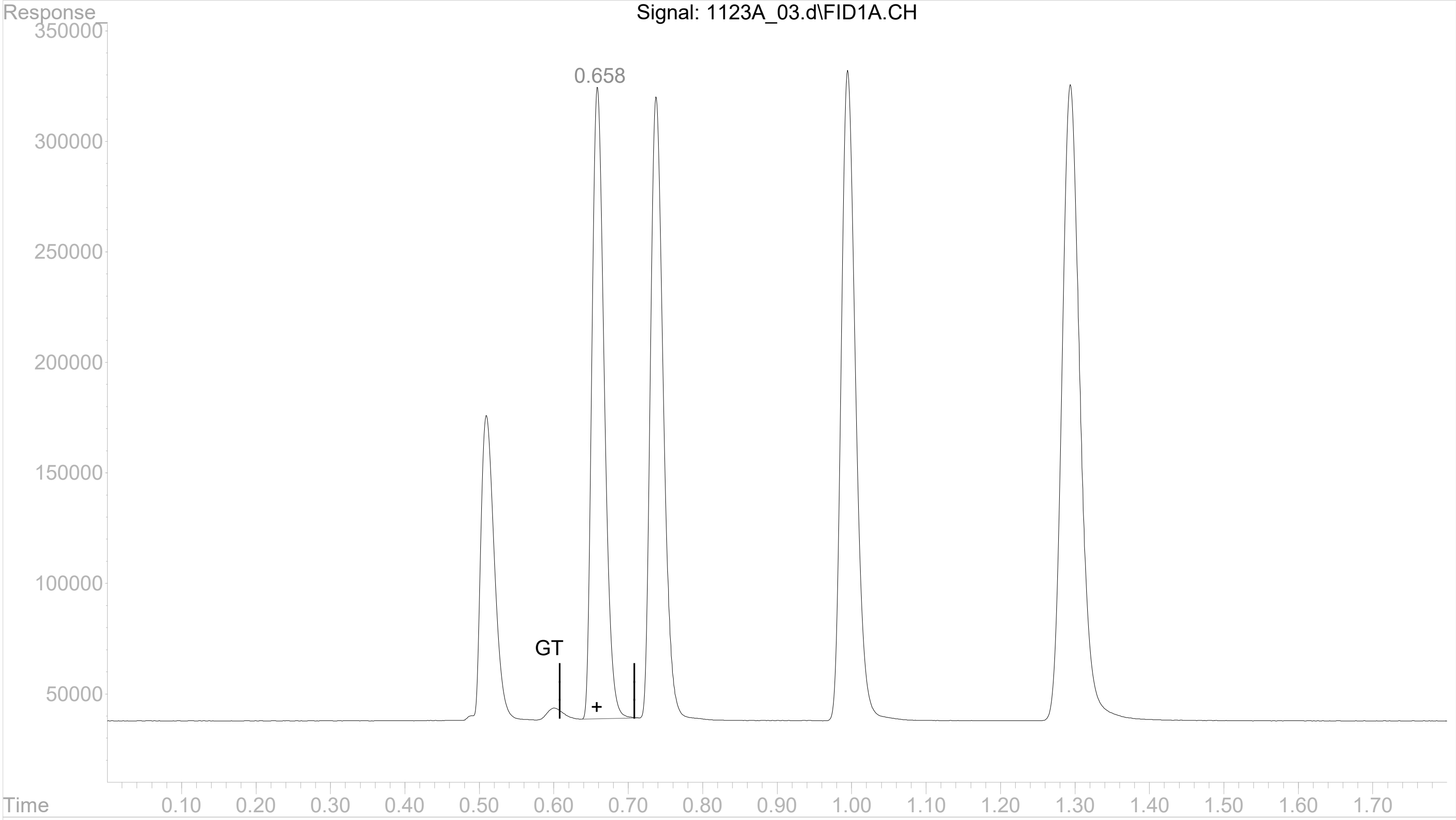
0.659min   0.1356870 mg/l

response   3369902

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:48:16 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(2) ethane

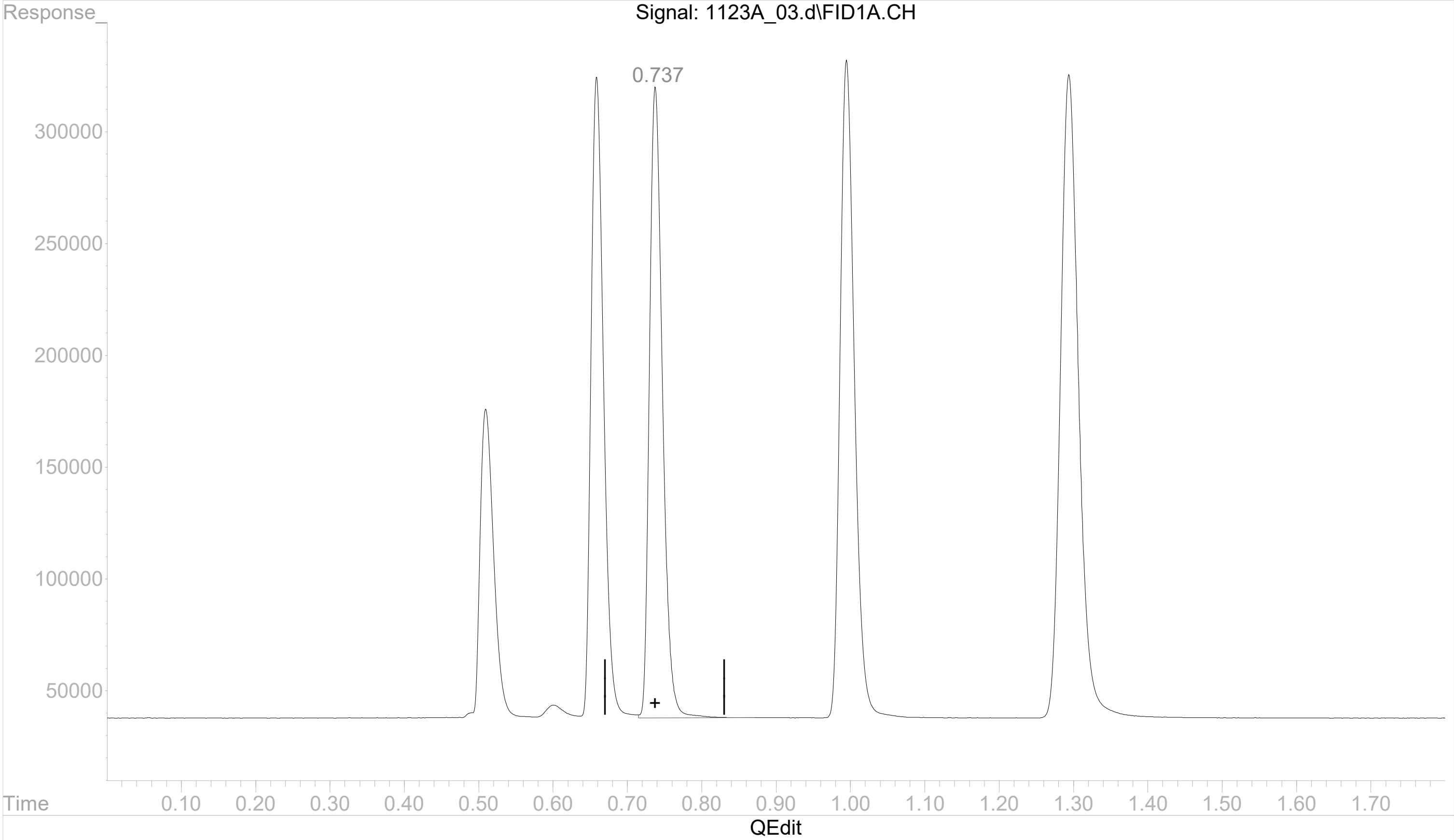
0.658min 0.1337626 mg/l m

response 3322107

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:48:16 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(3) ethene

0.738min 0.1357368 mg/l

response 3426893

(+) = Expected Retention Time

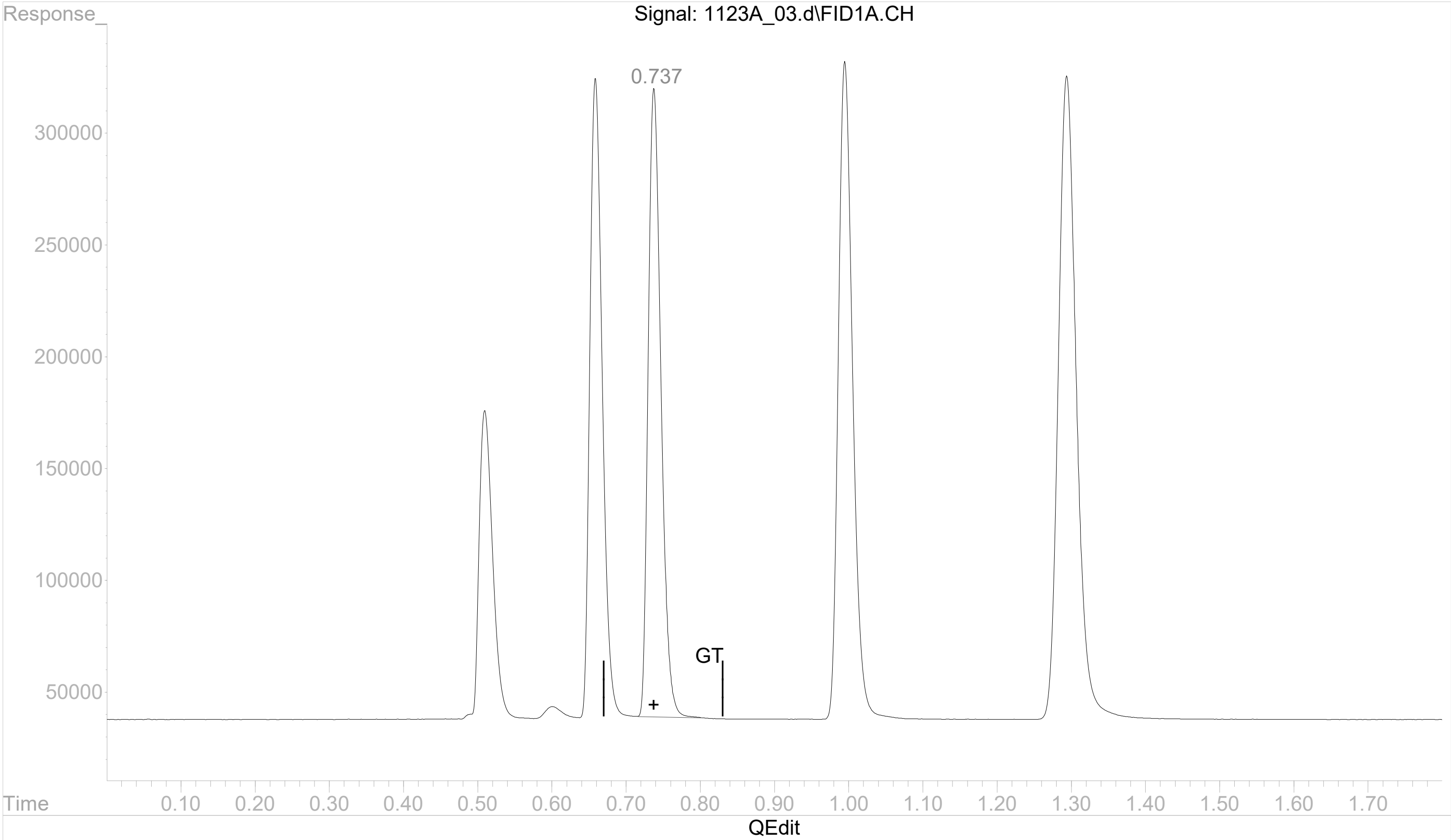
GC2RSKK23V.M Wed Nov 23 13:48:54 2022 Page: 1

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:48:16 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(3) ethene

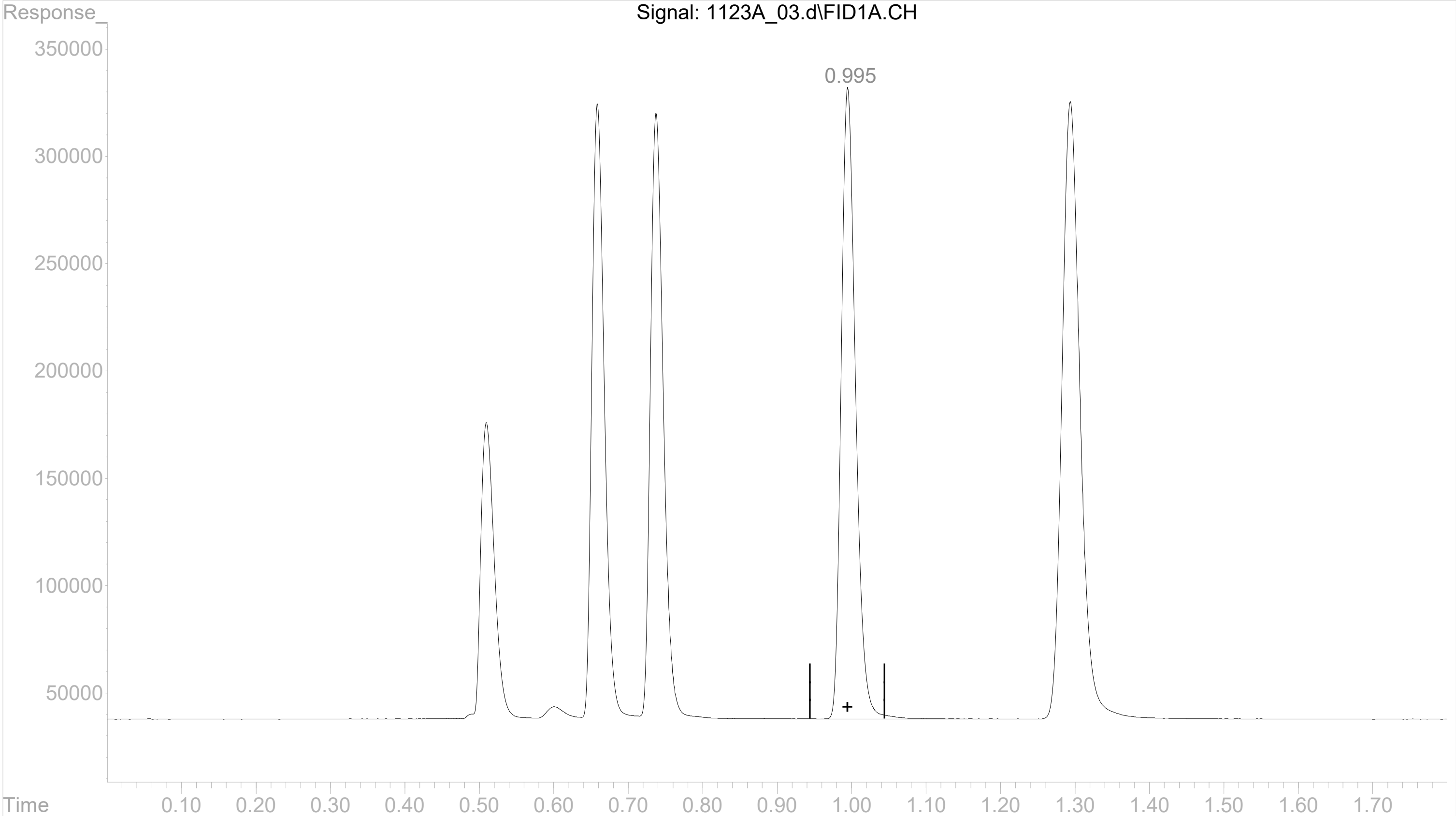
0.737min 0.1335866 mg/l m

response 3372607

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:48:16 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(4) acetylene

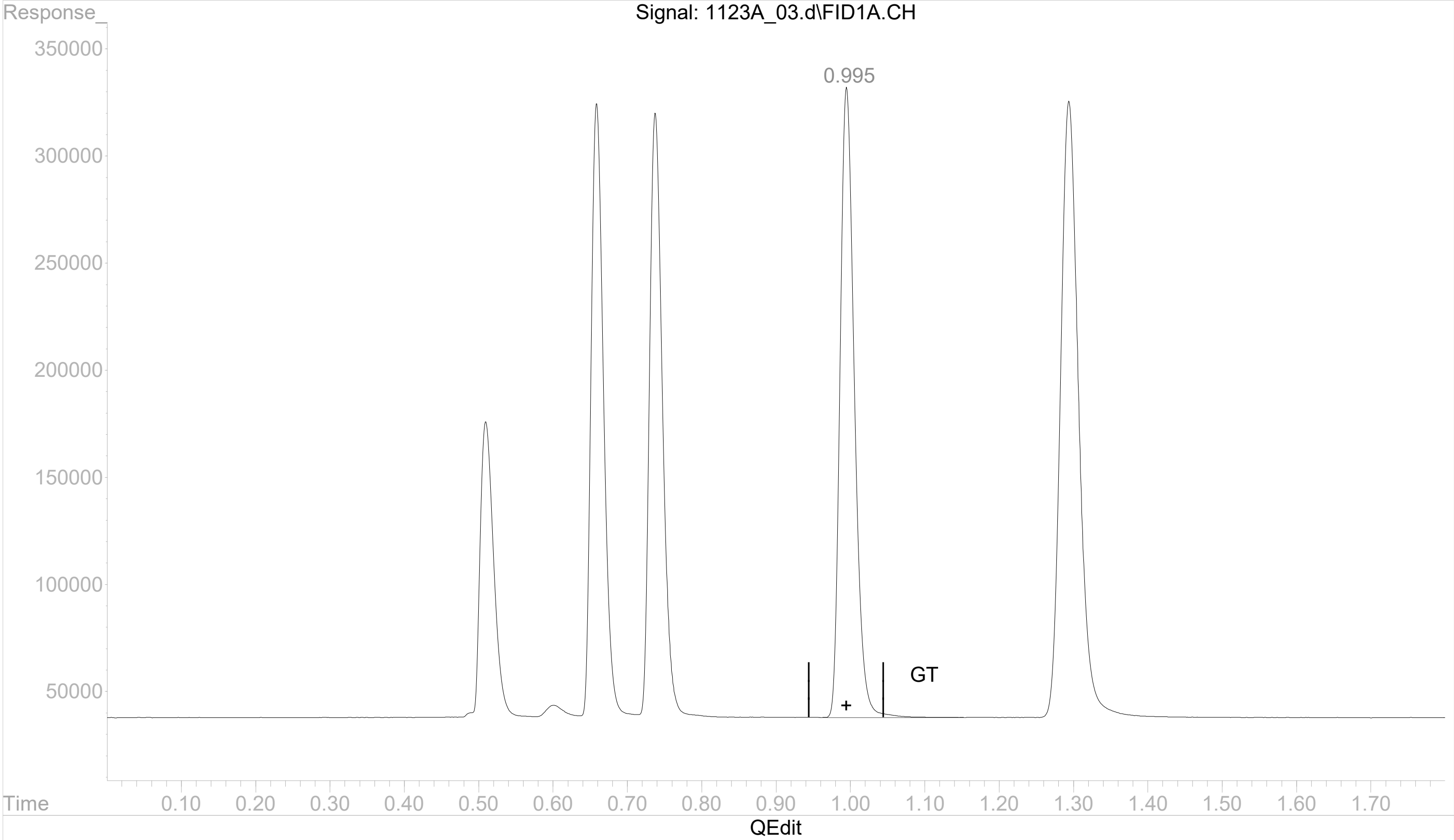
0.995min 0.2224856 mg/l

response 3961909

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:48:16 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(4) acetylene

0.995min 0.2227357 mg/l m

response 3966364

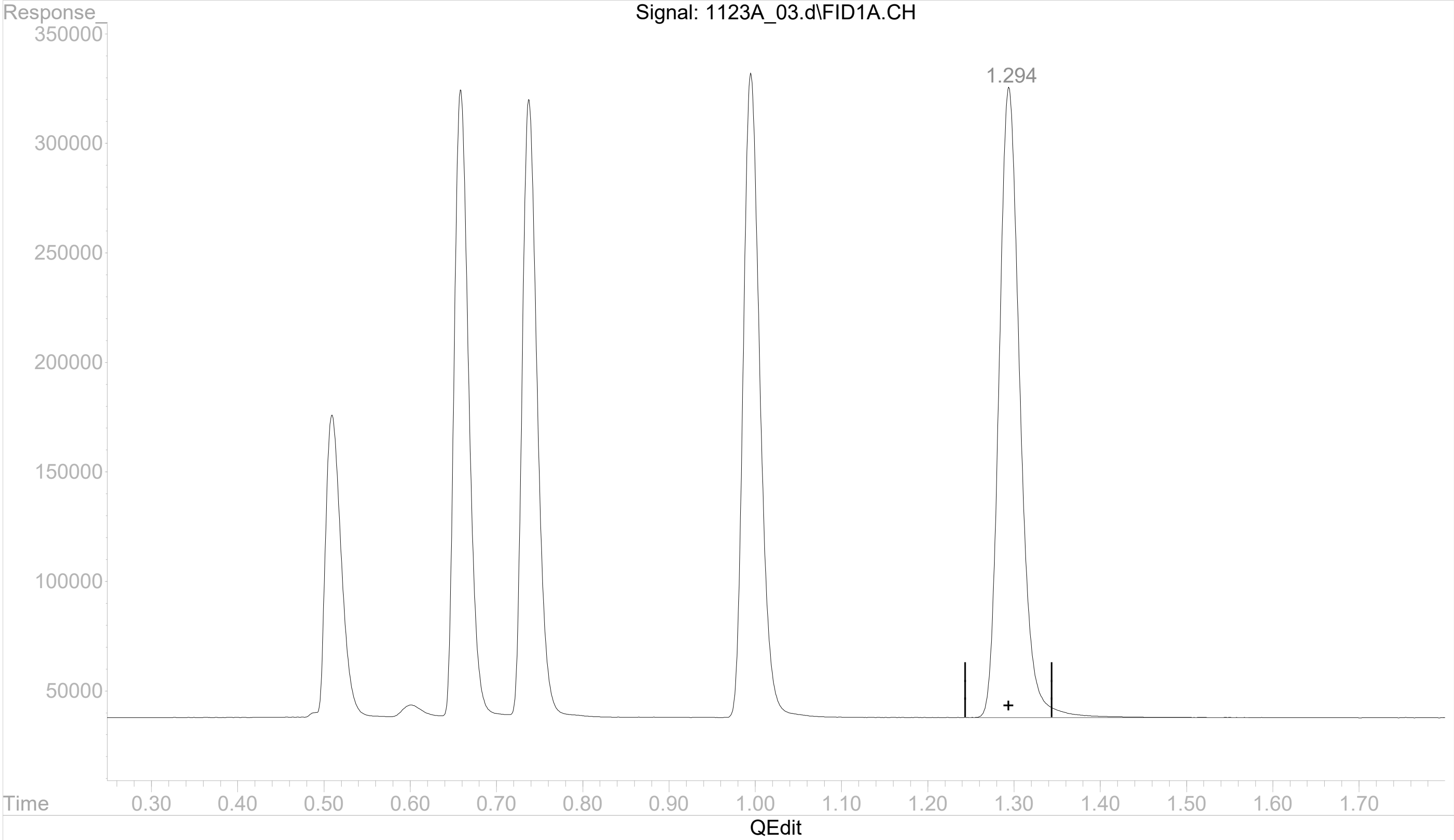
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:48:16 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(5) propane

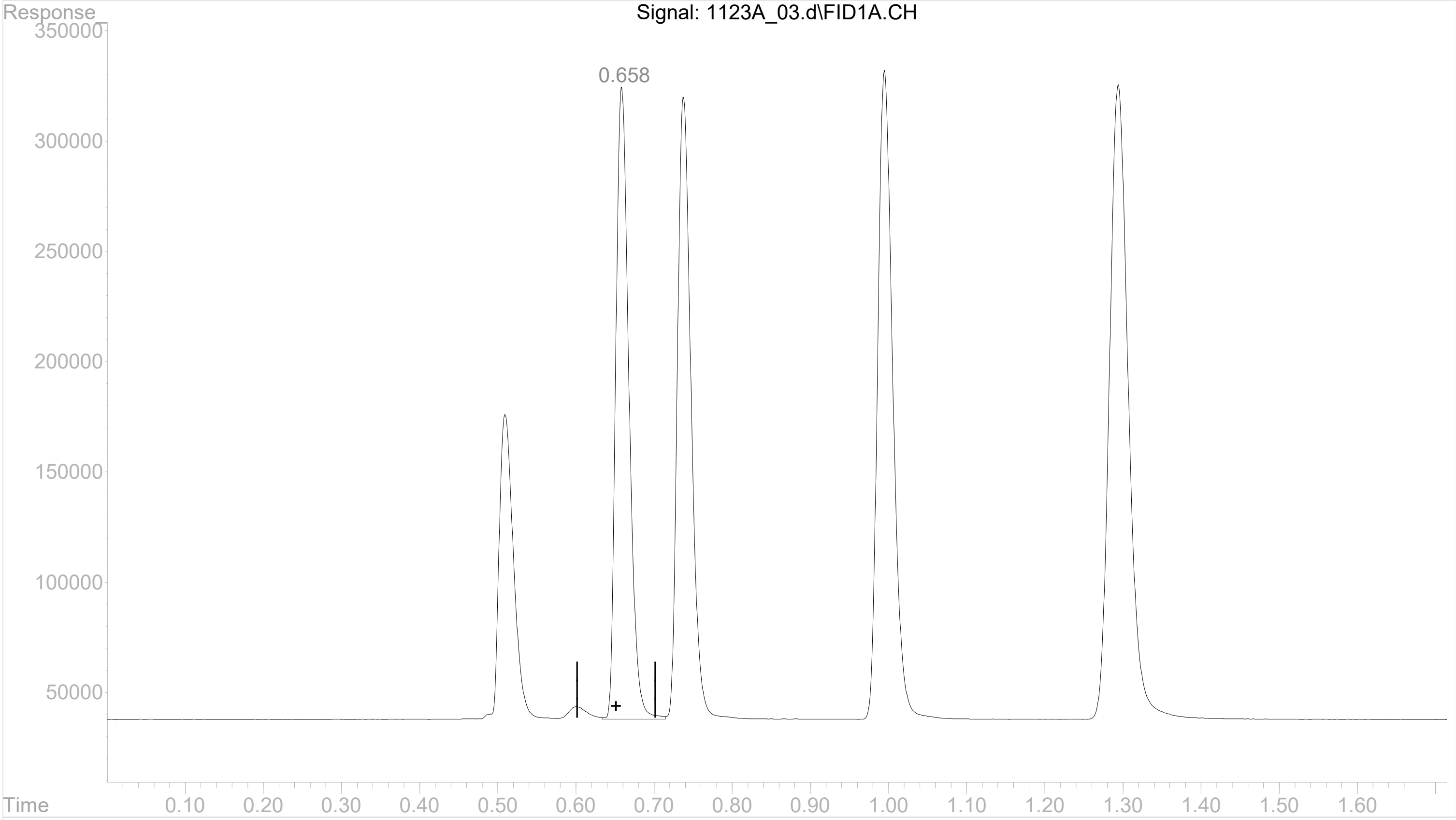
1.294min 0.1990582 mg/l

response 4872647

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:54:17 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:37:57 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(2) ethane

0.659min 0.0000000 mg/l

response 3369902

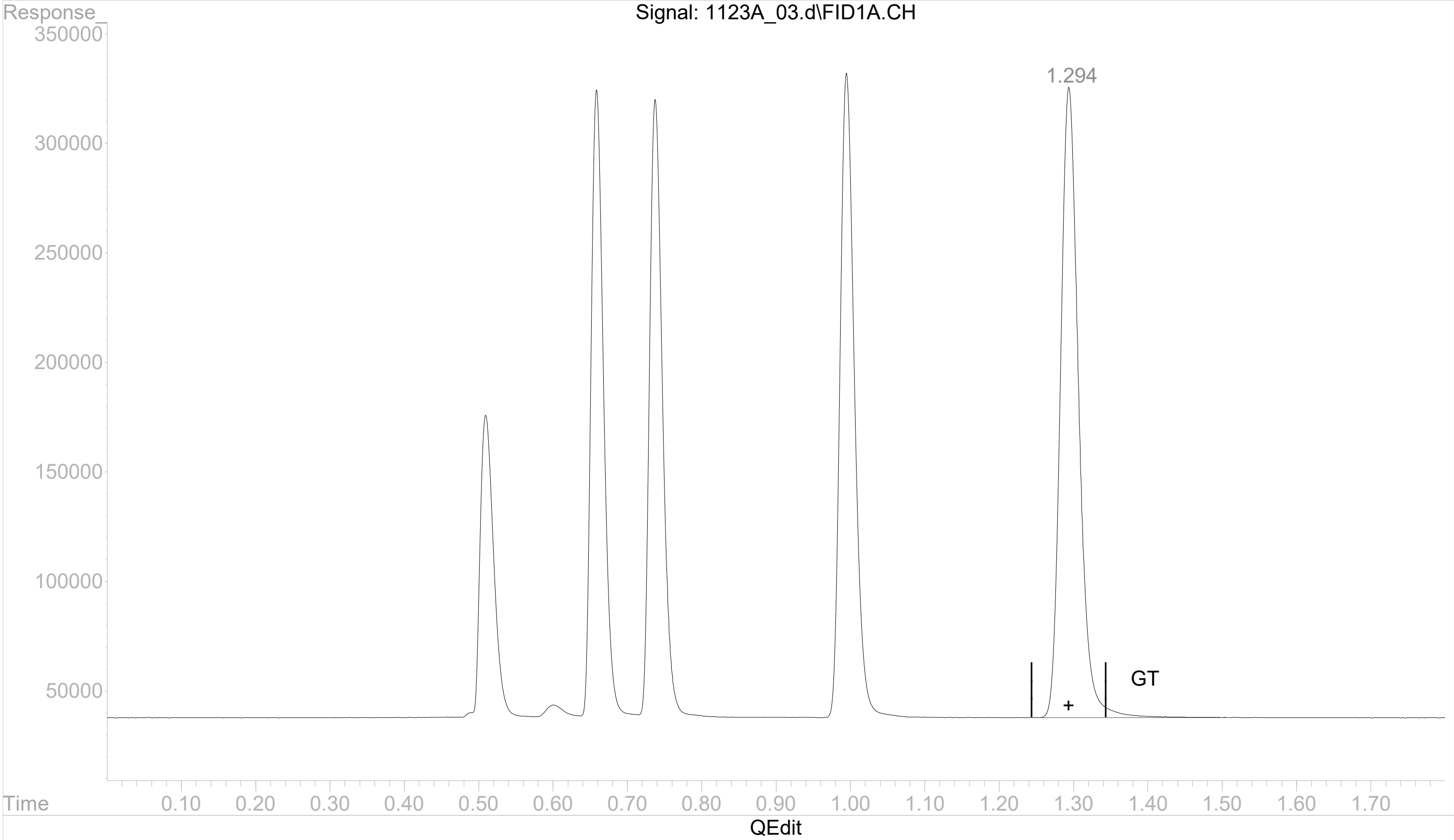
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:48:16 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(5) propane

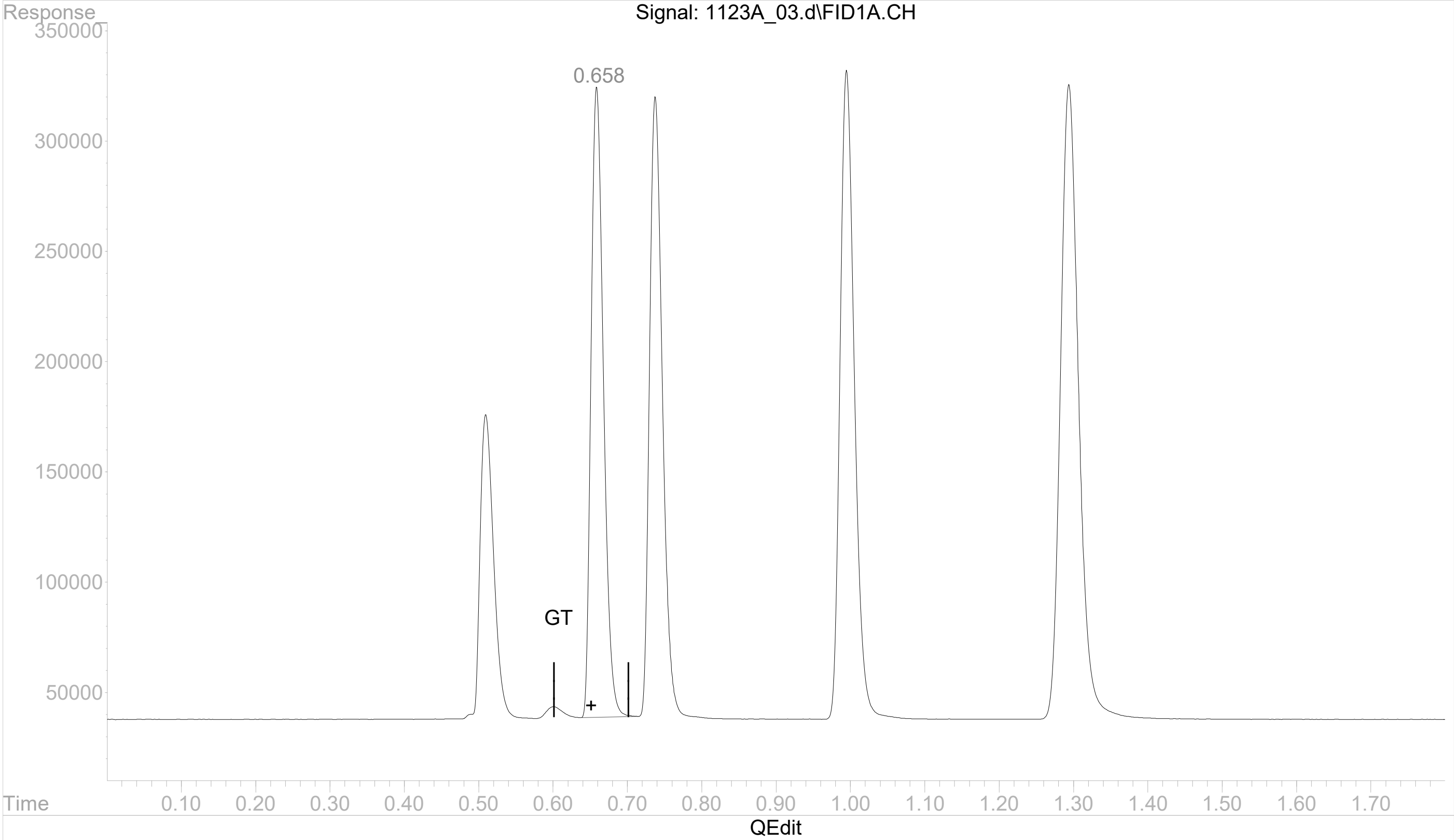
1.294min 0.1989238 mg/l m

response 4869358

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:54:17 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:37:57 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(2) ethane

0.658min 0.0000000 mg/l m

response 3322976

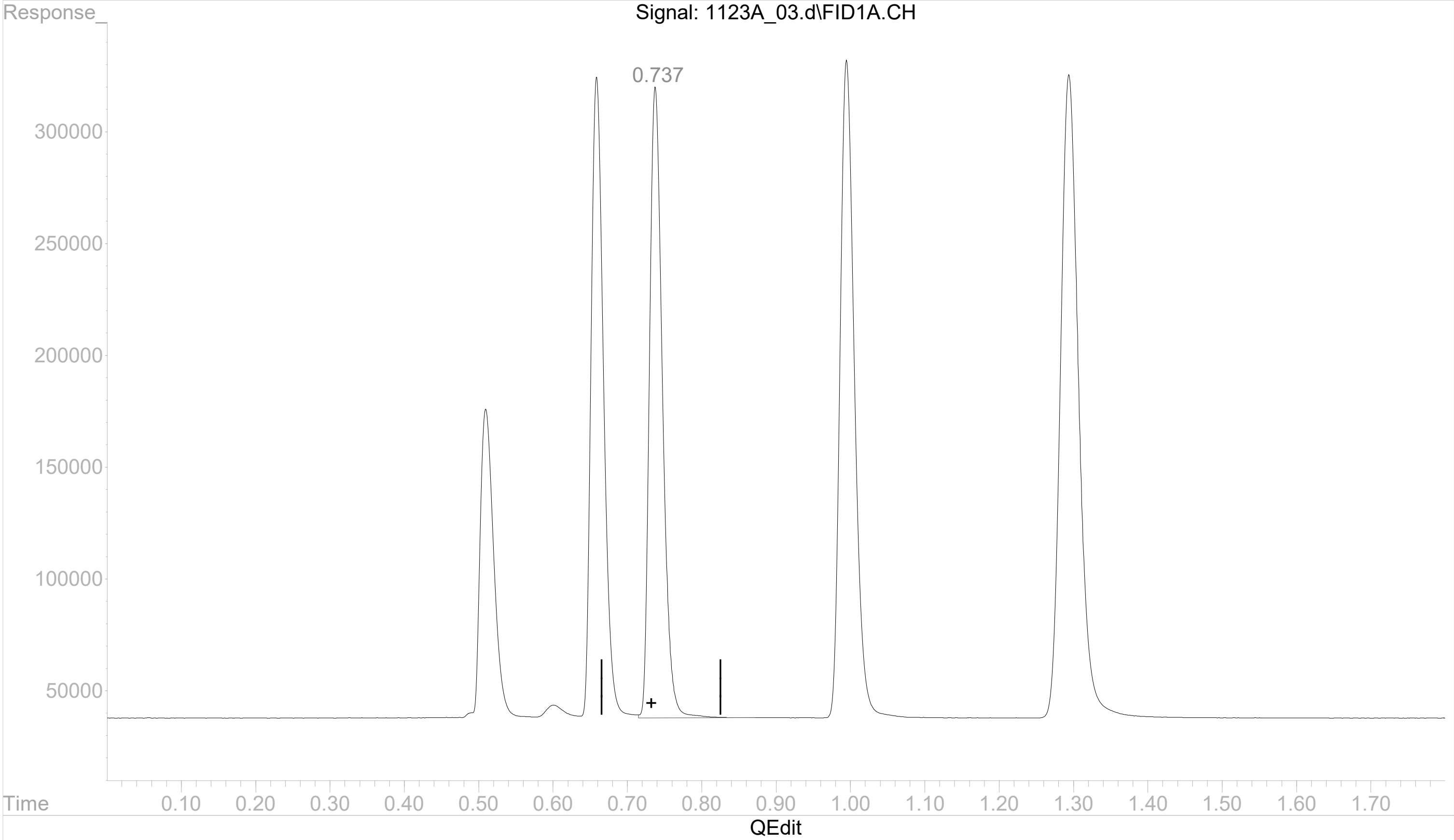
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:54:17 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:37:57 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(3) ethene

0.738min 0.0000000 mg/l

response 3426893

(+) = Expected Retention Time

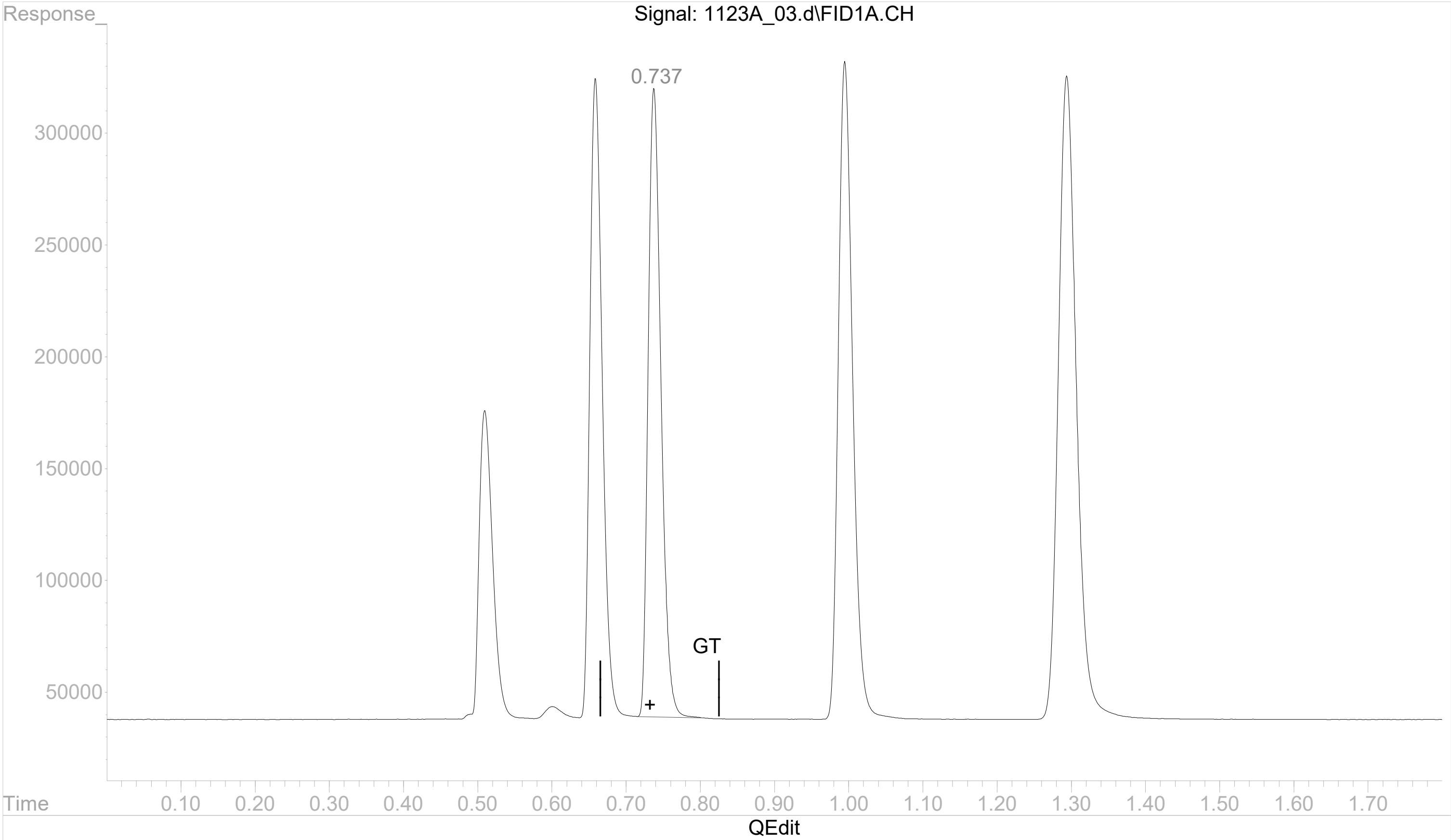
GC2RSKK23V.M Wed Nov 23 12:54:56 2022 Page: 1

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:54:17 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:37:57 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(3) ethene

0.737min 0.0000000 mg/l m

response 3371071

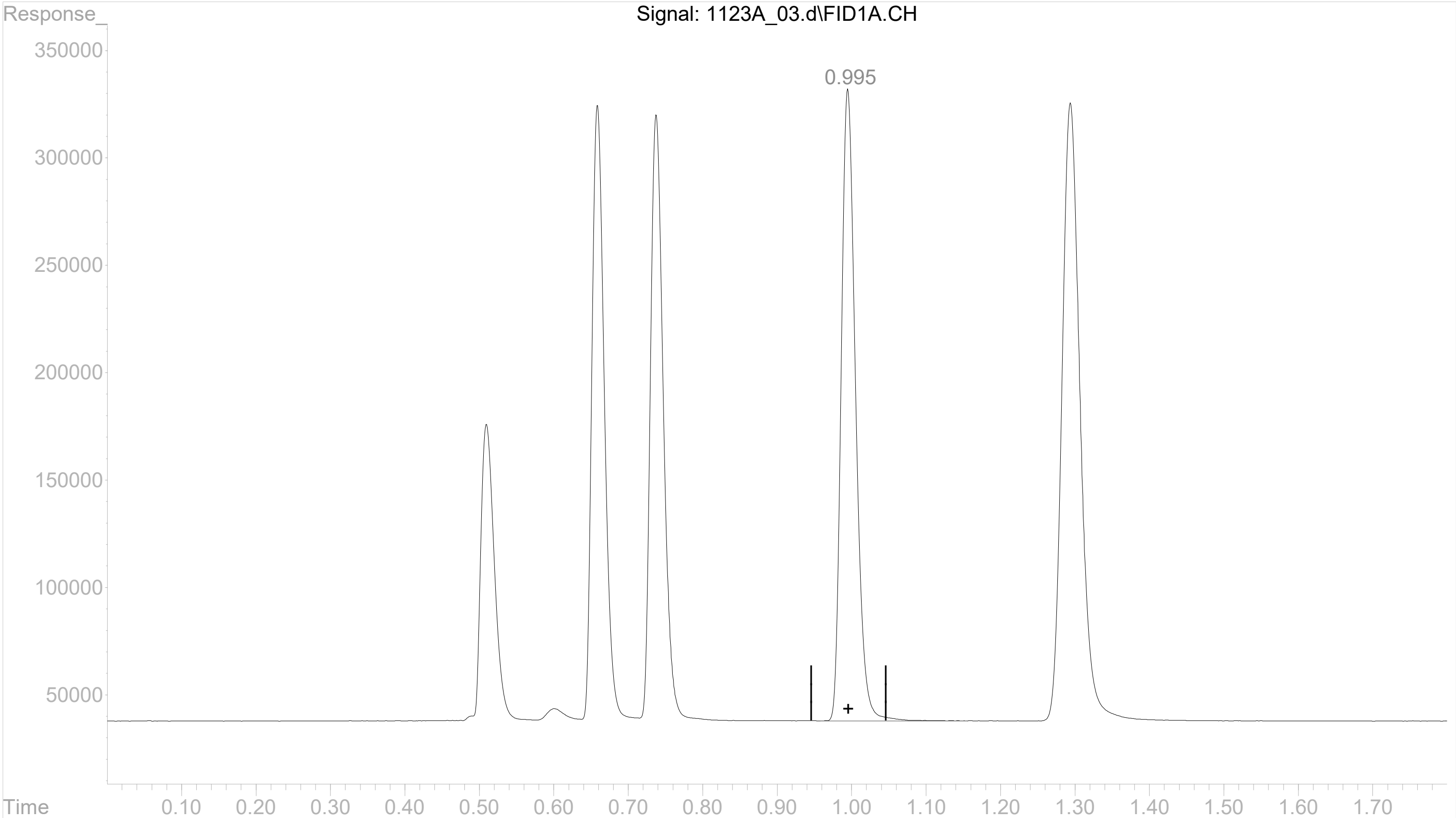
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On   : 23 Nov 2022 12:51 pm
Operator :
Sample   : STD AGC 3.0 RSK175
Misc   :
ALS Vial : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:54:17 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:37:57 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(4) acetylene

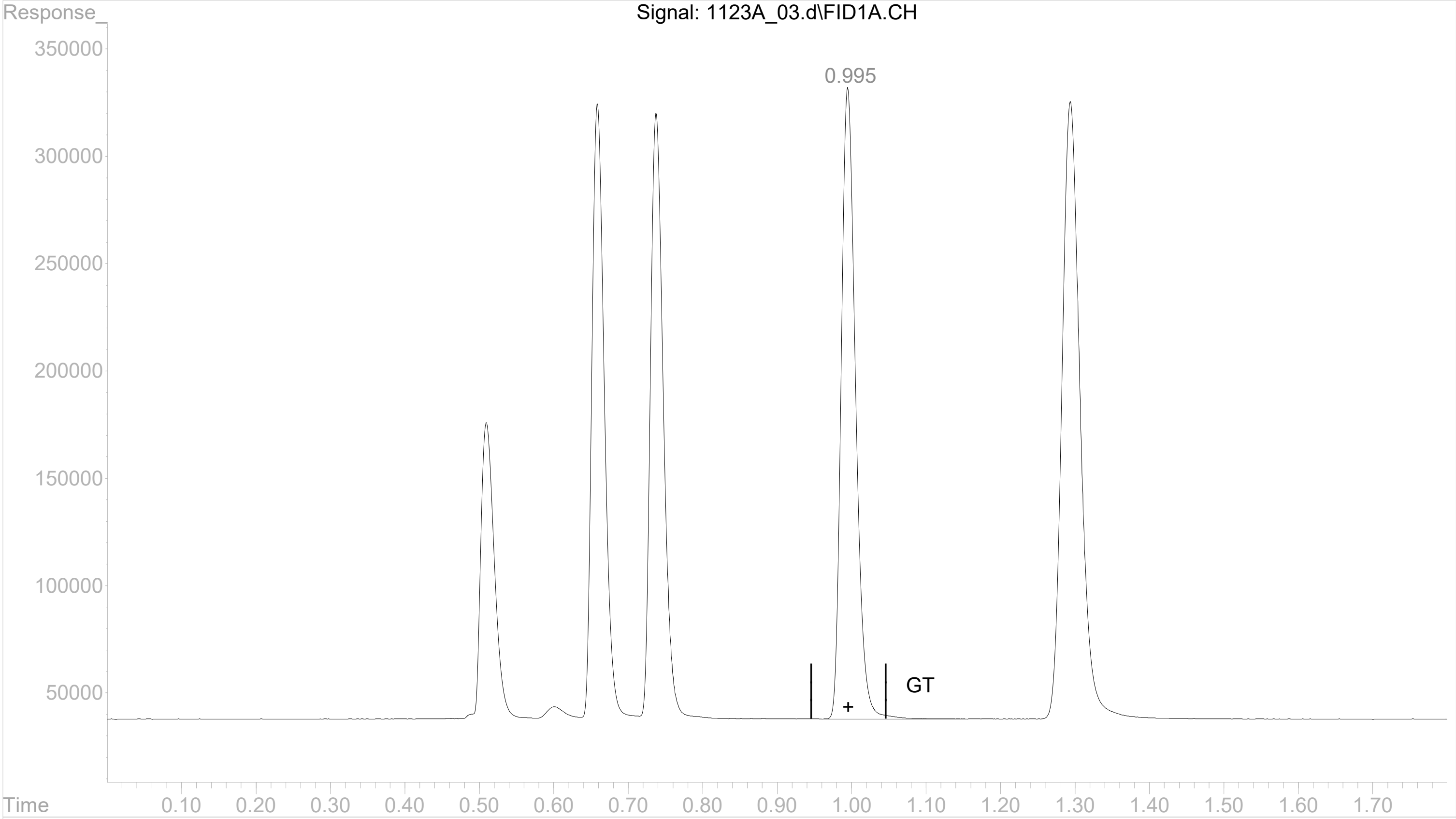
0.995min   0.0000000 mg/l

response   3961909

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  12:51 pm
Operator  :
Sample    : STD AGC 3.0 RSK175
Misc      :
ALS Vial  : 3   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:54:17 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 12:37:57 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(4) acetylene

0.995min 0.0000000 mg/l m

response 3966017

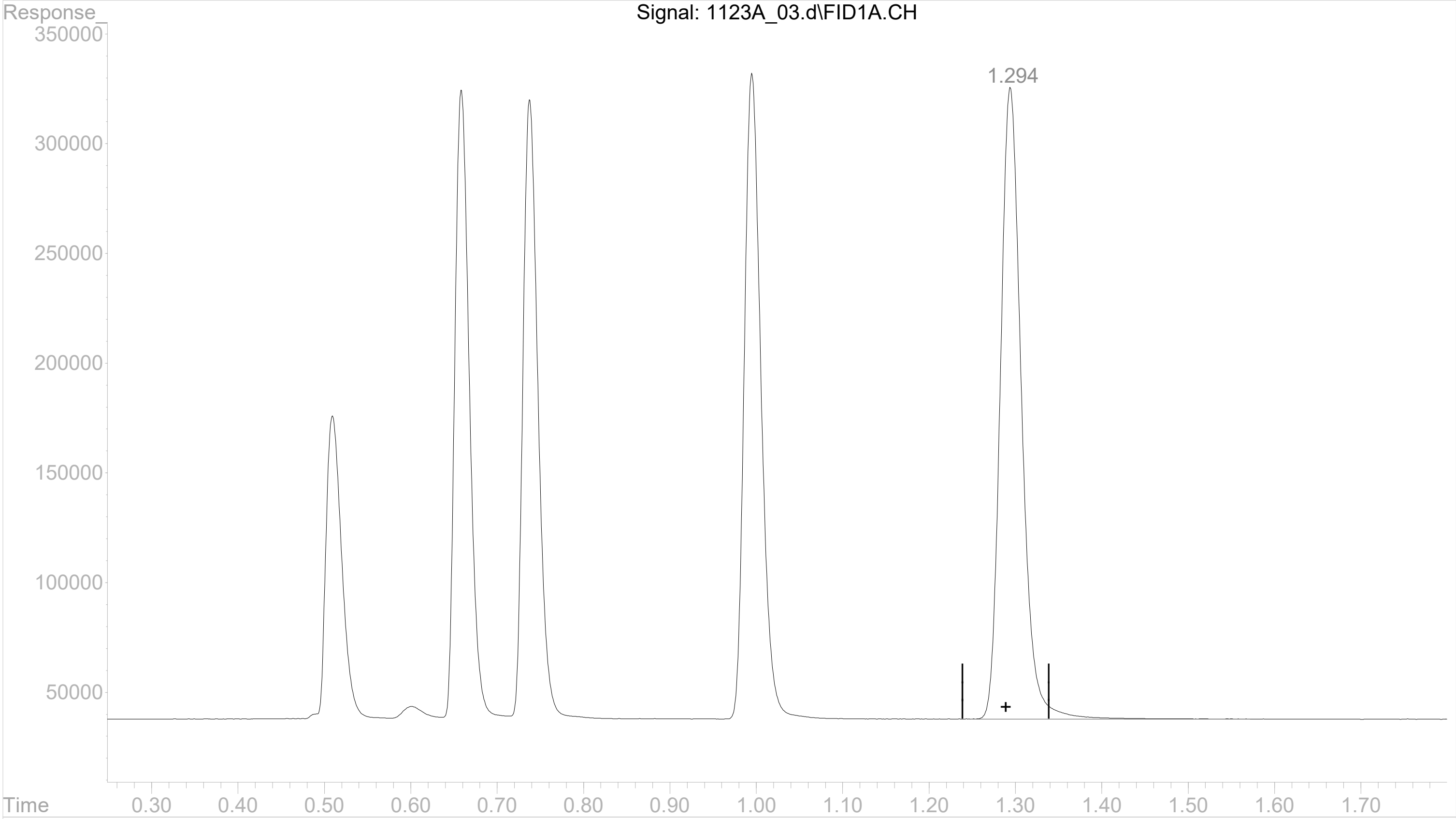
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_03.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 12:51 pm
Operator :
Sample : STD AGC 3.0 RSK175
Misc :
ALS Vial : 3 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 12:54:17 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 12:37:57 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(5) propane

1.294min 0.0000000 mg/l

response 4872647

(+) = Expected Retention Time

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_04.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:09 pm
Operator :
Sample : STD AGC 4 RSK175
Misc :
ALS Vial : 4 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:20:08 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:18:47 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	8551188	0.3475032 mg/l m
2)	ethane	0.66	17443854	0.6918713 mg/l m
3)	ethene	0.74	16838743	0.6329277 mg/l m
4)	acetylene	1.00	19467955	1.0581362 mg/l m
5)	propane	1.29	24627575	0.9751221 mg/l m

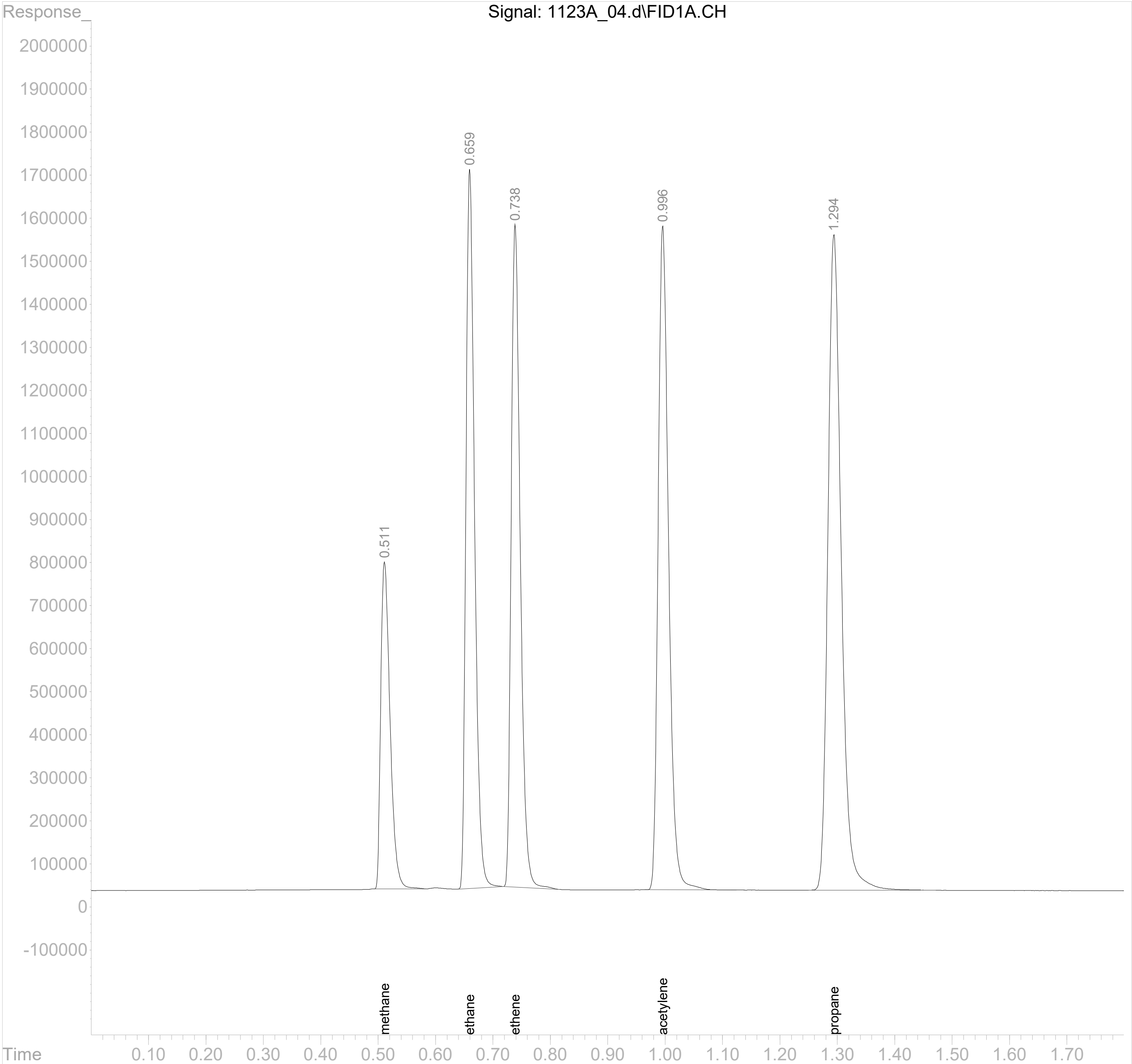
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report   (QT Reviewed)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_04.d
Signal(s) : FID1A.CH
Acq On   : 23 Nov 2022 13:09 pm
Operator :
Sample   : STD AGC 4 RSK175
Misc   :
ALS Vial : 4   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:20:08 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:18:47 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

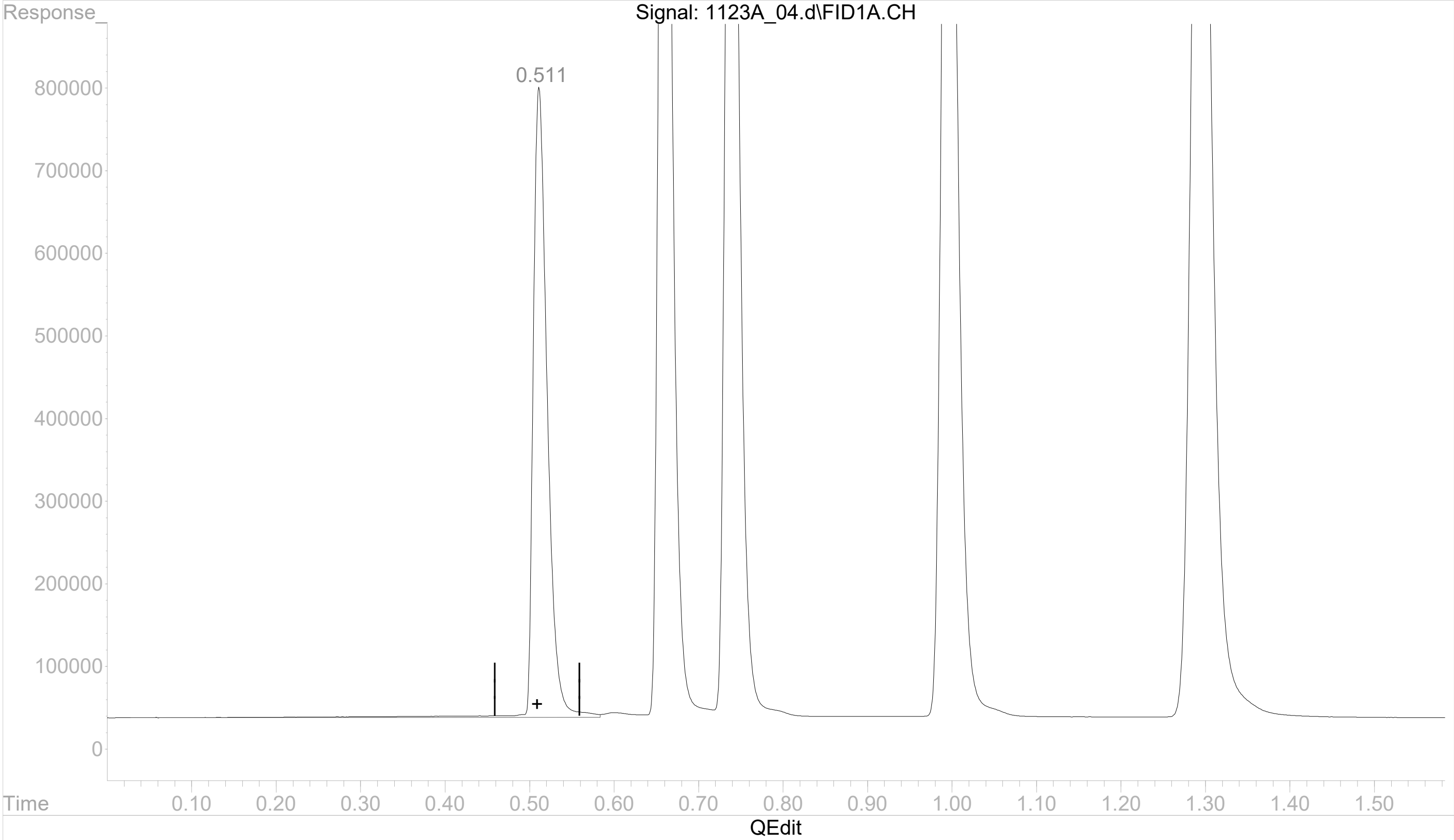


Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_04.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:09 pm
Operator :
Sample : STD AGC 4 RSK175
Misc :
ALS Vial : 4 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:19:01 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:18:47 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(1) methane

0.512min 0.3654256 mg/l

response 8992212

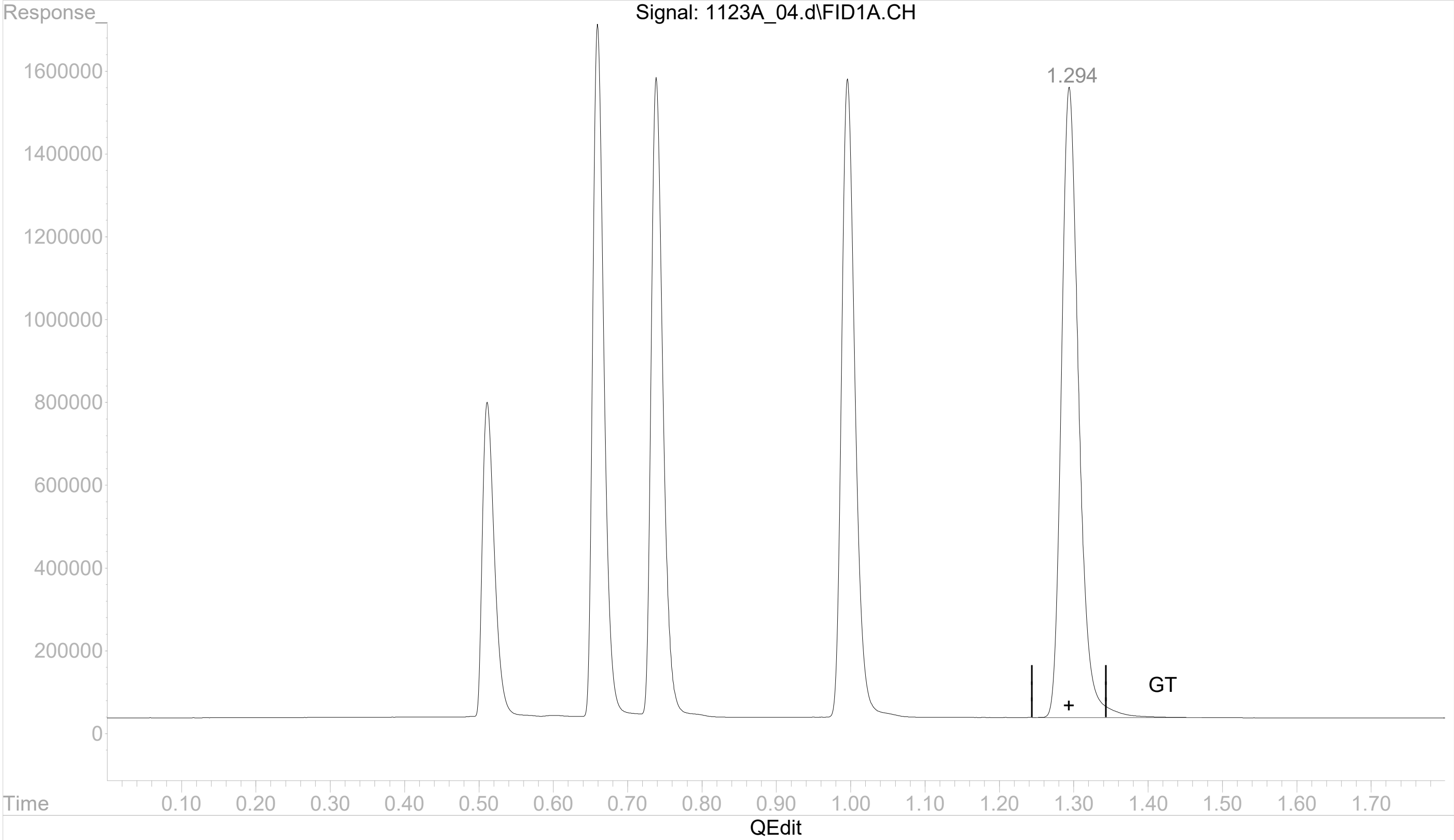
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_04.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:09 pm
Operator :
Sample : STD AGC 4 RSK175
Misc :
ALS Vial : 4 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:19:01 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:18:47 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(5) propane

1.294min 0.9751221 mg/l m

response 24627575

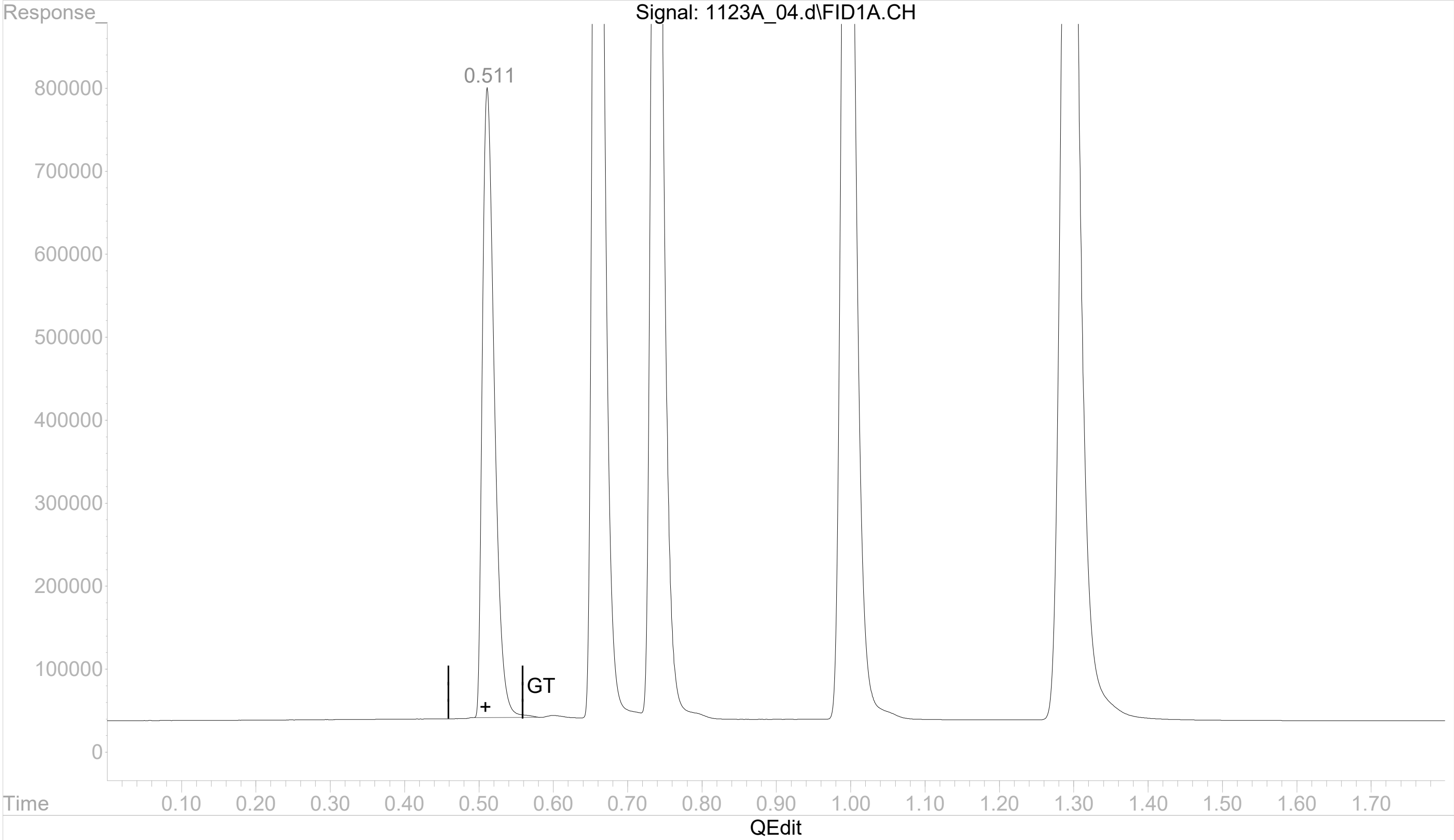
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_04.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:09 pm
Operator :
Sample : STD AGC 4 RSK175
Misc :
ALS Vial : 4 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:19:01 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:18:47 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(1) methane

0.511min 0.3475032 mg/l m

response 8551188

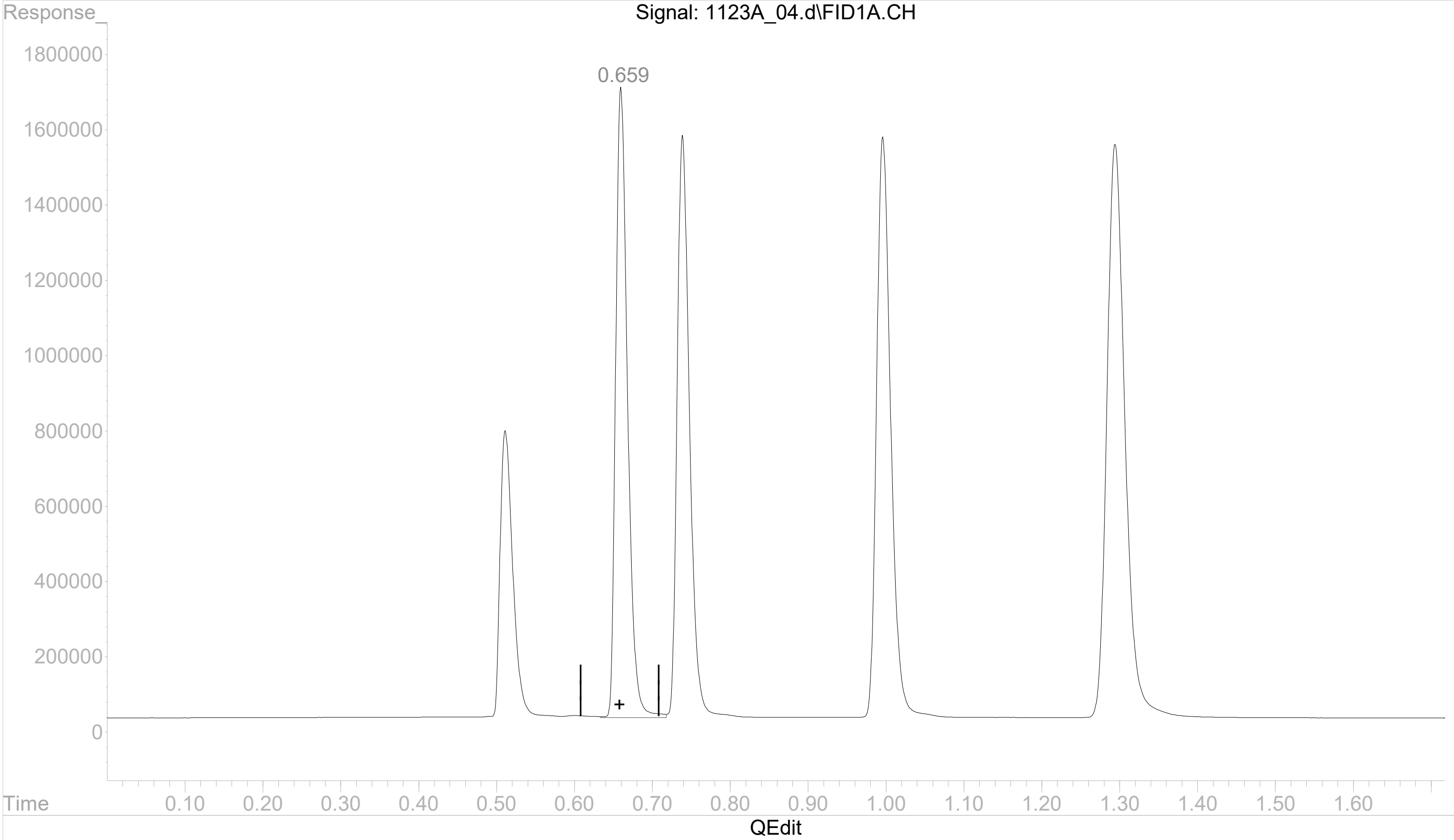
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_04.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:09 pm
Operator :
Sample : STD AGC 4 RSK175
Misc :
ALS Vial : 4 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:19:01 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:18:47 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(2) ethane

0.660min 0.7029332 mg/l

response 17722753

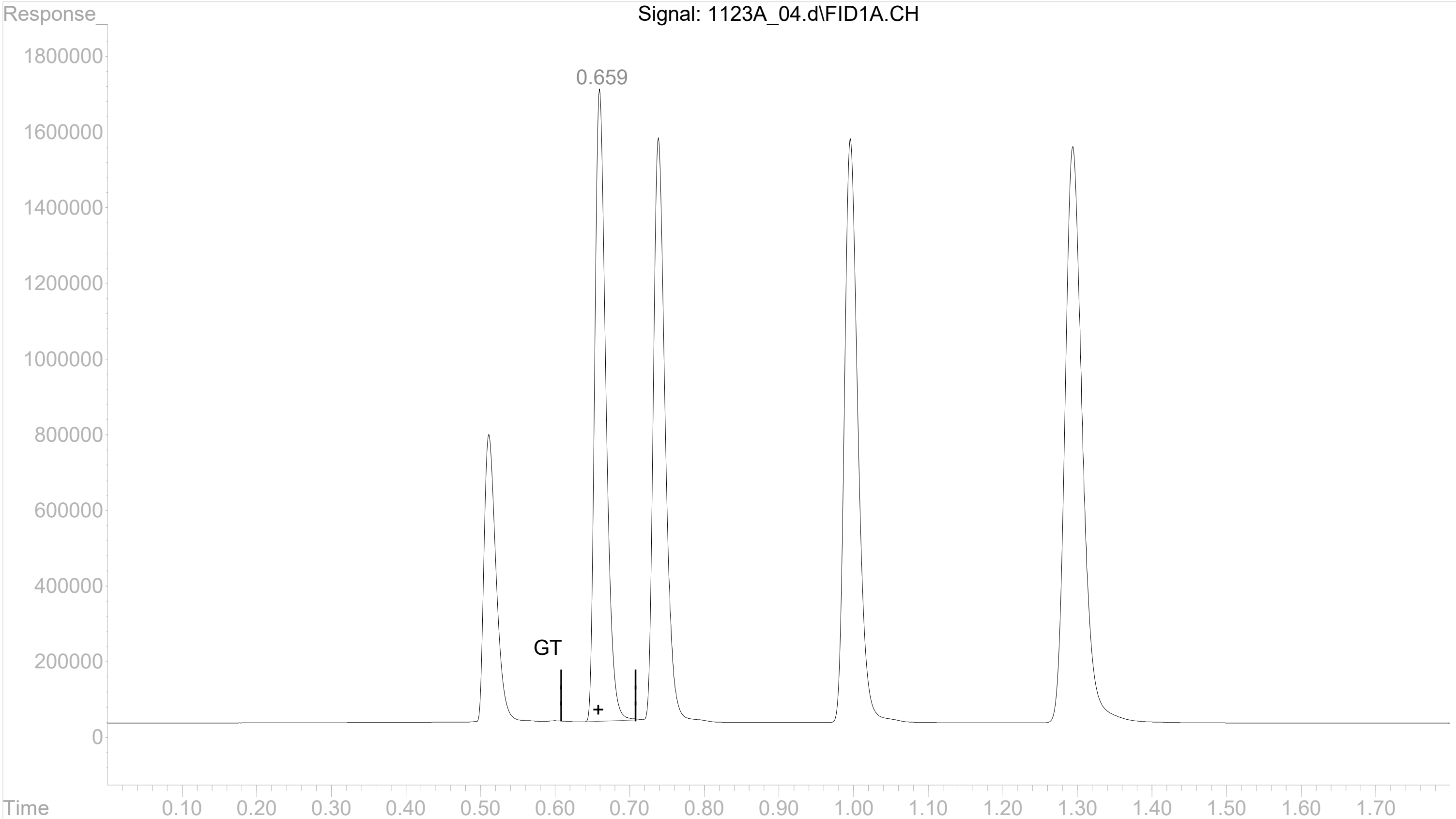
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_04.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:09 pm
Operator :
Sample : STD AGC 4 RSK175
Misc :
ALS Vial : 4 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:19:01 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:18:47 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(2) ethane

0.659min 0.6918713 mg/l m

response 17443854

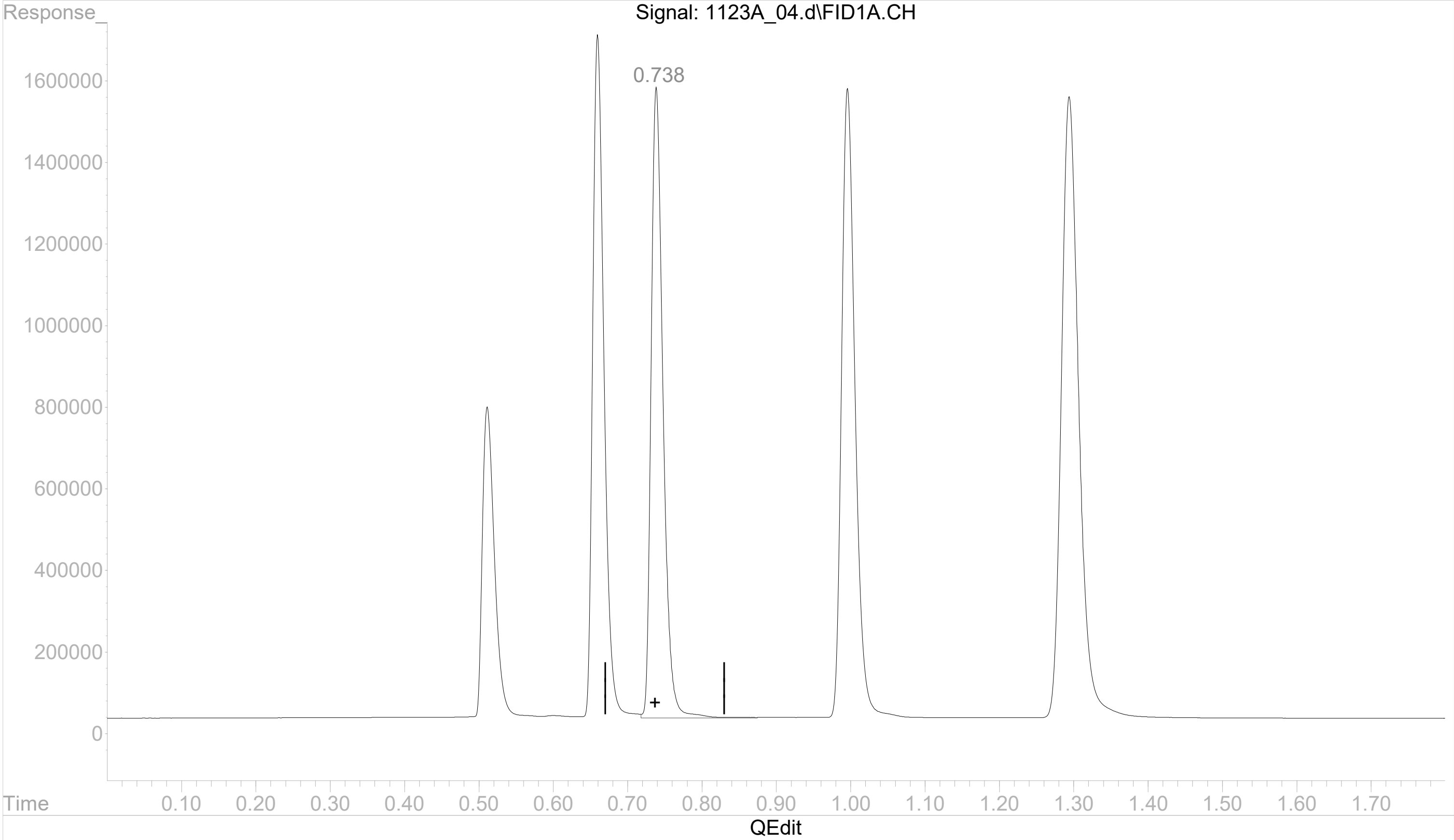
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_04.d
Signal(s) : FID1A.CH
Acq On   : 23 Nov 2022 13:09 pm
Operator :
Sample   : STD AGC 4 RSK175
Misc   :
ALS Vial : 4   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:19:01 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:18:47 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(3) ethene

0.739min   0.6462207 mg/l

response   17192398

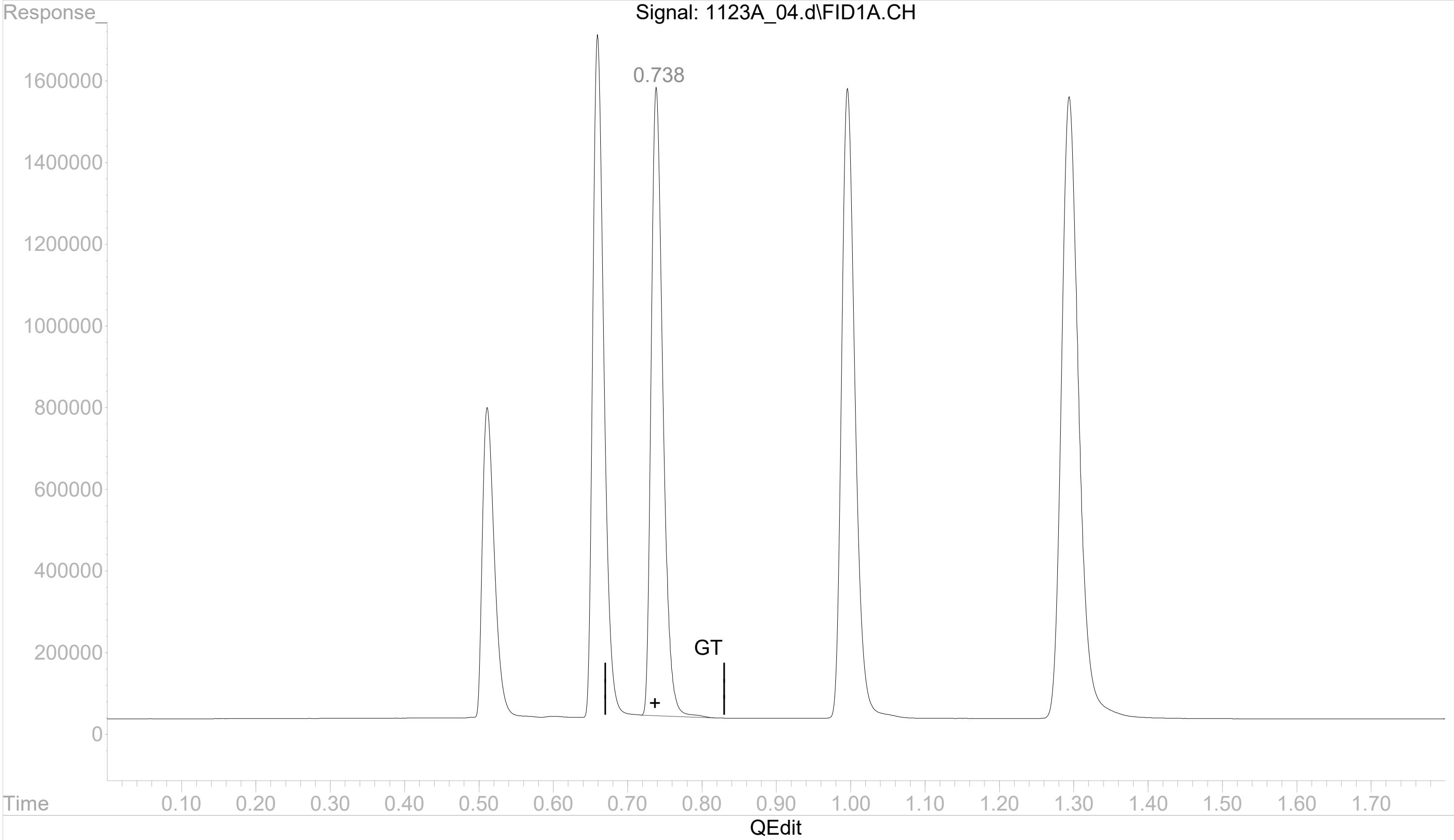
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_04.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:09 pm
Operator :
Sample : STD AGC 4 RSK175
Misc :
ALS Vial : 4 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:19:01 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:18:47 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(3) ethene

0.738min 0.6329277 mg/l m

response 16838743

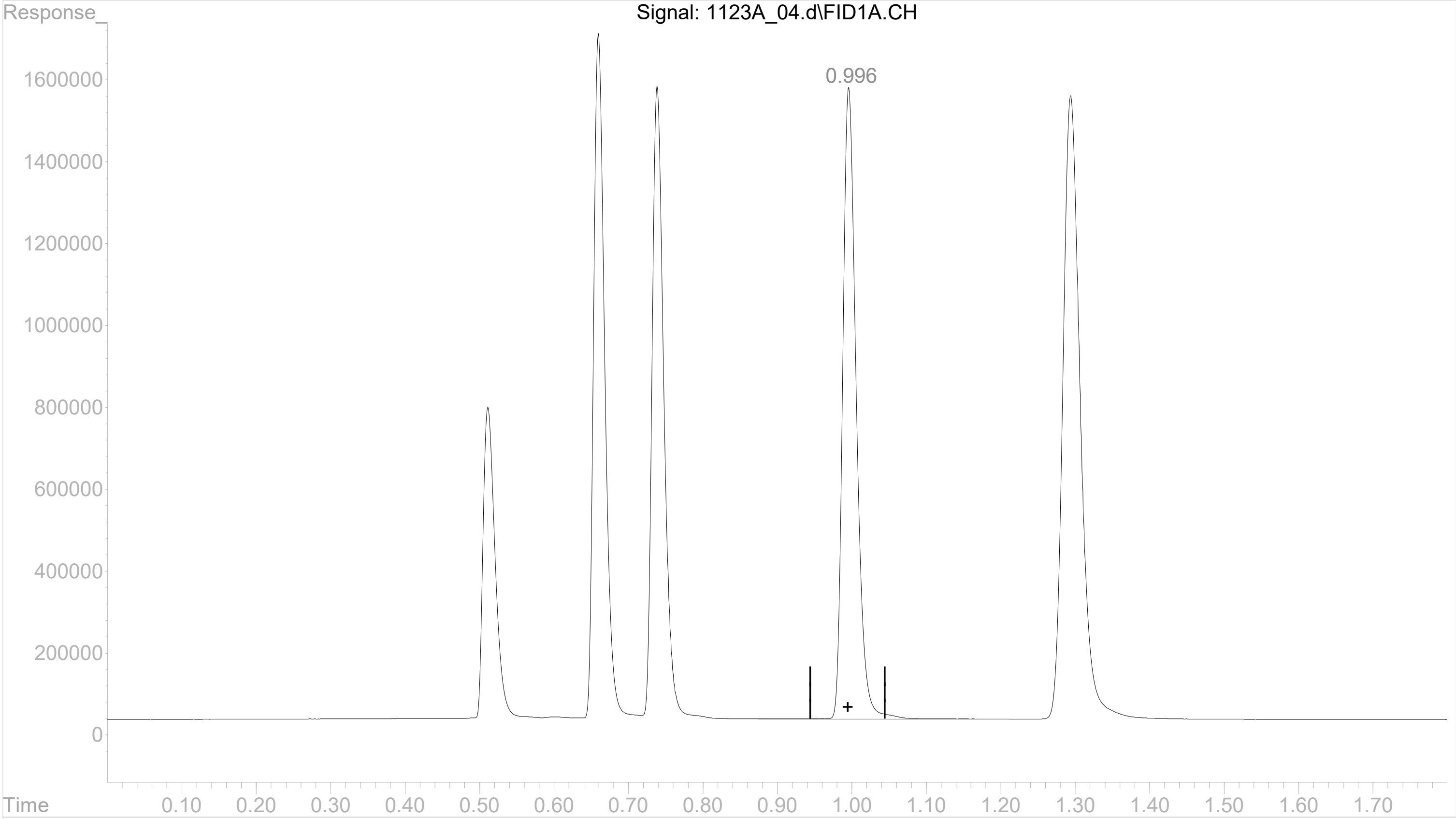
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_04.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:09 pm
Operator :
Sample : STD AGC 4 RSK175
Misc :
ALS Vial : 4 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:19:01 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:18:47 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(4) acetylene

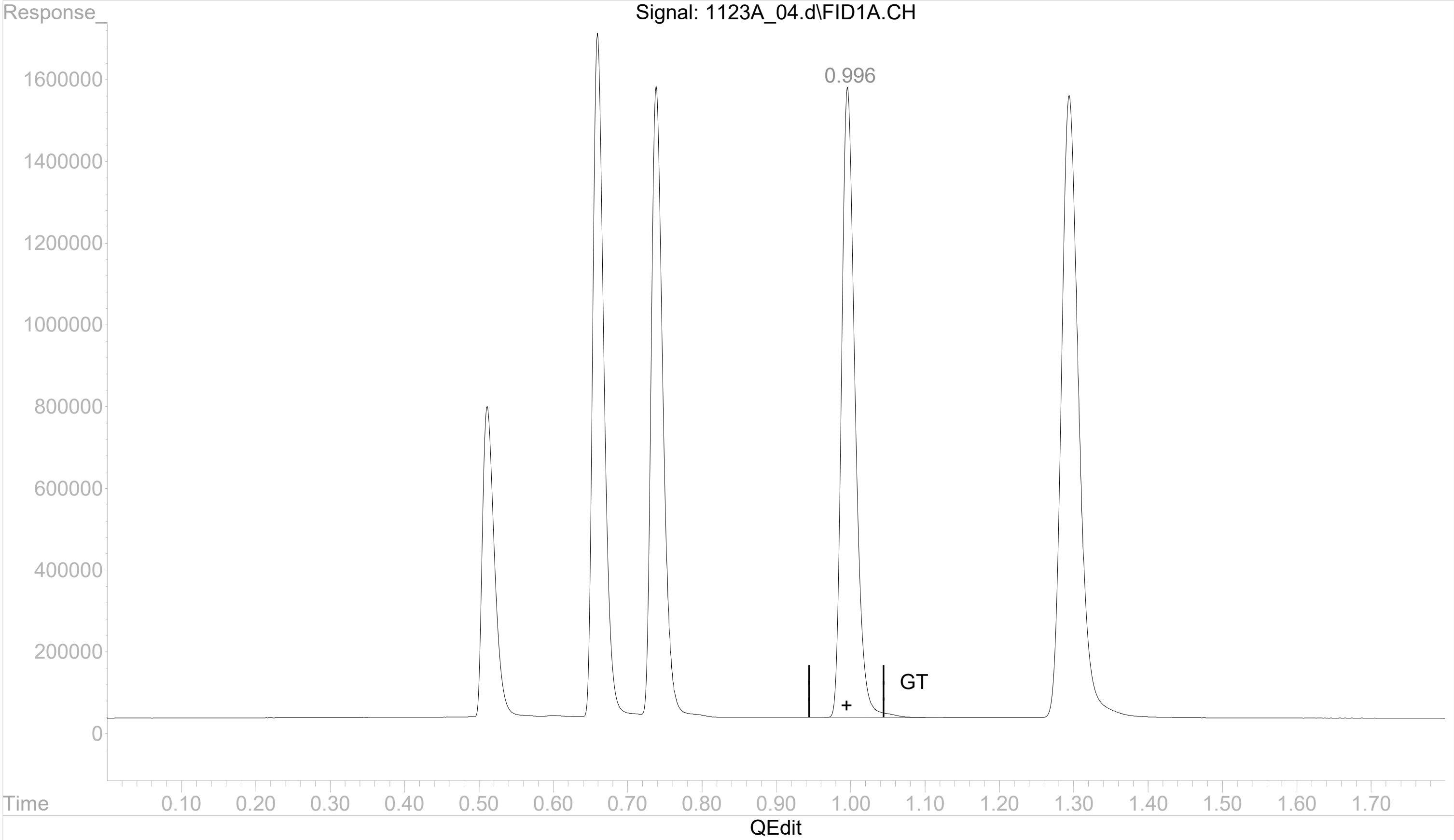
0.996min 1.0647487 mg/l

response 19589614

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_04.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:09 pm
Operator  :
Sample    : STD AGC 4 RSK175
Misc      :
ALS Vial  : 4   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:19:01 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:18:47 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(4) acetylene

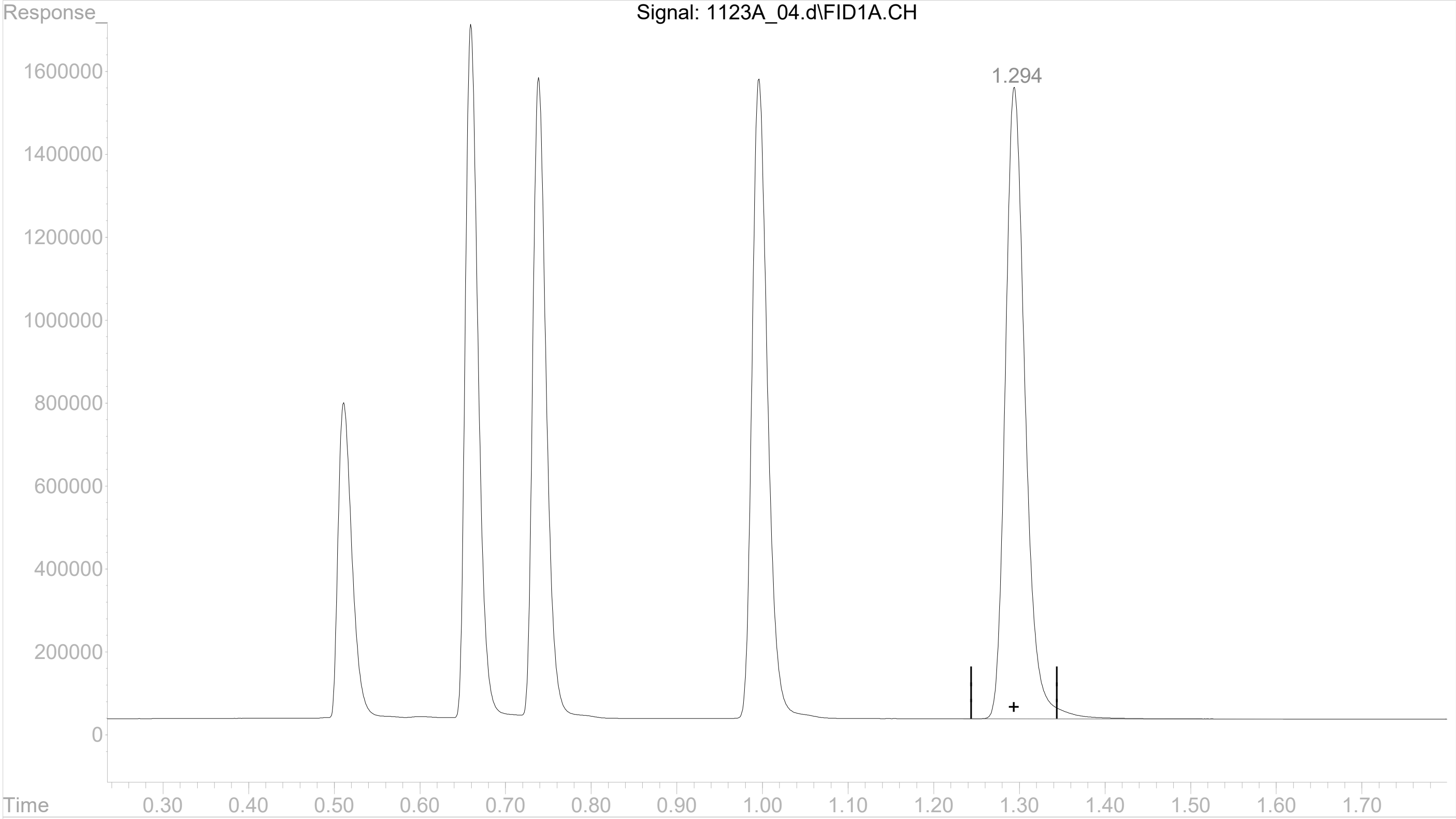
0.996min 1.0581362 mg/l m

response 19467955

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_04.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:09 pm
Operator  :
Sample    : STD AGC 4 RSK175
Misc      :
ALS Vial  : 4   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:19:01 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:18:47 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(5) propane

1.295min 0.9767073 mg/l

response 24667612

(+) = Expected Retention Time

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_05.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:20 pm
Operator  :
Sample    : STD AGC 5 RSK175
Misc      :
ALS Vial  : 5   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:24:23 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:20:19 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	15680134	0.6332384 mg/l m
2)	ethane	0.66	30998195	1.2075363 mg/l m
3)	ethene	0.74	29969968	1.1274185 mg/l m
4)	acetylene	1.00	34992885	1.8937024 mg/l m
5)	propane	1.29	43907564	1.7176733 mg/l m

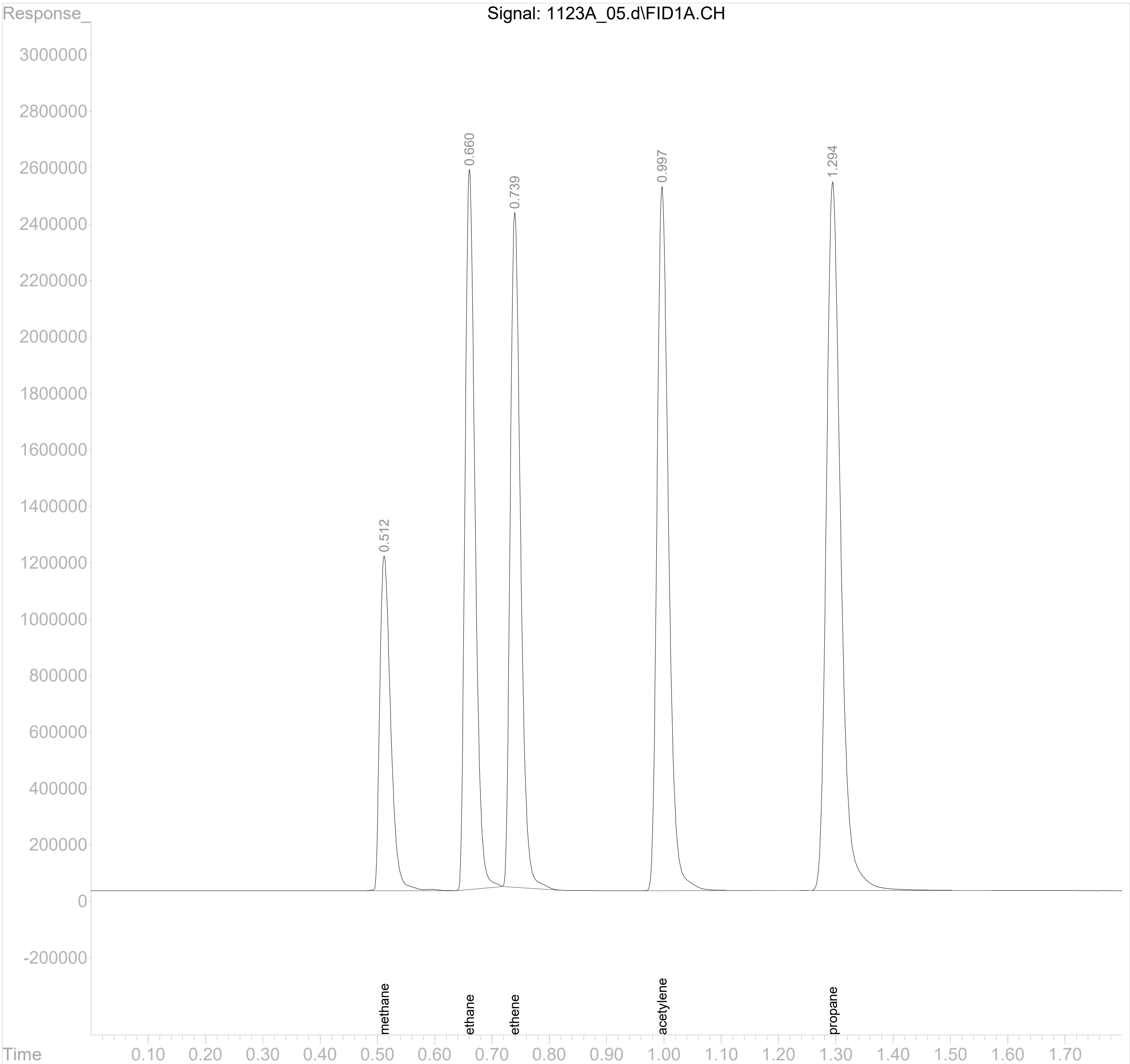
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_05.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:20 pm
Operator :
Sample : STD AGC 5 RSK175
Misc :
ALS Vial : 5 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:24:23 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:20:19 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

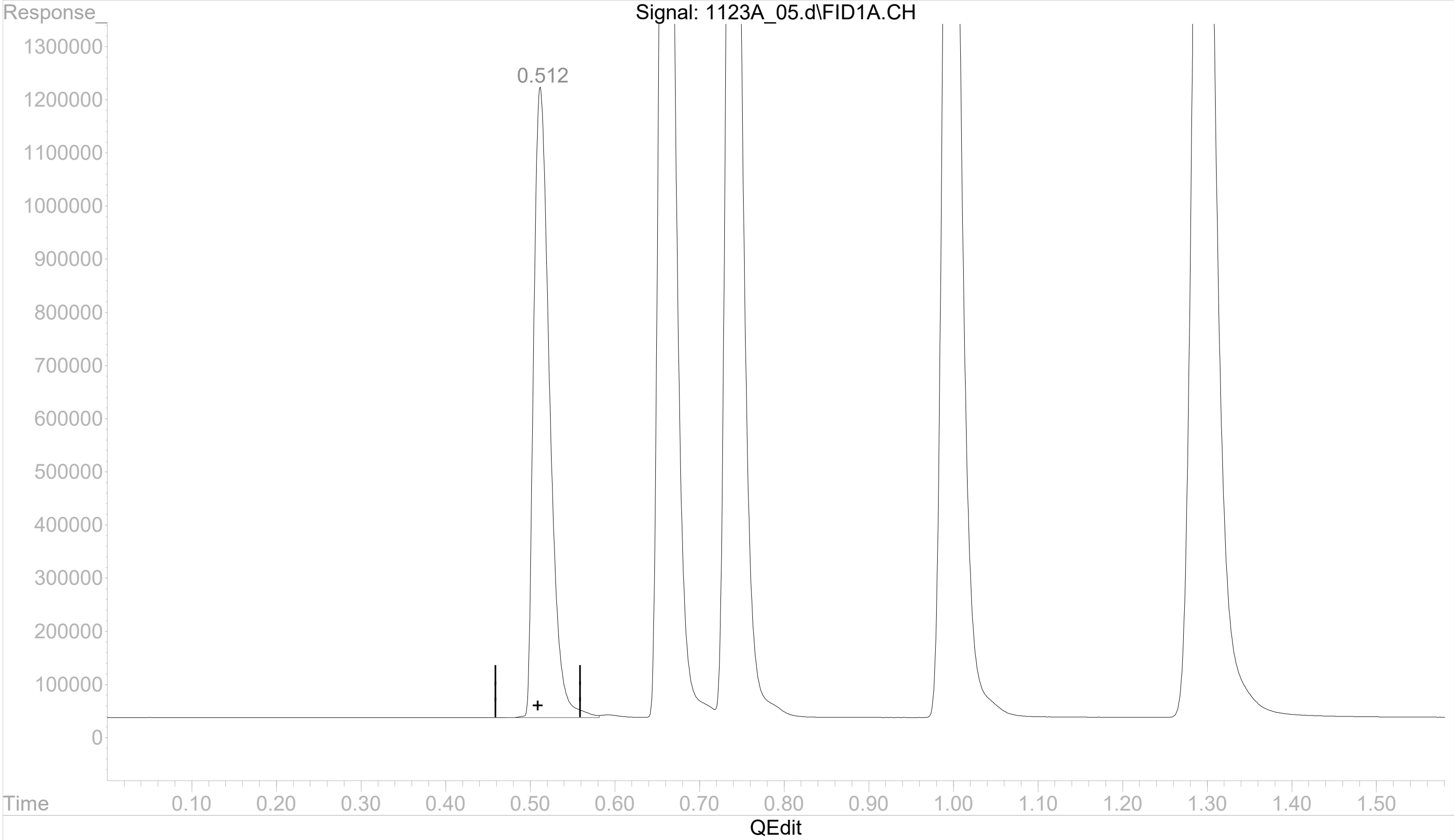


Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_05.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:20 pm
Operator :
Sample : STD AGC 5 RSK175
Misc :
ALS Vial : 5 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:23:04 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:20:19 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(1) methane

0.512min 0.6305179 mg/l

response 15612769

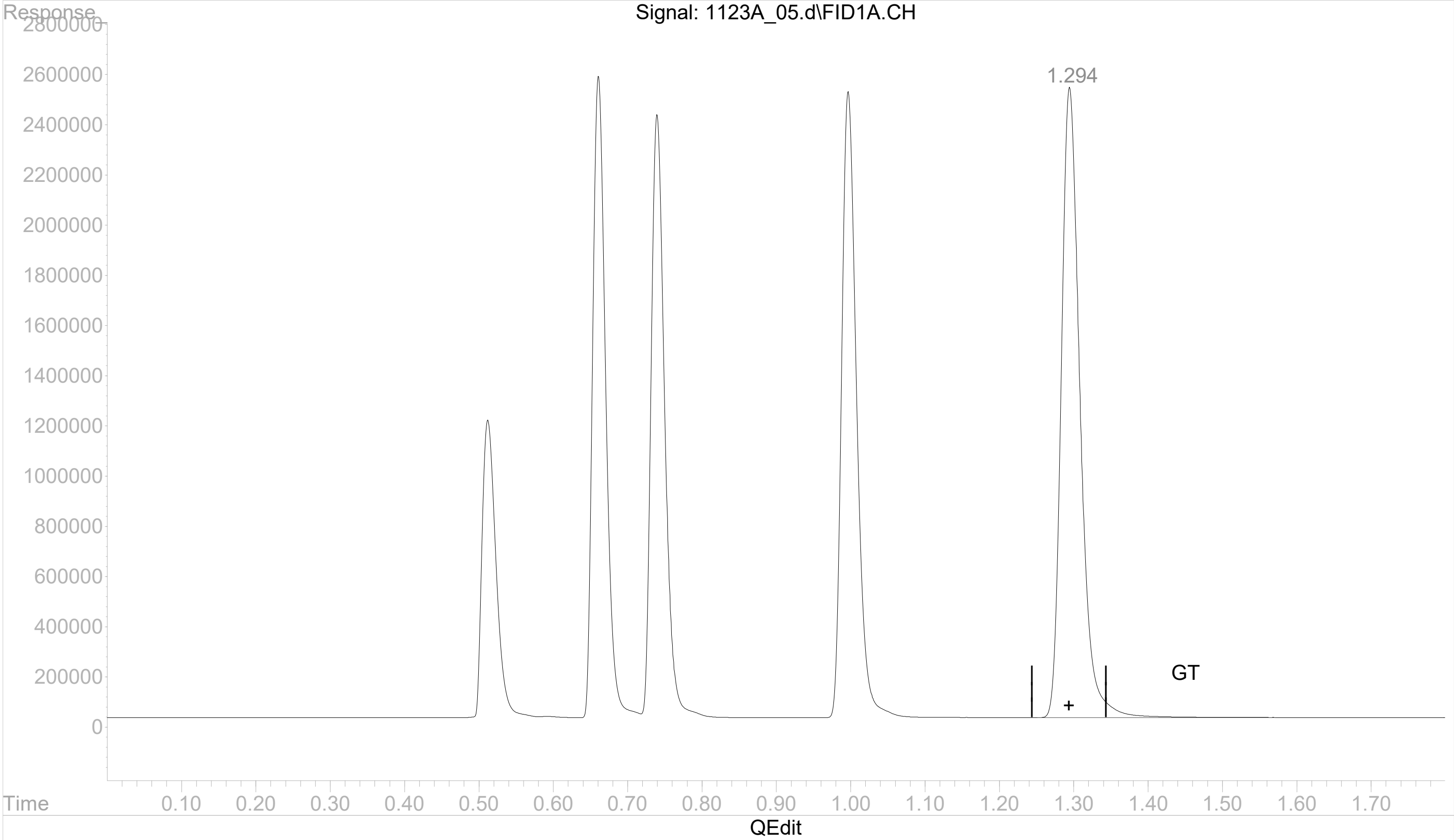
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_05.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:20 pm
Operator :
Sample : STD AGC 5 RSK175
Misc :
ALS Vial : 5 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:23:04 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:20:19 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(5) propane

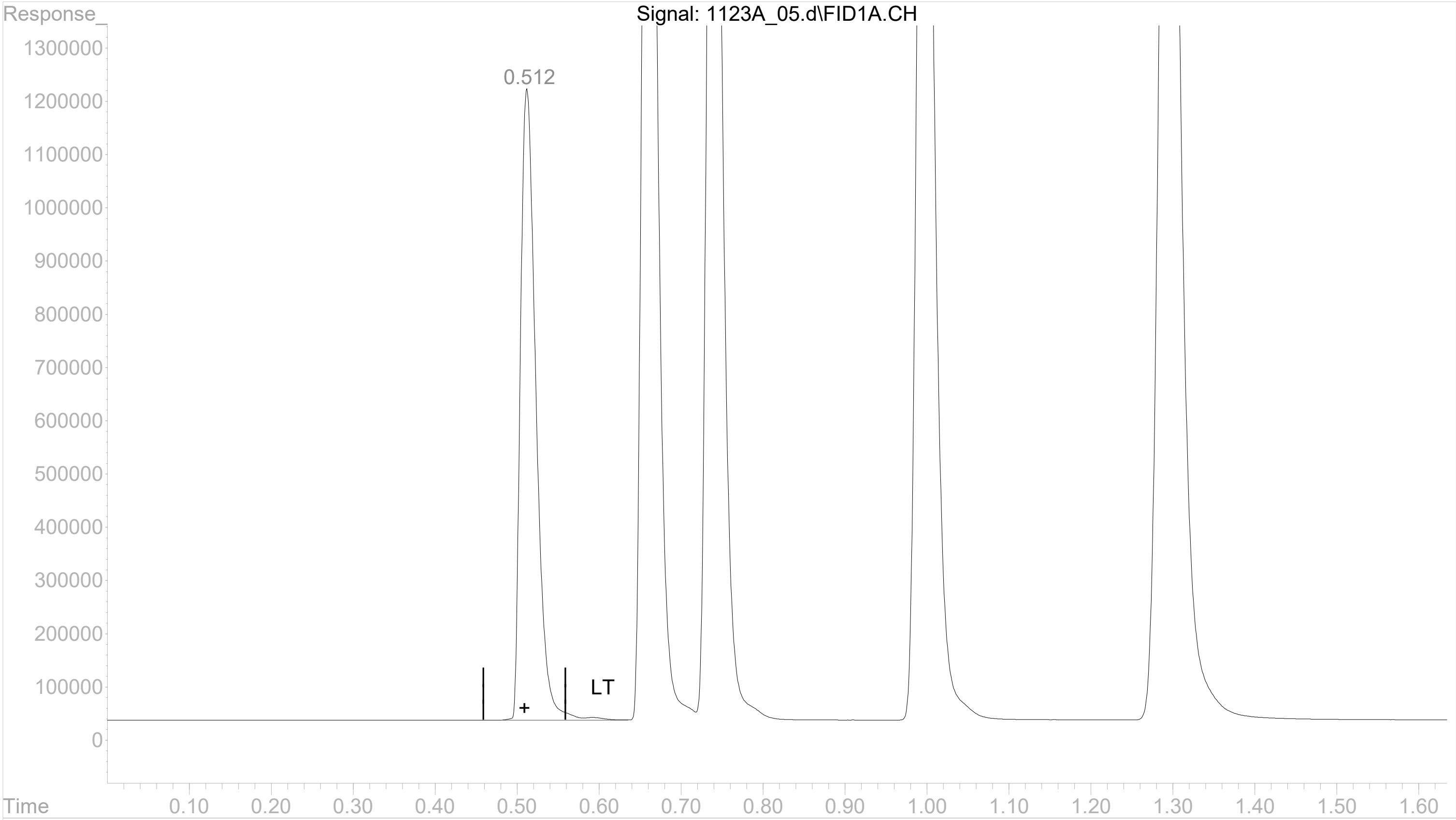
1.294min 1.7176733 mg/l m

response 43907564

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_05.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:20 pm
Operator  :
Sample    : STD AGC 5 RSK175
Misc      :
ALS Vial  : 5   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:23:04 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:20:19 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(1) methane

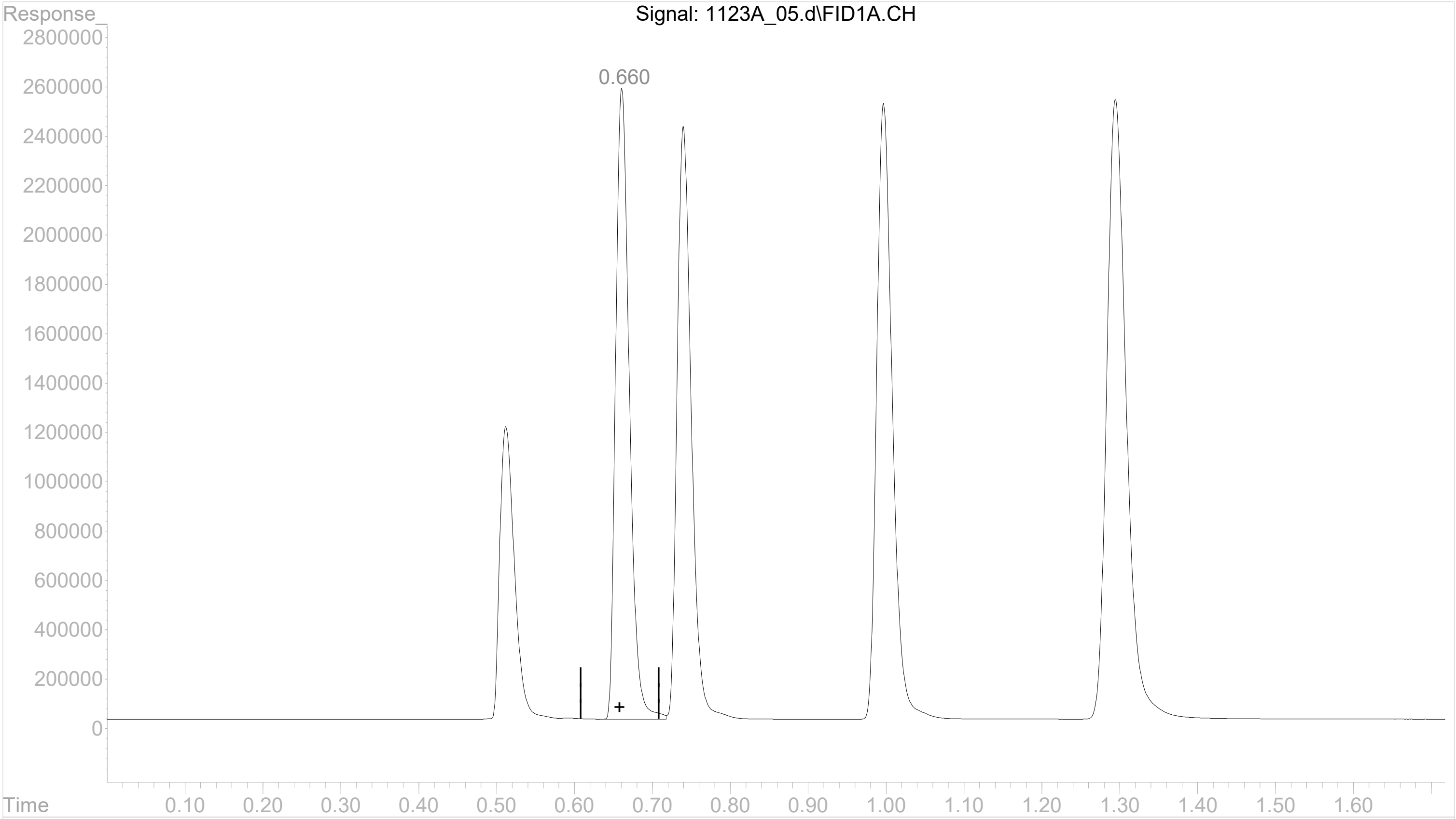
0.512min   0.6332384 mg/l m

response   15680134

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_05.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:20 pm
Operator  :
Sample    : STD AGC 5 RSK175
Misc      :
ALS Vial  : 5   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:23:04 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:20:19 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(2) ethane

0.661min 1.2213070 mg/l

response 31351698

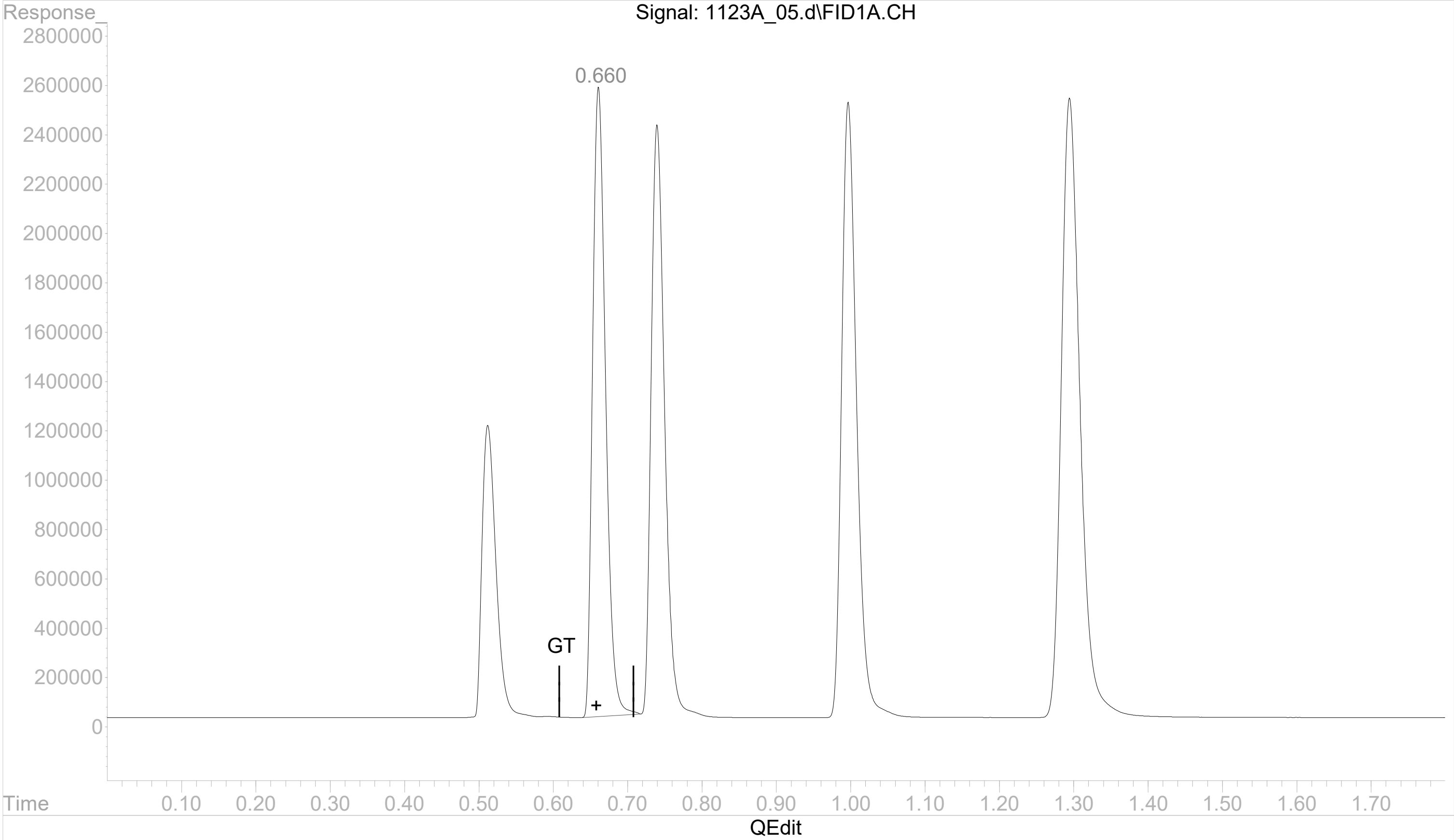
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_05.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:20 pm
Operator :
Sample : STD AGC 5 RSK175
Misc :
ALS Vial : 5 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:23:04 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:20:19 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(2) ethane

0.660min 1.2075363 mg/l m

response 30998195

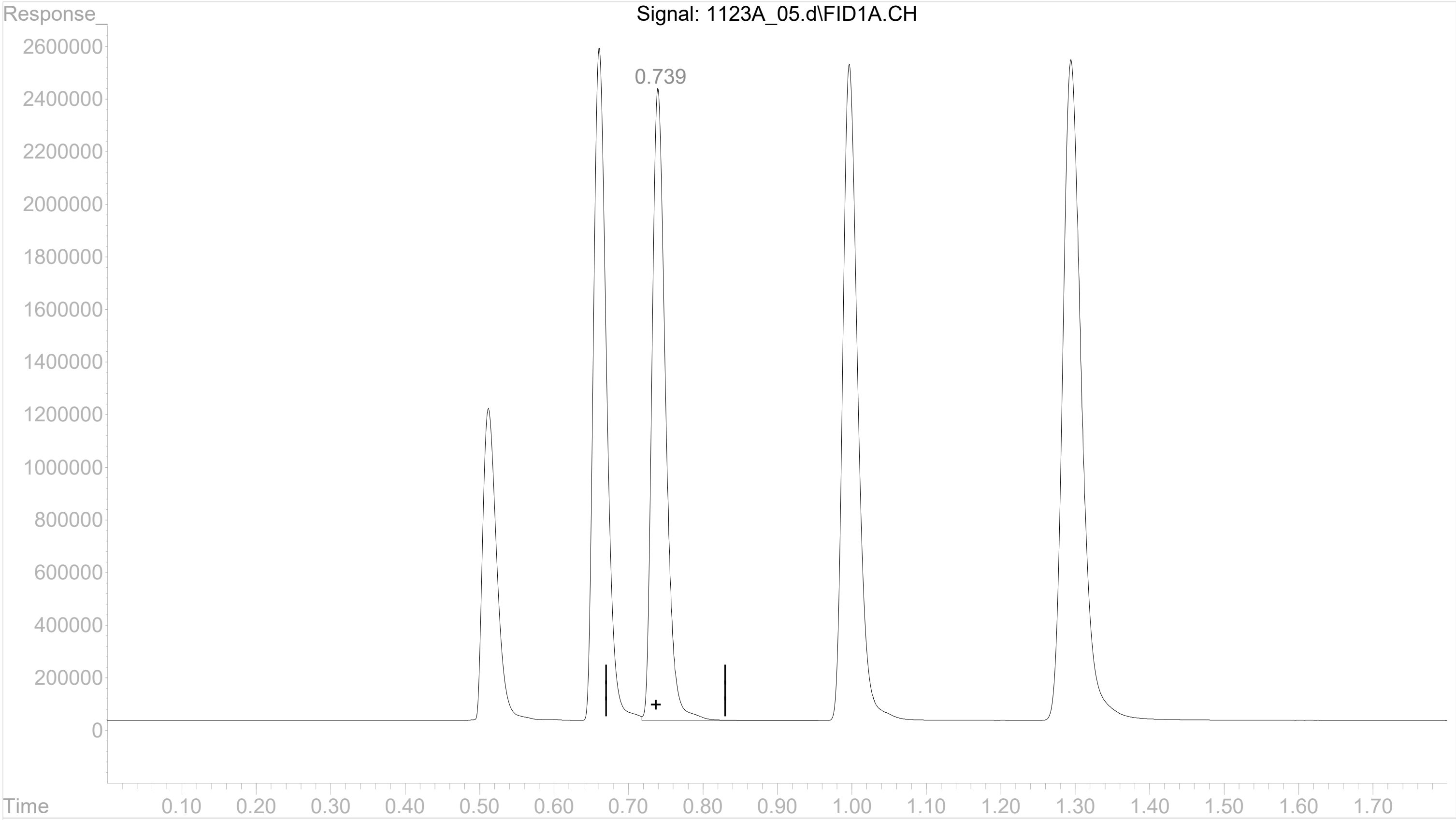
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_05.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:20 pm
Operator :
Sample : STD AGC 5 RSK175
Misc :
ALS Vial : 5 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:23:04 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:20:19 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(3) ethene

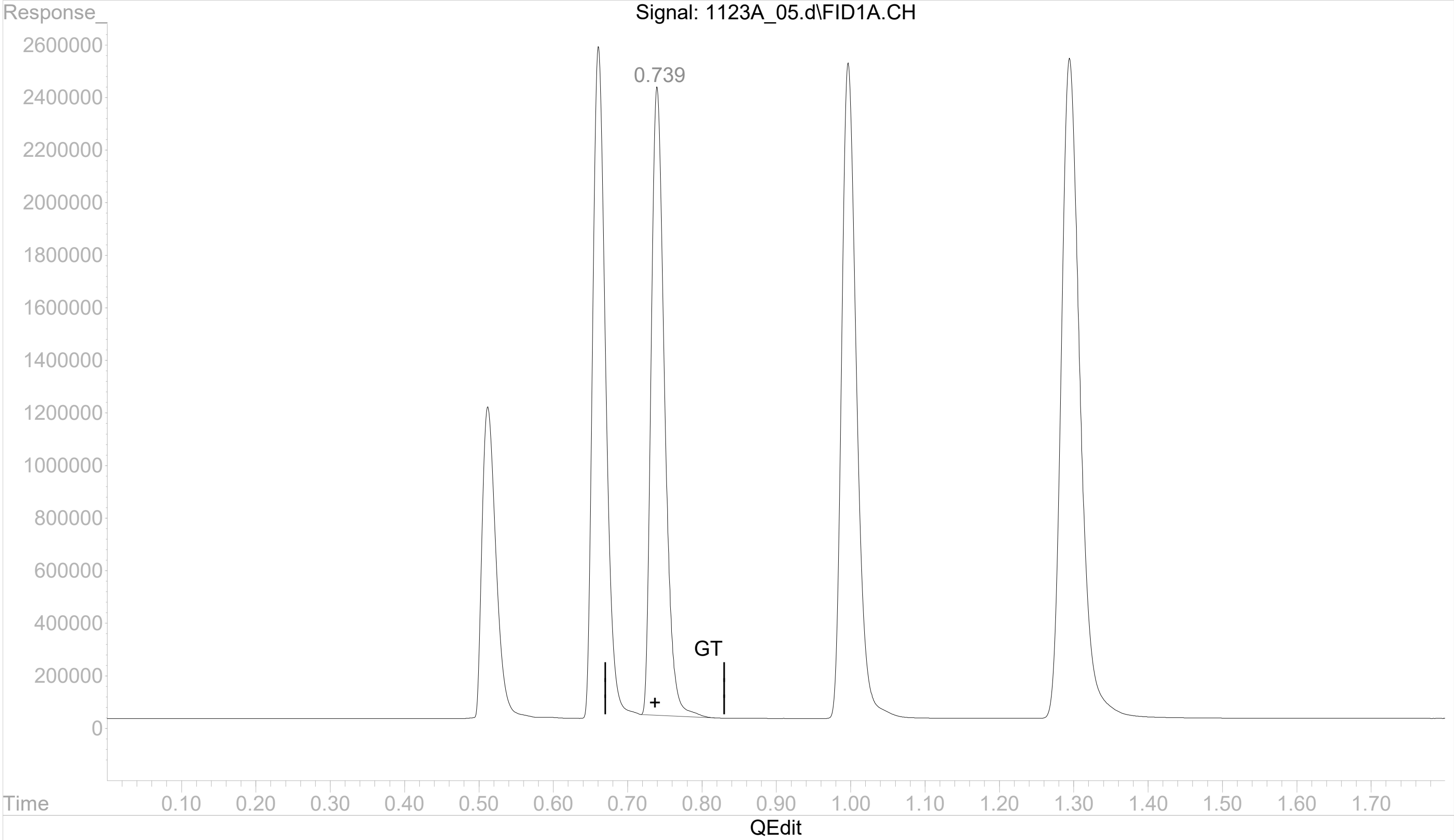
0.740min 1.1480322 mg/l

response 30517937

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_05.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:20 pm
Operator  :
Sample    : STD AGC 5 RSK175
Misc      :
ALS Vial  : 5   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:23:04 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:20:19 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(3) ethene

0.739min 1.1274185 mg/l m

response 29969968

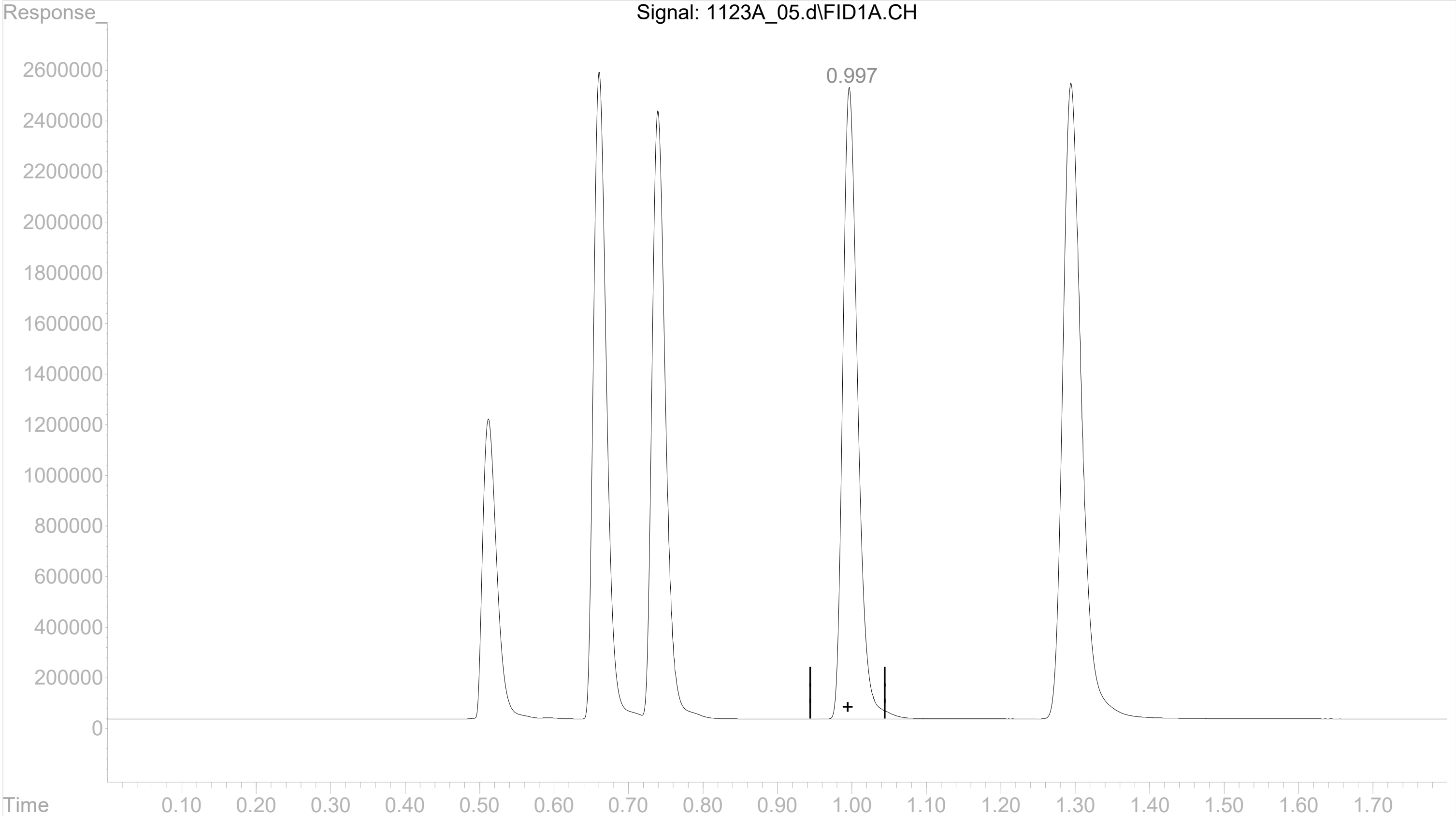
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_05.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:20 pm
Operator :
Sample : STD AGC 5 RSK175
Misc :
ALS Vial : 5 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:23:04 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:20:19 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(4) acetylene

0.997min 1.8949694 mg/l

response 35016298

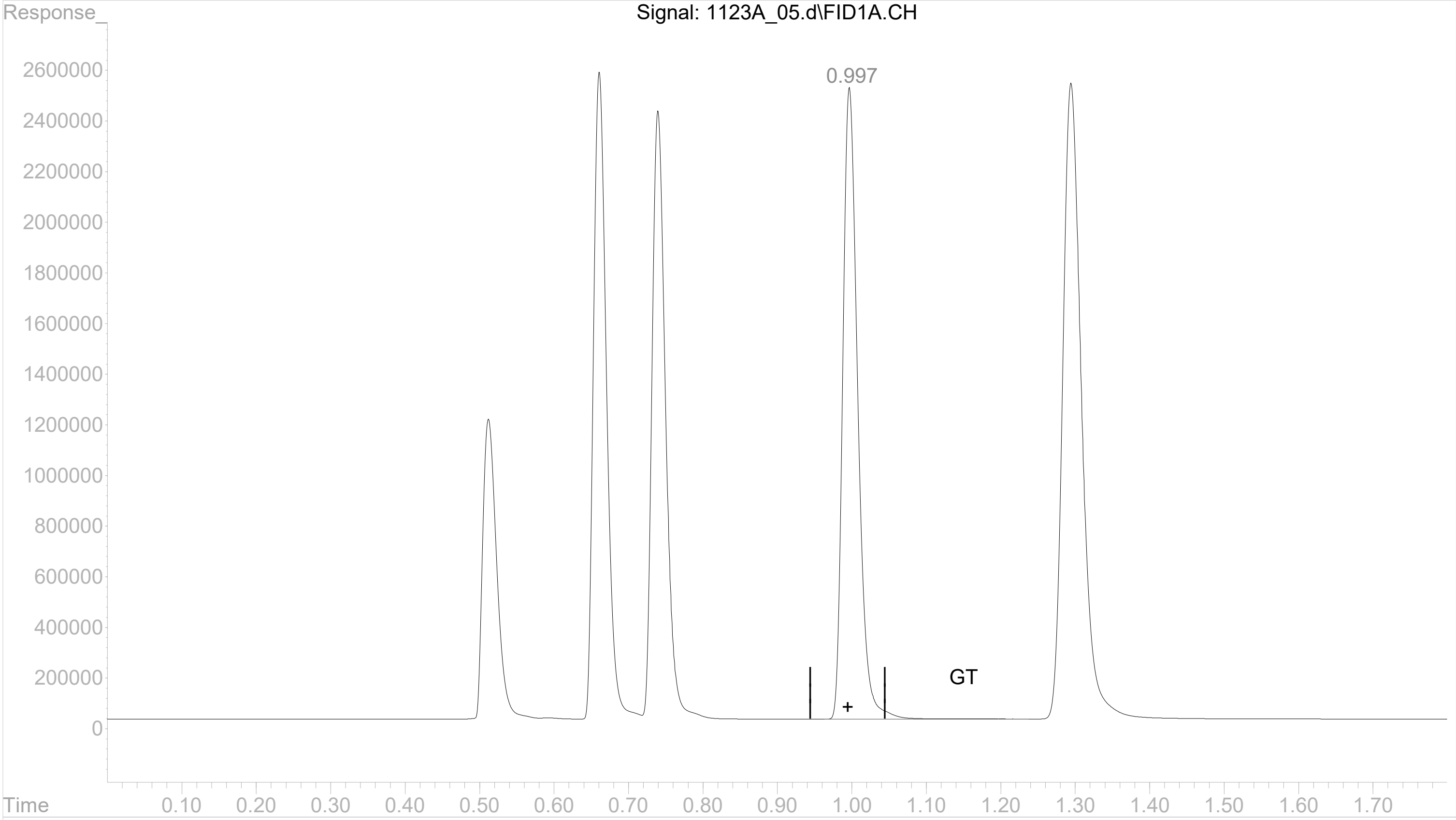
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_05.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:20 pm
Operator :
Sample : STD AGC 5 RSK175
Misc :
ALS Vial : 5 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:23:04 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:20:19 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(4) acetylene

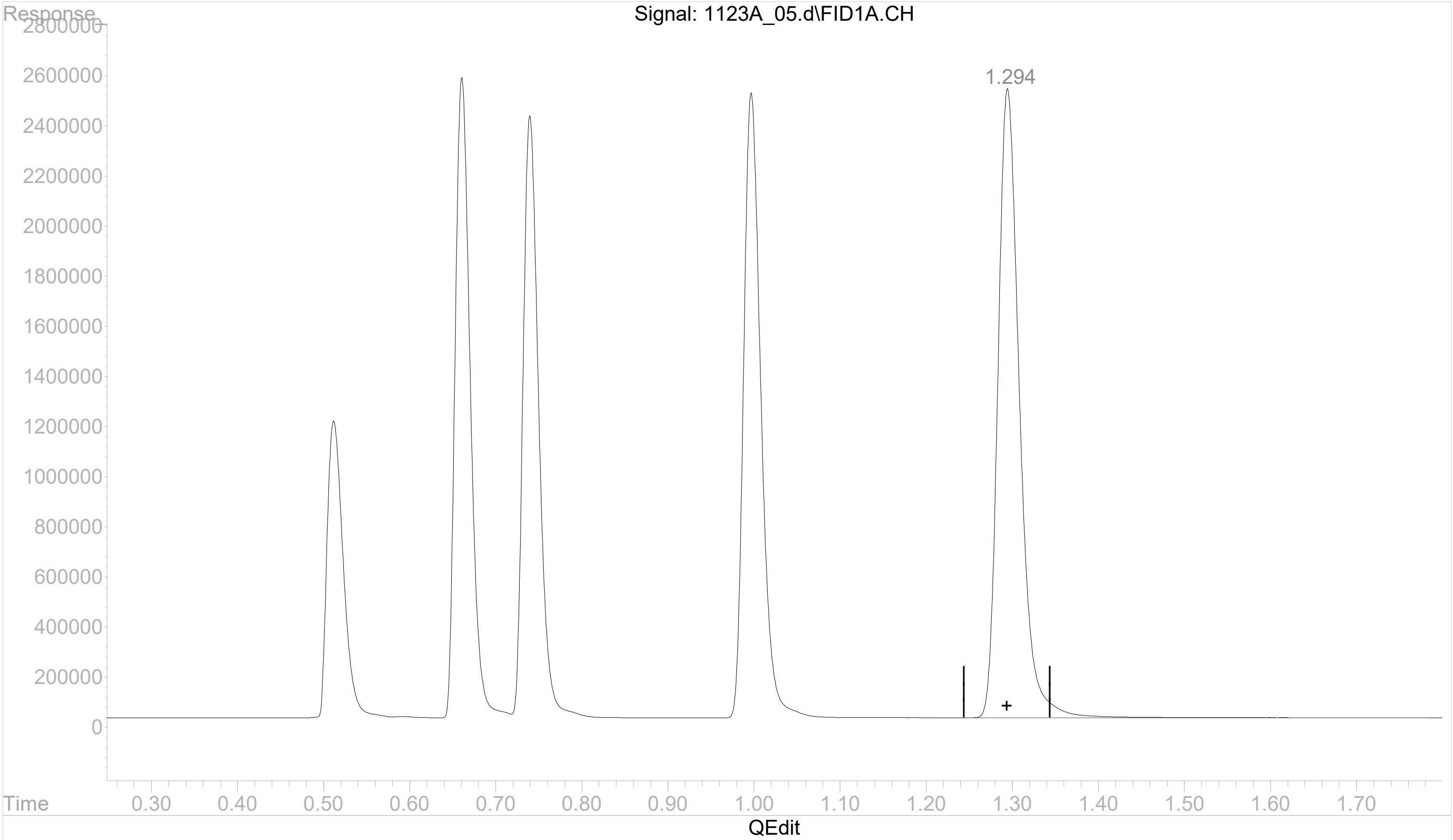
0.997min 1.8937024 mg/l m

response 34992885

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_05.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:20 pm
Operator  :
Sample    : STD AGC 5 RSK175
Misc      :
ALS Vial  : 5   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:23:04 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:20:19 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(5) propane

1.295min 1.7217476 mg/l

response 44011712

(+) = Expected Retention Time

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_06.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:26 pm
Operator  :
Sample    : STD AGC 6 RSK175
Misc      :
ALS Vial  : 6   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:30:02 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:24:42 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	76124411	3.1154018 mg/l m
2)	ethane	0.66	147166629	5.8071280 mg/l m
3)	ethene	0.74	143091228	5.5064868 mg/l m
4)	acetylene	0.99	172342247	9.4967251 mg/l m
5)	propane	1.29	207184005	8.2310501 mg/l m

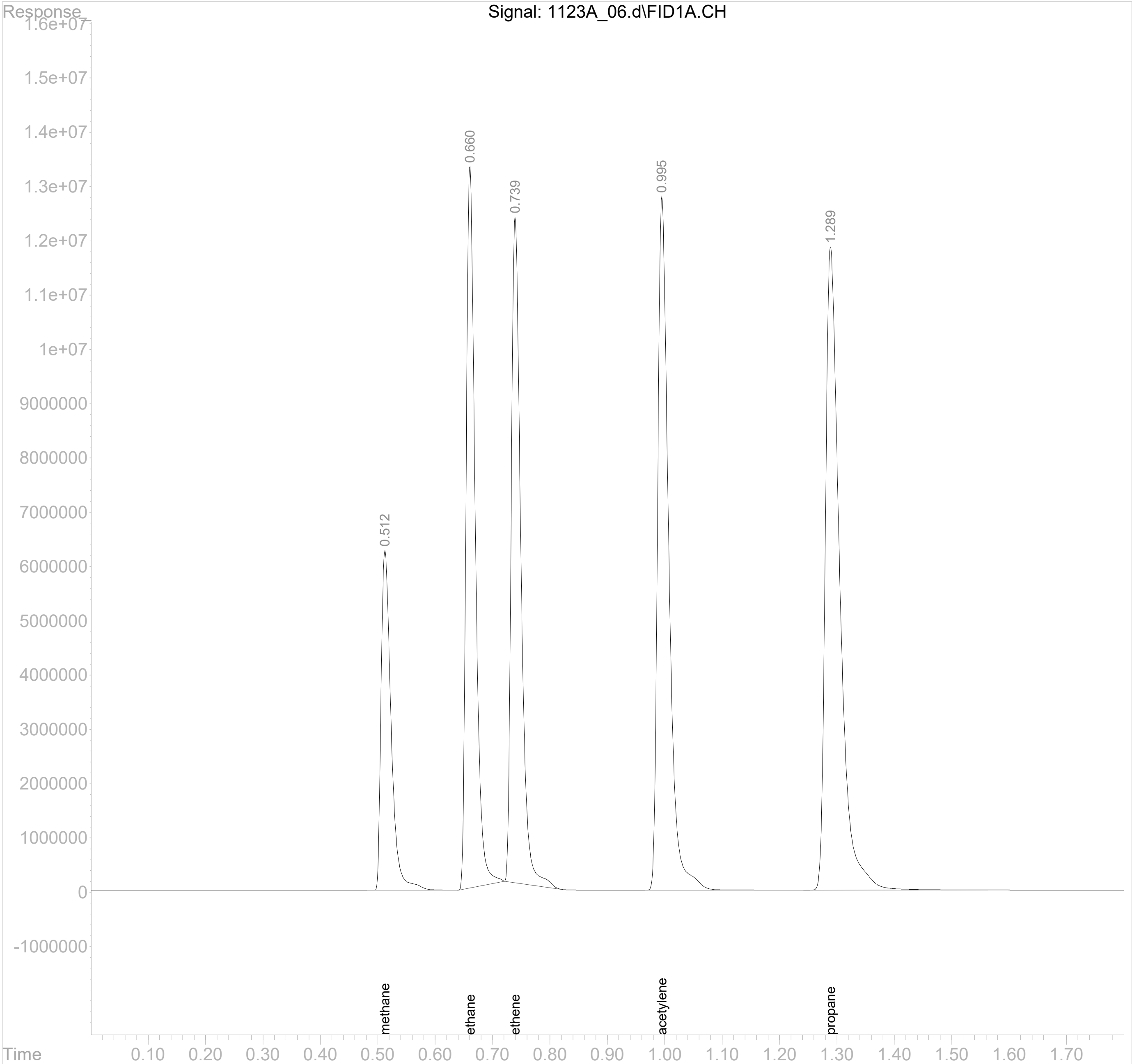
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report   (QT Reviewed)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_06.d
Signal(s) : FID1A.CH
Acq On   : 23 Nov 2022 13:26 pm
Operator :
Sample   : STD AGC 6 RSK175
Misc   :
ALS Vial : 6   Sample Multiplier: 1

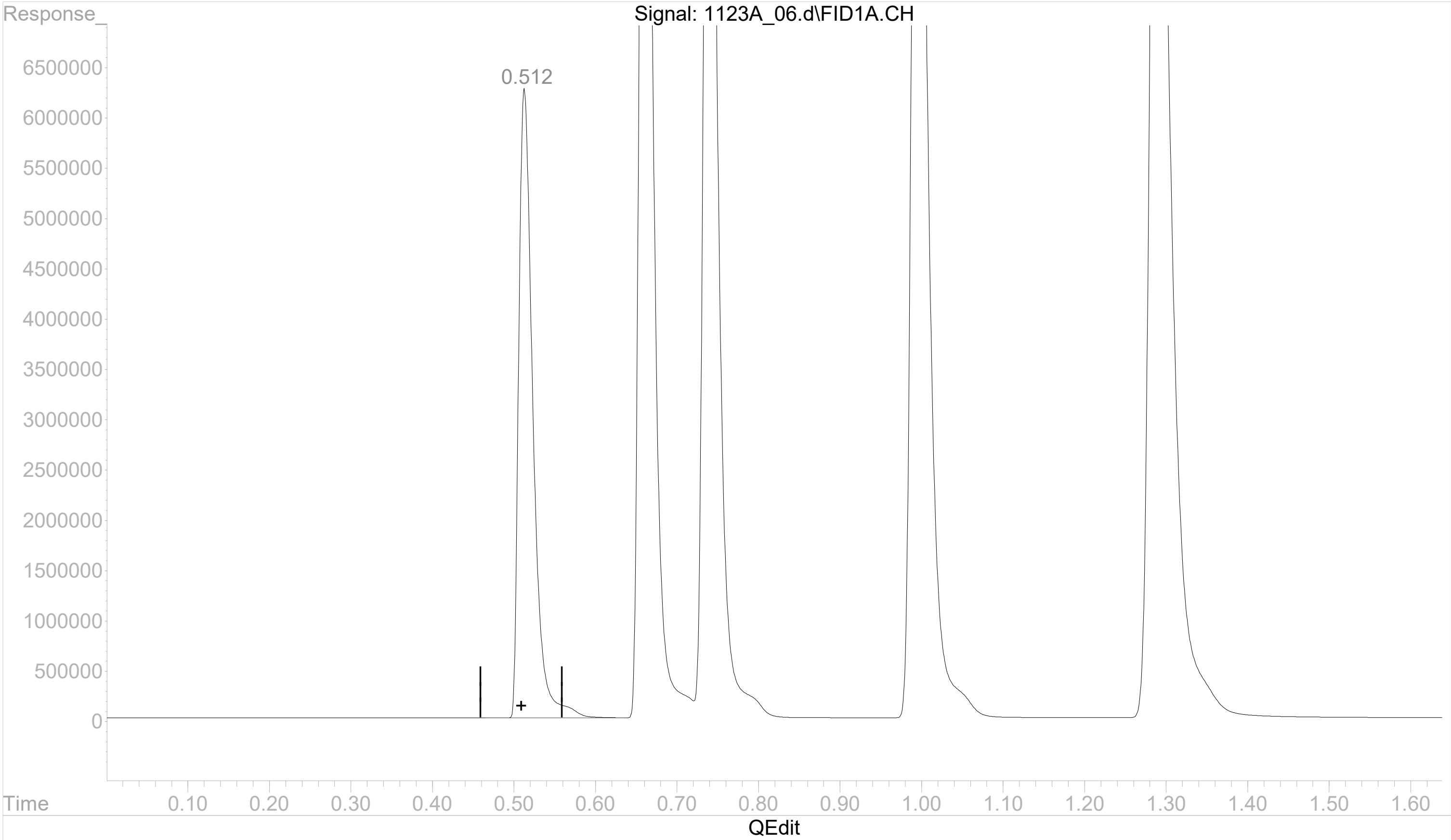
Integration File: autoint1.e
Quant Time: Nov 23 13:30:02 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:24:42 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_06.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:26 pm
Operator  :
Sample    : STD AGC 6 RSK175
Misc      :
ALS Vial  : 6   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:28:52 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:24:42 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(1) methane

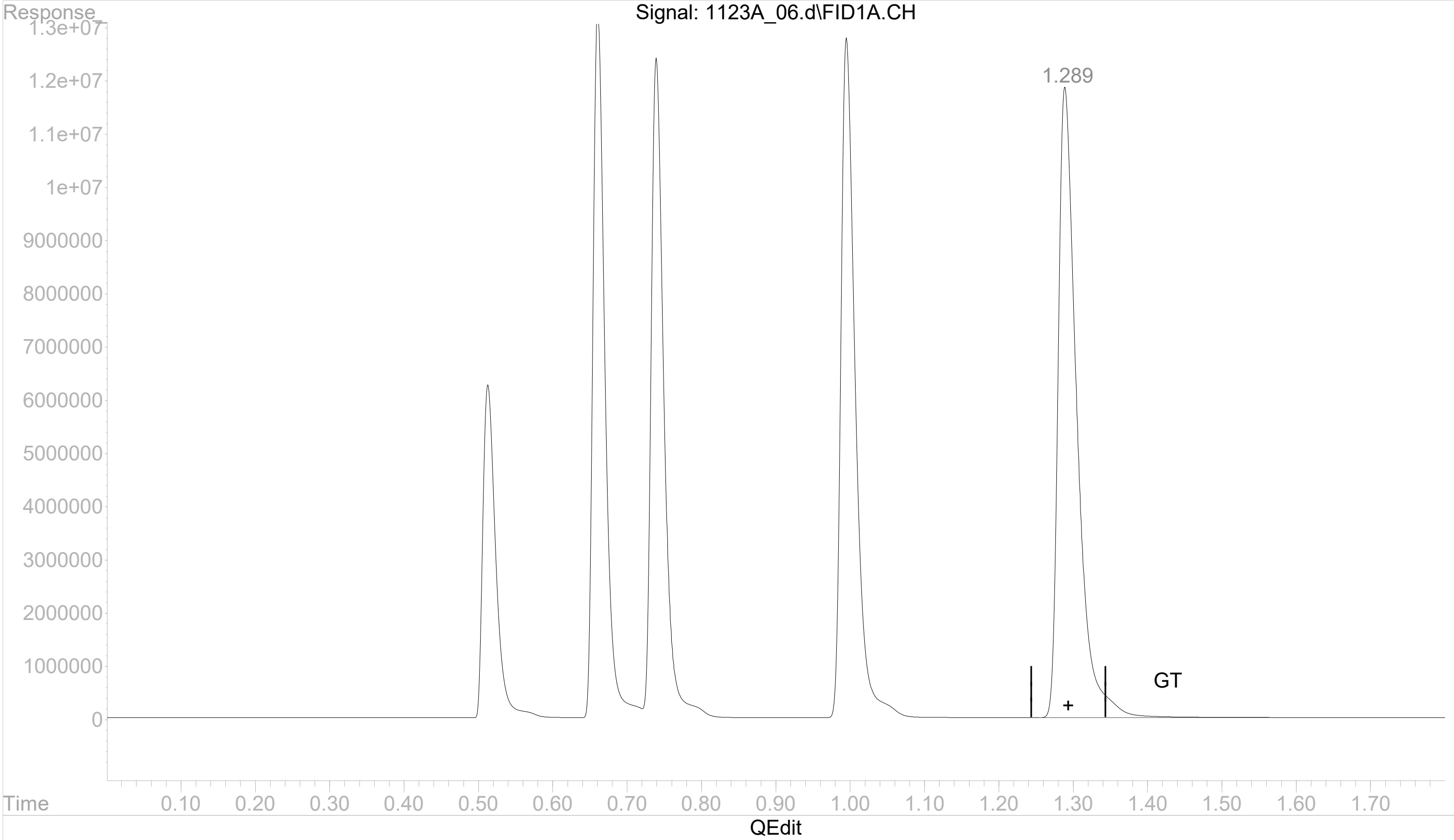
0.513min 3.1166380 mg/l

response 76154617

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_06.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:26 pm
Operator  :
Sample    : STD AGC 6 RSK175
Misc      :
ALS Vial  : 6   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:28:52 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:24:42 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(5) propane

1.289min 8.2310501 mg/l m

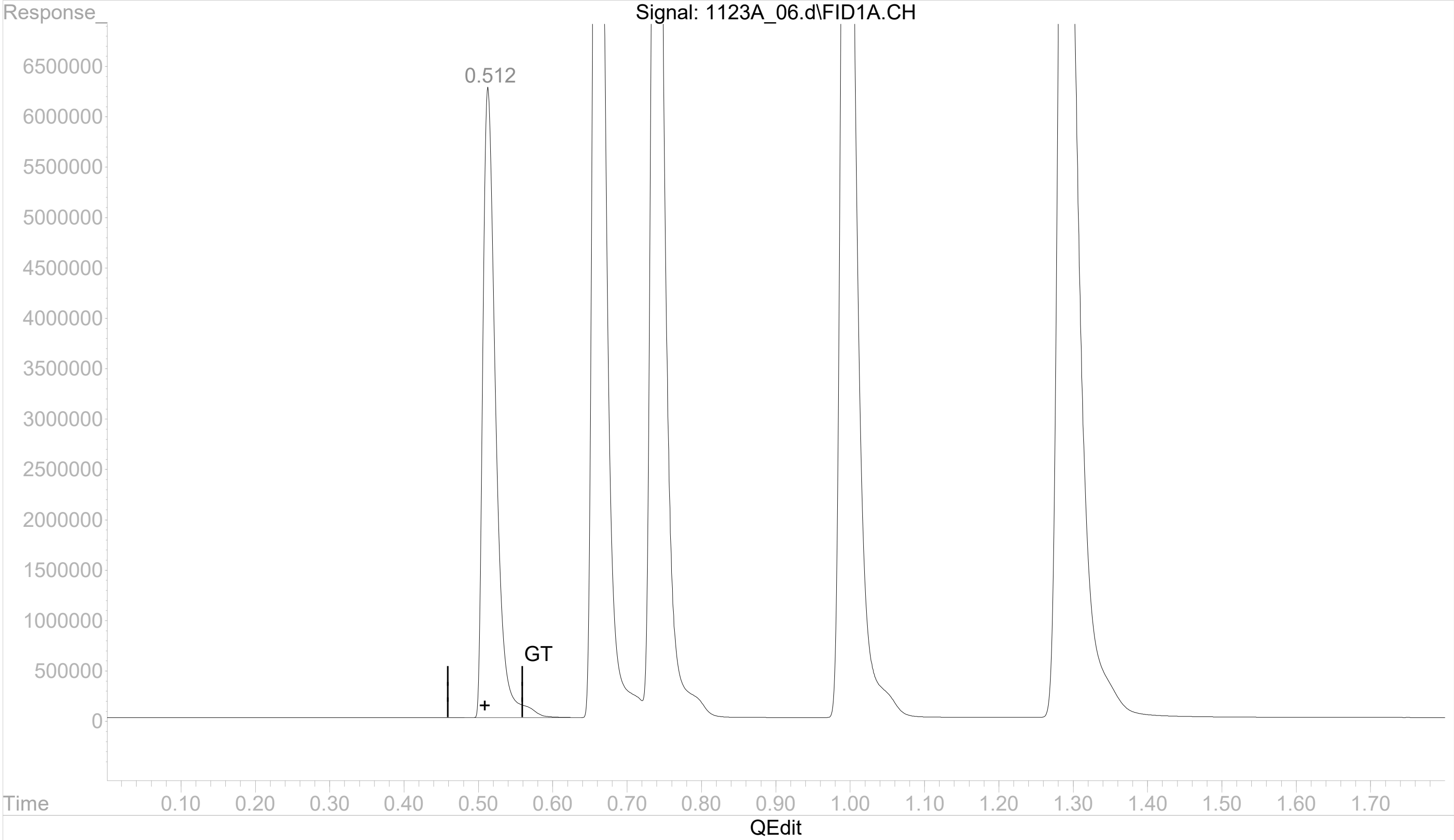
response 207184005

(+) = Expected Retention Time

GC2RSKK23V.M Wed Nov 23 13:30:06 2022 Page: 1

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_06.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:26 pm
Operator  :
Sample    : STD AGC 6 RSK175
Misc      :
ALS Vial  : 6   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:28:52 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:24:42 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(1) methane

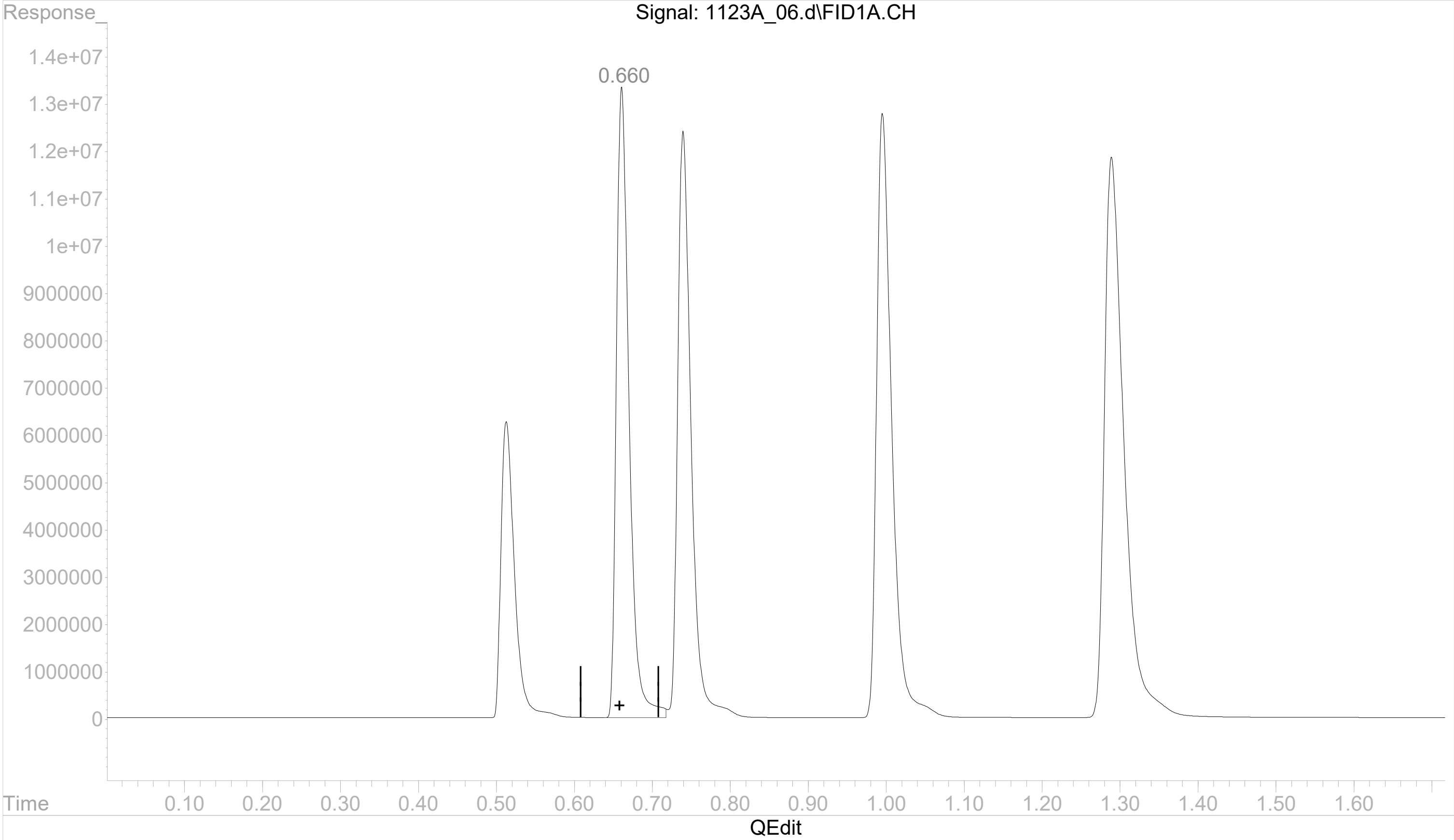
0.512min   3.1154018 mg/l m

response   76124411

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_06.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:26 pm
Operator  :
Sample    : STD AGC 6 RSK175
Misc      :
ALS Vial  : 6   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:28:52 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:24:42 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(2) ethane

0.661min 5.9469437 mg/l

response 150709896

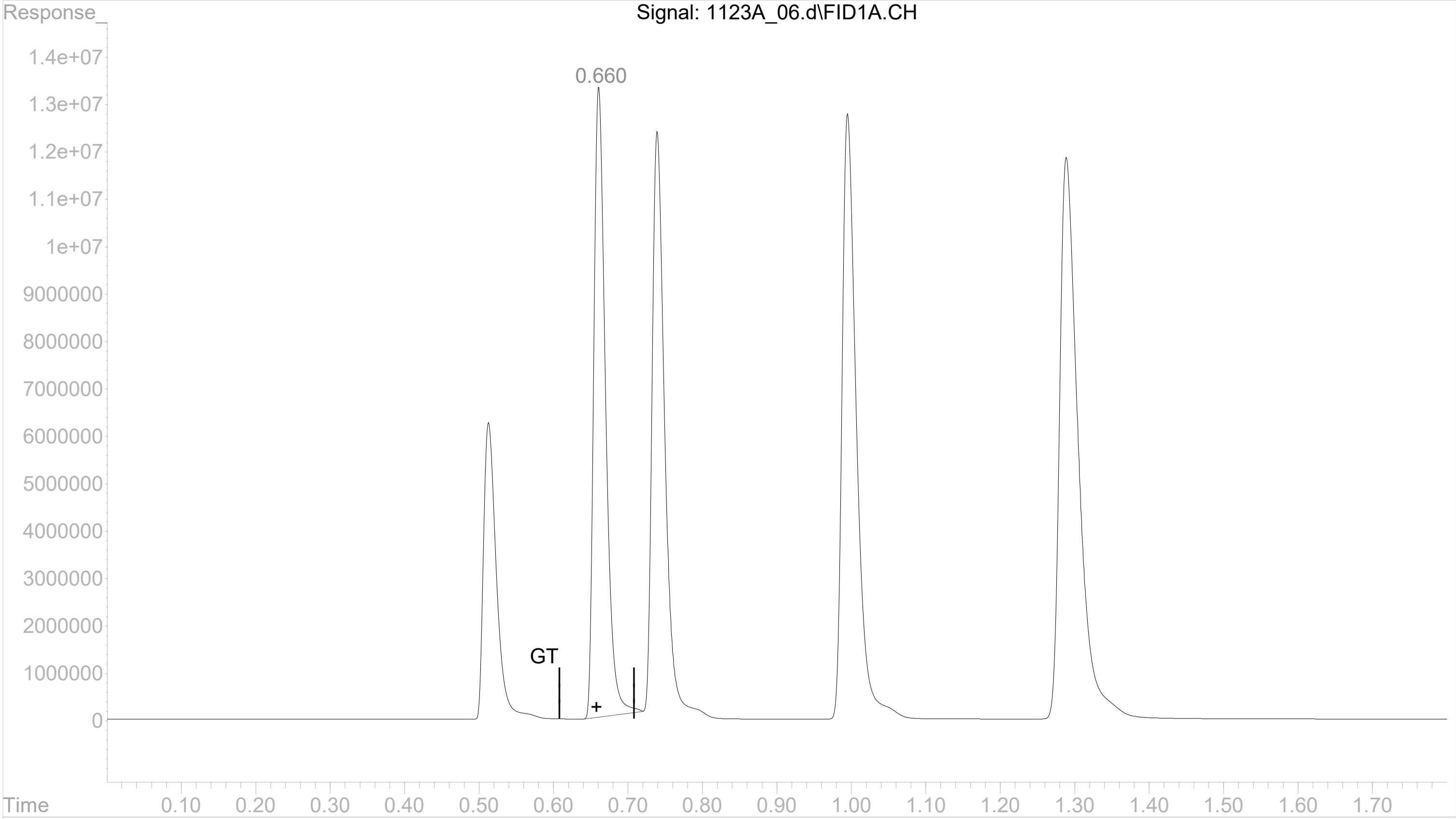
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_06.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:26 pm
Operator :
Sample : STD AGC 6 RSK175
Misc :
ALS Vial : 6 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:28:52 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:24:42 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



QEdit

(2) ethane

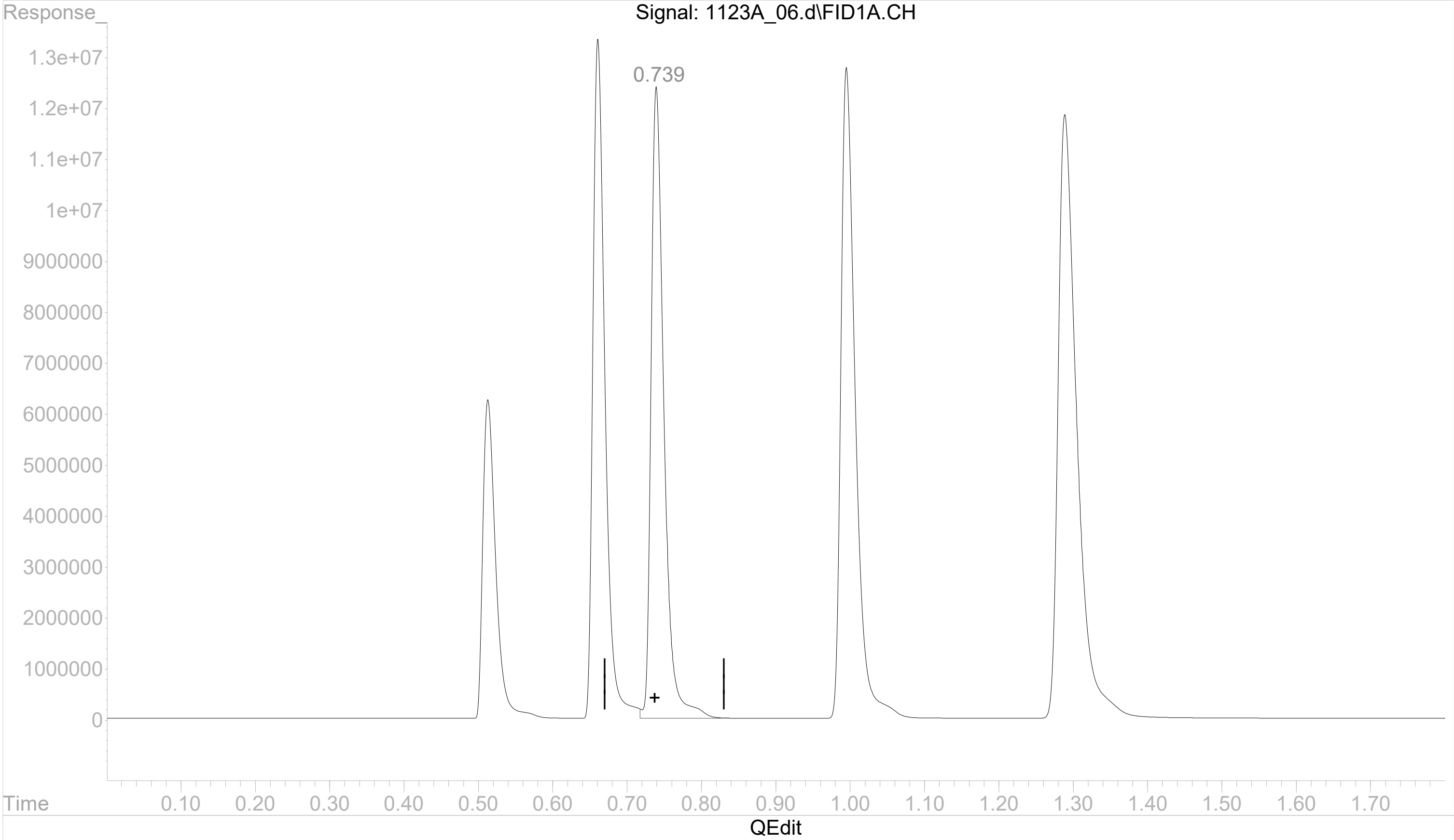
0.660min 5.8071280 mg/l m

response 147166629

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_06.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:26 pm
Operator  :
Sample    : STD AGC 6 RSK175
Misc      :
ALS Vial  : 6   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:28:52 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:24:42 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(3) ethene

0.740min 5.7306152 mg/l

response 148915414

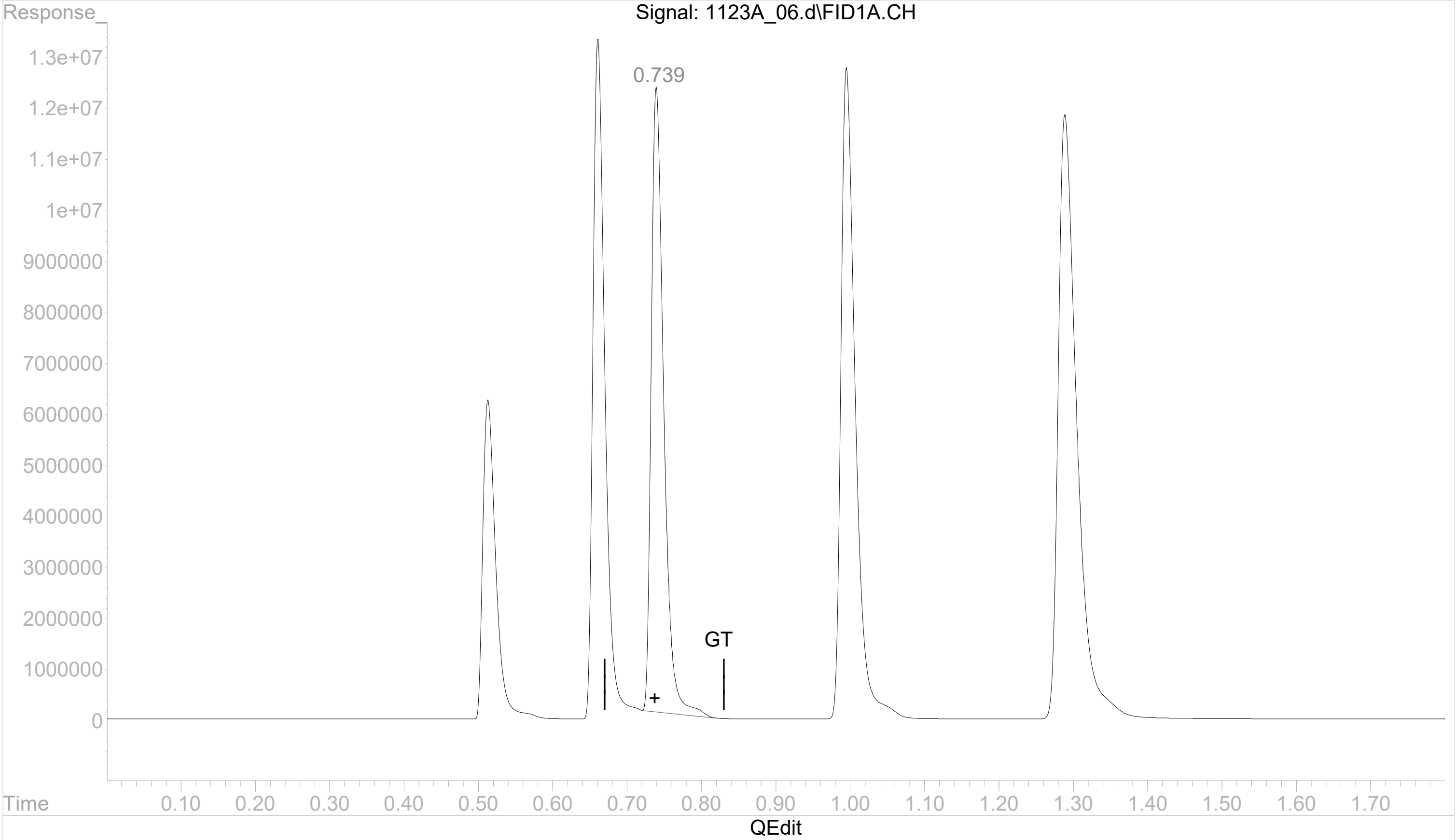
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_06.d
Signal(s) : FID1A.CH
Acq On   : 23 Nov 2022 13:26 pm
Operator :
Sample   : STD AGC 6 RSK175
Misc   :
ALS Vial : 6   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:28:52 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:24:42 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(3) ethene

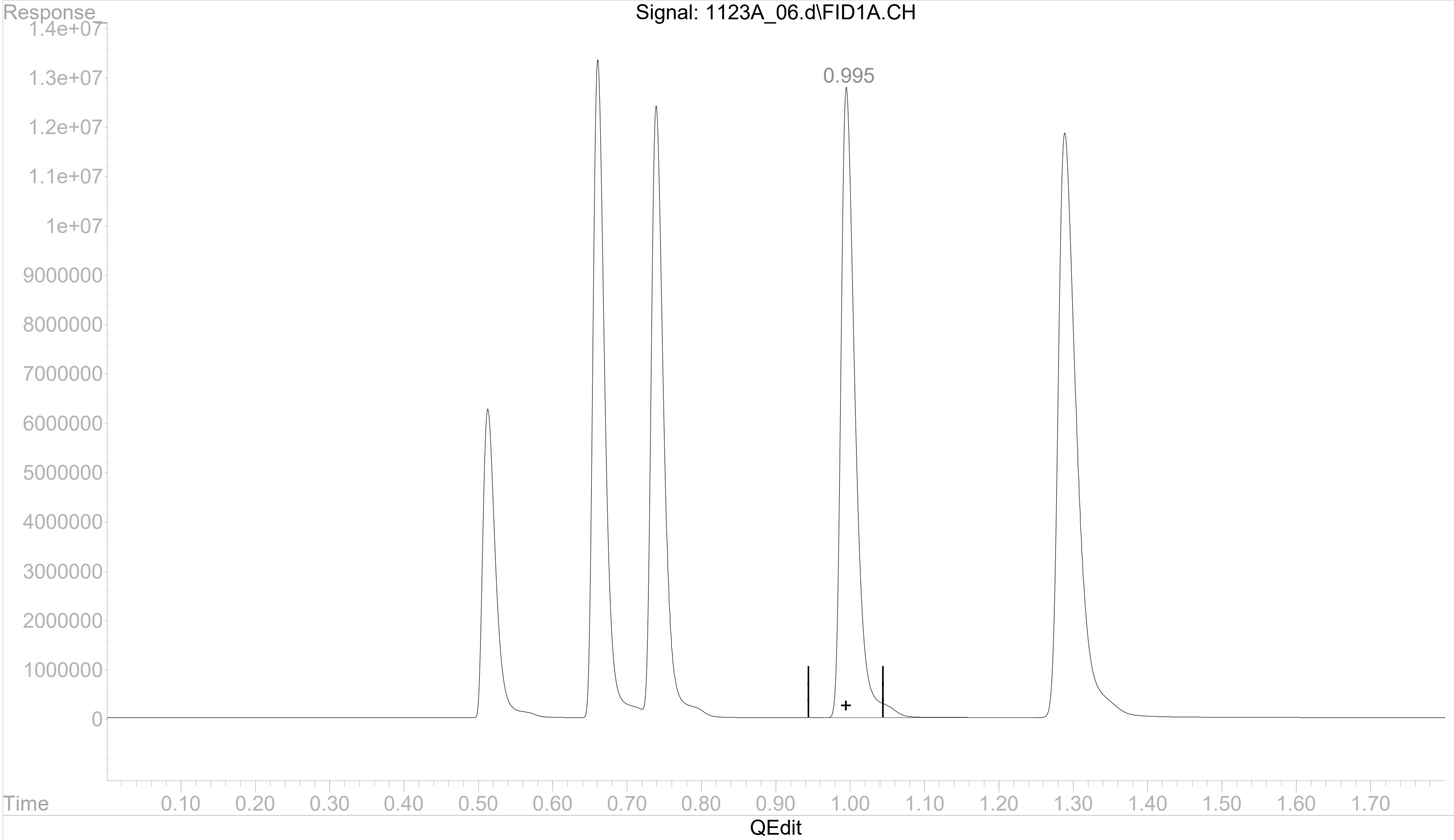
0.739min   5.5064868 mg/l m

response   143091228

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_06.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:26 pm
Operator  :
Sample    : STD AGC 6 RSK175
Misc      :
ALS Vial  : 6   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:28:52 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:24:42 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



(4) acetylene

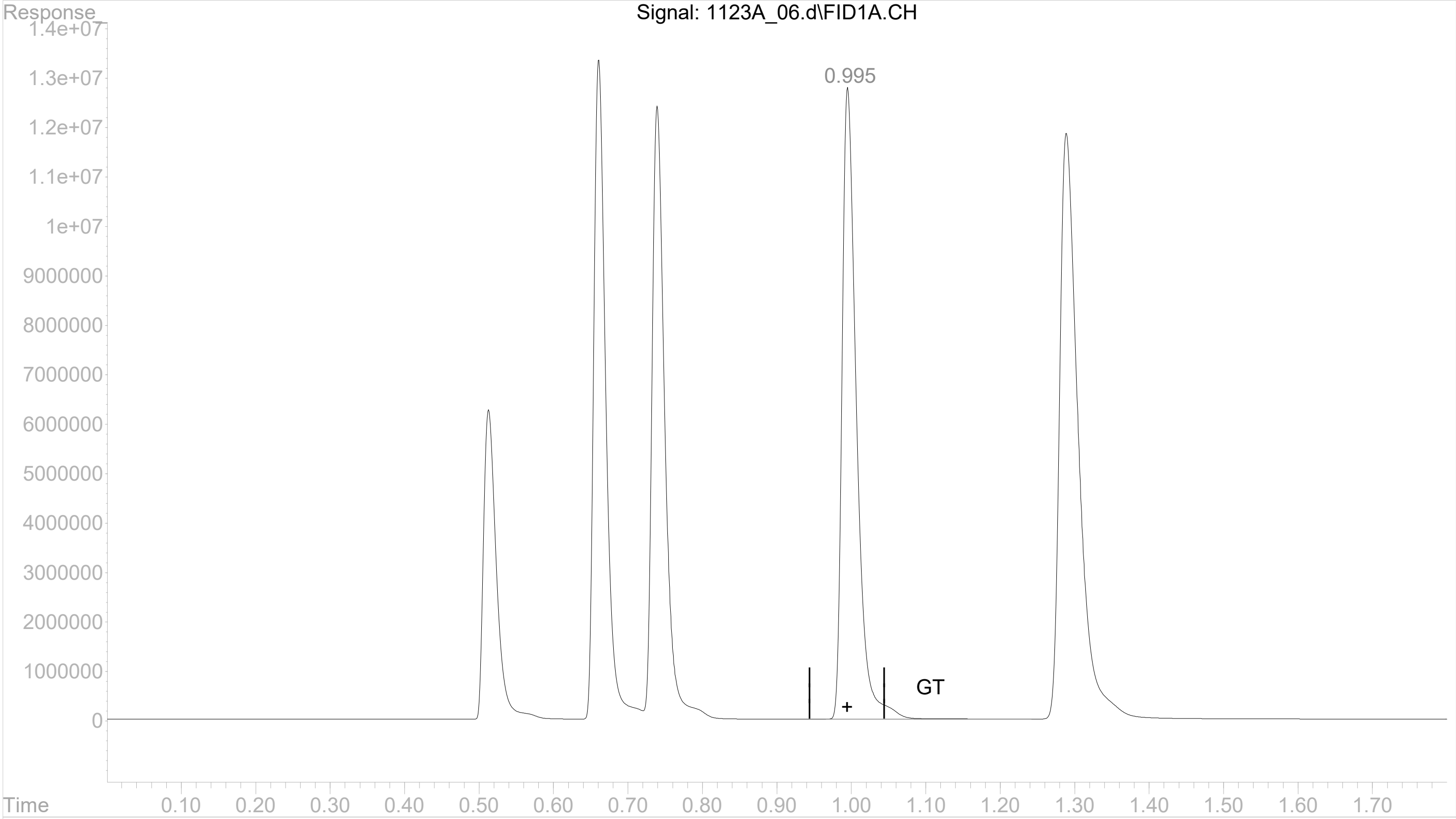
0.996min 9.5004147 mg/l

response 172409204

(+) = Expected Retention Time

Quantitation Report (Qedit)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_06.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:26 pm
Operator  :
Sample    : STD AGC 6 RSK175
Misc      :
ALS Vial  : 6   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:28:52 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:24:42 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



QEdit

(4) acetylene

0.995min 9.4967251 mg/l m

response 172342247

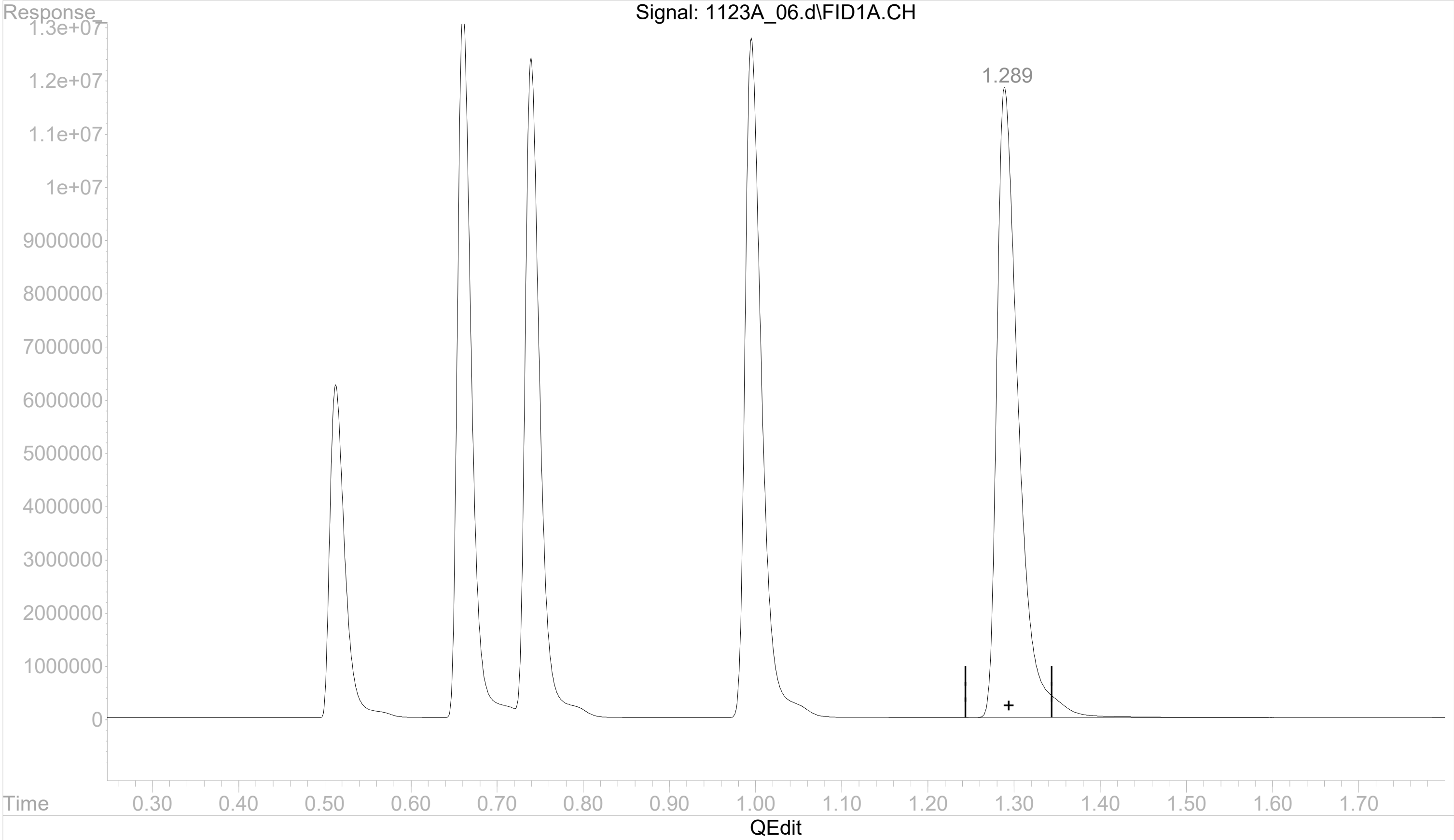
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_06.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:26 pm
Operator :
Sample : STD AGC 6 RSK175
Misc :
ALS Vial : 6 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:28:52 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:24:42 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(5) propane

1.290min 8.2477426 mg/l

response 207604173

(+) = Expected Retention Time

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_07.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:32 pm
Operator :
Sample : STD AGC 7 RSK175
Misc :
ALS Vial : 7 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:35:23 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:30:12 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

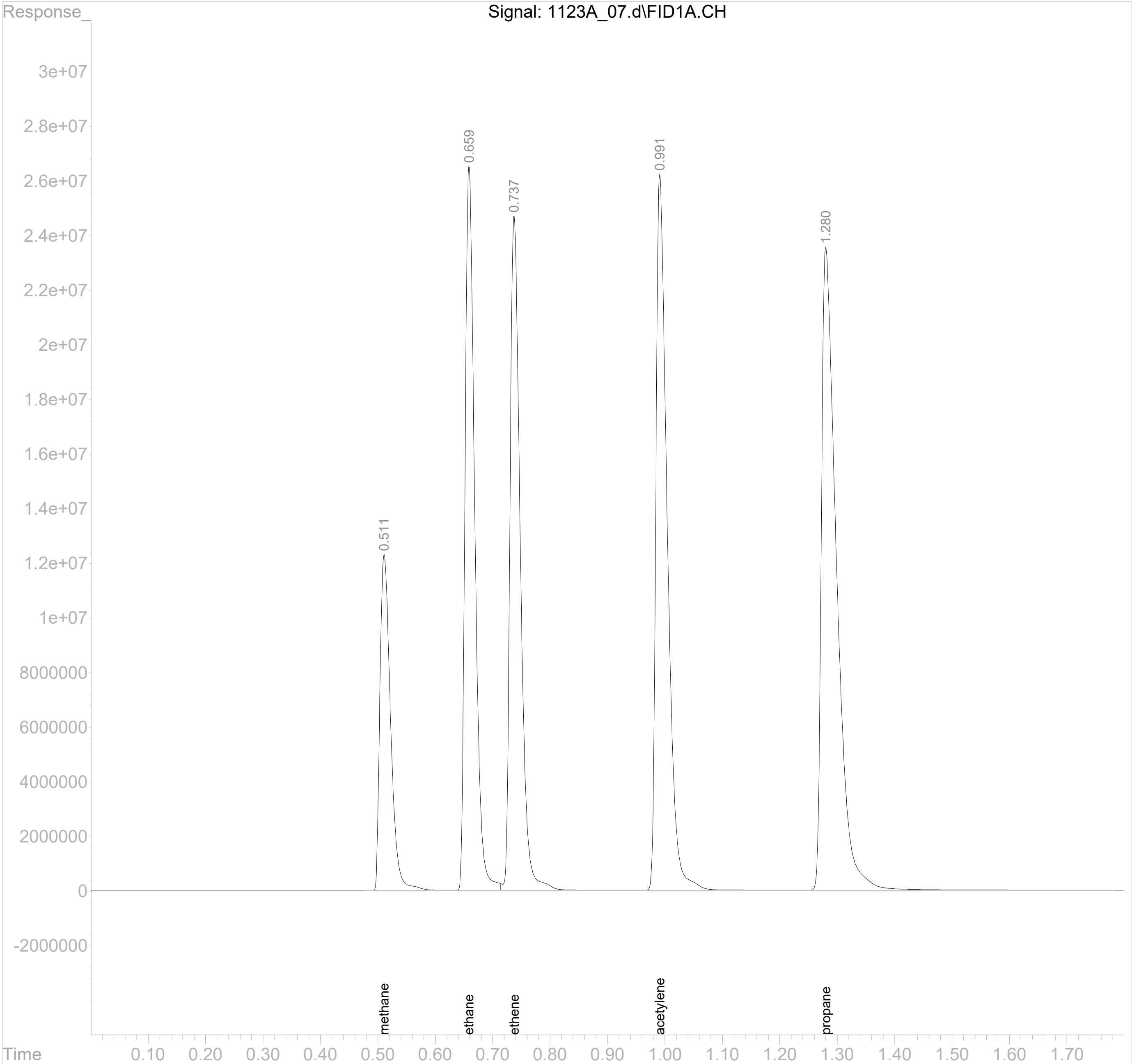
Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	158118043	6.5440398 mg/l
2)	ethane	0.66	313757417	12.5898633 mg/l
3)	ethene	0.74	308133428	12.1261653 mg/l
4)	acetylene	0.99	360741109	20.1702068 mg/l
5)	propane	1.28	431827121	17.4907871 mg/l

(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_07.d
Signal(s) : FID1A.CH
Acq On    : 23 Nov 2022  13:32 pm
Operator  :
Sample    : STD AGC 7 RSK175
Misc      :
ALS Vial  : 7   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:35:23 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:30:12 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



7C-OR

CONTINUING CALIBRATION VERIFICATION
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Lab File ID:	1123A_09-1	**Analysis date/time:**	11/23/22 13:43
Analytical Method:	RSK175	**Sample ID:**	SSCV

Analyte	**Avg. RRF**	**RRF**	**Min. RRF**	**Diff.**	**Max Diff.**	**True Value**	**Result**	**Result**	**Limits**
				%	*%*	*mg/l*	*mg/l*	*% Rec.*	*%*
ETHANE	24835840	23180730		6.66		0.1290	0.1204	93.30	80 - 120
ETHENE	25246600	24280390		3.83		0.1270	0.1221	96.10	80 - 120
METHANE	24042020	25357600		5.47		0.0678	0.07151	105	80 - 120

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_09.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:43 pm
Operator :
Sample : SSCV AGC 3.0 RSK175
Misc :
ALS Vial : 9 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:46:12 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
	Target Compounds			
1)	methane	0.51	1719245	0.0715100 mg/l m
2)	ethane	0.66	2990314	0.1204032 mg/l
3)	ethene	0.74	3083610	0.1221396 mg/l
4)	acetylene	1.00	3494304	0.1962267 mg/l
5)	propane	1.30	4326905	0.1767635 mg/l

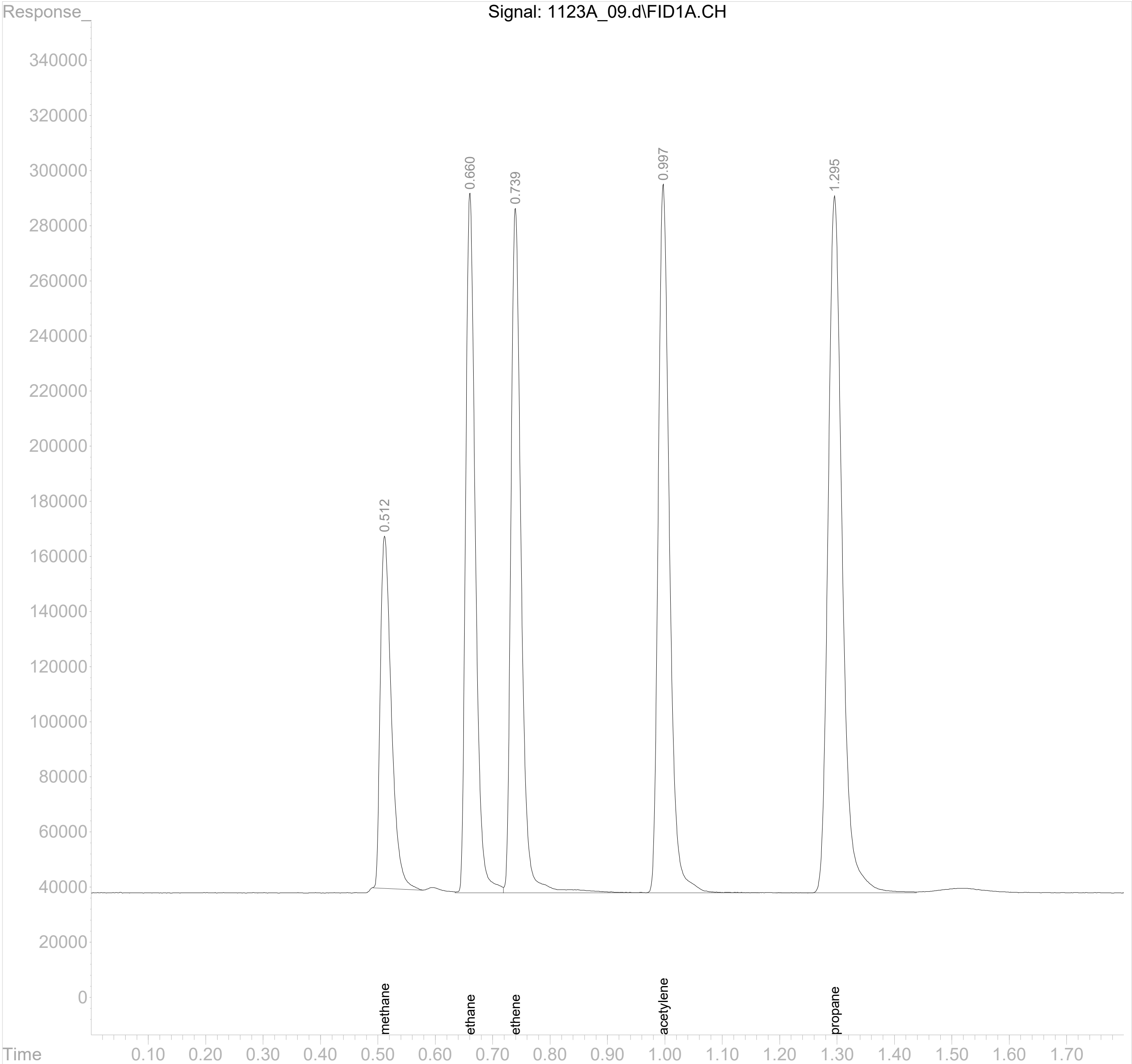
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_09.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:43 pm
Operator :
Sample : SSCV AGC 3.0 RSK175
Misc :
ALS Vial : 9 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:46:12 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

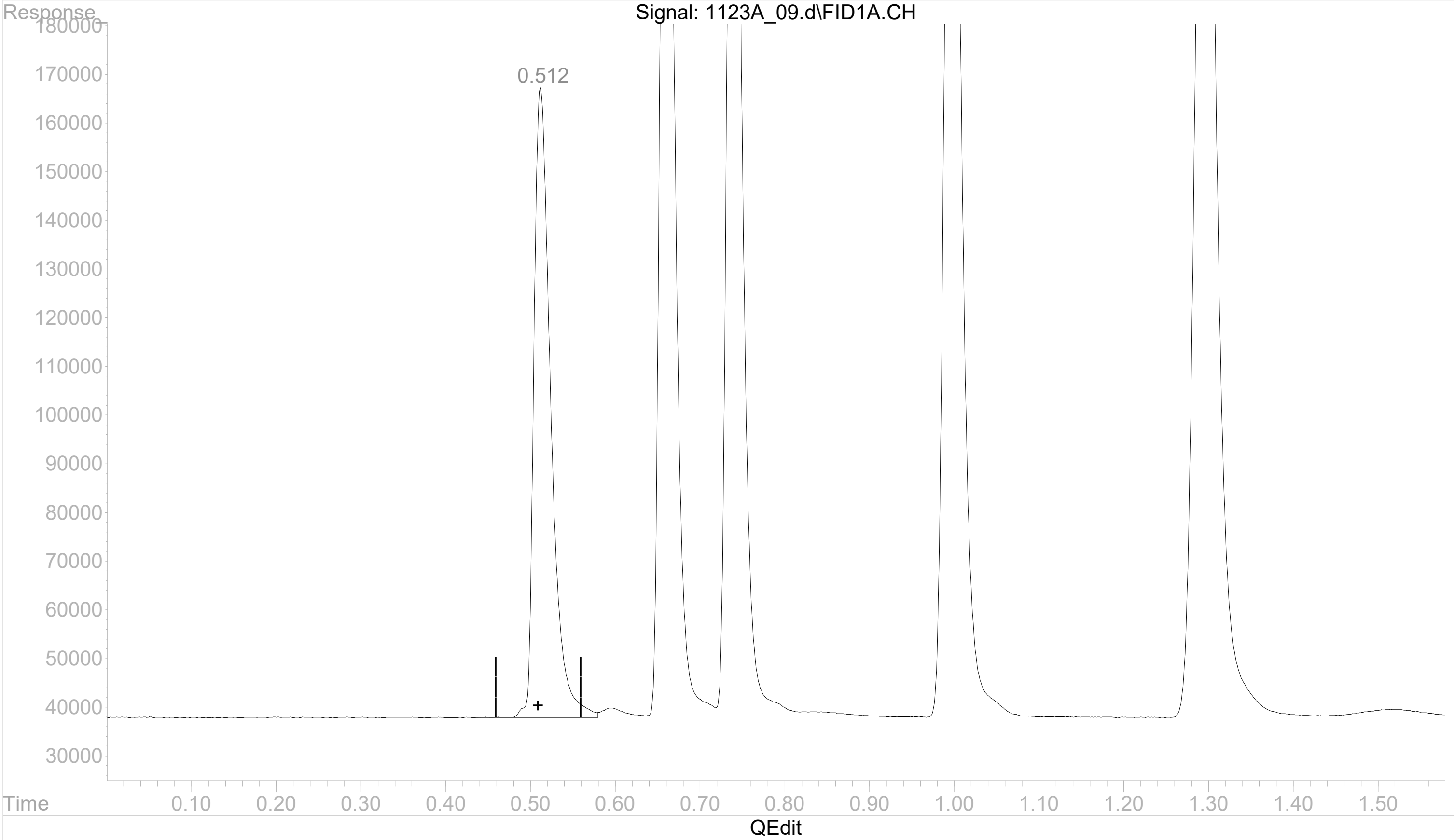


Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_09.d
Signal(s) : FID1A.CH
Acq On : 23 Nov 2022 13:43 pm
Operator :
Sample : SSCV AGC 3.0 RSK175
Misc :
ALS Vial : 9 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:46:03 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(1) methane

0.512min 0.0749961 mg/l

response 1803058

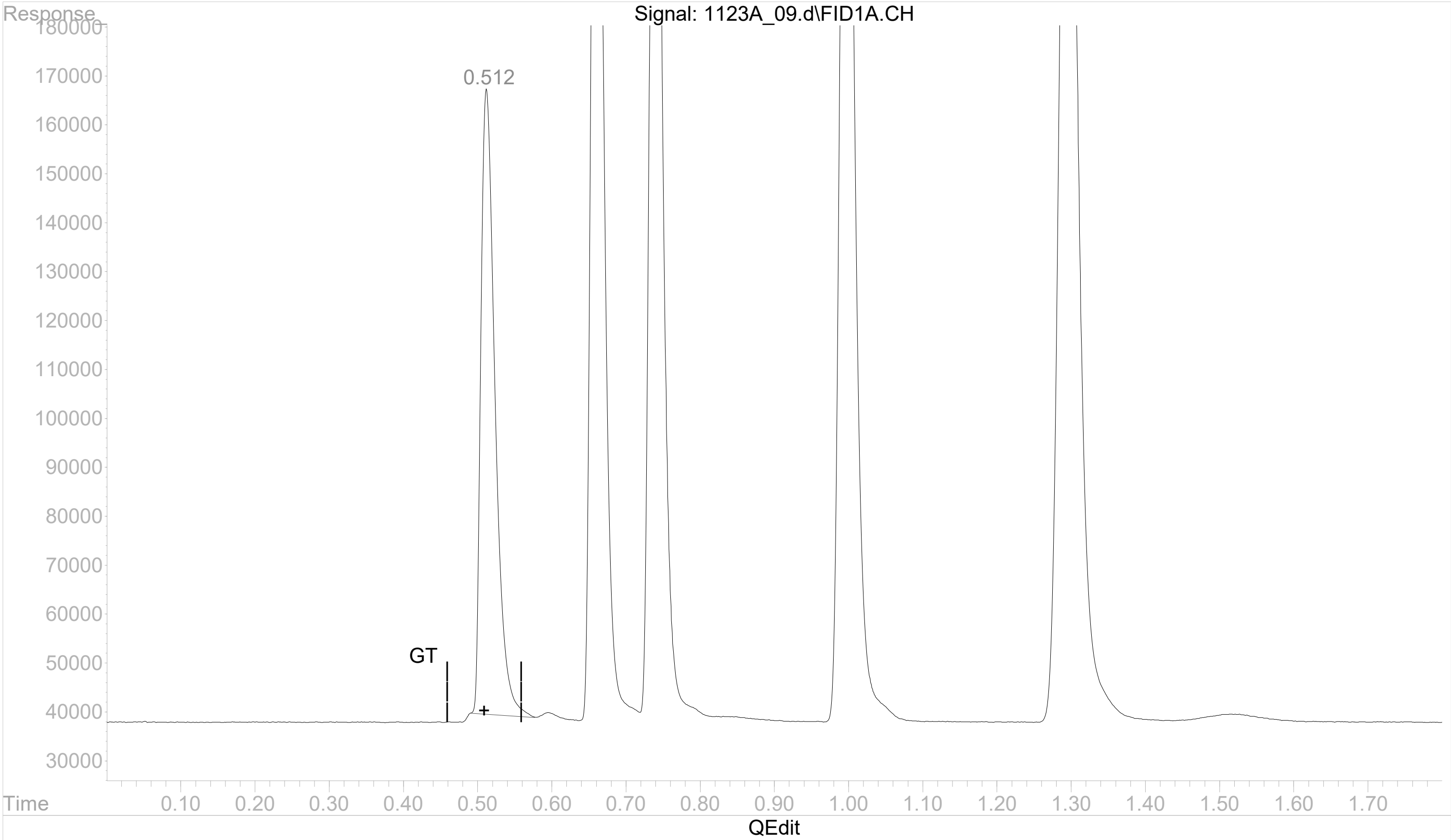
(+) = Expected Retention Time

Quantitation Report (Qedit)

Data Path : C:\msdchem\1\data\112322A\
Data File : 1123A_09.d
Signal(s) : FID1A.CH
Acq On   : 23 Nov 2022 13:43 pm
Operator :
Sample   : SSCV AGC 3.0 RSK175
Misc   :
ALS Vial : 9   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Nov 23 13:46:03 2022
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



(1) methane

0.512min   0.0715100 mg/l m

response   1719245

(+) = Expected Retention Time

7C-OR

CONTINUING CALIBRATION VERIFICATION
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Lab File ID:	0608_01-1	**Analysis date/time:**	06/08/23 09:45
Analytical Method:	RSK175	**Sample ID:**	ICV

Analyte	**Avg. RRF**	**RRF**	**Min. RRF**	**Diff.** *%*	**Max Diff.** *%*	**True Value** *mg/l*	**Result** *mg/l*	**Result** *% Rec.*	**Limits** *%*
ETHANE	24835840	22775530		8.30		0.1290	0.1183	91.70	80 - 120
ETHENE	25246600	23568040		6.65		0.1270	0.1186	93.40	80 - 120
METHANE	24042020	23759290		1.18		0.0678	0.06700	98.80	80 - 120

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_01.d
Signal(s) : FID1A.CH
Acq On : 08 Jun 2023 9:45 am
Operator :
Sample : ICVLCS AGC 3.0 RSK175
Misc :
ALS Vial : 1 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 09:47:35 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
	Target Compounds			
1)	methane	0.51	1610880	0.0670027 mg/l
2)	ethane	0.66	2938043	0.1182985 mg/l
3)	ethene	0.74	2993141	0.1185562 mg/l
4)	acetylene	0.99	3436165	0.1929618 mg/l
5)	propane	1.29	4262533	0.1741337 mg/l

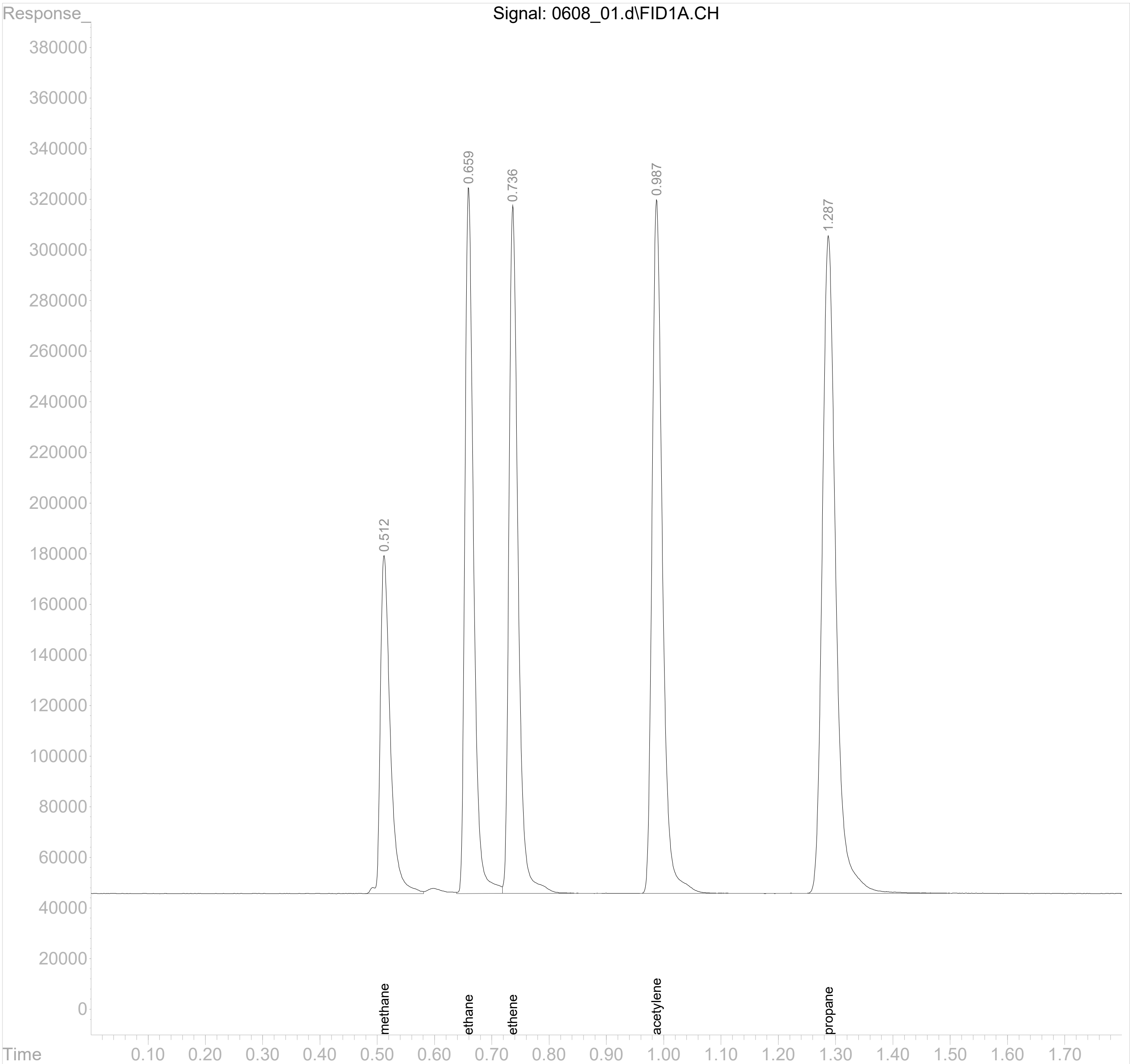
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_01.d
Signal(s) : FID1A.CH
Acq On : 08 Jun 2023 9:45 am
Operator :
Sample : ICVLCS AGC 3.0 RSK175
Misc :
ALS Vial : 1 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 09:47:35 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



7C-OR

CONTINUING CALIBRATION VERIFICATION
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Lab File ID:	0608_12-1	**Analysis date/time:**	06/08/23 10:26
Analytical Method:	RSK175	**Sample ID:**	CCV

Analyte	**Avg. RRF**	**RRF**	**Min. RRF**	**Diff.** *%*	**Max Diff.** *%*	**True Value** *mg/l*	**Result** *mg/l*	**Result** *% Rec.*	**Limits** *%*
ETHANE	24835840	22932530		7.66		0.1290	0.1191	92.30	80 - 120
ETHENE	25246600	23672500		6.23		0.1270	0.1191	93.80	80 - 120
METHANE	24042020	23879630		0.6750		0.0678	0.06734	99.30	80 - 120

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_12.d
Signal(s) : FID1A.CH
Acq On : 08 Jun 2023 10:26 am
Operator :
Sample : CCV AGC 3.0 RSK175
Misc :
ALS Vial : 12 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 10:31:01 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	1619039	0.0673421 mg/l
2)	ethane	0.66	2958296	0.1191140 mg/l
3)	ethene	0.74	3006408	0.1190817 mg/l
4)	acetylene	0.99	3523183	0.1978484 mg/l
5)	propane	1.29	4285695	0.1750800 mg/l

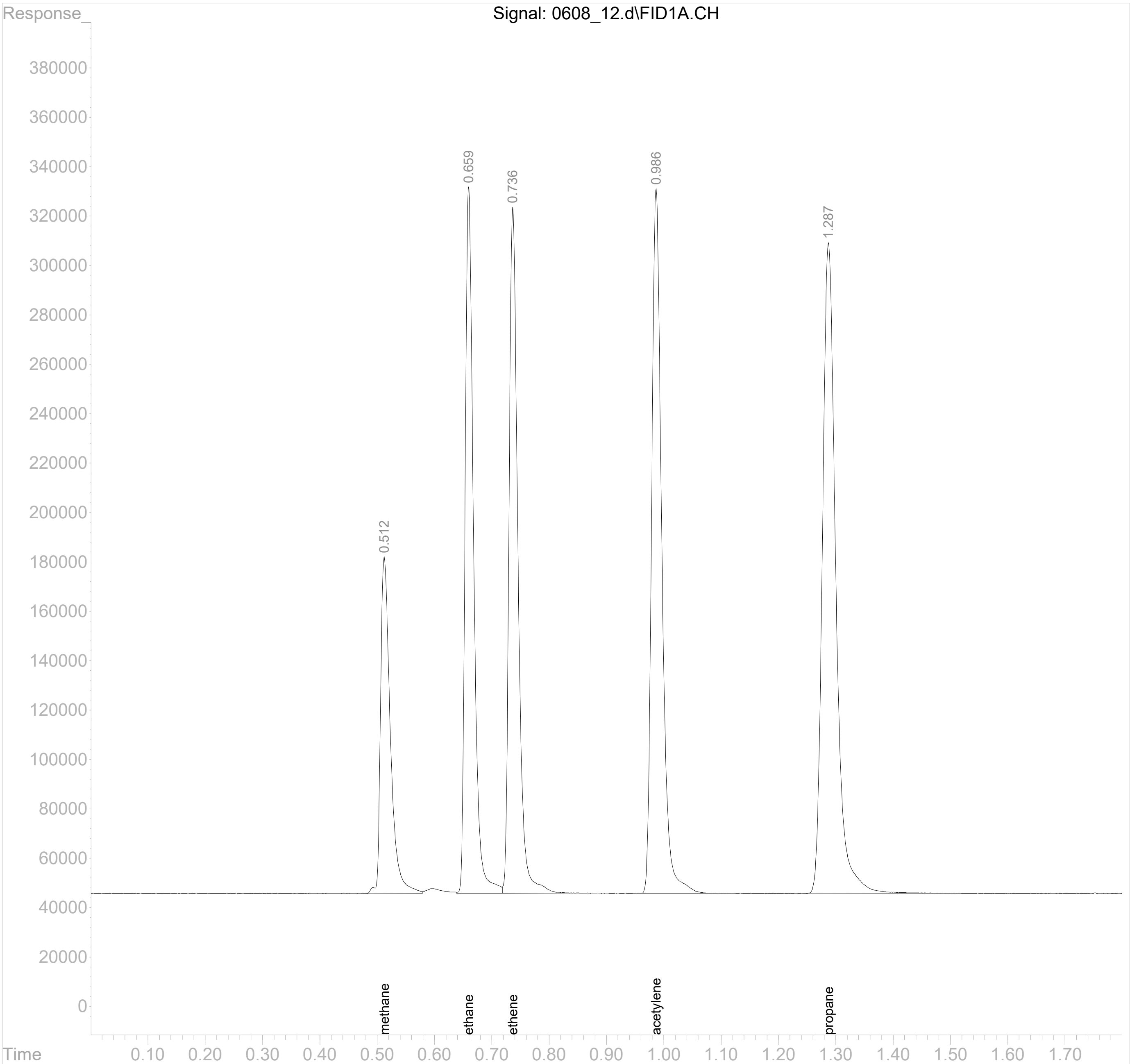
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report   (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_12.d
Signal(s) : FID1A.CH
Acq On   : 08 Jun 2023 10:26 am
Operator :
Sample   : CCV AGC 3.0 RSK175
Misc   :
ALS Vial : 12   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 10:31:01 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



CONTINUING CALIBRATION VERIFICATION
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Lab File ID:	0608_12-2	**Analysis date/time:**	06/08/23 10:26
Analytical Method:	RSK175	**Sample ID:**	CCV

Analyte	**Avg. RRF**	**RRF**	**Min. RRF**	**Diff.** *%*	**Max Diff.** *%*	**True Value** *mg/l*	**Result** *mg/l*	**Result** *% Rec.*	**Limits** *%*
ETHANE	24835840	22932530		7.66		0.1290	0.1191	92.30	80 - 120
ETHENE	25246600	23672500		6.23		0.1270	0.1191	93.80	80 - 120
METHANE	24042020	23879630		0.6750		0.0678	0.06734	99.30	80 - 120

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_12.d
Signal(s) : FID1A.CH
Acq On : 08 Jun 2023 10:26 am
Operator :
Sample : CCV AGC 3.0 RSK175
Misc :
ALS Vial : 12 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 10:31:01 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	1619039	0.0673421 mg/l
2)	ethane	0.66	2958296	0.1191140 mg/l
3)	ethene	0.74	3006408	0.1190817 mg/l
4)	acetylene	0.99	3523183	0.1978484 mg/l
5)	propane	1.29	4285695	0.1750800 mg/l

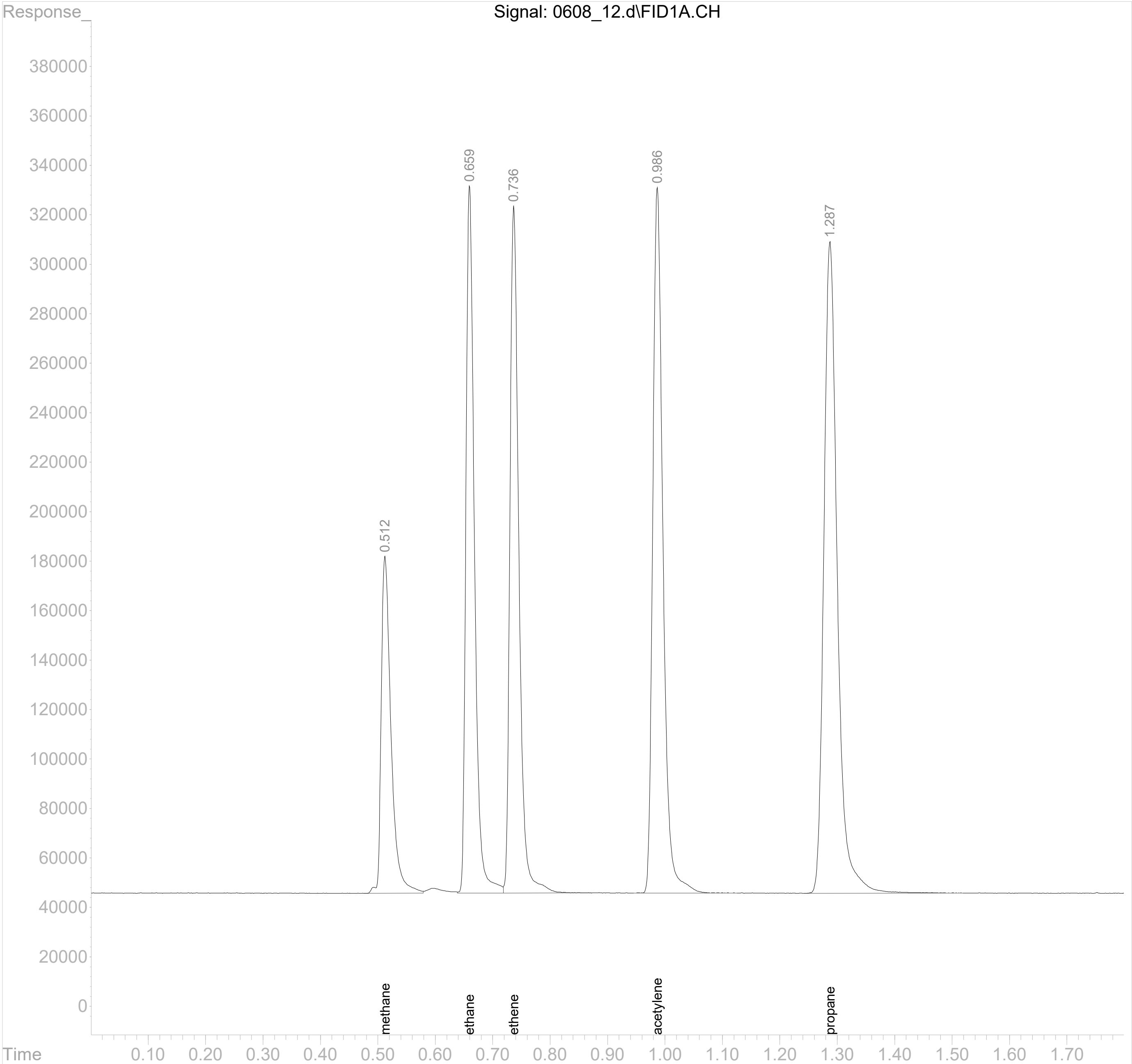
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_12.d
Signal(s) : FID1A.CH
Acq On : 08 Jun 2023 10:26 am
Operator :
Sample : CCV AGC 3.0 RSK175
Misc :
ALS Vial : 12 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 10:31:01 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



7C-OR

CONTINUING CALIBRATION VERIFICATION
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Lab File ID:	0608_23-1	**Analysis date/time:**	06/08/23 11:18
Analytical Method:	RSK175	**Sample ID:**	CCV

Analyte	**Avg. RRF**	**RRF**	**Min. RRF**	**Diff.** *%*	**Max Diff.** *%*	**True Value** *mg/l*	**Result** *mg/l*	**Result** *% Rec.*	**Limits** *%*
ETHANE	24835840	22651900		8.79		0.1290	0.1177	91.20	80 - 120
ETHENE	25246600	23663620		6.27		0.1270	0.1190	93.70	80 - 120
METHANE	24042020	23912420		0.5390		0.0678	0.06743	99.50	80 - 120

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060823\
Data File : 0608_23.d
Signal(s) : FID1A.CH
Acq On    : 08 Jun 2023  11:18 am
Operator  :
Sample    : CCV AGC 3.0 RSK175
Misc      :
ALS Vial  : 23   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:20:56 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```

	Compound	R.T.	Response	Conc Units
	Target Compounds			
1)	methane	0.51	1621262	0.0674345 mg/l
2)	ethane	0.66	2922095	0.1176564 mg/l
3)	ethene	0.74	3005280	0.1190370 mg/l
4)	acetylene	0.99	3451102	0.1938006 mg/l
5)	propane	1.29	4225157	0.1726068 mg/l

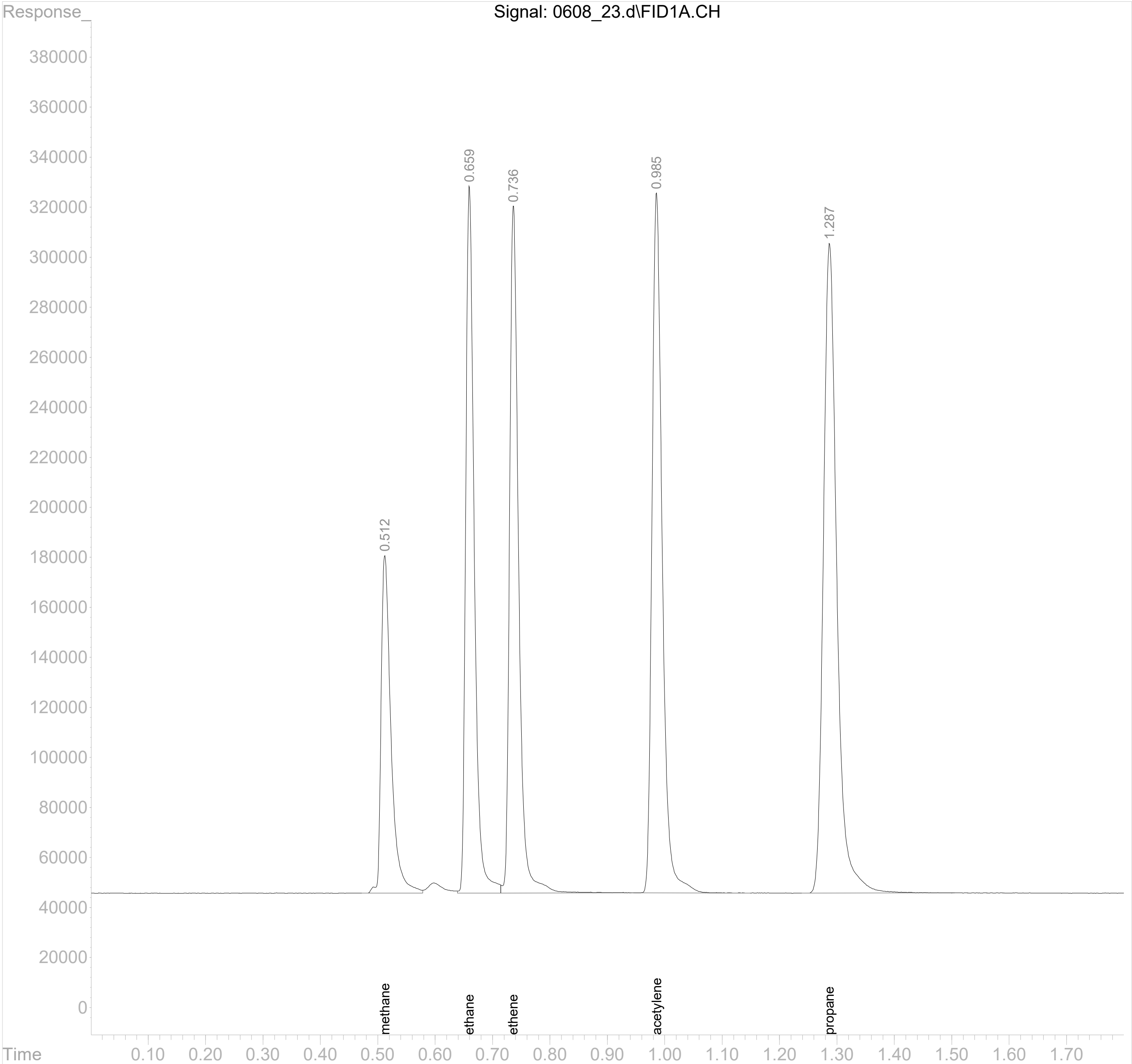
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report   (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_23.d
Signal(s) : FID1A.CH
Acq On   : 08 Jun 2023 11:18 am
Operator :
Sample   : CCV AGC 3.0 RSK175
Misc   :
ALS Vial : 23   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:20:56 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



CONTINUING CALIBRATION VERIFICATION
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Lab File ID:	0608_23-2	**Analysis date/time:**	06/08/23 11:18
Analytical Method:	RSK175	**Sample ID:**	CCV

Analyte	**Avg. RRF**	**RRF**	**Min. RRF**	**Diff.** *%*	**Max Diff.** *%*	**True Value** *mg/l*	**Result** *mg/l*	**Result** *% Rec.*	**Limits** *%*
ETHANE	24835840	22651900		8.79		0.1290	0.1177	91.20	80 - 120
ETHENE	25246600	23663620		6.27		0.1270	0.1190	93.70	80 - 120
METHANE	24042020	23912420		0.5390		0.0678	0.06743	99.50	80 - 120

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

Quantitation Report (QT Reviewed)

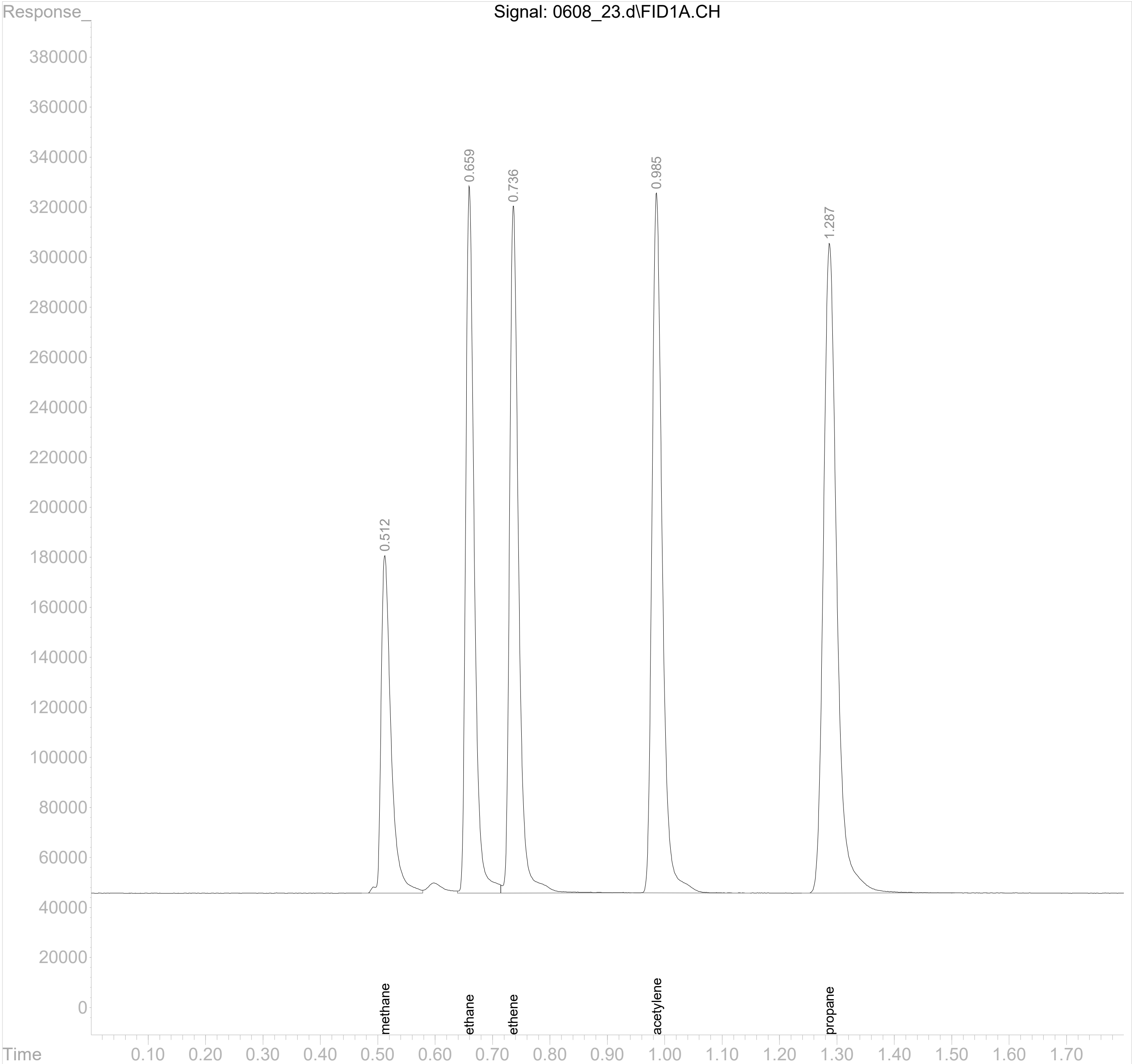
```
Data Path : C:\msdchem\1\data\060823\
Data File : 0608_23.d
Signal(s) : FID1A.CH
Acq On    : 08 Jun 2023  11:18 am
Operator  :
Sample    : CCV AGC 3.0 RSK175
Misc      :
ALS Vial  : 23   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:20:56 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```

	Compound	R.T.	Response	Conc Units
	Target Compounds			
1)	methane	0.51	1621262	0.0674345 mg/l
2)	ethane	0.66	2922095	0.1176564 mg/l
3)	ethene	0.74	3005280	0.1190370 mg/l
4)	acetylene	0.99	3451102	0.1938006 mg/l
5)	propane	1.29	4225157	0.1726068 mg/l

(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060823\
Data File : 0608_23.d
Signal(s) : FID1A.CH
Acq On    : 08 Jun 2023  11:18 am
Operator  :
Sample    : CCV AGC 3.0 RSK175
Misc      :
ALS Vial  : 23   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:20:56 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



7C-OR

CONTINUING CALIBRATION VERIFICATION
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Lab File ID:	0608_27A-1	**Analysis date/time:**	06/08/23 11:40
Analytical Method:	RSK175	**Sample ID:**	CCV

Analyte	**Avg. RRF**	**RRF**	**Min. RRF**	**Diff.**	**Max Diff.**	**True Value**	**Result**	**Result**	**Limits**
				%	*%*	*mg/l*	*mg/l*	*% Rec.*	*%*
ETHANE	24835840	22826460		8.09		0.1290	0.1186	91.90	80 - 120
ETHENE	25246600	23841570		5.57		0.1270	0.1199	94.40	80 - 120
METHANE	24042020	22858750		4.92		0.0678	0.06446	95.10	80 - 120

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_27A.d
Signal(s) : FID1A.CH
Acq On : 08 Jun 2023 11:40 am
Operator :
Sample : CCV AGC 3.0 RSK175
Misc :
ALS Vial : 27 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:43:25 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
	Target Compounds			
1)	methane	0.51	1549823	0.0644631 mg/l
2)	ethane	0.66	2944613	0.1185631 mg/l
3)	ethene	0.74	3027879	0.1199322 mg/l
4)	acetylene	0.99	3472421	0.1949978 mg/l
5)	propane	1.29	4244364	0.1733915 mg/l

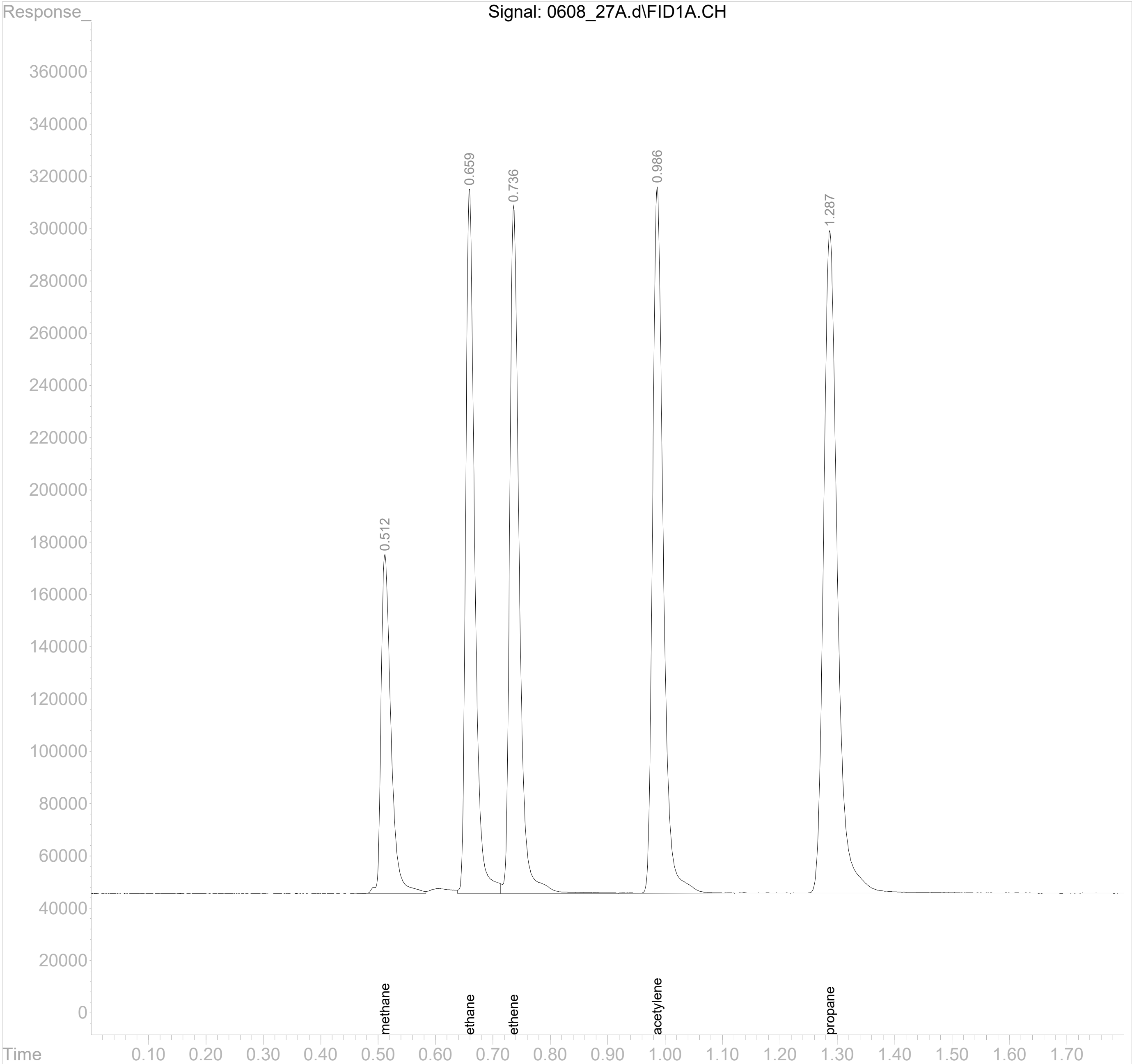
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report   (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_27A.d
Signal(s) : FID1A.CH
Acq On   : 08 Jun 2023 11:40 am
Operator :
Sample   : CCV AGC 3.0 RSK175
Misc   :
ALS Vial : 27   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:43:25 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



7C-OR

CONTINUING CALIBRATION VERIFICATION
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Lab File ID:	0609B_01-1	**Analysis date/time:**	06/09/23 10:53
Analytical Method:	RSK175	**Sample ID:**	ICV

Analyte	**Avg. RRF**	**RRF**	**Min. RRF**	**Diff.** *%*	**Max Diff.** *%*	**True Value** *mg/l*	**Result** *mg/l*	**Result** *% Rec.*	**Limits** *%*
ETHANE	24835840	22884750		7.86		0.1290	0.1189	92.20	80 - 120
ETHENE	25246600	23618880		6.45		0.1270	0.1188	93.50	80 - 120
METHANE	24042020	24871240		3.45		0.0678	0.07014	103	80 - 120

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

Quantitation Report (QT Reviewed)

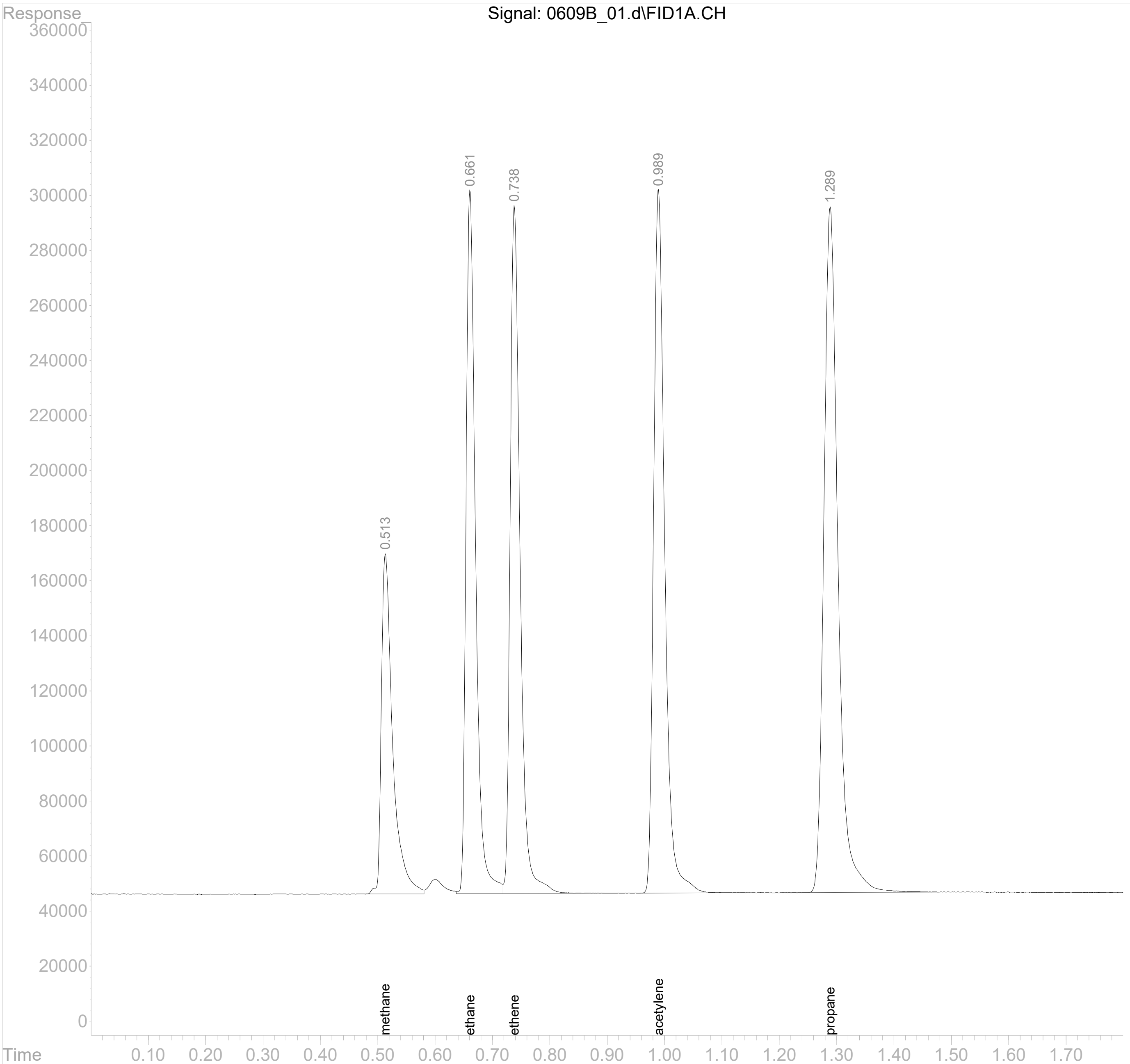
```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_01.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  10:53 am
Operator  :
Sample    : ICVLCS AGC 3.0 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 10:57:51 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	1686270	0.0701385 mg/l
2)	ethane	0.66	2952133	0.1188659 mg/l
3)	ethene	0.74	2999598	0.1188119 mg/l
4)	acetylene	0.99	3415635	0.1918089 mg/l
5)	propane	1.29	4262723	0.1741415 mg/l

(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_01.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  10:53 am
Operator  :
Sample    : ICVLCS AGC 3.0 RSK175
Misc      :
ALS Vial  : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 10:57:51 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



7C-OR

CONTINUING CALIBRATION VERIFICATION
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Lab File ID:	0609B_12-1	**Analysis date/time:**	06/09/23 11:29
Analytical Method:	RSK175	**Sample ID:**	CCV

Analyte	**Avg. RRF**	**RRF**	**Min. RRF**	**Diff.**	**Max Diff.**	**True Value**	**Result**	**Result**	**Limits**
				%	*%*	*mg/l*	*mg/l*	*% Rec.*	*%*
ETHANE	24835840	22367800		9.94		0.1290	0.1162	90.10	80 - 120
ETHENE	25246600	23255790		7.89		0.1270	0.1170	92.10	80 - 120
METHANE	24042020	23675780		1.52		0.0678	0.06677	98.50	80 - 120

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_12.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  11:29 am
Operator  :
Sample    : CCV AGC 3.0 RSK175
Misc      :
ALS Vial  : 12   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:32:11 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```

	Compound	R.T.	Response	Conc Units
	Target Compounds			
1)	methane	0.51	1605218	0.0667672 mg/l
2)	ethane	0.66	2885446	0.1161807 mg/l
3)	ethene	0.74	2953486	0.1169855 mg/l
4)	acetylene	0.99	3382643	0.1899562 mg/l
5)	propane	1.29	4217796	0.1723061 mg/l

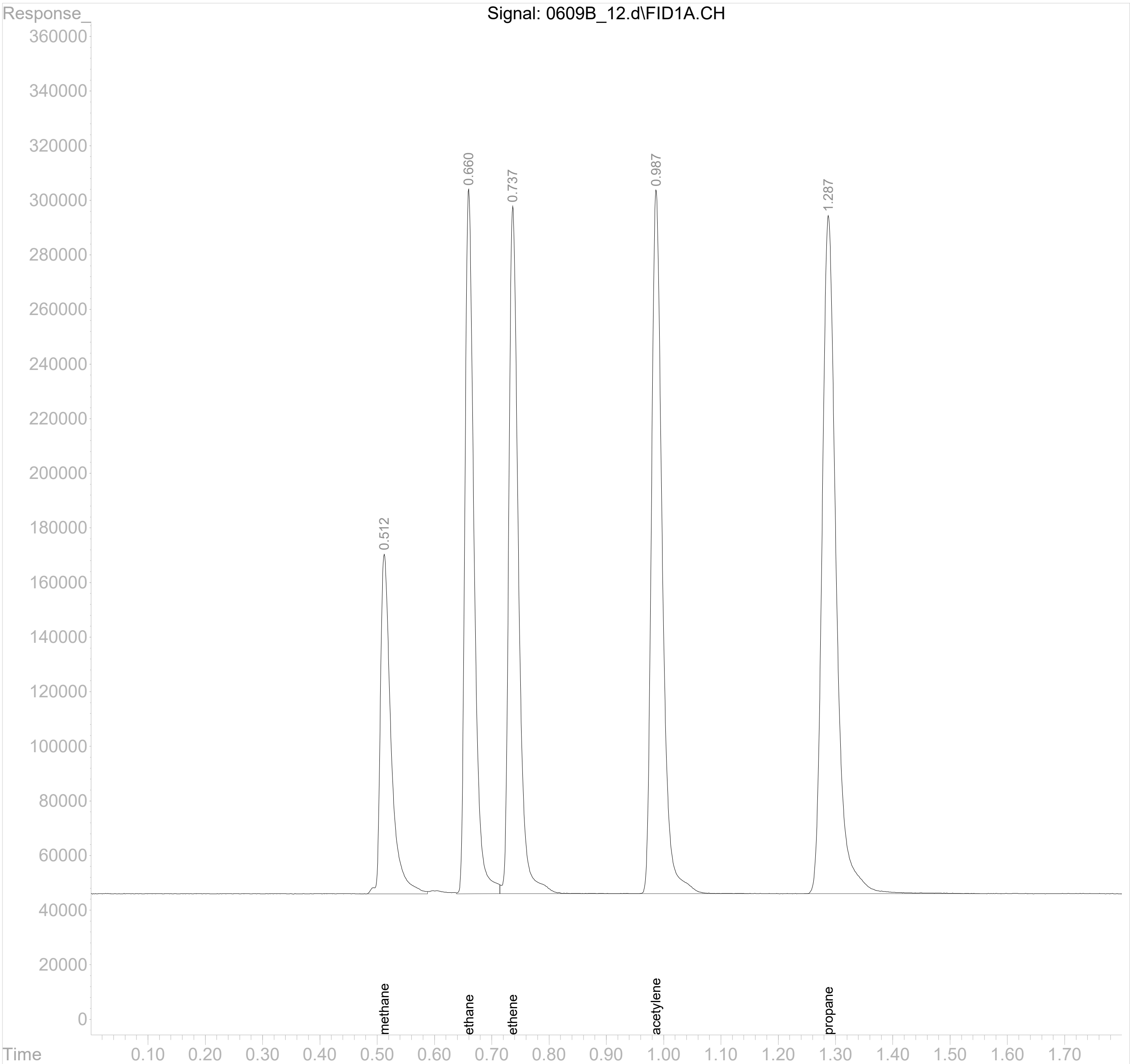
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_12.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 11:29 am
Operator :
Sample : CCV AGC 3.0 RSK175
Misc :
ALS Vial : 12 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:32:11 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



7C-OR

CONTINUING CALIBRATION VERIFICATION
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Lab File ID:	0609B_12-2	**Analysis date/time:**	06/09/23 11:29
Analytical Method:	RSK175	**Sample ID:**	CCV

Analyte	**Avg. RRF**	**RRF**	**Min. RRF**	**Diff.**	**Max Diff.**	**True Value**	**Result**	**Result**	**Limits**
				%	*%*	*mg/l*	*mg/l*	*% Rec.*	*%*
ETHANE	24835840	22367800		9.94		0.1290	0.1162	90.10	80 - 120
ETHENE	25246600	23255790		7.89		0.1270	0.1170	92.10	80 - 120
METHANE	24042020	23675780		1.52		0.0678	0.06677	98.50	80 - 120

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_12.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 11:29 am
Operator :
Sample : CCV AGC 3.0 RSK175
Misc :
ALS Vial : 12 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:32:11 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

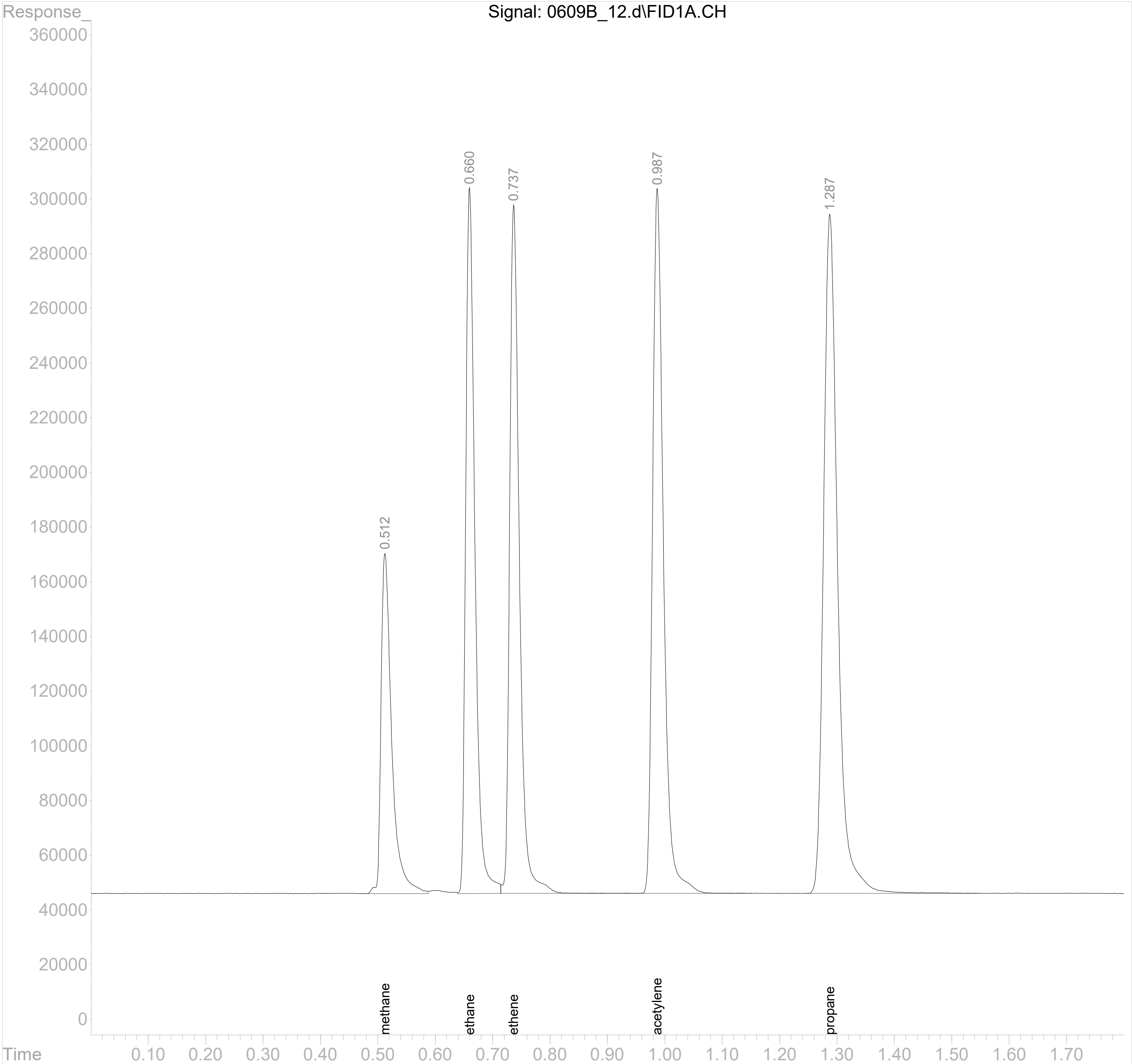
Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	1605218	0.0667672 mg/l
2)	ethane	0.66	2885446	0.1161807 mg/l
3)	ethene	0.74	2953486	0.1169855 mg/l
4)	acetylene	0.99	3382643	0.1899562 mg/l
5)	propane	1.29	4217796	0.1723061 mg/l

(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report   (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_12.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  11:29 am
Operator  :
Sample    : CCV AGC 3.0 RSK175
Misc      :
ALS Vial  : 12   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:32:11 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



7C-OR

CONTINUING CALIBRATION VERIFICATION
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Lab File ID:	0609B_23-1	**Analysis date/time:**	06/09/23 13:23
Analytical Method:	RSK175	**Sample ID:**	CCV

Analyte	**Avg. RRF**	**RRF**	**Min. RRF**	**Diff.**	**Max Diff.**	**True Value**	**Result**	**Result**	**Limits**
				%	*%*	*mg/l*	*mg/l*	*% Rec.*	*%*
ETHANE	24835840	22808060		8.16		0.1290	0.1185	91.90	80 - 120
ETHENE	25246600	23634720		6.38		0.1270	0.1189	93.60	80 - 120
METHANE	24042020	24617080		2.39		0.0678	0.06942	102	80 - 120

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_23.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 13:23 pm
Operator :
Sample : CCV AGC 3.0 RSK175
Misc :
ALS Vial : 23 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:26:50 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
	Target Compounds			
1)	methane	0.51	1669038	0.0694217 mg/l
2)	ethane	0.66	2942239	0.1184675 mg/l
3)	ethene	0.74	3001610	0.1188916 mg/l
4)	acetylene	0.98	3448141	0.1936343 mg/l
5)	propane	1.29	4283794	0.1750023 mg/l

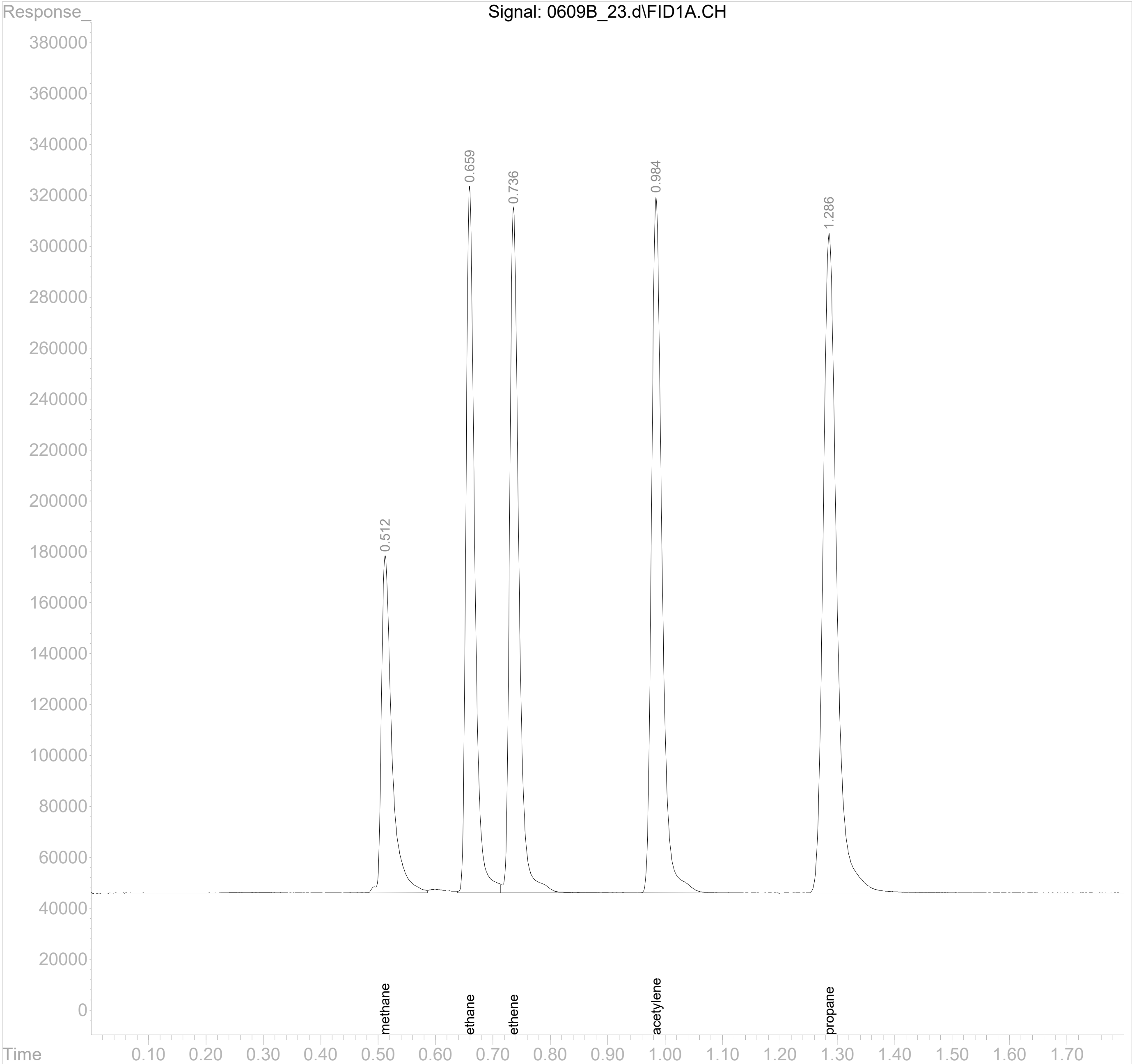
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report   (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_23.d
Signal(s) : FID1A.CH
Acq On   : 09 Jun 2023 13:23 pm
Operator :
Sample   : CCV AGC 3.0 RSK175
Misc   :
ALS Vial : 23   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:26:50 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



CONTINUING CALIBRATION VERIFICATION
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Lab File ID:	0609B_23-2	**Analysis date/time:**	06/09/23 13:23
Analytical Method:	RSK175	**Sample ID:**	CCV

Analyte	**Avg. RRF**	**RRF**	**Min. RRF**	**Diff.** *%*	**Max Diff.** *%*	**True Value** *mg/l*	**Result** *mg/l*	**Result** *% Rec.*	**Limits** *%*
ETHANE	24835840	22808060		8.16		0.1290	0.1185	91.90	80 - 120
ETHENE	25246600	23634720		6.38		0.1270	0.1189	93.60	80 - 120
METHANE	24042020	24617080		2.39		0.0678	0.06942	102	80 - 120

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_23.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 13:23 pm
Operator :
Sample : CCV AGC 3.0 RSK175
Misc :
ALS Vial : 23 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:26:50 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	1669038	0.0694217 mg/l
2)	ethane	0.66	2942239	0.1184675 mg/l
3)	ethene	0.74	3001610	0.1188916 mg/l
4)	acetylene	0.98	3448141	0.1936343 mg/l
5)	propane	1.29	4283794	0.1750023 mg/l

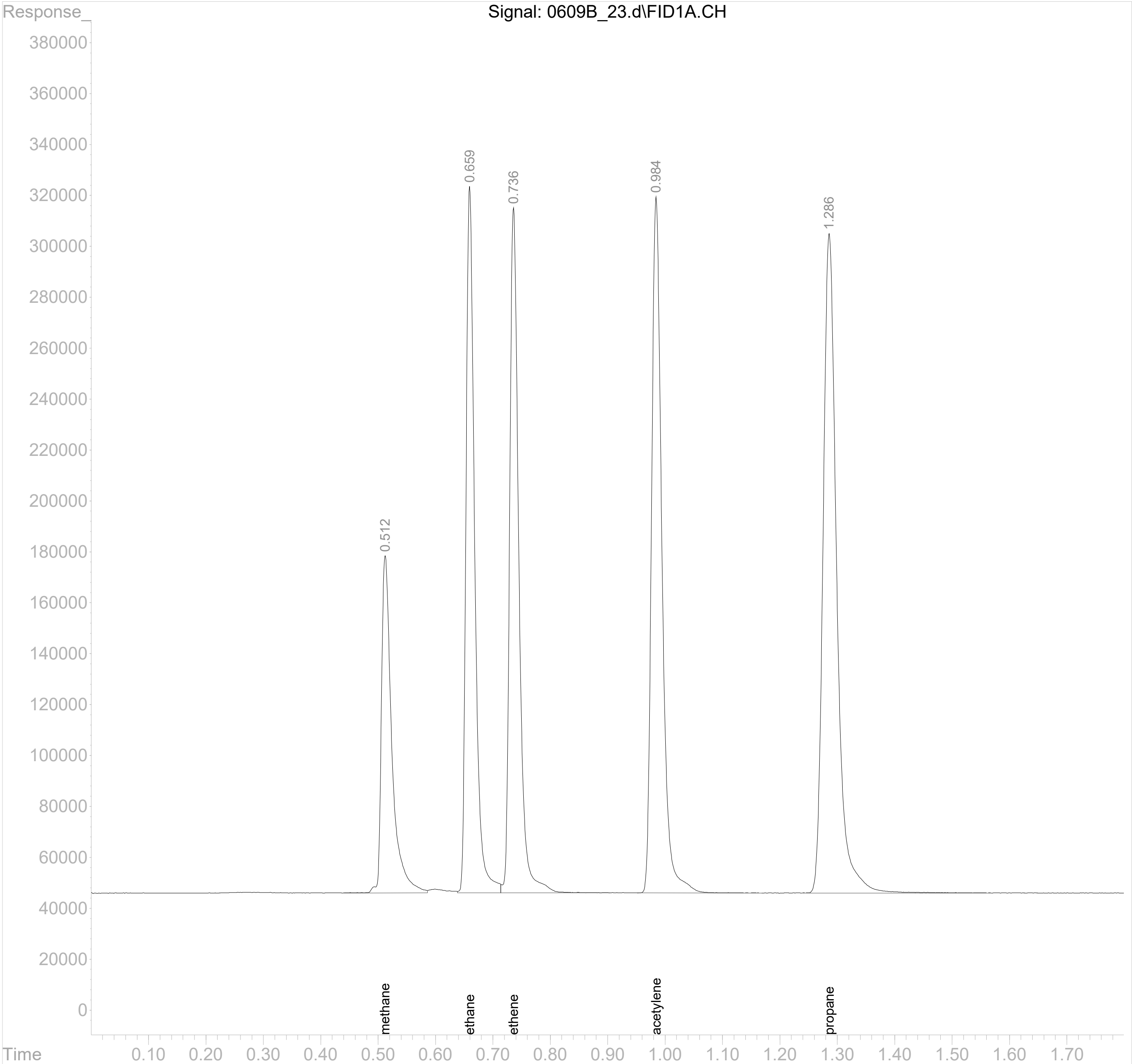
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_23.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 13:23 pm
Operator :
Sample : CCV AGC 3.0 RSK175
Misc :
ALS Vial : 23 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:26:50 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



7C-OR

CONTINUING CALIBRATION VERIFICATION
SINGLE COMPONENT ANALYTES

SDG:	L1623041	**Calibration (begin) date/time:**	11/23/22 12:44
Instrument ID:	AIRGC2	**Calibration (end) date/time:**	11/23/22 13:32
Lab File ID:	0609B_29a-1	**Analysis date/time:**	06/09/23 13:46
Analytical Method:	RSK175	**Sample ID:**	CCV

Analyte	**Avg. RRF**	**RRF**	**Min. RRF**	**Diff.**	**Max Diff.**	**True Value**	**Result**	**Result**	**Limits**
				%	*%*	*mg/l*	*mg/l*	*% Rec.*	*%*
ETHANE	24835840	22555090		9.18		0.1290	0.1172	90.90	80 - 120
ETHENE	25246600	23403160		7.30		0.1270	0.1177	92.70	80 - 120
METHANE	24042020	24580910		2.24		0.0678	0.06932	102	80 - 120

D: Surrogate recovery cannot be used for control limit evaluation due to dilution.

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_29a.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  13:46 pm
Operator  :
Sample    : CCV AGC 3.0 RSK175
Misc      :
ALS Vial  : 29   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:52:03 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	1666586	0.0693197 mg/l
2)	ethane	0.66	2909607	0.1171536 mg/l
3)	ethene	0.74	2972201	0.1177268 mg/l
4)	acetylene	0.99	3408360	0.1914004 mg/l
5)	propane	1.29	4241056	0.1732563 mg/l

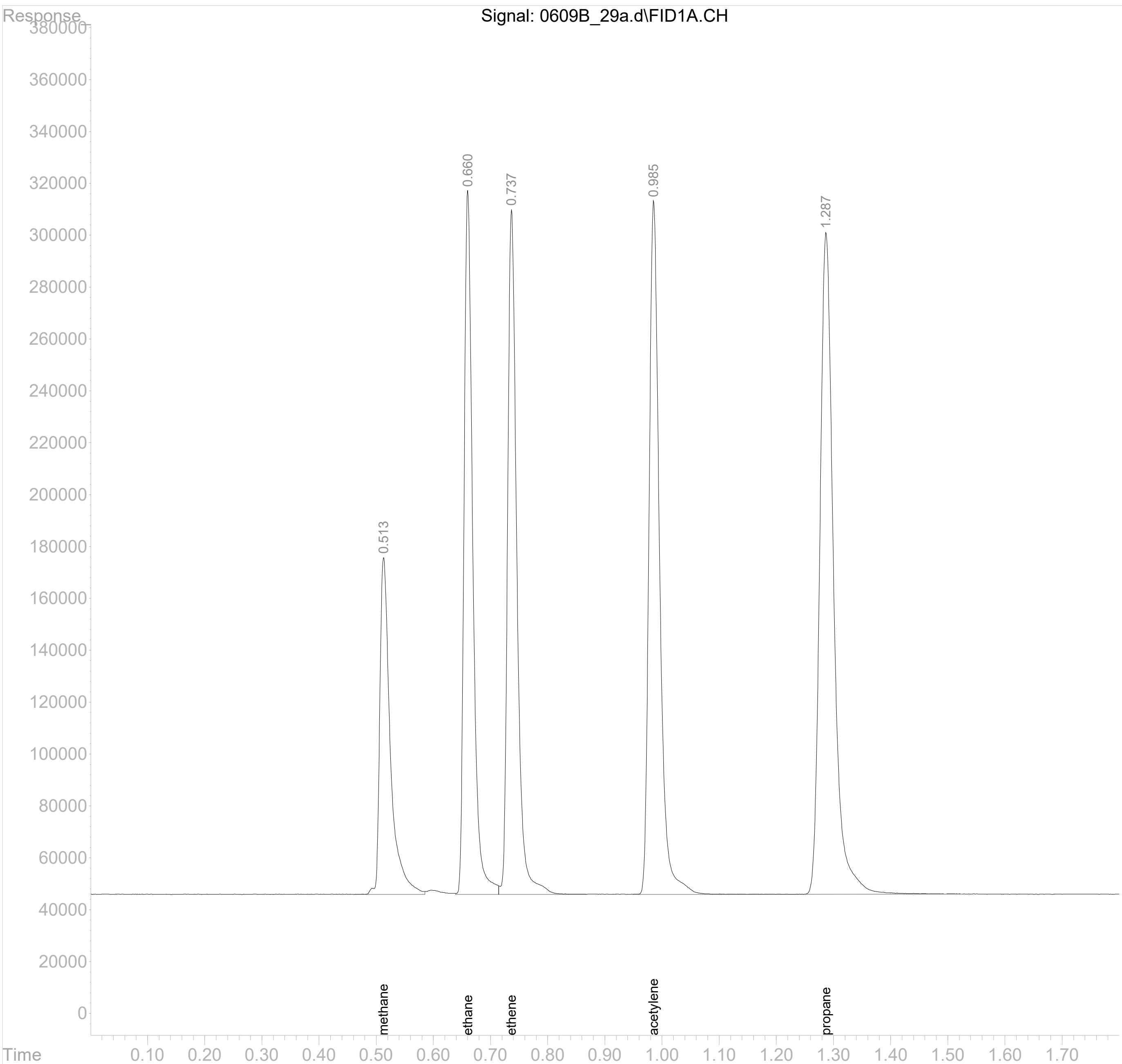
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_29a.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 13:46 pm
Operator :
Sample : CCV AGC 3.0 RSK175
Misc :
ALS Vial : 29 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:52:03 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



ANALYTICAL SEQUENCE

SDG:	L1623041	**Analytical Method:**	RSK175
Instrument ID:	AIRGC2	**Calibration Start Date:**	11/23/22 12:44
		Calibration End Date:	11/23/22 13:32

Client Sample ID	Lab Sample ID	File ID	Analysis Date Time	Dilution	Batch
CAL	1	1123A_01	11/23/22 12:44		
CAL	2	1123A_02	11/23/22 12:48		
CAL	3.0	1123A_03	11/23/22 12:51		
CAL	4	1123A_04	11/23/22 13:09		
CAL	5	1123A_05	11/23/22 13:20		
CAL	6	1123A_06	11/23/22 13:26		
CAL	7	1123A_07	11/23/22 13:32		
SSCV	AIRGC2112322A1123A_09-1605379	1123A_09-1	11/23/22 13:43		
ICV	AIRGC20608230608_01-1605379	0608_01-1	06/08/23 09:45		
LCS	R3934254-1	0608_01LCS	06/08/23 09:45	1	WG2073383
BLANK	R3934254-2	0608_02	06/08/23 09:48	1	WG2073383
OS	L1619855-03	0608_03	06/08/23 09:54		
CCV	AIRGC20608230608_12-1605379	0608_12-1	06/08/23 10:26		
CCV	AIRGC20608230608_12-2605379	0608_12-2	06/08/23 10:26		
DUP	R3934254-3	0608_14	06/08/23 10:35	1	WG2073383
OS	L1622529-13	0608_15	06/08/23 10:41		
OT07-EB-060123	L1623041-01	0608_21	06/08/23 11:10	1	WG2073383
OT07-MW02-060123	L1623041-02	0608_22	06/08/23 11:13	1	WG2073383
CCV	AIRGC20608230608_23-1605379	0608_23-1	06/08/23 11:18		
CCV	AIRGC20608230608_23-2605379	0608_23-2	06/08/23 11:18		
OT07-MW05-060123	L1623041-04	0608_24	06/08/23 11:22	1	WG2073383
OT07-MW06-060123	L1623041-05	0608_25	06/08/23 11:29	1	WG2073383
DUP	R3934254-4	0608_26	06/08/23 11:37	1	WG2073383
CCV	AIRGC20608230608_27A-1605379	0608_27A-1	06/08/23 11:40		
LCSD	R3934254-5	0608_27	06/08/23 11:40	1	WG2073383
ICV	AIRGC2060923B0609B_01-1605379	0609B_01-1	06/09/23 10:53		
LCS	R3934861-1	0609B_01LCS	06/09/23 10:53	1	WG2074059
BLANK	R3934861-2	0609B_02	06/09/23 11:03	1	WG2074059
OT07-MW04-060123	L1623041-03	0609B_03	06/09/23 11:06	1	WG2074059
OT07-MW04-060123MS	L1623041-06	0609B_04	06/09/23 11:08	1	WG2074059
OT07-MW04-060123MSD	L1623041-07	0609B_05	06/09/23 11:11	1	WG2074059
OT07-MW806-060123	L1623041-08	0609B_06	06/09/23 11:14	1	WG2074059
CCV	AIRGC2060923B0609B_12-1605379	0609B_12-1	06/09/23 11:29		
CCV	AIRGC2060923B0609B_12-2605379	0609B_12-2	06/09/23 11:29		
OS	L1623162-01	0609B_13	06/09/23 11:32		
DUP	R3934861-3	0609B_14	06/09/23 12:52	1	WG2074059
CCV	AIRGC2060923B0609B_23-1605379	0609B_23-1	06/09/23 13:23		
CCV	AIRGC2060923B0609B_23-2605379	0609B_23-2	06/09/23 13:23		
OS	L1623669-02	0609B_24	06/09/23 13:27		
DUP	R3934861-4	0609B_26	06/09/23 13:33	1	WG2074059
MS	R3934861-5	0609B_27	06/09/23 13:38	1	WG2074059
MSD	R3934861-6	0609B_28	06/09/23 13:42	1	WG2074059
CCV	AIRGC2060923B0609B_29a-1605379	0609B_29a-1	06/09/23 13:46		
LCSD	R3934861-7	0609B_29	06/09/23 13:46	1	WG2074059

DETECTION LIMIT SUMMARY

Lab Sample IDs:	L1623041-01,02,03,04,05,06,07,08	**Analytical Method:**	RSK175
Matrix:	GW	**Prep Method:**	RSK175

Analyte	CAS	MDL *mg/l*	RDL *mg/l*
Methane	74-82-8	0.002910	0.0120
Ethane	74-84-0	0.004070	0.02
Ethene	74-85-1	0.004260	0.02

SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
R3934254-2

Lab Sample ID:	R3934254-2	**SDG:**	L1623041
Client Sample ID:	BLANK	**Collected Date/Time:**	________________
Lab File ID:	0608_02	**Received Date/Time:**	________________
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/08/23 09:48
Analytical Batch:	WG2073383	**Analysis Date/Time:**	06/08/23 09:48
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

Analyte	CAS	RT	Result *ug/l*	Qualifier	DL *ug/l*	LOD *ug/l*	LOQ *ug/l*
Methane	74-82-8	0	6.00	U	2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report (QT Reviewed)

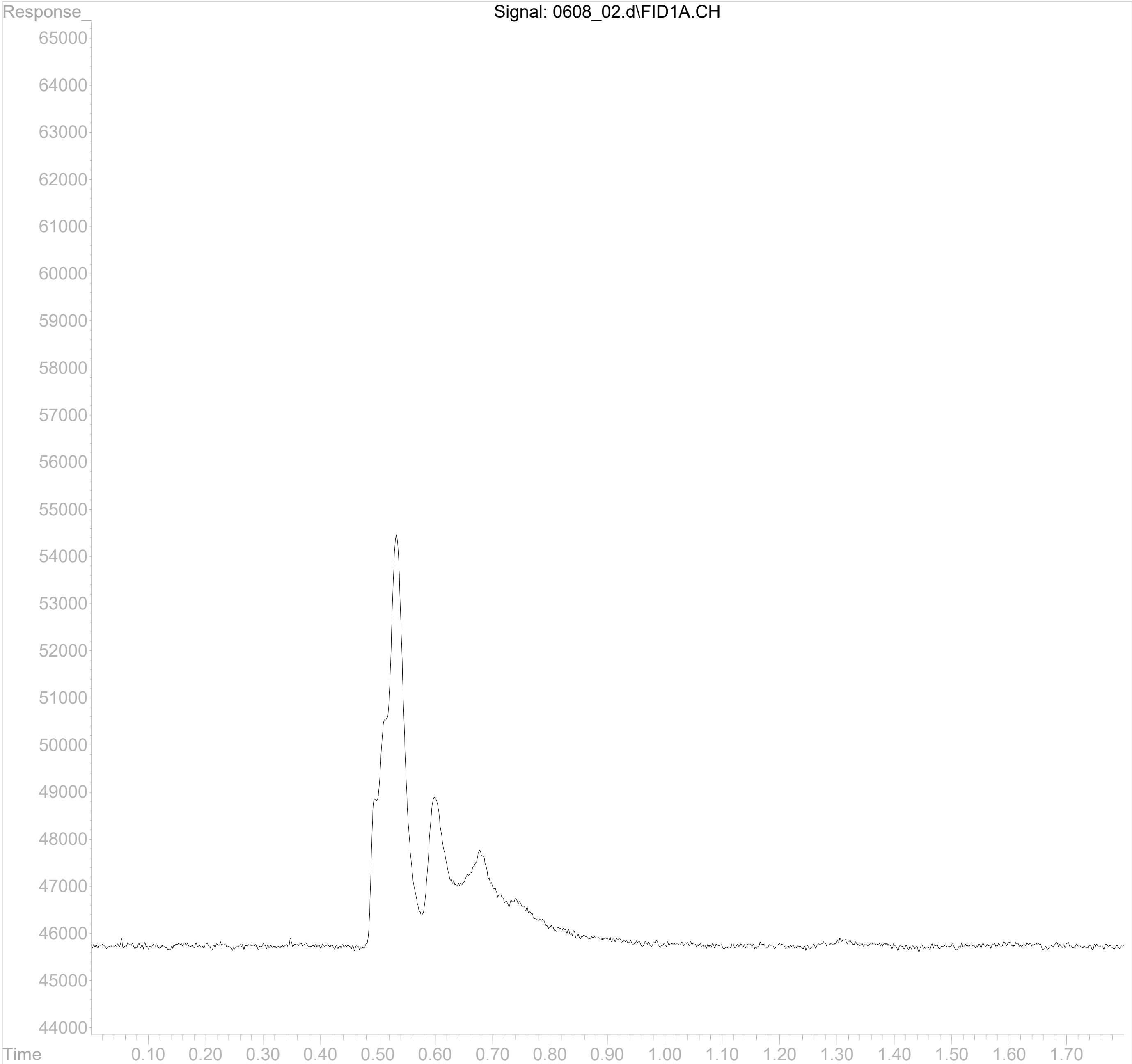
```
Data Path : C:\msdchem\1\data\060823\
Data File : 0608_02.d
Signal(s) : FID1A.CH
Acq On    : 08 Jun 2023   9:48 am
Operator  :
Sample    : BLANK 1x WG2073383
Misc      :
ALS Vial  : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 09:50:47 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :

        Compound                     R.T.       Response    Conc Units
 -------------------------------------------------------------------------------

 Target Compounds
 ---------------------------------------------------------------------------

(f)=RT Delta > 1/2 Window                                (m)=manual int.
```

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_02.d
Signal(s) : FID1A.CH
Acq On : 08 Jun 2023 9:48 am
Operator :
Sample : BLANK 1x WG2073383
Misc :
ALS Vial : 2 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 09:50:47 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



1A-OR

SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.: R3934861-2

Lab Sample ID:	R3934861-2	**SDG:**	L1623041
Client Sample ID:	BLANK	**Collected Date/Time:**	________
Lab File ID:	0609B_02	**Received Date/Time:**	________
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/09/23 11:03
Analytical Batch:	WG2074059	**Analysis Date/Time:**	06/09/23 11:03
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________
Total Solids (%):	________	**Final Wt/Vol:**	20 mL

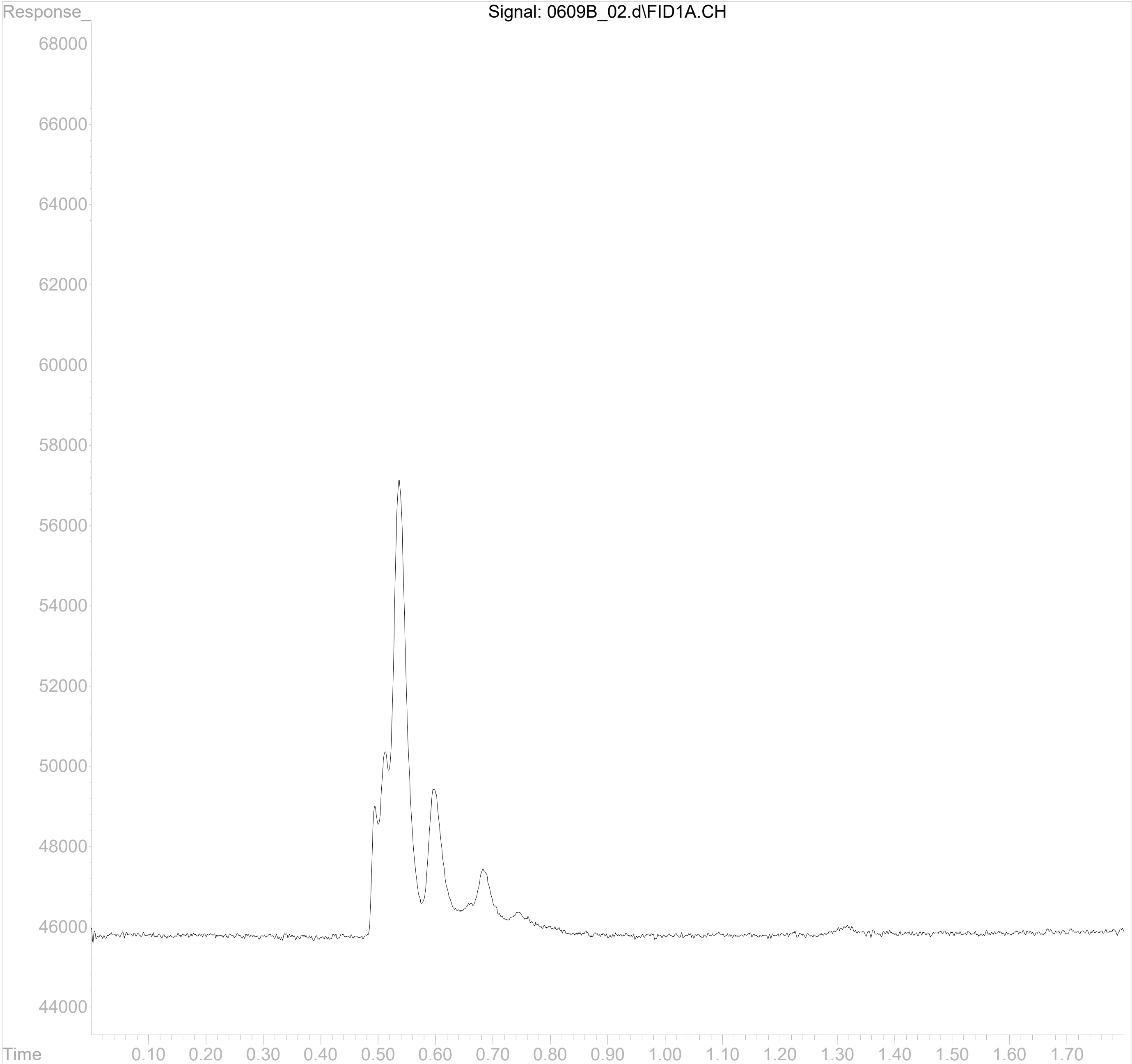
Analyte	CAS	RT	Result	Qualifier	DL	LOD	LOQ
			ug/l		*ug/l*	*ug/l*	*ug/l*
Methane	74-82-8	0	6.00	U	2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report   (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_02.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  11:03 am
Operator  :
Sample    : BLANK 1x WG2074059
Misc      :
ALS Vial  : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:07:51 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :

        Compound                     R.T.       Response    Conc Units
 --------------------------------------------------------------------------------

 Target Compounds
 ---------------------------------------------------------------------------

(f)=RT Delta > 1/2 Window                                (m)=manual int.
```

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_02.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  11:03 am
Operator  :
Sample    : BLANK 1x WG2074059
Misc      :
ALS Vial  : 2   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 11:07:51 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



1A-OR

SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
R3934254-3

Lab Sample ID:	R3934254-3	**SDG:**	L1623041
Client Sample ID:	DUP	**Collected Date/Time:**	05/22/23 13:29
Lab File ID:	0608_14	**Received Date/Time:**	05/25/23 08:45
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/08/23 10:35
Analytical Batch:	WG2073383	**Analysis Date/Time:**	06/08/23 10:35
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

Analyte	**CAS**	**RT**	**Result**	**Qualifier**	**DL**	**LOD**	**LOQ**
			ug/l		*ug/l*	*ug/l*	*ug/l*
Methane	74-82-8	0	6.00	U	2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report (QT Reviewed)

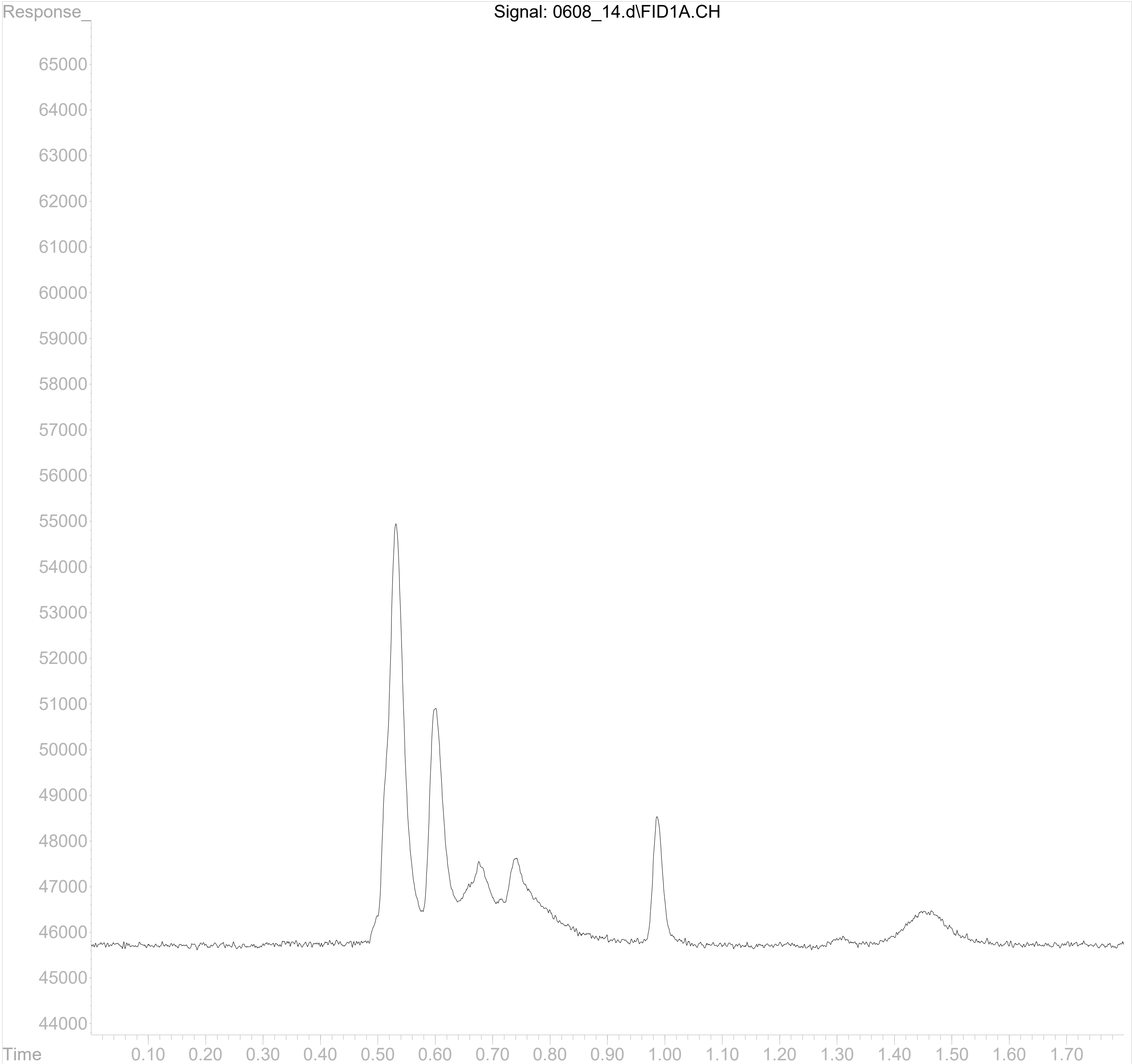
```
Data Path : C:\msdchem\1\data\060823\
Data File : 0608_14.d
Signal(s) : FID1A.CH
Acq On    : 08 Jun 2023  10:35 am
Operator  :
Sample    : DUP 1x WG2073383 L1619855-03
Misc      :
ALS Vial  : 14   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 10:38:38 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :

        Compound                    R.T.      Response    Conc Units
 -------------------------------------------------------------------------------

 Target Compounds
 ---------------------------------------------------------------------------

(f)=RT Delta > 1/2 Window                                (m)=manual int.
```

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_14.d
Signal(s) : FID1A.CH
Acq On : 08 Jun 2023 10:35 am
Operator :
Sample : DUP 1x WG2073383 L1619855-03
Misc :
ALS Vial : 14 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 10:38:38 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
R3934254-1

Lab Sample ID:	R3934254-1	**SDG:**	L1623041
Client Sample ID:	LCS	**Collected Date/Time:**	________________
Lab File ID:	0608_01LCS	**Received Date/Time:**	________________
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/08/23 09:45
Analytical Batch:	WG2073383	**Analysis Date/Time:**	06/08/23 09:45
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

Analyte	CAS	RT	Result *ug/l*	Qualifier	DL *ug/l*	LOD *ug/l*	LOQ *ug/l*
Methane	74-82-8	0.51	67.0		2.91	6.00	12.0
Ethane	74-84-0	0.66	118		4.07	10.0	20.0
Ethene	74-85-1	0.74	119		4.26	10.0	20.0

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_01.d
Signal(s) : FID1A.CH
Acq On : 08 Jun 2023 9:45 am
Operator :
Sample : ICVLCS AGC 3.0 RSK175
Misc :
ALS Vial : 1 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 09:47:35 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	1610880	0.0670027 mg/l
2)	ethane	0.66	2938043	0.1182985 mg/l
3)	ethene	0.74	2993141	0.1185562 mg/l
4)	acetylene	0.99	3436165	0.1929618 mg/l
5)	propane	1.29	4262533	0.1741337 mg/l

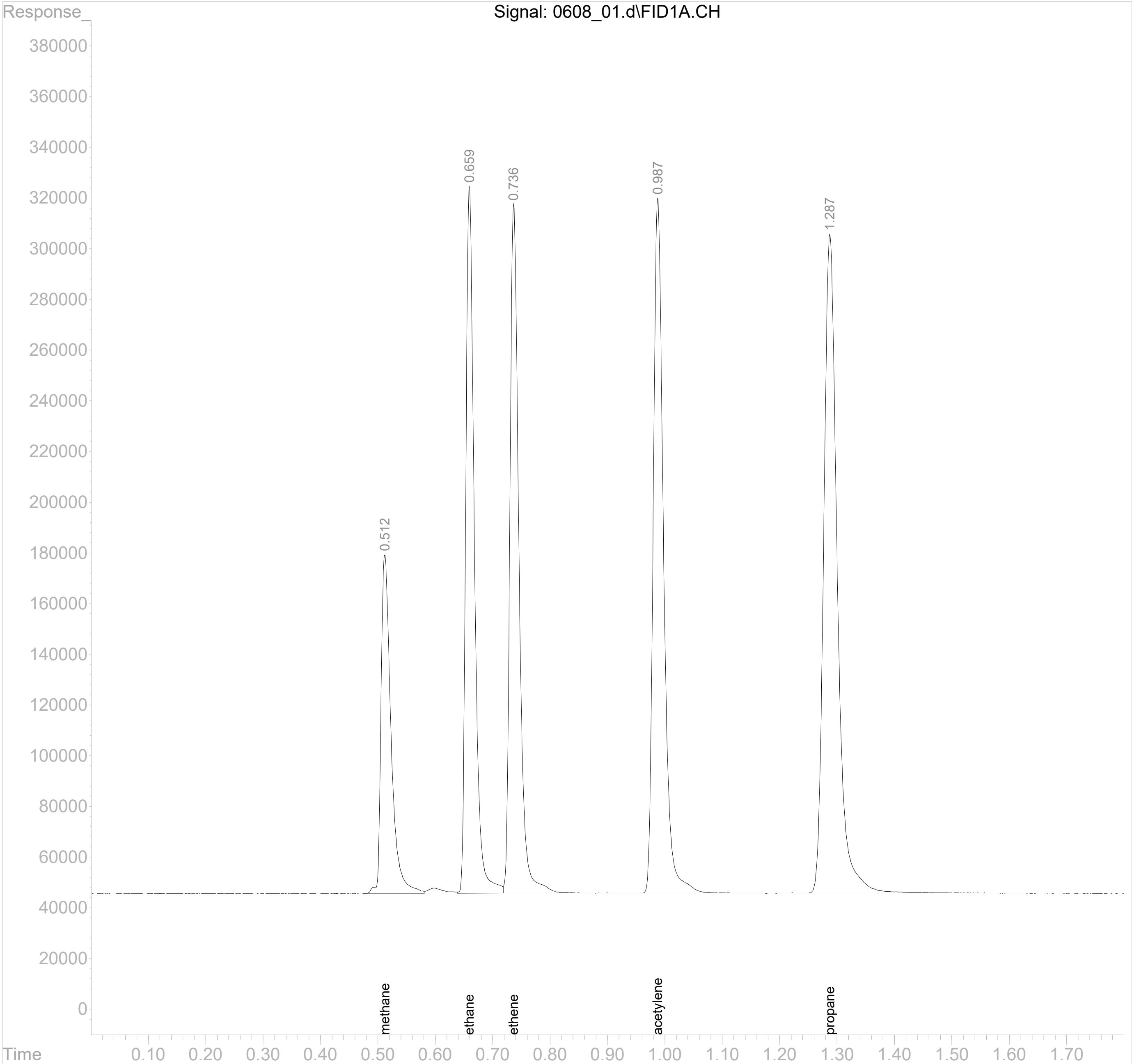
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_01.d
Signal(s) : FID1A.CH
Acq On : 08 Jun 2023 9:45 am
Operator :
Sample : ICVLCS AGC 3.0 RSK175
Misc :
ALS Vial : 1 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 09:47:35 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
R3934254-4

Lab Sample ID:	R3934254-4	**SDG:**	L1623041
Client Sample ID:	DUP	**Collected Date/Time:**	06/01/23 22:15
Lab File ID:	0608_26	**Received Date/Time:**	06/03/23 09:00
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/08/23 11:37
Analytical Batch:	WG2073383	**Analysis Date/Time:**	06/08/23 11:37
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

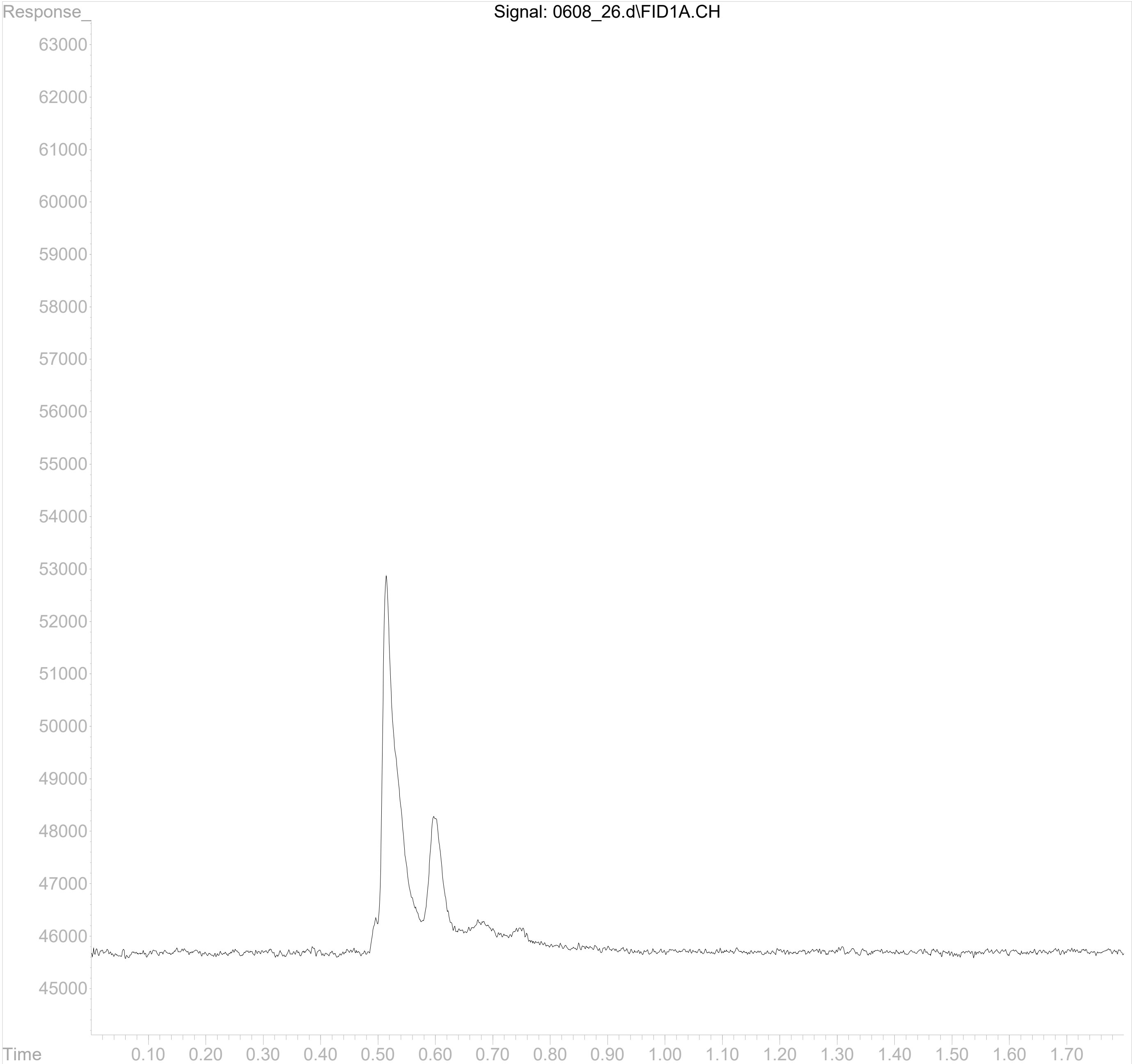
Analyte	**CAS**	**RT**	**Result** *ug/l*	**Qualifier**	**DL** *ug/l*	**LOD** *ug/l*	**LOQ** *ug/l*
Methane	74-82-8	0	6.00	U	2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060823\
Data File : 0608_26.d
Signal(s) : FID1A.CH
Acq On    : 08 Jun 2023  11:37 am
Operator  :
Sample    : DUP 1x WG2073383 L1622529-13
Misc      :
ALS Vial  : 26   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:39:45 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :

        Compound                     R.T.       Response    Conc Units
 -------------------------------------------------------------------------------

 Target Compounds
 ----------------------------------------------------------------------------

(f)=RT Delta > 1/2 Window                                (m)=manual int.
```

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060823\
Data File : 0608_26.d
Signal(s) : FID1A.CH
Acq On    : 08 Jun 2023  11:37 am
Operator  :
Sample    : DUP 1x WG2073383 L1622529-13
Misc      :
ALS Vial  : 26   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:39:45 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
R3934861-3

Lab Sample ID:	R3934861-3	**SDG:**	L1623041
Client Sample ID:	DUP	**Collected Date/Time:**	06/05/23 12:55
Lab File ID:	0609B_14	**Received Date/Time:**	06/06/23 15:55
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/09/23 12:52
Analytical Batch:	WG2074059	**Analysis Date/Time:**	06/09/23 12:52
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

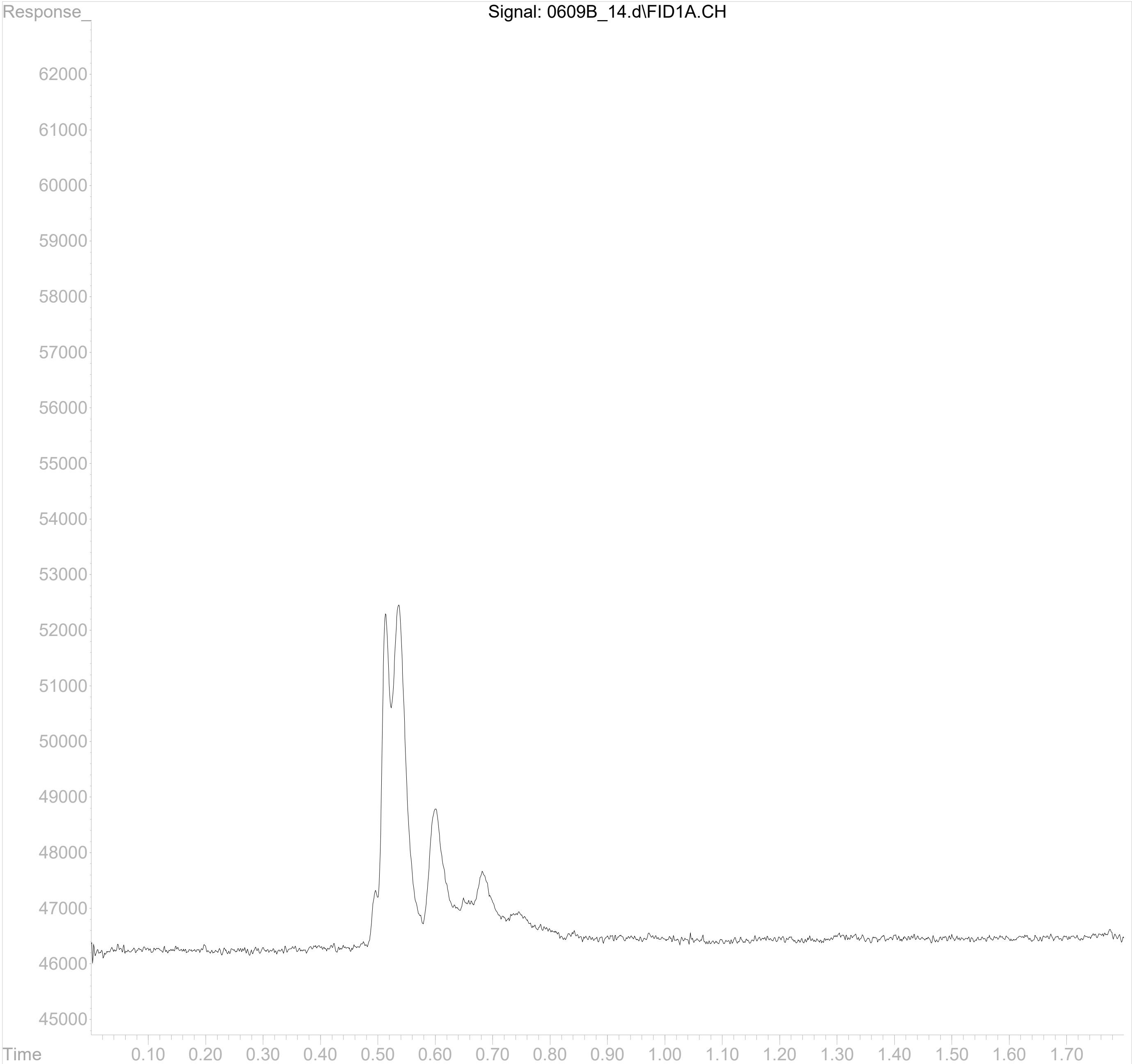
Analyte	CAS	RT	Result *ug/l*	Qualifier	DL *ug/l*	LOD *ug/l*	LOQ *ug/l*
Methane	74-82-8	0	6.00	U	2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_14.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  12:52 pm
Operator  :
Sample    : DUP 1x WG2074059 L1623162-01
Misc      :
ALS Vial  : 14   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 12:55:49 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :

        Compound                     R.T.       Response    Conc Units
 -------------------------------------------------------------------------------

 Target Compounds
 ---------------------------------------------------------------------------

(f)=RT Delta > 1/2 Window                                (m)=manual int.
```

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_14.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  12:52 pm
Operator  :
Sample    : DUP 1x WG2074059 L1623162-01
Misc      :
ALS Vial  : 14   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 12:55:49 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
R3934861-4

Lab Sample ID:	R3934861-4	**SDG:**	L1623041
Client Sample ID:	DUP	**Collected Date/Time:**	06/06/23 13:55
Lab File ID:	0609B_26	**Received Date/Time:**	06/07/23 16:00
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/09/23 13:33
Analytical Batch:	WG2074059	**Analysis Date/Time:**	06/09/23 13:33
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

Analyte	**CAS**	**RT**	**Result** *ug/l*	**Qualifier**	**DL** *ug/l*	**LOD** *ug/l*	**LOQ** *ug/l*
Methane	74-82-8	0.51	687		2.91	6.00	12.0
Ethane	74-84-0	0	10.0	U	4.07	10.0	20.0
Ethene	74-85-1	0	10.0	U	4.26	10.0	20.0

Quantitation Report (QT Reviewed)

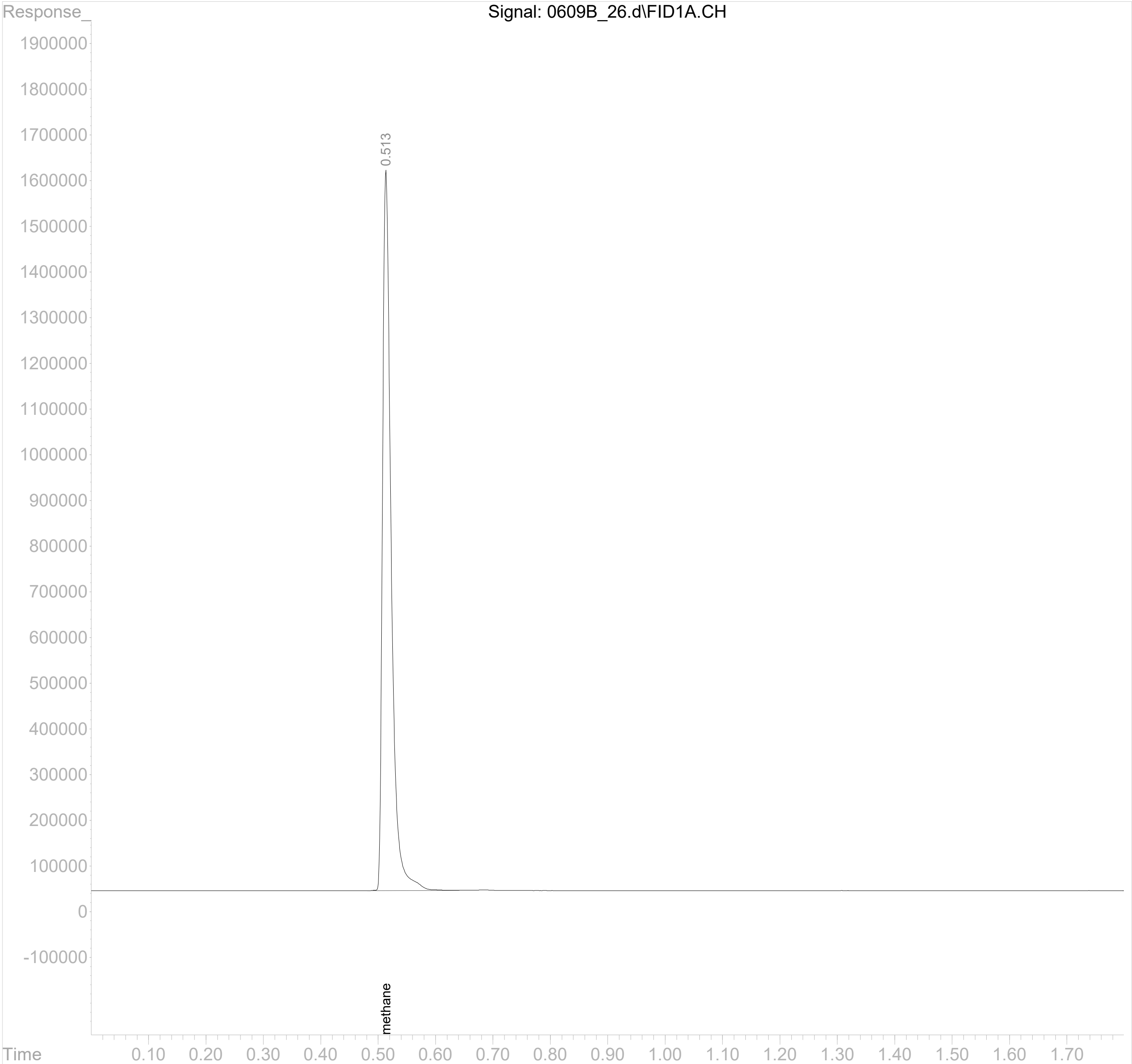
```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_26.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  13:33 pm
Operator  :
Sample    : DUP 1x WG2074059 L1623669-02
Misc      :
ALS Vial  : 26   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:36:43 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	16523004	0.6872553 mg/l

(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_26.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  13:33 pm
Operator  :
Sample    : DUP 1x WG2074059 L1623669-02
Misc      :
ALS Vial  : 26   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:36:43 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
R3934861-1

Lab Sample ID:	R3934861-1	**SDG:**	L1623041
Client Sample ID:	LCS	**Collected Date/Time:**	__________________
Lab File ID:	0609B_01LCS	**Received Date/Time:**	__________________
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/09/23 10:53
Analytical Batch:	WG2074059	**Analysis Date/Time:**	06/09/23 10:53
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	__________________
Total Solids (%):	__________________	**Final Wt/Vol:**	20 mL

Analyte	**CAS**	**RT**	**Result**	**Qualifier**	**DL**	**LOD**	**LOQ**
			ug/l		*ug/l*	*ug/l*	*ug/l*
Methane	74-82-8	0.51	70.1		2.91	6.00	12.0
Ethane	74-84-0	0.66	119		4.07	10.0	20.0
Ethene	74-85-1	0.74	119		4.26	10.0	20.0

Quantitation Report   (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_01.d
Signal(s) : FID1A.CH
Acq On   : 09 Jun 2023 10:53 am
Operator :
Sample   : ICVLCS AGC 3.0 RSK175
Misc   :
ALS Vial : 1   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 10:57:51 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
	Target Compounds			
1)	methane	0.51	1686270	0.0701385 mg/l
2)	ethane	0.66	2952133	0.1188659 mg/l
3)	ethene	0.74	2999598	0.1188119 mg/l
4)	acetylene	0.99	3415635	0.1918089 mg/l
5)	propane	1.29	4262723	0.1741415 mg/l

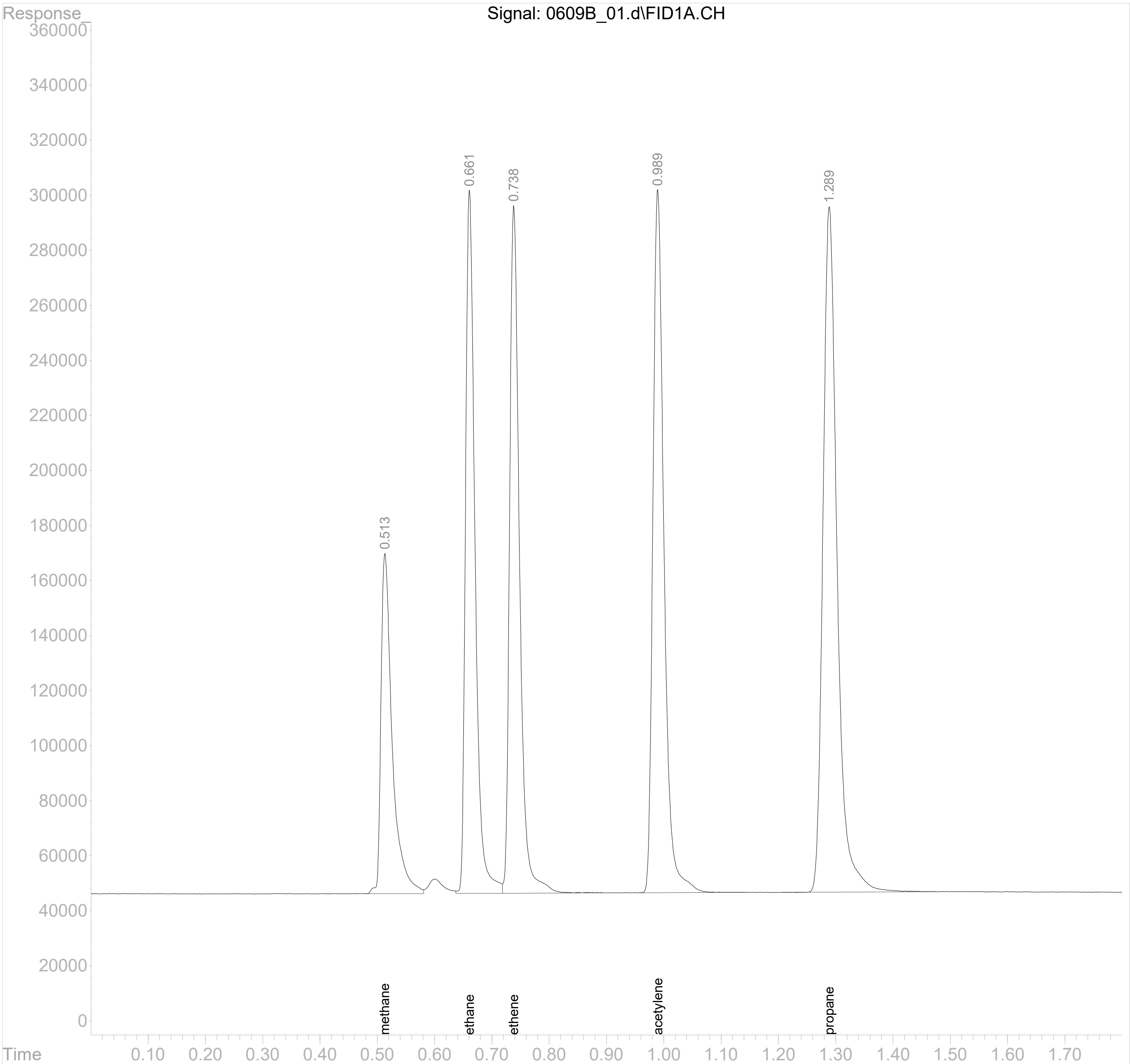
(f)=RT Delta > 1/2 Window   (m)=manual int.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_01.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 10:53 am
Operator :
Sample : ICVLCS AGC 3.0 RSK175
Misc :
ALS Vial : 1 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 10:57:51 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
R3934254-5

Lab Sample ID:	R3934254-5	**SDG:**	L1623041
Client Sample ID:	LCSD	**Collected Date/Time:**	________________
Lab File ID:	0608_27	**Received Date/Time:**	________________
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/08/23 11:40
Analytical Batch:	WG2073383	**Analysis Date/Time:**	06/08/23 11:40
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

Analyte	CAS	RT	Result	Qualifier	DL	LOD	LOQ
			ug/l		*ug/l*	*ug/l*	*ug/l*
Methane	74-82-8	0.51	64.5		2.91	6.00	12.0
Ethane	74-84-0	0.66	119		4.07	10.0	20.0
Ethene	74-85-1	0.74	120		4.26	10.0	20.0

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_27.d
Signal(s) : FID1A.CH
Acq On : 08 Jun 2023 11:40 am
Operator :
Sample : LCSD 1x WG2073383
Misc :
ALS Vial : 27 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:43:25 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

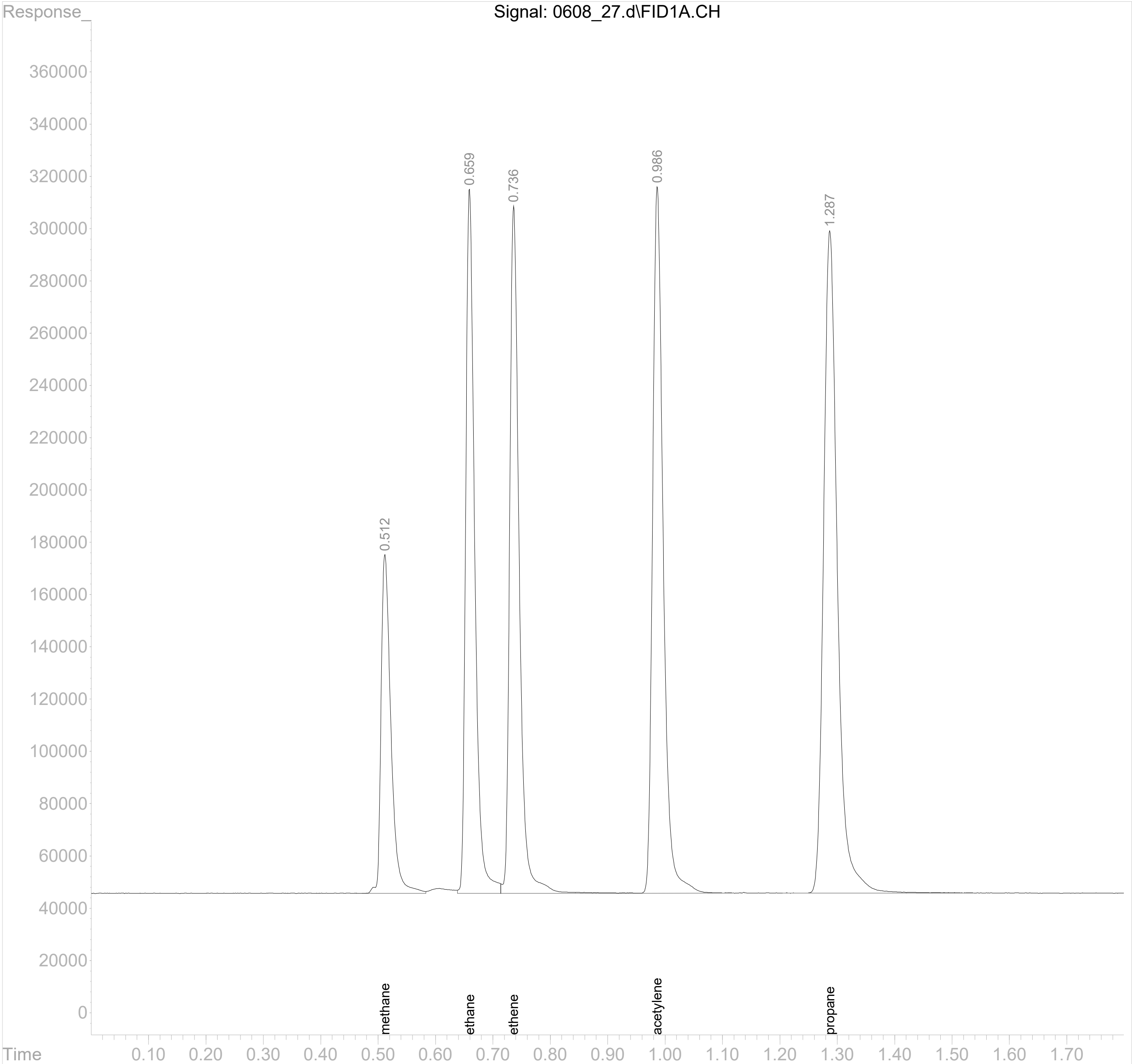
	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	1549823	0.0644631 mg/l
2)	ethane	0.66	2944613	0.1185631 mg/l
3)	ethene	0.74	3027879	0.1199322 mg/l
4)	acetylene	0.99	3472421	0.1949978 mg/l
5)	propane	1.29	4244364	0.1733915 mg/l

(f)=RT Delta > 1/2 Window (m)=manual int.

Data Path : C:\msdchem\1\data\060823\
Data File : 0608_27.d
Signal(s) : FID1A.CH
Acq On   : 08 Jun 2023 11:40 am
Operator :
Sample   : LCSD 1x WG2073383
Misc   :
ALS Vial : 27   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 08 11:43:25 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



1A-OR

SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
R3934861-7

Lab Sample ID:	R3934861-7	**SDG:**	L1623041
Client Sample ID:	LCSD	**Collected Date/Time:**	__________
Lab File ID:	0609B_29	**Received Date/Time:**	__________
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/09/23 13:46
Analytical Batch:	WG2074059	**Analysis Date/Time:**	06/09/23 13:46
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	__________
Total Solids (%):	__________	**Final Wt/Vol:**	20 mL

Analyte	CAS	RT	Result	Qualifier	DL	LOD	LOQ
			ug/l		*ug/l*	*ug/l*	*ug/l*
Methane	74-82-8	0.51	69.3		2.91	6.00	12.0
Ethane	74-84-0	0.66	117		4.07	10.0	20.0
Ethene	74-85-1	0.74	118		4.26	10.0	20.0

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_29.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 13:46 pm
Operator :
Sample : LCSD 1x WG2074059
Misc :
ALS Vial : 29 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:52:03 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.51	1666586	0.0693197 mg/l
2)	ethane	0.66	2909607	0.1171536 mg/l
3)	ethene	0.74	2972201	0.1177268 mg/l
4)	acetylene	0.99	3408360	0.1914004 mg/l
5)	propane	1.29	4241056	0.1732563 mg/l

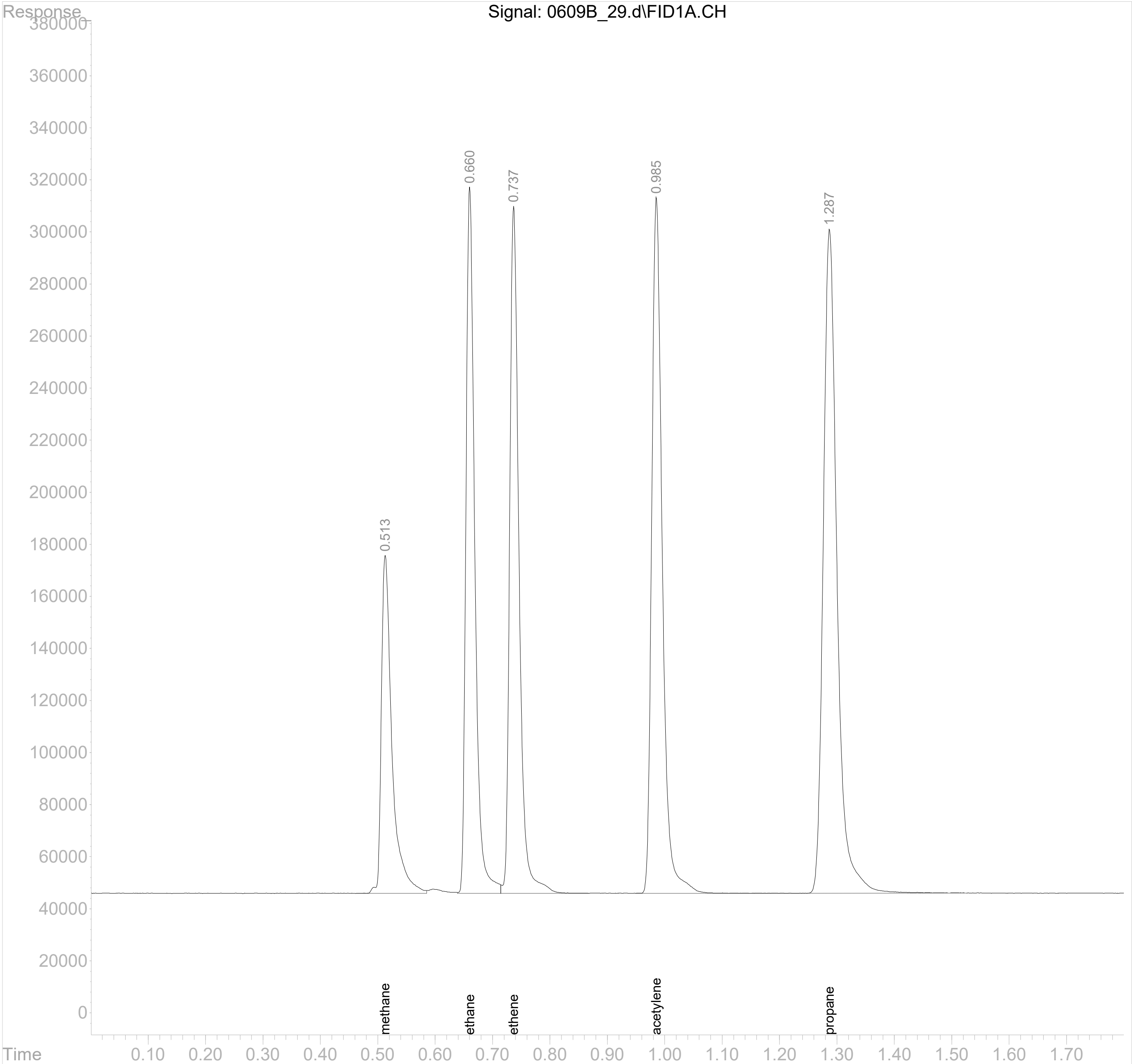
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_29.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 13:46 pm
Operator :
Sample : LCSD 1x WG2074059
Misc :
ALS Vial : 29 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:52:03 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
R3934861-5

Lab Sample ID:	R3934861-5	**SDG:**	L1623041
Client Sample ID:	MS	**Collected Date/Time:**	06/01/23 09:35
Lab File ID:	0609B_27	**Received Date/Time:**	06/06/23 09:00
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/09/23 13:38
Analytical Batch:	WG2074059	**Analysis Date/Time:**	06/09/23 13:38
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

Analyte	CAS	RT	Result *ug/l*	Qualifier	DL *ug/l*	LOD *ug/l*	LOQ *ug/l*
Methane	74-82-8	0.50	99.1		2.91	6.00	12.0
Ethane	74-84-0	0.65	123		4.07	10.0	20.0
Ethene	74-85-1	0.72	122		4.26	10.0	20.0

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_27.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 13:38 pm
Operator :
Sample : MS 1x WG2074059 L1623041-03
Misc :
ALS Vial : 27 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:44:25 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.50	2382657	0.0991039 mg/l
2)	ethane	0.65	3063522	0.1233508 mg/l
3)	ethene	0.72	3088307	0.1223256 mg/l
4)	acetylene	0.97	3579027	0.2009844 mg/l
5)	propane	1.28	4426884	0.1808478 mg/l

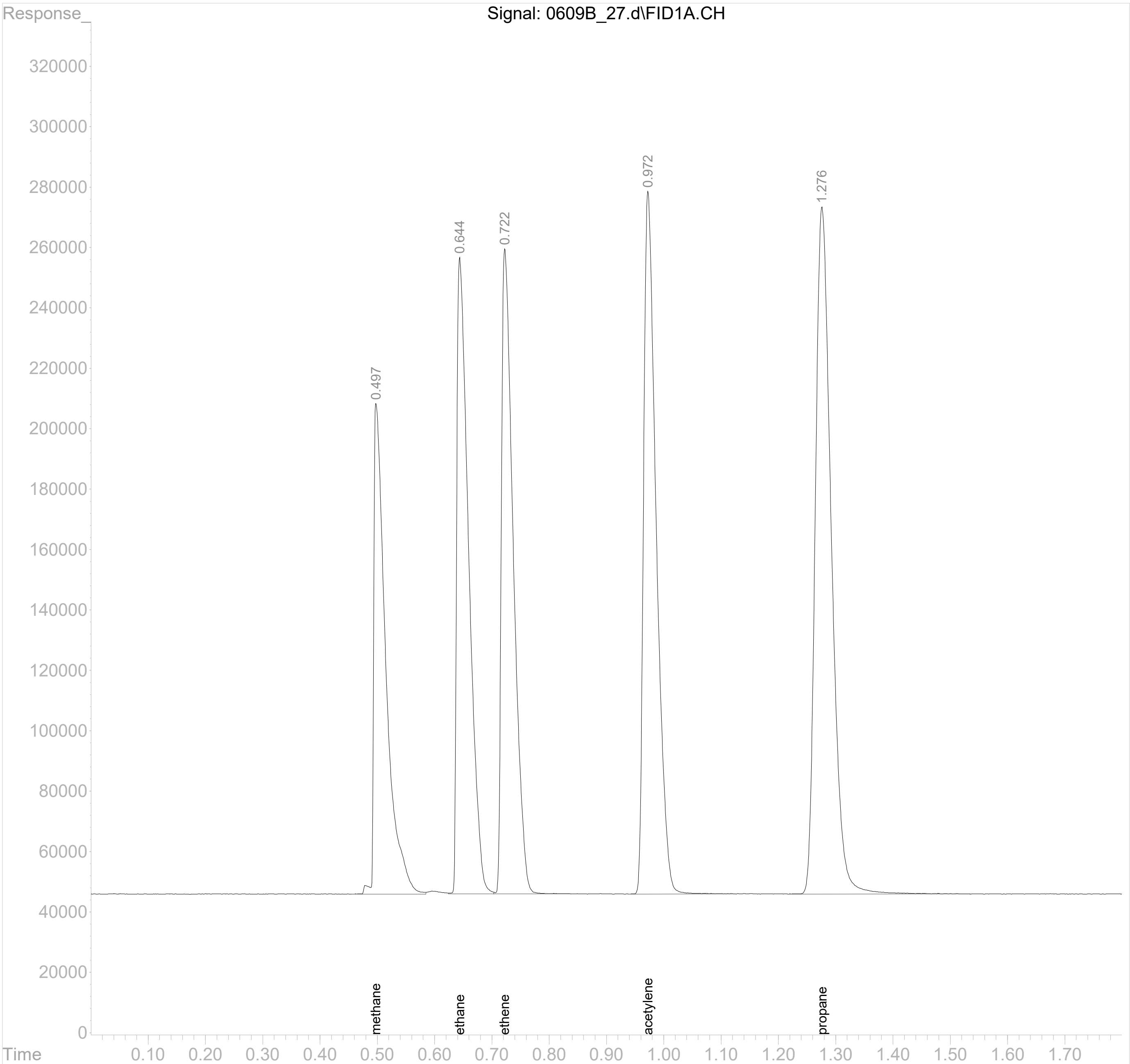
(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_27.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 13:38 pm
Operator :
Sample : MS 1x WG2074059 L1623041-03
Misc :
ALS Vial : 27 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:44:25 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

Volume Inj. :
Signal Phase :
Signal Info :



SAMPLE RESULT SUMMARY
ORGANIC ANALYSIS DATA SHEET

SAMPLE NO.:
R3934861-6

Lab Sample ID:	R3934861-6	**SDG:**	L1623041
Client Sample ID:	MSD	**Collected Date/Time:**	06/01/23 09:35
Lab File ID:	0609B_28	**Received Date/Time:**	06/06/23 09:00
Instrument ID:	AIRGC2	**Preparation Date/Time:**	06/09/23 13:42
Analytical Batch:	WG2074059	**Analysis Date/Time:**	06/09/23 13:42
Dilution Factor:	1	**Prep Method:**	RSK175
Analytical Method:	RSK175	**Sample Vol Used:**	20 mL
Matrix:	GW	**Initial Wt/Vol:**	________________
Total Solids (%):	________________	**Final Wt/Vol:**	20 mL

Analyte	**CAS**	**RT**	**Result**	**Qualifier**	**DL**	**LOD**	**LOQ**
			ug/l		*ug/l*	*ug/l*	*ug/l*
Methane	74-82-8	0.50	103		2.91	6.00	12.0
Ethane	74-84-0	0.65	130		4.07	10.0	20.0
Ethene	74-85-1	0.72	129		4.26	10.0	20.0

Quantitation Report (QT Reviewed)

Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_28.d
Signal(s) : FID1A.CH
Acq On : 09 Jun 2023 13:42 pm
Operator :
Sample : MSD 1x WG2074059 L1623041-03
Misc :
ALS Vial : 28 Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:45:10 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation 6890 Scale Mode: Large solvent peaks clipped

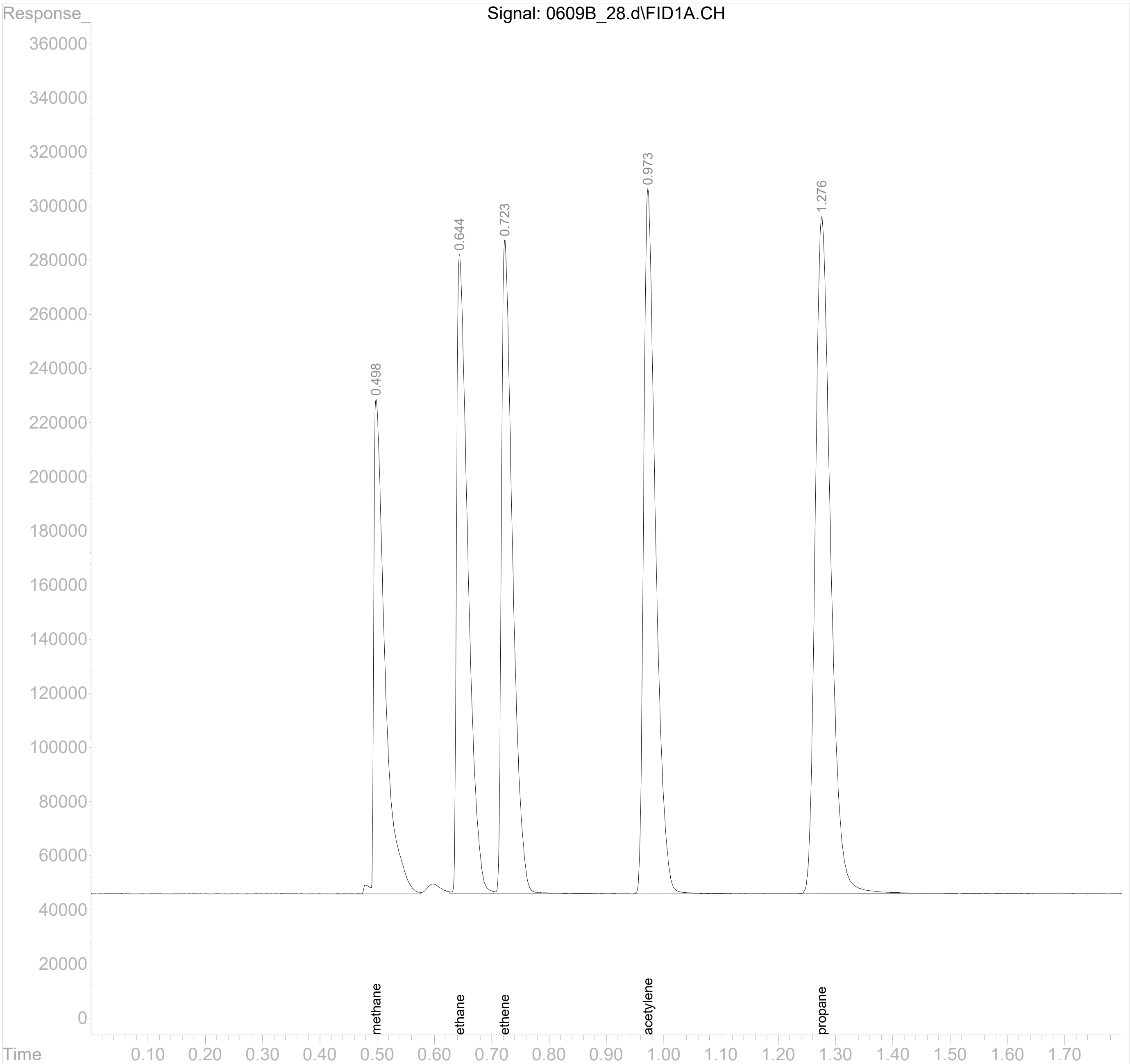
Volume Inj. :
Signal Phase :
Signal Info :

	Compound	R.T.	Response	Conc Units
Target Compounds				
1)	methane	0.50	2474048	0.1029052 mg/l
2)	ethane	0.65	3234540	0.1302368 mg/l
3)	ethene	0.72	3262839	0.1292387 mg/l
4)	acetylene	0.97	3762101	0.2112651 mg/l
5)	propane	1.28	4651240	0.1900133 mg/l

(f)=RT Delta > 1/2 Window (m)=manual int.

Quantitation Report (QT Reviewed)

```
Data Path : C:\msdchem\1\data\060923B\
Data File : 0609B_28.d
Signal(s) : FID1A.CH
Acq On    : 09 Jun 2023  13:42 pm
Operator  :
Sample    : MSD 1x WG2074059 L1623041-03
Misc      :
ALS Vial  : 28   Sample Multiplier: 1

Integration File: autoint1.e
Quant Time: Jun 09 13:45:10 2023
Quant Method : C:\msdchem\1\methods\GC2RSKK23V.M
Quant Title  :
QLast Update : Wed Nov 23 13:35:39 2022
Response via : Initial Calibration
Integrator: ChemStation   6890 Scale Mode: Large solvent peaks clipped

Volume Inj.  :
Signal Phase :
Signal Info  :
```



RSK175 Water VOCs Benchsheet

Batch: WG2073383

Analyst: BAW4010 **Batch Date/Time:** 06/07/23 15:51 **Prep End Date/Time:** 06/07/23 15:54 **40mL VOA Vial Lot#:** 44892 **SOP:** ENV-SOP-MTJL-0106
Method: RSK175 **pH Lot:** 10BDH4321

ICV Standard: 21L29533 Amt. Used: 100 µL Exp. Date:12/13/23 **LCS/D/MS/D Standard:** 21L29533 Amt. Used: 100 µL Exp. Date:12/13/23

Sample Number	Prep Flags	Analytical Dilution	Sample Vol Used (mL)	Final Volume (mL)	pH	Collection Date/Time	Prep Factor	Prep Ratio	Spike Factor	Surrogate Factor	Review Analyst	Review Date
BLANK		1	20	20	<2		1	1	1	1	BAW4010	06/07/23 15:54:41
LCS		1	20	20	<2		1	1	1	1	BAW4010	06/07/23 15:54:41
LCSD		1	20	20	<2		1	1	1	1	BAW4010	06/07/23 15:54:41
DUP(L1619855-03)		1	20	20	<2		1	1	1	1	CCM3548	06/08/23 11:46:29
DUP(L1622529-13)		1	20	20	<2		1	1	1	1	CCM3548	06/08/23 11:46:29
1. L1619855-03	T8	1	20	20	<2	05/22/23 13:29	1	1	1	1	BAW4010	06/07/23 15:54:41
2. L1622529-03		1	20	20	<2	06/01/23 19:25	1	1	1	1	BAW4010	06/07/23 15:54:41
3. L1622529-04		1	20	20	<2	06/01/23 19:55	1	1	1	1	BAW4010	06/07/23 15:54:41
4. L1622529-06		1	20	20	<2	06/01/23 22:15	1	1	1	1	BAW4010	06/07/23 15:54:41
5. L1622529-07		1	20	20	<2	06/01/23 21:25	1	1	1	1	BAW4010	06/07/23 15:54:41
6. L1622529-08		1	20	20	<2	06/01/23 21:00	1	1	1	1	BAW4010	06/07/23 15:54:41
7. L1622529-09		1	20	20	<2	06/01/23 21:50	1	1	1	1	BAW4010	06/07/23 15:54:41
8. L1622529-10		1	20	20	<2	06/01/23 21:40	1	1	1	1	BAW4010	06/07/23 15:54:41
9. L1622529-11		1	20	20	<2	06/01/23 20:30	1	1	1	1	BAW4010	06/07/23 15:54:41
10. L1622529-12		1	20	20	<2	06/01/23 22:50	1	1	1	1	BAW4010	06/07/23 15:54:41
11. L1622529-13		1	20	20	<2	06/01/23 23:30	1	1	1	1	BAW4010	06/07/23 15:54:41
12. L1622976-01		1	20	20	<2	06/05/23 07:34	1	1	1	1	BAW4010	06/07/23 15:54:41
13. L1622976-11		1	20	20	<2	06/05/23 11:38	1	1	1	1	BAW4010	06/07/23 15:54:41
14. L1623037-01	T8	1	20	20	7	05/23/23 09:45	1	1	1	1	BAW4010	06/07/23 15:54:41
15. L1623037-02	T8	1	20	20	<2	05/23/23 10:45	1	1	1	1	BAW4010	06/07/23 15:54:41
16. L1623037-03	T8	1	20	20	7	05/23/23 12:05	1	1	1	1	BAW4010	06/07/23 15:54:41
17. L1623041-01		1	20	20	<2	06/01/23 12:14	1	1	1	1	BAW4010	06/07/23 15:54:41
18. L1623041-02		1	20	20	<2	06/01/23 12:30	1	1	1	1	BAW4010	06/07/23 15:54:41
19. L1623041-04		1	20	20	<2	06/01/23 09:10	1	1	1	1	BAW4010	06/07/23 15:54:41
20. L1623041-05		1	20	20	<2	06/01/23 11:00	1	1	1	1	BAW4010	06/07/23 15:54:41

Comments:
20mL of headspace created using Helium.

Reviewed By:CCM3548 on 06/08/23 11:46:29

RSK175 Water VOCs Benchsheet

Batch: WG2074059

Analyst: BAW4010 **Batch Date/Time:** 06/08/23 14:23 **Prep End Date/Time:** 06/08/23 14:24 **40mL VOA Vial Lot#:** 44892 **SOP:** ENV-SOP-MTJL-0106
Method: RSK175 **pH Lot:** 10BDH4321

ICV Standard: 21L29533 Amt. Used: 100 µL Exp. Date:12/13/23 **LCS/D/MS/D Standard:** 21L29533 Amt. Used: 100 µL Exp. Date:12/13/23

Sample Number	Analytical Dilution	Sample Vol Used (mL)	Final Volume (mL)	pH	Collection Date/Time	Prep Factor	Prep Ratio	Spike Factor	Surrogate Factor	Review Analyst	Review Date
BLANK	1	20	20	<2		1	1	1	1	BAW4010	06/08/23 14:24:42
LCS	1	20	20	<2		1	1	1	1	BAW4010	06/08/23 14:24:42
LCSD	1	20	20	<2		1	1	1	1	BAW4010	06/08/23 14:24:42
MS(L1623041-03)	1	20	20	<2		1	1	1	1	BAW4010	06/09/23 13:47:42
MSD(L1623041-03)	1	20	20	<2		1	1	1	1	BAW4010	06/09/23 13:47:42
DUP(L1623162-01)	1	20	20	<2		1	1	1	1	BAW4010	06/09/23 13:47:42
DUP(L1623669-02)	1	20	20	<2		1	1	1	1	BAW4010	06/09/23 13:47:42
1. L1623041-03	1	20	20	<2	06/01/23 09:35	1	1	1	1	BAW4010	06/08/23 14:24:42
2. L1623041-06	1	20	20	<2	06/01/23 09:55	1	1	1	1	BAW4010	06/08/23 14:24:42
3. L1623041-07	1	20	20	<2	06/01/23 10:15	1	1	1	1	BAW4010	06/08/23 14:24:42
4. L1623041-08	1	20	20	<2	06/01/23 11:00	1	1	1	1	BAW4010	06/08/23 14:24:42
5. L1623094-01	1	20	20	<2	06/01/23 11:15	1	1	1	1	BAW4010	06/08/23 14:24:42
6. L1623094-02	1	20	20	<2	06/02/23 09:10	1	1	1	1	BAW4010	06/08/23 14:24:42
7. L1623094-03	1	20	20	<2	06/01/23 15:15	1	1	1	1	BAW4010	06/08/23 14:24:42
8. L1623094-05	1	20	20	<2	06/01/23 08:45	1	1	1	1	BAW4010	06/08/23 14:24:42
9. L1623094-06	1	20	20	<2	06/02/23 12:37	1	1	1	1	BAW4010	06/08/23 14:24:42
10. L1623162-01	1	20	20	<2	06/05/23 12:55	1	1	1	1	BAW4010	06/08/23 14:24:42
11. L1623162-02	1	20	20	<2	06/05/23 13:55	1	1	1	1	BAW4010	06/08/23 14:24:42
12. L1623162-03	1	20	20	<2	06/05/23 14:45	1	1	1	1	BAW4010	06/08/23 14:24:42
13. L1623162-04	1	20	20	<2	06/05/23 10:25	1	1	1	1	BAW4010	06/08/23 14:24:42
14. L1623162-05	1	20	20	<2	06/05/23 11:30	1	1	1	1	BAW4010	06/08/23 14:24:42
15. L1623162-06	1	20	20	<2	06/05/23 00:00	1	1	1	1	BAW4010	06/08/23 14:24:42
16. L1623162-07	1	20	20	<2	06/05/23 15:00	1	1	1	1	BAW4010	06/08/23 14:24:42
17. L1623424-01	1	20	20	<2	06/06/23 11:05	1	1	1	1	BAW4010	06/08/23 14:24:42
18. L1623669-01	1	20	20	<2	06/06/23 15:30	1	1	1	1	BAW4010	06/08/23 14:24:42
19. L1623669-02	1	20	20	<2	06/06/23 13:55	1	1	1	1	BAW4010	06/08/23 14:24:42
20. L1623669-03	1	20	20	<2	06/06/23 12:25	1	1	1	1	BAW4010	06/08/23 14:24:42

Comments:
20mL of headspace created using Helium.

Reviewed By:BAW4010 on 06/09/23 13:47:42

Guide to Reading and Understanding Your Laboratory Report

The information below is designed to better explain the various terms used in your report of analytical results from the Laboratory. This is not intended as a comprehensive explanation, and if you have additional questions please contact your project representative.

Results Disclaimer - Information that may be provided by the customer, and contained within this report, include Permit Limits, Project Name, Sample ID, Sample Matrix, Sample Preservation, Field Blanks, Field Spikes, Field Duplicates, On-Site Data, Sampling Collection Dates/Times, and Sampling Location. Results relate to the accuracy of this information provided, and as the samples are received.

Abbreviations and Definitions

COD	Coefficient of Determination.
DL	Detection Limit.
LOD	Limit of Detection.
LOQ	Limit of Quantitation.
MDL	Method Detection Limit.
RDL	Reported Detection Limit.
Rec.	Recovery.
RPD	Relative Percent Difference.
RRF	Relative Response Factor.
RT	Retention Time.
SDG	Sample Delivery Group.
Analyte	The name of the particular compound or analysis performed. Some Analyses and Methods will have multiple analytes reported.
Dilution	If the sample matrix contains an interfering material, the sample preparation volume or weight values differ from the standard, or if concentrations of analytes in the sample are higher than the highest limit of concentration that the laboratory can accurately report, the sample may be diluted for analysis. If a value different than 1 is used in this field, the result reported has already been corrected for this factor.
Limits	These are the target % recovery ranges or % difference value that the laboratory has historically determined as normal for the method and analyte being reported. Successful QC Sample analysis will target all analytes recovered or duplicated within these ranges.
Qualifier	This column provides a letter and/or number designation that corresponds to additional information concerning the result reported. If a Qualifier is present, a definition per Qualifier is provided within the Glossary and Definitions page and potentially a discussion of possible implications of the Qualifier in the Case Narrative if applicable.
Result	The actual analytical final result (corrected for any sample specific characteristics) reported for your sample. If there was no measurable result returned for a specific analyte, the result in this column may state "ND" (Not Detected) or "BDL" (Below Detectable Levels). The information in the results column should always be accompanied by either an MDL (Method Detection Limit) or RDL (Reporting Detection Limit) that defines the lowest value that the laboratory could detect or report for this analyte.
Uncertainty (Radiochemistry)	Confidence level of 2 sigma.
Case Narrative (Cn)	A brief discussion about the included sample results, including a discussion of any non-conformances to protocol observed either at sample receipt by the laboratory from the field or during the analytical process. If present, there will be a section in the Case Narrative to discuss the meaning of any data qualifiers used in the report.
Quality Control Summary (Qc)	This section of the report includes the results of the laboratory quality control analyses required by procedure or analytical methods to assist in evaluating the validity of the results reported for your samples. These analyses are not being performed on your samples typically, but on laboratory generated material.
Sample Chain of Custody (Sc)	This is the document created in the field when your samples were initially collected. This is used to verify the time and date of collection, the person collecting the samples, and the analyses that the laboratory is requested to perform. This chain of custody also documents all persons (excluding commercial shippers) that have had control or possession of the samples from the time of collection until delivery to the laboratory for analysis.
Sample Results (Sr)	This section of your report will provide the results of all testing performed on your samples. These results are provided by sample ID and are separated by the analyses performed on each sample. The header line of each analysis section for each sample will provide the name and method number for the analysis reported.
Sample Summary (Ss)	This section of the Analytical Report defines the specific analyses performed for each sample ID, including the dates and times of preparation and/or analysis.
NI	Manual Integration Code to indicate that the peak was not integrated at all by the computer software.
LT	Manual Integration Code to indicate that the peak in question was inappropriately integrated to an area less than what it should be (i.e., peak area was cut).
GT	Manual Integration Code to indicate that the peak in question was inappropriately integrated to an area greater than it should be (i.e., peak tailing).
BA	Manual Integration Code to indicate that the baseline had to be adjusted correctly by the analyst.
WP	Manual Integration Code to indicate that the wrong peak was chosen.
CO	Manual Integration Code to indicate that the analyst had to split two co-eluting peaks apart that were not (or could not be) separated by the computer system.
RT	Manual Integration Code to indicate that the retention time for the peak in question has shifted from the expected retention time.
INT	Manual Integration Code to indicate that there was electronic interference (i.e., noise).

Qualifier	Description
U	Below Detectable Limits: Indicates that the analyte was not detected.

Pace Analytical National 12065 Lebanon Rd Mount Juliet, TN 37122

Alabama	40660	Nebraska	NE-OS-15-05
Alaska	17-026	Nevada	TN000032021-1
Arizona	AZ0612	New Hampshire	2975
Arkansas	88-0469	New Jersey–NELAP	TN002
California	2932	New Mexico [1]	TN00003
Colorado	TN00003	New York	11742
Connecticut	PH-0197	North Carolina	Env375
Florida	E87487	North Carolina [1]	DW21704
Georgia	NELAP	North Carolina [3]	41
Georgia [1]	923	North Dakota	R-140
Idaho	TN00003	Ohio–VAP	CL0069
Illinois	200008	Oklahoma	9915
Indiana	C-TN-01	Oregon	TN200002
Iowa	364	Pennsylvania	68-02979
Kansas	E-10277	Rhode Island	LAO00356
Kentucky [1 6]	KY90010	South Carolina	84004002
Kentucky [2]	16	South Dakota	n/a
Louisiana	AI30792	Tennessee [1 4]	2006
Louisiana	LA018	Texas	T104704245-20-18
Maine	TN00003	Texas [5]	LAB0152
Maryland	324	Utah	TN000032021-11
Massachusetts	M-TN003	Vermont	VT2006
Michigan	9958	Virginia	110033
Minnesota	047-999-395	Washington	C847
Mississippi	TN00003	West Virginia	233
Missouri	340	Wisconsin	998093910
Montana	CERT0086	Wyoming	A2LA
A2LA – ISO 17025	1461.01	AIHA-LAP,LLC EMLAP	100789
A2LA – ISO 17025 [5]	1461.02	DOD	1461.01
Canada	1461.01	USDA	P330-15-00234
EPA–Crypto	TN00003		

[1] Drinking Water [2] Underground Storage Tanks [3] Aquatic Toxicity [4] Chemical/Microbiological [5] Mold [6] Wastewater n/a Accreditation not applicable

* Not all certifications held by the laboratory are applicable to the results reported in the attached report.

* Accreditation is only applicable to the test methods specified on each scope of accreditation held by Pace Analytical.

E085

Chain of Custody

PAGC Baton Rouge Laboratory

Pace Analytical www.pacelabs.com

Workorder: **Workorder Name:** Plexus Scott AFB **Results Requested By:** 6/22/23

Report / Invoice To:

PM: Carolyn Dicharry
Pace Analytical Gulf Coast
7979 Innovation Park Drive
Baton Rouge, LA 70820
Phone (225) 769-4900
Email: carolyn.dicharry@pacelabs.com

Pace National
12065 Lebanon Road
Mt. Juliet, TN 37122

State of Sample Origin: IL

L1623041

Item	Sample ID	Collect Date/Time	Lab ID	Matrix	Preserved Containers: HCl					Requested Analysis: RSK-175											LAB USE ONLY
1	OT07-EB-060123	6/1/2023 12:14	22306021001	W	x					X											-01
2	OT07-MW02-060123	6/1/2023 12:30	22306021002	W	x					X											-02
3	OT07-MW04-060123	6/1/2023 9:35	22306021003	W	x					X											-03
4	OT07-MW05-060123	6/1/2023 9:10	22306021004	W	x					X											-04
5	OT07-MW06-060123	6/1/2023 11:00	22306021005	W	x					X											-05
6	OT07-MW04-060123MS	6/1/2023 9:55	22306021006	W	x					X											-06
7	OT07-MW04-060123MSD	6/1/2023 10:15	22306021007	W	x					X											-07
8	OT07-MW806-060123	6/1/2023 11:00	22306021008	W	x					X											-08
9																					
10																					

Transfers	Released By	Date/Time	Reveived By	Date/Time	Comments
1	[signature]	6/5/23 17:00	FedEx		Level IV Report DOD
2		~~Hama M~~	Hama Murechiwa	6.6.23 0900	
3					

Cooler Temperature on Receipt ______ °C	Custody Seal Y or N	Received on Ice Y or N	Sample Intact Y or N

Sample Receipt Checklist

COC Seal Present/Intact: Y N
COC Signed/Accurate: Y N
Bottles arrive intact: Y N
Correct bottles used: Y N
Sufficient volume sent: Y N
RAD Screen <0.5 mR/hr: Y N

If Applicable
VOA Zero Headspace: Y N
Pres.Correct/Check: Y N

.8 +0 = .8

6484 9129 2128

L1623041

Pace Analytical®
www.pacelabs.com

Ship to :
PACE National
12065 Lebanon Road
Mt. Juliet, TN 37122

INTER_LABORATORY WORK ORDER # ____________

(To be complete by sending lab)

Sending Project No:	223060210
Receiving Project No:	
Check Box for Consolidated Invoice:	
Date Prepared:	6/2/2023
REQUESTED COMPLETION DATE:	6/23/2023

Sending Region	**IR24-Gulf Coast**	Sending Project Mgr.	Carolyn Dicharry
Receiving Region	IR80- Pace National	External Client	Plexus
State of Sample Origin	IL	QC Deliverable	**DOD Level IV**

All questions should be addressed to sending project manager.

Requested Reportable Units ____________ Report Wet or Dry Weight? ____________ Cert Needed: DOD

WORK REQUESTED

Method Description	Container Type		Quantity of containers	Preservative	Quantity of Samples	Unit Price	Amount
RSK175	40mL		2	HCl	8	$70.00	$560.00
						TOTAL	$560.00

Special Requirements: **DOD Level IV report,**

Receiving Region Dept.	Acctg. Code	Totals from above	Revenue Allocation: Receiving Region (80%)	Revenue Allocation: Client Services Dept. Sending Region (20%)
20-210		$560.00	$448.00	$112.00
		TOTAL	$448.00	$112.00

* Custom Revenue Allocation

FOR ANALYTICAL WORK COMPLETED THIS SECTION ALSO

Chain of Custody Included: [x] Yes [ ] No Return Samples to Sending Region: [ ] Yes [x] No

Matrix: [ ] Soil [x] Water [ ] **Air** [ ] Other (identity) ____________

CONFIRMATION OF WORK COMPLETED

Date Completed: ____________ Receiving Project Manager: ____________

Original sent to the receiving lab - Copy kept at the sending lab.

When work completed: Original sent to the ABM at the receiving laboratory. Copies are made to corporate as needed.

LELAP Certificate Number: 01955
A2LA Accredited (DoD ELAP-QSM 5.4) Certificate Number: 6429.01

ANALYTICAL RESULTS

PERFORMED BY

Pace Analytical Gulf Coast
7979 Innovation Park Dr.
Baton Rouge, LA 70820
(225) 769-4900

Report Date 06/29/2023

Report # 223060502

Project Scott AFB

Samples Collected 6/2/23

Deliver To	***Additional Recipients***
Patrick Reilley Plexus Scientific Corporation 5510 Cherokee Ave. Suite 350 Alexandria, VA 22312 (703) 989-8405	NONE

ACCREDITED
CERT # 6429.01

Laboratory Endorsement

Sample analysis was performed in accordance with approved methodologies provided by the Environmental Protection Agency or other recognized agencies. The samples and their corresponding extracts will be maintained for a period of 30 days unless otherwise arranged. Following this retention period the samples will be disposed in accordance with Pace Gulf Coast's Standard Operating Procedures.

Common Abbreviations that may be Utilized in this Report

ND	Indicates the result was Not Detected at the specified reporting limit
NO	Indicates the sample did not ignite when preliminary test performed for EPA Method 1030
DO	Indicates the result was Diluted Out
MI	Indicates the result was subject to Matrix Interference
TNTC	Indicates the result was Too Numerous To Count
SUBC	Indicates the analysis was Sub-Contracted
FLD	Indicates the analysis was performed in the Field
DL	Detection Limit
LOD	Limit of Detection
LOQ	Limit of Quantitation
RE	Re-analysis
CF	HPLC or GC Confirmation
00:01	Reported as a time equivalent to 12:00 AM

Reporting Flags that may be Utilized in this Report

J or I	Indicates the result is between the MDL and LOQ
J	DOD flag on analyte in the parent sample for MS/MSD outside acceptance criteria
U	Indicates the compound was analyzed for but not detected
B or V	Indicates the analyte was detected in the associated Method Blank
Q	Indicates a non-compliant QC Result (See Q Flag Application Report)
*****	Indicates a non-compliant or not applicable QC recovery or RPD – see narrative
E	Organics **-** The result is estimated because it exceeded the instrument calibration range
E	Metals - % diference for the serial dilution is > 10%
L	Reporting Limits adjusted to meet risk-based limit.
P	RPD between primary and confirmation result is greater than 40
DL	Diluted analysis – when appended to Client Sample ID

Sample receipt at Pace Gulf Coast is documented through the attached chain of custody. In accordance with NELAC, this report shall be reproduced only in full and with the written permission of Pace Gulf Coast. The results contained within this report relate only to the samples reported. The documented results are presented within this report.

This report pertains only to the samples listed in the Report Sample Summary and should be retained as a permanent record thereof. The results contained within this report are intended for the use of the client. Any unauthorized use of the information contained in this report is prohibited.

I certify that this data package is in compliance with The NELAC Institute (TNI) Standard 2009 and terms and conditions of the contract and Statement of Work both technically and for completeness, for other than the conditions in the case narrative. Release of the data contained in this hardcopy data package and in the computer readable data submitted has been authorized by the Quality Assurance Manager or his/her designee, as verified by the following signature.

Estimated uncertainty of measurement is available upon request. This report is in compliance with the DOD QSM as specified in the contract if applicable.

Calista P. Caigle

Authorized Signature
Pace Gulf Coast Report 223060502

Certifications

Certification	Certification Number
A2LA Accredited (DoD ELAP-QSM 5.4)	6429.01
Alabama	01955
Arkansas	88-0655
Colorado	01955
Delaware	01955
Florida	E87854
Georgia	01955
Hawaii	01955
Idaho	01955
Illinois	200048
Indiana	01955
Kansas	E-10354
Kentucky	95
Louisiana	01955
Maryland	01955
Massachusetts	01955
Michigan	01955
Mississippi	01955
Missouri	01955
Montana	N/A
Nebraska	01955
New Mexico	01955
North Carolina	618
North Dakota	R-195
Oklahoma	9403
South Carolina	73006001
South Dakota	01955
Tennessee	01955
Texas	T104704178
Vermont	01955
Virginia	460215
Washington	C929
USDA Soil Permit	P330-16-00234

Case Narrative

Client: Plexus Consultants **Report:** 223060502

Pace Analytical Gulf Coast received and analyzed the sample(s) listed
on the Report Sample Summary page of this report. Receipt of the sample(s) is documented
by the attached chain of custody. This applies only to the sample(s) listed in this report.
No sample integrity or quality control exceptions were identified unless noted below.

This report was completed in accordance with DOD QSM 5.3 as specified in the contract.

VOLATILES MASS SPECTROMETRY

In the EPA 8260B DOD analysis for analytical batch 767415, the MS/MSD exhibited recovery failures for 1,2,4-Trimethylbenzene. This is due to the high concentration of target analytes in the native sample.

In the EPA 8260B DOD analysis for analytical batch 767415, the LCS exhibited recovery failures for Chloroethane and Bromobenzene.

In the EPA 8260B DOD analysis for analytical batch 767415, the %D/%Drift is outside ± 20% for 1,2-Dibromo-3-chloropropane in the opening CCV. The analytes were not detected in the associated samples. The closing CCV was within limits for all compounds.

In the EPA 8260B DOD analysis for analytical batch 767415, the ICV exhibited recovery failures for Acetone and Chloromethane.

In the EPA Method 8260B DOD analysis, samples 22306050201 (OT07-BO-40-4-6-8) and 22306050204 (OT07-BO-840-4-6-8) had to be diluted due to the presence of non-target background and to bracket the concentration of target analytes within the calibration range of the instrument. The dilutions are reflected in the elevated detection limits.

In the EPA 8260B analysis, sample 22306050201 (parent) did not match 22306050204 (FLDDUP). The container labels were verified by analyst.

Sample Summary

Lab ID	Client ID	Matrix	Collect Date	Receive Date
22306050201	OT07-BO-40-4-6-8	Solid	6/02/23 09:05	6/03/23 10:50
22306050202	OT07-BO-40-4-6-8-MS	Solid	6/02/23 09:10	6/03/23 10:50
22306050203	OT07-BO-40-4-6-8-MSD	Solid	6/02/23 09:15	6/03/23 10:50
22306050204	OT07-BO-840-4-6-8	Solid	6/02/23 09:05	6/03/23 10:50

Test Summary

EPA 8260B DOD					
Lab ID	**Client ID**	**Matrix**	**Batch**	**Instru**	**RunDate**
22306050201	OT07-BO-40-4-6-8	Solid	767415	MSV15	6/12/23 18:23
22306050202	OT07-BO-40-4-6-8-MS	Solid	767415	MSV15	6/12/23 19:38
22306050203	OT07-BO-40-4-6-8-MSD	Solid	767415	MSV15	6/12/23 19:57
22306050204	OT07-BO-840-4-6-8	Solid	767415	MSV15	6/12/23 18:42

SM 2540 G-2011					
Lab ID	**Client ID**	**Matrix**	**Batch**	**Instru**	**RunDate**
22306050201	OT07-BO-40-4-6-8	Solid	767748	BAL15	6/19/23 14:00
22306050202	OT07-BO-40-4-6-8-MS	Solid	767748	BAL15	6/19/23 14:00
22306050203	OT07-BO-40-4-6-8-MSD	Solid	767748	BAL15	6/19/23 14:00
22306050204	OT07-BO-840-4-6-8	Solid	767748	BAL15	6/19/23 14:00

Manual Integrations

Lab ID	Client ID	Matrix	Method	Parameter
22306050201	OT07-BO-40-4-6-8	Solid	EPA 8260B DOD	4-Isopropyltoluene
22306050201	OT07-BO-40-4-6-8	Solid	EPA 8260B DOD	n-Butylbenzene
22306050204	OT07-BO-840-4-6-8	Solid	EPA 8260B DOD	4-Isopropyltoluene

Q Flag Summary

Client Sample ID: **OT07-BO-40-4-6-8** Lab Sample ID: **22306050201**

EPA 8260B DOD Analysis Date: 6/12/2023 6:23:00 PM						
CAS	**Analyte**	**CCV**	**LCS/D**	**SURR**	**ISTD**	**CLCCV**
96-12-8	1,2-Dibromo-3-chloropropane	X				
108-86-1	Bromobenzene		X			
75-00-3	Chloroethane		X			

Client Sample ID: **OT07-BO-840-4-6-8** Lab Sample ID: **22306050204**

EPA 8260B DOD Analysis Date: 6/12/2023 6:42:00 PM						
CAS	**Analyte**	**CCV**	**LCS/D**	**SURR**	**ISTD**	**CLCCV**
96-12-8	1,2-Dibromo-3-chloropropane	X				
108-86-1	Bromobenzene		X			
75-00-3	Chloroethane		X			

CCV=Initial or Continuing CCV out of limits
LCS/D=LCS/LCSD out of limits
SURR=Surrogate out of limits
ISTD=Internal Standard out of limits
CLCCV=Closing CCV out of limits

EPA 8260B DOD

Form 1A

Results

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060502			Client Sample ID:	OT07-BO-40-4-6-8
Collect Date:	06/02/23	Time:	0905	GCAL Sample ID:	22306050201
Matrix:	Solid	% Moisture:	22.5	Instrument ID:	MSV15
Sample Amt:	5.92	g		Lab File ID:	230612/c0942
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	250	Analyst:	KN1	Analytical Batch:	767415
Analysis Date:	06/12/23	Time:	1823	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/kg

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	136	U	68.1	136	1360
71-55-6	1,1,1-Trichloroethane	136	U	68.1	136	1360
79-34-5	1,1,2,2-Tetrachloroethane	136	U	68.1	136	1360
79-00-5	1,1,2-Trichloroethane	136	U	68.1	136	1360
75-34-3	1,1-Dichloroethane	136	U	68.1	136	1360
75-35-4	1,1-Dichloroethene	136	U	68.1	136	1360
563-58-6	1,1-Dichloropropene	136	U	68.1	136	1360
87-61-6	1,2,3-Trichlorobenzene	272	U	136	272	1360
96-18-4	1,2,3-Trichloropropane	272	U	136	272	1360
120-82-1	1,2,4-Trichlorobenzene	272	U	136	272	1360
95-63-6	1,2,4-Trimethylbenzene	1960	J	68.1	136	1360
96-12-8	1,2-Dibromo-3-chloropropane	545	UQ	136	545	1360
106-93-4	1,2-Dibromoethane	545	U	136	545	1360
95-50-1	1,2-Dichlorobenzene	136	U	68.1	136	1360
107-06-2	1,2-Dichloroethane	136	U	68.1	136	1360
540-59-0	1,2-Dichloroethene (total)	272	U	136	272	2720
78-87-5	1,2-Dichloropropane	136	U	68.1	136	1360
108-67-8	1,3,5-Trimethylbenzene	306	J	68.1	136	1360
541-73-1	1,3-Dichlorobenzene	136	U	68.1	136	1360
142-28-9	1,3-Dichloropropane	136	U	68.1	136	1360
106-46-7	1,4-Dichlorobenzene	136	U	68.1	136	1360
594-20-7	2,2-Dichloropropane	136	U	68.1	136	1360
78-93-3	2-Butanone	545	U	136	545	1360
95-49-8	2-Chlorotoluene	136	U	68.1	136	1360
591-78-6	2-Hexanone	545	U	136	545	1360
106-43-4	4-Chlorotoluene	136	U	68.1	136	1360
99-87-6	4-Isopropyltoluene	147	J	68.1	136	1360
108-10-1	4-Methyl-2-pentanone	136	U	68.1	136	1360
67-64-1	Acetone	545	U	136	545	6810
71-43-2	Benzene	136	U	68.1	136	1360
108-86-1	Bromobenzene	136	UQ	68.1	136	1360
74-97-5	Bromochloromethane	272	U	136	272	1360
75-27-4	Bromodichloromethane	136	U	68.1	136	1360
75-25-2	Bromoform	272	U	136	272	1360
74-83-9	Bromomethane	545	U	136	545	1360
75-15-0	Carbon disulfide	136	U	68.1	136	1360
56-23-5	Carbon tetrachloride	136	U	68.1	136	1360

FORM I VOA

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060502			Client Sample ID:	OT07-BO-40-4-6-8
Collect Date:	06/02/23	Time:	0905	GCAL Sample ID:	22306050201
Matrix:	Solid	% Moisture:	22.5	Instrument ID:	MSV15
Sample Amt:	5.92	g		Lab File ID:	230612/c0942
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	250	Analyst:	KN1	Analytical Batch:	767415
Analysis Date:	06/12/23	Time:	1823	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/kg

CAS	***ANALYTE***	***RESULT***	***Q***	***DL***	***LOD***	***LOQ***
108-90-7	Chlorobenzene	136	U	68.1	136	1360
75-00-3	Chloroethane	136	UQ	68.1	136	1360
67-66-3	Chloroform	136	U	68.1	136	1360
74-87-3	Chloromethane	545	U	136	545	1360
156-59-2	cis-1,2-Dichloroethene	136	U	68.1	136	1360
10061-01-5	cis-1,3-Dichloropropene	136	U	68.1	136	1360
124-48-1	Dibromochloromethane	136	U	68.1	136	1360
74-95-3	Dibromomethane	136	U	68.1	136	1360
75-71-8	Dichlorodifluoromethane	136	U	68.1	136	1360
100-41-4	Ethylbenzene	241	J	68.1	136	1360
87-68-3	Hexachlorobutadiene	272	U	136	272	1360
98-82-8	Isopropylbenzene (Cumene)	128	J	68.1	136	1360
136777-61-2	m,p-Xylene	310	J	68.1	272	2720
75-09-2	Methylene chloride	545	U	272	545	2720
91-20-3	Naphthalene	316	J	136	272	1360
104-51-8	n-Butylbenzene	146	J	68.1	136	1360
103-65-1	n-Propylbenzene	315	J	68.1	136	1360
95-47-6	o-Xylene	136	U	68.1	136	1360
135-98-8	sec-Butylbenzene	239	J	68.1	136	1360
100-42-5	Styrene	136	U	68.1	136	1360
1634-04-4	tert-Butyl methyl ether (MTBE)	136	U	68.1	136	1360
98-06-6	tert-Butylbenzene	272	U	136	272	1360
127-18-4	Tetrachloroethene	272	U	136	272	1360
108-88-3	Toluene	136	U	68.1	136	1360
156-60-5	trans-1,2-Dichloroethene	136	U	68.1	136	1360
10061-02-6	trans-1,3-Dichloropropene	136	U	68.1	136	1360
79-01-6	Trichloroethene	136	U	68.1	136	1360
75-69-4	Trichlorofluoromethane	136	U	68.1	136	1360
75-01-4	Vinyl chloride	136	U	68.1	136	1360

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230612.s.b/c0942.d
Lab Smp Id: 22306050201   Client Smp ID: 250X
Inj Date : 12-JUN-2023 18:23
Operator : KN1   Inst ID: msv15.i
Smp Info : 22306050201*C*250X
Misc Info : MSV~52948~*250*SMR
Comment   :
Method   : /var/chem/msv15.i/230612.s.b/8260bw15DOD.m
Meth Date : 13-Jun-2023 15:50 kmn   Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20   Cal File: c0234.d
Als bottle: 1
Dil Factor: 250.00000
Integrator: HP RTE   Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/(Ws*(100-M)/100))/5000 * CpndVariable

Name	Value	Description
DF	250.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Ws	5.92000	Weight of sample extracted (g)
M	0.00000	% Moisture (not decanted)
Va	100.00000	Volume of aliquot extract added (uL)
m	0.00000	

Cpnd Variable   Local Compound Variable

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	CONCENTRATIONS FINAL (UG/KG)	SIMILARITY
	22 Hexane	57	3.687	3.684	(0.627)	1751	0.56205	119	8370
	36 Cyclohexane	56	4.928	4.928	(0.838)	3429	1.30232	275	6211
$	42 Dibromofluoromethane	111	5.156	5.156	(0.876)	85693	54.3188	11500	6939(R)
	46 2,2,4 Trimethylpentane	57	5.401	5.398	(0.918)	81781	11.1355	2350	9598
$	50 1,2-Dichloroethane-d4	65	5.629	5.629	(0.957)	87145	53.0521	11200	9777(R)
*	55 FLUOROBENZENE	96	5.883	5.880	(1.000)	319452	50.0000		9825
	58 Methyl Cyclohexane	83	6.025	6.028	(1.024)	11292	3.65597	772	7748
$	74 Toluene-d8	98	7.221	7.221	(0.866)	310937	46.9773	9920	9866(R)
*	91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	141230	50.0000		9887
	93 Ethylbenzene +	106	8.365	8.369	(1.003)	2624	0.88485	187	6851(H)
	95 p,m-Xylene	106	8.462	8.465	(1.015)	4081	1.13828	240	9447
	97 o-Xylene	106	8.748	8.745	(1.049)	287	0.08596	18.2	4499

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (UG/KG)	SIMILARITY
104 Isopropylbenzene	105	8.944	8.944	(1.072)	4098	0.46894	99.0	8117
M 105 TOTAL XYLENE	106				4368	1.22424	258	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	107400	49.9633	10500	9484(R)
109 n-Propylbenzene	91	9.201	9.201	(0.939)	12932	1.15340	244	6598
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	8481	1.12460	237	7294
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	53270	7.19693	1520	9297
120 sec-Butylbenzene	105	9.619	9.623	(0.981)	8566	0.87809	185	7632
121 p-Isopropyltoluene	119	9.700	9.703	(0.990)	4646	0.53803	114	8360(M2)
* 124 1,4-DICHLOROBENZENE-D4	152	9.803	9.806	(1.000)	143566	50.0000		9676
126 n-Butylbenzene	91	9.954	9.954	(1.015)	6169	0.53723	113	8784(M2)
132 Naphthalene	128	11.285	11.291	(1.151)	6554	1.15911	245	9142

QC Flag Legend

R - Spike/Surrogate failed recovery limits.
M2- Compound response manually integrated because
Target system integrated incorrectly.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230612.s.b/c0942.d

Date : 12-JUN-2023 18:23

Client ID: 250X

Sample Info: 22306050201*C*250X

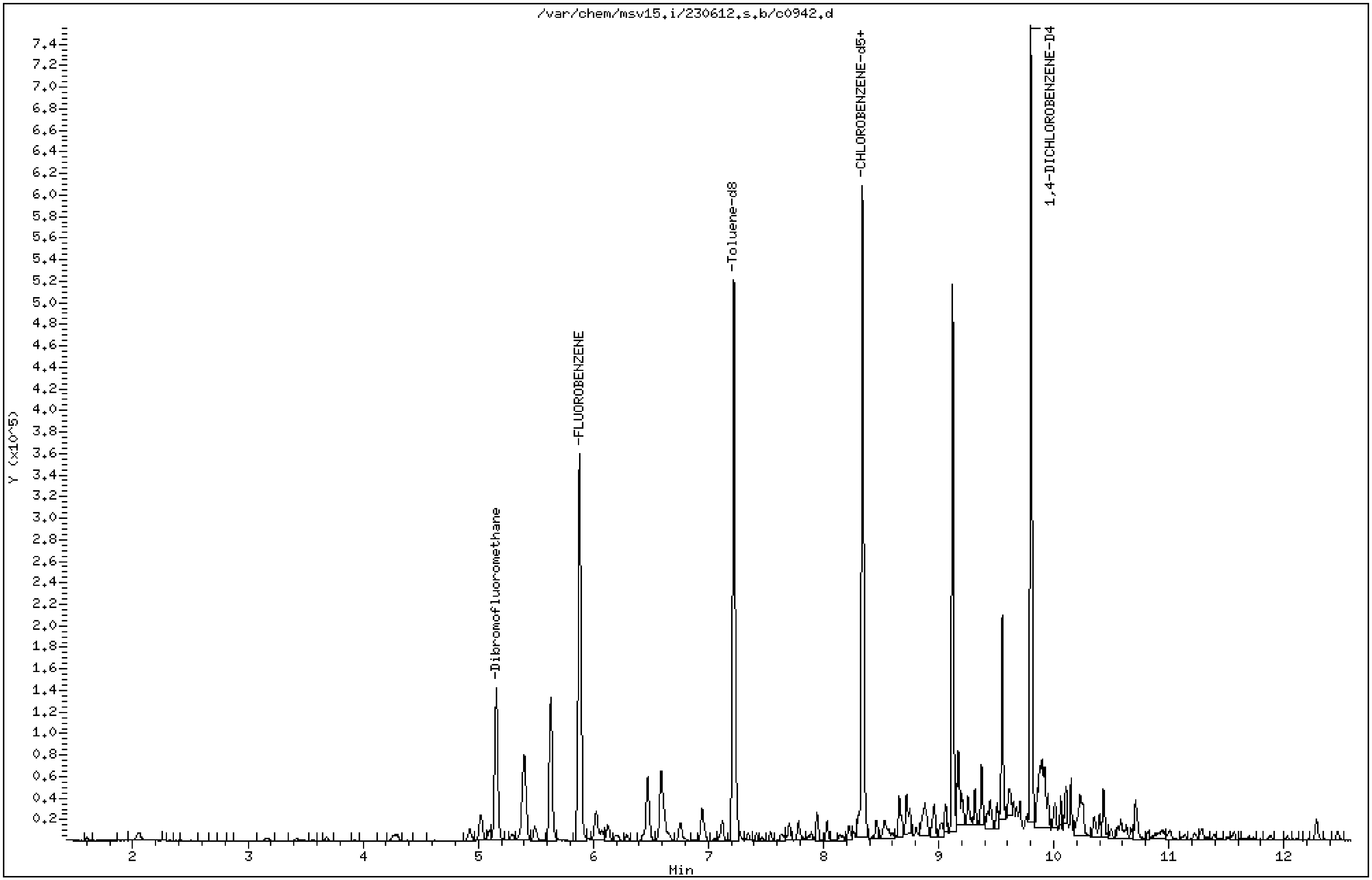
Purge Volume: 5.0

Column phase: RTX-VMS-30M

Instrument: msv15.i

Operator: KN1

Column diameter: 0.25



Data File: /var/chem/msv15.i/230612.s.b/c0942.d Page 2

Date : 12-JUN-2023 18:23

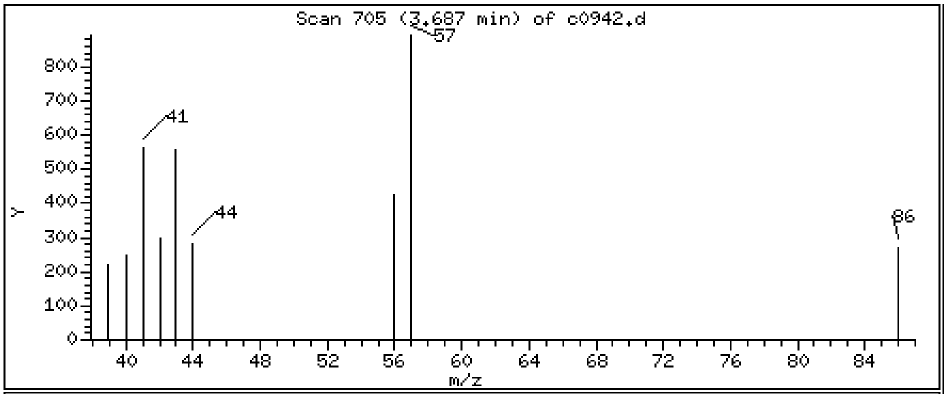
Client ID: 250X Instrument: msv15.i

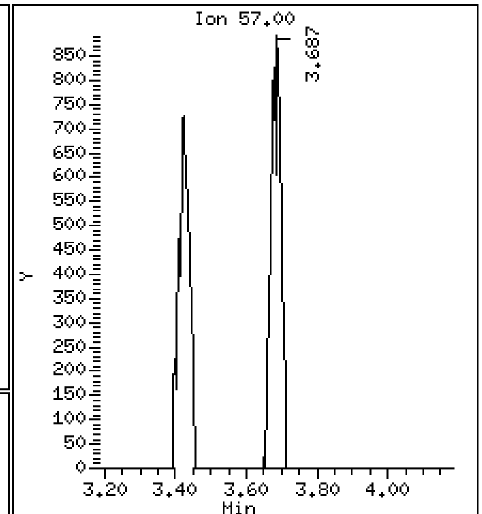
Sample Info: 22306050201*C*250X

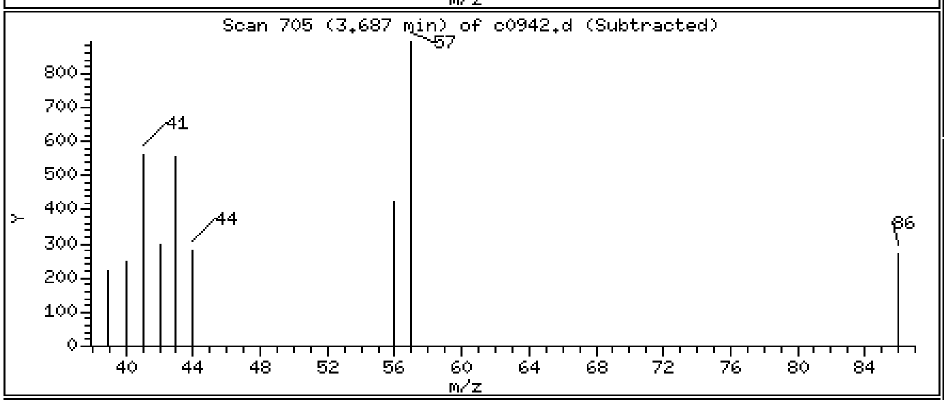
Purge Volume: 5.0 Operator: KN1

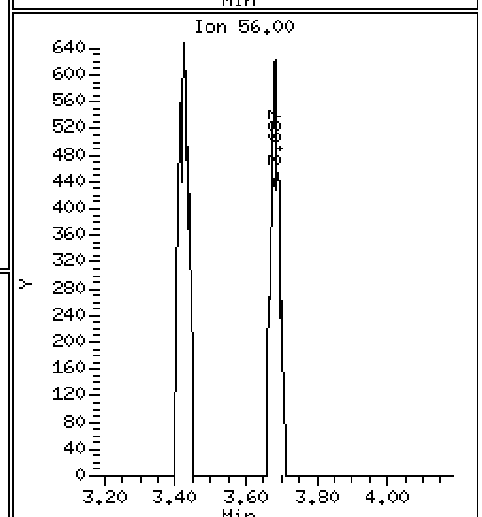
Column phase: RTX-VMS-30M Column diameter: 0.25

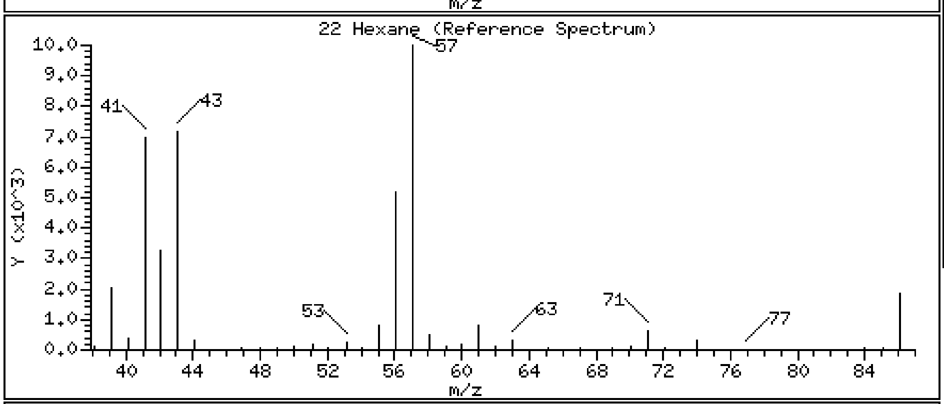
22 Hexane Concentration: 119 UG/KG

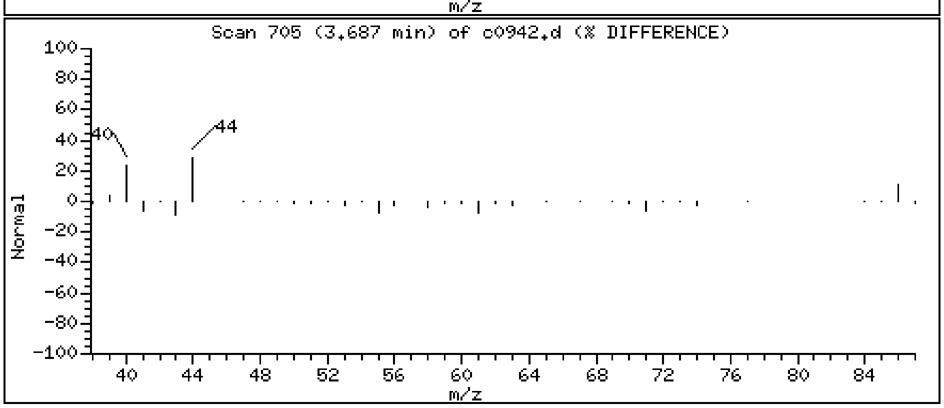












Data File: /var/chem/msv15.i/230612.s.b/c0942.d

Date : 12-JUN-2023 18:23

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050201*C*250X

Purge Volume: 5.0

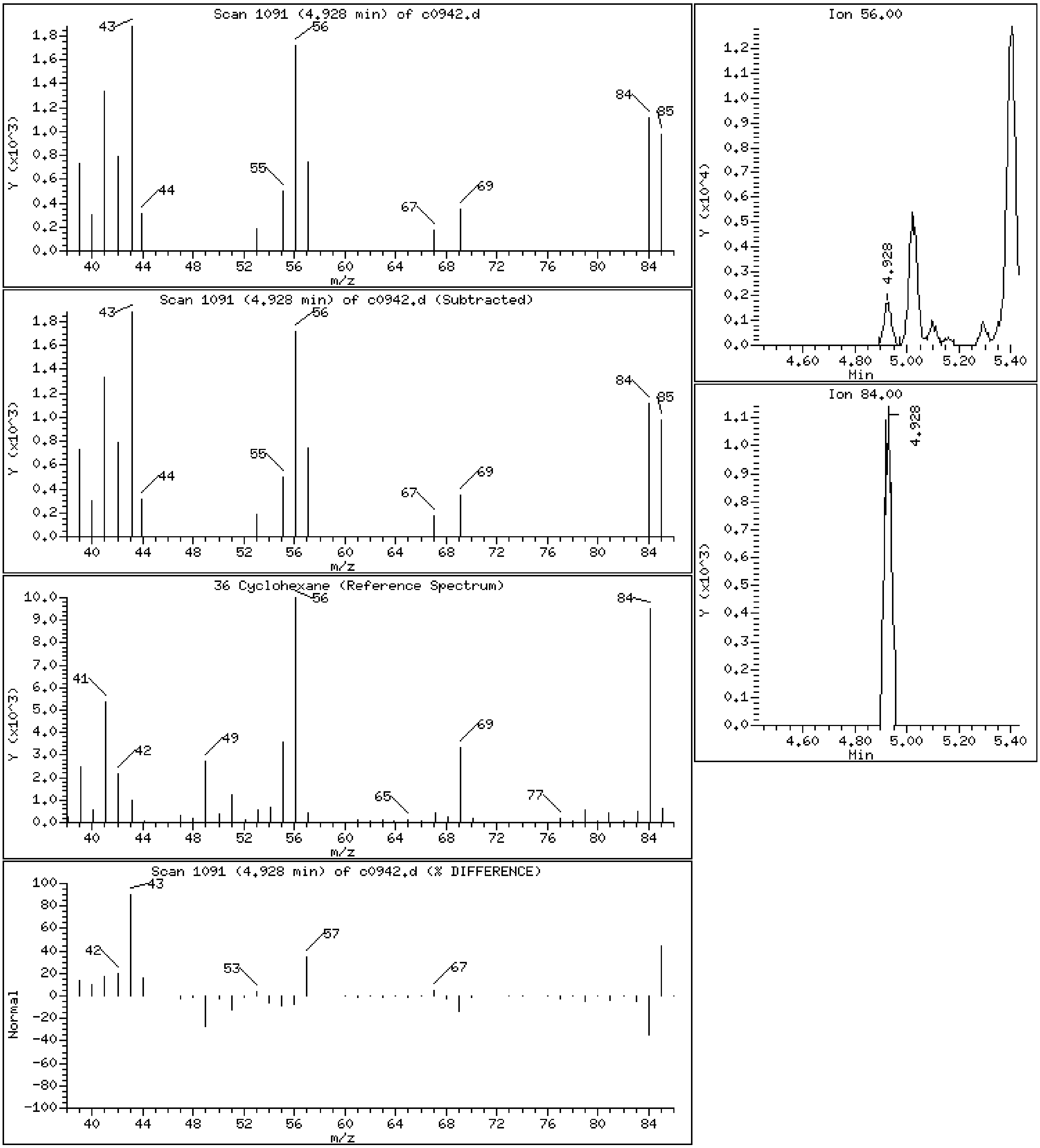
Operator: KN1

Column phase: RTX-VMS-30M

Column diameter: 0.25

36 Cyclohexane

Concentration: 275 UG/KG



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Date : 12-JUN-2023 18:23

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050201*C*250X

Purge Volume: 5.0

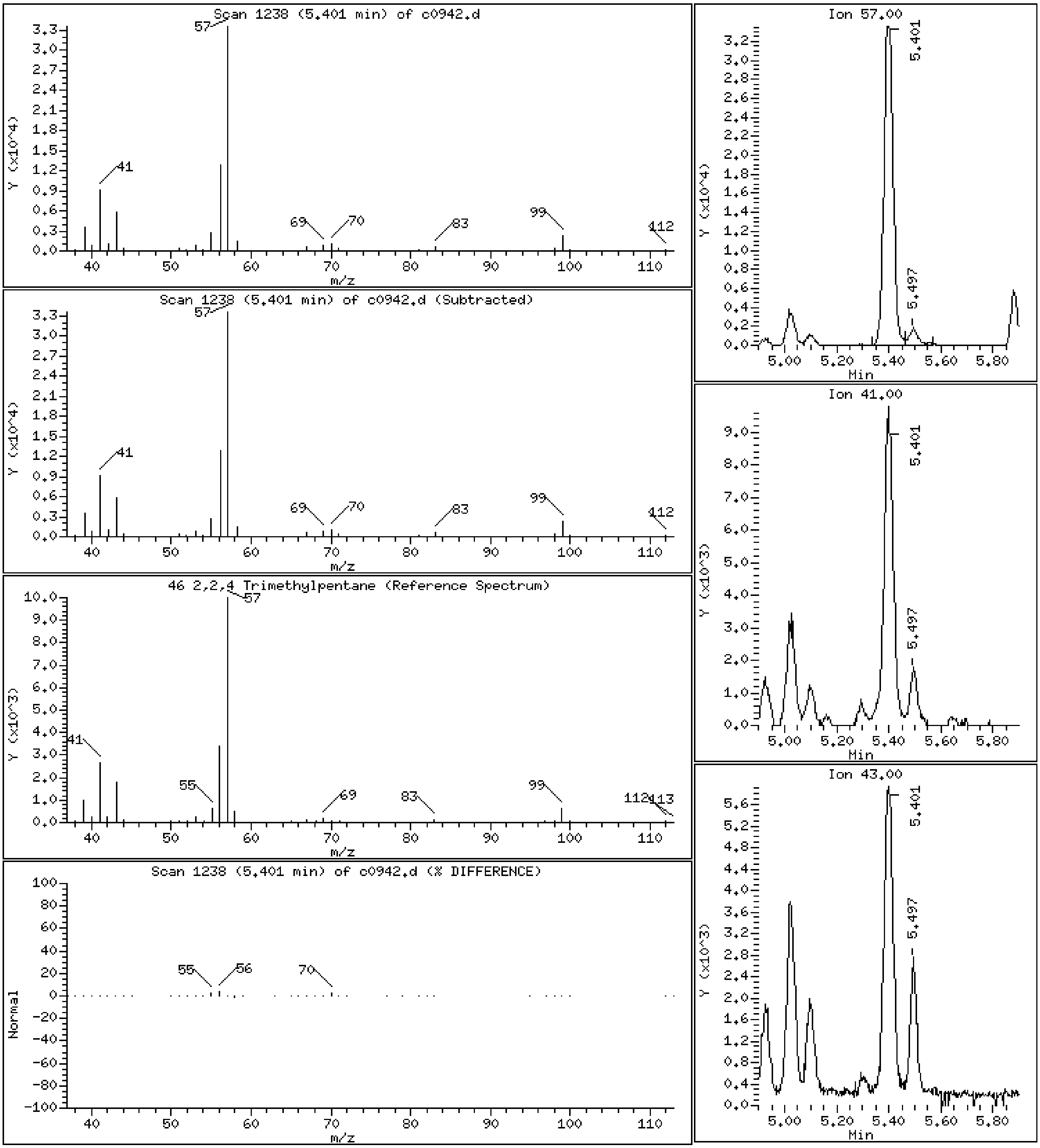
Operator: KN1

Column phase: RTX-VMS-30M

Column diameter: 0.25

46 2,2,4 Trimethylpentane

Concentration: 2350 UG/KG



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Date : 12-JUN-2023 18:23

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050201*C*250X

Purge Volume: 5.0

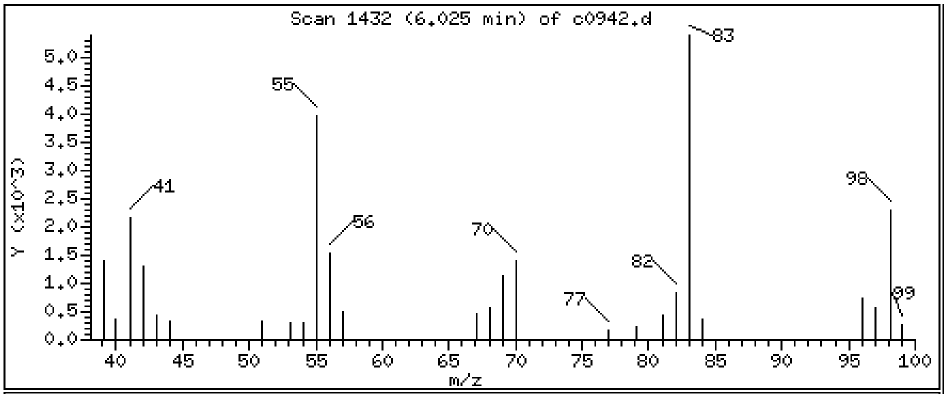
Operator: KN1

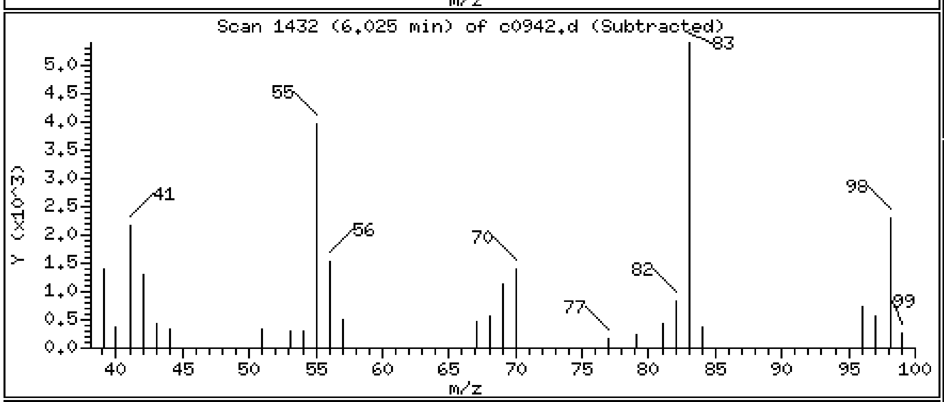
Column phase: RTX-VMS-30M

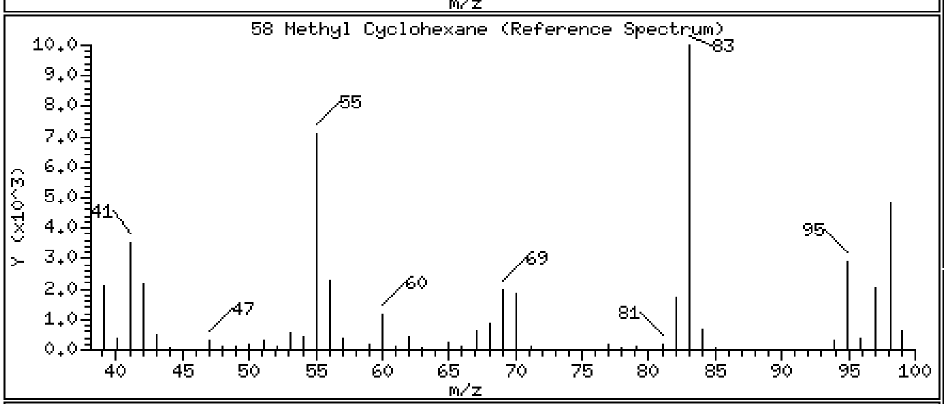
Column diameter: 0.25

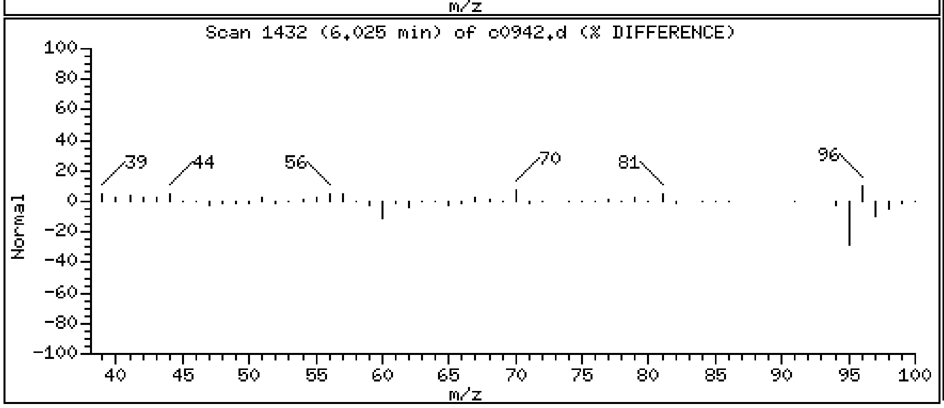
58 Methyl Cyclohexane

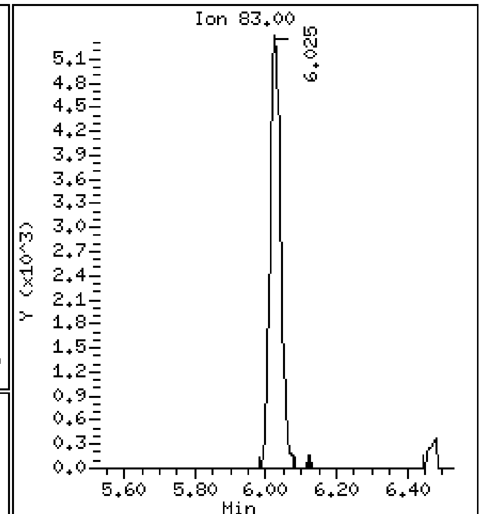
Concentration: 772 UG/KG

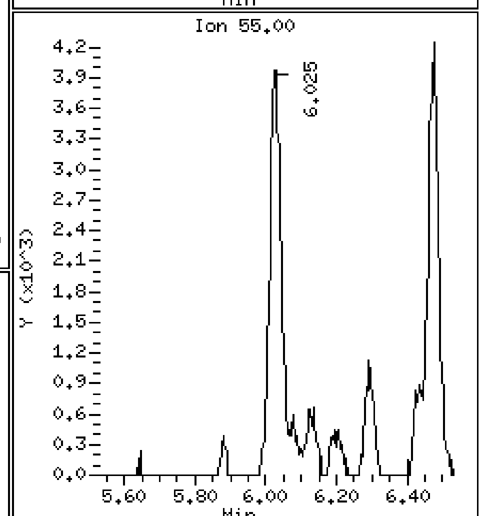


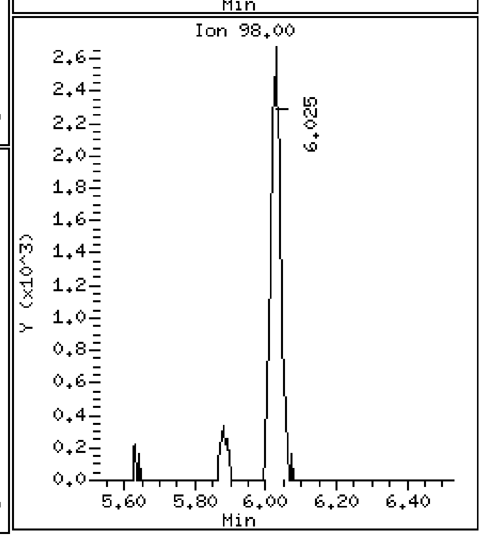












Date : 12-JUN-2023 18:23

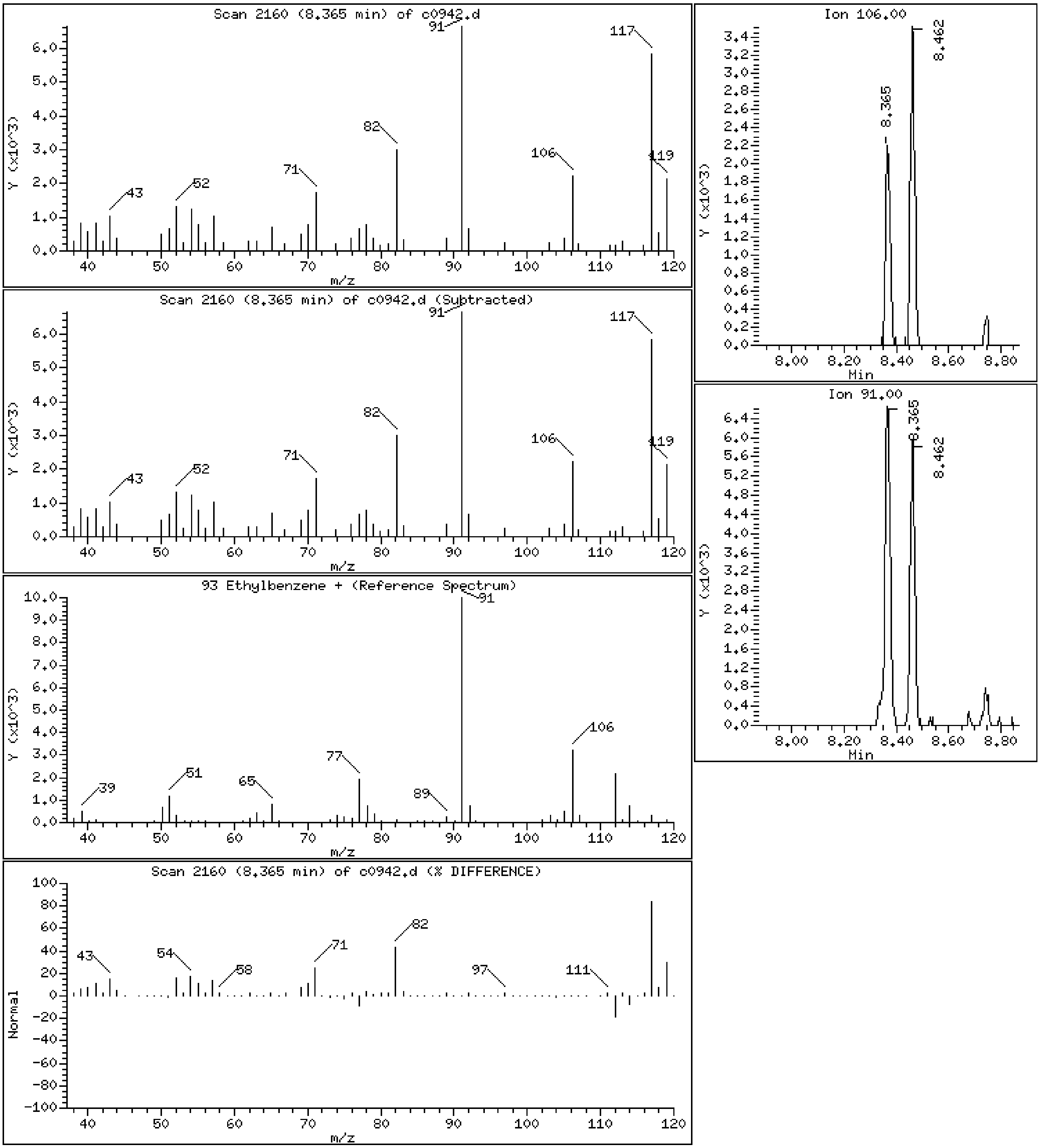
Client ID: 250X Instrument: msv15.i

Sample Info: 22306050201*C*250X

Purge Volume: 5.0 Operator: KN1

Column phase: RTX-VMS-30M Column diameter: 0.25

93 Ethylbenzene + Concentration: 187 UG/KG



Data File: /var/chem/msv15.i/230612.s.b/c0942.d

Date : 12-JUN-2023 18:23

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050201*C*250X

Purge Volume: 5.0

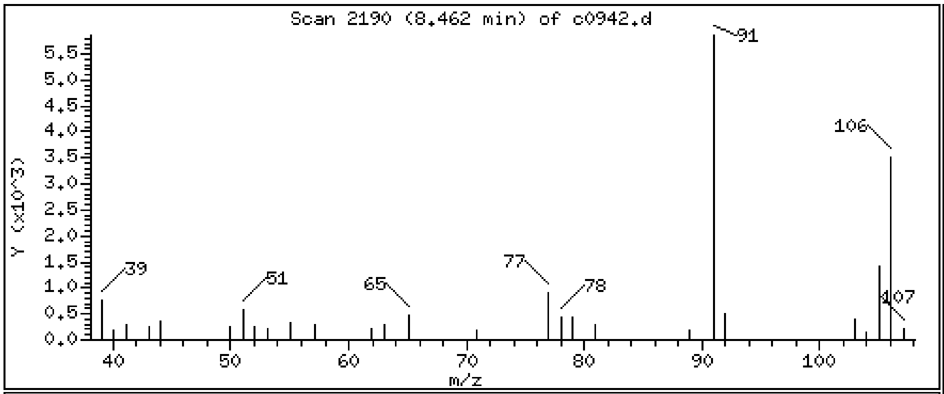
Operator: KN1

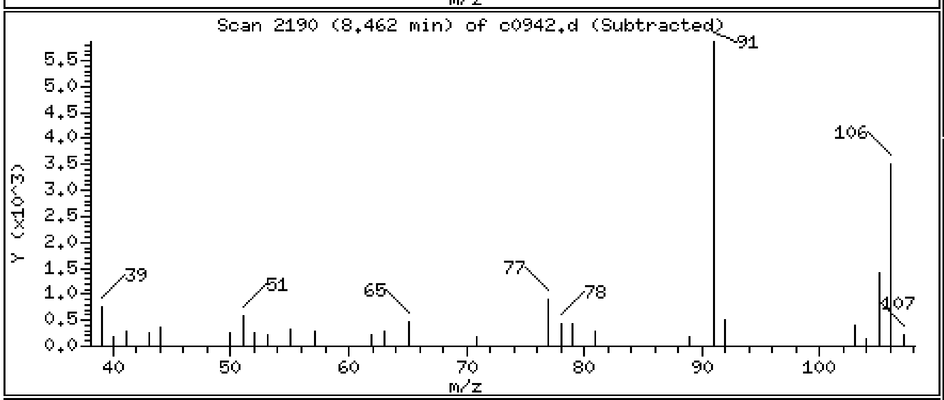
Column phase: RTX-VMS-30M

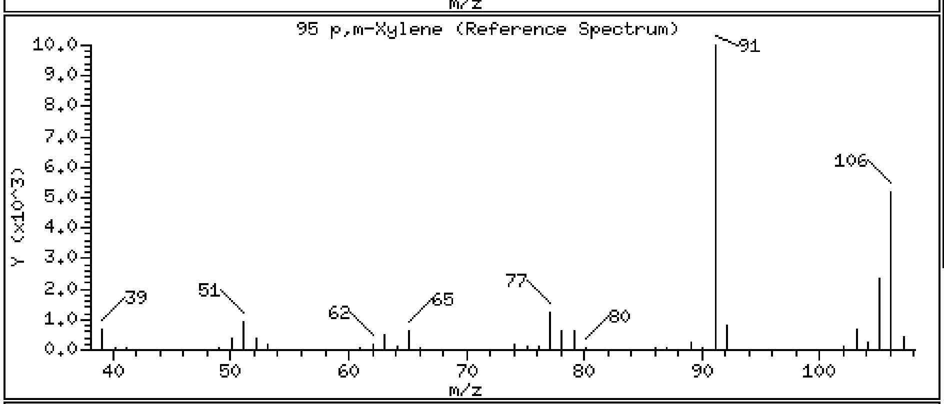
Column diameter: 0.25

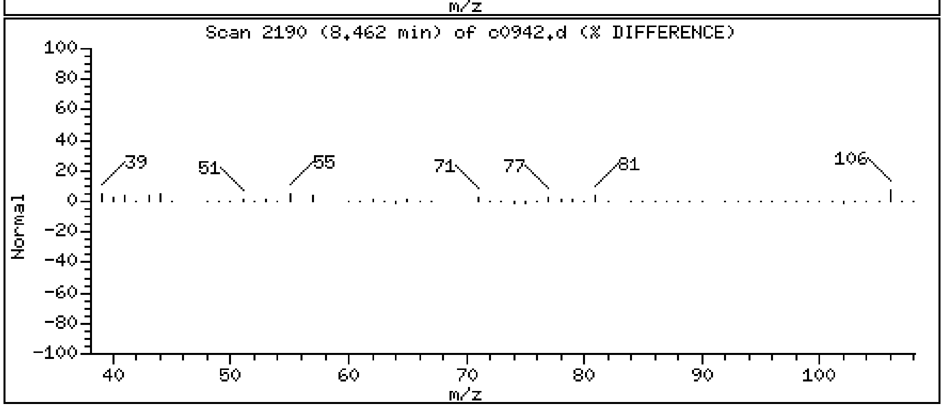
95 p,m-Xylene

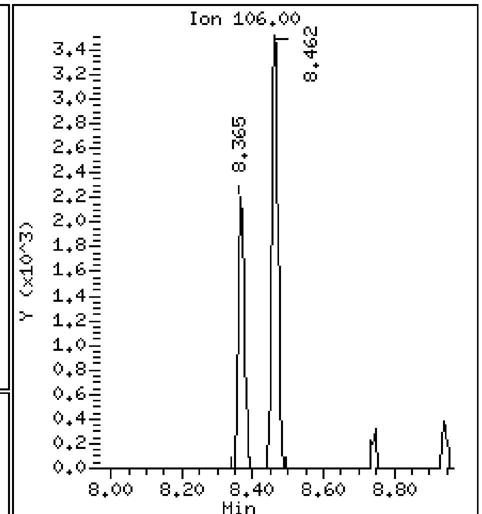
Concentration: 240 UG/KG

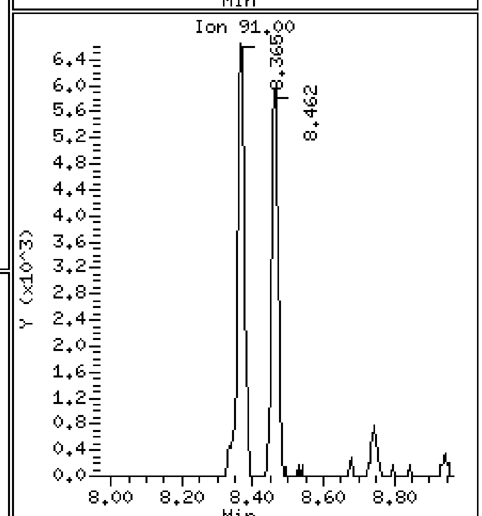












Data File: /var/chem/msv15.i/230612.s.b/c0942.d

Date : 12-JUN-2023 18:23

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050201*C*250X

Purge Volume: 5.0

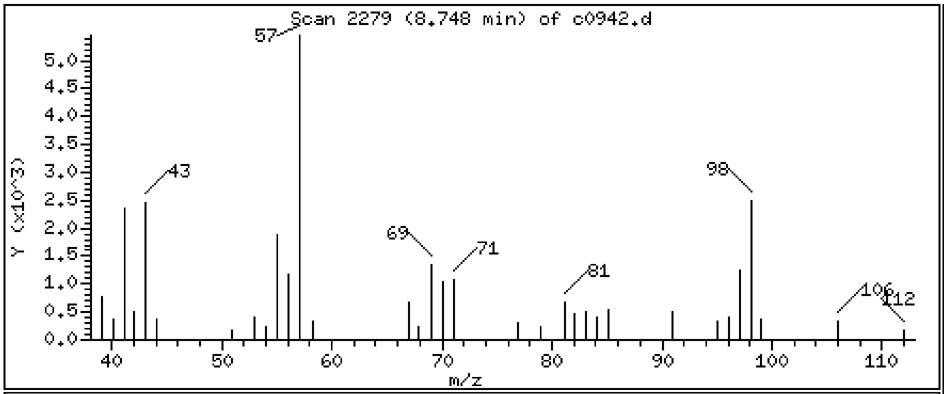
Operator: KN1

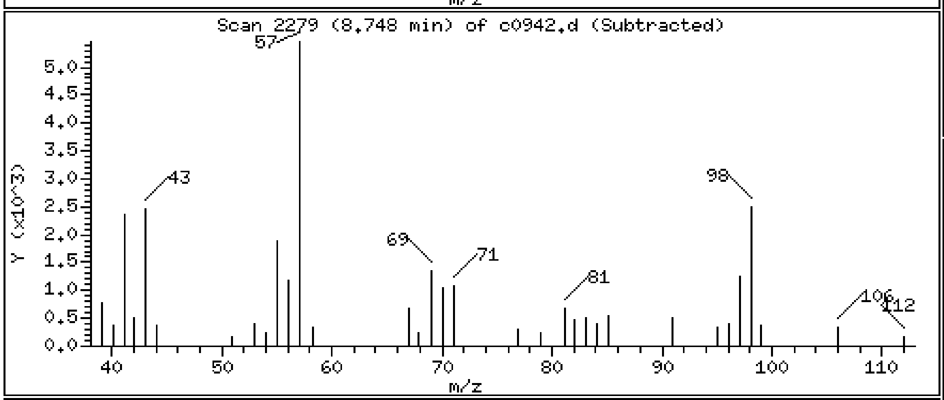
Column phase: RTX-VMS-30M

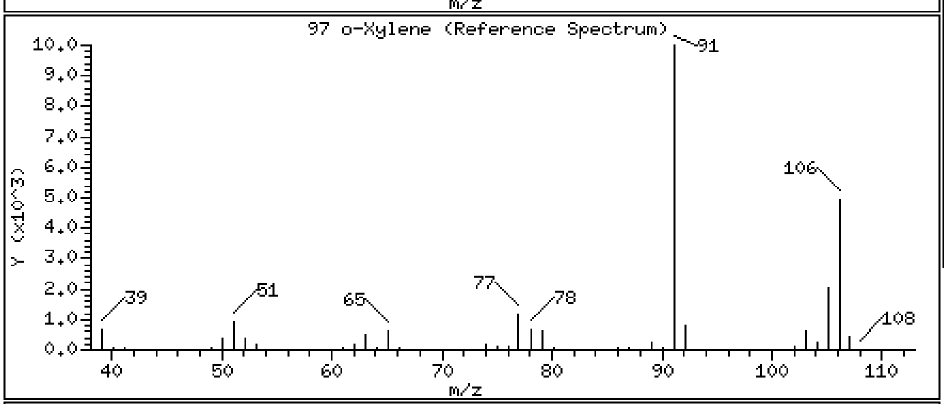
Column diameter: 0.25

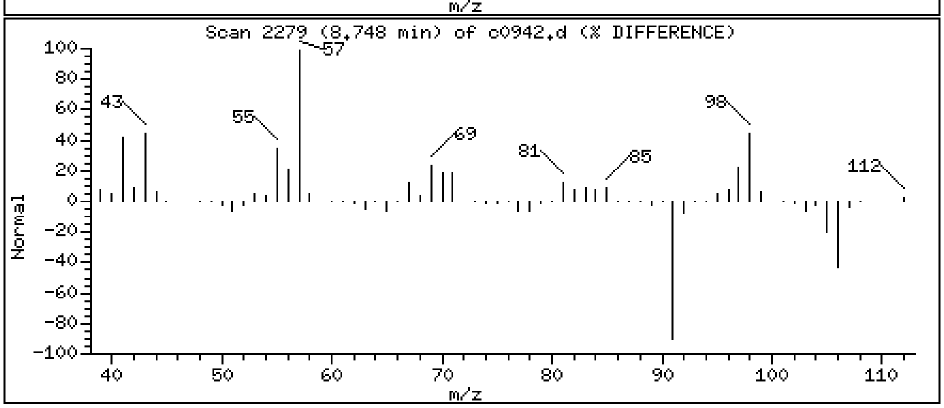
97 o-Xylene

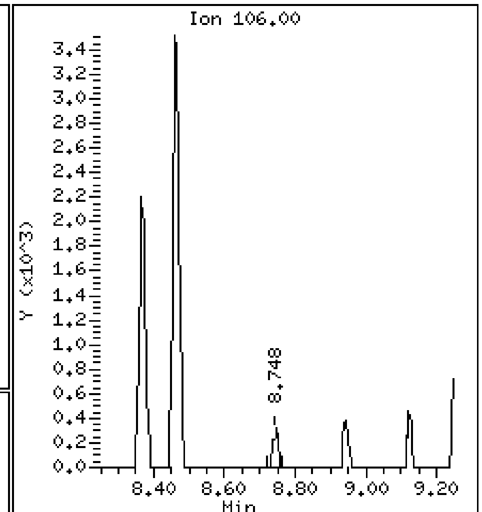
Concentration: 18.2 UG/KG

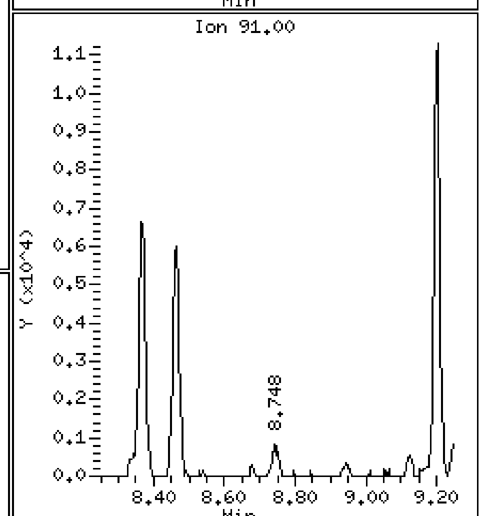












Data File: /var/chem/msv15.i/230612.s.b/c0942.d

Date : 12-JUN-2023 18:23

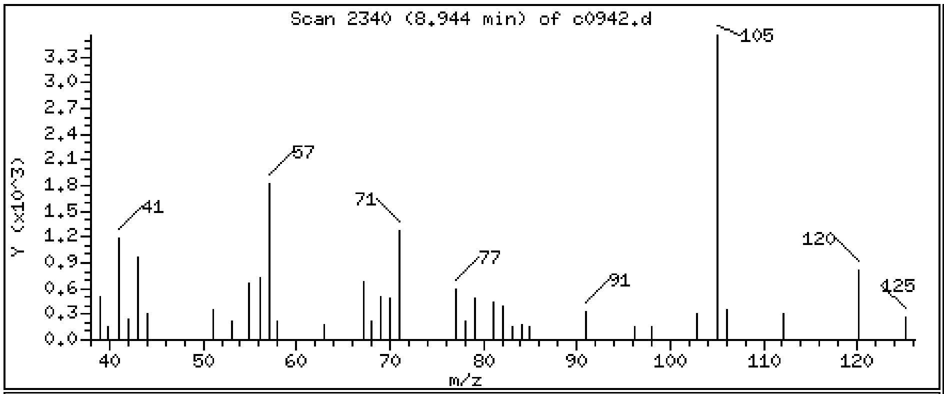
Client ID: 250X Instrument: msv15.i

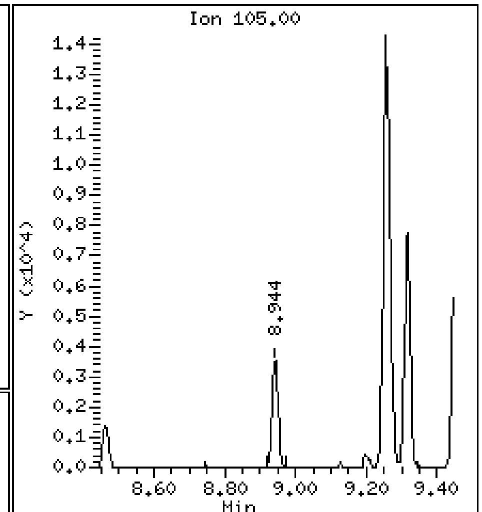
Sample Info: 22306050201*C*250X

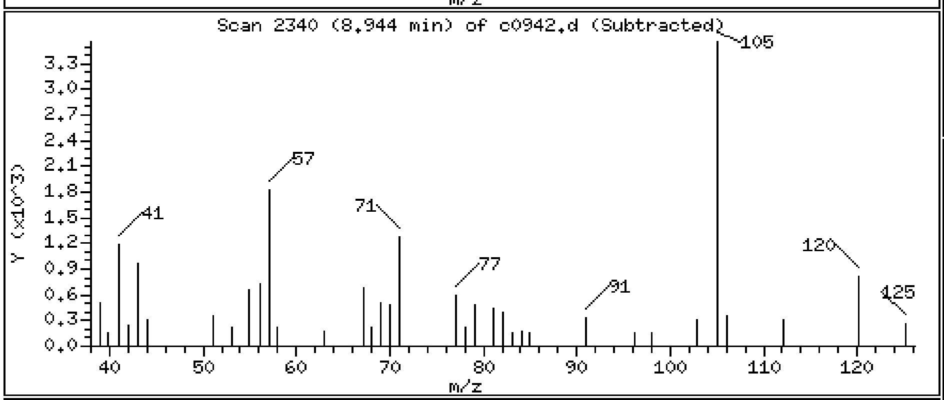
Purge Volume: 5.0 Operator: KN1

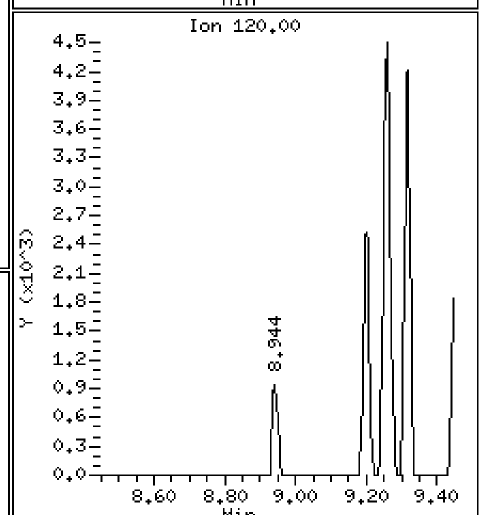
Column phase: RTX-VMS-30M Column diameter: 0.25

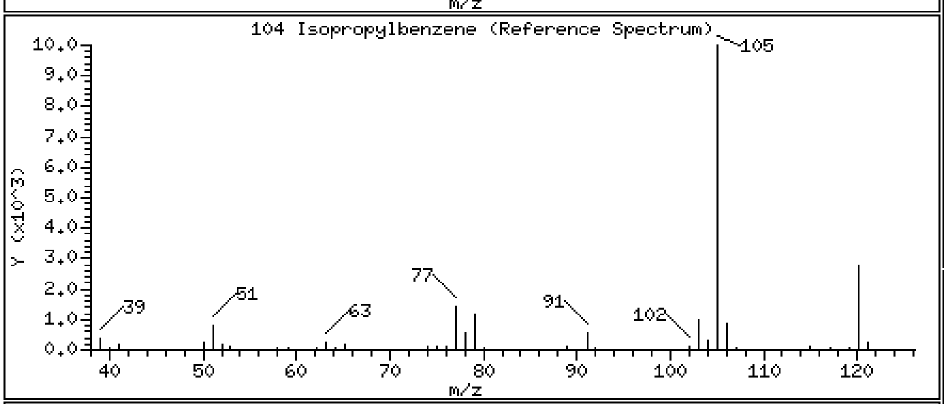
104 Isopropylbenzene Concentration: 99.0 UG/KG

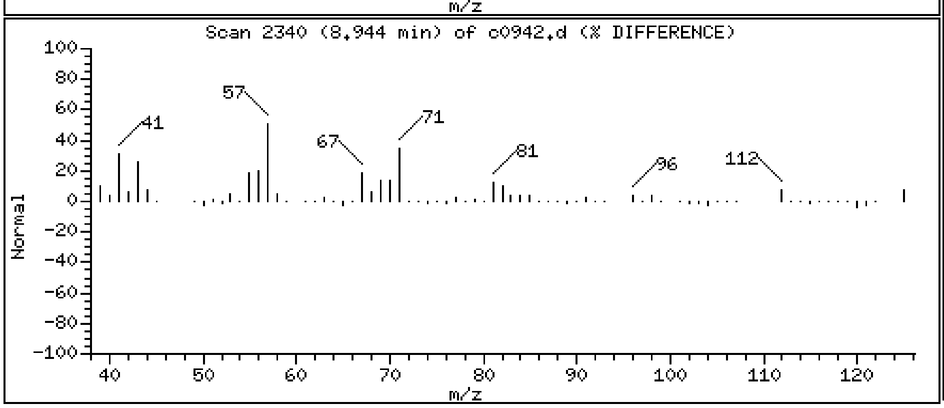


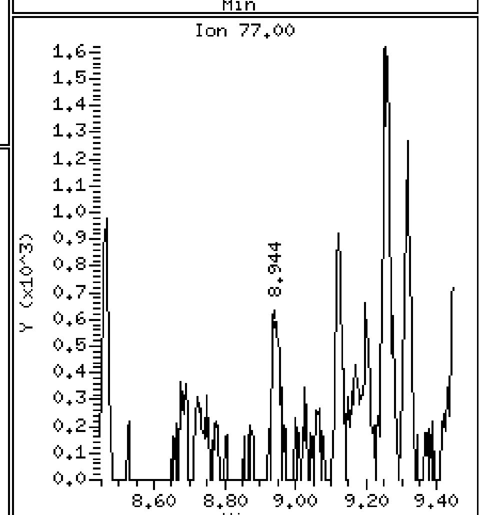












Data File: /var/chem/msv15.i/230612.s.b/c0942.d

Date : 12-JUN-2023 18:23

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050201*C*250X

Purge Volume: 5.0

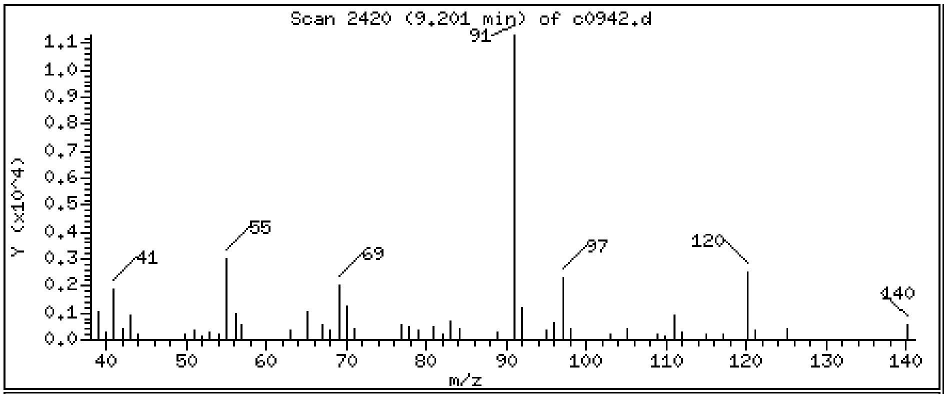
Operator: KN1

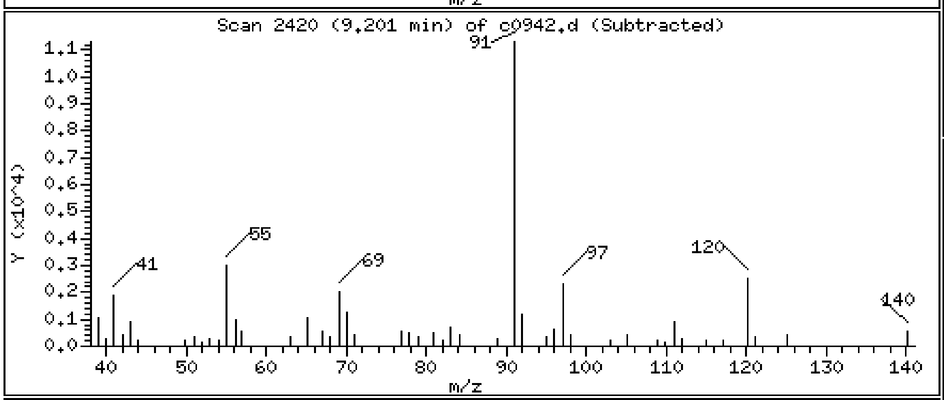
Column phase: RTX-VMS-30M

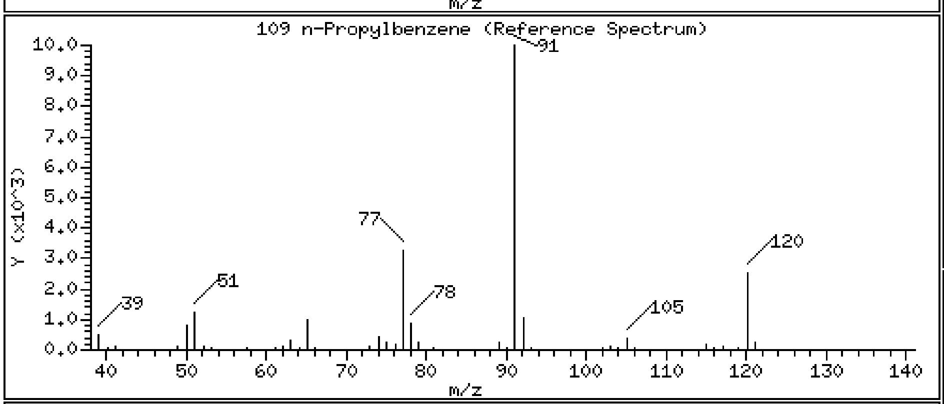
Column diameter: 0.25

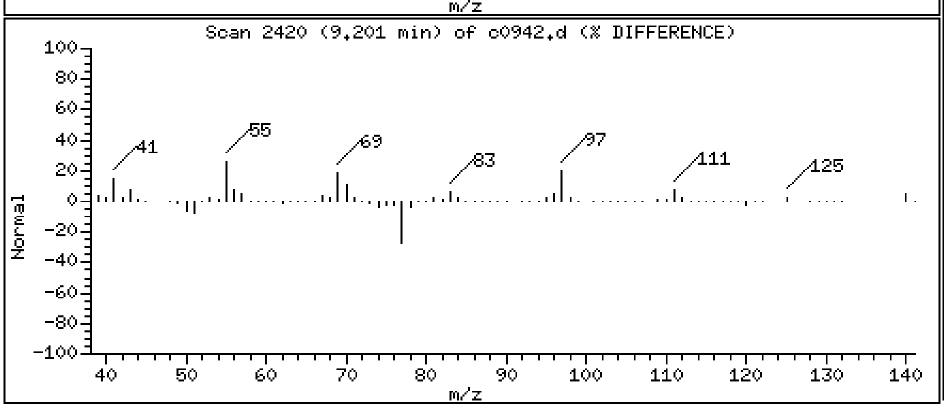
109 n-Propylbenzene

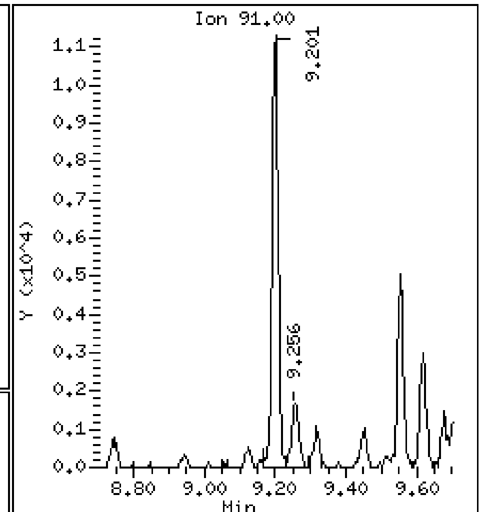
Concentration: 244 UG/KG

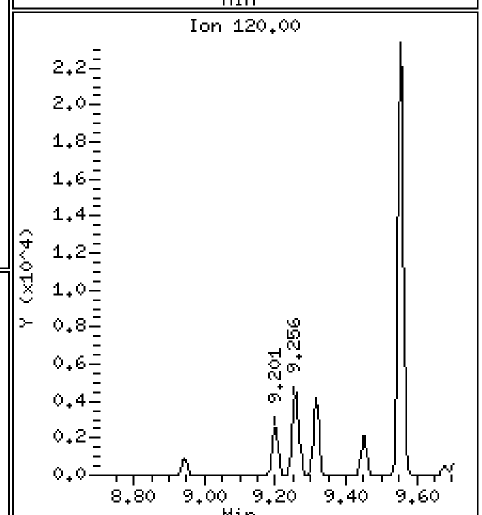


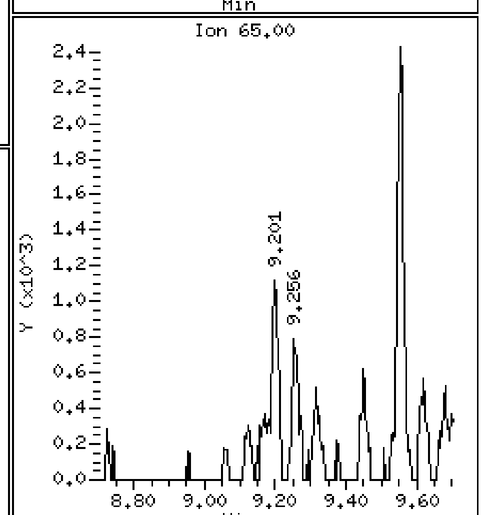












Data File: /var/chem/msv15.i/230612.s.b/c0942.d

Date : 12-JUN-2023 18:23

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050201*C*250X

Purge Volume: 5.0

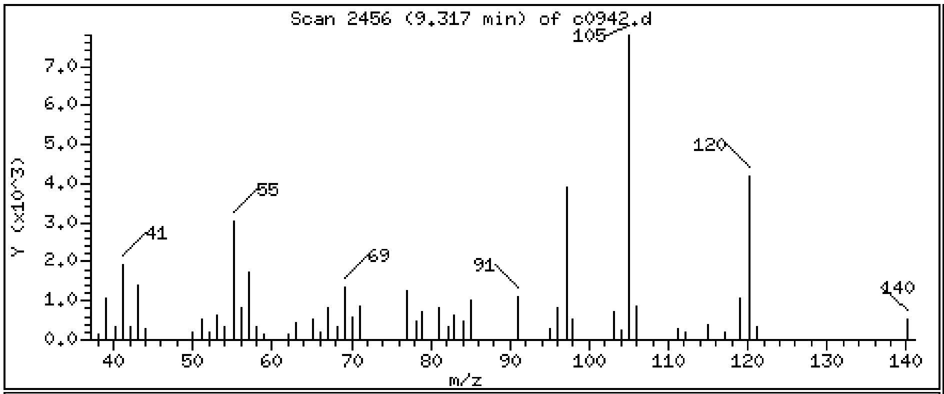
Operator: KN1

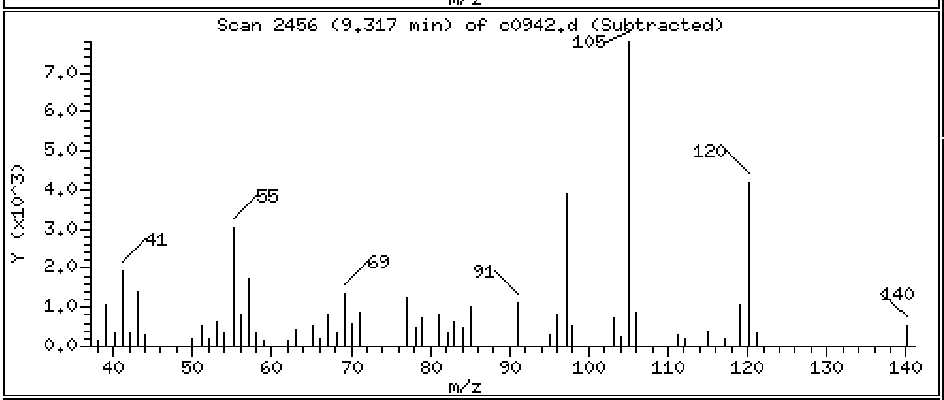
Column phase: RTX-VMS-30M

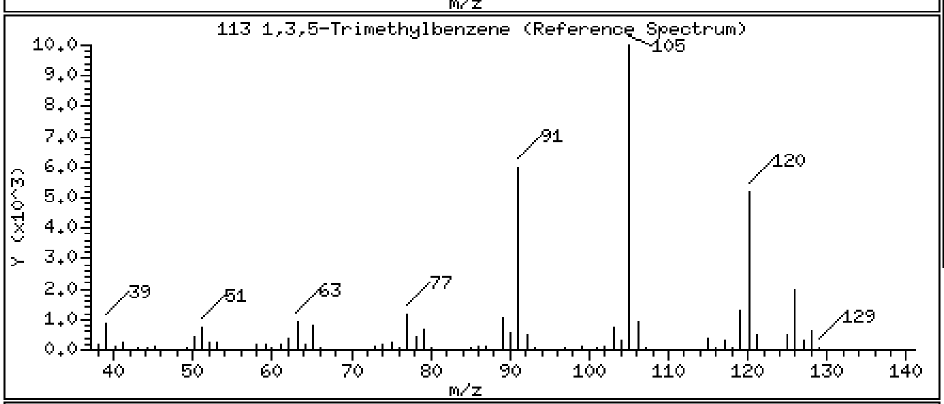
Column diameter: 0.25

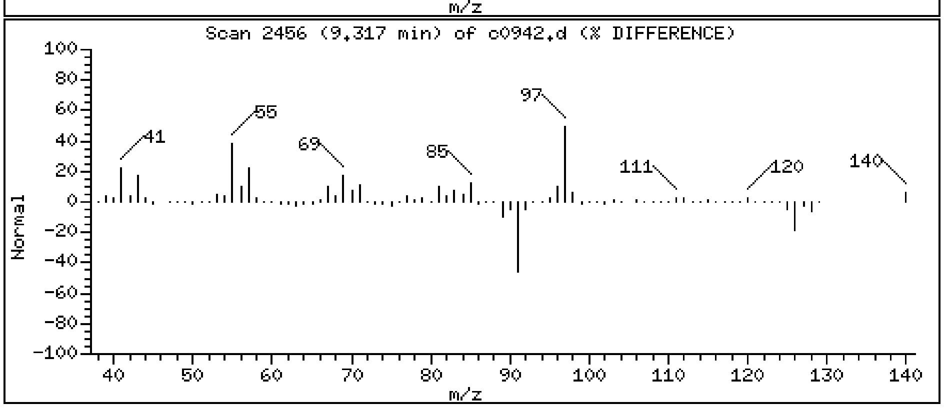
113 1,3,5-Trimethylbenzene

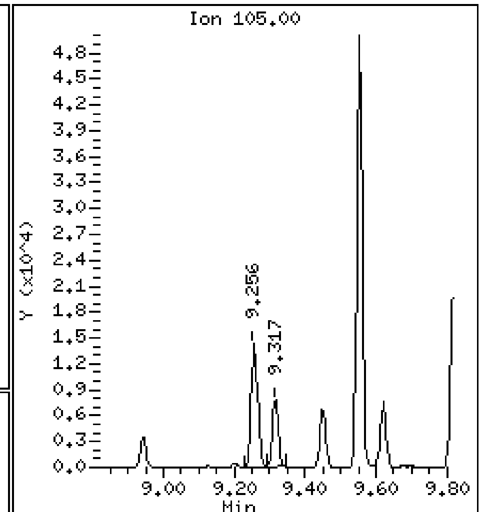
Concentration: 237 UG/KG

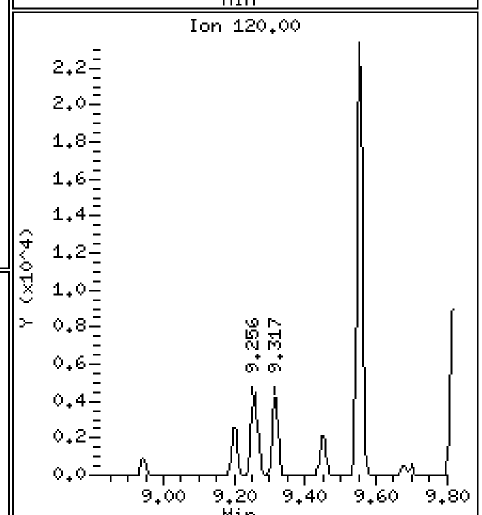












Data File: /var/chem/msv15.i/230612.s.b/c0942.d

Date : 12-JUN-2023 18:23

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050201*C*250X

Purge Volume: 5.0

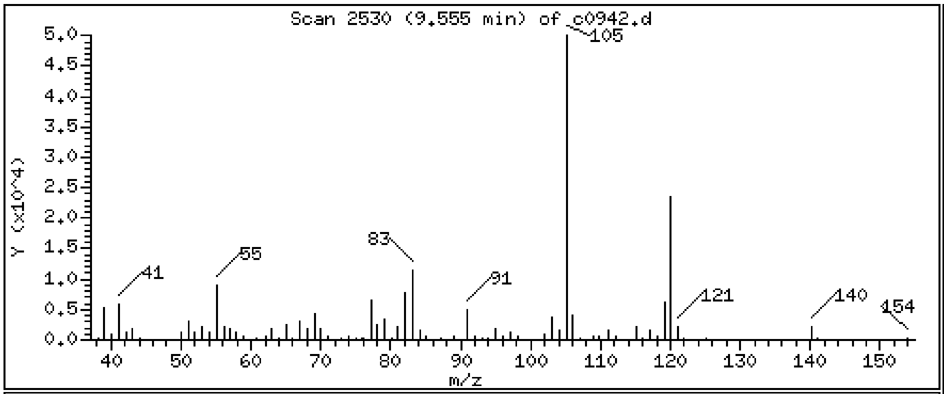
Operator: KN1

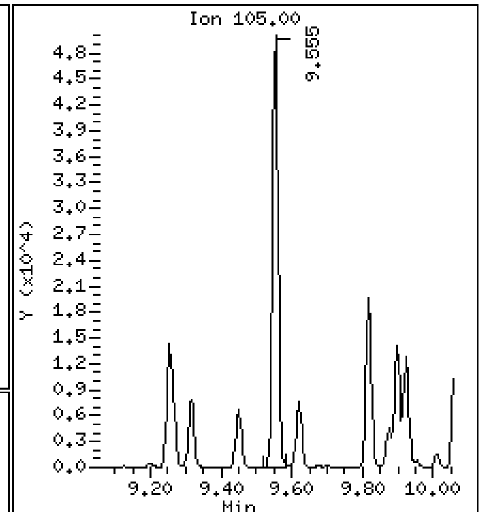
Column phase: RTX-VMS-30M

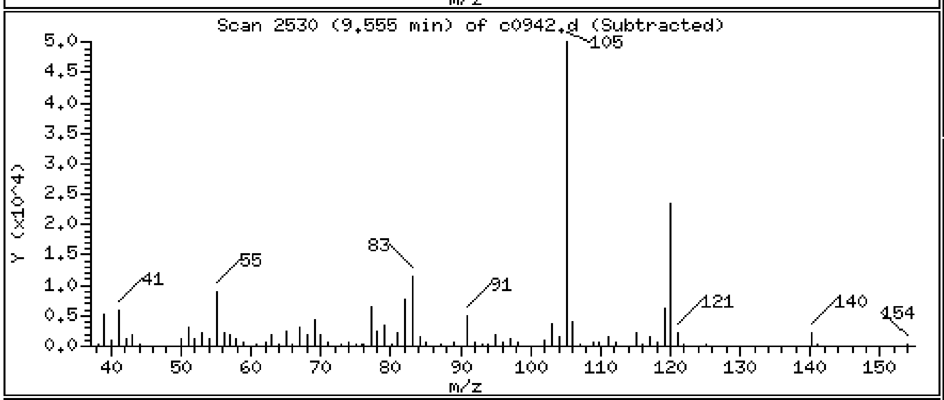
Column diameter: 0.25

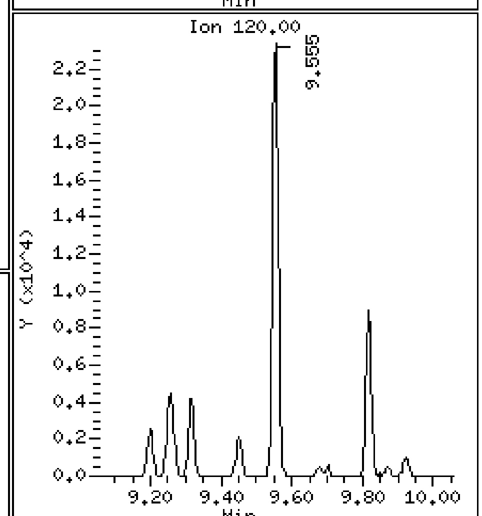
119 1,2,4-Trimethylbenzene

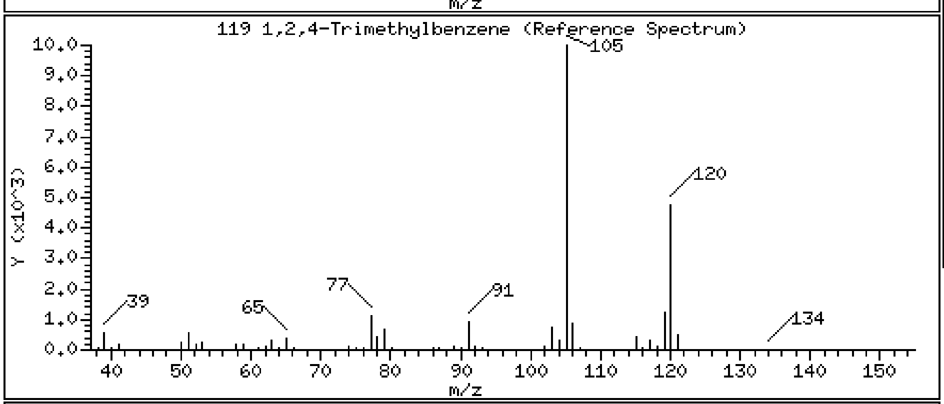
Concentration: 1520 UG/KG

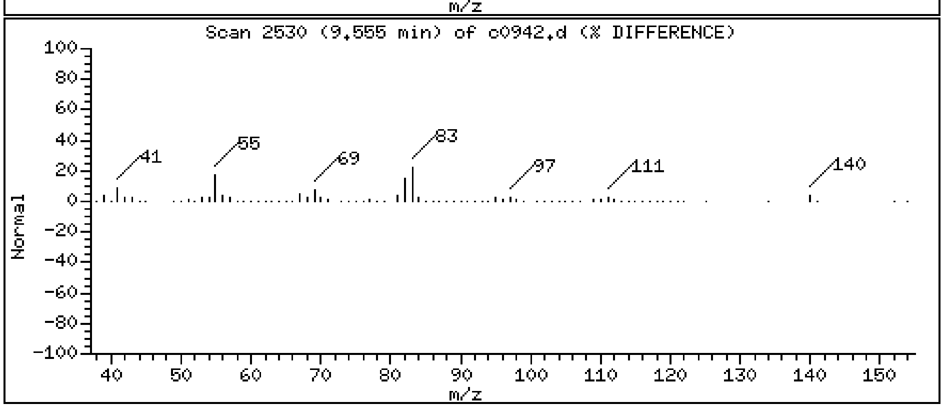












Date : 12-JUN-2023 18:23

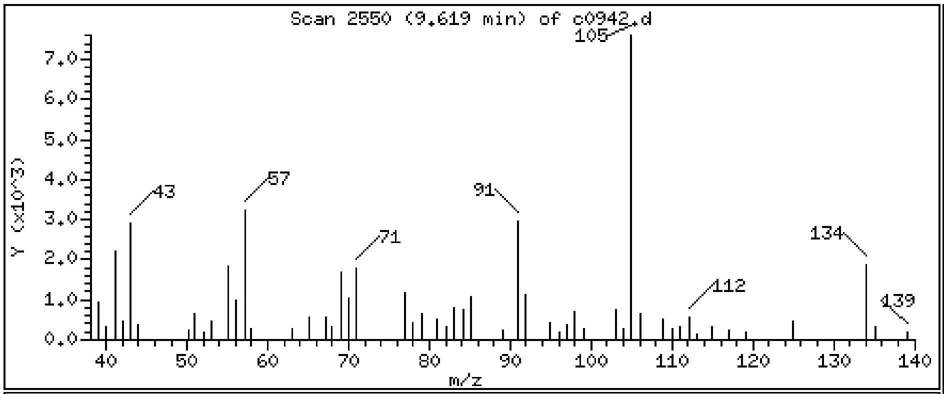
Client ID: 250X Instrument: msv15.i

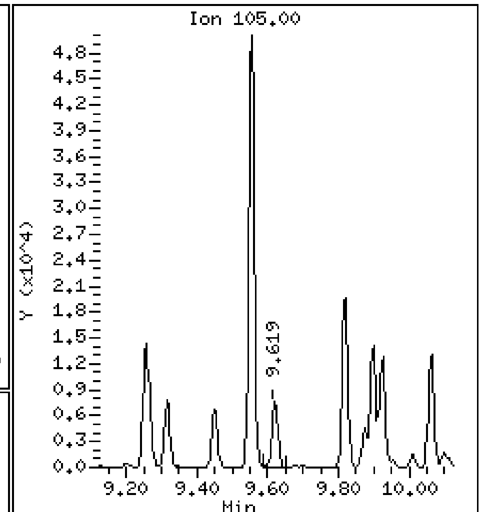
Sample Info: 22306050201*C*250X

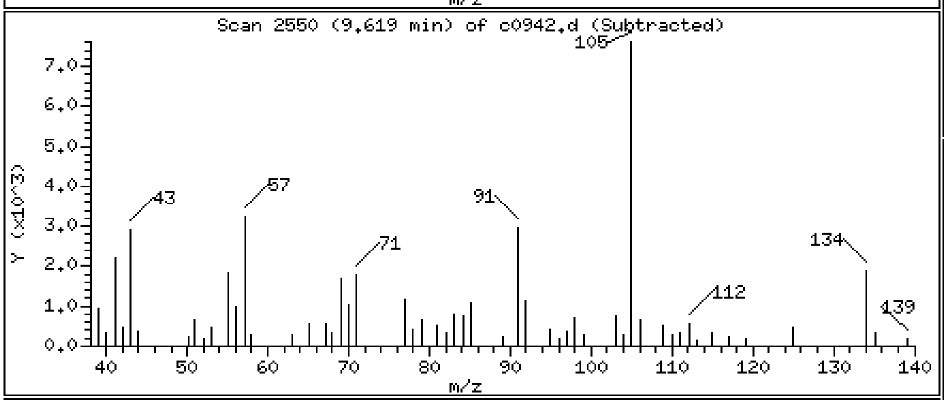
Purge Volume: 5.0 Operator: KN1

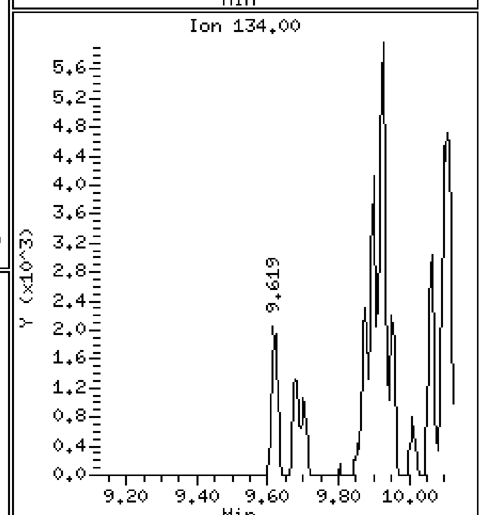
Column phase: RTX-VMS-30M Column diameter: 0.25

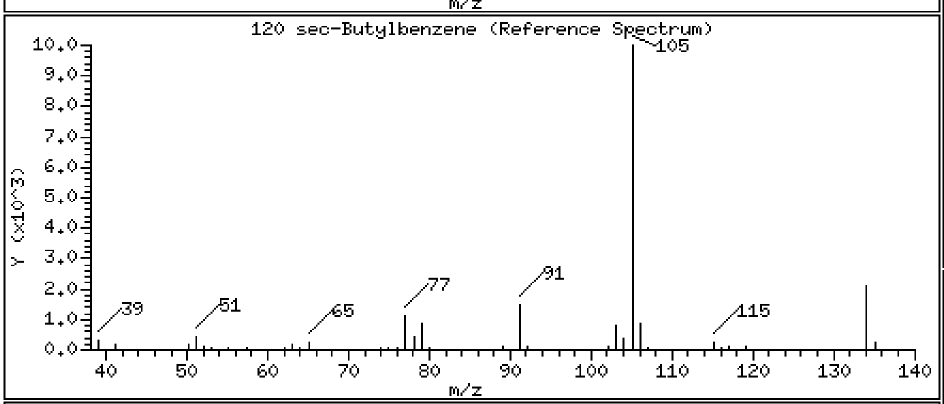
120 sec-Butylbenzene Concentration: 185 UG/KG

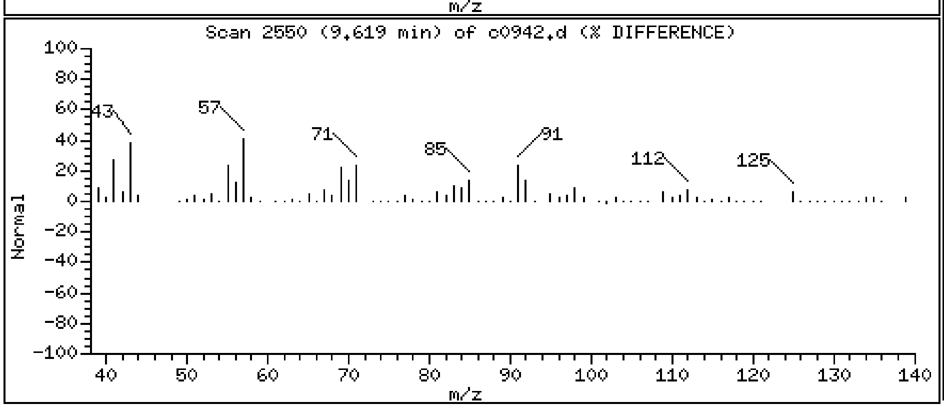


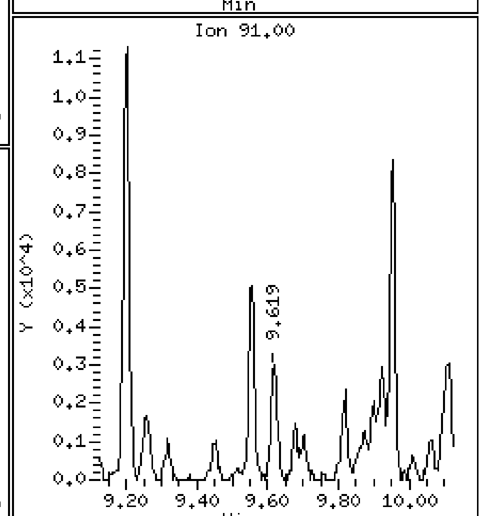












Data File: /var/chem/msv15.i/230612.s.b/c0942.d

Date : 12-JUN-2023 18:23

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050201*C*250X

Purge Volume: 5.0

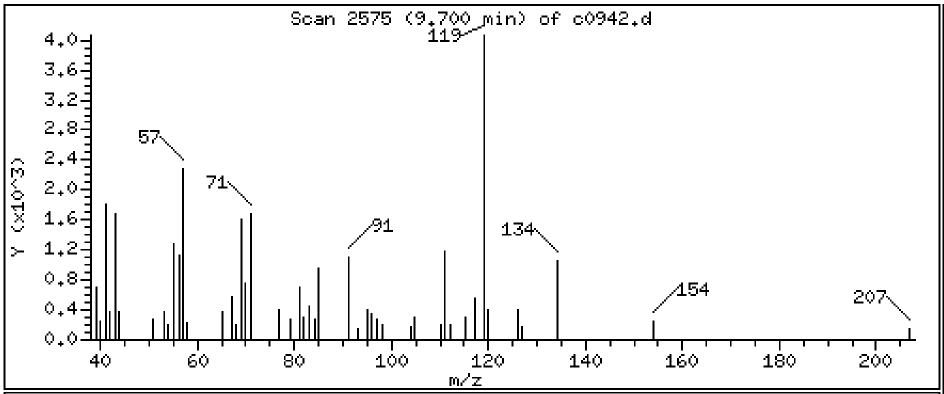
Operator: KN1

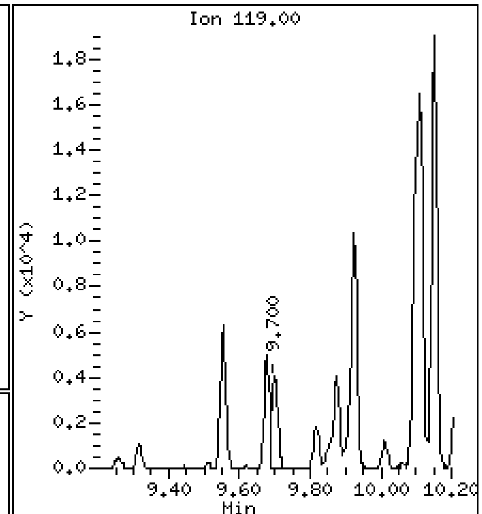
Column phase: RTX-VMS-30M

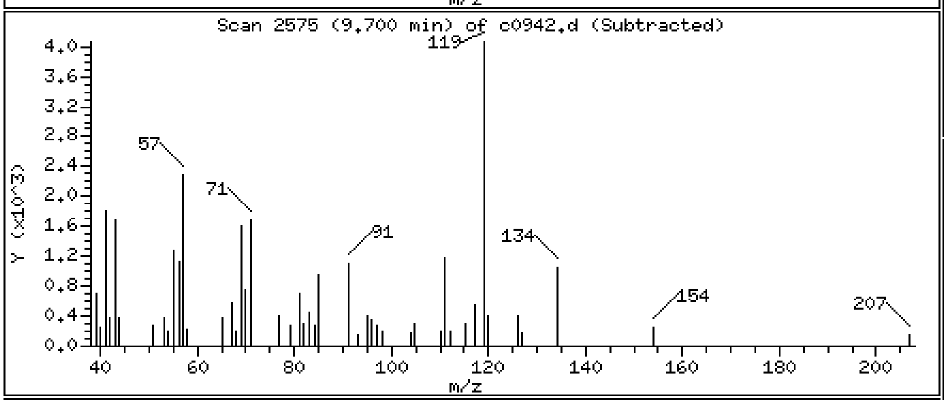
Column diameter: 0.25

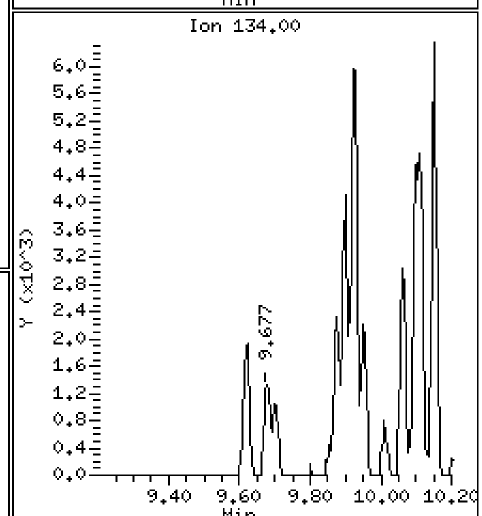
121 p-Isopropyltoluene

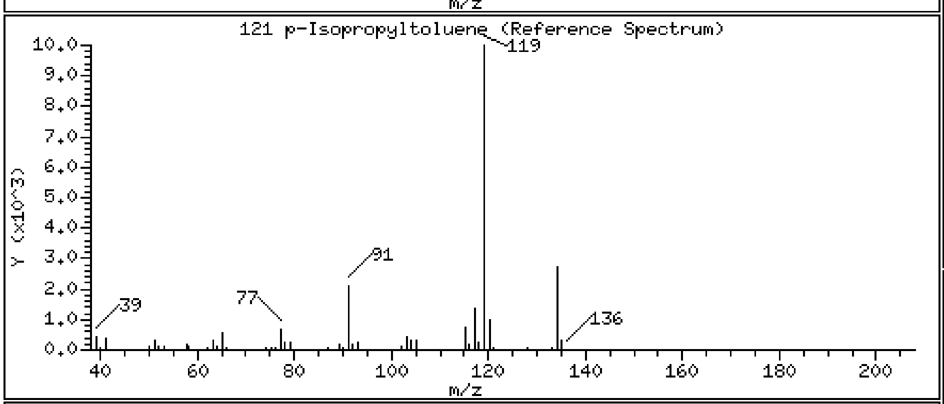
Concentration: 114 UG/KG

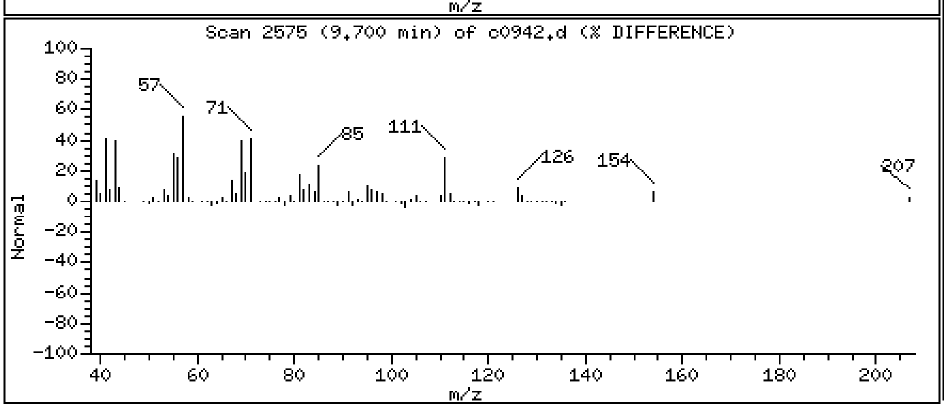


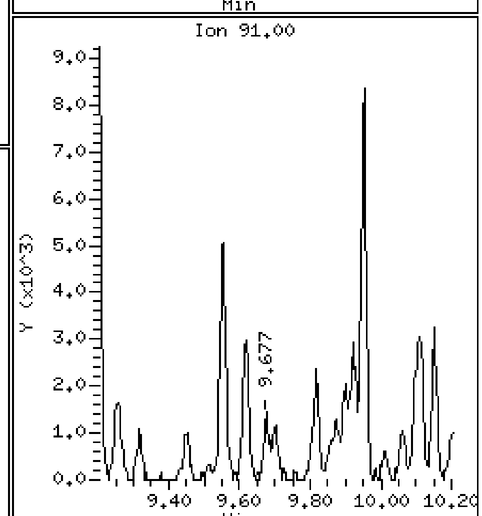












Date : 12-JUN-2023 18:23

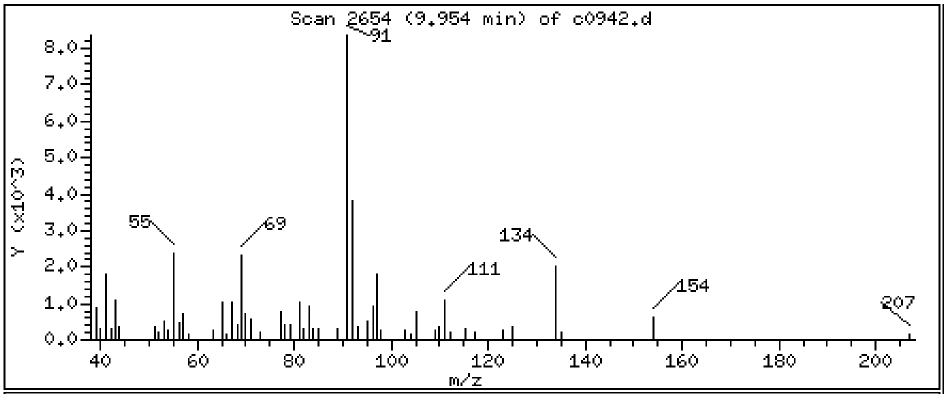
Client ID: 250X Instrument: msv15.i

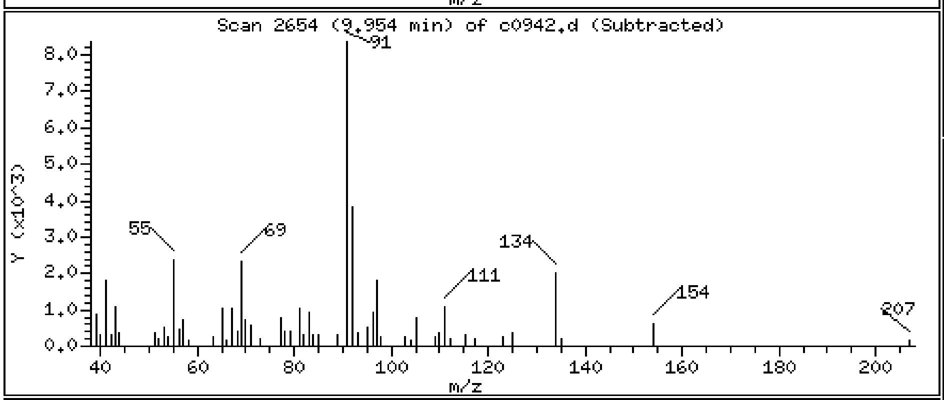
Sample Info: 22306050201*C*250X

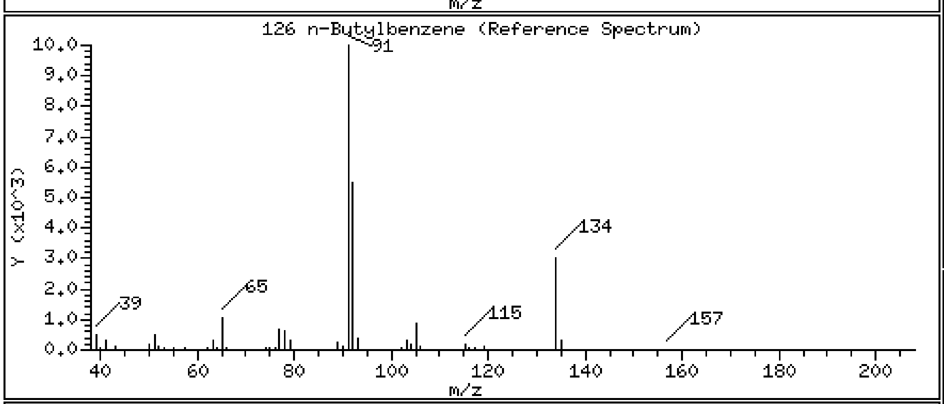
Purge Volume: 5.0 Operator: KN1

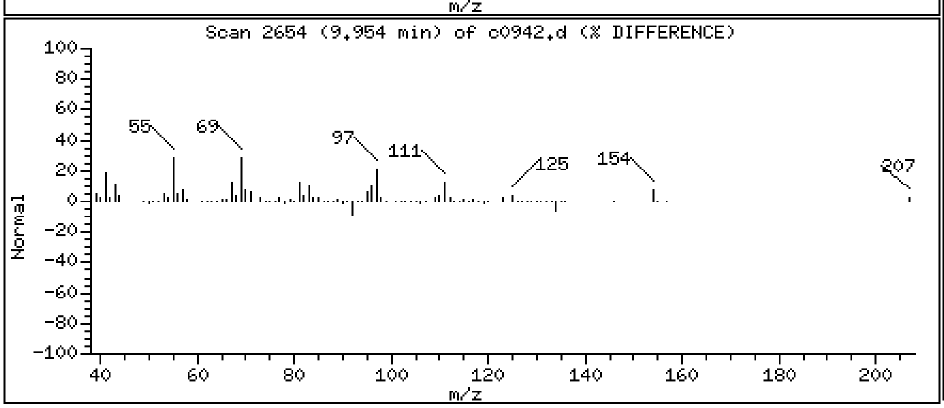
Column phase: RTX-VMS-30M Column diameter: 0.25

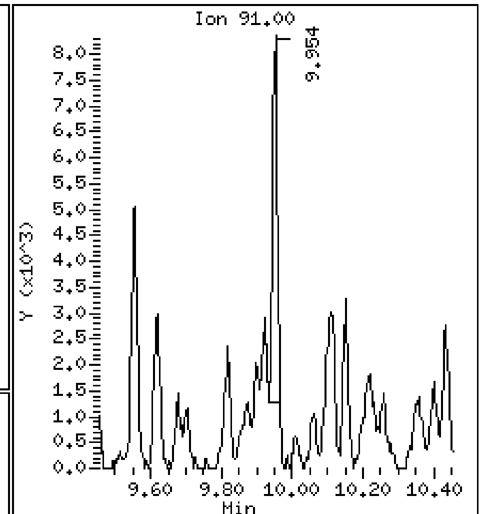
126 n-Butylbenzene Concentration: 113 UG/KG

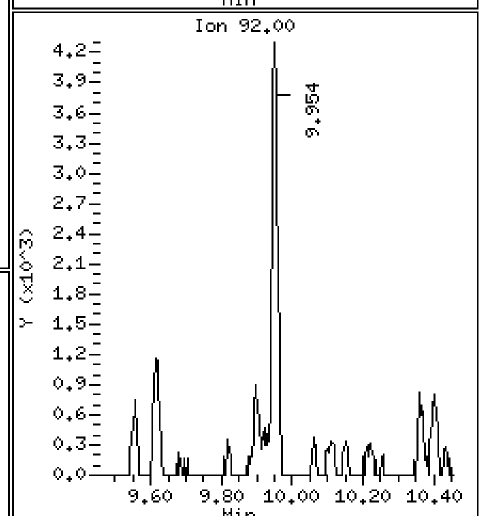


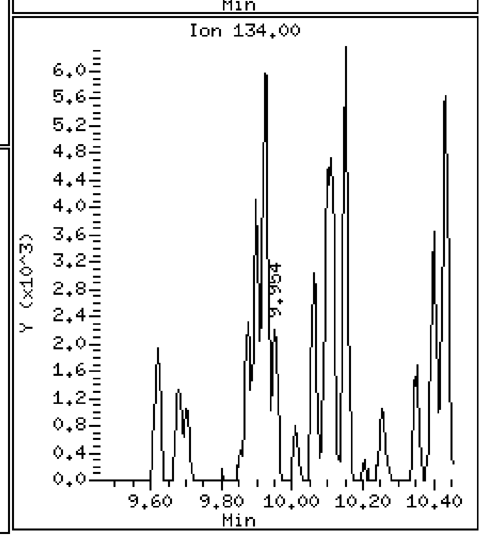












Data File: /var/chem/msv15.i/230612.s.b/c0942.d

Date : 12-JUN-2023 18:23

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050201*C*250X

Purge Volume: 5.0

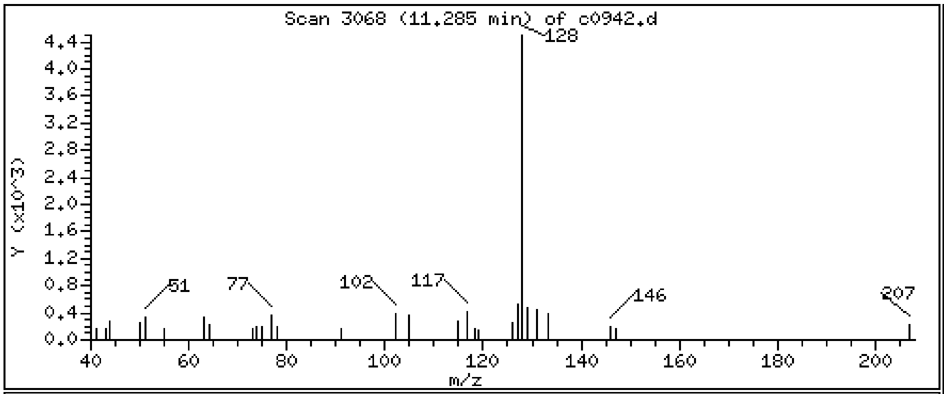
Operator: KN1

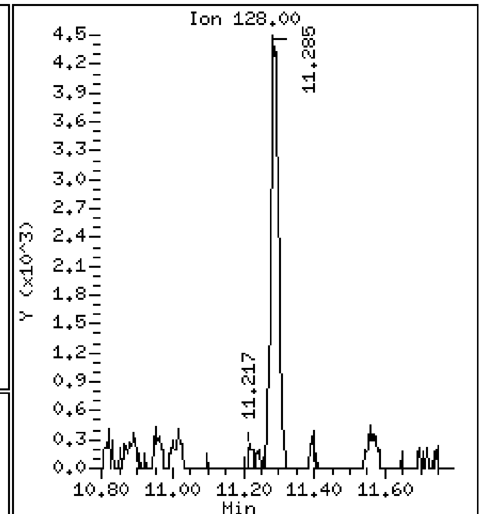
Column phase: RTX-VMS-30M

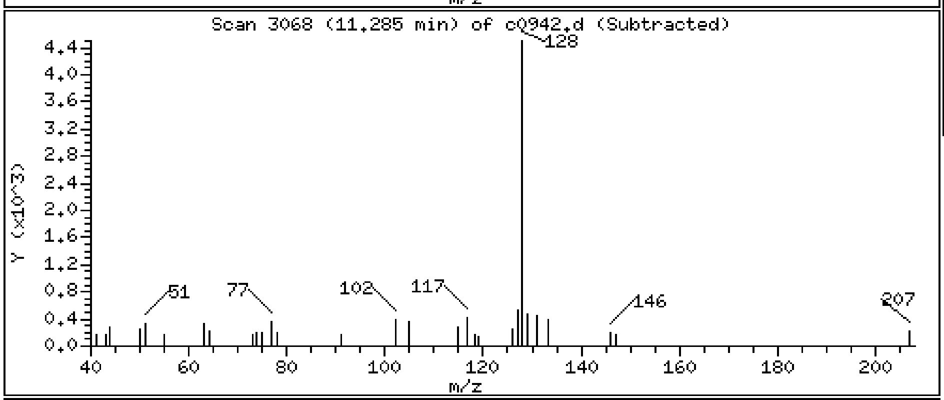
Column diameter: 0.25

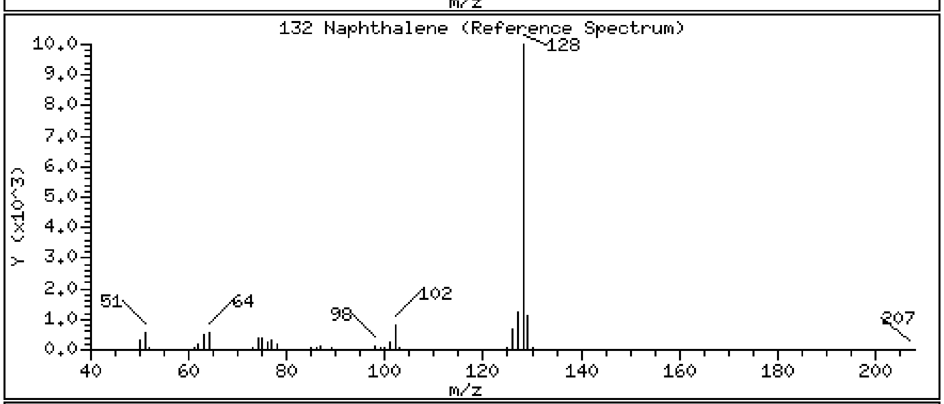
132 Naphthalene

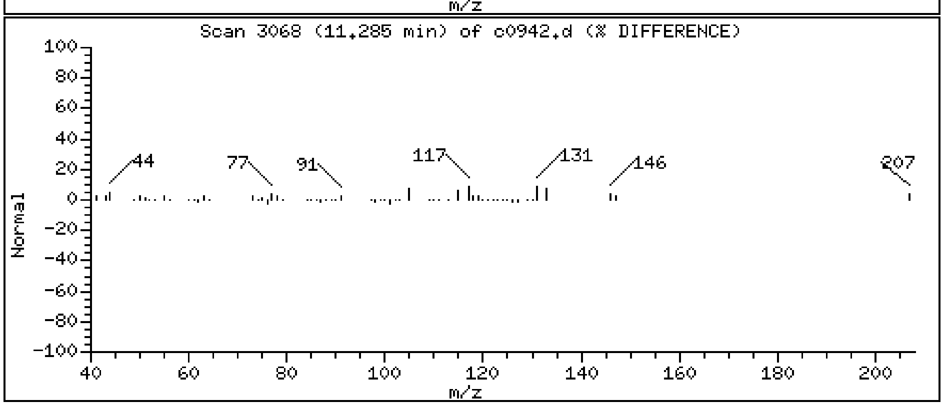
Concentration: 245 UG/KG







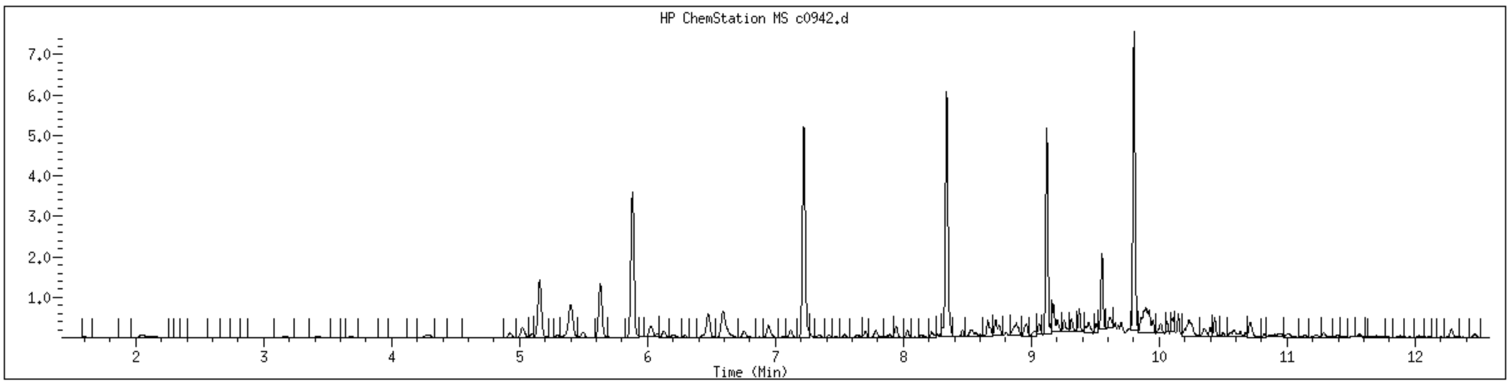




Report Date: 06/14/2023 13:30

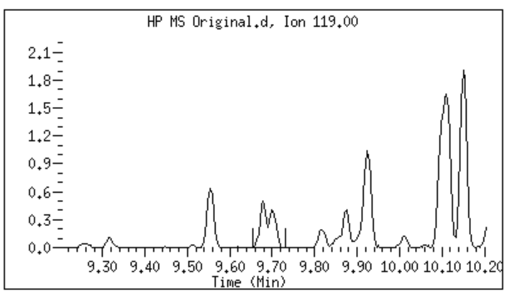
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID	: 22306050201	SampleType	: SAMPLE
Injection Date	: 06/12/2023 18:23	Instrument	: msv15.i
Operator	: KN1		
Sample Info	: 22306050201*C*250X		
Misc Info	: MSV~52948~*250*SMR		
Method	: /var/chem/msv15.i/230612.s.b/8260bw15DOD.m		
Dilution	: 250		
Matrix	: SOIL		
Integrator	: HP RTE	Compound Sublist	: 8260DOD



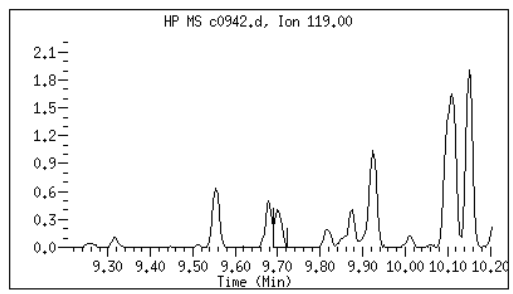
Original   Final

121 p-Isopropyltoluene   CAS#: 99-87-6   Reason: M2

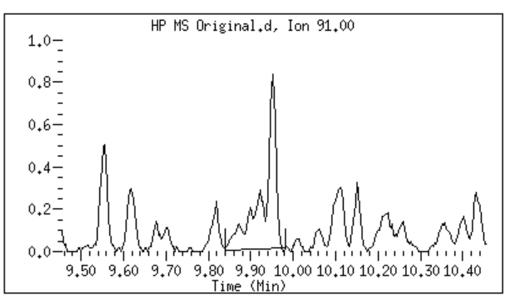


Electronic Signature Applied

User: kmn
Date: 06/14/2023 13:30

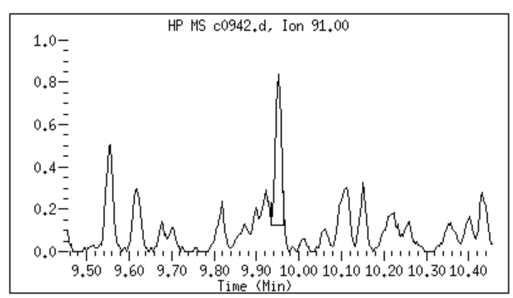


126 n-Butylbenzene   CAS#: 104-51-8   Reason: M2



Electronic Signature Applied

User: kmn
Date: 06/14/2023 13:30



M2 - Target system integrated incorrectly

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060502			Client Sample ID:	OT07-BO-40-4-6-8-MS		
Collect Date:	06/02/23	Time:	0910	GCAL Sample ID:	22306050202		
Matrix:	Solid	% Moisture:	22.5	Instrument ID:	MSV15		
Sample Amt:	5.63	g		Lab File ID:	230612/c0946		
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M	ID	.25 (mm)
Dilution Factor:	250	Analyst:	KN1	Analytical Batch:	767415		
Analysis Date:	06/12/23	Time:	1938	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/kg

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	13300		71.6	143	1430
71-55-6	1,1,1-Trichloroethane	16100		71.6	143	1430
79-34-5	1,1,2,2-Tetrachloroethane	15700		71.6	143	1430
79-00-5	1,1,2-Trichloroethane	15000		71.6	143	1430
75-34-3	1,1-Dichloroethane	16900		71.6	143	1430
75-35-4	1,1-Dichloroethene	17000		71.6	143	1430
563-58-6	1,1-Dichloropropene	16200		71.6	143	1430
87-61-6	1,2,3-Trichlorobenzene	14400		143	286	1430
96-18-4	1,2,3-Trichloropropane	12200		143	286	1430
120-82-1	1,2,4-Trichlorobenzene	14300		143	286	1430
95-63-6	1,2,4-Trimethylbenzene	33500		71.6	143	1430
96-12-8	1,2-Dibromo-3-chloropropane	12200		143	573	1430
106-93-4	1,2-Dibromoethane	13900		143	573	1430
95-50-1	1,2-Dichlorobenzene	12700		71.6	143	1430
107-06-2	1,2-Dichloroethane	15300		71.6	143	1430
540-59-0	1,2-Dichloroethene (total)	33100		143	286	2860
78-87-5	1,2-Dichloropropane	16900		71.6	143	1430
108-67-8	1,3,5-Trimethylbenzene	17200		71.6	143	1430
541-73-1	1,3-Dichlorobenzene	12500		71.6	143	1430
142-28-9	1,3-Dichloropropane	13900		71.6	143	1430
106-46-7	1,4-Dichlorobenzene	12500		71.6	143	1430
594-20-7	2,2-Dichloropropane	14400		71.6	143	1430
78-93-3	2-Butanone	91600		143	573	1430
95-49-8	2-Chlorotoluene	12700		71.6	143	1430
591-78-6	2-Hexanone	69800		143	573	1430
106-43-4	4-Chlorotoluene	12200		71.6	143	1430
99-87-6	4-Isopropyltoluene	18200		71.6	143	1430
108-10-1	4-Methyl-2-pentanone	73500		71.6	143	1430
67-64-1	Acetone	83700		143	573	7160
71-43-2	Benzene	16800		71.6	143	1430
108-86-1	Bromobenzene	12500		71.6	143	1430
74-97-5	Bromochloromethane	16500		143	286	1430
75-27-4	Bromodichloromethane	15200		71.6	143	1430
75-25-2	Bromoform	12800		143	286	1430
74-83-9	Bromomethane	14800		143	573	1430
75-15-0	Carbon disulfide	16200		71.6	143	1430
56-23-5	Carbon tetrachloride	16200		71.6	143	1430

FORM I VOA

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060502	Client Sample ID:	OT07-BO-40-4-6-8-MS
Collect Date:	06/02/23 Time: 0910	GCAL Sample ID:	22306050202
Matrix:	Solid % Moisture: 22.5	Instrument ID:	MSV15
Sample Amt:	5.63 g	Lab File ID:	230612/c0946
Injection Vol.:	1 (µL)	GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	250 Analyst: KN1	Analytical Batch:	767415
Analysis Date:	06/12/23 Time: 1938	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/kg

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	13700		71.6	143	1430
75-00-3	Chloroethane	17900		71.6	143	1430
67-66-3	Chloroform	15900		71.6	143	1430
74-87-3	Chloromethane	17200		143	573	1430
156-59-2	cis-1,2-Dichloroethene	16400		71.6	143	1430
10061-01-5	cis-1,3-Dichloropropene	16200		71.6	143	1430
124-48-1	Dibromochloromethane	14700		71.6	143	1430
74-95-3	Dibromomethane	16400		71.6	143	1430
75-71-8	Dichlorodifluoromethane	16600		71.6	143	1430
100-41-4	Ethylbenzene	16500		71.6	143	1430
87-68-3	Hexachlorobutadiene	16000		143	286	1430
98-82-8	Isopropylbenzene (Cumene)	15000		71.6	143	1430
136777-61-2	m,p-Xylene	31100		71.6	286	2860
75-09-2	Methylene chloride	16800		286	573	2860
91-20-3	Naphthalene	18100		143	286	1430
104-51-8	n-Butylbenzene	14400		71.6	143	1430
103-65-1	n-Propylbenzene	15200		71.6	143	1430
95-47-6	o-Xylene	14300		71.6	143	1430
135-98-8	sec-Butylbenzene	14800		71.6	143	1430
100-42-5	Styrene	15100		71.6	143	1430
1634-04-4	tert-Butyl methyl ether (MTBE)	16400		71.6	143	1430
98-06-6	tert-Butylbenzene	12200		143	286	1430
127-18-4	Tetrachloroethene	14100		143	286	1430
108-88-3	Toluene	14100		71.6	143	1430
156-60-5	trans-1,2-Dichloroethene	16800		71.6	143	1430
10061-02-6	trans-1,3-Dichloropropene	15700		71.6	143	1430
79-01-6	Trichloroethene	17200		71.6	143	1430
75-69-4	Trichlorofluoromethane	16500		71.6	143	1430
75-01-4	Vinyl chloride	17300		71.6	143	1430

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230612.s.b/c0946.d
Lab Smp Id: 22306050202 Client Smp ID: 250X
Inj Date : 12-JUN-2023 19:38
Operator : KN1 Inst ID: msv15.i
Smp Info : 22306050202*MS*C*250X
Misc Info : MSV~52948~*250*SMR
Comment :
Method : /var/chem/msv15.i/230612.s.b/8260bw15DOD.m
Meth Date : 13-Jun-2023 15:50 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 QC Sample: MS
Dil Factor: 250.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/(Ws*(100-M)/100))/5000 * CpndVariable

Name	Value	Description
DF	250.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Ws	5.63000	Weight of sample extracted (g)
M	0.00000	% Moisture (not decanted)
Va	100.00000	Volume of aliquot extract added (uL)
m	0.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	CONCENTRATIONS FINAL (UG/KG)	SIMILARITY
1 Dichlorodifluoromethane	85	1.465	1.468	(0.249)	104069	57.9942	12900	9786(R)
2 Chloromethane ++	50	1.648	1.649	(0.280)	96493	59.7070	13300	9823(R)
3 Vinyl Chloride +	62	1.716	1.719	(0.292)	137018	60.2098	13400	9835(R)
5 1-3 Butadiene	39	1.732	1.735	(0.294)	104398	61.6028	13700	9662
6 Bromomethane	94	2.012	2.015	(0.342)	99261	52.0137	11500	9764(R)
7 Chloroethane	64	2.121	2.137	(0.361)	104958	62.7461	13900	9291(R)
8 Trichlorofluoromethane	101	2.266	2.272	(0.385)	190803	57.5174	12800	9769(R)
11 1,1-Dichloroethene +	96	2.780	2.784	(0.473)	112575	59.6465	13200	8327(R)
12 Carbon Disulfide	76	2.809	2.816	(0.477)	328510	56.8953	12600	8231(R)
13 1,1,2Trichlotrifluoroethane	101	2.832	2.832	(0.481)	119680	58.4335	13000	9478(R)
15 Methyl Iodide	142	2.931	2.935	(0.498)	137678	60.6273	13500	9540(R)
16 Acrolein	56	3.166	3.150	(0.538)	15727	54.7211	12100	1680(R)

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (UG/KG)	SIMILARITY
	17 Allyl chloride	39	3.301	3.304	(0.561)	98449	59.3463	13200	9798
	18 Methylene Chloride	49	3.414	3.417	(0.580)	122238	58.5266	13000	4853(R)
	19 Acetone	43	3.468	3.465	(0.590)	250501	292.216	64900	5184(RH)
	20 trans-1,2-Dichloroethene	61	3.590	3.587	(0.610)	152889	58.4317	13000	8470(R)
	21 Methyl Acetate	43	3.610	3.610	(0.614)	297861	279.104	62000	9658(R)
	22 Hexane	57	3.684	3.684	(0.626)	263461	69.8882	15500	9293
	23 MTBE	73	3.716	3.713	(0.632)	290884	57.0013	12700	9659(R)
	24 tert-Butyl Alcohol	59	3.825	3.819	(0.650)	64027	284.427	63100	8099
	25 Acetonitrile	41	3.941	3.941	(0.670)	48960	322.589	71600	9511
	26 Isopropyl Ether	45	4.111	4.108	(0.699)	280540	61.6152	13700	9651
	27 Chloroprene	53	4.195	4.195	(0.713)	147431	57.4850	12800	8628
	28 1,1-Dichloroethane ++	63	4.214	4.214	(0.716)	206014	58.9634	13100	9520(R)
	29 Acrylonitrile	53	4.259	4.259	(0.724)	171232	287.408	63800	8626(R)
	31 Ethyl Tert-butyl Ether	59	4.471	4.468	(0.760)	293427	57.2617	12700	8513
	30 Vinyl Acetate	43	4.468	4.472	(0.760)	30740	10.9098	2420	1934(R)
	32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	143448	57.1087	12700	9832(R)
M	33 Total 1,2-Dichloroethene	61				296337	115.540	25700	0
	34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	156971	50.5456	11200	9549(R)
	35 Bromochloromethane	128	4.918	4.919	(0.836)	57039	57.8403	12800	8523(R)
	36 Cyclohexane	56	4.925	4.928	(0.837)	231445	72.6431	16100	9132(R)
	37 Chloroform +	83	4.992	4.996	(0.849)	198302	55.4822	12300	9653(R)
	39 Carbon Tetrachloride	117	5.118	5.121	(0.870)	163875	56.7178	12600	7796(R)
$	42 Dibromofluoromethane	111	5.156	5.156	(0.876)	101354	53.0938	11800	8110(R)
	43 1,1,1-Trichloroethane	97	5.182	5.179	(0.881)	186404	56.3183	12500	9067(R)
	45 1,1-Dichloropropene	75	5.291	5.291	(0.899)	166577	56.6082	12600	9434(R)
	44 2-Butanone	43	5.275	5.275	(0.897)	284539	319.789	71000	6192(R)
	46 2,2,4 Trimethylpentane	57	5.401	5.398	(0.918)	1759512	197.992	44000	9823
	48 Benzene	78	5.513	5.513	(0.937)	465302	58.5004	13000	8709(R)
	56 tert-amyl Methyl Ether	73	5.632	5.629	(0.957)	294696	56.4370	12500	7105
$	50 1,2-Dichloroethane-d4	65	5.632	5.629	(0.957)	99777	50.1982	11100	8264(R)
	52 1,2-Dichloroethane	62	5.690	5.690	(0.967)	134231	53.6479	11900	9752(R)
	51 Isobutyl Alcohol	43	5.725	5.726	(0.973)	11635	223.531	49600	3949
*	55 FLUOROBENZENE	96	5.883	5.880	(1.000)	386552	50.0000		9849
	58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	371483	99.3958	22100	9394(R)
	57 Trichloroethene	130	6.031	6.031	(1.025)	146326	59.9003	13300	7744(R)
	62 Dibromomethane	93	6.394	6.394	(1.087)	73059	57.0825	12700	9632(R)
	64 1,2-Dichloropropane +	63	6.481	6.481	(1.102)	116100	59.0594	13100	7162(R)
	65 Bromodichloromethane	83	6.536	6.539	(1.111)	141539	53.1548	11800	9788(R)
	67 1,4- Dioxane	88	6.722	6.719	(1.143)	19389	1330.18	295000	8245
	68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	147288	58.8104	13100	9315
	72 2-Chloroethyl vinyl ether	63	7.021	7.021	(1.193)	113342	203.216	45100	9761(R)
	73 cis-1,3-Dichloropropene	75	7.073	7.070	(1.202)	179444	56.9195	12600	9774(R)
$	74 Toluene-d8	98	7.221	7.221	(0.866)	382651	45.9499	10200	9824(R)
	75 Toluene +	91	7.262	7.262	(0.871)	510856	49.2720	10900	9577(R)
	77 4-methyl-2-pentanone	43	7.548	7.552	(0.905)	494919	256.862	57000	7526(R)
	78 Tetrachloroethene	164	7.558	7.558	(0.906)	115023	49.0699	10900	8062(R)
	79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	164617	55.0902	12200	8736(R)

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (UG/KG)	SIMILARITY
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	109868	52.1720	11600	9282(R)
82 Dibromochloromethane	129	7.825	7.825	(0.938)	122532	51.1552	11400	9678(R)
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	137876	49.7911	11100	9708
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	170459	48.6153	10800	9581(R)
M 83 1-3 Dichloropropene total	100				344061	112.010	24900	0
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	109627	48.6501	10800	9754(R)
89 2-Hexanone	43	8.143	8.143	(0.976)	385967	243.818	54100	9501(R)
90 1-Chlorohexane	91	8.333	8.333	(0.999)	185482	50.0438	11100	7869
* 91 CHLOROBENZENE-d5	82	8.339	8.340	(1.000)	177689	50.0000		9150
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	326805	47.6262	10600	9474(R)
93 Ethylbenzene +	106	8.368	8.369	(1.003)	214553	57.5051	12800	8918(R)
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	107152	46.4734	10300	9251(R)
95 p,m-Xylene	106	8.465	8.465	(1.015)	489687	108.559	24100	9919(R)
97 o-Xylene	106	8.745	8.745	(1.049)	209808	49.9491	11100	9794(R)
101 Styrene	104	8.777	8.777	(1.052)	284814	52.6600	11700	9864(R)
102 Bromoform ++	173	8.799	8.799	(1.055)	86613	44.8658	9960	8641(R)
104 Isopropylbenzene	105	8.944	8.944	(1.072)	576588	52.4420	11600	9825(R)
M 105 TOTAL XYLENE	106				699495	158.508	35200	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	136251	50.3794	11200	9402(R)
107 cis-1,4-dichloro-2-butene	53	9.159	9.159	(0.934)	50019	55.1904	12300	8946
109 n-Propylbenzene	91	9.201	9.201	(0.939)	743485	53.3486	11800	8373(R)
108 Bromobenzene	77	9.191	9.195	(0.938)	203219	43.8163	9730	9296(R)
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.243	(0.943)	150090	55.1263	12200	9602(R)
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	395502	44.5213	9880	9100(R)
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	560344	59.7785	13300	7905(R)
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	157674	42.6765	9480	7228(R)
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	40313	44.8576	9960	9241(R)
116 Cyclohexanone	55	9.365	9.365	(0.955)	125614	278.374	61800	8941
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	339740	42.6175	9460	9828(R)
118 tert-butylbenzene	91	9.513	9.516	(0.970)	223591	42.4845	9430	9741(R)
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	1077760	117.145	26000	9427(R)
120 sec-Butylbenzene	105	9.622	9.623	(0.982)	629750	51.9359	11500	9591(R)
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	683853	63.7067	14100	9026(R)
123 1,3-Dichlorobenzene	146	9.761	9.764	(0.996)	236479	43.5998	9680	9431(R)
125 1,4-Dichlorobenzene	146	9.812	9.815	(1.001)	243987	43.7367	9710	8381(RH)
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.806	(1.000)	178449	50.0000		9277
126 n-Butylbenzene	91	9.954	9.954	(1.015)	718819	50.3620	11200	9129(R)
127 1,2-Dichlorobenzene	146	10.063	10.066	(1.027)	218357	44.5022	9880	9483(R)
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.558	(1.077)	29267	42.6953	9480	6486(R)
130 Hexachlorobutadiene	225	10.998	10.999	(1.122)	83996	55.9991	12400	9468(R)
131 1,2,4-Trichlorobenzene	180	11.027	11.034	(1.125)	149608	49.8813	11100	9487(R)
132 Naphthalene	128	11.288	11.291	(1.152)	443632	63.1217	14000	9877(R)
133 1,2,3-Trichlorobenzene	180	11.442	11.449	(1.167)	130705	50.5210	11200	9563(R)

QC Flag Legend

R - Spike/Surrogate failed recovery limits.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230612.s.b/c0946.d

Page 1

Date : 12-JUN-2023 19:38

Client ID: 250X

Instrument: msv15.i

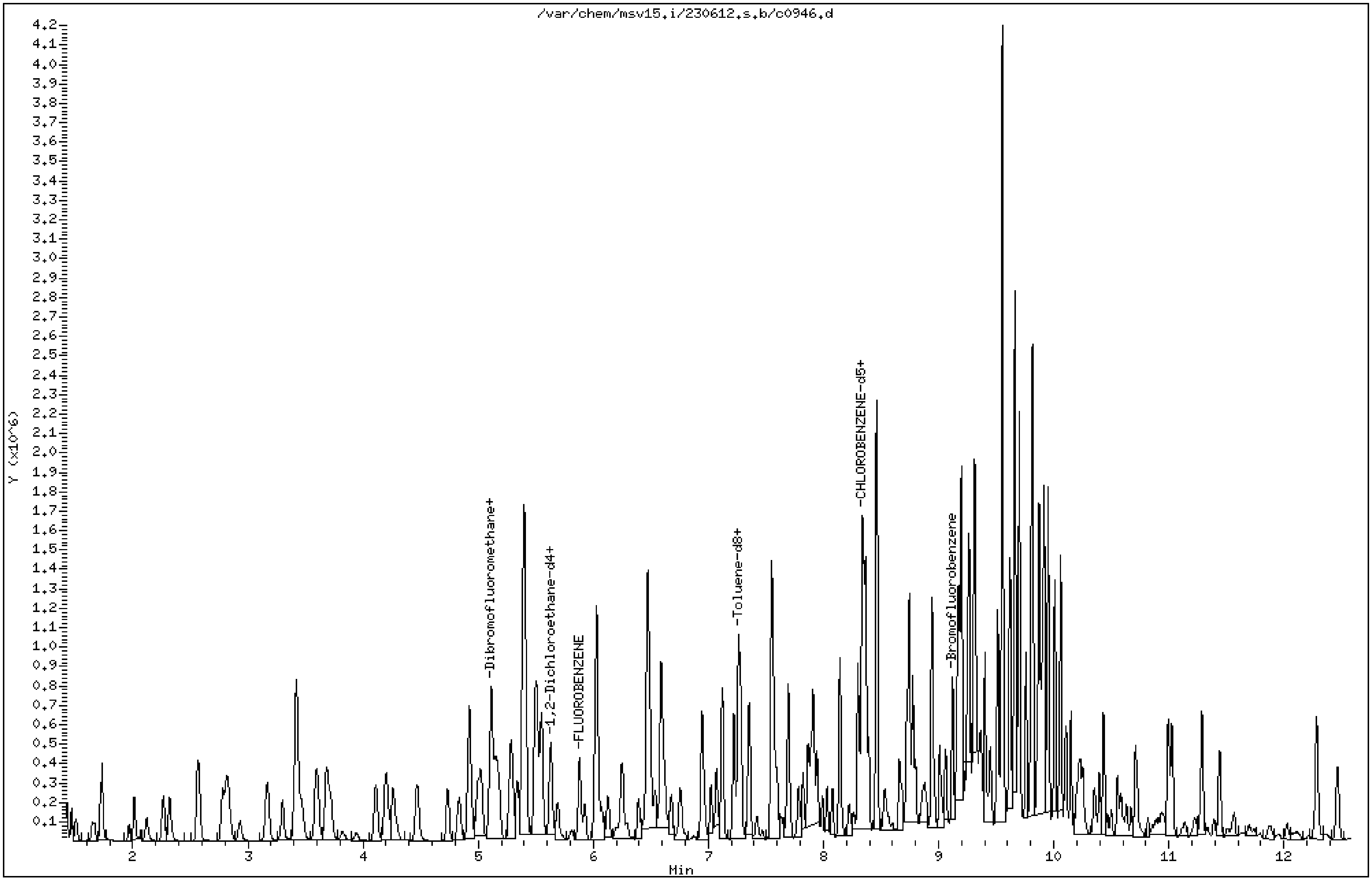
Sample Info: 22306050202*MS*C*250X

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

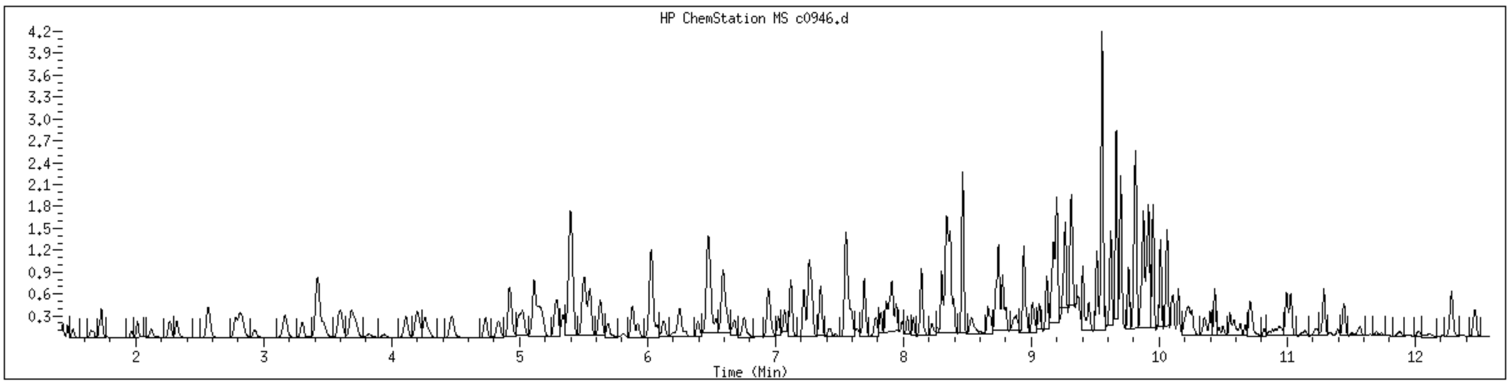
Column diameter: 0.25



Report Date: 06/14/2023 13:52

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 22306050202   SampleType   : MS
Injection Date: 06/12/2023 19:38   Instrument   : msv15.i
Operator   : KN1
Sample Info   : 22306050202*MS*C*250X
Misc Info   : MSV~52948~*250*SMR
Method   : /var/chem/msv15.i/230612.s.b/8260bw15DOD.m
Dilution   : 250
Matrix   : SOIL
Integrator   : HP RTE   Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060502			Client Sample ID:	OT07-BO-40-4-6-8-MSD
Collect Date:	06/02/23	Time:	0915	GCAL Sample ID:	22306050203
Matrix:	Solid	% Moisture:	22.5	Instrument ID:	MSV15
Sample Amt:	5.73	g		Lab File ID:	230612/c0947
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	250	Analyst:	KN1	Analytical Batch:	767415
Analysis Date:	06/12/23	Time:	1957	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/kg

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	12700		70.3	141	1410
71-55-6	1,1,1-Trichloroethane	15100		70.3	141	1410
79-34-5	1,1,2,2-Tetrachloroethane	14400		70.3	141	1410
79-00-5	1,1,2-Trichloroethane	14200		70.3	141	1410
75-34-3	1,1-Dichloroethane	16000		70.3	141	1410
75-35-4	1,1-Dichloroethene	16400		70.3	141	1410
563-58-6	1,1-Dichloropropene	15200		70.3	141	1410
87-61-6	1,2,3-Trichlorobenzene	13200		141	281	1410
96-18-4	1,2,3-Trichloropropane	11500		141	281	1410
120-82-1	1,2,4-Trichlorobenzene	12800		141	281	1410
95-63-6	1,2,4-Trimethylbenzene	30400		70.3	141	1410
96-12-8	1,2-Dibromo-3-chloropropane	11700		141	563	1410
106-93-4	1,2-Dibromoethane	13300		141	563	1410
95-50-1	1,2-Dichlorobenzene	11500		70.3	141	1410
107-06-2	1,2-Dichloroethane	14400		70.3	141	1410
540-59-0	1,2-Dichloroethene (total)	31200		141	281	2810
78-87-5	1,2-Dichloropropane	16000		70.3	141	1410
108-67-8	1,3,5-Trimethylbenzene	15500		70.3	141	1410
541-73-1	1,3-Dichlorobenzene	11300		70.3	141	1410
142-28-9	1,3-Dichloropropane	13200		70.3	141	1410
106-46-7	1,4-Dichlorobenzene	11100		70.3	141	1410
594-20-7	2,2-Dichloropropane	13800		70.3	141	1410
78-93-3	2-Butanone	88600		141	563	1410
95-49-8	2-Chlorotoluene	11500		70.3	141	1410
591-78-6	2-Hexanone	68100		141	563	1410
106-43-4	4-Chlorotoluene	11100		70.3	141	1410
99-87-6	4-Isopropyltoluene	16500		70.3	141	1410
108-10-1	4-Methyl-2-pentanone	71600		70.3	141	1410
67-64-1	Acetone	82800		141	563	7030
71-43-2	Benzene	15900		70.3	141	1410
108-86-1	Bromobenzene	11400		70.3	141	1410
74-97-5	Bromochloromethane	15600		141	281	1410
75-27-4	Bromodichloromethane	15500		70.3	141	1410
75-25-2	Bromoform	12400		141	281	1410
74-83-9	Bromomethane	14100		141	563	1410
75-15-0	Carbon disulfide	15500		70.3	141	1410
56-23-5	Carbon tetrachloride	15300		70.3	141	1410

FORM I VOA

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060502	Client Sample ID:	OT07-BO-40-4-6-8-MSD
Collect Date:	06/02/23 Time: 0915	GCAL Sample ID:	22306050203
Matrix:	Solid % Moisture: 22.5	Instrument ID:	MSV15
Sample Amt:	5.73 g	Lab File ID:	230612/c0947
Injection Vol.:	1 (µL)	GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	250 Analyst: KN1	Analytical Batch:	767415
Analysis Date:	06/12/23 Time: 1957	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/kg

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	12800		70.3	141	1410
75-00-3	Chloroethane	17500		70.3	141	1410
67-66-3	Chloroform	15100		70.3	141	1410
74-87-3	Chloromethane	16200		141	563	1410
156-59-2	cis-1,2-Dichloroethene	15300		70.3	141	1410
10061-01-5	cis-1,3-Dichloropropene	15200		70.3	141	1410
124-48-1	Dibromochloromethane	13900		70.3	141	1410
74-95-3	Dibromomethane	15500		70.3	141	1410
75-71-8	Dichlorodifluoromethane	16100		70.3	141	1410
100-41-4	Ethylbenzene	14800		70.3	141	1410
87-68-3	Hexachlorobutadiene	14600		141	281	1410
98-82-8	Isopropylbenzene (Cumene)	14200		70.3	141	1410
136777-61-2	m,p-Xylene	28900		70.3	281	2810
75-09-2	Methylene chloride	15700		281	563	2810
91-20-3	Naphthalene	17200		141	281	1410
104-51-8	n-Butylbenzene	13300		70.3	141	1410
103-65-1	n-Propylbenzene	13800		70.3	141	1410
95-47-6	o-Xylene	13400		70.3	141	1410
135-98-8	sec-Butylbenzene	13500		70.3	141	1410
100-42-5	Styrene	13500		70.3	141	1410
1634-04-4	tert-Butyl methyl ether (MTBE)	15700		70.3	141	1410
98-06-6	tert-Butylbenzene	11000		141	281	1410
127-18-4	Tetrachloroethene	13300		141	281	1410
108-88-3	Toluene	13000		70.3	141	1410
156-60-5	trans-1,2-Dichloroethene	15700		70.3	141	1410
10061-02-6	trans-1,3-Dichloropropene	14800		70.3	141	1410
79-01-6	Trichloroethene	16400		70.3	141	1410
75-69-4	Trichlorofluoromethane	16100		70.3	141	1410
75-01-4	Vinyl chloride	16600		70.3	141	1410

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230612.s.b/c0947.d
Lab Smp Id: 22306050203 Client Smp ID: 250X
Inj Date : 12-JUN-2023 19:57
Operator : KN1 Inst ID: msv15.i
Smp Info : 22306050203*MSD*C*250X
Misc Info : MSV~52948~*250*SMR
Comment :
Method : /var/chem/msv15.i/230612.s.b/8260bw15DOD.m
Meth Date : 13-Jun-2023 15:50 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 QC Sample: MSD
Dil Factor: 250.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/(Ws*(100-M)/100))/5000 * CpndVariable

Name	Value	Description
DF	250.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Ws	5.73000	Weight of sample extracted (g)
M	0.00000	% Moisture (not decanted)
Va	100.00000	Volume of aliquot extract added (uL)
m	0.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (UG/KG)	SIMILARITY
1 Dichlorodifluoromethane	85	1.465	1.468	(0.249)	104649	57.1584	12500	9802(R)
2 Chloromethane ++	50	1.652	1.649	(0.281)	95281	57.7854	12600	9824(R)
3 Vinyl Chloride +	62	1.716	1.719	(0.292)	136787	58.9137	12900	9818(R)
5 1-3 Butadiene	39	1.735	1.735	(0.295)	103310	59.7493	13000	9663
6 Bromomethane	94	2.015	2.015	(0.343)	97456	50.0529	10900	9766(R)
7 Chloroethane	64	2.118	2.137	(0.360)	106152	62.1987	13600	9459(R)
8 Trichlorofluoromethane	101	2.266	2.272	(0.385)	193949	57.3038	12500	9707(R)
11 1,1-Dichloroethene +	96	2.783	2.784	(0.473)	111971	58.1474	12700	8559(R)
12 Carbon Disulfide	76	2.809	2.816	(0.477)	323808	54.9664	12000	8233(R)
13 1,1,2Trichlotrifluoroethane	101	2.832	2.832	(0.481)	117859	56.4008	12300	9482(R)
15 Methyl Iodide	142	2.931	2.935	(0.498)	130410	56.4021	12300	9511(R)
16 Acrolein	56	3.166	3.150	(0.538)	14806	50.4927	11000	1661(R)

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (UG/KG)	SIMILARITY
	17 Allyl chloride	39	3.301	3.304	(0.561)	97791	57.7780	12600	9753
	18 Methylene Chloride	49	3.414	3.417	(0.580)	119191	55.9336	12200	4979(R)
	19 Acetone	43	3.468	3.465	(0.590)	257481	294.389	64200	5217(RH)
	20 trans-1,2-Dichloroethene	61	3.591	3.587	(0.610)	149727	56.0860	12200	8359(R)
	21 Methyl Acetate	43	3.607	3.610	(0.613)	312019	286.560	62500	9399(R)
	22 Hexane	57	3.684	3.684	(0.626)	252170	65.5636	14300	9296
	23 MTBE	73	3.719	3.713	(0.632)	290505	55.7957	12200	9747(R)
	24 tert-Butyl Alcohol	59	3.822	3.819	(0.650)	68159	296.765	64700	8112
	25 Acetonitrile	41	3.944	3.941	(0.670)	46819	302.352	66000	9573
	26 Isopropyl Ether	45	4.111	4.108	(0.699)	278052	59.8551	13100	9585
	27 Chloroprene	53	4.195	4.195	(0.713)	144511	55.2266	12000	8653
	28 1,1-Dichloroethane ++	63	4.218	4.214	(0.717)	202783	56.8852	12400	9650(R)
	29 Acrylonitrile	53	4.263	4.259	(0.725)	178066	292.939	63900	8707(R)
	31 Ethyl Tert-butyl Ether	59	4.468	4.468	(0.760)	291529	55.7607	12200	8485
	30 Vinyl Acetate	43	4.468	4.472	(0.760)	30954	10.7674	2350	2003(R)
	32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	140078	54.6587	11900	9815(R)
M	33 Total 1,2-Dichloroethene	61				289805	110.745	24200	0
	34 2,2-Dichloropropane	77	4.835	4.838	(0.822)	154999	48.9187	10700	9628(R)
	35 Bromochloromethane	128	4.922	4.919	(0.837)	55912	55.5706	12100	8339(R)
	36 Cyclohexane	56	4.925	4.928	(0.837)	218576	67.2406	14700	9147(R)
	37 Chloroform +	83	4.992	4.996	(0.849)	194830	53.4274	11700	9679(R)
	39 Carbon Tetrachloride	117	5.118	5.121	(0.870)	161088	54.6452	11900	7790(R)
$	42 Dibromofluoromethane	111	5.156	5.156	(0.876)	103212	52.9926	11600	8152(R)
	43 1,1,1-Trichloroethane	97	5.182	5.179	(0.881)	181539	53.7584	11700	9045(R)
	45 1,1-Dichloropropene	75	5.291	5.291	(0.899)	162612	54.1626	11800	9380(R)
	44 2-Butanone	43	5.275	5.275	(0.897)	286026	315.072	68700	6189(R)
	46 2,2,4 Trimethylpentane	57	5.398	5.398	(0.917)	1552106	171.182	37300	9826
	48 Benzene	78	5.513	5.513	(0.937)	457468	56.3724	12300	8821(R)
	56 tert-amyl Methyl Ether	73	5.632	5.629	(0.957)	292577	54.9176	12000	7183
$	50 1,2-Dichloroethane-d4	65	5.632	5.629	(0.957)	101488	50.0442	10900	8316(R)
	52 1,2-Dichloroethane	62	5.693	5.690	(0.968)	130636	51.1734	11200	9825(R)
	51 Isobutyl Alcohol	43	5.732	5.726	(0.974)	12622	237.674	51800	3113
*	55 FLUOROBENZENE	96	5.883	5.880	(1.000)	394390	50.0000		9847
	58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	347414	91.1084	19900	9236(R)
	57 Trichloroethene	130	6.031	6.031	(1.025)	144702	58.0583	12700	7880(R)
	62 Dibromomethane	93	6.394	6.394	(1.087)	71863	55.0322	12000	9612(R)
	64 1,2-Dichloropropane +	63	6.478	6.481	(1.101)	114408	57.0421	12400	6934(R)
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	149097	54.8804	12000	9875(R)
	67 1,4- Dioxane	88	6.722	6.719	(1.143)	19676	1323.04	289000	8388
	68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	144698	56.6280	12400	9347
	72 2-Chloroethyl vinyl ether	63	7.024	7.021	(1.194)	122911	215.715	47100	9785(R)
	73 cis-1,3-Dichloropropene	75	7.069	7.070	(1.202)	173276	53.8707	11800	9735(R)
$	74 Toluene-d8	98	7.224	7.221	(0.866)	388381	45.9071	10000	9866(R)
	75 Toluene +	91	7.262	7.262	(0.871)	487256	46.2593	10100	9557(R)
	77 4-methyl-2-pentanone	43	7.549	7.552	(0.905)	497637	254.253	55500	7471(R)
	78 Tetrachloroethene	164	7.558	7.558	(0.906)	111909	46.9932	10300	7993(R)
	79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	160725	52.7188	11500	8753(R)

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (UG/KG)	SIMILARITY
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	108209	50.5789	11000	9360(R)
82 Dibromochloromethane	129	7.825	7.825	(0.938)	119974	49.3023	10800	9693(R)
84 3,4-dichloro-1-butene	75	7.864	7.867	(0.943)	133964	47.6202	10400	9744
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	167270	46.9581	10200	9547(R)
M 83 1-3 Dichloropropene total	100				334001	106.590	23300	0
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	108070	47.2075	10300	9737(R)
89 2-Hexanone	43	8.143	8.143	(0.976)	389176	242.012	52800	9495(R)
90 1-Chlorohexane	91	8.333	8.333	(0.999)	178388	47.3755	10300	7840
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	180518	50.0000		9204
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	318260	45.6541	9960	9329(R)
93 Ethylbenzene +	106	8.368	8.369	(1.003)	199606	52.6606	11500	8968(R)
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	105281	44.9463	9810	9248(R)
95 p,m-Xylene	106	8.465	8.465	(1.015)	470644	102.702	22400	9931(R)
97 o-Xylene	106	8.745	8.745	(1.049)	203133	47.6021	10400	9826(R)
101 Styrene	104	8.777	8.777	(1.052)	264820	48.1959	10500	9843(R)
102 Bromoform ++	173	8.799	8.799	(1.055)	86152	43.9455	9590	8623(R)
104 Isopropylbenzene	105	8.944	8.944	(1.072)	562466	50.3559	11000	9843(R)
M 105 TOTAL XYLENE	106				673777	150.304	32800	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	138578	50.4368	11000	9368(R)
107 cis-1,4-dichloro-2-butene	53	9.159	9.159	(0.934)	50571	53.3833	11600	8949
109 n-Propylbenzene	91	9.201	9.201	(0.939)	716619	49.1942	10700	8412(R)
108 Bromobenzene	77	9.192	9.195	(0.938)	196434	40.5193	8840	9251(R)
110 1,1,2,2-Tetrachloroethane++	83	9.243	9.243	(0.943)	146505	51.4795	11200	9576(R)
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	380744	41.0040	8950	8939(R)
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	539225	55.0345	12000	7978(R)
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	158497	41.0416	8950	7153(R)
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	41313	43.9844	9600	9270(R)
116 Cyclohexanone	55	9.365	9.365	(0.955)	130370	276.413	60300	8978
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	329861	39.5865	8640	9820(R)
118 tert-butylbenzene	91	9.516	9.516	(0.971)	215172	39.1144	8530	9764(R)
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	1038930	108.035	23600	9368(R)
120 sec-Butylbenzene	105	9.622	9.623	(0.982)	611861	48.2755	10500	9582(R)
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	660269	58.8462	12800	8969(R)
123 1,3-Dichlorobenzene	146	9.761	9.764	(0.996)	228015	40.2189	8770	9433(R)
125 1,4-Dichlorobenzene	146	9.812	9.815	(1.001)	230391	39.5112	8620	8166(RH)
* 124 1,4-DICHLOROBENZENE-D4	152	9.803	9.806	(1.000)	186526	50.0000		9353
126 n-Butylbenzene	91	9.954	9.954	(1.015)	701111	46.9942	10300	9120(R)
127 1,2-Dichlorobenzene	146	10.066	10.066	(1.027)	209224	40.7944	8900	9583(R)
129 1,2-Dibromo-3-Chloropropane	157	10.558	10.558	(1.077)	29738	41.5272	9060	6501(R)
130 Hexachlorobutadiene	225	10.995	10.999	(1.122)	81021	51.6767	11300	9390(R)
131 1,2,4-Trichlorobenzene	180	11.028	11.034	(1.125)	142457	45.4403	9910	9452(R)
132 Naphthalene	128	11.288	11.291	(1.152)	446369	60.7610	13300	9803(R)
133 1,2,3-Trichlorobenzene	180	11.442	11.449	(1.167)	126048	46.6112	10200	9550(R)

QC Flag Legend

R - Spike/Surrogate failed recovery limits.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230612.s.b/c0947.d

Page 1

Date : 12-JUN-2023 19:57

Client ID: 250X

Instrument: msv15.i

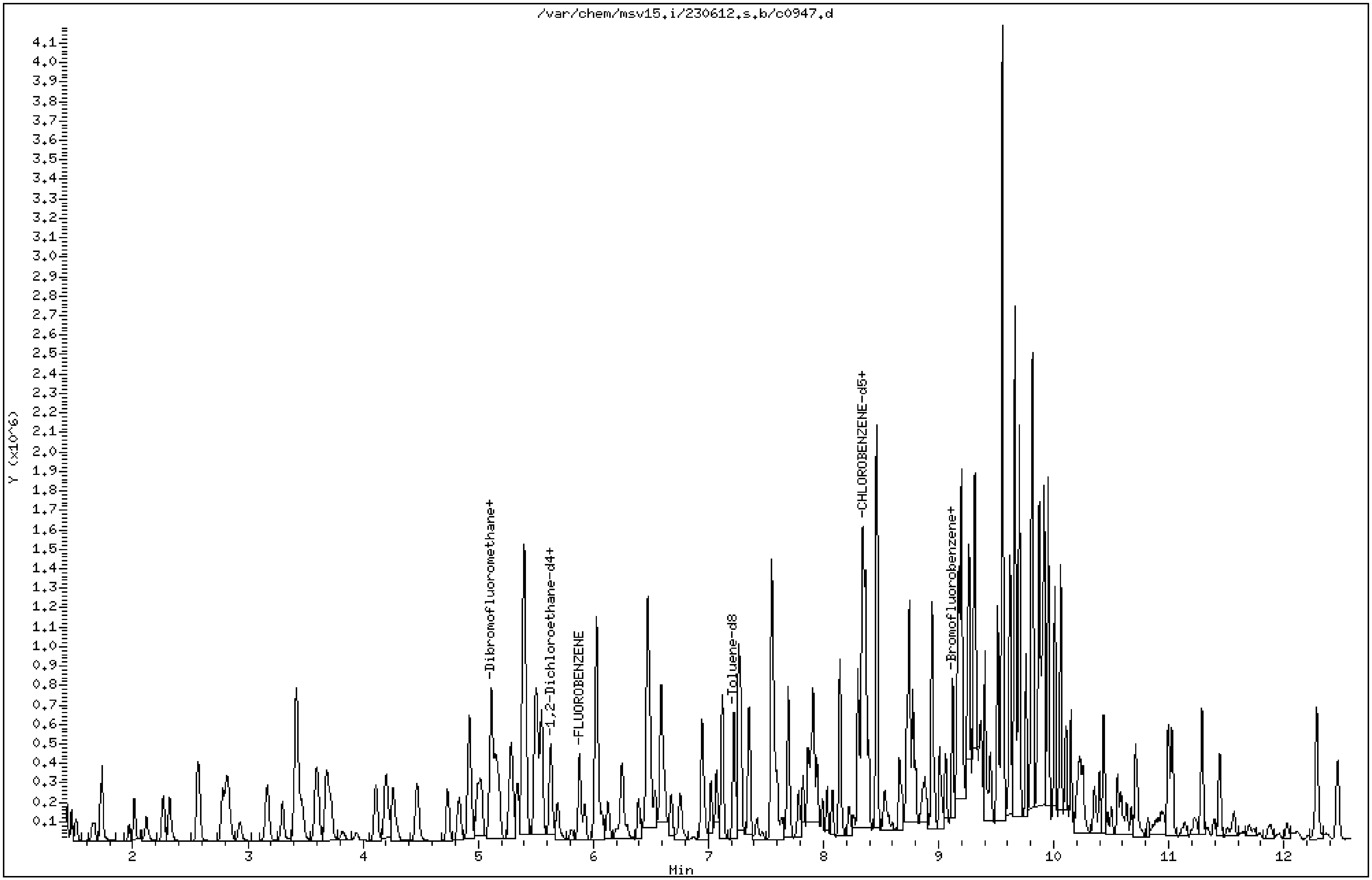
Sample Info: 22306050203*MSD*C*250X

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

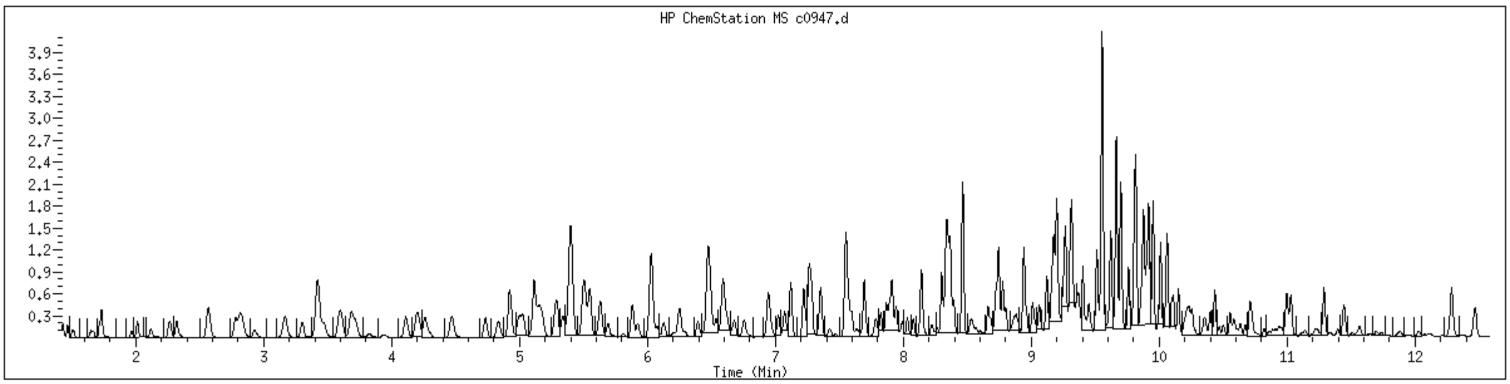
Column diameter: 0.25



Report Date: 06/14/2023 13:53

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID	: 22306050203	SampleType	: MSD
Injection Date	: 06/12/2023 19:57	Instrument	: msv15.i
Operator	: KN1		
Sample Info	: 22306050203*MSD*C*250X		
Misc Info	: MSV~52948~*250*SMR		
Method	: /var/chem/msv15.i/230612.s.b/8260bw15DOD.m		
Dilution	: 250		
Matrix	: SOIL		
Integrator	: HP RTE	Compound Sublist	: 8260DOD



NO MANUAL INTEGRATIONS

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060502			Client Sample ID:	OT07-BO-840-4-6-8
Collect Date:	06/02/23	Time:	0905	GCAL Sample ID:	22306050204
Matrix:	Solid	% Moisture:	20.2	Instrument ID:	MSV15
Sample Amt:	6.06	g		Lab File ID:	230612/c0943
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	250	Analyst:	KN1	Analytical Batch:	767415
Analysis Date:	06/12/23	Time:	1842	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/kg

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	129	U	64.6	129	1290
71-55-6	1,1,1-Trichloroethane	129	U	64.6	129	1290
79-34-5	1,1,2,2-Tetrachloroethane	129	U	64.6	129	1290
79-00-5	1,1,2-Trichloroethane	129	U	64.6	129	1290
75-34-3	1,1-Dichloroethane	129	U	64.6	129	1290
75-35-4	1,1-Dichloroethene	129	U	64.6	129	1290
563-58-6	1,1-Dichloropropene	129	U	64.6	129	1290
87-61-6	1,2,3-Trichlorobenzene	259	U	129	259	1290
96-18-4	1,2,3-Trichloropropane	259	U	129	259	1290
120-82-1	1,2,4-Trichlorobenzene	259	U	129	259	1290
95-63-6	1,2,4-Trimethylbenzene	28500		64.6	129	1290
96-12-8	1,2-Dibromo-3-chloropropane	517	UQ	129	517	1290
106-93-4	1,2-Dibromoethane	517	U	129	517	1290
95-50-1	1,2-Dichlorobenzene	129	U	64.6	129	1290
107-06-2	1,2-Dichloroethane	129	U	64.6	129	1290
540-59-0	1,2-Dichloroethene (total)	259	U	129	259	2590
78-87-5	1,2-Dichloropropane	129	U	64.6	129	1290
108-67-8	1,3,5-Trimethylbenzene	6080		64.6	129	1290
541-73-1	1,3-Dichlorobenzene	129	U	64.6	129	1290
142-28-9	1,3-Dichloropropane	129	U	64.6	129	1290
106-46-7	1,4-Dichlorobenzene	129	U	64.6	129	1290
594-20-7	2,2-Dichloropropane	129	U	64.6	129	1290
78-93-3	2-Butanone	517	U	129	517	1290
95-49-8	2-Chlorotoluene	129	U	64.6	129	1290
591-78-6	2-Hexanone	517	U	129	517	1290
106-43-4	4-Chlorotoluene	129	U	64.6	129	1290
99-87-6	4-Isopropyltoluene	2430		64.6	129	1290
108-10-1	4-Methyl-2-pentanone	7460		64.6	129	1290
67-64-1	Acetone	517	U	129	517	6460
71-43-2	Benzene	337	J	64.6	129	1290
108-86-1	Bromobenzene	129	UQ	64.6	129	1290
74-97-5	Bromochloromethane	259	U	129	259	1290
75-27-4	Bromodichloromethane	129	U	64.6	129	1290
75-25-2	Bromoform	259	U	129	259	1290
74-83-9	Bromomethane	517	U	129	517	1290
75-15-0	Carbon disulfide	129	U	64.6	129	1290
56-23-5	Carbon tetrachloride	129	U	64.6	129	1290

FORM I VOA

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060502			Client Sample ID:	OT07-BO-840-4-6-8		
Collect Date:	06/02/23	Time:	0905	GCAL Sample ID:	22306050204		
Matrix:	Solid	% Moisture:	20.2	Instrument ID:	MSV15		
Sample Amt:	6.06	g		Lab File ID:	230612/c0943		
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M	ID	.25 (mm)
Dilution Factor:	250	Analyst:	KN1	Analytical Batch:	767415		
Analysis Date:	06/12/23	Time:	1842	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: *ug/kg*

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	129	U	64.6	129	1290
75-00-3	Chloroethane	129	UQ	64.6	129	1290
67-66-3	Chloroform	129	U	64.6	129	1290
74-87-3	Chloromethane	517	U	129	517	1290
156-59-2	cis-1,2-Dichloroethene	129	U	64.6	129	1290
10061-01-5	cis-1,3-Dichloropropene	129	U	64.6	129	1290
124-48-1	Dibromochloromethane	129	U	64.6	129	1290
74-95-3	Dibromomethane	129	U	64.6	129	1290
75-71-8	Dichlorodifluoromethane	129	U	64.6	129	1290
100-41-4	Ethylbenzene	3150		64.6	129	1290
87-68-3	Hexachlorobutadiene	259	U	129	259	1290
98-82-8	Isopropylbenzene (Cumene)	1880		64.6	129	1290
136777-61-2	m,p-Xylene	4960		64.6	259	2590
75-09-2	Methylene chloride	517	U	259	517	2590
91-20-3	Naphthalene	4830		129	259	1290
104-51-8	n-Butylbenzene	2820		64.6	129	1290
103-65-1	n-Propylbenzene	4170		64.6	129	1290
95-47-6	o-Xylene	443	J	64.6	129	1290
135-98-8	sec-Butylbenzene	3250		64.6	129	1290
100-42-5	Styrene	129	U	64.6	129	1290
1634-04-4	tert-Butyl methyl ether (MTBE)	129	U	64.6	129	1290
98-06-6	tert-Butylbenzene	259	U	129	259	1290
127-18-4	Tetrachloroethene	259	U	129	259	1290
108-88-3	Toluene	129	U	64.6	129	1290
156-60-5	trans-1,2-Dichloroethene	129	U	64.6	129	1290
10061-02-6	trans-1,3-Dichloropropene	129	U	64.6	129	1290
79-01-6	Trichloroethene	129	U	64.6	129	1290
75-69-4	Trichlorofluoromethane	129	U	64.6	129	1290
75-01-4	Vinyl chloride	129	U	64.6	129	1290

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230612.s.b/c0943.d
Lab Smp Id: 22306050204 Client Smp ID: 250X
Inj Date : 12-JUN-2023 18:42
Operator : KN1 Inst ID: msv15.i
Smp Info : 22306050204*C*250X
Misc Info : MSV~52948~*250*SMR
Comment :
Method : /var/chem/msv15.i/230612.s.b/8260bw15DOD.m
Meth Date : 13-Jun-2023 15:50 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1
Dil Factor: 250.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/(Ws*(100-M)/100))/5000 * CpndVariable

Name	Value	Description
DF	250.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Ws	6.06000	Weight of sample extracted (g)
M	0.00000	% Moisture (not decanted)
Va	100.00000	Volume of aliquot extract added (uL)
m	0.00000	

Cpnd Variable Local Compound Variable

		QUANT SIG					CONCENTRATIONS ON-COLUMN	FINAL	
	Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(UG/KG)	SIMILARITY
	22 Hexane	57	3.687	3.684	(0.627)	39393	12.1029	2500	9145
	36 Cyclohexane	56	4.928	4.928	(0.838)	65207	23.7041	4890	6230
$	42 Dibromofluoromethane	111	5.156	5.156	(0.876)	87433	53.0470	10900	7440(R)
	46 2,2,4 Trimethylpentane	57	5.401	5.398	(0.918)	1771197	230.836	47600	9751(A)
	48 Benzene	78	5.517	5.513	(0.938)	8943	1.30224	269	2857
$	50 1,2-Dichloroethane-d4	65	5.632	5.629	(0.957)	89895	52.3812	10800	9372(R)
*	55 FLUOROBENZENE	96	5.883	5.880	(1.000)	333753	50.0000		9883
	58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	215972	66.9282	13800	7661
$	74 Toluene-d8	98	7.224	7.221	(0.866)	339048	46.9167	9680	9796(R)
	77 4-methyl-2-pentanone	43	7.542	7.552	(0.904)	44205	28.8469	5950	3842
*	91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	154197	50.0000		9773
	93 Ethylbenzene +	106	8.369	8.369	(1.003)	39394	12.1671	2510	8498

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (UG/KG)	SIMILARITY
95 p,m-Xylene	106	8.462	8.465	(1.015)	75113	19.1888	3960	9840
97 o-Xylene	106	8.745	8.745	(1.049)	6245	1.71326	353	6691
101 Styrene	104	8.774	8.777	(1.052)	989	0.21072	43.5	4464
104 Isopropylbenzene	105	8.944	8.944	(1.072)	69567	7.29125	1500	8918
M 105 TOTAL XYLENE	106				81358	20.9020	4310	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	117903	50.2369	10400	9167(R)
109 n-Propylbenzene	91	9.201	9.201	(0.939)	199662	16.1403	3330	6856
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	195556	23.5031	4850	7284(H)
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	900687	110.291	22700	9522
120 sec-Butylbenzene	105	9.619	9.623	(0.981)	135115	12.5536	2590	8108
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	89740	9.41835	1940	9408(M2)
* 124 1,4-DICHLOROBENZENE-D4	152	9.803	9.806	(1.000)	158398	50.0000		9738
126 n-Butylbenzene	91	9.954	9.954	(1.015)	138320	10.9177	2250	9556
132 Naphthalene	128	11.288	11.291	(1.152)	116299	18.6422	3850	9611

QC Flag Legend

A - Target compound detected but, quantitated amount exceeded maximum amount.
R - Spike/Surrogate failed recovery limits.
M2- Compound response manually integrated because Target system integrated incorrectly.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230612.s.b/c0943.d

Page 1

Date : 12-JUN-2023 18:42

Client ID: 250X

Instrument: msv15.i

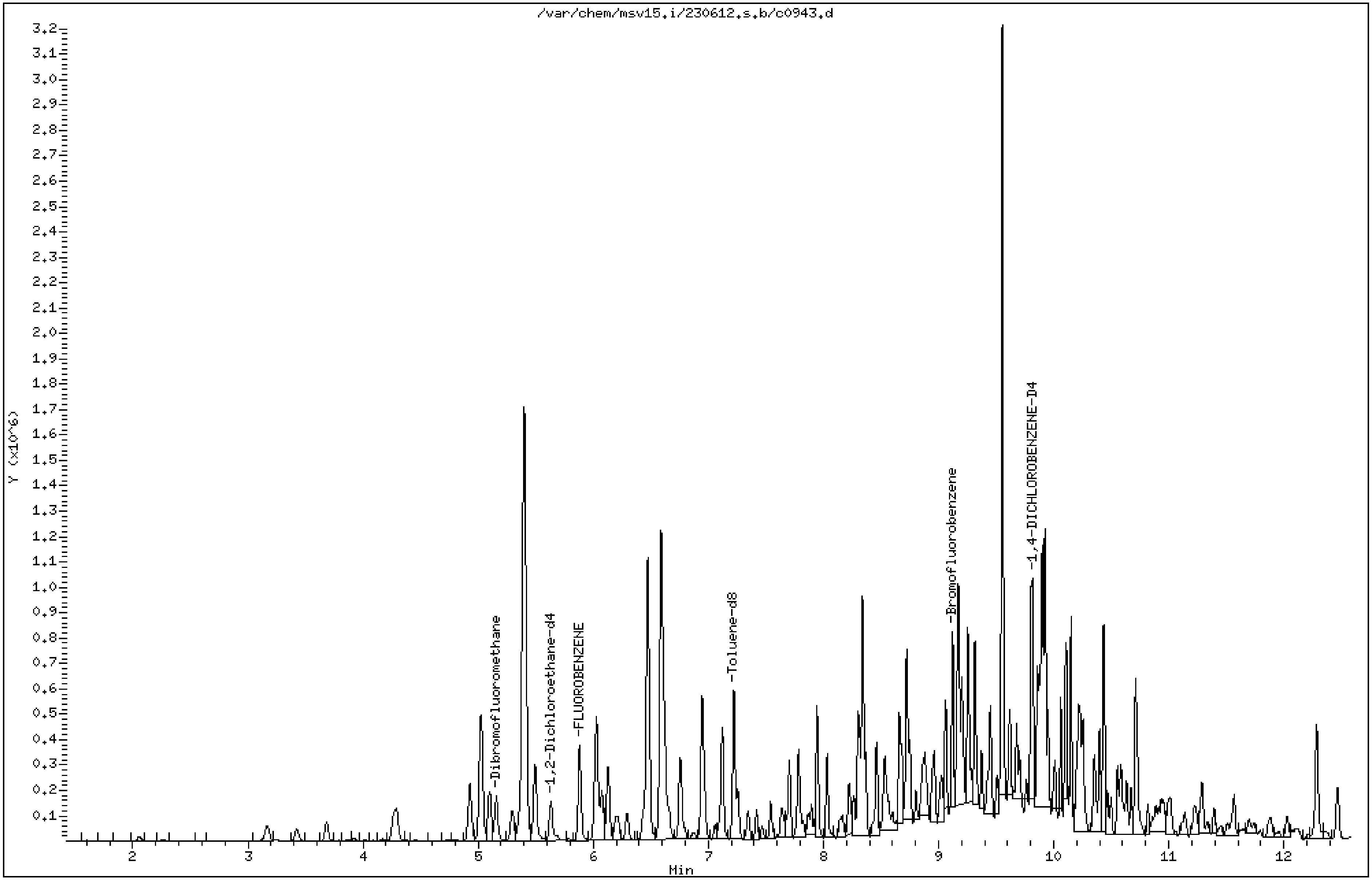
Sample Info: 22306050204*C*250X

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

Column diameter: 0.25



Data File: /var/chem/msv15.i/230612.s.b/c0943.d

Date : 12-JUN-2023 18:42

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0

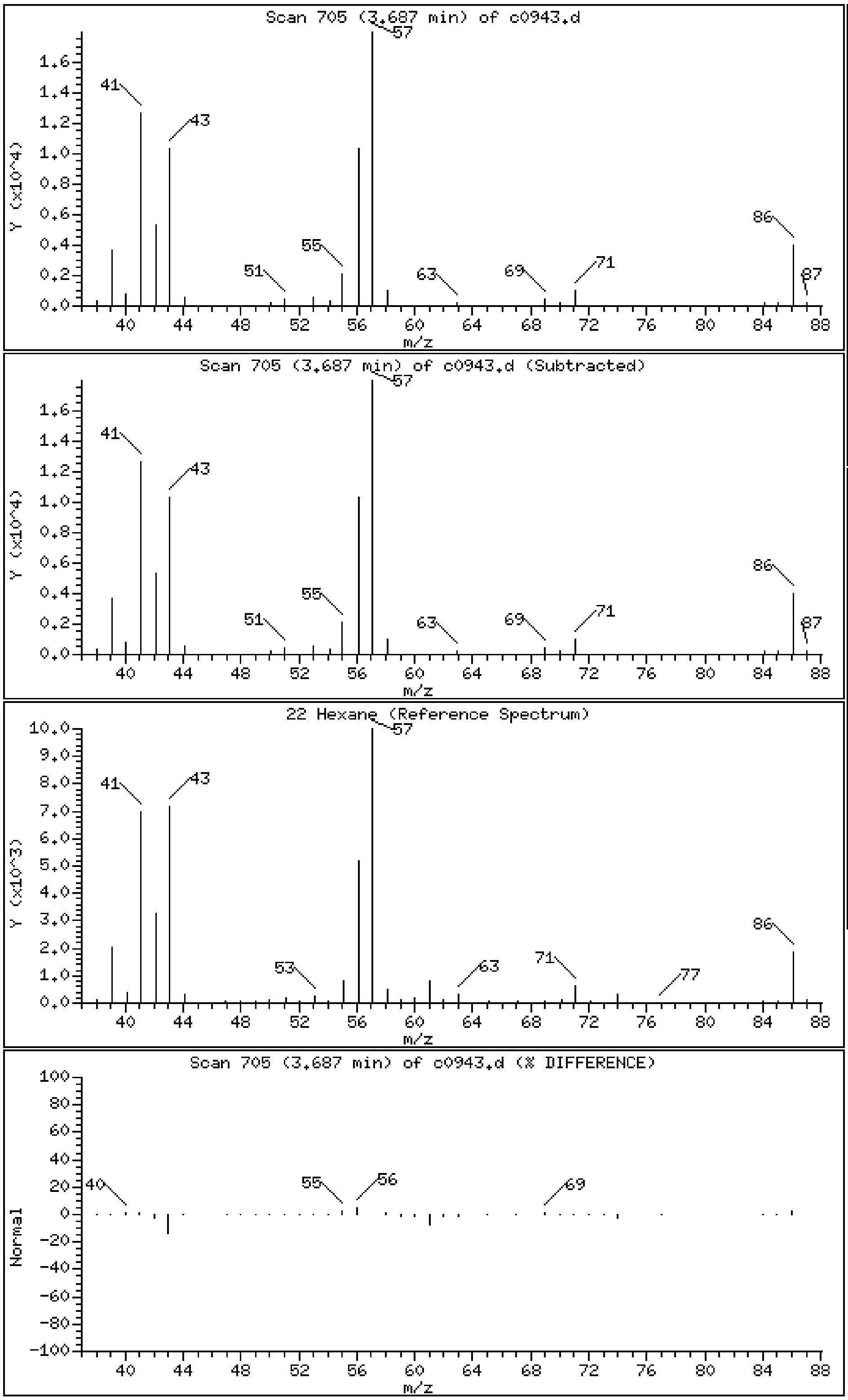
Operator: KN1

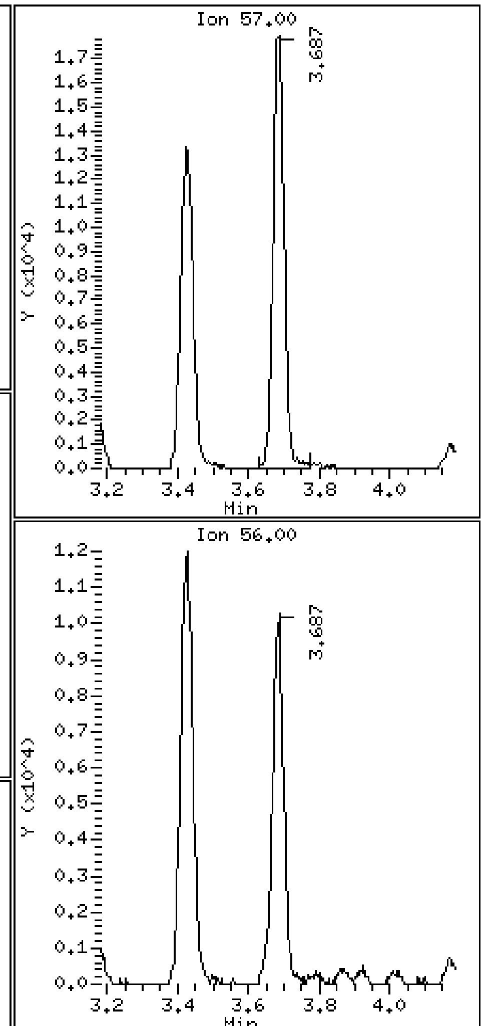
Column phase: RTX-VMS-30M

Column diameter: 0.25

22 Hexane

Concentration: 2500 UG/KG





Data File: /var/chem/msv15.i/230612.s.b/c0943.d Page 3

Date : 12-JUN-2023 18:42

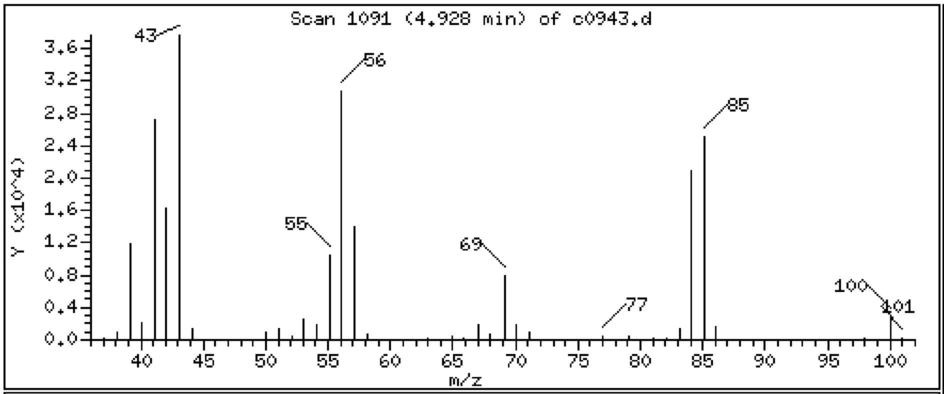
Client ID: 250X Instrument: msv15.i

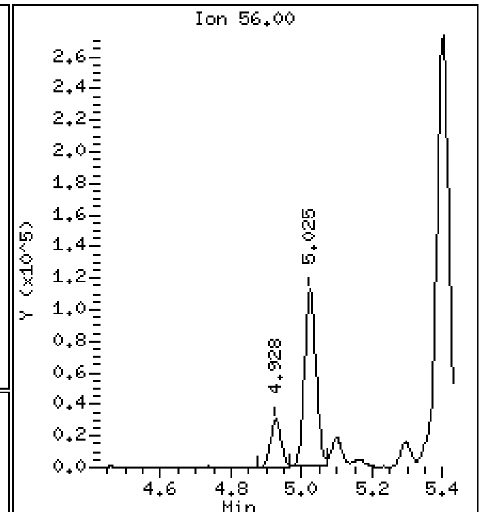
Sample Info: 22306050204*C*250X

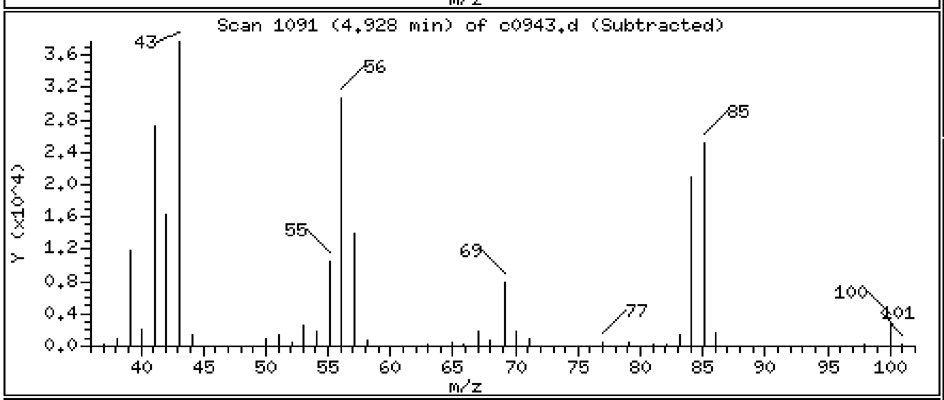
Purge Volume: 5.0 Operator: KN1

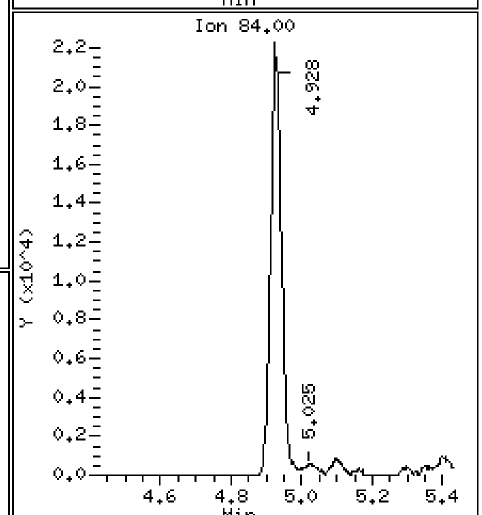
Column phase: RTX-VMS-30M Column diameter: 0.25

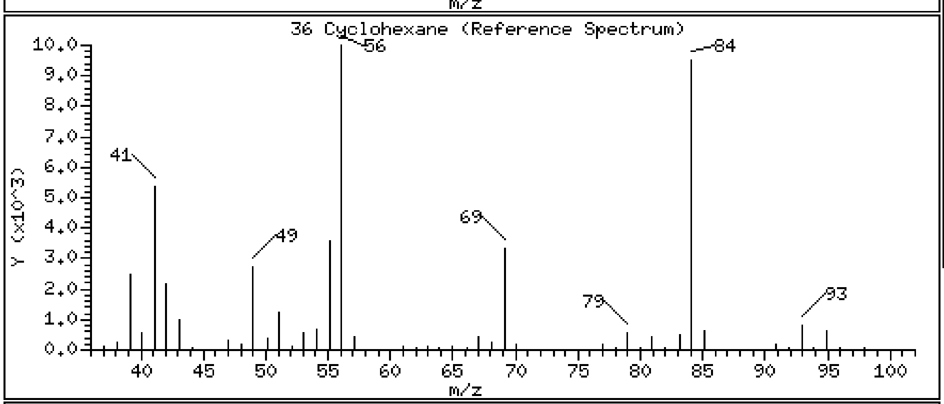
36 Cyclohexane Concentration: 4890 UG/KG

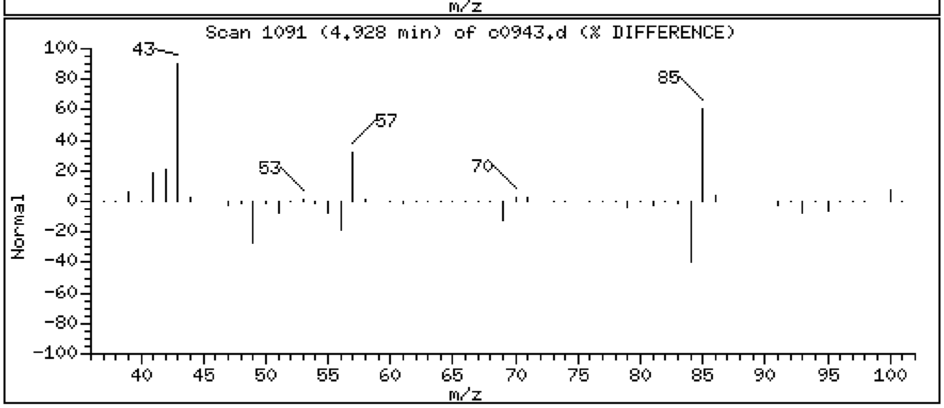












Data File: /var/chem/msv15.i/230612.s.b/c0943.d

Date : 12-JUN-2023 18:42

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0

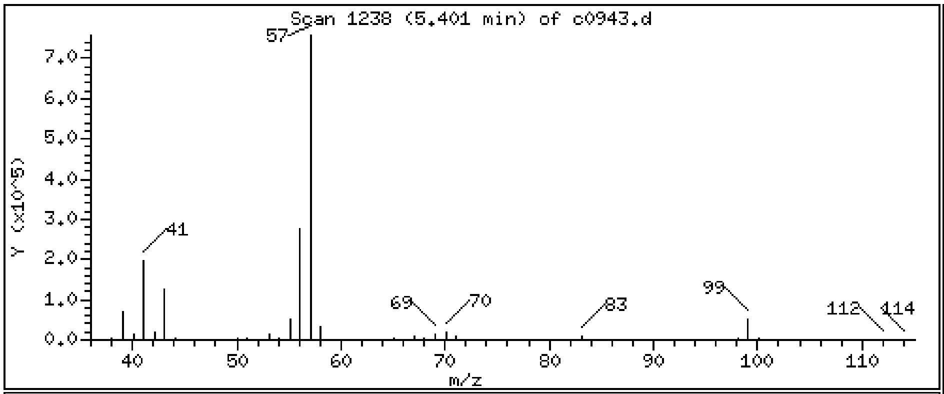
Operator: KN1

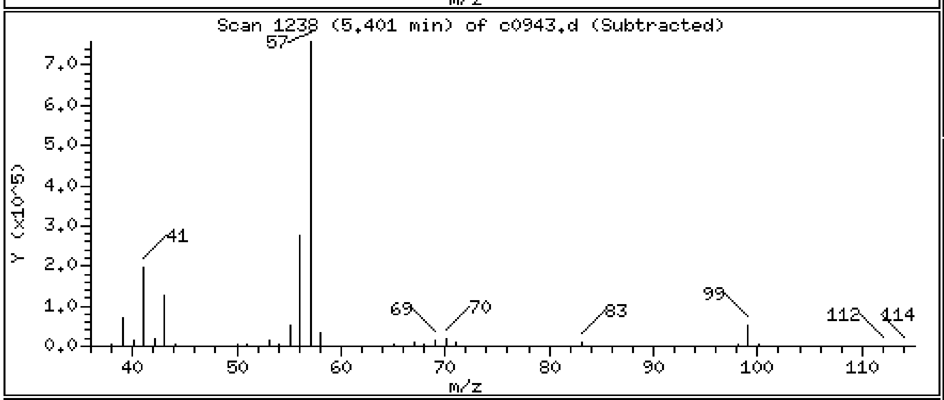
Column phase: RTX-VMS-30M

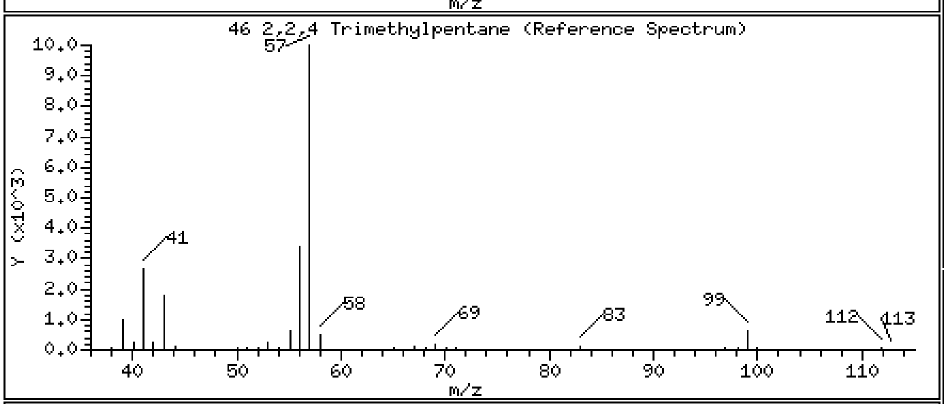
Column diameter: 0.25

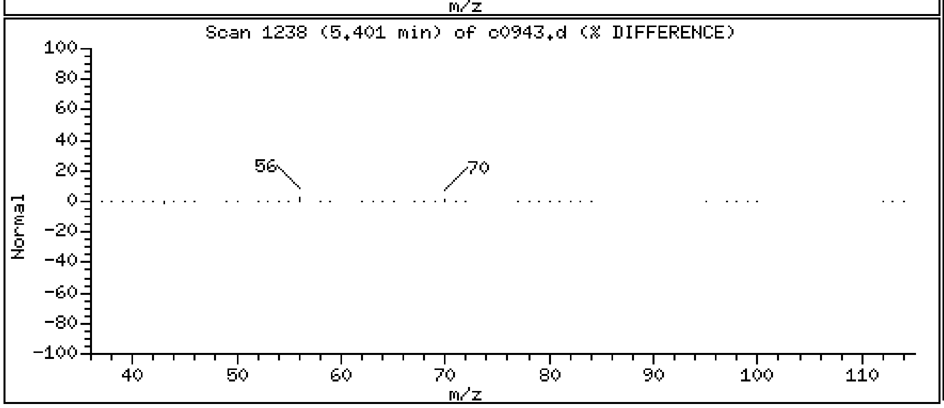
46 2,2,4 Trimethylpentane

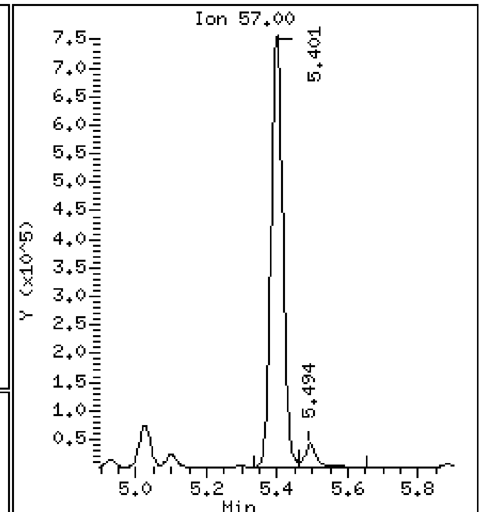
Concentration: 47600 UG/KG

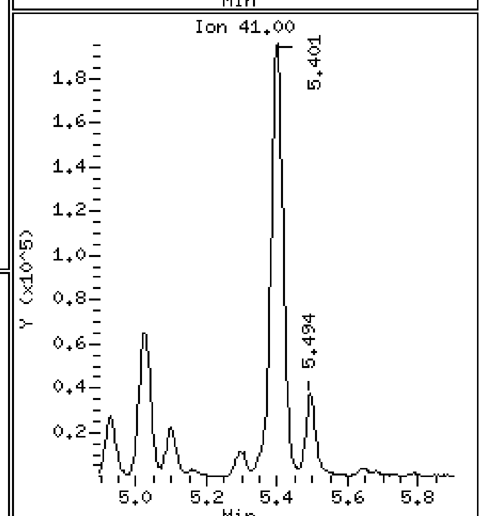


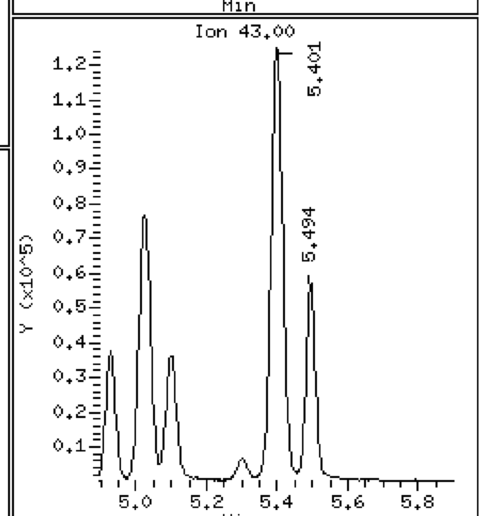












Data File: /var/chem/msv15.i/230612.s.b/c0943.d

Date : 12-JUN-2023 18:42

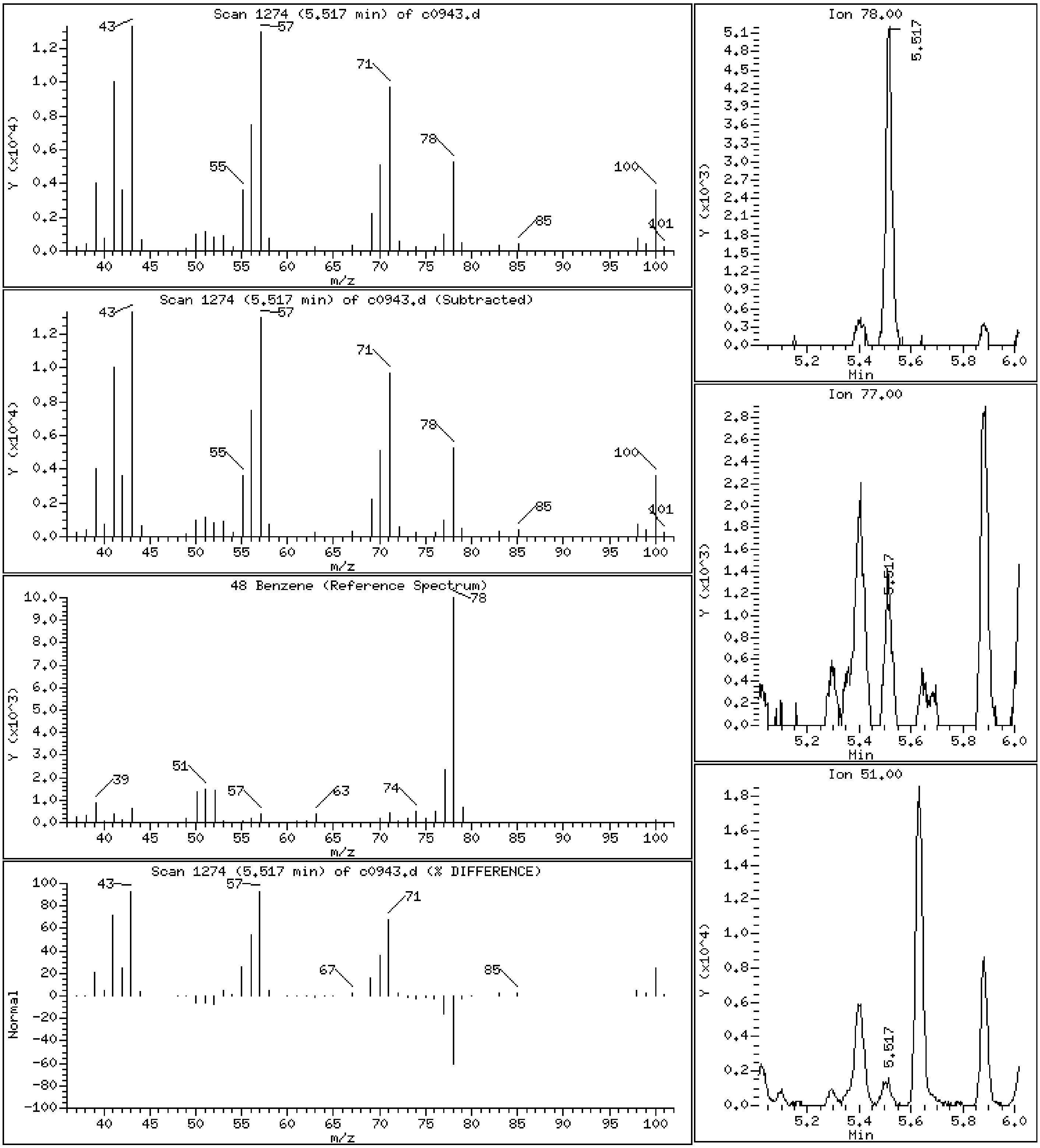
Client ID: 250X Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0 Operator: KN1

Column phase: RTX-VMS-30M Column diameter: 0.25

48 Benzene Concentration: 269 UG/KG



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Date : 12-JUN-2023 18:42

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0

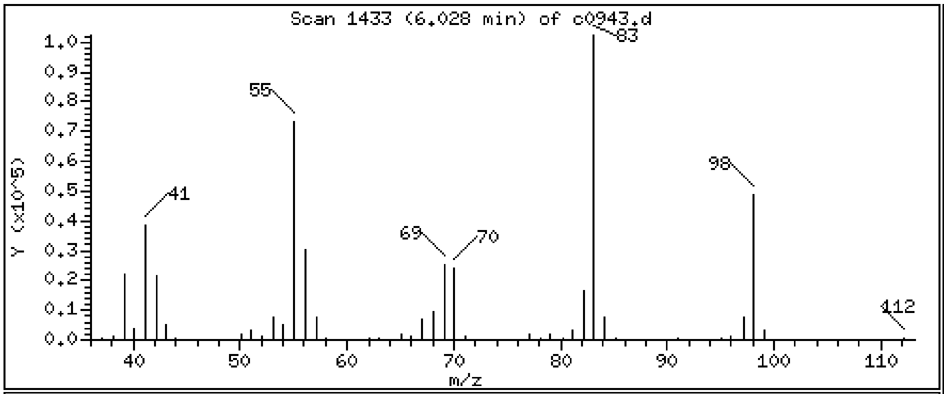
Operator: KN1

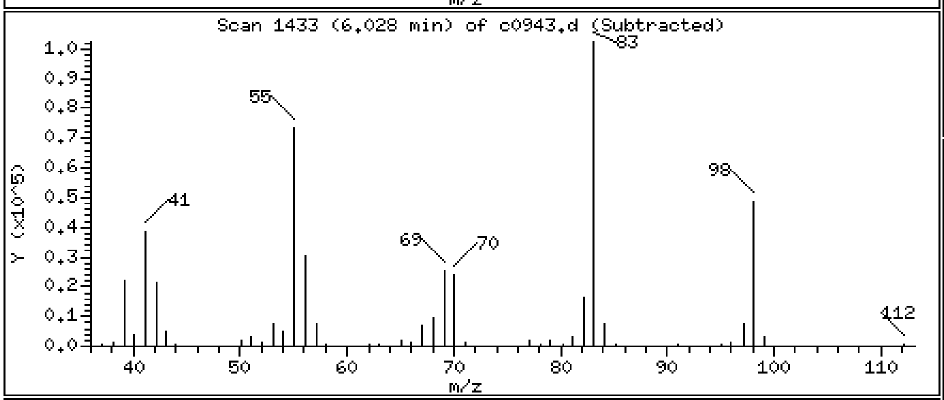
Column phase: RTX-VMS-30M

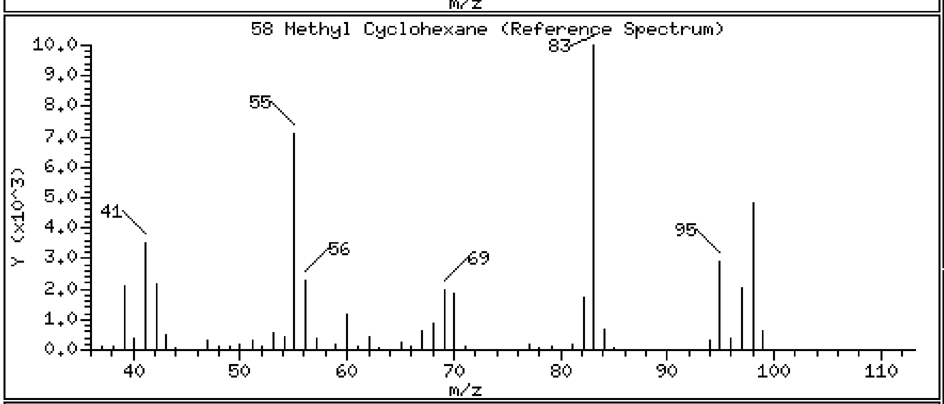
Column diameter: 0.25

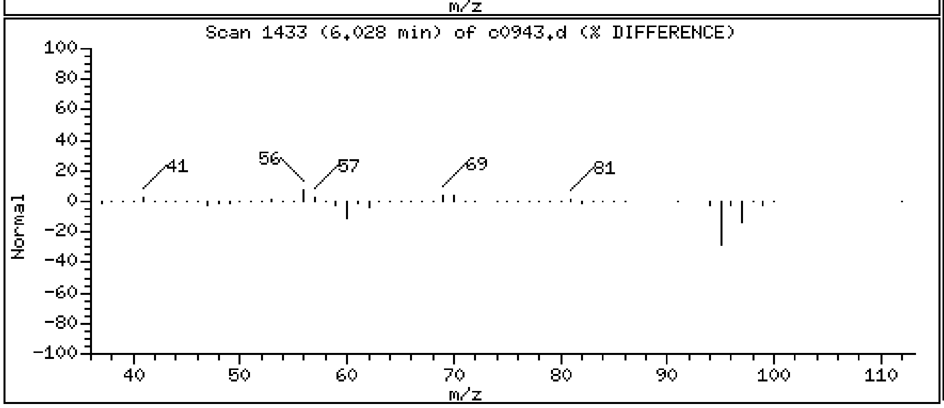
58 Methyl Cyclohexane

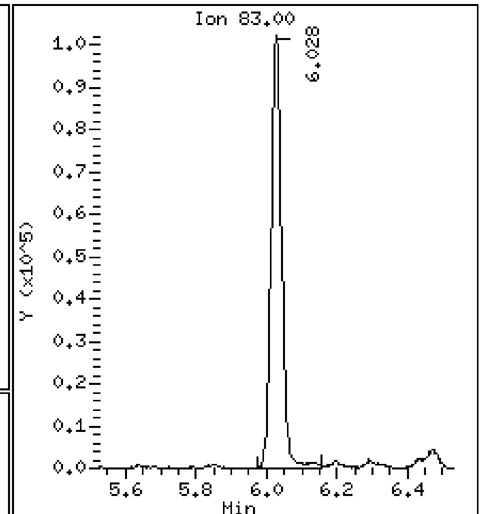
Concentration: 13800 UG/KG

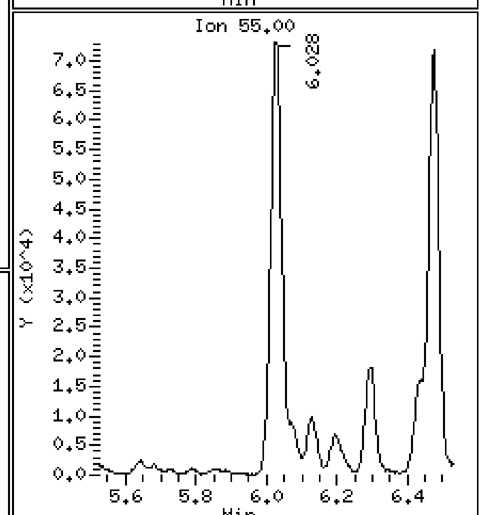


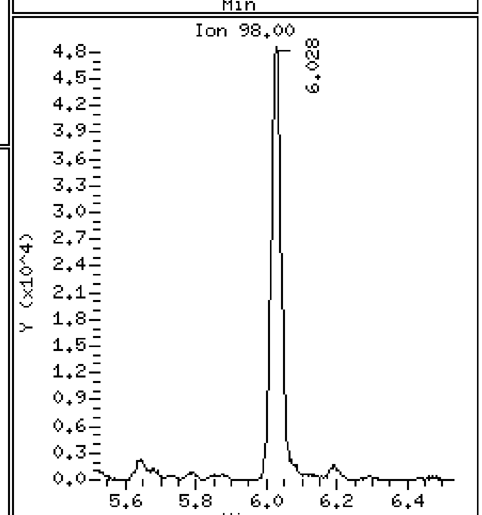












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Date : 12-JUN-2023 18:42

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0

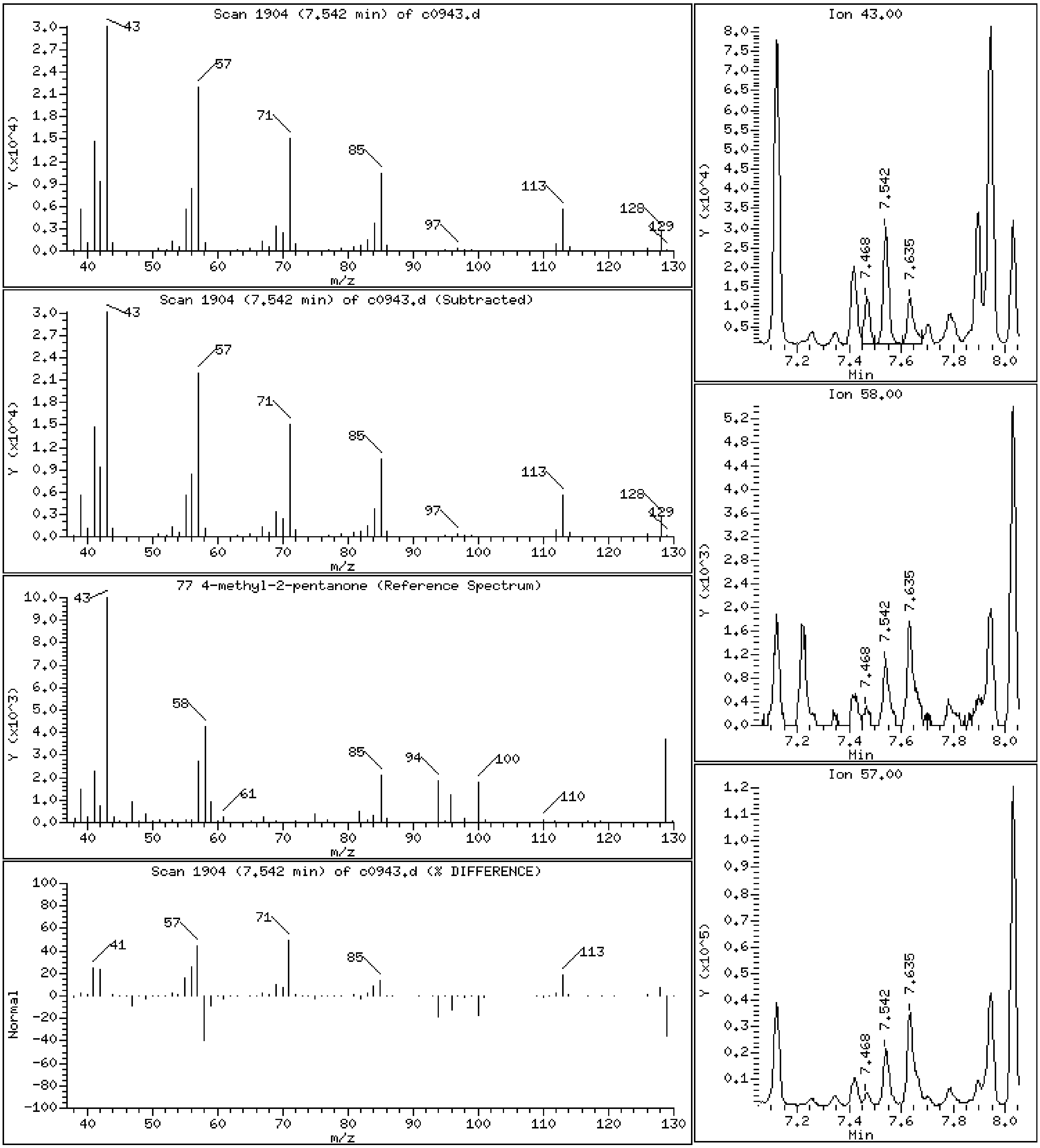
Operator: KN1

Column phase: RTX-VMS-30M

Column diameter: 0.25

77 4-methyl-2-pentanone

Concentration: 5950 UG/KG



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Date : 12-JUN-2023 18:42

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0

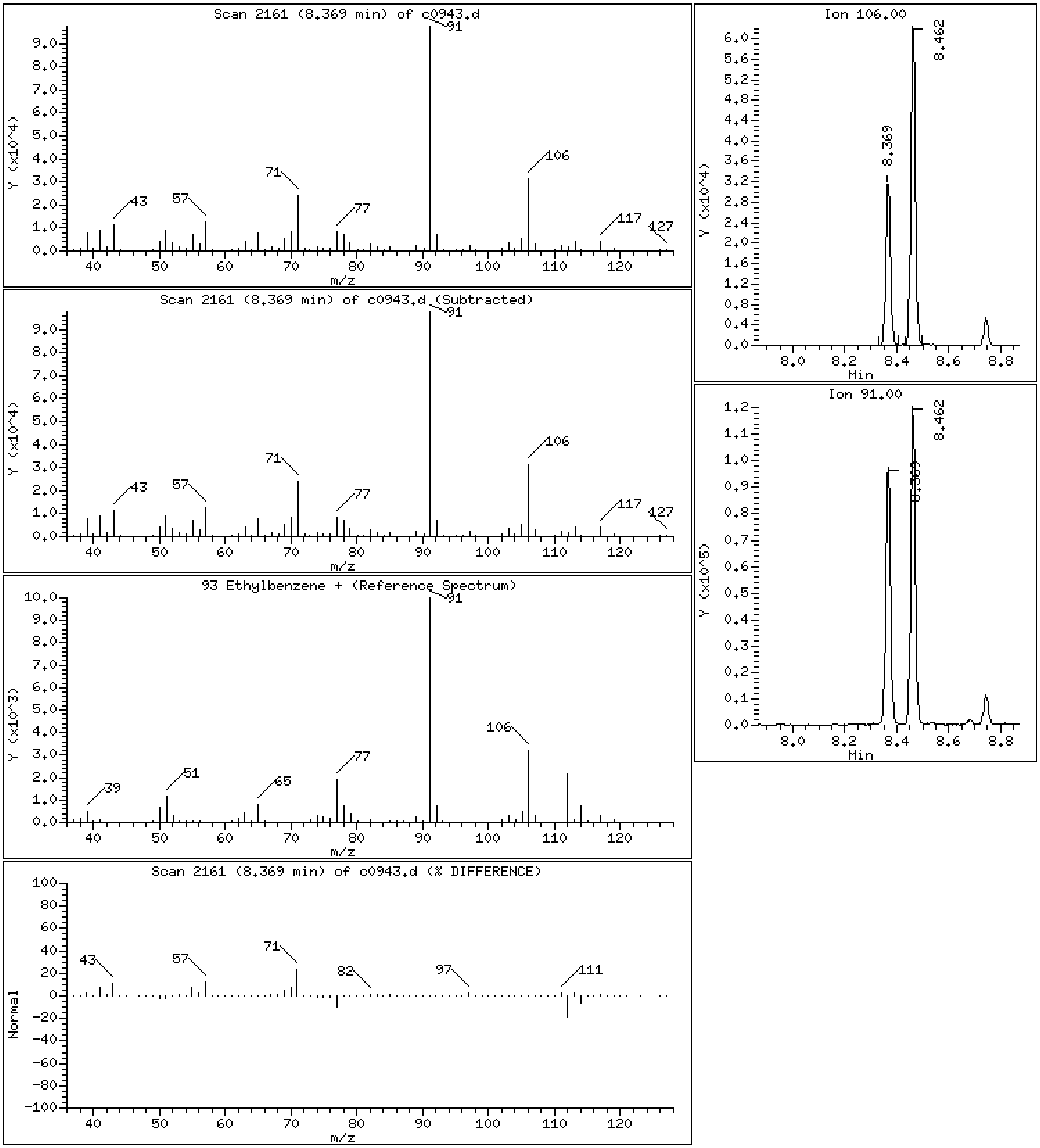
Operator: KN1

Column phase: RTX-VMS-30M

Column diameter: 0.25

93 Ethylbenzene +

Concentration: 2510 UG/KG



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Date : 12-JUN-2023 18:42

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0

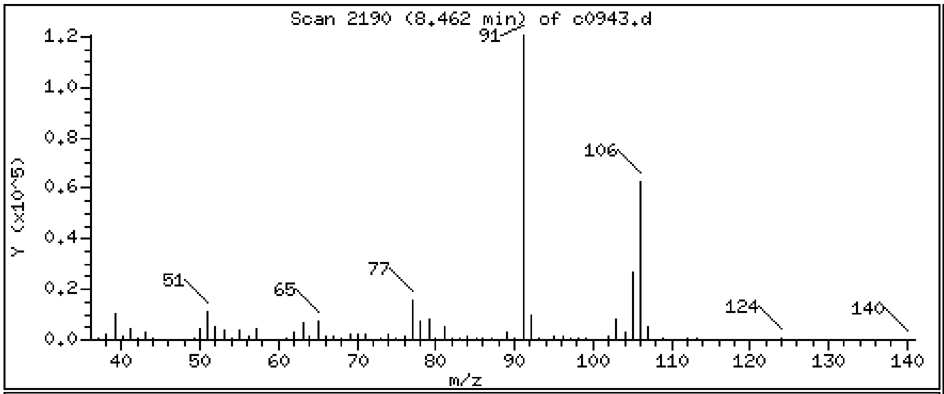
Operator: KN1

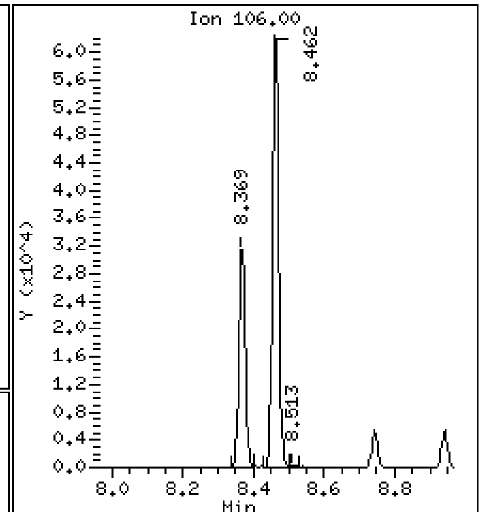
Column phase: RTX-VMS-30M

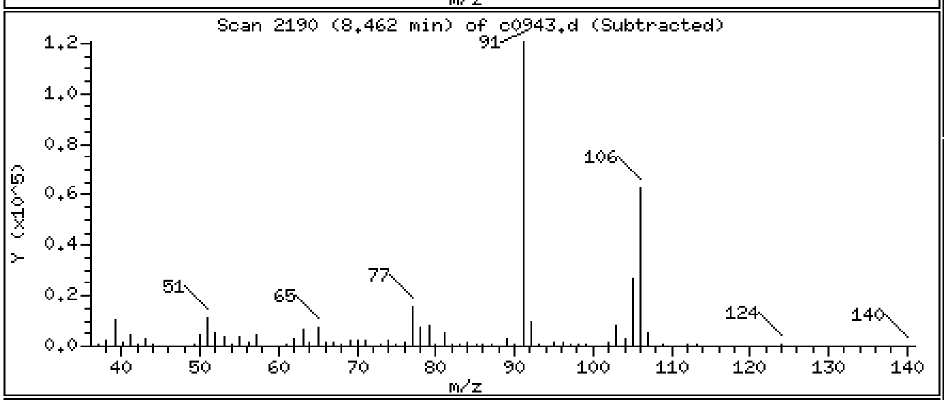
Column diameter: 0.25

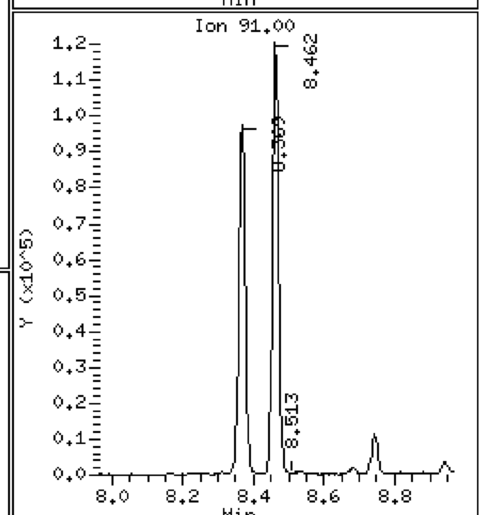
95 p,m-Xylene

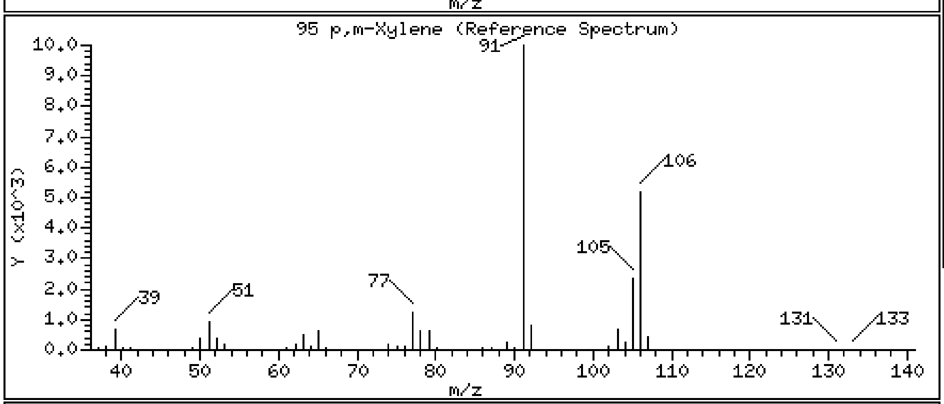
Concentration: 3960 UG/KG

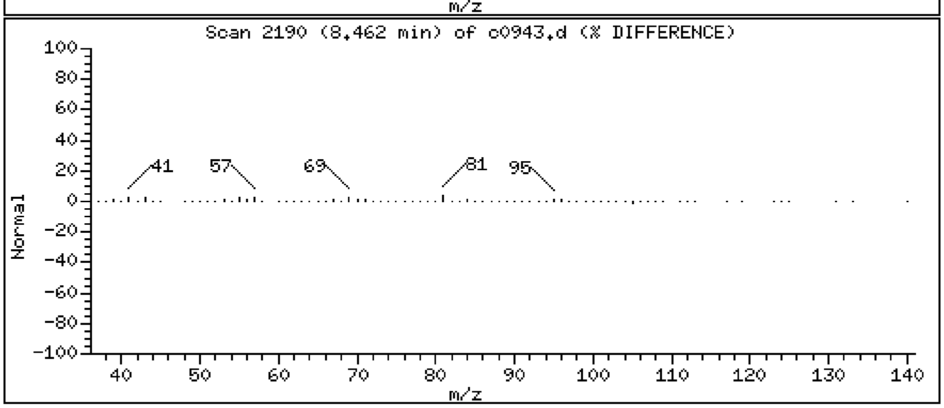












Data File: /var/chem/msv15.i/230612.s.b/c0943.d

Date : 12-JUN-2023 18:42

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0

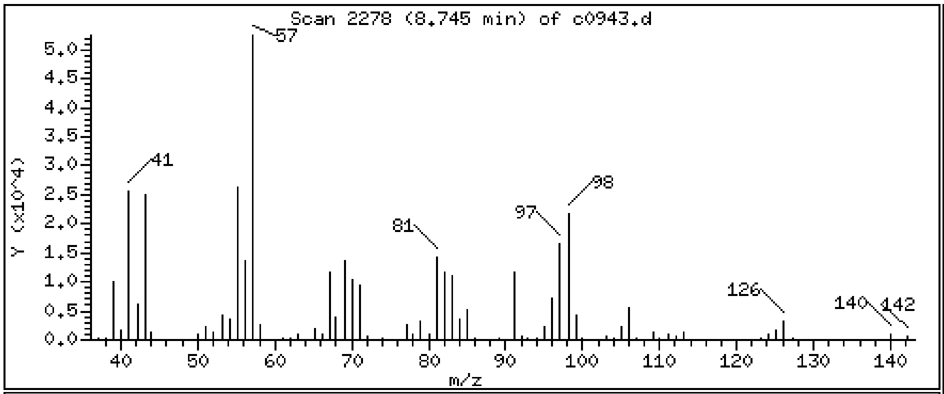
Operator: KN1

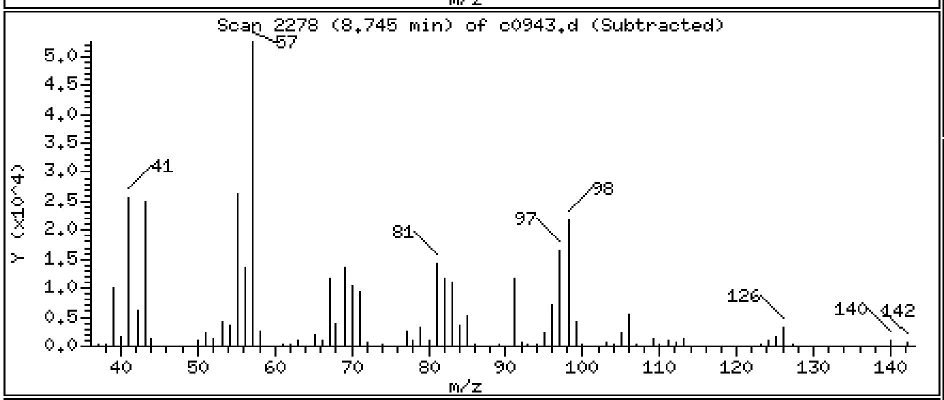
Column phase: RTX-VMS-30M

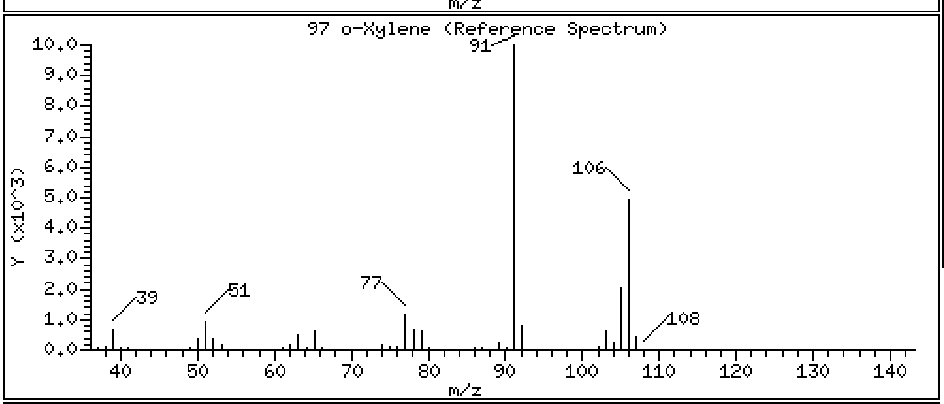
Column diameter: 0.25

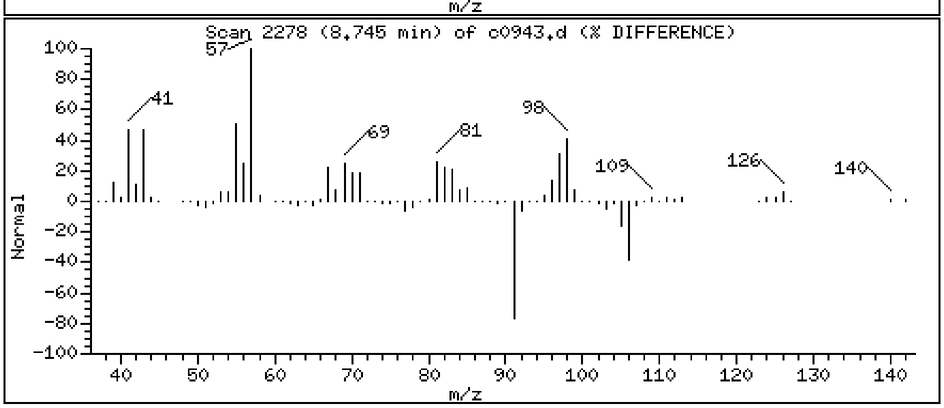
97 o-Xylene

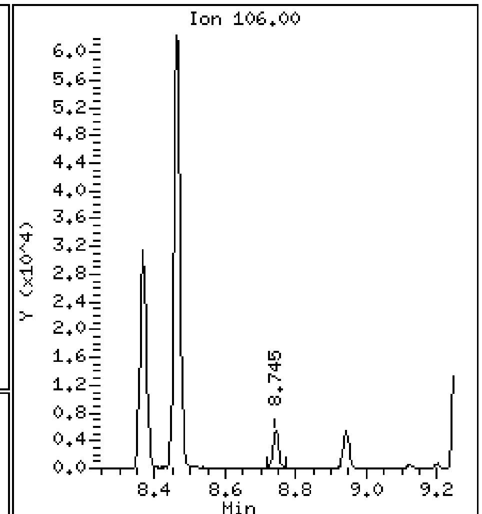
Concentration: 353 UG/KG

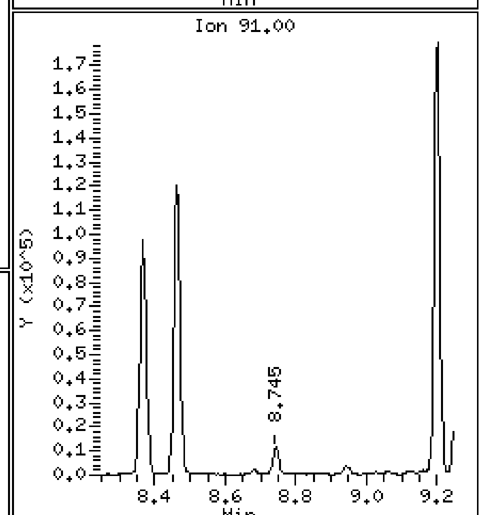












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Date : 12-JUN-2023 18:42

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0

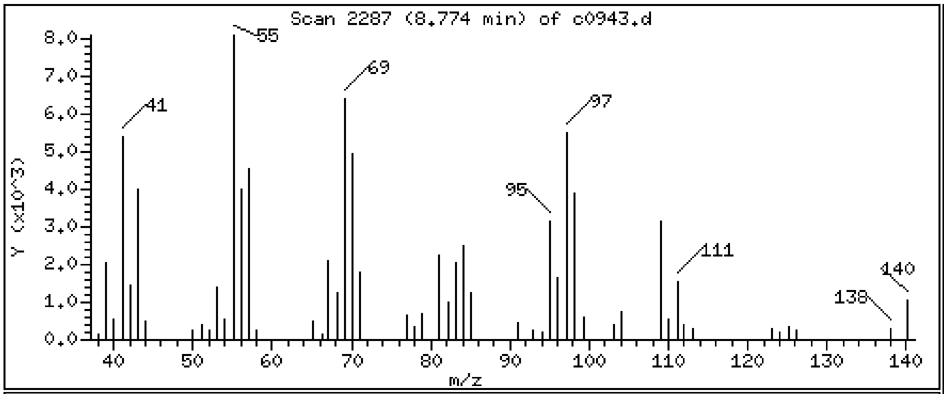
Operator: KN1

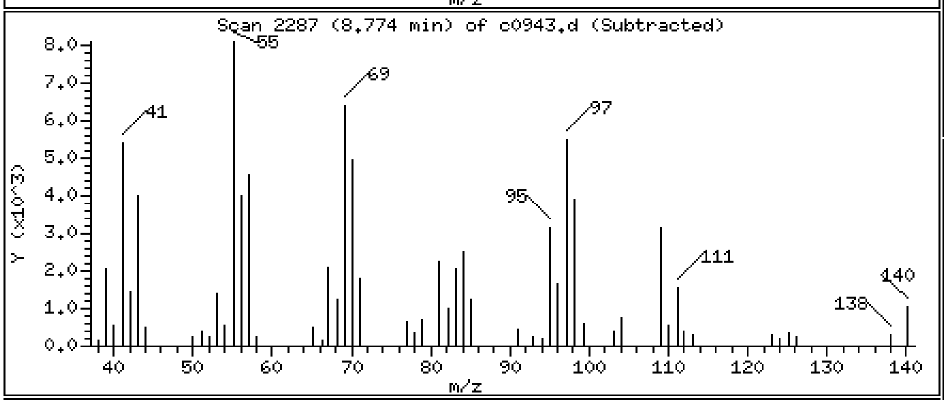
Column phase: RTX-VMS-30M

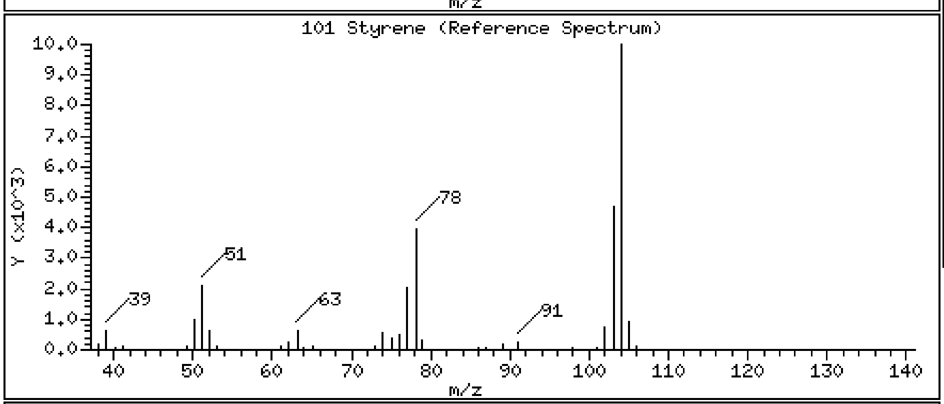
Column diameter: 0.25

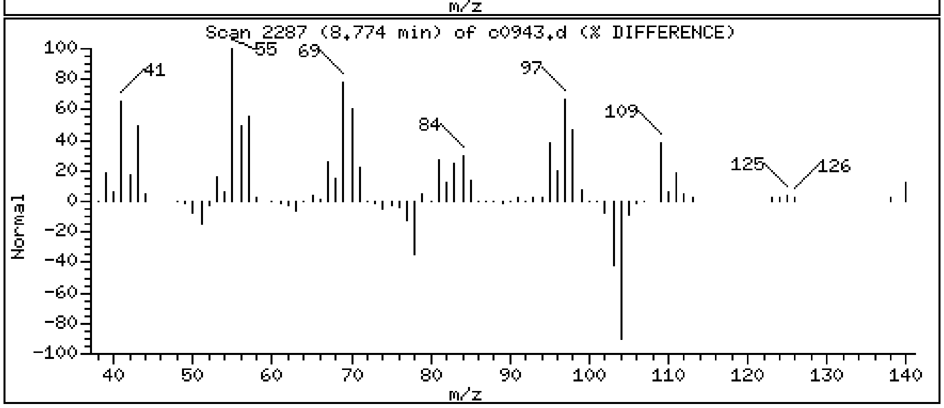
101 Styrene

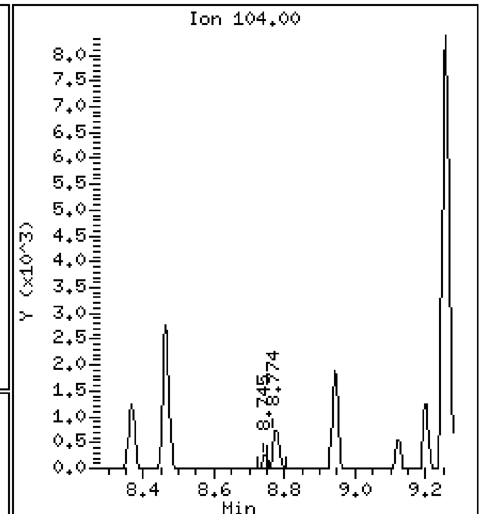
Concentration: 43.5 UG/KG

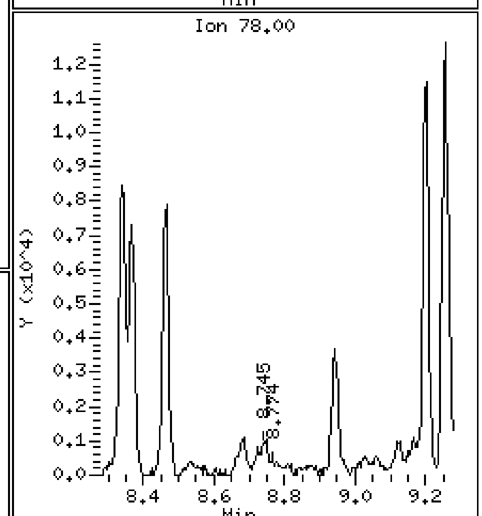












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Date : 12-JUN-2023 18:42

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0

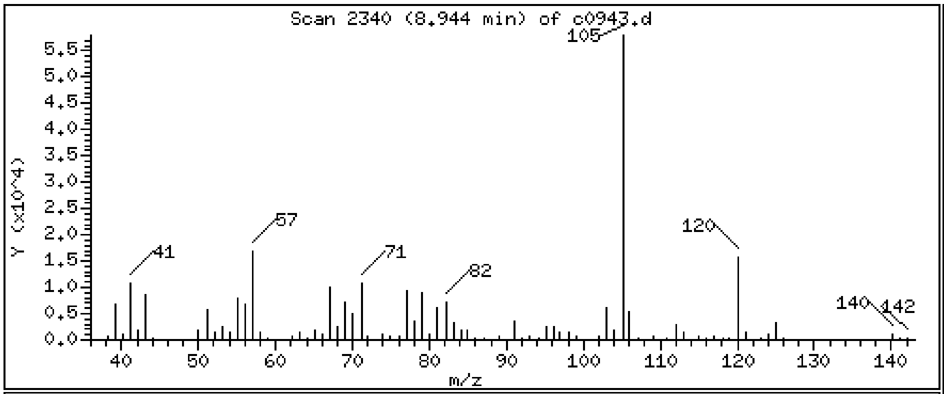
Operator: KN1

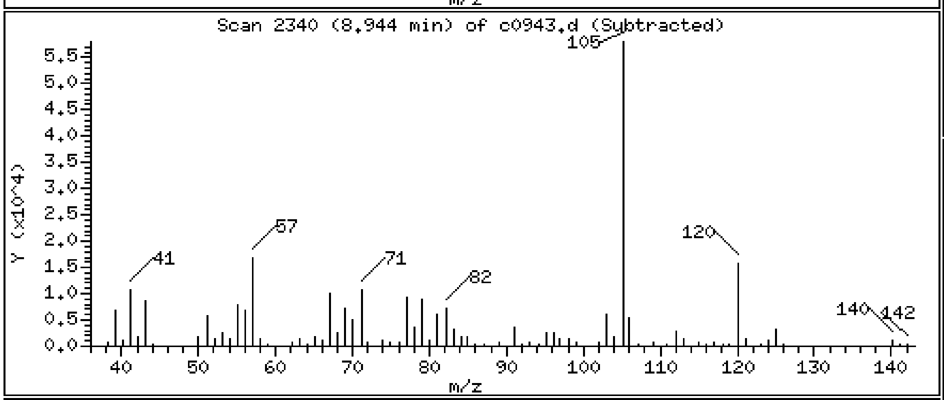
Column phase: RTX-VMS-30M

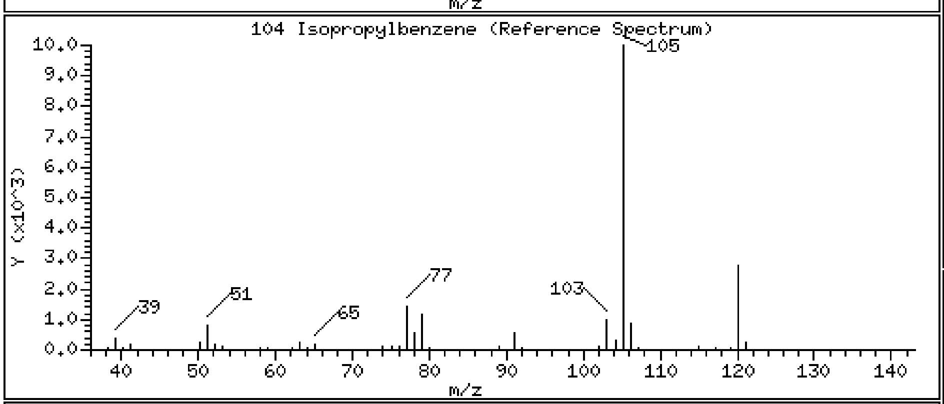
Column diameter: 0.25

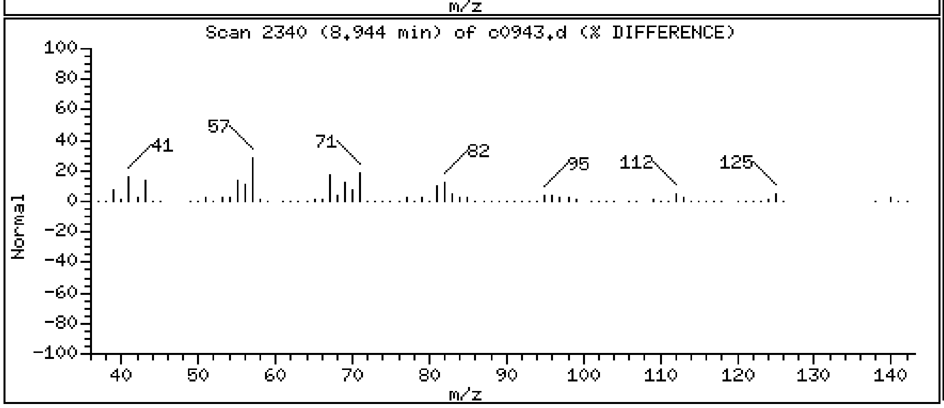
104 Isopropylbenzene

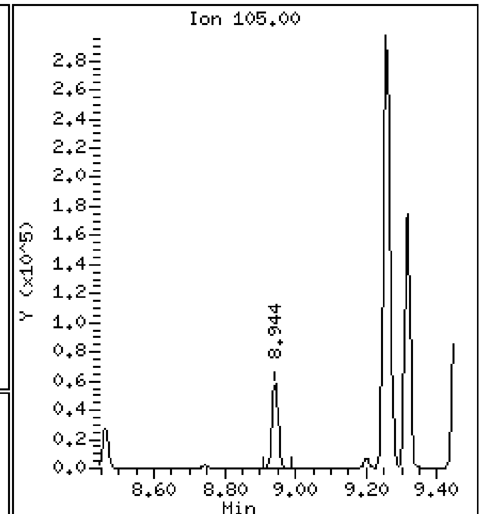
Concentration: 1500 UG/KG

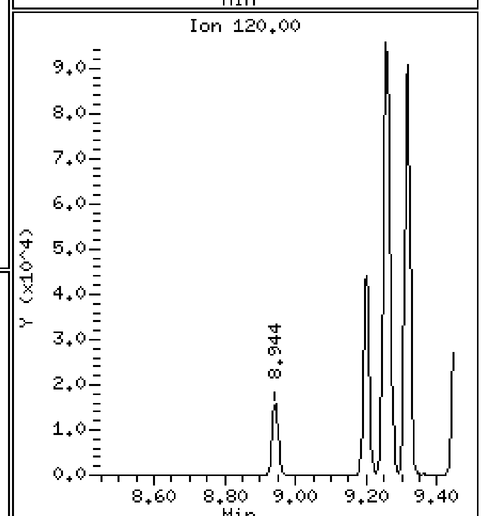


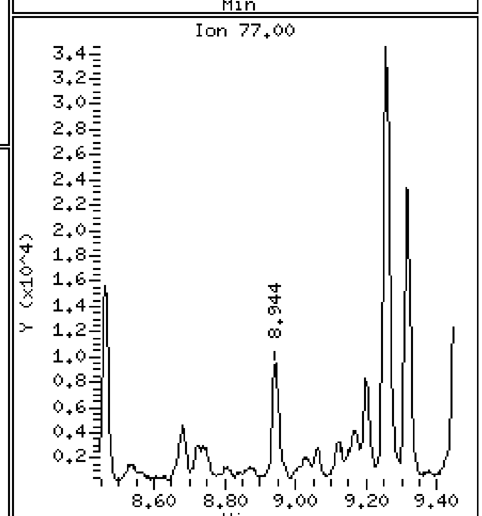












Date : 12-JUN-2023 18:42

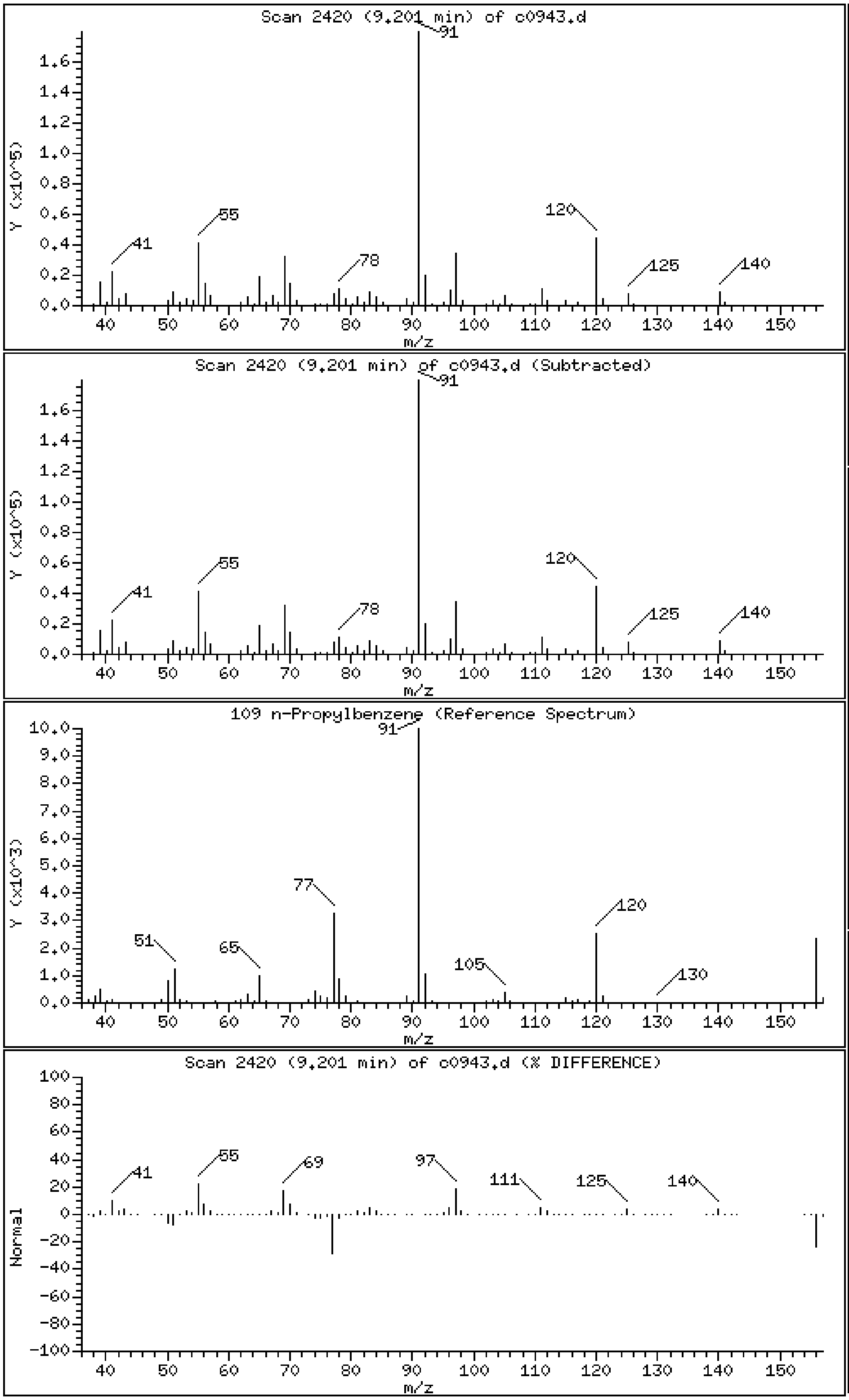
Client ID: 250X Instrument: msv15.i

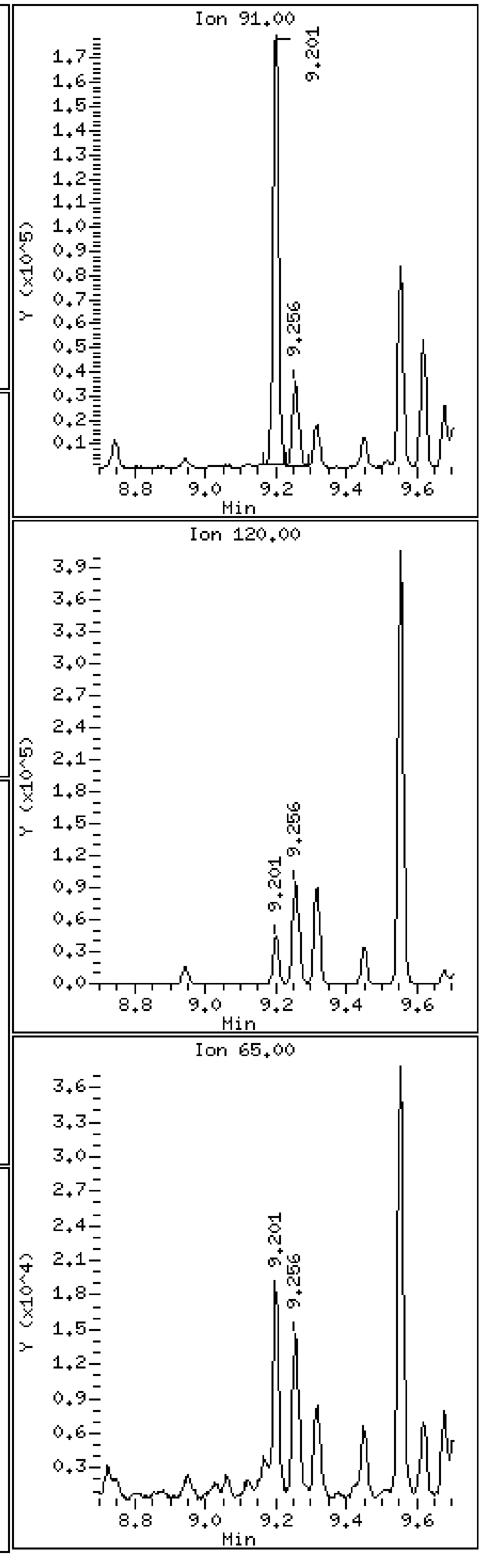
Sample Info: 22306050204*C*250X

Purge Volume: 5.0 Operator: KN1

Column phase: RTX-VMS-30M Column diameter: 0.25

109 n-Propylbenzene Concentration: 3330 UG/KG





Data File: /var/chem/msv15.i/230612.s.b/c0943.d   Page 14

Date : 12-JUN-2023 18:42

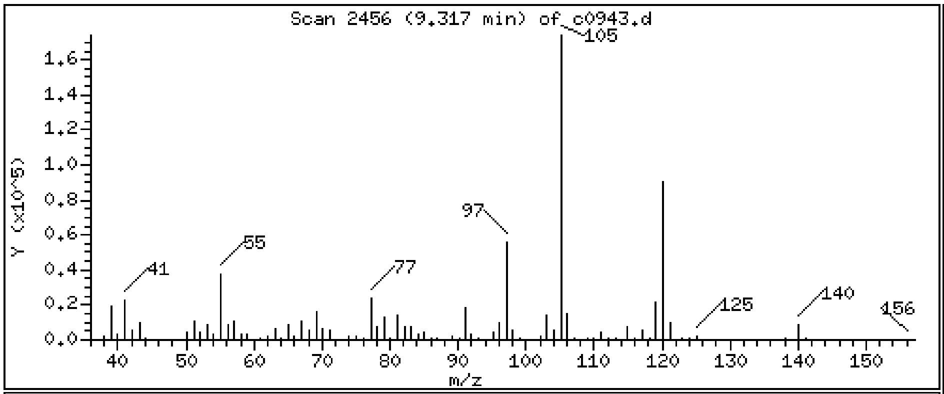
Client ID: 250X   Instrument: msv15.i

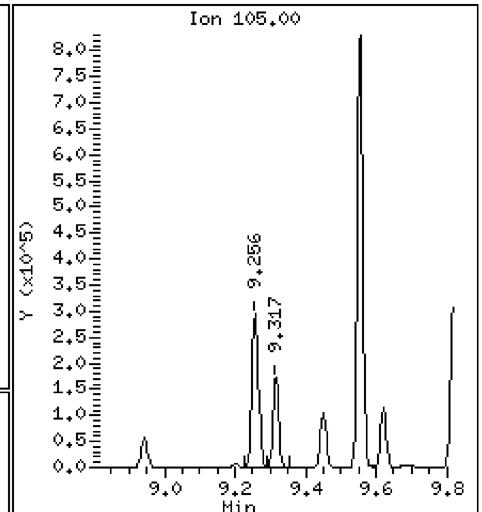
Sample Info: 22306050204*C*250X

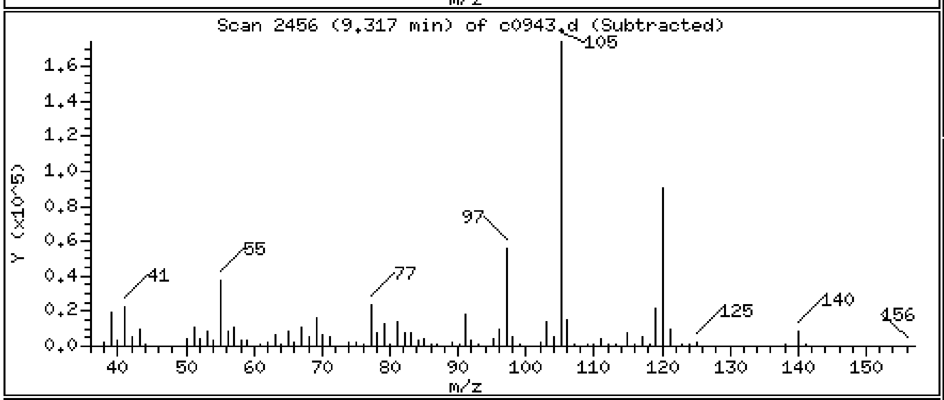
Purge Volume: 5.0   Operator: KN1

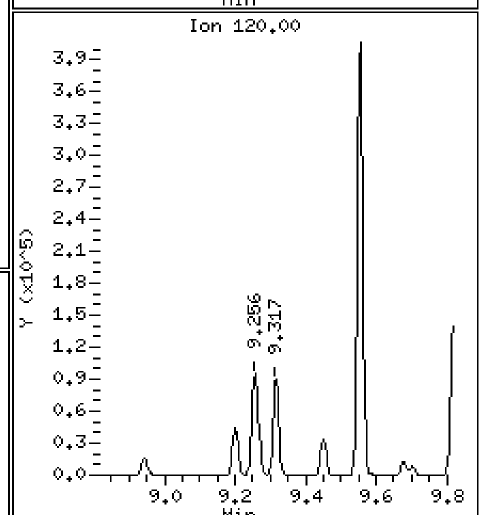
Column phase: RTX-VMS-30M   Column diameter: 0.25

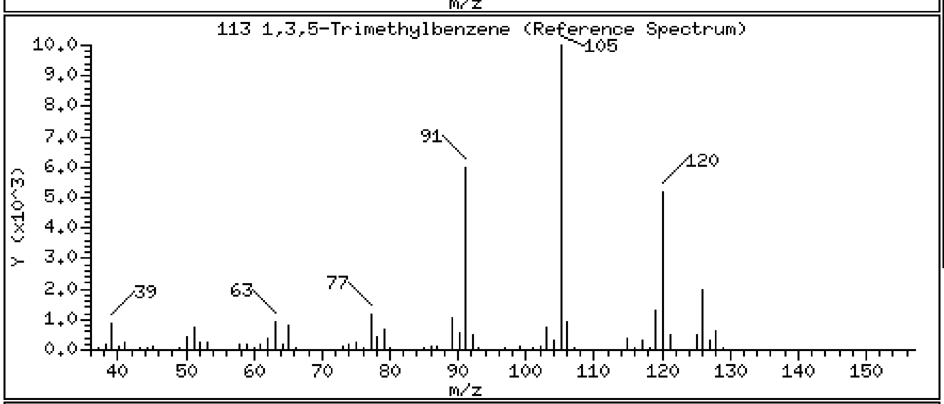
113 1,3,5-Trimethylbenzene   Concentration: 4850 UG/KG

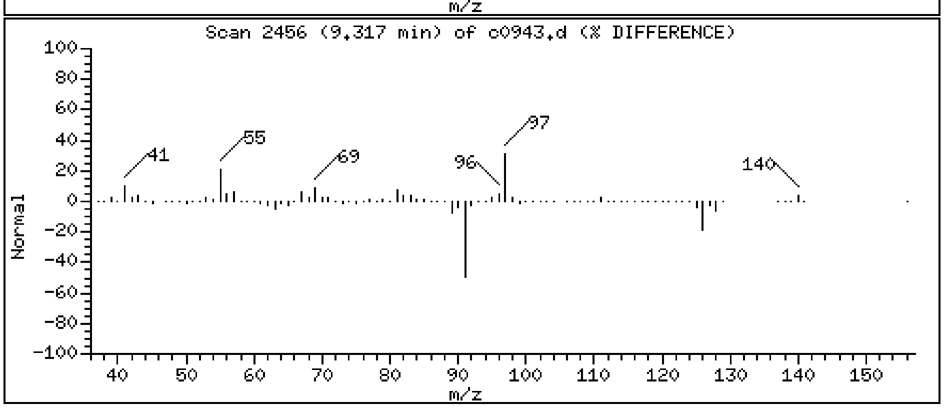












Data File: /var/chem/msv15.i/230612.s.b/c0943.d

Date : 12-JUN-2023 18:42

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0

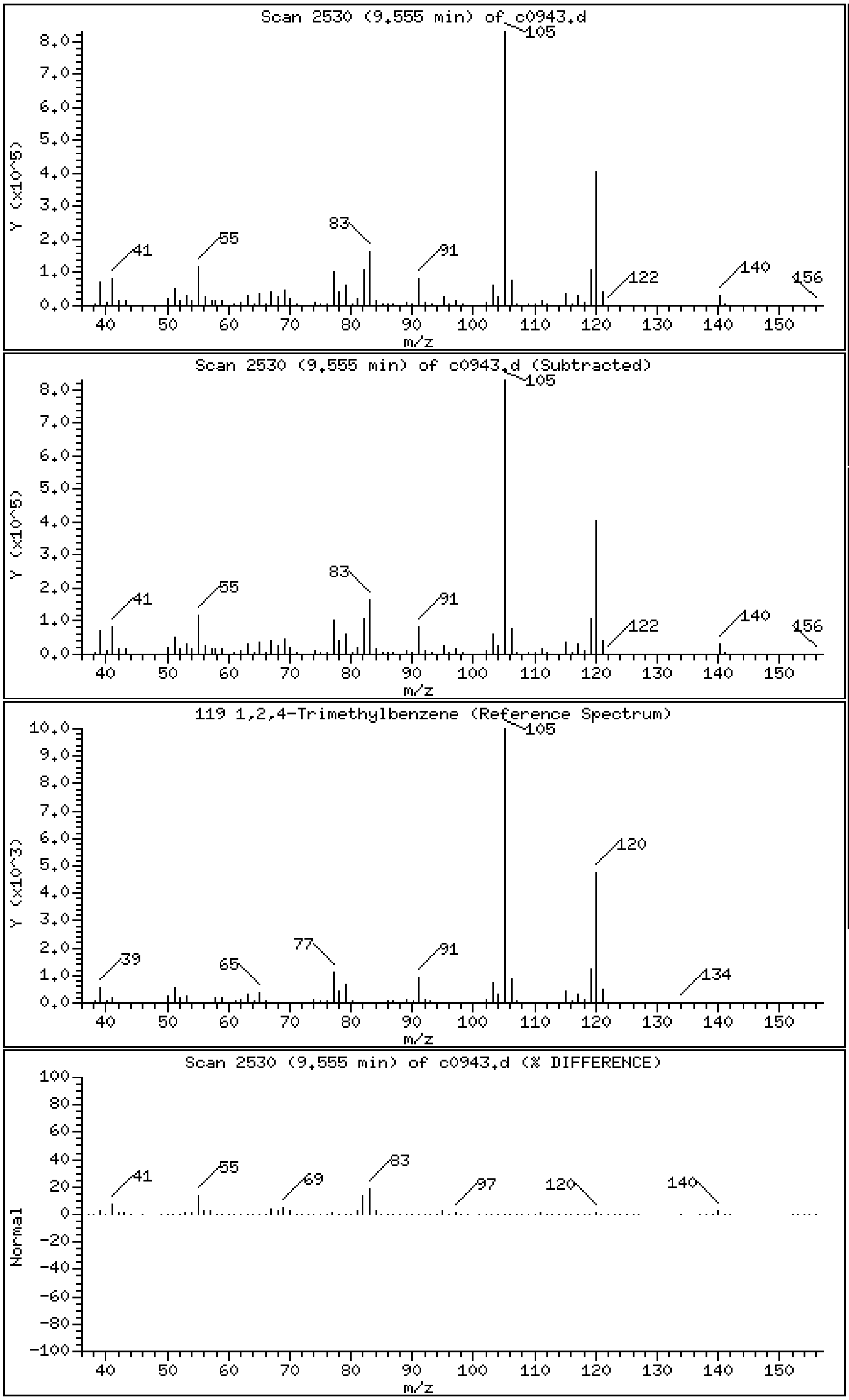
Operator: KN1

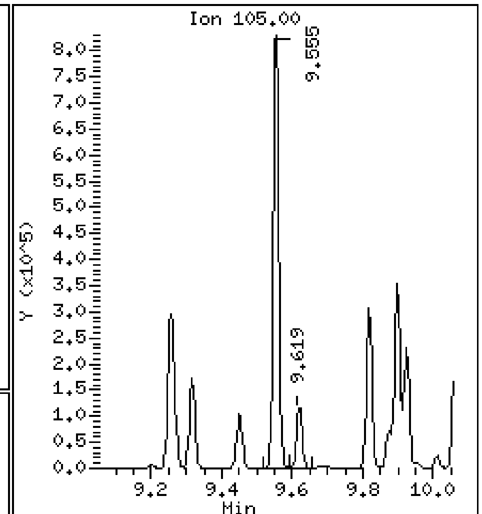
Column phase: RTX-VMS-30M

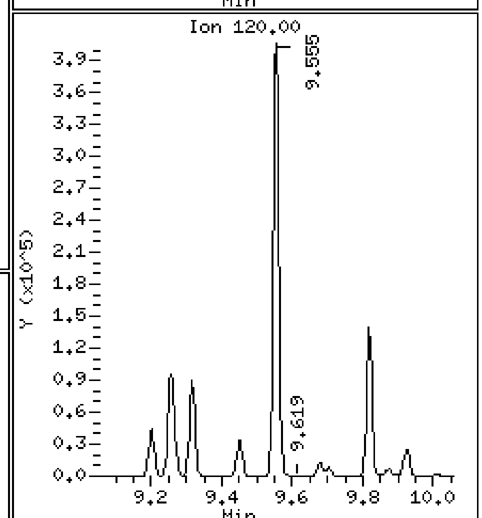
Column diameter: 0.25

119 1,2,4-Trimethylbenzene

Concentration: 22700 UG/KG







Data File: /var/chem/msv15.i/230612.s.b/c0943.d

Date : 12-JUN-2023 18:42

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0

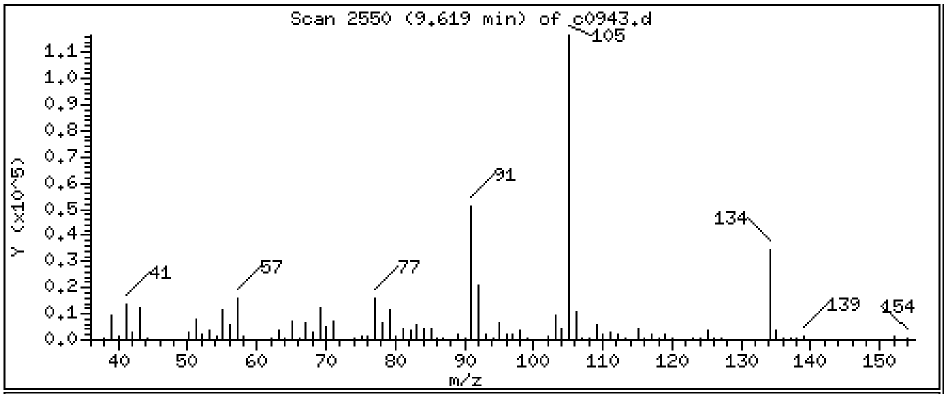
Operator: KN1

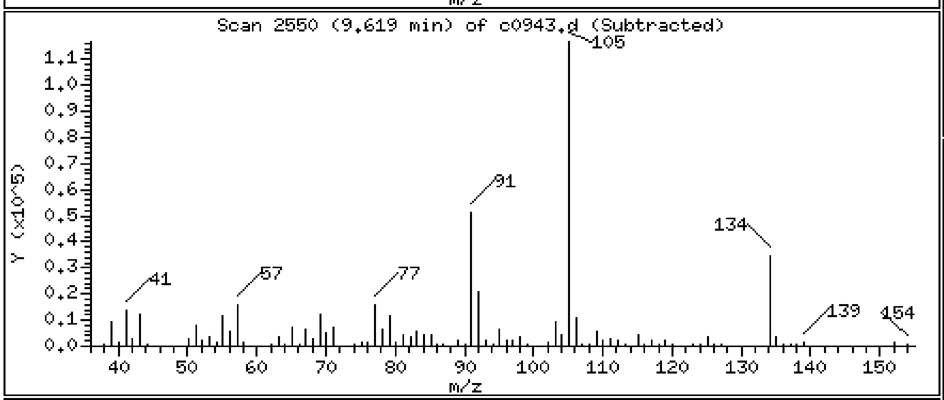
Column phase: RTX-VMS-30M

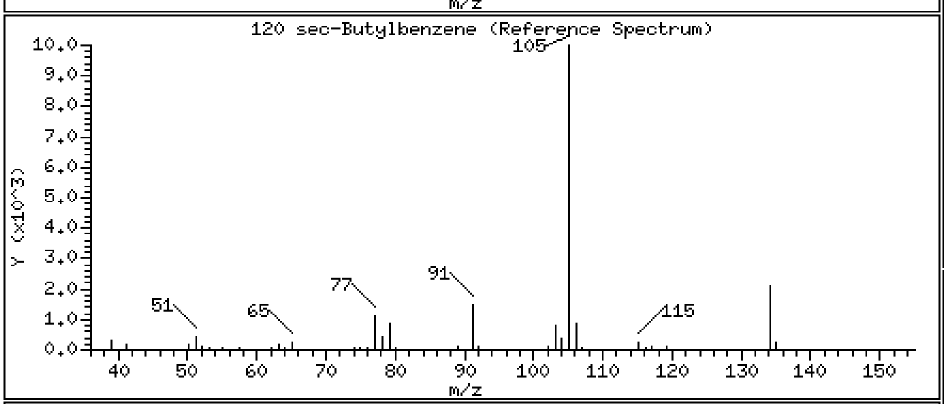
Column diameter: 0.25

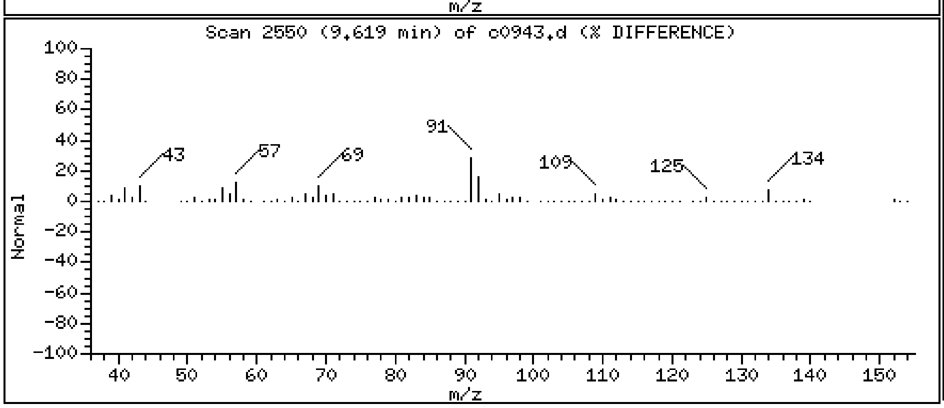
120 sec-Butylbenzene

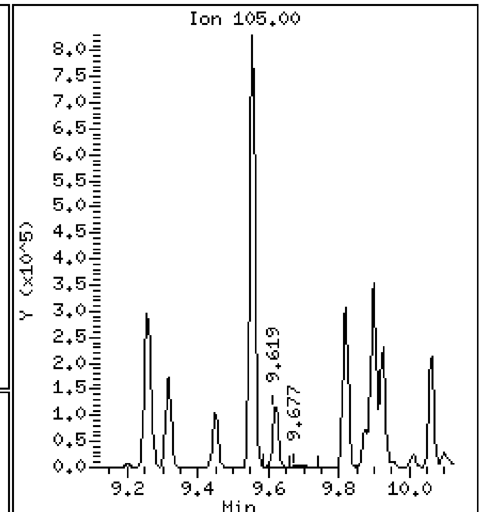
Concentration: 2590 UG/KG

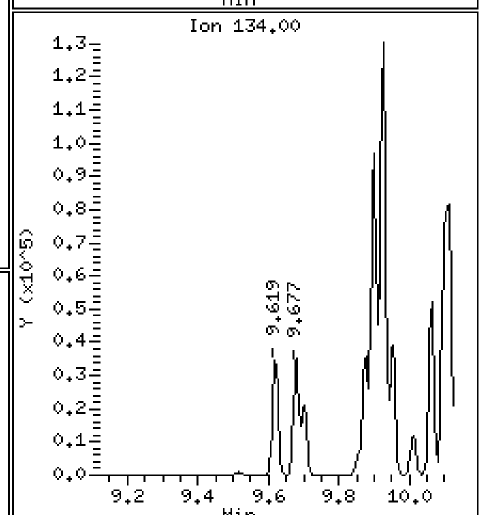


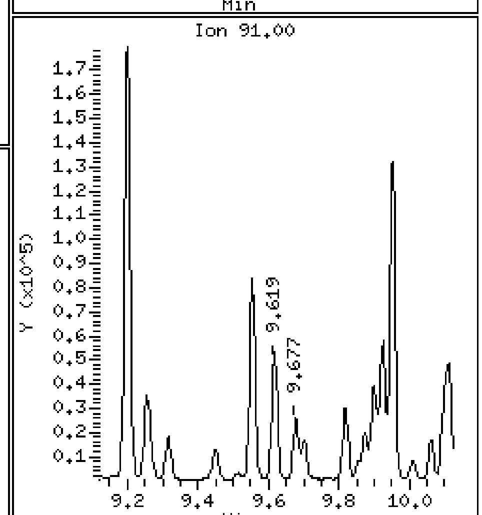












Date : 12-JUN-2023 18:42

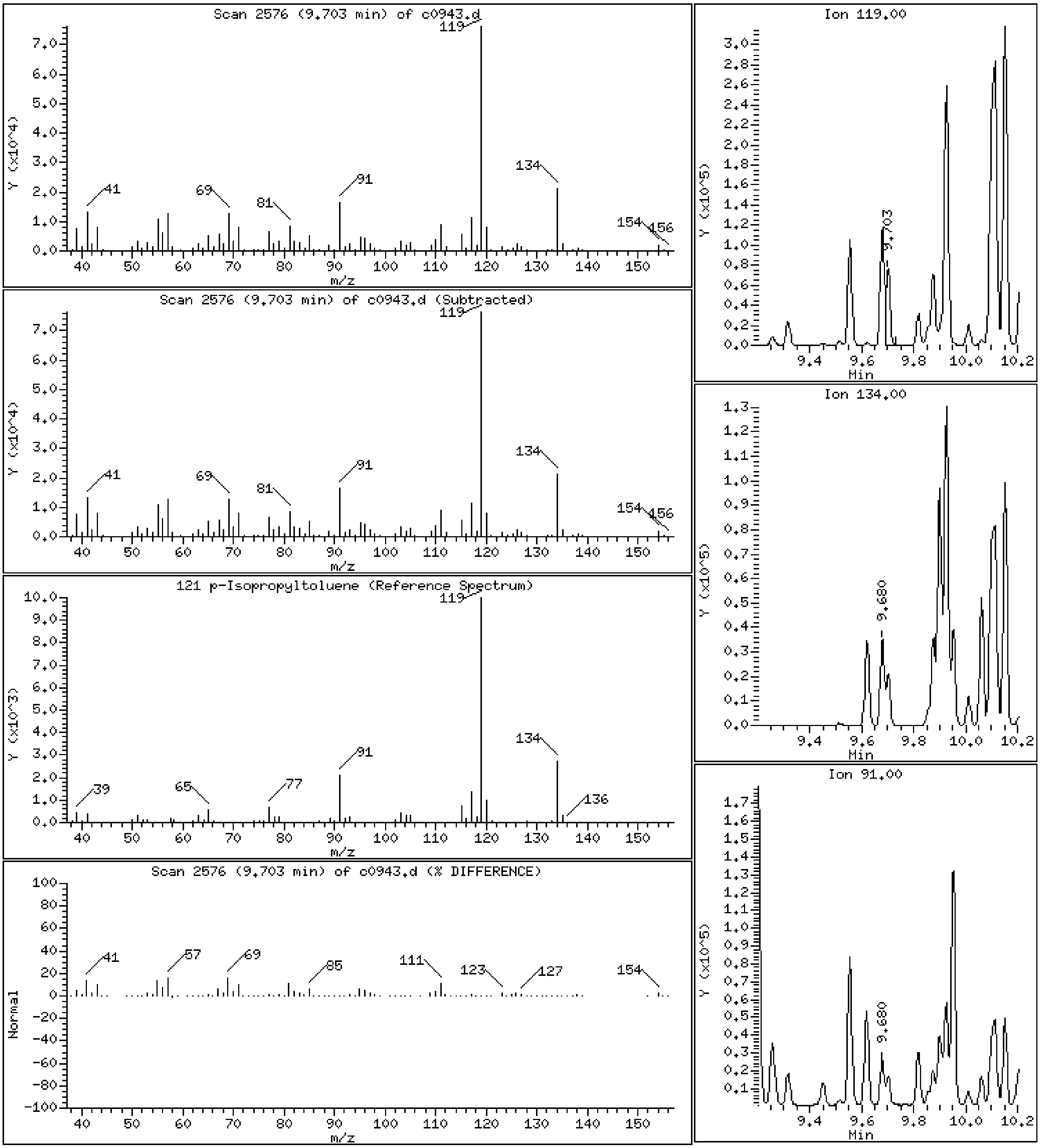
Client ID: 250X Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0 Operator: KN1

Column phase: RTX-VMS-30M Column diameter: 0.25

121 p-Isopropyltoluene Concentration: 1940 UG/KG



Date : 12-JUN-2023 18:42

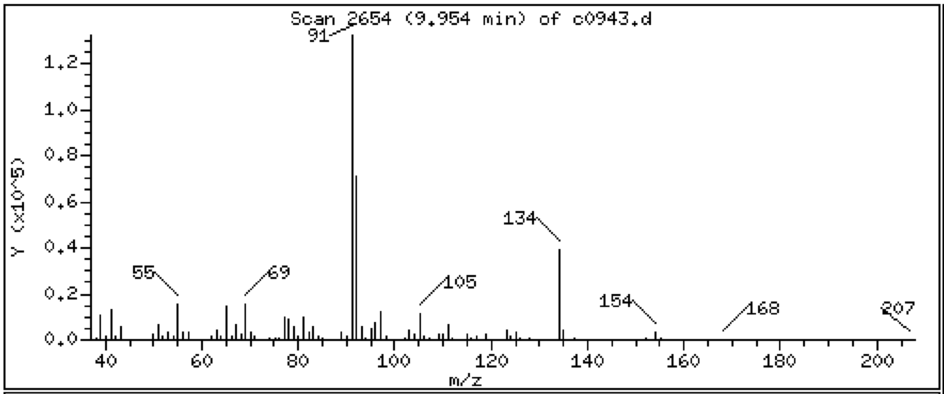
Client ID: 250X Instrument: msv15.i

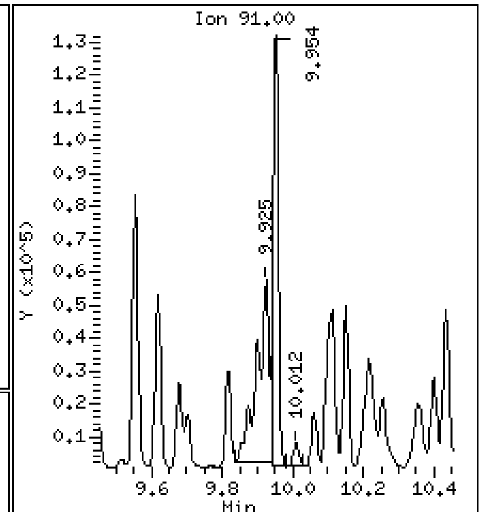
Sample Info: 22306050204*C*250X

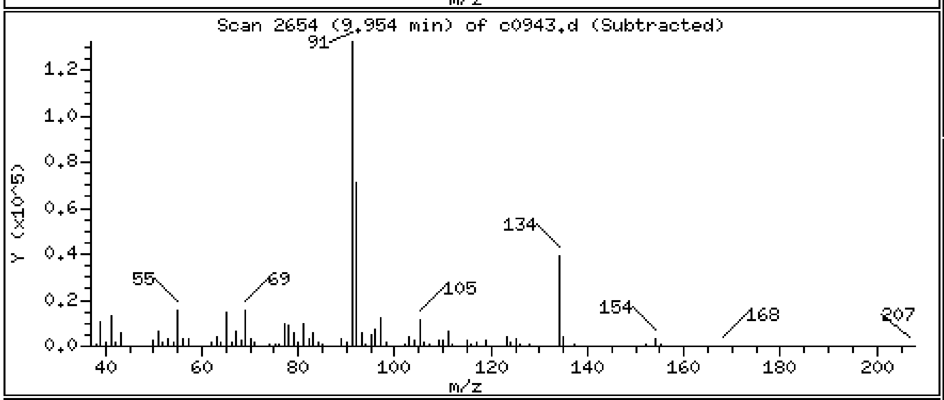
Purge Volume: 5.0 Operator: KN1

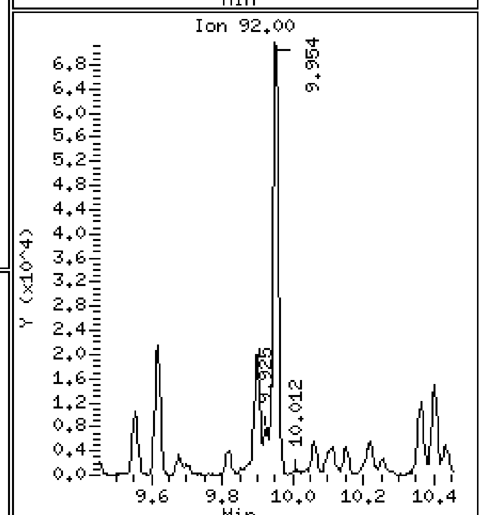
Column phase: RTX-VMS-30M Column diameter: 0.25

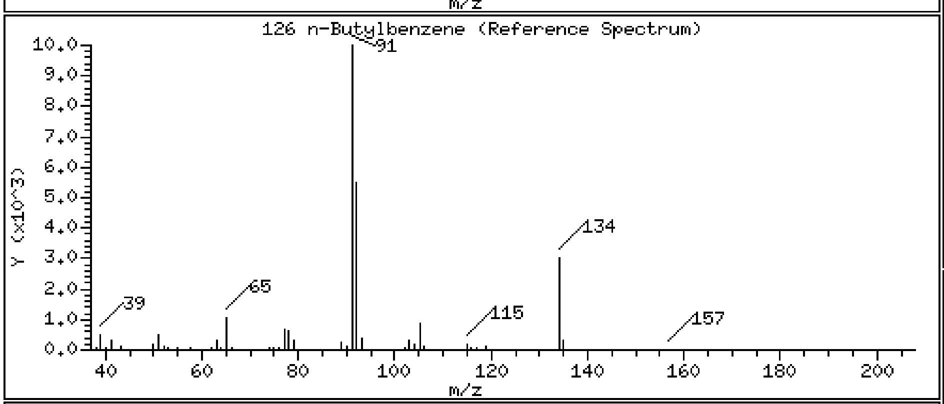
126 n-Butylbenzene Concentration: 2250 UG/KG

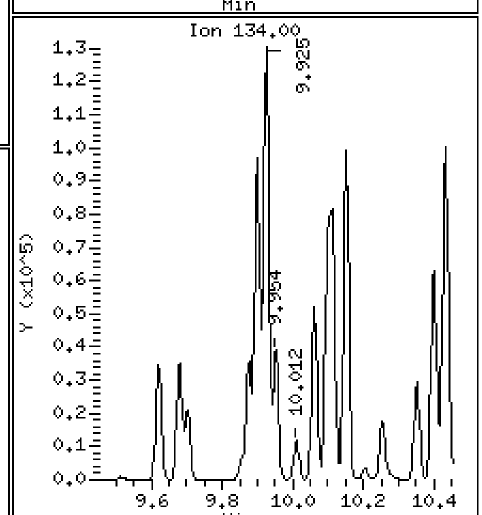


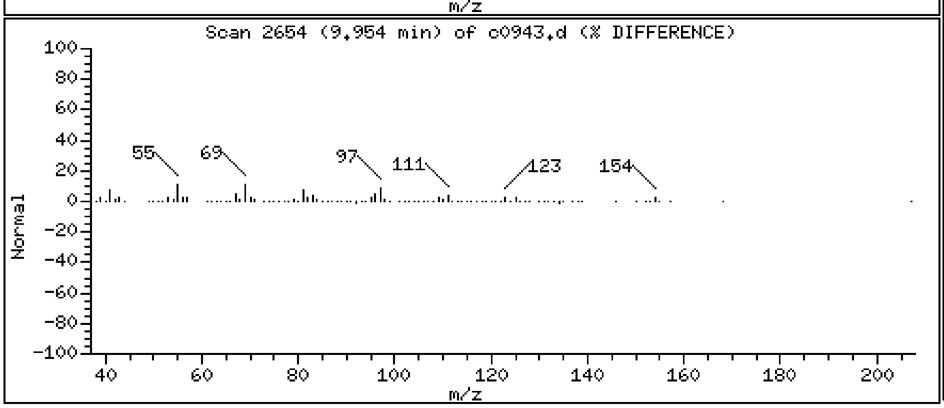












Data File: /var/chem/msv15.i/230612.s.b/c0943.d

Date : 12-JUN-2023 18:42

Client ID: 250X

Instrument: msv15.i

Sample Info: 22306050204*C*250X

Purge Volume: 5.0

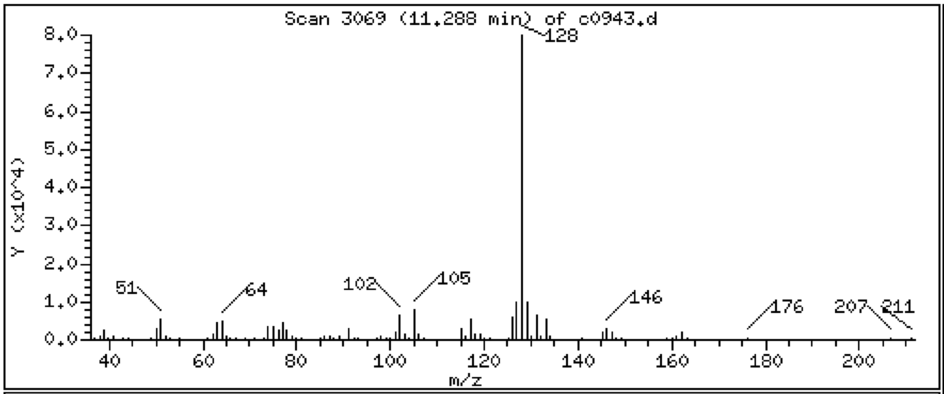
Operator: KN1

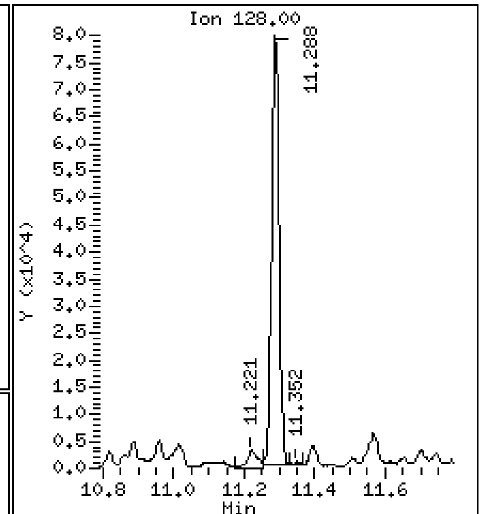
Column phase: RTX-VMS-30M

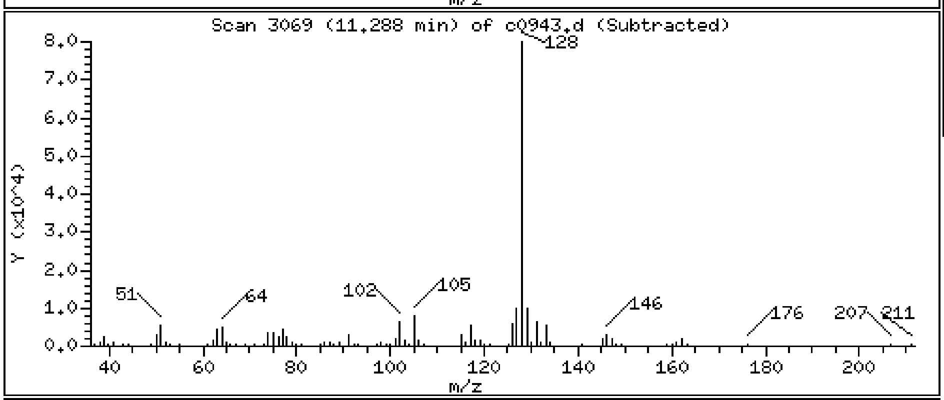
Column diameter: 0.25

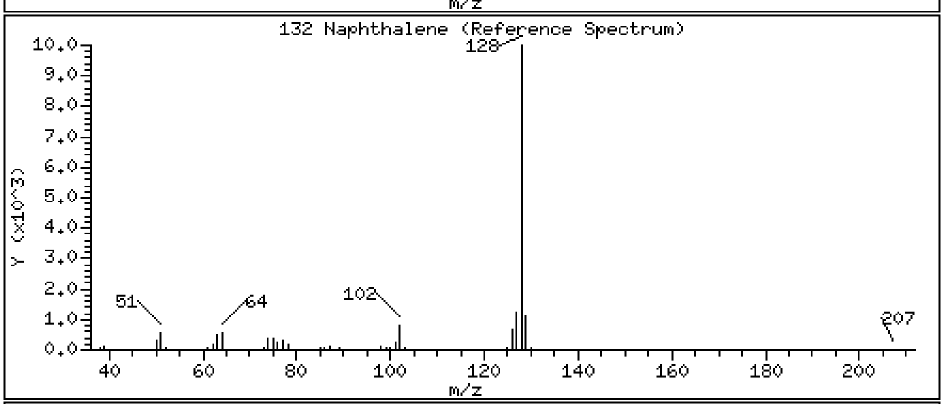
132 Naphthalene

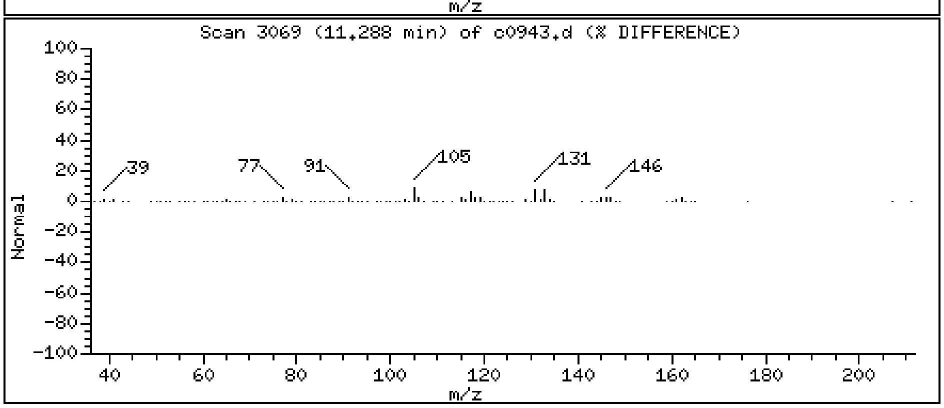
Concentration: 3850 UG/KG







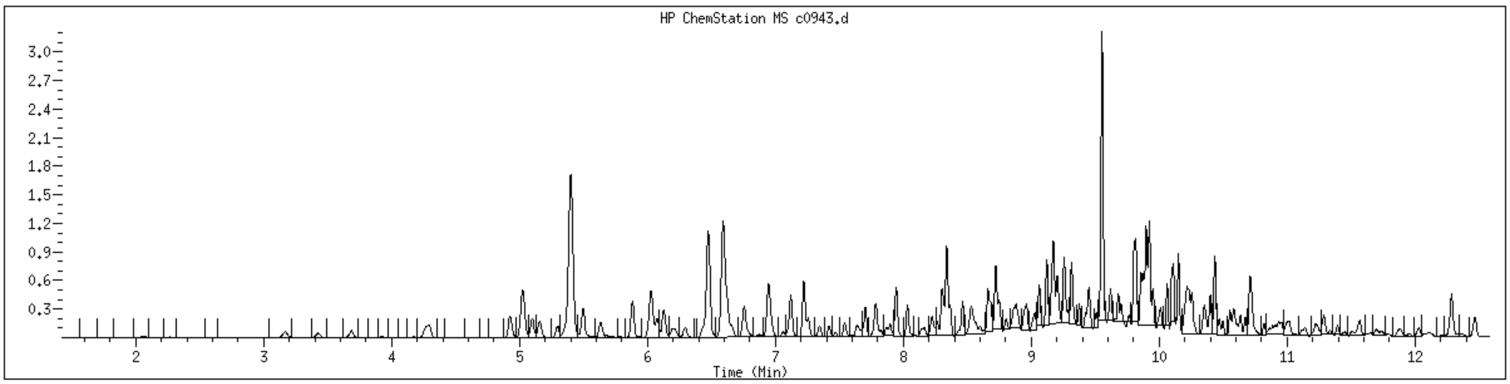




Report Date: 06/14/2023 13:50

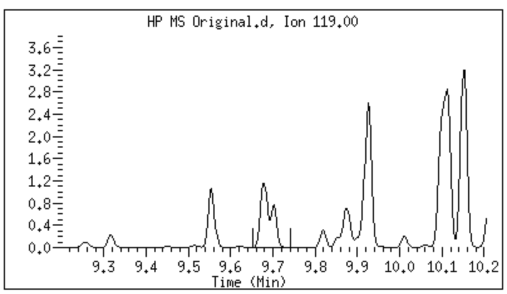
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 22306050204 SampleType : SAMPLE
Injection Date: 06/12/2023 18:42 Instrument : msv15.i
Operator : KN1
Sample Info : 22306050204*C*250X
Misc Info : MSV~52948~*250*SMR
Method : /var/chem/msv15.i/230612.s.b/8260bw15DOD.m
Dilution : 250
Matrix : SOIL
Integrator : HP RTE Compound Sublist: 8260DOD



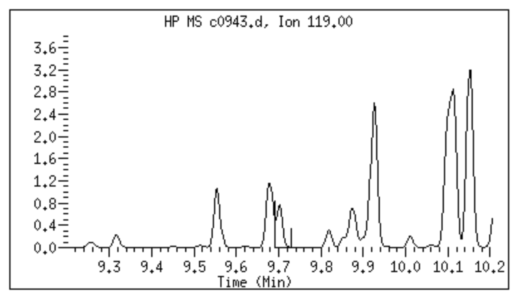
Original Final

121 p-Isopropyltoluene CAS#: 99-87-6 Reason: M2



Electronic Signature Applied

User: kmn
Date: 06/14/2023 13:45



M2 - Target system integrated incorrectly

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060502			Client Sample ID:	MB2491795
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2491795
Matrix:	Solid	% Moisture:	NA	Instrument ID:	MSV15
Sample Amt:	5	g		Lab File ID:	230612/c0922
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	50	Analyst:	KN1	Analytical Batch:	767415
Analysis Date:	06/12/23	Time:	1207	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/kg

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	25.0	U	12.5	25.0	250
71-55-6	1,1,1-Trichloroethane	25.0	U	12.5	25.0	250
79-34-5	1,1,2,2-Tetrachloroethane	25.0	U	12.5	25.0	250
79-00-5	1,1,2-Trichloroethane	25.0	U	12.5	25.0	250
75-34-3	1,1-Dichloroethane	25.0	U	12.5	25.0	250
75-35-4	1,1-Dichloroethene	25.0	U	12.5	25.0	250
563-58-6	1,1-Dichloropropene	25.0	U	12.5	25.0	250
87-61-6	1,2,3-Trichlorobenzene	50.0	U	25.0	50.0	250
96-18-4	1,2,3-Trichloropropane	50.0	U	25.0	50.0	250
120-82-1	1,2,4-Trichlorobenzene	50.0	U	25.0	50.0	250
95-63-6	1,2,4-Trimethylbenzene	25.0	U	12.5	25.0	250
96-12-8	1,2-Dibromo-3-chloropropane	100	U	25.0	100	250
106-93-4	1,2-Dibromoethane	100	U	25.0	100	250
95-50-1	1,2-Dichlorobenzene	25.0	U	12.5	25.0	250
107-06-2	1,2-Dichloroethane	25.0	U	12.5	25.0	250
540-59-0	1,2-Dichloroethene (total)	50.0	U	25.0	50.0	500
78-87-5	1,2-Dichloropropane	25.0	U	12.5	25.0	250
108-67-8	1,3,5-Trimethylbenzene	25.0	U	12.5	25.0	250
541-73-1	1,3-Dichlorobenzene	25.0	U	12.5	25.0	250
142-28-9	1,3-Dichloropropane	25.0	U	12.5	25.0	250
106-46-7	1,4-Dichlorobenzene	25.0	U	12.5	25.0	250
594-20-7	2,2-Dichloropropane	25.0	U	12.5	25.0	250
78-93-3	2-Butanone	100	U	25.0	100	250
95-49-8	2-Chlorotoluene	25.0	U	12.5	25.0	250
591-78-6	2-Hexanone	100	U	25.0	100	250
106-43-4	4-Chlorotoluene	25.0	U	12.5	25.0	250
99-87-6	4-Isopropyltoluene	25.0	U	12.5	25.0	250
108-10-1	4-Methyl-2-pentanone	25.0	U	12.5	25.0	250
67-64-1	Acetone	100	U	25.0	100	1250
71-43-2	Benzene	25.0	U	12.5	25.0	250
108-86-1	Bromobenzene	25.0	U	12.5	25.0	250
74-97-5	Bromochloromethane	50.0	U	25.0	50.0	250
75-27-4	Bromodichloromethane	25.0	U	12.5	25.0	250
75-25-2	Bromoform	50.0	U	25.0	50.0	250
74-83-9	Bromomethane	100	U	25.0	100	250
75-15-0	Carbon disulfide	25.0	U	12.5	25.0	250
56-23-5	Carbon tetrachloride	25.0	U	12.5	25.0	250

FORM I VOA

1A
VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060502			Client Sample ID:	MB2491795		
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2491795		
Matrix:	Solid	% Moisture:	NA	Instrument ID:	MSV15		
Sample Amt:	5	g		Lab File ID:	230612/c0922		
Injection Vol.:	1	(μL)		GC Column:	RTX-VMS-30M	ID .25	(mm)
Dilution Factor:	50	Analyst:	KN1	Analytical Batch:	767415		
Analysis Date:	06/12/23	Time:	1207	Analytical Method:	EPA 8260B		

CONCENTRATION UNITS: ug/kg

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	25.0	U	12.5	25.0	250
75-00-3	Chloroethane	25.0	U	12.5	25.0	250
67-66-3	Chloroform	25.0	U	12.5	25.0	250
74-87-3	Chloromethane	100	U	25.0	100	250
156-59-2	cis-1,2-Dichloroethene	25.0	U	12.5	25.0	250
10061-01-5	cis-1,3-Dichloropropene	25.0	U	12.5	25.0	250
124-48-1	Dibromochloromethane	25.0	U	12.5	25.0	250
74-95-3	Dibromomethane	25.0	U	12.5	25.0	250
75-71-8	Dichlorodifluoromethane	25.0	U	12.5	25.0	250
100-41-4	Ethylbenzene	25.0	U	12.5	25.0	250
87-68-3	Hexachlorobutadiene	50.0	U	25.0	50.0	250
98-82-8	Isopropylbenzene (Cumene)	25.0	U	12.5	25.0	250
136777-61-2	m,p-Xylene	15.4	J	12.5	50.0	500
75-09-2	Methylene chloride	100	U	50.0	100	500
91-20-3	Naphthalene	50.0	U	25.0	50.0	250
104-51-8	n-Butylbenzene	25.0	U	12.5	25.0	250
103-65-1	n-Propylbenzene	25.0	U	12.5	25.0	250
95-47-6	o-Xylene	25.0	U	12.5	25.0	250
135-98-8	sec-Butylbenzene	25.0	U	12.5	25.0	250
100-42-5	Styrene	25.0	U	12.5	25.0	250
1634-04-4	tert-Butyl methyl ether (MTBE)	25.0	U	12.5	25.0	250
98-06-6	tert-Butylbenzene	50.0	U	25.0	50.0	250
127-18-4	Tetrachloroethene	50.0	U	25.0	50.0	250
108-88-3	Toluene	25.0	U	12.5	25.0	250
156-60-5	trans-1,2-Dichloroethene	25.0	U	12.5	25.0	250
10061-02-6	trans-1,3-Dichloropropene	25.0	U	12.5	25.0	250
79-01-6	Trichloroethene	25.0	U	12.5	25.0	250
75-69-4	Trichlorofluoromethane	25.0	U	12.5	25.0	250
75-01-4	Vinyl chloride	25.0	U	12.5	25.0	250

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230612.s.b/c0922.d
Lab Smp Id: 2491795 Client Smp ID: 50X
Inj Date : 12-JUN-2023 12:07
Operator : KN1 Inst ID: msv15.i
Smp Info : SMB*50X
Misc Info : MSV~529~*50*SMR
Comment :
Method : /var/chem/msv15.i/230612.s.b/8260bw15DOD.m
Meth Date : 13-Jun-2023 15:50 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 QC Sample: LCS
Dil Factor: 50.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/(Ws*(100-M)/100))/5000 * CpndVariable

Name	Value	Description
DF	50.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Ws	5.00000	Weight of sample extracted (g)
M	0.00000	% Moisture (not decanted)
Va	100.00000	Volume of aliquot extract added (uL)
m	0.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (UG/KG)	SIMILARITY
6 Bromomethane	94	2.012	2.015	(0.342)	731	0.40580	20.3	4672(R)
15 Methyl Iodide	142	2.925	2.935	(0.497)	320	1.77320	88.7	3819(R)
18 Methylene Chloride	49	3.417	3.417	(0.581)	717	0.36368	18.2	7027(R)
$ 42 Dibromofluoromethane	111	5.159	5.156	(0.877)	93856	52.0861	2600	6907
$ 50 1,2-Dichloroethane-d4	65	5.629	5.629	(0.957)	95973	51.1521	2560	9733
* 55 FLUOROBENZENE	96	5.880	5.880	(1.000)	364881	50.0000		9874
$ 74 Toluene-d8	98	7.220	7.221	(0.866)	336343	46.6932	2330	9885
* 91 CHLOROBENZENE-d5	82	8.339	8.340	(1.000)	153699	50.0000		9879
95 p,m-Xylene	106	8.462	8.465	(1.015)	1204	0.30858	15.4	8067(R)
M 105 TOTAL XYLENE	106				1204	0.30858	15.4	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	115392	49.3263	2470	9594
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.806	(1.000)	152172	50.0000		9689

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (UG/KG)	SIMILARITY
132 Naphthalene	128	11.294	11.291	(1.152)	1570	0.26196	13.1	6191(R)
133 1,2,3-Trichlorobenzene	180	11.445	11.449	(1.168)	796	0.36080	18.0	6039(R)

QC Flag Legend

R - Spike/Surrogate failed recovery limits.

Data File: /var/chem/msv15.i/230612.s.b/c0922.d

Date : 12-JUN-2023 12:07

Client ID: 50X

Sample Info: SMB*50X

Purge Volume: 5.0

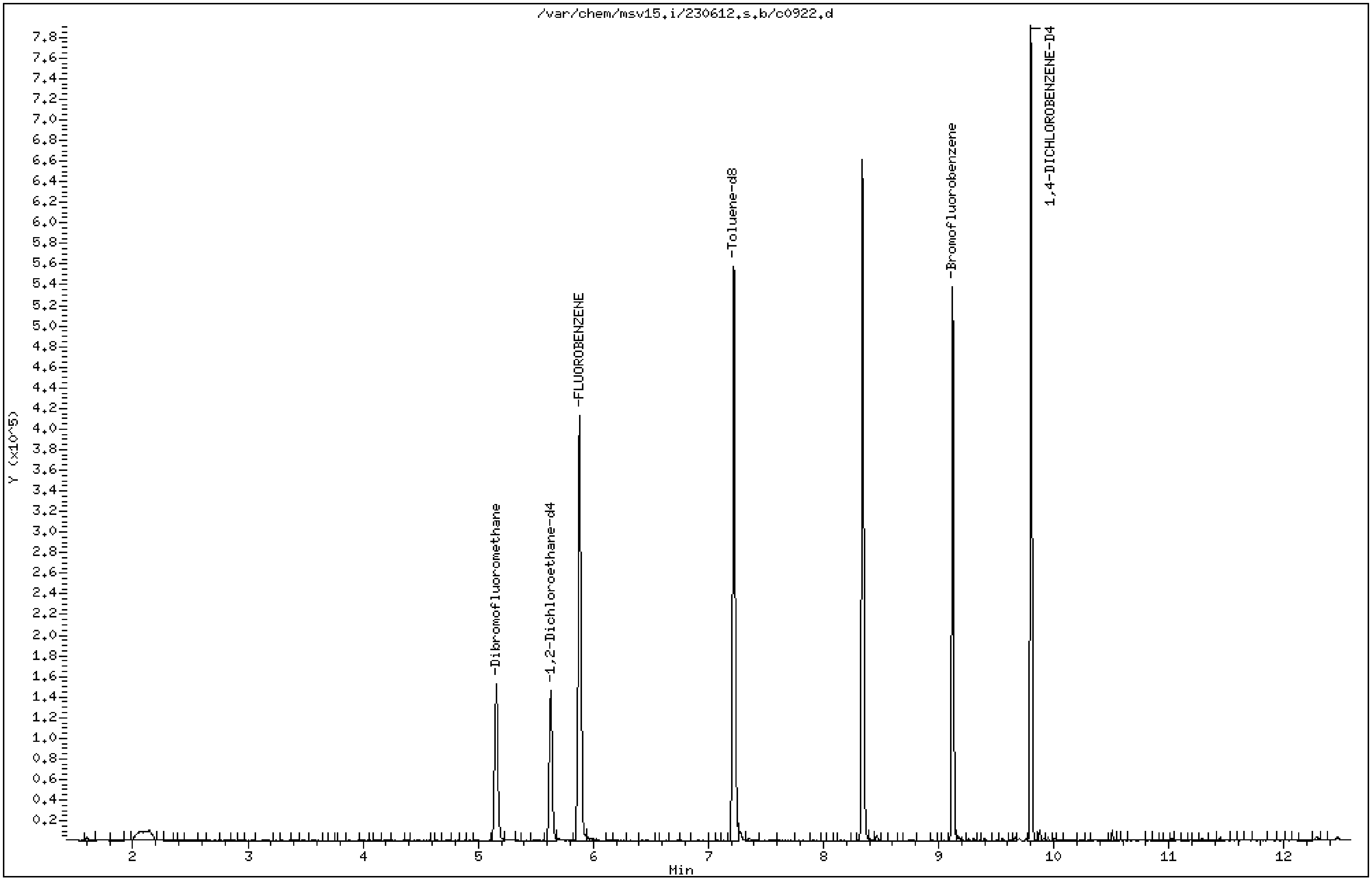
Column phase: RTX-VMS-30M

Page 1

Instrument: msv15.i

Operator: KN1

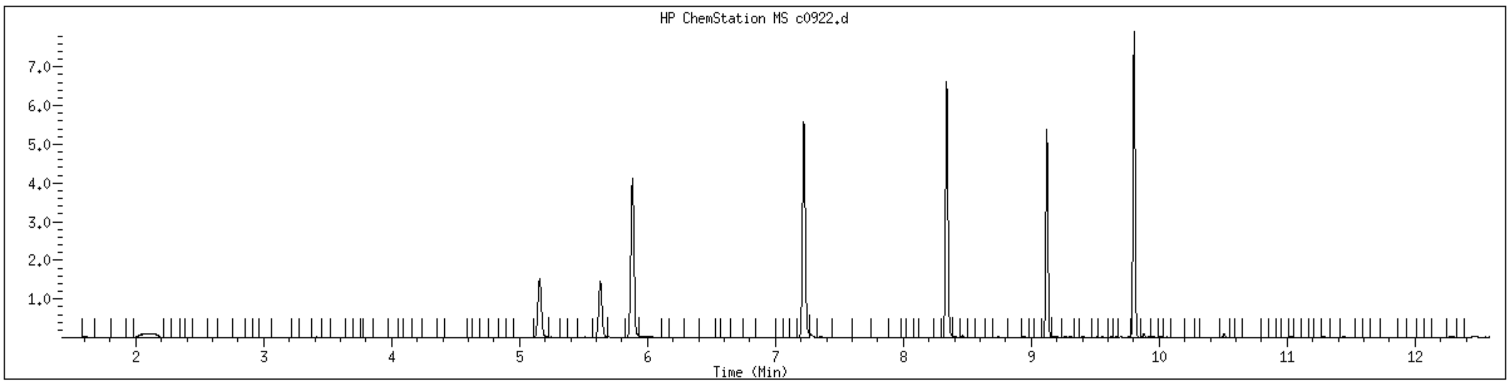
Column diameter: 0.25



Data file : /var/chem/msv15.i/230612.s.b/c0922.d Page: 1
Report Date: 06/13/2023 16:14

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 2491795 SampleType : LCS
Injection Date: 06/12/2023 12:07 Instrument : msv15.i
Operator : KN1
Sample Info : SMB*50X
Misc Info : MSV~529~*50*SMR
Method : /var/chem/msv15.i/230612.s.b/8260bw15DOD.m
Dilution : 50.0
Matrix : SOIL
Integrator : HP RTE Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060502	Client Sample ID:	LCS2491796
Collect Date:	NA Time: NA	GCAL Sample ID:	2491796
Matrix:	Solid % Moisture: NA	Instrument ID:	MSV15
Sample Amt:	5 g	Lab File ID:	230612/c0920
Injection Vol.:	1 (µL)	GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	50 Analyst: KN1	Analytical Batch:	767415
Analysis Date:	06/12/23 Time: 1129	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/kg

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
630-20-6	1,1,1,2-Tetrachloroethane	2120		12.5	25.0	250
71-55-6	1,1,1-Trichloroethane	2370		12.5	25.0	250
79-34-5	1,1,2,2-Tetrachloroethane	1930		12.5	25.0	250
79-00-5	1,1,2-Trichloroethane	2150		12.5	25.0	250
75-34-3	1,1-Dichloroethane	2540		12.5	25.0	250
75-35-4	1,1-Dichloroethene	2540		12.5	25.0	250
563-58-6	1,1-Dichloropropene	2470		12.5	25.0	250
87-61-6	1,2,3-Trichlorobenzene	2010		25.0	50.0	250
96-18-4	1,2,3-Trichloropropane	1830		25.0	50.0	250
120-82-1	1,2,4-Trichlorobenzene	2040		25.0	50.0	250
95-63-6	1,2,4-Trimethylbenzene	1990		12.5	25.0	250
96-12-8	1,2-Dibromo-3-chloropropane	1700		25.0	100	250
106-93-4	1,2-Dibromoethane	2180		25.0	100	250
95-50-1	1,2-Dichlorobenzene	1940		12.5	25.0	250
107-06-2	1,2-Dichloroethane	2290		12.5	25.0	250
540-59-0	1,2-Dichloroethene (total)	5010		25.0	50.0	500
78-87-5	1,2-Dichloropropane	2540		12.5	25.0	250
108-67-8	1,3,5-Trimethylbenzene	1990		12.5	25.0	250
541-73-1	1,3-Dichlorobenzene	1960		12.5	25.0	250
142-28-9	1,3-Dichloropropane	2200		12.5	25.0	250
106-46-7	1,4-Dichlorobenzene	1940		12.5	25.0	250
594-20-7	2,2-Dichloropropane	2320		12.5	25.0	250
78-93-3	2-Butanone	13200		25.0	100	250
95-49-8	2-Chlorotoluene	1920		12.5	25.0	250
591-78-6	2-Hexanone	10200		25.0	100	250
106-43-4	4-Chlorotoluene	1970		12.5	25.0	250
99-87-6	4-Isopropyltoluene	1980		12.5	25.0	250
108-10-1	4-Methyl-2-pentanone	9830		12.5	25.0	250
67-64-1	Acetone	12600		25.0	100	1250
71-43-2	Benzene	2480		12.5	25.0	250
108-86-1	Bromobenzene	1880		12.5	25.0	250
74-97-5	Bromochloromethane	2470		25.0	50.0	250
75-27-4	Bromodichloromethane	2400		12.5	25.0	250
75-25-2	Bromoform	1960		25.0	50.0	250
74-83-9	Bromomethane	1740		25.0	100	250
75-15-0	Carbon disulfide	2480		12.5	25.0	250
56-23-5	Carbon tetrachloride	2350		12.5	25.0	250

FORM I VOA

1A

VOLATILE ORGANICS ANALYSIS DATA SHEET

Report No:	223060502			Client Sample ID:	LCS2491796
Collect Date:	NA	Time:	NA	GCAL Sample ID:	2491796
Matrix:	Solid	% Moisture:	NA	Instrument ID:	MSV15
Sample Amt:	5	g		Lab File ID:	230612/c0920
Injection Vol.:	1	(µL)		GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	50	Analyst:	KN1	Analytical Batch:	767415
Analysis Date:	06/12/23	Time:	1129	Analytical Method:	EPA 8260B

CONCENTRATION UNITS: ug/kg

CAS	ANALYTE	RESULT	Q	DL	LOD	LOQ
108-90-7	Chlorobenzene	2160		12.5	25.0	250
75-00-3	Chloroethane	725		12.5	25.0	250
67-66-3	Chloroform	2440		12.5	25.0	250
74-87-3	Chloromethane	2430		25.0	100	250
156-59-2	cis-1,2-Dichloroethene	2450		12.5	25.0	250
10061-01-5	cis-1,3-Dichloropropene	2430		12.5	25.0	250
124-48-1	Dibromochloromethane	2280		12.5	25.0	250
74-95-3	Dibromomethane	2400		12.5	25.0	250
75-71-8	Dichlorodifluoromethane	2430		12.5	25.0	250
100-41-4	Ethylbenzene	2210		12.5	25.0	250
87-68-3	Hexachlorobutadiene	2110		25.0	50.0	250
98-82-8	Isopropylbenzene (Cumene)	2220		12.5	25.0	250
136777-61-2	m,p-Xylene	4440		12.5	50.0	500
75-09-2	Methylene chloride	2500		50.0	100	500
91-20-3	Naphthalene	1850		25.0	50.0	250
104-51-8	n-Butylbenzene	1970		12.5	25.0	250
103-65-1	n-Propylbenzene	1980		12.5	25.0	250
95-47-6	o-Xylene	2190		12.5	25.0	250
135-98-8	sec-Butylbenzene	2000		12.5	25.0	250
100-42-5	Styrene	2260		12.5	25.0	250
1634-04-4	tert-Butyl methyl ether (MTBE)	2420		12.5	25.0	250
98-06-6	tert-Butylbenzene	1890		25.0	50.0	250
127-18-4	Tetrachloroethene	2240		25.0	50.0	250
108-88-3	Toluene	2230		12.5	25.0	250
156-60-5	trans-1,2-Dichloroethene	2550		12.5	25.0	250
10061-02-6	trans-1,3-Dichloropropene	2390		12.5	25.0	250
79-01-6	Trichloroethene	2480		12.5	25.0	250
75-69-4	Trichlorofluoromethane	2520		12.5	25.0	250
75-01-4	Vinyl chloride	2350		12.5	25.0	250

FORM I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230612.s.b/c0920.d
Lab Smp Id: 2491796 Client Smp ID: 50X
Inj Date : 12-JUN-2023 11:29
Operator : KN1 Inst ID: msv15.i
Smp Info : LCS*50X
Misc Info : MSV~529~50*SMR
Comment :
Method : /var/chem/msv15.i/230612.s.b/8260bw15DOD.m
Meth Date : 13-Jun-2023 15:50 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 QC Sample: LCS
Dil Factor: 50.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/(Ws*(100-M)/100))/5000 * CpndVariable

Name	Value	Description
DF	50.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Ws	5.00000	Weight of sample extracted (g)
M	0.00000	% Moisture (not decanted)
Va	100.00000	Volume of aliquot extract added (uL)
m	0.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (UG/KG)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	85018	48.5753	2430	9683
2 Chloromethane ++	50	1.655	1.649	(0.281)	76596	48.5934	2430	9741
3 Vinyl Chloride +	62	1.716	1.719	(0.292)	104438	47.0532	2350	9840
5 1-3 Butadiene	39	1.735	1.735	(0.295)	83613	50.5852	2530	9573
6 Bromomethane	94	2.015	2.015	(0.343)	64946	34.8925	1740	9729
7 Chloroethane	64	2.105	2.137	(0.358)	23672	14.5093	725	8444(R)
8 Trichlorofluoromethane	101	2.250	2.272	(0.383)	162865	50.3365	2520	9740
11 1,1-Dichloroethene +	96	2.777	2.784	(0.472)	93583	50.8371	2540	8417
12 Carbon Disulfide	76	2.806	2.816	(0.477)	279224	49.5818	2480	8133
13 1,1,2Trichlotrifluoroethane	101	2.828	2.832	(0.481)	103411	51.7664	2590	9505
15 Methyl Iodide	142	2.928	2.935	(0.498)	99154	45.1926	2260	9590
16 Acrolein	56	3.150	3.150	(0.536)	71016	253.341	12700	4479

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (UG/KG)	SIMILARITY
	17 Allyl chloride	39	3.301	3.304	(0.561)	84085	51.9682	2600	9764
	18 Methylene Chloride	49	3.413	3.417	(0.581)	101888	50.0163	2500	4585
	19 Acetone	43	3.475	3.465	(0.591)	210384	251.622	12600	4846(H)
	20 trans-1,2-Dichloroethene	61	3.587	3.587	(0.610)	130209	51.0216	2550	9000
	21 Methyl Acetate	43	3.610	3.610	(0.614)	267035	256.544	12800	9322(R)
	22 Hexane	57	3.680	3.684	(0.626)	199638	54.2965	2710	9297
	23 MTBE	73	3.719	3.713	(0.632)	241150	48.4500	2420	9770
	24 tert-Butyl Alcohol	59	3.831	3.819	(0.652)	46453	211.574	10600	8181
	25 Acetonitrile	41	3.947	3.941	(0.671)	40192	271.512	13600	9518
	26 Isopropyl Ether	45	4.111	4.108	(0.699)	235821	53.1027	2660	9634
	27 Chloroprene	53	4.192	4.195	(0.713)	124745	49.8689	2490	8880
	28 1,1-Dichloroethane ++	63	4.214	4.214	(0.717)	173063	50.7845	2540	9579
	29 Acrylonitrile	53	4.262	4.259	(0.725)	146841	252.698	12600	9836
	31 Ethyl Tert-butyl Ether	59	4.468	4.468	(0.760)	243516	48.7229	2440	8046
	30 Vinyl Acetate	43	4.471	4.472	(0.760)	137788	50.1380	2510	5718(R)
	32 cis-1,2-Dichloroethene	61	4.732	4.735	(0.805)	120268	49.0907	2450	9838
M	33 Total 1,2-Dichloroethene	61				250477	100.112	5010	0
	34 2,2-Dichloropropane	77	4.835	4.838	(0.822)	140435	46.3640	2320	9592
	35 Bromochloromethane	128	4.921	4.919	(0.837)	47436	49.3183	2470	8834
	36 Cyclohexane	56	4.925	4.928	(0.838)	164478	52.9293	2650	9314
	37 Chloroform +	83	4.995	4.996	(0.850)	169986	48.7619	2440	9810
	39 Carbon Tetrachloride	117	5.118	5.121	(0.870)	132461	47.0041	2350	8252
$	42 Dibromofluoromethane	111	5.156	5.156	(0.877)	97354	52.2875	2610	8467
	43 1,1,1-Trichloroethane	97	5.179	5.179	(0.881)	153163	47.4449	2370	9565
	45 1,1-Dichloropropene	75	5.291	5.291	(0.900)	141851	49.4240	2470	9267
	44 2-Butanone	43	5.275	5.275	(0.897)	229302	264.224	13200	6484
	46 2,2,4 Trimethylpentane	57	5.397	5.398	(0.918)	352652	40.6857	2030	9428
	48 Benzene	78	5.513	5.513	(0.938)	385123	49.6437	2480	9595
	56 tert-amyl Methyl Ether	73	5.629	5.629	(0.957)	242700	47.6541	2380	7459
$	50 1,2-Dichloroethane-d4	65	5.629	5.629	(0.957)	95893	49.4636	2470	8597
	52 1,2-Dichloroethane	62	5.690	5.690	(0.968)	111964	45.8796	2290	9810
	51 Isobutyl Alcohol	43	5.741	5.726	(0.976)	10247	201.841	10100	2994
*	55 FLUOROBENZENE	96	5.880	5.880	(1.000)	377022	50.0000		9857
	58 Methyl Cyclohexane	83	6.024	6.028	(1.025)	186825	51.2513	2560	8605
	57 Trichloroethene	130	6.031	6.031	(1.026)	118208	49.6130	2480	8890
	62 Dibromomethane	93	6.394	6.394	(1.087)	59933	48.0105	2400	9599
	64 1,2-Dichloropropane +	63	6.481	6.481	(1.102)	97464	50.8326	2540	7993
	65 Bromodichloromethane	83	6.539	6.539	(1.112)	124783	48.0467	2400	9759
	67 1,4- Dioxane	88	6.728	6.719	(1.144)	15089	1061.34	53100	9414
	68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.181)	119798	49.0430	2450	9582
	72 2-Chloroethyl vinyl ether	63	7.024	7.021	(1.195)	115540	212.193	10600	9763
	73 cis-1,3-Dichloropropene	75	7.069	7.070	(1.202)	149734	48.6961	2430	9812
$	74 Toluene-d8	98	7.224	7.221	(0.866)	355632	46.5681	2330	9918
	75 Toluene +	91	7.262	7.262	(0.871)	424311	44.6264	2230	9659
	77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	346207	196.570	9830	8480
	78 Tetrachloroethene	164	7.558	7.558	(0.906)	96377	44.8343	2240	8270
	79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.288)	139387	47.8259	2390	8667

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (UG/KG)	SIMILARITY
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	83165	43.0639	2150	9558
82 Dibromochloromethane	129	7.825	7.825	(0.938)	99947	45.5005	2280	9696
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	112139	44.1597	2210	9666
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	141774	44.0916	2200	9599
M 83 1-3 Dichloropropene total	100				289121	96.5220	4830	0
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	90244	43.6707	2180	9774
89 2-Hexanone	43	8.143	8.143	(0.976)	295652	204.091	10200	9771
90 1-Chlorohexane	91	8.333	8.333	(0.999)	152371	44.8288	2240	7336
* 91 CHLOROBENZENE-d5	82	8.339	8.340	(1.000)	162950	50.0000		9277
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	272442	43.2950	2160	9264
93 Ethylbenzene +	106	8.368	8.369	(1.003)	151138	44.1725	2210	9085
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	89804	42.4723	2120	9272
95 p,m-Xylene	106	8.465	8.465	(1.015)	367515	88.8442	4440	9929
97 o-Xylene	106	8.744	8.745	(1.049)	169057	43.8879	2190	9918
101 Styrene	104	8.777	8.777	(1.052)	224135	45.1892	2260	9892
102 Bromoform ++	173	8.799	8.799	(1.055)	69095	39.1402	1960	8733
104 Isopropylbenzene	105	8.944	8.944	(1.072)	447962	44.4285	2220	9887
M 105 TOTAL XYLENE	106				536572	132.732	6640	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	127675	51.4785	2570	9652
107 cis-1,4-dichloro-2-butene	53	9.159	9.159	(0.934)	33487	39.1881	1960	9567
109 n-Propylbenzene	91	9.201	9.201	(0.939)	521441	39.6830	1980	8719
108 Bromobenzene	77	9.191	9.195	(0.938)	164671	37.6562	1880	8740(R)
110 1,1,2,2-Tetrachloroethane++	83	9.243	9.243	(0.943)	99337	38.6960	1930	9840
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	321396	38.3714	1920	8392(R)
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	352213	39.8515	1990	8259
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	127339	36.5543	1830	7558
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	31190	36.8519	1840	9428(M2)
116 Cyclohexanone	55	9.365	9.365	(0.955)	77370	182.305	9120	9736
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	296262	39.4154	1970	9855
118 tert-butylbenzene	91	9.516	9.516	(0.971)	187711	37.8281	1890	9866(R)
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	345551	39.8348	1990	9372
120 sec-Butylbenzene	105	9.622	9.623	(0.982)	457955	40.0563	2000	9896
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	400533	39.5740	1980	8930
123 1,3-Dichlorobenzene	146	9.761	9.764	(0.996)	199979	39.1044	1960	9793(R)
125 1,4-Dichlorobenzene	146	9.812	9.815	(1.001)	203916	38.7686	1940	8129(RH)
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.806	(1.000)	168254	50.0000		9387
126 n-Butylbenzene	91	9.953	9.954	(1.015)	531287	39.4785	1970	8982
127 1,2-Dichlorobenzene	146	10.066	10.066	(1.027)	179950	38.8969	1940	9798(R)
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.558	(1.077)	21866	34.0048	1700	9494
130 Hexachlorobutadiene	225	10.998	10.999	(1.122)	59823	42.2999	2110	9395
131 1,2,4-Trichlorobenzene	180	11.031	11.034	(1.125)	115347	40.7885	2040	9582
132 Naphthalene	128	11.291	11.291	(1.152)	245299	37.0169	1850	9862
133 1,2,3-Trichlorobenzene	180	11.445	11.449	(1.168)	97972	40.1634	2010	9598

QC Flag Legend

R - Spike/Surrogate failed recovery limits.
M2- Compound response manually integrated because
    Target system integrated incorrectly.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230612.s.b/c0920.d

Page 1

Date : 12-JUN-2023 11:29

Client ID: 50X

Instrument: msv15.i

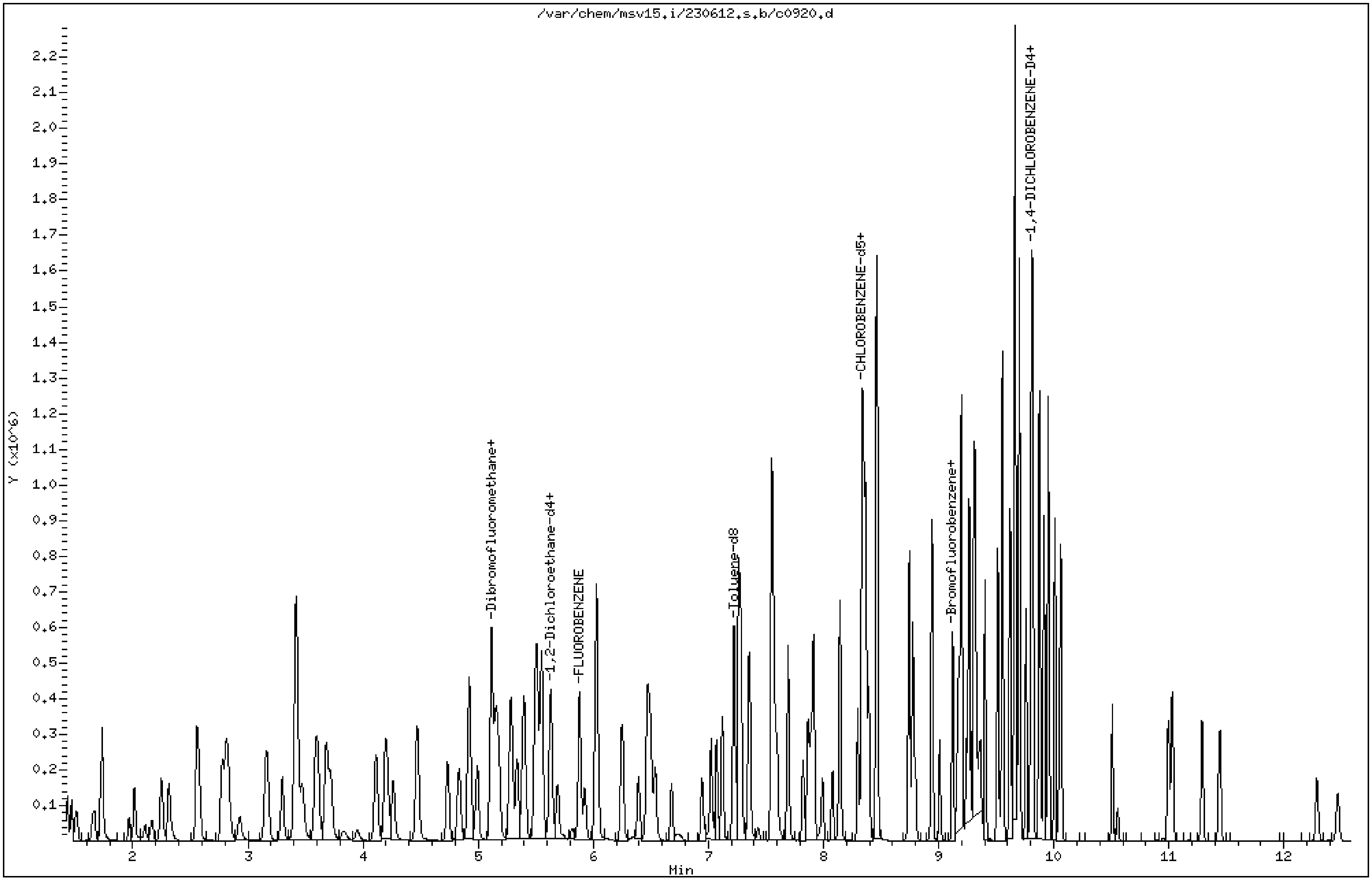
Sample Info: LCS*50X

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

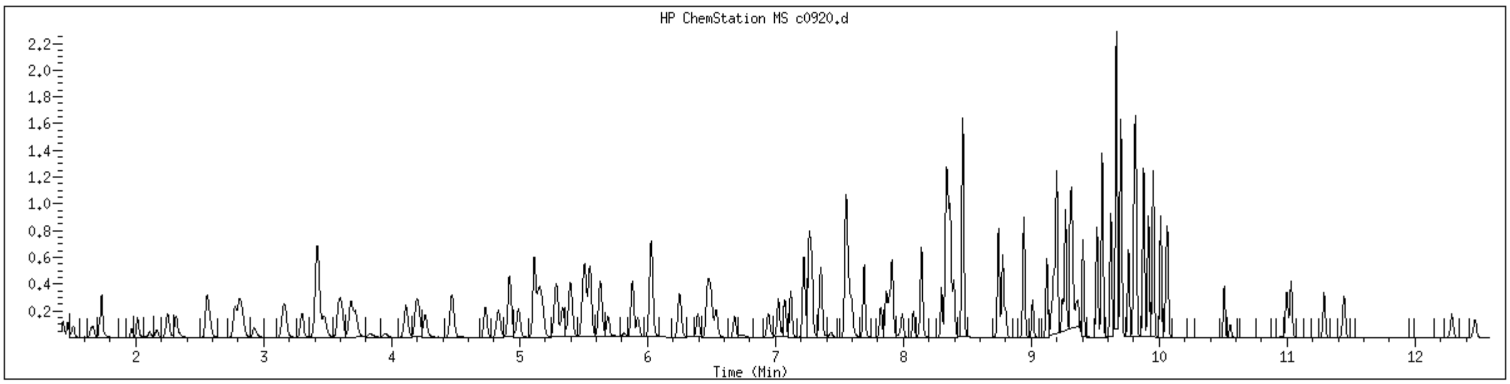
Column diameter: 0.25



Report Date: 06/13/2023 15:58

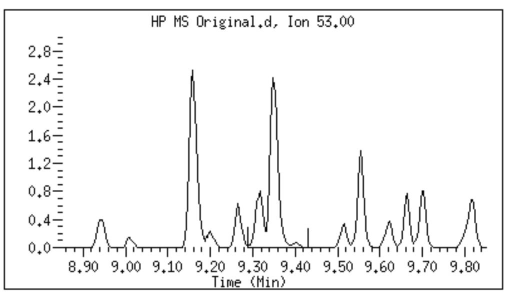
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 2491796 SampleType : LCS
Injection Date: 06/12/2023 11:29 Instrument : msv15.i
Operator : KN1
Sample Info : LCS*50X
Misc Info : MSV~529~50*SMR
Method : /var/chem/msv15.i/230612.s.b/8260bw15DOD.m
Dilution : 50.0
Matrix : SOIL
Integrator : HP RTE Compound Sublist: 8260DOD



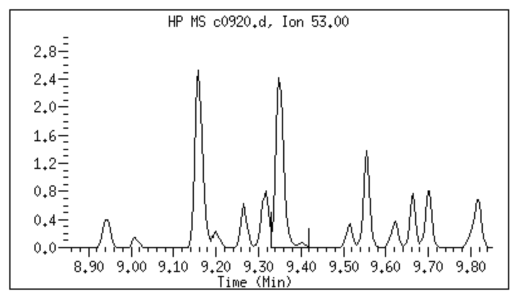
Original Final

115 trans-1,4-Dichloro-2-Butene CAS#: 110-57-6 Reason: M2



Electronic Signature Applied

User: kmn
Date: 06/13/2023 15:57



M2 - Target system integrated incorrectly

EPA 8260B DOD

Form 2B

Surrogates

Soil

2B

SOIL VOLATILE SYSTEM MONITORING COMPOUND RECOVERY

Report No: 223060502 Analytical Method: EPA 8260B

	Client Sample ID	GCALSample ID	SMC1	#	SMC2	#	SMC3	#	SMC4	#	TOT OUT
1 .	OT07-BO-40-4-6-8	22306050201	106		99		109		94		0
2 .	OT07-BO-40-4-6-8-MS	22306050202	100		101		106		92		0
3 .	OT07-BO-40-4-6-8-MSD	22306050203	100		101		106		92		0
4 .	OT07-BO-840-4-6-8	22306050204	105		101		106		94		0
5 .	MB2491795	2491795	102		99		104		93		0
6 .	LCS2491796	2491796	99		103		104		93		0

		QC LIMITS
SMC 1	1,2-Dichloroethane-d4	71 - 136
SMC 2	4-Bromofluorobenzene	79 - 119
SMC 3	Dibromofluoromethane	78 - 119
SMC 4	Toluene-d8	85 - 116

Column to be used to flag recovery values

* Values outside of QC limits

FORM II VOA-2

EPA 8260B DOD

Form 3B

Spikes

Soil

3B
SOIL VOLATILE MS/MSD RECOVERY

Report No: 223060502
Parent Sample ID: OT07-BO-40-4-6-8
Analytical Method: EPA 8260B
Analytical Batch: 767415

GCAL QC ID: 22306050202

ANALYTE	UNITS	SPIKE ADDED	SAMPLE RESULT	MS RESULT	MS % REC	#	QC LIMITS
1,1,1,2-Tetrachloroethane	ug/kg	14300	0	13300	93		78 - 125
1,1,1-Trichloroethane	ug/kg	14300	0	16100	113		73 - 130
1,1,2,2-Tetrachloroethane	ug/kg	14300	0	15700	110		70 - 124
1,1,2-Trichloroethane	ug/kg	14300	0	15000	104		78 - 121
1,1-Dichloroethane	ug/kg	14300	0	16900	118		76 - 125
1,1-Dichloroethene	ug/kg	14300	0	17000	119		70 - 131
1,1-Dichloropropene	ug/kg	14300	0	16200	114		76 - 125
1,2,3-Trichlorobenzene	ug/kg	14300	0	14400	101		66 - 130
1,2,3-Trichloropropane	ug/kg	14300	0	12200	85		73 - 125
1,2,4-Trichlorobenzene	ug/kg	14300	0	14300	100		67 - 129
1,2,4-Trimethylbenzene	ug/kg	14300	1960	33500	221	*	75 - 123
1,2-Dibromo-3-chloropropane	ug/kg	14300	0	12200	85		61 - 132
1,2-Dibromoethane	ug/kg	14300	0	13900	97		78 - 122
1,2-Dichlorobenzene	ug/kg	14300	0	12700	89		78 - 121
1,2-Dichloroethane	ug/kg	14300	0	15300	107		73 - 128
1,2-Dichloroethene (total)	ug/kg	28600	0	33100	116		78 - 122
1,2-Dichloropropane	ug/kg	14300	0	16900	118		76 - 123
1,3,5-Trimethylbenzene	ug/kg	14300	306	17200	118		73 - 124
1,3-Dichlorobenzene	ug/kg	14300	0	12500	87		77 - 121
1,3-Dichloropropane	ug/kg	14300	0	13900	97		77 - 121
1,4-Dichlorobenzene	ug/kg	14300	0	12500	87		75 - 120
2,2-Dichloropropane	ug/kg	14300	0	14400	101		67 - 133
2-Butanone	ug/kg	71600	0	91600	128		51 - 148
2-Chlorotoluene	ug/kg	14300	0	12700	89		75 - 122
2-Hexanone	ug/kg	71600	0	69800	97		53 - 145
4-Chlorotoluene	ug/kg	14300	0	12200	85		72 - 124
4-Isopropyltoluene	ug/kg	14300	147	18200	126		73 - 127
4-Methyl-2-pentanone	ug/kg	71600	0	73500	103		65 - 135
Acetone	ug/kg	71600	0	83700	117		36 - 164
Benzene	ug/kg	14300	0	16800	117		77 - 121
Bromobenzene	ug/kg	14300	0	12500	88		78 - 121
Bromochloromethane	ug/kg	14300	0	16500	115		78 - 125
Bromodichloromethane	ug/kg	14300	0	15200	106		75 - 127
Bromoform	ug/kg	14300	0	12800	90		67 - 132
Bromomethane	ug/kg	14300	0	14800	104		53 - 143
Carbon disulfide	ug/kg	14300	0	16200	114		63 - 132
Carbon tetrachloride	ug/kg	14300	0	16200	114		70 - 135
Chlorobenzene	ug/kg	14300	0	13700	95		79 - 120

RPD : 0 out of 66 outside limits

Spike Recovery: 2 out of 132 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III SV-1

3B

SOIL VOLATILE MS/MSD RECOVERY

Report No: 223060502

Parent Sample ID: OT07-BO-40-4-6-8

Analytical Method: EPA 8260B

Analytical Batch: 767415

Chloroethane	ug/kg	14300	0	17900	125		59 - 139
Chloroform	ug/kg	14300	0	15900	111		78 - 123
Chloromethane	ug/kg	14300	0	17200	120		50 - 136
Dibromochloromethane	ug/kg	14300	0	14700	103		74 - 126
Dibromomethane	ug/kg	14300	0	16400	114		78 - 125
Dichlorodifluoromethane	ug/kg	14300	0	16600	116		29 - 149
Ethylbenzene	ug/kg	14300	241	16500	114		76 - 122
Hexachlorobutadiene	ug/kg	14300	0	16000	112		61 - 135
Isopropylbenzene (Cumene)	ug/kg	14300	128	15000	104		68 - 134
Methylene chloride	ug/kg	14300	0	16800	117		70 - 128
Naphthalene	ug/kg	14300	316	18100	124		62 - 129
Styrene	ug/kg	14300	0	15100	105		76 - 124
Tetrachloroethene	ug/kg	14300	0	14100	98		73 - 128
Toluene	ug/kg	14300	0	14100	98		77 - 121
Trichloroethene	ug/kg	14300	0	17200	120		77 - 123
Trichlorofluoromethane	ug/kg	14300	0	16500	115		62 - 140
Vinyl chloride	ug/kg	14300	0	17300	121		77 - 124
cis-1,2-Dichloroethene	ug/kg	14300	0	16400	114		77 - 123
cis-1,3-Dichloropropene	ug/kg	14300	0	16200	114		74 - 126
m,p-Xylene	ug/kg	28600	310	31100	107		77 - 123
n-Butylbenzene	ug/kg	14300	146	14400	100		70 - 128
n-Propylbenzene	ug/kg	14300	315	15200	104		73 - 125
o-Xylene	ug/kg	14300	23.5	14300	100		77 - 123
sec-Butylbenzene	ug/kg	14300	239	14800	102		73 - 126
tert-Butyl methyl ether (MTBE)	ug/kg	14300	0	16400	114		73 - 125
tert-Butylbenzene	ug/kg	14300	0	12200	85		73 - 125
trans-1,2-Dichloroethene	ug/kg	14300	0	16800	117		74 - 125
trans-1,3-Dichloropropene	ug/kg	14300	0	15700	110		71 - 130

RPD : 0 out of 66 outside limits

Spike Recovery: 2 out of 132 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III SV-1

3B
SOIL VOLATILE MS/MSD RECOVERY

Report No: 223060502

Analytical Method: EPA 8260B

Parent Sample ID: OT07-BO-40-4-6-8

Analytical Batch: 767415

GCAL QC ID: 22306050203

ANALYTE	UNITS	SPIKE ADDED	MSD RESULT	MSD % REC	#	% RPD	#	QC LIMITS REC	QC LIMITS RPD
1,1,1,2-Tetrachloroethane	ug/kg	14100	12700	90		5		78 - 125	0 - 20
1,1,1-Trichloroethane	ug/kg	14100	15100	107		7		73 - 130	0 - 20
1,1,2,2-Tetrachloroethane	ug/kg	14100	14400	103		9		70 - 124	0 - 20
1,1,2-Trichloroethane	ug/kg	14100	14200	101		5		78 - 121	0 - 20
1,1-Dichloroethane	ug/kg	14100	16000	114		5		76 - 125	0 - 20
1,1-Dichloroethene	ug/kg	14100	16400	116		4		70 - 131	0 - 20
1,1-Dichloropropene	ug/kg	14100	15200	108		7		76 - 125	0 - 20
1,2,3-Trichlorobenzene	ug/kg	14100	13200	94		9		66 - 130	0 - 20
1,2,3-Trichloropropane	ug/kg	14100	11500	82		6		73 - 125	0 - 20
1,2,4-Trichlorobenzene	ug/kg	14100	12800	91		11		67 - 129	0 - 20
1,2,4-Trimethylbenzene	ug/kg	14100	30400	202	*	10		75 - 123	0 - 20
1,2-Dibromo-3-chloropropane	ug/kg	14100	11700	83		5		61 - 132	0 - 20
1,2-Dibromoethane	ug/kg	14100	13300	94		5		78 - 122	0 - 20
1,2-Dichlorobenzene	ug/kg	14100	11500	82		10		78 - 121	0 - 20
1,2-Dichloroethane	ug/kg	14100	14400	103		6		73 - 128	0 - 20
1,2-Dichloroethene (total)	ug/kg	28100	31200	111		6		78 - 122	0 - 20
1,2-Dichloropropane	ug/kg	14100	16000	114		5		76 - 123	0 - 20
1,3,5-Trimethylbenzene	ug/kg	14100	15500	108		10		73 - 124	0 - 20
1,3-Dichlorobenzene	ug/kg	14100	11300	80		10		77 - 121	0 - 20
1,3-Dichloropropane	ug/kg	14100	13200	94		6		77 - 121	0 - 20
1,4-Dichlorobenzene	ug/kg	14100	11100	79		12		75 - 120	0 - 20
2,2-Dichloropropane	ug/kg	14100	13800	98		5		67 - 133	0 - 20
2-Butanone	ug/kg	70300	88600	126		3		51 - 148	0 - 20
2-Chlorotoluene	ug/kg	14100	11500	82		10		75 - 122	0 - 20
2-Hexanone	ug/kg	70300	68100	97		2		53 - 145	0 - 20
4-Chlorotoluene	ug/kg	14100	11100	79		9		72 - 124	0 - 20
4-Isopropyltoluene	ug/kg	14100	16500	116		10		73 - 127	0 - 20
4-Methyl-2-pentanone	ug/kg	70300	71600	102		3		65 - 135	0 - 20
Acetone	ug/kg	70300	82800	118		1		36 - 164	0 - 20
Benzene	ug/kg	14100	15900	113		6		77 - 121	0 - 20
Bromobenzene	ug/kg	14100	11400	81		10		78 - 121	0 - 20
Bromochloromethane	ug/kg	14100	15600	111		6		78 - 125	0 - 20
Bromodichloromethane	ug/kg	14100	15500	110		2		75 - 127	0 - 20
Bromoform	ug/kg	14100	12400	88		4		67 - 132	0 - 20
Bromomethane	ug/kg	14100	14100	100		5		53 - 143	0 - 20
Carbon disulfide	ug/kg	14100	15500	110		5		63 - 132	0 - 20
Carbon tetrachloride	ug/kg	14100	15300	109		6		70 - 135	0 - 20
Chlorobenzene	ug/kg	14100	12800	91		6		79 - 120	0 - 20

RPD : 0 out of 66 outside limits

Spike Recovery: 2 out of 132 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III SV-1

3B

SOIL VOLATILE MS/MSD RECOVERY

Report No: 223060502

Parent Sample ID: OT07-BO-40-4-6-8

Analytical Method: EPA 8260B

Analytical Batch: 767415

Chloroethane	ug/kg	14100	17500	125		2		59 - 139	0 - 20
Chloroform	ug/kg	14100	15100	107		5		78 - 123	0 - 20
Chloromethane	ug/kg	14100	16200	116		5		50 - 136	0 - 20
Dibromochloromethane	ug/kg	14100	13900	99		5		74 - 126	0 - 20
Dibromomethane	ug/kg	14100	15500	110		6		78 - 125	0 - 20
Dichlorodifluoromethane	ug/kg	14100	16100	115		3		29 - 149	0 - 20
Ethylbenzene	ug/kg	14100	14800	104		11		76 - 122	0 - 20
Hexachlorobutadiene	ug/kg	14100	14600	104		9		61 - 135	0 - 20
Isopropylbenzene (Cumene)	ug/kg	14100	14200	100		5		68 - 134	0 - 20
Methylene chloride	ug/kg	14100	15700	112		6		70 - 128	0 - 20
Naphthalene	ug/kg	14100	17200	120		5		62 - 129	0 - 20
Styrene	ug/kg	14100	13500	96		11		76 - 124	0 - 20
Tetrachloroethene	ug/kg	14100	13300	94		6		73 - 128	0 - 20
Toluene	ug/kg	14100	13000	93		8		77 - 121	0 - 20
Trichloroethene	ug/kg	14100	16400	116		5		77 - 123	0 - 20
Trichlorofluoromethane	ug/kg	14100	16100	115		2		62 - 140	0 - 20
Vinyl chloride	ug/kg	14100	16600	118		4		77 - 124	0 - 20
cis-1,2-Dichloroethene	ug/kg	14100	15300	109		7		77 - 123	0 - 20
cis-1,3-Dichloropropene	ug/kg	14100	15200	108		7		74 - 126	0 - 20
m,p-Xylene	ug/kg	28100	28900	102		7		77 - 123	0 - 20
n-Butylbenzene	ug/kg	14100	13300	93		8		70 - 128	0 - 20
n-Propylbenzene	ug/kg	14100	13800	96		10		73 - 125	0 - 20
o-Xylene	ug/kg	14100	13400	95		7		77 - 123	0 - 20
sec-Butylbenzene	ug/kg	14100	13500	95		9		73 - 126	0 - 20
tert-Butyl methyl ether (MTBE)	ug/kg	14100	15700	112		4		73 - 125	0 - 20
tert-Butylbenzene	ug/kg	14100	11000	78		10		73 - 125	0 - 20
trans-1,2-Dichloroethene	ug/kg	14100	15700	112		6		74 - 125	0 - 20
trans-1,3-Dichloropropene	ug/kg	14100	14800	105		6		71 - 130	0 - 20

RPD : 0 out of 66 outside limits

Spike Recovery: 2 out of 132 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III SV-1

3B

SOIL VOLATILE LCS/LCSD RECOVERY

Report No: 223060502

Analytical Method: EPA 8260B

Analytical Batch: 767415

GCAL QC ID: 2491796

ANALYTE	***UNITS***	***SPIKE ADDED***	***SAMPLE RESULT***	***LCS RESULT***	***LCS % REC***	***#***	***QC LIMITS***
1,1,1,2-Tetrachloroethane	ug/kg	2500	0	2120	85		78 - 125
1,1,1-Trichloroethane	ug/kg	2500	0	2370	95		73 - 130
1,1,2,2-Tetrachloroethane	ug/kg	2500	0	1930	77		70 - 124
1,1,2-Trichloroethane	ug/kg	2500	0	2150	86		78 - 121
1,1-Dichloroethane	ug/kg	2500	0	2540	102		76 - 125
1,1-Dichloroethene	ug/kg	2500	0	2540	102		70 - 131
1,1-Dichloropropene	ug/kg	2500	0	2470	99		76 - 125
1,2,3-Trichlorobenzene	ug/kg	2500	0	2010	80		66 - 130
1,2,3-Trichloropropane	ug/kg	2500	0	1830	73		73 - 125
1,2,4-Trichlorobenzene	ug/kg	2500	0	2040	82		67 - 129
1,2,4-Trimethylbenzene	ug/kg	2500	0	1990	80		75 - 123
1,2-Dibromo-3-chloropropane	ug/kg	2500	0	1700	68		61 - 132
1,2-Dibromoethane	ug/kg	2500	0	2180	87		78 - 122
1,2-Dichlorobenzene	ug/kg	2500	0	1940	78		78 - 121
1,2-Dichloroethane	ug/kg	2500	0	2290	92		73 - 128
1,2-Dichloroethene (total)	ug/kg	5000	0	5010	100		78 - 122
1,2-Dichloropropane	ug/kg	2500	0	2540	102		76 - 123
1,3,5-Trimethylbenzene	ug/kg	2500	0	1990	80		73 - 124
1,3-Dichlorobenzene	ug/kg	2500	0	1960	78		77 - 121
1,3-Dichloropropane	ug/kg	2500	0	2200	88		77 - 121
1,4-Dichlorobenzene	ug/kg	2500	0	1940	78		75 - 120
2,2-Dichloropropane	ug/kg	2500	0	2320	93		67 - 133
2-Butanone	ug/kg	12500	0	13200	106		51 - 148
2-Chlorotoluene	ug/kg	2500	0	1920	77		75 - 122
2-Hexanone	ug/kg	12500	0	10200	82		53 - 145
4-Chlorotoluene	ug/kg	2500	0	1970	79		72 - 124
4-Isopropyltoluene	ug/kg	2500	0	1980	79		73 - 127
4-Methyl-2-pentanone	ug/kg	12500	0	9830	79		65 - 135
Acetone	ug/kg	12500	0	12600	101		36 - 164
Benzene	ug/kg	2500	0	2480	99		77 - 121
Bromobenzene	ug/kg	2500	0	1880	75	*	78 - 121
Bromochloromethane	ug/kg	2500	0	2470	99		78 - 125
Bromodichloromethane	ug/kg	2500	0	2400	96		75 - 127
Bromoform	ug/kg	2500	0	1960	78		67 - 132
Bromomethane	ug/kg	2500	0	1740	70		53 - 143
Carbon disulfide	ug/kg	2500	0	2480	99		63 - 132
Carbon tetrachloride	ug/kg	2500	0	2350	94		70 - 135
Chlorobenzene	ug/kg	2500	0	2160	86		79 - 120

RPD : 0 out of 0 outside limits

Spike Recovery: 2 out of 66 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III SV-1

3B
SOIL VOLATILE LCS/LCSD RECOVERY

Report No: 223060502

Analytical Method: EPA 8260B

Analytical Batch: 767415

Chloroethane	ug/kg	2500	0	725	29	*	59 - 139
Chloroform	ug/kg	2500	0	2440	98		78 - 123
Chloromethane	ug/kg	2500	0	2430	97		50 - 136
Dibromochloromethane	ug/kg	2500	0	2280	91		74 - 126
Dibromomethane	ug/kg	2500	0	2400	96		78 - 125
Dichlorodifluoromethane	ug/kg	2500	0	2430	97		29 - 149
Ethylbenzene	ug/kg	2500	0	2210	88		76 - 122
Hexachlorobutadiene	ug/kg	2500	0	2110	84		61 - 135
Isopropylbenzene (Cumene)	ug/kg	2500	0	2220	89		68 - 134
Methylene chloride	ug/kg	2500	0	2500	100		70 - 128
Naphthalene	ug/kg	2500	0	1850	74		62 - 129
Styrene	ug/kg	2500	0	2260	90		76 - 124
Tetrachloroethene	ug/kg	2500	0	2240	90		73 - 128
Toluene	ug/kg	2500	0	2230	89		77 - 121
Trichloroethene	ug/kg	2500	0	2480	99		77 - 123
Trichlorofluoromethane	ug/kg	2500	0	2520	101		62 - 140
Vinyl chloride	ug/kg	2500	0	2350	94		77 - 124
cis-1,2-Dichloroethene	ug/kg	2500	0	2450	98		77 - 123
cis-1,3-Dichloropropene	ug/kg	2500	0	2430	97		74 - 126
m,p-Xylene	ug/kg	5000	0	4440	89		77 - 123
n-Butylbenzene	ug/kg	2500	0	1970	79		70 - 128
n-Propylbenzene	ug/kg	2500	0	1980	79		73 - 125
o-Xylene	ug/kg	2500	0	2190	88		77 - 123
sec-Butylbenzene	ug/kg	2500	0	2000	80		73 - 126
tert-Butyl methyl ether (MTBE)	ug/kg	2500	0	2420	97		73 - 125
tert-Butylbenzene	ug/kg	2500	0	1890	76		73 - 125
trans-1,2-Dichloroethene	ug/kg	2500	0	2550	102		74 - 125
trans-1,3-Dichloropropene	ug/kg	2500	0	2390	96		71 - 130

RPD : 0 out of 0 outside limits

Spike Recovery: 2 out of 66 outside limits

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

FORM III SV-1

EPA 8260B DOD

Form 4A

Method Blanks

4A

VOLATILE METHOD BLANK SUMMARY

Report No:	223060502	Method Blank ID:	2491795
Matrix:	Solid	Instrument ID:	MSV15
Sample Amt:	5 g	Lab File ID:	230612/c0922
Injection Vol.:	1 (µL)	GC Column:	RTX-VMS-30M ID .25 (mm)
Dilution Factor:	50 Analyst: KN1	Analytical Batch:	767415
Analysis Date:	06/12/23 Time: 1207	Analytical Method:	EPA 8260B

THIS METHOD BLANK APPLIES TO THE FOLLOWING SAMPLES, MS AND MSD

	CLIENT SAMPLE ID	*GCAL SAMPLE ID*	*LAB FILE ID*	*DATE ANALYZED*	*TIME ANALYZED*
1 .	LCS2491796	2491796	230612/c0920	06/12/23	1129
2 .	OT07-BO-40-4-6-8	22306050201	230612/c0942	06/12/23	1823
3 .	OT07-BO-840-4-6-8	22306050204	230612/c0943	06/12/23	1842
4 .	OT07-BO-40-4-6-8-MS	22306050202	230612/c0946	06/12/23	1938
5 .	OT07-BO-40-4-6-8-MSD	22306050203	230612/c0947	06/12/23	1957

FORM IV VOA

EPA 8260B DOD

Form 5A

Tunes

5A
VOLATILE ORGANICS INSTRUMENT PERFORMANCE CHECK
BROMOFLUOROBENZENE (BFB)

Report No:	223060502	Tune ID:	1000
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230517/c0227t
Analyst:	CJR	Analytical Batch:	766139
Analysis Date:	05/17/23 Time: 1337	Analytical Method:	EPA 8260B

m / e	***ION ABUNDANCE CRITERIA***	***% Relative Abundance***		
50	15.0 - 40.0% of mass 95	17.86	( )	
75	30.0 - 66.0% of mass 95	50.49	( )	
95	Base Peak, 100% relative abundance	100	( )	
96	5.0 -9.0% of mass 95	6.84	( )	
173	Less than 2.0% of mass 174	.93	(1)	1
174	50.0 - 120.0% of mass 95	93.38	( )	
175	4.0 - 9.0% of mass 174	6.6	(7.07)	1
176	95.0 - 101.0% of mass 174	89.76	(96.13)	1
177	5.0 - 9.0% of mass 176	5.85	(6.52)	2

1- Value is % mass 174
2- Value is % mass 176

THIS CHECK APPLIES TO THE FOLLOWING SAMPLES, MS, MSD, BLANKS, AND STANDARDS:

	CLIENT SAMPLE ID	***GCAL SAMPLE ID***	***LAB FILE ID***	***DATE ANALYZED***	***TIME ANALYZED***
1 .	V15STD001	1203	230517/c0228	05/17/23	1428
2 .	V15STD005	1204	230517/c0229	05/17/23	1447
3 .	V15STD010	1205	230517/c0230	05/17/23	1505
4 .	V15STD020	1206	230517/c0231	05/17/23	1524
5 .	V15STD050	1207	230517/c0232	05/17/23	1543
6 .	V15STD100	1208	230517/c0233	05/17/23	1601
7 .	V15STD200	1209	230517/c0234	05/17/23	1620
8 .	V15ICV050	1600	230517/c0238	05/17/23	1735

FORM V VOA

Data File: /var/chem/msv15.i/230517.s.b/c0227t.d

Date : 17-MAY-2023 13:37

Client ID: V15BFB

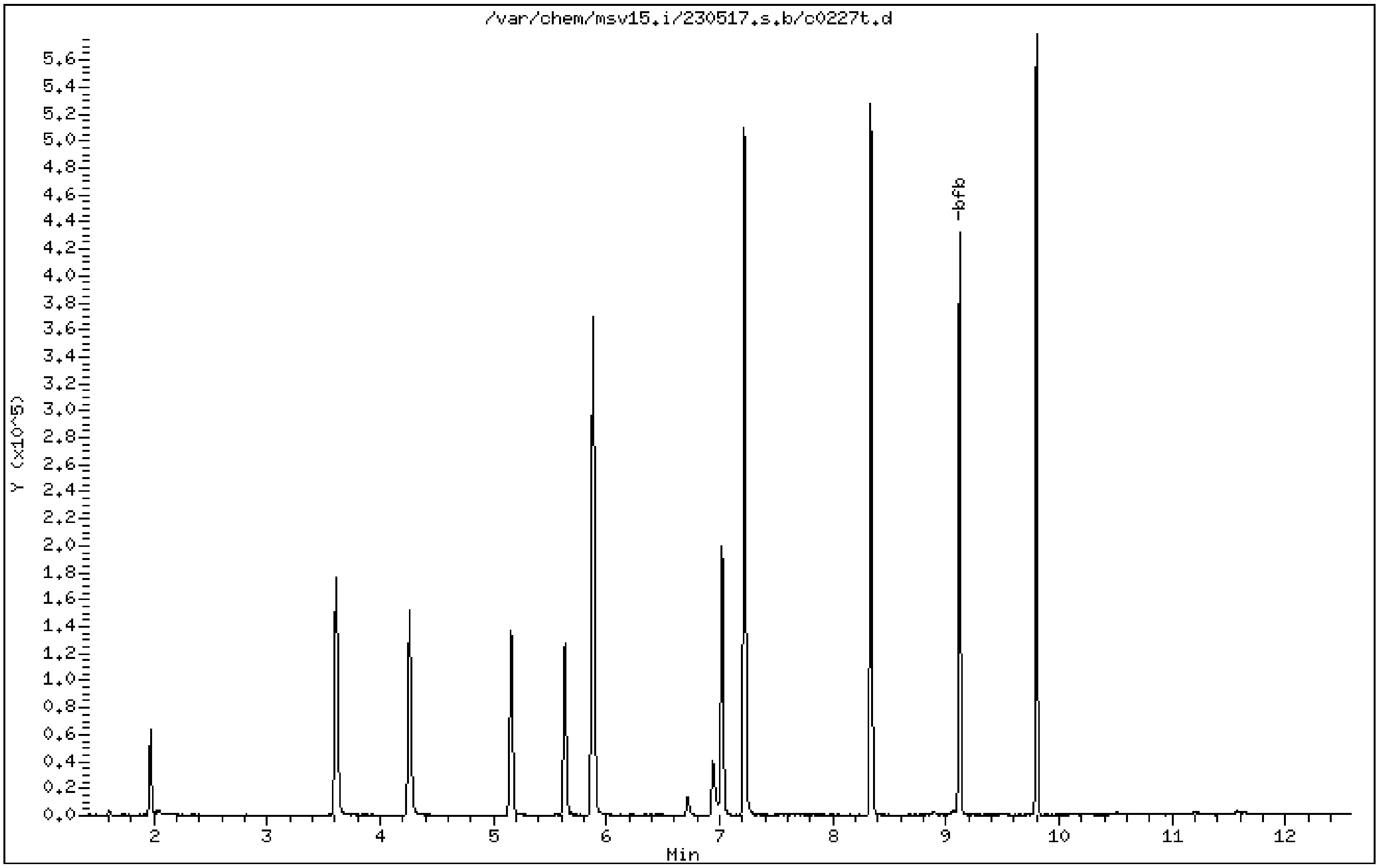
Instrument: msv15.i

Sample Info: HANDSPIKE*

Operator: CJR

Column phase: RTX-VMS-30M

Column diameter: 0.25



Data File: /var/chem/msv15.i/230517.s.b/c0227t.d

Date : 17-MAY-2023 13:37

Client ID: V15BFB

Instrument: msv15.i

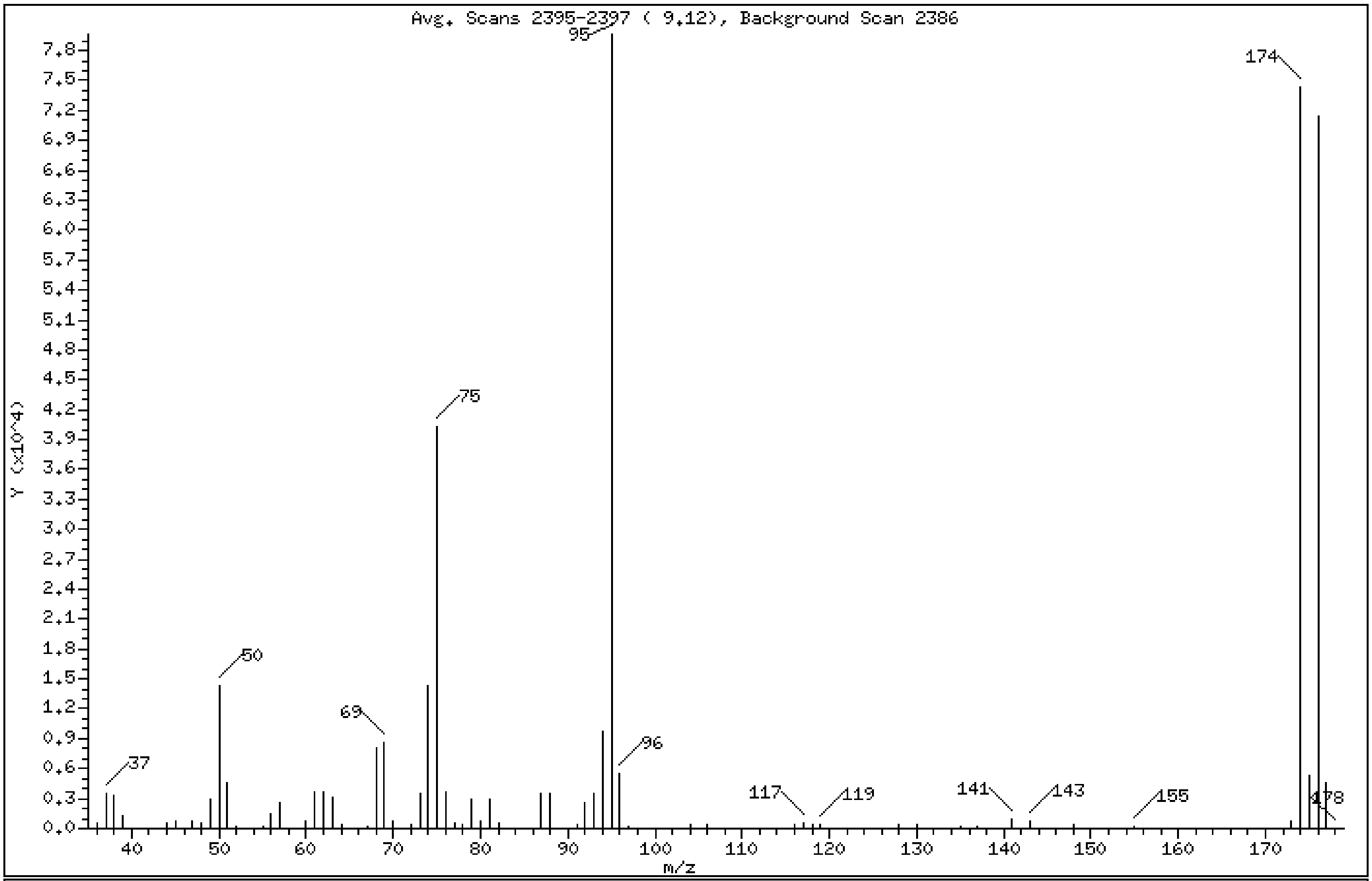
Sample Info: HANDSPIKE*

Operator: CJR

Column phase: RTX-VMS-30M

Column diameter: 0.25

1 bfb



m/e	ION ABUNDANCE CRITERIA	% RELATIVE ABUNDANCE
95	Base Peak, 100% relative abundance	100.00
50	15.00 - 40.00% of mass 95	17.86
75	30.00 - 60.00% of mass 95	50.49
96	5.00 - 9.00% of mass 95	6.84
173	Less than 2.00% of mass 174	0.94 (1.00)
174	50.00 - 120.00% of mass 95	93.38
175	4.00 - 9.00% of mass 174	6.60 (7.07)
176	95.00 - 101.00% of mass 174	89.76 (96.13)
177	5.00 - 9.00% of mass 176	5.86 (6.52)

Date : 17-MAY-2023 13:37

Client ID: V15BFB Instrument: msv15.i

Sample Info: HANDSPIKE*

Operator: CJR

Column phase: RTX-VMS-30M Column diameter: 0.25

Data File: c0227t.d
Spectrum: Avg. Scans 2395-2397 (9.12), Background Scan 2386
Location of Maximum: 95.00
Number of points: 71

m/z	Y	m/z	Y	m/z	Y	m/z	Y
36.00	583	62.00	3625	86.00	64	130.00	285
37.00	3521	63.00	3025	87.00	3493	131.00	53
38.00	3245	64.00	289	88.00	3484	135.00	111
39.00	1293	67.00	213	91.00	351	137.00	225
40.00	55	68.00	7996	92.00	2530	141.00	859
44.00	530	69.00	8577	93.00	3442	143.00	810
45.00	716	70.00	692	94.00	9736	146.00	76
47.00	769	72.00	403	95.00	79664	148.00	284
48.00	485	73.00	3476	96.00	5450	155.00	205
49.00	2961	74.00	14223	97.00	108	157.00	64
50.00	14227	75.00	40216	104.00	386	172.00	68
51.00	4506	76.00	3609	106.00	392	173.00	745
52.00	157	77.00	552	116.00	280	174.00	74392
55.00	251	78.00	386	117.00	554	175.00	5257
56.00	1394	79.00	2900	118.00	300	176.00	71512
57.00	2508	80.00	819	119.00	433	177.00	4665
60.00	808	81.00	2969	128.00	278	178.00	61
61.00	3669	82.00	602	129.00	58		

5A

VOLATILE ORGANICS INSTRUMENT PERFORMANCE CHECK

BROMOFLUOROBENZENE (BFB)

Report No:	223060502	Tune ID:	1000
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230612/c0915s
Analyst:	CJR	Analytical Batch:	767415
Analysis Date:	06/12/23 Time: 0921	Analytical Method:	EPA 8260B

m / e	***ION ABUNDANCE CRITERIA***	***% Relative Abundance***		
50	15.0 - 40.0% of mass 95	18.21	( )	
75	30.0 - 66.0% of mass 95	47.57	( )	
95	Base Peak, 100% relative abundance	100	( )	
96	5.0 -9.0% of mass 95	6.57	( )	
173	Less than 2.0% of mass 174	.73	(.83)	1
174	50.0 - 120.0% of mass 95	89.1	( )	
175	4.0 - 9.0% of mass 174	6.39	(7.18)	1
176	95.0 - 101.0% of mass 174	85.9	(96.41)	1
177	5.0 - 9.0% of mass 176	5.75	(6.7)	2

1- Value is % mass 174 2- Value is % mass 176

THIS CHECK APPLIES TO THE FOLLOWING SAMPLES, MS, MSD, BLANKS, AND STANDARDS:

	CLIENT SAMPLE ID	***GCAL SAMPLE ID***	***LAB FILE ID***	***DATE ANALYZED***	***TIME ANALYZED***
1 .	V15STD050	1400	230612/c0917s	06/12/23	1030
2 .	LCS2491796	2491796	230612/c0920	06/12/23	1129
3 .	MB2491795	2491795	230612/c0922	06/12/23	1207
4 .	OT07-BO-40-4-6-8	22306050201	230612/c0942	06/12/23	1823
5 .	OT07-BO-840-4-6-8	22306050204	230612/c0943	06/12/23	1842
6 .	OT07-BO-40-4-6-8-MS	22306050202	230612/c0946	06/12/23	1938
7 .	OT07-BO-40-4-6-8-MSD	22306050203	230612/c0947	06/12/23	1957
8 .	V15STD050	1440	230612/c0948s	06/12/23	2016

FORM V VOA

Data File: /var/chem/msv15.i/230612.s.b/c0915s.d

Date : 12-JUN-2023 09:21

Client ID: V15BFB

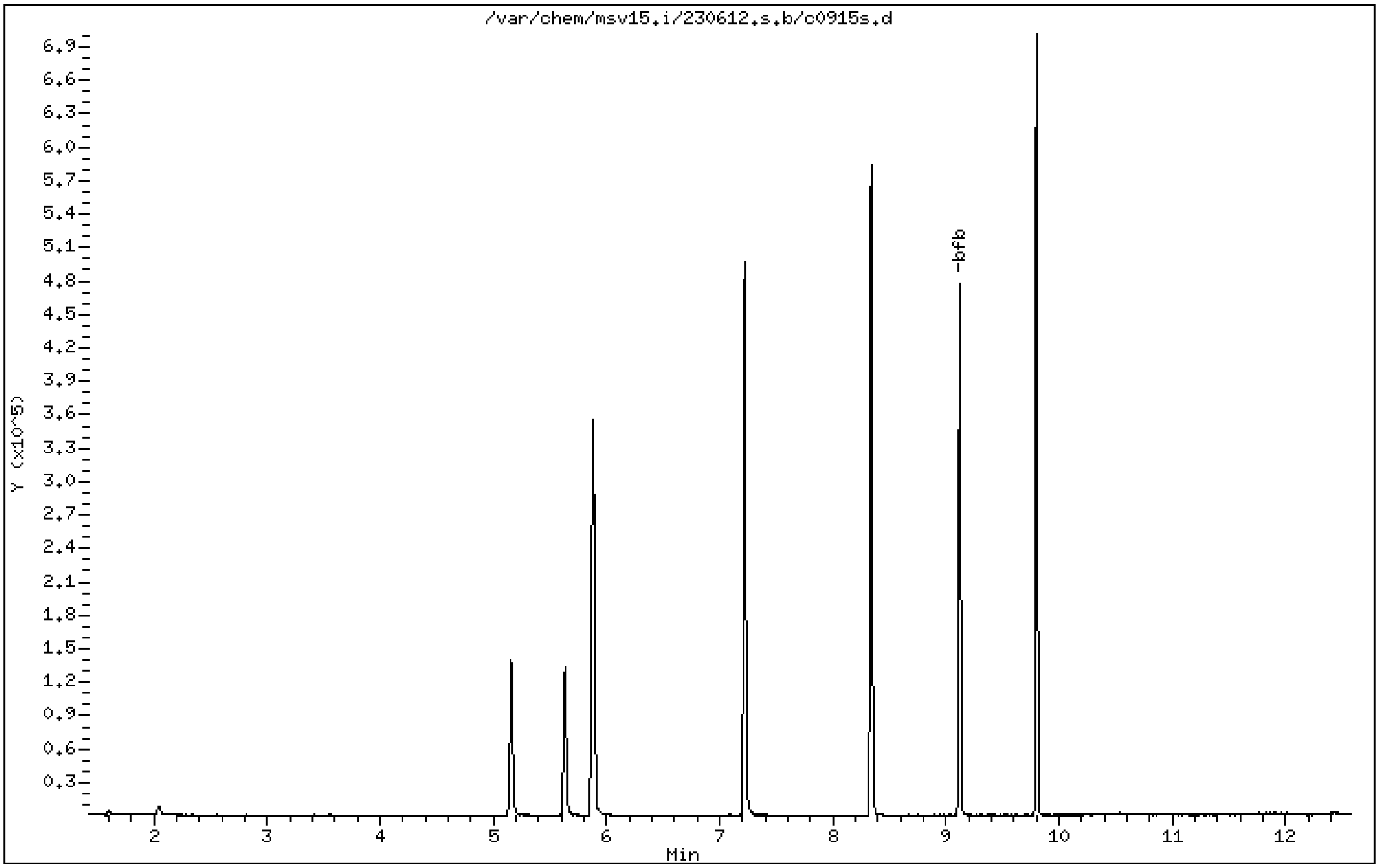
Instrument: msv15.i

Sample Info: BLANK*

Operator: CJR

Column phase: RTX-VMS-30M

Column diameter: 0.25



Data File: /var/chem/msv15.i/230612.s.b/c0915s.d

Date : 12-JUN-2023 09:21

Client ID: V15BFB

Instrument: msv15.i

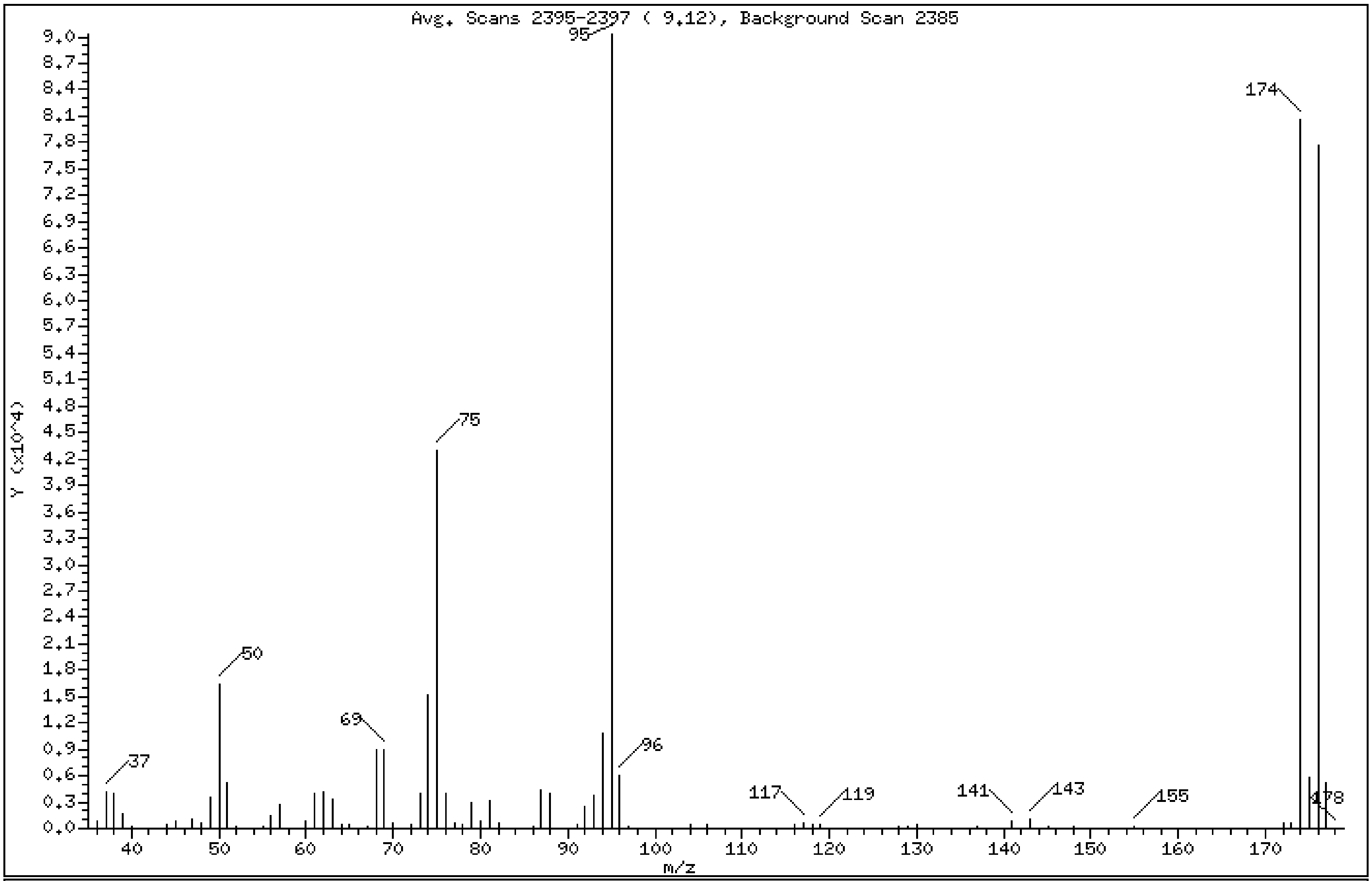
Sample Info: BLANK*

Operator: CJR

Column phase: RTX-VMS-30M

Column diameter: 0.25

1 bfb



m/e	ION ABUNDANCE CRITERIA	% RELATIVE ABUNDANCE
95	Base Peak, 100% relative abundance	100.00
50	15.00 - 40.00% of mass 95	18.21
75	30.00 - 60.00% of mass 95	47.57
96	5.00 - 9.00% of mass 95	6.57
173	Less than 2.00% of mass 174	0.74 (0.83)
174	50.00 - 120.00% of mass 95	89.10
175	4.00 - 9.00% of mass 174	6.40 (7.18)
176	95.00 - 101.00% of mass 174	85.90 (96.41)
177	5.00 - 9.00% of mass 176	5.76 (6.70)

Date : 12-JUN-2023 09:21

Client ID: V15BFB Instrument: msv15.i

Sample Info: BLANK*

Operator: CJR

Column phase: RTX-VMS-30M Column diameter: 0.25

Data File: c0915s.d

Spectrum: Avg. Scans 2395-2397 (9.12), Background Scan 2385

Location of Maximum: 95.00

Number of points: 76

m/z	Y	m/z	Y	m/z	Y	m/z	Y
36.00	794	62.00	4075	87.00	4463	137.00	125
37.00	4174	63.00	3298	88.00	3878	141.00	845
38.00	3874	64.00	400	91.00	340	143.00	1031
39.00	1640	65.00	356	92.00	2514	145.00	180
40.00	129	67.00	268	93.00	3796	146.00	53
43.00	72	68.00	8861	94.00	10773	148.00	252
44.00	456	69.00	8980	95.00	90320	153.00	51
45.00	781	70.00	706	96.00	5931	155.00	232
47.00	1112	72.00	489	97.00	114	157.00	51
48.00	520	73.00	3937	104.00	356	172.00	523
49.00	3511	74.00	15161	105.00	54	173.00	667
50.00	16440	75.00	42960	106.00	448	174.00	80472
51.00	5100	76.00	3943	116.00	330	175.00	5778
52.00	208	77.00	655	117.00	522	176.00	77584
55.00	301	78.00	454	118.00	325	177.00	5202
56.00	1419	79.00	2986	119.00	471	178.00	51
57.00	2601	80.00	898	128.00	280		
58.00	50	81.00	3089	129.00	201		
60.00	855	82.00	637	130.00	370		
61.00	3962	86.00	119	135.00	56		

EPA 8260B DOD

Form 6A

Calibrations

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

						LabQC ID - FileID - Conc	1203 ~ 230517/c0228 ~ 1
Report No:	223060502			Instrument ID:	MSV15	1204 ~ 230517/c0229 ~ 5	1205 ~ 230517/c0230 ~ 10
GC Column:	RTX-VMS-30M	ID	.25 (mm)	Analyst:	CJR	1206 ~ 230517/c0231 ~ 20	1207 ~ 230517/c0232 ~ 50
Calib. Date 1:	05/17/23	Time 1:	1428	Analytical Batch:	766139	1208 ~ 230517/c0233 ~ 100	1209 ~ 230517/c0234 ~ 200
Calib. Date 2:	05/17/23	Time 2:	1620	Analytical Method:	EPA 8260B		

ANALYTE	***1201***	***1202***	***1203***	***1204***	***1205***	***1206***	***1207***	***1208***	***1209***	***RF / m / A***	***b / B***	***C***	***FIT***	***TYPE***	
1,1,1,2-Tetrachloroethane			0.689	0.580	0.587	0.639	0.660	0.688	0.699	0.649			7.532	A	
1,1,1-Trichloroethane			0.370	0.406	0.400	0.446	0.449	0.463	0.462	0.428			8.438	A	
1,1,2,2-Tetrachloroethane			0.623	0.775	0.805	0.826	0.806	0.767	0.739	0.763			8.956	A	
1,1,2-Trichloroethane			0.498	0.574	0.586	0.609	0.621	0.627	0.634	0.593			7.930	A	
1,1-Dichloroethane			0.387	0.452	0.436	0.467	0.473	0.473	0.474	0.452			7.012	A	
1,1-Dichloroethene			0.228	0.231	0.227	0.249	0.255	0.260	0.259	0.244			6.049	A	
1,1-Dichloropropene			0.351	0.366	0.359	0.395	0.400	0.395	0.399	0.381			5.599	A	
1,2,3-Trichlorobenzene			0.563	0.674	0.710	0.755	0.767	0.785	0.820	0.725			11.87	A	
1,2,3-Trichloropropane			1.037	0.998	1.019	1.101	1.072	1.027	0.993	1.035			3.774	A	
1,2,4-Trichlorobenzene			0.701	0.800	0.794	0.864	0.905	0.887	0.932	0.840			9.520	A	
1,2,4-Trimethylbenzene			2.030	2.501	2.556	2.795	2.762	2.704	2.698	2.578			10.24	A	
1,2-Dibromo-3-chloropropan			159	1562	3400	8227	22062	51173	114746	0.196	0.017		0.998	W	
1,2-Dibromo-3-chloropropan			0.079	0.151	0.157	0.183	0.183	0.192	0.200						
1,2-Dibromoethane			0.492	0.620	0.633	0.656	0.672	0.684	0.681	0.634			10.57	A	
1,2-Dichlorobenzene			1.211	1.328	1.349	1.412	1.437	1.437	1.449	1.375			6.254	A	
1,2-Dichloroethane			0.291	0.318	0.324	0.336	0.333	0.335	0.330	0.324			4.858	A	
1,2-Dichloroethane-d4			0.268	0.233	0.242	0.266	0.261	0.266	0.264	0.257			5.296	A	
1,2-Dichloroethene (total)			0.298	0.317	0.322	0.344	0.346	0.346	0.349	0.332			5.867	A	
1,2-Dichloropropane			0.223	0.241	0.245	0.263	0.267	0.268	0.273	0.254			7.163	A	
1,3,5-Trimethylbenzene			2.180	2.566	2.599	2.836	2.777	2.749	2.677	2.626			8.337	A	
1,3-Butadiene			0.205	0.216	0.210	0.241	0.216	0.220	0.226	0.219			5.335	A	
1,3-Dichlorobenzene			1.405	1.475	1.487	1.562	1.571	1.563	1.576	1.520			4.297	A	
1,3-Dichloropropane			0.855	0.979	0.977	1.040	1.030	1.008	1.017	0.987			6.346	A	
1,3-Dichloropropylene			0.353	0.368	0.376	0.410	0.418	0.427	0.428	0.397			7.748	A	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines
to allow for evaluation against minimum RRF

RF = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

				LabQC ID - FileID - Conc	1203 ~ 230517/c0228 ~ 1
Report No:	223060502	Instrument ID:	MSV15	1204 ~ 230517/c0229 ~ 5	1205 ~ 230517/c0230 ~ 10
GC Column:	RTX-VMS-30M ID .25 (mm)	Analyst:	CJR	1206 ~ 230517/c0231 ~ 20	1207 ~ 230517/c0232 ~ 50
Calib. Date 1:	05/17/23 Time 1: 1428	Analytical Batch:	766139	1208 ~ 230517/c0233 ~ 100	1209 ~ 230517/c0234 ~ 200
Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Method:	EPA 8260B		

ANALYTE	1201	1202	1203	1204	1205	1206	1207	1208	1209	RF / m / A	b / B	C	FIT	TYPE	
1,4 Dioxane				0.002	0.002	0.002	0.002	0.002	0.002	0.002			8.418	A	
1,4-Dichlorobenzene			1.456	1.496	1.520	1.622	1.618	1.602	1.628	1.563			4.521	A	
1-Bromo-2-Chloroethane			0.283	0.310	0.317	0.335	0.338	0.345	0.339	0.324			6.761	A	
1-Chlorohexane			0.935	0.993	1.053	1.001	1.141	1.067	1.110	1.043			6.876	A	
1-Nitropropane			0.000	0.072	0.071	0.070	0.073	0.082	0.075	0.074			6.121	A	
2,2,4-Trimethylpentane			0.856	1.088	1.095	1.189	1.242	1.279	1.297	1.150			13.35	A	
2,2-Dichloropropane			0.354	0.395	0.391	0.419	0.417	0.421	0.415	0.402			6.083	A	
2,3-Dichloro-1-propene			0.354	0.390	0.386	0.420	0.442	0.448	0.459	0.414			9.328	A	
2-Butanone			0.082	0.111	0.111	0.125	0.123	0.127	0.126	0.115			13.88	A	
2-Chloroethylvinyl ether (RS			1244	8287	19577	35793	105023	237890	569730	0.074	0.089		0.991	W	
2-Chloroethylvinyl ether			0.042	0.054	0.063	0.056	0.063	0.070	0.078						
2-Chlorotoluene			2.240	2.509	2.503	2.651	2.591	2.484	2.446	2.489			5.210	A	
2-Hexanone (RSP)			2993	22444	46436	109907	287469	616069	1298993	0.450	0.052		0.999	W	
2-Hexanone			0.260	0.389	0.388	0.441	0.445	0.453	0.448						
2-Nitropropane (RSP)			147	1643	3719	8113	19389	44203	94852	0.163	0.012		0.999	W	
2-Nitropropane			0.064	0.143	0.156	0.163	0.150	0.162	0.164						
4-Bromofluorobenzene			0.843	0.716	0.700	0.744	0.761	0.769	0.794	0.761			6.324	A	
4-Chlorotoluene			1.953	2.229	2.308	2.411	2.317	2.246	2.172	2.234			6.512	A	
4-Isopropyltoluene			2.454	2.856	2.926	3.216	3.265	3.156	3.180	3.008			9.564	A	
4-Methyl-2-pentanone (RSP)			3673	26975	57708	132162	343899	740988	1596464	0.548	0.054		0.999	W	
4-Methyl-2-pentanone			0.319	0.468	0.483	0.530	0.532	0.545	0.551						
Acetone			0.091	0.112	0.102	0.115	0.117	0.125	0.114	0.111			9.992	A	
Acetonitrile			0.015	0.019	0.020	0.022	0.021	0.021	0.020	0.020			11.00	A	
Acrolein			0.031	0.035	0.036	0.039	0.039	0.040	0.040	0.037			9.334	A	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines to allow for evaluation against minimum RRF

$\overline{RF}$ = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

Report No:	223060502			Instrument ID:	MSV15	**LabQC ID - FileID - Conc**	1203 ~ 230517/c0228 ~ 1
						1204 ~ 230517/c0229 ~ 5	1205 ~ 230517/c0230 ~ 10
GC Column:	RTX-VMS-30M	ID	.25 (mm)	Analyst:	CJR	1206 ~ 230517/c0231 ~ 20	1207 ~ 230517/c0232 ~ 50
Calib. Date 1:	05/17/23	Time 1:	1428	Analytical Batch:	766139	1208 ~ 230517/c0233 ~ 100	1209 ~ 230517/c0234 ~ 200
Calib. Date 2:	05/17/23	Time 2:	1620	Analytical Method:	EPA 8260B		

ANALYTE	1201	1202	1203	1204	1205	1206	1207	1208	1209	RF / m / A	b / B	C	FIT	TYPE	
Acrylonitrile			0.068	0.072	0.073	0.081	0.079	0.084	0.081	0.077			7.748	A	
Allyl chloride (RSP)			889	6819	14165	30710	79708	157006	283118	0.215	0.000		0.990	W	
Allyl chloride			0.150	0.224	0.227	0.239	0.239	0.230	0.194						
Benzene			0.892	0.998	0.985	1.066	1.065	1.095	1.101	1.029			7.292	A	
Bromobenzene			1.234	1.364	1.281	1.377	1.336	1.267	1.237	1.300			4.559	A	
Bromochloromethane			0.111	0.130	0.132	0.137	0.132	0.124	0.125	0.128			6.591	A	
Bromodichloromethane			0.282	0.328	0.331	0.353	0.365	0.373	0.379	0.344			9.852	A	
Bromoform (RSP)			720	4668	9910	23618	64825	146620	332459	0.554	0.017		0.996	W	
Bromoform			0.313	0.405	0.414	0.474	0.501	0.539	0.574						
Bromomethane			0.197	0.222	0.227	0.242	0.266	0.291	0.283	0.247			14.01	A	
Carbon disulfide			0.724	0.726	0.698	0.767	0.762	0.772	0.778	0.747			4.073	A	
Carbon tetrachloride			0.319	0.340	0.345	0.381	0.395	0.407	0.429	0.374			10.66	A	
Chlorobenzene			1.703	1.851	1.869	1.967	1.989	2.072	2.065	1.931			6.835	A	
Chloroethane			0.189	0.204	0.215	0.225	0.222	0.225	0.234	0.216			7.027	A	
Chloroform			0.421	0.447	0.439	0.475	0.483	0.481	0.491	0.462			5.720	A	
Chloromethane			0.189	0.206	0.202	0.218	0.210	0.216	0.223	0.209			5.502	A	
Chloroprene			0.285	0.316	0.313	0.355	0.346	0.352	0.355	0.332			8.187	A	
Cyclohexane			0.358	0.391	0.387	0.440	0.432	0.436	0.442	0.412			8.047	A	
Cyclohexanone (RSP)			762	6175	13437	31285	80908	170231	352267	0.127	0.026		0.998	W	
Cyclohexanone			0.076	0.119	0.124	0.139	0.134	0.127	0.123						
Dibromochloromethane			0.507	0.620	0.629	0.699	0.728	0.755	0.780	0.674			14.11	A	
Dibromofluoromethane			0.218	0.229	0.233	0.253	0.260	0.264	0.272	0.247			8.180	A	
Dibromomethane			0.131	0.161	0.162	0.172	0.176	0.179	0.178	0.166			10.37	A	
Dichlorodifluoromethane			0.189	0.226	0.228	0.238	0.236	0.252	0.257	0.232			9.641	A	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines
to allow for evaluation against minimum RRF

$\overline{RF}$ = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

				LabQC ID - FileID - Conc	1203 ~ 230517/c0228 ~ 1
Report No:	223060502	Instrument ID:	MSV15	1204 ~ 230517/c0229 ~ 5	1205 ~ 230517/c0230 ~ 10
GC Column:	RTX-VMS-30M ID .25 (mm)	Analyst:	CJR	1206 ~ 230517/c0231 ~ 20	1207 ~ 230517/c0232 ~ 50
Calib. Date 1:	05/17/23 Time 1: 1428	Analytical Batch:	766139	1208 ~ 230517/c0233 ~ 100	1209 ~ 230517/c0234 ~ 200
Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Method:	EPA 8260B		

ANALYTE	***1201***	***1202***	***1203***	***1204***	***1205***	***1206***	***1207***	***1208***	***1209***	***RF / m / A***	***b / B***	***C***	***FIT***	***TYPE***	
Dicyclopentadiene			2.596	3.146	3.125	3.418	3.368	3.122	3.097	3.125			8.526	A	
Diethyl ether			0.132	0.161	0.163	0.174	0.177	0.183	0.187	0.168			10.97	A	
Ethyl acetate			0.152	0.193	0.199	0.212	0.213	0.223	0.223	0.202			12.28	A	
Ethyl methacrylate				0.367	0.376	0.425	0.423	0.425	0.429	0.408			6.879	A	
Ethyl tert-butyl ether (ETBE)			0.539	0.624	0.643	0.694	0.701	0.718	0.721	0.663			9.954	A	
Ethylbenzene			0.854	1.018	1.012	1.085	1.105	1.135	1.139	1.050			9.550	A	
Hexachlorobutadiene			0.325	0.409	0.395	0.436	0.441	0.446	0.489	0.420			12.25	A	
Isobutyl alcohol				0.006	0.007	0.007	0.007	0.007	0.007	0.007			6.979	A	
Isopropylbenzene (Cumene)			2.611	2.933	2.968	3.262	3.296	3.326	3.260	3.094			8.606	A	
Methacrylonitrile			0.087	0.113	0.116	0.122	0.122	0.129	0.129	0.117			12.30	A	
Methyl Acetate			0.125	0.128	0.135	0.144	0.145	0.149	0.141	0.138			6.660	A	
Methyl iodide (RSP)				5565	14998	37592	102973	212199	417991	0.302	0.033		0.997	W	
Methyl iodide				0.183	0.240	0.293	0.309	0.311	0.286						
Methyl methacrylate				0.145	0.160	0.179	0.183	0.196	0.193	0.176			11.25	A	
Methylcyclohexane			0.383	0.454	0.452	0.510	0.522	0.525	0.538	0.483			11.61	A	
Methylene chloride			0.258	0.265	0.259	0.278	0.275	0.276	0.279	0.270			3.396	A	
Naphthalene			1.450	1.776	1.925	2.079	2.122	2.176	2.257	1.969			14.21	A	
Propionitrile			0.019	0.028	0.029	0.031	0.030	0.031	0.031	0.028			14.77	A	
Styrene			1.151	1.366	1.439	1.574	1.668	1.718	1.737	1.522			14.12	A	
Tetrachloroethene			0.560	0.637	0.610	0.666	0.678	0.711	0.754	0.660			9.766	A	
Tetrahydrofuran				0.070	0.072	0.077	0.076	0.079	0.078	0.075			4.789	A	
Toluene			2.466	2.815	2.866	3.025	3.049	3.058	3.142	2.917			7.867	A	
Toluene-d8			2.108	2.223	2.238	2.436	2.449	2.464	2.486	2.343			6.411	A	
Trichloroethene			0.250	0.297	0.297	0.326	0.333	0.345	0.365	0.316			12.11	A	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines
to allow for evaluation against minimum RRF

$\overline{RF}$ = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

				LabQC ID - FileID - Conc	1203 ~ 230517/c0228 ~ 1
Report No:	223060502	Instrument ID:	MSV15	1204 ~ 230517/c0229 ~ 5	1205 ~ 230517/c0230 ~ 10
GC Column:	RTX-VMS-30M ID .25 (mm)	Analyst:	CJR	1206 ~ 230517/c0231 ~ 20	1207 ~ 230517/c0232 ~ 50
Calib. Date 1:	05/17/23 Time 1: 1428	Analytical Batch:	766139	1208 ~ 230517/c0233 ~ 100	1209 ~ 230517/c0234 ~ 200
Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Method:	EPA 8260B		

ANALYTE	1201	1202	1203	1204	1205	1206	1207	1208	1209	RF / m / A	b / B	C	FIT	TYPE	
Trichlorofluoromethane			0.391	0.406	0.407	0.447	0.446	0.456	0.450	0.429			6.255	A	
Trichlorotrifluoroethane			0.222	0.258	0.248	0.274	0.278	0.284	0.291	0.265			9.027	A	
Vinyl acetate			0.335	0.347	0.359	0.380	0.377	0.384	0.370	0.364			5.027	A	
Vinyl chloride			0.262	0.280	0.294	0.305	0.300	0.306	0.315	0.294			6.153	A	
Xylene (total)			1.018	1.173	1.202	1.284	1.309	1.351	1.345	1.240			9.615	A	
cis-1,2-Dichloroethene			0.301	0.312	0.318	0.336	0.336	0.335	0.336	0.325			4.524	A	
cis-1,3-Dichloropropene			0.376	0.381	0.383	0.422	0.428	0.430	0.435	0.408			6.502	A	
diisopropyl Ether (DIPE)			0.487	0.557	0.570	0.621	0.625	0.631	0.632	0.589			9.224	A	
m,p-Xylene			1.043	1.191	1.230	1.311	1.337	1.392	1.380	1.269			9.804	A	
n-Butyl alcohol (RSP)				406	1071	2315	6770	13877	29929	0.004	0.184		0.999	W	
n-Butyl alcohol				0.003	0.003	0.004	0.004	0.004	0.004						
n-Butylbenzene			3.111	3.881	3.920	4.292	4.350	4.230	4.209	3.999			10.77	A	
n-Hexane			0.395	0.468	0.475	0.519	0.519	0.518	0.518	0.488			9.511	A	
n-Propylbenzene			3.472	3.870	3.924	4.233	4.160	3.910	3.764	3.905			6.458	A	
o-Xylene			0.967	1.137	1.147	1.230	1.253	1.267	1.273	1.182			9.281	A	
sec-Butylbenzene			2.870	3.354	3.338	3.702	3.648	3.462	3.408	3.397			8.001	A	
t-Butanol (TBA)				0.025	0.025	0.030	0.030	0.032	0.033	0.029			11.96	A	
t-amyl methyl ether (TAME)			0.516	0.619	0.651	0.711	0.722	0.753	0.756	0.675			12.86	A	
tert-Butyl methyl ether (MTB			0.519	0.613	0.647	0.690	0.702	0.725	0.726	0.660			11.33	A	
tert-Butylbenzene			1.253	1.504	1.473	1.588	1.567	1.479	1.459	1.475			7.408	A	
tert-amyl alcohol (TAA)			0.081	0.091	0.094	0.104	0.108	0.111	0.112	0.100			11.52	A	
trans-1,2-Dichloroethene			0.296	0.323	0.326	0.352	0.355	0.356	0.361	0.338			7.187	A	
trans-1,3-Dichloropropene			0.331	0.356	0.368	0.398	0.408	0.424	0.421	0.387			9.208	A	
trans-1,4-Dichloro-2-butene (			295	2548	5589	12313	31798	67695	141123	0.253	0.005		0.998	W	

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines
to allow for evaluation against minimum RRF

RF = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

6A

VOLATILE ORGANICS INITIAL CALIBRATION DATA

				LabQC ID - FileID - Conc	1203 ~ 230517/c0228 ~ 1
Report No:	223060502	Instrument ID:	MSV15	1204 ~ 230517/c0229 ~ 5	1205 ~ 230517/c0230 ~ 10
GC Column:	RTX-VMS-30M ID .25 (mm)	Analyst:	CJR	1206 ~ 230517/c0231 ~ 20	1207 ~ 230517/c0232 ~ 50
Calib. Date 1:	05/17/23 Time 1: 1428	Analytical Batch:	766139	1208 ~ 230517/c0233 ~ 100	1209 ~ 230517/c0234 ~ 200
Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Method:	EPA 8260B		

ANALYTE	***1201***	***1202***	***1203***	***1204***	***1205***	***1206***	***1207***	***1208***	***1209***	***RF / m / A b / B***	***C***	***FIT***	***TYPE***
trans-1,4-Dichloro-2-butene			0.147	0.246	0.258	0.274	0.264	0.253	0.246				

FIT = %RSD for Average Curve and the Coefficient of Determination (r2) for Linear and Quadratic

Curve Types: A - Averaged, L - Linear Regression, W - Weighted Linear, Q - Quadratic

For curve types L and Q, the RRF and RSP (Response) are shown on separate lines
to allow for evaluation against minimum RRF

$\overline{RF}$ = Mean Response Factor For Average Curve

m,b = Slope and Intercept For Linear Curve

A,B,C = Coefficents For Quadratic Curve

* - %RSD > 15% or R^2 < 0.99

FORM V I VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0228.d
Lab Smp Id: 1203 Client Smp ID: V15STD001
Inj Date : 17-MAY-2023 14:28
Operator : CJR Inst ID: msv15.i
Smp Info : 1203*V15STD001
Misc Info : MSV~527~*1*SMR
Comment :
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr Quant Type: ISTD
Cal Date : 17-MAY-2023 14:28 Cal File: c0228.d
Als bottle: 1 Calibration Sample, Level: 3
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	1117	1.00000	0.813	7981
2 Chloromethane ++	50	1.649	1.649	(0.280)	1117	1.00000	0.902	8081
3 Vinyl Chloride +	62	1.719	1.719	(0.292)	1550	1.00000	0.889	7630
5 1-3 Butadiene	39	1.735	1.735	(0.295)	1215	1.00000	0.936	8374
6 Bromomethane	94	2.021	2.021	(0.344)	1167	1.00000	0.798	6338
7 Chloroethane	64	2.144	2.144	(0.364)	1121	1.00000	0.875	5692
8 Trichlorofluoromethane	101	2.276	2.276	(0.387)	2314	1.00000	0.911	8545
9 Isoprene	67	2.565	2.565	(0.436)	1820	1.00000	0.866	3894(a)
10 Ethyl Ether	59	2.584	2.584	(0.439)	783	1.00000	0.786	7309
11 1,1-Dichloroethene +	96	2.787	2.787	(0.474)	1352	1.00000	0.935	7941
12 Carbon Disulfide	76	2.812	2.812	(0.478)	4288	1.00000	0.969	6865
13 1,1,2Trichlotrifluoroethane	101	2.825	2.825	(0.480)	1314	1.00000	0.837	7000
15 Methyl Iodide	142	2.935	2.935	(0.499)	399	1.00000	1.85	7203
16 Acrolein	56	3.150	3.150	(0.535)	914	5.00000	4.15	4621

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
	17 Allyl chloride	39	3.311	3.311	(0.563)	889	1.00000	0.696	7439
	18 Methylene Chloride	49	3.420	3.420	(0.581)	1527	1.00000	0.954	4607
	19 Acetone	43	3.481	3.481	(0.592)	2691	5.00000	4.10	2977
	20 trans-1,2-Dichloroethene	61	3.587	3.587	(0.610)	1751	1.00000	0.874	8521
	21 Methyl Acetate	43	3.613	3.613	(0.614)	3689	5.00000	4.51	8424
	22 Hexane	57	3.674	3.674	(0.625)	2341	1.00000	0.811	8367
	23 MTBE	73	3.722	3.722	(0.633)	3072	1.00000	0.786	7791
	24 tert-Butyl Alcohol	59	3.825	3.825	(0.650)	141	5.00000	0.818	0(M1)
	25 Acetonitrile	41	3.941	3.941	(0.670)	449	5.00000	3.86	7055(M1)
	26 Isopropyl Ether	45	4.115	4.115	(0.699)	2883	1.00000	0.827	8475
	27 Chloroprene	53	4.198	4.198	(0.714)	1688	1.00000	0.859	0(M1)
	28 1,1-Dichloroethane ++	63	4.208	4.208	(0.715)	2295	1.00000	0.857	0(M1)
	29 Acrylonitrile	53	4.269	4.269	(0.726)	2016	5.00000	4.42	8501(M1)
	30 Vinyl Acetate	43	4.478	4.478	(0.761)	1983	1.00000	0.919	0(M1)
	31 Ethyl Tert-butyl Ether	59	4.475	4.475	(0.761)	3191	1.00000	0.813	7123
	32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	1781	1.00000	0.926	8626
M	33 Total 1,2-Dichloroethene	61				3532	2.00000	1.80	0
	34 2,2-Dichloropropane	77	4.832	4.832	(0.821)	2094	1.00000	0.880	8052
	35 Bromochloromethane	128	4.918	4.918	(0.836)	660	1.00000	0.874	8707
	36 Cyclohexane	56	4.922	4.922	(0.837)	2121	1.00000	0.869	8275
	37 Chloroform +	83	4.992	4.992	(0.849)	2493	1.00000	0.910	7638
	38 Ethyl Acetate	43	5.118	5.118	(0.870)	4501	5.00000	3.76	5435
	39 Carbon Tetrachloride	117	5.118	5.118	(0.870)	1892	1.00000	0.855	8652
	40 Tetrahydrofuran	42	5.166	5.166	(0.878)	1339	5.00000	3.00	7738
$	42 Dibromofluoromethane	111	5.160	5.160	(0.877)	1289	1.00000	0.881	6908
	43 1,1,1-Trichloroethane	97	5.176	5.176	(0.880)	2190	1.00000	0.864	7976
	44 2-Butanone	43	5.279	5.279	(0.897)	2441	5.00000	3.58	7013
	45 1,1-Dichloropropene	75	5.295	5.295	(0.900)	2076	1.00000	0.921	8493
	46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	5071	1.00000	0.745	8261(M1)
	47 Propionitrile	54	5.529	5.529	(0.940)	573	5.00000	3.40	0(M1)
	48 Benzene	78	5.510	5.510	(0.937)	5285	1.00000	0.867	8631
	49 Methylacrylonitrile	41	5.552	5.552	(0.944)	2580	5.00000	3.73	6827
	56 tert-amyl Methyl Ether	73	5.635	5.635	(0.958)	3055	1.00000	0.764	0(M1)
$	50 1,2-Dichloroethane-d4	65	5.629	5.629	(0.957)	1586	1.00000	1.04	7517
	52 1,2-Dichloroethane	62	5.690	5.690	(0.967)	1724	1.00000	0.899	0(M1)
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	296139	50.0000		9838
	57 Trichloroethene	130	6.037	6.037	(1.026)	1478	1.00000	0.790	8964
	58 Methyl Cyclohexane	83	6.021	6.021	(1.024)	2268	1.00000	0.792	8167
	59 Tert-amyl alcohol	59	6.246	6.246	(1.062)	2411	5.00000	4.06	6622
	60 n-Butanol	56	6.333	6.333	(1.077)	40	5.00000	10.8	0(M1)
	62 Dibromomethane	93	6.394	6.394	(1.087)	773	1.00000	0.788	8244
	63 2-3 Dichloro-1-Proprene	75	6.462	6.462	(1.098)	2096	1.00000	0.855	8662
	64 1,2-Dichloropropane +	63	6.478	6.478	(1.101)	1322	1.00000	0.878	7097
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	1669	1.00000	0.818	6990
	66 Methyl methacrylate	41	6.684	6.684	(1.136)	682	1.00000	0.654	7630
	67 1,4- Dioxane	88	6.725	6.725	(1.143)	1435	25.0000	129	7786
	68 1-Bromo-2-chloroethane	63	6.947	6.947	(1.181)	1677	1.00000	0.874	0(M1)

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	1244	5.00000	7.28	8497
73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	2225	1.00000	0.921	8076
$ 74 Toluene-d8	98	7.221	7.221	(0.866)	4852	1.00000	0.900	8965
75 Toluene +	91	7.262	7.262	(0.871)	5676	1.00000	0.845	9140
76 2-Nitropropane	43	7.436	7.436	(0.892)	147	1.00000	1.01	6280(M1)
77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	3673	5.00000	5.60	8940
78 Tetrachloroethene	164	7.552	7.552	(0.906)	1288	1.00000	0.848	7934
79 trans-1,3-Dichloropropene	75	7.571	7.571	(1.287)	1959	1.00000	0.856	7773
80 Ethyl Methacrylate	41	7.697	7.697	(0.923)	576	1.00000	0.614	0(M1)
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	1146	1.00000	0.840	7775
82 Dibromochloromethane	129	7.825	7.825	(0.938)	1166	1.00000	0.752	7123
M 83 1-3 Dichloropropene total	100				4184	2.00000	1.78	0
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	1374	1.00000	0.766	8212(a)
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	1969	1.00000	0.867	8499
88 1,2-Dibromoethane(EDB)	107	7.996	7.996	(0.959)	1133	1.00000	0.776	7873
89 2-Hexanone	43	8.147	8.147	(0.977)	2993	5.00000	5.51	8901
90 1-Chlorohexane	91	8.330	8.330	(0.999)	2151	1.00000	0.896	2514
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	115086	50.0000		9715
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	3920	1.00000	0.882	3030
93 Ethylbenzene +	106	8.368	8.368	(1.003)	1966	1.00000	0.814	8026
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	1585	1.00000	1.06	8215
95 p,m-Xylene	106	8.462	8.462	(1.015)	4800	2.00000	1.64	9540
97 o-Xylene	106	8.745	8.745	(1.049)	2226	1.00000	0.818	9362
101 Styrene	104	8.777	8.777	(1.052)	2650	1.00000	0.756	9109
102 Bromoform ++	173	8.799	8.799	(1.055)	720	1.00000	1.42	5747
104 Isopropylbenzene	105	8.944	8.944	(1.072)	6010	1.00000	0.844	9186
M 105 TOTAL XYLENE	106				7026	3.00000	2.46	0
$ 106 Bromofluorobenzene	174	9.121	9.121	(1.094)	1941	1.00000	1.11	8401
107 cis-1,4-dichloro-2-butene	53	9.156	9.156	(0.934)	319	1.00000	0.626	0(aM1)
109 n-Propylbenzene	91	9.198	9.198	(0.938)	6966	1.00000	0.889	9095
108 Bromobenzene	77	9.192	9.192	(0.937)	2475	1.00000	0.949	8775
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.942)	1249	1.00000	0.816	0(M1)
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	4494	1.00000	0.900	8094
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	4374	1.00000	0.830	7843
114 1,2,3-Trichloropropane	75	9.327	9.327	(0.951)	2080	1.00000	1.00	6425
115 trans-1,4-Dichloro-2-Butene	53	9.346	9.346	(0.953)	295	1.00000	0.820	6672
116 Cyclohexanone	55	9.368	9.368	(0.955)	762	5.00000	4.30	8114
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	3917	1.00000	0.874	9150
118 tert-butylbenzene	91	9.513	9.513	(0.970)	2514	1.00000	0.850	9368
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.974)	4072	1.00000	0.787	8689
120 sec-Butylbenzene	105	9.622	9.622	(0.981)	5758	1.00000	0.845	9251
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	4924	1.00000	0.816	8524
122 Dicyclopentadiene	66	9.700	9.700	(0.989)	5208	1.00000	0.831	7745
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.995)	2819	1.00000	0.925	8991
* 124 1,4-DICHLOROBENZENE-D4	152	9.806	9.806	(1.000)	100307	50.0000		9667
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	2920	1.00000	0.931	3199(H)
126 n-Butylbenzene	91	9.954	9.954	(1.015)	6242	1.00000	0.778	8690

	QUANT SIG					AMOUNTS		
						CAL-AMT	ON-COL	
Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
127 1,2-Dichlorobenzene	146	10.066	10.066	(1.027)	2430	1.00000	0.881	8254
129 1,2-Dibromo-3-Chloropropane	157	10.558	10.558	(1.077)	159	1.00000	1.24	0(M1)
130 Hexachlorobutadiene	225	10.999	10.999	(1.122)	652	1.00000	0.773	7245
131 1,2,4-Trichlorobenzene	180	11.031	11.031	(1.125)	1406	1.00000	0.834	8026
132 Naphthalene	128	11.288	11.288	(1.151)	2909	1.00000	0.736	8473
133 1,2,3-Trichlorobenzene	180	11.446	11.446	(1.167)	1129	1.00000	0.776	6978

QC Flag Legend

a - Target compound detected but, quantitated amount
    Below Limit Of Quantitation(BLOQ).
M1- Compound response manually integrated because
    Target system did not integrate.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0228.d

Page 1

Date : 17-MAY-2023 14:28

Client ID: V15STD001

Instrument: msv15.i

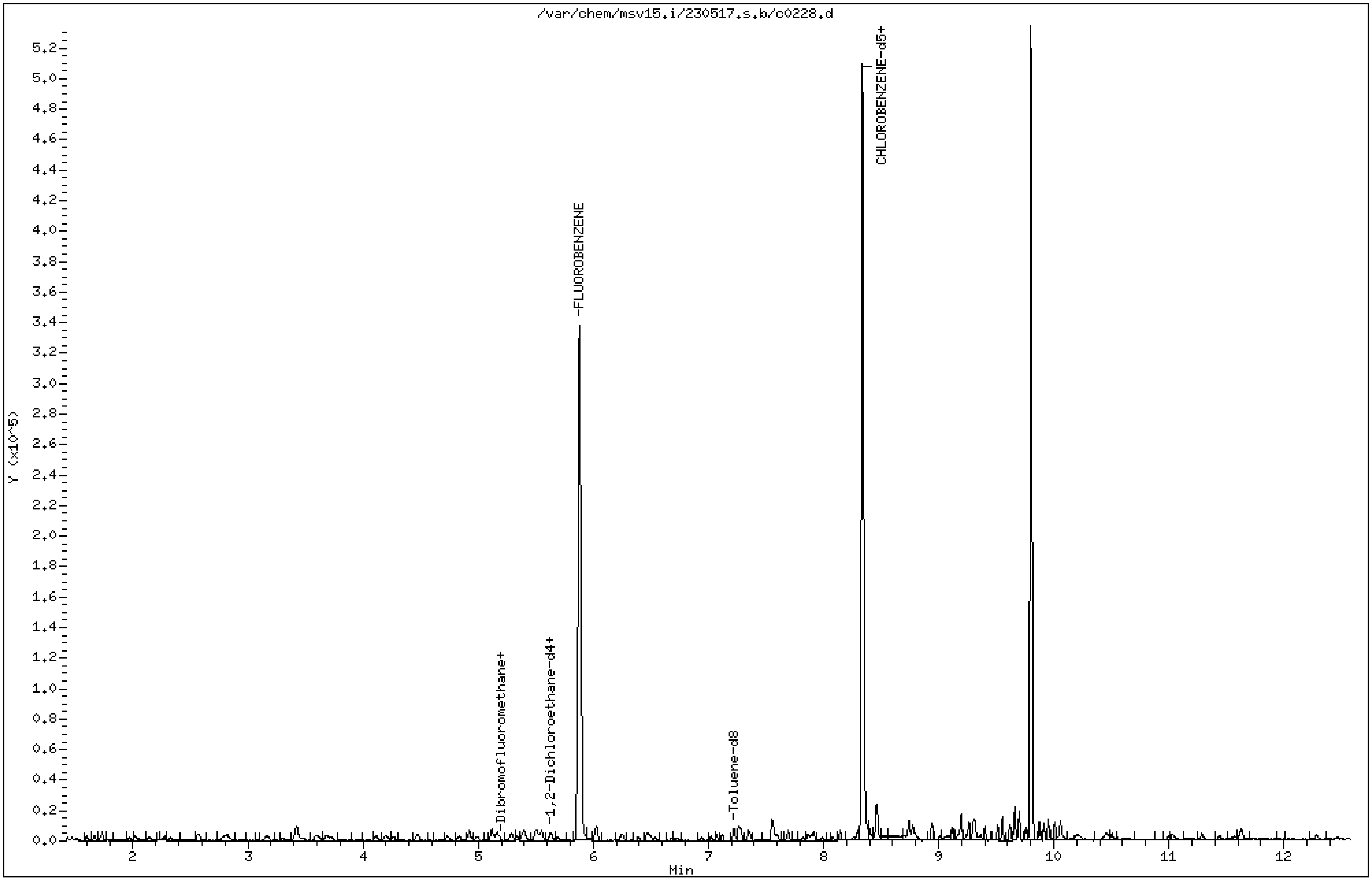
Sample Info: 1203*V15STD001

Purge Volume: 5.0

Operator: CJR

Column phase: RTX-VMS-30M

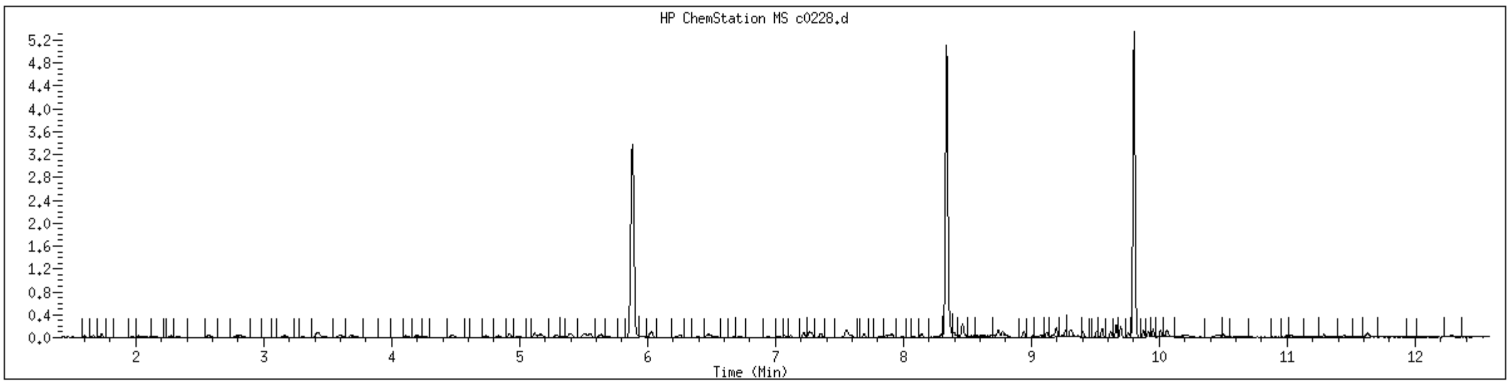
Column diameter: 0.25



Report Date: 05/23/2023 16:34

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1203
SampleType : CALIB_3
Injection Date: 05/17/2023 14:28
Instrument : msv15.i
Operator : CJR
Sample Info : 1203*V15STD001
Misc Info : MSV~527~*1*SMR
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE
Compound Sublist: 8260b+AppIXDOD



Original Final

==

29 Acrylonitrile CAS#: 107-13-1 Reason: M1

Electronic Signature
Applied

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0229.d
Lab Smp Id: 1204   Client Smp ID: V15STD005
Inj Date : 17-MAY-2023 14:47
Operator : CJR   Inst ID: msv15.i
Smp Info : 1204*V15STD005
Misc Info : MSV~527~*1*SMR
Comment   :
Method   : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr   Quant Type: ISTD
Cal Date : 17-MAY-2023 14:47   Cal File: c0229.d
Als bottle: 1   Calibration Sample, Level: 4
Dil Factor: 1.00000
Integrator: HP RTE   Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable   Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	6870	5.00000	4.86	9447
2 Chloromethane ++	50	1.652	1.652	(0.281)	6263	5.00000	4.92	9311
3 Vinyl Chloride +	62	1.719	1.719	(0.292)	8525	5.00000	4.75	9304
5 1-3 Butadiene	39	1.735	1.735	(0.295)	6564	5.00000	4.92	9510
6 Bromomethane	94	2.018	2.018	(0.343)	6749	5.00000	4.49	9197
7 Chloroethane	64	2.137	2.137	(0.363)	6205	5.00000	4.71	8861
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	12359	5.00000	4.73	9124
9 Isoprene	67	2.571	2.571	(0.437)	10068	5.00000	4.66	4501(a)
10 Ethyl Ether	59	2.587	2.587	(0.440)	4907	5.00000	4.79	8213
11 1,1-Dichloroethene +	96	2.787	2.787	(0.474)	7051	5.00000	4.74	8512
12 Carbon Disulfide	76	2.816	2.816	(0.479)	22125	5.00000	4.86	7846
13 1,1,2Trichlotrifluoroethane	101	2.841	2.841	(0.483)	7874	5.00000	4.88	8604
15 Methyl Iodide	142	2.935	2.935	(0.499)	5565	5.00000	4.65	8826
16 Acrolein	56	3.156	3.156	(0.537)	5341	25.0000	23.6	4526

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
	17 Allyl chloride	39	3.308	3.308	(0.562)	6819	5.00000	5.21	9512
	18 Methylene Chloride	49	3.420	3.420	(0.581)	8077	5.00000	4.91	4855
	19 Acetone	43	3.475	3.475	(0.591)	17117	25.0000	25.3	4888(H)
	20 trans-1,2-Dichloroethene	61	3.591	3.591	(0.610)	9832	5.00000	4.77	9142
	21 Methyl Acetate	43	3.619	3.619	(0.615)	19451	25.0000	23.1	8637
	22 Hexane	57	3.684	3.684	(0.626)	14269	5.00000	4.80	8905
	23 MTBE	73	3.729	3.729	(0.634)	18657	5.00000	4.64	9421
	24 tert-Butyl Alcohol	59	3.835	3.835	(0.652)	3739	25.0000	21.1	0(M1)
	25 Acetonitrile	41	3.944	3.944	(0.670)	2904	25.0000	24.3	9271
	26 Isopropyl Ether	45	4.115	4.115	(0.699)	16980	5.00000	4.73	9377
	27 Chloroprene	53	4.198	4.198	(0.714)	9634	5.00000	4.77	8675
	28 1,1-Dichloroethane ++	63	4.214	4.214	(0.716)	13766	5.00000	5.00	9244
	29 Acrylonitrile	53	4.266	4.266	(0.725)	10944	25.0000	23.3	9481
	30 Vinyl Acetate	43	4.475	4.475	(0.761)	10571	5.00000	4.76	5764
	31 Ethyl Tert-butyl Ether	59	4.478	4.478	(0.761)	19016	5.00000	4.71	8017
	32 cis-1,2-Dichloroethene	61	4.738	4.738	(0.805)	9499	5.00000	4.80	9504
M	33 Total 1,2-Dichloroethene	61				19331	10.0000	9.57	0
	34 2,2-Dichloropropane	77	4.835	4.835	(0.822)	12035	5.00000	4.92	9292
	35 Bromochloromethane	128	4.922	4.922	(0.837)	3963	5.00000	5.10	9439
	36 Cyclohexane	56	4.928	4.928	(0.838)	11895	5.00000	4.74	9099
	37 Chloroform +	83	4.996	4.996	(0.849)	13628	5.00000	4.84	9429
	38 Ethyl Acetate	43	5.121	5.121	(0.870)	29353	25.0000	23.8	5908
	41 sec-Butanol	45	5.179	5.179	(0.880)	218	5.00000	2.34	0(M1)
	39 Carbon Tetrachloride	117	5.118	5.118	(0.870)	10352	5.00000	4.55	8107
	40 Tetrahydrofuran	42	5.160	5.160	(0.877)	10647	25.0000	23.2	8163
$	42 Dibromofluoromethane	111	5.160	5.160	(0.877)	6988	5.00000	4.65	7654
	43 1,1,1-Trichloroethane	97	5.179	5.179	(0.880)	12365	5.00000	4.74	9516
	44 2-Butanone	43	5.279	5.279	(0.897)	16865	25.0000	24.1	6896
	45 1,1-Dichloropropene	75	5.295	5.295	(0.900)	11134	5.00000	4.80	9184
	46 2,2,4 Trimethylpentane	57	5.404	5.404	(0.919)	33148	5.00000	4.73	8645
	47 Propionitrile	54	5.536	5.536	(0.941)	4217	25.0000	24.3	8622
	48 Benzene	78	5.517	5.517	(0.938)	30390	5.00000	4.85	9547
	49 Methylacrylonitrile	41	5.555	5.555	(0.944)	17141	25.0000	24.1	7322
	56 tert-amyl Methyl Ether	73	5.635	5.635	(0.958)	18841	5.00000	4.58	0(M1)
$	50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	7109	5.00000	4.54	8209
	52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	9684	5.00000	4.91	9460
	51 Isobutyl Alcohol	43	5.732	5.732	(0.974)	886	25.0000	21.6	2591(H)
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	304598	50.0000		9849
	57 Trichloroethene	130	6.034	6.034	(1.026)	9041	5.00000	4.70	9137
	58 Methyl Cyclohexane	83	6.025	6.025	(1.024)	13832	5.00000	4.70	8749
	59 Tert-amyl alcohol	59	6.256	6.256	(1.063)	13852	25.0000	22.7	6786
	60 n-Butanol	56	6.336	6.336	(1.077)	406	25.0000	25.3	0(M1)
	62 Dibromomethane	93	6.394	6.394	(1.087)	4899	5.00000	4.86	9054
	63 2-3 Dichloro-1-Proprene	75	6.468	6.468	(1.099)	11872	5.00000	4.71	8806
	64 1,2-Dichloropropane +	63	6.484	6.484	(1.102)	7338	5.00000	4.74	8050
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	9982	5.00000	4.76	9358
	66 Methyl methacrylate	41	6.677	6.677	(1.135)	4413	5.00000	4.12	9007

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
67 1,4- Dioxane	88	6.722	6.722	(1.143)	1658	125.000	144	7925
68 1-Bromo-2-chloroethane	63	6.947	6.947	(1.181)	9449	5.00000	4.79	8920
72 2-Chloroethyl vinyl ether	63	7.021	7.021	(1.193)	8287	25.0000	22.9	9327
73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	11598	5.00000	4.67	9450
$ 74 Toluene-d8	98	7.224	7.224	(0.866)	25628	5.00000	4.74	9708
75 Toluene +	91	7.262	7.262	(0.871)	32455	5.00000	4.83	9594
76 2-Nitropropane	43	7.436	7.436	(0.892)	1643	5.00000	4.99	6263(M3)
77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	26975	25.0000	24.0	8386
78 Tetrachloroethene	164	7.558	7.558	(0.906)	7348	5.00000	4.83	8044
79 trans-1,3-Dichloropropene	75	7.578	7.578	(1.288)	10829	5.00000	4.60	9230
80 Ethyl Methacrylate	41	7.697	7.697	(0.923)	4235	5.00000	4.51	7480
81 1,1,2-Trichloroethane	97	7.690	7.690	(0.922)	6622	5.00000	4.85	9421
82 Dibromochloromethane	129	7.825	7.825	(0.938)	7142	5.00000	4.60	9365
M 83 1-3 Dichloropropene total	100				22427	10.0000	9.27	0
84 3,4-dichloro-1-butene	75	7.864	7.864	(0.943)	8232	5.00000	4.58	9669(a)
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	11280	5.00000	4.96	9428
87 1-Nitropropane	43	7.976	7.976	(0.956)	825	5.00000	4.85	6201(M3)
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	7147	5.00000	4.89	9183
89 2-Hexanone	43	8.143	8.143	(0.976)	22444	25.0000	24.2	9600
90 1-Chlorohexane	91	8.333	8.333	(0.999)	11448	5.00000	4.76	3265
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	115277	50.0000		9742
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	21340	5.00000	4.79	4912
93 Ethylbenzene +	106	8.368	8.368	(1.003)	11735	5.00000	4.85	9125
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	6687	5.00000	4.47	9145
95 p,m-Xylene	106	8.465	8.465	(1.015)	27464	10.0000	9.38	9801
97 o-Xylene	106	8.745	8.745	(1.049)	13105	5.00000	4.81	9625
101 Styrene	104	8.774	8.774	(1.052)	15750	5.00000	4.49	9736
102 Bromoform ++	173	8.799	8.799	(1.055)	4668	5.00000	4.51	8307
104 Isopropylbenzene	105	8.941	8.941	(1.072)	33812	5.00000	4.74	9580
M 105 TOTAL XYLENE	106				40569	15.0000	14.2	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	8258	5.00000	4.71	9335
107 cis-1,4-dichloro-2-butene	53	9.156	9.156	(0.934)	2427	5.00000	4.61	9285(a)
109 n-Propylbenzene	91	9.201	9.201	(0.939)	40109	5.00000	4.96	8649
108 Bromobenzene	77	9.192	9.192	(0.938)	14139	5.00000	5.25	9052
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.943)	8032	5.00000	5.08	9527
112 2-Chlorotoluene	91	9.301	9.301	(0.949)	26003	5.00000	5.04	7697
113 1,3,5-Trimethylbenzene	105	9.314	9.314	(0.950)	26594	5.00000	4.89	9014
114 1,2,3-Trichloropropane	75	9.327	9.327	(0.951)	10340	5.00000	4.82	6905
115 trans-1,4-Dichloro-2-Butene	53	9.346	9.346	(0.953)	2548	5.00000	5.10	9136
116 Cyclohexanone	55	9.362	9.362	(0.955)	6175	25.0000	24.8	9506
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	23095	5.00000	4.99	9575
118 tert-butylbenzene	91	9.513	9.513	(0.970)	15583	5.00000	5.10	9653
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	25913	5.00000	4.85	9319
120 sec-Butylbenzene	105	9.622	9.622	(0.982)	34756	5.00000	4.94	9665
121 p-Isopropyltoluene	119	9.700	9.700	(0.990)	29602	5.00000	4.75	8812
122 Dicyclopentadiene	66	9.700	9.700	(0.990)	32601	5.00000	5.03	7854
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.996)	15286	5.00000	4.85	9445

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.803	9.803	(1.000)	103631	50.0000		9619
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	15506	5.00000	4.79	4773(H)
126 n-Butylbenzene	91	9.954	9.954	(1.015)	40223	5.00000	4.85	8824
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	13759	5.00000	4.83	9317
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	1562	5.00000	4.68	8525
130 Hexachlorobutadiene	225	10.995	10.995	(1.122)	4238	5.00000	4.87	9004
131 1,2,4-Trichlorobenzene	180	11.028	11.028	(1.125)	8295	5.00000	4.76	9147
132 Naphthalene	128	11.288	11.288	(1.152)	18400	5.00000	4.51	9663
133 1,2,3-Trichlorobenzene	180	11.439	11.439	(1.167)	6986	5.00000	4.65	9151

QC Flag Legend

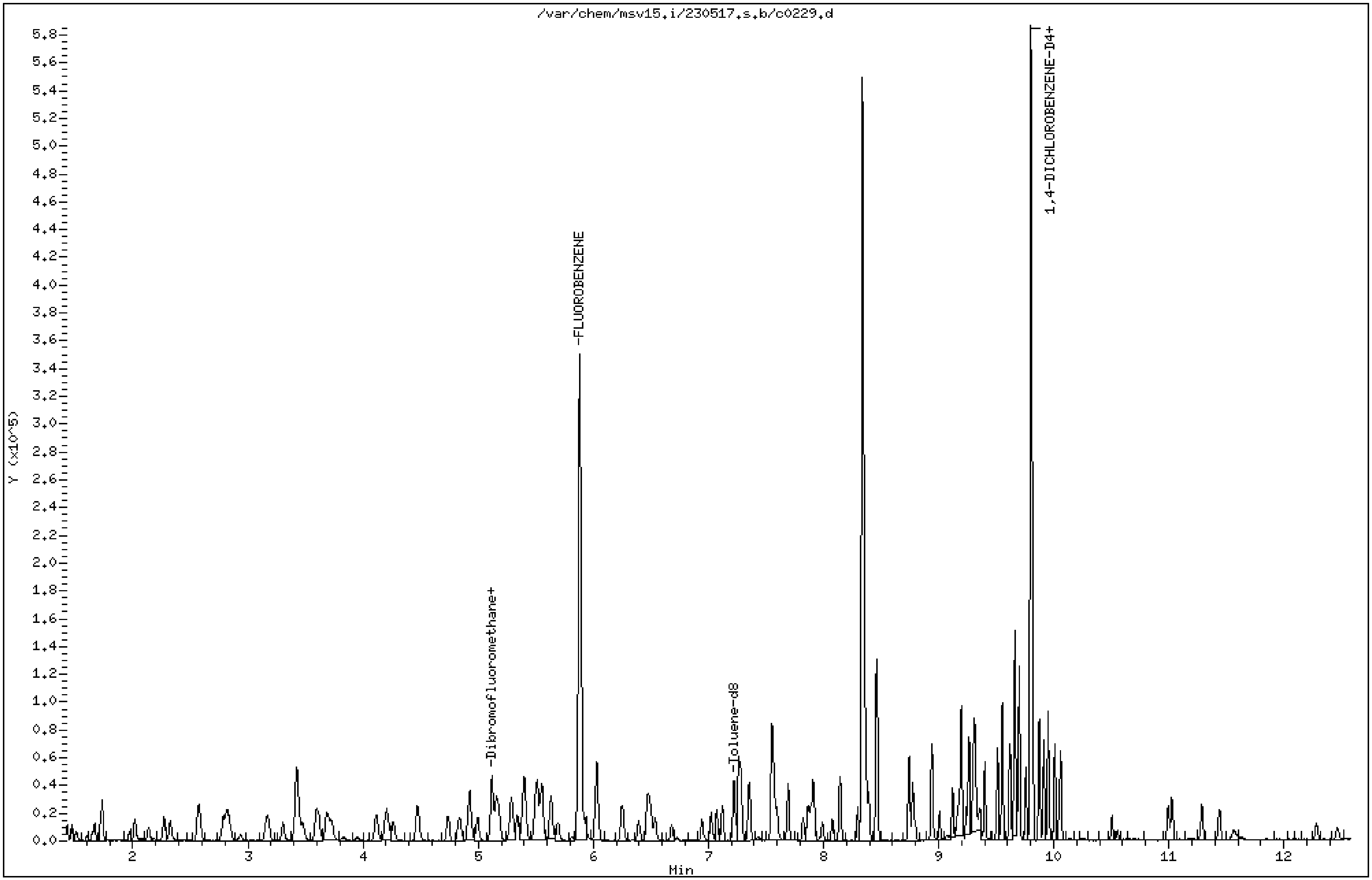
a - Target compound detected but, quantitated amount Below Limit Of Quantitation(BLOQ).
M1- Compound response manually integrated because Target system did not integrate.
M3- Compound response manually integrated because Target system integrated incorrect peak.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0229.d
Date : 17-MAY-2023 14:47
Client ID: V15STD005
Sample Info: 1204*V15STD005
Purge Volume: 5.0
Column phase: RTX-VMS-30M

Page 1

Instrument: msv15.i

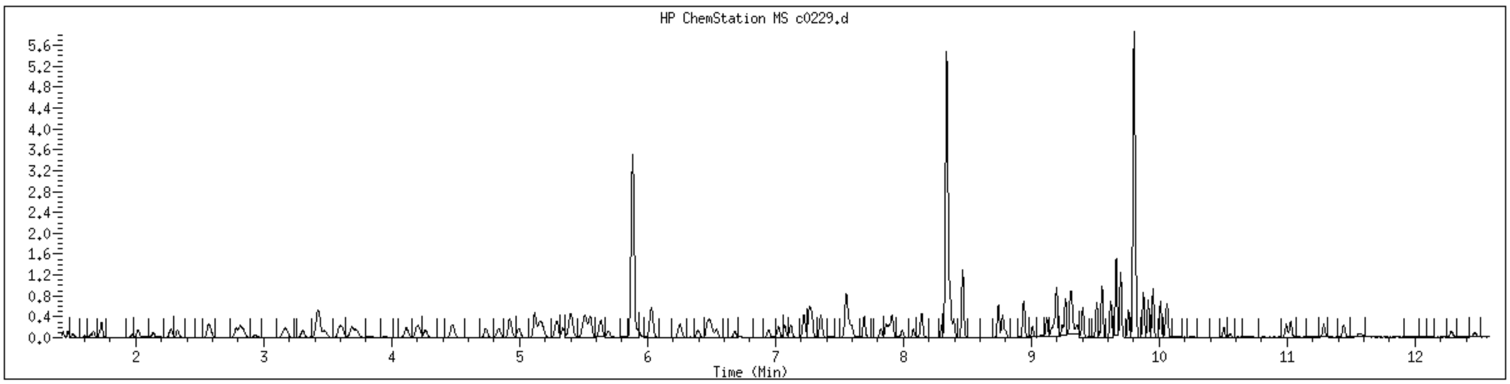
Operator: CJR
Column diameter: 0.25



Report Date: 05/23/2023 16:34

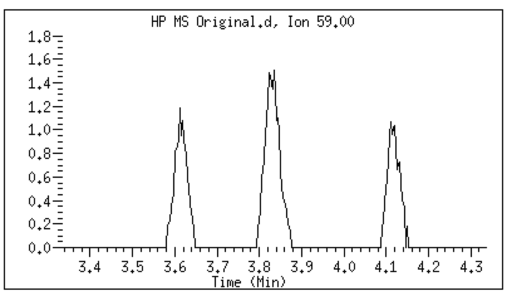
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID	: 1204	SampleType	: CALIB_4
Injection Date	: 05/17/2023 14:47	Instrument	: msv15.i
Operator	: CJR		
Sample Info	: 1204*V15STD005		
Misc Info	: MSV~527~*1*SMR		
Method	: /var/chem/msv15.i/230517.s.b/8260bw15DOD.m		
Dilution	: 1.00		
Matrix	: WATER		
Integrator	: HP RTE	Compound Sublist	: 8260b+AppIXDOD



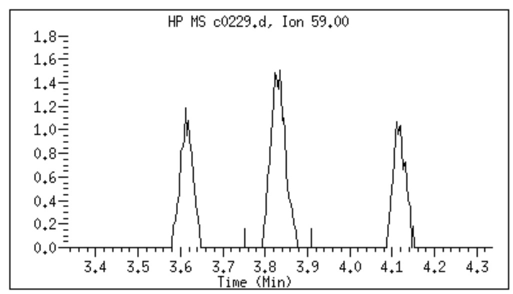
Original Final

24 tert-Butyl Alcohol CAS#: 75-65-0 Reason: M1

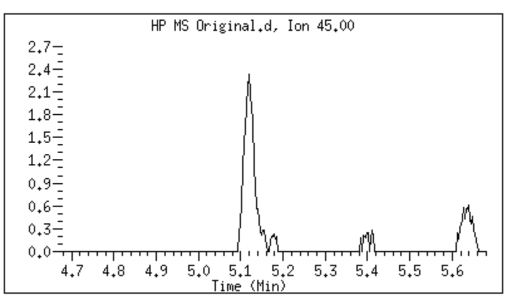


Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:01

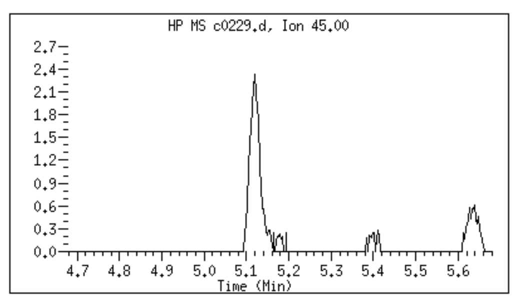


41 sec-Butanol CAS#: 78-92-2 Reason: M1



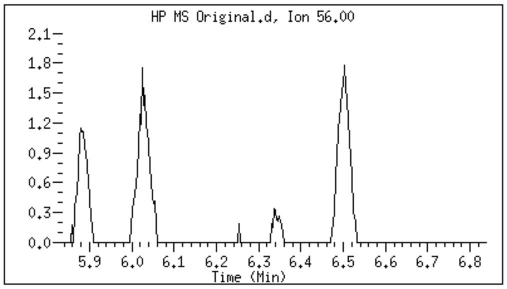
Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:10



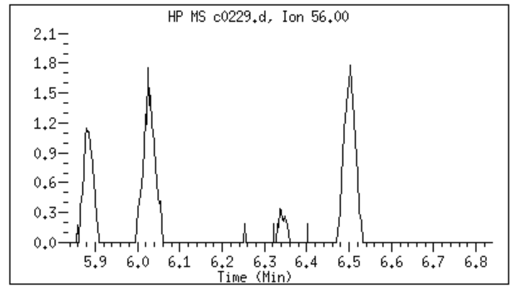
Original Final

60 n-Butanol CAS#: 71-36-3 Reason: M1

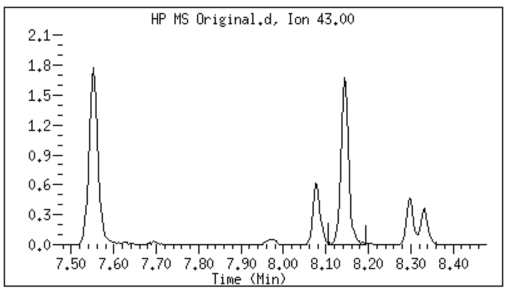


Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:11

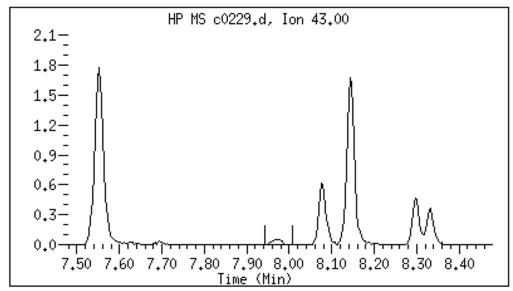


87 1-Nitropropane CAS#: 108-03-2 Reason: M3

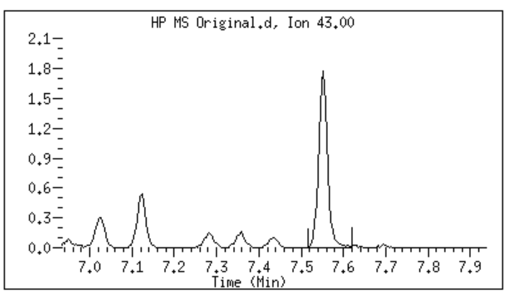


Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:11

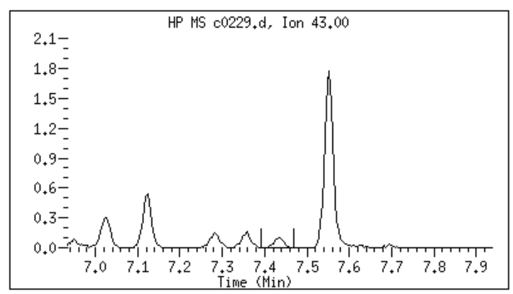


76 2-Nitropropane CAS#: 79-46-9 Reason: M3

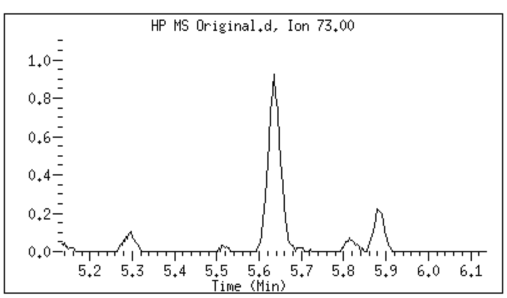


Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:11

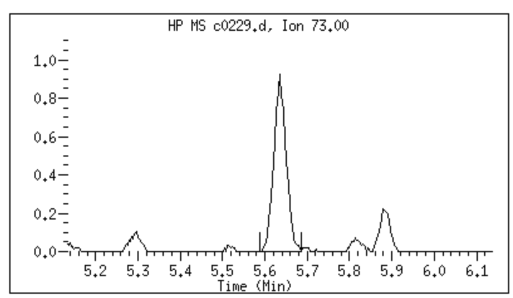


56 tert-amyl Methyl Ether CAS#: 994-05-8 Reason: M1



Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:11



M1 - Target system did not integrate
M3 - Target system integrated incorrect peak

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0230.d
Lab Smp Id: 1205 Client Smp ID: V15STD010
Inj Date : 17-MAY-2023 15:05
Operator : CJR Inst ID: msv15.i
Smp Info : 1205*V15STD010
Misc Info : MSV~527~*1*SMR
Comment :
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr Quant Type: ISTD
Cal Date : 17-MAY-2023 15:05 Cal File: c0230.d
Als bottle: 1 Calibration Sample, Level: 5
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

						AMOUNTS		
	QUANT SIG					CAL-AMT	ON-COL	
Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	14263	10.0000	9.83	9700
2 Chloromethane ++	50	1.655	1.655	(0.281)	12647	10.0000	9.68	9525
3 Vinyl Chloride +	62	1.719	1.719	(0.292)	18349	10.0000	9.97	9540
5 1-3 Butadiene	39	1.735	1.735	(0.295)	13136	10.0000	9.59	9546
6 Bromomethane	94	2.018	2.018	(0.343)	14184	10.0000	9.19	9485
7 Chloroethane	64	2.140	2.140	(0.364)	13449	10.0000	9.94	9099
8 Trichlorofluoromethane	101	2.275	2.275	(0.387)	25447	10.0000	9.49	9463
9 Isoprene	67	2.571	2.571	(0.437)	20734	10.0000	9.35	4545
10 Ethyl Ether	59	2.584	2.584	(0.439)	10191	10.0000	9.70	7856
11 1,1-Dichloroethene +	96	2.787	2.787	(0.474)	14180	10.0000	9.29	8431
12 Carbon Disulfide	76	2.816	2.816	(0.479)	43652	10.0000	9.35	8143
13 1,1,2Trichlotrifluoroethane	101	2.835	2.835	(0.482)	15524	10.0000	9.37	9161
15 Methyl Iodide	142	2.938	2.938	(0.499)	14998	10.0000	9.58	9298
16 Acrolein	56	3.153	3.153	(0.536)	11173	50.0000	48.1	5597

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
	==========================	====	==	======	======	========	=======	=======	==========
	17 Allyl chloride	39	3.304	3.304	(0.562)	14165	10.0000	10.6	9531
	18 Methylene Chloride	49	3.420	3.420	(0.581)	16213	10.0000	9.60	4808
	19 Acetone	43	3.475	3.475	(0.591)	31942	50.0000	46.1	4801(M3H)
	20 trans-1,2-Dichloroethene	61	3.594	3.594	(0.611)	20360	10.0000	9.62	8792
	21 Methyl Acetate	43	3.613	3.613	(0.614)	42193	50.0000	48.9	9581
	22 Hexane	57	3.687	3.687	(0.627)	29696	10.0000	9.74	9271
	23 MTBE	73	3.722	3.722	(0.633)	40428	10.0000	9.80	9613
	24 tert-Butyl Alcohol	59	3.832	3.832	(0.651)	7929	50.0000	43.6	0(M1)
	25 Acetonitrile	41	3.951	3.951	(0.672)	6105	50.0000	49.7	9201
	26 Isopropyl Ether	45	4.115	4.115	(0.699)	35627	10.0000	9.68	9608
	27 Chloroprene	53	4.198	4.198	(0.714)	19593	10.0000	9.45	8523
	28 1,1-Dichloroethane ++	63	4.218	4.218	(0.717)	27282	10.0000	9.66	9184
	29 Acrylonitrile	53	4.266	4.266	(0.725)	22963	50.0000	47.7	9705
	30 Vinyl Acetate	43	4.478	4.478	(0.761)	22447	10.0000	9.85	5566
	31 Ethyl Tert-butyl Ether	59	4.475	4.475	(0.761)	40182	10.0000	9.70	8160
	32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	19871	10.0000	9.78	9665
M	33 Total 1,2-Dichloroethene	61				40231	20.0000	19.4	0
	34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	24426	10.0000	9.73	9446
	35 Bromochloromethane	128	4.915	4.915	(0.835)	8274	10.0000	10.4	9691
	36 Cyclohexane	56	4.928	4.928	(0.838)	24198	10.0000	9.39	8982
	37 Chloroform +	83	4.992	4.992	(0.849)	27444	10.0000	9.50	9653
	38 Ethyl Acetate	43	5.118	5.118	(0.870)	62159	50.0000	49.2	5949
	41 sec-Butanol	45	5.172	5.172	(0.879)	846	10.0000	8.84	5494
	39 Carbon Tetrachloride	117	5.121	5.121	(0.870)	21594	10.0000	9.24	8162
	40 Tetrahydrofuran	42	5.156	5.156	(0.876)	22473	50.0000	47.8	8495
$	42 Dibromofluoromethane	111	5.156	5.156	(0.876)	14586	10.0000	9.45	7353
	43 1,1,1-Trichloroethane	97	5.182	5.182	(0.881)	25029	10.0000	9.35	9255
	44 2-Butanone	43	5.282	5.282	(0.898)	34712	50.0000	48.2	7410
	45 1,1-Dichloropropene	75	5.295	5.295	(0.900)	22455	10.0000	9.44	9417
	46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	68436	10.0000	9.52	8674
	47 Propionitrile	54	5.533	5.533	(0.940)	9033	50.0000	50.8	8496
	48 Benzene	78	5.516	5.516	(0.938)	61559	10.0000	9.57	9746
	49 Methylacrylonitrile	41	5.552	5.552	(0.944)	36232	50.0000	49.6	7428
	56 tert-amyl Methyl Ether	73	5.635	5.635	(0.958)	40710	10.0000	9.64	7197
$	50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	15141	10.0000	9.42	8410
	52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	20228	10.0000	10.0	9571
	51 Isobutyl Alcohol	43	5.729	5.729	(0.974)	2126	50.0000	50.5	2571(H)
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	312572	50.0000		9889
	57 Trichloroethene	130	6.034	6.034	(1.026)	18554	10.0000	9.39	9173
	58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	28236	10.0000	9.34	8375
	59 Tert-amyl alcohol	59	6.253	6.253	(1.063)	29410	50.0000	47.0	6748
	60 n-Butanol	56	6.336	6.336	(1.077)	1071	50.0000	50.5	7961
	62 Dibromomethane	93	6.394	6.394	(1.087)	10112	10.0000	9.77	9161
	63 2-3 Dichloro-1-Proprene	75	6.465	6.465	(1.099)	24100	10.0000	9.31	9084
	64 1,2-Dichloropropane +	63	6.481	6.481	(1.102)	15321	10.0000	9.64	7653
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	20686	10.0000	9.61	9655
	66 Methyl methacrylate	41	6.680	6.680	(1.136)	9989	10.0000	9.08	9633

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
67 1,4- Dioxane	88	6.725	6.725	(1.143)	2688	250.000	228	8111
68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	19847	10.0000	9.80	9201
72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	19577	50.0000	46.9	9712
73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	23948	10.0000	9.39	9684
$ 74 Toluene-d8	98	7.224	7.224	(0.866)	53506	10.0000	9.55	9711
75 Toluene +	91	7.266	7.266	(0.871)	68537	10.0000	9.83	9427
76 2-Nitropropane	43	7.436	7.436	(0.892)	3719	10.0000	10.2	6280(M3)
77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	57708	50.0000	46.7	8332
78 Tetrachloroethene	164	7.558	7.558	(0.906)	14595	10.0000	9.25	8090
79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	23020	10.0000	9.53	8768
80 Ethyl Methacrylate	41	7.693	7.693	(0.922)	8992	10.0000	9.23	7322
81 1,1,2-Trichloroethane	97	7.696	7.696	(0.923)	14000	10.0000	9.88	9406
82 Dibromochloromethane	129	7.822	7.822	(0.938)	15051	10.0000	9.34	9376
M 83 1-3 Dichloropropene total	100				46968	20.0000	18.9	0
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	17920	10.0000	9.62	9664
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	23361	10.0000	9.90	9471
87 1-Nitropropane	43	7.963	7.963	(0.955)	1689	10.0000	9.58	6165(M3)
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	15124	10.0000	9.98	9582
89 2-Hexanone	43	8.143	8.143	(0.976)	46436	50.0000	45.8	9665
90 1-Chlorohexane	91	8.330	8.330	(0.999)	25187	10.0000	10.1	4639
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	119551	50.0000		9744
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	44691	10.0000	9.68	6492
93 Ethylbenzene +	106	8.365	8.365	(1.003)	24200	10.0000	9.64	9513
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	14046	10.0000	9.05	9178
95 p,m-Xylene	106	8.462	8.462	(1.015)	58802	20.0000	19.4	9834
97 o-Xylene	106	8.745	8.745	(1.049)	27430	10.0000	9.71	9818
101 Styrene	104	8.777	8.777	(1.052)	34395	10.0000	9.45	9627
102 Bromoform ++	173	8.799	8.799	(1.055)	9910	10.0000	8.34	8431
104 Isopropylbenzene	105	8.941	8.941	(1.072)	70970	10.0000	9.59	9764
M 105 TOTAL XYLENE	106				86232	30.0000	29.1	0
$ 106 Bromofluorobenzene	174	9.121	9.121	(1.094)	16738	10.0000	9.20	9267
107 cis-1,4-dichloro-2-butene	53	9.159	9.159	(0.934)	5294	10.0000	9.64	9530
109 n-Propylbenzene	91	9.198	9.198	(0.938)	84880	10.0000	10.0	9156
108 Bromobenzene	77	9.195	9.195	(0.938)	27711	10.0000	9.86	9652
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.943)	17408	10.0000	10.5	9513
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	54141	10.0000	10.1	9036
113 1,3,5-Trimethylbenzene	105	9.314	9.314	(0.950)	56216	10.0000	9.90	8672
114 1,2,3-Trichloropropane	75	9.327	9.327	(0.951)	22041	10.0000	9.84	7091
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	5589	10.0000	10.4	9127
116 Cyclohexanone	55	9.365	9.365	(0.955)	13437	50.0000	50.2	9649
117 4-Chlorotoluene	91	9.401	9.401	(0.959)	49913	10.0000	10.3	9702
118 tert-butylbenzene	91	9.513	9.513	(0.970)	31856	10.0000	9.99	9734
119 1,2,4-Trimethylbenzene	105	9.552	9.552	(0.974)	55290	10.0000	9.92	9287
120 sec-Butylbenzene	105	9.619	9.619	(0.981)	72201	10.0000	9.82	9736
121 p-Isopropyltoluene	119	9.700	9.700	(0.990)	63300	10.0000	9.73	8748
122 Dicyclopentadiene	66	9.700	9.700	(0.990)	67596	10.0000	10.0	7818
123 1,3-Dichlorobenzene	146	9.757	9.757	(0.995)	32156	10.0000	9.78	9499

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.802	(1.000)	108151	50.0000		9497
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	32888	10.0000	9.73	6212(H)
126 n-Butylbenzene	91	9.954	9.954	(1.015)	84800	10.0000	9.80	8788
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	29173	10.0000	9.81	9455
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	3400	10.0000	8.86	9015
130 Hexachlorobutadiene	225	10.992	10.992	(1.121)	8551	10.0000	9.41	9039
131 1,2,4-Trichlorobenzene	180	11.028	11.028	(1.125)	17174	10.0000	9.45	9283
132 Naphthalene	128	11.288	11.288	(1.152)	41641	10.0000	9.78	9784
133 1,2,3-Trichlorobenzene	180	11.442	11.442	(1.167)	15358	10.0000	9.79	9253

QC Flag Legend

M1- Compound response manually integrated because
Target system did not integrate.
M3- Compound response manually integrated because
Target system integrated incorrect peak.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0230.d

Page 1

Date : 17-MAY-2023 15:05

Client ID: V15STD010

Instrument: msv15.i

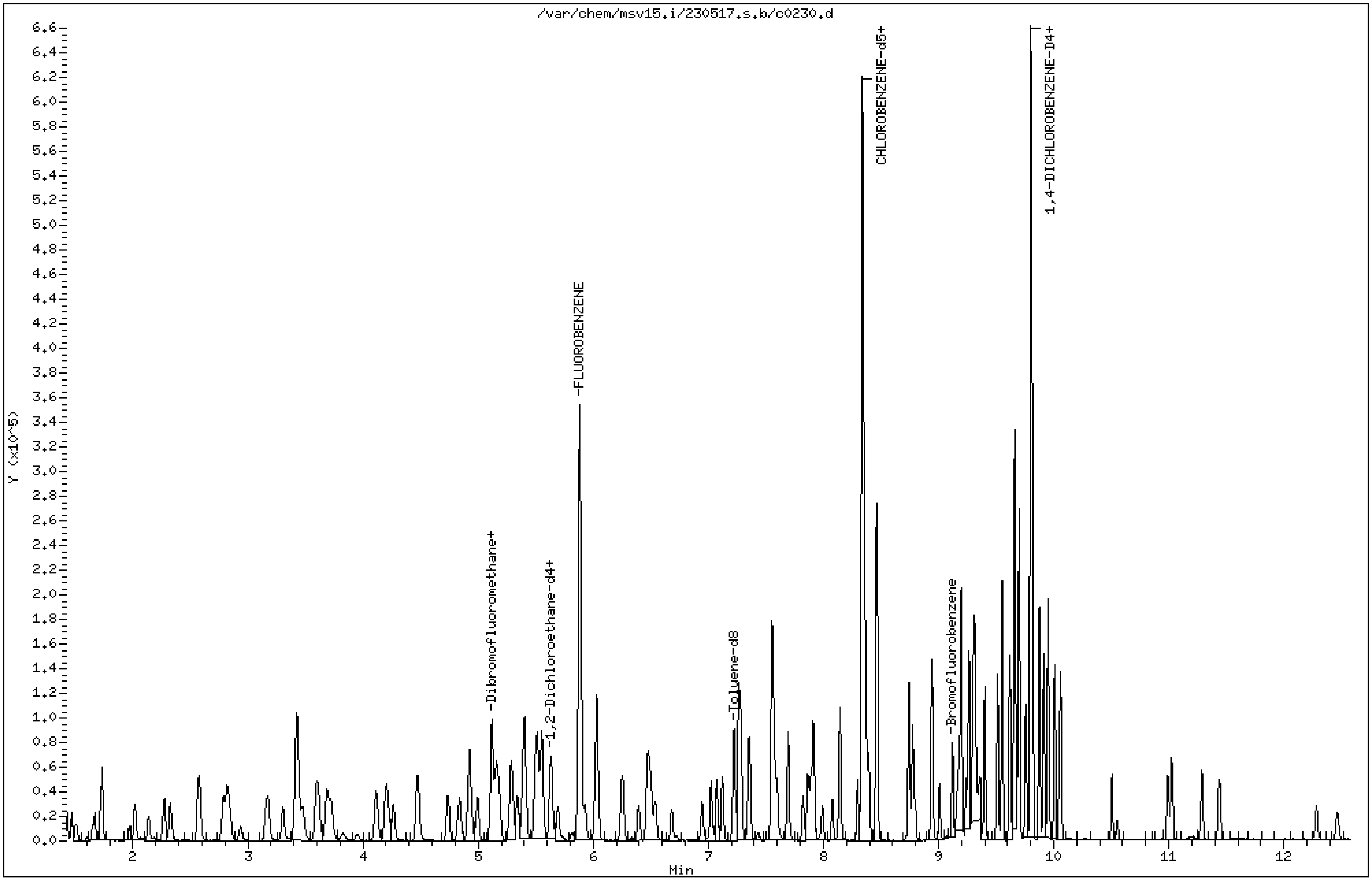
Sample Info: 1205*V15STD010

Purge Volume: 5.0

Operator: CJR

Column phase: RTX-VMS-30M

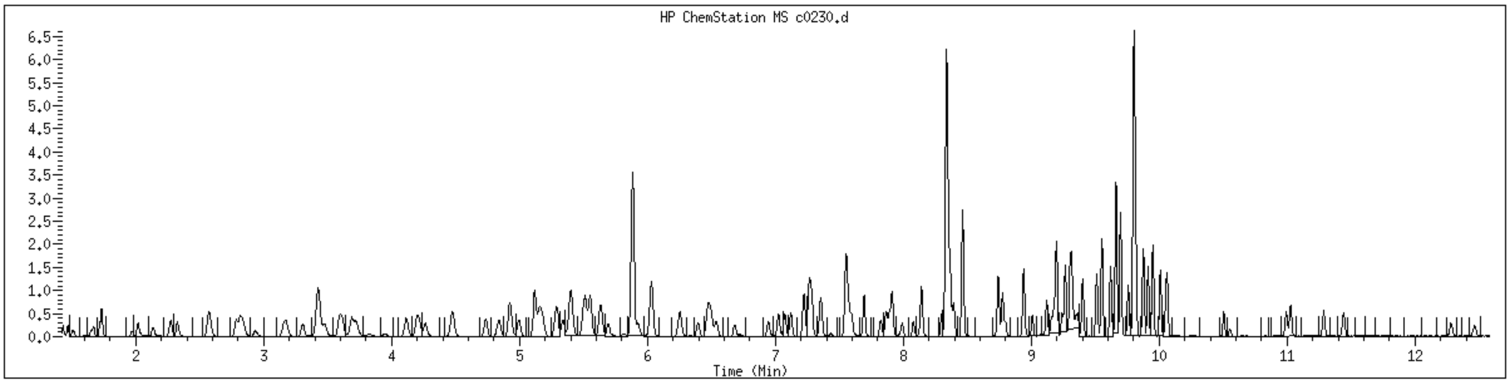
Column diameter: 0.25



Report Date: 05/23/2023 16:34

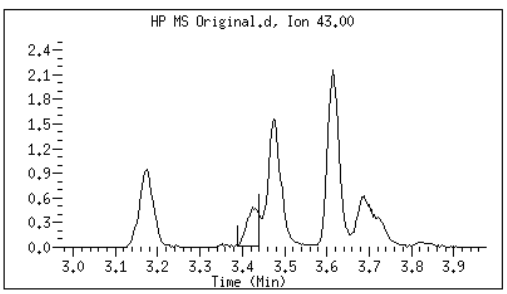
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1205
SampleType : CALIB_5
Injection Date: 05/17/2023 15:05
Instrument : msv15.i
Operator : CJR
Sample Info : 1205*V15STD010
Misc Info : MSV~527~*1*SMR
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE
Compound Sublist: 8260b+AppIXDOD



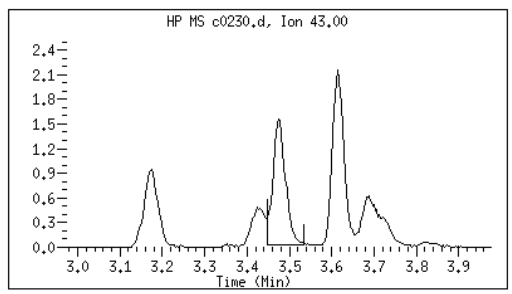
Original | Final

19 Acetone CAS#: 67-64-1 Reason: M3

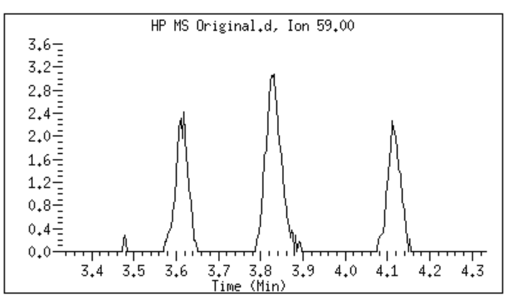


Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:09

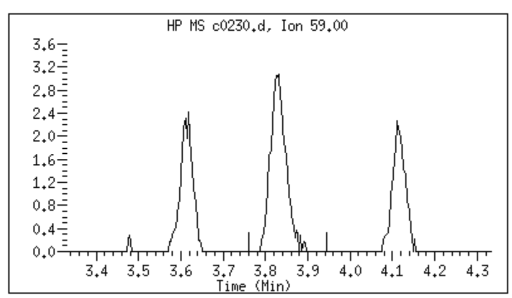


24 tert-Butyl Alcohol CAS#: 75-65-0 Reason: M1



Electronic Signature Applied

User: smr
Date: 05/17/2023 16:54



Original   Final

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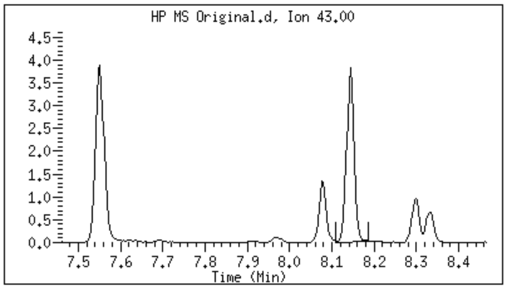
87 1-Nitropropane   CAS#: 108-03-2   Reason: M3



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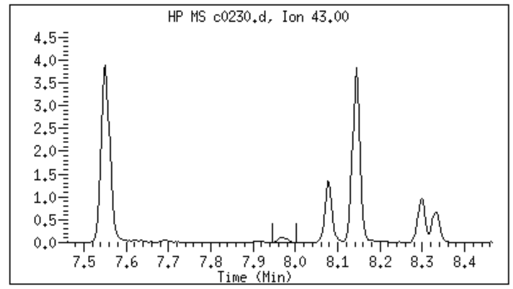
User: kmn
Date: 05/17/2023 16:09



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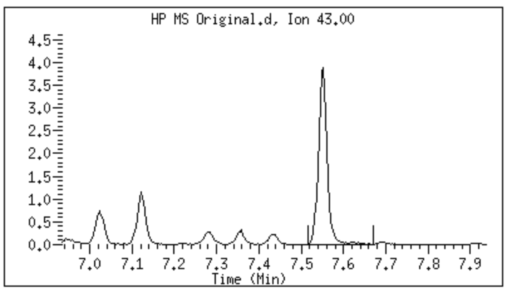
76 2-Nitropropane   CAS#: 79-46-9   Reason: M3



Electronic Signature
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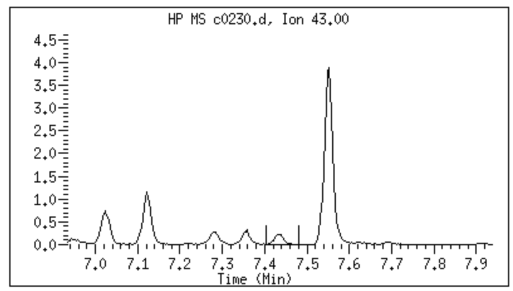
User: kmn
Date: 05/17/2023 16:09



M1 - Target system did not integrate
M3 - Target system integrated incorrect peak

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0231.d
Lab Smp Id: 1206 Client Smp ID: V15STD020
Inj Date : 17-MAY-2023 15:24
Operator : CJR Inst ID: msv15.i
Smp Info : 1206*V15STD020
Misc Info : MSV~527~*1*SMR
Comment :
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr Quant Type: ISTD
Cal Date : 17-MAY-2023 15:24 Cal File: c0231.d
Als bottle: 1 Calibration Sample, Level: 6
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	30602	20.0000	20.5	9579
2 Chloromethane ++	50	1.652	1.652	(0.281)	27979	20.0000	20.8	9764
3 Vinyl Chloride +	62	1.719	1.719	(0.292)	39220	20.0000	20.7	9633
5 1-3 Butadiene	39	1.735	1.735	(0.295)	30933	20.0000	22.0	9743
6 Bromomethane	94	2.018	2.018	(0.343)	31149	20.0000	19.6	9615
7 Chloroethane	64	2.137	2.137	(0.363)	28920	20.0000	20.8	9217
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	57432	20.0000	20.8	9556
9 Isoprene	67	2.571	2.571	(0.437)	46930	20.0000	20.6	4547
10 Ethyl Ether	59	2.587	2.587	(0.440)	22321	20.0000	20.7	8374
11 1,1-Dichloroethene +	96	2.787	2.787	(0.474)	31976	20.0000	20.4	8452
12 Carbon Disulfide	76	2.816	2.816	(0.479)	98564	20.0000	20.5	8107
13 1,1,2Trichlotrifluoroethane	101	2.835	2.835	(0.482)	35157	20.0000	20.7	9209
15 Methyl Iodide	142	2.938	2.938	(0.499)	37592	20.0000	21.0	9394
16 Acrolein	56	3.150	3.150	(0.535)	24838	100.000	104	6296

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
	17 Allyl chloride	39	3.304	3.304	(0.562)	30710	20.0000	22.3	9620
	18 Methylene Chloride	49	3.420	3.420	(0.581)	35774	20.0000	20.6	4893
	19 Acetone	43	3.472	3.472	(0.590)	73902	100.000	104	5274(H)
	20 trans-1,2-Dichloroethene	61	3.594	3.594	(0.611)	45226	20.0000	20.8	8649
	21 Methyl Acetate	43	3.616	3.616	(0.615)	92707	100.000	105	9190
	22 Hexane	57	3.690	3.690	(0.627)	66680	20.0000	21.3	9214
	23 MTBE	73	3.719	3.719	(0.632)	88622	20.0000	20.9	9656
	24 tert-Butyl Alcohol	59	3.829	3.829	(0.651)	19001	100.000	102	4488(M1)
	25 Acetonitrile	41	3.947	3.947	(0.671)	13943	100.000	111	9401
	26 Isopropyl Ether	45	4.118	4.118	(0.700)	79721	20.0000	21.1	9666
	27 Chloroprene	53	4.195	4.195	(0.713)	45592	20.0000	21.4	9048
	28 1,1-Dichloroethane ++	63	4.214	4.214	(0.716)	59997	20.0000	20.7	9338
	29 Acrylonitrile	53	4.263	4.263	(0.725)	52154	100.000	105	9726
	30 Vinyl Acetate	43	4.475	4.475	(0.761)	48754	20.0000	20.8	5654
	31 Ethyl Tert-butyl Ether	59	4.475	4.475	(0.761)	89151	20.0000	20.9	8078
	32 cis-1,2-Dichloroethene	61	4.738	4.738	(0.805)	43138	20.0000	20.7	9753
M	33 Total 1,2-Dichloroethene	61				88364	40.0000	41.5	0
	34 2,2-Dichloropropane	77	4.841	4.841	(0.823)	53854	20.0000	20.9	9473
	35 Bromochloromethane	128	4.922	4.922	(0.837)	17654	20.0000	21.5	9337
	36 Cyclohexane	56	4.928	4.928	(0.838)	56469	20.0000	21.3	9183
	37 Chloroform +	83	4.992	4.992	(0.849)	60968	20.0000	20.5	9692
	38 Ethyl Acetate	43	5.118	5.118	(0.870)	135862	100.000	105	5979
	41 sec-Butanol	45	5.176	5.176	(0.880)	1986	20.0000	20.2	3813(M2)
	39 Carbon Tetrachloride	117	5.121	5.121	(0.870)	48921	20.0000	20.4	8218
	40 Tetrahydrofuran	42	5.153	5.153	(0.876)	49466	100.000	102	8726
$	42 Dibromofluoromethane	111	5.156	5.156	(0.876)	32486	20.0000	20.5	7668
	43 1,1,1-Trichloroethane	97	5.182	5.182	(0.881)	57293	20.0000	20.8	9219
	44 2-Butanone	43	5.279	5.279	(0.897)	80056	100.000	108	6570
	45 1,1-Dichloropropene	75	5.291	5.291	(0.899)	50739	20.0000	20.8	9100
	46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	152803	20.0000	20.7	8849
	47 Propionitrile	54	5.529	5.529	(0.940)	19658	100.000	108	7888
	48 Benzene	78	5.517	5.517	(0.938)	136973	20.0000	20.7	9590
	49 Methylacrylonitrile	41	5.555	5.555	(0.944)	78410	100.000	105	7324
	56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	91354	20.0000	21.1	7198
$	50 1,2-Dichloroethane-d4	65	5.636	5.636	(0.958)	34198	20.0000	20.7	8488
	52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	43155	20.0000	20.8	9643
	51 Isobutyl Alcohol	43	5.726	5.726	(0.973)	4569	100.000	106	3872(H)
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	321153	50.0000		9858
	57 Trichloroethene	130	6.034	6.034	(1.026)	41878	20.0000	20.6	9179
	58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	65519	20.0000	21.1	8441
	59 Tert-amyl alcohol	59	6.250	6.250	(1.062)	66952	100.000	104	6639
	60 n-Butanol	56	6.336	6.336	(1.077)	2315	100.000	96.2	8107
	62 Dibromomethane	93	6.394	6.394	(1.087)	22119	20.0000	20.8	9312
	63 2-3 Dichloro-1-Proprene	75	6.465	6.465	(1.099)	53953	20.0000	20.3	9062
	64 1,2-Dichloropropane +	63	6.484	6.484	(1.102)	33848	20.0000	20.7	8182
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	45364	20.0000	20.5	9637
	66 Methyl methacrylate	41	6.684	6.684	(1.136)	23055	20.0000	20.4	9786

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
67 1,4- Dioxane	88	6.722	6.722	(1.143)	6140	500.000	507	8714
68 1-Bromo-2-chloroethane	63	6.947	6.947	(1.181)	43039	20.0000	20.7	9169
72 2-Chloroethyl vinyl ether	63	7.025	7.025	(1.194)	35793	100.000	80.0	9654
73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	54197	20.0000	20.7	9655
$ 74 Toluene-d8	98	7.221	7.221	(0.866)	121420	20.0000	20.8	9790
75 Toluene +	91	7.262	7.262	(0.871)	150767	20.0000	20.7	9583
76 2-Nitropropane	43	7.433	7.433	(0.891)	8113	20.0000	20.6	6241(M3)
77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	132162	100.000	99.5	8363
78 Tetrachloroethene	164	7.558	7.558	(0.906)	33221	20.0000	20.2	8128
79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	51089	20.0000	20.6	8725
80 Ethyl Methacrylate	41	7.693	7.693	(0.922)	21202	20.0000	20.9	7438
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	30378	20.0000	20.6	9445
82 Dibromochloromethane	129	7.825	7.825	(0.938)	34865	20.0000	20.8	9495
M 83 1-3 Dichloropropene total	100				105286	40.0000	41.3	0
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	39972	20.0000	20.6	9657
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	51849	20.0000	21.1	9495
87 1-Nitropropane	43	7.970	7.970	(0.956)	3514	20.0000	19.1	6124(M3)
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	32723	20.0000	20.7	9553
89 2-Hexanone	43	8.143	8.143	(0.976)	109907	100.000	101	9733
90 1-Chlorohexane	91	8.333	8.333	(0.999)	49908	20.0000	19.2	5571
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	124613	50.0000		9548
92 Chlorobenzene ++	112	8.349	8.349	(1.001)	98051	20.0000	20.4	6798
93 Ethylbenzene +	106	8.369	8.369	(1.003)	54089	20.0000	20.7	9039
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	31843	20.0000	19.7	9185
95 p,m-Xylene	106	8.462	8.462	(1.015)	130713	40.0000	41.3	9833
97 o-Xylene	106	8.741	8.741	(1.048)	61312	20.0000	20.8	9794
101 Styrene	104	8.777	8.777	(1.052)	78466	20.0000	20.7	9812
102 Bromoform ++	173	8.799	8.799	(1.055)	23618	20.0000	18.0	8549
104 Isopropylbenzene	105	8.941	8.941	(1.072)	162607	20.0000	21.1	9788
M 105 TOTAL XYLENE	106				192025	60.0000	62.1	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	37087	20.0000	19.6	9512
107 cis-1,4-dichloro-2-butene	53	9.156	9.156	(0.934)	12077	20.0000	21.1	9554
109 n-Propylbenzene	91	9.198	9.198	(0.938)	190472	20.0000	21.7	9186
108 Bromobenzene	77	9.192	9.192	(0.938)	61947	20.0000	21.2	9231
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.943)	37146	20.0000	21.6	9635
112 2-Chlorotoluene	91	9.301	9.301	(0.949)	119264	20.0000	21.3	7837
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	127607	20.0000	21.6	7926
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	49542	20.0000	21.3	8317
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	12313	20.0000	21.9	9166
116 Cyclohexanone	55	9.365	9.365	(0.955)	31285	100.000	111	9750
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	108488	20.0000	21.6	9724
118 tert-butylbenzene	91	9.513	9.513	(0.970)	71451	20.0000	21.5	9781
119 1,2,4-Trimethylbenzene	105	9.552	9.552	(0.974)	125754	20.0000	21.7	9318
120 sec-Butylbenzene	105	9.619	9.619	(0.981)	166584	20.0000	21.8	9815
121 p-Isopropyltoluene	119	9.700	9.700	(0.990)	144696	20.0000	21.4	8803
122 Dicyclopentadiene	66	9.700	9.700	(0.990)	153772	20.0000	21.9	7930
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.996)	70260	20.0000	20.5	9588

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.803	9.803	(1.000)	112488	50.0000		9420
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	72969	20.0000	20.8	7603(H)
126 n-Butylbenzene	91	9.950	9.950	(1.015)	193138	20.0000	21.5	8931
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	63555	20.0000	20.5	9518
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	8227	20.0000	19.5	9153
130 Hexachlorobutadiene	225	10.992	10.992	(1.121)	19628	20.0000	20.8	9018
131 1,2,4-Trichlorobenzene	180	11.028	11.028	(1.125)	38876	20.0000	20.6	9350
132 Naphthalene	128	11.288	11.288	(1.152)	93547	20.0000	21.1	9762
133 1,2,3-Trichlorobenzene	180	11.442	11.442	(1.167)	33962	20.0000	20.8	9324

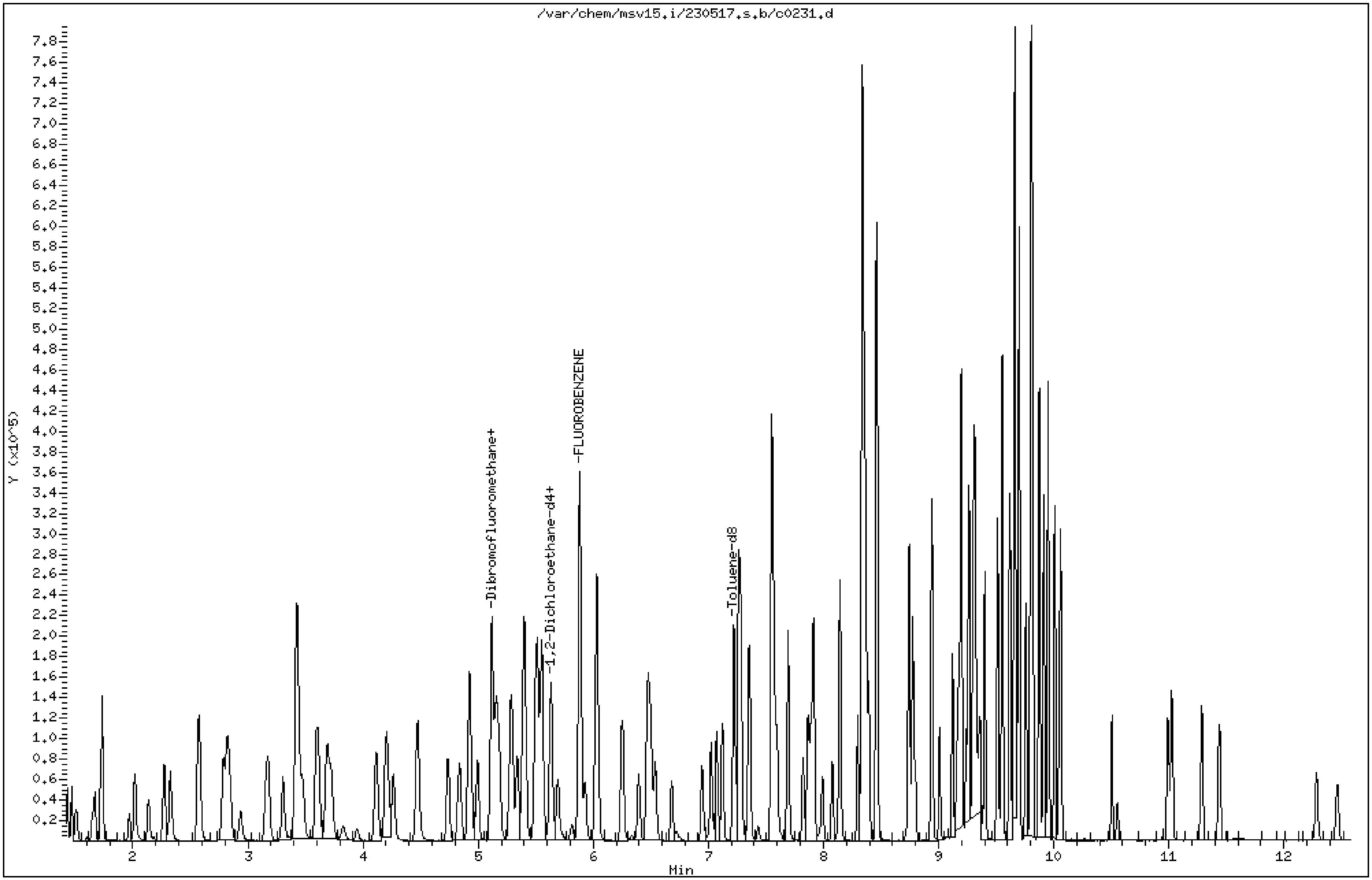
QC Flag Legend

M1- Compound response manually integrated because
Target system did not integrate.
M2- Compound response manually integrated because
Target system integrated incorrectly.
M3- Compound response manually integrated because
Target system integrated incorrect peak.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0231.d
Date : 17-MAY-2023 15:24
Client ID: V15STD020
Sample Info: 1206*V15STD020
Purge Volume: 5.0
Column phase: RTX-VMS-30M

Page 1

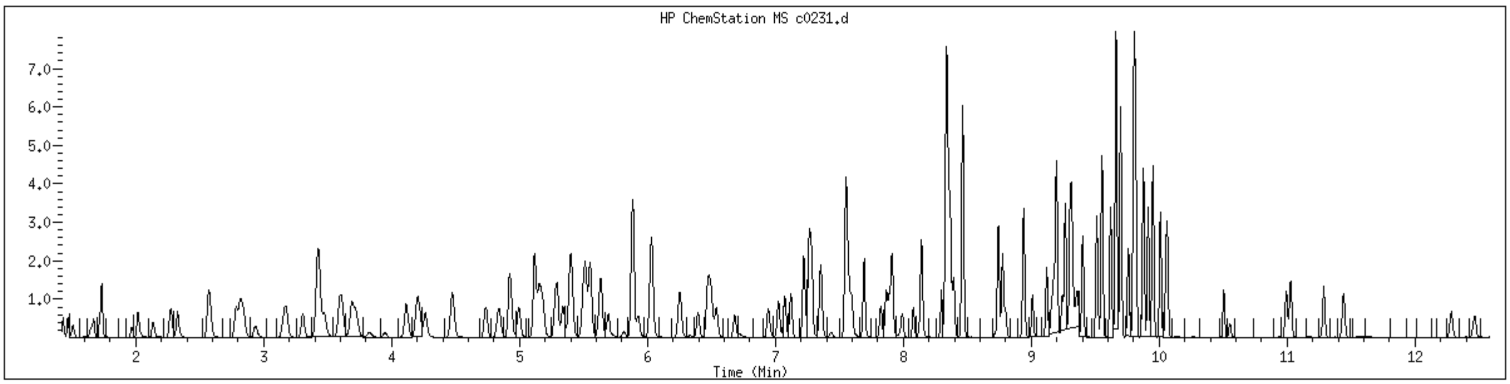
Instrument: msv15.i
Operator: CJR
Column diameter: 0.25



Report Date: 05/23/2023 16:34

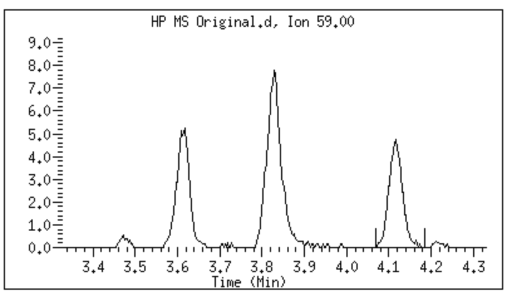
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1206 SampleType : CALIB_6
Injection Date: 05/17/2023 15:24 Instrument : msv15.i
Operator : CJR
Sample Info : 1206*V15STD020
Misc Info : MSV~527~*1*SMR
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260b+AppIXDOD



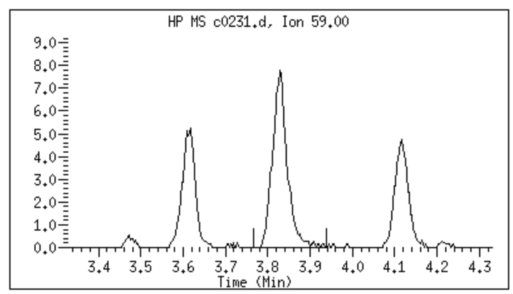
Original Final

24 tert-Butyl Alcohol CAS#: 75-65-0 Reason: M1

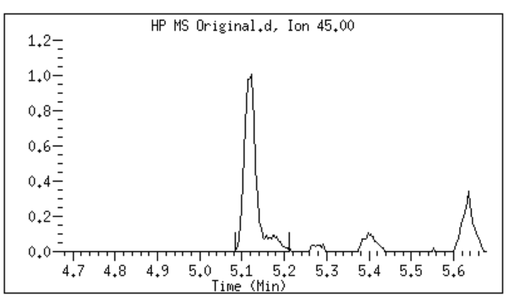


Electronic Signature Applied

User: smr
Date: 05/17/2023 16:54

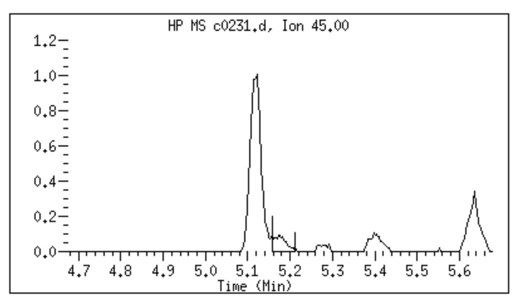


41 sec-Butanol CAS#: 78-92-2 Reason: M2



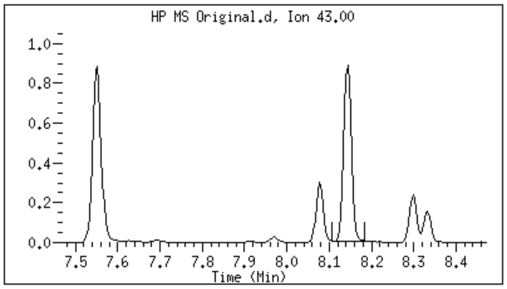
Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:07



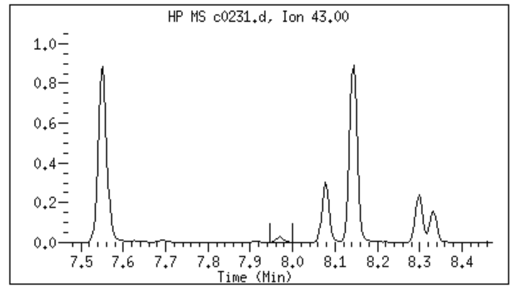
Original Final

87 1-Nitropropane CAS#: 108-03-2 Reason: M3

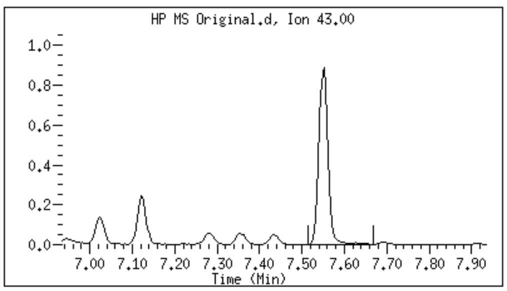


Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:08

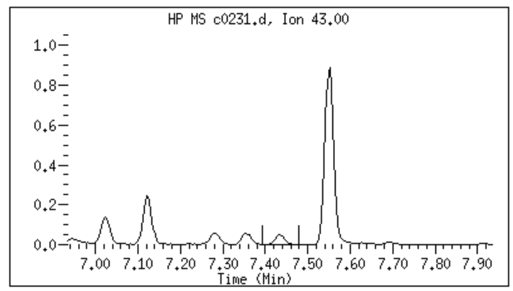


76 2-Nitropropane CAS#: 79-46-9 Reason: M3



Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:08



M1 - Target system did not integrate
M2 - Target system integrated incorrectly
M3 - Target system integrated incorrect peak

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0232.d
Lab Smp Id: 1207   Client Smp ID: V15STD050
Inj Date : 17-MAY-2023 15:43
Operator : CJR   Inst ID: msv15.i
Smp Info : 1207*V15STD050
Misc Info : MSV~527~*1*SMR
Comment   :
Method   : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr   Quant Type: ISTD
Cal Date : 17-MAY-2023 15:43   Cal File: c0232.d
Als bottle: 1   Calibration Sample, Level: 7
Dil Factor: 1.00000
Integrator: HP RTE   Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable   Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	78410	50.0000	50.7	9584
2 Chloromethane ++	50	1.649	1.649	(0.280)	69968	50.0000	50.3	9752
3 Vinyl Chloride +	62	1.716	1.716	(0.292)	99748	50.0000	50.9	9767
5 1-3 Butadiene	39	1.735	1.735	(0.295)	71997	50.0000	49.3	9711
6 Bromomethane	94	2.015	2.015	(0.343)	88397	50.0000	53.8	9715
7 Chloroethane	64	2.134	2.134	(0.363)	74021	50.0000	51.4	9207
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	148578	50.0000	52.0	9605
9 Isoprene	67	2.568	2.568	(0.437)	123576	50.0000	52.3	4543
10 Ethyl Ether	59	2.584	2.584	(0.439)	59003	50.0000	52.7	8054
11 1,1-Dichloroethene +	96	2.784	2.784	(0.473)	84852	50.0000	52.2	8264
12 Carbon Disulfide	76	2.816	2.816	(0.479)	253765	50.0000	51.0	7974
13 1,1,2Trichlotrifluoroethane	101	2.835	2.835	(0.482)	92398	50.0000	52.4	9297
15 Methyl Iodide	142	2.938	2.938	(0.499)	102973	50.0000	52.9	9533
16 Acrolein	56	3.150	3.150	(0.535)	65649	250.000	265	5940

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
17 Allyl chloride	39	3.304	3.304	(0.562)	79708	50.0000	55.8	9726
18 Methylene Chloride	49	3.420	3.420	(0.581)	91561	50.0000	50.9	4726
19 Acetone	43	3.472	3.472	(0.590)	194463	250.000	263	5202(H)
20 trans-1,2-Dichloroethene	61	3.591	3.591	(0.610)	118341	50.0000	52.5	8758
21 Methyl Acetate	43	3.613	3.613	(0.614)	241456	250.000	263	9514
22 Hexane	57	3.687	3.687	(0.627)	172742	50.0000	53.2	9240
23 MTBE	73	3.722	3.722	(0.633)	233809	50.0000	53.2	9696
24 tert-Butyl Alcohol	59	3.828	3.828	(0.651)	49824	250.000	257	4397(M1)
25 Acetonitrile	41	3.944	3.944	(0.670)	34560	250.000	264	9591
26 Isopropyl Ether	45	4.115	4.115	(0.699)	208071	50.0000	53.1	9642
27 Chloroprene	53	4.198	4.198	(0.714)	115317	50.0000	52.2	8500
28 1,1-Dichloroethane ++	63	4.218	4.218	(0.717)	157635	50.0000	52.4	9536
29 Acrylonitrile	53	4.259	4.259	(0.724)	131821	250.000	257	9793
30 Vinyl Acetate	43	4.475	4.475	(0.761)	125423	50.0000	51.7	5643
31 Ethyl Tert-butyl Ether	59	4.475	4.475	(0.761)	233451	50.0000	52.9	8063
32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	111972	50.0000	51.8	9712
M 33 Total 1,2-Dichloroethene	61				230313	100.000	104	0
34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	138874	50.0000	51.9	9556
35 Bromochloromethane	128	4.918	4.918	(0.836)	43994	50.0000	51.8	9323
36 Cyclohexane	56	4.928	4.928	(0.838)	143772	50.0000	52.4	9392
37 Chloroform +	83	4.996	4.996	(0.849)	160796	50.0000	52.2	9820
38 Ethyl Acetate	43	5.115	5.115	(0.869)	355035	250.000	264	5963
41 sec-Butanol	45	5.172	5.172	(0.879)	5762	50.0000	56.5	4882
39 Carbon Tetrachloride	117	5.121	5.121	(0.870)	131370	50.0000	52.8	8278
40 Tetrahydrofuran	42	5.144	5.144	(0.874)	125936	250.000	251	9684
$ 42 Dibromofluoromethane	111	5.160	5.160	(0.877)	86430	50.0000	52.6	8697
43 1,1,1-Trichloroethane	97	5.182	5.182	(0.881)	149626	50.0000	52.5	9183
44 2-Butanone	43	5.275	5.275	(0.897)	205452	250.000	268	5939
45 1,1-Dichloropropene	75	5.291	5.291	(0.899)	133125	50.0000	52.5	9458
46 2,2,4 Trimethylpentane	57	5.398	5.398	(0.917)	413577	50.0000	54.0	8947
47 Propionitrile	54	5.533	5.533	(0.940)	50566	250.000	267	8629
48 Benzene	78	5.513	5.513	(0.937)	354755	50.0000	51.8	9352
49 Methylacrylonitrile	41	5.552	5.552	(0.944)	203546	250.000	262	7394
56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	240391	50.0000	53.4	7175
$ 50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	86756	50.0000	50.7	8536
52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	110787	50.0000	51.4	9778
51 Isobutyl Alcohol	43	5.726	5.726	(0.973)	11454	250.000	255	3975(H)
* 55 FLUOROBENZENE	96	5.883	5.883	(1.000)	332949	50.0000		9863
57 Trichloroethene	130	6.031	6.031	(1.025)	110781	50.0000	52.7	8874
58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	173750	50.0000	54.0	8337
59 Tert-amyl alcohol	59	6.253	6.253	(1.063)	179846	250.000	270	6492
60 n-Butanol	56	6.333	6.333	(1.077)	6770	250.000	254	8815
62 Dibromomethane	93	6.394	6.394	(1.087)	58579	50.0000	53.1	9370
63 2-3 Dichloro-1-Proprene	75	6.465	6.465	(1.099)	147136	50.0000	53.4	9187
64 1,2-Dichloropropane +	63	6.484	6.484	(1.102)	88787	50.0000	52.4	8237
65 Bromodichloromethane	83	6.539	6.539	(1.111)	121677	50.0000	53.1	9706
66 Methyl methacrylate	41	6.680	6.680	(1.136)	61021	50.0000	52.1	9763

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
	67 1,4- Dioxane	88	6.722	6.722	(1.143)	14798	1250.00	1180	9142
	68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	112416	50.0000	52.1	9201
	72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	105023	250.000	218	9726
	73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	142505	50.0000	52.5	9741
$	74 Toluene-d8	98	7.221	7.221	(0.866)	316629	50.0000	52.2	9835
	75 Toluene +	91	7.262	7.262	(0.871)	394310	50.0000	52.3	9567
	76 2-Nitropropane	43	7.433	7.433	(0.891)	19389	50.0000	46.7	7539(M3)
	77 4-methyl-2-pentanone	43	7.549	7.549	(0.905)	343899	250.000	245	7731
	78 Tetrachloroethene	164	7.558	7.558	(0.906)	87675	50.0000	51.4	8245
	79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	135976	50.0000	52.8	8844
	80 Ethyl Methacrylate	41	7.693	7.693	(0.922)	54743	50.0000	51.9	7514
	81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	80250	50.0000	52.4	9466
	82 Dibromochloromethane	129	7.825	7.825	(0.938)	94202	50.0000	54.0	9607
M	83 1-3 Dichloropropene total	100				278481	100.000	105	0
	84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	109440	50.0000	54.3	9652
	86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	133214	50.0000	52.2	9499
	87 1-Nitropropane	43	7.970	7.970	(0.956)	9402	50.0000	49.3	6046(M3)
	88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	86960	50.0000	53.0	9621
	89 2-Hexanone	43	8.143	8.143	(0.976)	287469	250.000	249	9714
	90 1-Chlorohexane	91	8.330	8.330	(0.999)	147569	50.0000	54.7	8330
*	91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	129315	50.0000		9070
	92 Chlorobenzene ++	112	8.352	8.352	(1.002)	257209	50.0000	51.5	9673
	93 Ethylbenzene +	106	8.368	8.368	(1.003)	142956	50.0000	52.6	8968
	94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	85304	50.0000	50.8	9187
	95 p,m-Xylene	106	8.462	8.462	(1.015)	345906	100.000	105	9859
	97 o-Xylene	106	8.745	8.745	(1.049)	162009	50.0000	53.0	9868
	101 Styrene	104	8.777	8.777	(1.052)	215667	50.0000	54.8	9851
	102 Bromoform ++	173	8.799	8.799	(1.055)	64825	50.0000	46.1	8519
	104 Isopropylbenzene	105	8.941	8.941	(1.072)	426261	50.0000	53.3	9857
M	105 TOTAL XYLENE	106				507915	150.000	158	0
$	106 Bromofluorobenzene	174	9.124	9.124	(1.094)	98372	50.0000	50.0	9556
	107 cis-1,4-dichloro-2-butene	53	9.156	9.156	(0.934)	32211	50.0000	52.7	9669
	109 n-Propylbenzene	91	9.198	9.198	(0.938)	501136	50.0000	53.3	9127
	108 Bromobenzene	77	9.192	9.192	(0.938)	160928	50.0000	51.4	9297
	110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.943)	97142	50.0000	52.9	9701
	112 2-Chlorotoluene	91	9.301	9.301	(0.949)	312075	50.0000	52.0	7880
	113 1,3,5-Trimethylbenzene	105	9.314	9.314	(0.950)	334535	50.0000	52.9	8628
	114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	129079	50.0000	51.8	8535
	115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	31798	50.0000	52.4	9147
	116 Cyclohexanone	55	9.362	9.362	(0.955)	80908	250.000	266	9741
	117 4-Chlorotoluene	91	9.401	9.401	(0.959)	279122	50.0000	51.9	9711
	118 tert-butylbenzene	91	9.513	9.513	(0.970)	188790	50.0000	53.1	9779
	119 1,2,4-Trimethylbenzene	105	9.552	9.552	(0.974)	332703	50.0000	53.6	9348
	120 sec-Butylbenzene	105	9.619	9.619	(0.981)	439395	50.0000	53.7	9863
	121 p-Isopropyltoluene	119	9.700	9.700	(0.990)	393228	50.0000	54.3	8839
	122 Dicyclopentadiene	66	9.696	9.696	(0.989)	405727	50.0000	53.9	8017
	123 1,3-Dichlorobenzene	146	9.761	9.761	(0.996)	189196	50.0000	51.7	9736

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.803	9.803	(1.000)	120452	50.0000		9013
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	194889	50.0000	51.8	9052
126 n-Butylbenzene	91	9.950	9.950	(1.015)	523927	50.0000	54.4	8921
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	173109	50.0000	52.3	9590
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	22062	50.0000	47.6	9460
130 Hexachlorobutadiene	225	10.995	10.995	(1.122)	53170	50.0000	52.5	9129
131 1,2,4-Trichlorobenzene	180	11.028	11.028	(1.125)	108954	50.0000	53.8	9480
132 Naphthalene	128	11.285	11.285	(1.151)	255601	50.0000	53.9	9810
133 1,2,3-Trichlorobenzene	180	11.442	11.442	(1.167)	92445	50.0000	52.9	9351

QC Flag Legend

M1- Compound response manually integrated because
Target system did not integrate.
M3- Compound response manually integrated because
Target system integrated incorrect peak.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0232.d

Page 1

Date : 17-MAY-2023 15:43

Client ID: V15STD050

Instrument: msv15.i

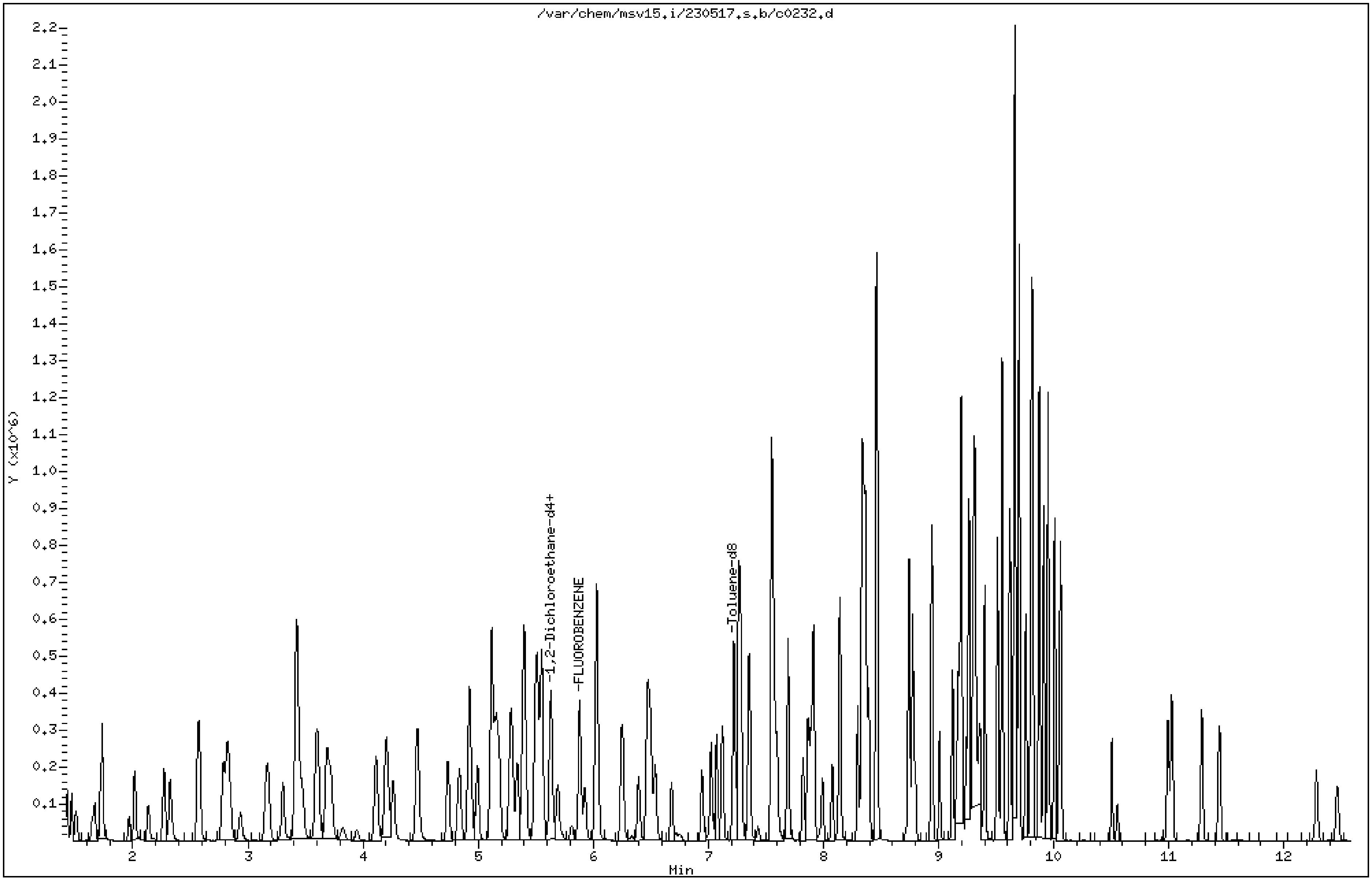
Sample Info: 1207*V15STD050

Purge Volume: 5.0

Operator: CJR

Column phase: RTX-VMS-30M

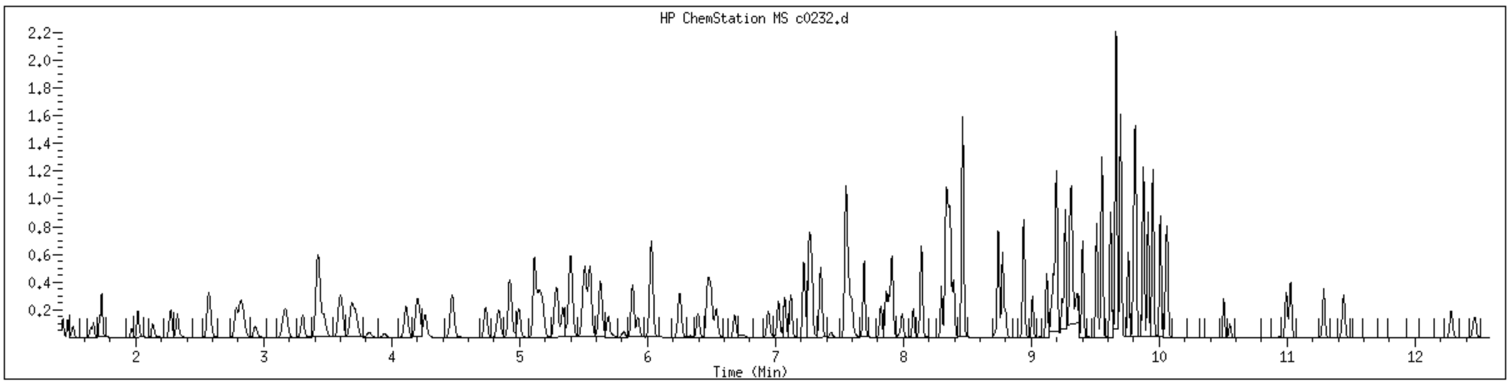
Column diameter: 0.25



Data file : /var/chem/msv15.i/230517.s.b/c0232.d   Page: 1
Report Date: 05/23/2023 16:34

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 1207   SampleType   : CALIB_7
Injection Date: 05/17/2023 15:43   Instrument   : msv15.i
Operator   : CJR
Sample Info   : 1207*V15STD050
Misc Info   : MSV~527~*1*SMR
Method   : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260b+AppIXDOD



Original   Final

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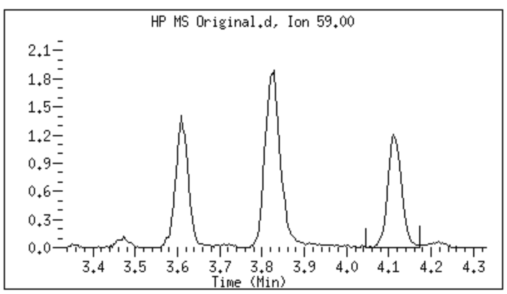
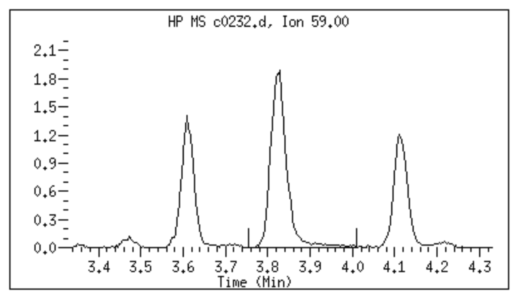
24 tert-Butyl Alcohol   CAS#: 75-65-0   Reason: M1



Electronic Signature
Applied

User: smr
Date: 05/17/2023 16:54



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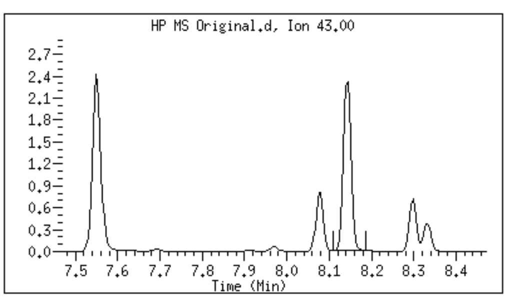
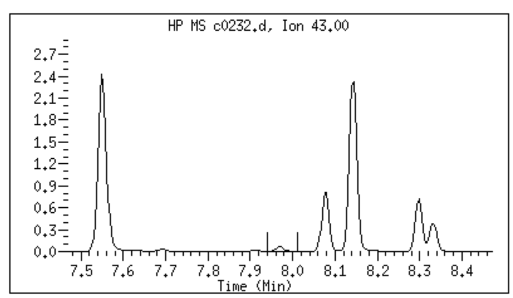
87 1-Nitropropane   CAS#: 108-03-2   Reason: M3



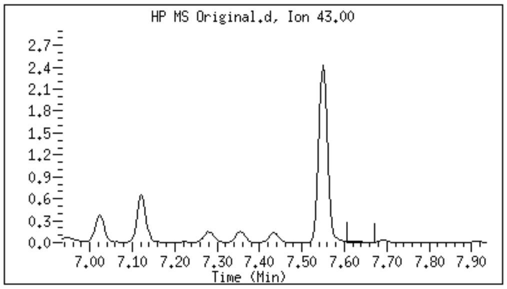
Electronic Signature
Applied

User: kmn
Date: 05/17/2023 16:06



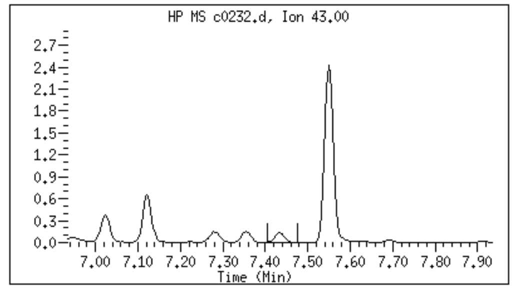
Original | Final

76 2-Nitropropane   CAS#: 79-46-9   Reason: M3



Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:05



M1 - Target system did not integrate
M3 - Target system integrated incorrect peak

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0233.d
Lab Smp Id: 1208 Client Smp ID: V15STD100
Inj Date : 17-MAY-2023 16:01
Operator : CJR Inst ID: msv15.i
Smp Info : 1208*V15STD100
Misc Info : MSV~527~*1*SMR
Comment :
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr Quant Type: ISTD
Cal Date : 17-MAY-2023 16:01 Cal File: c0233.d
Als bottle: 1 Calibration Sample, Level: 8
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	172031	100.000	109	9731
2 Chloromethane ++	50	1.652	1.652	(0.281)	147309	100.000	103	9792
3 Vinyl Chloride +	62	1.716	1.716	(0.292)	208727	100.000	104	9708
5 1-3 Butadiene	39	1.735	1.735	(0.295)	150376	100.000	101	9651
6 Bromomethane	94	2.015	2.015	(0.342)	198723	100.000	118	9684
7 Chloroethane	64	2.131	2.131	(0.362)	153830	100.000	104	9179
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	311532	100.000	106	9663
9 Isoprene	67	2.568	2.568	(0.436)	262217	100.000	108	4561
10 Ethyl Ether	59	2.584	2.584	(0.439)	124721	100.000	109	8162
11 1,1-Dichloroethene +	96	2.787	2.787	(0.474)	177451	100.000	107	8490
12 Carbon Disulfide	76	2.815	2.815	(0.479)	526778	100.000	103	7993
13 1,1,2Trichlotrifluoroethane	101	2.835	2.835	(0.482)	193690	100.000	107	9285
15 Methyl Iodide	142	2.934	2.934	(0.499)	212199	100.000	105	9496
16 Acrolein	56	3.150	3.150	(0.535)	137707	500.000	543	5928

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
	17 Allyl chloride	39	3.304	3.304	(0.562)	157006	100.000	107	9615
	18 Methylene Chloride	49	3.417	3.417	(0.581)	188254	100.000	102	5029
	19 Acetone	43	3.468	3.468	(0.590)	426163	500.000	563	5381(H)
	20 trans-1,2-Dichloroethene	61	3.590	3.590	(0.610)	243290	100.000	105	8636
	21 Methyl Acetate	43	3.613	3.613	(0.614)	507232	500.000	538	9509
	22 Hexane	57	3.687	3.687	(0.627)	353751	100.000	106	9204
	23 MTBE	73	3.716	3.716	(0.632)	494605	100.000	110	9663
	24 tert-Butyl Alcohol	59	3.825	3.825	(0.650)	110597	500.000	557	4363(M1)
	25 Acetonitrile	41	3.944	3.944	(0.670)	71541	500.000	534	9484
	26 Isopropyl Ether	45	4.111	4.111	(0.699)	430561	100.000	107	9614
	27 Chloroprene	53	4.195	4.195	(0.713)	239972	100.000	106	8963
	28 1,1-Dichloroethane ++	63	4.217	4.217	(0.717)	323020	100.000	105	9477
	29 Acrylonitrile	53	4.262	4.262	(0.725)	288059	500.000	548	9893
	30 Vinyl Acetate	43	4.471	4.471	(0.760)	262383	100.000	105	5563
	31 Ethyl Tert-butyl Ether	59	4.471	4.471	(0.760)	490115	100.000	108	8076
	32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	228949	100.000	103	9760
M	33 Total 1,2-Dichloroethene	61				472239	200.000	209	0
	34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	287486	100.000	105	9526
	35 Bromochloromethane	128	4.921	4.921	(0.837)	84873	100.000	97.5	8900
	36 Cyclohexane	56	4.928	4.928	(0.838)	297498	100.000	106	9469
	37 Chloroform +	83	4.992	4.992	(0.849)	327993	100.000	104	9797
	38 Ethyl Acetate	43	5.114	5.114	(0.869)	761216	500.000	552	5982
	41 sec-Butanol	45	5.169	5.169	(0.879)	10464	100.000	100	4608(M2)
	39 Carbon Tetrachloride	117	5.121	5.121	(0.870)	277815	100.000	109	8131
	40 Tetrahydrofuran	42	5.147	5.147	(0.875)	269211	500.000	524	9634
$	42 Dibromofluoromethane	111	5.156	5.156	(0.876)	180094	100.000	107	8340
	43 1,1,1-Trichloroethane	97	5.179	5.179	(0.880)	316198	100.000	108	9662
	44 2-Butanone	43	5.275	5.275	(0.897)	434935	500.000	554	5885
	45 1,1-Dichloropropene	75	5.294	5.294	(0.900)	269553	100.000	104	9464
	46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	872671	100.000	111	8987
	47 Propionitrile	54	5.536	5.536	(0.941)	107416	500.000	554	8501
	48 Benzene	78	5.516	5.516	(0.938)	747133	100.000	106	9713
	49 Methylacrylonitrile	41	5.552	5.552	(0.944)	438552	500.000	550	7375
	56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	514030	100.000	112	7143
$	50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	181227	100.000	103	8566
	52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	228401	100.000	103	9796
	51 Isobutyl Alcohol	43	5.725	5.725	(0.973)	24027	500.000	523	4027(H)
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	341246	50.0000		9856
	57 Trichloroethene	130	6.034	6.034	(1.026)	235220	100.000	109	9160
	58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	358295	100.000	109	8253
	59 Tert-amyl alcohol	59	6.249	6.249	(1.062)	377263	500.000	552	6377
	60 n-Butanol	56	6.336	6.336	(1.077)	13877	500.000	500	9175
	62 Dibromomethane	93	6.394	6.394	(1.087)	122260	100.000	108	9423
	63 2-3 Dichloro-1-Proprene	75	6.468	6.468	(1.099)	305492	100.000	108	8952
	64 1,2-Dichloropropane +	63	6.481	6.481	(1.102)	182874	100.000	105	7750
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	254840	100.000	108	9785
	66 Methyl methacrylate	41	6.680	6.680	(1.136)	133664	100.000	111	9798

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
67 1,4- Dioxane	88	6.719	6.719	(1.142)	31682	2500.00	2460	9397
68 1-Bromo-2-chloroethane	63	6.947	6.947	(1.181)	235395	100.000	106	9300
72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	237890	500.000	477	9722
73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	293449	100.000	105	9773
$ 74 Toluene-d8	98	7.224	7.224	(0.866)	670484	100.000	105	9863
75 Toluene +	91	7.262	7.262	(0.871)	832367	100.000	105	9644
76 2-Nitropropane	43	7.433	7.433	(0.891)	44203	100.000	100	9543
77 4-methyl-2-pentanone	43	7.548	7.548	(0.905)	740988	500.000	500	7634
78 Tetrachloroethene	164	7.558	7.558	(0.906)	193548	100.000	108	8247
79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	289421	100.000	110	8816
80 Ethyl Methacrylate	41	7.693	7.693	(0.922)	115542	100.000	104	7595
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	170515	100.000	106	9445
82 Dibromochloromethane	129	7.825	7.825	(0.938)	205410	100.000	112	9607
M 83 1-3 Dichloropropene total	100				582870	200.000	215	0
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	236353	100.000	111	9615
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	274252	100.000	102	9513
87 1-Nitropropane	43	7.966	7.966	(0.955)	22429	100.000	112	9565
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	186234	100.000	108	9676
89 2-Hexanone	43	8.143	8.143	(0.976)	616069	500.000	505	9678
90 1-Chlorohexane	91	8.333	8.333	(0.999)	290313	100.000	102	9047
* 91 CHLOROBENZENE-d5	82	8.339	8.339	(1.000)	136082	50.0000		8492
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	563797	100.000	107	9208
93 Ethylbenzene +	106	8.368	8.368	(1.003)	308926	100.000	108	8989
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	187313	100.000	106	9173
95 p,m-Xylene	106	8.465	8.465	(1.015)	757967	200.000	219	9923
97 o-Xylene	106	8.744	8.744	(1.049)	344807	100.000	107	9910
101 Styrene	104	8.777	8.777	(1.052)	467692	100.000	113	9871
102 Bromoform ++	173	8.799	8.799	(1.055)	146620	100.000	98.1	8533
104 Isopropylbenzene	105	8.944	8.944	(1.072)	905107	100.000	107	9904
M 105 TOTAL XYLENE	106				1102774	300.000	327	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	209299	100.000	101	9636
107 cis-1,4-dichloro-2-butene	53	9.156	9.156	(0.934)	69381	100.000	102	9591(M1)
109 n-Propylbenzene	91	9.201	9.201	(0.939)	1044472	100.000	100	8574
108 Bromobenzene	77	9.191	9.191	(0.938)	338565	100.000	97.5	9005
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.943)	204844	100.000	101	9750
112 2-Chlorotoluene	91	9.301	9.301	(0.949)	663599	100.000	99.8	7671
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	734421	100.000	105	8034
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	274378	100.000	99.2	8060
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	67695	100.000	100	9296
116 Cyclohexanone	55	9.365	9.365	(0.955)	170231	500.000	503	9679
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	600124	100.000	101	9858
118 tert-butylbenzene	91	9.513	9.513	(0.970)	395010	100.000	100	9883
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	722324	100.000	105	9387
120 sec-Butylbenzene	105	9.622	9.622	(0.982)	924935	100.000	102	9859
121 p-Isopropyltoluene	119	9.699	9.699	(0.990)	843229	100.000	105	8905
122 Dicyclopentadiene	66	9.699	9.699	(0.990)	834113	100.000	99.9	7872
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.996)	417578	100.000	103	9817

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.802	(1.000)	133580	50.0000		8572
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	427974	100.000	102	9389
126 n-Butylbenzene	91	9.953	9.953	(1.015)	1130200	100.000	106	8876
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	384022	100.000	105	9684
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	51173	100.000	98.6	9553
130 Hexachlorobutadiene	225	10.995	10.995	(1.122)	119125	100.000	106	9234
131 1,2,4-Trichlorobenzene	180	11.027	11.027	(1.125)	237015	100.000	106	9503
132 Naphthalene	128	11.288	11.288	(1.152)	581316	100.000	110	9846
133 1,2,3-Trichlorobenzene	180	11.439	11.439	(1.167)	209737	100.000	108	9447

QC Flag Legend

M1- Compound response manually integrated because
Target system did not integrate.
M2- Compound response manually integrated because
Target system integrated incorrectly.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0233.d

Page 1

Date : 17-MAY-2023 16:01

Client ID: V15STD100

Instrument: msv15.i

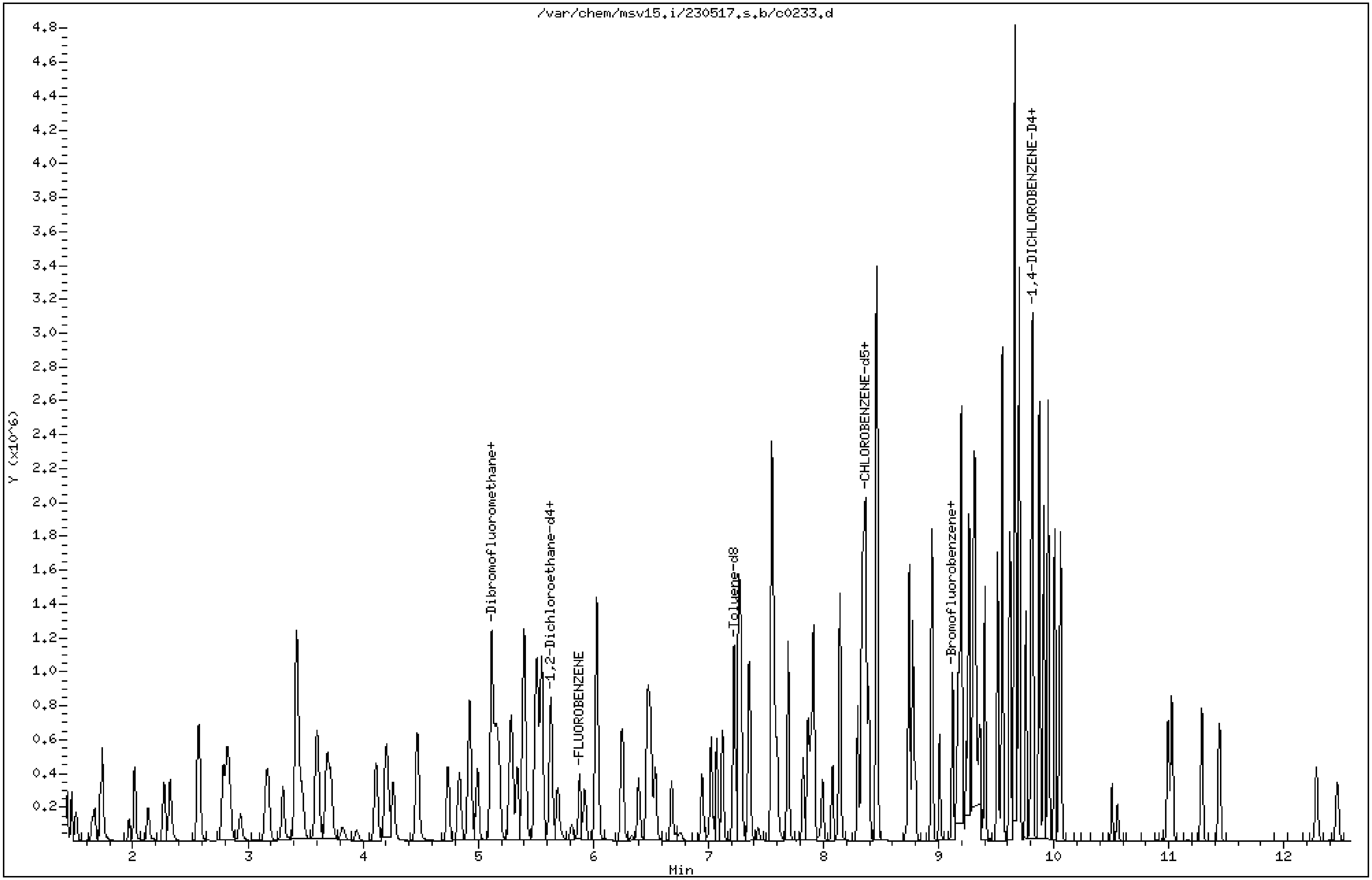
Sample Info: 1208*V15STD100

Purge Volume: 5.0

Operator: CJR

Column phase: RTX-VMS-30M

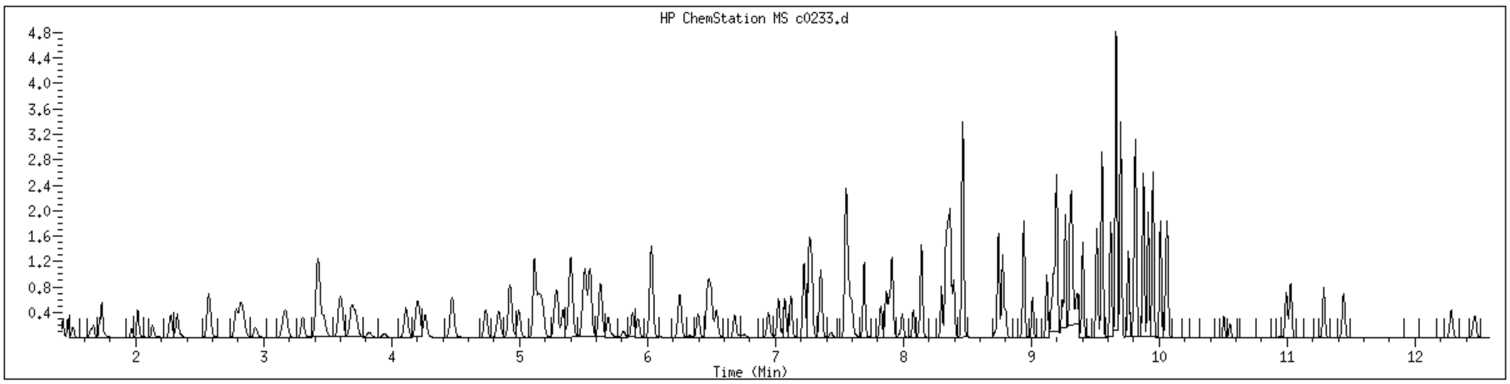
Column diameter: 0.25



Data file : /var/chem/msv15.i/230517.s.b/c0233.d   Page: 1
Report Date: 05/23/2023 16:34

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 1208   SampleType   : CALIB_8
Injection Date: 05/17/2023 16:01   Instrument   : msv15.i
Operator   : CJR
Sample Info   : 1208*V15STD100
Misc Info   : MSV~527~*1*SMR
Method   : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260b+AppIXDOD



Original   Final

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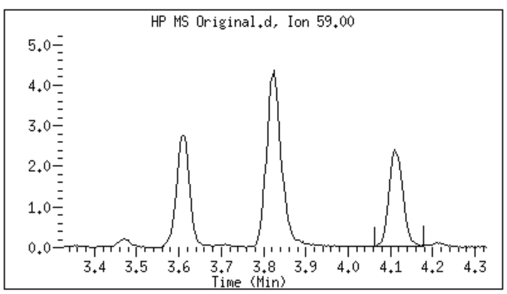
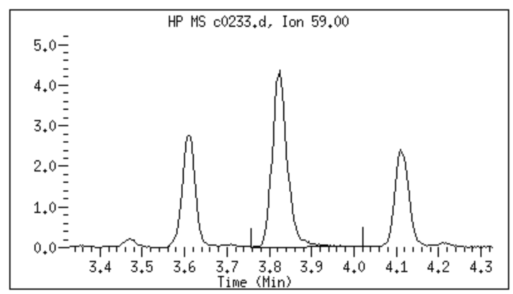
24 tert-Butyl Alcohol   CAS#: 75-65-0   Reason: M1



Electronic Signature
Applied

User: smr
Date: 05/17/2023 16:54



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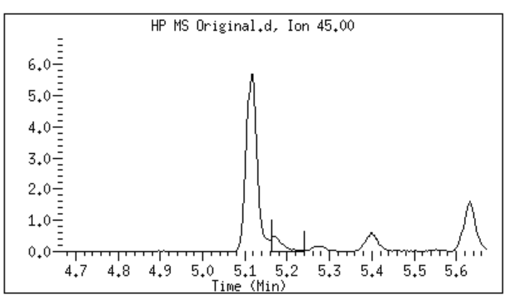
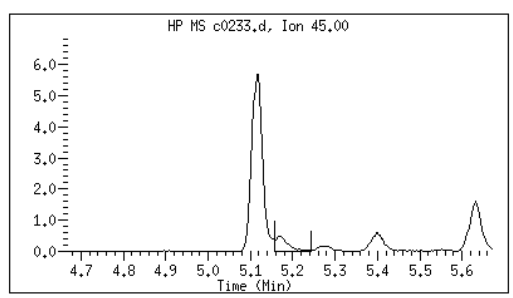
41 sec-Butanol   CAS#: 78-92-2   Reason: M2



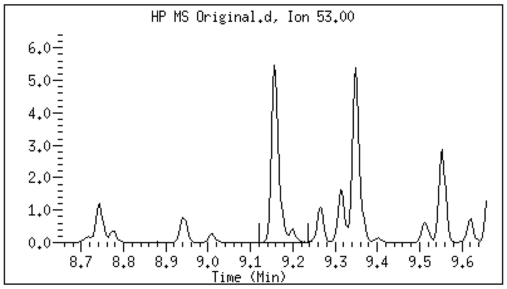
Electronic Signature
Applied

User: smr
Date: 05/17/2023 16:56



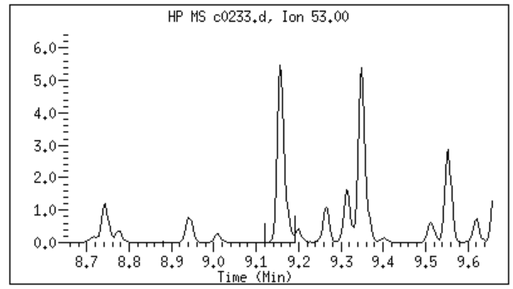
Original Final

107 cis-1,4-dichloro-2-butene CAS#: 1476-11-5 Reason: M1



Electronic Signature Applied

User: smr
Date: 05/17/2023 17:03



M1 - Target system did not integrate
M2 - Target system integrated incorrectly

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0234.d
Lab Smp Id: 1209 Client Smp ID: V15STD200
Inj Date : 17-MAY-2023 16:20
Operator : CJR Inst ID: msv15.i
Smp Info : 1209*V15STD200
Misc Info : MSV~527~*1*SMR
Comment :
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 Calibration Sample, Level: 9
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	374622	200.000	221	9697(A)
2 Chloromethane ++	50	1.651	1.651	(0.281)	325407	200.000	213	9715(A)
3 Vinyl Chloride +	62	1.716	1.716	(0.292)	459198	200.000	214	9835(A)
5 1-3 Butadiene	39	1.735	1.735	(0.295)	330256	200.000	206	9684(A)
6 Bromomethane	94	2.015	2.015	(0.342)	413368	200.000	230	9638(A)
7 Chloroethane	64	2.124	2.124	(0.361)	340914	200.000	216	9248(A)
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	657182	200.000	210	9737(A)
9 Isoprene	67	2.568	2.568	(0.436)	574359	200.000	222	4548(A)
10 Ethyl Ether	59	2.581	2.581	(0.439)	272599	200.000	222	7783
11 1,1-Dichloroethene +	96	2.786	2.786	(0.474)	377300	200.000	212	8635(A)
12 Carbon Disulfide	76	2.812	2.812	(0.478)	1135413	200.000	208	8986(A)
13 1,1,2Trichlotrifluoroethane	101	2.831	2.831	(0.481)	424355	200.000	220	9157(A)
15 Methyl Iodide	142	2.938	2.938	(0.499)	417991	200.000	191	0(M1)
16 Acrolein	56	3.150	3.150	(0.535)	292509	1000.00	1080	5736(A)

	Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
	=========================	====	==	======	======	========	=======	=======	==========
	17 Allyl chloride	39	3.304	3.304	(0.562)	283118	200.000	181	9676
	18 Methylene Chloride	49	3.417	3.417	(0.581)	407794	200.000	207	4968(A)
	19 Acetone	43	3.471	3.471	(0.590)	831654	1000.00	1030	5046(AH)
	20 trans-1,2-Dichloroethene	61	3.593	3.593	(0.611)	526882	200.000	213	8305(A)
	21 Methyl Acetate	43	3.613	3.613	(0.614)	1028831	1000.00	1020	9597(A)
	22 Hexane	57	3.687	3.687	(0.627)	756413	200.000	213	9320(A)
	23 MTBE	73	3.716	3.716	(0.632)	1059252	200.000	220	9677(A)
	24 tert-Butyl Alcohol	59	3.825	3.825	(0.650)	239814	1000.00	1130	4409(AM1)
	25 Acetonitrile	41	3.947	3.947	(0.671)	147588	1000.00	1030	9518(A)
	26 Isopropyl Ether	45	4.111	4.111	(0.699)	922362	200.000	215	9619(A)
	27 Chloroprene	53	4.195	4.195	(0.713)	517468	200.000	214	9019(A)
	28 1,1-Dichloroethane ++	63	4.217	4.217	(0.717)	691595	200.000	210	9485(A)
	29 Acrylonitrile	53	4.262	4.262	(0.725)	592835	1000.00	1050	9903(A)
	30 Vinyl Acetate	43	4.471	4.471	(0.760)	539371	200.000	203	5449(A)
	31 Ethyl Tert-butyl Ether	59	4.471	4.471	(0.760)	1051655	200.000	217	8094(A)
	32 cis-1,2-Dichloroethene	61	4.738	4.738	(0.805)	490815	200.000	207	9726(A)
M	33 Total 1,2-Dichloroethene	61				1017697	400.000	420	0
	34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	605510	200.000	207	9535(A)
	35 Bromochloromethane	128	4.921	4.921	(0.837)	182538	200.000	196	8720
	36 Cyclohexane	56	4.928	4.928	(0.838)	644703	200.000	214	9502(A)
	37 Chloroform +	83	4.995	4.995	(0.849)	716125	200.000	212	9803(A)
	38 Ethyl Acetate	43	5.114	5.114	(0.869)	1626356	1000.00	1100	5994(A)
	41 sec-Butanol	45	5.169	5.169	(0.879)	21713	200.000	194	4338(M2)
	39 Carbon Tetrachloride	117	5.118	5.118	(0.870)	625925	200.000	230	8202(A)
	40 Tetrahydrofuran	42	5.143	5.143	(0.874)	570538	1000.00	1040	9434(A)
$	42 Dibromofluoromethane	111	5.159	5.159	(0.877)	396531	200.000	220	9082(A)
	43 1,1,1-Trichloroethane	97	5.182	5.182	(0.881)	674327	200.000	216	9398(A)
	44 2-Butanone	43	5.275	5.275	(0.897)	918709	1000.00	1090	5783(A)
	45 1,1-Dichloropropene	75	5.291	5.291	(0.899)	582804	200.000	210	9452(A)
	46 2,2,4 Trimethylpentane	57	5.400	5.400	(0.918)	1892720	200.000	226	9026(A)
	47 Propionitrile	54	5.532	5.532	(0.940)	223502	1000.00	1080	8548(A)
	48 Benzene	78	5.516	5.516	(0.938)	1605982	200.000	214	9719(A)
	49 Methylacrylonitrile	41	5.552	5.552	(0.944)	939794	1000.00	1100	7432(A)
	56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	1103297	200.000	224	7099(A)
$	50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	385290	200.000	205	8615(A)
	52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	480930	200.000	204	9772(A)
	51 Isobutyl Alcohol	43	5.728	5.728	(0.974)	49197	1000.00	1000	3200(A)
*	55 FLUOROBENZENE	96	5.883	5.883	(1.000)	364826	50.0000		9873
	57 Trichloroethene	130	6.034	6.034	(1.026)	533090	200.000	231	9215(A)
	58 Methyl Cyclohexane	83	6.027	6.027	(1.025)	785744	200.000	223	8229(A)
	59 Tert-amyl alcohol	59	6.253	6.253	(1.063)	817261	1000.00	1120	6299(A)
	60 n-Butanol	56	6.333	6.333	(1.077)	29929	1000.00	999	9196
	62 Dibromomethane	93	6.394	6.394	(1.087)	260479	200.000	216	9493(A)
	63 2-3 Dichloro-1-Proprene	75	6.468	6.468	(1.099)	669179	200.000	222	9024(A)
	64 1,2-Dichloropropane +	63	6.484	6.484	(1.102)	397850	200.000	214	8183(A)
	65 Bromodichloromethane	83	6.539	6.539	(1.111)	552483	200.000	220	9781(A)
	66 Methyl methacrylate	41	6.680	6.680	(1.136)	280924	200.000	219	9697(A)

		QUANT SIG					AMOUNTS		
							CAL-AMT	ON-COL	
	Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
	67 1,4- Dioxane	88	6.719	6.719	(1.142)	68178	5000.00	4960	9417
	68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	495032	200.000	209	9411(A)
	72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	569730	1000.00	1060	9694(A)
	73 cis-1,3-Dichloropropene	75	7.072	7.072	(1.202)	634947	200.000	213	9786(A)
$	74 Toluene-d8	98	7.224	7.224	(0.866)	1440134	200.000	212	9897(A)
	75 Toluene +	91	7.262	7.262	(0.871)	1820208	200.000	215	9702(A)
	76 2-Nitropropane	43	7.436	7.436	(0.892)	94852	200.000	202	9563(A)
	77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	1596464	1000.00	1010	8326(A)
	78 Tetrachloroethene	164	7.558	7.558	(0.906)	436848	200.000	229	8108(A)
	79 trans-1,3-Dichloropropene	75	7.577	7.577	(1.288)	614230	200.000	218	9168(A)
	80 Ethyl Methacrylate	41	7.693	7.693	(0.922)	248686	200.000	211	7566(A)
	81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	367035	200.000	214	9419(A)
	82 Dibromochloromethane	129	7.825	7.825	(0.938)	451726	200.000	231	9679(A)
M	83 1-3 Dichloropropene total	100				1249177	400.000	431	0
	84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	508123	200.000	225	9574(A)
	86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	589359	200.000	206	9555(A)
	87 1-Nitropropane	43	7.970	7.970	(0.956)	43235	200.000	202	9621(A)
	88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	394224	200.000	215	9733(A)
	89 2-Hexanone	43	8.143	8.143	(0.976)	1298993	1000.00	999	9648
	90 1-Chlorohexane	91	8.333	8.333	(0.999)	643268	200.000	213	9631(A)
*	91 CHLOROBENZENE-d5	82	8.339	8.339	(1.000)	144815	50.0000		7544
	92 Chlorobenzene ++	112	8.352	8.352	(1.002)	1196185	200.000	214	8717(A)
	93 Ethylbenzene +	106	8.368	8.368	(1.003)	659880	200.000	217	8987(A)
	94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	404707	200.000	215	9155(A)
	95 p,m-Xylene	106	8.465	8.465	(1.015)	1599176	400.000	435	9904(A)
	97 o-Xylene	106	8.744	8.744	(1.049)	737290	200.000	215	9912(A)
	101 Styrene	104	8.777	8.777	(1.052)	1006134	200.000	228	9833(A)
	102 Bromoform ++	173	8.799	8.799	(1.055)	332459	200.000	208	8477(A)
	104 Isopropylbenzene	105	8.941	8.941	(1.072)	1888524	200.000	211	9879(A)
M	105 TOTAL XYLENE	106				2336466	600.000	650	0
$	106 Bromofluorobenzene	174	9.124	9.124	(1.094)	459747	200.000	209	9703(A)
	107 cis-1,4-dichloro-2-butene	53	9.159	9.159	(0.934)	142937	200.000	196	9533(M2)
	109 n-Propylbenzene	91	9.201	9.201	(0.939)	2159345	200.000	193	8677
	108 Bromobenzene	77	9.191	9.191	(0.938)	709893	200.000	190	8885
	110 1,1,2,2-Tetrachloroethane++	83	9.243	9.243	(0.943)	423864	200.000	194	9821
	112 2-Chlorotoluene	91	9.301	9.301	(0.949)	1403108	200.000	197	7581
	113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	1535851	200.000	204	8068(A)
	114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	569775	200.000	192	7911
	115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	141123	200.000	195	9319
	116 Cyclohexanone	55	9.365	9.365	(0.955)	352267	1000.00	968	9703
	117 4-Chlorotoluene	91	9.404	9.404	(0.959)	1246147	200.000	194	9826
	118 tert-butylbenzene	91	9.513	9.513	(0.970)	836925	200.000	198	9856
	119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	1547624	200.000	209	9326(A)
	120 sec-Butylbenzene	105	9.619	9.619	(0.981)	1955174	200.000	201	9850(A)
	121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	1824169	200.000	211	9040(A)
	122 Dicyclopentadiene	66	9.699	9.699	(0.990)	1776527	200.000	198	7839
	123 1,3-Dichlorobenzene	146	9.760	9.760	(0.996)	904099	200.000	207	9847(A)

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.802	(1.000)	143421	50.0000		7717
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	933693	200.000	208	9349(A)
126 n-Butylbenzene	91	9.953	9.953	(1.015)	2414411	200.000	210	8894(A)
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	831202	200.000	211	9752(A)
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	114746	200.000	205	9757(A)
130 Hexachlorobutadiene	225	10.995	10.995	(1.122)	280592	200.000	233	9361(A)
131 1,2,4-Trichlorobenzene	180	11.027	11.027	(1.125)	534470	200.000	222	9588(A)
132 Naphthalene	128	11.288	11.288	(1.152)	1294831	200.000	229	9873(A)
133 1,2,3-Trichlorobenzene	180	11.442	11.442	(1.167)	470431	200.000	226	9599(A)

QC Flag Legend

A - Target compound detected but, quantitated amount exceeded maximum amount.
M1- Compound response manually integrated because Target system did not integrate.
M2- Compound response manually integrated because Target system integrated incorrectly.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0234.d

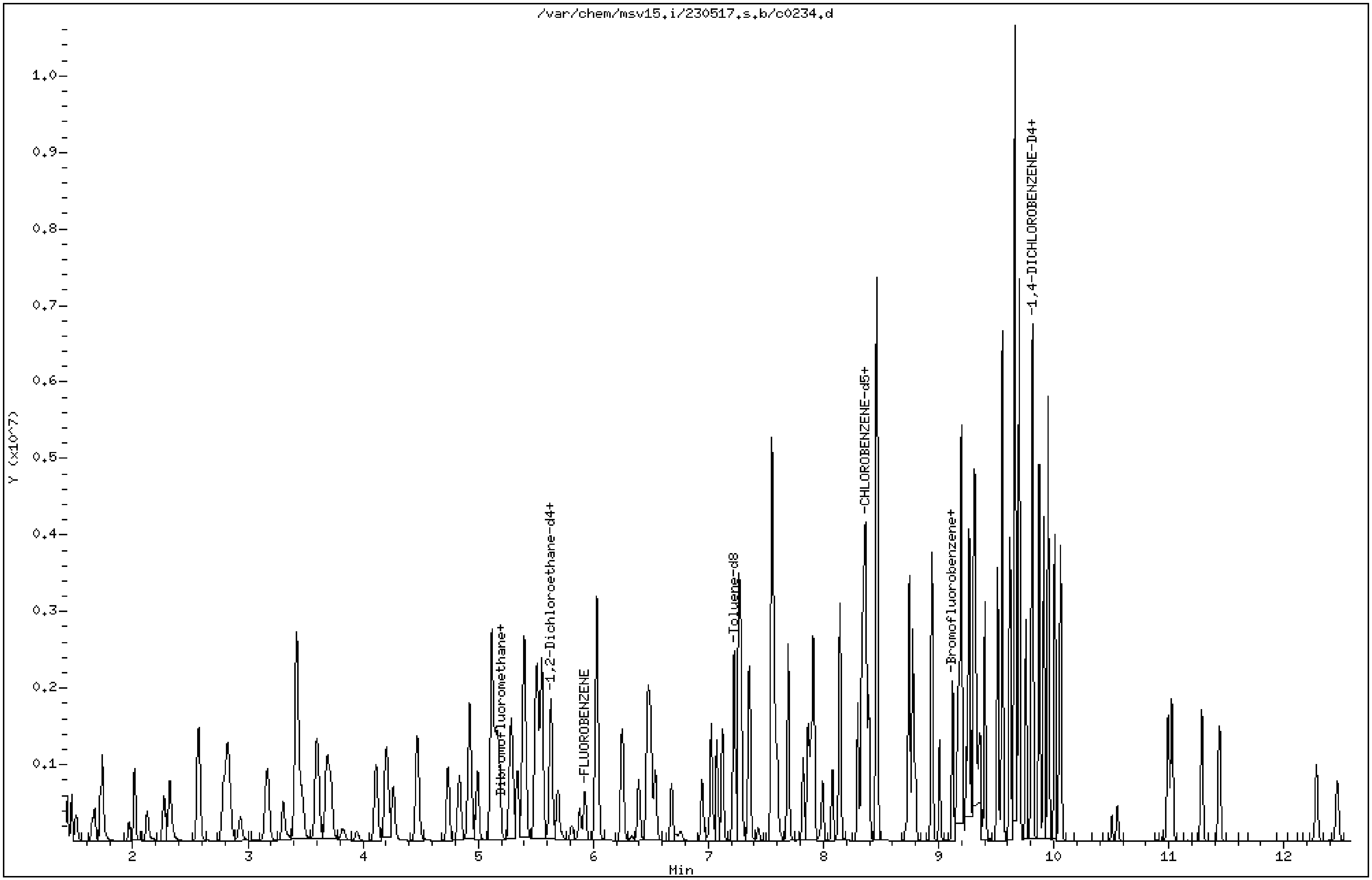
Date : 17-MAY-2023 16:20

Client ID: V15STD200 | Instrument: msv15.i

Sample Info: 1209*V15STD200

Purge Volume: 5.0 | Operator: CJR

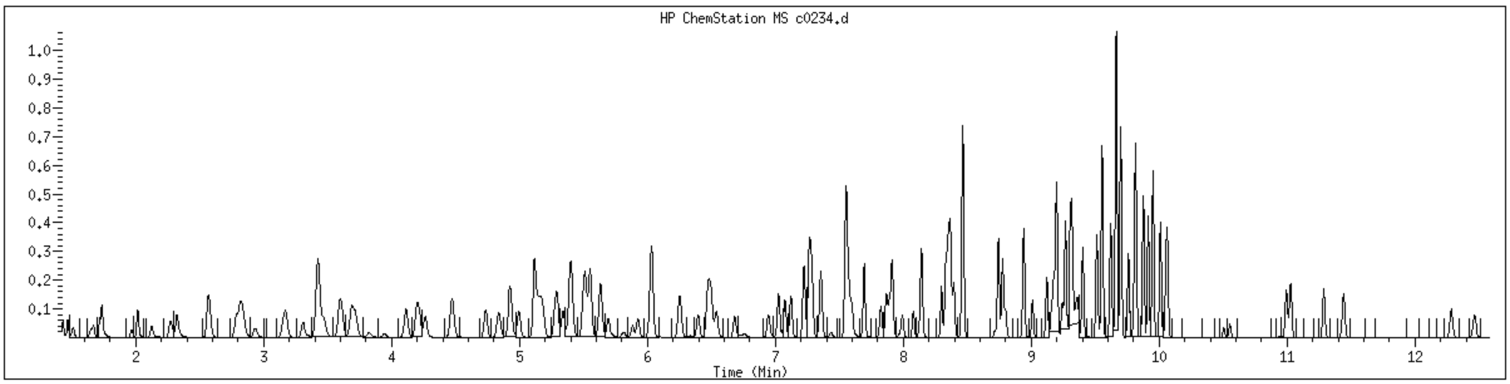
Column phase: RTX-VMS-30M | Column diameter: 0.25



Data file : /var/chem/msv15.i/230517.s.b/c0234.d   Page: 1
Report Date: 05/23/2023 16:34

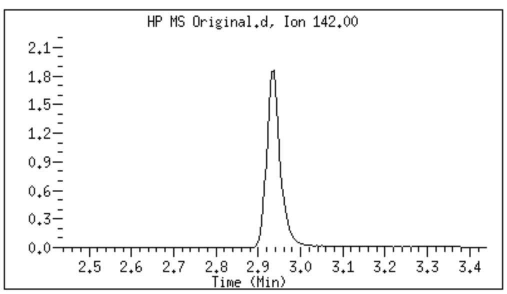
MANUAL INTEGRATION GRAPHIC REPORT

Lab ID	: 1209	SampleType	: CALIB_9
Injection Date	: 05/17/2023 16:20	Instrument	: msv15.i
Operator	: CJR		
Sample Info	: 1209*V15STD200		
Misc Info	: MSV~527~*1*SMR		
Method	: /var/chem/msv15.i/230517.s.b/8260bw15DOD.m		
Dilution	: 1.00		
Matrix	: WATER		
Integrator	: HP RTE	Compound Sublist:	8260b+AppIXDOD



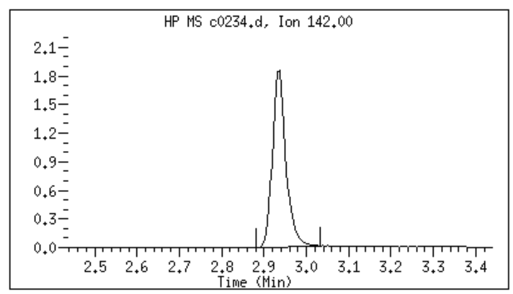
Original   Final

15 Methyl Iodide   CAS#: 74-88-4   Reason: M1



Electronic Signature Applied

User: kmn
Date: 05/17/2023 16:34



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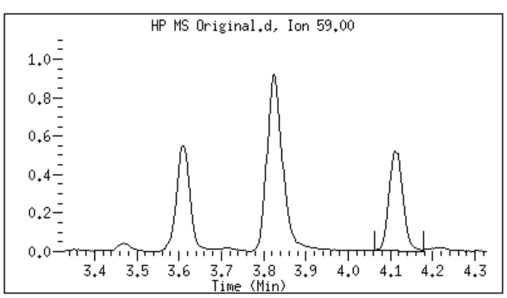
24 tert-Butyl Alcohol   CAS#: 75-65-0   Reason: M1



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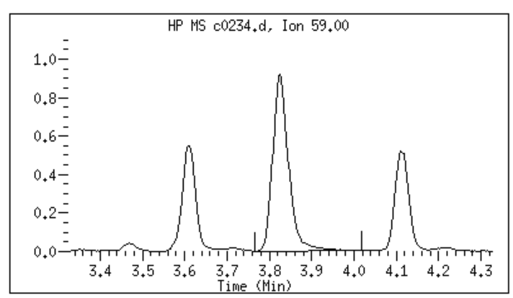

Original Final

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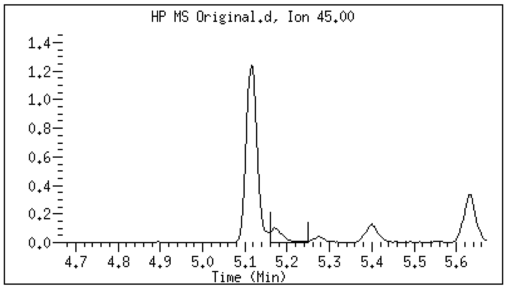
41 sec-Butanol CAS#: 78-92-2 Reason: M2



Electronic Signature Applied

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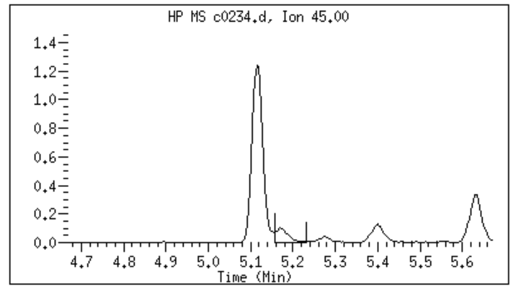
Date: 05/17/2023 16:58



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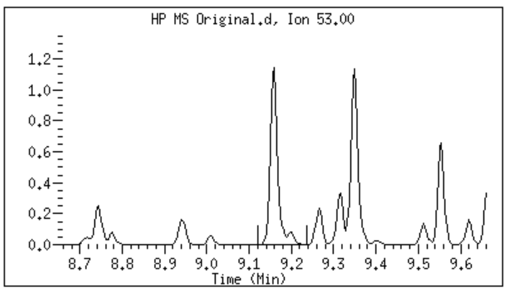
107 cis-1,4-dichloro-2-butene CAS#: 1476-11-5 Reason: M2



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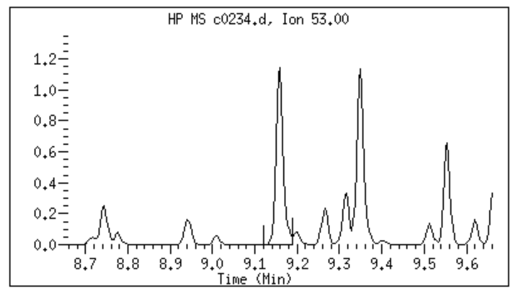


M1 - Target system did not integrate
M2 - Target system integrated incorrectly

EPA 8260B DOD

Form 6I

ICAL Verifications

6I

ORGANICS INITIAL CALIBRATION VERIFICATION

Report No:	223060502	Instrument ID:	MSV15
Analysis Date:	05/17/23 1735	Lab File ID:	230517/c0238
Analytical Method:	EPA 8260B DOD	Analytical Batch:	766139

ANALYTE	UNITS	TRUE	FOUND	%REC	LCL	UCL	Q
1,1,1,2-Tetrachloroethane	ug/L	50.0	48.4	97	80	120	
1,1,1-Trichloroethane	ug/L	50.0	48.9	98	80	120	
1,1,2,2-Tetrachloroethane	ug/L	50.0	49.7	99	80	120	
1,1,2-Trichloroethane	ug/L	50.0	48.6	97	80	120	
1,1-Dichloroethane	ug/L	50.0	48.1	96	80	120	
1,1-Dichloroethene	ug/L	50.0	48.1	96	80	120	
1,1-Dichloropropene	ug/L	50.0	47.7	95	80	120	
1,2,3-Trichlorobenzene	ug/L	50.0	52.0	104	80	120	
1,2,3-Trichloropropane	ug/L	50.0	47.8	96	80	120	
1,2,4-Trichlorobenzene	ug/L	50.0	51.4	103	80	120	
1,2,4-Trimethylbenzene	ug/L	50.0	48.8	98	80	120	
1,2-Dibromo-3-chloropropane	ug/L	50.0	48.4	97	80	120	
1,2-Dibromoethane	ug/L	50.0	50.3	101	80	120	
1,2-Dichlorobenzene	ug/L	50.0	48.9	98	80	120	
1,2-Dichloroethane	ug/L	50.0	47.4	95	80	120	
1,2-Dichloroethene(Total)	ug/L	100	95.4	95	80	120	
1,2-Dichloropropane	ug/L	50.0	48.7	97	80	120	
1,3,5-Trimethylbenzene	ug/L	50.0	48.0	96	80	120	
1,3-Dichlorobenzene	ug/L	50.0	48.0	96	80	120	
1,3-Dichloropropane	ug/L	50.0	46.9	94	80	120	
1,4-Dichlorobenzene	ug/L	50.0	48.3	97	80	120	
2,2-Dichloropropane	ug/L	50.0	49.8	100	80	120	
2-Butanone	ug/L	250	220	88	80	120	
2-Chlorotoluene	ug/L	50.0	47.3	95	80	120	
2-Hexanone	ug/L	250	215	86	80	120	
4-Chlorotoluene	ug/L	50.0	47.5	95	80	120	
4-Isopropyltoluene	ug/L	50.0	49.3	99	80	120	
4-Methyl-2-pentanone	ug/L	250	233	93	80	120	
Acetone	ug/L	250	189	76	80	120	*
Benzene	ug/L	50.0	47.7	95	80	120	
Bromobenzene	ug/L	50.0	49.0	98	80	120	
Bromochloromethane	ug/L	50.0	48.3	97	80	120	
Bromodichloromethane	ug/L	50.0	49.5	99	80	120	
Bromoform	ug/L	50.0	45.3	91	80	120	
Bromomethane	ug/L	50.0	52.7	105	80	120	
Carbon disulfide	ug/L	50.0	44.2	88	80	120	
Carbon tetrachloride	ug/L	50.0	48.3	97	80	120	
Chlorobenzene	ug/L	50.0	49.2	98	80	120	

FORM 6I - ORG

6I

ORGANICS INITIAL CALIBRATION VERIFICATION

Report No:	223060502	Instrument ID:	MSV15
Analysis Date:	05/17/23 1735	Lab File ID:	230517/c0238
Analytical Method:	EPA 8260B DOD	Analytical Batch:	766139

ANALYTE	*UNITS*	*TRUE*	*FOUND*	*%REC*	*LCL*	*UCL*	*Q*
Chloroethane	ug/L	50.0	48.0	96	80	120	
Chloroform	ug/L	50.0	48.4	97	80	120	
Chloromethane	ug/L	50.0	39.6	79	80	120	*
cis-1,2-Dichloroethene	ug/L	50.0	47.6	95	80	120	
cis-1,3-Dichloropropene	ug/L	50.0	49.1	98	80	120	
Dibromochloromethane	ug/L	50.0	51.6	103	80	120	
Dibromomethane	ug/L	50.0	50.2	100	80	120	
Dichlorodifluoromethane	ug/L	50.0	45.2	90	80	120	
Ethylbenzene	ug/L	50.0	49.7	99	80	120	
Hexachlorobutadiene	ug/L	50.0	50.8	102	80	120	
Isopropylbenzene (Cumene)	ug/L	50.0	50.0	100	80	120	
m,p-Xylene	ug/L	100	99.2	99	80	120	
Methylene chloride	ug/L	50.0	47.3	95	80	120	
Naphthalene	ug/L	50.0	52.4	105	80	120	
n-Butylbenzene	ug/L	50.0	50.0	100	80	120	
n-Propylbenzene	ug/L	50.0	47.4	95	80	120	
o-Xylene	ug/L	50.0	49.9	100	80	120	
sec-Butylbenzene	ug/L	50.0	48.1	96	80	120	
Styrene	ug/L	50.0	51.1	102	80	120	
tert-Butyl methyl ether (MTBE)	ug/L	50.0	51.1	102	80	120	
tert-Butylbenzene	ug/L	50.0	48.0	96	80	120	
Tetrachloroethene	ug/L	50.0	48.9	98	80	120	
Toluene	ug/L	50.0	48.2	96	80	120	
trans-1,2-Dichloroethene	ug/L	50.0	47.7	95	80	120	
trans-1,3-Dichloropropene	ug/L	50.0	49.9	100	80	120	
Trichloroethene	ug/L	50.0	48.2	96	80	120	
Trichlorofluoromethane	ug/L	50.0	48.7	97	80	120	
Vinyl chloride	ug/L	50.0	44.7	89	80	120	

FORM 6I - ORG

GCAL, Inc.

Data file : /var/chem/msv15.i/230517.s.b/c0238.d
Lab Smp Id: 1600 Client Smp ID: V15ICV050
Inj Date : 17-MAY-2023 17:35
Operator : CJR Inst ID: msv15.i
Smp Info : 1600*V15ICV050
Misc Info : MSV~527~*1*SMR
Comment :
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Meth Date : 23-May-2023 16:34 smr Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 QC Sample: LCS
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260b+AppIXDOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	CONCENTRATIONS FINAL (ug/L)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	77603	45.1510	45.2	9645
2 Chloromethane ++	50	1.651	1.651	(0.281)	61323	39.6168	39.6	9769(R)
3 Vinyl Chloride +	62	1.719	1.716	(0.292)	97449	44.7088	44.7	9748
5 1-3 Butadiene	39	1.735	1.735	(0.295)	73091	45.0296	45.0	9710
6 Bromomethane	94	2.018	2.015	(0.343)	96342	52.7085	52.7	9662
7 Chloroethane	64	2.134	2.124	(0.363)	76883	47.9874	48.0	9218
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	154711	48.6924	48.7	9629
9 Isoprene	67	2.568	2.568	(0.436)	127374	48.4794	48.5	4543
10 Ethyl Ether	59	2.584	2.581	(0.439)	62644	50.3212	50.3	8119
11 1,1-Dichloroethene +	96	2.783	2.786	(0.473)	87031	48.1441	48.1	8240
12 Carbon Disulfide	76	2.815	2.812	(0.479)	244627	44.2342	44.2	8010
13 1,1,2Trichlotrifluoroethane	101	2.835	2.831	(0.482)	94362	48.1020	48.1	9333
15 Methyl Iodide	142	2.934	2.938	(0.499)	92242	42.8982	42.9	9558
16 Acrolein	56	3.153	3.150	(0.536)	73042	265.343	265	5651

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
17 Allyl chloride	39	3.304	3.304	(0.562)	78905	49.6600	49.7	9653
18 Methylene Chloride	49	3.420	3.417	(0.581)	94560	47.2694	47.3	4846
19 Acetone	43	3.471	3.471	(0.590)	154872	188.623	189	5308(RH)
20 trans-1,2-Dichloroethene	61	3.590	3.593	(0.610)	119659	47.7467	47.7	8663
21 Methyl Acetate	43	3.613	3.613	(0.614)	289070	282.801	283	9378
22 Hexane	57	3.690	3.687	(0.627)	178006	49.3001	49.3	9241
23 MTBE	73	3.719	3.716	(0.632)	249745	51.0961	51.1	9736
24 tert-Butyl Alcohol	59	3.828	3.825	(0.651)	38701	179.496	179	7982(R)
25 Acetonitrile	41	3.944	3.947	(0.670)	39150	269.318	269	9584
26 Isopropyl Ether	45	4.111	4.111	(0.699)	212742	48.7834	48.8	9674
27 Chloroprene	53	4.198	4.195	(0.714)	117752	47.9358	47.9	8544
28 1,1-Dichloroethane ++	63	4.217	4.217	(0.717)	161086	48.1359	48.1	9450
29 Acrylonitrile	53	4.262	4.262	(0.725)	156392	274.065	274	9847
30 Vinyl Acetate	43	4.475	4.471	(0.761)	119847	44.4086	44.4	5352
31 Ethyl Tert-butyl Ether	59	4.475	4.471	(0.761)	244877	49.8928	49.9	8154
32 cis-1,2-Dichloroethene	61	4.738	4.738	(0.805)	114542	47.6100	47.6	9765
M 33 Total 1,2-Dichloroethene	61				234201	95.3567	95.4	0
34 2,2-Dichloropropane	77	4.835	4.838	(0.822)	148076	49.7823	49.8	9494
35 Bromochloromethane	128	4.921	4.921	(0.837)	45598	48.2758	48.3	9214
36 Cyclohexane	56	4.928	4.928	(0.838)	145283	47.6088	47.6	9276
37 Chloroform +	83	4.995	4.995	(0.849)	165566	48.3641	48.4	9818
38 Ethyl Acetate	43	5.114	5.114	(0.869)	365177	244.082	244	5844
41 sec-Butanol	45	5.169	5.169	(0.879)	6061	53.4915	53.5	4661
39 Carbon Tetrachloride	117	5.121	5.118	(0.870)	133664	48.3000	48.3	8663
40 Tetrahydrofuran	42	5.147	5.143	(0.875)	132874	238.436	238	9668
$ 42 Dibromofluoromethane	111	5.159	5.159	(0.877)	95490	52.2260	52.2	8503
43 1,1,1-Trichloroethane	97	5.179	5.182	(0.880)	154965	48.8826	48.9	9702
44 2-Butanone	43	5.275	5.275	(0.897)	187809	220.376	220	6124
45 1,1-Dichloropropene	75	5.291	5.291	(0.899)	134442	47.7008	47.7	9415
46 2,2,4 Trimethylpentane	57	5.401	5.400	(0.918)	271077	31.8473	31.8	9752(R)
47 Propionitrile	54	5.532	5.532	(0.940)	56466	268.203	268	8644
48 Benzene	78	5.516	5.516	(0.938)	363251	47.6822	47.7	9659
49 Methylacrylonitrile	41	5.552	5.552	(0.944)	218562	252.818	253	7411
56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	253152	50.6170	50.6	7287
$ 50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	95934	50.3913	50.4	8616
52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	113628	47.4145	47.4	9758
51 Isobutyl Alcohol	43	5.725	5.728	(0.973)	12910	258.954	259	3960
* 55 FLUOROBENZENE	96	5.883	5.883	(1.000)	370239	50.0000		9869
57 Trichloroethene	130	6.034	6.034	(1.026)	112727	48.1794	48.2	9113
58 Methyl Cyclohexane	83	6.028	6.027	(1.025)	175444	49.0110	49.0	8381
59 Tert-amyl alcohol	59	6.253	6.253	(1.063)	185786	250.439	250	6416
60 n-Butanol	56	6.330	6.333	(1.076)	8397	282.755	283	9117
62 Dibromomethane	93	6.397	6.394	(1.087)	61518	50.1831	50.2	9433
63 2-3 Dichloro-1-Proprene	75	6.468	6.468	(1.099)	151981	49.5895	49.6	8925
64 1,2-Dichloropropane +	63	6.484	6.484	(1.102)	91663	48.6830	48.7	8206
65 Bromodichloromethane	83	6.539	6.539	(1.111)	126290	49.5178	49.5	9738
66 Methyl methacrylate	41	6.680	6.680	(1.136)	64193	49.2674	49.3	9691

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
67 1,4- Dioxane	88	6.722	6.719	(1.143)	19360	1386.71	1390	9562
68 1-Bromo-2-chloroethane	63	6.947	6.944	(1.181)	116609	48.6121	48.6	9204
72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	112521	210.472	210	9677
73 cis-1,3-Dichloropropene	75	7.072	7.072	(1.202)	148184	49.0749	49.1	9747
$ 74 Toluene-d8	98	7.224	7.224	(0.866)	350079	50.9102	50.9	9881
75 Toluene +	91	7.262	7.262	(0.871)	412375	48.1671	48.2	9649
76 2-Nitropropane	43	7.433	7.436	(0.891)	22691	48.1291	48.1	9503
77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	369543	232.524	233	8204
78 Tetrachloroethene	164	7.558	7.558	(0.906)	94581	48.8642	48.9	8070
79 trans-1,3-Dichloropropene	75	7.574	7.577	(1.287)	142850	49.9121	49.9	8713
80 Ethyl Methacrylate	41	7.693	7.693	(0.922)	56383	47.1315	47.1	7547
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	84497	48.5919	48.6	9434
82 Dibromochloromethane	129	7.822	7.825	(0.938)	102055	51.5977	51.6	9577
M 83 1-3 Dichloropropene total	100				291034	98.9870	99.0	0
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	115834	50.6589	50.7	9630
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	135904	46.9398	46.9	9532
87 1-Nitropropane	43	7.970	7.970	(0.956)	12569	58.0825	58.1	9477
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	93654	50.3325	50.3	9662
89 2-Hexanone	43	8.143	8.143	(0.976)	280409	214.833	215	9661
90 1-Chlorohexane	91	8.333	8.333	(0.999)	145965	47.6929	47.7	7375
* 91 CHLOROBENZENE-d5	82	8.339	8.339	(1.000)	146725	50.0000		9164
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	278963	49.2335	49.2	9515
93 Ethylbenzene +	106	8.368	8.368	(1.003)	153222	49.7336	49.7	9003
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	92203	48.4290	48.4	9195
95 p,m-Xylene	106	8.462	8.465	(1.015)	369416	99.1790	99.2	9876
97 o-Xylene	106	8.741	8.744	(1.048)	173049	49.8921	49.9	9904
101 Styrene	104	8.777	8.777	(1.052)	228037	51.0600	51.1	9848
102 Bromoform ++	173	8.799	8.799	(1.055)	72295	45.3428	45.3	8561
104 Isopropylbenzene	105	8.941	8.941	(1.072)	453954	50.0014	50.0	9864
M 105 TOTAL XYLENE	106				542465	149.071	149	0
$ 106 Bromofluorobenzene	174	9.121	9.124	(1.094)	112552	50.3992	50.4	9478
107 cis-1,4-dichloro-2-butene	53	9.156	9.159	(0.934)	35844	49.5760	49.6	9651
109 n-Propylbenzene	91	9.198	9.201	(0.938)	527111	47.4110	47.4	9138
108 Bromobenzene	77	9.191	9.191	(0.938)	181205	48.9742	49.0	9236
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.243	(0.943)	108001	49.7234	49.7	9716
112 2-Chlorotoluene	91	9.301	9.301	(0.949)	335416	47.3292	47.3	7811
113 1,3,5-Trimethylbenzene	105	9.314	9.317	(0.950)	359160	48.0291	48.0	8699
114 1,2,3-Trichloropropane	75	9.326	9.330	(0.951)	140772	47.7607	47.8	6907
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	34815	48.5408	48.5	9207
116 Cyclohexanone	55	9.365	9.365	(0.955)	84106	233.849	234	9735
117 4-Chlorotoluene	91	9.400	9.404	(0.959)	301767	47.4503	47.5	9763
118 tert-butylbenzene	91	9.513	9.513	(0.970)	201507	47.9946	48.0	9862
119 1,2,4-Trimethylbenzene	105	9.552	9.555	(0.974)	358167	48.7993	48.8	9381
120 sec-Butylbenzene	105	9.619	9.619	(0.981)	465691	48.1419	48.1	9891
121 p-Isopropyltoluene	119	9.699	9.703	(0.990)	422559	49.3442	49.3	8922
122 Dicyclopentadiene	66	9.699	9.699	(0.990)	421000	47.3238	47.3	7799
123 1,3-Dichlorobenzene	146	9.761	9.760	(0.996)	207739	48.0105	48.0	9805

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	CONCENTRATIONS ON-COLUMN (ppb)	FINAL (ug/L)	SIMILARITY
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.802	(1.000)	142360	50.0000		9109
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	214779	48.2611	48.3	8952
126 n-Butylbenzene	91	9.953	9.953	(1.015)	569717	50.0344	50.0	8827
127 1,2-Dichlorobenzene	146	10.063	10.063	(1.027)	191245	48.8574	48.9	9694
129 1,2-Dibromo-3-Chloropropane	157	10.555	10.555	(1.077)	26530	48.4000	48.4	9538
130 Hexachlorobutadiene	225	10.995	10.995	(1.122)	60750	50.7686	50.8	9257
131 1,2,4-Trichlorobenzene	180	11.027	11.027	(1.125)	122925	51.3747	51.4	9528
132 Naphthalene	128	11.285	11.288	(1.151)	293974	52.4313	52.4	9820
133 1,2,3-Trichlorobenzene	180	11.442	11.442	(1.167)	107422	52.0474	52.0	9471

QC Flag Legend

R - Spike/Surrogate failed recovery limits.
H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230517.s.b/c0238.d

Page 1

Date : 17-MAY-2023 17:35

Client ID: V15ICV050

Instrument: msv15.i

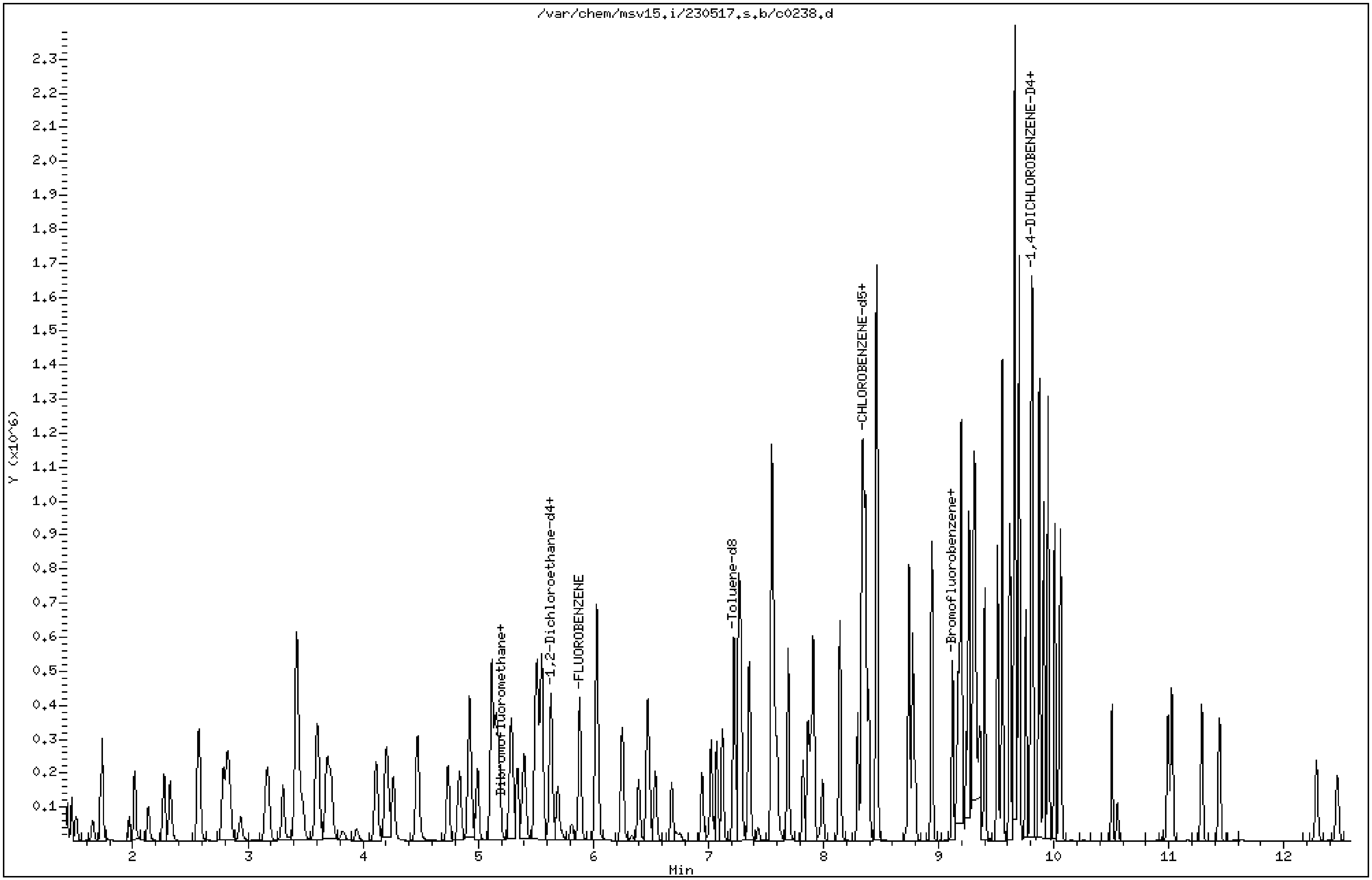
Sample Info: 1600*V15ICV050

Purge Volume: 5.0

Operator: CJR

Column phase: RTX-VMS-30M

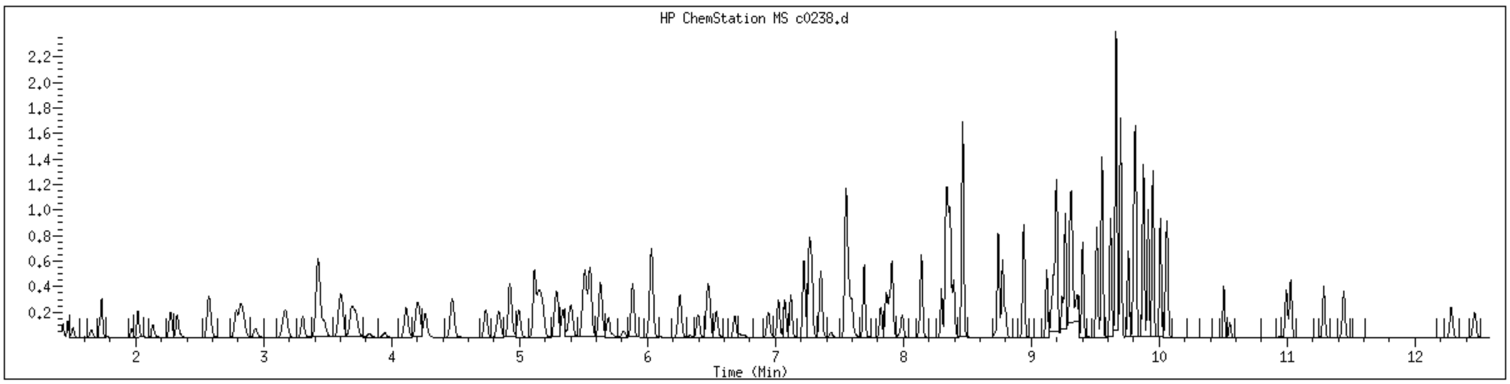
Column diameter: 0.25



Data file : /var/chem/msv15.i/230517.s.b/c0238.d Page: 1
Report Date: 05/23/2023 16:34

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1600 SampleType : LCS
Injection Date: 05/17/2023 17:35 Instrument : msv15.i
Operator : CJR
Sample Info : 1600*V15ICV050
Misc Info : MSV~527~*1*SMR
Method : /var/chem/msv15.i/230517.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260b+AppIXDOD



NO MANUAL INTEGRATIONS

EPA 8260B DOD

Form 7A

CCAL Verifications

7A
VOLATILE CONTINUING CALIBRATION CHECK

Report No:	223060502	CCAL ID:	1400
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230612/c0917s
Init. Calib. Date 1:	05/17/23 Time 1: 1428	Analyst:	KN1
Init. Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Batch:	767415
Analysis Date:	06/12/23 Time: 1030	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
1,1,1,2-Tetrachloroethane	50.0	50.0	0.649	0.649	.01	.08	20	A	
1,1,1-Trichloroethane	50.0	49.3	0.428	0.422	.01	-1.48	20	A	
1,1,2,2-Tetrachloroethane	50.0	47.9	0.763	0.731	.3	-4.22	20	A	
1,1,2-Trichloroethane	50.0	50.4	0.593	0.597	.01	.74	20	A	
1,1-Dichloroethane	50.0	50.8	0.452	0.459	.1	1.64	20	A	
1,1-Dichloroethene	50.0	51.6	0.244	0.252	.01	3.2	20	A	
1,1-Dichloropropene	50.0	50.3	0.381	0.383	.01	.64	20	A	
1,2,3-Trichlorobenzene	50.0	49.8	0.725	0.721	.01	-.5	20	A	
1,2,3-Trichloropropane	50.0	45.1	1.035	0.934	.01	-9.82	20	A	
1,2,4-Trichlorobenzene	50.0	50.8	0.840	0.854	.01	1.64	20	A	
1,2,4-Trimethylbenzene	50.0	49.8	2.578	2.568	.01	-.37	20	A	
1,2-Dibromo-3-chloropropane	50.0	39.1	0.196	0.150	.01	-21.8	20	W	*
1,2-Dibromoethane	50.0	50.3	0.634	0.638	.01	.59	20	A	
1,2-Dichlorobenzene	50.0	49.2	1.375	1.354	.01	-1.51	20	A	
1,2-Dichloroethane	50.0	46.4	0.324	0.300	.01	-7.21	20	A	
1,2-Dichloroethane-d4	50.0	47.0	0.257	0.242	.01	-6.06	20	A	
1,2-Dichloroethene (total)	100.0	102.0	0.332	0.340	.01	2.41	20	A	
1,2-Dichloropropane	50.0	52.5	0.254	0.267	.01	4.94	20	A	
1,3,5-Trimethylbenzene	50.0	49.8	2.626	2.615	.01	-.45	20	A	
1,3-Dichlorobenzene	50.0	49.4	1.520	1.503	.01	-1.13	20	A	
1,3-Dichloropropane	50.0	49.7	0.987	0.980	.01	-.65	20	A	
1,4-Dichlorobenzene	50.0	49.1	1.563	1.536	.01	-1.73	20	A	
2,2-Dichloropropane	50.0	42.7	0.402	0.343	.01	-14.6	20	A	
2-Butanone	250.0	250.0	0.115	0.115	.01	-.13	20	A	
2-Chlorotoluene	50.0	48.4	2.489	2.412	.01	-3.11	20	A	
2-Hexanone	250.0	236.0	0.450	0.420	.01	-5.6	20	W	
4-Bromofluorobenzene	50.0	51.8	0.761	0.788	.01	3.54	20	A	
4-Chlorotoluene	50.0	49.8	2.234	2.223	.01	-.48	20	A	
4-Isopropyltoluene	50.0	49.5	3.008	2.981	.01	-.9	20	A	
4-Methyl-2-pentanone	250.0	217.0	0.548	0.469	.01	-13.2	20	W	
Acetone	250.0	251.0	0.111	0.111	.01	.53	20	A	
Benzene	50.0	51.2	1.029	1.054	.01	2.41	20	A	
Bromobenzene	50.0	49.2	1.300	1.279	.01	-1.59	20	A	
Bromochloromethane	50.0	51.5	0.128	0.131	.01	3.04	20	A	
Bromodichloromethane	50.0	50.7	0.344	0.349	.01	1.32	20	A	
Bromoform	50.0	45.8	0.554	0.498	.1	-8.4	20	W	
Bromomethane	50.0	41.3	0.247	0.204	.01	-17.3	20	A	
Carbon disulfide	50.0	51.8	0.747	0.773	.01	3.54	20	A	
Carbon tetrachloride	50.0	49.1	0.374	0.367	.01	-1.87	20	A	

FORM V II VOA

7A
VOLATILE CONTINUING CALIBRATION CHECK

Report No:	223060502			CCAL ID:	1400
GC Column:	RTX-VMS-30M	ID	.25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0		(µL)	Lab File ID:	230612/c0917s
Init. Calib. Date 1:	05/17/23	Time 1:	1428	Analyst:	KN1
Init. Calib. Date 2:	05/17/23	Time 2:	1620	Analytical Batch:	767415
Analysis Date:	06/12/23	Time:	1030	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
Chlorobenzene	50.0	50.3	1.931	1.942	.3	.58	20	A	
Chloroethane	50.0	48.2	0.216	0.209	.01	-3.62	20	A	
Chloroform	50.0	49.1	0.462	0.454	.01	-1.87	20	A	
Chloromethane	50.0	58.0	0.209	0.242	.1	16	20	A	
Dibromochloromethane	50.0	53.5	0.674	0.721	.01	7.03	20	A	
Dibromofluoromethane	50.0	50.8	0.247	0.251	.01	1.59	20	A	
Dibromomethane	50.0	48.7	0.166	0.161	.01	-2.64	20	A	
Dichlorodifluoromethane	50.0	53.4	0.232	0.248	.01	6.82	20	A	
Ethylbenzene	50.0	50.7	1.050	1.065	.01	1.46	20	A	
Hexachlorobutadiene	50.0	52.9	0.420	0.445	.01	5.85	20	A	
Isopropylbenzene (Cumene)	50.0	51.2	3.094	3.167	.01	2.36	20	A	
Methylene chloride	50.0	51.2	0.270	0.277	.01	2.44	20	A	
Naphthalene	50.0	44.7	1.969	1.760	.01	-10.6	20	A	
Styrene	50.0	53.8	1.522	1.639	.01	7.67	20	A	
Tetrachloroethene	50.0	51.6	0.660	0.680	.01	3.1	20	A	
Toluene	50.0	50.8	2.917	2.966	.01	1.66	20	A	
Toluene-d8	50.0	51.0	2.343	2.391	.01	2.04	20	A	
Trichloroethene	50.0	51.4	0.316	0.325	.01	2.71	20	A	
Trichlorofluoromethane	50.0	48.0	0.429	0.412	.01	-4.04	20	A	
Vinyl chloride	50.0	50.9	0.294	0.300	.01	1.82	20	A	
cis-1,2-Dichloroethene	50.0	50.6	0.325	0.329	.01	1.14	20	A	
cis-1,3-Dichloropropene	50.0	49.8	0.408	0.406	.01	-.49	20	A	
m,p-Xylene	100.0	102.0	1.269	1.301	.01	2.49	20	A	
n-Butylbenzene	50.0	48.8	3.999	3.902	.01	-2.42	20	A	
n-Propylbenzene	50.0	49.8	3.905	3.891	.01	-.35	20	A	
o-Xylene	50.0	51.5	1.182	1.217	.01	2.98	20	A	
sec-Butylbenzene	50.0	49.8	3.397	3.385	.01	-.36	20	A	
tert-Butyl methyl ether (MTBE)	50.0	41.2	0.660	0.544	.01	-17.6	20	A	
tert-Butylbenzene	50.0	48.1	1.475	1.420	.01	-3.71	20	A	
trans-1,2-Dichloroethene	50.0	51.8	0.338	0.351	.01	3.63	20	A	
trans-1,3-Dichloropropene	50.0	49.0	0.387	0.378	.01	-2.09	20	A	

FORM V II VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230612.s.b/c0917s.d
Lab Smp Id: 1400 Client Smp ID: V15STD050
Inj Date : 12-JUN-2023 10:30
Operator : KN1 Inst ID: msv15.i
Smp Info : 1400*V15STD050*51363
Misc Info : MSV~529~*1*SMR
Comment :
Method : /var/chem/msv15.i/230612.s.b/8260bw15DOD.m
Meth Date : 13-Jun-2023 15:50 kmn Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 Continuing Calibration Sample
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	AMOUNTS ON-COL (ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	85115	50.0000	53.4	9739
2 Chloromethane ++	50	1.649	1.649	(0.280)	83228	50.0000	58.0	9794
3 Vinyl Chloride +	62	1.719	1.719	(0.292)	102886	50.0000	50.9	9847
5 1-3 Butadiene	39	1.735	1.735	(0.295)	78432	50.0000	52.1	9635
6 Bromomethane	94	2.015	2.015	(0.343)	70004	50.0000	41.3	9800
7 Chloroethane	64	2.137	2.137	(0.363)	71585	50.0000	48.2	9224
8 Trichlorofluoromethane	101	2.272	2.272	(0.386)	141348	50.0000	48.0	9755
11 1,1-Dichloroethene +	96	2.784	2.784	(0.473)	86488	50.0000	51.6	8322
12 Carbon Disulfide	76	2.816	2.816	(0.479)	265456	50.0000	51.8	7906
13 1,1,2Trichlotrifluoroethane	101	2.832	2.832	(0.482)	96298	50.0000	52.9	9104
15 Methyl Iodide	142	2.935	2.935	(0.499)	87383	50.0000	43.8	9513
16 Acrolein	56	3.150	3.150	(0.536)	55482	250.000	217	4922
17 Allyl chloride	39	3.304	3.304	(0.562)	75096	50.0000	51.0	9821
18 Methylene Chloride	49	3.417	3.417	(0.581)	95004	50.0000	51.2	4730

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
19 Acetone	43	3.465	3.465	(0.589)	191316	250.000	251	5731(H)
20 trans-1,2-Dichloroethene	61	3.587	3.587	(0.610)	120393	50.0000	51.8	9006
21 Methyl Acetate	43	3.610	3.610	(0.614)	229236	250.000	242	9639
22 Hexane	57	3.684	3.684	(0.626)	175709	50.0000	52.5	9279
23 MTBE	73	3.713	3.713	(0.631)	186689	50.0000	41.2	9319
24 tert-Butyl Alcohol	59	3.819	3.819	(0.649)	25929	250.000	130	8197
25 Acetonitrile	41	3.941	3.941	(0.670)	35938	250.000	267	9516
26 Isopropyl Ether	45	4.108	4.108	(0.699)	217837	50.0000	53.9	9657
27 Chloroprene	53	4.195	4.195	(0.713)	114858	50.0000	50.4	8769
28 1,1-Dichloroethane ++	63	4.214	4.214	(0.717)	157678	50.0000	50.8	9542
29 Acrylonitrile	53	4.259	4.259	(0.724)	130474	250.000	247	9875
31 Ethyl Tert-butyl Ether	59	4.468	4.468	(0.760)	212865	50.0000	46.8	8209
30 Vinyl Acetate	43	4.472	4.472	(0.760)	109214	50.0000	43.6	5519
32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	112796	50.0000	50.6	9816
M 33 Total 1,2-Dichloroethene	61				233189	100.000	102	0
34 2,2-Dichloropropane	77	4.838	4.838	(0.823)	117729	50.0000	42.7	9565
35 Bromochloromethane	128	4.919	4.919	(0.836)	45116	50.0000	51.5	9280
36 Cyclohexane	56	4.928	4.928	(0.838)	153052	50.0000	54.1	9541
37 Chloroform +	83	4.996	4.996	(0.850)	155731	50.0000	49.1	9789
39 Carbon Tetrachloride	117	5.121	5.121	(0.871)	125890	50.0000	49.1	8389
$ 42 Dibromofluoromethane	111	5.156	5.156	(0.877)	86112	50.0000	50.8	8672
43 1,1,1-Trichloroethane	97	5.179	5.179	(0.881)	144793	50.0000	49.3	9442
45 1,1-Dichloropropene	75	5.291	5.291	(0.900)	131500	50.0000	50.3	9515
44 2-Butanone	43	5.275	5.275	(0.897)	197280	250.000	250	6387
46 2,2,4 Trimethylpentane	57	5.398	5.398	(0.918)	291312	50.0000	36.9	9812
48 Benzene	78	5.513	5.513	(0.938)	361677	50.0000	51.2	9497
56 tert-amyl Methyl Ether	73	5.629	5.629	(0.957)	213358	50.0000	46.0	7397
$ 50 1,2-Dichloroethane-d4	65	5.629	5.629	(0.957)	82910	50.0000	47.0	8569
52 1,2-Dichloroethane	62	5.690	5.690	(0.968)	103086	50.0000	46.4	9828
51 Isobutyl Alcohol	43	5.726	5.726	(0.974)	8055	250.000	174	3822
* 55 FLUOROBENZENE	96	5.880	5.880	(1.000)	343271	50.0000		9873
58 Methyl Cyclohexane	83	6.028	6.028	(1.025)	173651	50.0000	52.3	8289
57 Trichloroethene	130	6.031	6.031	(1.026)	111404	50.0000	51.4	8880
62 Dibromomethane	93	6.394	6.394	(1.087)	55327	50.0000	48.7	9615
64 1,2-Dichloropropane +	63	6.481	6.481	(1.102)	91594	50.0000	52.5	8083
65 Bromodichloromethane	83	6.539	6.539	(1.112)	119787	50.0000	50.7	9762
67 1,4- Dioxane	88	6.719	6.719	(1.143)	13880	1250.00	1070	9226
68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.181)	115806	50.0000	52.1	8925
72 2-Chloroethyl vinyl ether	63	7.021	7.021	(1.194)	14279	250.000	32.6	9627
73 cis-1,3-Dichloropropene	75	7.070	7.070	(1.202)	139288	50.0000	49.8	9784
$ 74 Toluene-d8	98	7.221	7.221	(0.866)	316846	50.0000	51.0	9878
75 Toluene +	91	7.262	7.262	(0.871)	393027	50.0000	50.8	9617
77 4-methyl-2-pentanone	43	7.552	7.552	(0.906)	310884	250.000	217	8602
78 Tetrachloroethene	164	7.558	7.558	(0.906)	90117	50.0000	51.6	8522
79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.288)	129903	50.0000	49.0	8797
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	79110	50.0000	50.4	9542
82 Dibromochloromethane	129	7.825	7.825	(0.938)	95599	50.0000	53.5	9717

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	105417	50.0000	51.0	9663
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	129889	50.0000	49.7	9616
M 83 1-3 Dichloropropene total	100				269191	100.000	98.7	0
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	84521	50.0000	50.3	9793
89 2-Hexanone	43	8.143	8.143	(0.976)	278314	250.000	236	9794
90 1-Chlorohexane	91	8.333	8.333	(0.999)	142814	50.0000	51.7	7907
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	132516	50.0000		9188
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	257365	50.0000	50.3	9262
93 Ethylbenzene +	106	8.369	8.369	(1.003)	141151	50.0000	50.7	9141
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	86041	50.0000	50.0	9228
95 p,m-Xylene	106	8.465	8.465	(1.015)	344782	100.000	102	9920
97 o-Xylene	106	8.745	8.745	(1.049)	161304	50.0000	51.5	9890
101 Styrene	104	8.777	8.777	(1.052)	217155	50.0000	53.8	9875
102 Bromoform ++	173	8.799	8.799	(1.055)	65958	50.0000	45.8	8752
104 Isopropylbenzene	105	8.944	8.944	(1.072)	419659	50.0000	51.2	9898
M 105 TOTAL XYLENE	106				506086	150.000	154	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	104413	50.0000	51.8	9601
107 cis-1,4-dichloro-2-butene	53	9.159	9.159	(0.934)	31414	50.0000	48.3	9577
109 n-Propylbenzene	91	9.201	9.201	(0.938)	498572	50.0000	49.8	8921
108 Bromobenzene	77	9.195	9.195	(0.938)	163865	50.0000	49.2	9575
110 1,1,2,2-Tetrachloroethane++	83	9.243	9.243	(0.943)	93626	50.0000	47.9	9822
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	309005	50.0000	48.4	8137
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	335014	50.0000	49.8	8369
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.951)	119618	50.0000	45.1	7281
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.953)	29027	50.0000	45.0	9449
116 Cyclohexanone	55	9.365	9.365	(0.955)	70763	250.000	219	9733
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	284826	50.0000	49.8	9857
118 tert-butylbenzene	91	9.516	9.516	(0.970)	181942	50.0000	48.1	9906
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.974)	329090	50.0000	49.8	9339
120 sec-Butylbenzene	105	9.623	9.623	(0.981)	433759	50.0000	49.8	9877
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	381901	50.0000	49.5	8896
123 1,3-Dichlorobenzene	146	9.764	9.764	(0.996)	192521	50.0000	49.4	9824
125 1,4-Dichlorobenzene	146	9.815	9.815	(1.001)	196817	50.0000	49.1	9026
* 124 1,4-DICHLOROBENZENE-D4	152	9.806	9.806	(1.000)	128132	50.0000		9061
126 n-Butylbenzene	91	9.954	9.954	(1.015)	500023	50.0000	48.8	9023
127 1,2-Dichlorobenzene	146	10.066	10.066	(1.027)	173488	50.0000	49.2	9690
129 1,2-Dibromo-3-Chloropropane	157	10.558	10.558	(1.077)	19198	50.0000	39.1	9447
130 Hexachlorobutadiene	225	10.999	10.999	(1.122)	57000	50.0000	52.9	9292
131 1,2,4-Trichlorobenzene	180	11.034	11.034	(1.125)	109444	50.0000	50.8	9687
132 Naphthalene	128	11.291	11.291	(1.151)	225518	50.0000	44.7	9777
133 1,2,3-Trichlorobenzene	180	11.449	11.449	(1.168)	92419	50.0000	49.8	9678

QC Flag Legend

H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230612.s.b/c0917s.d

Date : 12-JUN-2023 10:30

Client ID: V15STD050

Sample Info: 1400*V15STD050*51363

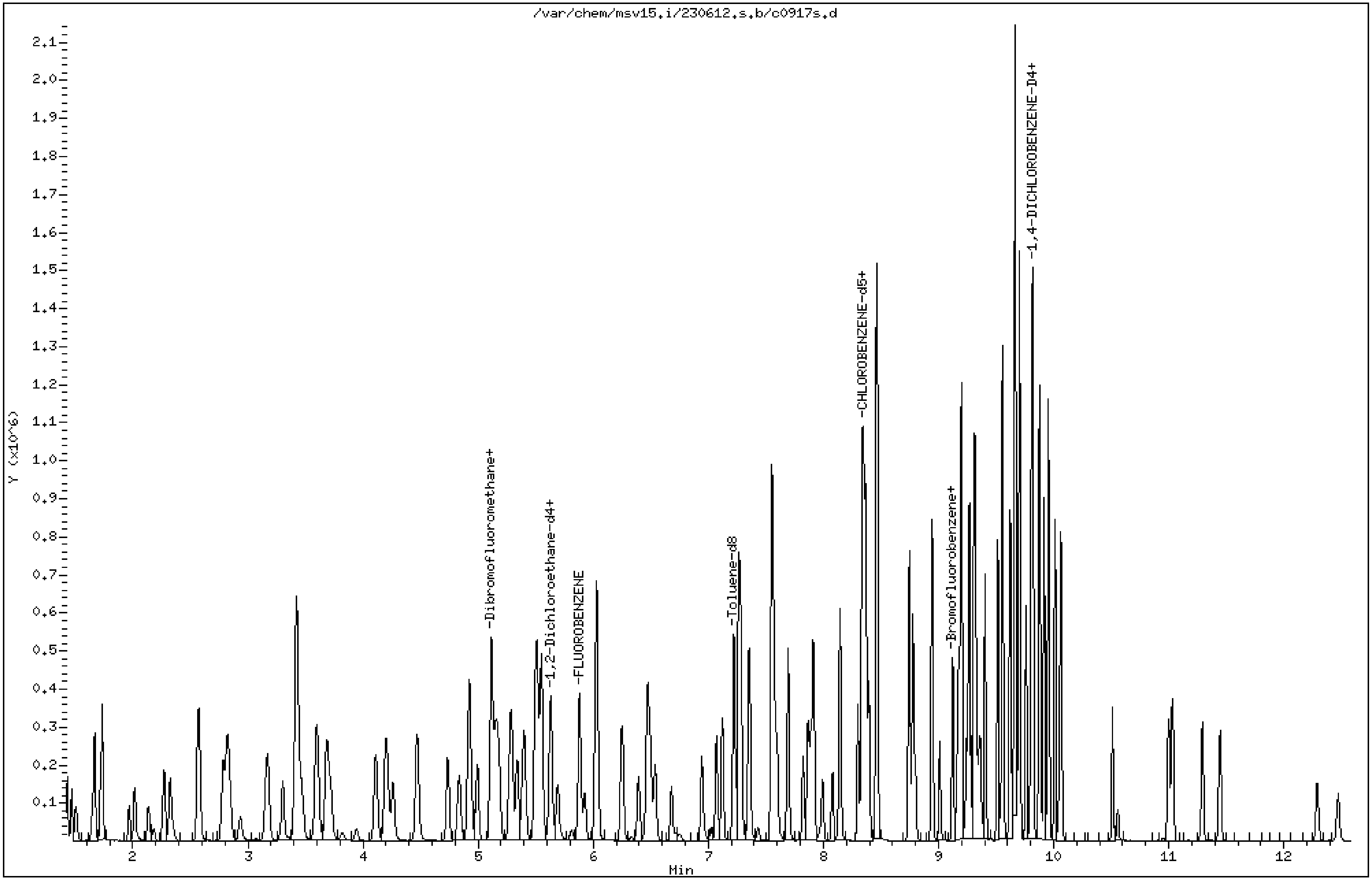
Purge Volume: 5.0

Column phase: RTX-VMS-30M

Instrument: msv15.i

Operator: KN1

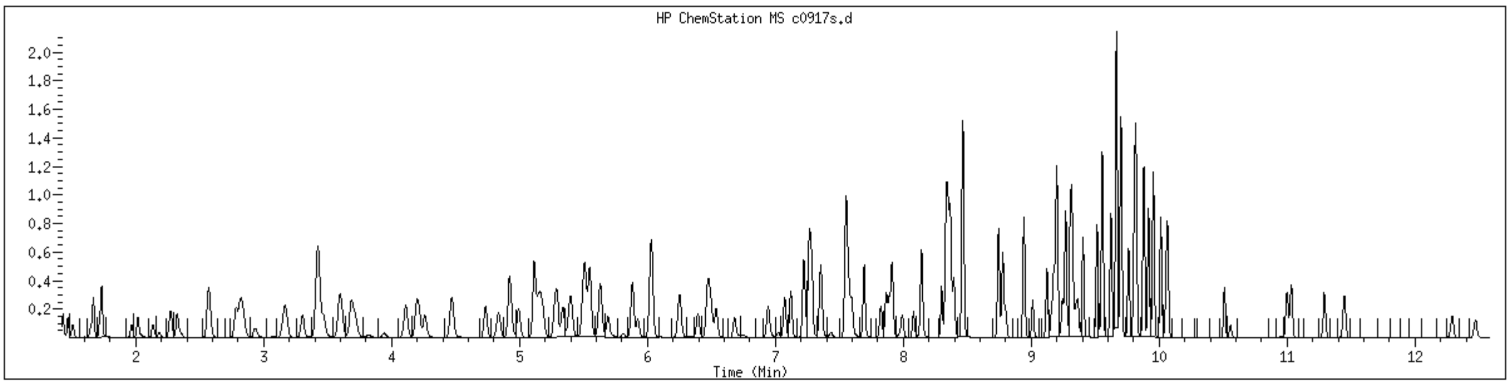
Column diameter: 0.25



Report Date: 06/13/2023 15:50

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID : 1400 SampleType : CCALIB_7
Injection Date: 06/12/2023 10:30 Instrument : msv15.i
Operator : KN1
Sample Info : 1400*V15STD050*51363
Misc Info : MSV~529~*1*SMR
Method : /var/chem/msv15.i/230612.s.b/8260bw15DOD.m
Dilution : 1.00
Matrix : WATER
Integrator : HP RTE Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

7A
VOLATILE CLOSING CALIBRATION CHECK

Report No:	223060502	CCAL ID:	1440
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230612/c0948s
Init. Calib. Date 1:	05/17/23 Time 1: 1428	Analyst:	KN1
Init. Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Batch:	767415
Analysis Date:	06/12/23 Time: 2016	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
1,1,1,2-Tetrachloroethane	50.0	49.7	0.649	0.645	.01	-.65	50	A	
1,1,1-Trichloroethane	50.0	47.9	0.428	0.410	.01	-4.24	50	A	
1,1,2,2-Tetrachloroethane	50.0	51.1	0.763	0.780	.3	2.3	50	A	
1,1,2-Trichloroethane	50.0	51.1	0.593	0.605	.01	2.15	50	A	
1,1-Dichloroethane	50.0	50.2	0.452	0.454	.1	.49	50	A	
1,1-Dichloroethene	50.0	51.2	0.244	0.250	.01	2.41	50	A	
1,1-Dichloropropene	50.0	48.1	0.381	0.366	.01	-3.83	50	A	
1,2,3-Trichlorobenzene	50.0	52.5	0.725	0.761	.01	5.04	50	A	
1,2,3-Trichloropropane	50.0	47.8	1.035	0.990	.01	-4.36	50	A	
1,2,4-Trichlorobenzene	50.0	50.8	0.840	0.854	.01	1.63	50	A	
1,2,4-Trimethylbenzene	50.0	48.1	2.578	2.479	.01	-3.82	50	A	
1,2-Dibromo-3-chloropropane	50.0	48.4	0.196	0.186	.01	-3.2	50	W	
1,2-Dibromoethane	50.0	52.1	0.634	0.660	.01	4.13	50	A	
1,2-Dichlorobenzene	50.0	47.9	1.375	1.318	.01	-4.16	50	A	
1,2-Dichloroethane	50.0	46.1	0.324	0.299	.01	-7.71	50	A	
1,2-Dichloroethane-d4	50.0	46.2	0.257	0.237	.01	-7.65	50	A	
1,2-Dichloroethene (total)	100.0	99.4	0.332	0.330	.01	-.58	50	A	
1,2-Dichloropropane	50.0	50.4	0.254	0.256	.01	.87	50	A	
1,3,5-Trimethylbenzene	50.0	47.4	2.626	2.491	.01	-5.17	50	A	
1,3-Dichlorobenzene	50.0	47.9	1.520	1.456	.01	-4.17	50	A	
1,3-Dichloropropane	50.0	51.1	0.987	1.008	.01	2.14	50	A	
1,4-Dichlorobenzene	50.0	46.8	1.563	1.463	.01	-6.38	50	A	
2,2-Dichloropropane	50.0	44.2	0.402	0.355	.01	-11.5	50	A	
2-Butanone	250.0	284.0	0.115	0.131	.01	13.6	50	A	
2-Chlorotoluene	50.0	46.2	2.489	2.302	.01	-7.5	50	A	
2-Hexanone	250.0	269.0	0.450	0.481	.01	7.6	50	W	
4-Bromofluorobenzene	50.0	51.3	0.761	0.781	.01	2.66	50	A	
4-Chlorotoluene	50.0	47.0	2.234	2.099	.01	-6.02	50	A	
4-Isopropyltoluene	50.0	46.5	3.008	2.797	.01	-6.99	50	A	
4-Methyl-2-pentanone	250.0	260.0	0.548	0.565	.01	4	50	W	
Acetone	250.0	271.0	0.111	0.120	.01	8.28	50	A	
Benzene	50.0	49.6	1.029	1.020	.01	-.89	50	A	
Bromobenzene	50.0	47.0	1.300	1.222	.01	-5.98	50	A	
Bromochloromethane	50.0	50.0	0.128	0.128	.01	-.04	50	A	
Bromodichloromethane	50.0	49.0	0.344	0.337	.01	-2.05	50	A	
Bromoform	50.0	48.5	0.554	0.527	.1	-3	50	W	
Bromomethane	50.0	45.1	0.247	0.223	.01	-9.77	50	A	
Carbon disulfide	50.0	49.9	0.747	0.746	.01	-.14	50	A	
Carbon tetrachloride	50.0	48.5	0.374	0.363	.01	-3	50	A	

FORM V II VOA

7A
VOLATILE CLOSING CALIBRATION CHECK

Report No:	223060502	CCAL ID:	1440
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230612/c0948s
Init. Calib. Date 1:	05/17/23 Time 1: 1428	Analyst:	KN1
Init. Calib. Date 2:	05/17/23 Time 2: 1620	Analytical Batch:	767415
Analysis Date:	06/12/23 Time: 2016	Analytical Method:	EPA 8260B

ANALYTE	TRUE	CONC	RRF	RRF CCV	Min RRF	%D/%Drift	Max %D/ %Drift	TYPE	Q
Chlorobenzene	50.0	49.5	1.931	1.913	.3	-.92	50	A	
Chloroethane	50.0	48.8	0.216	0.211	.01	-2.42	50	A	
Chloroform	50.0	48.0	0.462	0.444	.01	-3.94	50	A	
Chloromethane	50.0	50.2	0.209	0.210	.1	.33	50	A	
Dibromochloromethane	50.0	54.8	0.674	0.738	.01	9.53	50	A	
Dibromofluoromethane	50.0	49.6	0.247	0.245	.01	-.89	50	A	
Dibromomethane	50.0	49.6	0.166	0.164	.01	-.86	50	A	
Dichlorodifluoromethane	50.0	52.5	0.232	0.244	.01	5.02	50	A	
Ethylbenzene	50.0	50.6	1.050	1.063	.01	1.23	50	A	
Hexachlorobutadiene	50.0	49.8	0.420	0.419	.01	-.35	50	A	
Isopropylbenzene (Cumene)	50.0	50.4	3.094	3.122	.01	.9	50	A	
Methylene chloride	50.0	49.7	0.270	0.269	.01	-.59	50	A	
Naphthalene	50.0	54.3	1.969	2.138	.01	8.59	50	A	
Styrene	50.0	52.3	1.522	1.590	.01	4.5	50	A	
Tetrachloroethene	50.0	50.4	0.660	0.665	.01	.85	50	A	
Toluene	50.0	50.1	2.917	2.924	.01	.24	50	A	
Toluene-d8	50.0	51.1	2.343	2.397	.01	2.29	50	A	
Trichloroethene	50.0	50.9	0.316	0.322	.01	1.86	50	A	
Trichlorofluoromethane	50.0	48.8	0.429	0.419	.01	-2.36	50	A	
Vinyl chloride	50.0	50.4	0.294	0.297	.01	.89	50	A	
cis-1,2-Dichloroethene	50.0	49.0	0.325	0.319	.01	-1.9	50	A	
cis-1,3-Dichloropropene	50.0	49.1	0.408	0.401	.01	-1.72	50	A	
m,p-Xylene	100.0	100.0	1.269	1.271	.01	.15	50	A	
n-Butylbenzene	50.0	45.8	3.999	3.664	.01	-8.37	50	A	
n-Propylbenzene	50.0	46.9	3.905	3.665	.01	-6.14	50	A	
o-Xylene	50.0	50.8	1.182	1.201	.01	1.57	50	A	
sec-Butylbenzene	50.0	47.2	3.397	3.210	.01	-5.52	50	A	
tert-Butyl methyl ether (MTBE)	50.0	51.0	0.660	0.673	.01	1.94	50	A	
tert-Butylbenzene	50.0	45.3	1.475	1.337	.01	-9.35	50	A	
trans-1,2-Dichloroethene	50.0	50.3	0.338	0.341	.01	.7	50	A	
trans-1,3-Dichloropropene	50.0	48.4	0.387	0.374	.01	-3.13	50	A	

FORM V II VOA

GCAL, Inc.

Data file : /var/chem/msv15.i/230612.s.b/c0948s.d
Lab Smp Id: 1440 Client Smp ID: V15STD050
Inj Date : 12-JUN-2023 20:16
Operator : KN1 Inst ID: msv15.i
Smp Info : 1440*V15STD050
Misc Info : MSV~52949~*1*SMR
Comment :
Method : /var/chem/msv15.i/230612.s.b/8260bw15DODclos.m
Meth Date : 21-Jun-2023 16:16 smr Quant Type: ISTD
Cal Date : 17-MAY-2023 16:20 Cal File: c0234.d
Als bottle: 1 Continuing Calibration Sample
Dil Factor: 1.00000
Integrator: HP RTE Compound Sublist: 8260DOD.sub
Target Version: 3.50
Processing Host: org.gcal.com

Concentration Formula: Amt * DF * Uf/Vo * CpndVariable

Name	Value	Description
DF	1.00000	Dilution Factor
Uf	5.00000	ng unit correction factor
Vo	5.00000	Sample Volume purged (mL)
DF	1.00000	

Cpnd Variable Local Compound Variable

						AMOUNTS		
	QUANT SIG					CAL-AMT	ON-COL	
Compounds	MASS	RT	EXP RT	REL RT	RESPONSE	(ppb)	(ppb)	SIMILARITY
1 Dichlorodifluoromethane	85	1.468	1.468	(0.250)	97136	50.0000	52.5	9789
2 Chloromethane ++	50	1.655	1.655	(0.281)	83568	50.0000	50.2	9738
3 Vinyl Chloride +	62	1.716	1.716	(0.292)	118340	50.0000	50.4	9819
5 1-3 Butadiene	39	1.735	1.735	(0.295)	94822	50.0000	54.3	9635
6 Bromomethane	94	2.015	2.015	(0.343)	88751	50.0000	45.1	9776
7 Chloroethane	64	2.131	2.131	(0.362)	84127	50.0000	48.8	9252
8 Trichlorofluoromethane	101	2.269	2.269	(0.386)	166940	50.0000	48.8	9761
11 1,1-Dichloroethene +	96	2.783	2.783	(0.473)	99619	50.0000	51.2	8368
12 Carbon Disulfide	76	2.812	2.812	(0.478)	297169	50.0000	49.9	8147
13 1,1,2Trichlotrifluoroethane	101	2.835	2.835	(0.482)	106253	50.0000	50.3	9521
15 Methyl Iodide	142	2.931	2.931	(0.498)	119218	50.0000	51.2	9514
16 Acrolein	56	3.147	3.147	(0.535)	73814	250.000	249	6222
17 Allyl chloride	39	3.301	3.301	(0.561)	88375	50.0000	51.7	9777
18 Methylene Chloride	49	3.417	3.417	(0.581)	107010	50.0000	49.7	4947

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
19 Acetone	43	3.468	3.468	(0.590)	239200	250.000	271	5228(H)
20 trans-1,2-Dichloroethene	61	3.591	3.591	(0.610)	135800	50.0000	50.3	8437
21 Methyl Acetate	43	3.610	3.610	(0.614)	323975	250.000	294	9653
22 Hexane	57	3.687	3.687	(0.627)	197950	50.0000	50.9	9384
23 MTBE	73	3.716	3.716	(0.632)	268127	50.0000	51.0	9657
24 tert-Butyl Alcohol	59	3.822	3.822	(0.650)	63987	250.000	276	7987
25 Acetonitrile	41	3.944	3.944	(0.670)	43946	250.000	281	9468
26 Isopropyl Ether	45	4.111	4.111	(0.699)	249582	50.0000	53.2	9599
27 Chloroprene	53	4.195	4.195	(0.713)	127790	50.0000	48.3	8761
28 1,1-Dichloroethane ++	63	4.214	4.214	(0.716)	180967	50.0000	50.2	9399
29 Acrylonitrile	53	4.259	4.259	(0.724)	172547	250.000	281	9860
31 Ethyl Tert-butyl Ether	59	4.472	4.472	(0.760)	264290	50.0000	50.0	8283
30 Vinyl Acetate	43	4.472	4.472	(0.760)	109768	50.0000	37.8	4948
32 cis-1,2-Dichloroethene	61	4.735	4.735	(0.805)	127004	50.0000	49.0	9786
M 33 Total 1,2-Dichloroethene	61				262804	100.000	99.4	0
34 2,2-Dichloropropane	77	4.838	4.838	(0.822)	141643	50.0000	44.2	9570
35 Bromochloromethane	128	4.918	4.918	(0.836)	50809	50.0000	50.0	9296
36 Cyclohexane	56	4.925	4.925	(0.837)	168617	50.0000	51.3	9106
37 Chloroform +	83	4.992	4.992	(0.849)	176962	50.0000	48.0	9812
39 Carbon Tetrachloride	117	5.121	5.121	(0.870)	144455	50.0000	48.5	8092
$ 42 Dibromofluoromethane	111	5.156	5.156	(0.876)	97511	50.0000	49.6	8140
43 1,1,1-Trichloroethane	97	5.179	5.179	(0.880)	163362	50.0000	47.9	9652
45 1,1-Dichloropropene	75	5.295	5.295	(0.900)	145855	50.0000	48.1	9455
44 2-Butanone	43	5.275	5.275	(0.897)	260404	250.000	284	5858
46 2,2,4 Trimethylpentane	57	5.401	5.401	(0.918)	341672	50.0000	37.3	9463
48 Benzene	78	5.516	5.516	(0.938)	406290	50.0000	49.6	9789
56 tert-amyl Methyl Ether	73	5.632	5.632	(0.957)	266491	50.0000	49.5	7209
$ 50 1,2-Dichloroethane-d4	65	5.632	5.632	(0.957)	94611	50.0000	46.2	8556
52 1,2-Dichloroethane	62	5.693	5.693	(0.968)	119023	50.0000	46.1	9785
51 Isobutyl Alcohol	43	5.729	5.729	(0.974)	14494	250.000	270	3247
* 55 FLUOROBENZENE	96	5.883	5.883	(1.000)	398469	50.0000		9828
58 Methyl Cyclohexane	83	6.024	6.024	(1.024)	189774	50.0000	49.3	8522
57 Trichloroethene	130	6.031	6.031	(1.025)	128250	50.0000	50.9	8999
62 Dibromomethane	93	6.394	6.394	(1.087)	65402	50.0000	49.6	9695
64 1,2-Dichloropropane +	63	6.481	6.481	(1.102)	102203	50.0000	50.4	7952
65 Bromodichloromethane	83	6.539	6.539	(1.111)	134424	50.0000	49.0	9844
67 1,4- Dioxane	88	6.722	6.722	(1.143)	20249	1250.00	1350	9420
68 1-Bromo-2-chloroethane	63	6.944	6.944	(1.180)	130943	50.0000	50.7	9453
72 2-Chloroethyl vinyl ether	63	7.024	7.024	(1.194)	122982	250.000	214	9754
73 cis-1,3-Dichloropropene	75	7.073	7.073	(1.202)	159692	50.0000	49.1	9763
$ 74 Toluene-d8	98	7.221	7.221	(0.866)	363035	50.0000	51.1	9926
75 Toluene +	91	7.262	7.262	(0.871)	442920	50.0000	50.1	9623
77 4-methyl-2-pentanone	43	7.549	7.549	(0.905)	427657	250.000	260	7591
78 Tetrachloroethene	164	7.558	7.558	(0.906)	100747	50.0000	50.4	8080
79 trans-1,3-Dichloropropene	75	7.574	7.574	(1.287)	149198	50.0000	48.4	8817
81 1,1,2-Trichloroethane	97	7.693	7.693	(0.922)	91677	50.0000	51.1	9471
82 Dibromochloromethane	129	7.825	7.825	(0.938)	111805	50.0000	54.8	9748

Compounds	QUANT SIG MASS	RT	EXP RT	REL RT	RESPONSE	AMOUNTS CAL-AMT (ppb)	ON-COL (ppb)	SIMILARITY
84 3,4-dichloro-1-butene	75	7.867	7.867	(0.943)	121240	50.0000	51.4	9694
86 1,3-Dichloropropane	76	7.889	7.889	(0.946)	152623	50.0000	51.1	9589
M 83 1-3 Dichloropropene total	100				308890	100.000	97.6	0
88 1,2-Dibromoethane(EDB)	107	7.992	7.992	(0.958)	99997	50.0000	52.1	9762
89 2-Hexanone	43	8.143	8.143	(0.976)	363973	250.000	269	9757
90 1-Chlorohexane	91	8.333	8.333	(0.999)	159622	50.0000	50.5	7558
* 91 CHLOROBENZENE-d5	82	8.340	8.340	(1.000)	151453	50.0000		9195
92 Chlorobenzene ++	112	8.352	8.352	(1.002)	289733	50.0000	49.5	9465
93 Ethylbenzene +	106	8.368	8.368	(1.003)	160961	50.0000	50.6	9062
94 1,1,1,2-Tetrachloroethane	133	8.394	8.394	(1.007)	97627	50.0000	49.7	9361
95 p,m-Xylene	106	8.462	8.462	(1.015)	385061	100.000	100	9914
97 o-Xylene	106	8.745	8.745	(1.049)	181826	50.0000	50.8	9891
101 Styrene	104	8.777	8.777	(1.052)	240871	50.0000	52.3	9841
102 Bromoform ++	173	8.799	8.799	(1.055)	79848	50.0000	48.5	8657
104 Isopropylbenzene	105	8.944	8.944	(1.072)	472765	50.0000	50.4	9897
M 105 TOTAL XYLENE	106				566887	150.000	151	0
$ 106 Bromofluorobenzene	174	9.124	9.124	(1.094)	118328	50.0000	51.3	9678
107 cis-1,4-dichloro-2-butene	53	9.156	9.156	(0.934)	38368	50.0000	51.2	9574
109 n-Propylbenzene	91	9.201	9.201	(0.939)	541129	50.0000	46.9	8705
108 Bromobenzene	77	9.192	9.192	(0.938)	180390	50.0000	47.0	8764
110 1,1,2,2-Tetrachloroethane++	83	9.240	9.240	(0.943)	115223	50.0000	51.1	9814
112 2-Chlorotoluene	91	9.304	9.304	(0.949)	339922	50.0000	46.2	8638
113 1,3,5-Trimethylbenzene	105	9.317	9.317	(0.950)	367721	50.0000	47.4	8135
114 1,2,3-Trichloropropane	75	9.330	9.330	(0.952)	146187	50.0000	47.8	8137
115 trans-1,4-Dichloro-2-Butene	53	9.349	9.349	(0.954)	36333	50.0000	48.8	9284
116 Cyclohexanone	55	9.365	9.365	(0.955)	106150	250.000	284	9675
117 4-Chlorotoluene	91	9.404	9.404	(0.959)	309951	50.0000	47.0	9863
118 tert-butylbenzene	91	9.513	9.513	(0.970)	197355	50.0000	45.3	9842
119 1,2,4-Trimethylbenzene	105	9.555	9.555	(0.975)	366057	50.0000	48.1	9346
120 sec-Butylbenzene	105	9.622	9.622	(0.982)	473947	50.0000	47.2	9881
121 p-Isopropyltoluene	119	9.703	9.703	(0.990)	413012	50.0000	46.5	8981
123 1,3-Dichlorobenzene	146	9.761	9.761	(0.996)	215021	50.0000	47.9	9803
125 1,4-Dichlorobenzene	146	9.812	9.812	(1.001)	216055	50.0000	46.8	8615(H)
* 124 1,4-DICHLOROBENZENE-D4	152	9.802	9.802	(1.000)	147646	50.0000		9251
126 n-Butylbenzene	91	9.954	9.954	(1.015)	541038	50.0000	45.8	8907
127 1,2-Dichlorobenzene	146	10.066	10.066	(1.027)	194540	50.0000	47.9	9806
129 1,2-Dibromo-3-Chloropropane	157	10.558	10.558	(1.077)	27497	50.0000	48.4	9557
130 Hexachlorobutadiene	225	10.999	10.999	(1.122)	61832	50.0000	49.8	9442
131 1,2,4-Trichlorobenzene	180	11.031	11.031	(1.125)	126104	50.0000	50.8	9612
132 Naphthalene	128	11.288	11.288	(1.152)	315723	50.0000	54.3	9856
133 1,2,3-Trichlorobenzene	180	11.442	11.442	(1.167)	112421	50.0000	52.5	9564

QC Flag Legend

H - Operator selected an alternate compound hit.

Data File: /var/chem/msv15.i/230612.s.b/c0948s.d

Page 1

Date : 12-JUN-2023 20:16

Client ID: V15STD050

Instrument: msv15.i

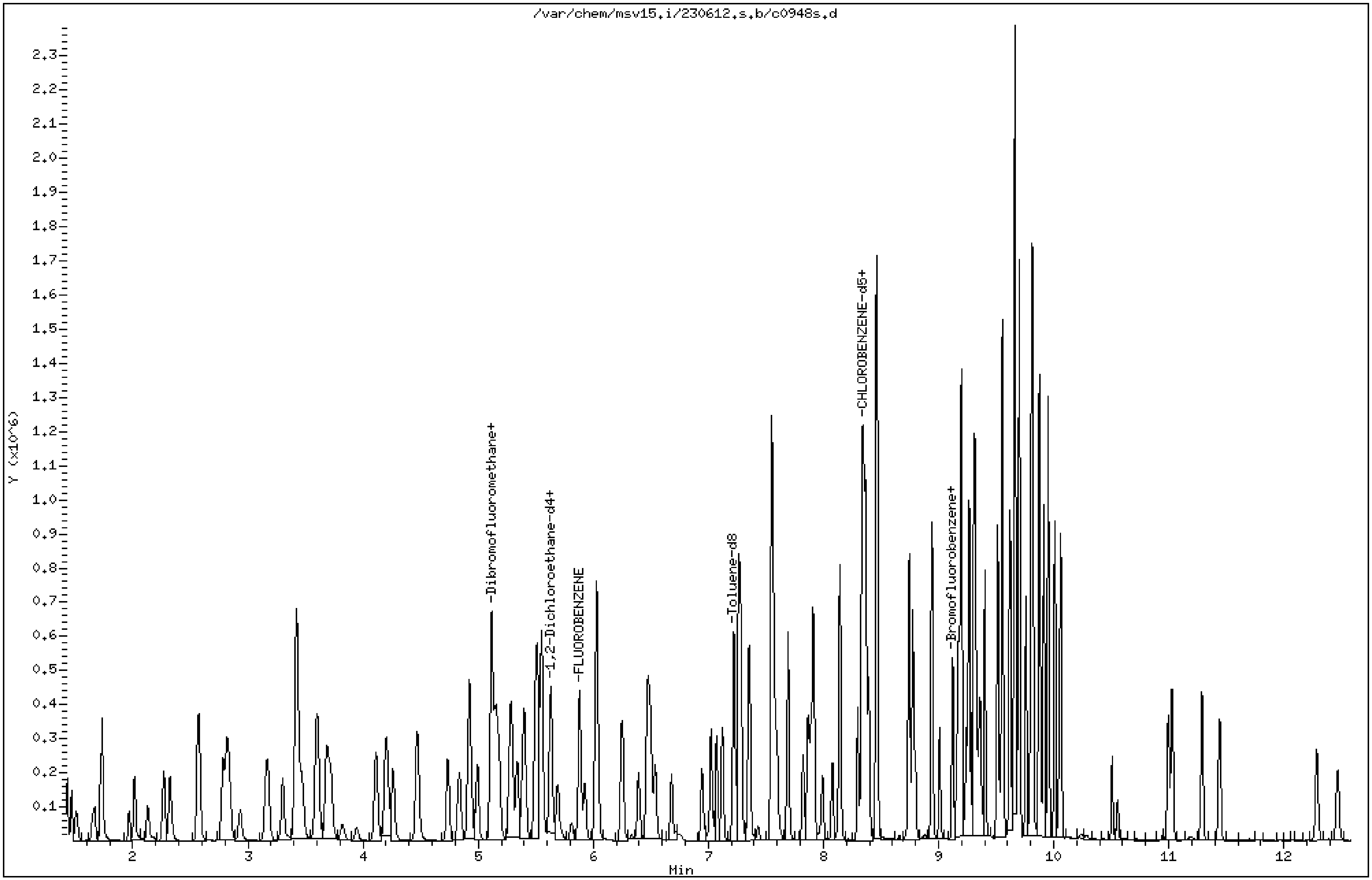
Sample Info: 1440*V15STD050

Purge Volume: 5.0

Operator: KN1

Column phase: RTX-VMS-30M

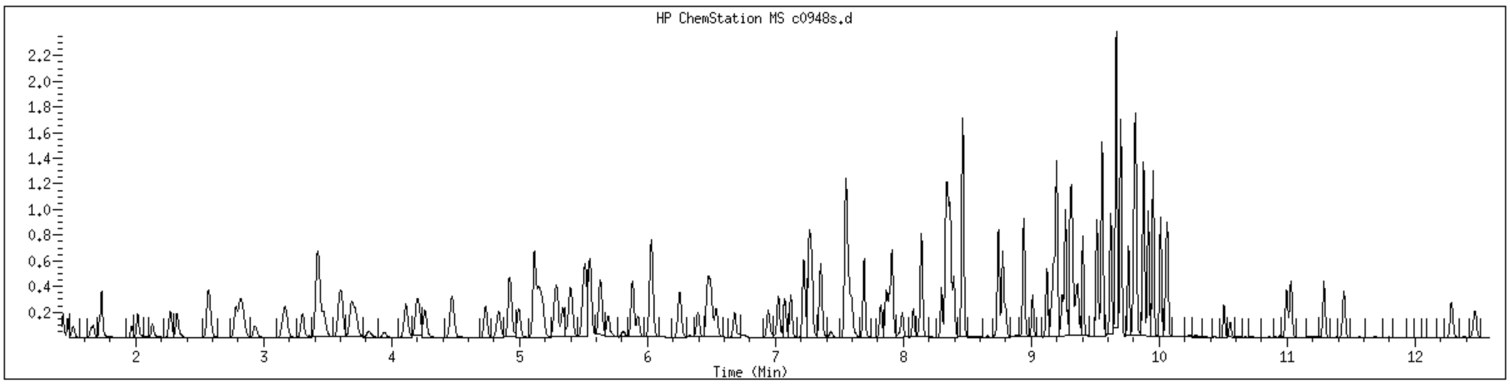
Column diameter: 0.25



Data file : /var/chem/msv15.i/230612.s.b/c0948s.d   Page: 1
Report Date: 06/21/2023 16:16

MANUAL INTEGRATION GRAPHIC REPORT

Lab ID   : 1440   SampleType   : CCALIB_7
Injection Date: 06/12/2023 20:16   Instrument   : msv15.i
Operator   : KN1
Sample Info   : 1440*V15STD050
Misc Info   : MSV~52949~*1*SMR
Method   : /var/chem/msv15.i/230612.s.b/8260bw15DODclos.m
Dilution   : 1.00
Matrix   : WATER
Integrator   : HP RTE   Compound Sublist: 8260DOD



NO MANUAL INTEGRATIONS

EPA 8260B DOD

Form 8A

Internal Standards

8A

VOLATILE INTERNAL STANDARD AREA AND RT SUMMARY

Report No:	223060502	Standard ID:	1400
GC Column:	RTX-VMS-30M ID .25 (mm)	Instrument ID:	MSV15
Injection Vol.:	1.0 (µL)	Lab File ID:	230612/c0917s
Analyst:	KN1	Analytical Batch:	767415
Analysis Date:	06/12/23 Time: 1030	Analytical Method:	EPA 8260B

	IS 1 Area	RT	IS 2 Area	RT	IS 3 Area	RT
STANDARD	132516	8.34	128132	9.81	343271	5.88

CLIENT SAMPLE ID	***GCAL SAMP ID***	IS 1 Area	#	RT	#	IS 2 Area	#	RT	#	IS 3 Area	#	RT	#
LCS2491796	2491796	162950		8.34		168254		9.8		377022		5.88	
MB2491795	2491795	153699		8.34		152172		9.8		364881		5.88	
OT07-BO-40-4-6-8	22306050201	141230		8.34		143566		9.8		319452		5.88	
OT07-BO-840-4-6-8	22306050204	154197		8.34		158398		9.8		333753		5.88	
OT07-BO-40-4-6-8-MS	22306050202	177689		8.34		178449		9.8		386552		5.88	
OT07-BO-40-4-6-8-MSD	22306050203	180518		8.34		186526		9.8		394390		5.88	

IS 1 ID : Chlorobenzene-d5

IS 2 ID : 1,4-Dichlorobenzene-d4

IS 3 ID : Fluorobenzene

Column used to flag values outside QC limits with an asterisk

* Value outside of QC limits

AREA UPPER LIMIT = +100% of internal standard area

AREA LOWER LIMIT = -50% of internal standard area

RT UPPER LIMIT = +0.17 minutes of internal standard RT

RT LOWER LIMIT = -0.17 minutes of internal standard RT

FORM V III VOA

EPA 8260B DOD

RunLogs

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 17-MAY-2023
Instrument: msv15.i
Analyst(s): CJR

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
BLANK	8260b+App	c0221b.d	5.00 ml	17-MAY-2023 09:56	1.0	CJR	
1000	all	c0221.d	0.00 ml	17-MAY-2023 09:56	1.0	CJR	
BLANK	8260b+App	c0222.d	5.00 ml	17-MAY-2023 10:42	1.0	KN1	
BLANK	8260b+App	c0223.d	5.00 ml	17-MAY-2023 11:25	1.0	CJR	
BLANK	all	c0223t.d	0.00 ml	17-MAY-2023 11:25	1.0	CJR	
BLANK	8260b+App	c0224.d	5.00 ml	17-MAY-2023 11:44	1.0	CJR	
BLANK	all	c0224t.d	0.00 ml	17-MAY-2023 11:44	1.0	CJR	
BLANK	8260b+App	c0225.d	5.00 ml	17-MAY-2023 12:42	1.0	CJR	
BLANK	all	c0225t.d	0.00 ml	17-MAY-2023 12:42	1.0	CJR	
BLANK	8260b+App	c0226.d	5.00 ml	17-MAY-2023 13:15	1.0	CJR	
BLANK	all	c0226t.d	0.00 ml	17-MAY-2023 13:15	1.0	CJR	
HANDSPIKE	8260b+App	c0227.d	5.00 ml	17-MAY-2023 13:37	1.0	CJR	
HANDSPIKE	all	c0227t.d	0.00 ml	17-MAY-2023 13:37	1.0	CJR	
1203	8260b+App	c0228.d	5.00 ml	17-MAY-2023 14:28	1.0	CJR	
1203	8260b+App	c0228llc.d	5.00 ml	17-MAY-2023 14:28	1.0	CJR	
1204	8260b+App	c0229.d	5.00 ml	17-MAY-2023 14:47	1.0	CJR	
1204	8260b+App	c0229llc.d	5.00 ml	17-MAY-2023 14:47	1.0	CJR	
1205	8260b+App	c0230.d	5.00 ml	17-MAY-2023 15:05	1.0	CJR	
1206	8260b+App	c0231.d	5.00 ml	17-MAY-2023 15:24	1.0	CJR	
1207	8260b+App	c0232.d	5.00 ml	17-MAY-2023 15:43	1.0	CJR	
1208	8260b+App	c0233.d	5.00 ml	17-MAY-2023 16:01	1.0	CJR	
1209	8260b+App	c0234.d	5.00 ml	17-MAY-2023 16:20	1.0	CJR	

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 17-MAY-2023
Instrument: msv15.i
Analyst(s): CJR

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
BLANK	8260b+App	c0235.d	5.00 ml	17-MAY-2023 16:39	1.0	CJR	
1600	8260b+App	c0238.d	5.00 ml	17-MAY-2023 17:35	1.0	CJR	
1600	all	c0238t.d	0.00 ml	17-MAY-2023 17:35	1.0	CJR	

Legend:
- Peak was not identified (NI)
- Peak integration was less area than it should be (LT)
- Peak Integration was more area than it should be (GT)
- Baseline adjustment required (BA)
- Wrong peak identified (WP)
- Split co-eluting peaks (CO)
- Retention time shift (RT)
- Interference correction (INT)
- No spectral match (for deletion of analytes, where capabilities exist) (SM)

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 12-JUN-2023
Instrument: msv15.i
Analyst(s): KN1

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
BLANK	8260b+App	c0914.d	5.00 ml	12-JUN-2023 09:02	1.0	KN1	
1000	all	c0915.d	0.00 ml	12-JUN-2023 09:21	1.0	CJR	
1000	all	c0915s.d	0.00 ml	12-JUN-2023 09:21	1.0	CJR	
1400	8260b+App	c0916.d	5.00 ml	12-JUN-2023 09:53	1.0	KN1	
1400	8260DOD	c0917.d	5.00 ml	12-JUN-2023 10:30	1.0	KN1	
2491799	8260DOD	c0917L.d	5.00 ml	12-JUN-2023 10:30	1.0	KN1	
1400	8260DOD	c0917s.d	5.00 ml	12-JUN-2023 10:30	1.0	KN1	
1400	8260DOD	c0918.d	5.00 ml	12-JUN-2023 10:49	1.0	KN1	
BLANK	8260DOD	c0919.d	5.00 ml	12-JUN-2023 11:07	1.0	KN1	
2491796	8260DOD	c0920.d	5.00 g	12-JUN-2023 11:29	50.0	KN1	115-CO
BLANK	8260DOD	c0921.d	5.00 ml	12-JUN-2023 11:48	1.0	KN1	
2491795	8260DOD	c0922.d	5.00 g	12-JUN-2023 12:07	50.0	KN1	
2491798	8260DOD	c0923.d	5.00 ml	12-JUN-2023 12:26	1.0	KN1	
22306083001	8260DOD	c0924.d	5.00 ml	12-JUN-2023 12:45	1.0	KN1	
22306085301	8260DOD	c0925.d	5.00 ml	12-JUN-2023 13:03	1.0	KN1	
22306083015	8260DOD	c0926.d	5.00 ml	12-JUN-2023 13:22	1.0	KN1	
22306085302	8260DOD	c0927.d	5.00 ml	12-JUN-2023 13:41	1.0	KN1	
22306085303	8260DOD	c0928.d	5.00 ml	12-JUN-2023 14:00	1.0	KN1	
22306085304	8260DOD	c0929.d	5.00 ml	12-JUN-2023 14:18	1.0	KN1	
22306085305	8260DOD	c0930.d	5.00 ml	12-JUN-2023 14:37	1.0	KN1	
22306083005	8260DOD	c0931.d	5.00 ml	12-JUN-2023 14:56	1.0	KN1	
22306083009	8260DOD	c0932.d	5.00 ml	12-JUN-2023 15:15	1.0	KN1	

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 12-JUN-2023
Instrument: msv15.i
Analyst(s): KN1

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
22306083011	8260DOD	c0933.d	5.00 ml	12-JUN-2023 15:34	1.0	KN1	
22306083012	8260DOD	c0934.d	5.00 ml	12-JUN-2023 15:52	1.0	KN1	
22306083014	8260DOD	c0935.d	5.00 ml	12-JUN-2023 16:11	1.0	KN1	
22306083004	8260DOD	c0936.d	5.00 ml	12-JUN-2023 16:30	2.0	KN1	
22306083013	8260DOD	c0937.d RR@1X	5.00 ml	12-JUN-2023 16:49	2.0	KN1	OD
22306083008	8260DOD	c0938.d	5.00 ml	12-JUN-2023 17:08	5.0	KN1	
22306083010	8260DOD	c0939.d RR@2X	5.00 ml	12-JUN-2023 17:26	5.0	KN1	OD
22306083002	8260DOD	c0940.d	5.00 ml	12-JUN-2023 17:45	10.0	KN1	
22306083003	8260DOD	c0941.d	5.00 ml	12-JUN-2023 18:04	10.0	KN1	
22306050201	8260DOD	c0942.d	5.92 g	12-JUN-2023 18:23	250.0	KN1	121-CO [SEAN SAID TO REPORT THESE RESULTS]
22306050204	8260DOD	c0943.d	6.06 g	12-JUN-2023 18:42	250.0	KN1	121-CO
22306083006	8260DOD	c0944.d	5.00 ml	12-JUN-2023 19:01	1.0	KN1	
22306083007	8260DOD	c0945.d	5.00 ml	12-JUN-2023 19:20	1.0	KN1	
22306050202	8260DOD	c0946.d	5.63 g	12-JUN-2023 19:38	250.0	KN1	
22306050203	8260DOD	c0947.d	5.73 g	12-JUN-2023 19:57	250.0	KN1	
1440	8260DOD	c0948.d	5.00 ml	12-JUN-2023 20:16	1.0	KN1	
1440	8260DOD	c0948s.d	5.00 ml	12-JUN-2023 20:16	1.0	KN1	
BLANK	8260DOD	c0949.d	5.00 ml	12-JUN-2023 20:35	1.0	KN1	
BLANK	8260DOD	c0950.d	5.00 ml	12-JUN-2023 20:53	1.0	KN1	
BLANK	8260DOD	c0951.d	5.00 ml	12-JUN-2023 21:12	1.0	KN1	
BLANK	8260DOD	c0952.d	5.00 ml	12-JUN-2023 21:31	1.0	KN1	
BLANK	8260DOD	c0953.d	5.00 ml	12-JUN-2023 21:50	1.0	KN1	

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LABORATORY CHRONICLE: MSV DEPARTMENT

Date: 12-JUN-2023
Instrument: msv15.i
Analyst(s): KN1

	Standard	Conc	ID	EXP
	8260 IS/SS	50		
1000 (BFB)	BFB IS/SS	50		
1400 (CCV)	8260	250		
	Ac/Ac/VA	MC		
	CVE	250		

Sample ID	Sublist	DataFile	Wgt/Vol	Injection Time	Dil	Analyst	Comments/MI
BLANK	8260DOD	c0954.d	5.00 ml	12-JUN-2023 22:09	1.0	KN1	

Legend:

- Peak was not identified (NI)
- Peak integration was less area than it should be (LT)
- Peak Integration was more area than it should be (GT)
- Baseline adjustment required (BA)
- Wrong peak identified (WP)
- Split co-eluting peaks (CO)
- Retention time shift (RT)
- Interference correction (INT)
- No spectral match (for deletion of analytes, where capabilities exist) (SM)

MS VOA Sample Weights

Container ID	Vial	Inital Wgt	Final Wgt	Sample Wgt	Analyst	Date	Bal ID	Preservative	Comments
22306050201-B	B	35.72	41.59	5.87	KMN	06/05/2023 13:33	BAL24	SODIUM BISULFATE	VIAL LOT# 022023-3
22306050201-C	C	33.42	39.34	5.92	KMN	06/05/2023 13:34	BAL24	METHANOL	VIAL LOT# 022023-3
22306050201-D	A	35.72	41.58	5.86	KMN	06/05/2023 13:33	BAL24	SODIUM BISULFATE	VIAL LOT# 022023-3
22306050202-B	B	35.77	41.50	5.73	KMN	06/05/2023 13:38	BAL24	SODIUM BISULFATE	VIAL LOT# 022023-3
22306050202-C	C	33.74	39.37	5.63	KMN	06/05/2023 13:38	BAL24	METHANOL	VIAL LOT# 022023-3
22306050202-D	A	35.70	41.44	5.74	KMN	06/05/2023 13:37	BAL24	SODIUM BISULFATE	VIAL LOT# 022023-3
22306050203-B	B	35.77	40.22	4.45	KMN	06/05/2023 13:31	BAL24	SODIUM BISULFATE	VIAL LOT# 022023-3
22306050203-C	C	33.52	39.25	5.73	KMN	06/05/2023 13:31	BAL24	METHANOL	VIAL LOT# 022023-3
22306050203-D	A	35.67	41.51	5.84	KMN	06/05/2023 13:30	BAL24	SODIUM BISULFATE	VIAL LOT# 022023-3
22306050204-B	B	35.82	41.65	5.83	KMN	06/05/2023 13:35	BAL24	SODIUM BISULFATE	VIAL LOT# 022023-3
22306050204-C	C	33.66	39.72	6.06	KMN	06/05/2023 13:36	BAL24	METHANOL	VIAL LOT# 022023-3
22306050204-D	A	35.77	41.97	6.20	KMN	06/05/2023 13:35	BAL24	SODIUM BISULFATE	VIAL LOT# 022023-3

Dry Weight/Percent Moisture

767748

ANALYST/TECH	AMH	START DATE/TIME	6/19/2023 14:00	END DATE/TIME	6/20/2023 10:59	BATCH	767748

#	LAB ID	Pan ID	Pan Weight (g)	Initial Wgt (g) (Sample+Pan)	Final Wgt #1 (g) (Sample+Pan)	Final Wgt #2 (g) (Sample+Pan)	Diff (g)	Initial Wgt (g) (LessPan)	Final Wgt (g) (LessPan)	Total Solids (%)	Total Moisture (%)
1	22306050201	1	1.0043	11.0006	8.7558			9.9963	7.7515	77.54	22.46
2	22306050202	1	1.0043	11.0006	8.7558			9.9963	7.7515	77.54	22.46
3	22306050203	1	1.0043	11.0006	8.7558			9.9963	7.7515	77.54	22.46
4	22306050204	2	1.0019	11.0261	8.9980			10.0242	7.9961	79.77	20.23
5	2493616	3	1.0045	10.9386	8.8051			9.9341	7.8006	78.52	21.48
6											
7											
8											
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30											

EQUIPMENT\CONDITIONS

BALANCE ID		
BAL15		

NOTES

TS % = ((Final Sample Mass - Initial Sample Mass) x 100) / Initial Sample Mass;

Client ID: 4887 - Plexus Consultants

SDG: 223060502

PM: CAM

Pace

CHAIN-OF-CUSTODY / Analytical Request Do

The Chain-of-Custody is a LEGAL DOCUMENT. All relevant fields must

Submitting a sample via this chain of custody constitutes acknowledgment and acceptance of the Pace Terms and Conditions found at https://info.pacela

Section A Required Client Information:	Section B Required Project Information:	Section C Invoice Information:
Company: PLEXUS CONSULTANTS	Report To: Patrick Reilley	Attention: Same as Section A
Address: 5510 Cherokee Ave.	Copy To:	Company Name:
Suite 350, Alexandria, VA 22312		Address:
Email: preilley@plexsci.com	Purchase Order #:	Pace Quote:
Phone: (703) 989-8405 Fax:	Project Name: Scott AFB - OT007 Soil	Pace Project Manager: carolyn.dicharry@pacelabs.com
Requested Due Date:	Project #: [illegible]	Pace Profile #: 271212

Regulatory Agency	State / Location
	IL

MATRIX / CODE: Drinking Water DW; Water WT; Waste Water WW; Product P; Soil/Solid SL; Oil OL; Wipe WP; Air AR; Other OT; Tissue TS

SAMPLE ID — One Character per box. (A-Z, 0-9 / , -) Sample Ids must be unique

Requested Analysis Filtered (Y/N)

ITEM #	SAMPLE ID	MATRIX CODE (see valid codes to left)	SAMPLE TYPE (G=GRAB C=COMP)	COLLECTED START DATE	START TIME	END DATE	END TIME	SAMPLE TEMP AT COLLECTION	# OF CONTAINERS	Unpreserved	H2SO4	HNO3	HCl	NaOH	Na2S2O3	Methanol	Other	Analyses Test Y/N	8260 Volatile Organic Comp	Residual Chlorine (Y/N)
1	OT07-BO-40-4-6-8	SL	G	6/2/23	9:05	6/2/23	9:05		4	1						1	2		X	
2	~~OT07-BO-MS~~ OT07-BO-40-4-6-8 MS	SL	G	6/2/23	9:10	6/2/23	9:10		4	1						1	2		X	
3	~~OT07-BO-MSD~~ OT07-BO-40-4-6-8 MSD	SL	G	6/2/23	9:15	6/2/23	9:15		4	1						1	2		X	
4	~~OT07-BO-Dup~~ OT07-BO-840-4-6-8	SL	G	6/2/23	9:20	6/2/23	9:05		4	1						1	2		X	
5		SL																	X	
6																				
7																				
8																				
9																				
10																				
11																				
12																				

ADDITIONAL COMMENTS	RELINQUISHED BY / AFFILIATION	DATE	TIME	ACCEPTED BY / AFFILIATION	DATE	TIME	SAMPLE CONDITIONS
	Plexus	6/2/23	15:15	Fed Ex	6/2/23	15:15	
	FedEx	6/3/23	10:50	Dollie McCune	6/3/23	10:50	2.9 +0.0 E113

6484 9129 1544

SAMPLER NAME AND SIGNATURE

PRINT Name of SAMPLER: John Filson

SIGNATURE of SAMPLER: [signature] DATE Signed: 6/2/23

TEMP in C	Received on Ice (Y/N)	Custody Sealed Cooler (Y/N)	Samples Intact (Y/N)

SAMPLE RECEIVING CHECKLIST

SAMPLE DELIVERY GROUP	**223060502**
Client **PM** CAM 4887 - Flexus Consultants	**Transport Method** FEDEX
Profile Number 271212	**Received By** McCune, Dodie N.
Line Item(s) 34 - S - 7 Day Full VOA	**Receive Date(s)** 06/03/23

CHECKLIST	YES	NO
Samples received with proper thermal preservation?	☑	☐
Radioactivity is <1600 cpm? If no, record cpm value in notes section.	☐	☐
COC relinquished and complete (including sampleIDs, collect times, and sampler)?	☑	☐
All containers received in good condition and within hold time?	☑	☐
All sample labels and containers received match the chain of custody?	☑	☐
Preservative added to any containers?	☐	☑
If received, was headspace for VOC water containers < 6mm?	☑	☐
Samples collected in containers provided by Pace Gulf Coast?	☑	☐

COOLERS

Airbill Thermometer ID: 443	Temp °C
648491291544	2.9

DISCREPANCIES

None

LAB PRESERVATIONS

None

NOTES

Revision 1.6

SAMPLE RECEIVING CHECKLIST

* 2 2 3 0 6 0 5 0 2 *

SAMPLE DELIVERY GROUP 223060502

Client **PM** CAM 4887 - Plexus Consultants	**Transport Method** FEDEX
Profile Number 271212	**Received By** McCune, Dodie N.
Line Item(s) 34 - S - 7 Day Full VOA	**Receive Date(s)** 06/03/23

CHECKLIST	YES	NO
Samples received with proper thermal preservation?	☑	☐
Radioactivity is <1600 cpm? If no, record cpm value in notes section.	☐	☐
COC relinquished and complete (including sampleIDs, collect times, and sampler)?	☑	☐
All containers received in good condition and within hold time?	☑	☐
All sample labels and containers received match the chain of custody?	☑	☐
Preservative added to any containers?	☐	☑
If received, was headspace for VOC water containers < 6mm?	☑	☐
Samples collected in containers provided by Pace Gulf Coast?	☑	☐

COOLERS

Airbill Thermometer ID: 443	Temp °C
648491291544	2.9

DISCREPANCIES

None

LAB PRESERVATIONS

None

NOTES

Revision 1.6

APPENDIX C

Data Quality Validation Reports

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Project: Scott AFB OT007
Data Package: 223060210
Laboratory: Pace Analytical Gulf Coast
Report Date 09/25/2023

A Level 2B data review was performed on data package 223060210 for groundwater samples collected at Scott AFB site OT007 in June 2023. The data analysis was completed by Pace Analytical Gulf Coast in Baton Rouge, LA. The Dissolved Gases analysis was subcontracted by Pace Analytical in Mount Juliet, TN. The samples and analyses reviewed in this document are shown in Table 1 below.

Table 1: Samples and Analyses

Lab Sample ID	Client Sample ID	Date Collected	VOCs 8260B	TOC 9060A	Total Alkalinity SM2023B	Anions 9056A	Metals 6020B	Dissolved gases RSK-175
22306021001	OT07-EB-060123	06/01/23	X	X	X	X	X	X
22306021002	OT07-MW02-060123	06/01/23	X	X	X	X	X	X
22306021003	OT07-MW04-060123	06/01/23	X	X	X	X	X	X
22306021004	OT07-MW05-060123	06/01/23	X	X	X	X	X	X
22306021005	OT07-MW06-060123	06/01/23	X	X	X	X	X	X
22306021006	OT07-MW04-060123MS	06/01/23	X	X	X	X	X	X
22306021007	OT07-MW04-060123MSD	06/01/23	X	X	X	X	X	X
22306021008	OT07-MW-806-060123	06/01/23	X	X	X	X	X	X
22306021009	OT07-TB-01-060124	06/01/23	X					

Four (4) field samples, one (1) field duplicates, one (1) matrix spike/matrix spike duplicate (MS/MSD) pair, and one (1) trip blank were submitted as part of this data package.

The data have been validated using the quality control requirements and criteria specified in the EPA methods and the following guidance:

- Final Uniform Federal Policy Quality Assurance Project Plan, Scott AFB (UFP-QAPP), March 2018.
- Department of Defense Quality Systems Manual for Environmental Laboratories, Final Version 5.3, May 2019.
- USEPA Contract Laboratory Program National Functional Guidelines for Inorganic Superfund Methods Data Review, November 2020.
- USEPA Contract Laboratory Program National Functional Guidelines for Organic Superfund Methods Data Review, November 2020.

Table 2. The qualifiers used for validation are defined below:

Qualifier	Definition
J	The value is a quantitative estimate.
R	The result is rejected. The result should not be used for qualitative or quantitative purposes.
U	The analyte was not detected above the detection limit (DL). The value represents the limit of detection (LOD).
UJ	The analyte was not detected above the detection limit (DL). The value represents the limit of detection (LOD).

The parameters evaluated are discussed in the following sections:

Documentation:

All samples were received in acceptable condition. The cooler was received at a temperature of 2.1^{o}C, which was within the acceptable range of 4±2^{o}C.

Holding Times:
All preparation and analytical holding times were met.

Dilutions:
Samples were analyzed at a 1x dilution with the following exceptions:

Method 9056A: Samples were analyzed for sulfate at a dilution of 2x (OT07-MW05-060123), 5x (OT07-MW02-060123, OT07-MW06-060123, and OT07-MW-806-060123) and 50x (OT07-MW04-060123, OT07-MW04-060123 MS, and OT07-MW04-060123 MSD).

Instrument Performance:
All parameters were within acceptance criteria.

Laboratory Control Samples/ Laboratory Control Sample Duplicates (LCS/LCSD):
LCS results were within acceptance criteria with the following exceptions:

EPA 8260B: The LCS/LCSD RPD for 1,2-dibromo-3-chloropropane, 2-butanone, 4-methyl-2-pentanone, and tert-butyl methyl ether (MTBE) were above the control limit. No qualification was necessary.

Matrix Spike/Matrix Spike Duplicates (MS/MSD):
All MS/MSD results were within acceptance criteria with the following exceptions:

EPA 8260B: The MS/MSD RPD was above acceptance criteria for dichlorodifluoromethane. No qualifications were necessary.

Calibrations:
All Initial Calibration and Continuing Calibration Verification Blanks were within the acceptable method criteria with the following exceptions:

EPA 8260B: In the analysis for analytical batches 767028 and 767081, the ICV recovery is outside the control limits for acetone and chloromethane. In the analysis for analytical batch 766942, the ICV recoveries were outside the control limits for acetone, 2-butanone, 2-hexanone and chloromethane. In the analysis for analytical batch 767028, the %Drift is outside ± 20% for dichlorodifluoromethane and bromomethane in the opening CCV. In the analysis for analytical batch 767081, the %Drift is outside ± 50% for acetone in the closing CCV. Samples OT07-EB-060123, OT07-MW02-060123, OT07-MW04-060123, OT07-MW05-060123, and OT07-MW06-060123 were qualified as "UJ" for acetone, 2-butanone, 2-hexanone and chloromethane. Sample OT07-MW-806-060123 was qualified as "UJ" for acetone, chloromethane, dichlorodifluoromethane and bromomethane. Sample OT07-TB-01-060124 was qualified as "J" for acetone and "UJ" for chloromethane.

Internal Standards:
All internal standards were within the acceptable criteria.

Surrogate Recovery:
All surrogate recoveries were within acceptance criteria with the following exception:

EPA 8260B: The surrogate dibromofluoromethane was above the upper control limit for sample OT07-MW04-060123. The sample was non-detect for all compounds; no qualification was necessary.

Laboratory/Field Blanks:

No positive detections were received in trip blank, equipment blank, or laboratory blanks with the following exceptions:

EPA 8260B: The equipment blank contained detections of acetone, methylene chloride, and chloride at concentrations less than the LOQ. The method blank contained a detection of manganese at a concentration less than the LOQ. The trip blank contained acetone at a concentration less than the LOQ. All acetone, methylene chloride, chloride, and manganese results were non-detect or detections greater than 10 times the blank result and no qualifiers were necessary.

Manual Integrations:
Manual integrations were reviewed and were found to be performed and recorded appropriately.

Field Duplicates:
One field duplicate was analyzed and the relative percent differences (RPDs) were calculated.

Table 4. Field Duplicate Results

Analyte	Units	Primary	Duplicate	RPD	Action
		OT07-MW06-060123	**OT07-MW806-060123**		
Arsenic	ug/L	4.36	11.7	91	J
Iron	ug/L	5250	4790	9.2	none
Manganese	ug/L	861	829	3.8	none
1,2-Dichloroethene(Total)	ug/L	14.3	13.5	5.7	none
Benzene	ug/L	1.71	1.57	8.5	none
Vinyl chloride	ug/L	94.6	88.5	6.7	none
cis-1,2-Dichloroethene	ug/L	14.3	13.5	5.8	none
Chloride	mg/L	1.5	2.11	34	J
Nitrate	mg/L	0.135 J	ND	NC	none
Sulfate	mg/L	33.5	37.9	12	none
Total Organic Carbon	mg/L	10.7	8.93	18	none
Total Alkalinity	mg/L CaCO3	146	168	14	none

ND- not detected
NC- not calculated

Conclusions:
Sample results were reported using DoD QSM, version 5.3, May 2019. QSMv5.3 requires the reporting the high level limit of quantitation (LOQ), the mid-level limit of detection (LOD), and the low level detection limit (DL). All detections above the DL but below the LOQ are qualified "J" to indicate that they are quantitative estimates. Non-detect results are reported to the LOD. Overall, the analyses were performed in accordance with the requirements of the methods.

Zero (0) data were rejected and 100% of the data are usable. The overall data quality appears to be acceptable. The quality of the data is sufficient for qualitative and quantitative evaluations.

Data Reviewer:

Claire Costello
Chemical Engineer
Plexus Scientific Corporation
September 26, 2023
Revised November 19, 2023

Project: Scott AFB OT007
Data Package: 223060502
Laboratory: Pace Analytical Gulf Coast
Report Date: 06/29/2023

A Level 2B data review was performed on data package 223060502 for soil samples collected at Scott AFB site OT007 in June 2023. The data analysis was completed by Pace Analytical Gulf Coast in Baton Rouge, LA. The samples and analyses reviewed in this document are shown in Table 1 below.

Table 1: Samples and Analyses

Lab Sample ID	Client Sample ID	Date Collected	VOCs 8260B
22306050201	OT07-BO-40-4-6-8	06/02/23	X
22306050202	OT07-BO-40-4-6-8-MS	06/02/23	X
22306050203	OT07-BO-40-4-6-8-MSD	06/02/23	X
22306050204	OT07-BO-840-4-6-8	06/02/23	X

One (1) field sample, one (1) field duplicate, and one (1) matrix spike/matrix spike duplicate (MS/MSD) pair were submitted as part of this data package.

The data have been validated using the quality control requirements and criteria specified in the EPA methods and the following guidance:

- Final Uniform Federal Policy Quality Assurance Project Plan, Scott AFB (UFP-QAPP), March 2018.
- Department of Defense Quality Systems Manual for Environmental Laboratories, Final Version 5.3, May 2019.
- USEPA Contract Laboratory Program National Functional Guidelines for Inorganic Superfund Methods Data Review, November 2020.
- USEPA Contract Laboratory Program National Functional Guidelines for Organic Superfund Methods Data Review, November 2020.

Table 2. The qualifiers used for validation are defined below:

Qualifier	Definition
J	The value is a quantitative estimate.
R	The result is rejected. The result should not be used for qualitative or quantitative purposes.
U	The analyte was not detected above the detection limit (DL). The value represents the limit of detection (LOD).
UJ	The analyte was not detected above the detection limit (DL). The value represents the limit of detection (LOD).

The parameters evaluated are discussed in the following sections:

Documentation:

All samples were received in acceptable condition.

The cooler was received at a temperature of 2.9°C, which was within the acceptable range of 4±2°C.

Holding Times:
All preparation and analytical holding times were met.

Dilutions:
Samples were analyzed at a 1x dilution with the following exceptions:

Method 8260B: Samples were analyzed at a dilution of 250x in all samples. Dilutions result in elevated detection limits and should be considered when evaluating data usability.

Instrument Performance:
All parameters were within acceptance criteria.

Laboratory Control Samples/ Laboratory Control Sample Duplicates (LCS/LCSD):
LCS results were within acceptance criteria with the following exceptions:

EPA 8260B: In the analysis for analytical batch 767415, the LCS recovery for chloroethane and bromobenzene were below control limits. The samples were qualified as "UJ" for chloroethane and bromobenzene.

Matrix Spike/Matrix Spike Duplicates (MS/MSD):
All MS/MSD results were within acceptance criteria with the following exceptions:

EPA 8260B: The MS and MSD recoveries were below the control limits for sample OT07-BO-40-4-6-8-MS and OT07-BO-40-4-6-8-MSD for 1,2,4-trimethylbenzene. The LCS had an acceptable recovery. The results for parent sample OT07-BO-40-4-6-8 were qualified as "J" for 1,2,4-trimethylbenzene.

Calibrations:
All Initial Calibration and Continuing Calibration Verification Blanks were within the acceptable method criteria with the following exceptions:

EPA 8260B: In the analysis for analytical batch 767415, the % Drift is outside ± 20% for 1,2-dibromo-3-chloropropane in the opening CCV. Also, there were ICV recovery failures for acetone and chloromethane, The samples were qualified as "UJ" for 1,2-dibromo-3-chloropropane.

Internal Standards:
All internal standards were within the acceptable criteria.

Surrogate Recovery:
All surrogate recoveries were within acceptance criteria.

Laboratory/Field Blanks:

No positive detections were received in trip or laboratory blanks with the following exceptions:

EPA 8260B: The method blank contained a detection of m,p-Xylene at a concentration less than the LOQ. All results are more than 10 times the blank result; no qualifiers were applied.

Manual Integrations:
Manual integrations were reviewed and were found to be performed and recorded appropriately.

Field Duplicates:
One field duplicate was analyzed and the relative percent differences (RPDs) were calculated. The analytes with RPDs were greater than 50% are qualified as "J".

Table 4. Field Duplicate Results

Analyte	Units	Primary	Duplicate	RPD	Action
		OT07-BO-40-4-6-8	**OT07-BO-840-4-6-8**		
1,2,4-Trimethylbenzene	ug/Kg	1960 J	28500	174	J
1,3,5-Trimethylbenzene	ug/Kg	306 J	6080	180	J
4-Isopropyltoluene	ug/Kg	147 J	2430	177	J
4-Methyl-2-pentanone	ug/Kg	ND	7460	NC	none
Benzene	ug/Kg	ND	337 J	NC	none
Ethylbenzene	ug/Kg	241 J	3150	172	J
Isopropylbenzene (Cumene)	ug/Kg	128 J	1880	174	J
Naphthalene	ug/Kg	316 J	4830	175	J
m,p-Xylene	ug/Kg	310 J	4960	176	J
n-Butylbenzene	ug/Kg	146 J	2820	180	J
n-Propylbenzene	ug/Kg	315 J	4170	171	J
o-Xylene	ug/Kg	ND	443 J	NC	none
sec-Butylbenzene	ug/Kg	239 J	3250	172	J

Conclusions:

Sample results were reported using DoD QSM, version 5.3, May 2019. QSMv5.3 requires the reporting the high level limit of quantitation (LOQ), the mid-level limit of detection (LOD), and the low level detection limit (DL). All detections above the DL but below the LOQ are qualified "J" to indicate that they are quantitative estimates. Non-detect results are reported to the LOD. Overall, the analyses were performed in accordance with the requirements of the methods.

Zero (0) data were rejected and 100% of the data are usable. The overall data quality appears to be acceptable. The quality of the data is sufficient for qualitative and quantitative evaluations.

Data Reviewer:

Claire Costello
Chemical Engineer
Plexus Scientific Corporation
July 3, 2023
Revised November 19, 2023

APPENDIX D

Waste Disposal Documentation

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Please print or type.

Form Approved. OMB No. 2050-0039

UNIFORM HAZARDOUS WASTE MANIFEST

1. Generator ID Number: I L 7 5 7 0 0 2 4 1 7 7

2. Page 1 of 1

3. Emergency Response Phone: 800-424-9300

4. Manifest Tracking Number: 025657382 JJK

5. Generator's Name and Mailing Address

375 CES/CEIE
701 HANGER ROAD (BUILDING 3306)
SCOTT AIR FORCE BASE IL 62225

Att: RODGER ALLISON

Generator's Phone: 256 - 2925

Generator's Site Address (if different than mailing address)

6. Transporter 1 Company Name: ILLINI ENVIRONMENTAL LLC — U.S. EPA ID Number: I L R 0 0 0 1 0 7 0 8 6

7. Transporter 2 Company Name — U.S. EPA ID Number

8. Designated Facility Name and Site Address — U.S. EPA ID Number: I L R 0 0 0 1 0 7 0 8 6

ILLINI ENVIRONMENTAL LLC
8895 CALIFORNIA DRIVE
CASEYVILLE IL 62232

Facility's Phone: 618 397-1234

9a. HM	9b. U.S. DOT Description (including Proper Shipping Name, Hazard Class, ID Number, and Packing Group (if any))	10. Containers No.	Type	11. Total Quantity	12. Unit Wt./Vol.	13. Waste Codes		
	1. NON-HAZARDOUS, NON-RCRA, NON-DOT, (VEGETABLE OIL EMULSION)	2	DM	110	G	~~IL44~~		
	2. NON-HAZARDOUS, NON-RCRA, NON-DOT (DECON WATER)	1	DM	55	G	~~IL44~~		
	3. NON-HAZARDOUS, NON-RCRA, NON-DOT (DECON WATER)	1	DM	55	G	~~IL44~~		
	4. NON-HAZARDOUS, NON-RCRA, NON-DOT (SOIL CUTTING)	1	DM	55	G	~~IL45~~		

14. Special Handling Instructions and Additional Information

APPROVAL (1) 23-255-1 LFD (33014) (2) 23-255-4 LFD 5A026/33029 (3) 23-255-3 OT007 (4) 23-255-2 LFD OT007
IL TRANS # 5256 IL SITE # 1630255011 INVOICE (PLEXUS)

15. **GENERATOR'S/OFFEROR'S CERTIFICATION:** I hereby declare that the contents of this consignment are fully and accurately described above by the proper shipping name, and are classified, packaged, marked and labeled/placarded, and are in all respects in proper condition for transport according to applicable international and national governmental regulations. If export shipment and I am the Primary Exporter, I certify that the contents of this consignment conform to the terms of the attached EPA Acknowledgment of Consent.
I certify that the waste minimization statement identified in 40 CFR 262.27(a) (if I am a large quantity generator) or (b) (if I am a small quantity generator) is true.

Generator's/Offeror's Printed/Typed Name: JOE WEBER — Signature: Joe Weber — Month 10 Day 5 Year 23

16. International Shipments ☐ Import to U.S. ☐ Export from U.S. Port of entry/exit:
Transporter signature (for exports only): Date leaving U.S.:

17. Transporter Acknowledgment of Receipt of Materials

Transporter 1 Printed/Typed Name: Darrell White — Signature: Darrell White — Month 10 Day 5 Year 23

Transporter 2 Printed/Typed Name — Signature — Month Day Year

18. Discrepancy

18a. Discrepancy Indication Space ☐ Quantity ☐ Type ☐ Residue ☐ Partial Rejection ☐ Full Rejection

Manifest Reference Number:

18b. Alternate Facility (or Generator) — U.S. EPA ID Number

Facility's Phone:

18c. Signature of Alternate Facility (or Generator) — Month Day Year

19. Hazardous Waste Report Management Method Codes (i.e., codes for hazardous waste treatment, disposal, and recycling systems)

1. 2. 3. 4.

20. Designated Facility Owner or Operator: Certification of receipt of hazardous materials covered by the manifest except as noted in Item 18a

Printed/Typed Name: Bradley Lax — Signature: [signature] — Month 10 Day 5 Year 23

GENERATOR — INT'L — TRANSPORTER — DESIGNATED FACILITY

Illini Environmental, Inc.

Incoming Waste Analysis Form

Date Material Received: 10-5-23 Date Sample Processed: 10-6-23

Transported By: Darrel Processed By: PL

Generator Name: 375 CES

Description of Waste: Vegetable oil

Physical Characteristics:

Sample Size: 250 ml

Color: Yellow/white

Odor: Mild

Composition:

Solid:

Sludge:

Water:

Oil:

Other: 100

Disposal Information:

Offload Location: Storage

Container: DM

Quantity: 2

Testing Information:

pH: 4

Specific Gravity: NT

Flashpoint: >140

Compatibility: P

Contain at least 50% oil (Y/N) N

Considered Used Oil (Y/N) N

Halogen: (If either above is Y) NT

PCB SW9079 (If either above is Y) NT

Key:

ND – Non Detect	ppm – Parts Per Million
NT – Not Tested	> - Greater Than
NA – Not Applicable	< - Lesser Than

Notes:

Approved: Y N

Approved By: [signature]

Illini Environmental, Inc.

Incoming Waste Analysis Form

Date Material Received: 10-5-23 Date Sample Processed: 10-6-23

Transported By: Darrel Processed By: PA

Generator Name: 375 CES

Description of Waste: Decon water

Physical Characteristics:

Sample Size: 250 ml

Color: Clear

Odor: Mild

Composition:

Solid:

Sludge:

Water: 95

Oil:

Other: 5

Disposal Information:

Offload Location: Storage

Container: DM

Quantity: 1

Testing Information:

pH: 7

Specific Gravity: NT

Flashpoint: >140

Compatibility: P

Contain at least 50% oil (Y/N) N

Considered Used Oil (Y/N) N

Halogen: (If either above is Y) NT

PCB SW9079 (If either above is Y) NT

Key:

ND – Non Detect

NT – Not Tested

NA – Not Applicable

ppm – Parts Per Million

> – Greater Than

< – Lesser Than

Notes:

Approved: Y N

Approved By: [signature]

Illini Environmental, Inc.

Incoming Waste Analysis Form

Date Material Received: 10-5-23 Date Sample Processed: 10-6-23

Transported By: Darrel Processed By: PL

Generator Name: 375 CES

Description of Waste: Soil cutting

Physical Characteristics:

Sample Size: 200 ml

Color: Brown

Odor: Mild

Composition:

Solid: 100

Sludge:

Water:

Oil:

Other:

Disposal Information:

Offload Location: Storage

Container: DM

Quantity: 1

Testing Information:

pH: NT

Specific Gravity: NT

Flashpoint: ? 140

Compatibility: P

Contain at least 50% oil (Y/N) N

Considered Used Oil (Y/N) N

Halogen: (If either above is Y) NT

PCB SW9079 (If either above is Y) NT

Key:

ND – Non Detect	ppm – Parts Per Million
NT – Not Tested	> – Greater Than
NA – Not Applicable	< – Lesser Than

Notes:

Approved: Y N

Approved By: [signature]

Illini Environmental, Inc.

Incoming Waste Analysis Form

Date Material Received: 10-5-23 Date Sample Processed: 10-6-23

Transported By: Darrel Processed By: Pa

Generator Name: 375 CES

Description of Waste: Decon water

Physical Characteristics:

Sample Size: 200-1

Color: Brown

Odor: mild

Composition:

Solid:

Sludge:

Water: 95

Oil:

Other: 5

Disposal Information:

Offload Location: Storage

Container: Dm

Quantity: 1

Testing Information:

pH: 8

Specific Gravity: NT

Flashpoint: >140

Compatibility: P

Contain at least 50% oil (Y/N) N

Considered Used Oil (Y/N) N

Halogen: (If either above is Y) NT

PCB SW9079 (If either above is Y) NT

Key:	
ND – Non Detect	ppm – Parts Per Million
NT – Not Tested	> – Greater Than
NA – Not Applicable	< – Lesser Than

Notes:

Approved: Y N

Approved By: [signature]

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APPENDIX E

Mann-Kendall Analysis Supporting Documentation

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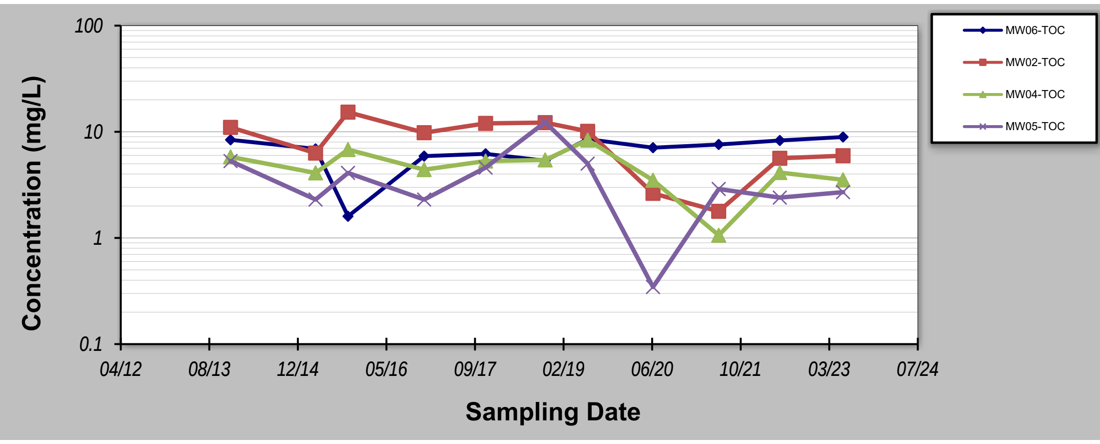
GSI MANN-KENDALL TOOLKIT
for Constituent Trend Analysis

Evaluation Date:	**1-Sep-23**	Job ID:	**9999**
Facility Name:	**OT007**	Constituent:	**TOC**
Conducted By:	**Plexus**	Concentration Units:	**mg/L**

Sampling Point ID:		**MW06-TOC**	**MW02-TOC**	**MW04-TOC**	**MW05-TOC**			
Sampling Event	Sampling Date	TOC CONCENTRATION (mg/L)						
1	12-Dec-13	8.4	11	5.8	5.3			
2	6-Apr-15	6.9	6.3	4.1	2.3			
3	6-Oct-15	1.6	15.3	6.8	4.1			
4	9-Dec-16	5.9	9.8	4.4	2.3			
5	21-Nov-17	6.2	12	5.3	4.6			
6	23-Oct-18	5.35	12.2	5.43	12.3			
7	20-Jun-19	8.53	10.1	8.36	5.01			
8	23-Jun-20	7.08	2.63	3.5	0.345			
9	29-Jun-21	7.59	1.78	1.06	2.89			
10	9-Jun-22	8.28	5.63	4.13	2.4			
11	1-Jun-23	8.93	5.95	3.53	2.7			
12								
13								
14								
15								
16								
17								
18								
19								
20								
Coefficient of Variation:		**0.31**	**0.51**	**0.40**	**0.77**			
Mann-Kendall Statistic (S):		**21**	**-21**	**-17**	**-8**			
Confidence Factor:		**94.0%**	**94.0%**	**89.1%**	**70.3%**			
Concentration Trend:		Prob. Increasing	Prob. Decreasing	Stable	Stable			



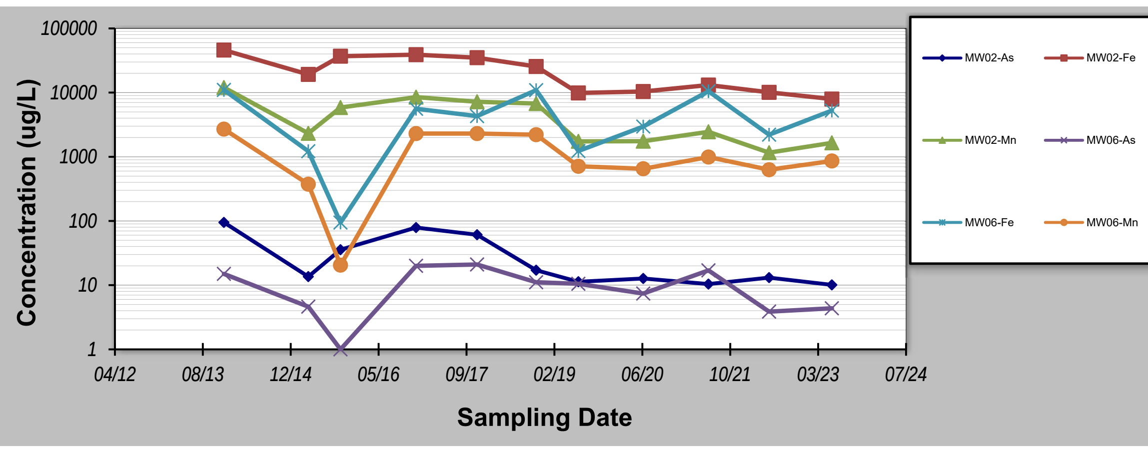
Notes:

1. At least four independent sampling events per well are required for calculating the trend. *Methodology is valid for 4 to 40 samples.*
2. Confidence in Trend = Confidence (in percent) that constituent concentration is increasing (S>0) or decreasing (S<0): >95% = Increasing or Decreasing; ≥ 90% = Probably Increasing or Probably Decreasing; < 90% and S>0 = No Trend; < 90%, S≤0, and COV ≥ 1 = No Trend; < 90% and COV < 1 = Stable.
3. Methodology based on "MAROS: A Decision Support System for Optimizing Monitoring Plans", J.J. Aziz, M. Ling, H.S. Rifai, C.J. Newell, and J.R. Gonzales, *Ground Water* , 41(3):355-367, 2003.

GSI MANN-KENDALL TOOLKIT
for Constituent Trend Analysis

Evaluation Date: **1-Sep-23** Job ID: **9999**
Facility Name: **OT007** Constituent: **(Multiple Metals) Total**
Conducted By: **Plexus** Concentration Units: **ug/L**

Sampling Event	Sampling Date	MW02-As	MW02-Fe	MW02-Mn	MW06-As	MW06-Fe	MW06-Mn	
		(MULTIPLE METALS) TOTAL CONCENTRATION (ug/L)						
1	12-Dec-13	95	46000	12000	15	11000	2700	
2	6-Apr-15	13.6	19200	2320	4.62	1220	374	
3	6-Oct-15	35.7	36900	5850	1	94.5	20.5	
4	9-Dec-16	79	39000	8500	20	5600	2300	
5	21-Nov-17	61	35000	7200	21	4300	2300	
6	23-Oct-18	17.1	25500	6750	11.1	10900	2210	
7	20-Jun-19	11.3	9870	1740	10.5	1230	712	
8	23-Jun-20	12.7	10400	1750	7.4	2960	652	
9	29-Jun-21	10.4	13100	2450	17	10500	991	
10	9-Jun-22	13.1	10100	1160	3.87	2210	629	
11	1-Jun-23	10.1	7940	1650	4.36	5250	861	
12								
13								
14								
15								
16								
17								
18								
19								
20								
Coefficient of Variation:		0.95	0.61	0.77	0.66	0.81	0.75	
Mann-Kendall Statistic (S):		-35	-37	-31	-11	1	-12	
Confidence Factor:		99.7%	99.8%	99.2%	77.7%	50.0%	79.9%	
Concentration Trend:		Decreasing	Decreasing	Decreasing	Stable	No Trend	Stable	



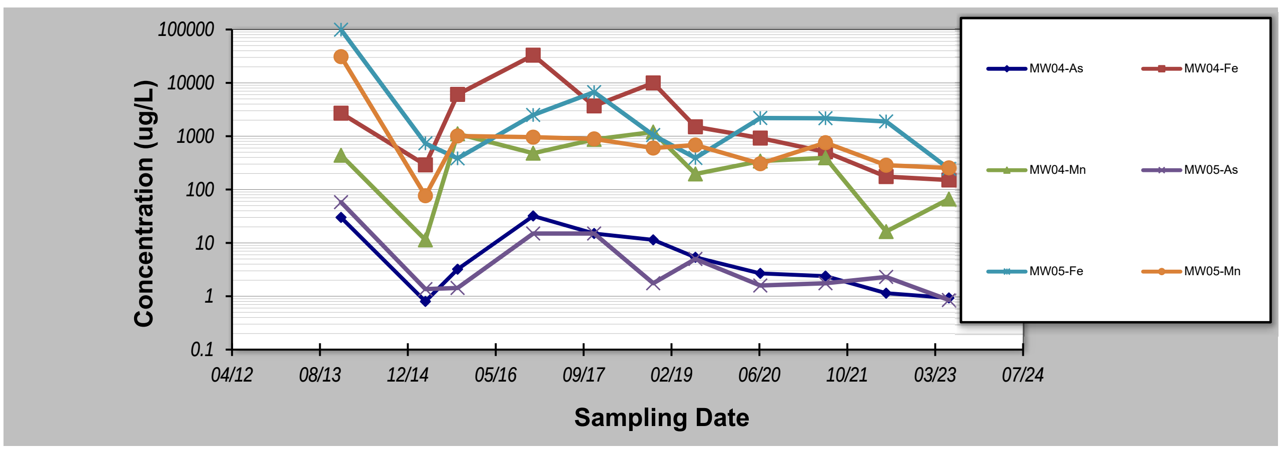
Notes:

1. At least four independent sampling events per well are required for calculating the trend. *Methodology is valid for 4 to 40 samples.*
2. Confidence in Trend = Confidence (in percent) that constituent concentration is increasing (S>0) or decreasing (S<0): >95% = Increasing or Decreasing; ≥ 90% = Probably Increasing or Probably Decreasing; < 90% and S>0 = No Trend; < 90%, S≤0, and COV ≥ 1 = No Trend; < 90% and COV < 1 = Stable.
3. Methodology based on "MAROS: A Decision Support System for Optimizing Monitoring Plans", J.J. Aziz, M. Ling, H.S. Rifai, C.J. Newell, and J.R. Gonzales, *Ground Water* , 41(3):355-367, 2003.

GSI MANN-KENDALL TOOLKIT
for Constituent Trend Analysis

Evaluation Date:	1-Sep-23	Job ID:	9999
Facility Name:	OT007	Constituent:	(Multiple Metals) Total
Conducted By:	Plexus	Concentration Units:	ug/L

Sampling Event	Sampling Point ID: / Sampling Date	MW04-As	MW04-Fe	MW04-Mn	MW05-As	MW05-Fe	MW05-Mn	
		(MULTIPLE METALS) TOTAL CONCENTRATION (ug/L)						
1	12-Dec-13	30	2700	440	58	99000	31000	
2	6-Apr-15	0.8	290	11.4	1.36	734	76.7	
3	6-Oct-15	3.21	6070	1110	1.43	384	1010	
4	9-Dec-16	32	33000	480	15	2500	960	
5	21-Nov-17	15	3700	870	15	6700	890	
6	23-Oct-18	11.4	9960	1200	1.75	1050	600	
7	20-Jun-19	5.37	1500	196	5	391	685	
8	23-Jun-20	2.68	923	342	1.59	2190	305	
9	29-Jun-21	2.39	508	394	1.75	2170	756	
10	9-Jun-22	1.14	175	16.3	2.3	1890	286	
11	1-Jun-23	0.93	151	66.5	0.85	246	255	
12								
13								
14								
15								
16								
17								
18								
19								
20								
	Coefficient of Variation:	1.21	1.80	0.91	1.79	2.75	2.74	
	Mann-Kendall Statistic (S):	-27	-27	-13	-13	-15	-29	
	Confidence Factor:	98.0%	98.0%	82.1%	82.1%	85.9%	98.7%	
	Concentration Trend:	Decreasing	Decreasing	Stable	No Trend	No Trend	Decreasing	



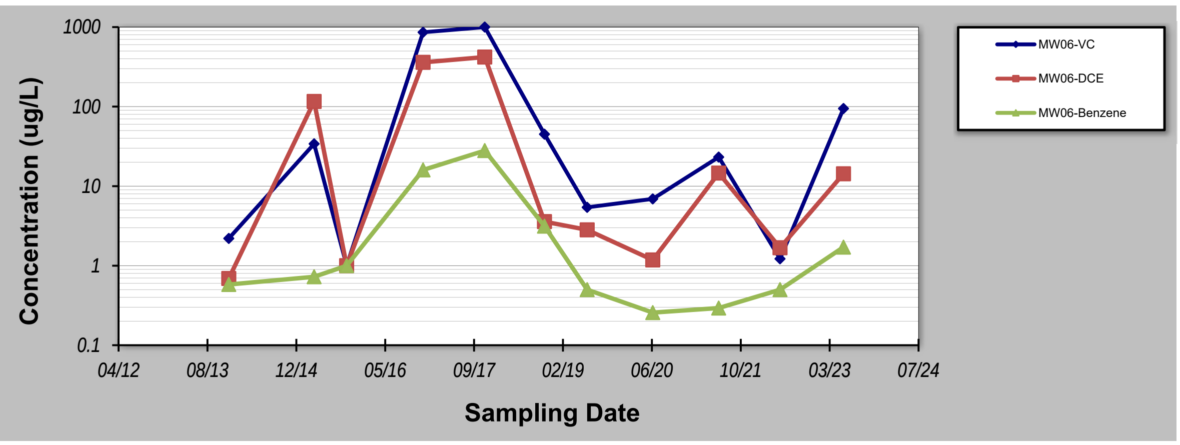
Notes:

1. At least four independent sampling events per well are required for calculating the trend. *Methodology is valid for 4 to 40 samples.*
2. Confidence in Trend = Confidence (in percent) that constituent concentration is increasing (S>0) or decreasing (S<0): >95% = Increasing or Decreasing; ≥ 90% = Probably Increasing or Probably Decreasing; < 90% and S>0 = No Trend; < 90%, S≤0, and COV ≥ 1 = No Trend; < 90% and COV < 1 = Stable.
3. Methodology based on "MAROS: A Decision Support System for Optimizing Monitoring Plans", J.J. Aziz, M. Ling, H.S. Rifai, C.J. Newell, and J.R. Gonzales, *Ground Water*, 41(3):355-367, 2003.

GSI MANN-KENDALL TOOLKIT
for Constituent Trend Analysis

Evaluation Date:	**1-Sep-23**	Job ID:	**9999**
Facility Name:	**OT007**	Constituent:	**(Multiple Constituent)**
Conducted By:	**Plexus**	Concentration Units:	**ug/L**

	Sampling Point ID:	MW06-VC	MW06-DCE	MW06-Benzene				
Sampling Event	Sampling Date	(MULTIPLE CONSTITUENT) CONCENTRATION (ug/L)						
1	12-Dec-13	2.2	0.69	0.58				
2	6-Apr-15	34	116	0.725				
3	6-Oct-15	1	1	1				
4	9-Dec-16	860	360	16				
5	21-Nov-17	1000	420	28				
6	23-Oct-18	45	3.58	3.14				
7	20-Jun-19	5.41	2.82	0.5				
8	23-Jun-20	6.91	1.18	0.257				
9	29-Jun-21	23.2	14.6	0.293				
10	9-Jun-22	1.22	1.68	0.5				
11	1-Jun-23	94.6	14.3	1.71				
12								
13								
14								
15								
16								
17								
18								
19								
20								
	Coefficient of Variation:	1.96	1.82	1.87				
	Mann-Kendall Statistic (S):	3	3	-8				
	Confidence Factor:	56.0%	56.0%	70.3%				
	Concentration Trend:	No Trend	No Trend	No Trend				



Notes:

1. At least four independent sampling events per well are required for calculating the trend. *Methodology is valid for 4 to 40 samples.*
2. Confidence in Trend = Confidence (in percent) that constituent concentration is increasing (S>0) or decreasing (S<0): >95% = Increasing or Decreasing; ≥ 90% = Probably Increasing or Probably Decreasing; < 90% and S>0 = No Trend; < 90%, S≤0, and COV ≥ 1 = No Trend; < 90% and COV < 1 = Stable.
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